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CSI API ETABS v1

Introduction

This document illustrates how to connect to ETABS using the CSI Application Programming Interface (API) from a variety of clients.

The API is defined in the form of a .NET 4.7.1 dynamic link library, and is implemented by ETABS. It can be accessed from .NET 4.7.1 and COM based clients.

Please note that ETABS must be installed both on the API client development machine, and on the client end-user's machine.

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Release Notes

This topic contains the following sections:

- [Current Version](#)
- [Select Notes from Previous Versions](#)
- [See Also](#)

▲ Current Version

Remote API

The ETABS API can now be used to start and/or connect to a running instance of ETABS on a Remote Computer, allowing for models with many load cases to be analyzed in a parallel, distributed manner. Please refer to the page, [Remote API](#)

ETABSV17 Deprecated

ETABSV17, ETABS 2016, ETABS 2015 and ETABS 2013 are no longer supported. It is recommended that all client applications be upgraded to reference the latest version of ETABSV1.DLL or CSiAPIv1.DLL .

▲ Select Notes from Previous Versions

ETABSV1.DLL

Beginning with Version 18 of ETABS, the API library no longer has the program version as part of its name. So, while the name of the API library for ETABS version 17 was ETABSV17.DLL , the name of the API library for ETABS version 18 is **ETABSV1.DLL** . The name of the API library will remain ETABSV1.DLL , even as new major versions of ETABS are released. A new API function, `cHelper.GetOAPIVersionNumber` , has been added. This API version number will increment as new API functions are added, but the API library name will remain ETABSV1.DLL .

Once users reference the new ETABSV1.DLL in their client applications, they will no longer need to update with every major release. The ETABSV1.DLL reference in their client application will automatically use the latest edition of ETABSV1.DLL that is registered with each product installation. Even so, developers should always use `cHelper.GetOAPIVersionNumber` to check which version of the API library they are using.

CSiAPIv1.DLL

Beginning with Version 18 of ETABS, a new API library, **CSiAPIv1.DLL**, has been introduced. This library is compatible with SAP2000, CSiBridge, and ETABS. It will be available with all new versions of each product. Developers can now create API client applications that reference CSiAPIv1.DLL , and connect to either SAP2000, CSiBridge, or ETABS, without any code changes required. Similar to the new ETABSV1.DLL , the assembly name CSiAPIv1.DLL will not change, even as new major versions of SAP2000, CSiBridge, and ETABS are released. CSiAPIv1.DLL also exposes

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cHelper.GetOAPIVersionNumber , which should be used by developers to check which version of the API library they are using. As the version increases and new functionality is added, compatibility with previous versions will be preserved, and client applications will not require recompilation.

The latest version of CSiAPIv1.DLL included with ETABS version 18 is strong-named, making it incompatible for .NET clients developed with previously released versions of CSiAPIv1.DLL that were released with SAP2000 or CSiBridge v21.0.0 - v21.0.2 . .NET clients that reference previous versions of CSiAPIv1.DLL can be recompiled with the latest assembly included with ETABS version 18 in order for the client to be compatible with all products.

Interactive Database

A powerful new feature, the Interactive Database, has been added to the API in ETABS version 18. This feature allows the API user to retrieve all available data in the program, including analysis and design results. It also allows the user to set nearly all model parameters programmatically. Please refer to the page, [Interactive Database](#)

Attaching to a Running Instance

API client applications can connect to a manually started instance of ETABS. Please refer to the page, [Attaching to a Manually Started Instance of ETABS](#)

▲ See Also

Other Resources

[Remote API](#)

[Interactive Database](#)

[Attaching to a Manually Started Instance of ETABS](#)

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Attaching to a Manually Started Instance of ETABS

These instructions document the steps necessary to attach to instances of ETABS that were started manually. The example code is in VB.NET.

1. Start ETABS manually, e.g. by clicking on the program shortcut
2. Instead of creating a new ETABSObject, attach to the instance of the ETABSObject that was created and added to the Running Object Table by ETABS.

The code below creates an ETABSObject and starts the program:

VB

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```
Dim myETABSObject As ETABSv1.cOAPI
Dim myHelper as ETABSv1.cHelper = New ETABSv1.Helper

myETABSObject = myHelper.CreateObject("C:\Program Files (x86)\Computers and Structures\ETA

ret = myETABSObject.ApplicationStart()

Dim mySapModel As ETABSv1.cSapModel = myETABSObject.SapModel
```

Replace the code above with the code below to attach to the existing ETABSObject. Since the program is already started, there is no need to call ETABSObject.ApplicationStart:

VB

Copy

```
Dim myETABSObject As ETABSv1.cOAPI
Dim myHelper as ETABSv1.cHelper = New ETABSv1.Helper

myETABSObject = myHelper.GetObject("CSI.ETABS.API.ETABSObject")

Dim mySapModel As ETABSv1.cSapModel = myETABSObject.SapModel
```

3. Other calls can then be made normally. The program can be ended manually or by calling ETABSObject.ApplicationExit
4. Please note that if multiple instances of ETABS are manually started, an API client can only attach to the instance that was started first.

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Information for Plugin Developers

This topic contains the following sections:

- [Plugin Considerations](#)
- [The cPlugin Class](#)
- [Adding Plugins](#)

Plugins can be developed to make use of the ETABS API from inside the program.

No license is required to use the API for plugin development, beyond having a valid license for ETABS. However, technical support for developing with the API is not included. For qualified users and third-party developers who would like technical support to help them build solutions and integrations with ETABS using the ETABS API, Computers and Structures, Inc. has created a subscription-based service, CSI Developer Network (CSIDN) (see <http://www.csiamerica.com/support/csidsn>). Developers must subscribe to CSIDN to be eligible for technical support for the ETABS API.

Plugins can be developed in a .NET compatible language (e.g. VB.NET, C#, etc.) as a .NET DLL, or by using any language capable of creating a COM server compiled as a DLL. An example plugin project can be found at:

<https://wiki.csiamerica.com/display/kb/Sample+Plugin+1>

■ Plugin Considerations

The plugin must reference the ETABSV1 API library (either ETABSV1.DLL or ETABSV1.TLB), which must be registered on the developer's and the user's systems. This registration takes place during the installation of ETABS. The ETABSV1.DLL is compiled as AnyCPU, hence it can be referenced by both 64-bit and AnyCPU-compiled plugin projects.

Please note that COM plugins must be registered for COM on the user's system, before they can be used in ETABS. This will require that the ETABSV1.DLL be present in the directory where the COM plugin is being registered. It is the plugin developer's responsibility to instruct users how to install the plugin. Plugins should be installed only after ETABS has been installed. COM Plugin developers should consider that registering assemblies usually requires administrative permissions, which many users may not have.

Plugins compiled as 32-bit will not work with 64-bit ETABS.

We recommend that developers create .NET plugins that are compiled as 64-bit or AnyCPU . This will allow your plugin to be called from both 64-bit ETABS, and will not require registration.

Please make sure your plugin is stable, handles errors well, and does not cause any unintended changes to the user's model.

We will attempt to maintain a stable interface in ETABS, however, that cannot be guaranteed, and updates to your plugin may be required for future versions of ETABS.

■ The cPlugin Class

In order for ETABS to be able to launch your plugin, your plugin assembly must contain a class called cPlugin that implements the cPluginContract interface in the ETABS API.

Please refer to the cPluginContract page for an explanation of its methods. They are also described below.

Class cPlugin must contain two methods, cPlugin.Info and cPlugin.Main

VB

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```
Public Function Info(ByRef Text As String) As Integer
```

The function cPlugin.Info has one reference argument, Text, and returns a 32-bit signed integer. The return value should be zero if successful. The string argument should be filled in by the function, and may be plain text or rich text. This string will be displayed when the user first adds the plugin to ETABS. You can use this string to tell the user the purpose and author of the plugin. This is in addition to any information you may provide when the user executes the plugin.

VB

Copy

```
Public Sub Main(ByRef SapModel As ETABSV1.cSapModel, ByRef ISapPlugin As ETABSV1.cPluginCallback)
```

The subroutine cPlugin.Main is the entry point to launch your plugin. All functionality in your plugin will proceed from this method. If your plugin has an initial form, it should be opened within this method.

cPlugin.Main has two reference arguments of types ETABSV1.cSapModel and ETABSV1.cPluginCallback:

The SapModel argument is a reference to the SapModel object upon which all operations will be performed

For the ISapPlugin argument, please refer to the cPluginCallback page for an explanation of its properties. Its only method is very important and explained below.

VB

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```
ETABSV1.cPluginCallback.Finish(ByVal iVal As Integer)
```

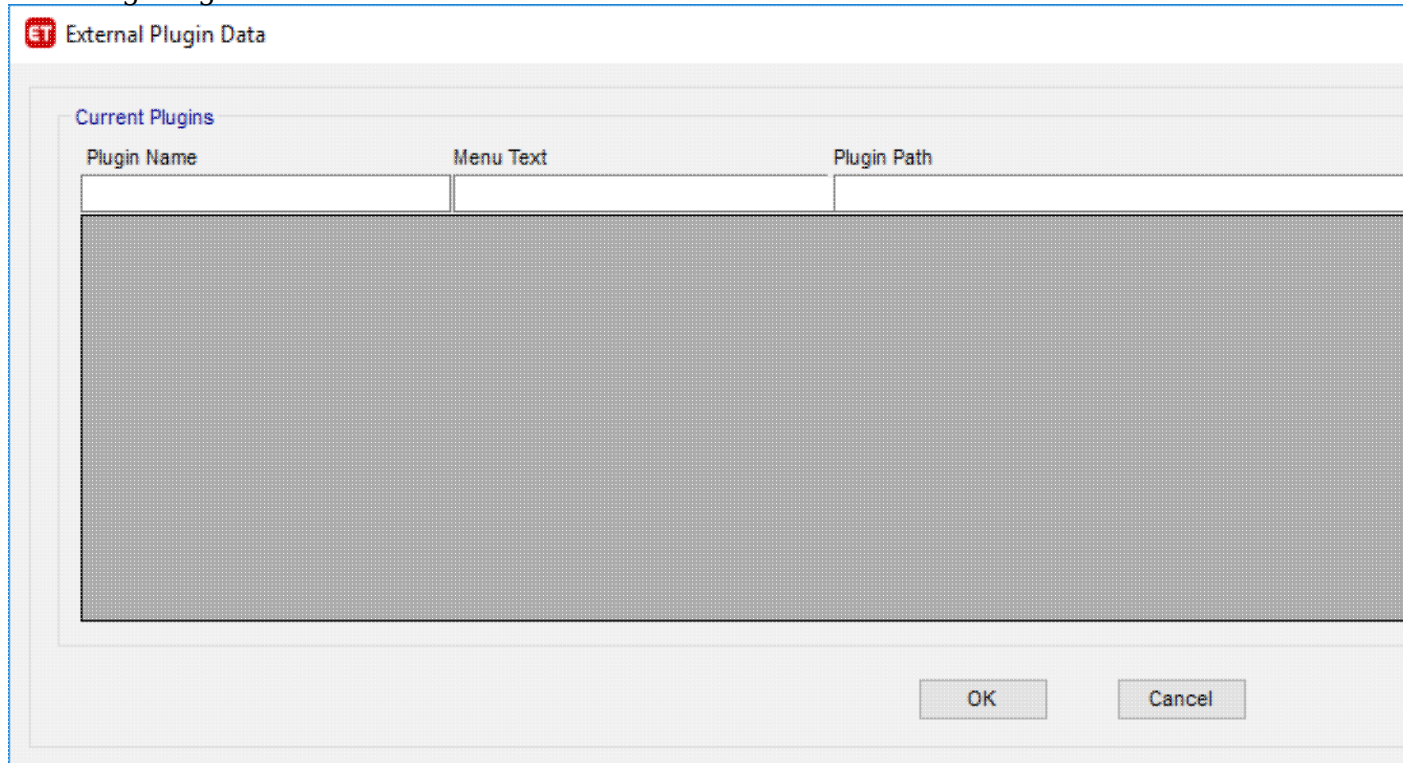
ETABSV1.cPluginCallback contains a Finish subroutine that must be called immediately before the plugin closes (e.g., if the plugin has a single main window, at the end of the close event of that form). Its one argument, iVal, is an error flag to let ETABS know if the operation was successful or not. Zero indicates no error. ETABS will wait indefinitely for ETABSV1.cPluginCallback.Finish to be called, so the plugin programmer must make sure that it is called when the plugin completes.

It is OK for cPlugin.Main to return before the actual work is completed (e.g., return after displaying a form where the functionality implemented in the plugin can be accessed through different command buttons). However, it is imperative to call ETABSv1.cPluginCallback.Finish to return control back to ETABS when the plugin is ready to close.

Your plugin should include options for the user to obtain information about the product, developer, and technical support. Support for your plugin will not be provided by Computers and Structures, Inc.

As currently implemented, the cPlugin object will be destroyed between invocations from the ETABS Tools menu command that calls it, so data cannot be saved.

Adding Plugins



Plugin Name	Menu Text	Plugin Path
-------------	-----------	-------------

External Plugin Data Form

Adding a COM Plugin

To add a COM plugin to the Tools menu in ETABS, users will select "Add/Show Plugins" from the Tools menu. In the External Plugin Data form, they will type the name of the COM server DLL in the "Plugin Name" field.

Please note that ETABS will look for the plugin by Type Library name (usually the name of the plugin project), which you define when developing your COM server DLL. We suggest using a unique name such as ETABSPlugIn_xxx_yyy, where xxx is your (company) name, and yyy is the name to distinguish the plugin from other plugins that you develop, e.g. ETABSPlugIn_OmniCorp_Robo1.

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The "Plugin Name" must exactly match the COM server DLL name. However, the plugin will appear under the Tools menu under the "Menu Text" name. That can be whatever the user likes, but it should be different for each plugin added.

The "Plugin Path" field should be left blank for COM plugins.

After clicking the "Add" button, the "Status" field will indicate whether the plugin was successfully located in the registry and loaded.

Adding a .NET Plugin

The process for adding a .NET plugin is much simpler. In the External Plugin Data form, the user should simply browse for and select the plugin .NET DLL and click the "Add" button. The rest of the fields will be automatically populated. It is still recommended that .NET plugins be given unique names.

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Interactive Database

Beginning with Version 18, ETABS includes a new set of database tables that allow significantly better interactive editing capability (similar to other CSI programs, such as SAP2000 and SAFE). For API users who wish to take advantage of these interactive tables, a new class has been added, `cDatabaseTables`. It is recommended that API users first familiarize themselves with the new database tables using the ETABS GUI, to understand how data can be requested, viewed, modified, and applied to the model.

Generally, the API user will start with a call to `GetAvailableTables`. They can decide which tables they want to retrieve data from, or edit and apply to the model. A call to `GetAllFieldsInTable` will show the user which columns the table has, and which columns can be edited and imported.

If the user simply wants to retrieve data, they can use one of the `GetTableForDisplay...` functions. Functions are provided to set the Load Cases, Load Combinations, and Load Patterns for which data is desired. The `SetOutputOptionsForDisplay` function can be used to set other options for display.

If the user would like to edit a table and import it into the model, they should start by calling one of the `GetTableForEditing...` functions. This will retrieve the table data in one of several formats. The user can then edit that data, but they must ensure that the format of the data is not altered. Then they can import the edited table data with the respective `SetTableForEditing...` function. These functions only operate on one table at a time, but any number of edited tables can be imported by calling them successively. Finally, to apply the edited tables to the model, the user will call `ApplyEditedTables`. If for any reason the user would like to clear the internal list of edited tables set using the `SetTableForEditing...` functions, they can call the `CancelTableEditing` function.

For thorough descriptions of database table operations, please refer to the function documentation in `cDatabaseTables`. For detailed examples of interactive database editing using the API, please visit the CSI Wiki page at: <https://wiki.csiamerica.com> and search for "interactive database API".

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Launching the Installed Version of ETABS Automatically

With the release of ETABS 2016 v16.1.0, new functionality has been added to the cHelper interface to allow users to launch the application without supplying the path to the ETABS.exe . The example code is in VB.NET.

1. Previously, users would launch the program with the function cHelper.CreateObject , providing the full path to the ETABS.exe as an argument, as below:

VB

Copy

```
Dim myETABSObject As ETABSV1.COAPI
```

```
Dim myHelper as ETABSV1.cHelper = New ETABSV1.Helper
```

```
myETABSObject = myHelper.CreateObject("C:\Program Files (x86)\Computers and Structures\ETA
```

```
ret = myETABSObject.ApplicationStart()
```

```
Dim mySapModel As ETABSV1.cSapModel = myETABSObject.SapModel
```

2. While the code above is still available, a simpler method has been added to cHelper . The CreateObjectProgID function takes the Program ID as an argument, and automatically launches the most recently installed version of ETABS:

VB

Copy

```
Dim myETABSObject As ETABSV1.COAPI
```

```
Dim myHelper as ETABSV1.cHelper = New ETABSV1.Helper
```

```
myETABSObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

```
ret = myETABSObject.ApplicationStart()
```

```
Dim mySapModel As ETABSV1.cSapModel = myETABSObject.SapModel
```

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Programming Concepts Used in the CSI API

This topic contains the following sections:

- Option Base
- Fixed-Size and Dynamic Arrays
- Optional Arguments
- ByVal and ByRef

Some of the programming concepts and definitions that apply to the CSI API are explained herein.

▲ Option Base

ETABS uses a lower bound of 0 for all arrays. Any program that accesses ETABS through the API should also use a lower bound of 0 for its arrays.

▲ Fixed-Size and Dynamic Arrays

Arrays can be used to refer to a series of variables by the same name and to use a number (an index) to distinguish them. Visual Basic has two types of arrays: fixed-size and dynamic. A fixed-size array always remains the same size. A dynamic array can change its size while the program is running.

A fixed-size array is declared with the size indicated. For example, the following line declares MyFixedArray dimensioned to 2.

VB

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```
Dim MyFixedArray(2) as Double
```

Dimensioning the array to 2 means that it holds three data items:

VB

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```
MyFixedArray(0) = first data item
MyFixedArray(1) = second data item
MyFixedArray(2) = third data item
```

Dynamic arrays are declared with no size indicated as shown here:

VB

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```
Dim MyDynamicArray() as Double
```

Dynamic arrays are dimensioned sometime after they are declared using a statement such as the following:

VB

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```
ReDim MyDynamicArray(2)
```

Any array that is dimensioned inside ETABS must be declared as a dynamic array so that ETABS can redimension it. It is probably a good idea to declare all arrays as dynamic arrays for simplicity. As an example, the analysis results obtained through the CSI API are stored in arrays that are defined as dynamic arrays by the user and then dimensioned and filled inside of ETABS.

Optional Arguments

Some of the CSI API functions have optional arguments. For example, the CountLoadDispl function has two optional arguments: Name and LoadPat. It is not necessary to include the optional arguments when calling this function. All four of the following calls are valid.

VB

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```
ret = SapModel.PointObj.CountLoadDispl(Count)
```

```
ret = SapModel.PointObj.CountLoadDispl(Count, Name)
```

```
ret = SapModel.PointObj.CountLoadDispl(Count, , LoadPat)
```

```
ret = SapModel.PointObj.CountLoadDispl(Count, Name, LoadPat)
```

Note in the third example, the first optional item is not included and the second optional item is included. In that case, commas must be used to delineate the missing arguments.

ByVal and ByRef

Variables are passed to the CSI API functions using the ByRef or the ByVal keyword.

ByVal means that the variable is passed by value. This allows the CSI API to access a copy of the variable but not the original variable. This means the value of the variable in another application can not be changed by the API.

ByRef means the argument is passed by reference. This passes the address of the variable to the CSI API instead of passing a copy of the value. It allows the CSI API to access the actual variable, and, as a result, allows ETABS to change the variable's actual value in an application.

Variables are passed ByRef when data needs to be returned in them from ETABS to your application. In addition, the API requires that all arrays be passed ByRef.

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Remote API

The ETABS API can be used to start and/or connect to a running instance of ETABS on a remote computer that is running the API Service. This can be particularly useful if you need to run many load cases (e.g., earthquakes for performance-based design), and there are multiple machines available for running analyses simultaneously.

Simultaneous runs can be started on multiple Remote Computers using an API script or plug-in, and results can be merged to the Main Computer programmatically, without user intervention, as they become available. Other applications could include using distributed processing to run a large parameter study or Monte Carlo simulation.

Terminology

- **Main Computer** - Your primary computer
- **Remote Computer** - Any other computer that you have access to (physically or over the network) and that is available for running analyses and/or API scripts
- **API Service** - `CSiAPIService.exe` command line utility running on a remote computer that enables Remote API
- **TCP Port** - A TCP (Transmission Control Protocol) port is an integer between 0 to 65535 used to identify which service is to receive a packet/message sent to a Remote Computer

Requirements and Limitations

- Depending on the type of firewall installed on a Remote Computer, you may have to create a firewall exception on that Remote Computer to allow the API Service to communicate on your network. This is done automatically for Windows Firewall during the installation of ETABS, but it may have to be manually done for other firewalls and will require administrative privileges. This is typically a one-time operation.
- The API Service has two modes:

◆ Product-Specific:

- ◊ Listens to connections made from ETABSV1.dll at default TCP port 11647
- ◊ This is the default mode and can be started using the following command:

`CSiAPIService.exe` or `CSiAPIService.exe --api A`

◆ Cross-Product:

- ◊ Listens to connections made from CSiAPIv1.dll at default TCP port 11646
- ◊ This can be started using the following command:

`CSiAPIService.exe --api C`

- Multiple API Service instances can be run simultaneously on a Remote Computer as long as they do not use the same TCP port. In case of TCP port conflicts with other programs, it is possible to assign a specific TCP port to each API Service instance using the following command:

◆ `CSiAPIService.exe --port [portNumber]`

where [portNumber] is between 1024 and 49151

It is also possible to override the default TCP port for Product-Specific and Cross-Product API modes using the following Windows environment variables:

◆ Product-Specific: `ETABSV1_COAPI_DEFAULT_PORT`

◆ Cross-Product: `CSiAPIv1_COAPI_DEFAULT_PORT`

▲ Procedure

- Install the ETABS program on the Main and Remote Computers
- On each Remote Computer, open a command prompt and run `CSiAPIService.exe`, located in the ETABS installation folder, to start the API Service
 - ◆ **Tip:** Type `CSiAPIService.exe --help` to view a detailed list of options where you can set
 - ◇ The TCP port to use
 - ◇ API mode: Product-specific or Cross-product
 - ◆ **Tip:** You can set the API Service to run automatically when your computer starts. See [Change which apps run automatically at startup in Windows 10](#) for details
- On your Main Computer, run a script, program, or ETABS Plug-in that uses one of the following API calls to start (`CreateObject`) or connect to (`GetObject`) an instance of ETABS on a Remote Computer:
 - ◆ `cHelper.CreateObjectHost( ByVal hostName As String, ByVal fullPath As String)`
 - ◆ `cHelper.CreateObjectHostPort( ByVal hostName As String, ByVal portNumber As Integer, ByVal fullPath As String)`
 - ◆ `cHelper.CreateObjectProgIDHost( ByVal hostName As String, ByVal progID As String)`
 - ◆ `cHelper.CreateObjectProgIDHostPort( ByVal hostName As String, ByVal portNumber As Integer, ByVal progID As String)`
 - ◆ `cHelper.GetObjectHost( ByVal hostName As String, ByVal progID As String)`
 - ◆ `cHelper.GetObjectHostPort( ByVal hostName As String, ByVal portNumber As Integer, ByVal progID As String)`

The above API calls receive the name of the Remote Computer (e.g. `hostName = "myserver"`), and optionally a TCP port number, in addition to the arguments of the regular `cHelper.CreateObject()` or `GetObject()` API calls. Upon successful instantiation at a Remote Computer, subsequent calls to the returned `cOAPI` object will execute on the Remote Computer. API calls can be used to open models, modify them, run analysis and design, extract results, and merge

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results back to identical models on the Main Computer.

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Unit Abbreviations

In the documentation of each CSI API function, parameters that have units associated with them are followed by one of the following abbreviations, to indicate the units for those parameters.

[L] = Length

[F] = Force, [F] = [ML/s²]

[M] = Mass

[s] = Time, seconds

[T] = Temperature

[cyc] = Cycles

[rad] = Radians (angle measurement)

[deg] = Degrees (angle measurement)

Combinations of these abbreviations are used in many cases. For example, moments are indicated as [FL] and stresses are indicated as [F/L²].

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Overview

The steps that follow provide a general overview of the procedure for connecting to ETABS through the API.

1. Early bind to the interface: `ETABSV1.dll` for .NET 4.5 clients and `ETABSV1.tlb` for COM clients.
2. Late bind to `ETABS.exe`, create an instance of the "ETABSObject", and get a reference to the `cOAPI` interface.
3. Use the `cOAPI` interface to start/end the program and get a reference to the `cSapModel` interface to access all API classes and functions.
4. **This step is only necessary when ETABS.exe is located on a network share.**

Find the configuration file for the API client ([clientname].exe.config) under its installation folder and add the following to it:

XML

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```
<configuration>
  <runtime>
    <loadFromRemoteSources enabled="true" />
  </runtime>
</configuration>
```

If the configuration file is not present, create it with the above contents.

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Visual Basic for Applications (VBA) via Excel

1. Create a new VBA project.
2. Add a reference to `ETABSv1.tlb` to the VBA project. This early binds to the API using the COM interface. `ETABSv1.tlb` can be found in the directory where ETABS is installed.
3. Add a new module to the VBA project and paste in the following, ready-to-run code.

VB

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Sub Main()

```

' set the following flag to True to attach to an existing instance of the program
' otherwise a new instance of the program will be started
Dim AttachToInstance As Boolean
AttachToInstance = False

' set the following flag to True to manually specify the path to ETABS.exe
' this allows for a connection to a version of ETABS other than the latest installation
' otherwise the latest installed version of ETABS will be launched
Dim SpecifyPath As Boolean
SpecifyPath = False

' if the above flag is set to True, specify the path to ETABS below
Dim ProgramPath As String
ProgramPath = "C:\Program Files\Computers and Structures\ETABS 19\ETABS.exe"

' full path to the model
' set it to the desired path of your model
Dim ModelDirectory As String
ModelDirectory = "C:\CSi_ETABS_API_Example"
If Len(Dir(ModelDirectory, vbDirectory)) = 0 Then
    MkDir ModelDirectory
End If

Dim ModelName As String
ModelName = "ETABS_API_Example.edb"

Dim ModelPath As String
ModelPath = ModelDirectory & Application.PathSeparator & ModelName

' create API helper object
Dim myHelper As ETABSv1.cHelper
Set myHelper = New ETABSv1.Helper

' dimension the ETABS Object as c0API type
Dim myETABSObject As ETABSv1.c0API
Set myETABSObject = Nothing

' use ret to check return values of API calls
Dim ret As Long

If AttachToInstance Then
    ' attach to a running instance of ETABS
    ' get the active ETABS object

```

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```
Set myETABSObject = myHelper.GetObject("CSI.ETABS.API.ETABSObject")
Else
  If SpecifyPath Then
    'create an instance of the ETABS object from the specified path
    Set myETABSObject = myHelper.CreateObject(ProgramPath)
  Else
    'create an instance of the ETABS object from the latest installed ETABS
    Set myETABSObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
  End If
  'start ETABS application
  myETABSObject.ApplicationStart
End If

'get a reference to cSapModel to access all API classes and functions
Dim mySapModel As ETABSv1.cSapModel
Set mySapModel = myETABSObject.SapModel

'initialize model
ret = mySapModel.InitializeNewModel()

'create steel deck template model
ret = mySapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'save model
ret = mySapModel.File.Save(ModelPath)

'run analysis
ret = mySapModel.Analyze.RunAnalysis

'close ETABS
myETABSObject.ApplicationExit (False)

'clean up variables
Set mySapModel = Nothing
Set myETABSObject = Nothing

If ret = 0 Then
  MsgBox "API script completed successfully."
Else
  MsgBox "API script FAILED to complete."
End If

Exit Sub

ErrorHandler:
  MsgBox "Cannot run API script: " & Err.Description

End Sub
```

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CSI API ETABS v1

Visual Basic or C# 2012 (.NET 4.0)

1. Create a new project of type Windows Form Application
2. Add a reference to â ETABSv1.dllâ to the .NET project. This early binds to the API using the .NET interface.
3. Add a button to the form, double-click the button, then paste in the following, ready-to-run code.

C#VBC++F#Copy

```

using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Windows.Forms;

namespace WindowsFormsApplication1
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
        }

        private void button1_Click(object sender, EventArgs e)
        {
            //set the following flag to true to attach to an existing instance of the prog
            //otherwise a new instance of the program will be started
            bool AttachToInstance;
            AttachToInstance = false;

            //set the following flag to true to manually specify the path to ETABS.exe
            //this allows for a connection to a version of ETABS other than the latest ins
            //otherwise the latest installed version of ETABS will be launched
            bool SpecifyPath;
            SpecifyPath = false;

            //if the above flag is set to true, specify the path to ETABS below
            string ProgramPath;
            ProgramPath = @"C:\Program Files\Computers and Structures\ETABS 19\ETABS.exe";

            //full path to the model
            //set it to an already existing folder
            string ModelDirectory = @"C:\CSi_ETABS_API_Example";
            try
            {
                System.IO.Directory.CreateDirectory(ModelDirectory);
            }
            catch (Exception ex)

```


Introduction

```
{
    MessageBox.Show("Could not create directory: " + ModelDirectory);
}

string ModelName = "ETABS_API_Example.edb";
string ModelPath = ModelDirectory + System.IO.Path.DirectorySeparatorChar + ModelName;

//dimension the ETABS Object as cOAPI type
ETABSV1.cOAPI myETABSObject = null;

//Use ret to check if functions return successfully (ret = 0) or fail (ret = -1)
int ret = 0;

//create API helper object
ETABSV1.cHelper myHelper;
try
{
    myHelper = new ETABSV1.Helper();
}
catch (Exception ex)
{
    MessageBox.Show("Cannot create an instance of the Helper object");
    return;
}

if (AttachToInstance)
{
    //attach to a running instance of ETABS
    try
    {
        //get the active ETABS object
        myETABSObject = myHelper.GetObject("CSI.ETABS.API.ETABSObject");
    }
    catch (Exception ex)
    {
        MessageBox.Show("No running instance of the program found or failed to attach");
        return;
    }
}
else
{
    if (SpecifyPath)
    {
        //create an instance of the ETABS object from the specified path
        try
        {
            //create ETABS object
            myETABSObject = myHelper.CreateObject(ProgramPath);
        }
        catch (Exception ex)
        {
            MessageBox.Show("Cannot start a new instance of the program from " + ProgramPath);
            return;
        }
    }
    else
    {
        //create an instance of the ETABS object from the latest installed ETABS
        try
        {
            //create ETABS object
            myETABSObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject");
        }
        catch (Exception ex)
        {
            MessageBox.Show("Cannot start a new instance of the program from the latest installed ETABS");
            return;
        }
    }
}
```

Introduction

```
    }
    catch (Exception ex)
    {
        MessageBox.Show("Cannot start a new instance of the program.");
        return;
    }
}
//start ETABS application
ret = myETABSObject.ApplicationStart();
}

//Get a reference to cSapModel to access all API classes and functions
ETABSV1.cSapModel mySapModel = default(ETABSV1.cSapModel);
mySapModel = myETABSObject.SapModel;

//Initialize model
ret = mySapModel.InitializeNewModel();

//Create steel deck template model
ret = mySapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24);

//Save model
ret = mySapModel.File.Save(ModelPath);

//Run analysis
ret = mySapModel.Analyze.RunAnalysis();

//Close ETABS
myETABSObject.ApplicationExit(false);

//Clean up variables
mySapModel = null;
myETABSObject = null;

//Check ret value
if (ret == 0)
{
    MessageBox.Show("API script completed successfully.");
}
else
{
    MessageBox.Show("API script FAILED to complete.");
}
}
}
```

Public Class Form1

Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click

'set the following flag to True to attach to a running instance of the program
'otherwise a new instance of the program will be started
Dim AttachToInstance As Boolean
AttachToInstance = True

'set the following flag to True to manually specify the path to ETABS.exe
'this allows for a connection to a version of ETABS other than the latest installa
'otherwise the latest installed version of ETABS will be launched
Dim SpecifyPath As Boolean
SpecifyPath = False

Introduction

```
'if the above flag is set to True, specify the path to ETABS below
Dim ProgramPath As String
'ProgramPath = "C:\Program Files (x86)\Computers and Structures\ETABS 19\ETABS.exe"

'full path to the model
'set it to an already existing folder
Dim ModelDirectory As String = "C:\CSi_ETABS_API_Example"
Try
    System.IO.Directory.CreateDirectory(ModelDirectory)
Catch ex As Exception
    MsgBox("Could not create directory: " + ModelDirectory)
End Try

Dim ModelName As String = "ETABS_API_Example.edb"
Dim ModelPath As String = ModelDirectory + System.IO.Path.DirectorySeparatorChar + ModelName

'dimension the ETABS Object as cOAPI type
Dim myETABSObject As ETABSV1.cOAPI = Nothing

'Use ret to check if functions return successfully (ret = 0) or fail (ret = nonzero)
Dim ret As Integer

'create API helper object
Dim myHelper As ETABSV1.cHelper
Try
    myHelper = New ETABSV1.Helper
Catch ex As Exception
    MsgBox("Cannot create an instance of the Helper object")
End Try

If AttachToInstance Then
    'attach to a running instance of ETABS
    Try
        'get the active ETABS object
        myETABSObject = myHelper.GetObject("CSI.ETABS.API.ETABSObject")
    Catch ex As Exception
        MsgBox("No running instance of the program found or failed to attach.")
        Return
    End Try
Else
    If SpecifyPath Then
        Try
            'create an instance of the ETABS object from the specified path
            myETABSObject = myHelper.CreateObject(ProgramPath)
        Catch ex As Exception
            MsgBox("Cannot start a new instance of the program from " + ProgramPath)
            Return
        End Try
    Else
        Try
            'create an instance of the ETABS object from the latest installed ETABS
            myETABSObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
        Catch ex As Exception
            MsgBox("Cannot start a new instance of the program.")
            Return
        End Try
    End If

    'start ETABS application
    ret = myETABSObject.ApplicationStart()
End If
```

Introduction

```
'Get a reference to cSapModel to access all API classes and functions
Dim mySapModel As ETABSV1.cSapModel
mySapModel = myETABSObject.SapModel

'Initialize model
ret = mySapModel.InitializeNewModel()

'Create steel deck template model
ret = mySapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'Save model
ret = mySapModel.File.Save(ModelPath)

'Run analysis
ret = mySapModel.Analyze.RunAnalysis()

'Close ETABS
myETABSObject.ApplicationExit(False)

'Clean up variables
mySapModel = Nothing
myETABSObject = Nothing

'Check ret value
If ret = 0 Then
    MsgBox("API script completed successfully.")
Else
    MsgBox("API script FAILED to complete.")
End If

End Sub
End Class
```

No code example is currently available or this language may not be supported.

No code example is currently available or this language may not be supported.

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CSI API ETABS v1

Visual C++ 2012

This example was created and tested in Microsoft Visual C++ 2012.

1. Create a Visual C++ Win32 project using the template Win32 Console Application.
2. In the Win32 Application Wizard, add common header files for ATL under the Application Settings tab.
3. Copy the ETABSv1.tlb file from the installation folder to the project directory. It should be copied to the same level as your client application's project file.
4. Open the "stdafx.h" file generated by the wizard by double clicking on it and add the following line to the end of the document:

C++

Copy

```
#include <atlsafe.h>
```

5. Open the .cpp file generated by the wizard by double-clicking on it and paste in the following ready-to-run code:

C++

Copy

```
#include "stdafx.h"
```

```
#import "ETABSv1.tlb" no_dual_interfaces
```

```
using namespace std;
```

```
bool CheckHRESULT(HRESULT hRes, const wchar_t* msg)
```

```
{
    if (hRes == S_FALSE || FAILED(hRes)) {
        MessageBox(0, msg, L"ERROR!", MB_SETFOREGROUND);
        return (false);
    }
    return (true);
}
```

```
int _tmain(int argc, _TCHAR* argv[])
```

```
{
    // set the following flag to true to attach to an existing instance of the program
    // otherwise a new instance of the program will be started
    bool bAttachToInstance;
    bAttachToInstance = false;

    // set the following flag to true to manually specify the path to ETABS.exe
    // this allows for a connection to a version of ETABS other than the latest installation
    // otherwise the latest installed version of ETABS will be launched
    bool bSpecifyPath;
    bSpecifyPath = false;

    // if the above flag is set to true, specify the path to ETABS below
    const wchar_t* ProgramPath;
    ProgramPath = L"C:\\Program Files\\Computers and Structures\\ETABS 19\\ETABS.exe";

    // use res to check if functions return successfully (res = 0) or fail (res = nonzero)
```

Introduction

```
HRESULT hRes = 0;
int res = 0;

// initialize COM
hRes = CoInitialize(NULL);
if (!CheckHRESULT(hRes, L"Error initializing COM subsystem!")) return (hRes);

// ETABSObject pointer
CComPtr<ETABSV1::cOAPI> pETABSObject;

try {
    // create Helper
    CComPtr<ETABSV1::cHelper> pHelper;
    hRes = pHelper.CoCreateInstance(L"ETABSV1.Helper", NULL, CLSCTX_INPROC_SERVER);
    if (!CheckHRESULT(hRes, L"Cannot instantiate helper!")) return (hRes);

    if (bAttachToInstance) {
        // attach to a running instance of ETABS

        // get the active ETABS object
        pETABSObject = pHelper->GetObject(L"CSI.ETABS.API.ETABSObject");
        hRes = ((pETABSObject) ? S_OK : S_FALSE);
        if (!CheckHRESULT(hRes, L"Cannot attach to ETABSObject!")) return (hRes);
    }
    else {
        // start a new instance of ETABS

        if (bSpecifyPath) {
            // create an instance of the ETABS object from the specified path
            pETABSObject = pHelper->CreateObject(ProgramPath);
            hRes = ((pETABSObject) ? S_OK : S_FALSE);
            if (!CheckHRESULT(hRes, L"Cannot instantiate ETABSObject!")) return (hRes);
        }
        else {
            // create an instance of the ETABS object from the latest installed ETABS
            pETABSObject = pHelper->CreateObjectProgID(L"CSI.ETABS.API.ETABSObject");
            hRes = ((pETABSObject) ? S_OK : S_FALSE);
            if (!CheckHRESULT(hRes, L"Cannot instantiate ETABSObject!")) return (hRes);
        }
        // start ETABS application
        res = pETABSObject->ApplicationStart();
        if (!CheckHRESULT(hRes, L"ETABSObject.ApplicationStart failed!")) return (hRes);
    }

    // Get a reference to SapModel to access all API classes and functions
    CComPtr<ETABSV1::cSapModel> pSapModel = pETABSObject->SapModel;

    // initialize model
    res = pSapModel->InitializeNewModel(ETABSV1::eUnits_kip_in_F);

    // create steel deck template model
    res = pSapModel->File->NewSteelDeck(4, 12, 12, 4, 4, 24, 24);

    // save model;
    CreateDirectory(L"C:\\\\CSi_ETABS_API_example\\\\", NULL);
    res = pSapModel->File->Save(L"C:\\\\CSi_ETABS_API_example\\\\example.edb");

    // run model (this will create the analysis model)
    res = pSapModel->Analyze->RunAnalysis();
}
```

Introduction

```
// clear all case and combo output selections
res = pSapModel->Results->Setup->DeselectAllCasesAndCombosForOutput();

// set case and combo output selections
res = pSapModel->Results->Setup->SetCaseSelectedForOutput("DEAD", VARIANT_TRUE);

// get frame forces for all frame objects
long NumberResults;
CComSafeArray<BSTR> Obj(1);
Obj[0] = ::SysAllocString(L "");
CComSafeArray<double> ObjSta(1);
CComSafeArray<BSTR> Elm(1);
Elm[0] = ::SysAllocString(L "");
CComSafeArray<double> ElmSta(1);
CComSafeArray<BSTR> LoadCase(1);
LoadCase[0] = ::SysAllocString(L "");
CComSafeArray<BSTR> StepType(1);
StepType[0] = ::SysAllocString(L "");
CComSafeArray<double> StepNum(1);
CComSafeArray<double> P(1);
CComSafeArray<double> V2(1);
CComSafeArray<double> V3(1);
CComSafeArray<double> T(1);
CComSafeArray<double> M2(1);
CComSafeArray<double> M3(1);

LPSAFEARRAY pObj = Obj.Detach();
LPSAFEARRAY pObjSta = ObjSta.Detach();
LPSAFEARRAY pElm = Elm.Detach();
LPSAFEARRAY pElmSta = ElmSta.Detach();
LPSAFEARRAY pLoadCase = LoadCase.Detach();
LPSAFEARRAY pStepType = StepType.Detach();
LPSAFEARRAY pStepNum = StepNum.Detach();
LPSAFEARRAY pP = P.Detach();
LPSAFEARRAY pV2 = V2.Detach();
LPSAFEARRAY pV3 = V3.Detach();
LPSAFEARRAY pT = T.Detach();
LPSAFEARRAY pM2 = M2.Detach();
LPSAFEARRAY pM3 = M3.Detach();

res = pSapModel->Results->FrameForce("All", ETABSv1::eItemTypeElm_GroupElm, &NumberResults);

Obj.Attach(pObj);
ObjSta.Attach(pObjSta);
Elm.Attach(pElm);
ElmSta.Attach(pElmSta);
LoadCase.Attach(pLoadCase);
StepType.Attach(pStepType);
StepNum.Attach(pStepNum);
P.Attach(pP);
V2.Attach(pV2);
V3.Attach(pV3);
T.Attach(pT);
M2.Attach(pM2);
M3.Attach(pM3);

// close ETABS
CComVariant vFalse(false);
res = pETABSObject->ApplicationExit(false);

// uninitialized COM
CoUninitialize();
```

Introduction

```
        // we're done!  
        return (res);  
    }  
    catch (_com_error& ex) {  
        CheckHRESULT(ex.Error(), ex.ErrorMessage());  
  
        // close ETABS  
        CComVariant vRes = pETABSObject->ApplicationExit(false);  
  
        // uninitialized COM  
        CoUninitialize();  
  
        return (-1);  
    }  
}
```

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CSI API ETABS v1

MATLAB

This example was created and tested using MATLAB R2016a.

1. Create a new MATLAB ".m" file
2. Paste in the following, ready-to-run code:

Copy

```
%% clean-up the workspace & command window
clear;
clc;

%%set the following flag to true to attach to an existing instance of the program otherwise
AttachToInstance = false(); % true(); %

%% set the following flag to true to manually specify the path to ETABS.exe
%% this allows for a connection to a version of ETABS other than the latest installation
%% otherwise the latest installed version of ETABS will be launched
SpecifyPath = false(); % true(); %

%% if the above flag is set to true, specify the path to ETABS below
ProgramPath = 'C:\Program Files\Computers and Structures\ETABS 18\ETABS.exe';

%% full path to API dll
%% set it to the installation folder
APIDLLPath = 'C:\Program Files\Computers and Structures\ETABS 18\ETABSv1.dll';

%% full path to the model
%% set it to the desired path of your model
ModelDirectory = 'C:\Csi_ETABS_API_Example';
if ~exist(ModelDirectory, 'dir')
    mkdir(ModelDirectory);
end
ModelName = 'API_1-001.edb';
ModelPath = strcat(ModelDirectory, filesep, ModelName);

%% create API helper object
a = NET.addAssembly(APIDLLPath);
helper = ETABSv1.Helper;
helper = NET.explicitCast(helper, 'ETABSv1.cHelper');

if AttachToInstance
    %% attach to a running instance of ETABS
    ETABSObject = helper.GetObject('CSI.ETABS.API.ETABSObject');
    ETABSObject = NET.explicitCast(ETABSObject, 'ETABSv1.cOAPI');
else
    if SpecifyPath
        %% create an instance of the ETABS object from the specified path
        ETABSObject = helper.CreateObject(ProgramPath);
    else
        %% create an instance of the ETABS object from the latest installed ETABS
        ETABSObject = helper.CreateObjectProgID('CSI.ETABS.API.ETABSObject');
    end
    ETABSObject = NET.explicitCast(ETABSObject, 'ETABSv1.cOAPI');

    %% start ETABS application
    ETABSObject.ApplicationStart;
end
```

Introduction

```
helper = 0;

%% create SapModel object
SapModel = NET.explicitCast(ETABSObject.SapModel, 'ETABSV1.cSapModel');

%% initialize model
ret = SapModel.InitializeNewModel;

%% create new blank model
File = NET.explicitCast(SapModel.File, 'ETABSV1.cFile');
ret = File.NewBlank;

%% define material property
PropMaterial = NET.explicitCast(SapModel.PropMaterial, 'ETABSV1.cPropMaterial');
ret = PropMaterial.SetMaterial('CONC', ETABSV1.eMatType.Concrete);

%% assign isotropic mechanical properties to material
ret = PropMaterial.SetMPIsotropic('CONC', 3600, 0.2, 0.0000055);

%% define rectangular frame section property
PropFrame = NET.explicitCast(SapModel.PropFrame, 'ETABSV1.cPropFrame');
ret = PropFrame.SetRectangle('R1', 'CONC', 12, 12);

%% define frame section property modifiers
ModValue = NET.createArray('System.Double', 8);
for i = 1 : 8
    ModValue(i) = 1;
end
ModValue(1) = 1000;
ModValue(2) = 0;
ModValue(3) = 0;
ret = PropFrame.SetModifiers('R1', ModValue);

%% switch to k-ft units
ret = SapModel.SetPresentUnits(ETABSV1.eUnits.kip_ft_F);

%% add frame object by coordinates
FrameObj = NET.explicitCast(SapModel.FrameObj, 'ETABSV1.cFrameObj');
FrameName1 = System.String(' ');
FrameName2 = System.String(' ');
FrameName3 = System.String(' ');
[ret, FrameName1] = FrameObj.AddByCoord(0, 0, 0, 0, 0, 10, FrameName1, 'R1', '1', 'Global');
[ret, FrameName2] = FrameObj.AddByCoord(0, 0, 10, 8, 0, 16, FrameName2, 'R1', '2', 'Global');
[ret, FrameName3] = FrameObj.AddByCoord(-4, 0, 10, 0, 0, 10, FrameName3, 'R1', '3', 'Global');

%% assign point object restraint at base
PointObj = NET.explicitCast(SapModel.PointObj, 'ETABSV1.cPointObj');
PointName1 = System.String(' ');
PointName2 = System.String(' ');
Restraint = NET.createArray('System.Boolean', 6);
for i = 1 : 4
    Restraint(i) = true();
end
for i = 5 : 6
    Restraint(i) = false();
end
[ret, PointName1, PointName2] = FrameObj.GetPoints(FrameName1, PointName1, PointName2);
ret = PointObj.SetRestraint(PointName1, Restraint);

%% assign point object restraint at top
for i = 1 : 2
    Restraint(i) = true();
```

Introduction

```
end
for i = 3 : 6
    Restraint(i) = false();
end
[ret, PointName1, PointName2] = FrameObj.GetPoints(FrameName2, PointName1, PointName2);
ret = PointObj.SetRestraint(PointName2, Restraint);

%% refresh view, update (initialize) zoom
View = NET.explicitCast(SapModel.View, 'ETABSv1.cView');
ret = View.RefreshView(0, false());

%% add load patterns
LoadPatterns = NET.explicitCast(SapModel.LoadPatterns, 'ETABSv1.cLoadPatterns');
ret = LoadPatterns.Add('1', ETABSv1.eLoadPatternType.Other, 1, true());
ret = LoadPatterns.Add('2', ETABSv1.eLoadPatternType.Other, 0, true());
ret = LoadPatterns.Add('3', ETABSv1.eLoadPatternType.Other, 0, true());
ret = LoadPatterns.Add('4', ETABSv1.eLoadPatternType.Other, 0, true());
ret = LoadPatterns.Add('5', ETABSv1.eLoadPatternType.Other, 0, true());
ret = LoadPatterns.Add('6', ETABSv1.eLoadPatternType.Other, 0, true());
ret = LoadPatterns.Add('7', ETABSv1.eLoadPatternType.Other, 0, true());

%% assign loading for load pattern 2
[ret, PointName1, PointName2] = FrameObj.GetPoints(FrameName3, PointName1, PointName2);
PointLoadValue = NET.createArray('System.Double', 6);
for i = 1 : 6
    PointLoadValue(i) = 0.0;
end
PointLoadValue(3) = -10;
ret = PointObj.SetLoadForce(PointName1, '2', PointLoadValue);
ret = FrameObj.SetLoadDistributed(FrameName3, '2', 1, 10, 0, 1, 1.8, 1.8);

%% assign loading for load pattern 3
[ret, PointName1, PointName2] = FrameObj.GetPoints(FrameName3, PointName1, PointName2);
for i = 1 : 6
    PointLoadValue(i) = 0.0;
end
PointLoadValue(3) = -17.2;
PointLoadValue(5) = -54.4;
ret = PointObj.SetLoadForce(PointName2, '3', PointLoadValue);

%% assign loading for load pattern 4
ret = FrameObj.SetLoadDistributed(FrameName2, '4', 1, 11, 0, 1, 2, 2);

%% assign loading for load pattern 5
ret = FrameObj.SetLoadDistributed(FrameName1, '5', 1, 2, 0, 1, 2, 2, 'Local');
ret = FrameObj.SetLoadDistributed(FrameName2, '5', 1, 2, 0, 1, -2, -2, 'Local');

%% assign loading for load pattern 6
ret = FrameObj.SetLoadDistributed(FrameName1, '6', 1, 2, 0, 1, 0.9984, 0.3744, 'Local');
ret = FrameObj.SetLoadDistributed(FrameName2, '6', 1, 2, 0, 1, -0.3744, 0, 'Local');

%% assign loading for load pattern 7
ret = FrameObj.SetLoadPoint(FrameName2, '7', 1, 2, 0.5, -15, 'Local');

%% switch to k-in units
ret = SapModel.SetPresentUnits(ETABSv1.eUnits.kip_in_F);

%% save model
ret = File.Save(ModelPath);

%% run model (this will create the analysis model)
Analyze = NET.explicitCast(SapModel.Analyze, 'ETABSv1.cAnalyze');
```

Introduction

```

ret = Analyze.RunAnalysis();

%% initialize for ETABS results
ETABSResult = zeros(7,1,'double');
[ret, PointName1, PointName2] = FrameObj.GetPoints(FrameName2, PointName1, PointName2);

%% get ETABS results for load cases 1 through 7
AnalysisResults = NET.explicitCast(SapModel.Results,'ETABSV1.cAnalysisResults');
AnalysisResultsSetup = NET.explicitCast(AnalysisResults.Setup,'ETABSV1.cAnalysisResultsSetup');

for i = 1 : 7
    NumberResults = 0;
    Obj = NET.createArray('System.String',2);
    Elm = NET.createArray('System.String',2);
    ACase = NET.createArray('System.String',2);
    StepType = NET.createArray('System.String',2);
    StepNum = NET.createArray('System.Double',2);
    U1 = NET.createArray('System.Double',2);
    U2 = NET.createArray('System.Double',2);
    U3 = NET.createArray('System.Double',2);
    R1 = NET.createArray('System.Double',2);
    R2 = NET.createArray('System.Double',2);
    R3 = NET.createArray('System.Double',2);
    ret = AnalysisResultsSetup.DeselectAllCasesAndCombosForOutput;
    ret = AnalysisResultsSetup.SetCaseSelectedForOutput(int2str(i));
    if i <= 4
        [ret, NumberResults, Obj, Elm, ACase, StepType, StepNum, U1, U2, U3, R1, R2, R3] = ...
            ETABSResult(i) = U3(1);
    else
        [ret, NumberResults, Obj, Elm, ACase, StepType, StepNum, U1, U2, U3, R1, R2, R3] = ...
            ETABSResult(i) = U1(1);
    end
end

%% close ETABS
ret = ETABSObject.ApplicationExit(false());
File = 0;
PropMaterial = 0;
PropFrame = 0;
FrameObj = 0;
PointObj = 0;
View = 0;
LoadPatterns = 0;
Analyze = 0;
AnalysisResults = 0;
AnalysisResultsSetup = 0;
SapModel = 0;
ETABSObject = 0;

%% fill independent results
IndResult = zeros(7,1,'double');
IndResult(1) = -0.02639;
IndResult(2) = 0.06296;
IndResult(3) = 0.06296;
IndResult(4) = -0.2963;
IndResult(5) = 0.3125;
IndResult(6) = 0.11556;
IndResult(7) = 0.00651;

%% fill percent difference
PercentDiff = zeros(7,1,'double');
for i = 1 : 7

```

Introduction

```
        PercentDiff(i) = (ETABSResult(i) / IndResult(i)) - 1;  
end  
  
%% display results  
ETABSResult  
IndResult  
PercentDiff
```

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CSI API ETABS v1

Python (COM)

This example was created using Python 3.4.3

1. Download and install Python 3.4.3 or higher for Windows. Python is freely available at python.org
2. Install the Python package "comtypes".

If you are using Python 3.4 or higher, you may install by opening a command prompt with administrative privileges and entering the command:

```
C:\>python -m pip install comtypes
```

Please note that your computer must be connected to the internet for the above command to work.

3. Create a Python .py file using IDLE or any text editor and paste in the following code. Please pay attention to the comments in this code block; they contain important information about running the script.

Python

Copy

```
import os
import sys
import comtypes.client

#set the following flag to True to attach to an existing instance of the program
#otherwise a new instance of the program will be started
AttachToInstance = True

#set the following flag to True to manually specify the path to ETABS.exe
#this allows for a connection to a version of ETABS other than the latest installation
#otherwise the latest installed version of ETABS will be launched
SpecifyPath = False

#if the above flag is set to True, specify the path to ETABS below
ProgramPath = "C:\Program Files\Computers and Structures\ETABS 19\ETABS.exe"

#full path to the model
#set it to the desired path of your model
APIPath = 'C:\CSi_ETABS_API_Example'
if not os.path.exists(APIPath):
    try:
        os.makedirs(APIPath)
    except OSError:
        pass
ModelPath = APIPath + os.sep + 'API_1-001.edb'

#create API helper object
helper = comtypes.client.CreateObject('ETABSV1.Helper')
helper = helper.QueryInterface(comtypes.gen.ETABSV1.cHelper)

if AttachToInstance:
    #attach to a running instance of ETABS
    try:
        #get the active ETABS object
```

Introduction

```
myETABSObject = helper.GetObject("CSI.ETABS.API.ETABSObject")
except (OSError, comtypes.COMError):
    print("No running instance of the program found or failed to attach.")
    sys.exit(-1)
else:
    if SpecifyPath:
        try:
            #create an instance of the ETABS object from the specified path
            myETABSObject = helper.CreateObject(ProgramPath)
        except (OSError, comtypes.COMError):
            print("Cannot start a new instance of the program from " + ProgramPath)
            sys.exit(-1)
    else:
        try:
            #create an instance of the ETABS object from the latest installed ETABS
            myETABSObject = helper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
        except (OSError, comtypes.COMError):
            print("Cannot start a new instance of the program.")
            sys.exit(-1)

    #start ETABS application
    myETABSObject.ApplicationStart()

#create SapModel object
SapModel = myETABSObject.SapModel

#initialize model
SapModel.InitializeNewModel()

#create new blank model
ret = SapModel.File.NewBlank()

#define material property
MATERIAL_CONCRETE = 2
ret = SapModel.PropMaterial.SetMaterial('CONC', MATERIAL_CONCRETE)

#assign isotropic mechanical properties to material
ret = SapModel.PropMaterial.SetMPIsotropic('CONC', 3600, 0.2, 0.0000055)

#define rectangular frame section property
ret = SapModel.PropFrame.SetRectangle('R1', 'CONC', 12, 12)

#define frame section property modifiers
ModValue = [1000, 0, 0, 1, 1, 1, 1, 1]
ret = SapModel.PropFrame.SetModifiers('R1', ModValue)

#switch to k-ft units
kip_ft_F = 4
ret = SapModel.SetPresentUnits(kip_ft_F)

#add frame object by coordinates
FrameName1 = ' '
FrameName2 = ' '
FrameName3 = ' '
[FrameName1, ret] = SapModel.FrameObj.AddByCoord(0, 0, 0, 0, 0, 10, FrameName1, 'R1', '1',
[FrameName2, ret] = SapModel.FrameObj.AddByCoord(0, 0, 10, 8, 0, 16, FrameName2, 'R1', '2',
[FrameName3, ret] = SapModel.FrameObj.AddByCoord(-4, 0, 10, 0, 0, 10, FrameName3, 'R1', '3'

#assign point object restraint at base
PointName1 = ' '
PointName2 = ' '
Restraint = [True, True, True, True, False, False]
```

Introduction

```
[PointName1, PointName2, ret] = SapModel.FrameObj.GetPoints(FrameName1, PointName1, PointName2)
ret = SapModel.PointObj.SetRestraint(PointName1, Restraint)

#assign point object restraint at top
Restraint = [True, True, False, False, False, False]
[PointName1, PointName2, ret] = SapModel.FrameObj.GetPoints(FrameName2, PointName1, PointName2)
ret = SapModel.PointObj.SetRestraint(PointName2, Restraint)

#refresh view, update (initialize) zoom
ret = SapModel.View.RefreshView(0, False)

#add load patterns
LTYPE_OTHER = 8
ret = SapModel.LoadPatterns.Add('1', LTYPE_OTHER, 1, True)
ret = SapModel.LoadPatterns.Add('2', LTYPE_OTHER, 0, True)
ret = SapModel.LoadPatterns.Add('3', LTYPE_OTHER, 0, True)
ret = SapModel.LoadPatterns.Add('4', LTYPE_OTHER, 0, True)
ret = SapModel.LoadPatterns.Add('5', LTYPE_OTHER, 0, True)
ret = SapModel.LoadPatterns.Add('6', LTYPE_OTHER, 0, True)
ret = SapModel.LoadPatterns.Add('7', LTYPE_OTHER, 0, True)

#assign loading for load pattern 2
[PointName1, PointName2, ret] = SapModel.FrameObj.GetPoints(FrameName3, PointName1, PointName2)
PointLoadValue = [0,0,-10,0,0,0]
ret = SapModel.PointObj.SetLoadForce(PointName1, '2', PointLoadValue)
ret = SapModel.FrameObj.SetLoadDistributed(FrameName3, '2', 1, 10, 0, 1, 1.8, 1.8)

#assign loading for load pattern 3
[PointName1, PointName2, ret] = SapModel.FrameObj.GetPoints(FrameName3, PointName1, PointName2)
PointLoadValue = [0,0,-17.2,0,-54.4,0]
ret = SapModel.PointObj.SetLoadForce(PointName2, '3', PointLoadValue)

#assign loading for load pattern 4
ret = SapModel.FrameObj.SetLoadDistributed(FrameName2, '4', 1, 11, 0, 1, 2, 2)

#assign loading for load pattern 5
ret = SapModel.FrameObj.SetLoadDistributed(FrameName1, '5', 1, 2, 0, 1, 2, 2, 'Local')
ret = SapModel.FrameObj.SetLoadDistributed(FrameName2, '5', 1, 2, 0, 1, -2, -2, 'Local')

#assign loading for load pattern 6
ret = SapModel.FrameObj.SetLoadDistributed(FrameName1, '6', 1, 2, 0, 1, 0.9984, 0.3744, 'Local')
ret = SapModel.FrameObj.SetLoadDistributed(FrameName2, '6', 1, 2, 0, 1, -0.3744, 0, 'Local')

#assign loading for load pattern 7
ret = SapModel.FrameObj.SetLoadPoint(FrameName2, '7', 1, 2, 0.5, -15, 'Local')

#switch to k-in units
kip_in_F = 3
ret = SapModel.SetPresentUnits(kip_in_F)

#save model
ret = SapModel.File.Save(ModelPath)

#run model (this will create the analysis model)
ret = SapModel.Analyze.RunAnalysis()

#initialize for results
ProgramResult = [0,0,0,0,0,0,0]
[PointName1, PointName2, ret] = SapModel.FrameObj.GetPoints(FrameName2, PointName1, PointName2)

#get results for load cases 1 through 7
for i in range(0,7):
```


Introduction

```
NumberResults = 0
Obj = []
Elm = []
ACase = []
StepType = []
StepNum = []
U1 = []
U2 = []
U3 = []
R1 = []
R2 = []
R3 = []
ObjectElm = 0
ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()
ret = SapModel.Results.Setup.SetCaseSelectedForOutput(str(i + 1))
if i <= 3:
    [NumberResults, Obj, Elm, ACase, StepType, StepNum, U1, U2, U3, R1, R2, R3, ret]
    ProgramResult[i] = U3[0]
else:
    [NumberResults, Obj, Elm, ACase, StepType, StepNum, U1, U2, U3, R1, R2, R3, ret]
    ProgramResult[i] = U1[0]

#close the program
ret = myETABSObject.ApplicationExit(False)
SapModel = None
myETABSObject = None

#fill independent results
IndResult = [0,0,0,0,0,0,0]
IndResult[0] = -0.02639
IndResult[1] = 0.06296
IndResult[2] = 0.06296
IndResult[3] = -0.2963
IndResult[4] = 0.3125
IndResult[5] = 0.11556
IndResult[6] = 0.00651

#fill percent difference
PercentDiff = [0,0,0,0,0,0,0]
for i in range(0,7):
    PercentDiff[i] = (ProgramResult[i] / IndResult[i]) - 1

#display results
for i in range(0,7):
    print()
    print(ProgramResult[i])
    print(IndResult[i])
    print(PercentDiff[i])
```

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CSI API ETABS v1

Python (NET)

This example was created using Python 3.8.6

1. Download and install a Python version between 3.4.3 and 3.8.6 for Windows. At the time of this writing (Jan 2021), a required library for this example will not work with Python 3.9. Python is freely available at python.org
2. Install the Python package "pythonnet".

If you are using Python 3.4 or higher, you may install by opening a command prompt with administrative privileges and entering the command:

```
C:\>python -m pip install pythonnet
```

Please note that your computer must be connected to the internet for the above command to work.

3. Create a Python .py file using IDLE or any text editor and paste in the following code. Please pay attention to the comments in this code block; they contain important information about running the script. This example includes an option for executing on a remote computer.

Python

Copy

```
import os
import sys
import clr
```

```
clr.AddReference("System.Runtime.InteropServices")
from System.Runtime.InteropServices import Marshal
```

```
#set the following path to the installed ETABS program directory
clr.AddReference(R'C:\Program Files\Computers and Structures\ETABS 19\ETABSv1.dll')
from ETABSv1 import *
```

```
#set the following flag to True to execute on a remote computer
Remote = False
```

```
#if the above flag is True, set the following variable to the hostname of the remote computer
#remember that the remote computer must have ETABS installed and be running the CSiAPIServ
RemoteComputer = "SpareComputer-DT"
```

```
#set the following flag to True to attach to an existing instance of the program
#otherwise a new instance of the program will be started
AttachToInstance = False
```

```
#set the following flag to True to manually specify the path to ETABS.exe
#this allows for a connection to a version of ETABS other than the latest installation
#otherwise the latest installed version of ETABS will be launched
SpecifyPath = False
```

```
#if the above flag is set to True, specify the path to ETABS below
ProgramPath = "C:\Program Files\Computers and Structures\ETABS 19\ETABS.exe"
```

```
#full path to the model
#set it to the desired path of your model
```

Introduction

```
APIPath = 'C:\CSi_ETABS_API_Example'
if not os.path.exists(APIPath):
    try:
        os.makedirs(APIPath)
    except OSError:
        pass
ModelPath = APIPath + os.sep + 'API_1-001.edb'

#create API helper object
helper = cHelper(Help())

if AttachToInstance:
    #attach to a running instance of ETABS
    try:
        #get the active ETABS object
        if Remote:
            myETABSObject = cOAPI(helper.GetObjectHost(RemoteComputer, "CSI.ETABS.API.ETABSObject"))
        else:
            myETABSObject = cOAPI(helper.GetObject("CSI.ETABS.API.ETABSObject"))
    except:
        print("No running instance of the program found or failed to attach.")
        sys.exit(-1)

else:
    if SpecifyPath:
        try:
            #create an instance of the ETABS object from the specified path
            if Remote:
                myETABSObject = cOAPI(helper.CreateObjectHost(RemoteComputer, ProgramPath))
            else:
                myETABSObject = cOAPI(helper.CreateObject(ProgramPath))
        except :
            print("Cannot start a new instance of the program from " + ProgramPath)
            sys.exit(-1)
    else:
        try:
            #create an instance of the ETABS object from the latest installed ETABS
            if Remote:
                myETABSObject = cOAPI(helper.CreateObjectProgIDHost(RemoteComputer, "CSI.ETABS.API.ETABSObject"))
            else:
                myETABSObject = cOAPI(helper.CreateObjectProgID("CSI.ETABS.API.ETABSObject"))
        except:
            print("Cannot start a new instance of the program.")
            sys.exit(-1)

    #start ETABS application
    myETABSObject.ApplicationStart()

#create SapModel object
SapModel = cSapModel(myETABSObject.SapModel)

#initialize model
SapModel.InitializeNewModel()

#create new blank model
File = cFile(SapModel.File)
ret = File.NewBlank()

#define material property
MATERIAL_CONCRETE = 2
```

Introduction

```
PropMaterial = cPropMaterial(SapModel.PropMaterial)
ret = PropMaterial.SetMaterial('CONC', MATERIAL_CONCRETE)

#assign isotropic mechanical properties to material
ret = PropMaterial.SetMPIsotropic('CONC', 3600, 0.2, 0.0000055)

#define rectangular frame section property
PropFrame = cPropFrame(SapModel.PropFrame)
ret = PropFrame.SetRectangle('R1', 'CONC', 12, 12)

#define frame section property modifiers
ModValue = [1000, 0, 0, 1, 1, 1, 1, 1]
ret = PropFrame.SetModifiers('R1', ModValue)

#switch to k-ft units
kip_ft_F = 4
ret = SapModel.SetPresentUnits(kip_ft_F)

#add frame object by coordinates
FrameObj = cFrameObj(SapModel.FrameObj)
FrameName1 = ' '
FrameName2 = ' '
FrameName3 = ' '
[ret, FrameName1] = FrameObj.AddByCoord(0, 0, 0, 0, 0, 10, FrameName1, 'R1', '1', 'Global')
[ret, FrameName2] = FrameObj.AddByCoord(0, 0, 10, 8, 0, 16, FrameName2, 'R1', '2', 'Global')
[ret, FrameName3] = FrameObj.AddByCoord(-4, 0, 10, 0, 0, 10, FrameName3, 'R1', '3', 'Global')

#assign point object restraint at base
PointObj = cPointObj(SapModel.PointObj)
PointName1 = ' '
PointName2 = ' '
Restraint = [True, True, True, True, False, False]
[ret, PointName1, PointName2] = FrameObj.GetPoints(FrameName1, PointName1, PointName2)
ret = PointObj.SetRestraint(PointName1, Restraint)

#assign point object restraint at top
Restraint = [True, True, False, False, False, False]
[ret, PointName1, PointName2] = FrameObj.GetPoints(FrameName2, PointName1, PointName2)
ret = PointObj.SetRestraint(PointName2, Restraint)

#refresh view, update (initialize) zoom
View = cView(SapModel.View)
ret = View.RefreshView(0, False)

#add load patterns
LTYPE_OTHER = 8
LoadPatterns = cLoadPatterns(SapModel.LoadPatterns)
ret = LoadPatterns.Add('1', LTYPE_OTHER, 1, True)
ret = LoadPatterns.Add('2', LTYPE_OTHER, 0, True)
ret = LoadPatterns.Add('3', LTYPE_OTHER, 0, True)
ret = LoadPatterns.Add('4', LTYPE_OTHER, 0, True)
ret = LoadPatterns.Add('5', LTYPE_OTHER, 0, True)
ret = LoadPatterns.Add('6', LTYPE_OTHER, 0, True)
ret = LoadPatterns.Add('7', LTYPE_OTHER, 0, True)

#assign loading for load pattern 2
[ret, PointName1, PointName2] = FrameObj.GetPoints(FrameName3, PointName1, PointName2)
PointLoadValue = [0,0,-10,0,0,0]
ret = PointObj.SetLoadForce(PointName1, '2', PointLoadValue)
ret = FrameObj.SetLoadDistributed(FrameName3, '2', 1, 10, 0, 1, 1.8, 1.8)

#assign loading for load pattern 3
```

Introduction

```
[ret, PointName1, PointName2] = FrameObj.GetPoints(FrameName3, PointName1, PointName2)
PointLoadValue = [0,0,-17.2,0,-54.4,0]
ret = PointObj.SetLoadForce(PointName2, '3', PointLoadValue)

#assign loading for load pattern 4
ret = FrameObj.SetLoadDistributed(FrameName2, '4', 1, 11, 0, 1, 2, 2)

#assign loading for load pattern 5
ret = FrameObj.SetLoadDistributed(FrameName1, '5', 1, 2, 0, 1, 2, 2, 'Local')
ret = FrameObj.SetLoadDistributed(FrameName2, '5', 1, 2, 0, 1, -2, -2, 'Local')

#assign loading for load pattern 6
ret = FrameObj.SetLoadDistributed(FrameName1, '6', 1, 2, 0, 1, 0.9984, 0.3744, 'Local')
ret = FrameObj.SetLoadDistributed(FrameName2, '6', 1, 2, 0, 1, -0.3744, 0, 'Local')

#assign loading for load pattern 7
ret = FrameObj.SetLoadPoint(FrameName2, '7', 1, 2, 0.5, -15, 'Local')

#switch to k-in units
kip_in_F = 3
ret = SapModel.SetPresentUnits(kip_in_F)

#save model
File = cFile(SapModel.File)
ret = File.Save(ModelPath)

#run model (this will create the analysis model)
Analyze = cAnalyze(SapModel.Analyze)
ret = Analyze.RunAnalysis()

#initialize for results
ProgramResult = [0,0,0,0,0,0,0]
[ret, PointName1, PointName2] = FrameObj.GetPoints(FrameName2, PointName1, PointName2)

#get results for load cases 1 through 7
Results = cAnalysisResults(SapModel.Results)
Setup = cAnalysisResultsSetup(Results.Setup)
for i in range(0,7):
    NumberResults = 0
    Obj = []
    Elm = []
    ACase = []
    StepType = []
    StepNum = []
    U1 = []
    U2 = []
    U3 = []
    R1 = []
    R2 = []
    R3 = []
    ObjectElm = 0
    ret = Setup.DeselectAllCasesAndCombosForOutput()
    ret = Setup.SetCaseSelectedForOutput(str(i + 1))
    if i <= 3:
        [ret, NumberResults, Obj, Elm, ACase, StepType, StepNum, U1, U2, U3, R1, R2, R3]
        ProgramResult[i] = U3[0]
    else:
        [ret, NumberResults, Obj, Elm, ACase, StepType, StepNum, U1, U2, U3, R1, R2, R3]
        ProgramResult[i] = U1[0]

#close the program
ret = myETABSObj.ApplicationExit(False)
```

Introduction

```
SapModel = None
myETABSObject = None

#fill independent results
IndResult = [0,0,0,0,0,0,0]
IndResult[0] = -0.02639
IndResult[1] = 0.06296
IndResult[2] = 0.06296
IndResult[3] = -0.2963
IndResult[4] = 0.3125
IndResult[5] = 0.11556
IndResult[6] = 0.00651

#fill percent difference
PercentDiff = [0,0,0,0,0,0,0]
for i in range(0,7):
    PercentDiff[i] = (ProgramResult[i] / IndResult[i]) - 1

#display results
for i in range(0,7):
    print()
    print(ProgramResult[i])
    print(IndResult[i])
    print(PercentDiff[i])
```

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CSI API ETABS v1

ETABSV1 Namespace

^

Interfaces

^	Interface	Description
→	<u>cAnalysisResults</u>	
→	<u>cAnalysisResultsSetup</u>	
→	<u>cAnalyze</u>	
→	<u>cAreaElm</u>	
→	<u>cAreaObj</u>	
→	<u>cAutoSeismic</u>	
→	<u>cCaseDirectHistoryLinear</u>	
→	<u>cCaseDirectHistoryNonlinear</u>	
→	<u>cCaseHyperStatic</u>	
→	<u>cCaseModalEigen</u>	
→	<u>cCaseModalHistoryLinear</u>	
→	<u>cCaseModalHistoryNonlinear</u>	
→	<u>cCaseModalRitz</u>	
→	<u>cCaseResponseSpectrum</u>	
→	<u>cCaseStaticLinear</u>	
→	<u>cCaseStaticNonlinear</u>	
→	<u>cCaseStaticNonlinearStaged</u>	
→	<u>cCombo</u>	
→	<u>cConstraint</u>	
→	<u>cDatabaseTables</u>	
→	<u>cDCoACI318_08_IBC2009</u>	
→	<u>cDCoACI318_11</u>	
→	<u>cDCoACI318_14</u>	
→	<u>cDCoACI318_19</u>	
→	<u>cDCoAS_3600_09</u>	
→	<u>cDCoAS_3600_2018</u>	
→	<u>cDCoBS8110_97</u>	
→	<u>cDCoChinese_2010</u>	
→	<u>cDCoEurocode_2_2004</u>	
→	<u>cDCoHong_Kong_CP_2013</u>	
→	<u>cDCoIndian_IS_456_2000</u>	
→	<u>cDCoItalianNTC2008C</u>	
→	<u>cDCoMexican_RCDF_2004</u>	
→	<u>cDCoMexican_RCDF_2017</u>	
→	<u>cDConcShellEurocode_2_2004</u>	
→	<u>cDConcSlabACI318_14</u>	

- ↪ cDConcSlabACI318_19
- ↪ cDCoNZS_3101_2006
- ↪ cDCoSP63133302011
- ↪ cDCoTS_500_2000
- ↪ cDCoTS_500_2000_R2018
- ↪ cDesignCompositeBeam
- ↪ cDesignConcrete
- ↪ cDesignConcreteShell
- ↪ cDesignConcreteSlab
- ↪ cDesignForces
- ↪ cDesignResults
- ↪ cDesignShearWall
- ↪ cDesignSteel
- ↪ cDesignStrip
- ↪ cDetailing
- ↪ cDiaphragm
- ↪ cDStAISC_ASD89
- ↪ cDStAISC_LRFD93
- ↪ cDStAISC360_05_IBC2006
- ↪ cDStAISC360_10
- ↪ cDStAustralian_AS4100_98
- ↪ cDStBS5950_2000
- ↪ cDStCanadian_S16_09
- ↪ cDStCanadian_S16_14
- ↪ cDStCanadian_S16_19
- ↪ cDStChinese_2010
- ↪ cDStChinese_2018
- ↪ cDStEurocode_3_2005
- ↪ cDStIndian_IS_800_2007
- ↪ cDStItalianNTC2008S
- ↪ cDStItalianNTC2018S
- ↪ cDStNewZealand_NZS3404_97
- ↪ cDStSP16_13330_2011
- ↪ cDStSP16_13330_2017
- ↪ cEditFrame
- ↪ cEditGeneral
- ↪ cFile
- ↪ cFrameObj
- ↪ cFunction
- ↪ cFunctionRS
- ↪ cGenDispl
- ↪ cGridSys

- [cGroup](#)
- [cHelper](#)
- [cLineElm](#)
- [cLinkObj](#)
- [cLoadCases](#)
- [cLoadPatterns](#)
- [cOAPI](#)
- [cPierLabel](#)
- [cPluginCallback](#)
- [cPluginContract](#)
- [cPointElm](#)
- [cPointObj](#)
- [cPropArea](#)
- [cPropAreaSpring](#)
- [cPropFrame](#)
- [cPropFrameSDShape](#)
- [cPropLineSpring](#)
- [cPropLink](#)
- [cPropMaterial](#)
- [cPropPointSpring](#)
- [cPropRebar](#)
- [cPropTendon](#)
- [cSapModel](#)
- [cSelect](#)
- [cSpandrelLabel](#)
- [cStory](#)
- [cTendonObj](#)
- [cTower](#)
- [cView](#)

Enumerations

Ä	Enumeration	Description
☞	<u>e2DFrameType</u>	
☞	<u>e3DFrameType</u>	
☞	<u>eAreaDesignOrientation</u>	The possible design orientations for a shell.
☞	<u>eCNameType</u>	
☞	<u>eConstraintAxis</u>	
☞	<u>eConstraintType</u>	
☞	<u>eDeckType</u>	The possible deck types.
☞	<u>eDesignActionType</u>	
☞	<u>eDiaphragmOption</u>	
☞	<u>eForce</u>	The force units that can be specified for the model.

Introduction

 <u>eFrameDesignOrientation</u>	The possible design orientations for a frame.
 <u>eFramePropType</u>	
 <u>eHingeLocationType</u>	
 <u>eItemType</u>	
 <u>eItemTypeElm</u>	
 <u>eLength</u>	The length units that can be specified for the model.
 <u>eLinkPropType</u>	
 <u>eLoadCaseType</u>	
 <u>eLoadPatternType</u>	
 <u>eMatCoupledType</u>	
 <u>eMatType</u>	
 <u>eMatTypeAluminum</u>	
 <u>eMatTypeColdFormed</u>	
 <u>eMatTypeConcrete</u>	
 <u>eMatTypeRebar</u>	
 <u>eMatTypeSteel</u>	
 <u>eMatTypeTendon</u>	
 <u>eNamedSetType</u>	
 <u>eObjType</u>	
 <u>eReturnCode</u>	
 <u>eShellType</u>	The possible shell types.
 <u>eSlabType</u>	The possible slab types.
 <u>eSuperObjectClass</u>	
 <u>eTemperature</u>	The temperature units that can be specified for the model.
 <u>eTemplateType</u>	
 <u>eUnits</u>	The units that can be specified for the model.
 <u>eWallPierRebarLayerType</u>	The pier rebar types.
 <u>eWallPropType</u>	The wall property type.
 <u>eWallSpandrelRebarLayerType</u>	The pier rebar types.

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CSI API ETABS v1

cAnalysisResults Interface

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
public interface cAnalysisResults
```

```
Public Interface cAnalysisResults
```

```
Dim instance As cAnalysisResults
```

```
public interface class cAnalysisResults
```

```
type cAnalysisResults = interface end
```

The cAnalysisResults type exposes the following members.

▲ Properties

$\hat{A}$ Name Description

 Setup

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▲ Methods

$\hat{A}$

Name

Description

 AreaForceShell

Reports the area forces for the specified area elements that are assigned shell section properties (not plane or asolid properties). Note that the forces reported are per unit of in-plane length.

 AreaJointForceShell

Reports the area joint forces for the point elements at each corner of the specified area elements that have shell-type properties

 AreaStrainShell

 AreaStrainShellLayered

 AreaStressShell

Reports the area stresses for the specified area elements that are assigned shell section properties. Stresses are reported at each point element associated with the area element

 AreaStressShellLayered

Reports the area stresses for the specified area elements that are assigned layered shell section properties

 AssembledJointMass

Reports the assembled joint masses for the specified point elements

Introduction

⇒ <u>BaseReact</u>	Reports the structure total base reactions
⇒ <u>BaseReactWithCentroid</u>	Reports the structure total base reactions and includes information on the centroid of the translational reaction forces
⇒ <u>BucklingFactor</u>	Reports buckling factors obtained from buckling load cases
⇒ <u>FrameForce</u>	Reports the frame forces for the specified line elements
⇒ <u>FrameJointForce</u>	Reports the frame joint forces for the point elements at each end of the specified line elements
⇒ <u>GeneralizedDispl</u>	Reports the displacement values for the specified generalized displacements
⇒ <u>JointAcc</u>	Reports the joint accelerations for the specified point elements. The accelerations reported by this function are relative accelerations.
⇒ <u>JointAccAbs</u>	Reports the joint absolute accelerations for the specified point elements. Absolute and relative accelerations are the same, except when reported for time history load cases subjected to acceleration loading
⇒ <u>JointDispl</u>	Reports the joint displacements for the specified point elements. The displacements reported by this function are relative displacements
⇒ <u>JointDisplAbs</u>	Reports the absolute joint displacements for the specified point elements. Absolute and relative displacements are the same except when reported for time history load cases subjected to acceleration loading
⇒ <u>JointDrifts</u>	Reports the joint drifts
⇒ <u>JointReact</u>	Reports the joint reactions for the specified point elements. The reactions reported are from restraints, springs and grounded (one-joint) links.
⇒ <u>JointVel</u>	Reports the joint velocities for the specified point elements. The velocities reported by this function are relative velocities
⇒ <u>JointVelAbs</u>	Reports the joint absolute velocities for the specified point elements. Absolute and relative velocities are the same, except when reported for time history load cases subjected to acceleration loading
⇒ <u>LinkDeformation</u>	Reports the link internal deformations
⇒ <u>LinkForce</u>	Reports the link forces at the point elements at the ends of the specified link elements
⇒ <u>LinkJointForce</u>	Reports the joint forces at the point elements at the ends of the specified link elements
⇒ <u>ModalLoadParticipationRatios</u>	Reports the modal load participation ratios for each selected modal analysis case

Introduction

- ⇒ [ModalParticipatingMassRatios](#) Reports the modal participating mass ratios for each mode of each selected modal analysis case
- ⇒ [ModalParticipationFactors](#) Reports the modal participation factors for each mode of each selected modal analysis case
- ⇒ [ModalPeriod](#) Reports the modal period, cyclic frequency, circular frequency and eigenvalue for each selected modal load case
- ⇒ [ModeShape](#) Reports the modal displacements (mode shapes) for the specified point elements
- ⇒ [PanelZoneDeformation](#) Reports the panel zone (link) internal deformations
- ⇒ [PanelZoneForce](#) Reports the panel zone (link) forces at the point elements at the ends of the specified panel zone (link) elements
- ⇒ [PierForce](#) Retrieves pier forces for any defined pier objects in the model
- ⇒ [SectionCutAnalysis](#) Reports the section cut force for sections cuts that are specified to have an Analysis (F1, F2, F3, M1, M2, M3) result type
- ⇒ [SectionCutDesign](#) Reports the section cut force for sections cuts that are specified to have a Design (P, V2, V3, T, M2, M3) result type
- ⇒ [SpandrelForce](#) Retrieves spandrel forces for any defined spandrel objects in the model
- ⇒ [StoryDrifts](#) Reports the story drifts

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▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalysisResults Properties

The cAnalysisResults type exposes the following members.

▀ Properties

Â **Name Description**



Setup

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▀ See Also

Reference

cAnalysisResults Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LST5B91A9B_0?cpp=51hu=.");S Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cAnalysisResultsSetup Setup { get; }
```

```
ReadOnly Property Setup As cAnalysisResultsSetup
Get
```

```
Dim instance As cAnalysisResults
Dim value As cAnalysisResultsSetup
```

```
value = instance.Setup
```

```
property cAnalysisResultsSetup^ Setup {
cAnalysisResultsSetup^ get ();
}
```

```
abstract Setup : cAnalysisResultsSetup with get
```

Property Value

Type: [cAnalysisResultsSetup](#)

▲ See Also

Reference

[cAnalysisResults Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalysisResults Methods

The cAnalysisResults type exposes the following members.

Methods

Á	Name	Description
⇒	<u>AreaForceShell</u>	Reports the area forces for the specified area elements that are assigned shell section properties (not plane or asolid properties). Note that the forces reported are per unit of in-plane length.
⇒	<u>AreaJointForceShell</u>	Reports the area joint forces for the point elements at each corner of the specified area elements that have shell-type properties
⇒	<u>AreaStrainShell</u>	
⇒	<u>AreaStrainShellLayered</u>	
⇒	<u>AreaStressShell</u>	Reports the area stresses for the specified area elements that are assigned shell section properties. Stresses are reported at each point element associated with the area element
⇒	<u>AreaStressShellLayered</u>	Reports the area stresses for the specified area elements that are assigned layered shell section properties
⇒	<u>AssembledJointMass</u>	Reports the assembled joint masses for the specified point elements
⇒	<u>BaseReact</u>	Reports the structure total base reactions
⇒	<u>BaseReactWithCentroid</u>	Reports the structure total base reactions and includes information on the centroid of the translational reaction forces
⇒	<u>BucklingFactor</u>	Reports buckling factors obtained from buckling load cases
⇒	<u>FrameForce</u>	Reports the frame forces for the specified line elements
⇒	<u>FrameJointForce</u>	Reports the frame joint forces for the point elements at each end of the specified line elements
⇒	<u>GeneralizedDispl</u>	Reports the displacement values for the specified generalized displacements
⇒	<u>JointAcc</u>	Reports the joint accelerations for the specified point elements. The accelerations reported by this function are relative accelerations.
⇒	<u>JointAccAbs</u>	Reports the joint absolute accelerations for the specified point elements. Absolute and relative accelerations are the same, except when reported for time history load cases subjected to acceleration loading
⇒	<u>JointDispl</u>	

	Reports the joint displacements for the specified point elements. The displacements reported by this function are relative displacements
⇒ <u>JointDisplAbs</u>	Reports the absolute joint displacements for the specified point elements. Absolute and relative displacements are the same except when reported for time history load cases subjected to acceleration loading
⇒ <u>JointDrifts</u>	Reports the joint drifts
⇒ <u>JointReact</u>	Reports the joint reactions for the specified point elements. The reactions reported are from restraints, springs and grounded (one-joint) links.
⇒ <u>JointVel</u>	Reports the joint velocities for the specified point elements. The velocities reported by this function are relative velocities
⇒ <u>JointVelAbs</u>	Reports the joint absolute velocities for the specified point elements. Absolute and relative velocities are the same, except when reported for time history load cases subjected to acceleration loading
⇒ <u>LinkDeformation</u>	Reports the link internal deformations
⇒ <u>LinkForce</u>	Reports the link forces at the point elements at the ends of the specified link elements
⇒ <u>LinkJointForce</u>	Reports the joint forces at the point elements at the ends of the specified link elements
⇒ <u>ModalLoadParticipationRatios</u>	Reports the modal load participation ratios for each selected modal analysis case
⇒ <u>ModalParticipatingMassRatios</u>	Reports the modal participating mass ratios for each mode of each selected modal analysis case
⇒ <u>ModalParticipationFactors</u>	Reports the modal participation factors for each mode of each selected modal analysis case
⇒ <u>ModalPeriod</u>	Reports the modal period, cyclic frequency, circular frequency and eigenvalue for each selected modal load case
⇒ <u>ModeShape</u>	Reports the modal displacements (mode shapes) for the specified point elements
⇒ <u>PanelZoneDeformation</u>	Reports the panel zone (link) internal deformations
⇒ <u>PanelZoneForce</u>	Reports the panel zone (link) forces at the point elements at the ends of the specified panel zone (link) elements
⇒ <u>PierForce</u>	Retrieves pier forces for any defined pier objects in the model
⇒ <u>SectionCutAnalysis</u>	Reports the section cut force for sections cuts that are specified to have an Analysis (F1, F2, F3, M1, M2, M3) result type
⇒ <u>SectionCutDesign</u>	Reports the section cut force for sections cuts that are specified to have a Design (P, V2, V3, T, M2,

Introduction

M3) result type

Retrieves spandrel forces for any defined spandrel objects in the model

Reports the story drifts

 [SpandrelForce](#)

 [StoryDrifts](#)

[Top](#)

 See Also

Reference

[cAnalysisResults Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LSTBCB27CA5_0?c5p5=::|nu=.

Method

Reports the area forces for the specified area elements that are assigned shell section properties (not plane or asolid properties). Note that the forces reported are per unit of in-plane length.

Namespace: Â ETABSV1

Assembly: Â ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int AreaForceShell(
    string Name,
    eItemTypeElm ItemTypeElm,
    ref int NumberResults,
    ref string[] Obj,
    ref string[] Elm,
    ref string[] PointElm,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] F11,
    ref double[] F22,
    ref double[] F12,
    ref double[] FMax,
    ref double[] FMin,
    ref double[] FAngle,
    ref double[] FVM,
    ref double[] M11,
    ref double[] M22,
    ref double[] M12,
    ref double[] MMax,
    ref double[] MMin,
    ref double[] MAngle,
    ref double[] V13,
    ref double[] V23,
    ref double[] VMax,
    ref double[] VAngle
)
```

```
Function AreaForceShell (
    Name As String,
    ItemTypeElm As eItemTypeElm,
    ByRef NumberResults As Integer,
    ByRef Obj As String(),
    ByRef Elm As String(),
    ByRef PointElm As String(),
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
```

cAnalysisResultsspan id="LSTBCB27CA5_0"AddLanguageSpecificTextSet("LSTBCB27CA5_0?c5p5=::|nu=.

Introduction

```
ByRef F11 As Double(),  
ByRef F22 As Double(),  
ByRef F12 As Double(),  
ByRef FMax As Double(),  
ByRef FMin As Double(),  
ByRef FAngle As Double(),  
ByRef FVM As Double(),  
ByRef M11 As Double(),  
ByRef M22 As Double(),  
ByRef M12 As Double(),  
ByRef MMax As Double(),  
ByRef MMin As Double(),  
ByRef MAngle As Double(),  
ByRef V13 As Double(),  
ByRef V23 As Double(),  
ByRef VMax As Double(),  
ByRef VAngle As Double()  
) As Integer
```

```
Dim instance As cAnalysisResults  
Dim Name As String  
Dim ItemTypeElm As eItemTypeElm  
Dim NumberResults As Integer  
Dim Obj As String()  
Dim Elm As String()  
Dim PointElm As String()  
Dim LoadCase As String()  
Dim StepType As String()  
Dim StepNum As Double()  
Dim F11 As Double()  
Dim F22 As Double()  
Dim F12 As Double()  
Dim FMax As Double()  
Dim FMin As Double()  
Dim FAngle As Double()  
Dim FVM As Double()  
Dim M11 As Double()  
Dim M22 As Double()  
Dim M12 As Double()  
Dim MMax As Double()  
Dim MMin As Double()  
Dim MAngle As Double()  
Dim V13 As Double()  
Dim V23 As Double()  
Dim VMax As Double()  
Dim VAngle As Double()  
Dim returnValue As Integer
```

```
returnValue = instance.AreaForceShell(Name,  
ItemTypeElm, NumberResults, Obj,  
Elm, PointElm, LoadCase, StepType,  
StepNum, F11, F22, F12, FMax, FMin,  
FAngle, FVM, M11, M22, M12, MMax, MMin,  
MAngle, V13, V23, VMax, VAngle)
```

```
int AreaForceShell(  
String^ Name,  
eItemTypeElm ItemTypeElm,  
int% NumberResults,  
array<String^>^% Obj,  
array<String^>^% Elm,  
array<String^>^% PointElm,
```

```

array<String>^% LoadCase,
array<String>^% StepType,
array<double>^% StepNum,
array<double>^% F11,
array<double>^% F22,
array<double>^% F12,
array<double>^% FMax,
array<double>^% FMin,
array<double>^% FAngle,
array<double>^% FVM,
array<double>^% M11,
array<double>^% M22,
array<double>^% M12,
array<double>^% MMax,
array<double>^% MMin,
array<double>^% MAngle,
array<double>^% V13,
array<double>^% V23,
array<double>^% VMax,
array<double>^% VAngle
)

abstract AreaForceShell :
    Name : string *
    ItemTypeElm : eItemTypeElm *
    NumberResults : int byref *
    Obj : string[] byref *
    Elm : string[] byref *
    PointElm : string[] byref *
    LoadCase : string[] byref *
    StepType : string[] byref *
    StepNum : float[] byref *
    F11 : float[] byref *
    F22 : float[] byref *
    F12 : float[] byref *
    FMax : float[] byref *
    FMin : float[] byref *
    FAngle : float[] byref *
    FVM : float[] byref *
    M11 : float[] byref *
    M22 : float[] byref *
    M12 : float[] byref *
    MMax : float[] byref *
    MMin : float[] byref *
    MAngle : float[] byref *
    V13 : float[] byref *
    V23 : float[] byref *
    VMax : float[] byref *
    VAngle : float[] byref -> int

```

Parameters

Name

Type: SystemString

The name of an existing area object, area element or group of objects, depending on the value of the ItemTypeElm item

ItemTypeElm

Type: ETABSv1eItemTypeElm

This is one of the following items in the eItemTypeElm enumeration.

Introduction

If this item is ObjectElm, the result request is for the area elements corresponding to the area object specified by the Name item.

If this item is Element, the result request is for the area element specified by the Name item.

If this item is GroupElm, the result request is for the area elements corresponding to all area objects included in the group specified by the Name item.

If this item is SelectionElm, the result request is for area elements corresponding to all selected area objects and the Name item is ignored.

NumberResults

Type:Â SystemInt32

The total number of results returned by the program

Obj

Type:Â SystemString

This is an array that includes the area object name associated with each result, if any

Elm

Type:Â SystemString

This is an array that includes the area element name associated with each result

PointElm

Type:Â SystemString

This is an array that includes the name of the point element where the results are reported

LoadCase

Type:Â SystemString

This is an array that includes the name of the analysis case or load combination associated with each result

StepType

Type:Â SystemString

This is an array that includes the step type, if any, for each result

StepNum

Type:Â SystemDouble

This is an array that includes the step number, if any, for each result

F11

Type:Â SystemDouble

The area element internal F11 membrane direct force per length reported in the area element local coordinate system. [F/L]

F22

Type:Â SystemDouble

The area element internal F22 membrane direct force per length reported in the area element local coordinate system. [F/L]

F12

Type:Â SystemDouble

The area element internal F12 membrane shear force per length reported in the area element local coordinate system. [F/L]

FMax

Type:Â SystemDouble

The maximum principal membrane force per length. [F/L]

FMin

Type: SystemDouble

The minimum principal membrane force per length. [F/L]

FAngle

Type: SystemDouble

The angle measured counter clockwise (when the local 3 axis is pointing toward you) from the area local 1 axis to the direction of the maximum principal membrane force. [deg]

FVM

Type: SystemDouble

The area element internal Von Mises membrane force per length. [F/L]

M11

Type: SystemDouble

The area element internal M11 plate bending moment per length reported in the area element local coordinate system. This item is only reported for area elements with properties that allow plate bending behavior. [FL/L]

M22

Type: SystemDouble

The area element internal M22 plate bending moment per length reported in the area element local coordinate system. This item is only reported for area elements with properties that allow plate bending behavior. [FL/L]

M12

Type: SystemDouble

The area element internal M12 plate twisting moment per length reported in the area element local coordinate system. This item is only reported for area elements with properties that allow plate bending behavior. [FL/L]

MMax

Type: SystemDouble

The maximum principal plate moment per length. This item is only reported for area elements with properties that allow plate bending behavior. [FL/L]

MMin

Type: SystemDouble

The minimum principal plate moment per length. This item is only reported for area elements with properties that allow plate bending behavior. [FL/L]

MAngle

Type: SystemDouble

The angle measured counter clockwise (when the local 3 axis is pointing toward you) from the area local 1 axis to the direction of the maximum principal plate moment. This item is only reported for area elements with properties that allow plate bending behavior. [deg]

V13

Type: SystemDouble

The area element internal V13 plate transverse shear force per length reported in the area element local coordinate system. This item is only reported for area elements with properties that allow plate bending behavior. [F/L]

V23

Type: SystemDouble

The area element internal V23 plate transverse shear force per length reported in the area element local coordinate system. This item is only reported for area elements with properties that allow plate bending behavior. [F/L]

VMax

Type: SystemDouble

The maximum plate transverse shear force. It is equal to the square root of the sum of the squares of V13 and V23. This item is only reported for area elements with properties that allow plate bending behavior. [F/L]

VAngle

Type: SystemDouble

The angle measured counter clockwise (when the local 3 axis is pointing toward you) from the area local 1 axis to the direction of Vmax. This item is only reported for area elements with properties that allow plate bending behavior. [deg]

Return Value

Type: Int32

Returns zero if the forces are successfully recovered, otherwise it returns a nonzero value

Remarks

See Results for more information.

Examples

VB

Copy

```
Public Sub Example()

    'dimension variables
    Dim SapObject as cOAPI
    Dim SapModel As cSapModel
    Dim ret As Integer = -1
    Dim NumberResults As Integer
    Dim Obj() As String
    Dim Elm() As String
    Dim PointElm() As String
    Dim LoadCase() As String
    Dim StepType() As String
    Dim StepNum() As Double
    Dim F11() As Double
    Dim F22() As Double
    Dim F12() As Double
    Dim FMax() As Double
    Dim FMin() As Double
    Dim FAngle() As Double
    Dim FVM() As Double
    Dim M11() As Double
    Dim M22() As Double
    Dim M12() As Double
    Dim MMax() As Double
    Dim MMin() As Double
    Dim MAngle() As Double
    Dim V13() As Double
    Dim V23() As Double
    Dim VMax() As Double
    Dim VAngle() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()
```

Return Value

Introduction

```
'create SapModel object
    SapModel = EtabsObject.SapModel

'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis()

'deselect all cases and combos
    ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()

'set case selected for output
    ret = SapModel.Results.Setup.SetCaseSelectedForOutput("DEAD")

'get area forces for area object "1"
    ret = SapModel.Results.AreaForceShell("1", eItemTypeElm.ObjectElm, NumberResults, Obj, Elm)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

➤ See Also

Reference

[cAnalysisResults Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LSTF22BCCA_0?cpp62nu=.");

Method

Reports the area joint forces for the point elements at each corner of the specified area elements that have shell-type properties

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int AreaJointForceShell(
    string Name,
    eItemTypeElm ItemTypeElm,
    ref int NumberResults,
    ref string[] Obj,
    ref string[] Elm,
    ref string[] PointElm,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] F1,
    ref double[] F2,
    ref double[] F3,
    ref double[] M1,
    ref double[] M2,
    ref double[] M3
)
```

```
Function AreaJointForceShell (
    Name As String,
    ItemTypeElm As eItemTypeElm,
    ByRef NumberResults As Integer,
    ByRef Obj As String(),
    ByRef Elm As String(),
    ByRef PointElm As String(),
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef F1 As Double(),
    ByRef F2 As Double(),
    ByRef F3 As Double(),
    ByRef M1 As Double(),
    ByRef M2 As Double(),
    ByRef M3 As Double()
) As Integer
```

```
Dim instance As cAnalysisResults
Dim Name As String
Dim ItemTypeElm As eItemTypeElm
Dim NumberResults As Integer
```

cAnalysisResultsspan id="LSTF22BCCA_0"AddLanguageSpecificTextSet("LSTF22BCCA_0?cpp62nu=.");

```

Dim Obj As String()
Dim Elm As String()
Dim PointElm As String()
Dim LoadCase As String()
Dim StepType As String()
Dim StepNum As Double()
Dim F1 As Double()
Dim F2 As Double()
Dim F3 As Double()
Dim M1 As Double()
Dim M2 As Double()
Dim M3 As Double()
Dim returnValue As Integer

returnValue = instance.AreaJointForceShell(Name,
ItemtypeElm, NumberResults, Obj,
Elm, PointElm, LoadCase, StepType,
StepNum, F1, F2, F3, M1, M2, M3)

```

```

int AreaJointForceShell(
String^ Name,
eItemTypeElm ItemTypeElm,
int% NumberResults,
array<String^>^% Obj,
array<String^>^% Elm,
array<String^>^% PointElm,
array<String^>^% LoadCase,
array<String^>^% StepType,
array<double>^% StepNum,
array<double>^% F1,
array<double>^% F2,
array<double>^% F3,
array<double>^% M1,
array<double>^% M2,
array<double>^% M3
)

```

```

abstract AreaJointForceShell :
    Name : string *
    ItemTypeElm : eItemTypeElm *
    NumberResults : int byref *
    Obj : string[] byref *
    Elm : string[] byref *
    PointElm : string[] byref *
    LoadCase : string[] byref *
    StepType : string[] byref *
    StepNum : float[] byref *
    F1 : float[] byref *
    F2 : float[] byref *
    F3 : float[] byref *
    M1 : float[] byref *
    M2 : float[] byref *
    M3 : float[] byref -> int

```

Parameters

Name

Type: SystemString

The name of an existing area object, area element or group of objects, depending on the value of the ItemTypeElm item

ItemTypeElm

Type:Â ETABSv1eItemTypeElm

This is one of the following items in the eItemTypeElm enumeration.

If this item is ObjectElm, the result request is for the area elements corresponding to the area object specified by the Name item.

If this item is Element, the result request is for the area element specified by the Name item.

If this item is GroupElm, the result request is for the area elements corresponding to all area objects included in the group specified by the Name item.

If this item is SelectionElm, the result request is for area elements corresponding to all selected area objects and the Name item is ignored.

NumberResults

Type:Â SystemInt32

The total number of results returned by the program

Obj

Type:Â SystemString

This is an array that includes the area object name associated with each result, if any

Elm

Type:Â SystemString

This is an array that includes the area element name associated with each result

PointElm

Type:Â SystemString

This is an array that includes the name of the point element where the results are reported

LoadCase

Type:Â SystemString

This is an array that includes the name of the analysis case or load combination associated with each result

StepType

Type:Â SystemString

This is an array that includes the step type, if any, for each result

StepNum

Type:Â SystemDouble

This is an array that includes the step number, if any, for each result

F1

Type:Â SystemDouble

This is an array that contains the joint force component in the point element local 1 axis direction. [F].

F2

Type:Â SystemDouble

This is an array that contains the joint force component in the point element local 2 axis direction. [F].

F3

Type:Â SystemDouble

This is an array that contains the joint force component in the point element

local 3 axis direction. [F].

M1

Type: SystemDouble

This is an array that contains the joint moment component about the point element local 1 axis direction. [FL].

M2

Type: SystemDouble

This is an array that contains the joint moment component about the point element local 2 axis direction. [FL].

M3

Type: SystemDouble

This is an array that contains the joint moment component about the point element local 3 axis direction. [FL].

Return Value

Type: Int32

Returns zero if the forces are successfully recovered, otherwise it returns a nonzero value.

Remarks

See Results for more information.

Examples

VB

Copy

```
Public Sub Example()
```

```
    'dimension variables
    Dim SapObject as cOAPI
    Dim SapModel As cSapModel
    Dim ret As Integer = -1
    Dim NumberResults As Integer
    Dim Obj() As String
    Dim Elm() As String
    Dim PointElm() As String
    Dim LoadCase() As String
    Dim StepType() As String
    Dim StepNum() As Double
    Dim F1() As Double
    Dim F2() As Double
    Dim F3() As Double
    Dim M1() As Double
    Dim M2() As Double
    Dim M3() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()
```

Introduction

```
'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis()

'deselect all cases and combos
    ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()

'set case selected for output
    ret = SapModel.Results.Setup.SetCaseSelectedForOutput("DEAD")

'get area joint forces for area object "1"
    ret = SapModel.Results.AreaJointForceShell("1", eItemTypeElm.ObjectElm, NumberResults, Obj

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

🔗 See Also

Reference

[cAnalysisResults Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LSTF7 Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int AreaStrainShell(
    string Name,
    eItemTypeElm ItemTypeElm,
    ref int NumberResults,
    ref string[] obj,
    ref string[] elm,
    ref string[] PointElm,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] e11top,
    ref double[] e22top,
    ref double[] g12top,
    ref double[] emaxtop,
    ref double[] emintop,
    ref double[] eangletop,
    ref double[] evmtop,
    ref double[] e11bot,
    ref double[] e22bot,
    ref double[] g12bot,
    ref double[] emaxbot,
    ref double[] eminbot,
    ref double[] eanglebot,
    ref double[] evmbot,
    ref double[] g13avg,
    ref double[] g23avg,
    ref double[] gmaxavg,
    ref double[] gangleavg
)
```

```
Function AreaStrainShell (
    Name As String,
    ItemTypeElm As eItemTypeElm,
    ByRef NumberResults As Integer,
    ByRef obj As String(),
    ByRef elm As String(),
    ByRef PointElm As String(),
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef e11top As Double(),
    ByRef e22top As Double(),
    ByRef g12top As Double(),
```

cAnalysisResultsspan id="LSTF7FC7A01_0"AddLanguageSpecificTextSet("LSTF7FC7A01_0?cp67::|nu=.")

Introduction

```
ByRef emaxtop As Double(),  
ByRef emintop As Double(),  
ByRef eangletop As Double(),  
ByRef evmtop As Double(),  
ByRef e11bot As Double(),  
ByRef e22bot As Double(),  
ByRef g12bot As Double(),  
ByRef emaxbot As Double(),  
ByRef eminbot As Double(),  
ByRef eanglebot As Double(),  
ByRef evmbot As Double(),  
ByRef g13avg As Double(),  
ByRef g23avg As Double(),  
ByRef gmaxavg As Double(),  
ByRef gangleavg As Double()  
) As Integer
```

```
Dim instance As cAnalysisResults  
Dim Name As String  
Dim ItemTypeElm As eItemTypeElm  
Dim NumberResults As Integer  
Dim obj As String()  
Dim elm As String()  
Dim PointElm As String()  
Dim LoadCase As String()  
Dim StepType As String()  
Dim StepNum As Double()  
Dim e11top As Double()  
Dim e22top As Double()  
Dim g12top As Double()  
Dim emaxtop As Double()  
Dim emintop As Double()  
Dim eangletop As Double()  
Dim evmtop As Double()  
Dim e11bot As Double()  
Dim e22bot As Double()  
Dim g12bot As Double()  
Dim emaxbot As Double()  
Dim eminbot As Double()  
Dim eanglebot As Double()  
Dim evmbot As Double()  
Dim g13avg As Double()  
Dim g23avg As Double()  
Dim gmaxavg As Double()  
Dim gangleavg As Double()  
Dim returnValue As Integer
```

```
returnValue = instance.AreaStrainShell(Name,  
ItemTypeElm, NumberResults, obj,  
elm, PointElm, LoadCase, StepType,  
StepNum, e11top, e22top, g12top, emaxtop,  
emintop, eangletop, evmtop, e11bot,  
e22bot, g12bot, emaxbot, eminbot,  
eanglebot, evmbot, g13avg, g23avg,  
gmaxavg, gangleavg)
```

```
int AreaStrainShell(  
String^ Name,  
eItemTypeElm ItemTypeElm,  
int% NumberResults,  
array<String^>^% obj,  
array<String^>^% elm,
```



```

array<String>^% PointElm,
array<String>^% LoadCase,
array<String>^% StepType,
array<double>^% StepNum,
array<double>^% e11top,
array<double>^% e22top,
array<double>^% g12top,
array<double>^% emaxtop,
array<double>^% emintop,
array<double>^% eangletop,
array<double>^% evmtop,
array<double>^% e11bot,
array<double>^% e22bot,
array<double>^% g12bot,
array<double>^% emaxbot,
array<double>^% eminbot,
array<double>^% eanglebot,
array<double>^% evmbot,
array<double>^% g13avg,
array<double>^% g23avg,
array<double>^% gmaxavg,
array<double>^% gangleavg
)

```

```

abstract AreaStrainShell :
  Name : string *
  ItemTypeElm : eItemTypeElm *
  NumberResults : int byref *
  obj : string[] byref *
  elm : string[] byref *
  PointElm : string[] byref *
  LoadCase : string[] byref *
  StepType : string[] byref *
  StepNum : float[] byref *
  e11top : float[] byref *
  e22top : float[] byref *
  g12top : float[] byref *
  emaxtop : float[] byref *
  emintop : float[] byref *
  eangletop : float[] byref *
  evmtop : float[] byref *
  e11bot : float[] byref *
  e22bot : float[] byref *
  g12bot : float[] byref *
  emaxbot : float[] byref *
  eminbot : float[] byref *
  eanglebot : float[] byref *
  evmbot : float[] byref *
  g13avg : float[] byref *
  g23avg : float[] byref *
  gmaxavg : float[] byref *
  gangleavg : float[] byref -> int

```

Parameters

Name

Type:Â SystemString

ItemTypeElm

Type:Â ETABSv1eItemTypeElm

NumberResults

Type:Â SystemInt32
 obj
 Type:Â SystemString
 elm
 Type:Â SystemString
 PointElm
 Type:Â SystemString
 LoadCase
 Type:Â SystemString
 StepType
 Type:Â SystemString
 StepNum
 Type:Â SystemDouble
 e11top
 Type:Â SystemDouble
 e22top
 Type:Â SystemDouble
 g12top
 Type:Â SystemDouble
 emaxtop
 Type:Â SystemDouble
 emintop
 Type:Â SystemDouble
 eangletop
 Type:Â SystemDouble
 evmtop
 Type:Â SystemDouble
 e11bot
 Type:Â SystemDouble
 e22bot
 Type:Â SystemDouble
 g12bot
 Type:Â SystemDouble
 emaxbot
 Type:Â SystemDouble
 eminbot
 Type:Â SystemDouble
 eanglebot
 Type:Â SystemDouble
 evmbot
 Type:Â SystemDouble
 g13avg
 Type:Â SystemDouble
 g23avg
 Type:Â SystemDouble
 gmaxavg
 Type:Â SystemDouble
 gangleavg
 Type:Â SystemDouble

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cAnalysisResults Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LSTF7 Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int AreaStrainShellLayered(
    string Name,
    eItemTypeElm ItemTypeElm,
    ref int NumberResults,
    ref string[] obj,
    ref string[] elm,
    ref string[] Layer,
    ref int[] IntPtNum,
    ref double[] IntPtLoc,
    ref string[] PointElm,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] E11,
    ref double[] E22,
    ref double[] G12,
    ref double[] EMax,
    ref double[] EMin,
    ref double[] EAngle,
    ref double[] EVM,
    ref double[] G13avg,
    ref double[] G23avg,
    ref double[] GMaxavg,
    ref double[] GAngleavg
)
```

```
Function AreaStrainShellLayered (
    Name As String,
    ItemTypeElm As eItemTypeElm,
    ByRef NumberResults As Integer,
    ByRef obj As String(),
    ByRef elm As String(),
    ByRef Layer As String(),
    ByRef IntPtNum As Integer(),
    ByRef IntPtLoc As Double(),
    ByRef PointElm As String(),
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef E11 As Double(),
    ByRef E22 As Double(),
    ByRef G12 As Double(),
    ByRef EMax As Double(),
```

cAnalysisResultsspan id="LSTF79B4D03_0"AddLanguageSpecificTextSet("LSTF79B4D03_0?cp72:|nu=.")

Introduction

```
ByRef EMin As Double(),  
ByRef EAngle As Double(),  
ByRef EVM As Double(),  
ByRef G13avg As Double(),  
ByRef G23avg As Double(),  
ByRef GMaxavg As Double(),  
ByRef GAngleavg As Double()  
) As Integer
```

```
Dim instance As cAnalysisResults  
Dim Name As String  
Dim ItemTypeElm As eItemTypeElm  
Dim NumberResults As Integer  
Dim obj As String()  
Dim elm As String()  
Dim Layer As String()  
Dim IntPtNum As Integer()  
Dim IntPtLoc As Double()  
Dim PointElm As String()  
Dim LoadCase As String()  
Dim StepType As String()  
Dim StepNum As Double()  
Dim E11 As Double()  
Dim E22 As Double()  
Dim G12 As Double()  
Dim EMax As Double()  
Dim EMin As Double()  
Dim EAngle As Double()  
Dim EVM As Double()  
Dim G13avg As Double()  
Dim G23avg As Double()  
Dim GMaxavg As Double()  
Dim GAngleavg As Double()  
Dim returnValue As Integer
```

```
returnValue = instance.AreaStrainShellLayered(Name,  
ItemTypeElm, NumberResults, obj,  
elm, Layer, IntPtNum, IntPtLoc, PointElm,  
LoadCase, StepType, StepNum, E11,  
E22, G12, EMax, EMin, EAngle, EVM, G13avg,  
G23avg, GMaxavg, GAngleavg)
```

```
int AreaStrainShellLayered(  
String^ Name,  
eItemTypeElm ItemTypeElm,  
int% NumberResults,  
array<String^>^% obj,  
array<String^>^% elm,  
array<String^>^% Layer,  
array<int>^% IntPtNum,  
array<double>^% IntPtLoc,  
array<String^>^% PointElm,  
array<String^>^% LoadCase,  
array<String^>^% StepType,  
array<double>^% StepNum,  
array<double>^% E11,  
array<double>^% E22,  
array<double>^% G12,  
array<double>^% EMax,  
array<double>^% EMin,  
array<double>^% EAngle,  
array<double>^% EVM,
```

```

array<double>^% G13avg,
array<double>^% G23avg,
array<double>^% GMaxavg,
array<double>^% GAngleavg
)

```

```

abstract AreaStrainShellLayered :
    Name : string *
    ItemTypeElm : eItemTypeElm *
    NumberResults : int byref *
    obj : string[] byref *
    elm : string[] byref *
    Layer : string[] byref *
    IntPtNum : int[] byref *
    IntPtLoc : float[] byref *
    PointElm : string[] byref *
    LoadCase : string[] byref *
    StepType : string[] byref *
    StepNum : float[] byref *
    E11 : float[] byref *
    E22 : float[] byref *
    G12 : float[] byref *
    EMax : float[] byref *
    EMin : float[] byref *
    EAngle : float[] byref *
    EVM : float[] byref *
    G13avg : float[] byref *
    G23avg : float[] byref *
    GMaxavg : float[] byref *
    GAngleavg : float[] byref -> int

```

Parameters

Name
Type:Â SystemString

ItemTypeElm
Type:Â ETABSv1eItemTypeElm

NumberResults
Type:Â SystemInt32

obj
Type:Â SystemString

elm
Type:Â SystemString

Layer
Type:Â SystemString

IntPtNum
Type:Â SystemInt32

IntPtLoc
Type:Â SystemDouble

PointElm
Type:Â SystemString

LoadCase
Type:Â SystemString

StepType
Type:Â SystemString

StepNum

E11 Type:Â [SystemDouble](#)
 E22 Type:Â [SystemDouble](#)
 G12 Type:Â [SystemDouble](#)
 EMax Type:Â [SystemDouble](#)
 EMin Type:Â [SystemDouble](#)
 EAngle Type:Â [SystemDouble](#)
 EVM Type:Â [SystemDouble](#)
 G13avg Type:Â [SystemDouble](#)
 G23avg Type:Â [SystemDouble](#)
 GMaxavg Type:Â [SystemDouble](#)
 GAngleavg Type:Â [SystemDouble](#)

Return Value

Type:Â [Int32](#)

▀ See Also

Reference

[cAnalysisResults Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LST92287A83_0")

Method

Reports the area stresses for the specified area elements that are assigned shell section properties. Stresses are reported at each point element associated with the area element

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int AreaStressShell(
    string Name,
    eItemTypeElm ItemTypeElm,
    ref int NumberResults,
    ref string[] Obj,
    ref string[] Elm,
    ref string[] PointElm,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] S11Top,
    ref double[] S22Top,
    ref double[] S12Top,
    ref double[] SMaxTop,
    ref double[] SMinTop,
    ref double[] SAngleTop,
    ref double[] SVMTop,
    ref double[] S11Bot,
    ref double[] S22Bot,
    ref double[] S12Bot,
    ref double[] SMaxBot,
    ref double[] SMinBot,
    ref double[] SAngleBot,
    ref double[] SVMBot,
    ref double[] S13Avg,
    ref double[] S23Avg,
    ref double[] SMaxAvg,
    ref double[] SAngleAvg
)
```

```
Function AreaStressShell (
    Name As String,
    ItemTypeElm As eItemTypeElm,
    ByRef NumberResults As Integer,
    ByRef Obj As String(),
    ByRef Elm As String(),
    ByRef PointElm As String(),
    ByRef LoadCase As String(),
    ByRef StepType As String(),
```

cAnalysisResultsspan id="LST92287A83_0"AddLanguageSpecificTextSet("LST92287A83_0?cpp76|nu=.");

Introduction

```
ByRef StepNum As Double(),  
ByRef S11Top As Double(),  
ByRef S22Top As Double(),  
ByRef S12Top As Double(),  
ByRef SMaxTop As Double(),  
ByRef SMinTop As Double(),  
ByRef SAngleTop As Double(),  
ByRef SVMTop As Double(),  
ByRef S11Bot As Double(),  
ByRef S22Bot As Double(),  
ByRef S12Bot As Double(),  
ByRef SMaxBot As Double(),  
ByRef SMinBot As Double(),  
ByRef SAngleBot As Double(),  
ByRef SVMBot As Double(),  
ByRef S13Avg As Double(),  
ByRef S23Avg As Double(),  
ByRef SMaxAvg As Double(),  
ByRef SAngleAvg As Double()  
) As Integer
```

```
Dim instance As cAnalysisResults  
Dim Name As String  
Dim ItemTypeElm As eItemTypeElm  
Dim NumberResults As Integer  
Dim Obj As String()  
Dim Elm As String()  
Dim PointElm As String()  
Dim LoadCase As String()  
Dim StepType As String()  
Dim StepNum As Double()  
Dim S11Top As Double()  
Dim S22Top As Double()  
Dim S12Top As Double()  
Dim SMaxTop As Double()  
Dim SMinTop As Double()  
Dim SAngleTop As Double()  
Dim SVMTop As Double()  
Dim S11Bot As Double()  
Dim S22Bot As Double()  
Dim S12Bot As Double()  
Dim SMaxBot As Double()  
Dim SMinBot As Double()  
Dim SAngleBot As Double()  
Dim SVMBot As Double()  
Dim S13Avg As Double()  
Dim S23Avg As Double()  
Dim SMaxAvg As Double()  
Dim SAngleAvg As Double()  
Dim returnValue As Integer
```

```
returnValue = instance.AreaStressShell(Name,  
ItemTypeElm, NumberResults, Obj,  
Elm, PointElm, LoadCase, StepType,  
StepNum, S11Top, S22Top, S12Top, SMaxTop,  
SMinTop, SAngleTop, SVMTop, S11Bot,  
S22Bot, S12Bot, SMaxBot, SMinBot,  
SAngleBot, SVMBot, S13Avg, S23Avg,  
SMaxAvg, SAngleAvg)
```

```
int AreaStressShell(  
String^ Name,
```

cAnalysisResultsspan id="LST92287A83_0"AddLanguageSpecificTextSet("LST92287A83_0?cpp77|nu=.");

```

eItemTypeElm ItemTypeElm,
int% NumberResults,
array<String^>^% Obj,
array<String^>^% Elm,
array<String^>^% PointElm,
array<String^>^% LoadCase,
array<String^>^% StepType,
array<double>^% StepNum,
array<double>^% S11Top,
array<double>^% S22Top,
array<double>^% S12Top,
array<double>^% SMaxTop,
array<double>^% SMinTop,
array<double>^% SAngleTop,
array<double>^% SVMTop,
array<double>^% S11Bot,
array<double>^% S22Bot,
array<double>^% S12Bot,
array<double>^% SMaxBot,
array<double>^% SMinBot,
array<double>^% SAngleBot,
array<double>^% SVMBot,
array<double>^% S13Avg,
array<double>^% S23Avg,
array<double>^% SMaxAvg,
array<double>^% SAngleAvg
)

abstract AreaStressShell :
    Name : string *
    ItemTypeElm : eItemTypeElm *
    NumberResults : int byref *
    Obj : string[] byref *
    Elm : string[] byref *
    PointElm : string[] byref *
    LoadCase : string[] byref *
    StepType : string[] byref *
    StepNum : float[] byref *
    S11Top : float[] byref *
    S22Top : float[] byref *
    S12Top : float[] byref *
    SMaxTop : float[] byref *
    SMinTop : float[] byref *
    SAngleTop : float[] byref *
    SVMTop : float[] byref *
    S11Bot : float[] byref *
    S22Bot : float[] byref *
    S12Bot : float[] byref *
    SMaxBot : float[] byref *
    SMinBot : float[] byref *
    SAngleBot : float[] byref *
    SVMBot : float[] byref *
    S13Avg : float[] byref *
    S23Avg : float[] byref *
    SMaxAvg : float[] byref *
    SAngleAvg : float[] byref -> int

```

Parameters

Name

Type:Â SystemString

The name of an existing area object, area element or group of objects, depending on the value of the ItemTypeElm item

ItemTypeElm

Type:Â ETABSv1eItemTypeElm

This is one of the following items in the eItemTypeElm enumeration.

If this item is ObjectElm, the result request is for the area elements corresponding to the area object specified by the Name item.

If this item is Element, the result request is for the area element specified by the Name item.

If this item is GroupElm, the result request is for the area elements corresponding to all area objects included in the group specified by the Name item.

If this item is SelectionElm, the result request is for area elements corresponding to all selected area objects and the Name item is ignored.

NumberResults

Type:Â SystemInt32

The total number of results returned by the program

Obj

Type:Â SystemString

This is an array that includes the area object name associated with each result, if any

Elm

Type:Â SystemString

This is an array that includes the area element name associated with each result

PointElm

Type:Â SystemString

This is an array that includes the name of the point element where the results are reported

LoadCase

Type:Â SystemString

This is an array that includes the name of the analysis case or load combination associated with each result

StepType

Type:Â SystemString

This is an array that includes the step type, if any, for each result

StepNum

Type:Â SystemDouble

This is an array that includes the step number, if any, for each result

S11Top

Type:Â SystemDouble

The area element internal S11 at the top of the specified area element, at the specified point element location, reported in the area element local coordinate system. [F/L2]

S22Top

Type:Â SystemDouble

The area element internal S22 at the top of the specified area element, at the

specified point element location, reported in the area element local coordinate system. [F/L2]

S12Top

Type: SystemDouble

The area element internal S12 at the top of the specified area element, at the specified point element location, reported in the area element local coordinate system. [F/L2]

SMaxTop

Type: SystemDouble

The area element maximum principal stress at the top of the specified area element, at the specified point element location. [F/L2]

SMinTop

Type: SystemDouble

The area element minimum principal stress at the top of the specified area element, at the specified point element location. [F/L2]

SAngleTop

Type: SystemDouble

The angle measured counter clockwise (when the local 3 axis is pointing toward you) from the area local 1 axis to the direction of the maximum principal stress, at the top of the specified area element. [deg]

SVMTop

Type: SystemDouble

The area element internal top Von Mises stress at the specified point element. [F/L2]

S11Bot

Type: SystemDouble

The area element internal S11 at the bottom of the specified area element, at the specified point element location, reported in the area element local coordinate system. [F/L2]

S22Bot

Type: SystemDouble

The area element internal S22 at the bottom of the specified area element, at the specified point element location, reported in the area element local coordinate system. [F/L2]

S12Bot

Type: SystemDouble

The area element internal S12 at the bottom of the specified area element, at the specified point element location, reported in the area element local coordinate system. [F/L2]

SMaxBot

Type: SystemDouble

The area element maximum principal stress at the bottom of the specified area element, at the specified point element location. [F/L2]

SMinBot

Type: SystemDouble

The area element minimum principal stress at the bottom of the specified area element, at the specified point element location. [F/L2]

SAngleBot

Type: SystemDouble

The angle measured counter clockwise (when the local 3 axis is pointing toward you) from the area local 1 axis to the direction of the maximum principal stress,

at the bottom of the specified area element. [deg]

SVMBot

Type: SystemDouble

The area element internal bottom Von Mises stress at the specified point element. [F/L2]

S13Avg

Type: SystemDouble

The area element average S13 out-of-plane shear stress at the specified point element. This item is only reported for area elements with properties that allow plate bending behavior. [F/L2]

S23Avg

Type: SystemDouble

The area element average S23 out-of-plane shear stress at the specified point element. This item is only reported for area elements with properties that allow plate bending behavior. [F/L2]

SMaxAvg

Type: SystemDouble

The area element maximum average out-of-plane shear stress. It is equal to the square root of the sum of the squares of S13Avg and S23Avg. This item is only reported for area elements with properties that allow plate bending behavior. [F/L2]

SAngleAvg

Type: SystemDouble

The angle measured counter clockwise (when the local 3 axis is pointing toward you) from the area local 1 axis to the direction of SMaxAvg. This item is only reported for area elements with properties that allow plate bending behavior. [deg]

Return Value

Type: Int32

Returns zero if the stresses are successfully recovered, otherwise it returns a nonzero value.

Remarks

See Results for more information.

Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim NumberResults As Integer  
    Dim Obj() As String  
    Dim ObjSta() As Double  
    Dim Elm() As String  
    Dim ElmSta() As Double  
    Dim LoadCase() As String  
    Dim StepType() As String  
    Dim StepNum() As Double  
    Dim S11Top() As Double  
    Dim S22Top() As Double  
    Dim S12Top() As Double  
    Dim SMaxTop() As Double
```

Introduction

```
Dim SMinTop() As Double
Dim SAngleTop() As Double
Dim SVMTop() As Double
Dim S11Bot() As Double
Dim S22Bot() As Double
Dim S12Bot() As Double
Dim SMaxBot() As Double
Dim SMinBot() As Double
Dim SAngleBot() As Double
Dim SVMBot() As Double
Dim S13Avg() As Double
Dim S23Avg() As Double
Dim SMaxAvg() As Double
Dim SAngleAvg() As Double

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'run analysis
System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
ret = SapModel.Analyze.RunAnalysis()

'deselect all cases and combos
ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()

'set case selected for output
ret = SapModel.Results.Setup.SetCaseSelectedForOutput("DEAD")

'get area stresses for area object "1"
ret = SapModel.Results.AreaStressShell("1", eItemTypeElm.ObjectElm, NumberResults, Obj, ET)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cAnalysisResults Interface

ETABSv1 Namespace

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»

CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LST3F02C79C_0")

Method

Reports the area stresses for the specified area elements that are assigned layered shell section properties

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int AreaStressShellLayered(
    string Name,
    eItemTypeElm ItemTypeElm,
    ref int NumberResults,
    ref string[] Obj,
    ref string[] Elm,
    ref string[] Layer,
    ref int[] IntPtNum,
    ref double[] IntPtLoc,
    ref string[] PointElm,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] S11,
    ref double[] S22,
    ref double[] S12,
    ref double[] SMax,
    ref double[] SMin,
    ref double[] SAngle,
    ref double[] SVM,
    ref double[] S13Avg,
    ref double[] S23Avg,
    ref double[] SMaxAvg,
    ref double[] SAngleAvg
)
```

```
Function AreaStressShellLayered (
    Name As String,
    ItemTypeElm As eItemTypeElm,
    ByRef NumberResults As Integer,
    ByRef Obj As String(),
    ByRef Elm As String(),
    ByRef Layer As String(),
    ByRef IntPtNum As Integer(),
    ByRef IntPtLoc As Double(),
    ByRef PointElm As String(),
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef S11 As Double(),
```

cAnalysisResultsspan id="LST3F02C79C_0"AddLanguageSpecificTextSet("LST3F02C79C_0?cp64::|nu=.")

Introduction

```
ByRef S22 As Double(),  
ByRef S12 As Double(),  
ByRef SMax As Double(),  
ByRef SMin As Double(),  
ByRef SAngle As Double(),  
ByRef SVM As Double(),  
ByRef S13Avg As Double(),  
ByRef S23Avg As Double(),  
ByRef SMaxAvg As Double(),  
ByRef SAngleAvg As Double()  
) As Integer
```

```
Dim instance As cAnalysisResults  
Dim Name As String  
Dim ItemTypeElm As eItemTypeElm  
Dim NumberResults As Integer  
Dim Obj As String()  
Dim Elm As String()  
Dim Layer As String()  
Dim IntPtNum As Integer()  
Dim IntPtLoc As Double()  
Dim PointElm As String()  
Dim LoadCase As String()  
Dim StepType As String()  
Dim StepNum As Double()  
Dim S11 As Double()  
Dim S22 As Double()  
Dim S12 As Double()  
Dim SMax As Double()  
Dim SMin As Double()  
Dim SAngle As Double()  
Dim SVM As Double()  
Dim S13Avg As Double()  
Dim S23Avg As Double()  
Dim SMaxAvg As Double()  
Dim SAngleAvg As Double()  
Dim returnValue As Integer
```

```
returnValue = instance.AreaStressShellLayered(Name,  
ItemTypeElm, NumberResults, Obj,  
Elm, Layer, IntPtNum, IntPtLoc, PointElm,  
LoadCase, StepType, StepNum, S11,  
S22, S12, SMax, SMin, SAngle, SVM, S13Avg,  
S23Avg, SMaxAvg, SAngleAvg)
```

```
int AreaStressShellLayered(  
String^ Name,  
eItemTypeElm ItemTypeElm,  
int% NumberResults,  
array<String^>^% Obj,  
array<String^>^% Elm,  
array<String^>^% Layer,  
array<int>^% IntPtNum,  
array<double>^% IntPtLoc,  
array<String^>^% PointElm,  
array<String^>^% LoadCase,  
array<String^>^% StepType,  
array<double>^% StepNum,  
array<double>^% S11,  
array<double>^% S22,  
array<double>^% S12,  
array<double>^% SMax,
```

cAnalysisResultsspan id="LST3F02C79C_0"AddLanguageSpecificTextSet("LST3F02C79C_0?cp65::|nu=.")

```

array<double>^% SMin,
array<double>^% SAngle,
array<double>^% SVM,
array<double>^% S13Avg,
array<double>^% S23Avg,
array<double>^% SMaxAvg,
array<double>^% SAngleAvg
)

```

```

abstract AreaStressShellLayered :
    Name : string *
    ItemTypeElm : eItemTypeElm *
    NumberResults : int byref *
    Obj : string[] byref *
    Elm : string[] byref *
    Layer : string[] byref *
    IntPtNum : int[] byref *
    IntPtLoc : float[] byref *
    PointElm : string[] byref *
    LoadCase : string[] byref *
    StepType : string[] byref *
    StepNum : float[] byref *
    S11 : float[] byref *
    S22 : float[] byref *
    S12 : float[] byref *
    SMax : float[] byref *
    SMin : float[] byref *
    SAngle : float[] byref *
    SVM : float[] byref *
    S13Avg : float[] byref *
    S23Avg : float[] byref *
    SMaxAvg : float[] byref *
    SAngleAvg : float[] byref -> int

```

Parameters

Name

Type: SystemString

The name of an existing area object, area element or group of objects, depending on the value of the ItemTypeElm item

ItemTypeElm

Type: ETABSv1eItemTypeElm

This is one of the following items in the eItemTypeElm enumeration.

If this item is ObjectElm, the result request is for the area elements corresponding to the area object specified by the Name item.

If this item is Element, the result request is for the area element specified by the Name item.

If this item is GroupElm, the result request is for the area elements corresponding to all area objects included in the group specified by the Name item.

If this item is SelectionElm, the result request is for area elements corresponding to all selected area objects and the Name item is ignored.

NumberResults

Parameters

	Type:Â <u>SystemInt32</u> The total number of results returned by the program
Obj	Type:Â <u>SystemString</u> This is an array that includes the area object name associated with each result, if any
Elm	Type:Â <u>SystemString</u> This is an array that includes the area element name associated with each result
Layer	Type:Â <u>SystemString</u> This is an array that includes the layer name associated with each result
IntPtNum	Type:Â <u>SystemInt32</u> This is an array that includes the integration point number within the specified layer of the area element
IntPtLoc	Type:Â <u>SystemDouble</u> This is an array that includes the integration point relative location within the specified layer of the area element. The location is between -1 (bottom of layer) and +1 (top of layer), inclusive. The midheight of the layer is at a value of 0
PointElm	Type:Â <u>SystemString</u> This is an array that includes the name of the point element where the results are reported
LoadCase	Type:Â <u>SystemString</u> This is an array that includes the name of the analysis case or load combination associated with each result
StepType	Type:Â <u>SystemString</u> This is an array that includes the step type, if any, for each result
StepNum	Type:Â <u>SystemDouble</u> This is an array that includes the step number, if any, for each result
S11	Type:Â <u>SystemDouble</u> The area element internal S11 stress at the specified point element location, for the specified layer and layer integration point, reported in the area element local coordinate system. [F/L2]
S22	Type:Â <u>SystemDouble</u> The area element internal S22 stress at the specified point element location, for the specified layer and layer integration point, reported in the area element local coordinate system. [F/L2]
S12	Type:Â <u>SystemDouble</u> The area element internal S12 stress at the specified point element location, for the specified layer and layer integration point, reported in the area element local coordinate system. [F/L2]
SMax	

Type: SystemDouble

The area element maximum principal stress at the specified point element location, for the specified layer and layer integration point. [F/L2]

SMin

Type: SystemDouble

The area element minimum principal stress at the specified point element location, for the specified layer and layer integration point. [F/L2]

SAngle

Type: SystemDouble

The angle measured counter clockwise (when the local 3 axis is pointing toward you) from the area local 1 axis to the direction of the maximum principal stress. [deg]

SVM

Type: SystemDouble

The area element internal Von Mises stress at the specified point element location, for the specified layer and layer integration point. [F/L2]

S13Avg

Type: SystemDouble

The area element average S13 out-of-plane shear stress at the specified point element location, for the specified layer and layer integration point. [F/L2]

S23Avg

Type: SystemDouble

The area element average S23 out-of-plane shear stress at the specified point element location, for the specified layer and layer integration point. [F/L2]

SMaxAvg

Type: SystemDouble

The area element maximum average out-of-plane shear stress for the specified layer and layer integration point. It is equal to the square root of the sum of the squares of S13Avg and S23Avg. [F/L2]

SAngleAvg

Type: SystemDouble

The angle measured counter clockwise (when the local 3 axis is pointing toward you) from the area local 1 axis to the direction of SMaxAvg. [deg]

Return Value

Type: Int32

Returns zero if the stresses are successfully recovered, otherwise it returns a nonzero value

Remarks

See [Results](#) for more information.

Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
  
    Dim MyNumberLayers As Integer  
    Dim MyLayerName() As String  
    Dim MyDist() As Double
```

Introduction

```
Dim MyThickness() As Double
Dim MyType() As Integer
Dim MyNumIntegrationPts() As Integer
Dim MyMatProp() As String
Dim MyMatAng() As Double
Dim MyS11Type() As Integer
Dim MyS22Type() As Integer
Dim MyS12Type() As Integer

Dim NumberResults As Integer
Dim Obj() As String
Dim ObjSta() As Double
Dim Elm() As String
Dim ElmSta() As Double
Dim LoadCase() As String
Dim StepType() As String
Dim StepNum() As Double
Dim S11() As Double
Dim S22() As Double
Dim S12() As Double
Dim SMax() As Double
Dim SMin() As Double
Dim SAngle() As Double
Dim SVM() As Double
Dim SVMBot() As Double
Dim S13Avg() As Double
Dim S23Avg() As Double
Dim SMaxAvg() As Double
Dim SAngleAvg() As Double

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new area property
ret = SapModel.PropArea.SetSlab("A1", eSlabType.Slab, eShellType.Layered, "4000Psi", 16.0)

'add A615Gr60 rebar material
ret = SapModel.PropMaterial.AddMaterial(Name, eMatType.Rebar, "United States", "ASTM A615")

'set area property layer parameters
MyNumberLayers = 5
ReDim MyLayerName(MyNumberLayers - 1)
ReDim MyDist(MyNumberLayers - 1)
ReDim MyThickness(MyNumberLayers - 1)
ReDim MyType(MyNumberLayers - 1)
ReDim MyNumIntegrationPts(MyNumberLayers - 1)
ReDim MyMatProp(MyNumberLayers - 1)
ReDim MyMatAng(MyNumberLayers - 1)
ReDim MyS11Type(MyNumberLayers - 1)
```

Introduction

```
ReDim MyS22Type(MyNumberLayers - 1)
ReDim MyS12Type(MyNumberLayers - 1)

MyLayerName(0) = "Concrete"
MyDist(0) = 0
MyThickness(0) = 16
MyType(0) = 1
MyNumIntegrationPts(0) = 2
MyMatProp(0) = "4000Psi"
MyMatAng(0) = 0
MyS11Type(0) = 1
MyS22Type(0) = 1
MyS12Type(0) = 1

MyLayerName(1) = "Top Bar 1"
MyDist(1) = 6
MyThickness(1) = 0.03
MyType(1) = 1
MyNumIntegrationPts(1) = 1
MyMatProp(1) = Name
MyMatAng(1) = 0
MyS11Type(1) = 1
MyS22Type(1) = 1
MyS12Type(1) = 1

MyLayerName(2) = "Top Bar 2"
MyDist(2) = 6
MyThickness(2) = 0.03
MyType(2) = 1
MyNumIntegrationPts(2) = 1
MyMatProp(2) = Name
MyMatAng(2) = 90
MyS11Type(2) = 1
MyS22Type(2) = 1
MyS12Type(2) = 1

MyLayerName(3) = "Bot Bar 1"
MyDist(3) = -6
MyThickness(3) = 0.03
MyType(3) = 1
MyNumIntegrationPts(3) = 1
MyMatProp(3) = Name
MyMatAng(3) = 0
MyS11Type(3) = 1
MyS22Type(3) = 1
MyS12Type(3) = 1

MyLayerName(4) = "Bot Bar 2"
MyDist(4) = -6
MyThickness(4) = 0.03
MyType(4) = 1
MyNumIntegrationPts(4) = 1
MyMatProp(4) = Name
MyMatAng(4) = 90
MyS11Type(4) = 1
MyS22Type(4) = 1
MyS12Type(4) = 1

ret = SapModel.PropArea.SetShellLayer_1("A1", MyNumberLayers, MyLayerName, MyDist, MyThick

'set area property
ret = SapModel.AreaObj.SetProperty("4", "A1")
```

Introduction

```
'run analysis
  System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
  ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
  ret = SapModel.Analyze.RunAnalysis()

'deselect all cases and combos
  ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()

'set case selected for output
  ret = SapModel.Results.Setup.SetCaseSelectedForOutput("DEAD")

'get area stresses for area object "4"
  ret = SapModel.Results.AreaStressShellLayered("4", eItemTypeElm.ObjectElm, NumberResults,

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cAnalysisResults Interface](#)

[ETABSv1 Namespace](#)

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»

CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LSTE5113990_0?cpp92|nu=.");

Method

Reports the assembled joint masses for the specified point elements

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int AssembledJointMass(
    string Name,
    eItemTypeElm ItemTypeElm,
    ref int NumberResults,
    ref string[] PointElm,
    ref double[] U1,
    ref double[] U2,
    ref double[] U3,
    ref double[] R1,
    ref double[] R2,
    ref double[] R3
)
```

```
Function AssembledJointMass (
    Name As String,
    ItemTypeElm As eItemTypeElm,
    ByRef NumberResults As Integer,
    ByRef PointElm As String(),
    ByRef U1 As Double(),
    ByRef U2 As Double(),
    ByRef U3 As Double(),
    ByRef R1 As Double(),
    ByRef R2 As Double(),
    ByRef R3 As Double()
) As Integer
```

```
Dim instance As cAnalysisResults
Dim Name As String
Dim ItemTypeElm As eItemTypeElm
Dim NumberResults As Integer
Dim PointElm As String()
Dim U1 As Double()
Dim U2 As Double()
Dim U3 As Double()
Dim R1 As Double()
Dim R2 As Double()
Dim R3 As Double()
Dim returnValue As Integer
```

```
returnValue = instance.AssembledJointMass(Name,
    ItemTypeElm, NumberResults, PointElm,
```

cAnalysisResultsspan id="LSTE5113990_0"AddLanguageSpecificTextSet("LSTE5113990_0?cpp92|nu=.");

U1, U2, U3, R1, R2, R3)

```
int AssembledJointMass(
String^ Name,
eItemTypeElm ItemTypeElm,
int% NumberResults,
array<String^>^% PointElm,
array<double>^% U1,
array<double>^% U2,
array<double>^% U3,
array<double>^% R1,
array<double>^% R2,
array<double>^% R3
)
```

```
abstract AssembledJointMass :
    Name : string *
    ItemTypeElm : eItemTypeElm *
    NumberResults : int byref *
    PointElm : string[] byref *
    U1 : float[] byref *
    U2 : float[] byref *
    U3 : float[] byref *
    R1 : float[] byref *
    R2 : float[] byref *
    R3 : float[] byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing point object, point element, or group of objects depending on the value of the ItemTypeElm item.

ItemTypeElm

Type:Â ETABSv1eItemTypeElm

This is one of the following items in the eItemTypeElm enumeration.

If this item is ObjectElm, the result request is for the point element corresponding to the point object specified by the Name item.

If this item is Element, the result request is for the point element specified by the Name item.

If this item is GroupElm, the result request is for all point elements directly or indirectly specified in the group specified by the Name item.

If this item is SelectionElm, the result request is for all point elements directly or indirectly selected and the Name item is ignored.

NumberResults

Type:Â SystemInt32

The total number of results returned by the program.

PointElm

Type:Â SystemString

This is an array that includes the point element name associated with each result

U1

Type: SystemDouble

This array contains the translational mass in the point element local 1 direction for each result. [M]

U2

Type: SystemDouble

This array contains the translational mass in the point element local 2 direction for each result. [M]

U3

Type: SystemDouble

This array contains the translational mass in the point element local 3 direction for each result. [M]

R1

Type: SystemDouble

This array contains the rotational mass moment of inertia about the point element local 1 axis for each result. [ML2]

R2

Type: SystemDouble

This array contains the rotational mass moment of inertia about the point element local 2 axis for each result. [ML2]

R3

Type: SystemDouble

This array contains the rotational mass moment of inertia about the point element local 3 axis for each result. [ML2]

Return Value

Type: Int32

Returns zero if the masses are successfully recovered, otherwise it returns a nonzero value

Remarks

See [Results](#) for more information.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberResults As Integer
    Dim Obj() As String
    Dim Elm() As String
    Dim PointElm() As String
    Dim U1() As Double
    Dim U2() As Double
    Dim U3() As Double
    Dim R1() As Double
    Dim R2() As Double
    Dim R3() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

Introduction

```
'start ETABS application
    ret = EtabsObject.ApplicationStart()

'create SapModel object
    SapModel = EtabsObject.SapModel

'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis()

'get assembled joint mass for all point elements
    ret = SapModel.Results.AssembledJointMass("ALL", eItemTypeElm.GroupElm, NumberResults, Poi

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cAnalysisResults Interface

ETABSv1 Namespace

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»

CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LSTAE68B025_0") Method

Reports the structure total base reactions

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int BaseReact(
    ref int NumberResults,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] FX,
    ref double[] FY,
    ref double[] FZ,
    ref double[] MX,
    ref double[] ParamMy,
    ref double[] MZ,
    ref double GX,
    ref double GY,
    ref double GZ
)
```

```
Function BaseReact (
    ByRef NumberResults As Integer,
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef FX As Double(),
    ByRef FY As Double(),
    ByRef FZ As Double(),
    ByRef MX As Double(),
    ByRef ParamMy As Double(),
    ByRef MZ As Double(),
    ByRef GX As Double,
    ByRef GY As Double,
    ByRef GZ As Double
) As Integer
```

```
Dim instance As cAnalysisResults
Dim NumberResults As Integer
Dim LoadCase As String()
Dim StepType As String()
Dim StepNum As Double()
Dim FX As Double()
Dim FY As Double()
Dim FZ As Double()
Dim MX As Double()
```

```

Dim ParamMy As Double()
Dim MZ As Double()
Dim GX As Double
Dim GY As Double
Dim GZ As Double
Dim returnValue As Integer

returnValue = instance.BaseReact(NumberResults,
LoadCase, StepType, StepNum, FX, FY,
FZ, MX, ParamMy, MZ, GX, GY, GZ)

int BaseReact(
int% NumberResults,
array<String^>^% LoadCase,
array<String^>^% StepType,
array<double>^% StepNum,
array<double>^% FX,
array<double>^% FY,
array<double>^% FZ,
array<double>^% MX,
array<double>^% ParamMy,
array<double>^% MZ,
double% GX,
double% GY,
double% GZ
)

abstract BaseReact :
    NumberResults : int byref *
    LoadCase : string[] byref *
    StepType : string[] byref *
    StepNum : float[] byref *
    FX : float[] byref *
    FY : float[] byref *
    FZ : float[] byref *
    MX : float[] byref *
    ParamMy : float[] byref *
    MZ : float[] byref *
    GX : float byref *
    GY : float byref *
    GZ : float byref -> int

```

Parameters

NumberResults

Type:Â SystemInt32

The total number of results returned by the program

LoadCase

Type:Â SystemString

This is an array that includes the name of the analysis case or load combination associated with each result

StepType

Type:Â SystemString

This is an array that includes the step type, if any, for each result

StepNum

Type:Â SystemDouble

This is an array that includes the step number, if any, for each result

FX

	Type: <u>SystemDouble</u> This array contains the base reaction force in the global X direction for each result. [F]
FY	Type: <u>SystemDouble</u> This array contains the base reaction force in the global Y direction for each result. [F]
FZ	Type: <u>SystemDouble</u> This array contains the base reaction force in the global Z direction for each result. [F]
MX	Type: <u>SystemDouble</u> This array contains the base reaction moment about the global X axis for each result. [FL]
ParamMy	Type: <u>SystemDouble</u> This array contains the base reaction moment about the global Y axis for each result. [FL]
MZ	Type: <u>SystemDouble</u> This array contains the base reaction moment about the global Z axis for each result. [FL]
GX	Type: <u>SystemDouble</u> The global X coordinate of the point at which the base reactions are reported. [L]
GY	Type: <u>SystemDouble</u> The global Y coordinate of the point at which the base reactions are reported. [L]
GZ	Type: <u>SystemDouble</u> The global Z coordinate of the point at which the base reactions are reported. [L]

Return Value

Type: Int32

Returns zero if the reactions are successfully recovered, otherwise it returns a nonzero value

Remarks

See [Results](#) for more information.

Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim NumberResults As Integer  
    Dim LoadCase() As String
```

Introduction

```
Dim StepType() As String
Dim StepNum() As Double
Dim Fx() As Double
Dim Fy() As Double
Dim Fz() As Double
Dim Mx() As Double
Dim ParamMy() As Double
Dim Mz() As Double
Dim gx as Double
Dim gy as Double
Dim gz as Double

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'run analysis
System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
ret = SapModel.Analyze.RunAnalysis()

'deselect all cases and combos
ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()

'set case selected for output
ret = SapModel.Results.Setup.SetCaseSelectedForOutput("DEAD")

'get base reactions
ret = SapModel.Results.BaseReact(NumberResults, LoadCase, StepType, StepNum, Fx, Fy, Fz, Mx, My, Mz)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

See Also

Reference

[cAnalysisResults Interface](#)

[ETABSv1 Namespace](#)

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Introduction

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CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LST323958BB_0?cp101:|nu=.")

Method

Reports the structure total base reactions and includes information on the centroid of the translational reaction forces

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int BaseReactWithCentroid(
    ref int NumberResults,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] FX,
    ref double[] FY,
    ref double[] FZ,
    ref double[] MX,
    ref double[] ParamMy,
    ref double[] MZ,
    ref double GX,
    ref double GY,
    ref double GZ,
    ref double[] XCentroidForFX,
    ref double[] YCentroidForFX,
    ref double[] ZCentroidForFX,
    ref double[] XCentroidForFY,
    ref double[] YCentroidForFY,
    ref double[] ZCentroidForFY,
    ref double[] XCentroidForFZ,
    ref double[] YCentroidForFZ,
    ref double[] ZCentroidForFZ
)
```

```
Function BaseReactWithCentroid (
    ByRef NumberResults As Integer,
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef FX As Double(),
    ByRef FY As Double(),
    ByRef FZ As Double(),
    ByRef MX As Double(),
    ByRef ParamMy As Double(),
    ByRef MZ As Double(),
    ByRef GX As Double,
    ByRef GY As Double,
    ByRef GZ As Double,
    ByRef XCentroidForFX As Double(),
```

Introduction

```
ByRef YCentroidForFX As Double(),  
ByRef ZCentroidForFX As Double(),  
ByRef XCentroidForFY As Double(),  
ByRef YCentroidForFY As Double(),  
ByRef ZCentroidForFY As Double(),  
ByRef XCentroidForFZ As Double(),  
ByRef YCentroidForFZ As Double(),  
ByRef ZCentroidForFZ As Double()  
) As Integer
```

```
Dim instance As cAnalysisResults  
Dim NumberResults As Integer  
Dim LoadCase As String()  
Dim StepType As String()  
Dim StepNum As Double()  
Dim FX As Double()  
Dim FY As Double()  
Dim FZ As Double()  
Dim MX As Double()  
Dim ParamMy As Double()  
Dim MZ As Double()  
Dim GX As Double  
Dim GY As Double  
Dim GZ As Double  
Dim XCentroidForFX As Double()  
Dim YCentroidForFX As Double()  
Dim ZCentroidForFX As Double()  
Dim XCentroidForFY As Double()  
Dim YCentroidForFY As Double()  
Dim ZCentroidForFY As Double()  
Dim XCentroidForFZ As Double()  
Dim YCentroidForFZ As Double()  
Dim ZCentroidForFZ As Double()  
Dim returnValue As Integer
```

```
returnValue = instance.BaseReactWithCentroid(NumberResults,  
LoadCase, StepType, StepNum, FX, FY,  
FZ, MX, ParamMy, MZ, GX, GY, GZ, XCentroidForFX,  
YCentroidForFX, ZCentroidForFX,  
XCentroidForFY, YCentroidForFY,  
ZCentroidForFY, XCentroidForFZ,  
YCentroidForFZ, ZCentroidForFZ)
```

```
int BaseReactWithCentroid(  
int% NumberResults,  
array<String^>^% LoadCase,  
array<String^>^% StepType,  
array<double>^% StepNum,  
array<double>^% FX,  
array<double>^% FY,  
array<double>^% FZ,  
array<double>^% MX,  
array<double>^% ParamMy,  
array<double>^% MZ,  
double% GX,  
double% GY,  
double% GZ,  
array<double>^% XCentroidForFX,  
array<double>^% YCentroidForFX,  
array<double>^% ZCentroidForFX,  
array<double>^% XCentroidForFY,  
array<double>^% YCentroidForFY,
```

```

array<double>^% ZCentroidForFY,
array<double>^% XCentroidForFZ,
array<double>^% YCentroidForFZ,
array<double>^% ZCentroidForFZ
)

abstract BaseReactWithCentroid :
    NumberResults : int byref *
    LoadCase : string[] byref *
    StepType : string[] byref *
    StepNum : float[] byref *
    FX : float[] byref *
    FY : float[] byref *
    FZ : float[] byref *
    MX : float[] byref *
    ParamMy : float[] byref *
    MZ : float[] byref *
    GX : float byref *
    GY : float byref *
    GZ : float byref *
    XCentroidForFX : float[] byref *
    YCentroidForFX : float[] byref *
    ZCentroidForFX : float[] byref *
    XCentroidForFY : float[] byref *
    YCentroidForFY : float[] byref *
    ZCentroidForFY : float[] byref *
    XCentroidForFZ : float[] byref *
    YCentroidForFZ : float[] byref *
    ZCentroidForFZ : float[] byref -> int

```

Parameters

NumberResults

Type:Â SystemInt32

The total number of results returned by the program

LoadCase

Type:Â SystemString

This is an array that includes the name of the analysis case or load combination associated with each result

StepType

Type:Â SystemString

This is an array that includes the step type, if any, for each result

StepNum

Type:Â SystemDouble

This is an array that includes the step number, if any, for each result

FX

Type:Â SystemDouble

This array contains the base reaction force in the global X direction for each result. [F]

FY

Type:Â SystemDouble

This array contains the base reaction force in the global Y direction for each result. [F]

FZ

Type:Â SystemDouble

This array contains the base reaction force in the global Z direction for each

result. [F]

MX
Type: SystemDouble
This array contains the base reaction moment about the global X axis for each result. [FL]

ParamMy
Type: SystemDouble
This array contains the base reaction moment about the global Y axis for each result. [FL]

MZ
Type: SystemDouble
This array contains the base reaction moment about the global Z axis for each result. [FL]

GX
Type: SystemDouble
The global X coordinate of the point at which the base reactions are reported. [L]

GY
Type: SystemDouble
The global Y coordinate of the point at which the base reactions are reported. [L]

GZ
Type: SystemDouble
The global Z coordinate of the point at which the base reactions are reported. [L]

XCentroidForFX
Type: SystemDouble
This is an array of the global X coordinates of the centroid of all global X-direction translational reaction forces for each result. [L]

YCentroidForFX
Type: SystemDouble
This is an array of the global Y coordinates of the centroid of all global X-direction translational reaction forces for each result. [L]

ZCentroidForFX
Type: SystemDouble
This is an array of the global Z coordinates of the centroid of all global X-direction translational reaction forces for each result. [L]

XCentroidForFY
Type: SystemDouble
This is an array of the global X coordinates of the centroid of all global Y-direction translational reaction forces for each result. [L]

YCentroidForFY
Type: SystemDouble
This is an array of the global Y coordinates of the centroid of all global Y-direction translational reaction forces for each result. [L]

ZCentroidForFY
Type: SystemDouble
This is an array of the global Z coordinates of the centroid of all global Y-direction translational reaction forces for each result. [L]

XCentroidForFZ

Type: SystemDouble

This is an array of the global X coordinates of the centroid of all global Z-direction translational reaction forces for each result. [L]

YCentroidForFZ

Type: SystemDouble

This is an array of the global Y coordinates of the centroid of all global Z-direction translational reaction forces for each result. [L]

ZCentroidForFZ

Type: SystemDouble

This is an array of the global Z coordinates of the centroid of all global Z-direction translational reaction forces for each result. [L]

Return Value

Type: Int32

Returns zero if the reactions are successfully recovered, otherwise it returns a nonzero value

Remarks

See Results for more information. Note that the reported base reaction centroids are not the same as the centroid of the applied loads

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberResults As Integer
    Dim LoadCase() As String
    Dim StepType() As String
    Dim StepNum() As Double
    Dim Fx() As Double
    Dim Fy() As Double
    Dim Fz() As Double
    Dim Mx() As Double
    Dim ParamMy() As Double
    Dim Mz() As Double
    Dim gx As Double
    Dim gy As Double
    Dim gz As Double
    Dim XCentroidForFx() As Double
    Dim YCentroidForFx() As Double
    Dim ZCentroidForFx() As Double
    Dim XCentroidForFy() As Double
    Dim YCentroidForFy() As Double
    Dim ZCentroidForFy() As Double
    Dim XCentroidForFz() As Double
    Dim YCentroidForFz() As Double
    Dim ZCentroidForFz() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()
```

Introduction

```
'create SapModel object
    SapModel = EtabsObject.SapModel

'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis()

'deselect all cases and combos
    ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()

'set case selected for output
    ret = SapModel.Results.Setup.SetCaseSelectedForOutput("DEAD")

'get base reactions with centroids
    ret = SapModel.Results.BaseReactWithCentroid(NumberResults, LoadCase, StepType, StepNum, F

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

[cAnalysisResults Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LSTF3 Method

Reports buckling factors obtained from buckling load cases

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int BucklingFactor(
    ref int NumberResults,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] Factor
)
```

```
Function BucklingFactor (
    ByRef NumberResults As Integer,
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef Factor As Double()
) As Integer
```

```
Dim instance As cAnalysisResults
Dim NumberResults As Integer
Dim LoadCase As String()
Dim StepType As String()
Dim StepNum As Double()
Dim Factor As Double()
Dim returnValue As Integer
```

```
returnValue = instance.BucklingFactor(NumberResults,
    LoadCase, StepType, StepNum, Factor)
```

```
int BucklingFactor(
    int% NumberResults,
    array<String^>^% LoadCase,
    array<String^>^% StepType,
    array<double>^% StepNum,
    array<double>^% Factor
)
```

```
abstract BucklingFactor :
    NumberResults : int byref *
    LoadCase : string[] byref *
    StepType : string[] byref *
    StepNum : float[] byref *
    Factor : float[] byref -> int
```

cAnalysisResultsspan id="LSTF34E86BD_0"AddLanguageSpecificTextSet("LSTF34E86BD_0?cp07::|nu=".

Parameters**NumberResults**Type: SystemInt32

The total number of results returned by the program

LoadCaseType: SystemString

This is an array that includes the name of the analysis case or load combination associated with each result

StepTypeType: SystemString

This is an array that includes the step type for each result. For buckling factors, the step type is always Mode

StepNumType: SystemDouble

This is an array that includes the step number for each result. For buckling factors, the step number is always the buckling mode number

FactorType: SystemDouble

This is an array that includes the buckling factors

Return ValueType: Int32

Returns zero if the factors are successfully recovered, otherwise it returns a nonzero value

▲ **Remarks**See Results for more information.▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberResults As Integer
    Dim LoadCase() As String
    Dim StepType() As String
    Dim StepNum() As Double
    Dim Factor() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
```


Introduction

```
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'run analysis
System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
ret = SapModel.Analyze.RunAnalysis()

'deselect all cases and combos
ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()

'set case selected for output
ret = SapModel.Results.Setup.SetCaseSelectedForOutput("DEAD")

'get buckling factors
ret = SapModel.Results.BucklingFactor(NumberResults, LoadCase, StepType, StepNum, Factor)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cAnalysisResults Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LSTE88ADBAD_0") Method

Reports the frame forces for the specified line elements

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int FrameForce(
    string Name,
    eItemTypeElm ItemTypeElm,
    ref int NumberResults,
    ref string[] Obj,
    ref double[] ObjSta,
    ref string[] Elm,
    ref double[] ElmSta,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] P,
    ref double[] V2,
    ref double[] V3,
    ref double[] T,
    ref double[] M2,
    ref double[] M3
)
```

```
Function FrameForce (
    Name As String,
    ItemTypeElm As eItemTypeElm,
    ByRef NumberResults As Integer,
    ByRef Obj As String(),
    ByRef ObjSta As Double(),
    ByRef Elm As String(),
    ByRef ElmSta As Double(),
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef P As Double(),
    ByRef V2 As Double(),
    ByRef V3 As Double(),
    ByRef T As Double(),
    ByRef M2 As Double(),
    ByRef M3 As Double()
) As Integer
```

```
Dim instance As cAnalysisResults
Dim Name As String
Dim ItemTypeElm As eItemTypeElm
```

cAnalysisResultsspan id="LSTE88ADBAD_0"AddLanguageSpecificTextSet("LSTE88ADBAD_0?b10=::|nu=

Introduction

```
Dim NumberResults As Integer
Dim Obj As String()
Dim ObjSta As Double()
Dim Elm As String()
Dim ElmSta As Double()
Dim LoadCase As String()
Dim StepType As String()
Dim StepNum As Double()
Dim P As Double()
Dim V2 As Double()
Dim V3 As Double()
Dim T As Double()
Dim M2 As Double()
Dim M3 As Double()
Dim returnValue As Integer

returnValue = instance.FrameForce(Name,
ItemtypeElm, NumberResults, Obj,
ObjSta, Elm, ElmSta, LoadCase, StepType,
StepNum, P, V2, V3, T, M2, M3)

int FrameForce(
String^ Name,
eItemTypeElm ItemTypeElm,
int% NumberResults,
array<String^>^% Obj,
array<double>^% ObjSta,
array<String^>^% Elm,
array<double>^% ElmSta,
array<String^>^% LoadCase,
array<String^>^% StepType,
array<double>^% StepNum,
array<double>^% P,
array<double>^% V2,
array<double>^% V3,
array<double>^% T,
array<double>^% M2,
array<double>^% M3
)

abstract FrameForce :
    Name : string *
    ItemTypeElm : eItemTypeElm *
    NumberResults : int byref *
    Obj : string[] byref *
    ObjSta : float[] byref *
    Elm : string[] byref *
    ElmSta : float[] byref *
    LoadCase : string[] byref *
    StepType : string[] byref *
    StepNum : float[] byref *
    P : float[] byref *
    V2 : float[] byref *
    V3 : float[] byref *
    T : float[] byref *
    M2 : float[] byref *
    M3 : float[] byref -> int
```

Parameters**Name**

Type:Â SystemString

The name of an existing line object, line element or group of objects, depending on the value of the ItemTypeElm item.

ItemTypeElm

Type:Â ETABSv1eItemTypeElm

This is one of the following items in the eItemTypeElm enumeration.

If this item is ObjectElm, the result request is for the line elements corresponding to the line object specified by the Name item.

If this item is Element, the result request is for the line element specified by the Name item.

If this item is GroupElm, the result request is for the line elements corresponding to all line objects included in the group specified by the Name item.

If this item is SelectionElm, the result request is for line elements corresponding to all selected line objects and the Name item is ignored.

NumberResults

Type:Â SystemInt32

The total number of results returned by the program

Obj

Type:Â SystemString

This is an array that includes the line object name associated with each result, if any

ObjSta

Type:Â SystemDouble

This is an array that includes the distance measured from the I-end of the line object to the result location

Elm

Type:Â SystemString

This is an array that includes the line element name associated with each result

ElmSta

Type:Â SystemDouble

This is an array that includes the distance measured from the I-end of the line element to the result location

LoadCase

Type:Â SystemString

This is an array that includes the name of the analysis case or load combination associated with each result

StepType

Type:Â SystemString

This is an array that includes the step type, if any, for each result

StepNum

Type:Â SystemDouble

This is an array that includes the step number, if any, for each result

P

- Type: SystemDouble
This is a one dimensional array of the axial force for each result. [F]
- V2
Type: SystemDouble
This is a one dimensional array of the shear force in the local 2 direction for each result. [F]
- V3
Type: SystemDouble
This is a one dimensional array of the shear force in the local 3 direction for each result. [F]
- T
Type: SystemDouble
This is a one dimensional array of the torsion for each result. [FL]
- M2
Type: SystemDouble
This is a one dimensional array of the moment about the local 2-axis for each result. [FL]
- M3
Type: SystemDouble
This is a one dimensional array of the moment about the local 3-axis for each result. [FL]

Return Value

Type: Int32

Returns zero if the forces are successfully recovered, otherwise it returns a nonzero value

Remarks

See Results for more information.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberResults As Integer
    Dim Obj() As String
    Dim ObjSta() As Double
    Dim Elm() As String
    Dim ElmSta() As Double
    Dim LoadCase() As String
    Dim StepType() As String
    Dim StepNum() As Double
    Dim P() As Double
    Dim V2() As Double
    Dim V3() As Double
    Dim T() As Double
    Dim M2() As Double
    Dim M3() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

Introduction

```
'start ETABS application
    ret = EtabsObject.ApplicationStart()

'create SapModel object
    SapModel = EtabsObject.SapModel

'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis()

'deselect all cases and combos
    ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()

'set case selected for output
    ret = SapModel.Results.Setup.SetCaseSelectedForOutput("DEAD")

'get frame forces
    ret = SapModel.Results.FrameForce("1", eItemTypeElm.ObjectElm, NumberResults, Obj, ObjSta, ObjSta2)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cAnalysisResults Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LST8A Method

Reports the frame joint forces for the point elements at each end of the specified line elements

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int FrameJointForce(
    string Name,
    eItemTypeElm ItemTypeElm,
    ref int NumberResults,
    ref string[] Obj,
    ref string[] Elm,
    ref string[] PointElm,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] F1,
    ref double[] F2,
    ref double[] F3,
    ref double[] M1,
    ref double[] M2,
    ref double[] M3
)
```

```
Function FrameJointForce (
    Name As String,
    ItemTypeElm As eItemTypeElm,
    ByRef NumberResults As Integer,
    ByRef Obj As String(),
    ByRef Elm As String(),
    ByRef PointElm As String(),
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef F1 As Double(),
    ByRef F2 As Double(),
    ByRef F3 As Double(),
    ByRef M1 As Double(),
    ByRef M2 As Double(),
    ByRef M3 As Double()
) As Integer
```

```
Dim instance As cAnalysisResults
Dim Name As String
Dim ItemTypeElm As eItemTypeElm
Dim NumberResults As Integer
```

cAnalysisResultsspan id="LST8AF6F3DC_0"AddLanguageSpecificTextSet("LST8AF6F3DC_0?c15=::|nu=.

```

Dim Obj As String()
Dim Elm As String()
Dim PointElm As String()
Dim LoadCase As String()
Dim StepType As String()
Dim StepNum As Double()
Dim F1 As Double()
Dim F2 As Double()
Dim F3 As Double()
Dim M1 As Double()
Dim M2 As Double()
Dim M3 As Double()
Dim returnValue As Integer

returnValue = instance.FrameJointForce(Name,
ItemtypeElm, NumberResults, Obj,
Elm, PointElm, LoadCase, StepType,
StepNum, F1, F2, F3, M1, M2, M3)

```

```

int FrameJointForce(
String^ Name,
eItemTypeElm ItemTypeElm,
int% NumberResults,
array<String^>^% Obj,
array<String^>^% Elm,
array<String^>^% PointElm,
array<String^>^% LoadCase,
array<String^>^% StepType,
array<double>^% StepNum,
array<double>^% F1,
array<double>^% F2,
array<double>^% F3,
array<double>^% M1,
array<double>^% M2,
array<double>^% M3
)

```

```

abstract FrameJointForce :
    Name : string *
    ItemTypeElm : eItemTypeElm *
    NumberResults : int byref *
    Obj : string[] byref *
    Elm : string[] byref *
    PointElm : string[] byref *
    LoadCase : string[] byref *
    StepType : string[] byref *
    StepNum : float[] byref *
    F1 : float[] byref *
    F2 : float[] byref *
    F3 : float[] byref *
    M1 : float[] byref *
    M2 : float[] byref *
    M3 : float[] byref -> int

```

Parameters

Name

Type: SystemString

The name of an existing line object, line element or group of objects, depending on the value of the ItemTypeElm item.

ItemTypeElm

Type:Â ETABSv1eItemTypeElm

This is one of the following items in the eItemTypeElm enumeration.

If this item is ObjectElm, the result request is for the line elements corresponding to the line object specified by the Name item.

If this item is Element, the result request is for the line element specified by the Name item.

If this item is GroupElm, the result request is for the line elements corresponding to all line objects included in the group specified by the Name item.

If this item is SelectionElm, the result request is for line elements corresponding to all selected line objects and the Name item is ignored.

NumberResults

Type:Â SystemInt32

The total number of results returned by the program

Obj

Type:Â SystemString

This is an array that includes the line object name associated with each result, if any

Elm

Type:Â SystemString

This is an array that includes the line element name associated with each result

PointElm

Type:Â SystemString

This is an array that includes the point element name associated with each result

LoadCase

Type:Â SystemString

This is an array that includes the name of the analysis case or load combination associated with each result

StepType

Type:Â SystemString

This is an array that includes the step type, if any, for each result

StepNum

Type:Â SystemDouble

This is an array that includes the step number, if any, for each result

F1

Type:Â SystemDouble

These is a one dimensional array that includes the joint force component in the point element local 1-axis direction [F]

F2

Type:Â SystemDouble

These is a one dimensional array that includes the joint force component in the point element local 2-axis direction [F]

F3

Type:Â SystemDouble

These is a one dimensional array that includes the joint force component in the

point element local 3-axis direction [F]

M1

Type: SystemDouble

These is a one dimensional array that includes the joint moment component about the point element local 1-axis. [FL]

M2

Type: SystemDouble

These is a one dimensional array that includes the joint moment component about the point element local 2-axis. [FL]

M3

Type: SystemDouble

These is a one dimensional array that includes the joint moment component about the point element local 3-axis. [FL]

Return Value

Type: Int32

Returns zero if the forces are successfully recovered, otherwise it returns a nonzero value

Remarks

See Results for more information.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberResults As Integer
    Dim Obj() As String
    Dim Elm() As String
    Dim PointElm() As String
    Dim LoadCase() As String
    Dim StepType() As String
    Dim StepNum() As Double
    Dim F1() As Double
    Dim F2() As Double
    Dim F3() As Double
    Dim M1() As Double
    Dim M2() As Double
    Dim M3() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)
```

Introduction

```
'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis()

'deselect all cases and combos
    ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()

'set case selected for output
    ret = SapModel.Results.Setup.SetCaseSelectedForOutput("DEAD")

'get frame joint forces for line object "1"
    ret = SapModel.Results.FrameJointForce("1", eItemTypeElm.ObjectElm, NumberResults, Obj, El)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cAnalysisResults Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LST95D3B2B2_0?cp120::|nu=")

Method

Reports the displacement values for the specified generalized displacements

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GeneralizedDispl(
    string Name,
    ref int NumberResults,
    ref string[] GD,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref string[] DType,
    ref double[] Value
)
```

```
Function GeneralizedDispl (
    Name As String,
    ByRef NumberResults As Integer,
    ByRef GD As String(),
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef DType As String(),
    ByRef Value As Double()
) As Integer
```

```
Dim instance As cAnalysisResults
Dim Name As String
Dim NumberResults As Integer
Dim GD As String()
Dim LoadCase As String()
Dim StepType As String()
Dim StepNum As Double()
Dim DType As String()
Dim Value As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GeneralizedDispl(Name,
    NumberResults, GD, LoadCase, StepType,
    StepNum, DType, Value)
```

```
int GeneralizedDispl(
    String^ Name,
    int% NumberResults,
    array<String^>^% GD,
```

cAnalysisResultsspan id="LST95D3B2B2_0"AddLanguageSpecificTextSet("LST95D3B2B2_0?cp120::|nu=")

```

array<String>^% LoadCase,
array<String>^% StepType,
array<double>^% StepNum,
array<String>^% DType,
array<double>^% Value
)

abstract GeneralizedDispl :
    Name : string *
    NumberResults : int byref *
    GD : string[] byref *
    LoadCase : string[] byref *
    StepType : string[] byref *
    StepNum : float[] byref *
    DType : string[] byref *
    Value : float[] byref -> int

```

Parameters

Name

Type:Â SystemString

The name of an existing generalized displacement for which results are returned.

If the program does not recognize this name as a defined generalized displacement, it returns results for all selected generalized displacements, if any. For example, entering a blank string (i.e., "") for the name will prompt the program to return results for all selected generalized displacements

NumberResults

Type:Â SystemInt32

The total number of results returned by the program

GD

Type:Â SystemString

This is an array that includes the generalized displacement name associated with each result

LoadCase

Type:Â SystemString

This is an array that includes the name of the analysis case or load combination associated with each result

StepType

Type:Â SystemString

This is an array that includes the step type, if any, for each result

StepNum

Type:Â SystemDouble

This is an array that includes the step number, if any, for each result

DType

Type:Â SystemString

This is an array that includes the generalized displacement type for each result. It is either Translation or Rotation

Value

Type:Â SystemDouble

This is an array of the generalized displacement values for each result. [L] when DType is Translation , [rad] when DType is Rotation

Return Value

Type: $\hat{A}$ Int32

Returns zero if the displacements are successfully recovered, otherwise it returns a nonzero value

▲ **Remarks**

See Results for more information.

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberResults As Integer
    Dim GD() As String
    Dim LoadCase() As String
    Dim StepType() As String
    Dim StepNum() As Double
    Dim DType() As String
    Dim Value() As Double
    Dim SF() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis()

    'deselect all cases and combos
    ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()

    'set case selected for output
    ret = SapModel.Results.Setup.SetCaseSelectedForOutput("DEAD")

    'add generalized displacement
    ret = SapModel.GDispl.Add("GD1", 1)

    'add point to generalized displacement
    ReDim SF(5)
    SF(0) = 0.5
    ret = SapModel.GDispl.SetPoint("GD1", "3", SF)

    'get generalized displacement results
    ret = SapModel.Results.GeneralizedDispl("GD1", NumberResults, GD, LoadCase, StepType, StepNum, DType, Value, SF)
```

```
'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cAnalysisResults Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LST3A Method

Reports the joint accelerations for the specified point elements. The accelerations reported by this function are relative accelerations.

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int JointAcc(
    string Name,
    eItemTypeElm ItemTypeElm,
    ref int NumberResults,
    ref string[] Obj,
    ref string[] Elm,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] U1,
    ref double[] U2,
    ref double[] U3,
    ref double[] R1,
    ref double[] R2,
    ref double[] R3
)
```

```
Function JointAcc (
    Name As String,
    ItemTypeElm As eItemTypeElm,
    ByRef NumberResults As Integer,
    ByRef Obj As String(),
    ByRef Elm As String(),
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef U1 As Double(),
    ByRef U2 As Double(),
    ByRef U3 As Double(),
    ByRef R1 As Double(),
    ByRef R2 As Double(),
    ByRef R3 As Double()
) As Integer
```

```
Dim instance As cAnalysisResults
Dim Name As String
Dim ItemTypeElm As eItemTypeElm
Dim NumberResults As Integer
Dim Obj As String()
Dim Elm As String()
```

cAnalysisResultsspan id="LST3AA45527_0"AddLanguageSpecificTextSet("LST3AA45527_0?cp124;|nu=").")


```

Dim LoadCase As String()
Dim StepType As String()
Dim StepNum As Double()
Dim U1 As Double()
Dim U2 As Double()
Dim U3 As Double()
Dim R1 As Double()
Dim R2 As Double()
Dim R3 As Double()
Dim returnValue As Integer

```

```

returnValue = instance.JointAcc(Name,
ItemTypeId, NumberResults, Obj,
Elm, LoadCase, StepType, StepNum,
U1, U2, U3, R1, R2, R3)

```

```

int JointAcc(
String^ Name,
eItemTypeId ItemTypeId,
int% NumberResults,
array<String^>^% Obj,
array<String^>^% Elm,
array<String^>^% LoadCase,
array<String^>^% StepType,
array<double>^% StepNum,
array<double>^% U1,
array<double>^% U2,
array<double>^% U3,
array<double>^% R1,
array<double>^% R2,
array<double>^% R3
)

```

```

abstract JointAcc :
    Name : string *
    ItemTypeId : eItemTypeId *
    NumberResults : int byref *
    Obj : string[] byref *
    Elm : string[] byref *
    LoadCase : string[] byref *
    StepType : string[] byref *
    StepNum : float[] byref *
    U1 : float[] byref *
    U2 : float[] byref *
    U3 : float[] byref *
    R1 : float[] byref *
    R2 : float[] byref *
    R3 : float[] byref -> int

```

Parameters

Name

Type: SystemString

The name of an existing point object, point element, or group of objects depending on the value of the ItemTypeId item.

ItemTypeId

Type: ETABSv1eItemTypeId

This is one of the following items in the eItemTypeId enumeration.

Introduction

If this item is ObjectElm, the result request is for the point element corresponding to the point object specified by the Name item.

If this item is Element, the result request is for the point element specified by the Name item.

If this item is GroupElm, the result request is for all point elements directly or indirectly specified in the group specified by the Name item.

If this item is SelectionElm, the result request is for all point elements directly or indirectly selected and the Name item is ignored.

NumberResults

Type:Â SystemInt32

The total number of results returned by the program.

Obj

Type:Â SystemString

This is an array that includes the point object name associated with each result, if any. Some results will have no point object associated with them. For those cases, this item will be blank.

Elm

Type:Â SystemString

This is an array that includes the point element name associated with each result.

LoadCase

Type:Â SystemString

This is an array that includes the name of the analysis case or load combination associated with each result.

StepType

Type:Â SystemString

This is an array that includes the step type, if any, for each result.

StepNum

Type:Â SystemDouble

This is an array that includes the step number, if any, for each result.

U1

Type:Â SystemDouble

This is a one dimensional array that includes the translational acceleration in the point element local 1 direction for each result. [L/s²]

U2

Type:Â SystemDouble

This is a one dimensional array that includes the translational acceleration in the point element local 2 direction for each result. [L/s²]

U3

Type:Â SystemDouble

This is a one dimensional array that includes the translational acceleration in the point element local 3 direction for each result. [L/s²]

R1

Type:Â SystemDouble

This is a one dimensional array that includes the rotational acceleration about the point element local 1 direction for each result. [rad/s²]

R2

Type: SystemDouble

This is a one dimensional array that includes the rotational acceleration about the point element local 2 direction for each result. [rad/s2]

R3

Type: SystemDouble

This is a one dimensional array that includes the rotational acceleration about the point element local 3 direction for each result. [rad/s2]

Return Value

Type: Int32

Returns zero if the accelerations are successfully recovered, otherwise it returns a nonzero value

Remarks

See Results for more information.

Examples

VB

Copy

```
Public Sub Example()
    Dim ret As Integer = -1

    Dim MyLoadType() As String
    Dim MyLoadName() As String
    Dim MyFunc() As String
    Dim MySF() As Double
    Dim MyTF() As Double
    Dim MyAT() As Double
    Dim MyCSys() As String
    Dim MyAng() As Double

    Dim NumberResults As Integer
    Dim Obj() As String
    Dim Elm() As String
    Dim LoadCase() As String
    Dim StepType() As String
    Dim StepNum() As Double
    Dim U1() As Double
    Dim U2() As Double
    Dim U3() As Double
    Dim R1() As Double
    Dim R2() As Double
    Dim R3() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)
```

Introduction

```
'add linear modal history load case
    ret = SapModel.LoadCases.ModHistLinear.SetCase("MyMHLCASE1")

'set load data
    ReDim MyLoadType(1)
    ReDim MyLoadName(1)
    ReDim MyFunc(1)
    ReDim MySF(1)
    ReDim MyTF(1)
    ReDim MyAT(1)
    ReDim MyCSys(1)
    ReDim MyAng(1)
    MyLoadType(0) = "Load"
    MyLoadName(0) = "DEAD"
    MyFunc(0) = "RAMPPTH"
    MySF(0) = 1
    MyTF(0) = 1
    MyAT(0) = 0
    MyCSys(0) = "Global"
    MyAng(0) = 0
    MyLoadType(1) = "Accel"
    MyLoadName(1) = "U2"
    MyFunc(1) = "UnifTH"
    MySF(1) = 2
    MyTF(1) = 1.5
    MyAT(1) = 10
    MyCSys(1) = "Global"
    MyAng(1) = 10
    ret = SapModel.LoadCases.ModHistLinear.SetLoads("MyMHLCASE1", 2, MyLoadType, MyLoadName, MyFu

'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis()

'deselect all cases and combos
    ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()

'set case selected for output
    ret = SapModel.Results.Setup.SetCaseSelectedForOutput("MyMHLCASE1")

'set modal history output option to step-by-step
    ret = SapModel.Results.Setup.SetOptionModalHist(2)

'get joint acceleration
    ret = SapModel.Results.JointAcc("All", eItemTypeElm.GroupElm, NumberResults, Obj, Elm, LoadCa

'check
    If NumberResults > 0 Then ret = 0 Else ret = -1

'close application
    mySapObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    mySapObject = Nothing
End Sub
```

▲ See Also

Reference

cAnalysisResults Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LST3C11481C_0") Method

Reports the joint absolute accelerations for the specified point elements. Absolute and relative accelerations are the same, except when reported for time history load cases subjected to acceleration loading

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int JointAccAbs(
    string Name,
    eItemTypeElm ItemTypeElm,
    ref int NumberResults,
    ref string[] Obj,
    ref string[] Elm,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] U1,
    ref double[] U2,
    ref double[] U3,
    ref double[] R1,
    ref double[] R2,
    ref double[] R3
)
```

```
Function JointAccAbs (
    Name As String,
    ItemTypeElm As eItemTypeElm,
    ByRef NumberResults As Integer,
    ByRef Obj As String(),
    ByRef Elm As String(),
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef U1 As Double(),
    ByRef U2 As Double(),
    ByRef U3 As Double(),
    ByRef R1 As Double(),
    ByRef R2 As Double(),
    ByRef R3 As Double()
) As Integer
```

```
Dim instance As cAnalysisResults
Dim Name As String
Dim ItemTypeElm As eItemTypeElm
Dim NumberResults As Integer
Dim Obj As String()
```

cAnalysisResultsspan id="LST3C11481C_0"AddLanguageSpecificTextSet("LST3C11481C_0?c139::|nu=.")

```

Dim Elm As String()
Dim LoadCase As String()
Dim StepType As String()
Dim StepNum As Double()
Dim U1 As Double()
Dim U2 As Double()
Dim U3 As Double()
Dim R1 As Double()
Dim R2 As Double()
Dim R3 As Double()
Dim returnValue As Integer

```

```

returnValue = instance.JointAccAbs(Name,
ItemTypeId, NumberResults, Obj,
Elm, LoadCase, StepType, StepNum,
U1, U2, U3, R1, R2, R3)

```

```

int JointAccAbs(
String^ Name,
eItemTypeId ItemTypeId,
int% NumberResults,
array<String^>^% Obj,
array<String^>^% Elm,
array<String^>^% LoadCase,
array<String^>^% StepType,
array<double>^% StepNum,
array<double>^% U1,
array<double>^% U2,
array<double>^% U3,
array<double>^% R1,
array<double>^% R2,
array<double>^% R3
)

```

```

abstract JointAccAbs :
    Name : string *
    ItemTypeId : eItemTypeId *
    NumberResults : int byref *
    Obj : string[] byref *
    Elm : string[] byref *
    LoadCase : string[] byref *
    StepType : string[] byref *
    StepNum : float[] byref *
    U1 : float[] byref *
    U2 : float[] byref *
    U3 : float[] byref *
    R1 : float[] byref *
    R2 : float[] byref *
    R3 : float[] byref -> int

```

Parameters

Name

Type: SystemString

The name of an existing point object, point element, or group of objects depending on the value of the ItemTypeId item.

ItemTypeId

Type: ETABSv1eItemTypeId

This is one of the following items in the eItemTypeId enumeration.

Introduction

If this item is ObjectElm, the result request is for the point element corresponding to the point object specified by the Name item.

If this item is Element, the result request is for the point element specified by the Name item.

If this item is GroupElm, the result request is for all point elements directly or indirectly specified in the group specified by the Name item.

If this item is SelectionElm, the result request is for all point elements directly or indirectly selected and the Name item is ignored.

NumberResults

Type:Â SystemInt32

The total number of results returned by the program.

Obj

Type:Â SystemString

This is an array that includes the point object name associated with each result, if any. Some results will have no point object associated with them. For those cases, this item will be blank.

Elm

Type:Â SystemString

This is an array that includes the point element name associated with each result.

LoadCase

Type:Â SystemString

This is an array that includes the name of the analysis case or load combination associated with each result.

StepType

Type:Â SystemString

This is an array that includes the step type, if any, for each result.

StepNum

Type:Â SystemDouble

This is an array that includes the step number, if any, for each result.

U1

Type:Â SystemDouble

This is a one dimensional array that includes the translational acceleration in the point element local 1 direction for each result. [L/s²]

U2

Type:Â SystemDouble

This is a one dimensional array that includes the translational acceleration in the point element local 2 direction for each result. [L/s²]

U3

Type:Â SystemDouble

This is a one dimensional array that includes the translational acceleration in the point element local 3 direction for each result. [L/s²]

R1

Type:Â SystemDouble

This is a one dimensional array that includes the rotational acceleration about the point element local 1 direction for each result. [rad/s²]

R2

Type: SystemDouble

This is a one dimensional array that includes the rotational acceleration about the point element local 2 direction for each result. [rad/s2]

R3

Type: SystemDouble

This is a one dimensional array that includes the rotational acceleration about the point element local 3 direction for each result. [rad/s2]

Return Value

Type: Int32

Returns zero if the accelerations are successfully recovered, otherwise it returns a nonzero value

Remarks

See Results for more information.

Examples

VB

Copy

```
Public Sub Example()
    Dim ret As Integer = -1

    Dim MyLoadType() As String
    Dim MyLoadName() As String
    Dim MyFunc() As String
    Dim MySF() As Double
    Dim MyTF() As Double
    Dim MyAT() As Double
    Dim MyCSys() As String
    Dim MyAng() As Double

    Dim NumberResults As Integer
    Dim Obj() As String
    Dim Elm() As String
    Dim LoadCase() As String
    Dim StepType() As String
    Dim StepNum() As Double
    Dim U1() As Double
    Dim U2() As Double
    Dim U3() As Double
    Dim R1() As Double
    Dim R2() As Double
    Dim R3() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)
```

Introduction

```
'add linear modal history load case
    ret = SapModel.LoadCases.ModHistLinear.SetCase("MyMHLCASE1")

'set load data
    ReDim MyLoadType(1)
    ReDim MyLoadName(1)
    ReDim MyFunc(1)
    ReDim MySF(1)
    ReDim MyTF(1)
    ReDim MyAT(1)
    ReDim MyCSys(1)
    ReDim MyAng(1)
    MyLoadType(0) = "Load"
    MyLoadName(0) = "DEAD"
    MyFunc(0) = "RAMPPTH"
    MySF(0) = 1
    MyTF(0) = 1
    MyAT(0) = 0
    MyCSys(0) = "Global"
    MyAng(0) = 0
    MyLoadType(1) = "Accel"
    MyLoadName(1) = "U2"
    MyFunc(1) = "UnifTH"
    MySF(1) = 2
    MyTF(1) = 1.5
    MyAT(1) = 10
    MyCSys(1) = "Global"
    MyAng(1) = 10
    ret = SapModel.LoadCases.ModHistLinear.SetLoads("MyMHLCASE1", 2, MyLoadType, MyLoadName, MyFu

'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis()

'deselect all cases and combos
    ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()

'set case selected for output
    ret = SapModel.Results.Setup.SetCaseSelectedForOutput("MyMHLCASE1")

'set modal history output option to step-by-step
    ret = SapModel.Results.Setup.SetOptionModalHist(2)

'get joint absolute acceleration
    ret = SapModel.Results.JointAccAbs("All", eItemTypeElm.GroupElm, NumberResults, Obj, Elm, Loa

'check
    If NumberResults > 0 Then ret = 0 Else ret = -1

'close application
    mySapObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    mySapObject = Nothing
End Sub
```

▲ See Also

Reference

cAnalysisResults Interface

ETABSv1 Namespace

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i»j

CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LSTF7776B28_0")

Method

Reports the joint displacements for the specified point elements. The displacements reported by this function are relative displacements

Namespace: Â ETABSv1

Assembly: Â ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int JointDispl(
    string Name,
    eItemTypeElm ItemTypeElm,
    ref int NumberResults,
    ref string[] Obj,
    ref string[] Elm,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] U1,
    ref double[] U2,
    ref double[] U3,
    ref double[] R1,
    ref double[] R2,
    ref double[] R3
)
```

```
Function JointDispl (
    Name As String,
    ItemTypeElm As eItemTypeElm,
    ByRef NumberResults As Integer,
    ByRef Obj As String(),
    ByRef Elm As String(),
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef U1 As Double(),
    ByRef U2 As Double(),
    ByRef U3 As Double(),
    ByRef R1 As Double(),
    ByRef R2 As Double(),
    ByRef R3 As Double()
) As Integer
```

```
Dim instance As cAnalysisResults
Dim Name As String
Dim ItemTypeElm As eItemTypeElm
Dim NumberResults As Integer
Dim Obj As String()
Dim Elm As String()
```

cAnalysisResultsspan id="LSTF7776B28_0"AddLanguageSpecificTextSet("LSTF7776B28_0?cp126|nu=");

```

Dim LoadCase As String()
Dim StepType As String()
Dim StepNum As Double()
Dim U1 As Double()
Dim U2 As Double()
Dim U3 As Double()
Dim R1 As Double()
Dim R2 As Double()
Dim R3 As Double()
Dim returnValue As Integer

```

```

returnValue = instance.JointDispl(Name,
ItemTypeId, NumberResults, Obj,
Elm, LoadCase, StepType, StepNum,
U1, U2, U3, R1, R2, R3)

```

```

int JointDispl(
String^ Name,
eItemTypeId ItemTypeId,
int% NumberResults,
array<String^>^% Obj,
array<String^>^% Elm,
array<String^>^% LoadCase,
array<String^>^% StepType,
array<double>^% StepNum,
array<double>^% U1,
array<double>^% U2,
array<double>^% U3,
array<double>^% R1,
array<double>^% R2,
array<double>^% R3
)

```

```

abstract JointDispl :
    Name : string *
    ItemTypeId : eItemTypeId *
    NumberResults : int byref *
    Obj : string[] byref *
    Elm : string[] byref *
    LoadCase : string[] byref *
    StepType : string[] byref *
    StepNum : float[] byref *
    U1 : float[] byref *
    U2 : float[] byref *
    U3 : float[] byref *
    R1 : float[] byref *
    R2 : float[] byref *
    R3 : float[] byref -> int

```

Parameters

Name

Type: SystemString

The name of an existing point object, point element, or group of objects depending on the value of the ItemTypeId item.

ItemTypeId

Type: ETABSv1eItemTypeId

This is one of the following items in the eItemTypeId enumeration.

Introduction

If this item is ObjectElm, the result request is for the point element corresponding to the point object specified by the Name item.

If this item is Element, the result request is for the point element specified by the Name item.

If this item is GroupElm, the result request is for all point elements directly or indirectly specified in the group specified by the Name item.

If this item is SelectionElm, the result request is for all point elements directly or indirectly selected and the Name item is ignored.

NumberResults

Type:Â SystemInt32

The total number of results returned by the program.

Obj

Type:Â SystemString

This is an array that includes the point object name associated with each result, if any. Some results will have no point object associated with them. For those cases, this item will be blank.

Elm

Type:Â SystemString

This is an array that includes the point element name associated with each result.

LoadCase

Type:Â SystemString

This is an array that includes the name of the analysis case or load combination associated with each result.

StepType

Type:Â SystemString

This is an array that includes the step type, if any, for each result.

StepNum

Type:Â SystemDouble

This is an array that includes the step number, if any, for each result.

U1

Type:Â SystemDouble

This is a one dimensional array that includes the displacement in the point element local 1 direction for each result. [L]

U2

Type:Â SystemDouble

This is a one dimensional array that includes the displacement in the point element local 2 direction for each result. [L]

U3

Type:Â SystemDouble

This is a one dimensional array that includes the displacement in the point element local 3 direction for each result. [L]

R1

Type:Â SystemDouble

This is a one dimensional array that includes the rotation about the point element local 1 direction for each result. [rad]

R2

Type: SystemDouble

This is a one dimensional array that includes the rotation about the point element local 2 direction for each result. [rad]

R3

Type: SystemDouble

This is a one dimensional array that includes the rotation about the point element local 3 direction for each result. [rad]

Return Value

Type: Int32

Returns zero if the displacements are successfully recovered, otherwise it returns a nonzero value

Remarks

See Results for more information.

Examples

VB

Copy

```
Public Sub Example()
    Dim ret As Integer = -1

    Dim MyLoadType() As String
    Dim MyLoadName() As String
    Dim MyFunc() As String
    Dim MySF() As Double
    Dim MyTF() As Double
    Dim MyAT() As Double
    Dim MyCSys() As String
    Dim MyAng() As Double

    Dim NumberResults As Integer
    Dim Obj() As String
    Dim Elm() As String
    Dim LoadCase() As String
    Dim StepType() As String
    Dim StepNum() As Double
    Dim U1() As Double
    Dim U2() As Double
    Dim U3() As Double
    Dim R1() As Double
    Dim R2() As Double
    Dim R3() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)
```

Introduction

```
'add linear modal history load case
    ret = SapModel.LoadCases.ModHistLinear.SetCase("MyMHLCASE1")

'set load data
    ReDim MyLoadType(1)
    ReDim MyLoadName(1)
    ReDim MyFunc(1)
    ReDim MySF(1)
    ReDim MyTF(1)
    ReDim MyAT(1)
    ReDim MyCSys(1)
    ReDim MyAng(1)
    MyLoadType(0) = "Load"
    MyLoadName(0) = "DEAD"
    MyFunc(0) = "RAMPPTH"
    MySF(0) = 1
    MyTF(0) = 1
    MyAT(0) = 0
    MyCSys(0) = "Global"
    MyAng(0) = 0
    MyLoadType(1) = "Accel"
    MyLoadName(1) = "U2"
    MyFunc(1) = "UnifTH"
    MySF(1) = 2
    MyTF(1) = 1.5
    MyAT(1) = 10
    MyCSys(1) = "Global"
    MyAng(1) = 10
    ret = SapModel.LoadCases.ModHistLinear.SetLoads("MyMHLCASE1", 2, MyLoadType, MyLoadName, MyFu

'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis()

'deselect all cases and combos
    ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()

'set case selected for output
    ret = SapModel.Results.Setup.SetCaseSelectedForOutput("MyMHLCASE1")

'set modal history output option to step-by-step
    ret = SapModel.Results.Setup.SetOptionModalHist(2)

'get joint displacements
    ret = SapModel.Results.JointDispl("ALL", GroupElm, NumberResults, Obj, Elm, LoadCase, StepTyp

'check
    If NumberResults > 0 Then ret = 0 Else ret = -1

'close application
    mySapObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    mySapObject = Nothing
End Sub
```

▲ See Also

Reference

cAnalysisResults Interface

ETABSv1 Namespace

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i»j

CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LSTFA Method

Reports the absolute joint displacements for the specified point elements. Absolute and relative displacements are the same except when reported for time history load cases subjected to acceleration loading

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int JointDisplAbs(
    string Name,
    eItemTypeElm ItemTypeElm,
    ref int NumberResults,
    ref string[] Obj,
    ref string[] Elm,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] U1,
    ref double[] U2,
    ref double[] U3,
    ref double[] R1,
    ref double[] R2,
    ref double[] R3
)
```

```
Function JointDisplAbs (
    Name As String,
    ItemTypeElm As eItemTypeElm,
    ByRef NumberResults As Integer,
    ByRef Obj As String(),
    ByRef Elm As String(),
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef U1 As Double(),
    ByRef U2 As Double(),
    ByRef U3 As Double(),
    ByRef R1 As Double(),
    ByRef R2 As Double(),
    ByRef R3 As Double()
) As Integer
```

```
Dim instance As cAnalysisResults
Dim Name As String
Dim ItemTypeElm As eItemTypeElm
Dim NumberResults As Integer
Dim Obj As String()
```

cAnalysisResultsspan id="LSTFAAA6B20_0"AddLanguageSpecificTextSet("LSTFAAA6B20_0?cp42::|nu=".

```

Dim Elm As String()
Dim LoadCase As String()
Dim StepType As String()
Dim StepNum As Double()
Dim U1 As Double()
Dim U2 As Double()
Dim U3 As Double()
Dim R1 As Double()
Dim R2 As Double()
Dim R3 As Double()
Dim returnValue As Integer

```

```

returnValue = instance.JointDisplAbs(Name,
ItemtypeElm, NumberResults, Obj,
Elm, LoadCase, StepType, StepNum,
U1, U2, U3, R1, R2, R3)

```

```

int JointDisplAbs(
String^ Name,
eItemTypeElm ItemTypeElm,
int% NumberResults,
array<String^>^% Obj,
array<String^>^% Elm,
array<String^>^% LoadCase,
array<String^>^% StepType,
array<double>^% StepNum,
array<double>^% U1,
array<double>^% U2,
array<double>^% U3,
array<double>^% R1,
array<double>^% R2,
array<double>^% R3
)

```

```

abstract JointDisplAbs :
    Name : string *
    ItemTypeElm : eItemTypeElm *
    NumberResults : int byref *
    Obj : string[] byref *
    Elm : string[] byref *
    LoadCase : string[] byref *
    StepType : string[] byref *
    StepNum : float[] byref *
    U1 : float[] byref *
    U2 : float[] byref *
    U3 : float[] byref *
    R1 : float[] byref *
    R2 : float[] byref *
    R3 : float[] byref -> int

```

Parameters

Name

Type: SystemString

The name of an existing point object, point element, or group of objects depending on the value of the ItemTypeElm item.

ItemTypeElm

Type: ETABSv1eItemTypeElm

This is one of the following items in the eItemTypeElm enumeration.

Introduction

If this item is ObjectElm, the result request is for the point element corresponding to the point object specified by the Name item.

If this item is Element, the result request is for the point element specified by the Name item.

If this item is GroupElm, the result request is for all point elements directly or indirectly specified in the group specified by the Name item.

If this item is SelectionElm, the result request is for all point elements directly or indirectly selected and the Name item is ignored.

NumberResults

Type:Â SystemInt32

The total number of results returned by the program.

Obj

Type:Â SystemString

This is an array that includes the point object name associated with each result, if any. Some results will have no point object associated with them. For those cases, this item will be blank.

Elm

Type:Â SystemString

This is an array that includes the point element name associated with each result.

LoadCase

Type:Â SystemString

This is an array that includes the name of the analysis case or load combination associated with each result.

StepType

Type:Â SystemString

This is an array that includes the step type, if any, for each result.

StepNum

Type:Â SystemDouble

This is an array that includes the step number, if any, for each result.

U1

Type:Â SystemDouble

This is a one dimensional array that includes the displacement in the point element local 1 direction for each result. [L]

U2

Type:Â SystemDouble

This is a one dimensional array that includes the displacement in the point element local 2 direction for each result. [L]

U3

Type:Â SystemDouble

This is a one dimensional array that includes the displacement in the point element local 3 direction for each result. [L]

R1

Type:Â SystemDouble

This is a one dimensional array that includes the rotation about the point element local 1 direction for each result. [rad]

R2

Type: SystemDouble

This is a one dimensional array that includes the rotation about the point element local 2 direction for each result. [rad]

R3

Type: SystemDouble

This is a one dimensional array that includes the rotation about the point element local 3 direction for each result. [rad]

Return Value

Type: Int32

Returns zero if the displacements are successfully recovered, otherwise it returns a nonzero value

Remarks

See Results for more information.

Examples

VB

Copy

```
Public Sub Example()
    Dim ret As Integer = -1

    Dim MyLoadType() As String
    Dim MyLoadName() As String
    Dim MyFunc() As String
    Dim MySF() As Double
    Dim MyTF() As Double
    Dim MyAT() As Double
    Dim MyCSys() As String
    Dim MyAng() As Double

    Dim NumberResults As Integer
    Dim Obj() As String
    Dim Elm() As String
    Dim LoadCase() As String
    Dim StepType() As String
    Dim StepNum() As Double
    Dim U1() As Double
    Dim U2() As Double
    Dim U3() As Double
    Dim R1() As Double
    Dim R2() As Double
    Dim R3() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)
```

Introduction

```
'add linear modal history load case
    ret = SapModel.LoadCases.ModHistLinear.SetCase("MyMHLCASE1")

'set load data
    ReDim MyLoadType(1)
    ReDim MyLoadName(1)
    ReDim MyFunc(1)
    ReDim MySF(1)
    ReDim MyTF(1)
    ReDim MyAT(1)
    ReDim MyCSys(1)
    ReDim MyAng(1)
    MyLoadType(0) = "Load"
    MyLoadName(0) = "DEAD"
    MyFunc(0) = "RAMPPTH"
    MySF(0) = 1
    MyTF(0) = 1
    MyAT(0) = 0
    MyCSys(0) = "Global"
    MyAng(0) = 0
    MyLoadType(1) = "Accel"
    MyLoadName(1) = "U2"
    MyFunc(1) = "UnifTH"
    MySF(1) = 2
    MyTF(1) = 1.5
    MyAT(1) = 10
    MyCSys(1) = "Global"
    MyAng(1) = 10
    ret = SapModel.LoadCases.ModHistLinear.SetLoads("MyMHLCASE1", 2, MyLoadType, MyLoadName, MyFu

'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis()

'deselect all cases and combos
    ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()

'set case selected for output
    ret = SapModel.Results.Setup.SetCaseSelectedForOutput("MyMHLCASE1")

'set modal history output option to step-by-step
    ret = SapModel.Results.Setup.SetOptionModalHist(2)

'get joint absolute displacements
    ret = SapModel.Results.JointDisplAbs("ALL", GroupElm, NumberResults, Obj, Elm, LoadCase, Step

'check
    If NumberResults > 0 Then ret = 0 Else ret = -1

'close application
    mySapObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    mySapObject = Nothing
End Sub
```

▲ See Also

Reference

cAnalysisResults Interface

ETABSv1 Namespace

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»

CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LSTA0913002_0?cpp148|nu=.");

Method

Reports the joint drifts

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int JointDrifts(
    ref int NumberResults,
    ref string[] Story,
    ref string[] Label,
    ref string[] Name,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] DisplacementX,
    ref double[] DisplacementY,
    ref double[] DriftX,
    ref double[] DriftY
)
```

```
Function JointDrifts (
    ByRef NumberResults As Integer,
    ByRef Story As String(),
    ByRef Label As String(),
    ByRef Name As String(),
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef DisplacementX As Double(),
    ByRef DisplacementY As Double(),
    ByRef DriftX As Double(),
    ByRef DriftY As Double()
) As Integer
```

```
Dim instance As cAnalysisResults
Dim NumberResults As Integer
Dim Story As String()
Dim Label As String()
Dim Name As String()
Dim LoadCase As String()
Dim StepType As String()
Dim StepNum As Double()
Dim DisplacementX As Double()
Dim DisplacementY As Double()
Dim DriftX As Double()
Dim DriftY As Double()
Dim returnValue As Integer
```

cAnalysisResultsspan id="LSTA0913002_0"AddLanguageSpecificTextSet("LSTA0913002_0?cpp148|nu=.");


```
returnValue = instance.JointDrifts(NumberResults,
Story, Label, Name, LoadCase, StepType,
StepNum, DisplacementX, DisplacementY,
DriftX, DriftY)
```

```
int JointDrifts(
int% NumberResults,
array<String^>^% Story,
array<String^>^% Label,
array<String^>^% Name,
array<String^>^% LoadCase,
array<String^>^% StepType,
array<double>^% StepNum,
array<double>^% DisplacementX,
array<double>^% DisplacementY,
array<double>^% DriftX,
array<double>^% DriftY
)
```

```
abstract JointDrifts :
    NumberResults : int byref *
    Story : string[] byref *
    Label : string[] byref *
    Name : string[] byref *
    LoadCase : string[] byref *
    StepType : string[] byref *
    StepNum : float[] byref *
    DisplacementX : float[] byref *
    DisplacementY : float[] byref *
    DriftX : float[] byref *
    DriftY : float[] byref -> int
```

Parameters

NumberResults

Type:Â SystemInt32

The total number of results returned by the program

Story

Type:Â SystemString

This is an array of the story levels associated with each result

Label

Type:Â SystemString

This is an array of the point labels for each result

Name

Type:Â SystemString

This is an array of the unique point names for each result

LoadCase

Type:Â SystemString

This is an array of the names of the analysis case or load combination associated with each result

StepType

Type:Â SystemString

This is an array of the step types, if any, for each result

StepNum

Type: SystemDouble

This is an array of the step numbers, if any, for each result

DisplacementX

Type: SystemDouble

This is an array of the displacements in the X direction [L]

DisplacementY

Type: SystemDouble

This is an array of the displacements in the Y direction [L]

DriftX

Type: SystemDouble

This is an array of the drifts in the X direction [L]

DriftY

Type: SystemDouble

This is an array of the drifts in the Y direction [L]

Return Value

Type: Int32

Returns zero if the results are successfully recovered, otherwise it returns a nonzero value

▀ Remarks

See Results for more information.

▀ See Also

Reference

cAnalysisResults Interface

ETABSv1 Namespace

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i»j

CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LST4DDB8F6E_0?151=::|nu=. Method

Reports the joint reactions for the specified point elements. The reactions reported are from restraints, springs and grounded (one-joint) links.

Namespace: Â ETABSv1

Assembly: Â ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

⚡ Syntax

C#

VB

C++

F#

Copy

```
int JointReact(
    string Name,
    eItemTypeElm ItemTypeElm,
    ref int NumberResults,
    ref string[] Obj,
    ref string[] Elm,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] F1,
    ref double[] F2,
    ref double[] F3,
    ref double[] M1,
    ref double[] M2,
    ref double[] M3
)
```

```
Function JointReact (
    Name As String,
    ItemTypeElm As eItemTypeElm,
    ByRef NumberResults As Integer,
    ByRef Obj As String(),
    ByRef Elm As String(),
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef F1 As Double(),
    ByRef F2 As Double(),
    ByRef F3 As Double(),
    ByRef M1 As Double(),
    ByRef M2 As Double(),
    ByRef M3 As Double()
) As Integer
```

```
Dim instance As cAnalysisResults
Dim Name As String
Dim ItemTypeElm As eItemTypeElm
Dim NumberResults As Integer
Dim Obj As String()
Dim Elm As String()
```

cAnalysisResultsspan id="LST4DDB8F6E_0"AddLanguageSpecificTextSet("LST4DDB8F6E_0?151=::|nu=.
151

```

Dim LoadCase As String()
Dim StepType As String()
Dim StepNum As Double()
Dim F1 As Double()
Dim F2 As Double()
Dim F3 As Double()
Dim M1 As Double()
Dim M2 As Double()
Dim M3 As Double()
Dim returnValue As Integer

```

```

returnValue = instance.JointReact(Name,
ItemtypeElm, NumberResults, Obj,
Elm, LoadCase, StepType, StepNum,
F1, F2, F3, M1, M2, M3)

```

```

int JointReact(
String^ Name,
eItemtypeElm ItemTypeElm,
int% NumberResults,
array<String^>^% Obj,
array<String^>^% Elm,
array<String^>^% LoadCase,
array<String^>^% StepType,
array<double>^% StepNum,
array<double>^% F1,
array<double>^% F2,
array<double>^% F3,
array<double>^% M1,
array<double>^% M2,
array<double>^% M3
)

```

```

abstract JointReact :
    Name : string *
    ItemTypeElm : eItemTypeElm *
    NumberResults : int byref *
    Obj : string[] byref *
    Elm : string[] byref *
    LoadCase : string[] byref *
    StepType : string[] byref *
    StepNum : float[] byref *
    F1 : float[] byref *
    F2 : float[] byref *
    F3 : float[] byref *
    M1 : float[] byref *
    M2 : float[] byref *
    M3 : float[] byref -> int

```

Parameters

Name

Type: SystemString

The name of an existing point object, point element, or group of objects depending on the value of the ItemTypeElm item.

ItemTypeElm

Type: ETABSv1eItemTypeElm

This is one of the following items in the eItemTypeElm enumeration.

Introduction

If this item is ObjectElm, the result request is for the point element corresponding to the point object specified by the Name item.

If this item is Element, the result request is for the point element specified by the Name item.

If this item is GroupElm, the result request is for all point elements directly or indirectly specified in the group specified by the Name item.

If this item is SelectionElm, the result request is for all point elements directly or indirectly selected and the Name item is ignored.

NumberResults

Type:Â SystemInt32

The total number of results returned by the program.

Obj

Type:Â SystemString

This is an array that includes the point object name associated with each result, if any. Some results will have no point object associated with them. For those cases, this item will be blank.

Elm

Type:Â SystemString

This is an array that includes the point element name associated with each result.

LoadCase

Type:Â SystemString

This is an array that includes the name of the analysis case or load combination associated with each result.

StepType

Type:Â SystemString

This is an array that includes the step type, if any, for each result.

StepNum

Type:Â SystemDouble

This is an array that includes the step number, if any, for each result.

F1

Type:Â SystemDouble

This is a one dimensional array that includes the reaction forces in the point element local 1 axis direction. [F].

F2

Type:Â SystemDouble

This is a one dimensional array that includes the reaction forces in the point element local 2 axis direction. [F].

F3

Type:Â SystemDouble

This is a one dimensional array that includes the reaction forces in the point element local 3 axis direction. [F].

M1

Type:Â SystemDouble

This is a one dimensional array that includes the reaction moments about the point element local 1 axis direction. [FL].

M2

Type: SystemDouble

This is a one dimensional array that includes the reaction moments about the point element local 2 axis direction. [FL].

M3

Type: SystemDouble

This is a one dimensional array that includes the reaction moments about the point element local 3 axis direction. [FL].

Return Value

Type: Int32

Returns zero if the reactions are successfully recovered; otherwise it returns a nonzero value.

Remarks

See Results for more information.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberResults As Integer
    Dim Obj() As String
    Dim Elm() As String
    Dim LoadCase() As String
    Dim StepType() As String
    Dim StepNum() As Double
    Dim F1() As Double
    Dim F2() As Double
    Dim F3() As Double
    Dim M1() As Double
    Dim M2() As Double
    Dim M3() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis()

    'deselect all cases and combos
    ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()
```

Introduction

```
'set case selected for output
  ret = SapModel.Results.Setup.SetCaseSelectedForOutput("DEAD")

'get joint reactions
  ret = SapModel.Results.JointReact("1", eItemTypeElm.Element, NumberResults, Obj, Elm, LoadCase)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

📌 See Also

Reference

cAnalysisResults Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LSTF8 Method

Reports the joint velocities for the specified point elements. The velocities reported by this function are relative velocities

Namespace: `ETABSv1`

Assembly: `ETABSv1 (in ETABSv1.dll)` Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int JointVel(
    string Name,
    eItemTypeElm ItemTypeElm,
    ref int NumberResults,
    ref string[] Obj,
    ref string[] Elm,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] U1,
    ref double[] U2,
    ref double[] U3,
    ref double[] R1,
    ref double[] R2,
    ref double[] R3
)
```

```
Function JointVel (
    Name As String,
    ItemTypeElm As eItemTypeElm,
    ByRef NumberResults As Integer,
    ByRef Obj As String(),
    ByRef Elm As String(),
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef U1 As Double(),
    ByRef U2 As Double(),
    ByRef U3 As Double(),
    ByRef R1 As Double(),
    ByRef R2 As Double(),
    ByRef R3 As Double()
) As Integer
```

```
Dim instance As cAnalysisResults
Dim Name As String
Dim ItemTypeElm As eItemTypeElm
Dim NumberResults As Integer
Dim Obj As String()
Dim Elm As String()
```

cAnalysisResultsspan id="LSTF88A5226_0"AddLanguageSpecificTextSet("LSTF88A5226_0?cp156|nu=");


```

Dim LoadCase As String()
Dim StepType As String()
Dim StepNum As Double()
Dim U1 As Double()
Dim U2 As Double()
Dim U3 As Double()
Dim R1 As Double()
Dim R2 As Double()
Dim R3 As Double()
Dim returnValue As Integer

```

```

returnValue = instance.JointVel(Name,
ItemtypeElm, NumberResults, Obj,
Elm, LoadCase, StepType, StepNum,
U1, U2, U3, R1, R2, R3)

```

```

int JointVel(
String^ Name,
eItemtypeElm ItemtypeElm,
int% NumberResults,
array<String^>^% Obj,
array<String^>^% Elm,
array<String^>^% LoadCase,
array<String^>^% StepType,
array<double>^% StepNum,
array<double>^% U1,
array<double>^% U2,
array<double>^% U3,
array<double>^% R1,
array<double>^% R2,
array<double>^% R3
)

```

```

abstract JointVel :
    Name : string *
    ItemTypeElm : eItemTypeElm *
    NumberResults : int byref *
    Obj : string[] byref *
    Elm : string[] byref *
    LoadCase : string[] byref *
    StepType : string[] byref *
    StepNum : float[] byref *
    U1 : float[] byref *
    U2 : float[] byref *
    U3 : float[] byref *
    R1 : float[] byref *
    R2 : float[] byref *
    R3 : float[] byref -> int

```

Parameters

Name

Type:Â SystemString

The name of an existing point object, point element, or group of objects depending on the value of the ItemTypeElm item.

ItemTypeElm

Type:Â ETABSv1eItemTypeElm

This is one of the following items in the eItemTypeElm enumeration.

Introduction

If this item is ObjectElm, the result request is for the point element corresponding to the point object specified by the Name item.

If this item is Element, the result request is for the point element specified by the Name item.

If this item is GroupElm, the result request is for all point elements directly or indirectly specified in the group specified by the Name item.

If this item is SelectionElm, the result request is for all point elements directly or indirectly selected and the Name item is ignored.

NumberResults

Type:Â SystemInt32

The total number of results returned by the program.

Obj

Type:Â SystemString

This is an array that includes the point object name associated with each result, if any. Some results will have no point object associated with them. For those cases, this item will be blank.

Elm

Type:Â SystemString

This is an array that includes the point element name associated with each result.

LoadCase

Type:Â SystemString

This is an array that includes the name of the analysis case or load combination associated with each result.

StepType

Type:Â SystemString

This is an array that includes the step type, if any, for each result.

StepNum

Type:Â SystemDouble

This is an array that includes the step number, if any, for each result.

U1

Type:Â SystemDouble

This is a one dimensional array that includes the translational velocity in the point element local 1 direction for each result. [L/s]

U2

Type:Â SystemDouble

This is a one dimensional array that includes the translational velocity in the point element local 2 direction for each result. [L/s]

U3

Type:Â SystemDouble

This is a one dimensional array that includes the translational velocity in the point element local 3 direction for each result. [L/s]

R1

Type:Â SystemDouble

This is a one dimensional array that includes the rotational velocity about the point element local 1 direction for each result. [rad/s]

R2

Type: SystemDouble

This is a one dimensional array that includes the rotational velocity about the point element local 2 direction for each result. [rad/s]

R3

Type: SystemDouble

This is a one dimensional array that includes the rotational velocity about the point element local 3 direction for each result. [rad/s]

Return Value

Type: Int32

Returns zero if the velocities are successfully recovered, otherwise it returns a nonzero value

Remarks

See Results for more information.

Examples

VB

Copy

```
Public Sub Example()
    Dim ret As Integer = -1

    Dim MyLoadType() As String
    Dim MyLoadName() As String
    Dim MyFunc() As String
    Dim MySF() As Double
    Dim MyTF() As Double
    Dim MyAT() As Double
    Dim MyCSys() As String
    Dim MyAng() As Double

    Dim NumberResults As Integer
    Dim Obj() As String
    Dim Elm() As String
    Dim LoadCase() As String
    Dim StepType() As String
    Dim StepNum() As Double
    Dim U1() As Double
    Dim U2() As Double
    Dim U3() As Double
    Dim R1() As Double
    Dim R2() As Double
    Dim R3() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)
```

Introduction

```
'add linear modal history load case
    ret = SapModel.LoadCases.ModHistLinear.SetCase("MyMHLCASE1")

'set load data
    ReDim MyLoadType(1)
    ReDim MyLoadName(1)
    ReDim MyFunc(1)
    ReDim MySF(1)
    ReDim MyTF(1)
    ReDim MyAT(1)
    ReDim MyCSys(1)
    ReDim MyAng(1)
    MyLoadType(0) = "Load"
    MyLoadName(0) = "DEAD"
    MyFunc(0) = "RAMPPTH"
    MySF(0) = 1
    MyTF(0) = 1
    MyAT(0) = 0
    MyCSys(0) = "Global"
    MyAng(0) = 0
    MyLoadType(1) = "Accel"
    MyLoadName(1) = "U2"
    MyFunc(1) = "UnifTH"
    MySF(1) = 2
    MyTF(1) = 1.5
    MyAT(1) = 10
    MyCSys(1) = "Global"
    MyAng(1) = 10
    ret = SapModel.LoadCases.ModHistLinear.SetLoads("MyMHLCASE1", 2, MyLoadType, MyLoadName, MyFu

'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis()

'deselect all cases and combos
    ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()

'set case selected for output
    ret = SapModel.Results.Setup.SetCaseSelectedForOutput("MyMHLCASE1")

'set modal history output option to step-by-step
    ret = SapModel.Results.Setup.SetOptionModalHist(2)

'get joint velocity
    ret = SapModel.Results.JointVel("ALL", GroupElm, NumberResults, Obj, Elm, LoadCase, StepType,

'check
    If NumberResults > 0 Then ret = 0 Else ret = -1

'close application
    mySapObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    mySapObject = Nothing
End Sub
```

▲ See Also

Reference

cAnalysisResults Interface

ETABSv1 Namespace

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i»j

CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LST63

Method

Reports the joint absolute velocities for the specified point elements. Absolute and relative velocities are the same, except when reported for time history load cases subjected to acceleration loading

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int JointVelAbs(
    string Name,
    eItemTypeElm ItemTypeElm,
    ref int NumberResults,
    ref string[] Obj,
    ref string[] Elm,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] U1,
    ref double[] U2,
    ref double[] U3,
    ref double[] R1,
    ref double[] R2,
    ref double[] R3
)
```

```
Function JointVelAbs (
    Name As String,
    ItemTypeElm As eItemTypeElm,
    ByRef NumberResults As Integer,
    ByRef Obj As String(),
    ByRef Elm As String(),
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef U1 As Double(),
    ByRef U2 As Double(),
    ByRef U3 As Double(),
    ByRef R1 As Double(),
    ByRef R2 As Double(),
    ByRef R3 As Double()
) As Integer
```

```
Dim instance As cAnalysisResults
Dim Name As String
Dim ItemTypeElm As eItemTypeElm
Dim NumberResults As Integer
Dim Obj As String()
```

cAnalysisResultsspan id="LST634644C5_0"AddLanguageSpecificTextSet("LST634644C5_0?cp162|nu=.");

```

Dim Elm As String()
Dim LoadCase As String()
Dim StepType As String()
Dim StepNum As Double()
Dim U1 As Double()
Dim U2 As Double()
Dim U3 As Double()
Dim R1 As Double()
Dim R2 As Double()
Dim R3 As Double()
Dim returnValue As Integer

```

```

returnValue = instance.JointVelAbs(Name,
ItemtypeElm, NumberResults, Obj,
Elm, LoadCase, StepType, StepNum,
U1, U2, U3, R1, R2, R3)

```

```

int JointVelAbs(
String^ Name,
eItemTypeElm ItemTypeElm,
int% NumberResults,
array<String^>^% Obj,
array<String^>^% Elm,
array<String^>^% LoadCase,
array<String^>^% StepType,
array<double>^% StepNum,
array<double>^% U1,
array<double>^% U2,
array<double>^% U3,
array<double>^% R1,
array<double>^% R2,
array<double>^% R3
)

```

```

abstract JointVelAbs :
    Name : string *
    ItemTypeElm : eItemTypeElm *
    NumberResults : int byref *
    Obj : string[] byref *
    Elm : string[] byref *
    LoadCase : string[] byref *
    StepType : string[] byref *
    StepNum : float[] byref *
    U1 : float[] byref *
    U2 : float[] byref *
    U3 : float[] byref *
    R1 : float[] byref *
    R2 : float[] byref *
    R3 : float[] byref -> int

```

Parameters

Name

Type: SystemString

The name of an existing point object, point element, or group of objects depending on the value of the ItemTypeElm item.

ItemTypeElm

Type: ETABSv1eItemTypeElm

This is one of the following items in the eItemTypeElm enumeration.

Introduction

If this item is ObjectElm, the result request is for the point element corresponding to the point object specified by the Name item.

If this item is Element, the result request is for the point element specified by the Name item.

If this item is GroupElm, the result request is for all point elements directly or indirectly specified in the group specified by the Name item.

If this item is SelectionElm, the result request is for all point elements directly or indirectly selected and the Name item is ignored.

NumberResults

Type:Â SystemInt32

The total number of results returned by the program.

Obj

Type:Â SystemString

This is an array that includes the point object name associated with each result, if any. Some results will have no point object associated with them. For those cases, this item will be blank.

Elm

Type:Â SystemString

This is an array that includes the point element name associated with each result.

LoadCase

Type:Â SystemString

This is an array that includes the name of the analysis case or load combination associated with each result.

StepType

Type:Â SystemString

This is an array that includes the step type, if any, for each result.

StepNum

Type:Â SystemDouble

This is an array that includes the step number, if any, for each result.

U1

Type:Â SystemDouble

This is a one dimensional array that includes the translational velocity in the point element local 1 direction for each result. [L/s]

U2

Type:Â SystemDouble

This is a one dimensional array that includes the translational velocity in the point element local 2 direction for each result. [L/s]

U3

Type:Â SystemDouble

This is a one dimensional array that includes the translational velocity in the point element local 3 direction for each result. [L/s]

R1

Type:Â SystemDouble

This is a one dimensional array that includes the rotational velocity about the point element local 1 direction for each result. [rad/s]

R2

Type: SystemDouble

This is a one dimensional array that includes the rotational velocity about the point element local 2 direction for each result. [rad/s]

R3

Type: SystemDouble

This is a one dimensional array that includes the rotational velocity about the point element local 3 direction for each result. [rad/s]

Return Value

Type: Int32

Returns zero if the velocities are successfully recovered, otherwise it returns a nonzero value

Remarks

See Results for more information.

Examples

VB

Copy

```
Public Sub Example()
    Dim ret As Integer = -1

    Dim MyLoadType() As String
    Dim MyLoadName() As String
    Dim MyFunc() As String
    Dim MySF() As Double
    Dim MyTF() As Double
    Dim MyAT() As Double
    Dim MyCSys() As String
    Dim MyAng() As Double

    Dim NumberResults As Integer
    Dim Obj() As String
    Dim Elm() As String
    Dim LoadCase() As String
    Dim StepType() As String
    Dim StepNum() As Double
    Dim U1() As Double
    Dim U2() As Double
    Dim U3() As Double
    Dim R1() As Double
    Dim R2() As Double
    Dim R3() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)
```

Introduction

```
'add linear modal history load case
    ret = SapModel.LoadCases.ModHistLinear.SetCase("MyMHLCASE1")

'set load data
    ReDim MyLoadType(1)
    ReDim MyLoadName(1)
    ReDim MyFunc(1)
    ReDim MySF(1)
    ReDim MyTF(1)
    ReDim MyAT(1)
    ReDim MyCSys(1)
    ReDim MyAng(1)
    MyLoadType(0) = "Load"
    MyLoadName(0) = "DEAD"
    MyFunc(0) = "RAMPPTH"
    MySF(0) = 1
    MyTF(0) = 1
    MyAT(0) = 0
    MyCSys(0) = "Global"
    MyAng(0) = 0
    MyLoadType(1) = "Accel"
    MyLoadName(1) = "U2"
    MyFunc(1) = "UnifTH"
    MySF(1) = 2
    MyTF(1) = 1.5
    MyAT(1) = 10
    MyCSys(1) = "Global"
    MyAng(1) = 10
    ret = SapModel.LoadCases.ModHistLinear.SetLoads("MyMHLCASE1", 2, MyLoadType, MyLoadName, MyFu

'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis()

'deselect all cases and combos
    ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()

'set case selected for output
    ret = SapModel.Results.Setup.SetCaseSelectedForOutput("MyMHLCASE1")

'set modal history output option to step-by-step
    ret = SapModel.Results.Setup.SetOptionModalHist(2)

'get joint absolute velocity
    ret = SapModel.Results.JointVelAbs("ALL", GroupElm, NumberResults, Obj, Elm, LoadCase, StepTy

'check
    If NumberResults > 0 Then ret = 0 Else ret = -1

'close application
    mySapObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    mySapObject = Nothing
End Sub
```

▲ See Also

Reference

cAnalysisResults Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LSTD65BD0B8_0?cnu=.")

Method

Reports the link internal deformations

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int LinkDeformation(
    string Name,
    eItemTypeElm ItemTypeElm,
    ref int NumberResults,
    ref string[] Obj,
    ref string[] Elm,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] U1,
    ref double[] U2,
    ref double[] U3,
    ref double[] R1,
    ref double[] R2,
    ref double[] R3
)
```

```
Function LinkDeformation (
    Name As String,
    ItemTypeElm As eItemTypeElm,
    ByRef NumberResults As Integer,
    ByRef Obj As String(),
    ByRef Elm As String(),
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef U1 As Double(),
    ByRef U2 As Double(),
    ByRef U3 As Double(),
    ByRef R1 As Double(),
    ByRef R2 As Double(),
    ByRef R3 As Double()
) As Integer
```

```
Dim instance As cAnalysisResults
Dim Name As String
Dim ItemTypeElm As eItemTypeElm
Dim NumberResults As Integer
Dim Obj As String()
Dim Elm As String()
Dim LoadCase As String()
```

cAnalysisResultsspan id="LSTD65BD0B8_0"AddLanguageSpecificTextSet("LSTD65BD0B8_0?cnu=.")

```

Dim StepType As String()
Dim StepNum As Double()
Dim U1 As Double()
Dim U2 As Double()
Dim U3 As Double()
Dim R1 As Double()
Dim R2 As Double()
Dim R3 As Double()
Dim returnValue As Integer

returnValue = instance.LinkDeformation(Name,
ItemtypeElm, NumberResults, Obj,
Elm, LoadCase, StepType, StepNum,
U1, U2, U3, R1, R2, R3)

int LinkDeformation(
String^ Name,
eItemtypeElm ItemTypeElm,
int% NumberResults,
array<String^>^% Obj,
array<String^>^% Elm,
array<String^>^% LoadCase,
array<String^>^% StepType,
array<double>^% StepNum,
array<double>^% U1,
array<double>^% U2,
array<double>^% U3,
array<double>^% R1,
array<double>^% R2,
array<double>^% R3
)

abstract LinkDeformation :
    Name : string *
    ItemTypeElm : eItemTypeElm *
    NumberResults : int byref *
    Obj : string[] byref *
    Elm : string[] byref *
    LoadCase : string[] byref *
    StepType : string[] byref *
    StepNum : float[] byref *
    U1 : float[] byref *
    U2 : float[] byref *
    U3 : float[] byref *
    R1 : float[] byref *
    R2 : float[] byref *
    R3 : float[] byref -> int

```

Parameters

Name

Type: [SystemString](#)

The name of an existing link object, link element, or group of objects depending on the value of the ItemTypeElm item.

ItemTypeElm

Type: [ETABSv1eItemTypeElm](#)

This is one of the following items in the [eItemTypeElm](#) enumeration.

If this item is ObjectElm, the result request is for the link element corresponding to the point object specified by the Name item.

If this item is Element, the result request is for the link element specified by the Name item.

If this item is GroupElm, the result request is for all link elements directly or indirectly specified in the group specified by the Name item.

If this item is SelectionElm, the result request is for all link elements directly or indirectly selected and the Name item is ignored.

NumberResults

Type:Â SystemInt32

The total number of results returned by the program.

Obj

Type:Â SystemString

This is an array that includes the link object name associated with each result, if any.

Elm

Type:Â SystemString

This is an array that includes the link element name associated with each result.

LoadCase

Type:Â SystemString

This is an array that includes the name of the analysis case or load combination associated with each result.

StepType

Type:Â SystemString

This is an array that includes the step type, if any, for each result.

StepNum

Type:Â SystemDouble

This is an array that includes the step number, if any, for each result.

U1

Type:Â SystemDouble

This is an array that includes the internal translational deformation of the link in the link element local 1-axis direction [L]

U2

Type:Â SystemDouble

This is an array that includes the internal translational deformation of the link in the link element local 2-axis direction [L]

U3

Type:Â SystemDouble

This is an array that includes the internal translational deformation of the link in the link element local 3-axis direction [L]

R1

Type:Â SystemDouble

This is an array that includes the internal rotational deformation of the link about the link element local 1-axis direction [L]

R2

Type:Â SystemDouble

This is an array that includes the internal rotational deformation of the link about the link element local 2-axis direction [L]

R3

Type: SystemDouble

This is an array that includes the internal rotational deformation of the link about the link element local 3-axis direction [L]

Return Value

Type: Int32

Returns zero if the deformations are successfully recovered, otherwise it returns a nonzero value

Remarks

See Results for more information.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim LinkObjName As String
    Dim NumberResults As Integer
    Dim Obj() As String
    Dim Elm() As String
    Dim LoadCase() As String
    Dim StepType() As String
    Dim StepNum() As Double
    Dim U1() As Double
    Dim U2() As Double
    Dim U3() As Double
    Dim R1() As Double
    Dim R2() As Double
    Dim R3() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link object by points
    ret = SapModel.LinkObj.AddByPoint("1", "2", LinkObjName)

    'set link property
    ret = SapModel.LinkObj.SetProperty(LinkObjName, "Link1")

    'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis()
```

Return Value

Introduction

```
'deselect all cases and combos
    ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()

'set case selected for output
    ret = SapModel.Results.Setup.SetCaseSelectedForOutput("DEAD")

'get link deformations for link object
    ret = SapModel.Results.LinkDeformation(LinkObjName, eItemTypeElm.ObjectElm, NumberResults,

'check
    If NumberResults > 0 Then ret = 0 Else ret = -1

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

[cAnalysisResults Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LST24198FA0_0") Method

Reports the link forces at the point elements at the ends of the specified link elements

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int LinkForce(
    string Name,
    eItemTypeElm ItemTypeElm,
    ref int NumberResults,
    ref string[] Obj,
    ref string[] Elm,
    ref string[] PointElm,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] P,
    ref double[] V2,
    ref double[] V3,
    ref double[] T,
    ref double[] M2,
    ref double[] M3
)
```

```
Function LinkForce (
    Name As String,
    ItemTypeElm As eItemTypeElm,
    ByRef NumberResults As Integer,
    ByRef Obj As String(),
    ByRef Elm As String(),
    ByRef PointElm As String(),
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef P As Double(),
    ByRef V2 As Double(),
    ByRef V3 As Double(),
    ByRef T As Double(),
    ByRef M2 As Double(),
    ByRef M3 As Double()
) As Integer
```

```
Dim instance As cAnalysisResults
Dim Name As String
Dim ItemTypeElm As eItemTypeElm
Dim NumberResults As Integer
Dim Obj As String()
```

```

Dim Elm As String()
Dim PointElm As String()
Dim LoadCase As String()
Dim StepType As String()
Dim StepNum As Double()
Dim P As Double()
Dim V2 As Double()
Dim V3 As Double()
Dim T As Double()
Dim M2 As Double()
Dim M3 As Double()
Dim returnValue As Integer

returnValue = instance.LinkForce(Name,
ItemtypeElm, NumberResults, Obj,
Elm, PointElm, LoadCase, StepType,
StepNum, P, V2, V3, T, M2, M3)

int LinkForce(
String^ Name,
eItemtypeElm ItemTypeElm,
int% NumberResults,
array<String^>^% Obj,
array<String^>^% Elm,
array<String^>^% PointElm,
array<String^>^% LoadCase,
array<String^>^% StepType,
array<double>^% StepNum,
array<double>^% P,
array<double>^% V2,
array<double>^% V3,
array<double>^% T,
array<double>^% M2,
array<double>^% M3
)

abstract LinkForce :
    Name : string *
    ItemTypeElm : eItemTypeElm *
    NumberResults : int byref *
    Obj : string[] byref *
    Elm : string[] byref *
    PointElm : string[] byref *
    LoadCase : string[] byref *
    StepType : string[] byref *
    StepNum : float[] byref *
    P : float[] byref *
    V2 : float[] byref *
    V3 : float[] byref *
    T : float[] byref *
    M2 : float[] byref *
    M3 : float[] byref -> int

```

Parameters

Name

Type: SystemString

The name of an existing link object, link element, or group of objects depending on the value of the ItemTypeElm item.

ItemTypeElm

Parameters

Type:Â ETABSv1eItemTypeElm

This is one of the following items in the eItemTypeElm enumeration.

If this item is ObjectElm, the result request is for the link element corresponding to the point object specified by the Name item.

If this item is Element, the result request is for the link element specified by the Name item.

If this item is GroupElm, the result request is for all link elements directly or indirectly specified in the group specified by the Name item.

If this item is SelectionElm, the result request is for all link elements directly or indirectly selected and the Name item is ignored.

NumberResults

Type:Â SystemInt32

The total number of results returned by the program.

Obj

Type:Â SystemString

This is an array that includes the link object name associated with each result, if any.

Elm

Type:Â SystemString

This is an array that includes the link element name associated with each result.

PointElm

Type:Â SystemString

This is an array that includes the point element name associated with each result.

LoadCase

Type:Â SystemString

This is an array that includes the name of the analysis case or load combination associated with each result.

StepType

Type:Â SystemString

This is an array that includes the step type, if any, for each result.

StepNum

Type:Â SystemDouble

This is an array that includes the step number, if any, for each result.

P

Type:Â SystemDouble

This is an array that includes the link axial force (in the link local 1-axis direction) at the specified point element. [F]

V2

Type:Â SystemDouble

This is an array that includes the link shear force component in the link element local 2-axis direction. [F]

V3

Type:Â SystemDouble

This is an array that includes the link shear force component in the link element local 3-axis direction. [F]

T

Type: SystemDouble

This is an array that includes the link torsion (about the link local 1-axis) at the specified point element. [FL]

M2

Type: SystemDouble

This is an array that includes the link moment component about the link element local 2-axis. [FL]

M3

Type: SystemDouble

This is an array that includes the link moment component about the link element local 3-axis. [FL]

Return Value

Type: Int32

Returns zero if the forces are successfully recovered, otherwise it returns a nonzero value

Remarks

See Results for more information.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim LinkObjName As String
    Dim NumberResults As Integer
    Dim Obj() As String
    Dim Elm() As String
    Dim LoadCase() As String
    Dim StepType() As String
    Dim StepNum() As Double
    Dim P() As Double
    Dim V2() As Double
    Dim V3() As Double
    Dim T() As Double
    Dim M2() As Double
    Dim M3() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link object by points
```

Introduction

```
ret = SapModel.LinkObj.AddByPoint("1", "2", LinkObjName)

'set link property
ret = SapModel.LinkObj.SetProperty(LinkObjName, "Link1")

'run analysis
System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
ret = SapModel.Analyze.RunAnalysis()

'deselect all cases and combos
ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()

'set case selected for output
ret = SapModel.Results.Setup.SetCaseSelectedForOutput("DEAD")

'get link forces for link object
ret = SapModel.Results.LinkForce(LinkObjName, ObjectElm, NumberResults, Obj, Elm, PointElm)

'check
If NumberResults > 0 Then ret = 0 Else ret = -1

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

See Also

Reference

[cAnalysisResults Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LSTC37E357_0") Method

Reports the joint forces at the point elements at the ends of the specified link elements

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int LinkJointForce(
    string Name,
    eItemTypeElm ItemTypeElm,
    ref int NumberResults,
    ref string[] Obj,
    ref string[] Elm,
    ref string[] PointElm,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] F1,
    ref double[] F2,
    ref double[] F3,
    ref double[] M1,
    ref double[] M2,
    ref double[] M3
)
```

```
Function LinkJointForce (
    Name As String,
    ItemTypeElm As eItemTypeElm,
    ByRef NumberResults As Integer,
    ByRef Obj As String(),
    ByRef Elm As String(),
    ByRef PointElm As String(),
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef F1 As Double(),
    ByRef F2 As Double(),
    ByRef F3 As Double(),
    ByRef M1 As Double(),
    ByRef M2 As Double(),
    ByRef M3 As Double()
) As Integer
```

```
Dim instance As cAnalysisResults
Dim Name As String
Dim ItemTypeElm As eItemTypeElm
Dim NumberResults As Integer
Dim Obj As String()
```

```

Dim Elm As String()
Dim PointElm As String()
Dim LoadCase As String()
Dim StepType As String()
Dim StepNum As Double()
Dim F1 As Double()
Dim F2 As Double()
Dim F3 As Double()
Dim M1 As Double()
Dim M2 As Double()
Dim M3 As Double()
Dim returnValue As Integer

returnValue = instance.LinkJointForce(Name,
ItemtypeElm, NumberResults, Obj,
Elm, PointElm, LoadCase, StepType,
StepNum, F1, F2, F3, M1, M2, M3)

```

```

int LinkJointForce(
String^ Name,
eItemTypeElm ItemTypeElm,
int% NumberResults,
array<String^>^% Obj,
array<String^>^% Elm,
array<String^>^% PointElm,
array<String^>^% LoadCase,
array<String^>^% StepType,
array<double>^% StepNum,
array<double>^% F1,
array<double>^% F2,
array<double>^% F3,
array<double>^% M1,
array<double>^% M2,
array<double>^% M3
)

```

```

abstract LinkJointForce :
    Name : string *
    ItemTypeElm : eItemTypeElm *
    NumberResults : int byref *
    Obj : string[] byref *
    Elm : string[] byref *
    PointElm : string[] byref *
    LoadCase : string[] byref *
    StepType : string[] byref *
    StepNum : float[] byref *
    F1 : float[] byref *
    F2 : float[] byref *
    F3 : float[] byref *
    M1 : float[] byref *
    M2 : float[] byref *
    M3 : float[] byref -> int

```

Parameters

Name

Type: SystemString

The name of an existing link object, link element, or group of objects depending on the value of the ItemTypeElm item.

ItemTypeElm

Type:Â ETABSv1eItemTypeElm

This is one of the following items in the eItemTypeElm enumeration.

If this item is ObjectElm, the result request is for the link element corresponding to the point object specified by the Name item.

If this item is Element, the result request is for the link element specified by the Name item.

If this item is GroupElm, the result request is for all link elements directly or indirectly specified in the group specified by the Name item.

If this item is SelectionElm, the result request is for all link elements directly or indirectly selected and the Name item is ignored.

NumberResults

Type:Â SystemInt32

The total number of results returned by the program.

Obj

Type:Â SystemString

This is an array that includes the link object name associated with each result, if any.

Elm

Type:Â SystemString

This is an array that includes the link element name associated with each result.

PointElm

Type:Â SystemString

This is an array that includes the point element name associated with each result.

LoadCase

Type:Â SystemString

This is an array that includes the name of the analysis case or load combination associated with each result.

StepType

Type:Â SystemString

This is an array that includes the step type, if any, for each result.

StepNum

Type:Â SystemDouble

This is an array that includes the step number, if any, for each result.

F1

Type:Â SystemDouble

This is an array that includes the joint force component in the point element local 1-axis direction. [F]

F2

Type:Â SystemDouble

This is an array that includes the joint force component in the point element local 2-axis direction. [F]

F3

Type:Â SystemDouble

This is an array that includes the joint force component in the point element local 3-axis direction. [F]

M1

Type: SystemDouble

This is an array that includes the joint moment component about the point element local 1-axis. [FL]

M2

Type: SystemDouble

This is an array that includes the joint moment component about the point element local 2-axis. [FL]

M3

Type: SystemDouble

This is an array that includes the joint moment component about the point element local 3-axis. [FL]

Return Value

Type: Int32

Returns zero if the forces are successfully recovered, otherwise it returns a nonzero value

Remarks

See Results for more information.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim LinkObjName As String
    Dim NumberResults As Integer
    Dim Obj() As String
    Dim Elm() As String
    Dim LoadCase() As String
    Dim StepType() As String
    Dim StepNum() As Double
    Dim F1() As Double
    Dim F2() As Double
    Dim F3() As Double
    Dim M1() As Double
    Dim M2() As Double
    Dim M3() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link object by points
```

Introduction

```
ret = SapModel.LinkObj.AddByPoint("1", "2", LinkObjName)

'set link property
ret = SapModel.LinkObj.SetProperty(LinkObjName, "Link1")

'run analysis
System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
ret = SapModel.Analyze.RunAnalysis()

'deselect all cases and combos
ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()

'set case selected for output
ret = SapModel.Results.Setup.SetCaseSelectedForOutput("DEAD")

'get link joint forces for link object
ret = SapModel.Results.LinkJointForce(LinkObjName, ObjectElm, NumberResults, Obj, Elm, Poi

'check
If NumberResults > 0 Then ret = 0 Else ret = -1

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

See Also

Reference

[cAnalysisResults Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LSTB3192730_0?cpp183|nu=.");

Method

Reports the modal load participation ratios for each selected modal analysis case

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int ModalLoadParticipationRatios(
    ref int NumberResults,
    ref string[] LoadCase,
    ref string[] ItemType,
    ref string[] Item,
    ref double[] Stat,
    ref double[] Dyn
)
```

```
Function ModalLoadParticipationRatios (
    ByRef NumberResults As Integer,
    ByRef LoadCase As String(),
    ByRef ItemType As String(),
    ByRef Item As String(),
    ByRef Stat As Double(),
    ByRef Dyn As Double()
) As Integer
```

```
Dim instance As cAnalysisResults
Dim NumberResults As Integer
Dim LoadCase As String()
Dim ItemType As String()
Dim Item As String()
Dim Stat As Double()
Dim Dyn As Double()
Dim returnValue As Integer
```

```
returnValue = instance.ModalLoadParticipationRatios(NumberResults,
    LoadCase, ItemType, Item, Stat, Dyn)
```

```
int ModalLoadParticipationRatios(
    int% NumberResults,
    array<String^>^% LoadCase,
    array<String^>^% ItemType,
    array<String^>^% Item,
    array<double>^% Stat,
    array<double>^% Dyn
)
```

```
abstract ModalLoadParticipationRatios :
    NumberResults : int byref *
```

cAnalysisResultsspan id="LSTB3192730_0"AddLanguageSpecificTextSet("LSTB3192730_0?cpp183|nu=.");

```

LoadCase : string[] byref *
ItemType : string[] byref *
Item : string[] byref *
Stat : float[] byref *
Dyn : float[] byref -> int

```

Parameters

NumberResults

Type: SystemInt32

The total number of results returned by the program

LoadCase

Type: SystemString

This is an array that includes the name of the modal load case associated with each result

ItemType

Type: SystemString

This is an array that includes Load Pattern, Acceleration, Link or Panel Zone. It specifies the type of item for which the modal load participation is reported

Item

Type: SystemString

This is an array whose values depend on the ItemType. If the ItemType is Load Pattern, this is the name of the load pattern.

If the ItemType is Acceleration, this is UX, UY, UZ, RX, RY, or RZ, indicating the acceleration direction.

If the ItemType is Link, this is the name of the link followed by U1, U2, U3, R1, R2, or R3 (in parenthesis), indicating the link degree of freedom for which the output is reported.

If the ItemType is Panel Zone, this is the name of the joint to which the panel zone is assigned, followed by U1, U2, U3, R1, R2, or R3 (in parenthesis), indicating the degree of freedom for which the output is reported.

Stat

Type: SystemDouble

This is an array that includes the percent static load participation ratio

Dyn

Type: SystemDouble

This is an array that includes the percent dynamic load participation ratio

Return Value

Type: Int32

Returns zero if the data is successfully recovered; otherwise it returns a nonzero value

Remarks

See [Results](#) for more information.

Examples

VB

Copy

```
Public Sub Example()
```

Introduction

```
'dimension variables
Dim SapObject as c0API
Dim SapModel As cSapModel
Dim ret As Integer = -1
Dim NumberResults As Integer
Dim LoadCase() As String
Dim ItemType() As String
Dim Item() As String
Dim Stat() As Double
Dim Dyn() As Double

'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
    ret = EtabsObject.ApplicationStart()

'create SapModel object
    SapModel = EtabsObject.SapModel

'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis()

'deselect all cases and combos
    ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()

'set case selected for output
    ret = SapModel.Results.Setup.SetCaseSelectedForOutput("MODAL")

'get modal load participation ratios
    ret = SapModel.Results.ModalLoadParticipationRatios(NumberResults, LoadCase, ItemType, Item)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

[cAnalysisResults Interface](#)

[ETABSv1 Namespace](#)

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Introduction

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CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LST9D Method

Reports the modal participating mass ratios for each mode of each selected modal analysis case

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int ModalParticipatingMassRatios(
    ref int NumberResults,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] Period,
    ref double[] UX,
    ref double[] UY,
    ref double[] UZ,
    ref double[] SumUX,
    ref double[] SumUY,
    ref double[] SumUZ,
    ref double[] RX,
    ref double[] RY,
    ref double[] RZ,
    ref double[] SumRX,
    ref double[] SumRY,
    ref double[] SumRZ
)
```

```
Function ModalParticipatingMassRatios (
    ByRef NumberResults As Integer,
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef Period As Double(),
    ByRef UX As Double(),
    ByRef UY As Double(),
    ByRef UZ As Double(),
    ByRef SumUX As Double(),
    ByRef SumUY As Double(),
    ByRef SumUZ As Double(),
    ByRef RX As Double(),
    ByRef RY As Double(),
    ByRef RZ As Double(),
    ByRef SumRX As Double(),
    ByRef SumRY As Double(),
    ByRef SumRZ As Double()
) As Integer
```

cAnalysisResultsspan id="LST9DFC350A_0"AddLanguageSpecificTextSet("LST9DFC350A_0?c16Z=:|nu=".

Introduction

```
Dim instance As cAnalysisResults
Dim NumberResults As Integer
Dim LoadCase As String()
Dim StepType As String()
Dim StepNum As Double()
Dim Period As Double()
Dim UX As Double()
Dim UY As Double()
Dim UZ As Double()
Dim SumUX As Double()
Dim SumUY As Double()
Dim SumUZ As Double()
Dim RX As Double()
Dim RY As Double()
Dim RZ As Double()
Dim SumRX As Double()
Dim SumRY As Double()
Dim SumRZ As Double()
Dim returnValue As Integer
```

```
returnValue = instance.ModalParticipatingMassRatios(NumberResults,
LoadCase, StepType, StepNum, Period,
UX, UY, UZ, SumUX, SumUY, SumUZ, RX,
RY, RZ, SumRX, SumRY, SumRZ)
```

```
int ModalParticipatingMassRatios(
int% NumberResults,
array<String^>^% LoadCase,
array<String^>^% StepType,
array<double>^% StepNum,
array<double>^% Period,
array<double>^% UX,
array<double>^% UY,
array<double>^% UZ,
array<double>^% SumUX,
array<double>^% SumUY,
array<double>^% SumUZ,
array<double>^% RX,
array<double>^% RY,
array<double>^% RZ,
array<double>^% SumRX,
array<double>^% SumRY,
array<double>^% SumRZ
)
```

```
abstract ModalParticipatingMassRatios :
    NumberResults : int byref *
    LoadCase : string[] byref *
    StepType : string[] byref *
    StepNum : float[] byref *
    Period : float[] byref *
    UX : float[] byref *
    UY : float[] byref *
    UZ : float[] byref *
    SumUX : float[] byref *
    SumUY : float[] byref *
    SumUZ : float[] byref *
    RX : float[] byref *
    RY : float[] byref *
    RZ : float[] byref *
    SumRX : float[] byref *
    SumRY : float[] byref *
```


SumRZ : float[] byref -> int

Parameters

NumberResults

Type:Â SystemInt32

The total number of results returned by the program

LoadCase

Type:Â SystemString

This is an array that includes the name of the modal load case associated with each result

StepType

Type:Â SystemString

This is an array that includes the step type, if any, for each result. For modal results, this will always be Mode

StepNum

Type:Â SystemDouble

This is an array that includes the step number for each result. For modal results, this is always the mode number

Period

Type:Â SystemDouble

This is an array that includes the period for each result. [s]

UX

Type:Â SystemDouble

This is an array that includes the modal participating mass ratio for the structure Ux degree of freedom. The ratio applies to the specified mode

UY

Type:Â SystemDouble

This is an array that includes the modal participating mass ratio for the structure Uy degree of freedom. The ratio applies to the specified mode

UZ

Type:Â SystemDouble

This is an array that includes the modal participating mass ratio for the structure Uz degree of freedom. The ratio applies to the specified mode

SumUX

Type:Â SystemDouble

This is an array that includes the cumulative sum of the modal participating mass ratios for the structure Ux degree of freedom

SumUY

Type:Â SystemDouble

This is an array that includes the cumulative sum of the modal participating mass ratios for the structure Uy degree of freedom

SumUZ

Type:Â SystemDouble

This is an array that includes the cumulative sum of the modal participating mass ratios for the structure Uz degree of freedom

RX

Type:Â SystemDouble

This is an array that includes the modal participating mass ratio for the structure Rx degree of freedom. The ratio applies to the specified mode

RY

Type: SystemDouble

This is an array that includes the modal participating mass ratio for the structure Ry degree of freedom. The ratio applies to the specified mode

RZ

Type: SystemDouble

This is an array that includes the modal participating mass ratio for the structure Rz degree of freedom. The ratio applies to the specified mode

SumRX

Type: SystemDouble

This is an array that includes the cumulative sum of the modal participating mass ratios for the structure Rx degree of freedom

SumRY

Type: SystemDouble

This is an array that includes the cumulative sum of the modal participating mass ratios for the structure Ry degree of freedom

SumRZ

Type: SystemDouble

This is an array that includes the cumulative sum of the modal participating mass ratios for the structure Rz degree of freedom

Return Value

Type: Int32

Returns zero if the data is successfully recovered; otherwise it returns a nonzero value

Remarks

See [Results](#) for more information.

Examples

VB

Copy

```
Public Sub Example()
```

```
    'dimension variables
    Dim SapObject as cOAPI
    Dim SapModel As cSapModel
    Dim ret As Integer = -1
    Dim NumberResults As Integer
    Dim LoadCase() As String
    Dim StepType() As String
    Dim StepNum() As Double
    Dim Period() As Double
    Dim Ux() As Double
    Dim Uy() As Double
    Dim Uz() As Double
    Dim SumUx() As Double
    Dim SumUy() As Double
    Dim SumUz() As Double
    Dim Rx() As Double
    Dim Ry() As Double
    Dim Rz() As Double
    Dim SumRx() As Double
    Dim SumRy() As Double
    Dim SumRz() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

Introduction

```
'start ETABS application
    ret = EtabsObject.ApplicationStart()

'create SapModel object
    SapModel = EtabsObject.SapModel

'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis()

'deselect all cases and combos
    ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()

'set case selected for output
    ret = SapModel.Results.Setup.SetCaseSelectedForOutput("MODAL")

'get modal participating mass ratios
    ret = SapModel.Results.ModalParticipatingMassRatios(NumberResults, LoadCase, StepType, Ste

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

🔗 See Also

Reference

[cAnalysisResults Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LST76 Method

Reports the modal participation factors for each mode of each selected modal analysis case

Namespace: `ETABSv1`

Assembly: `ETABSv1 (in ETABSv1.dll)` Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int ModalParticipationFactors(
    ref int NumberResults,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] Period,
    ref double[] UX,
    ref double[] UY,
    ref double[] UZ,
    ref double[] RX,
    ref double[] RY,
    ref double[] RZ,
    ref double[] ModalMass,
    ref double[] ModalStiff
)
```

```
Function ModalParticipationFactors (
    ByRef NumberResults As Integer,
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef Period As Double(),
    ByRef UX As Double(),
    ByRef UY As Double(),
    ByRef UZ As Double(),
    ByRef RX As Double(),
    ByRef RY As Double(),
    ByRef RZ As Double(),
    ByRef ModalMass As Double(),
    ByRef ModalStiff As Double()
) As Integer
```

```
Dim instance As cAnalysisResults
Dim NumberResults As Integer
Dim LoadCase As String()
Dim StepType As String()
Dim StepNum As Double()
Dim Period As Double()
Dim UX As Double()
Dim UY As Double()
```

cAnalysisResultsspan id="LST7673A124_0"AddLanguageSpecificTextSet("LST7673A124_0?cpp192|nu=.");

Introduction

```
Dim UZ As Double()
Dim RX As Double()
Dim RY As Double()
Dim RZ As Double()
Dim ModalMass As Double()
Dim ModalStiff As Double()
Dim returnValue As Integer

returnValue = instance.ModalParticipationFactors(NumberResults,
LoadCase, StepType, StepNum, Period,
UX, UY, UZ, RX, RY, RZ, ModalMass, ModalStiff)

int ModalParticipationFactors(
int% NumberResults,
array<String^>^% LoadCase,
array<String^>^% StepType,
array<double>^% StepNum,
array<double>^% Period,
array<double>^% UX,
array<double>^% UY,
array<double>^% UZ,
array<double>^% RX,
array<double>^% RY,
array<double>^% RZ,
array<double>^% ModalMass,
array<double>^% ModalStiff
)

abstract ModalParticipationFactors :
    NumberResults : int byref *
    LoadCase : string[] byref *
    StepType : string[] byref *
    StepNum : float[] byref *
    Period : float[] byref *
    UX : float[] byref *
    UY : float[] byref *
    UZ : float[] byref *
    RX : float[] byref *
    RY : float[] byref *
    RZ : float[] byref *
    ModalMass : float[] byref *
    ModalStiff : float[] byref -> int
```

Parameters

NumberResults

Type:Â [SystemInt32](#)

The total number of results returned by the program

LoadCase

Type:Â [SystemString](#)

This is an array that includes the name of the modal load case associated with each result

StepType

Type:Â [SystemString](#)

This is an array that includes the step type, if any, for each result. For modal results, this will always be Mode

StepNum

Type: SystemDouble

This is an array that includes the step number for each result. For modal results, this is always the mode number

Period

Type: SystemDouble

This is an array that includes the period for each result. [s]

UX

Type: SystemDouble

This is an array that includes the modal participation factor for the structure Ux degree of freedom. The ratio applies to the specified mode. [Fs2]

UY

Type: SystemDouble

This is an array that includes the modal participation factor for the structure Uy degree of freedom. The ratio applies to the specified mode. [Fs2]

UZ

Type: SystemDouble

This is an array that includes the modal participation factor for the structure Uz degree of freedom. The ratio applies to the specified mode. [Fs2]

RX

Type: SystemDouble

This is an array that includes the modal participation factor for the structure Rx degree of freedom. The ratio applies to the specified mode. [FLs2]

RY

Type: SystemDouble

This is an array that includes the modal participation factor for the structure Ry degree of freedom. The ratio applies to the specified mode. [FLs2]

RZ

Type: SystemDouble

This is an array that includes the modal participation factor for the structure Rz degree of freedom. The ratio applies to the specified mode. [FLs2]

ModalMass

Type: SystemDouble

This is an array that includes the modal mass for the specified mode. This is a measure of the kinetic energy in the structure as it is deforming in the specified mode. [FLs2]

ModalStiff

Type: SystemDouble

This is an array that includes the modal stiffness for the specified mode. This is a measure of the strain energy in the structure as it is deforming in the specified mode. [FL]

Return Value

Type: Int32

Returns zero if the data is successfully recovered; otherwise it returns a nonzero value

▲ Remarks

See [Results](#) for more information.

▲ Examples

VB

Copy

```
Public Sub Example()
```

Introduction

```
'dimension variables
Dim SapObject as cOAPI
Dim SapModel As cSapModel
Dim ret As Integer = -1
Dim NumberResults As Integer
Dim LoadCase() As String
Dim StepType() As String
Dim StepNum() As Double
Dim Period() As Double
Dim Ux() As Double
Dim Uy() As Double
Dim Uz() As Double
Dim Rx() As Double
Dim Ry() As Double
Dim Rz() As Double
Dim ModalMass() As Double
Dim ModalStiff() As Double

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'run analysis
System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
ret = SapModel.Analyze.RunAnalysis()

'deselect all cases and combos
ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()

'set case selected for output
ret = SapModel.Results.Setup.SetCaseSelectedForOutput("MODAL")

'get modal participation factors
ret = SapModel.Results.ModalParticipationFactors(NumberResults, LoadCase, StepType, StepNum)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

See Also

Reference

cAnalysisResults Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LST2D417329_0?cpu=197|nu=.") Method

Reports the modal period, cyclic frequency, circular frequency and eigenvalue for each selected modal load case

Namespace: `ETABSv1`

Assembly: `ETABSv1 (in ETABSv1.dll)` Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int ModalPeriod(
    ref int NumberResults,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] Period,
    ref double[] Frequency,
    ref double[] CircFreq,
    ref double[] EigenValue
)
```

```
Function ModalPeriod (
    ByRef NumberResults As Integer,
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef Period As Double(),
    ByRef Frequency As Double(),
    ByRef CircFreq As Double(),
    ByRef EigenValue As Double()
) As Integer
```

```
Dim instance As cAnalysisResults
Dim NumberResults As Integer
Dim LoadCase As String()
Dim StepType As String()
Dim StepNum As Double()
Dim Period As Double()
Dim Frequency As Double()
Dim CircFreq As Double()
Dim EigenValue As Double()
Dim returnValue As Integer
```

```
returnValue = instance.ModalPeriod(NumberResults,
    LoadCase, StepType, StepNum, Period,
    Frequency, CircFreq, EigenValue)
```

```
int ModalPeriod(
    int% NumberResults,
    array<String^>% LoadCase,
```

cAnalysisResultsspan id="LST2D417329_0"AddLanguageSpecificTextSet("LST2D417329_0?cpu=197|nu=.");

```

array<String>^% StepType,
array<double>^% StepNum,
array<double>^% Period,
array<double>^% Frequency,
array<double>^% CircFreq,
array<double>^% EigenValue
)

abstract ModalPeriod :
    NumberResults : int byref *
    LoadCase : string[] byref *
    StepType : string[] byref *
    StepNum : float[] byref *
    Period : float[] byref *
    Frequency : float[] byref *
    CircFreq : float[] byref *
    EigenValue : float[] byref -> int

```

Parameters

NumberResults

Type: SystemInt32

The total number of results returned by the program

LoadCase

Type: SystemString

This is an array that includes the name of the modal load case associated with each result

StepType

Type: SystemString

This is an array that includes the step type, if any, for each result. For modal results, this will always be Mode

StepNum

Type: SystemDouble

This is an array that includes the step number for each result. For modal results, this is always the mode number

Period

Type: SystemDouble

This is an array that includes the period for each result. [s]

Frequency

Type: SystemDouble

This is an array that includes the cyclic frequency for each result. [1/s]

CircFreq

Type: SystemDouble

This is an array that includes the circular frequency for each result. [rad/s]

EigenValue

Type: SystemDouble

This is an array that includes the eigenvalue for the specified mode for each result. [rad²/s²]

Return Value

Type: Int32

Returns zero if the data is successfully recovered; otherwise it returns a nonzero value

Remarks

See [Results](#) for more information.

Examples

VB

Copy

```
Public Sub Example()

    'dimension variables
    Dim SapObject as cOAPI
    Dim SapModel As cSapModel
    Dim ret As Integer = -1
    Dim NumberResults As Integer
    Dim LoadCase() As String
    Dim StepType() As String
    Dim StepNum() As Double
    Dim Period() As Double
    Dim Frequency() As Double
    Dim CircFreq() As Double
    Dim EigenValue() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis()

    'deselect all cases and combos
    ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()

    'set case selected for output
    ret = SapModel.Results.Setup.SetCaseSelectedForOutput("MODAL")

    'get modal period
    ret = SapModel.Results.ModalPeriod(NumberResults, LoadCase, StepType, StepNum, Period, Frequency, CircFreq, EigenValue)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

cAnalysisResults Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LST59 Method

Reports the modal displacements (mode shapes) for the specified point elements

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int ModeShape(
    string Name,
    eItemTypeElm ItemTypeElm,
    ref int NumberResults,
    ref string[] Obj,
    ref string[] Elm,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] U1,
    ref double[] U2,
    ref double[] U3,
    ref double[] R1,
    ref double[] R2,
    ref double[] R3
)
```

```
Function ModeShape (
    Name As String,
    ItemTypeElm As eItemTypeElm,
    ByRef NumberResults As Integer,
    ByRef Obj As String(),
    ByRef Elm As String(),
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef U1 As Double(),
    ByRef U2 As Double(),
    ByRef U3 As Double(),
    ByRef R1 As Double(),
    ByRef R2 As Double(),
    ByRef R3 As Double()
) As Integer
```

```
Dim instance As cAnalysisResults
Dim Name As String
Dim ItemTypeElm As eItemTypeElm
Dim NumberResults As Integer
Dim Obj As String()
Dim Elm As String()
Dim LoadCase As String()
```

cAnalysisResultsspan id="LST59F47D0D_0"AddLanguageSpecificTextSet("LST59F47D0D_0?capi::|nu=")

```

Dim StepType As String()
Dim StepNum As Double()
Dim U1 As Double()
Dim U2 As Double()
Dim U3 As Double()
Dim R1 As Double()
Dim R2 As Double()
Dim R3 As Double()
Dim returnValue As Integer

returnValue = instance.ModeShape(Name,
ItemTypesElm, NumberResults, Obj,
Elm, LoadCase, StepType, StepNum,
U1, U2, U3, R1, R2, R3)

```

```

int ModeShape(
String^ Name,
eItemTypesElm ItemTypesElm,
int% NumberResults,
array<String^>^% Obj,
array<String^>^% Elm,
array<String^>^% LoadCase,
array<String^>^% StepType,
array<double>^% StepNum,
array<double>^% U1,
array<double>^% U2,
array<double>^% U3,
array<double>^% R1,
array<double>^% R2,
array<double>^% R3
)

```

```

abstract ModeShape :
    Name : string *
    ItemTypesElm : eItemTypesElm *
    NumberResults : int byref *
    Obj : string[] byref *
    Elm : string[] byref *
    LoadCase : string[] byref *
    StepType : string[] byref *
    StepNum : float[] byref *
    U1 : float[] byref *
    U2 : float[] byref *
    U3 : float[] byref *
    R1 : float[] byref *
    R2 : float[] byref *
    R3 : float[] byref -> int

```

Parameters

Name

Type: [SystemString](#)

The name of an existing point object, point element, or group of objects depending on the value of the ItemTypesElm item.

ItemTypesElm

Type: [ETABSv1eItemTypesElm](#)

This is one of the following items in the [eItemTypesElm](#) enumeration.

Introduction

If this item is ObjectElm, the result request is for the point element corresponding to the point object specified by the Name item.

If this item is Element, the result request is for the point element specified by the Name item.

If this item is GroupElm, the result request is for all point elements directly or indirectly specified in the group specified by the Name item.

If this item is SelectionElm, the result request is for all point elements directly or indirectly selected and the Name item is ignored.

NumberResults

Type:Â SystemInt32

The total number of results returned by the program.

Obj

Type:Â SystemString

This is an array that includes the point object name associated with each result, if any. Some results will have no point object associated with them. For those cases, this item will be blank.

Elm

Type:Â SystemString

This is an array that includes the point element name associated with each result.

LoadCase

Type:Â SystemString

This is an array that includes the name of the analysis case or load combination associated with each result.

StepType

Type:Â SystemString

This is an array that includes the step type, if any, for each result.

StepNum

Type:Â SystemDouble

This is an array that includes the step number, if any, for each result.

U1

Type:Â SystemDouble

This is a one dimensional array that includes the displacement in the point element local 1-axis direction for each result. [L]

U2

Type:Â SystemDouble

This is a one dimensional array that includes the displacement in the point element local 2-axis direction for each result. [L]

U3

Type:Â SystemDouble

This is a one dimensional array that includes the displacement in the point element local 3-axis direction for each result. [L]

R1

Type:Â SystemDouble

This is a one dimensional array that includes the rotation about the point element local 1-axis for each result. [rad]

R2

Type: SystemDouble

This is a one dimensional array that includes the rotation about the point element local 2-axis for each result. [rad]

R3

Type: SystemDouble

This is a one dimensional array that includes the rotation about the point element local 3-axis for each result. [rad]

Return Value

Type: Int32

Returns zero if the displacements are successfully recovered, otherwise it returns a nonzero value

Remarks

See Results for more information.

Examples

VB

Copy

```
Public Sub Example()

    'dimension variables
    Dim SapObject as cOAPI
    Dim SapModel As cSapModel
    Dim ret As Integer = -1
    Dim NumberResults As Integer
    Dim LoadCase() As String
    Dim StepType() As String
    Dim StepNum() As Double
    Dim U1() As Double
    Dim U2() As Double
    Dim U3() As Double
    Dim R1() As Double
    Dim R2() As Double
    Dim R3() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis()

    'deselect all cases and combos
    ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()
```


Introduction

```
'set case selected for output
  ret = SapModel.Results.Setup.SetCaseSelectedForOutput("MODAL")

'get mode shape
  ret = SapModel.Results.ModeShape("ALL", GroupElm, NumberResults, Obj, Elm, LoadCase, Step)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

cAnalysisResults Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LST9394533A_0?cp206|nu=.");

Method

Reports the panel zone (link) internal deformations

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int PanelZoneDeformation(
    string Name,
    eItemTypeElm ItemTypeElm,
    ref int NumberResults,
    ref string[] Elm,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] U1,
    ref double[] U2,
    ref double[] U3,
    ref double[] R1,
    ref double[] R2,
    ref double[] R3
)
```

```
Function PanelZoneDeformation (
    Name As String,
    ItemTypeElm As eItemTypeElm,
    ByRef NumberResults As Integer,
    ByRef Elm As String(),
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef U1 As Double(),
    ByRef U2 As Double(),
    ByRef U3 As Double(),
    ByRef R1 As Double(),
    ByRef R2 As Double(),
    ByRef R3 As Double()
) As Integer
```

```
Dim instance As cAnalysisResults
Dim Name As String
Dim ItemTypeElm As eItemTypeElm
Dim NumberResults As Integer
Dim Elm As String()
Dim LoadCase As String()
Dim StepType As String()
Dim StepNum As Double()
Dim U1 As Double()
```

cAnalysisResultsspan id="LST9394533A_0"AddLanguageSpecificTextSet("LST9394533A_0?cp206|nu=.");

```

Dim U2 As Double()
Dim U3 As Double()
Dim R1 As Double()
Dim R2 As Double()
Dim R3 As Double()
Dim returnValue As Integer

returnValue = instance.PanelZoneDeformation(Name,
ItemtypeElm, NumberResults, Elm,
LoadCase, StepType, StepNum, U1, U2,
U3, R1, R2, R3)

int PanelZoneDeformation(
String^ Name,
eItemTypeElm ItemTypeElm,
int% NumberResults,
array<String^>^% Elm,
array<String^>^% LoadCase,
array<String^>^% StepType,
array<double>^% StepNum,
array<double>^% U1,
array<double>^% U2,
array<double>^% U3,
array<double>^% R1,
array<double>^% R2,
array<double>^% R3
)

abstract PanelZoneDeformation :
    Name : string *
    ItemTypeElm : eItemTypeElm *
    NumberResults : int byref *
    Elm : string[] byref *
    LoadCase : string[] byref *
    StepType : string[] byref *
    StepNum : float[] byref *
    U1 : float[] byref *
    U2 : float[] byref *
    U3 : float[] byref *
    R1 : float[] byref *
    R2 : float[] byref *
    R3 : float[] byref -> int

```

Parameters

Name

Type: SystemString

The name of an existing link object, link element, or group of objects depending on the value of the ItemTypeElm item.

ItemTypeElm

Type: ETABSv1eItemTypeElm

This is one of the following items in the eItemTypeElm enumeration.

If this item is ObjectElm, the result request is for the panel zone (link) element corresponding to the panel zone assignment to the point object specified by the Name item.

If this item is Element, the result request is for the panel zone (link) element specified by the Name item.

If this item is GroupElm, the result request is for all panel zone (link) elements directly or indirectly specified in the group specified by the Name item.

If this item is SelectionElm, the result request is for all panel zone (link) elements directly or indirectly selected and the Name item is ignored.

For GroupElm and SelectionElm a panel zone (link) element may be indirectly specified through point objects that have panel zone assignments.

NumberResults

Type:Â SystemInt32

The total number of results returned by the program.

Elm

Type:Â SystemString

This is an array that includes the panel zone (link) element name associated with each result.

LoadCase

Type:Â SystemString

This is an array that includes the name of the analysis case or load combination associated with each result.

StepType

Type:Â SystemString

This is an array that includes the step type, if any, for each result.

StepNum

Type:Â SystemDouble

This is an array that includes the step number, if any, for each result.

U1

Type:Â SystemDouble

This is a one dimensional array that includes the internal translational deformation of the panel zone (link) in the link element local 1-axis direction. [L]

U2

Type:Â SystemDouble

This is a one dimensional array that includes the internal translational deformation of the panel zone (link) in the link element local 2-axis direction. [L]

U3

Type:Â SystemDouble

This is a one dimensional array that includes the internal translational deformation of the panel zone (link) in the link element local 3-axis direction. [L]

R1

Type:Â SystemDouble

This is a one dimensional array that includes the internal rotational deformation of the panel zone (link) about the link element local 1-axis. [L]

R2

Type:Â SystemDouble

This is a one dimensional array that includes the internal rotational deformation of the panel zone (link) about the link element local 2-axis. [L]

R3

Type:Â SystemDouble

This is a one dimensional array that includes the internal rotational deformation

of the panel zone (link) about the link element local 3-axis. [L]

Return Value

Type: $\hat{A}$ Int32

Returns zero if the deformations are successfully recovered, otherwise it returns a nonzero value

Remarks

See Results for more information.

Examples

VB

Copy

```
Public Sub Example()

    'dimension variables
    Dim SapObject as cOAPI
    Dim SapModel As cSapModel
    Dim ret As Integer = -1
    Dim NumberResults As Integer
    Dim Elm() As String
    Dim LoadCase() As String
    Dim StepType() As String
    Dim StepNum() As Double
    Dim U1() As Double
    Dim U2() As Double
    Dim U3() As Double
    Dim R1() As Double
    Dim R2() As Double
    Dim R3() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'assign panel zone to point object "3"
    ret = SapModel.PointObj.SetPanelZone("3", 1, 2, 0, 0, "", 0, 0, 0)

    'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis()

    'deselect all cases and combos
    ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()

    'set case selected for output
    ret = SapModel.Results.Setup.SetCaseSelectedForOutput("MODAL")
```

Introduction

```
'get panel zone deformation for point object "3"
  ret = SapModel.Results.PanelZoneDeformation("3", eItemTypeElm.ObjectElm, NumberResults, ET

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

[cAnalysisResults Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LST292F718_0") Method

Reports the panel zone (link) forces at the point elements at the ends of the specified panel zone (link) elements

Namespace: `ETABSv1`

Assembly: `ETABSv1 (in ETABSv1.dll)` Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int PanelZoneForce(
    string Name,
    eItemTypeElm ItemTypeElm,
    ref int NumberResults,
    ref string[] Elm,
    ref string[] PointElm,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] P,
    ref double[] V2,
    ref double[] V3,
    ref double[] T,
    ref double[] M2,
    ref double[] M3
)
```

```
Function PanelZoneForce (
    Name As String,
    ItemTypeElm As eItemTypeElm,
    ByRef NumberResults As Integer,
    ByRef Elm As String(),
    ByRef PointElm As String(),
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef P As Double(),
    ByRef V2 As Double(),
    ByRef V3 As Double(),
    ByRef T As Double(),
    ByRef M2 As Double(),
    ByRef M3 As Double()
) As Integer
```

```
Dim instance As cAnalysisResults
Dim Name As String
Dim ItemTypeElm As eItemTypeElm
Dim NumberResults As Integer
Dim Elm As String()
Dim PointElm As String()
```

cAnalysisResultsspan id="LST2992F718_0"AddLanguageSpecificTextSet("LST2992F718_0?cp241|nu=.");l

```

Dim LoadCase As String()
Dim StepType As String()
Dim StepNum As Double()
Dim P As Double()
Dim V2 As Double()
Dim V3 As Double()
Dim T As Double()
Dim M2 As Double()
Dim M3 As Double()
Dim returnValue As Integer

returnValue = instance.PanelZoneForce(Name,
ItemTypeId, NumberResults, Elm,
PointId, LoadCase, StepType, StepNum,
P, V2, V3, T, M2, M3)

```

```

int PanelZoneForce(
String^ Name,
eItemTypeId ItemTypeId,
int% NumberResults,
array<String^>^% Elm,
array<String^>^% PointId,
array<String^>^% LoadCase,
array<String^>^% StepType,
array<double>^% StepNum,
array<double>^% P,
array<double>^% V2,
array<double>^% V3,
array<double>^% T,
array<double>^% M2,
array<double>^% M3
)

```

```

abstract PanelZoneForce :
    Name : string *
    ItemTypeId : eItemTypeId *
    NumberResults : int byref *
    Elm : string[] byref *
    PointId : string[] byref *
    LoadCase : string[] byref *
    StepType : string[] byref *
    StepNum : float[] byref *
    P : float[] byref *
    V2 : float[] byref *
    V3 : float[] byref *
    T : float[] byref *
    M2 : float[] byref *
    M3 : float[] byref -> int

```

Parameters

Name

Type: SystemString

The name of an existing point object, point element, or group of objects depending on the value of the ItemTypeId item.

ItemTypeId

Type: ETABSv1eItemTypeId

This is one of the following items in the eItemTypeId enumeration.

Introduction

If this item is ObjectElm, the result request is for the panel zone (link) element corresponding to the panel zone assignment to the point object specified by the Name item.

If this item is Element, the result request is for the panel zone (link) element specified by the Name item.

If this item is GroupElm, the result request is for all panel zone (link) elements directly or indirectly specified in the group specified by the Name item.

If this item is SelectionElm, the result request is for all panel zone (link) elements directly or indirectly selected and the Name item is ignored.

For GroupElm and SelectionElm a panel zone (link) element may be indirectly specified through point objects that have panel zone assignments.

NumberResults

Type:Â SystemInt32

The total number of results returned by the program.

Elm

Type:Â SystemString

This is an array that includes the panel zone (link) element name associated with each result.

PointElm

Type:Â SystemString

This is an array that includes the point element name associated with each result.

LoadCase

Type:Â SystemString

This is an array that includes the name of the analysis case or load combination associated with each result.

StepType

Type:Â SystemString

This is an array that includes the step type, if any, for each result.

StepNum

Type:Â SystemDouble

This is an array that includes the step number, if any, for each result.

P

Type:Â SystemDouble

This is an array that includes the panel zone (link) axial force (in the link local 1-axis direction) at the specified point element. [F]

V2

Type:Â SystemDouble

This is an array that includes the panel zone (link) shear force component in the link element local 2-axis direction. [F]

V3

Type:Â SystemDouble

This is an array that includes the panel zone (link) shear force component in the link element local 3-axis direction. [F]

T

Type:Â SystemDouble

This is an array that includes the panel zone (link) torsion (about the link local

1-axis) at the specified point element. [FL]

M2

Type: SystemDouble

This is an array that includes the panel zone (link) moment component about the link element local 2-axis. [F]

M3

Type: SystemDouble

This is an array that includes the panel zone (link) moment component about the link element local 3-axis. [F]

Return Value

Type: Int32

Returns zero if the forces are successfully recovered, otherwise it returns a nonzero value

Remarks

See Results for more information.

Examples

VB

Copy

```
Public Sub Example()

    'dimension variables
    Dim SapObject as c0API
    Dim SapModel As cSapModel
    Dim ret As Integer = -1
    Dim NumberResults As Integer
    Dim Elm() As String
    Dim PointElm() As String
    Dim LoadCase() As String
    Dim StepType() As String
    Dim StepNum() As Double
    Dim P() As Double
    Dim V2() As Double
    Dim V3() As Double
    Dim T() As Double
    Dim M2() As Double
    Dim M3() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'assign panel zone to point object "3"
    ret = SapModel.PointObj.SetPanelZone("3", 1, 2, 0, 0, "", 0, 0, 0)
```

Introduction

```
'run analysis
  System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
  ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
  ret = SapModel.Analyze.RunAnalysis()

'deselect all cases and combos
  ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()

'set case selected for output
  ret = SapModel.Results.Setup.SetCaseSelectedForOutput("MODAL")

'get panel zone force for point object "3"
  ret = SapModel.Results.PanelZoneForce("3", eItemTypeElm.ObjectElm, NumberResults, Elm, Poi

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cAnalysisResults Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LST278848F0_0?cpu=.") Method

Retrieves pier forces for any defined pier objects in the model

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int PierForce(
    ref int NumberResults,
    ref string[] StoryName,
    ref string[] PierName,
    ref string[] LoadCase,
    ref string[] Location,
    ref double[] P,
    ref double[] V2,
    ref double[] V3,
    ref double[] T,
    ref double[] M2,
    ref double[] M3
)
```

```
Function PierForce (
    ByRef NumberResults As Integer,
    ByRef StoryName As String(),
    ByRef PierName As String(),
    ByRef LoadCase As String(),
    ByRef Location As String(),
    ByRef P As Double(),
    ByRef V2 As Double(),
    ByRef V3 As Double(),
    ByRef T As Double(),
    ByRef M2 As Double(),
    ByRef M3 As Double()
) As Integer
```

```
Dim instance As cAnalysisResults
Dim NumberResults As Integer
Dim StoryName As String()
Dim PierName As String()
Dim LoadCase As String()
Dim Location As String()
Dim P As Double()
Dim V2 As Double()
Dim V3 As Double()
Dim T As Double()
Dim M2 As Double()
Dim M3 As Double()
Dim returnValue As Integer
```

cAnalysisResultsspan id="LST278848F0_0"AddLanguageSpecificTextSet("LST278848F0_0?cpu=.")>

```
returnValue = instance.PierForce(NumberResults,
StoryName, PierName, LoadCase, Location,
P, V2, V3, T, M2, M3)
```

```
int PierForce(
int% NumberResults,
array<String^>% StoryName,
array<String^>% PierName,
array<String^>% LoadCase,
array<String^>% Location,
array<double>% P,
array<double>% V2,
array<double>% V3,
array<double>% T,
array<double>% M2,
array<double>% M3
)
```

```
abstract PierForce :
    NumberResults : int byref *
    StoryName : string[] byref *
    PierName : string[] byref *
    LoadCase : string[] byref *
    Location : string[] byref *
    P : float[] byref *
    V2 : float[] byref *
    V3 : float[] byref *
    T : float[] byref *
    M2 : float[] byref *
    M3 : float[] byref -> int
```

Parameters

NumberResults

Type:Â SystemInt32

The total number of results returned by the program

StoryName

Type:Â SystemString

The story name of the pier object

PierName

Type:Â SystemString

The name of the pier object

LoadCase

Type:Â SystemString

The names of the load case

Location

Type:Â SystemString

The location on the pier, either "Top" or "Bottom", of the result being reported

P

Type:Â SystemDouble

The axial force [F]

V2

Type:Â SystemDouble

The shear force in the local 2 direction [F]

V3

T	Type:Â <u>SystemDouble</u>
	The shear force in the local 3 direction [F]
M2	Type:Â <u>SystemDouble</u>
	The torsion [FL]
M3	Type:Â <u>SystemDouble</u>
	The moment about the local 3 axis [FL]

Return Value

Type:Â Int32

Returns zero if the results are successfully retrieved; otherwise returns nonzero

Remarks

See Results for more information.

See Also

Reference

cAnalysisResults Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LSTE5 Method

Reports the section cut force for sections cuts that are specified to have an Analysis (F1, F2, F3, M1, M2, M3) result type

Namespace: `ETABSV1`

Assembly: `ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)`

Syntax

C#

VB

C++

F#

Copy

```
int SectionCutAnalysis(
    ref int NumberResults,
    ref string[] SCut,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] F1,
    ref double[] F2,
    ref double[] F3,
    ref double[] M1,
    ref double[] M2,
    ref double[] M3
)
```

```
Function SectionCutAnalysis (
    ByRef NumberResults As Integer,
    ByRef SCut As String(),
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef F1 As Double(),
    ByRef F2 As Double(),
    ByRef F3 As Double(),
    ByRef M1 As Double(),
    ByRef M2 As Double(),
    ByRef M3 As Double()
) As Integer
```

```
Dim instance As cAnalysisResults
Dim NumberResults As Integer
Dim SCut As String()
Dim LoadCase As String()
Dim StepType As String()
Dim StepNum As Double()
Dim F1 As Double()
Dim F2 As Double()
Dim F3 As Double()
Dim M1 As Double()
Dim M2 As Double()
Dim M3 As Double()
```

cAnalysisResultsspan id="LSTE5A39D7E_0"AddLanguageSpecificTextSet("LSTE5A39D7E_0?cpe::|nu="

```
Dim returnValue As Integer
```

```
returnValue = instance.SectionCutAnalysis(NumberResults,  
SCut, LoadCase, StepType, StepNum,  
F1, F2, F3, M1, M2, M3)
```

```
int SectionCutAnalysis(  
int% NumberResults,  
array<String^>^% SCut,  
array<String^>^% LoadCase,  
array<String^>^% StepType,  
array<double>^% StepNum,  
array<double>^% F1,  
array<double>^% F2,  
array<double>^% F3,  
array<double>^% M1,  
array<double>^% M2,  
array<double>^% M3  
)
```

```
abstract SectionCutAnalysis :  
    NumberResults : int byref *  
    SCut : string[] byref *  
    LoadCase : string[] byref *  
    StepType : string[] byref *  
    StepNum : float[] byref *  
    F1 : float[] byref *  
    F2 : float[] byref *  
    F3 : float[] byref *  
    M1 : float[] byref *  
    M2 : float[] byref *  
    M3 : float[] byref -> int
```

Parameters

NumberResults

Type:Â SystemInt32

The number total of results returned by the program

SCut

Type:Â SystemString

This is an array that includes the name of the section cut associated with each result

LoadCase

Type:Â SystemString

This is an array that includes the name of the analysis case or load combination associated with each result.

StepType

Type:Â SystemString

This is an array that includes the step type, if any, for each result.

StepNum

Type:Â SystemDouble

This is an array that includes the step number, if any, for each result.

F1

Type:Â SystemDouble

This is an array that includes the force in the section cut local 1-axis directions for each result. [F]

F2

Type: SystemDouble

This is an array that includes the force in the section cut local 2-axis directions for each result. [F]

F3

Type: SystemDouble

This is an array that includes the force in the section cut local 3-axis directions for each result. [F]

M1

Type: SystemDouble

This is an array that includes the moment about the section cut local 1-axis for each result. [FL]

M2

Type: SystemDouble

This is an array that includes the moment about the section cut local 1-axis for each result. [FL]

M3

Type: SystemDouble

This is an array that includes the moment about the section cut local 1-axis for each result. [FL]

Return Value

Type: Int32

Returns zero if the section cut forces are successfully recovered, otherwise it returns a nonzero value

Remarks

See [Results](#) for more information.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberResults As Integer
    Dim SCut() As String
    Dim LoadCase() As String
    Dim StepType() As String
    Dim StepNum() As Double
    Dim F1() As Double
    Dim F2() As Double
    Dim F3() As Double
    Dim M1() As Double
    Dim M2() As Double
    Dim M3() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
```

Introduction

```
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'run analysis
System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
ret = SapModel.Analyze.RunAnalysis()

'deselect all cases and combos
ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()

'set case selected for output
ret = SapModel.Results.Setup.SetCaseSelectedForOutput("DEAD")

'get section cut forces with analysis output convention
ret = SapModel.Results.SectionCutAnalysis(NumberResults, SCut, LoadCase, StepType, StepNum)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

🔗 See Also

Reference

[cAnalysisResults Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LST9377EF91_0") Method

Reports the section cut force for sections cuts that are specified to have a Design (P, V2, V3, T, M2, M3) result type

Namespace: `ETABSv1`

Assembly: `ETABSv1 (in ETABSv1.dll)` Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SectionCutDesign(
    ref int NumberResults,
    ref string[] SCut,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref double[] P,
    ref double[] V2,
    ref double[] V3,
    ref double[] T,
    ref double[] M2,
    ref double[] M3
)
```

```
Function SectionCutDesign (
    ByRef NumberResults As Integer,
    ByRef SCut As String(),
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef P As Double(),
    ByRef V2 As Double(),
    ByRef V3 As Double(),
    ByRef T As Double(),
    ByRef M2 As Double(),
    ByRef M3 As Double()
) As Integer
```

```
Dim instance As cAnalysisResults
Dim NumberResults As Integer
Dim SCut As String()
Dim LoadCase As String()
Dim StepType As String()
Dim StepNum As Double()
Dim P As Double()
Dim V2 As Double()
Dim V3 As Double()
Dim T As Double()
Dim M2 As Double()
Dim M3 As Double()
```

cAnalysisResultsspan id="LST9377EF91_0"AddLanguageSpecificTextSet("LST9377EF91_0?cp23|nu=.");

```
Dim returnValue As Integer
```

```
returnValue = instance.SectionCutDesign(NumberResults,  
SCut, LoadCase, StepType, StepNum,  
P, V2, V3, T, M2, M3)
```

```
int SectionCutDesign(  
int% NumberResults,  
array<String^>^% SCut,  
array<String^>^% LoadCase,  
array<String^>^% StepType,  
array<double>^% StepNum,  
array<double>^% P,  
array<double>^% V2,  
array<double>^% V3,  
array<double>^% T,  
array<double>^% M2,  
array<double>^% M3  
)
```

```
abstract SectionCutDesign :  
    NumberResults : int byref *  
    SCut : string[] byref *  
    LoadCase : string[] byref *  
    StepType : string[] byref *  
    StepNum : float[] byref *  
    P : float[] byref *  
    V2 : float[] byref *  
    V3 : float[] byref *  
    T : float[] byref *  
    M2 : float[] byref *  
    M3 : float[] byref -> int
```

Parameters

NumberResults

Type: System.Int32

The number total of results returned by the program

SCut

Type: System.String

This is an array that includes the name of the section cut associated with each result

LoadCase

Type: System.String

This is an array that includes the name of the analysis case or load combination associated with each result.

StepType

Type: System.String

This is an array that includes the step type, if any, for each result.

StepNum

Type: System.Double

This is an array that includes the step number, if any, for each result.

P

Type: System.Double

This is a one dimensional array of the axial force for each result. [F]

V2

Type: SystemDouble

This is a one dimensional array of the shear force in the section cut local 2 direction for each result. [F]

V3

Type: SystemDouble

This is a one dimensional array of the shear force in the section cut local 3 direction for each result. [F]

T

Type: SystemDouble

This is a one dimensional array of the torsion for each result. [FL]

M2

Type: SystemDouble

This is a one dimensional array of the moment about the section cut local 2-axis for each result. [FL]

M3

Type: SystemDouble

This is a one dimensional array of the moment about the section cut local 3-axis for each result. [FL]

Return Value

Type: Int32

Returns zero if the section cut forces are successfully recovered, otherwise it returns a nonzero value

Remarks

See [Results](#) for more information.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberResults As Integer
    Dim SCut() As String
    Dim LoadCase() As String
    Dim StepType() As String
    Dim StepNum() As Double
    Dim P() As Double
    Dim V2() As Double
    Dim V3() As Double
    Dim T() As Double
    Dim M2() As Double
    Dim M3() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
```

Introduction

```
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'run analysis
System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
ret = SapModel.Analyze.RunAnalysis()

'deselect all cases and combos
ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput()

'set case selected for output
ret = SapModel.Results.Setup.SetCaseSelectedForOutput("DEAD")

'get section cut forces with design output convention
ret = SapModel.Results.SectionCutDesign(NumberResults, SCut, LoadCase, StepType, StepNum,

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

See Also

Reference

cAnalysisResults Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LST32059737_0") Method

Retrieves spandrel forces for any defined spandrel objects in the model

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SpandrelForce(
    ref int NumberResults,
    ref string[] StoryName,
    ref string[] SpandrelName,
    ref string[] LoadCase,
    ref string[] Location,
    ref double[] P,
    ref double[] V2,
    ref double[] V3,
    ref double[] T,
    ref double[] M2,
    ref double[] M3
)
```

```
Function SpandrelForce (
    ByRef NumberResults As Integer,
    ByRef StoryName As String(),
    ByRef SpandrelName As String(),
    ByRef LoadCase As String(),
    ByRef Location As String(),
    ByRef P As Double(),
    ByRef V2 As Double(),
    ByRef V3 As Double(),
    ByRef T As Double(),
    ByRef M2 As Double(),
    ByRef M3 As Double()
) As Integer
```

```
Dim instance As cAnalysisResults
Dim NumberResults As Integer
Dim StoryName As String()
Dim SpandrelName As String()
Dim LoadCase As String()
Dim Location As String()
Dim P As Double()
Dim V2 As Double()
Dim V3 As Double()
Dim T As Double()
Dim M2 As Double()
Dim M3 As Double()
Dim returnValue As Integer
```

Introduction

```
returnValue = instance.SpandrelForce(NumberResults,  
StoryName, SpandrelName, LoadCase,  
Location, P, V2, V3, T, M2, M3)
```

```
int SpandrelForce(  
int% NumberResults,  
array<String^>^% StoryName,  
array<String^>^% SpandrelName,  
array<String^>^% LoadCase,  
array<String^>^% Location,  
array<double>^% P,  
array<double>^% V2,  
array<double>^% V3,  
array<double>^% T,  
array<double>^% M2,  
array<double>^% M3  
)
```

```
abstract SpandrelForce :  
    NumberResults : int byref *  
    StoryName : string[] byref *  
    SpandrelName : string[] byref *  
    LoadCase : string[] byref *  
    Location : string[] byref *  
    P : float[] byref *  
    V2 : float[] byref *  
    V3 : float[] byref *  
    T : float[] byref *  
    M2 : float[] byref *  
    M3 : float[] byref -> int
```

Parameters

NumberResults

Type:Â SystemInt32

The total number of results returned by the program

StoryName

Type:Â SystemString

The story name of the spandrel object

SpandrelName

Type:Â SystemString

The name of the spandrel object

LoadCase

Type:Â SystemString

The names of the load case

Location

Type:Â SystemString

The location on the spandrel, either "Left" or "Right", of the result being reported

P

Type:Â SystemDouble

The axial force [F]

V2

Type:Â SystemDouble

The shear force in the local 2 direction [F]

V3	Type: <u>SystemDouble</u> The shear force in the local 3 direction [F]
T	Type: <u>SystemDouble</u> The torsion [FL]
M2	Type: <u>SystemDouble</u> The moment about the local 2 axis [FL]
M3	Type: <u>SystemDouble</u> The moment about the local 3 axis [FL]

Return ValueType: Int32

Returns zero if the results are successfully retrieved; otherwise returns nonzero

▲ **Remarks**See Results for more information.▲ **See Also****Reference**cAnalysisResults InterfaceETABSv1 Namespace

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CSI API ETABS v1

cAnalysisResultsAddLanguageSpecificTextSet("LST7E4EFB4_0?cpp230nu=.");S

Method

Reports the story drifts

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int StoryDrifts(
    ref int NumberResults,
    ref string[] Story,
    ref string[] LoadCase,
    ref string[] StepType,
    ref double[] StepNum,
    ref string[] Direction,
    ref double[] Drift,
    ref string[] Label,
    ref double[] X,
    ref double[] Y,
    ref double[] Z
)
```

```
Function StoryDrifts (
    ByRef NumberResults As Integer,
    ByRef Story As String(),
    ByRef LoadCase As String(),
    ByRef StepType As String(),
    ByRef StepNum As Double(),
    ByRef Direction As String(),
    ByRef Drift As Double(),
    ByRef Label As String(),
    ByRef X As Double(),
    ByRef Y As Double(),
    ByRef Z As Double()
) As Integer
```

```
Dim instance As cAnalysisResults
Dim NumberResults As Integer
Dim Story As String()
Dim LoadCase As String()
Dim StepType As String()
Dim StepNum As Double()
Dim Direction As String()
Dim Drift As Double()
Dim Label As String()
Dim X As Double()
Dim Y As Double()
Dim Z As Double()
Dim returnValue As Integer
```

cAnalysisResultsspan id="LST7E4EFB4_0"AddLanguageSpecificTextSet("LST7E4EFB4_0?cpp230nu=.");S

```
returnValue = instance.StoryDrifts(NumberResults,
Story, LoadCase, StepType, StepNum,
Direction, Drift, Label, X, Y, Z)
```

```
int StoryDrifts(
int% NumberResults,
array<String^>^% Story,
array<String^>^% LoadCase,
array<String^>^% StepType,
array<double>^% StepNum,
array<String^>^% Direction,
array<double>^% Drift,
array<String^>^% Label,
array<double>^% X,
array<double>^% Y,
array<double>^% Z
)
```

```
abstract StoryDrifts :
    NumberResults : int byref *
    Story : string[] byref *
    LoadCase : string[] byref *
    StepType : string[] byref *
    StepNum : float[] byref *
    Direction : string[] byref *
    Drift : float[] byref *
    Label : string[] byref *
    X : float[] byref *
    Y : float[] byref *
    Z : float[] byref -> int
```

Parameters

NumberResults

Type:Â [SystemInt32](#)

The total number of results returned by the program

Story

Type:Â [SystemString](#)

This is an array of the story levels for each result

LoadCase

Type:Â [SystemString](#)

This is an array of the names of the analysis case or load combination associated with each result

StepType

Type:Â [SystemString](#)

This is an array of the step types, if any, for each result

StepNum

Type:Â [SystemDouble](#)

This is an array of the step numbers, if any, for each result

Direction

Type:Â [SystemString](#)

This is an array of the directions of the Drift values

Drift

Type:Â [SystemDouble](#)

This is an array of the drift values

Label

Type: [SystemString](#)

This is an array of the point labels for each result

X

Type: [SystemDouble](#)

This is an array of the displacements in the X direction [L]

Y

Type: [SystemDouble](#)

This is an array of the displacements in the Y direction [L]

Z

Type: [SystemDouble](#)

This is an array of the displacements in the Z direction [L]

Return Value

Type: [Int32](#)

Returns zero if the results are successfully recovered, otherwise it returns a nonzero value

▲ **Remarks**

See [Results](#) for more information.

▲ **See Also**

Reference

[cAnalysisResults Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalysisResultsSetup Interface

Namespace: Â ETABSv1

Assembly: Â ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
public interface cAnalysisResultsSetup
```

```
Public Interface cAnalysisResultsSetup
```

```
Dim instance As cAnalysisResultsSetup
```

```
public interface class cAnalysisResultsSetup
```

```
type cAnalysisResultsSetup = interface end
```

The cAnalysisResultsSetup type exposes the following members.

▲ Methods

Â	Name	Description
⇒	<u>DeselectAllCasesAndCombosForOutput</u>	Deselects all load cases and response combinations for output.
⇒	<u>GetCaseSelectedForOutput</u>	Checks if a load case is selected for output.
⇒	<u>GetComboSelectedForOutput</u>	Checks if a load combination is selected for output
⇒	<u>GetOptionBaseReactLoc</u>	Retrieves the global coordinates of the location at which the base reactions are reported
⇒	<u>GetOptionBucklingMode</u>	
⇒	<u>GetOptionDirectHist</u>	Retrieves the output option for direct history results.
⇒	<u>GetOptionModalHist</u>	Retrieves the output option for modal history results.
⇒	<u>GetOptionModeShape</u>	
⇒	<u>GetOptionMultiStepStatic</u>	
⇒	<u>GetOptionMultiValuedCombo</u>	
⇒	<u>GetOptionNLStatic</u>	
⇒	<u>SetCaseSelectedForOutput</u>	Sets a load case selected for output flag.
⇒	<u>SetComboSelectedForOutput</u>	Sets a load combination selected for output flag

- ⇒ [SetOptionBaseReactLoc](#)
- ⇒ [SetOptionBucklingMode](#)
- ⇒ [SetOptionDirectHist](#)
- ⇒ [SetOptionModalHist](#)
- ⇒ [SetOptionModeShape](#)
- ⇒ [SetOptionMultiStepStatic](#)
- ⇒ [SetOptionMultiValuedCombo](#)
- ⇒ [SetOptionNLStatic](#)

Sets the output option for modal history results.

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalysisResultsSetup Methods

The cAnalysisResultsSetup type exposes the following members.

Methods

⌘	Name	Description
⇒	<u>DeselectAllCasesAndCombosForOutput</u>	Deselects all load cases and response combinations for output.
⇒	<u>GetCaseSelectedForOutput</u>	Checks if a load case is selected for output.
⇒	<u>GetComboSelectedForOutput</u>	Checks if a load combination is selected for output
⇒	<u>GetOptionBaseReactLoc</u>	Retrieves the global coordinates of the location at which the base reactions are reported
⇒	<u>GetOptionBucklingMode</u>	
⇒	<u>GetOptionDirectHist</u>	Retrieves the output option for direct history results.
⇒	<u>GetOptionModalHist</u>	Retrieves the output option for modal history results.
⇒	<u>GetOptionModeShape</u>	
⇒	<u>GetOptionMultiStepStatic</u>	
⇒	<u>GetOptionMultiValuedCombo</u>	
⇒	<u>GetOptionNLStatic</u>	
⇒	<u>SetCaseSelectedForOutput</u>	Sets a load case selected for output flag.
⇒	<u>SetComboSelectedForOutput</u>	Sets a load combination selected for output flag
⇒	<u>SetOptionBaseReactLoc</u>	
⇒	<u>SetOptionBucklingMode</u>	
⇒	<u>SetOptionDirectHist</u>	
⇒	<u>SetOptionModalHist</u>	Sets the output option for modal history results.
⇒	<u>SetOptionModeShape</u>	
⇒	<u>SetOptionMultiStepStatic</u>	
⇒	<u>SetOptionMultiValuedCombo</u>	
⇒	<u>SetOptionNLStatic</u>	

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⌘ See Also

Reference

cAnalysisResultsSetup Interface

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CSI API ETABS v1

cAnalysisResultsSetupAddLanguageSpecificTextSet("Method

Deselects all load cases and response combinations for output.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int DeselectAllCasesAndCombosForOutput()
```

```
Function DeselectAllCasesAndCombosForOutput As Integer
```

```
Dim instance As cAnalysisResultsSetup
```

```
Dim returnValue As Integer
```

```
returnValue = instance.DeselectAllCasesAndCombosForOutput()
```

```
int DeselectAllCasesAndCombosForOutput()
```

```
abstract DeselectAllCasesAndCombosForOutput : unit -> int
```

Return Value

Type: Int32

Returns zero if the cases and combos are successfully deselected; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Selected As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
```

Introduction

```
ret = SapModel.InitializeNewModel()  
  
'create steel deck template model  
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)  
  
'run analysis  
System.IO.Directory.CreateDirectory("c:\CSI_API_temp")  
ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")  
ret = SapModel.Analyze.RunAnalysis  
  
'deselect all cases and combos  
ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput  
  
'set case selected for output  
ret = SapModel.Results.Setup.SetCaseSelectedForOutput("DEAD")  
  
'check if case is selected  
ret = SapModel.Results.Setup.GetCaseSelectedForOutput("DEAD", Selected)  
  
'close ETABS  
EtabsObject.ApplicationExit(False)  
  
'clean up variables  
SapModel = Nothing  
EtabsObject = Nothing  
End Sub
```

▲ See Also

Reference

[cAnalysisResultsSetup Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalysisResultsSetupAddLanguageSpecificTextSet("Method

Checks if a load case is selected for output.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetCaseSelectedForOutput(
    string Name,
    ref bool Selected
)
```

```
Function GetCaseSelectedForOutput (
    Name As String,
    ByRef Selected As Boolean
) As Integer
```

```
Dim instance As cAnalysisResultsSetup
Dim Name As String
Dim Selected As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetCaseSelectedForOutput(Name,
    Selected)
```

```
int GetCaseSelectedForOutput(
    String^ Name,
    bool% Selected
)
```

```
abstract GetCaseSelectedForOutput :
    Name : string *
    Selected : bool byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing load case.

Selected

Type: SystemBoolean

This item is True if the specified load case is to be selected for output, otherwise it is False.

Return Value

Type: Int32

Returns zero if the selected flag is successfully retrieved; otherwise it returns nonzero.

▀ **Remarks**

▀ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Selected As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis

    'deselect all cases and combos
    ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput

    'set case selected for output
    ret = SapModel.Results.Setup.SetCaseSelectedForOutput("DEAD")

    'check if case is selected
    ret = SapModel.Results.Setup.GetCaseSelectedForOutput("DEAD", Selected)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▀ **See Also**

Reference

[cAnalysisResultsSetup Interface](#)
[ETABSv1 Namespace](#)

Introduction

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CSI API ETABS v1

cAnalysisResultsSetupAddLanguageSpecificTextSet("Method

Checks if a load combination is selected for output

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetComboSelectedForOutput(
    string Name,
    ref bool Selected
)
```

```
Function GetComboSelectedForOutput (
    Name As String,
    ByRef Selected As Boolean
) As Integer
```

```
Dim instance As cAnalysisResultsSetup
Dim Name As String
Dim Selected As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetComboSelectedForOutput(Name,
    Selected)
```

```
int GetComboSelectedForOutput(
    String^ Name,
    bool% Selected
)
```

```
abstract GetComboSelectedForOutput :
    Name : string *
    Selected : bool byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing load combination

Selected

Type: SystemBoolean

This item is True if the specified load combination is selected for output

Return ValueType: $\hat{A}$ Int32

Returns 0 if the selected flag is successfully retrieved, otherwise it returns nonzero

▲ **Remarks**▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Selected As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add combo
    ret = SapModel.RespCombo.Add("COMB1", 0)

    'add load case to combo
    ret = SapModel.RespCombo.SetCaseList("COMB1", eCNameType.LoadCase, "DEAD", 1.4)

    'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis

    'deselect all cases and combos
    ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput

    'set combo selected for output
    ret = SapModel.Results.Setup.SetComboSelectedForOutput("COMB1")

    'check if combo is selected
    ret = SapModel.Results.Setup.GetComboSelectedForOutput("COMB1", Selected)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cAnalysisResultsSetup Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAnalysisResultsSetupAddLanguageSpecificTextSet("Method

Retrieves the global coordinates of the location at which the base reactions are reported

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetOptionBaseReactLoc(
    ref double GX,
    ref double GY,
    ref double GZ
)
```

```
Function GetOptionBaseReactLoc (
    ByRef GX As Double,
    ByRef GY As Double,
    ByRef GZ As Double
) As Integer
```

```
Dim instance As cAnalysisResultsSetup
Dim GX As Double
Dim GY As Double
Dim GZ As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetOptionBaseReactLoc(GX,
    GY, GZ)
```

```
int GetOptionBaseReactLoc(
    double% GX,
    double% GY,
    double% GZ
)
```

```
abstract GetOptionBaseReactLoc :
    GX : float byref *
    GY : float byref *
    GZ : float byref -> int
```

Parameters

GX

Type: $\hat{A}$ SystemDouble

The global X coordinate of the location at which the base reactions are reported

GY

Type: SystemDouble

The global Y coordinate of the location at which the base reactions are reported
GZ

Type: SystemDouble

The global Z coordinate of the location at which the base reactions are reported

Return Value

Type: Int32

Returns zero if the coordinates are successfully retrieved, otherwise it returns nonzero

▀ Remarks

▀ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Selected As Boolean
    Dim gx As Double
    Dim gy As Double
    Dim gz As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'get base reaction location
    ret = SapModel.Results.Setup.GetOptionBaseReactLoc(gx, gy, gz)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▀ See Also

Reference

cAnalysisResultsSetup Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAnalysisResultsSetupAddLanguageSpecificTextSet("Method

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetOptionBucklingMode(
    ref int BuckModeStart,
    ref int BuckModeEnd,
    ref bool BuckModeAll
)
```

```
Function GetOptionBucklingMode (
    ByRef BuckModeStart As Integer,
    ByRef BuckModeEnd As Integer,
    ByRef BuckModeAll As Boolean
) As Integer
```

```
Dim instance As cAnalysisResultsSetup
Dim BuckModeStart As Integer
Dim BuckModeEnd As Integer
Dim BuckModeAll As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOptionBucklingMode(BuckModeStart,
    BuckModeEnd, BuckModeAll)
```

```
int GetOptionBucklingMode(
    int% BuckModeStart,
    int% BuckModeEnd,
    bool% BuckModeAll
)
```

```
abstract GetOptionBucklingMode :
    BuckModeStart : int byref *
    BuckModeEnd : int byref *
    BuckModeAll : bool byref -> int
```

Parameters

BuckModeStart

Type: [System.Int32](#)

BuckModeEnd

Type: [System.Int32](#)

BuckModeAll

Type: [System.Boolean](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cAnalysisResultsSetup Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalysisResultsSetupAddLanguageSpecificTextSet("Method

Retrieves the output option for direct history results.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetOptionDirectHist(
    ref int Value
)
```

```
Function GetOptionDirectHist (
    ByRef Value As Integer
) As Integer
```

```
Dim instance As cAnalysisResultsSetup
Dim Value As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetOptionDirectHist(Value)
```

```
int GetOptionDirectHist(
    int% Value
)
```

```
abstract GetOptionDirectHist :
    Value : int byref -> int
```

Parameters

Value

Type: [SystemInt32](#)

This item is 1, 2 or 3

1. Envelopes
2. Step-by-Step
3. Last Step

Return Value

Type: [Int32](#)

Returns zero if the output option is successfully retrieved, otherwise it returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Selected As Boolean
    Dim Value As Integer

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'get output option
    ret = SapModel.Results.Setup.GetOptionDirectHist(Value)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

cAnalysisResultsSetup Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAnalysisResultsSetupAddLanguageSpecificTextSet("Method

Retrieves the output option for modal history results.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetOptionModalHist(
    ref int Value
)
```

```
Function GetOptionModalHist (
    ByRef Value As Integer
) As Integer
```

```
Dim instance As cAnalysisResultsSetup
Dim Value As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetOptionModalHist(Value)
```

```
int GetOptionModalHist(
    int% Value
)
```

```
abstract GetOptionModalHist :
    Value : int byref -> int
```

Parameters

Value

Type: [System.Int32](#)

This item is 1, 2 or 3

1. Envelopes
2. Step-by-Step
3. Last Step

Return Value

Type: [Int32](#)

Returns zero if the output option is successfully retrieved, otherwise it returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Selected As Boolean
    Dim Value As Integer

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'get output option
    ret = SapModel.Results.Setup.GetOptionModalHist(Value)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

🔗 See Also

Reference

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CSI API ETABS v1

cAnalysisResultsSetupAddLanguageSpecificTextSet("Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetOptionModeShape(
    ref int ModeShapeStart,
    ref int ModeShapeEnd,
    ref bool ModeShapesAll
)
```

```
Function GetOptionModeShape (
    ByRef ModeShapeStart As Integer,
    ByRef ModeShapeEnd As Integer,
    ByRef ModeShapesAll As Boolean
) As Integer
```

```
Dim instance As cAnalysisResultsSetup
Dim ModeShapeStart As Integer
Dim ModeShapeEnd As Integer
Dim ModeShapesAll As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOptionModeShape(ModeShapeStart,
    ModeShapeEnd, ModeShapesAll)
```

```
int GetOptionModeShape(
    int% ModeShapeStart,
    int% ModeShapeEnd,
    bool% ModeShapesAll
)
```

```
abstract GetOptionModeShape :
    ModeShapeStart : int byref *
    ModeShapeEnd : int byref *
    ModeShapesAll : bool byref -> int
```

Parameters

ModeShapeStart

Type: [System.Int32](#)

ModeShapeEnd

Type: [System.Int32](#)

ModeShapesAll

Type: [System.Boolean](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cAnalysisResultsSetup Interface](#)

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CSI API ETABS v1

cAnalysisResultsSetupAddLanguageSpecificTextSet("Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetOptionMultiStepStatic(
    ref int Value
)
```

```
Function GetOptionMultiStepStatic (
    ByRef Value As Integer
) As Integer
```

```
Dim instance As cAnalysisResultsSetup
Dim Value As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetOptionMultiStepStatic(Value)
```

```
int GetOptionMultiStepStatic(
    int% Value
)
```

```
abstract GetOptionMultiStepStatic :
    Value : int byref -> int
```

Parameters

Value

Type: [System.Int32](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cAnalysisResultsSetup Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalysisResultsSetupAddLanguageSpecificTextSet("Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetOptionMultiValuedCombo(
    ref int Value
)
```

```
Function GetOptionMultiValuedCombo (
    ByRef Value As Integer
) As Integer
```

```
Dim instance As cAnalysisResultsSetup
Dim Value As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetOptionMultiValuedCombo(Value)
```

```
int GetOptionMultiValuedCombo(
    int% Value
)
```

```
abstract GetOptionMultiValuedCombo :
    Value : int byref -> int
```

Parameters

Value

Type: [SystemInt32](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cAnalysisResultsSetup Interface](#)

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CSI API ETABS v1

cAnalysisResultsSetupAddLanguageSpecificTextSet("Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetOptionNLStatic(
    ref int Value
)
```

```
Function GetOptionNLStatic (
    ByRef Value As Integer
) As Integer
```

```
Dim instance As cAnalysisResultsSetup
Dim Value As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetOptionNLStatic(Value)
```

```
int GetOptionNLStatic(
    int% Value
)
```

```
abstract GetOptionNLStatic :
    Value : int byref -> int
```

Parameters

Value

Type: [SystemInt32](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cAnalysisResultsSetup Interface](#)

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CSI API ETABS v1

cAnalysisResultsSetupAddLanguageSpecificTextSet("Method

Sets a load case selected for output flag.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetCaseSelectedForOutput(
    string Name,
    bool Selected = true
)
```

```
Function SetCaseSelectedForOutput (
    Name As String,
    Optional Selected As Boolean = true
) As Integer
```

```
Dim instance As cAnalysisResultsSetup
Dim Name As String
Dim Selected As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetCaseSelectedForOutput(Name,
    Selected)
```

```
int SetCaseSelectedForOutput(
    String^ Name,
    bool Selected = true
)
```

```
abstract SetCaseSelectedForOutput :
    Name : string *
    ?Selected : bool
(* Defaults:
    let _Selected = defaultArg Selected true
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing load case.

Selected (Optional)

Type: SystemBoolean

This item is True if the specified load case is to be selected for output,

otherwise it is False.

Return Value

Type: Int32

Returns zero if the selected flag is successfully set; otherwise it returns nonzero.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis

    'deselect all cases and combos
    ret = SapModel.Results.Setup.DeselectAllCasesAndCombosForOutput

    'set case selected for output
    ret = SapModel.Results.Setup.SetCaseSelectedForOutput("DEAD")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cAnalysisResultsSetup Interface

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CSI API ETABS v1

cAnalysisResultsSetupAddLanguageSpecificTextSet("Method

Sets a load combination selected for output flag

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetComboSelectedForOutput(
    string Name,
    bool Selected = true
)
```

```
Function SetComboSelectedForOutput (
    Name As String,
    Selected As Boolean = true
) As Integer
```

```
Dim instance As cAnalysisResultsSetup
Dim Name As String
Dim Selected As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetComboSelectedForOutput(Name,
    Selected)
```

```
int SetComboSelectedForOutput(
    String^ Name,
    bool Selected = true
)
```

```
abstract SetComboSelectedForOutput :
    Name : string *
    ?Selected : bool
(* Defaults:
    let _Selected = defaultArg Selected true
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing load combination

Selected (Optional)

Type: SystemBoolean

This item is True if the specified load combination is to be selected for output,

otherwise it is False

Return Value

Type: Int32

Returns 0 if the selected flag is successfully set, otherwise it returns nonzero

▀ Remarks

▀ See Also

Reference

cAnalysisResultsSetup Interface

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CSI API ETABS v1

cAnalysisResultsSetupAddLanguageSpecificTextSet("Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetOptionBaseReactLoc(
    double GX,
    double GY,
    double GZ
)
```

```
Function SetOptionBaseReactLoc (
    GX As Double,
    GY As Double,
    GZ As Double
) As Integer
```

```
Dim instance As cAnalysisResultsSetup
Dim GX As Double
Dim GY As Double
Dim GZ As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetOptionBaseReactLoc(GX,
    GY, GZ)
```

```
int SetOptionBaseReactLoc(
    double GX,
    double GY,
    double GZ
)
```

```
abstract SetOptionBaseReactLoc :
    GX : float *
    GY : float *
    GZ : float -> int
```

Parameters

GX	Type: SystemDouble
GY	Type: SystemDouble
GZ	Type: SystemDouble

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cAnalysisResultsSetup Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalysisResultsSetupAddLanguageSpecificTextSet("Method

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetOptionBucklingMode(
int BuckModeStart,
int BuckModeEnd,
bool BuckModeAll = false
)
```

```
Function SetOptionBucklingMode (
BuckModeStart As Integer,
BuckModeEnd As Integer,
    BuckModeAll As Boolean = false
) As Integer
```

```
Dim instance As cAnalysisResultsSetup
Dim BuckModeStart As Integer
Dim BuckModeEnd As Integer
Dim BuckModeAll As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetOptionBucklingMode(BuckModeStart,
BuckModeEnd, BuckModeAll)
```

```
int SetOptionBucklingMode(
int BuckModeStart,
int BuckModeEnd,
bool BuckModeAll = false
)
```

```
abstract SetOptionBucklingMode :
    BuckModeStart : int *
    BuckModeEnd : int *
    ?BuckModeAll : bool
(* Defaults:
    let _BuckModeAll = defaultArg BuckModeAll false
*)
-> int
```

Parameters

BuckModeStart

Type: [System.Int32](#)

BuckModeEnd

cAnalysisResultsSetupAddLanguageSpecificTextSet("LST6961CAC6_0?cpp=::

Type: SystemInt32
BuckModeAll (Optional)
Type: SystemBoolean

Return Value

Type: Int32

■ See Also

Reference

cAnalysisResultsSetup Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAnalysisResultsSetupAddLanguageSpecificTextSet("Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetOptionDirectHist(
    int Value
)
```

```
Function SetOptionDirectHist (
    Value As Integer
) As Integer
```

```
Dim instance As cAnalysisResultsSetup
Dim Value As Integer
Dim returnValue As Integer
```

```
returnValue = instance.SetOptionDirectHist(Value)
```

```
int SetOptionDirectHist(
    int Value
)
```

```
abstract SetOptionDirectHist :
    Value : int -> int
```

Parameters

Value

Type: [SystemInt32](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cAnalysisResultsSetup Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalysisResultsSetupAddLanguageSpecificTextSet("Method

Sets the output option for modal history results.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetOptionModalHist(
int Value
)
```

```
Function SetOptionModalHist (
Value As Integer
) As Integer
```

```
Dim instance As cAnalysisResultsSetup
Dim Value As Integer
Dim returnValue As Integer
```

```
returnValue = instance.SetOptionModalHist(Value)
```

```
int SetOptionModalHist(
int Value
)
```

```
abstract SetOptionModalHist :
    Value : int -> int
```

Parameters

Value

Type: [SystemInt32](#)

This item is 1, 2 or 3

1. Envelopes
2. Step-by-Step
3. Last Step

Return Value

Type: [Int32](#)

Returns 0 if the output option is successfully set, otherwise it returns nonzero.

▲ Remarks

▲ See Also

Reference

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CSI API ETABS v1

cAnalysisResultsSetupAddLanguageSpecificTextSet("Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetOptionModeShape(
    int ModeShapeStart,
    int ModeShapeEnd,
    bool ModeShapesAll = false
)
```

```
Function SetOptionModeShape (
    ModeShapeStart As Integer,
    ModeShapeEnd As Integer,
    ModeShapesAll As Boolean = false
) As Integer
```

```
Dim instance As cAnalysisResultsSetup
Dim ModeShapeStart As Integer
Dim ModeShapeEnd As Integer
Dim ModeShapesAll As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetOptionModeShape(ModeShapeStart,
    ModeShapeEnd, ModeShapesAll)
```

```
int SetOptionModeShape(
    int ModeShapeStart,
    int ModeShapeEnd,
    bool ModeShapesAll = false
)
```

```
abstract SetOptionModeShape :
    ModeShapeStart : int *
    ModeShapeEnd : int *
    ?ModeShapesAll : bool
(* Defaults:
    let _ModeShapesAll = defaultArg ModeShapesAll false
*)
-> int
```

Parameters

ModeShapeStart

Type: [System.Int32](#)

ModeShapeEnd

Type: SystemInt32
ModeShapesAll (Optional)
Type: SystemBoolean

Return Value

Type: Int32

■ See Also

Reference

cAnalysisResultsSetup Interface

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CSI API ETABS v1

cAnalysisResultsSetupAddLanguageSpecificTextSet("Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetOptionMultiStepStatic(
int Value
)
```

```
Function SetOptionMultiStepStatic (
Value As Integer
) As Integer
```

```
Dim instance As cAnalysisResultsSetup
Dim Value As Integer
Dim returnValue As Integer
```

```
returnValue = instance.SetOptionMultiStepStatic(Value)
```

```
int SetOptionMultiStepStatic(
int Value
)
```

```
abstract SetOptionMultiStepStatic :
    Value : int -> int
```

Parameters

Value

Type: [SystemInt32](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cAnalysisResultsSetup Interface](#)

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CSI API ETABS v1

cAnalysisResultsSetupAddLanguageSpecificTextSet("Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetOptionMultiValuedCombo(
int Value
)
```

```
Function SetOptionMultiValuedCombo (
Value As Integer
) As Integer
```

```
Dim instance As cAnalysisResultsSetup
Dim Value As Integer
Dim returnValue As Integer
```

```
returnValue = instance.SetOptionMultiValuedCombo(Value)
```

```
int SetOptionMultiValuedCombo(
int Value
)
```

```
abstract SetOptionMultiValuedCombo :
    Value : int -> int
```

Parameters

Value

Type: [SystemInt32](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cAnalysisResultsSetup Interface](#)

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CSI API ETABS v1

cAnalysisResultsSetupAddLanguageSpecificTextSet("Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetOptionNLStatic(
int Value
)
```

```
Function SetOptionNLStatic (
Value As Integer
) As Integer
```

```
Dim instance As cAnalysisResultsSetup
Dim Value As Integer
Dim returnValue As Integer
```

```
returnValue = instance.SetOptionNLStatic(Value)
```

```
int SetOptionNLStatic(
int Value
)
```

```
abstract SetOptionNLStatic :
    Value : int -> int
```

Parameters

Value

Type: [SystemInt32](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cAnalysisResultsSetup Interface](#)

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CSI API ETABS v1

cAnalyze Interface

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cAnalyze
```

```
Public Interface cAnalyze
```

```
Dim instance As cAnalyze
```

```
public interface class cAnalyze
```

```
type cAnalyze = interface end
```

The cAnalyze type exposes the following members.

Methods

A

Name

Description

 [CreateAnalysisModel](#)

Creates the analysis model. If the analysis model is already created and current, nothing is done.

 [DeleteResults](#)

Deletes results for load cases.

 [GetActiveDOF](#)

Retrieves the model global degrees of freedom.

 [GetCaseStatus](#)

Retrieves the status for all load cases.

 [GetDesignResponseOption](#)

Retrieves the design and response recovery options.

 [GetRunCaseFlag](#)

Retrieves the run flags for all analysis cases.

 [GetSolverOption](#)

DEPRECATED. Please see [GetSolverOption_2\(Int32, String\)](#) or [GetSolverOption_3\(Int32, Int32, Int32, String\)](#)

 [GetSolverOption_1](#)

DEPRECATED. Please see [GetSolverOption_2\(Int32, String\)](#) or [GetSolverOption_3\(Int32, Int32, Int32, String\)](#)

 [GetSolverOption_2](#)

Retrieves the model solver options.

 [GetSolverOption_3](#)

Retrieves the model solver options.

 [MergeAnalysisResults](#)

Merges analysis results from a given model.

 [ModifyUndeformedGeometry](#)

Modifies the undeformed geometry based on displacements obtained from a specified load case

 [ModifyUndeformedGeometryModeShape](#)

Modifies the undeformed geometry based on the mode shapes of a specified mode

 [RunAnalysis](#)

Runs the analysis.

- ⇒ [SetActiveDOF](#) Sets the model global degrees of freedom.
- ⇒ [SetDesignResponseOption](#) Sets the design and response recovery options.
- ⇒ [SetRunCaseFlag](#) Sets the run flag for load cases.
- ⇒ [SetSolverOption](#) DEPRECATED. Please see [SetSolverOption_2\(Int32, Int32, Int32, String\)](#) or [SetSolverOption_3\(Int32, Int32, Int32, String\)](#)
- ⇒ [SetSolverOption_1](#) DEPRECATED. Please see [SetSolverOption_2\(Int32, Int32, Int32, String\)](#) or [SetSolverOption_3\(Int32, Int32, Int32, String\)](#)
- ⇒ [SetSolverOption_2](#) Sets the model solver options.
- ⇒ [SetSolverOption_3](#) Sets the model solver options.

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalyze Methods

The [cAnalyze](#) type exposes the following members.

Methods

Á	Name	Description
⇒	CreateAnalysisModel	Creates the analysis model. If the analysis model already exists and current, nothing is done.
⇒	DeleteResults	Deletes results for load cases.
⇒	GetActiveDOF	Retrieves the model global degrees of freedom.
⇒	GetCaseStatus	Retrieves the status for all load cases.
⇒	GetDesignResponseOption	Retrieves the design and response recovery options.
⇒	GetRunCaseFlag	Retrieves the run flags for all analysis cases.
⇒	GetSolverOption	DEPRECATED. Please see GetSolverOption_2(Int32, Int32, Int32, String) or GetSolverOption_3(Int32, Int32, Int32, String)
⇒	GetSolverOption_1	DEPRECATED. Please see GetSolverOption_2(Int32, Int32, Int32, String) or GetSolverOption_3(Int32, Int32, Int32, String)
⇒	GetSolverOption_2	Retrieves the model solver options.
⇒	GetSolverOption_3	Retrieves the model solver options.
⇒	MergeAnalysisResults	Merges analysis results from a given model.
⇒	ModifyUndeformedGeometry	Modifies the undeformed geometry based on displacements obtained from a specified load case
⇒	ModifyUndeformedGeometryModeShape	Modifies the undeformed geometry based on the mode shapes of a specified mode
⇒	RunAnalysis	Runs the analysis.
⇒	SetActiveDOF	Sets the model global degrees of freedom.
⇒	SetDesignResponseOption	Sets the design and response recovery options.
⇒	SetRunCaseFlag	Sets the run flag for load cases.
⇒	SetSolverOption	DEPRECATED. Please see SetSolverOption_2(Int32, Int32, Int32, String) or SetSolverOption_3(Int32, Int32, Int32, String)
⇒	SetSolverOption_1	DEPRECATED. Please see SetSolverOption_2(Int32, Int32, Int32, String) or SetSolverOption_3(Int32, Int32, Int32, String)
⇒	SetSolverOption_2	Sets the model solver options.
⇒	SetSolverOption_3	Sets the model solver options.

[Top](#)

See Also

Reference

cAnalyze Interface

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CSI API ETABS v1

cAnalyzeAddLanguageSpecificTextSet("LST2B502D76_0") Method

Creates the analysis model. If the analysis model is already created and current, nothing is done.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int CreateAnalysisModel()
```

```
Function CreateAnalysisModel As Integer
```

```
Dim instance As cAnalyze
```

```
Dim returnValue As Integer
```

```
returnValue = instance.CreateAnalysisModel()
```

```
int CreateAnalysisModel()
```

```
abstract CreateAnalysisModel : unit -> int
```

Return Value

Type: [Int32](#)

Returns zero if the analysis model is successfully created or it already exists and is current, otherwise it returns a nonzero value.

▲ Remarks

It is not necessary to call this function before running an analysis. The analysis model is automatically created, if necessary, when the model is run.

▲ Examples

[VB](#)

[Copy](#)

```
Public Sub Example()
```

```
    Dim SapModel As cSapModel
```

```
    Dim EtabsObject As cOAPI
```

```
    Dim ret As Integer = -1
```

```
    'create ETABS object
```

```
    Dim myHelper as cHelper = New Helper
```

```
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

```
    'start ETABS application
```

```
    ret = EtabsObject.ApplicationStart()
```

```
    'create SapModel object
```

cAnalyzeAddLanguageSpecificTextSet("LST2B502D76_0?cpp=::|nu[287](#)Create

Introduction

```
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'create the analysis model
ret = SapModel.Analyze.CreateAnalysisModel

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cAnalyze Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalyzeAddLanguageSpecificTextSet("LST1AE922AB_0") Method

Deletes results for load cases.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int DeleteResults(
    string Name,
    bool All = false
)
```

```
Function DeleteResults (
    Name As String,
    All As Boolean = false
) As Integer
```

```
Dim instance As cAnalyze
Dim Name As String
Dim All As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.DeleteResults(Name,
All)
```

```
int DeleteResults(
    String^ Name,
    bool All = false
)
```

```
abstract DeleteResults :
    Name : string *
    ?All : bool
(* Defaults:
    let _All = defaultArg All false
*)
-> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing load case that is to have its results deleted.

This item is ignored when the All item is True.

All (Optional)

cAnalyze.AddLanguageSpecificTextSet("LST1AE922AB_0?cpp=::|nu289;Delete

Type: System.Boolean

If this item is True, the results are deleted for all load cases, and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the results are successfully deleted; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis

    'delete load case results
    ret = SapModel.Analyze.DeleteResults("MODAL")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cAnalyze Interface

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Namespace: ETABSV1

▲ Syntax

C#

VBC++

F#

Copy

```
int GetActiveDOF(
ref bool[] DOF
)
```

```
Function GetActiveDOF (
ByRef DOF As Boolean()
) As Integer
```

```
Dim instance As cAnalyze
Dim DOF As Boolean()
Dim returnValue As Integer
```

```
returnValue = instance.GetActiveDOF(DOF)
```

```
int GetActiveDOF(
array<bool>^% DOF
)
```

```
abstract GetActiveDOF :
    DOF : bool[] byref -> int
```

DOF

Type: SystemBooleanType:Â Int32

Returns zero if the degrees of freedom are successfully retrieved; otherwise it returns a nonzero value.

Remarks

Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI
```

cAnalyzespan id="LST6CDFB02F_0"AddLanguageSpecificTextSet("LST6CDFB02F_0?cpp=::|n292);GetAc

Introduction

```
Dim ret As Integer = -1
Dim DOf() As Boolean

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'get model degrees of freedom
ret = SapModel.Analyze.GetActiveDOF(DOf)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

🔗 See Also

Reference

[cAnalyze Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cAnalyzeAddLanguageSpecificTextSet("LST711F6396_0?cpp=::|nu=294;GetCaseStatus") Method

Retrieves the status for all load cases.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetCaseStatus(
    ref int NumberItems,
    ref string[] CaseName,
    ref int[] Status
)
```

```
Function GetCaseStatus (
    ByRef NumberItems As Integer,
    ByRef CaseName As String(),
    ByRef Status As Integer()
) As Integer
```

```
Dim instance As cAnalyze
Dim NumberItems As Integer
Dim CaseName As String()
Dim Status As Integer()
Dim returnValue As Integer
```

```
returnValue = instance.GetCaseStatus(NumberItems,
    CaseName, Status)
```

```
int GetCaseStatus(
    int% NumberItems,
    array<String^>^% CaseName,
    array<int>^% Status
)
```

```
abstract GetCaseStatus :
    NumberItems : int byref *
    CaseName : string[] byref *
    Status : int[] byref -> int
```

Parameters

NumberItems

Type: [System.Int32](#)

The number of load cases for which the status is reported.

CaseName

Type: SystemString

This is an array that includes the name of each analysis case for which the status is reported.

Status

Type: SystemInt32

This is an array containing integers from 1 to 4, indicating the load case status.

1. Not run
2. Could not start
3. Not finished
4. Finished

Return Value

Type: Int32

Returns zero if the status is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberItems As Integer
    Dim CaseName() As String
    Dim Status() As Integer

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set load case run flag
    ret = SapModel.Analyze.SetRunCaseFlag("MODAL", False)

    'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis

    'get load case status
    ret = SapModel.Analyze.GetCaseStatus(NumberItems, CaseName, Status)

    'close ETABS
    EtabsObject.ApplicationExit(False)
```

```
'clean up variables  
  SapModel = Nothing  
  EtabsObject = Nothing  
End Sub
```

🔍 See Also

Reference

cAnalyze Interface

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CSI API ETABS v1

cAnalyzeAddLanguageSpecificTextSet("LST9F9E0209_0") Method

Retrieves the design and response recovery options.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetDesignResponseOption(
    ref int NumberDesignThreads,
    ref int NumberResponseRecoveryThreads,
    ref int UseMemoryMappedFilesForResponseRecovery,
    ref bool ModelDifferencesOKWhenMergingResults
)
```

```
Function GetDesignResponseOption (
    ByRef NumberDesignThreads As Integer,
    ByRef NumberResponseRecoveryThreads As Integer,
    ByRef UseMemoryMappedFilesForResponseRecovery As Integer,
    ByRef ModelDifferencesOKWhenMergingResults As Boolean
) As Integer
```

```
Dim instance As cAnalyze
Dim NumberDesignThreads As Integer
Dim NumberResponseRecoveryThreads As Integer
Dim UseMemoryMappedFilesForResponseRecovery As Integer
Dim ModelDifferencesOKWhenMergingResults As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetDesignResponseOption(NumberDesignThreads,
    NumberResponseRecoveryThreads,
    UseMemoryMappedFilesForResponseRecovery,
    ModelDifferencesOKWhenMergingResults)
```

```
int GetDesignResponseOption(
    int% NumberDesignThreads,
    int% NumberResponseRecoveryThreads,
    int% UseMemoryMappedFilesForResponseRecovery,
    bool% ModelDifferencesOKWhenMergingResults
)
```

```
abstract GetDesignResponseOption :
    NumberDesignThreads : int byref *
    NumberResponseRecoveryThreads : int byref *
    UseMemoryMappedFilesForResponseRecovery : int byref *
    ModelDifferencesOKWhenMergingResults : bool byref -> int
```

Parameters

NumberDesignThreads

Type:Â SystemInt32

Number of threads that design can use. Positive if user specified, negative if program determined.

NumberResponseRecoveryThreads

Type:Â SystemInt32

Number of threads that response recovery can use. Positive if user specified, negative if program determined.

UseMemoryMappedFilesForResponseRecovery

Type:Â SystemInt32

Flag for using memory mapped files for response recovery.

◊ -2 = Not using memory mapped files (Program Determined).

◊ -1 = Not using memory mapped files (user specified).

◊ 1 = Using memory mapped files (user specified).

◊ 2 = Using memory mapped files (Program Determined).

ModelDifferencesOKWhenMergingResults

Type:Â SystemBoolean

Flag for merging results in presence of any model differences. True if results from two models that are not identical are OK to merge, false otherwise.

Return ValueType:Â Int32

Returns zero if the options are successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim myHelper As cHelper
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    Dim NumberDesignThreads As Integer
    Dim NumberResponseRecoveryThreads As Integer
    Dim UseMemoryMappedFilesForResponseRecovery As Integer
    Dim ModelDifferencesOKWhenMergingResults As Boolean

    'create ETABS object
    myHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()
```

Introduction

```
'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'get model solver options
ret = SapModel.Analyze.GetDesignResponseOption(NumberDesignThreads, NumberResponseRecoveryTh

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
myHelper = Nothing
End Sub
```

🔍 See Also

Reference

cAnalyze Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAnalyzeAddLanguageSpecificTextSet("LSTCC9B853B0") Method

Retrieves the run flags for all analysis cases.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetRunCaseFlag(
    ref int NumberItems,
    ref string[] CaseName,
    ref bool[] Run
)
```

```
Function GetRunCaseFlag (
    ByRef NumberItems As Integer,
    ByRef CaseName As String(),
    ByRef Run As Boolean()
) As Integer
```

```
Dim instance As cAnalyze
Dim NumberItems As Integer
Dim CaseName As String()
Dim Run As Boolean()
Dim returnValue As Integer
```

```
returnValue = instance.GetRunCaseFlag(NumberItems,
    CaseName, Run)
```

```
int GetRunCaseFlag(
    int% NumberItems,
    array<String^>^% CaseName,
    array<bool>^% Run
)
```

```
abstract GetRunCaseFlag :
    NumberItems : int byref *
    CaseName : string[] byref *
    Run : bool[] byref -> int
```

Parameters

NumberItems

Type: [System.Int32](#)

The number of load cases for which the run flag is reported.

CaseName

Type: SystemString

This is an array that includes the name of each analysis case for which the run flag is reported.

Run

Type: SystemBoolean

This is an array of boolean values indicating if the specified load case is to be run.

Return Value

Type: Int32

Returns zero if the flags are successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberItems As Integer
    Dim CaseName() As String
    Dim Run() As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set load case run flag
    ret = SapModel.Analyze.SetRunCaseFlag("MODAL", False)

    'get load case run flags
    ret = SapModel.Analyze.GetRunCaseFlag(NumberItems, CaseName, Run)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cAnalyze Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAnalyzeAddLanguageSpecificTextSet("LST530E329A_0") Method

DEPRECATED. Please see [GetSolverOption_2\(Int32, Int32, Int32, String\)](#) or [GetSolverOption_3\(Int32, Int32, Int32, Int32, Int32, String\)](#)

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetSolverOption(
    ref int SolverType,
    ref bool Force32BitSolver,
    ref string StiffCase
)
```

```
Function GetSolverOption (
    ByRef SolverType As Integer,
    ByRef Force32BitSolver As Boolean,
    ByRef StiffCase As String
) As Integer
```

```
Dim instance As cAnalyze
Dim SolverType As Integer
Dim Force32BitSolver As Boolean
Dim StiffCase As String
Dim returnValue As Integer
```

```
returnValue = instance.GetSolverOption(SolverType,
    Force32BitSolver, StiffCase)
```

```
int GetSolverOption(
    int% SolverType,
    bool% Force32BitSolver,
    String^% StiffCase
)
```

```
abstract GetSolverOption :
    SolverType : int byref *
    Force32BitSolver : bool byref *
    StiffCase : string byref -> int
```

Parameters

SolverType

Type: [System.Int32](#)

Force32BitSolver

Type: [System.Boolean](#)

cAnalyzespan id="LST530E329A_0">AddLanguageSpecificTextSet("LST530E329A_0?cpp=::|nu=303;GetSol

StiffCase

Type: SystemString

Return Value

Type: Int32

▲ See Also

Reference

cAnalyze Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAnalyzeAddLanguageSpecificTextSet("LST48EFAFF5_0") Method

DEPRECATED. Please see [GetSolverOption_2\(Int32, Int32, Int32, String\)](#) or [GetSolverOption_3\(Int32, Int32, Int32, Int32, Int32, String\)](#)

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetSolverOption_1(
    ref int SolverType,
    ref int SolverProcessType,
    ref bool Force32BitSolver,
    ref string StiffCase
)
```

```
Function GetSolverOption_1 (
    ByRef SolverType As Integer,
    ByRef SolverProcessType As Integer,
    ByRef Force32BitSolver As Boolean,
    ByRef StiffCase As String
) As Integer
```

```
Dim instance As cAnalyze
Dim SolverType As Integer
Dim SolverProcessType As Integer
Dim Force32BitSolver As Boolean
Dim StiffCase As String
Dim returnValue As Integer
```

```
returnValue = instance.GetSolverOption_1(SolverType,
    SolverProcessType, Force32BitSolver,
    StiffCase)
```

```
int GetSolverOption_1(
    int% SolverType,
    int% SolverProcessType,
    bool% Force32BitSolver,
    String^% StiffCase
)
```

```
abstract GetSolverOption_1 :
    SolverType : int byref *
    SolverProcessType : int byref *
    Force32BitSolver : bool byref *
    StiffCase : string byref -> int
```

Parameters

SolverType

Type:Â [SystemInt32](#)

SolverProcessType

Type:Â [SystemInt32](#)

Force32BitSolver

Type:Â [SystemBoolean](#)

StiffCase

Type:Â [SystemString](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cAnalyze Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalyzeAddLanguageSpecificTextSet("LSTE1520482_0") Method

Retrieves the model solver options.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetSolverOption_2(
    ref int SolverType,
    ref int SolverProcessType,
    ref int NumberParallelRuns,
    ref string StiffCase
)
```

```
Function GetSolverOption_2 (
    ByRef SolverType As Integer,
    ByRef SolverProcessType As Integer,
    ByRef NumberParallelRuns As Integer,
    ByRef StiffCase As String
) As Integer
```

```
Dim instance As cAnalyze
Dim SolverType As Integer
Dim SolverProcessType As Integer
Dim NumberParallelRuns As Integer
Dim StiffCase As String
Dim returnValue As Integer
```

```
returnValue = instance.GetSolverOption_2(SolverType,
    SolverProcessType, NumberParallelRuns,
    StiffCase)
```

```
int GetSolverOption_2(
    int% SolverType,
    int% SolverProcessType,
    int% NumberParallelRuns,
    String^% StiffCase
)
```

```
abstract GetSolverOption_2 :
    SolverType : int byref *
    SolverProcessType : int byref *
    NumberParallelRuns : int byref *
    StiffCase : string byref -> int
```

Parameters**SolverType**Type: SystemInt32

This is 0, 1 or 2, indicating the solver type.

- ◊ 0 = Standard solver
- ◊ 1 = Advanced solver
- ◊ 2 = Multi-threaded solver

SolverProcessTypeType: SystemInt32

This is 0, 1 or 2, indicating the process the analysis is run.

- ◊ 0 = Auto (program determined)
- ◊ 1 = GUI process
- ◊ 2 = Separate process

NumberParallelRunsType: SystemInt32

This is an integer not including -1 or 0.

- ◊ Less than -1 = The negative of the program determined value when the assigned value is 0 = Auto parallel (use up to all physical cores - limited by license).
- ◊ 1 = Serial.
- ◊ Greater than 1 = User defined parallel (use up to this fixed number of cores - limited by license).

StiffCaseType: SystemString

The name of the load case used when outputting the mass and stiffness matrices to text files. If this item is blank, no matrices are output.

Return ValueType: Int32

Returns zero if the options are successfully retrieved; otherwise it returns a nonzero value.

▲ **Remarks**▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim SolverType As Integer
    Dim SolverProcessType As Integer
    Dim NumberParallelRuns As Integer
    Dim StiffCase As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
```


Introduction

```
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'get model solver options
ret = SapModel.Analyze.GetSolverOption_2(SolverType, SolverProcessType, NumberParallelRuns)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

[cAnalyze Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalyzeAddLanguageSpecificTextSet("LST25E7C0A1_0") Method

Retrieves the model solver options.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetSolverOption_3(
    ref int SolverType,
    ref int SolverProcessType,
    ref int NumberParallelRuns,
    ref int ResponseFileSizeMaxMB,
    ref int NumberAnalysisThreads,
    ref string StiffCase
)
```

```
Function GetSolverOption_3 (
    ByRef SolverType As Integer,
    ByRef SolverProcessType As Integer,
    ByRef NumberParallelRuns As Integer,
    ByRef ResponseFileSizeMaxMB As Integer,
    ByRef NumberAnalysisThreads As Integer,
    ByRef StiffCase As String
) As Integer
```

```
Dim instance As cAnalyze
Dim SolverType As Integer
Dim SolverProcessType As Integer
Dim NumberParallelRuns As Integer
Dim ResponseFileSizeMaxMB As Integer
Dim NumberAnalysisThreads As Integer
Dim StiffCase As String
Dim returnValue As Integer
```

```
returnValue = instance.GetSolverOption_3(SolverType,
    SolverProcessType, NumberParallelRuns,
    ResponseFileSizeMaxMB, NumberAnalysisThreads,
    StiffCase)
```

```
int GetSolverOption_3(
    int% SolverType,
    int% SolverProcessType,
    int% NumberParallelRuns,
    int% ResponseFileSizeMaxMB,
    int% NumberAnalysisThreads,
    String^% StiffCase
)
```

cAnalyzeAddLanguageSpecificTextSet("LST25E7C0A1_0?cpp=::|nu310;GetSol

```

abstract GetSolverOption_3 :
    SolverType : int byref *
    SolverProcessType : int byref *
    NumberParallelRuns : int byref *
    ResponseFileSizeMaxMB : int byref *
    NumberAnalysisThreads : int byref *
    StiffCase : string byref -> int

```

Parameters

SolverType

Type: SystemInt32

This is 0, 1 or 2, indicating the solver type.

- ◇ 0 = Standard solver
- ◇ 1 = Advanced solver
- ◇ 2 = Multi-threaded solver

SolverProcessType

Type: SystemInt32

This is 0, 1 or 2, indicating the process the analysis is run.

- ◇ 0 = Auto (program determined)
- ◇ 1 = GUI process
- ◇ 2 = Separate process

NumberParallelRuns

Type: SystemInt32

This is an integer not including -1 or 0.

- ◇ Less than -1 = The negative of the program determined value when the assigned value is 0 = Auto parallel (use up to all physical cores - limited by license).
- ◇ 1 = Serial.
- ◇ Greater than 1 = User defined parallel (use up to this fixed number of cores - limited by license).

ResponseFileSizeMaxMB

Type: SystemInt32

The maximum size of a response file in MB before a new response file is created. Positive if user specified, negative if program determined.

NumberAnalysisThreads

Type: SystemInt32

Number of threads that the analysis can use. Positive if user specified, negative if program determined.

StiffCase

Type: SystemString

The name of the load case used when outputting the mass and stiffness matrices to text files. If this item is blank, no matrices are output.

Return Value

Type: Int32

Returns zero if the options are successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim SolverType As Integer
    Dim SolverProcessType As Integer
    Dim NumberParallelRuns As Integer
    Dim StiffCase As String
    Dim ResponseFileSizeMaxMB As Integer
    Dim NumberAnalysisThreads As Integer

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'get model solver options
    ret = SapModel.Analyze.GetSolverOption_3(SolverType, SolverProcessType, NumberParallelRuns)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub

```

 See Also

ReferencecAnalyze InterfaceETABSv1 Namespace

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CSI API ETABS v1

cAnalyzeAddLanguageSpecificTextSet("LST80A9194C_0")

Method

Merges analysis results from a given model.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int MergeAnalysisResults(
    string SourceFileName
)
```

```
Function MergeAnalysisResults (
    SourceFileName As String
) As Integer
```

```
Dim instance As cAnalyze
Dim SourceFileName As String
Dim returnValue As Integer
```

```
returnValue = instance.MergeAnalysisResults(SourceFileName)
```

```
int MergeAnalysisResults(
    String^ SourceFileName
)
```

```
abstract MergeAnalysisResults :
    SourceFileName : string -> int
```

Parameters

SourceFileName

Type: [SystemString](#)

Full path to the model file to import results from.

Return Value

Type: [Int32](#)

Returns zero if the results are successfully merged; otherwise it returns a nonzero value.

▲ Remarks

The analysis model is automatically created as part of this function.

See [Merging Analysis Results](#) section in program help file for requirements and limitations.

cAnalyzeid="LST80A9194C_0"AddLanguageSpecificTextSet("LST80A9194C_0?cpp=::|nu3.13MergeA

IMPORTANT NOTE: Your model must have a file path defined before merging analysis results. If the model is opened from an existing file, a file path is defined. If the model is created from scratch, the File.Save function must be called with a file name before merging analysis results. Saving the file creates the file path.

Examples

VB

Copy

```
Public Sub Example()
    Dim myHelper As cHelper
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    myHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp\source")
    ret = SapModel.File.Save("C:\CSI_API_temp\source\model.edb")
    ret = SapModel.Analyze.RunAnalysis

    'initialize a new model
    ret = SapModel.InitializeNewModel()

    'create the same steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'merge analysis results
    ret = SapModel.Analyze.SetSolverOption_2(1, 2, 0)
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp\target")
    ret = SapModel.File.Save("C:\CSI_API_temp\target\model.edb")
    ret = SapModel.Analyze.MergeAnalysisResults("C:\CSI_API_temp\source\model.edb")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
    myHelper = Nothing
End Sub
```

See Also

Reference

cAnalyze Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAnalyzeAddLanguageSpecificTextSet("LST8DFA66A1 Method

Modifies the undeformed geometry based on displacements obtained from a specified load case

Namespace: `ETABSv1`

Assembly: `ETABSv1` (in `ETABSv1.dll`) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int ModifyUndeformedGeometry(
    string CaseName,
    double SF,
    int Stage = -1,
    bool Original = false
)
```

```
Function ModifyUndeformedGeometry (
    CaseName As String,
    SF As Double,
    Stage As Integer = -1,
    Original As Boolean = false
) As Integer
```

```
Dim instance As cAnalyze
Dim CaseName As String
Dim SF As Double
Dim Stage As Integer
Dim Original As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.ModifyUndeformedGeometry(CaseName,
    SF, Stage, Original)
```

```
int ModifyUndeformedGeometry(
    String^ CaseName,
    double SF,
    int Stage = -1,
    bool Original = false
)
```

```
abstract ModifyUndeformedGeometry :
    CaseName : string *
    SF : float *
    ?Stage : int *
    ?Original : bool
(* Defaults:
    let _Stage = defaultArg Stage -1
    let _Original = defaultArg Original false
*)
```

cAnalyzeAddLanguageSpecificTextSet("LST8DFA66A1_0?cpp=::|nu316;Modify

-> int

Parameters

CaseName

Type:Â SystemString

The name of the static load case from which displacements are obtained.

SF

Type:Â SystemDouble

The scale factor applied to the displacements.

Stage (Optional)

Type:Â SystemInt32

This item applies only when the specified load case is a staged construction load case. It is the stage number from which the displacements are obtained.

Specifying a -1 for this item means to use the last run stage.

Original (Optional)

Type:Â SystemBoolean

If this item is True, all other input items in this function are ignored and the original undeformed geometry data is reinstated.

Return Value

Type:Â Int32

Returns zero if it is successful; otherwise it returns a nonzero value

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'assign point object restraints
    ReDim Value(5)
    For i = 0 To 5
        Value(i) = True
    Next i
    ret = SapModel.PointObj.SetRestraint("1", Value)
```

Introduction

```
'run analysis
  System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
  ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
  ret = SapModel.Analyze.RunAnalysis

'modify undeformed geometry
  ret = SapModel.Analyze.ModifyUndeformedGeometry("DEAD", 1)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

See Also

Reference

cAnalyze Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAnalyzeAddLanguageSpecificTextSet("LSTC803A37_0") Method

Modifies the undeformed geometry based on the shape of a specified mode

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int ModifyUndeformedGeometryModeShape(
    string CaseName,
    int Mode,
    double MaxDispl,
    int Direction,
    bool Original = false
)
```

```
Function ModifyUndeformedGeometryModeShape (
    CaseName As String,
    Mode As Integer,
    MaxDispl As Double,
    Direction As Integer,
    Optional Original As Boolean = false
) As Integer
```

```
Dim instance As cAnalyze
Dim CaseName As String
Dim Mode As Integer
Dim MaxDispl As Double
Dim Direction As Integer
Dim Original As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.ModifyUndeformedGeometryModeShape(CaseName,
    Mode, MaxDispl, Direction, Original)
```

```
int ModifyUndeformedGeometryModeShape(
    String^ CaseName,
    int Mode,
    double MaxDispl,
    int Direction,
    bool Original = false
)
```

```
abstract ModifyUndeformedGeometryModeShape :
    CaseName : string *
    Mode : int *
    MaxDispl : float *
    Direction : int *
    ?Original : bool
```

cAnalyzeAddLanguageSpecificTextSet("LSTC803A37_0?cpp=::|nu=."[BIM](#)[ModifyUndeformedGeometryModeShape](#)

```
(* Defaults:
    let _Original = defaultArg Original false
*)
-> int
```

Parameters

CaseName

Type:Â SystemString

The name of a load case

Mode

Type:Â SystemInt32

The mode shape

MaxDispl

Type:Â SystemDouble

The maximum displacement to which the mode shape will be scaled

Direction

Type:Â SystemInt32

The direction in which to apply the geometry modification

1. X

2. Y

3. Z

4. Resultant

Original (Optional)

Type:Â SystemBoolean

If this item is True, all other input items in this function are ignored and the original undeformed geometry data is reinstated.

Return Value

Type:Â Int32

Returns zero if it is successful; otherwise it returns a nonzero value

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
```

Introduction

```
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'assign point object restraints
ReDim Value(5)
For i = 0 To 5
    Value(i) = True
Next i
ret = SapModel.PointObj.SetRestraint("1", Value)

'run analysis
System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
ret = SapModel.Analyze.RunAnalysis

'modify undeformed geometry
ret = SapModel.Analyze.ModifyUndeformedGeometryModeShape("Modal", 1, 1.0, 2)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

📌 See Also

Reference

[cAnalyze Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalyzeAddLanguageSpecificTextSet("LST7E890C17_0") Method

Runs the analysis.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int RunAnalysis()
```

```
Function RunAnalysis As Integer
```

```
Dim instance As cAnalyze
```

```
Dim returnValue As Integer
```

```
returnValue = instance.RunAnalysis()
```

```
int RunAnalysis()
```

```
abstract RunAnalysis : unit -> int
```

Return Value

Type: [Int32](#)

Returns zero if the analysis model is successfully run; otherwise it returns a nonzero value.

▲ Remarks

The analysis model is automatically created as part of this function.

IMPORTANT NOTE: Your model must have a file path defined before running the analysis. If the model is opened from an existing file, a file path is defined. If the model is created from scratch, the File.Save function must be called with a file name before running the analysis. Saving the file creates the file path.

▲ Examples

[VB](#)

[Copy](#)

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1
```

```
    'create ETABS object
```

```
    Dim myHelper as cHelper = New Helper
```

```
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

```
cAnalyze.RunAnalysis("LST7E890C17_0")
```

```
'start ETABS application
    ret = EtabsObject.ApplicationStart()

'create SapModel object
    SapModel = EtabsObject.SapModel

'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

🔗 See Also

Reference

[cAnalyze Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalyzeAddLanguageSpecificTextSet("LST284D793B_0?cpp=::|nu-324;SetActiveDof")

Method

Sets the model global degrees of freedom.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▀ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetActiveDof(
    ref bool[] DOF
)
```

```
Function SetActiveDof (
    ByRef DOF As Boolean()
) As Integer
```

```
Dim instance As cAnalyze
Dim DOF As Boolean()
Dim returnValue As Integer
```

```
returnValue = instance.SetActiveDof(DOF)
```

```
int SetActiveDof(
    array<bool>^% DOF
)
```

```
abstract SetActiveDof :
    DOF : bool[] byref -> int
```

Parameters

DOF

Type: [System.Boolean](#)

Return Value

Type: [Int32](#)

Returns zero if the degrees of freedom are successfully set; otherwise it returns a nonzero value.

▀ Remarks

▀ Examples

[VB](#)

[Copy](#)

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
```

cAnalyzeAddLanguageSpecificTextSet("LST284D793B_0?cpp=::|nu-324;SetActiveDof")

Introduction

```
Dim ret As Integer = -1
Dim DOF() As Boolean

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set model degrees of freedom
ReDim DOF(5)
DOF(0) = True
DOF(1) = True
DOF(2) = True
ret = SapModel.Analyze.SetActiveDOF(DOF)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cAnalyze Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalyzeAddLanguageSpecificTextSet("LSTE623EEBD Method

Sets the design and response recovery options.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetDesignResponseOption(
    int NumberDesignThreads,
    int NumberResponseRecoveryThreads,
    int UseMemoryMappedFilesForResponseRecovery,
    bool ModelDifferencesOKWhenMergingResults
)
```

```
Function SetDesignResponseOption (
    NumberDesignThreads As Integer,
    NumberResponseRecoveryThreads As Integer,
    UseMemoryMappedFilesForResponseRecovery As Integer,
    ModelDifferencesOKWhenMergingResults As Boolean
) As Integer
```

```
Dim instance As cAnalyze
Dim NumberDesignThreads As Integer
Dim NumberResponseRecoveryThreads As Integer
Dim UseMemoryMappedFilesForResponseRecovery As Integer
Dim ModelDifferencesOKWhenMergingResults As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetDesignResponseOption(NumberDesignThreads,
    NumberResponseRecoveryThreads,
    UseMemoryMappedFilesForResponseRecovery,
    ModelDifferencesOKWhenMergingResults)
```

```
int SetDesignResponseOption(
    int NumberDesignThreads,
    int NumberResponseRecoveryThreads,
    int UseMemoryMappedFilesForResponseRecovery,
    bool ModelDifferencesOKWhenMergingResults
)
```

```
abstract SetDesignResponseOption :
    NumberDesignThreads : int *
    NumberResponseRecoveryThreads : int *
    UseMemoryMappedFilesForResponseRecovery : int *
    ModelDifferencesOKWhenMergingResults : bool -> int
```

cAnalyzeAddLanguageSpecificTextSet("LSTE623EEBD_0?cpp=::|n026");SetDe

Parameters

NumberDesignThreads

Type: SystemInt32

Number of threads that design can use. Set to positive for user specified, non-positive for program determined.

NumberResponseRecoveryThreads

Type: SystemInt32

Number of threads that response recovery can use. Set to positive for user specified, non-positive for program determined.

UseMemoryMappedFilesForResponseRecovery

Type: SystemInt32

Flag for using memory mapped files for response recovery. Set to

◊ -1 = Do not use memory mapped files.

◊ 0 = Program determined.

◊ 1 = Use memory mapped files.

ModelDifferencesOKWhenMergingResults

Type: SystemBoolean

Flag for merging results in presence of any model differences. Set to true to enable merging of results from two models that are not identical, to false otherwise.

Return ValueType: Int32

Returns zero if the options are successfully set; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set model solver options
    ret = SapModel.Analyze.SetDesignResponseOption(4, 4, 0, False)

    'close ETABS
    EtabsObject.ApplicationExit(False)
```

```
'clean up variables  
  SapModel = Nothing  
  EtabsObject = Nothing  
End Sub
```

📌 See Also

Reference

cAnalyze Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAnalyzeAddLanguageSpecificTextSet("LST3863E913_0") Method

Sets the run flag for load cases.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetRunCaseFlag(
    string Name,
    bool Run,
    bool All = false
)
```

```
Function SetRunCaseFlag (
    Name As String,
    Run As Boolean,
    All As Boolean = false
) As Integer
```

```
Dim instance As cAnalyze
Dim Name As String
Dim Run As Boolean
Dim All As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetRunCaseFlag(Name,
    Run, All)
```

```
int SetRunCaseFlag(
    String^ Name,
    bool Run,
    bool All = false
)
```

```
abstract SetRunCaseFlag :
    Name : string *
    Run : bool *
    ?All : bool
(* Defaults:
    let _All = defaultArg All false
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing load case that is to have its run flag set.

This item is ignored when the All item is True.

Run

Type:Â SystemBoolean

If this item is True, the specified load case is to be run.

All (Optional)

Type:Â SystemBoolean

If this item is True, the run flag is set as specified by the Run item for all load cases, and the Name item is ignored.

Return Value

Type:Â Int32

Returns zero if the flag is successfully set; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set load case run flag
    ret = SapModel.Analyze.SetRunCaseFlag("MODAL", False)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cAnalyze Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAnalyzeAddLanguageSpecificTextSet("LST101AE555_0") Method

DEPRECATED. Please see [SetSolverOption_2\(Int32, Int32, Int32, String\)](#) or [SetSolverOption_3\(Int32, Int32, Int32, Int32, Int32, String\)](#)

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetSolverOption(
    int SolverType,
    bool Force32BitSolver,
    string StiffCase = ""
)
```

```
Function SetSolverOption (
    SolverType As Integer,
    Force32BitSolver As Boolean,
    StiffCase As String = ""
) As Integer
```

```
Dim instance As cAnalyze
Dim SolverType As Integer
Dim Force32BitSolver As Boolean
Dim StiffCase As String
Dim returnValue As Integer
```

```
returnValue = instance.SetSolverOption(SolverType,
    Force32BitSolver, StiffCase)
```

```
int SetSolverOption(
    int SolverType,
    bool Force32BitSolver,
    String^ StiffCase = L""
)
```

```
abstract SetSolverOption :
    SolverType : int *
    Force32BitSolver : bool *
    ?StiffCase : string
(* Defaults:
    let _StiffCase = defaultArg StiffCase ""
*)
-> int
```


Parameters

SolverType

Type: [SystemInt32](#)

Force32BitSolver

Type: [SystemBoolean](#)

StiffCase (Optional)

Type: [SystemString](#)

Return Value

Type: [Int32](#)

See Also

Reference

[cAnalyze Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAnalyzeAddLanguageSpecificTextSet("LSTFA6BEE69_0") Method

DEPRECATED. Please see [SetSolverOption_2\(Int32, Int32, Int32, String\)](#) or [SetSolverOption_3\(Int32, Int32, Int32, Int32, Int32, String\)](#)

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetSolverOption_1(
    int SolverType,
    int SolverProcessType,
    bool Force32BitSolver,
    string StiffCase = ""
)
```

```
Function SetSolverOption_1 (
    SolverType As Integer,
    SolverProcessType As Integer,
    Force32BitSolver As Boolean,
    StiffCase As String = ""
) As Integer
```

```
Dim instance As cAnalyze
Dim SolverType As Integer
Dim SolverProcessType As Integer
Dim Force32BitSolver As Boolean
Dim StiffCase As String
Dim returnValue As Integer
```

```
returnValue = instance.SetSolverOption_1(SolverType,
    SolverProcessType, Force32BitSolver,
    StiffCase)
```

```
int SetSolverOption_1(
    int SolverType,
    int SolverProcessType,
    bool Force32BitSolver,
    String^ StiffCase = L""
)
```

```
abstract SetSolverOption_1 :
    SolverType : int *
    SolverProcessType : int *
    Force32BitSolver : bool *
    ?StiffCase : string
(* Defaults:
    let _StiffCase = defaultArg StiffCase ""
*)
```

cAnalyzeAddLanguageSpecificTextSet("LSTFA6BEE69_0?cpp=::|num=34");SetSol

-> int

Parameters

SolverType

Type: SystemInt32

SolverProcessType

Type: SystemInt32

Force32BitSolver

Type: SystemBoolean

StiffCase (Optional)

Type: SystemString

Return Value

Type: Int32

▲ See Also

Reference

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CSI API ETABS v1

cAnalyzeAddLanguageSpecificTextSet("LSTDF8FB267_0?cpp=::|nu336;SetSol

Method

Sets the model solver options.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetSolverOption_2(
    int SolverType,
    int SolverProcessType,
    int NumberParallelRuns,
    string StiffCase = ""
)
```

```
Function SetSolverOption_2 (
    SolverType As Integer,
    SolverProcessType As Integer,
    NumberParallelRuns As Integer,
    StiffCase As String = ""
) As Integer
```

```
Dim instance As cAnalyze
Dim SolverType As Integer
Dim SolverProcessType As Integer
Dim NumberParallelRuns As Integer
Dim StiffCase As String
Dim returnValue As Integer
```

```
returnValue = instance.SetSolverOption_2(SolverType,
    SolverProcessType, NumberParallelRuns,
    StiffCase)
```

```
int SetSolverOption_2(
    int SolverType,
    int SolverProcessType,
    int NumberParallelRuns,
    String^ StiffCase = L""
)
```

```
abstract SetSolverOption_2 :
    SolverType : int *
    SolverProcessType : int *
    NumberParallelRuns : int *
    ?StiffCase : string
(* Defaults:
    let _StiffCase = defaultArg StiffCase ""
*)
-> int
```

cAnalyzeAddLanguageSpecificTextSet("LSTDF8FB267_0?cpp=::|nu336;SetSol

Parameters

SolverType

Type: SystemInt32

This is 0, 1 or 2, indicating the solver type.

- ◊ 0 = Standard solver
- ◊ 1 = Advanced solver
- ◊ 2 = Multi-threaded solver

SolverProcessType

Type: SystemInt32

This is 0, 1 or 2, indicating the process the analysis is run.

- ◊ 0 = Auto (program determined)
- ◊ 1 = GUI process
- ◊ 2 = Separate process

NumberParallelRuns

Type: SystemInt32

This is an integer not including -1.

- ◊ -Less than -1 = Auto parallel (use up to all physical cores - limited by license). Treated the same as 0.
- ◊ -1 = Illegal value; will return an error
- ◊ 0 = Auto parallel (use up to all physical cores)
- ◊ 1 = Serial
- ◊ Greater than 1 = User defined parallel (use up to this fixed number of cores - limited by license)

StiffCase (Optional)

Type: SystemString

The name of the load case used when outputting the mass and stiffness matrices to text files. If this item is blank, no matrices are output.

Return Value

Type: Int32

Returns zero if the options are successfully set; otherwise it returns a nonzero value.

▲Remarks

'''

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
```

Introduction

```
ret = SapModel.InitializeNewModel()  
  
'create steel deck template model  
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)  
  
'set model solver options  
ret = SapModel.Analyze.SetSolverOption_2(1, 1, 3, "DEAD")  
  
'close ETABS  
EtabsObject.ApplicationExit(False)  
  
'clean up variables  
SapModel = Nothing  
EtabsObject = Nothing  
End Sub
```

🔗 See Also

Reference

cAnalyze Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAnalyzeAddLanguageSpecificTextSet("LSTA7D104BC0") Method

Sets the model solver options.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetSolverOption_3(
    int SolverType,
    int SolverProcessType,
    int NumberParallelRuns,
    int ResponseFileSizeMaxMB,
    int NumberAnalysisThreads,
    string StiffCase = ""
)
```

```
Function SetSolverOption_3 (
    SolverType As Integer,
    SolverProcessType As Integer,
    NumberParallelRuns As Integer,
    ResponseFileSizeMaxMB As Integer,
    NumberAnalysisThreads As Integer,
    StiffCase As String = ""
) As Integer
```

```
Dim instance As cAnalyze
Dim SolverType As Integer
Dim SolverProcessType As Integer
Dim NumberParallelRuns As Integer
Dim ResponseFileSizeMaxMB As Integer
Dim NumberAnalysisThreads As Integer
Dim StiffCase As String
Dim returnValue As Integer
```

```
returnValue = instance.SetSolverOption_3(SolverType,
    SolverProcessType, NumberParallelRuns,
    ResponseFileSizeMaxMB, NumberAnalysisThreads,
    StiffCase)
```

```
int SetSolverOption_3(
    int SolverType,
    int SolverProcessType,
    int NumberParallelRuns,
    int ResponseFileSizeMaxMB,
    int NumberAnalysisThreads,
    String^ StiffCase = L""
)
```

cAnalyzeAddLanguageSpecificTextSet("LSTA7D104BC_0?cpp=::|n039);SetSo

```

abstract SetSolverOption_3 :
  SolverType : int *
  SolverProcessType : int *
  NumberParallelRuns : int *
  ResponseFileSizeMaxMB : int *
  NumberAnalysisThreads : int *
  ?StiffCase : string
(* Defaults:
   let _StiffCase = defaultArg StiffCase ""
*)
-> int

```

Parameters

SolverType

Type: SystemInt32

This is 0, 1 or 2, indicating the solver type.

- ◊ 0 = Standard solver
- ◊ 1 = Advanced solver
- ◊ 2 = Multi-threaded solver

SolverProcessType

Type: SystemInt32

This is 0, 1 or 2, indicating the process the analysis is run.

- ◊ 0 = Auto (program determined)
- ◊ 1 = GUI process
- ◊ 2 = Separate process

NumberParallelRuns

Type: SystemInt32

This is an integer not including -1.

- ◊ Less than -1 = Auto parallel (use up to all physical cores - limited by license). Treated the same as 0.
- ◊ -1 = Illegal value; will return an error
- ◊ 0 = Auto parallel (use up to all physical cores - limited by license)
- ◊ 1 = Serial
- ◊ Greater than 1 = User defined parallel (use up to this fixed number of cores - limited by license)

ResponseFileSizeMaxMB

Type: SystemInt32

The maximum size of a response file in MB before a new response file is created. Non-positive means program determined.

NumberAnalysisThreads

Type: SystemInt32

Number of threads that the analysis can use. Non-positive means program determined.

StiffCase (Optional)

Type: SystemString

The name of the load case used when outputting the mass and stiffness matrices to text files. If this item is blank, no matrices are output.

Return Value

Type: Int32

Returns zero if the options are successfully set; otherwise it returns a nonzero value.

Remarks

'''

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set model solver options
    ret = SapModel.Analyze.SetSolverOption_3(1, 1, 3, 0, 0, "DEAD")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

cAnalyze Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAreaElm Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cAreaElm
```

```
Public Interface cAreaElm
```

```
Dim instance As cAreaElm
```

```
public interface class cAreaElm
```

```
type cAreaElm = interface end
```

The cAreaElm type exposes the following members.

▲ Methods

▲	Name	Description
⇒	Count	
⇒	GetLoadTemperature	
⇒	GetLoadUniform	
⇒	GetLocalAxes	
⇒	GetMaterialOverwrite	
⇒	GetModifiers	
⇒	GetNameList	
⇒	GetObj	
⇒	GetOffsets	
⇒	GetPoints	
⇒	GetProperty	
⇒	GetThickness	
⇒	GetTransformationMatrix	

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAreaElm Methods

The [cAreaElm](#) type exposes the following members.

Methods

	Name	Description
⇒	Count	
⇒	GetLoadTemperature	
⇒	GetLoadUniform	
⇒	GetLocalAxes	
⇒	GetMaterialOverwrite	
⇒	GetModifiers	
⇒	GetNameList	
⇒	GetObj	
⇒	GetOffsets	
⇒	GetPoints	
⇒	GetProperty	
⇒	GetThickness	
⇒	GetTransformationMatrix	

[Top](#)

See Also

Reference

[cAreaElm Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAreaElmAddLanguageSpecificTextSet("LSTEB646556 Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

int Count()

Function Count As Integer

Dim instance As cAreaElm

Dim returnValue As Integer

returnValue = instance.Count()

int Count()

abstract Count : unit -> int

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cAreaElm Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAreaElmAddLanguageSpecificTextSet("LST5556A0EC0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetLoadTemperature(
    string Name,
    ref int NumberItems,
    ref string[] AreaName,
    ref string[] LoadPat,
    ref int[] MyType,
    ref double[] Value,
    ref string[] PatternName,
    eItemTypeElm ItemTypeElm = eItemTypeElm.Element
)
```

```
Function GetLoadTemperature (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef AreaName As String(),
    ByRef LoadPat As String(),
    ByRef MyType As Integer(),
    ByRef Value As Double(),
    ByRef PatternName As String(),
    ByRef ItemTypeElm As eItemTypeElm = eItemTypeElm.Element
) As Integer
```

```
Dim instance As cAreaElm
Dim Name As String
Dim NumberItems As Integer
Dim AreaName As String()
Dim LoadPat As String()
Dim MyType As Integer()
Dim Value As Double()
Dim PatternName As String()
Dim ItemTypeElm As eItemTypeElm
Dim returnValue As Integer
```

```
returnValue = instance.GetLoadTemperature(Name,
    NumberItems, AreaName, LoadPat, MyType,
    Value, PatternName, ItemTypeElm)
```

```
int GetLoadTemperature(
    String^ Name,
    int% NumberItems,
    array<String^>^% AreaName,
    array<String^>^% LoadPat,
```

```

array<int>^% MyType,
array<double>^% Value,
array<String^>^% PatternName,
eItemTypeElm ItemTypeElm = eItemTypeElm::Element
)

abstract GetLoadTemperature :
    Name : string *
    NumberItems : int byref *
    AreaName : string[] byref *
    LoadPat : string[] byref *
    MyType : int[] byref *
    Value : float[] byref *
    PatternName : string[] byref *
    ?ItemTypeElm : eItemTypeElm
(* Defaults:
    let _ItemTypeElm = defaultArg ItemTypeElm eItemTypeElm.Element
*)
-> int

```

Parameters

Name
Type:Â [SystemString](#)

NumberItems
Type:Â [SystemInt32](#)

AreaName
Type:Â [SystemString](#)

LoadPat
Type:Â [SystemString](#)

MyType
Type:Â [SystemInt32](#)

Value
Type:Â [SystemDouble](#)

PatternName
Type:Â [SystemString](#)

ItemTypeElm (Optional)
Type:Â [ETABSv1eItemTypeElm](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cAreaElm Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAreaElmAddLanguageSpecificTextSet("LSTDBFCBDE Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetLoadUniform(
    string Name,
    ref int NumberItems,
    ref string[] AreaName,
    ref string[] LoadPat,
    ref string[] CSys,
    ref int[] Dir,
    ref double[] Value,
    eItemTypeElm ItemTypeElm = eItemTypeElm.Element
)
```

```
Function GetLoadUniform (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef AreaName As String(),
    ByRef LoadPat As String(),
    ByRef CSys As String(),
    ByRef Dir As Integer(),
    ByRef Value As Double(),
    ByRef ItemTypeElm As eItemTypeElm = eItemTypeElm.Element
) As Integer
```

```
Dim instance As cAreaElm
Dim Name As String
Dim NumberItems As Integer
Dim AreaName As String()
Dim LoadPat As String()
Dim CSys As String()
Dim Dir As Integer()
Dim Value As Double()
Dim ItemTypeElm As eItemTypeElm
Dim returnValue As Integer
```

```
returnValue = instance.GetLoadUniform(Name,
    NumberItems, AreaName, LoadPat, CSys,
    Dir, Value, ItemTypeElm)
```

```
int GetLoadUniform(
    String^ Name,
    int% NumberItems,
    array<String^>^% AreaName,
    array<String^>^% LoadPat,
```

cAreaElmspan id="LSTDBFCBDBD_0"AddLanguageSpecificTextSet("LSTDBFCBDBD_0?cpp=:340=");Get


```

array<String>^% CSys,
array<int>^% Dir,
array<double>^% Value,
eItemTypeElm ItemTypeElm = eItemTypeElm::Element
)

abstract GetLoadUniform :
    Name : string *
    NumberItems : int byref *
    AreaName : string[] byref *
    LoadPat : string[] byref *
    CSys : string[] byref *
    Dir : int[] byref *
    Value : float[] byref *
    ?ItemTypeElm : eItemTypeElm
(* Defaults:
    let _ItemTypeElm = defaultArg ItemTypeElm eItemTypeElm.Element
*)
-> int

```

Parameters

Name
Type: [SystemString](#)

NumberItems
Type: [SystemInt32](#)

AreaName
Type: [SystemString](#)

LoadPat
Type: [SystemString](#)

CSys
Type: [SystemString](#)

Dir
Type: [SystemInt32](#)

Value
Type: [SystemDouble](#)

ItemTypeElm (Optional)
Type: [ETABSv1eItemTypeElm](#)

Return Value

Type: [Int32](#)

See Also

Reference

[cAreaElm Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAreaElmAddLanguageSpecificTextSet("LST871CD9A4_0?cpp=::|n350");GetLo

Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetLocalAxes(
    string Name,
    ref double Ang
)
```

```
Function GetLocalAxes (
    Name As String,
    ByRef Ang As Double
) As Integer
```

```
Dim instance As cAreaElm
Dim Name As String
Dim Ang As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetLocalAxes(Name,
    Ang)
```

```
int GetLocalAxes(
    String^ Name,
    double% Ang
)
```

```
abstract GetLocalAxes :
    Name : string *
    Ang : float byref -> int
```

Parameters

Name

Type: SystemString

Ang

Type: SystemDouble

Return Value

Type: Int32

▲ See Also

Reference

cAreaElm Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAreaElmAddLanguageSpecificTextSet("LST840538A2_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetMaterialOverwrite(
    string Name,
    ref string PropName
)
```

```
Function GetMaterialOverwrite (
    Name As String,
    ByRef PropName As String
) As Integer
```

```
Dim instance As cAreaElm
Dim Name As String
Dim PropName As String
Dim returnValue As Integer
```

```
returnValue = instance.GetMaterialOverwrite(Name,
    PropName)
```

```
int GetMaterialOverwrite(
    String^ Name,
    String^% PropName
)
```

```
abstract GetMaterialOverwrite :
    Name : string *
    PropName : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

PropName

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cAreaElm Interface

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CSI API ETABS v1

cAreaElmAddLanguageSpecificTextSet("LST3900F5F1_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetModifiers(
    string Name,
    ref double[] Value
)
```

```
Function GetModifiers (
    Name As String,
    ByRef Value As Double()
) As Integer
```

```
Dim instance As cAreaElm
Dim Name As String
Dim Value As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetModifiers(Name,
    Value)
```

```
int GetModifiers(
    String^ Name,
    array<double>^% Value
)
```

```
abstract GetModifiers :
    Name : string *
    Value : float[] byref -> int
```

Parameters

Name

Type: [SystemString](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cAreaElm Interface

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CSI API ETABS v1

cAreaElmAddLanguageSpecificTextSet("LST44128A78") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNameList(
    ref int NumberNames,
    ref string[] MyName
)
```

```
Function GetNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cAreaElm
Dim NumberNames As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetNameList(NumberNames,
    MyName)
```

```
int GetNameList(
    int% NumberNames,
    array<String^>^% MyName
)
```

```
abstract GetNameList :
    NumberNames : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberNames

Type: [System.Int32](#)

MyName

Type: [System.String](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cAreaElm Interface

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CSI API ETABS v1

cAreaElmAddLanguageSpecificTextSet("LST9E683408_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetObj(
    string Name,
    ref string Obj
)
```

```
Function GetObj (
    Name As String,
    ByRef Obj As String
) As Integer
```

```
Dim instance As cAreaElm
Dim Name As String
Dim Obj As String
Dim returnValue As Integer
```

```
returnValue = instance.GetObj(Name, Obj)
```

```
int GetObj(
    String^ Name,
    String^% Obj
)
```

```
abstract GetObj :
    Name : string *
    Obj : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

Obj

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cAreaElm Interface

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CSI API ETABS v1

cAreaElmAddLanguageSpecificTextSet("LSTE5B6C21E_0?cpp=::|n060");GetOffset

Method

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetOffsets(
    string Name,
    ref int OffsetType,
    ref string OffsetPattern,
    ref double OffsetPatternSF,
    ref double[] Offset
)
```

```
Function GetOffsets (
    Name As String,
    ByRef OffsetType As Integer,
    ByRef OffsetPattern As String,
    ByRef OffsetPatternSF As Double,
    ByRef Offset As Double()
) As Integer
```

```
Dim instance As cAreaElm
Dim Name As String
Dim OffsetType As Integer
Dim OffsetPattern As String
Dim OffsetPatternSF As Double
Dim Offset As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetOffsets(Name,
    OffsetType, OffsetPattern, OffsetPatternSF,
    Offset)
```

```
int GetOffsets(
    String^ Name,
    int% OffsetType,
    String^% OffsetPattern,
    double% OffsetPatternSF,
    array<double>^% Offset
)
```

```
abstract GetOffsets :
    Name : string *
    OffsetType : int byref *
    OffsetPattern : string byref *
    OffsetPatternSF : float byref *
    Offset : float[] byref -> int
```

cAreaElmspan id="LSTE5B6C21E_0"AddLanguageSpecificTextSet("LSTE5B6C21E_0?cpp=::|n060");GetOffset

Parameters

Name

Type: SystemString

OffsetType

Type: SystemInt32

OffsetPattern

Type: SystemString

OffsetPatternSF

Type: SystemDouble

Offset

Type: SystemDouble

Return Value

Type: Int32

See Also

Reference

cAreaElm Interface

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CSI API ETABS v1

cAreaElmAddLanguageSpecificTextSet("LST26D692D8_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPoints(
    string Name,
    ref int NumberPoints,
    ref string[] Point
)
```

```
Function GetPoints (
    Name As String,
    ByRef NumberPoints As Integer,
    ByRef Point As String()
) As Integer
```

```
Dim instance As cAreaElm
Dim Name As String
Dim NumberPoints As Integer
Dim Point As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetPoints(Name,
    NumberPoints, Point)
```

```
int GetPoints(
    String^ Name,
    int% NumberPoints,
    array<String^>^% Point
)
```

```
abstract GetPoints :
    Name : string *
    NumberPoints : int byref *
    Point : string[] byref -> int
```

Parameters

Name

Type: [SystemString](#)

NumberPoints

Type: [SystemInt32](#)

Point

Type: [SystemString](#)

cAreaElmAddLanguageSpecificTextSet("LST26D692D8_0?cpp=::|nu362;GetPo

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cAreaElm Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAreaElmAddLanguageSpecificTextSet("LSTC479D31F Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetProperty(
    string Name,
    ref string PropName
)
```

```
Function GetProperty (
    Name As String,
    ByRef PropName As String
) As Integer
```

```
Dim instance As cAreaElm
Dim Name As String
Dim PropName As String
Dim returnValue As Integer
```

```
returnValue = instance.GetProperty(Name,
    PropName)
```

```
int GetProperty(
    String^ Name,
    String^% PropName
)
```

```
abstract GetProperty :
    Name : string *
    PropName : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

PropName

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cAreaElm Interface

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CSI API ETABS v1

cAreaElmAddLanguageSpecificTextSet("LSTB1949E5B Method

Namespace: Â ETABSv1

Assembly: Â ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetThickness(
    string Name,
    ref int ThicknessType,
    ref string ThicknessPattern,
    ref double ThicknessPatternSF,
    ref double[] Thickness
)
```

```
Function GetThickness (
    Name As String,
    ByRef ThicknessType As Integer,
    ByRef ThicknessPattern As String,
    ByRef ThicknessPatternSF As Double,
    ByRef Thickness As Double()
) As Integer
```

```
Dim instance As cAreaElm
Dim Name As String
Dim ThicknessType As Integer
Dim ThicknessPattern As String
Dim ThicknessPatternSF As Double
Dim Thickness As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetThickness(Name,
    ThicknessType, ThicknessPattern,
    ThicknessPatternSF, Thickness)
```

```
int GetThickness(
    String^ Name,
    int% ThicknessType,
    String^% ThicknessPattern,
    double% ThicknessPatternSF,
    array<double>^% Thickness
)
```

```
abstract GetThickness :
    Name : string *
    ThicknessType : int byref *
    ThicknessPattern : string byref *
    ThicknessPatternSF : float byref *
    Thickness : float[] byref -> int
```

Parameters

Name

Type: SystemString

ThicknessType

Type: SystemInt32

ThicknessPattern

Type: SystemString

ThicknessPatternSF

Type: SystemDouble

Thickness

Type: SystemDouble

Return Value

Type: Int32

See Also

Reference

cAreaElm Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAreaElmAddLanguageSpecificTextSet("LST7E49F76B Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetTransformationMatrix(
    string Name,
    ref double[] Value
)
```

```
Function GetTransformationMatrix (
    Name As String,
    ByRef Value As Double()
) As Integer
```

```
Dim instance As cAreaElm
Dim Name As String
Dim Value As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetTransformationMatrix(Name,
Value)
```

```
int GetTransformationMatrix(
    String^ Name,
    array<double>^% Value
)
```

```
abstract GetTransformationMatrix :
    Name : string *
    Value : float[] byref -> int
```

Parameters

Name

Type: [SystemString](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cAreaElm Interface

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CSI API ETABS v1

cAreaObj Interface

Namespace: Â ETABSV1

Assembly: Â ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
public interface cAreaObj
```

```
Public Interface cAreaObj
```

```
Dim instance As cAreaObj
```

```
public interface class cAreaObj
```

```
type cAreaObj = interface end
```

The cAreaObj type exposes the following members.

▲ Methods

Â	Name	Description
⇒	<u>AddByCoord</u>	Adds a new area object, defining points at the specified coordinates.
⇒	<u>AddByPoint</u>	Adds a new area object whose defining points are specified by name.
⇒	<u>ChangeName</u>	
⇒	<u>Count</u>	
⇒	<u>Delete</u>	Deletes area objects.
⇒	<u>DeleteLoadTemperature</u>	
⇒	<u>DeleteLoadUniform</u>	Deletes the uniform load assignments to the specified area objects for the specified load pattern.
⇒	<u>DeleteLoadUniformToFrame</u>	
⇒	<u>DeleteLoadWindPressure</u>	
⇒	<u>DeleteMass</u>	
⇒	<u>DeleteModifiers</u>	
⇒	<u>DeleteSpring</u>	
⇒	<u>GetAllAreas</u>	Retrieves select data for all area objects in the model
⇒	<u>GetCurvedEdges</u>	Retrieves curve data for all edges of an area object
⇒	<u>GetDesignOrientation</u>	Retrieves the design orientation of an area object.
⇒	<u>GetDiaphragm</u>	Retrieves the diaphragm assignment to the specified area object

⇒ <u>GetEdgeConstraint</u>	
⇒ <u>GetElm</u>	
⇒ <u>GetGroupAssign</u>	Retrieves the groups to which an area object is assigned.
⇒ <u>GetGUID</u>	
⇒ <u>GetLabelFromName</u>	Retrieves the label and story for a unique area object name
⇒ <u>GetLabelNameList</u>	Retrieves the names and labels of all defined area objects.
⇒ <u>GetLoadTemperature</u>	Retrieves the temperature load assignments to area objects.
⇒ <u>GetLoadUniform</u>	Retrieves the uniform load assignments to area objects.
⇒ <u>GetLoadUniformToFrame</u>	Retrieves the uniform to frame load assignments to area objects.
⇒ <u>GetLoadWindPressure</u>	Retrieves the wind pressure load assignments to area objects.
⇒ <u>GetLocalAxes</u>	
⇒ <u>GetMass</u>	
⇒ <u>GetMaterialOverwrite</u>	
⇒ <u>GetModifiers</u>	
⇒ <u>GetNameFromLabel</u>	Retrieves the unique name of an area object, given the label and story level
⇒ <u>GetNameList</u>	Retrieves the names of all defined area objects.
⇒ <u>GetNameListOnStory</u>	Retrieves the names of the defined area objects on a given story.
⇒ <u>GetOffsets3</u>	
⇒ <u>GetOpening</u>	Retrieves whether the specified area object is an opening.
⇒ <u>GetPier</u>	Retrieves the pier label assignments of an area object
⇒ <u>GetPoints</u>	Retrieves the names of the point objects that define an area object.
⇒ <u>GetProperty</u>	Retrieves the area property assigned to an area object.
⇒ <u>GetRebarDataPier</u>	Retrieves rebar data for an area pier object
⇒ <u>GetRebarDataSpandrel</u>	Retrieves rebar data for an area spandrel object
⇒ <u>GetSelected</u>	
⇒ <u>GetSelectedEdge</u>	
⇒ <u>GetSpandrel</u>	Retrieves the Spandrel label assignments of an area object
⇒ <u>GetSpringAssignment</u>	Retrieves the named area spring property assignment for an area object
⇒ <u>GetTransformationMatrix</u>	

Introduction

⇒ <u>SetDiaphragm</u>	Assigns a diaphragm to a specified area object
⇒ <u>SetEdgeConstraint</u>	Makes generated edge constraint assignments to area objects.
⇒ <u>SetGroupAssign</u>	
⇒ <u>SetGUID</u>	
⇒ <u>SetLoadTemperature</u>	
⇒ <u>SetLoadUniform</u>	Assigns uniform loads to area objects.
⇒ <u>SetLoadUniformToFrame</u>	
⇒ <u>SetLoadWindPressure</u>	
⇒ <u>SetLocalAxes</u>	Assigns a local axis angle to area objects.
⇒ <u>SetMass</u>	
⇒ <u>SetMaterialOverwrite</u>	
⇒ <u>SetModifiers</u>	
⇒ <u>SetOpening</u>	Designates an area object(s) as an opening.
⇒ <u>SetPier</u>	Sets the pier label assignment of one or more area objects
⇒ <u>SetProperty</u>	Assigns an area property to area objects.
⇒ <u>SetSelected</u>	
⇒ <u>SetSelectedEdge</u>	
⇒ <u>SetSpandrel</u>	Sets the Spandrel label assignment of one or more area objects
⇒ <u>SetSpringAssignment</u>	Assigns an existing named area spring property to area objects

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▲ See Also

Reference

ETABSv1 Namespace

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CSI API ETABS v1

cAreaObj Methods

The cAreaObj type exposes the following members.

Methods

Á	Name	Description
⇒	<u>AddByCoord</u>	Adds a new area object, defining points at the specified coordinates.
⇒	<u>AddByPoint</u>	Adds a new area object whose defining points are specified by name.
⇒	<u>ChangeName</u>	
⇒	<u>Count</u>	
⇒	<u>Delete</u>	Deletes area objects.
⇒	<u>DeleteLoadTemperature</u>	
⇒	<u>DeleteLoadUniform</u>	Deletes the uniform load assignments to the specified area objects for the specified load pattern.
⇒	<u>DeleteLoadUniformToFrame</u>	
⇒	<u>DeleteLoadWindPressure</u>	
⇒	<u>DeleteMass</u>	
⇒	<u>DeleteModifiers</u>	
⇒	<u>DeleteSpring</u>	
⇒	<u>GetAllAreas</u>	Retrieves select data for all area objects in the model
⇒	<u>GetCurvedEdges</u>	Retrieves curve data for all edges of an area object
⇒	<u>GetDesignOrientation</u>	Retrieves the design orientation of an area object.
⇒	<u>GetDiaphragm</u>	Retrieves the diaphragm assignment to the specified area object
⇒	<u>GetEdgeConstraint</u>	
⇒	<u>GetElm</u>	
⇒	<u>GetGroupAssign</u>	Retrieves the groups to which an area object is assigned.
⇒	<u>GetGUID</u>	
⇒	<u>GetLabelFromName</u>	Retrieves the label and story for a unique area object name
⇒	<u>GetLabelNameList</u>	Retrieves the names and labels of all defined area objects.
⇒	<u>GetLoadTemperature</u>	Retrieves the temperature load assignments to area objects.
⇒	<u>GetLoadUniform</u>	Retrieves the uniform load assignments to area objects.
⇒	<u>GetLoadUniformToFrame</u>	Retrieves the uniform to frame load assignments to area objects.
⇒	<u>GetLoadWindPressure</u>	Retrieves the wind pressure load assignments to area objects.

⇒ <u>GetLocalAxes</u>	
⇒ <u>GetMass</u>	
⇒ <u>GetMaterialOverwrite</u>	
⇒ <u>GetModifiers</u>	
⇒ <u>GetNameFromLabel</u>	Retrieves the unique name of an area object, given the label and story level
⇒ <u>GetNameList</u>	Retrieves the names of all defined area objects.
⇒ <u>GetNameListOnStory</u>	Retrieves the names of the defined area objects on a given story.
⇒ <u>GetOffsets3</u>	
⇒ <u>GetOpening</u>	Retrieves whether the specified area object is an opening.
⇒ <u>GetPier</u>	Retrieves the pier label assignments of an area object
⇒ <u>GetPoints</u>	Retrieves the names of the point objects that define an area object.
⇒ <u>GetProperty</u>	Retrieves the area property assigned to an area object.
⇒ <u>GetRebarDataPier</u>	Retrieves rebar data for an area pier object
⇒ <u>GetRebarDataSpandrel</u>	Retrieves rebar data for an area spandrel object
⇒ <u>GetSelected</u>	
⇒ <u>GetSelectedEdge</u>	
⇒ <u>GetSpandrel</u>	Retrieves the Spandrel label assignments of an area object
⇒ <u>GetSpringAssignment</u>	Retrieves the named area spring property assignment for an area object
⇒ <u>GetTransformationMatrix</u>	
⇒ <u>SetDiaphragm</u>	Assigns a diaphragm to a specified area object
⇒ <u>SetEdgeConstraint</u>	Makes generated edge constraint assignments to area objects.
⇒ <u>SetGroupAssign</u>	
⇒ <u>SetGUID</u>	
⇒ <u>SetLoadTemperature</u>	
⇒ <u>SetLoadUniform</u>	Assigns uniform loads to area objects.
⇒ <u>SetLoadUniformToFrame</u>	
⇒ <u>SetLoadWindPressure</u>	
⇒ <u>SetLocalAxes</u>	Assigns a local axis angle to area objects.
⇒ <u>SetMass</u>	
⇒ <u>SetMaterialOverwrite</u>	
⇒ <u>SetModifiers</u>	
⇒ <u>SetOpening</u>	Designates an area object(s) as an opening.
⇒ <u>SetPier</u>	Sets the pier label assignment of one or more area objects

Introduction

 <u> SetProperty </u>	Assigns an area property to area objects.
 <u> SetSelected </u>	
 <u> SetSelectedEdge </u>	
 <u> SetSpandrel </u>	Sets the Spandrel label assignment of one or more area objects
 <u> SetSpringAssignment </u>	Assigns an existing named area spring property to area objects

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▀ See Also

Reference

[cAreaObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LSTFAF2997F_0") Method

Adds a new area object, defining points at the specified coordinates.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int AddByCoord(
int NumberPoints,
ref double[] X,
ref double[] Y,
ref double[] Z,
ref string Name,
string PropName = "Default",
string UserName = "",
string CSys = "Global"
)
```

```
Function AddByCoord (
NumberPoints As Integer,
ByRef X As Double(),
ByRef Y As Double(),
ByRef Z As Double(),
ByRef Name As String,
    PropName As String = "Default",
    UserName As String = "",
    CSys As String = "Global"
) As Integer
```

```
Dim instance As cAreaObj
Dim NumberPoints As Integer
Dim X As Double()
Dim Y As Double()
Dim Z As Double()
Dim Name As String
Dim PropName As String
Dim UserName As String
Dim CSys As String
Dim returnValue As Integer
```

```
returnValue = instance.AddByCoord(NumberPoints,
X, Y, Z, Name, PropName, UserName, CSys)
```

```
int AddByCoord(
int NumberPoints,
array<double>^% X,
array<double>^% Y,
array<double>^% Z,
```

cAreaObjid="LSTFAF2997F_0"AddLanguageSpecificTextSet("LSTFAF2997F_0?cpp=::|nu376;AddBy

```

String^% Name,
String^ PropName = L"Default",
String^ UserName = L"",
String^ CSys = L"Global"
)

abstract AddByCoord :
    NumberPoints : int *
    X : float[] byref *
    Y : float[] byref *
    Z : float[] byref *
    Name : string byref *
    ?PropName : string *
    ?UserName : string *
    ?CSys : string
(* Defaults:
    let _PropName = defaultArg PropName "Default"
    let _UserName = defaultArg UserName ""
    let _CSys = defaultArg CSys "Global"
*)
-> int

```

Parameters

NumberPoints

Type:Â SystemInt32

The number of points in the area object.

X

Type:Â SystemDouble

This is the array of x coordinates for the corner points of the area object. The coordinates are in the coordinate system defined by the CSys item. The coordinates should be ordered to run clockwise or counter clockwise around the area object.

Y

Type:Â SystemDouble

This is the array of y coordinates for the corner points of the area object. The coordinates are in the coordinate system defined by the CSys item. The coordinates should be ordered to run clockwise or counter clockwise around the area object.

Z

Type:Â SystemDouble

This is the array of z coordinates for the corner points of the area object. The coordinates are in the coordinate system defined by the CSys item. The coordinates should be ordered to run clockwise or counter clockwise around the area object.

Name

Type:Â SystemString

This is the name that the program ultimately assigns to the area object. If no userName is specified, the program assigns a default name to the area object. If a userName is specified and that name is not used for another area object, the userName is assigned to the area object; otherwise a default name is assigned to the area object.

PropName (Optional)

Type:Â SystemString

This is Default, None or the name of a defined area property.

If it is Default, the program assigns a default area property to the area object. If it is None, no area property is assigned to the area object. If it is the name of a defined area property, that property is assigned to the area object.

UserName (Optional)

Type: SystemString

CSys (Optional)

Type: SystemString

The name of the coordinate system in which the area object point coordinates are defined.

Return Value

Type: Int32

Returns zero if the area object is successfully added, otherwise it returns a nonzero value.

▀ Remarks

▀ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim x() As Double
    Dim y() As Double
    Dim z() As Double
    Dim Name As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create blank template model
    ret = SapModel.File.NewBlank()

    'add area object by coordinates
    ReDim x(5)
    ReDim y(5)
    ReDim z(5)
    x(0) = 50: y(0) = 0
    x(1) = 100: y(1) = 0
    x(2) = 150: y(2) = 40
    x(3) = 100: y(3) = 80
    x(4) = 50: y(4) = 80
    x(5) = 0: y(5) = 40
    ret = SapModel.AreaObj.AddByCoord(6, x, y, z, Name)
```

Introduction

```
'refresh view
    ret = SapModel.View.RefreshView(0, False)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

🔍 See Also

Reference

cAreaObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST5AC18E56_0") Method

Adds a new area object whose defining points are specified by name.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int AddByPoint(
    int NumberPoints,
    ref string[] Point,
    ref string Name,
    string PropName = "Default",
    string UserName = ""
)
```

```
Function AddByPoint (
    NumberPoints As Integer,
    ByRef Point As String(),
    ByRef Name As String,
    PropName As String = "Default",
    UserName As String = ""
) As Integer
```

```
Dim instance As cAreaObj
Dim NumberPoints As Integer
Dim Point As String()
Dim Name As String
Dim PropName As String
Dim UserName As String
Dim returnValue As Integer
```

```
returnValue = instance.AddByPoint(NumberPoints,
    Point, Name, PropName, UserName)
```

```
int AddByPoint(
    int NumberPoints,
    array<String^>^% Point,
    String^% Name,
    String^ PropName = L"Default",
    String^ UserName = L""
)
```

```
abstract AddByPoint :
    NumberPoints : int *
    Point : string[] byref *
    Name : string byref *
    ?PropName : string *
    ?UserName : string
```

cAreaObjid="LST5AC18E56_0"AddLanguageSpecificTextSet("LST5AC18E56_0?cpp=::|nu380);AddBy


```
(* Defaults:
    let _PropName = defaultArg PropName "Default"
    let _UserName = defaultArg UserName ""
*)
-> int
```

Parameters

NumberPoints

Type: [SystemInt32](#)

The number of points in the area object.

Point

Type: [SystemString](#)

This is an array containing the names of the point objects that define the added area object. The point object names should be ordered to run clockwise or counter clockwise around the area object.

Name

Type: [SystemString](#)

This is the name that the program ultimately assigns to the area object. If no userName is specified, the program assigns a default name to the area object. If a userName is specified and that name is not used for another area object, the userName is assigned to the area object; otherwise a default name is assigned to the area object.

PropName (Optional)

Type: [SystemString](#)

This is Default, None or the name of a defined area property.

If it is Default, the program assigns a default area property to the area object. If it is None, no area property is assigned to the area object. If it is the name of a defined area property, that property is assigned to the area object.

UserName (Optional)

Type: [SystemString](#)

Return Value

Type: [Int32](#)

Returns zero if the area object is successfully added; otherwise it returns a nonzero value.

Remarks

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Point() As String
    Dim Name As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
```

Introduction

```
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'add area object by points
Redim Point(3)
Point(0) = "8"
Point(1) = "2"
Point(2) = "65"
Point(3) = "68"
ret = SapModel.AreaObj.AddByPoint(4, Point, Name)

'refresh view
ret = SapModel.View.RefreshView(0, False)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

📌 See Also

Reference

[cAreaObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST882ABAAD_0?cpp=::|n883");Chang Method

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int ChangeName(
    string Name,
    string NewName
)
```

```
Function ChangeName (
    Name As String,
    NewName As String
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim NewName As String
Dim returnValue As Integer
```

```
returnValue = instance.ChangeName(Name,
    NewName)
```

```
int ChangeName(
    String^ Name,
    String^ NewName
)
```

```
abstract ChangeName :
    Name : string *
    NewName : string -> int
```

Parameters

Name

Type: [SystemString](#)

NewName

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

cAreaObjspan id="LST882ABAAD_0"AddLanguageSpecificTextSet("LST882ABAAD_0?cpp=::|n883");Chang

Reference

cAreaObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST86656344_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

int Count()

Function Count As Integer

```
Dim instance As cAreaObj
Dim returnValue As Integer
```

```
returnValue = instance.Count()
```

```
int Count()
```

```
abstract Count : unit -> int
```

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cAreaObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST1B61C519_0")

Method

Deletes area objects.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int Delete(
    string Name,
    eItemType ItemType = eItemType.Objects
)
```

```
Function Delete (
    Name As String,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.Delete(Name, ItemType)
```

```
int Delete(
    String^ Name,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract Delete :
    Name : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing area object or group, depending on the value of the ItemType item.

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

cAreaObjid="LST1B61C519_0"AddLanguageSpecificTextSet("LST1B61C519_0?cpp=::|nu386;Delete

- ◇ Object = 0
- ◇ Group = 1
- ◇ SelectedObjects = 2

If this item is Objects, the area object specified by the Name item is deleted.

If this item is Group, all of the area objects in the group specified by the Name item are deleted.

If this item is SelectedObjects, all of the selected area objects are deleted, and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the area objects are successfully deleted; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'delete area object
    ret = SapModel.AreaObj.Delete("2")

    'refresh view
    ret = SapModel.View.RefreshView(0, False)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cAreaObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST56CD51F7") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int DeleteLoadTemperature(
    string Name,
    string LoadPat,
    eItemType ItemType = eItemType.Objects
)
```

```
Function DeleteLoadTemperature (
    Name As String,
    LoadPat As String,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim LoadPat As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.DeleteLoadTemperature(Name,
    LoadPat, ItemType)
```

```
int DeleteLoadTemperature(
    String^ Name,
    String^ LoadPat,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract DeleteLoadTemperature :
    Name : string *
    LoadPat : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: [SystemString](#)

LoadPat

cAreaObjspan id="LST56CD51F7_0"AddLanguageSpecificTextSet("LST56CD51F7_0?cpp=::|nu389;Delete

Type: SystemString
ItemType (Optional)
Type: ETABSv1eItemType

Return Value

Type: Int32

See Also

Reference

cAreaObj Interface

ETABSv1 Namespace

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»

CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LSTABAC581B_0?cpp=::|n091");Delete

Method

Deletes the uniform load assignments to the specified area objects for the specified load pattern.

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int DeleteLoadUniform(
    string Name,
    string LoadPat,
    eItemType ItemType = eItemType.Objects
)
```

```
Function DeleteLoadUniform (
    Name As String,
    LoadPat As String,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim LoadPat As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.DeleteLoadUniform(Name,
    LoadPat, ItemType)
```

```
int DeleteLoadUniform(
    String^ Name,
    String^ LoadPat,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract DeleteLoadUniform :
    Name : string *
    LoadPat : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

cAreaObjid="LSTABAC581B_0"AddLanguageSpecificTextSet("LSTABAC581B_0?cpp=::|n091");Delete

Parameters

Name

Type: SystemString

The name of an existing area object or group, depending on the value of the ItemType item.

LoadPat

Type: SystemString

The name of a defined load pattern.

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the load assignments are deleted for the area object specified by the Name item.

If this item is Group, the load assignments are deleted for all area objects in the group specified by the Name item.

If this item is SelectedObjects, the load assignments are deleted for all selected area objects, and the Name item is ignored.

Return ValueType: Int32

Returns zero if the load assignments are successfully deleted, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'assign area object uniform loads
```

Introduction

```
ret = SapModel.AreaObj.SetLoadUniform("ALL", "DEAD", -0.01, 2, False, "Local", eItemType.Group)

'delete area object uniform load
ret = SapModel.AreaObj.DeleteLoadUniform("3", "DEAD")

'refresh view
ret = SapModel.View.RefreshView(0, False)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

🔗 See Also

Reference

[cAreaObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LSTE75E73E6_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int DeleteLoadUniformToFrame(
    string Name,
    string LoadPat,
    eItemType ItemType = eItemType.Objects
)
```

```
Function DeleteLoadUniformToFrame (
    Name As String,
    LoadPat As String,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim LoadPat As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.DeleteLoadUniformToFrame(Name,
    LoadPat, ItemType)
```

```
int DeleteLoadUniformToFrame(
    String^ Name,
    String^ LoadPat,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract DeleteLoadUniformToFrame :
    Name : string *
    LoadPat : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: [SystemString](#)

LoadPat

cAreaObjspan id="LSTE75E73E6_0"AddLanguageSpecificTextSet("LSTE75E73E6_0?cpp=::|nu394;Delete

Type: SystemString
ItemType (Optional)
Type: ETABSv1eItemType

Return Value

Type: Int32

See Also

Reference

cAreaObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST7C493849_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int DeleteLoadWindPressure(
    string Name,
    string LoadPat,
    eItemType ItemType = eItemType.Objects
)
```

```
Function DeleteLoadWindPressure (
    Name As String,
    LoadPat As String,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim LoadPat As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.DeleteLoadWindPressure(Name,
    LoadPat, ItemType)
```

```
int DeleteLoadWindPressure(
    String^ Name,
    String^ LoadPat,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract DeleteLoadWindPressure :
    Name : string *
    LoadPat : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: [SystemString](#)

LoadPat

cAreaObjspan id="LST7C493849_0"AddLanguageSpecificTextSet("LST7C493849_0?cpp=::|nu=396DeleteL

Type: SystemString
ItemType (Optional)
Type: ETABSv1eItemType

Return Value

Type: Int32

See Also

Reference

cAreaObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LSTA231C24D Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int DeleteMass(
    string Name,
    eItemType ItemType = eItemType.Objects
)
```

```
Function DeleteMass (
    Name As String,
    Optional ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.DeleteMass(Name,
    ItemType)
```

```
int DeleteMass(
    String^ Name,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract DeleteMass :
    Name : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: [SystemString](#)

ItemType (Optional)

Type: [ETABSv1eItemType](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cAreaObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LSTC84B1F75_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int DeleteModifiers(
    string Name,
    eItemType ItemType = eItemType.Objects
)
```

```
Function DeleteModifiers (
    Name As String,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.DeleteModifiers(Name,
    ItemType)
```

```
int DeleteModifiers(
    String^ Name,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract DeleteModifiers :
    Name : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: [SystemString](#)

ItemType (Optional)

Type: [ETABSv1eItemType](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cAreaObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LSTC3711729_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int DeleteSpring(
    string Name,
    eItemType ItemType = eItemType.Objects
)
```

```
Function DeleteSpring (
    Name As String,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.DeleteSpring(Name,
    ItemType)
```

```
int DeleteSpring(
    String^ Name,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract DeleteSpring :
    Name : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: [SystemString](#)

ItemType (Optional)

Type: [ETABSv1eItemType](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cAreaObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST834E121F_0") Method

Retrieves select data for all area objects in the model

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetAllAreas(
    ref int NumberNames,
    ref string[] MyName,
    ref eAreaDesignOrientation[] DesignOrientation,
    ref int NumberBoundaryPts,
    ref int[] PointDelimiter,
    ref string[] PointNames,
    ref double[] PointX,
    ref double[] PointY,
    ref double[] PointZ
)
```

```
Function GetAllAreas (
    ByRef NumberNames As Integer,
    ByRef MyName As String(),
    ByRef DesignOrientation As eAreaDesignOrientation(),
    ByRef NumberBoundaryPts As Integer,
    ByRef PointDelimiter As Integer(),
    ByRef PointNames As String(),
    ByRef PointX As Double(),
    ByRef PointY As Double(),
    ByRef PointZ As Double()
) As Integer
```

```
Dim instance As cAreaObj
Dim NumberNames As Integer
Dim MyName As String()
Dim DesignOrientation As eAreaDesignOrientation()
Dim NumberBoundaryPts As Integer
Dim PointDelimiter As Integer()
Dim PointNames As String()
Dim PointX As Double()
Dim PointY As Double()
Dim PointZ As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetAllAreas(NumberNames,
    MyName, DesignOrientation, NumberBoundaryPts,
    PointDelimiter, PointNames, PointX,
    PointY, PointZ)
```



```

int GetAllAreas(
int% NumberNames,
array<String^>^% MyName,
array<eAreaDesignOrientation>^% DesignOrientation,
int% NumberBoundaryPts,
array<int>^% PointDelimiter,
array<String^>^% PointNames,
array<double>^% PointX,
array<double>^% PointY,
array<double>^% PointZ
)

abstract GetAllAreas :
    NumberNames : int byref *
    MyName : string[] byref *
    DesignOrientation : eAreaDesignOrientation[] byref *
    NumberBoundaryPts : int byref *
    PointDelimiter : int[] byref *
    PointNames : string[] byref *
    PointX : float[] byref *
    PointY : float[] byref *
    PointZ : float[] byref -> int

```

Parameters

NumberNames
Type:Â [SystemInt32](#)

MyName
Type:Â [SystemString](#)

DesignOrientation
Type:Â [ETABSv1eAreaDesignOrientation](#)

NumberBoundaryPts
Type:Â [SystemInt32](#)

PointDelimiter
Type:Â [SystemInt32](#)

PointNames
Type:Â [SystemString](#)

PointX
Type:Â [SystemDouble](#)

PointY
Type:Â [SystemDouble](#)

PointZ
Type:Â [SystemDouble](#)

Return Value

Type:Â [Int32](#)

▲ Remarks

▲ See Also

Reference

[cAreaObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST479396FD_0") Method

Retrieves curve data for all edges of an area object

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetCurvedEdges(
    string Name,
    ref int NumEdges,
    ref int[] CurveType,
    ref double[] Tension,
    ref int[] NumPoints,
    ref double[] gx,
    ref double[] gy,
    ref double[] gz
)
```

```
Function GetCurvedEdges (
    Name As String,
    ByRef NumEdges As Integer,
    ByRef CurveType As Integer(),
    ByRef Tension As Double(),
    ByRef NumPoints As Integer(),
    ByRef gx As Double(),
    ByRef gy As Double(),
    ByRef gz As Double()
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim NumEdges As Integer
Dim CurveType As Integer()
Dim Tension As Double()
Dim NumPoints As Integer()
Dim gx As Double()
Dim gy As Double()
Dim gz As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetCurvedEdges(Name,
    NumEdges, CurveType, Tension, NumPoints,
    gx, gy, gz)
```

```
int GetCurvedEdges(
    String^ Name,
    int% NumEdges,
    array<int>^% CurveType,
```

cAreaObjspan id="LST479396FD_0"AddLanguageSpecificTextSet("LST479396FD_0?cpp=::|nu-407;GetCur

```

array<double>^% Tension,
array<int>^% NumPoints,
array<double>^% gx,
array<double>^% gy,
array<double>^% gz
)

abstract GetCurvedEdges :
    Name : string *
    NumEdges : int byref *
    CurveType : int[] byref *
    Tension : float[] byref *
    NumPoints : int[] byref *
    gx : float[] byref *
    gy : float[] byref *
    gz : float[] byref -> int

```

Parameters

Name

Type:Â SystemString

The name of an existing area object with curved edges

NumEdges

Type:Â SystemInt32

The number of edges of the area object

CurveType

Type:Â SystemInt32

This is an array of size NumEdges. Contains the curve type for each edge, an integer from the following list:

- ◊ Straight = 0
- ◊ Circular Curve = 1
- ◊ Multilinear Curve = 2
- ◊ Bezier Curve = 3
- ◊ Spline Curve = 4

Tension

Type:Â SystemDouble

This is an array of size NumEdges. Used for spline, a value that specifies the amount that the curve bends between control points. Values greater than 1 produce unpredictable results

NumPoints

Type:Â SystemInt32

This is an array of size NumEdges. The number of points along the curve for each edge

gx

Type:Â SystemDouble

This array has a size equal to the sum of all the items in the NumPoints array. Contains the global x coordinates Use the values in the NumPoints array to keep track of which points apply to which curve.

gy

Type:Â SystemDouble

This array has a size equal to the sum of all the items in the NumPoints array. Contains the global y coordinates Use the values in the NumPoints array to keep track of which points apply to which curve.

gz

Type: SystemDouble

This array has a size equal to the sum of all the items in the NumPoints array. Contains the global z coordinates Use the values in the NumPoints array to keep track of which points apply to which curve.

Return Value

Type: Int32

Returns zero if the data is successfully retrieved, otherwise it returns a nonzero value

▲ See Also

Reference

cAreaObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LSTEE272340_0")

Method

Retrieves the design orientation of an area object.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetDesignOrientation(
    string Name,
    ref eAreaDesignOrientation DesignOrientation
)
```

```
Function GetDesignOrientation (
    Name As String,
    ByRef DesignOrientation As eAreaDesignOrientation
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim DesignOrientation As eAreaDesignOrientation
Dim returnValue As Integer
```

```
returnValue = instance.GetDesignOrientation(Name,
    DesignOrientation)
```

```
int GetDesignOrientation(
    String^ Name,
    eAreaDesignOrientation% DesignOrientation
)
```

```
abstract GetDesignOrientation :
    Name : string *
    DesignOrientation : eAreaDesignOrientation byref -> int
```

Parameters

Name

Type: SystemString

The name of a defined area object.

DesignOrientation

Type: ETABSv1eAreaDesignOrientation

This is one of the items in the eAreaDesignOrientation enumeration.

Return ValueType: Int32

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim DesignOrientation As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'get area property
    ret = SapModel.AreaObj.GetDesignOrientation("1", DesignOrientation)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

ReferencecAreaObj InterfaceETABSv1 Namespace

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST9F3A69F0_0") Method

Retrieves the diaphragm assignment to the specified area object

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetDiaphragm(
    string Name,
    ref string DiaphragmName
)
```

```
Function GetDiaphragm (
    Name As String,
    ByRef DiaphragmName As String
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim DiaphragmName As String
Dim returnValue As Integer
```

```
returnValue = instance.GetDiaphragm(Name,
    DiaphragmName)
```

```
int GetDiaphragm(
    String^ Name,
    String^% DiaphragmName
)
```

```
abstract GetDiaphragm :
    Name : string *
    DiaphragmName : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of a defined area object.

DiaphragmName

Type: [SystemString](#)

The name of the diaphragm assigned to the area object. This item is "None" if no diaphragm is assigned to the area object.

Return ValueType: Int32

Returns zero if the diaphragm is successfully retrieved; otherwise it returns a nonzero value.

▀ **Remarks**

▀ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim DiaphragmName As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 4, 4, 24, 24)

    'define a new diaphragm
    ret = SapModel.Diaphragm.SetDiaphragm("MyDiaph1A", SemiRigid:=True)

    'assign diaphragm to area
    ret = SapModel.AreaObj.SetDiaphragm("4", "MyDiaph1A")

    'get area diaphragm assignment
    ret = SapModel.AreaObj.GetDiaphragm("4", DiaphragmName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▀ **See Also**

Reference

cAreaObj Interface

ETABSv1 Namespace

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Introduction

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LSTBEC71E6C0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetEdgeConstraint(
    string Name,
    ref bool ConstraintExists
)
```

```
Function GetEdgeConstraint (
    Name As String,
    ByRef ConstraintExists As Boolean
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim ConstraintExists As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetEdgeConstraint(Name,
    ConstraintExists)
```

```
int GetEdgeConstraint(
    String^ Name,
    bool% ConstraintExists
)
```

```
abstract GetEdgeConstraint :
    Name : string *
    ConstraintExists : bool byref -> int
```

Parameters

Name

Type: [SystemString](#)

ConstraintExists

Type: [SystemBoolean](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cAreaObj Interface

ETABSv1 Namespace

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»

CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LSTF8B39EA2_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetElm(
    string Name,
    ref int NElm,
    ref string[] Elm
)
```

```
Function GetElm (
    Name As String,
    ByRef NElm As Integer,
    ByRef Elm As String()
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim NElm As Integer
Dim Elm As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetElm(Name, NElm,
    Elm)
```

```
int GetElm(
    String^ Name,
    int% NElm,
    array<String^>^% Elm
)
```

```
abstract GetElm :
    Name : string *
    NElm : int byref *
    Elm : string[] byref -> int
```

Parameters

Name

Type: SystemString

NElm

Type: SystemInt32

Elm

Type: SystemString

cAreaObjspan id="LSTF8B39EA2_0"AddLanguageSpecificTextSet("LSTF8B39EA2_0?cpp=::|nu417);GetElm

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cAreaObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST6BE04EDC_0") Method

Retrieves the groups to which an area object is assigned.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetGroupAssign(
    string Name,
    ref int NumberGroups,
    ref string[] Groups
)
```

```
Function GetGroupAssign (
    Name As String,
    ByRef NumberGroups As Integer,
    ByRef Groups As String()
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim NumberGroups As Integer
Dim Groups As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetGroupAssign(Name,
    NumberGroups, Groups)
```

```
int GetGroupAssign(
    String^ Name,
    int% NumberGroups,
    array<String^>^% Groups
)
```

```
abstract GetGroupAssign :
    Name : string *
    NumberGroups : int byref *
    Groups : string[] byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing area object.

NumberGroups

cAreaObjspan id="LST6BE04EDC_0"AddLanguageSpecificTextSet("LST6BE04EDC_0?cpp=::|n=19");GetG

Type: SystemInt32

The number of group names retrieved.

Groups

Type: SystemString

This is an array of the names of the groups to which the area object is assigned.

Return Value

Type: Int32

Returns zero if the group assignments are successfully retrieved, otherwise it returns a nonzero value.

Remarks

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberGroups As Integer
    Dim Groups() as String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new groups
    ret = SapModel.GroupDef.SetGroup("Group1")
    ret = SapModel.GroupDef.SetGroup("Group2")

    'add area object to groups
    ret = SapModel.AreaObj.SetGroupAssign("1", "Group1")
    ret = SapModel.AreaObj.SetGroupAssign("1", "Group2")

    'get area object groups
    ret = SapModel.AreaObj.GetGroupAssign("1", NumberGroups, Groups)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

cAreaObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST4B1EBFFD0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetGUID(
    string Name,
    ref string GUID
)
```

```
Function GetGUID (
    Name As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.GetGUID(Name, GUID)
```

```
int GetGUID(
    String^ Name,
    String^% GUID
)
```

```
abstract GetGUID :
    Name : string *
    GUID : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

GUID

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cAreaObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST552DAB56 Method

Retrieves the label and story for a unique area object name

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetLabelFromName(
    string Name,
    ref string Label,
    ref string Story
)
```

```
Function GetLabelFromName (
    Name As String,
    ByRef Label As String,
    ByRef Story As String
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim Label As String
Dim Story As String
Dim returnValue As Integer
```

```
returnValue = instance.GetLabelFromName(Name,
    Label, Story)
```

```
int GetLabelFromName(
    String^ Name,
    String^% Label,
    String^% Story
)
```

```
abstract GetLabelFromName :
    Name : string *
    Label : string byref *
    Story : string byref -> int
```

Parameters

Name

Type: SystemString

The name of the area object

Label

Type:Â SystemString
 The area object label

Story

Type:Â SystemString
 The area object story level

Return Value

Type:Â Int32
 Returns zero if the data is successfully retrieved, otherwise it returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberNames As Integer
    Dim Label As String
    Dim Story As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'get area object data
    ret = SapModel.AreaObj.GetLabelFromName("3", Label, Story)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cAreaObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LSTDBD2DBC Method

Retrieves the names and labels of all defined area objects.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetLabelNameList(
    ref int NumberNames,
    ref string[] MyName,
    ref string[] MyLabel,
    ref string[] MyStory
)
```

```
Function GetLabelNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String(),
    ByRef MyLabel As String(),
    ByRef MyStory As String()
) As Integer
```

```
Dim instance As cAreaObj
Dim NumberNames As Integer
Dim MyName As String()
Dim MyLabel As String()
Dim MyStory As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetLabelNameList(NumberNames,
    MyName, MyLabel, MyStory)
```

```
int GetLabelNameList(
    int% NumberNames,
    array<String^>^% MyName,
    array<String^>^% MyLabel,
    array<String^>^% MyStory
)
```

```
abstract GetLabelNameList :
    NumberNames : int byref *
    MyName : string[] byref *
    MyLabel : string[] byref *
    MyStory : string[] byref -> int
```

Parameters**NumberNames**Type: SystemInt32

The number of area object names retrieved by the program.

MyNameType: SystemString

This is a one-dimensional array of area object names. The MyName array is created as a dynamic, zero-based, array by the API user:

```
Dim MyName() as String
```

The array is dimensioned to (NumberNames + 1) inside the program, filled with the names, and returned to the API user.

MyLabelType: SystemString

This is a one-dimensional array of area object labels. The MyLabel array is created as a dynamic, zero-based, array by the API user:

```
Dim MyLabel() as String
```

The array is dimensioned to (NumberNames + 1) inside the program, filled with the labels, and returned to the API user.

MyStoryType: SystemString

This is a one-dimensional array of the story levels of the area objects. The MyStory array is created as a dynamic, zero-based, array by the API user:

```
Dim MyStory() as String
```

The array is dimensioned to (NumberNames + 1) inside the program, filled with the story levels, and returned to the API user.

Return ValueType: Int32

Returns zero if the data is successfully retrieved, otherwise it returns nonzero.

▲ **Remarks**▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberNames As Integer
    Dim MyName() As String
    Dim MyLabel() As String
    Dim MyStory() As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```



```
'start ETABS application
  ret = EtabsObject.ApplicationStart()

'create SapModel object
  SapModel = EtabsObject.SapModel

'initialize model
  ret = SapModel.InitializeNewModel()

'create steel deck template model
  ret = SapModel.File.NewSteelDeck(4, 12, 4, 4, 24, 24)

'get area object data
  ret = SapModel.AreaObj.GetLabelNameList(NumberNames, MyName, MyLabel, MyStory)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
```

End Sub

▲ See Also

Reference

[cAreaObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LSTA737EC5C0") Method

Retrieves the temperature load assignments to area objects.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetLoadTemperature(
    string Name,
    ref int NumberItems,
    ref string[] AreaName,
    ref string[] LoadPat,
    ref int[] MyType,
    ref double[] Value,
    ref string[] PatternName,
    eItemType ItemType = eItemType.Objects
)
```

```
Function GetLoadTemperature (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef AreaName As String(),
    ByRef LoadPat As String(),
    ByRef MyType As Integer(),
    ByRef Value As Double(),
    ByRef PatternName As String(),
    ByRef ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim NumberItems As Integer
Dim AreaName As String()
Dim LoadPat As String()
Dim MyType As Integer()
Dim Value As Double()
Dim PatternName As String()
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.GetLoadTemperature(Name,
    NumberItems, AreaName, LoadPat, MyType,
    Value, PatternName, ItemType)
```

```
int GetLoadTemperature(
    String^ Name,
    int% NumberItems,
    array<String^>^% AreaName,
```

cAreaObjspan id="LSTA737EC5C_0"AddLanguageSpecificTextSet("LSTA737EC5C_0?cpp=::|n430");GetLo

```

array<String>^% LoadPat,
array<int>^% MyType,
array<double>^% Value,
array<String>^% PatternName,
eItemType ItemType = eItemType::Objects
)

abstract GetLoadTemperature :
    Name : string *
    NumberItems : int byref *
    AreaName : string[] byref *
    LoadPat : string[] byref *
    MyType : int[] byref *
    Value : float[] byref *
    PatternName : string[] byref *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of an existing area object or group, depending on the value of the ItemType item.

NumberItems

Type:Â SystemInt32

The total number of temperature loads retrieved for the specified area objects.

AreaName

Type:Â SystemString

This is an array that includes the name of the area object associated with each temperature load.

LoadPat

Type:Â SystemString

This is an array that includes the name of the load pattern associated with each temperature load.

MyType

Type:Â SystemInt32

This is an array that includes either 1 or 3, indicating the type of temperature load.

◊ 1 = Temperature

◊ 3 = Temperature gradient along local 3 axis

Value

Type:Â SystemDouble

This is an array that includes the temperature load value. [T] for MyType = 1 and [T/L] for MyType= 3.

PatternName

Type:Â SystemString

This is an array that includes the joint pattern name, if any, used to specify the temperature load.

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the assignments are retrieved for the area object specified by the Name item.

If this item is Group, the assignments are retrieved for all area objects in the group specified by the Name item.

If this item is SelectedObjects, assignments are retrieved for is made to all selected area objects, and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the load assignments are successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ See Also

Reference

cAreaObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST791F4BC1_0")

Method

Retrieves the uniform load assignments to area objects.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetLoadUniform(
    string Name,
    ref int NumberItems,
    ref string[] AreaName,
    ref string[] LoadPat,
    ref string[] CSys,
    ref int[] Dir,
    ref double[] Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function GetLoadUniform (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef AreaName As String(),
    ByRef LoadPat As String(),
    ByRef CSys As String(),
    ByRef Dir As Integer(),
    ByRef Value As Double(),
    ByRef ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim NumberItems As Integer
Dim AreaName As String()
Dim LoadPat As String()
Dim CSys As String()
Dim Dir As Integer()
Dim Value As Double()
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.GetLoadUniform(Name,
    NumberItems, AreaName, LoadPat, CSys,
    Dir, Value, ItemType)
```

```
int GetLoadUniform(
    String^ Name,
    int% NumberItems,
    array<String^>^% AreaName,
```

cAreaObjspan id="LST791F4BC1_0"AddLanguageSpecificTextSet("LST791F4BC1_0?cpp=::|nu433;GetLoa

```

array<String>^% LoadPat,
array<String>^% CSys,
array<int>^% Dir,
array<double>^% Value,
eItemType ItemType = eItemType::Objects
)

abstract GetLoadUniform :
    Name : string *
    NumberItems : int byref *
    AreaName : string[] byref *
    LoadPat : string[] byref *
    CSys : string[] byref *
    Dir : int[] byref *
    Value : float[] byref *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of an existing area object or group, depending on the value of the ItemType item.

NumberItems

Type:Â SystemInt32

The total number of uniform loads retrieved for the specified area objects.

AreaName

Type:Â SystemString

This is an array that includes the name of the area object associated with each uniform load.

LoadPat

Type:Â SystemString

This is an array that includes the name of the load pattern associated with each uniform load.

CSys

Type:Â SystemString

This is an array that includes the name of the coordinate system associated with each uniform load.

Dir

Type:Â SystemInt32

Value

Type:Â SystemDouble

The uniform load value. [F/L²]

ItemType (Optional)

Type:Â ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

- ◊ Object = 0
- ◊ Group = 1
- ◊ SelectedObjects = 2

If this item is Objects, the assignments are retrieved for the area object specified by the Name item.

If this item is Group, the assignments are retrieved for all area objects in the group specified by the Name item.

If this item is SelectedObjects, assignments are retrieved for is made to all selected area objects, and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the load assignments are successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberItems As Integer
    Dim AreaName() As String
    Dim LoadPat() As String
    Dim CSys() As String
    Dim Dir() As Integer
    Dim Value() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'assign area object uniform loads
    ret = SapModel.AreaObj.SetLoadUniform("ALL", "DEAD", -0.01, 2, False, "Local", eItemType.Group)

    'get area object uniform load
    ret = SapModel.AreaObj.GetLoadUniform("3", NumberItems, AreaName, LoadPat, CSys, Dir, Value)

    'refresh view
    ret = SapModel.View.RefreshView(0, False)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
```

```
SapModel = Nothing  
EtabsObject = Nothing
```

End Sub

▲ See Also

Reference

cAreaObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST8D67DF34_0") Method

Retrieves the uniform to frame load assignments to area objects.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetLoadUniformToFrame(
    string Name,
    ref int NumberItems,
    ref string[] AreaName,
    ref string[] LoadPat,
    ref string[] CSys,
    ref int[] Dir,
    ref double[] Value,
    ref int[] DistType,
    eItemType ItemType = eItemType.Objects
)
```

```
Function GetLoadUniformToFrame (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef AreaName As String(),
    ByRef LoadPat As String(),
    ByRef CSys As String(),
    ByRef Dir As Integer(),
    ByRef Value As Double(),
    ByRef DistType As Integer(),
    ByRef ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim NumberItems As Integer
Dim AreaName As String()
Dim LoadPat As String()
Dim CSys As String()
Dim Dir As Integer()
Dim Value As Double()
Dim DistType As Integer()
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.GetLoadUniformToFrame(Name,
    NumberItems, AreaName, LoadPat, CSys,
    Dir, Value, DistType, ItemType)
```

```
int GetLoadUniformToFrame(
```

cAreaObjid="LST8D67DF34_0"AddLanguageSpecificTextSet("LST8D67DF34_0?cpp=::|nu437);GetLoa

```

String^ Name,
int% NumberItems,
array<String^>^% AreaName,
array<String^>^% LoadPat,
array<String^>^% CSys,
array<int>^% Dir,
array<double>^% Value,
array<int>^% DistType,
eItemType ItemType = eItemType::Objects
)

abstract GetLoadUniformToFrame :
    Name : string *
    NumberItems : int byref *
    AreaName : string[] byref *
    LoadPat : string[] byref *
    CSys : string[] byref *
    Dir : int[] byref *
    Value : float[] byref *
    DistType : int[] byref *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of an existing area object or group, depending on the value of the ItemType item.

NumberItems

Type:Â SystemInt32

The total number of uniform to frame loads retrieved for the specified area objects.

AreaName

Type:Â SystemString

This is an array that includes the name of the area object associated with each uniform to frame load.

LoadPat

Type:Â SystemString

This is an array that includes the name of the load pattern associated with each uniform to frame load.

CSys

Type:Â SystemString

This is an array that includes the name of the coordinate system associated with each uniform to frame load.

Dir

Type:Â SystemInt32

Value

Type:Â SystemDouble

The uniform load value. [F/L²]

DistType

Type: SystemInt32

This is either 1 or 2, indicating the load distribution type.

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the assignments are retrieved for the area object specified by the Name item.

If this item is Group, the assignments are retrieved for all area objects in the group specified by the Name item.

If this item is SelectedObjects, assignments are retrieved for is made to all selected area objects, and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the load assignments are successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ See Also

Reference

cAreaObj Interface

ETABSv1 Namespace

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»

CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST66E250E0_0") Method

Retrieves the wind pressure load assignments to area objects.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetLoadWindPressure(
    string Name,
    ref int NumberItems,
    ref string[] AreaName,
    ref string[] LoadPat,
    ref int[] MyType,
    ref double[] Cp,
    eItemType ItemType = eItemType.Objects
)
```

```
Function GetLoadWindPressure (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef AreaName As String(),
    ByRef LoadPat As String(),
    ByRef MyType As Integer(),
    ByRef Cp As Double(),
    ByRef ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim NumberItems As Integer
Dim AreaName As String()
Dim LoadPat As String()
Dim MyType As Integer()
Dim Cp As Double()
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.GetLoadWindPressure(Name,
    NumberItems, AreaName, LoadPat, MyType,
    Cp, ItemType)
```

```
int GetLoadWindPressure(
    String^ Name,
    int% NumberItems,
    array<String^>^% AreaName,
    array<String^>^% LoadPat,
    array<int>^% MyType,
    array<double>^% Cp,
```

cAreaObjspan id="LST66E250E0_0"AddLanguageSpecificTextSet("LST66E250E0_0?cpp=::|nu-4.4.0)GetLoa

```

eItemType ItemType = eItemType::Objects
)

abstract GetLoadWindPressure :
    Name : string *
    NumberItems : int byref *
    AreaName : string[] byref *
    LoadPat : string[] byref *
    MyType : int[] byref *
    Cp : float[] byref *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int

```

Parameters

Name

Type: SystemString

The name of an existing area object or group, depending on the value of the ItemType item.

NumberItems

Type: SystemInt32

The total number of wind pressure loads retrieved for the specified area objects.

AreaName

Type: SystemString

This is an array that includes the name of the area object associated with each wind pressure load.

LoadPat

Type: SystemString

This is an array that includes the name of the load pattern associated with each wind pressure load.

MyType

Type: SystemInt32

This is an array that includes either 1 or 2, indicating the wind pressure type.

1. Windward, pressure varies over height
2. Other, pressure is constant over height

Cp

Type: SystemDouble

This is an array that includes the wind pressure coefficient value.

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

- ◊ Object = 0
- ◊ Group = 1
- ◊ SelectedObjects = 2

If this item is Objects, the assignments are retrieved for the area object specified by the Name item.

If this item is Group, the assignments are retrieved for all area objects in the group specified by the Name item.

If this item is SelectedObjects, assignments are retrieved for is made to all selected area objects, and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the load assignments are successfully retrieved; it returns a nonzero value.

▲ Remarks

▲ See Also

Reference

cAreaObj Interface

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST9D1165AB Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetLocalAxes(
    string Name,
    ref double Ang,
    ref bool Advanced
)
```

```
Function GetLocalAxes (
    Name As String,
    ByRef Ang As Double,
    ByRef Advanced As Boolean
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim Ang As Double
Dim Advanced As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetLocalAxes(Name,
    Ang, Advanced)
```

```
int GetLocalAxes(
    String^ Name,
    double% Ang,
    bool% Advanced
)
```

```
abstract GetLocalAxes :
    Name : string *
    Ang : float byref *
    Advanced : bool byref -> int
```

Parameters

Name

Type: SystemString

Ang

Type: SystemDouble

Advanced

Type: SystemBoolean

cAreaObjspan id="LST9D1165AB_0"AddLanguageSpecificTextSet("LST9D1165AB_0?cpp=::|nu443;GetLoc

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cAreaObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST96155EE_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetMass(
    string Name,
    ref double MassOverL2
)
```

```
Function GetMass (
    Name As String,
    ByRef MassOverL2 As Double
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim MassOverL2 As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetMass(Name, MassOverL2)
```

```
int GetMass(
    String^ Name,
    double% MassOverL2
)
```

```
abstract GetMass :
    Name : string *
    MassOverL2 : float byref -> int
```

Parameters

Name

Type: [SystemString](#)

MassOverL2

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cAreaObj Interface

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST606A546E_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetMaterialOverwrite(
    string Name,
    ref string PropName
)
```

```
Function GetMaterialOverwrite (
    Name As String,
    ByRef PropName As String
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim PropName As String
Dim returnValue As Integer
```

```
returnValue = instance.GetMaterialOverwrite(Name,
    PropName)
```

```
int GetMaterialOverwrite(
    String^ Name,
    String^% PropName
)
```

```
abstract GetMaterialOverwrite :
    Name : string *
    PropName : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

PropName

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cAreaObj Interface

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LSTF51E0B32_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetModifiers(
    string Name,
    ref double[] Value
)
```

```
Function GetModifiers (
    Name As String,
    ByRef Value As Double()
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim Value As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetModifiers(Name,
    Value)
```

```
int GetModifiers(
    String^ Name,
    array<double>^% Value
)
```

```
abstract GetModifiers :
    Name : string *
    Value : float[] byref -> int
```

Parameters

Name

Type: [SystemString](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cAreaObj Interface

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST71E54D8C Method

Retrieves the unique name of an area object, given the label and story level

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetNameFromLabel(
    string Label,
    string Story,
    ref string Name
)
```

```
Function GetNameFromLabel (
    Label As String,
    Story As String,
    ByRef Name As String
) As Integer
```

```
Dim instance As cAreaObj
Dim Label As String
Dim Story As String
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.GetNameFromLabel(Label,
    Story, Name)
```

```
int GetNameFromLabel(
    String^ Label,
    String^ Story,
    String^% Name
)
```

```
abstract GetNameFromLabel :
    Label : string *
    Story : string *
    Name : string byref -> int
```

Parameters

Label

Type: SystemString

The area object label

Story

Type: SystemString

The area object story level

Name

Type: SystemString

The unique name of the area object

Return Value

Type: Int32

Returns zero if the data is successfully retrieved, otherwise it returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberNames As Integer
    Dim Name As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'get area object data
    ret = SapModel.AreaObj.GetNameFromLabel("F1", "Story4", Name)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cAreaObj Interface

ETABSV1 Namespace

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LSTC65908A8_0") Method

Retrieves the names of all defined area objects.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNameList(
    ref int NumberNames,
    ref string[] MyName
)
```

```
Function GetNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cAreaObj
Dim NumberNames As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetNameList(NumberNames,
    MyName)
```

```
int GetNameList(
    int% NumberNames,
    array<String^>^% MyName
)
```

```
abstract GetNameList :
    NumberNames : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberNames

Type: [System.Int32](#)

The number of area object names retrieved by the program.

MyName

Type: [System.String](#)

This is a one-dimensional array of area object names. The MyName array is created as a dynamic, zero-based, array by the API user:

```
Dim MyName() as String
```

cAreaObjAddLanguageSpecificTextSet("LSTC65908A8_0?cpp=::|nu454;GetNar

The array is dimensioned to (NumberNames + 1) inside the ETABS program, filled with the names, and returned to the API user.

Return Value

Type: Int32

Returns zero if the names are successfully retrieved, otherwise it returns nonzero.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberNames As Integer
    Dim MyName() As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'get area object names
    ret = SapModel.AreaObj.GetNameList(NumberNames, MyName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ **See Also**

Reference

cAreaObj Interface

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST4AFDCEAE_0")

Method

Retrieves the names of the defined area objects on a given story.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetNameListOnStory(
    string StoryName,
    ref int NumberNames,
    ref string[] MyName
)
```

```
Function GetNameListOnStory (
    StoryName As String,
    ByRef NumberNames As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cAreaObj
Dim StoryName As String
Dim NumberNames As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetNameListOnStory(StoryName,
    NumberNames, MyName)
```

```
int GetNameListOnStory(
    String^ StoryName,
    int% NumberNames,
    array<String^>^% MyName
)
```

```
abstract GetNameListOnStory :
    StoryName : string *
    NumberNames : int byref *
    MyName : string[] byref -> int
```

Parameters

StoryName

Type: SystemString

The name of an existing story.

NumberNames

cAreaObjid="LST4AFDCEAE_0"AddLanguageSpecificTextSet("LST4AFDCEAE_0?cpp=::|456.");GetN

Type: SystemInt32

The number of area object names retrieved by the program.

MyName

Type: SystemString

This is a one-dimensional array of area object names. The MyName array is created as a dynamic, zero-based, array by the API user:

```
Dim MyName() as String
```

The array is dimensioned to (NumberNames + 1) inside the ETABS program, filled with the names, and returned to the API user.

Return Value

Type: Int32

Returns zero if the names are successfully retrieved, otherwise it returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberNames As Integer
    Dim MyName() As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'get area object names
    ret = SapModel.AreaObj.GetNameListOnStory("Story2", NumberNames, MyName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cAreaObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST3B9F70C6_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetOffsets3(
    string Name,
    ref int NumberPoints,
    ref double[] Offsets
)
```

```
Function GetOffsets3 (
    Name As String,
    ByRef NumberPoints As Integer,
    ByRef Offsets As Double()
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim NumberPoints As Integer
Dim Offsets As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetOffsets3(Name,
    NumberPoints, Offsets)
```

```
int GetOffsets3(
    String^ Name,
    int% NumberPoints,
    array<double>^% Offsets
)
```

```
abstract GetOffsets3 :
    Name : string *
    NumberPoints : int byref *
    Offsets : float[] byref -> int
```

Parameters

Name

Type: [SystemString](#)

NumberPoints

Type: [SystemInt32](#)

Offsets

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cAreaObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST2770883C_0") Method

Retrieves whether the specified area object is an opening.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetOpening(
    string Name,
    ref bool IsOpening
)
```

```
Function GetOpening (
    Name As String,
    ByRef IsOpening As Boolean
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim IsOpening As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOpening(Name,
    IsOpening)
```

```
int GetOpening(
    String^ Name,
    bool% IsOpening
)
```

```
abstract GetOpening :
    Name : string *
    IsOpening : bool byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of a defined area object.

IsOpening

Type: [SystemBoolean](#)

This item is True if the specified area object is an opening, otherwise it is False.

Return Value

Type: Int32

The function returns zero if the assignment is successfully retrieved; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim IsOpening as Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'designate an area object as an opening
    ret = SapModel.AreaObj.SetOpening("3", True)

    'retrieve whether area object is an opening
    ret = SapModel.AreaObj.GetOpening("3", IsOpening)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ **See Also**

Reference

cAreaObj Interface

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST91839761_0") Method

Retrieves the pier label assignments of an area object

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPier(
    string Name,
    ref string PierName
)
```

```
Function GetPier (
    Name As String,
    ByRef PierName As String
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim PierName As String
Dim returnValue As Integer
```

```
returnValue = instance.GetPier(Name, PierName)
```

```
int GetPier(
    String^ Name,
    String^% PierName
)
```

```
abstract GetPier :
    Name : string *
    PierName : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing area object

PierName

Type: [SystemString](#)

The name of the pier assignment, if any, or "None"

Return Value

Type: Int32

Returns zero if the assignment is successfully retrieved; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim PierName as String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new pier label
    ret = SapModel.PierLabel.SetPier("MyPier")

    'set assignment
    ret = SapModel.AreaObj.SetPier("3", "MyPier")

    'get assignment
    ret = SapModel.AreaObj.GetPier("3", PierName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cAreaObj Interface

ETABSv1 Namespace

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Introduction

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST1371D4FA_0") Method

Retrieves the names of the point objects that define an area object.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPoints(
    string Name,
    ref int NumberPoints,
    ref string[] Point
)
```

```
Function GetPoints (
    Name As String,
    ByRef NumberPoints As Integer,
    ByRef Point As String()
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim NumberPoints As Integer
Dim Point As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetPoints(Name,
    NumberPoints, Point)
```

```
int GetPoints(
    String^ Name,
    int% NumberPoints,
    array<String^>^% Point
)
```

```
abstract GetPoints :
    Name : string *
    NumberPoints : int byref *
    Point : string[] byref -> int
```

Parameters

Name

Type: [System.String](#)

The name of a defined area object.

NumberPoints

Type: SystemInt32

The number of point objects that define the area object.

Point

Type: SystemString

Return Value

Type: Int32

Returns zero if the point object names are successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberPoints As Integer
    Dim point() As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'get names of points
    ret = SapModel.AreaObj.GetPoints("1", NumberPoints, Point)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cAreaObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LSTC43F8FE5_0") Method

Retrieves the area property assigned to an area object.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetProperty(
    string Name,
    ref string PropName
)
```

```
Function GetProperty (
    Name As String,
    ByRef PropName As String
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim PropName As String
Dim returnValue As Integer
```

```
returnValue = instance.GetProperty(Name,
    PropName)
```

```
int GetProperty(
    String^ Name,
    String^% PropName
)
```

```
abstract GetProperty :
    Name : string *
    PropName : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of a defined area object.

PropName

Type: [SystemString](#)

The name of the area property assigned to the area object. This item is None if no area property is assigned to the area object.

Return ValueType: Int32

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim PropName As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'get area property
    ret = SapModel.AreaObj.GetProperty("1", PropName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

ReferencecAreaObj InterfaceETABSv1 Namespace

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LSTB10CA052 Method

Retrieves rebar data for an area pier object

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetRebarDataPier(
    string Name,
    ref int NumberRebarLayers,
    ref string[] LayerID,
    ref eWallPierRebarLayerType[] LayerType,
    ref double[] ClearCover,
    ref string[] BarSizeName,
    ref double[] BarArea,
    ref double[] BarSpacing,
    ref int[] NumberBars,
    ref bool[] Confined,
    ref double[] EndZoneLength,
    ref double[] EndZoneThickness,
    ref double[] EndZoneOffset
)
```

```
Function GetRebarDataPier (
    Name As String,
    ByRef NumberRebarLayers As Integer,
    ByRef LayerID As String(),
    ByRef LayerType As eWallPierRebarLayerType(),
    ByRef ClearCover As Double(),
    ByRef BarSizeName As String(),
    ByRef BarArea As Double(),
    ByRef BarSpacing As Double(),
    ByRef NumberBars As Integer(),
    ByRef Confined As Boolean(),
    ByRef EndZoneLength As Double(),
    ByRef EndZoneThickness As Double(),
    ByRef EndZoneOffset As Double()
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim NumberRebarLayers As Integer
Dim LayerID As String()
Dim LayerType As eWallPierRebarLayerType()
Dim ClearCover As Double()
Dim BarSizeName As String()
Dim BarArea As Double()
Dim BarSpacing As Double()
```

cAreaObjspan id="LSTB10CA052_0"AddLanguageSpecificTextSet("LSTB10CA052_0?cpp=::|nu471");GetRe

```

Dim NumberBars As Integer()
Dim Confined As Boolean()
Dim EndZoneLength As Double()
Dim EndZoneThickness As Double()
Dim EndZoneOffset As Double()
Dim returnValue As Integer

returnValue = instance.GetRebarDataPier(Name,
NumberRebarLayers, LayerID, LayerType,
ClearCover, BarSizeName, BarArea,
BarSpacing, NumberBars, Confined,
EndZoneLength, EndZoneThickness,
EndZoneOffset)

int GetRebarDataPier(
String^ Name,
int% NumberRebarLayers,
array<String^>^% LayerID,
array<eWallPierRebarLayerType>^% LayerType,
array<double>^% ClearCover,
array<String^>^% BarSizeName,
array<double>^% BarArea,
array<double>^% BarSpacing,
array<int>^% NumberBars,
array<bool>^% Confined,
array<double>^% EndZoneLength,
array<double>^% EndZoneThickness,
array<double>^% EndZoneOffset
)

abstract GetRebarDataPier :
    Name : string *
    NumberRebarLayers : int byref *
    LayerID : string[] byref *
    LayerType : eWallPierRebarLayerType[] byref *
    ClearCover : float[] byref *
    BarSizeName : string[] byref *
    BarArea : float[] byref *
    BarSpacing : float[] byref *
    NumberBars : int[] byref *
    Confined : bool[] byref *
    EndZoneLength : float[] byref *
    EndZoneThickness : float[] byref *
    EndZoneOffset : float[] byref -> int

```

Parameters

Name

Type:Â SystemString

The name of an existing area object

NumberRebarLayers

Type:Â SystemInt32

LayerID

Type:Â SystemString

LayerType

Type:Â ETABSv1eWallPierRebarLayerType

ClearCover

Type:Â SystemDouble

BarSizeName

Type:Â SystemString

BarArea

Type:Â SystemDouble

BarSpacing

Type:Â SystemDouble

NumberBars

Type:Â SystemInt32

Confined

Type:Â SystemBoolean

EndZoneLength

Type:Â SystemDouble

EndZoneThickness

Type:Â SystemDouble

EndZoneOffset

Type:Â SystemDouble

Return Value

Type:Â Int32

▲ Remarks

▲ See Also

Reference

cAreaObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST279861DB_0")

Method

Retrieves rebar data for an area spandrel object

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetRebarDataSpandrel(
    string Name,
    ref int NumberRebarLayers,
    ref string[] LayerID,
    ref eWallSpandrelRebarLayerType[] LayerType,
    ref double[] ClearCover,
    ref int[] BarSizeIndex,
    ref double[] BarArea,
    ref double[] BarSpacing,
    ref int[] NumberBars,
    ref bool[] Confined
)
```

```
Function GetRebarDataSpandrel (
    Name As String,
    ByRef NumberRebarLayers As Integer,
    ByRef LayerID As String(),
    ByRef LayerType As eWallSpandrelRebarLayerType(),
    ByRef ClearCover As Double(),
    ByRef BarSizeIndex As Integer(),
    ByRef BarArea As Double(),
    ByRef BarSpacing As Double(),
    ByRef NumberBars As Integer(),
    ByRef Confined As Boolean()
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim NumberRebarLayers As Integer
Dim LayerID As String()
Dim LayerType As eWallSpandrelRebarLayerType()
Dim ClearCover As Double()
Dim BarSizeIndex As Integer()
Dim BarArea As Double()
Dim BarSpacing As Double()
Dim NumberBars As Integer()
Dim Confined As Boolean()
Dim returnValue As Integer
```

```
returnValue = instance.GetRebarDataSpandrel(Name,
    NumberRebarLayers, LayerID, LayerType,
```

cAreaObjspan id="LST279861DB_0"AddLanguageSpecificTextSet("LST279861DB_0?cpp=::|nu474;GetReb

ClearCover, BarSizeIndex, BarArea,
BarSpacing, NumberBars, Confined)

```
int GetRebarDataSpandrel(
String^ Name,
int% NumberRebarLayers,
array<String^>^% LayerID,
array<eWallSpandrelRebarLayerType>^% LayerType,
array<double>^% ClearCover,
array<int>^% BarSizeIndex,
array<double>^% BarArea,
array<double>^% BarSpacing,
array<int>^% NumberBars,
array<bool>^% Confined
)
```

```
abstract GetRebarDataSpandrel :
    Name : string *
    NumberRebarLayers : int byref *
    LayerID : string[] byref *
    LayerType : eWallSpandrelRebarLayerType[] byref *
    ClearCover : float[] byref *
    BarSizeIndex : int[] byref *
    BarArea : float[] byref *
    BarSpacing : float[] byref *
    NumberBars : int[] byref *
    Confined : bool[] byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing area object

NumberRebarLayers

Type:Â SystemInt32

LayerID

Type:Â SystemString

LayerType

Type:Â ETABSv1eWallSpandrelRebarLayerType

ClearCover

Type:Â SystemDouble

BarSizeIndex

Type:Â SystemInt32

BarArea

Type:Â SystemDouble

BarSpacing

Type:Â SystemDouble

NumberBars

Type:Â SystemInt32

Confined

Type:Â SystemBoolean

Return Value

Type: Int32

▲ Remarks

▲ See Also

Reference

cAreaObj Interface

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LSTB393D3FF Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetSelected(
    string Name,
    ref bool Selected
)
```

```
Function GetSelected (
    Name As String,
    ByRef Selected As Boolean
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim Selected As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetSelected(Name,
    Selected)
```

```
int GetSelected(
    String^ Name,
    bool% Selected
)
```

```
abstract GetSelected :
    Name : string *
    Selected : bool byref -> int
```

Parameters

Name

Type: [SystemString](#)

Selected

Type: [SystemBoolean](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST9BCAFD4B_0?cpp=::|n79") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetSelectedEdge(
    string Name,
    ref int NumberEdges,
    ref bool[] Selected
)
```

```
Function GetSelectedEdge (
    Name As String,
    ByRef NumberEdges As Integer,
    ByRef Selected As Boolean()
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim NumberEdges As Integer
Dim Selected As Boolean()
Dim returnValue As Integer
```

```
returnValue = instance.GetSelectedEdge(Name,
    NumberEdges, Selected)
```

```
int GetSelectedEdge(
    String^ Name,
    int% NumberEdges,
    array<bool>^% Selected
)
```

```
abstract GetSelectedEdge :
    Name : string *
    NumberEdges : int byref *
    Selected : bool[] byref -> int
```

Parameters

Name

Type: SystemString

NumberEdges

Type: SystemInt32

Selected

Type: SystemBoolean

cAreaObjspan id="LST9BCAFD4B_0"AddLanguageSpecificTextSet("LST9BCAFD4B_0?cpp=::|n79");GetS

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cAreaObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LSTD8E8ECD1_0?cpp=::|181.");GetS

Method

Retrieves the Spandrel label assignments of an area object

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetSpandrel(
    string Name,
    ref string SpandrelName
)
```

```
Function GetSpandrel (
    Name As String,
    ByRef SpandrelName As String
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim SpandrelName As String
Dim returnValue As Integer
```

```
returnValue = instance.GetSpandrel(Name,
    SpandrelName)
```

```
int GetSpandrel(
    String^ Name,
    String^% SpandrelName
)
```

```
abstract GetSpandrel :
    Name : string *
    SpandrelName : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing area object

SpandrelName

Type: [SystemString](#)

The name of the Spandrel assignment, if any, or "None"

Return Value

Type: Int32

Returns zero if the assignment is successfully retrieved; otherwise it returns a nonzero value.

▀ **Remarks**

▀ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim SpandrelName As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 4, 4, 24, 24)

    'define new Spandrel label
    ret = SapModel.SpandrelLabel.SetSpandrel("MySpandrel")

    'set assignment
    ret = SapModel.AreaObj.SetSpandrel("3", "MySpandrel")

    'get assignment
    ret = SapModel.AreaObj.GetSpandrel("3", SpandrelName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▀ **See Also**

Reference

cAreaObj Interface

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LSTE6D7F653_0")

Method

Retrieves the named area spring property assignment for an area object

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetSpringAssignment(
    string Name,
    ref string SpringProp
)
```

```
Function GetSpringAssignment (
    Name As String,
    ByRef SpringProp As String
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim SpringProp As String
Dim returnValue As Integer
```

```
returnValue = instance.GetSpringAssignment(Name,
    SpringProp)
```

```
int GetSpringAssignment(
    String^ Name,
    String^% SpringProp
)
```

```
abstract GetSpringAssignment :
    Name : string *
    SpringProp : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing area object

SpringProp

Type: [SystemString](#)

The name of an existing area spring property

Return Value

Type: Int32

Returns zero if the assignment is successfully retrieved, otherwise it returns a nonzero value

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim SpringProp As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 4, 4, 24, 24)

    'define new spring property
    ret = SapModel.PropAreaSpring.SetAreaSpringProp("mySpringProp1", 0, 0, 100, 0)

    'assign property
    ret = SapModel.AreaObj.SetSpringAssignment("1", "mySpringProp1")

    'get assigned property
    Dim prop as String
    ret = SapModel.AreaObj.GetSpringAssignment("1", prop)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ **See Also**

Reference

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LSTD1A237E4") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetTransformationMatrix(
    string Name,
    ref double[] Value,
    bool IsGlobal = true
)
```

```
Function GetTransformationMatrix (
    Name As String,
    ByRef Value As Double(),
    Optional IsGlobal As Boolean = true
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim Value As Double()
Dim IsGlobal As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetTransformationMatrix(Name,
    Value, IsGlobal)
```

```
int GetTransformationMatrix(
    String^ Name,
    array<double>^% Value,
    bool IsGlobal = true
)
```

```
abstract GetTransformationMatrix :
    Name : string *
    Value : float[] byref *
    ?IsGlobal : bool
(* Defaults:
    let _IsGlobal = defaultArg IsGlobal true
*)
-> int
```

Parameters

Name

Type: [SystemString](#)

Value

cAreaObjspan id="LSTD1A237E4_0"AddLanguageSpecificTextSet("LSTD1A237E4_0?cpp=::|nu487);GetTra

Type: SystemDouble
IsGlobal (Optional)
Type: SystemBoolean

Return Value

Type: Int32

See Also

Reference

cAreaObj Interface

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST5061722_0")

Method

Assigns a diaphragm to a specified area object

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetDiaphragm(
    string Name,
    string DiaphragmName
)
```

```
Function SetDiaphragm (
    Name As String,
    DiaphragmName As String
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim DiaphragmName As String
Dim returnValue As Integer
```

```
returnValue = instance.SetDiaphragm(Name,
    DiaphragmName)
```

```
int SetDiaphragm(
    String^ Name,
    String^ DiaphragmName
)
```

```
abstract SetDiaphragm :
    Name : string *
    DiaphragmName : string -> int
```

Parameters

Name

Type: SystemString

The name of an existing area object

DiaphragmName

Type: SystemString

This the name of an existing diaphragm to be assigned to the specified area object. If it is "None", no diaphragm will be assigned to the area object.

Return ValueType: Int32

Returns zero if the diaphragm is successfully assigned; otherwise it returns a nonzero value.

▀ **Remarks**

▀ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim DiaphragmName As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define a new diaphragm
    ret = SapModel.Diaphragm.SetDiaphragm("MyDiaph1A", SemiRigid:=True)

    'assign diaphragm to area
    ret = SapModel.AreaObj.SetDiaphragm("4", "MyDiaph1A")

    'get area diaphragm assignment
    ret = SapModel.AreaObj.GetDiaphragm("4", DiaphragmName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▀ **See Also**

Reference

cAreaObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LSTC8D75288_0")

Method

Makes generated edge constraint assignments to area objects.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetEdgeConstraint(
    string Name,
    bool ConstraintExists,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetEdgeConstraint (
    Name As String,
    ConstraintExists As Boolean,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim ConstraintExists As Boolean
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetEdgeConstraint(Name,
    ConstraintExists, ItemType)
```

```
int SetEdgeConstraint(
    String^ Name,
    bool ConstraintExists,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetEdgeConstraint :
    Name : string *
    ConstraintExists : bool *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing area object or group, depending on the value of the ItemType item.

ConstraintExists

Type:Â SystemBoolean

This item is True if an automatic edge constraint is generated by the program for the area object in the analysis model.

ItemType (Optional)

Type:Â ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the assignment is made to the area object specified by the Name item.

If this item is Group, the assignment is made to all area objects in the group specified by the Name item.

If this item is SelectedObjects, assignment is made to all selected area objects, and the Name item is ignored.

Return Value

Type:Â Int32

Returns zero if the edge constraint option is successfully assigned; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'assign auto edge constraint option
    ret = SapModel.AreaObj.SetEdgeConstraint("ALL", True, eItemType.Group)
    ret = SapModel.AreaObj.SetEdgeConstraint("2", False)
```

```
'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

📌 See Also

Reference

[cAreaObj Interface](#)

[ETABSv1 Namespace](#)

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»

CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LSTBEDB5968_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetGroupAssign(
    string Name,
    string GroupName,
    bool Remove = false,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetGroupAssign (
    Name As String,
    GroupName As String,
    Remove As Boolean = false,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim GroupName As String
Dim Remove As Boolean
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetGroupAssign(Name,
    GroupName, Remove, ItemType)
```

```
int SetGroupAssign(
    String^ Name,
    String^ GroupName,
    bool Remove = false,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetGroupAssign :
    Name : string *
    GroupName : string *
    ?Remove : bool *
    ?ItemType : eItemType
(* Defaults:
    let _Remove = defaultArg Remove false
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

cAreaObjAddLanguageSpecificTextSet("LSTBEDB5968_0?cpp=::|n495");SetGr

Parameters

Name

Type: SystemString

GroupName

Type: SystemString

Remove (Optional)

Type: SystemBoolean

ItemType (Optional)

Type: ETABSv1eItemType

Return Value

Type: Int32

▲ See Also

Reference

cAreaObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LSTF203E09C_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetGUID(
    string Name,
    string GUID = ""
)
```

```
Function SetGUID (
    Name As String,
    GUID As String = ""
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.SetGUID(Name, GUID)
```

```
int SetGUID(
    String^ Name,
    String^ GUID = L""
)
```

```
abstract SetGUID :
    Name : string *
    ?GUID : string
(* Defaults:
    let _GUID = defaultArg GUID ""
*)
-> int
```

Parameters

Name

Type: [SystemString](#)

GUID (Optional)

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cAreaObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LSTBC2B2EA8_0?cpp=::|n499");SetLo

Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetLoadTemperature(
    string Name,
    string LoadPat,
    int MyType,
    double Value,
    string PatternName = "",
    bool Replace = true,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetLoadTemperature (
    Name As String,
    LoadPat As String,
    MyType As Integer,
    Value As Double,
    PatternName As String = "",
    Replace As Boolean = true,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim LoadPat As String
Dim MyType As Integer
Dim Value As Double
Dim PatternName As String
Dim Replace As Boolean
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetLoadTemperature(Name,
    LoadPat, MyType, Value, PatternName,
    Replace, ItemType)
```

```
int SetLoadTemperature(
    String^ Name,
    String^ LoadPat,
    int MyType,
    double Value,
    String^ PatternName = L"",
    bool Replace = true,
    eItemType ItemType = eItemType::Objects
)
```

cAreaObjAddLanguageSpecificTextSet("LSTBC2B2EA8_0?cpp=::|n499");SetLo

```

)

abstract SetLoadTemperature :
    Name : string *
    LoadPat : string *
    MyType : int *
    Value : float *
    ?PatternName : string *
    ?Replace : bool *
    ?ItemType : eItemType
(* Defaults:
    let _PatternName = defaultArg PatternName ""
    let _Replace = defaultArg Replace true
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int

```

Parameters

Name
Type:Â [SystemString](#)

LoadPat
Type:Â [SystemString](#)

MyType
Type:Â [SystemInt32](#)

Value
Type:Â [SystemDouble](#)

PatternName (Optional)
Type:Â [SystemString](#)

Replace (Optional)
Type:Â [SystemBoolean](#)

ItemType (Optional)
Type:Â [ETABSv1eItemType](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cAreaObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LSTF06FCA4A_0?cpp=::|n59.1");SetLo

Method

Assigns uniform loads to area objects.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetLoadUniform(
    string Name,
    string LoadPat,
    double Value,
    int Dir,
    bool Replace = true,
    string CSys = "Global",
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetLoadUniform (
    Name As String,
    LoadPat As String,
    Value As Double,
    Dir As Integer,
    Replace As Boolean = true,
    CSys As String = "Global",
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim LoadPat As String
Dim Value As Double
Dim Dir As Integer
Dim Replace As Boolean
Dim CSys As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetLoadUniform(Name,
    LoadPat, Value, Dir, Replace, CSys,
    ItemType)
```

```
int SetLoadUniform(
    String^ Name,
    String^ LoadPat,
    double Value,
    int Dir,
    bool Replace = true,
    String^ CSys = L"Global",
```

cAreaObjspan id="LSTF06FCA4A_0"AddLanguageSpecificTextSet("LSTF06FCA4A_0?cpp=::|n59.1");SetLo

```

eItemType ItemType = eItemType::Objects
)

abstract SetLoadUniform :
  Name : string *
  LoadPat : string *
  Value : float *
  Dir : int *
  ?Replace : bool *
  ?CSys : string *
  ?ItemType : eItemType
(* Defaults:
  let _Replace = defaultArg Replace true
  let _CSys = defaultArg CSys "Global"
  let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int

```

Parameters

Name

Type: SystemString

The name of an existing area object or group, depending on the value of the ItemType item.

LoadPat

Type: SystemString

The name of a defined load pattern.

Value

Type: SystemDouble

The uniform load value. [F/L2]

Dir

Type: SystemInt32

Replace (Optional)

Type: SystemBoolean

If this item is True, all previous uniform loads, if any, assigned to the specified area object(s), in the specified load pattern, are deleted before making the new assignment.

CSys (Optional)

Type: SystemString

This is Local or the name of a defined coordinate system, indicating the coordinate system in which the uniform load is specified.

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the assignment is made to the area object specified by the Name item.

If this item is Group, the assignment is made to all area objects in the group specified by the Name item.

If this item is SelectedObjects, assignment is made to all selected area objects, and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the loads are successfully assigned; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'assign area object uniform loads
    ret = SapModel.AreaObj.SetLoadUniform("ALL", "DEAD", -0.01, 2, False, "Local", eItemType.Group)

    'refresh view
    ret = SapModel.View.RefreshView(0, False)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cAreaObj Interface

ETABSv1 Namespace

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Introduction

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LSTA2DA0EEA_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetLoadUniformToFrame(
    string Name,
    string LoadPat,
    double Value,
    int Dir,
    int DistType,
    bool Replace = true,
    string CSys = "Global",
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetLoadUniformToFrame (
    Name As String,
    LoadPat As String,
    Value As Double,
    Dir As Integer,
    DistType As Integer,
    Replace As Boolean = true,
    CSys As String = "Global",
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim LoadPat As String
Dim Value As Double
Dim Dir As Integer
Dim DistType As Integer
Dim Replace As Boolean
Dim CSys As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetLoadUniformToFrame(Name,
    LoadPat, Value, Dir, DistType, Replace,
    CSys, ItemType)
```

```
int SetLoadUniformToFrame(
    String^ Name,
    String^ LoadPat,
    double Value,
    int Dir,
```

cAreaObjid="LSTA2DA0EEA_0"AddLanguageSpecificTextSet("LSTA2DA0EEA_0?cpp=::|505");SetL

```

int DistType,
bool Replace = true,
String^ CSys = L"Global",
eItemType ItemType = eItemType::Objects
)

abstract SetLoadUniformToFrame :
    Name : string *
    LoadPat : string *
    Value : float *
    Dir : int *
    DistType : int *
    ?Replace : bool *
    ?CSys : string *
    ?ItemType : eItemType
(* Defaults:
    let _Replace = defaultArg Replace true
    let _CSys = defaultArg CSys "Global"
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int

```

Parameters

Name
Type: [SystemString](#)

LoadPat
Type: [SystemString](#)

Value
Type: [SystemDouble](#)

Dir
Type: [SystemInt32](#)

DistType
Type: [SystemInt32](#)

Replace (Optional)
Type: [SystemBoolean](#)

CSys (Optional)
Type: [SystemString](#)

ItemType (Optional)
Type: [ETABSv1eItemType](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cAreaObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST86467CAA_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetLoadWindPressure(
    string Name,
    string LoadPat,
    int MyType,
    double Cp,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetLoadWindPressure (
    Name As String,
    LoadPat As String,
    MyType As Integer,
    Cp As Double,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim LoadPat As String
Dim MyType As Integer
Dim Cp As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetLoadWindPressure(Name,
    LoadPat, MyType, Cp, ItemType)
```

```
int SetLoadWindPressure(
    String^ Name,
    String^ LoadPat,
    int MyType,
    double Cp,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetLoadWindPressure :
    Name : string *
    LoadPat : string *
    MyType : int *
    Cp : float *
    ?ItemType : eItemType
(* Defaults:
```

cAreaObjid="LST86467CAA_0"AddLanguageSpecificTextSet("LST86467CAA_0?cpp=::|nu508;SetLoa

Introduction

```
let _ItemType = defaultArg ItemType eItemType.Objects  
*)  
-> int
```

Parameters

Name

Type: SystemString

LoadPat

Type: SystemString

MyType

Type: SystemInt32

Cp

Type: SystemDouble

ItemType (Optional)

Type: ETABSv1eItemType

Return Value

Type: Int32

▲ See Also

Reference

cAreaObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST4C343407_0") Method

Assigns a local axis angle to area objects.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetLocalAxes(
    string Name,
    double Ang,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetLocalAxes (
    Name As String,
    Ang As Double,
    Optional ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim Ang As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetLocalAxes(Name,
    Ang, ItemType)
```

```
int SetLocalAxes(
    String^ Name,
    double Ang,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetLocalAxes :
    Name : string *
    Ang : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing area object or group, depending on the value of the ItemType item.

Ang

Type:Â SystemDouble

This is the angle that the local 1 and 2 axes are rotated about the positive local 3 axis from the default orientation. The rotation for a positive angle appears counter clockwise when the local +3 axis is pointing toward you. [deg]

ItemType (Optional)

Type:Â ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the assignment is made to the area object specified by the Name item.

If this item is Group, the assignment is made to all area objects in the group specified by the Name item.

If this item is SelectedObjects, assignment is made to all selected area objects, and the Name item is ignored.

Return Value

Type:Â Int32

Returns zero if the local axis angle is successfully assigned; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'assign area object local axis angle
    ret = SapModel.AreaObj.SetLocalAxes("3", 30)
```

```
'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

📌 See Also

Reference

cAreaObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST5292A524_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetMass(
    string Name,
    double MassOverL2,
    bool Replace = false,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetMass (
    Name As String,
    MassOverL2 As Double,
    Replace As Boolean = false,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim MassOverL2 As Double
Dim Replace As Boolean
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetMass(Name, MassOverL2,
    Replace, ItemType)
```

```
int SetMass(
    String^ Name,
    double MassOverL2,
    bool Replace = false,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetMass :
    Name : string *
    MassOverL2 : float *
    ?Replace : bool *
    ?ItemType : eItemType
(* Defaults:
    let _Replace = defaultArg Replace false
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

cAreaObjAddLanguageSpecificTextSet("LST5292A524_0?cpp=::|nu=5").SetMas

Parameters

Name

Type: SystemString

MassOverL2

Type: SystemDouble

Replace (Optional)

Type: SystemBoolean

ItemType (Optional)

Type: ETABSv1eItemType

Return Value

Type: Int32

▲ See Also

Reference

cAreaObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST6FF1C9D70") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetMaterialOverwrite(
    string Name,
    string PropName,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetMaterialOverwrite (
    Name As String,
    PropName As String,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim PropName As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetMaterialOverwrite(Name,
    PropName, ItemType)
```

```
int SetMaterialOverwrite(
    String^ Name,
    String^ PropName,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetMaterialOverwrite :
    Name : string *
    PropName : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: [SystemString](#)

PropName

cAreaObjspan id="LST6FF1C9D7_0"AddLanguageSpecificTextSet("LST6FF1C9D7_0?cpp=::|nu54.5);SetMa

Type: SystemString
ItemType (Optional)
Type: ETABSv1eItemType

Return Value

Type: Int32

See Also

Reference

cAreaObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST8C51573B_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetModifiers(
    string Name,
    ref double[] Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetModifiers (
    Name As String,
    ByRef Value As Double(),
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim Value As Double()
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetModifiers(Name,
    Value, ItemType)
```

```
int SetModifiers(
    String^ Name,
    array<double>^% Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetModifiers :
    Name : string *
    Value : float[] byref *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: [SystemString](#)

Value

cAreaObjspan id="LST8C51573B_0"AddLanguageSpecificTextSet("LST8C51573B_0?cpp=::|nu5.17;SetMod

Type: SystemDouble
ItemType (Optional)
Type: ETABSv1eItemType

Return Value

Type: Int32

■ See Also

Reference

cAreaObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST2CF890B_0") Method

Designates an area object(s) as an opening.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetOpening(
    string Name,
    bool IsOpening,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOpening (
    Name As String,
    IsOpening As Boolean,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim IsOpening As Boolean
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOpening(Name,
    IsOpening, ItemType)
```

```
int SetOpening(
    String^ Name,
    bool IsOpening,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOpening :
    Name : string *
    IsOpening : bool *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing area object or group, depending on the value of the ItemType item.

IsOpening

Type:Â SystemBoolean

This item should be set to True if the specified area object(s) is to be an opening, otherwise it should be False.

ItemType (Optional)

Type:Â ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the assignment is made to the area object specified by the Name item.

If this item is Group, the assignment is made to all area objects in the group specified by the Name item.

If this item is SelectedObjects, the assignment is made to all selected area objects, and the Name item is ignored.

Return Value

Type:Â Int32

Returns zero if the opening is successfully assigned; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'designate an area object as an opening
    ret = SapModel.AreaObj.SetOpening("3", True)
```

Introduction

```
'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

🔍 See Also

Reference

[cAreaObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST219D31B2_0")

Method

Sets the pier label assignment of one or more area objects

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetPier(
    string Name,
    string PierName,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetPier (
    Name As String,
    PierName As String,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim PierName As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetPier(Name, PierName,
    ItemType)
```

```
int SetPier(
    String^ Name,
    String^ PierName,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetPier :
    Name : string *
    PierName : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing area object or group, depending on the value of the ItemType item.

PierName

Type: SystemString

The name of the pier assignment, or "None", to unset any assignment

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the assignment is made to the area object specified by the Name item.

If this item is Group, the assignment is made to all area objects in the group specified by the Name item.

If this item is SelectedObjects, assignment is made to all selected area objects, and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the assignment is successful; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim PierName as String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 4, 4, 24, 24)

    'define new pier label
    ret = SapModel.PierLabel.SetPier("MyPier")

    'set assignment
    ret = SapModel.AreaObj.SetPier("3", "MyPier")
```

Parameters

```
'get assignment
    ret = SapModel.AreaObj.GetPier("3", PierName)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cAreaObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST4DD66AD4_0?cpp=::|n525");SetPr

Method

Assigns an area property to area objects.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetProperty(
    string Name,
    string PropName,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetProperty (
    Name As String,
    PropName As String,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim PropName As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetProperty(Name,
    PropName, ItemType)
```

```
int SetProperty(
    String^ Name,
    String^ PropName,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetProperty :
    Name : string *
    PropName : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

cAreaObjid="LST4DD66AD4_0"AddLanguageSpecificTextSet("LST4DD66AD4_0?cpp=::|n525");SetPr

Type:Â SystemString

The name of an existing area object or group, depending on the value of the ItemType item.

PropName

Type:Â SystemString

This is None or the name of a area property to be assigned to the specified area object(s). None means that no property is assigned to the area object.

ItemType (Optional)

Type:Â ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the assignment is made to the area object specified by the Name item.

If this item is Group, the assignment is made to all area objects in the group specified by the Name item.

If this item is SelectedObjects, assignment is made to all selected area objects, and the Name item is ignored.

Return Value

Type:Â Int32

Returns zero if the property is successfully assigned; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set area property
    ret = SapModel.AreaObj.SetProperty("4", "None")
```

Introduction

```
'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

🔍 See Also

Reference

[cAreaObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST69503999_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetSelected(
    string Name,
    bool Selected,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetSelected (
    Name As String,
    Selected As Boolean,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim Selected As Boolean
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetSelected(Name,
    Selected, ItemType)
```

```
int SetSelected(
    String^ Name,
    bool Selected,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetSelected :
    Name : string *
    Selected : bool *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: [SystemString](#)

Selected

Type: SystemBoolean
ItemType (Optional)
Type: ETABSv1eItemType

Return Value

Type: Int32

■ See Also

Reference

cAreaObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LST8B8DB848 Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetSelectedEdge(
    string Name,
    int EdgeNum,
    bool Selected
)
```

```
Function SetSelectedEdge (
    Name As String,
    EdgeNum As Integer,
    Selected As Boolean
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim EdgeNum As Integer
Dim Selected As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetSelectedEdge(Name,
    EdgeNum, Selected)
```

```
int SetSelectedEdge(
    String^ Name,
    int EdgeNum,
    bool Selected
)
```

```
abstract SetSelectedEdge :
    Name : string *
    EdgeNum : int *
    Selected : bool -> int
```

Parameters

Name

Type: SystemString

EdgeNum

Type: SystemInt32

Selected

Type: SystemBoolean

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cAreaObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LSTC2C4510B_0")

Method

Sets the Spandrel label assignment of one or more area objects

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetSpandrel(
    string Name,
    string SpandrelName,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetSpandrel (
    Name As String,
    SpandrelName As String,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim SpandrelName As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetSpandrel(Name,
    SpandrelName, ItemType)
```

```
int SetSpandrel(
    String^ Name,
    String^ SpandrelName,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetSpandrel :
    Name : string *
    SpandrelName : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing area object or group, depending on the value of the ItemType item.

SpandrelName

Type: SystemString

The name of the Spandrel assignment, or "None", to unset any assignment

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the assignment is made to the area object specified by the Name item.

If this item is Group, the assignment is made to all area objects in the group specified by the Name item.

If this item is SelectedObjects, assignment is made to all selected area objects, and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the assignment is successful; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim SpandrelName as String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 4, 4, 24, 24)

    'define new Spandrel label
    ret = SapModel.SpandrelLabel.SetSpandrel("MySpandrel")

    'set assignment
    ret = SapModel.AreaObj.SetSpandrel("3", "MySpandrel")
```

Parameters

Introduction

```
'get assignment
    ret = SapModel.AreaObj.GetSpandrel("3", SpandrelName)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cAreaObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAreaObjAddLanguageSpecificTextSet("LSTDAA74E5F0") Method

Assigns an existing named area spring property to area objects

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetSpringAssignment(
    string Name,
    string SpringProp,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetSpringAssignment (
    Name As String,
    SpringProp As String,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cAreaObj
Dim Name As String
Dim SpringProp As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetSpringAssignment(Name,
    SpringProp, ItemType)
```

```
int SetSpringAssignment(
    String^ Name,
    String^ SpringProp,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetSpringAssignment :
    Name : string *
    SpringProp : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

cAreaObjspan id="LSTDAA74E5F_0"AddLanguageSpecificTextSet("LSTDAA74E5F_0?cpp=::|n535");SetSp

Type: SystemString

The name of an existing area object or group depending on the value of the ItemType item.

SpringProp

Type: SystemString

The name of an existing area spring property

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the property assignment is made to the area object specified by the Name item.

If this item is Group, the property assignment is made to all area objects in the group specified by the Name item.

If this item is SelectedObjects, the property assignment is made to all selected area objects and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the property is successfully assigned, otherwise it returns a nonzero value

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim SpringProp As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new spring property
    ret = SapModel.PropAreaSpring.SetAreaSpringProp("mySpringProp1", 0, 0, 100, 0)

    'assign property
```

Parameters

Introduction

```
ret = SapModel.AreaObj.SetSpringAssignment("1", "mySpringProp1")

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

📌 See Also

Reference

[cAreaObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAutoSeismic Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cAutoSeismic
```

```
Public Interface cAutoSeismic
```

```
Dim instance As cAutoSeismic
```

```
public interface class cAutoSeismic
```

```
type cAutoSeismic = interface end
```

The cAutoSeismic type exposes the following members.

▲ Methods

▲	Name	Description
---	------	-------------

🔗	GetIBC2006	
---	----------------------------	--

🔗	SetIBC2006	
---	----------------------------	--

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cAutoSeismic Methods

The [cAutoSeismic](#) type exposes the following members.

Methods

	Name	Description
--	------	-------------

	GetIBC2006	
--	----------------------------	--

	SetIBC2006	
--	----------------------------	--

[Top](#)

See Also

Reference

[cAutoSeismic Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cAutoSeismicAddLanguageSpecificTextSet("LST5C876E29_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetIBC2006(
    string Name,
    ref int DirFlag,
    ref double Eccen,
    ref int PeriodFlag,
    ref int CtType,
    ref double UserT,
    ref bool UserZ,
    ref double TopZ,
    ref double BottomZ,
    ref double R,
    ref double Omega,
    ref double Cd,
    ref double I,
    ref int IBC2006Option,
    ref double Latitude,
    ref double Longitude,
    ref string ZipCode,
    ref double Ss,
    ref double S1,
    ref double Tl,
    ref int SiteClass,
    ref double Fa,
    ref double Fv
)
```

```
Function GetIBC2006 (
    Name As String,
    ByRef DirFlag As Integer,
    ByRef Eccen As Double,
    ByRef PeriodFlag As Integer,
    ByRef CtType As Integer,
    ByRef UserT As Double,
    ByRef UserZ As Boolean,
    ByRef TopZ As Double,
    ByRef BottomZ As Double,
    ByRef R As Double,
    ByRef Omega As Double,
    ByRef Cd As Double,
    ByRef I As Double,
    ByRef IBC2006Option As Integer,
    ByRef Latitude As Double,
    ByRef Longitude As Double,
```

cAutoSeismicAddLanguageSpecificTextSet("LST5C876E29_0?cpp=540=.");Ge

Introduction

```
ByRef ZipCode As String,  
ByRef Ss As Double,  
ByRef S1 As Double,  
ByRef Tl As Double,  
ByRef SiteClass As Integer,  
ByRef Fa As Double,  
ByRef Fv As Double  
) As Integer
```

```
Dim instance As cAutoSeismic  
Dim Name As String  
Dim DirFlag As Integer  
Dim Eccen As Double  
Dim PeriodFlag As Integer  
Dim CtType As Integer  
Dim UserT As Double  
Dim UserZ As Boolean  
Dim TopZ As Double  
Dim BottomZ As Double  
Dim R As Double  
Dim Omega As Double  
Dim Cd As Double  
Dim I As Double  
Dim IBC2006Option As Integer  
Dim Latitude As Double  
Dim Longitude As Double  
Dim ZipCode As String  
Dim Ss As Double  
Dim S1 As Double  
Dim Tl As Double  
Dim SiteClass As Integer  
Dim Fa As Double  
Dim Fv As Double  
Dim returnValue As Integer
```

```
returnValue = instance.GetIBC2006(Name,  
DirFlag, Eccen, PeriodFlag, CtType,  
UserT, UserZ, TopZ, BottomZ, R, Omega,  
Cd, I, IBC2006Option, Latitude, Longitude,  
ZipCode, Ss, S1, Tl, SiteClass, Fa,  
Fv)
```

```
int GetIBC2006(  
String^ Name,  
int% DirFlag,  
double% Eccen,  
int% PeriodFlag,  
int% CtType,  
double% UserT,  
bool% UserZ,  
double% TopZ,  
double% BottomZ,  
double% R,  
double% Omega,  
double% Cd,  
double% I,  
int% IBC2006Option,  
double% Latitude,  
double% Longitude,  
String^ ZipCode,  
double% Ss,  
double% S1,
```

cAutoSeismicAddLanguageSpecificTextSet("LST5C876E29_0?cpp=541u=.");Ge

```

double% Tl,
int% SiteClass,
double% Fa,
double% Fv
)

abstract GetIBC2006 :
    Name : string *
    DirFlag : int byref *
    Eccen : float byref *
    PeriodFlag : int byref *
    CtType : int byref *
    UserT : float byref *
    UserZ : bool byref *
    TopZ : float byref *
    BottomZ : float byref *
    R : float byref *
    Omega : float byref *
    Cd : float byref *
    I : float byref *
    IBC2006Option : int byref *
    Latitude : float byref *
    Longitude : float byref *
    ZipCode : string byref *
    Ss : float byref *
    S1 : float byref *
    Tl : float byref *
    SiteClass : int byref *
    Fa : float byref *
    Fv : float byref -> int

```

Parameters

Name	Type:Â <u>SystemString</u>
DirFlag	Type:Â <u>SystemInt32</u>
Eccen	Type:Â <u>SystemDouble</u>
PeriodFlag	Type:Â <u>SystemInt32</u>
CtType	Type:Â <u>SystemInt32</u>
UserT	Type:Â <u>SystemDouble</u>
UserZ	Type:Â <u>SystemBoolean</u>
TopZ	Type:Â <u>SystemDouble</u>
BottomZ	Type:Â <u>SystemDouble</u>
R	Type:Â <u>SystemDouble</u>
Omega	Type:Â <u>SystemDouble</u>
Cd	

I Type:Â SystemDouble
 Type:Â SystemDouble
 IBC2006Option
 Type:Â SystemInt32
 Latitude
 Type:Â SystemDouble
 Longitude
 Type:Â SystemDouble
 ZipCode
 Type:Â SystemString
 Ss
 Type:Â SystemDouble
 S1
 Type:Â SystemDouble
 Tl
 Type:Â SystemDouble
 SiteClass
 Type:Â SystemInt32
 Fa
 Type:Â SystemDouble
 Fv
 Type:Â SystemDouble

Return Value

Type:Â Int32

▀ See Also

Reference

cAutoSeismic Interface

ETABSv1 Namespace

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CSI API ETABS v1

cAutoSeismicAddLanguageSpecificTextSet("LST18EEB299_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetIBC2006(
    string Name,
    int DirFlag,
    double Eccen,
    int PeriodFlag,
    int CtType,
    double UserT,
    bool UserZ,
    double TopZ,
    double BottomZ,
    double R,
    double Omega,
    double Cd,
    double I,
    int IBC2006Option,
    double Latitude,
    double Longitude,
    string ZipCode,
    double Ss,
    double S1,
    double Tl,
    int SiteClass,
    double Fa,
    double Fv
)
```

```
Function SetIBC2006 (
    Name As String,
    DirFlag As Integer,
    Eccen As Double,
    PeriodFlag As Integer,
    CtType As Integer,
    UserT As Double,
    UserZ As Boolean,
    TopZ As Double,
    BottomZ As Double,
    R As Double,
    Omega As Double,
    Cd As Double,
    I As Double,
    IBC2006Option As Integer,
    Latitude As Double,
    Longitude As Double,
```

cAutoSeismicAddLanguageSpecificTextSet("LST18EEB299_0?cpp=544u=.");Se

Introduction

```
ZipCode As String,  
Ss As Double,  
S1 As Double,  
Tl As Double,  
SiteClass As Integer,  
Fa As Double,  
Fv As Double  
) As Integer
```

```
Dim instance As cAutoSeismic  
Dim Name As String  
Dim DirFlag As Integer  
Dim Eccen As Double  
Dim PeriodFlag As Integer  
Dim CtType As Integer  
Dim UserT As Double  
Dim UserZ As Boolean  
Dim TopZ As Double  
Dim BottomZ As Double  
Dim R As Double  
Dim Omega As Double  
Dim Cd As Double  
Dim I As Double  
Dim IBC2006Option As Integer  
Dim Latitude As Double  
Dim Longitude As Double  
Dim ZipCode As String  
Dim Ss As Double  
Dim S1 As Double  
Dim Tl As Double  
Dim SiteClass As Integer  
Dim Fa As Double  
Dim Fv As Double  
Dim returnValue As Integer
```

```
returnValue = instance.SetIBC2006(Name,  
DirFlag, Eccen, PeriodFlag, CtType,  
UserT, UserZ, TopZ, BottomZ, R, Omega,  
Cd, I, IBC2006Option, Latitude, Longitude,  
ZipCode, Ss, S1, Tl, SiteClass, Fa,  
Fv)
```

```
int SetIBC2006(  
String^ Name,  
int DirFlag,  
double Eccen,  
int PeriodFlag,  
int CtType,  
double UserT,  
bool UserZ,  
double TopZ,  
double BottomZ,  
double R,  
double Omega,  
double Cd,  
double I,  
int IBC2006Option,  
double Latitude,  
double Longitude,  
String^ ZipCode,  
double Ss,  
double S1,
```

```
double Tl,
int SiteClass,
double Fa,
double Fv
)
```

```
abstract SetIBC2006 :
    Name : string *
    DirFlag : int *
    Eccen : float *
    PeriodFlag : int *
    CtType : int *
    UserT : float *
    UserZ : bool *
    TopZ : float *
    BottomZ : float *
    R : float *
    Omega : float *
    Cd : float *
    I : float *
    IBC2006Option : int *
    Latitude : float *
    Longitude : float *
    ZipCode : string *
    Ss : float *
    S1 : float *
    Tl : float *
    SiteClass : int *
    Fa : float *
    Fv : float -> int
```

Parameters

Name

Type:Â SystemString

DirFlag

Type:Â SystemInt32

Eccen

Type:Â SystemDouble

PeriodFlag

Type:Â SystemInt32

CtType

Type:Â SystemInt32

UserT

Type:Â SystemDouble

UserZ

Type:Â SystemBoolean

TopZ

Type:Â SystemDouble

BottomZ

Type:Â SystemDouble

R

Type:Â SystemDouble

Omega

Type:Â SystemDouble

Cd

I Type:Â SystemDouble
 Type:Â SystemDouble
 IBC2006Option
 Type:Â SystemInt32
 Latitude
 Type:Â SystemDouble
 Longitude
 Type:Â SystemDouble
 ZipCode
 Type:Â SystemString
 Ss
 Type:Â SystemDouble
 S1
 Type:Â SystemDouble
 Tl
 Type:Â SystemDouble
 SiteClass
 Type:Â SystemInt32
 Fa
 Type:Â SystemDouble
 Fv
 Type:Â SystemDouble

Return Value

Type:Â Int32

▀ See Also

Reference

cAutoSeismic Interface

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CSI API ETABS v1

cCaseDirectHistoryLinear Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cCaseDirectHistoryLinear
```

```
Public Interface cCaseDirectHistoryLinear
```

```
Dim instance As cCaseDirectHistoryLinear
```

```
public interface class cCaseDirectHistoryLinear
```

```
type cCaseDirectHistoryLinear = interface end
```

The cCaseDirectHistoryLinear type exposes the following members.

▲ Methods

▲	Name	Description
---	------	-------------

🔗	GetLoads	Retrieves the load data for the specified load case
---	--------------------------	---

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▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cCaseDirectHistoryLinear Methods

The cCaseDirectHistoryLinear type exposes the following members.

Methods

Ã	Name	Description
ð	<u>GetLoads</u>	Retrieves the load data for the specified load case

Top

See Also

Reference

cCaseDirectHistoryLinear Interface

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CSI API ETABS v1

cCaseDirectHistoryLinearAddLanguageSpecificTextSet Method

Retrieves the load data for the specified load case

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetLoads(
    string Name,
    ref int NumberLoads,
    ref string[] LoadType,
    ref string[] LoadName,
    ref string[] Func,
    ref double[] SF,
    ref double[] Tf,
    ref double[] At,
    ref string[] CSys,
    ref double[] Ang
)
```

```
Function GetLoads (
    Name As String,
    ByRef NumberLoads As Integer,
    ByRef LoadType As String(),
    ByRef LoadName As String(),
    ByRef Func As String(),
    ByRef SF As Double(),
    ByRef Tf As Double(),
    ByRef At As Double(),
    ByRef CSys As String(),
    ByRef Ang As Double()
) As Integer
```

```
Dim instance As cCaseDirectHistoryLinear
Dim Name As String
Dim NumberLoads As Integer
Dim LoadType As String()
Dim LoadName As String()
Dim Func As String()
Dim SF As Double()
Dim Tf As Double()
Dim At As Double()
Dim CSys As String()
Dim Ang As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetLoads(Name,
    NumberLoads, LoadType, LoadName,
```

Func, SF, Tf, At, CSys, Ang)

```
int GetLoads(
String^ Name,
int% NumberLoads,
array<String^>^% LoadType,
array<String^>^% LoadName,
array<String^>^% Func,
array<double>^% SF,
array<double>^% Tf,
array<double>^% At,
array<String^>^% CSys,
array<double>^% Ang
)
```

```
abstract GetLoads :
    Name : string *
    NumberLoads : int byref *
    LoadType : string[] byref *
    LoadName : string[] byref *
    Func : string[] byref *
    SF : float[] byref *
    Tf : float[] byref *
    At : float[] byref *
    CSys : string[] byref *
    Ang : float[] byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing linear direct integration time history load case

NumberLoads

Type:Â SystemInt32

The number of loads assigned to the specified analysis case

LoadType

Type:Â SystemString

This is an array that includes Load or Accel, indicating the type of each load assigned to the load case

LoadName

Type:Â SystemString

This is an array that includes the name of each load assigned to the load case.

If the LoadType item is Load, this item is the name of a defined load pattern.

If the LoadType item is Accel, this item is U1, U2, U3, R1, R2 or R3, indicating the direction of the load.

Func

Type:Â SystemString

This is an array that includes the name of the time history function associated with each load

SF

Type:Â SystemDouble

This is an array that includes the scale factor of each load assigned to the load case. [L/s²] for U1 U2 and U3; otherwise unitless

Tf

Type: SystemDouble

This is an array that includes the time scale factor of each load assigned to the load case

At

Type: SystemDouble

This is an array that includes the arrival time of each load assigned to the load case

CSys

Type: SystemString

This is an array that includes the name of the coordinate system associated with each load. If this item is a blank string, the Global coordinate system is assumed.

This item applies only when the LoadType item is Accel.

Ang

Type: SystemDouble

This is an array that includes the angle between the acceleration local 1 axis and the +X-axis of the coordinate system specified by the CSys item. The rotation is about the Z-axis of the specified coordinate system. [deg]

Return Value

Type: Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ See Also

Reference

cCaseDirectHistoryLinear Interface

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CSI API ETABS v1

cCaseDirectHistoryNonlinear Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cCaseDirectHistoryNonlinear
```

```
Public Interface cCaseDirectHistoryNonlinear
```

```
Dim instance As cCaseDirectHistoryNonlinear
```

```
public interface class cCaseDirectHistoryNonlinear
```

```
type cCaseDirectHistoryNonlinear = interface end
```

The cCaseDirectHistoryNonlinear type exposes the following members.

▲ Methods

▲	Name	Description
---	------	-------------

🔗	GetLoads	Retrieves the load data for the specified load case
---	--------------------------	---

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cCaseDirectHistoryNonlinear Methods

The cCaseDirectHistoryNonlinear type exposes the following members.

Methods

⌕ Name	Description
🔍 <u>GetLoads</u>	Retrieves the load data for the specified load case

Top

🔍 See Also

Reference

cCaseDirectHistoryNonlinear Interface

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CSI API ETABS v1

cCaseDirectHistoryNonlinearAddLanguageSpecificTextMethod

Retrieves the load data for the specified load case

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetLoads(
    string Name,
    ref int NumberLoads,
    ref string[] LoadType,
    ref string[] LoadName,
    ref string[] Func,
    ref double[] SF,
    ref double[] Tf,
    ref double[] At,
    ref string[] CSys,
    ref double[] Ang
)
```

```
Function GetLoads (
    Name As String,
    ByRef NumberLoads As Integer,
    ByRef LoadType As String(),
    ByRef LoadName As String(),
    ByRef Func As String(),
    ByRef SF As Double(),
    ByRef Tf As Double(),
    ByRef At As Double(),
    ByRef CSys As String(),
    ByRef Ang As Double()
) As Integer
```

```
Dim instance As cCaseDirectHistoryNonlinear
Dim Name As String
Dim NumberLoads As Integer
Dim LoadType As String()
Dim LoadName As String()
Dim Func As String()
Dim SF As Double()
Dim Tf As Double()
Dim At As Double()
Dim CSys As String()
Dim Ang As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetLoads(Name,
    NumberLoads, LoadType, LoadName,
```

Func, SF, Tf, At, CSys, Ang)

```
int GetLoads(
String^ Name,
int% NumberLoads,
array<String^>^% LoadType,
array<String^>^% LoadName,
array<String^>^% Func,
array<double>^% SF,
array<double>^% Tf,
array<double>^% At,
array<String^>^% CSys,
array<double>^% Ang
)
```

```
abstract GetLoads :
    Name : string *
    NumberLoads : int byref *
    LoadType : string[] byref *
    LoadName : string[] byref *
    Func : string[] byref *
    SF : float[] byref *
    Tf : float[] byref *
    At : float[] byref *
    CSys : string[] byref *
    Ang : float[] byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing nonlinear direct integration time history load case

NumberLoads

Type:Â SystemInt32

The number of loads assigned to the specified analysis case

LoadType

Type:Â SystemString

This is an array that includes Load or Accel, indicating the type of each load assigned to the load case

LoadName

Type:Â SystemString

This is an array that includes the name of each load assigned to the load case.

If the LoadType item is Load, this item is the name of a defined load pattern.

If the LoadType item is Accel, this item is U1, U2, U3, R1, R2 or R3, indicating the direction of the load.

Func

Type:Â SystemString

This is an array that includes the name of the time history function associated with each load

SF

Type:Â SystemDouble

This is an array that includes the scale factor of each load assigned to the load case. [L/s²] for U1 U2 and U3; otherwise unitless

Tf

Type: SystemDouble

This is an array that includes the time scale factor of each load assigned to the load case

At

Type: SystemDouble

This is an array that includes the arrival time of each load assigned to the load case

CSys

Type: SystemString

This is an array that includes the name of the coordinate system associated with each load. If this item is a blank string, the Global coordinate system is assumed.

This item applies only when the LoadType item is Accel.

Ang

Type: SystemDouble

This is an array that includes the angle between the acceleration local 1 axis and the +X-axis of the coordinate system specified by the CSys item. The rotation is about the Z-axis of the specified coordinate system. [deg]

Return Value

Type: Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ See Also

Reference

cCaseDirectHistoryNonlinear Interface

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CSI API ETABS v1

cCaseHyperStatic Interface

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cCaseHyperStatic
```

```
Public Interface cCaseHyperStatic
```

```
Dim instance As cCaseHyperStatic
```

```
public interface class cCaseHyperStatic
```

```
type cCaseHyperStatic = interface end
```

The cCaseHyperStatic type exposes the following members.

▲ Methods

▲	Name	Description
🔗	GetBaseCase	Retrieves the base case for the specified hyperstatic load case
🔗	SetBaseCase	Sets the base case for the specified hyperstatic load case
🔗	SetCase	Initializes a hyperstatic load case. If this function is called for an existing load case, all items for the case are reset to their default value

[Top](#)

▲ See Also

Reference

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cCaseHyperStatic Methods

The [cCaseHyperStatic](#) type exposes the following members.

Methods

⚡	Name	Description
⇒	GetBaseCase	Retrieves the base case for the specified hyperstatic load case
⇒	SetBaseCase	Sets the base case for the specified hyperstatic load case
⇒	SetCase	Initializes a hyperstatic load case. If this function is called for an existing load case, all items for the case are reset to their default value

[Top](#)

⚡ See Also

Reference

[cCaseHyperStatic Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cCaseHyperStaticAddLanguageSpecificTextSet("LST57D81BA2_0") Method

Retrieves the base case for the specified hyperstatic load case

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetBaseCase(
    string Name,
    ref string HyperStaticCase
)
```

```
Function GetBaseCase (
    Name As String,
    ByRef HyperStaticCase As String
) As Integer
```

```
Dim instance As cCaseHyperStatic
Dim Name As String
Dim HyperStaticCase As String
Dim returnValue As Integer
```

```
returnValue = instance.GetBaseCase(Name,
HyperStaticCase)
```

```
int GetBaseCase(
    String^ Name,
    String^% HyperStaticCase
)
```

```
abstract GetBaseCase :
    Name : string *
    HyperStaticCase : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing hyperstatic load case

HyperStaticCase

Type: [SystemString](#)

The name of an existing static linear load case that is the base case for the specified hyperstatic load case

Return Value

Type: Int32

Returns zero if the base case is successfully retrieved; otherwise it returns a nonzero value

▀ Remarks

▀ See Also

Reference

cCaseHyperStatic Interface

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CSI API ETABS v1

cCaseHyperStaticAddLanguageSpecificTextSet("LST299AEEAD_0") Method

Sets the base case for the specified hyperstatic load case

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetBaseCase(
    string Name,
    string HyperStaticCase
)
```

```
Function SetBaseCase (
    Name As String,
    HyperStaticCase As String
) As Integer
```

```
Dim instance As cCaseHyperStatic
Dim Name As String
Dim HyperStaticCase As String
Dim returnValue As Integer
```

```
returnValue = instance.SetBaseCase(Name,
HyperStaticCase)
```

```
int SetBaseCase(
    String^ Name,
    String^ HyperStaticCase
)
```

```
abstract SetBaseCase :
    Name : string *
    HyperStaticCase : string -> int
```

Parameters

Name

Type: SystemString

The name of an existing hyperstatic load case

HyperStaticCase

Type: SystemString

The name of an existing static linear load case that is the base case for the specified hyperstatic load case

Return Value

Type: Int32

Returns zero if the base case is successfully set; otherwise it returns a nonzero value

▲ Remarks

▲ See Also

Reference

cCaseHyperStatic Interface

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CSI API ETABS v1

cCaseHyperStaticAddLanguageSpecificTextSet("LSTF Method

Initializes a hyperstatic load case. If this function is called for an existing load case, all items for the case are reset to their default value

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetCase(
    string Name
)
```

```
Function SetCase (
    Name As String
) As Integer
```

```
Dim instance As cCaseHyperStatic
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.SetCase(Name)
```

```
int SetCase(
    String^ Name
)
```

```
abstract SetCase :
    Name : string -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing or new load case. If this is an existing case, that case is modified; otherwise, a new case is added

Return Value

Type: [Int32](#)

Returns zero if the load case is successfully initialized; otherwise it returns a nonzero value

Remarks

See Also

Reference

cCaseHyperStatic Interface

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CSI API ETABS v1

cCaseModalEigen Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cCaseModalEigen
```

```
Public Interface cCaseModalEigen
```

```
Dim instance As cCaseModalEigen
```

```
public interface class cCaseModalEigen
```

```
type cCaseModalEigen = interface end
```

The cCaseModalEigen type exposes the following members.

▲ Methods

▲	Name	Description
	GetInitialCase	This function retrieves the initial condition assumed for the specified load case.
	GetLoads	This function retrieves the load data for the specified load case.
	GetNumberModes	This function retrieves the number of modes requested for the specified load case.
	GetParameters	This function retrieves various parameters for the specified load case.
	SetCase	This function initializes a modal Eigen load case. If this function is called for an existing load case, all items for the case are reset to their default value.
	SetInitialCase	This function sets the initial condition for the specified load case.
	SetLoads	This function sets the load data for the specified load case.
	SetNumberModes	This function sets the number of modes requested for the specified load case.
	SetParameters	This function sets various parameters for the specified load case.

[Top](#)

▲ See Also

Reference

ETABSv1 Namespace

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CSI API ETABS v1

cCaseModalEigen Methods

The [cCaseModalEigen](#) type exposes the following members.

Methods

⌕	Name	Description
🔍	GetInitialCase	This function retrieves the initial condition assumed for the specified load case.
🔍	GetLoads	This function retrieves the load data for the specified load case.
🔍	GetNumberModes	This function retrieves the number of modes requested for the specified load case.
🔍	GetParameters	This function retrieves various parameters for the specified load case.
🔍	SetCase	This function initializes a modal Eigen load case. If this function is called for an existing load case, all items for the case are reset to their default value.
🔍	SetInitialCase	This function sets the initial condition for the specified load case.
🔍	SetLoads	This function sets the load data for the specified load case.
🔍	SetNumberModes	This function sets the number of modes requested for the specified load case.
🔍	SetParameters	This function sets various parameters for the specified load case.

[Top](#)

🔍 See Also

Reference

[cCaseModalEigen Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cCaseModalEigenAddLanguageSpecificTextSet("LST5542BB3_0") Method

This function retrieves the initial condition assumed for the specified load case.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetInitialCase(
    string Name,
    ref string InitialCase
)
```

```
Function GetInitialCase (
    Name As String,
    ByRef InitialCase As String
) As Integer
```

```
Dim instance As cCaseModalEigen
Dim Name As String
Dim InitialCase As String
Dim returnValue As Integer
```

```
returnValue = instance.GetInitialCase(Name,
    InitialCase)
```

```
int GetInitialCase(
    String^ Name,
    String^% InitialCase
)
```

```
abstract GetInitialCase :
    Name : string *
    InitialCase : string byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing modal Eigen load case.

InitialCase

Type: SystemString

This is blank, None, or the name of an existing analysis case. This item specifies if the load case starts from zero initial conditions, that is, an unstressed state, or if it starts using the stiffness that occurs at the end of a nonlinear static or nonlinear direct integration time history load case.

If the specified initial case is a nonlinear static or nonlinear direct integration time history load case, the stiffness at the end of that case is used. If the initial case is anything else, zero initial conditions are assumed.

Return Value

Type: $\hat{A}$ [Int32](#)

Returns zero if the initial condition is successfully retrieved; otherwise it returns a nonzero value.

▲ [Remarks](#)

▲ [Examples](#)

VB

[Copy](#)

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim InitialCase As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'add static nonlinear load case
    ret = SapModel.LoadCases.StaticNonlinear.SetCase("SN1")

    'add modal Eigen load case
    ret = SapModel.LoadCases.ModalEigen.SetCase("LCASE1")

    'set initial condition
    ret = SapModel.LoadCases.ModalEigen.SetInitialCase("LCASE1", "SN1")

    'get initial condition
    ret = SapModel.LoadCases.ModalEigen.GetInitialCase("LCASE1", InitialCase)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ [See Also](#)

Reference

cCaseModalEigen Interface

ETABSv1 Namespace

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CSI API ETABS v1

cCaseModalEigenAddLanguageSpecificTextSet("LST60DD382A_0") Method

This function retrieves the load data for the specified load case.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetLoads(
    string Name,
    ref int NumberLoads,
    ref string[] LoadType,
    ref string[] LoadName,
    ref double[] TargetPar,
    ref bool[] StaticCorrect
)
```

```
Function GetLoads (
    Name As String,
    ByRef NumberLoads As Integer,
    ByRef LoadType As String(),
    ByRef LoadName As String(),
    ByRef TargetPar As Double(),
    ByRef StaticCorrect As Boolean()
) As Integer
```

```
Dim instance As cCaseModalEigen
Dim Name As String
Dim NumberLoads As Integer
Dim LoadType As String()
Dim LoadName As String()
Dim TargetPar As Double()
Dim StaticCorrect As Boolean()
Dim returnValue As Integer
```

```
returnValue = instance.GetLoads(Name,
    NumberLoads, LoadType, LoadName,
    TargetPar, StaticCorrect)
```

```
int GetLoads(
    String^ Name,
    int% NumberLoads,
    array<String^>^% LoadType,
    array<String^>^% LoadName,
    array<double>^% TargetPar,
    array<bool>^% StaticCorrect
)
```

abstract GetLoads :

cCaseModalEigenAddLanguageSpecificTextSet("LST60DD382A_0?579=::|nu=


```

Name : string *
NumberLoads : int byref *
LoadType : string[] byref *
LoadName : string[] byref *
TargetPar : float[] byref *
StaticCorrect : bool[] byref -> int

```

Parameters

Name

Type:Â SystemString

The name of an existing modal Eigen load case.

NumberLoads

Type:Â SystemInt32

The number of loads assigned to the specified analysis case.

LoadType

Type:Â SystemString

This is an array that includes Load, Accel or Link, indicating the type of each load assigned to the load case.

LoadName

Type:Â SystemString

This is an array that includes the name of each load assigned to the load case. If the LoadType item is Load, this item is the name of a defined load pattern. If the LoadType item is Accel, this item is UX, UY, UZ, RX, RY or RZ, indicating the direction of the load. If the LoadType item is Link, this item is not used.

TargetPar

Type:Â SystemDouble

This is an array that includes the target mass participation ratio.

StaticCorrect

Type:Â SystemBoolean

This is an array that includes either 0 or 1, indicating if static correction modes are to be calculated.

Return Value

Type:Â Int32

The function returns zero if the data is successfully retrieved; otherwise it returns a nonzero value.

Remarks

Examples

VB

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim InitialCase As String
    Dim MyLoadType() As String
    Dim MyLoadName() As String
    Dim MyTargetPar() As Double
    Dim MyStaticCorrect() As Boolean
    Dim NumberLoads As Integer
    Dim LoadType() As String
    Dim LoadName() As String

```

Introduction

```
Dim TargetPar() As Double
Dim StaticCorrect() As Boolean

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'add modal Eigen load case
ret = SapModel.LoadCases.ModalEigen.SetCase("LCASE1")

'set load data
ReDim MyLoadType(2)
ReDim MyLoadName(2)
ReDim MyTargetPar(2)
ReDim MyStaticCorrect(2)
MyLoadType(0) = "Load"
MyLoadName(0) = "DEAD"
MyTargetPar(0) = 99
MyStaticCorrect(0) = 1
MyLoadType(1) = "Accel"
MyLoadName(1) = "UZ"
MyTargetPar(1) = 99
MyStaticCorrect(1) = 0
MyLoadType(2) = "Link"
MyTargetPar(2) = 99
MyStaticCorrect(2) = 0
ret = mySapModel.LoadCases.ModalEigen.SetLoads("LCASE1", 3, MyLoadType, MyLoadName, MyTargetPar)

'get load data
ret = mySapModel.LoadCases.ModalEigen.GetLoads("LCASE1", NumberLoads, LoadType, LoadName, TargetPar)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

[cCaseModalEigen Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cCaseModalEigenAddLanguageSpecificTextSet("LST415FDF42_0?576") Method

This function retrieves the number of modes requested for the specified load case.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNumberModes(
    string Name,
    ref int MaxModes,
    ref int MinModes
)
```

```
Function GetNumberModes (
    Name As String,
    ByRef MaxModes As Integer,
    ByRef MinModes As Integer
) As Integer
```

```
Dim instance As cCaseModalEigen
Dim Name As String
Dim MaxModes As Integer
Dim MinModes As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetNumberModes(Name,
    MaxModes, MinModes)
```

```
int GetNumberModes(
    String^ Name,
    int% MaxModes,
    int% MinModes
)
```

```
abstract GetNumberModes :
    Name : string *
    MaxModes : int byref *
    MinModes : int byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing modal Eigen load case.

MaxModes

Type: SystemInt32

The maximum number of modes requested.

MinModes

Type: SystemInt32

The minimum number of modes requested.

Return Value

Type: Int32

The function returns zero if the number of modes is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MaxModes As Integer
    Dim MinModes As Integer

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'add modal Eigen load case
    ret = SapModel.LoadCases.ModalEigen.SetCase("LCASE1")

    'get load data
    ret = mySapModel.LoadCases.ModalEigen.GetNumberModes("LCASE1", MaxModes, MinModes)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cCaseModalEigen Interface

ETABSv1 Namespace

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CSI API ETABS v1

cCaseModalEigenAddLanguageSpecificTextSet("LST6300068B_0")

Method

This function retrieves various parameters for the specified load case.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetParameters(
    string Name,
    ref double EigenShiftFreq,
    ref double EigenCutOff,
    ref double EigenTol,
    ref int AllowAutoFreqShift
)
```

```
Function GetParameters (
    Name As String,
    ByRef EigenShiftFreq As Double,
    ByRef EigenCutOff As Double,
    ByRef EigenTol As Double,
    ByRef AllowAutoFreqShift As Integer
) As Integer
```

```
Dim instance As cCaseModalEigen
Dim Name As String
Dim EigenShiftFreq As Double
Dim EigenCutOff As Double
Dim EigenTol As Double
Dim AllowAutoFreqShift As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetParameters(Name,
    EigenShiftFreq, EigenCutOff, EigenTol,
    AllowAutoFreqShift)
```

```
int GetParameters(
    String^ Name,
    double% EigenShiftFreq,
    double% EigenCutOff,
    double% EigenTol,
    int% AllowAutoFreqShift
)
```

```
abstract GetParameters :
    Name : string *
    EigenShiftFreq : float byref *
    EigenCutOff : float byref *
    EigenTol : float byref *
```

AllowAutoFreqShift : int byref -> int

Parameters

Name

Type:Â SystemString

The name of an existing modal Eigen load case.

EigenShiftFreq

Type:Â SystemDouble

The Eigenvalue shift frequency. [cyc/s]

EigenCutOff

Type:Â SystemDouble

The Eigencutoff frequency radius. [cyc/s]

EigenTol

Type:Â SystemDouble

The relative convergence tolerance for Eigenvalues.

AllowAutoFreqShift

Type:Â SystemInt32

This is either 0 or 1, indicating if automatic frequency shifting is allowed. 0 = Automatic frequency shifting is NOT allowed 1 = Automatic frequency shifting Is allowed

Return Value

Type:Â Int32

The function returns zero if the parameters are successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim EigenShiftFreq As Double
    Dim EigenCutOff As Double
    Dim EigenTol As Double
    Dim AllowAutoFreqShift As Integer

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)
```


Introduction

```
'add modal Eigen load case
    ret = SapModel.LoadCases.ModalEigen.SetCase("LCASE1")

'get load data
    ret = mySapModel.LoadCases.ModalEigen.GetParameters("LCASE1", EigenShiftFreq, EigenCutOff, Eig

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

📌 See Also

Reference

cCaseModalEigen Interface

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CSI API ETABS v1

cCaseModalEigenAddLanguageSpecificTextSet("LSTC Method

This function initializes a modal Eigen load case. If this function is called for an existing load case, all items for the case are reset to their default value.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetCase(
    string Name
)
```

```
Function SetCase (
    Name As String
) As Integer
```

```
Dim instance As cCaseModalEigen
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.SetCase(Name)
```

```
int SetCase(
    String^ Name
)
```

```
abstract SetCase :
    Name : string -> int
```

Parameters

Name

Type: SystemString

The name of an existing or new load case. If this is an existing case, that case is modified; otherwise, a new case is added.

Return Value

Type: Int32

The function returns zero if the load case is successfully initialized; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

cCaseModalEigenspan id="LSTCC7DB888_0"AddLanguageSpecificTextSet("LSTCC7DB888_0582p=::|nu=

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'add modal Eigen load case
    ret = SapModel.LoadCases.ModalEigen.SetCase("LCASE1")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

See Also

Reference

cCaseModalEigen Interface

ETABSv1 Namespace

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CSI API ETABS v1

cCaseModalEigenAddLanguageSpecificTextSet("LST20079D90_0")

Method

This function sets the initial condition for the specified load case.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetInitialCase(
    string Name,
    string InitialCase
)
```

```
Function SetInitialCase (
    Name As String,
    InitialCase As String
) As Integer
```

```
Dim instance As cCaseModalEigen
Dim Name As String
Dim InitialCase As String
Dim returnValue As Integer
```

```
returnValue = instance.SetInitialCase(Name,
    InitialCase)
```

```
int SetInitialCase(
    String^ Name,
    String^ InitialCase
)
```

```
abstract SetInitialCase :
    Name : string *
    InitialCase : string -> int
```

Parameters

Name

Type: SystemString

The name of an existing modal Eigen load case.

InitialCase

Type: SystemString

This is blank, None, or the name of an existing analysis case. This item specifies if the load case starts from zero initial conditions, that is, an unstressed state, or if it starts using the stiffness that occurs at the end of a nonlinear static or nonlinear direct integration time history load case.

If the specified initial case is a nonlinear static or nonlinear direct integration time history load case, the stiffness at the end of that case is used. If the initial case is anything else, zero initial conditions are assumed.

Return Value

Type: [Int32](#)

Returns zero if the initial condition is successfully retrieved; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

[Copy](#)

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim InitialCase As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'add static nonlinear load case
    ret = SapModel.LoadCases.StaticNonlinear.SetCase("SN1")

    'add modal Eigen load case
    ret = SapModel.LoadCases.ModalEigen.SetCase("LCASE1")

    'set initial condition
    ret = SapModel.LoadCases.ModalEigen.SetInitialCase("LCASE1", "SN1")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ **See Also**

Reference

cCaseModalEigen Interface

ETABSv1 Namespace

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CSI API ETABS v1

cCaseModalEigenAddLanguageSpecificTextSet("LSTA Method

This function sets the load data for the specified load case.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetLoads(
    string Name,
    int NumberLoads,
    ref string[] LoadType,
    ref string[] LoadName,
    ref double[] TargetPar,
    ref bool[] StaticCorrect
)
```

```
Function SetLoads (
    Name As String,
    NumberLoads As Integer,
    ByRef LoadType As String(),
    ByRef LoadName As String(),
    ByRef TargetPar As Double(),
    ByRef StaticCorrect As Boolean()
) As Integer
```

```
Dim instance As cCaseModalEigen
Dim Name As String
Dim NumberLoads As Integer
Dim LoadType As String()
Dim LoadName As String()
Dim TargetPar As Double()
Dim StaticCorrect As Boolean()
Dim returnValue As Integer
```

```
returnValue = instance.SetLoads(Name,
    NumberLoads, LoadType, LoadName,
    TargetPar, StaticCorrect)
```

```
int SetLoads(
    String^ Name,
    int NumberLoads,
    array<String^>^% LoadType,
    array<String^>^% LoadName,
    array<double>^% TargetPar,
    array<bool>^% StaticCorrect
)
```

abstract SetLoads :

cCaseModalEigenspan id="LSTA8A01C44_0"AddLanguageSpecificTextSet("LSTA8A01C44_0?587=::|nu=.

```

Name : string *
NumberLoads : int *
LoadType : string[] byref *
LoadName : string[] byref *
TargetPar : float[] byref *
StaticCorrect : bool[] byref -> int

```

Parameters

Name

Type:Â SystemString

The name of an existing modal Eigen load case.

NumberLoads

Type:Â SystemInt32

The number of loads assigned to the specified analysis case.

LoadType

Type:Â SystemString

This is an array that includes Load, Accel or Link, indicating the type of each load assigned to the load case.

LoadName

Type:Â SystemString

This is an array that includes the name of each load assigned to the load case. If the LoadType item is Load, this item is the name of a defined load pattern. If the LoadType item is Accel, this item is UX, UY, UZ, RX, RY or RZ, indicating the direction of the load. If the LoadType item is Link, this item is not used.

TargetPar

Type:Â SystemDouble

This is an array that includes the target mass participation ratio.

StaticCorrect

Type:Â SystemBoolean

This is an array that includes either 0 or 1, indicating if static correction modes are to be calculated.

Return Value

Type:Â Int32

The function returns zero if the data is successfully set; otherwise it returns a nonzero value.

Remarks

Examples

VB

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim InitialCase As String
    Dim MyLoadType() As String
    Dim MyLoadName() As String
    Dim MyTargetPar() As Double
    Dim MyStaticCorrect() As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper

```


Introduction

```
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'add modal Eigen load case
ret = SapModel.LoadCases.ModalEigen.SetCase("LCASE1")

'set load data
ReDim MyLoadType(2)
ReDim MyLoadName(2)
ReDim MyTargetPar(2)
ReDim MyStaticCorrect(2)
MyLoadType(0) = "Load"
MyLoadName(0) = "DEAD"
MyTargetPar(0) = 99
MyStaticCorrect(0) = 1
MyLoadType(1) = "Accel"
MyLoadName(1) = "UZ"
MyTargetPar(1) = 99
MyStaticCorrect(1) = 0
MyLoadType(2) = "Link"
MyTargetPar(2) = 99
MyStaticCorrect(2) = 0

'add modal Eigen load case
ret = SapModel.LoadCases.ModalEigen.SetLoads("LCASE1", 3, MyLoadType, MyLoadName, MyTargetPar)

'set load data
ReDim MyLoadType(2)
ReDim MyLoadName(2)
ReDim MyTargetPar(2)
ReDim MyStaticCorrect(2)
MyLoadType(0) = "Load"
MyLoadName(0) = "DEAD"
MyTargetPar(0) = 99
MyStaticCorrect(0) = 1
MyLoadType(1) = "Accel"
MyLoadName(1) = "UZ"
MyTargetPar(1) = 99
MyStaticCorrect(1) = 0
MyLoadType(2) = "Link"
MyTargetPar(2) = 99
MyStaticCorrect(2) = 0
ret = mySapModel.LoadCases.ModalEigen.SetLoads("LCASE1", 3, MyLoadType, MyLoadName, MyTargetPar)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
```

End Sub

▲ See Also

Reference

cCaseModalEigen Interface

ETABSv1 Namespace

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CSI API ETABS v1

cCaseModalEigenAddLanguageSpecificTextSet("LST864F68ED_0") Method

This function sets the number of modes requested for the specified load case.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetNumberModes(
    string Name,
    int MaxModes,
    int MinModes
)
```

```
Function SetNumberModes (
    Name As String,
    MaxModes As Integer,
    MinModes As Integer
) As Integer
```

```
Dim instance As cCaseModalEigen
Dim Name As String
Dim MaxModes As Integer
Dim MinModes As Integer
Dim returnValue As Integer
```

```
returnValue = instance.SetNumberModes(Name,
    MaxModes, MinModes)
```

```
int SetNumberModes(
    String^ Name,
    int MaxModes,
    int MinModes
)
```

```
abstract SetNumberModes :
    Name : string *
    MaxModes : int *
    MinModes : int -> int
```

Parameters

Name

Type: SystemString

The name of an existing modal Eigen load case.

MaxModes

Type: [SystemInt32](#)

The maximum number of modes requested.

MinModes

Type: [SystemInt32](#)

The minimum number of modes requested.

Return Value

Type: [Int32](#)

The function returns zero if the number of modes is successfully set; otherwise it returns a nonzero value.

▀ Remarks

▀ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'add modal Eigen load case
    ret = SapModel.LoadCases.ModalEigen.SetCase("LCASE1")

    'get load data
    ret = mySapModel.LoadCases.ModalEigen.SetNumberModes("LCASE1", 10, 2)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▀ See Also

Reference

[cCaseModalEigen Interface](#)

[ETABSv1 Namespace](#)

Introduction

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CSI API ETABS v1

cCaseModalEigenAddLanguageSpecificTextSet("LSTD Method

This function sets various parameters for the specified load case.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetParameters(
    string Name,
    double EigenShiftFreq,
    double EigenCutOff,
    double EigenTol,
    int AllowAutoFreqShift
)
```

```
Function SetParameters (
    Name As String,
    EigenShiftFreq As Double,
    EigenCutOff As Double,
    EigenTol As Double,
    AllowAutoFreqShift As Integer
) As Integer
```

```
Dim instance As cCaseModalEigen
Dim Name As String
Dim EigenShiftFreq As Double
Dim EigenCutOff As Double
Dim EigenTol As Double
Dim AllowAutoFreqShift As Integer
Dim returnValue As Integer
```

```
returnValue = instance.SetParameters(Name,
    EigenShiftFreq, EigenCutOff, EigenTol,
    AllowAutoFreqShift)
```

```
int SetParameters(
    String^ Name,
    double EigenShiftFreq,
    double EigenCutOff,
    double EigenTol,
    int AllowAutoFreqShift
)
```

```
abstract SetParameters :
    Name : string *
    EigenShiftFreq : float *
    EigenCutOff : float *
    EigenTol : float *
```

cCaseModalEigenspan id="LSTDD089042_0"AddLanguageSpecificTextSet("LSTDD089042_0?594=::|nu=.

AllowAutoFreqShift : int -> int

Parameters

Name

Type:Â SystemString

The name of an existing modal Eigen load case.

EigenShiftFreq

Type:Â SystemDouble

The Eigenvalue shift frequency. [cyc/s]

EigenCutOff

Type:Â SystemDouble

The Eigencutoff frequency radius. [cyc/s]

EigenTol

Type:Â SystemDouble

The relative convergence tolerance for Eigenvalues.

AllowAutoFreqShift

Type:Â SystemInt32

This is either 0 or 1, indicating if automatic frequency shifting is allowed. 0 = Automatic frequency shifting is NOT allowed 1 = Automatic frequency shifting Is allowed

Return Value

Type:Â Int32

The function returns zero if the parameters are successfully set; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'add modal Eigen load case
    ret = SapModel.LoadCases.ModalEigen.SetCase("LCASE1")

    'get load data
```

Parameters

Introduction

```
ret = mySapModel.LoadCases.ModalEigen.SetParameters("LCASE1", 0.05, 0.0001, 1E-10, 1)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

📌 See Also

Reference

cCaseModalEigen Interface

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CSI API ETABS v1

cCaseModalHistoryLinear Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cCaseModalHistoryLinear
```

```
Public Interface cCaseModalHistoryLinear
```

```
Dim instance As cCaseModalHistoryLinear
```

```
public interface class cCaseModalHistoryLinear
```

```
type cCaseModalHistoryLinear = interface end
```

The cCaseModalHistoryLinear type exposes the following members.

▲ Methods

▲	Name	Description
---	------	-------------

🔗	GetLoads	Retrieves the load data for the specified load case
---	--------------------------	---

🔗	SetCase	Initializes a linear modal history analysis case.
---	-------------------------	---

🔗	SetLoads	
---	--------------------------	--

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cCaseModalHistoryLinear Methods

The cCaseModalHistoryLinear type exposes the following members.

Methods

⌕	Name	Description
⇒	<u>GetLoads</u>	Retrieves the load data for the specified load case
⇒	<u>SetCase</u>	Initializes a linear modal history analysis case.
⇒	<u>SetLoads</u>	

Top

⌕ See Also

Reference

cCaseModalHistoryLinear Interface

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CSI API ETABS v1

cCaseModalHistoryLinearAddLanguageSpecificTextSet Method

Retrieves the load data for the specified load case

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetLoads(
    string Name,
    ref int NumberLoads,
    ref string[] LoadType,
    ref string[] LoadName,
    ref string[] Func,
    ref double[] SF,
    ref double[] Tf,
    ref double[] At,
    ref string[] CSys,
    ref double[] Ang
)
```

```
Function GetLoads (
    Name As String,
    ByRef NumberLoads As Integer,
    ByRef LoadType As String(),
    ByRef LoadName As String(),
    ByRef Func As String(),
    ByRef SF As Double(),
    ByRef Tf As Double(),
    ByRef At As Double(),
    ByRef CSys As String(),
    ByRef Ang As Double()
) As Integer
```

```
Dim instance As cCaseModalHistoryLinear
Dim Name As String
Dim NumberLoads As Integer
Dim LoadType As String()
Dim LoadName As String()
Dim Func As String()
Dim SF As Double()
Dim Tf As Double()
Dim At As Double()
Dim CSys As String()
Dim Ang As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetLoads(Name,
    NumberLoads, LoadType, LoadName,
```

cCaseModalHistoryLinearspan id="LSTD2BB391_0"AddLanguageSpecificTextSet("LSTD2BB391_0?cpp=::

Func, SF, Tf, At, CSys, Ang)

```
int GetLoads(
String^ Name,
int% NumberLoads,
array<String^>^% LoadType,
array<String^>^% LoadName,
array<String^>^% Func,
array<double>^% SF,
array<double>^% Tf,
array<double>^% At,
array<String^>^% CSys,
array<double>^% Ang
)
```

```
abstract GetLoads :
    Name : string *
    NumberLoads : int byref *
    LoadType : string[] byref *
    LoadName : string[] byref *
    Func : string[] byref *
    SF : float[] byref *
    Tf : float[] byref *
    At : float[] byref *
    CSys : string[] byref *
    Ang : float[] byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing linear modal history analysis case

NumberLoads

Type:Â SystemInt32

The number of loads assigned to the specified analysis case

LoadType

Type:Â SystemString

This is an array that includes Load or Accel, indicating the type of each load assigned to the load case

LoadName

Type:Â SystemString

This is an array that includes the name of each load assigned to the load case.

If the LoadType item is Load, this item is the name of a defined load pattern.

If the LoadType item is Accel, this item is U1, U2, U3, R1, R2 or R3, indicating the direction of the load.

Func

Type:Â SystemString

This is an array that includes the name of the time history function associated with each load

SF

Type:Â SystemDouble

This is an array that includes the scale factor of each load assigned to the load case. [L/s²] for U1 U2 and U3; otherwise unitless

Tf

Type: SystemDouble

This is an array that includes the time scale factor of each load assigned to the load case

At

Type: SystemDouble

This is an array that includes the arrival time of each load assigned to the load case

CSys

Type: SystemString

This is an array that includes the name of the coordinate system associated with each load. If this item is a blank string, the Global coordinate system is assumed.

This item applies only when the LoadType item is Accel.

Ang

Type: SystemDouble

This is an array that includes the angle between the acceleration local 1 axis and the +X-axis of the coordinate system specified by the CSys item. The rotation is about the Z-axis of the specified coordinate system. [deg]

This item applies only when the LoadType item is Accel.

Return Value

Type: Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ See Also

Reference

cCaseModalHistoryLinear Interface

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CSI API ETABS v1

cCaseModalHistoryLinearAddLanguageSpecificTextSet Method

Initializes a linear modal history analysis case.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetCase(
    string Name
)
```

```
Function SetCase (
    Name As String
) As Integer
```

```
Dim instance As cCaseModalHistoryLinear
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.SetCase(Name)
```

```
int SetCase(
    String^ Name
)
```

```
abstract SetCase :
    Name : string -> int
```

Parameters

Name

Type: SystemString

The name of an existing or new load case. If this is an existing case, that case is modified; otherwise a new case is added.

Return Value

Type: Int32

Returns zero if the load case is successfully initialized; otherwise it returns a nonzero value.

▲ Remarks

If this function is called for an existing load case then all items for the case are reset to their default value.

▲ Examples

cCaseModalHistoryLinearid="LSTB7811A5_0"AddLanguageSpecificTextSet("LSTB7811A5_02cpp=::|

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add linear modal history load case
    ret = SapModel.LoadCases.ModHistLinear.SetCase("MHLCASE1")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

 See Also
Reference[cCaseModalHistoryLinear Interface](#)[ETABSV1 Namespace](#)

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CSI API ETABS v1

cCaseModalHistoryLinearAddLanguageSpecificTextSet Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetLoads(
    string Name,
    int NumberLoads,
    ref string[] LoadType,
    ref string[] LoadName,
    ref string[] Func,
    ref double[] SF,
    ref double[] Tf,
    ref double[] At,
    ref string[] CSys,
    ref double[] Ang
)
```

```
Function SetLoads (
    Name As String,
    NumberLoads As Integer,
    ByRef LoadType As String(),
    ByRef LoadName As String(),
    ByRef Func As String(),
    ByRef SF As Double(),
    ByRef Tf As Double(),
    ByRef At As Double(),
    ByRef CSys As String(),
    ByRef Ang As Double()
) As Integer
```

```
Dim instance As cCaseModalHistoryLinear
Dim Name As String
Dim NumberLoads As Integer
Dim LoadType As String()
Dim LoadName As String()
Dim Func As String()
Dim SF As Double()
Dim Tf As Double()
Dim At As Double()
Dim CSys As String()
Dim Ang As Double()
Dim returnValue As Integer
```

```
returnValue = instance.SetLoads(Name,
    NumberLoads, LoadType, LoadName,
    Func, SF, Tf, At, CSys, Ang)
```



```

int SetLoads(
String^ Name,
int NumberLoads,
array<String^>^% LoadType,
array<String^>^% LoadName,
array<String^>^% Func,
array<double>^% SF,
array<double>^% Tf,
array<double>^% At,
array<String^>^% CSys,
array<double>^% Ang
)

abstract SetLoads :
    Name : string *
    NumberLoads : int *
    LoadType : string[] byref *
    LoadName : string[] byref *
    Func : string[] byref *
    SF : float[] byref *
    Tf : float[] byref *
    At : float[] byref *
    CSys : string[] byref *
    Ang : float[] byref -> int

```

Parameters

Name
Type:Â [SystemString](#)

NumberLoads
Type:Â [SystemInt32](#)

LoadType
Type:Â [SystemString](#)

LoadName
Type:Â [SystemString](#)

Func
Type:Â [SystemString](#)

SF
Type:Â [SystemDouble](#)

Tf
Type:Â [SystemDouble](#)

At
Type:Â [SystemDouble](#)

CSys
Type:Â [SystemString](#)

Ang
Type:Â [SystemDouble](#)

Return Value

Type:Â [Int32](#)
 ▲ See Also

Reference

cCaseModalHistoryLinear Interface

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CSI API ETABS v1

cCaseModalHistoryNonlinear Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cCaseModalHistoryNonlinear
```

```
Public Interface cCaseModalHistoryNonlinear
```

```
Dim instance As cCaseModalHistoryNonlinear
```

```
public interface class cCaseModalHistoryNonlinear
```

```
type cCaseModalHistoryNonlinear = interface end
```

The cCaseModalHistoryNonlinear type exposes the following members.

▲ Methods

▲	Name	Description
---	------	-------------

🔗	GetLoads	Retrieves the load data for the specified load case
---	--------------------------	---

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cCaseModalHistoryNonlinear Methods

The cCaseModalHistoryNonlinear type exposes the following members.

Methods

Â	Name	Description
🔗	<u>GetLoads</u>	Retrieves the load data for the specified load case

Top

🔗 See Also

Reference

cCaseModalHistoryNonlinear Interface

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CSI API ETABS v1

cCaseModalHistoryNonlinearAddLanguageSpecificTextMethod

Retrieves the load data for the specified load case

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetLoads(
    string Name,
    ref int NumberLoads,
    ref string[] LoadType,
    ref string[] LoadName,
    ref string[] Func,
    ref double[] SF,
    ref double[] Tf,
    ref double[] At,
    ref string[] CSys,
    ref double[] Ang
)
```

```
Function GetLoads (
    Name As String,
    ByRef NumberLoads As Integer,
    ByRef LoadType As String(),
    ByRef LoadName As String(),
    ByRef Func As String(),
    ByRef SF As Double(),
    ByRef Tf As Double(),
    ByRef At As Double(),
    ByRef CSys As String(),
    ByRef Ang As Double()
) As Integer
```

```
Dim instance As cCaseModalHistoryNonlinear
Dim Name As String
Dim NumberLoads As Integer
Dim LoadType As String()
Dim LoadName As String()
Dim Func As String()
Dim SF As Double()
Dim Tf As Double()
Dim At As Double()
Dim CSys As String()
Dim Ang As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetLoads(Name,
    NumberLoads, LoadType, LoadName,
```

Func, SF, Tf, At, CSys, Ang)

```
int GetLoads(
String^ Name,
int% NumberLoads,
array<String^>^% LoadType,
array<String^>^% LoadName,
array<String^>^% Func,
array<double>^% SF,
array<double>^% Tf,
array<double>^% At,
array<String^>^% CSys,
array<double>^% Ang
)
```

```
abstract GetLoads :
    Name : string *
    NumberLoads : int byref *
    LoadType : string[] byref *
    LoadName : string[] byref *
    Func : string[] byref *
    SF : float[] byref *
    Tf : float[] byref *
    At : float[] byref *
    CSys : string[] byref *
    Ang : float[] byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing nonlinear modal history analysis case

NumberLoads

Type:Â SystemInt32

The number of loads assigned to the specified analysis case

LoadType

Type:Â SystemString

This is an array that includes Load or Accel, indicating the type of each load assigned to the load case

LoadName

Type:Â SystemString

This is an array that includes the name of each load assigned to the load case.

If the LoadType item is Load, this item is the name of a defined load pattern.

If the LoadType item is Accel, this item is U1, U2, U3, R1, R2 or R3, indicating the direction of the load.

Func

Type:Â SystemString

This is an array that includes the name of the time history function associated with each load

SF

Type:Â SystemDouble

This is an array that includes the scale factor of each load assigned to the load case. [L/s²] for U1 U2 and U3; otherwise unitless

Tf

Type: SystemDouble

This is an array that includes the time scale factor of each load assigned to the load case

At

Type: SystemDouble

This is an array that includes the arrival time of each load assigned to the load case

CSys

Type: SystemString

This is an array that includes the name of the coordinate system associated with each load. If this item is a blank string, the Global coordinate system is assumed.

This item applies only when the LoadType item is Accel.

Ang

Type: SystemDouble

This is an array that includes the angle between the acceleration local 1 axis and the +X-axis of the coordinate system specified by the CSys item. The rotation is about the Z-axis of the specified coordinate system. [deg]

This item applies only when the LoadType item is Accel.

Return Value

Type: Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ See Also

Reference

cCaseModalHistoryNonlinear Interface

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CSI API ETABS v1

cCaseModalRitz Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cCaseModalRitz
```

Public Interface cCaseModalRitz

```
Dim instance As cCaseModalRitz
```

```
public interface class cCaseModalRitz
```

```
type cCaseModalRitz = interface end
```

The cCaseModalRitz type exposes the following members.

▲ Methods

▲	Name	Description
➤	GetInitialCase	This function retrieves the initial condition assumed for the specified load case.
➤	GetLoads	This function retrieves the load data for the specified load case.
➤	GetNumberModes	This function retrieves the number of modes requested for the specified load case.
➤	SetCase	This function initializes a modal Ritz load case. If this function is called for an existing load case, all items for the case are reset to their default value.
➤	SetInitialCase	This function sets the initial condition for the specified load case.
➤	SetLoads	This function sets the load data for the specified load case.
➤	SetNumberModes	This function sets the number of modes requested for the specified load case.

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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Introduction

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CSI API ETABS v1

cCaseModalRitz Methods

The cCaseModalRitz type exposes the following members.

Methods

⌚	Name	Description
⇒	<u>GetInitialCase</u>	This function retrieves the initial condition assumed for the specified load case.
⇒	<u>GetLoads</u>	This function retrieves the load data for the specified load case.
⇒	<u>GetNumberModes</u>	This function retrieves the number of modes requested for the specified load case.
⇒	<u>SetCase</u>	This function initializes a modal Ritz load case. If this function is called for an existing load case, all items for the case are reset to their default value.
⇒	<u>SetInitialCase</u>	This function sets the initial condition for the specified load case.
⇒	<u>SetLoads</u>	This function sets the load data for the specified load case.
⇒	<u>SetNumberModes</u>	This function sets the number of modes requested for the specified load case.

Top

⌚ See Also

Reference

cCaseModalRitz Interface

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»

CSI API ETABS v1

cCaseModalRitzAddLanguageSpecificTextSet("LST4CF24935_0?cpp645|nu=."); Method

This function retrieves the initial condition assumed for the specified load case.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetInitialCase(
    string Name,
    ref string InitialCase
)
```

```
Function GetInitialCase (
    Name As String,
    ByRef InitialCase As String
) As Integer
```

```
Dim instance As cCaseModalRitz
Dim Name As String
Dim InitialCase As String
Dim returnValue As Integer
```

```
returnValue = instance.GetInitialCase(Name,
    InitialCase)
```

```
int GetInitialCase(
    String^ Name,
    String^% InitialCase
)
```

```
abstract GetInitialCase :
    Name : string *
    InitialCase : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing modal Ritz load case.

InitialCase

Type: [SystemString](#)

This is blank, None, or the name of an existing analysis case. This item specifies if the load case starts from zero initial conditions, that is, an unstressed state, or if it starts using the stiffness that occurs at the end of a nonlinear static or nonlinear direct integration time history load case.

cCaseModalRitzspan id="LST4CF24935_0"AddLanguageSpecificTextSet("LST4CF24935_0?cpp645|nu=.");

If the specified initial case is a nonlinear static or nonlinear direct integration time history load case, the stiffness at the end of that case is used. If the initial case is anything else, zero initial conditions are assumed.

Return Value

Type: [Int32](#)

Returns zero if the initial condition is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

[Copy](#)

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim InitialCase As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'add static nonlinear load case
    ret = SapModel.LoadCases.StaticNonlinear.SetCase("SN1")

    'add modal Ritz load case
    ret = SapModel.LoadCases.ModalRitz.SetCase("LCASE1")

    'set initial condition
    ret = SapModel.LoadCases.ModalRitz.SetInitialCase("LCASE1", "SN1")

    'get initial condition
    ret = SapModel.LoadCases.ModalRitz.GetInitialCase("LCASE1", InitialCase)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cCaseModalRitz Interface

ETABSv1 Namespace

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CSI API ETABS v1

cCaseModalRitzAddLanguageSpecificTextSet("LST70B0A51C_0?cp18:|nu=.") Method

This function retrieves the load data for the specified load case.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetLoads(
    string Name,
    ref int NumberLoads,
    ref string[] LoadType,
    ref string[] LoadName,
    ref int[] RitzMaxCyc,
    ref double[] TargetPar
)
```

```
Function GetLoads (
    Name As String,
    ByRef NumberLoads As Integer,
    ByRef LoadType As String(),
    ByRef LoadName As String(),
    ByRef RitzMaxCyc As Integer(),
    ByRef TargetPar As Double()
) As Integer
```

```
Dim instance As cCaseModalRitz
Dim Name As String
Dim NumberLoads As Integer
Dim LoadType As String()
Dim LoadName As String()
Dim RitzMaxCyc As Integer()
Dim TargetPar As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetLoads(Name,
    NumberLoads, LoadType, LoadName,
    RitzMaxCyc, TargetPar)
```

```
int GetLoads(
    String^ Name,
    int% NumberLoads,
    array<String^>^% LoadType,
    array<String^>^% LoadName,
    array<int>^% RitzMaxCyc,
    array<double>^% TargetPar
)
```

abstract GetLoads :

cCaseModalRitzAddLanguageSpecificTextSet("LST70B0A51C_0?cp18:|nu=.")

```

Name : string *
NumberLoads : int byref *
LoadType : string[] byref *
LoadName : string[] byref *
RitzMaxCyc : int[] byref *
TargetPar : float[] byref -> int

```

Parameters

Name

Type:Â SystemString

The name of an existing modal Ritz load case.

NumberLoads

Type:Â SystemInt32

The number of loads assigned to the specified analysis case.

LoadType

Type:Â SystemString

This is an array that includes Load, Accel or Link, indicating the type of each load assigned to the load case.

LoadName

Type:Â SystemString

This is an array that includes the name of each load assigned to the load case. If the LoadType item is Load, this item is the name of a defined load pattern. If the LoadType item is Accel, this item is UX, UY, UZ, RX, RY or RZ, indicating the direction of the load. If the LoadType item is Link, this item is not used.

RitzMaxCyc

Type:Â SystemInt32

This is an array that includes the maximum number of generation cycles to be performed for the specified ritz starting vector. A value of 0 means there is no limit on the number of cycles.

TargetPar

Type:Â SystemDouble

This is an array that includes the target mass participation ratio.

Return Value

Type:Â Int32

The function returns zero if the data is successfully retrieved; otherwise it returns a nonzero value.

Remarks

Examples

VB

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim InitialCase As String
    Dim MyLoadType() As String
    Dim MyLoadName() As String
    Dim MyRitzMaxCyc() As Integer
    Dim MyTargetPar() As Double
    Dim NumberLoads As Integer
    Dim LoadType() As String

```

Introduction

```
Dim LoadName() As String
Dim RitzMaxCyc() As Integer
Dim TargetPar() As Double

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'add modal Ritz load case
ret = SapModel.LoadCases.ModalRitz.SetCase("LCASE1")

'set load data
ReDim MyLoadType(2)
ReDim MyLoadName(2)
ReDim MyRitzMaxCyc(2)
ReDim MyTargetPar(2)
MyLoadType(0) = "Load"
MyLoadName(0) = "DEAD"
MyRitzMaxCyc(0) = 0
MyTargetPar(0) = 99
MyLoadType(1) = "Accel"
MyLoadName(1) = "UZ"
MyRitzMaxCyc(1) = 0
MyTargetPar(1) = 99
MyLoadType(2) = "Link"
MyRitzMaxCyc(2) = 0
MyTargetPar(2) = 99
ret = mySapModel.LoadCases.ModalRitz.SetLoads("LCASE1", 3, MyLoadType, MyLoadName, MyRitzMaxCyc)

'get load data
ret = mySapModel.LoadCases.ModalRitz.GetLoads("LCASE1", NumberLoads, LoadType, LoadName, RitzMaxCyc)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cCaseModalRitz Interface

ETABSV1 Namespace

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CSI API ETABS v1

cCaseModalRitzAddLanguageSpecificTextSet("LSTC516B6CD_0") Method

This function retrieves the number of modes requested for the specified load case.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNumberModes(
    string Name,
    ref int MaxModes,
    ref int MinModes
)
```

```
Function GetNumberModes (
    Name As String,
    ByRef MaxModes As Integer,
    ByRef MinModes As Integer
) As Integer
```

```
Dim instance As cCaseModalRitz
Dim Name As String
Dim MaxModes As Integer
Dim MinModes As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetNumberModes(Name,
    MaxModes, MinModes)
```

```
int GetNumberModes(
    String^ Name,
    int% MaxModes,
    int% MinModes
)
```

```
abstract GetNumberModes :
    Name : string *
    MaxModes : int byref *
    MinModes : int byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing modal Ritz load case.

MaxModes

Type: SystemInt32

The maximum number of modes requested.

MinModes

Type: SystemInt32

The minimum number of modes requested.

Return Value

Type: Int32

The function returns zero if the number of modes is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MaxModes As Integer
    Dim MinModes As Integer

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'add modal Ritz load case
    ret = SapModel.LoadCases.ModalRitz.SetCase("LCASE1")

    'get load data
    ret = mySapModel.LoadCases.ModalRitz.GetNumberModes("LCASE1", MaxModes, MinModes)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cCaseModalRitz Interface

ETABSv1 Namespace

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CSI API ETABS v1

cCaseModalRitzAddLanguageSpecificTextSet("LSTEFD39A95_0?comp25:|nu=.") Method

This function initializes a modal Ritz load case. If this function is called for an existing load case, all items for the case are reset to their default value.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetCase(
    string Name
)
```

```
Function SetCase (
    Name As String
) As Integer
```

```
Dim instance As cCaseModalRitz
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.SetCase(Name)
```

```
int SetCase(
    String^ Name
)
```

```
abstract SetCase :
    Name : string -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing or new load case. If this is an existing case, that case is modified; otherwise, a new case is added.

Return Value

Type: [Int32](#)

The function returns zero if the load case is successfully initialized; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

cCaseModalRitzAddLanguageSpecificTextSet("LSTEFD39A95_0?comp25:|nu=.")

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'add modal Ritz load case
    ret = SapModel.LoadCases.ModalRitz.SetCase("LCASE1")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub

```

▲ See Also

ReferencecCaseModalRitz InterfaceETABSv1 Namespace

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CSI API ETABS v1

cCaseModalRitzAddLanguageSpecificTextSet("LSTE8929C76_0?cp627|nu=.") Method

This function sets the initial condition for the specified load case.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetInitialCase(
    string Name,
    string InitialCase
)
```

```
Function SetInitialCase (
    Name As String,
    InitialCase As String
) As Integer
```

```
Dim instance As cCaseModalRitz
Dim Name As String
Dim InitialCase As String
Dim returnValue As Integer
```

```
returnValue = instance.SetInitialCase(Name,
    InitialCase)
```

```
int SetInitialCase(
    String^ Name,
    String^ InitialCase
)
```

```
abstract SetInitialCase :
    Name : string *
    InitialCase : string -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing modal Ritz load case.

InitialCase

Type: [SystemString](#)

This is blank, None, or the name of an existing analysis case. This item specifies if the load case starts from zero initial conditions, that is, an unstressed state, or if it starts using the stiffness that occurs at the end of a nonlinear static or nonlinear direct integration time history load case.

cCaseModalRitzspan id="LSTE8929C76_0"AddLanguageSpecificTextSet("LSTE8929C76_0?cp627|nu=.")

If the specified initial case is a nonlinear static or nonlinear direct integration time history load case, the stiffness at the end of that case is used. If the initial case is anything else, zero initial conditions are assumed.

Return Value

Type: $\hat{A}$ [Int32](#)

Returns zero if the initial condition is successfully retrieved; otherwise it returns a nonzero value.

▲ [Remarks](#)

▲ [Examples](#)

VB

[Copy](#)

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim InitialCase As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'add static nonlinear load case
    ret = SapModel.LoadCases.StaticNonlinear.SetCase("SN1")

    'add modal Ritz load case
    ret = SapModel.LoadCases.ModalRitz.SetCase("LCASE1")

    'set initial condition
    ret = SapModel.LoadCases.ModalRitz.SetInitialCase("LCASE1", "SN1")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ [See Also](#)

Reference

cCaseModalRitz Interface

ETABSv1 Namespace

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CSI API ETABS v1

cCaseModalRitzAddLanguageSpecificTextSet("LST31E06942_0?cpu=.") Method

This function sets the load data for the specified load case.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetLoads(
    string Name,
    int NumberLoads,
    ref string[] LoadType,
    ref string[] LoadName,
    ref int[] RitzMaxCyc,
    ref double[] TargetPar
)
```

```
Function SetLoads (
    Name As String,
    NumberLoads As Integer,
    ByRef LoadType As String(),
    ByRef LoadName As String(),
    ByRef RitzMaxCyc As Integer(),
    ByRef TargetPar As Double()
) As Integer
```

```
Dim instance As cCaseModalRitz
Dim Name As String
Dim NumberLoads As Integer
Dim LoadType As String()
Dim LoadName As String()
Dim RitzMaxCyc As Integer()
Dim TargetPar As Double()
Dim returnValue As Integer
```

```
returnValue = instance.SetLoads(Name,
    NumberLoads, LoadType, LoadName,
    RitzMaxCyc, TargetPar)
```

```
int SetLoads(
    String^ Name,
    int NumberLoads,
    array<String^>^% LoadType,
    array<String^>^% LoadName,
    array<int>^% RitzMaxCyc,
    array<double>^% TargetPar
)
```

abstract SetLoads :

cCaseModalRitzAddLanguageSpecificTextSet("LST31E06942_0?cpu=.");S

```

Name : string *
NumberLoads : int *
LoadType : string[] byref *
LoadName : string[] byref *
RitzMaxCyc : int[] byref *
TargetPar : float[] byref -> int

```

Parameters

Name

Type:Â SystemString

The name of an existing modal Ritz load case.

NumberLoads

Type:Â SystemInt32

The number of loads assigned to the specified analysis case.

LoadType

Type:Â SystemString

This is an array that includes Load, Accel or Link, indicating the type of each load assigned to the load case.

LoadName

Type:Â SystemString

This is an array that includes the name of each load assigned to the load case. If the LoadType item is Load, this item is the name of a defined load pattern. If the LoadType item is Accel, this item is UX, UY, UZ, RX, RY or RZ, indicating the direction of the load. If the LoadType item is Link, this item is not used.

RitzMaxCyc

Type:Â SystemInt32

This is an array that includes the maximum number of generation cycles to be performed for the specified ritz starting vector. A value of 0 means there is no limit on the number of cycles.

TargetPar

Type:Â SystemDouble

This is an array that includes the target mass participation ratio.

Return Value

Type:Â Int32

The function returns zero if the data is successfully set; otherwise it returns a nonzero value.

Remarks

Examples

VB

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim InitialCase As String
    Dim MyLoadType() As String
    Dim MyLoadName() As String
    Dim MyRitzMaxCyc() As Integer
    Dim MyTargetPar() As Double

```

```

'create ETABS object

```

Introduction

```
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'add modal Ritz load case
ret = SapModel.LoadCases.ModalRitz.SetCase("LCASE1")

'set load data
ReDim MyLoadType(2)
ReDim MyLoadName(2)
ReDim MyRitzMaxCyc(2)
ReDim MyTargetPar(2)
MyLoadType(0) = "Load"
MyLoadName(0) = "DEAD"
MyRitzMaxCyc(0) = 0
MyTargetPar(0) = 99
MyLoadType(1) = "Accel"
MyLoadName(1) = "UZ"
MyRitzMaxCyc(1) = 0
MyTargetPar(1) = 99
MyLoadType(2) = "Link"
MyRitzMaxCyc(2) = 0
MyTargetPar(2) = 99

'add modal Ritz load case
ret = SapModel.LoadCases.ModalRitz.SetLoads("LCASE1", 3, MyLoadType, MyLoadName, MyRitzMaxCyc)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cCaseModalRitz Interface

ETABSv1 Namespace

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CSI API ETABS v1

cCaseModalRitzAddLanguageSpecificTextSet("LST7B67D8FE_0") Method

This function sets the number of modes requested for the specified load case.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetNumberModes(
    string Name,
    int MaxModes,
    int MinModes
)
```

```
Function SetNumberModes (
    Name As String,
    MaxModes As Integer,
    MinModes As Integer
) As Integer
```

```
Dim instance As cCaseModalRitz
Dim Name As String
Dim MaxModes As Integer
Dim MinModes As Integer
Dim returnValue As Integer
```

```
returnValue = instance.SetNumberModes(Name,
    MaxModes, MinModes)
```

```
int SetNumberModes(
    String^ Name,
    int MaxModes,
    int MinModes
)
```

```
abstract SetNumberModes :
    Name : string *
    MaxModes : int *
    MinModes : int -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing modal Ritz load case.

MaxModes

Type: [SystemInt32](#)

The maximum number of modes requested.

MinModes

Type: [SystemInt32](#)

The minimum number of modes requested.

Return Value

Type: [Int32](#)

The function returns zero if the number of modes is successfully set; otherwise it returns a nonzero value.

Remarks

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'add modal Ritz load case
    ret = SapModel.LoadCases.ModalRitz.SetCase("LCASE1")

    'get load data
    ret = mySapModel.LoadCases.ModalRitz.SetNumberModes("LCASE1", 10, 2)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

[cCaseModalRitz Interface](#)

[ETABSv1 Namespace](#)

Introduction

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CSI API ETABS v1

cCaseResponseSpectrum Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cCaseResponseSpectrum
```

```
Public Interface cCaseResponseSpectrum
```

```
Dim instance As cCaseResponseSpectrum
```

```
public interface class cCaseResponseSpectrum
```

```
type cCaseResponseSpectrum = interface end
```

The cCaseResponseSpectrum type exposes the following members.

▲ Methods

▲	Name	Description
	GetDampConstant	
	GetDampInterpolated	
	GetDampOverrides	
	GetDampProportional	
	GetDampType	
	GetDiaphragmEccentricityOverride	
	GetDirComb	
	GetEccentricity	
	GetLoads	
	GetModalCase	
	GetModalComb	
	GetModalComb_1	
	SetCase	
	SetEccentricity	
	SetLoads	
	SetModalCase	

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▲ See Also

Reference

ETABSv1 Namespace

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CSI API ETABS v1

cCaseResponseSpectrum Methods

The [cCaseResponseSpectrum](#) type exposes the following members.

Methods

⌕	Name	Description
⇒	GetDampConstant	
⇒	GetDampInterpolated	
⇒	GetDampOverrides	
⇒	GetDampProportional	
⇒	GetDampType	
⇒	GetDiaphragmEccentricityOverride	
⇒	GetDirComb	
⇒	GetEccentricity	
⇒	GetLoads	
⇒	GetModalCase	
⇒	GetModalComb	
⇒	GetModalComb_1	
⇒	SetCase	
⇒	SetEccentricity	
⇒	SetLoads	
⇒	SetModalCase	

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CSI API ETABS v1

cCaseResponseSpectrumAddLanguageSpecificTextSet Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetDampConstant(
    string Name,
    ref double Damp
)
```

```
Function GetDampConstant (
    Name As String,
    ByRef Damp As Double
) As Integer
```

```
Dim instance As cCaseResponseSpectrum
Dim Name As String
Dim Damp As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetDampConstant(Name,
    Damp)
```

```
int GetDampConstant(
    String^ Name,
    double% Damp
)
```

```
abstract GetDampConstant :
    Name : string *
    Damp : float byref -> int
```

Parameters

Name

Type: [SystemString](#)

Damp

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cCaseResponseSpectrum Interface

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CSI API ETABS v1

cCaseResponseSpectrumAddLanguageSpecificTextSet Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetDampInterpolated(
    string Name,
    ref int DampType,
    ref int NumberItems,
    ref double[] Time,
    ref double[] Damp
)
```

```
Function GetDampInterpolated (
    Name As String,
    ByRef DampType As Integer,
    ByRef NumberItems As Integer,
    ByRef Time As Double(),
    ByRef Damp As Double()
) As Integer
```

```
Dim instance As cCaseResponseSpectrum
Dim Name As String
Dim DampType As Integer
Dim NumberItems As Integer
Dim Time As Double()
Dim Damp As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetDampInterpolated(Name,
    DampType, NumberItems, Time, Damp)
```

```
int GetDampInterpolated(
    String^ Name,
    int% DampType,
    int% NumberItems,
    array<double>^% Time,
    array<double>^% Damp
)
```

```
abstract GetDampInterpolated :
    Name : string *
    DampType : int byref *
    NumberItems : int byref *
    Time : float[] byref *
    Damp : float[] byref -> int
```

Parameters

Name

Type: SystemString

DampType

Type: SystemInt32

NumberItems

Type: SystemInt32

Time

Type: SystemDouble

Damp

Type: SystemDouble

Return Value

Type: Int32

See Also

Reference

cCaseResponseSpectrum Interface

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cCaseResponseSpectrumAddLanguageSpecificTextSet Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetDampOverrides(
    string Name,
    ref int NumberItems,
    ref int[] Mode,
    ref double[] Damp
)
```

```
Function GetDampOverrides (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef Mode As Integer(),
    ByRef Damp As Double()
) As Integer
```

```
Dim instance As cCaseResponseSpectrum
Dim Name As String
Dim NumberItems As Integer
Dim Mode As Integer()
Dim Damp As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetDampOverrides(Name,
    NumberItems, Mode, Damp)
```

```
int GetDampOverrides(
    String^ Name,
    int% NumberItems,
    array<int>^% Mode,
    array<double>^% Damp
)
```

```
abstract GetDampOverrides :
    Name : string *
    NumberItems : int byref *
    Mode : int[] byref *
    Damp : float[] byref -> int
```

Parameters

Name

Type: SystemString

cCaseResponseSpectrumspan id="LST29703570_0"AddLanguageSpecificTextSet("LST29703570_30?cpp=

NumberItems

Type: SystemInt32

Mode

Type: SystemInt32

Damp

Type: SystemDouble

Return Value

Type: Int32

▲ See Also

Reference

cCaseResponseSpectrum Interface

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cCaseResponseSpectrumAddLanguageSpecificTextSet Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetDampProportional(
    string Name,
    ref int DampType,
    ref double DampA,
    ref double DampB,
    ref double DampF1,
    ref double DampF2,
    ref double DampD1,
    ref double DampD2
)
```

```
Function GetDampProportional (
    Name As String,
    ByRef DampType As Integer,
    ByRef DampA As Double,
    ByRef DampB As Double,
    ByRef DampF1 As Double,
    ByRef DampF2 As Double,
    ByRef DampD1 As Double,
    ByRef DampD2 As Double
) As Integer
```

```
Dim instance As cCaseResponseSpectrum
Dim Name As String
Dim DampType As Integer
Dim DampA As Double
Dim DampB As Double
Dim DampF1 As Double
Dim DampF2 As Double
Dim DampD1 As Double
Dim DampD2 As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetDampProportional(Name,
    DampType, DampA, DampB, DampF1, DampF2,
    DampD1, DampD2)
```

```
int GetDampProportional(
    String^ Name,
    int% DampType,
    double% DampA,
    double% DampB,
```

cCaseResponseSpectrumspan id="LST EAF97B9E_0"AddLanguageSpecificTextSet("LST EAF97B9E_0?cp

```

double% DampF1,
double% DampF2,
double% DampD1,
double% DampD2
)

abstract GetDampProportional :
    Name : string *
    DampType : int byref *
    DampA : float byref *
    DampB : float byref *
    DampF1 : float byref *
    DampF2 : float byref *
    DampD1 : float byref *
    DampD2 : float byref -> int

```

Parameters

Name

Type:Â [SystemString](#)

DampType

Type:Â [SystemInt32](#)

DampA

Type:Â [SystemDouble](#)

DampB

Type:Â [SystemDouble](#)

DampF1

Type:Â [SystemDouble](#)

DampF2

Type:Â [SystemDouble](#)

DampD1

Type:Â [SystemDouble](#)

DampD2

Type:Â [SystemDouble](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cCaseResponseSpectrum Interface](#)

[ETABSv1 Namespace](#)

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cCaseResponseSpectrumAddLanguageSpecificTextSet Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetDampType(
    string Name,
    ref int DampType
)
```

```
Function GetDampType (
    Name As String,
    ByRef DampType As Integer
) As Integer
```

```
Dim instance As cCaseResponseSpectrum
Dim Name As String
Dim DampType As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetDampType(Name,
    DampType)
```

```
int GetDampType(
    String^ Name,
    int% DampType
)
```

```
abstract GetDampType :
    Name : string *
    DampType : int byref -> int
```

Parameters

Name

Type: [SystemString](#)

DampType

Type: [SystemInt32](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cCaseResponseSpectrum Interface

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cCaseResponseSpectrumAddLanguageSpecificTextSet Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetDiaphragmEccentricityOverride(
    string Name,
    ref int Num,
    ref string[] Diaph,
    ref double[] Eccen
)
```

```
Function GetDiaphragmEccentricityOverride (
    Name As String,
    ByRef Num As Integer,
    ByRef Diaph As String(),
    ByRef Eccen As Double()
) As Integer
```

```
Dim instance As cCaseResponseSpectrum
Dim Name As String
Dim Num As Integer
Dim Diaph As String()
Dim Eccen As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetDiaphragmEccentricityOverride(Name,
    Num, Diaph, Eccen)
```

```
int GetDiaphragmEccentricityOverride(
    String^ Name,
    int% Num,
    array<String^>^% Diaph,
    array<double>^% Eccen
)
```

```
abstract GetDiaphragmEccentricityOverride :
    Name : string *
    Num : int byref *
    Diaph : string[] byref *
    Eccen : float[] byref -> int
```

Parameters

Name

Type: [System.String](#)

Num

Type: SystemInt32

Diaph

Type: SystemString

Eccen

Type: SystemDouble

Return Value

Type: Int32

■ See Also

Reference

cCaseResponseSpectrum Interface

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CSI API ETABS v1

cCaseResponseSpectrumAddLanguageSpecificTextSet Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetDirComb(
    string Name,
    ref int MyType,
    ref double SF
)
```

```
Function GetDirComb (
    Name As String,
    ByRef MyType As Integer,
    ByRef SF As Double
) As Integer
```

```
Dim instance As cCaseResponseSpectrum
Dim Name As String
Dim MyType As Integer
Dim SF As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetDirComb(Name,
    MyType, SF)
```

```
int GetDirComb(
    String^ Name,
    int% MyType,
    double% SF
)
```

```
abstract GetDirComb :
    Name : string *
    MyType : int byref *
    SF : float byref -> int
```

Parameters

Name

Type: SystemString

MyType

Type: SystemInt32

SF

Type: SystemDouble

cCaseResponseSpectrumspan id="LST7C319708_0"AddLanguageSpecificTextSet("LST7C319708_0?cpp=

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cCaseResponseSpectrum Interface](#)

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cCaseResponseSpectrumAddLanguageSpecificTextSet Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetEccentricity(
    string Name,
    ref double Eccen
)
```

```
Function GetEccentricity (
    Name As String,
    ByRef Eccen As Double
) As Integer
```

```
Dim instance As cCaseResponseSpectrum
Dim Name As String
Dim Eccen As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetEccentricity(Name,
    Eccen)
```

```
int GetEccentricity(
    String^ Name,
    double% Eccen
)
```

```
abstract GetEccentricity :
    Name : string *
    Eccen : float byref -> int
```

Parameters

Name

Type: SystemString

Eccen

Type: SystemDouble

Return Value

Type: Int32

▲ See Also

Reference

cCaseResponseSpectrum Interface

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cCaseResponseSpectrumAddLanguageSpecificTextSet Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetLoads(
    string Name,
    ref int NumberLoads,
    ref string[] LoadName,
    ref string[] Func,
    ref double[] SF,
    ref string[] CSys,
    ref double[] Ang
)
```

```
Function GetLoads (
    Name As String,
    ByRef NumberLoads As Integer,
    ByRef LoadName As String(),
    ByRef Func As String(),
    ByRef SF As Double(),
    ByRef CSys As String(),
    ByRef Ang As Double()
) As Integer
```

```
Dim instance As cCaseResponseSpectrum
Dim Name As String
Dim NumberLoads As Integer
Dim LoadName As String()
Dim Func As String()
Dim SF As Double()
Dim CSys As String()
Dim Ang As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetLoads(Name,
    NumberLoads, LoadName, Func, SF, CSys,
    Ang)
```

```
int GetLoads(
    String^ Name,
    int% NumberLoads,
    array<String^>^% LoadName,
    array<String^>^% Func,
    array<double>^% SF,
    array<String^>^% CSys,
    array<double>^% Ang
```

)

```

abstract GetLoads :
    Name : string *
    NumberLoads : int byref *
    LoadName : string[] byref *
    Func : string[] byref *
    SF : float[] byref *
    CSys : string[] byref *
    Ang : float[] byref -> int

```

Parameters

Name

Type:Â [SystemString](#)

NumberLoads

Type:Â [SystemInt32](#)

LoadName

Type:Â [SystemString](#)

Func

Type:Â [SystemString](#)

SF

Type:Â [SystemDouble](#)

CSys

Type:Â [SystemString](#)

Ang

Type:Â [SystemDouble](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cCaseResponseSpectrum Interface](#)

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cCaseResponseSpectrumAddLanguageSpecificTextSet Method

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Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetModalCase(
    string Name,
    ref string ModalCase
)
```

```
Function GetModalCase (
    Name As String,
    ByRef ModalCase As String
) As Integer
```

```
Dim instance As cCaseResponseSpectrum
Dim Name As String
Dim ModalCase As String
Dim returnValue As Integer
```

```
returnValue = instance.GetModalCase(Name,
    ModalCase)
```

```
int GetModalCase(
    String^ Name,
    String^% ModalCase
)
```

```
abstract GetModalCase :
    Name : string *
    ModalCase : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

ModalCase

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cCaseResponseSpectrum Interface

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CSI API ETABS v1

cCaseResponseSpectrumAddLanguageSpecificTextSet Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetModalComb(
    string Name,
    ref int MyType,
    ref double F1,
    ref double F2,
    ref double Td
)
```

```
Function GetModalComb (
    Name As String,
    ByRef MyType As Integer,
    ByRef F1 As Double,
    ByRef F2 As Double,
    ByRef Td As Double
) As Integer
```

```
Dim instance As cCaseResponseSpectrum
Dim Name As String
Dim MyType As Integer
Dim F1 As Double
Dim F2 As Double
Dim Td As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetModalComb(Name,
    MyType, F1, F2, Td)
```

```
int GetModalComb(
    String^ Name,
    int% MyType,
    double% F1,
    double% F2,
    double% Td
)
```

```
abstract GetModalComb :
    Name : string *
    MyType : int byref *
    F1 : float byref *
    F2 : float byref *
    Td : float byref -> int
```

Parameters

Name

Type: SystemString

MyType

Type: SystemInt32

F1

Type: SystemDouble

F2

Type: SystemDouble

Td

Type: SystemDouble

Return Value

Type: Int32

See Also

Reference

cCaseResponseSpectrum Interface

ETABSv1 Namespace

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CSI API ETABS v1

cCaseResponseSpectrumAddLanguageSpecificTextSet Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetModalComb_1(
    string Name,
    ref int MyType,
    ref double F1,
    ref double F2,
    ref int PeriodicRigidCombType,
    ref double Td
)
```

```
Function GetModalComb_1 (
    Name As String,
    ByRef MyType As Integer,
    ByRef F1 As Double,
    ByRef F2 As Double,
    ByRef PeriodicRigidCombType As Integer,
    ByRef Td As Double
) As Integer
```

```
Dim instance As cCaseResponseSpectrum
Dim Name As String
Dim MyType As Integer
Dim F1 As Double
Dim F2 As Double
Dim PeriodicRigidCombType As Integer
Dim Td As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetModalComb_1(Name,
    MyType, F1, F2, PeriodicRigidCombType,
    Td)
```

```
int GetModalComb_1(
    String^ Name,
    int% MyType,
    double% F1,
    double% F2,
    int% PeriodicRigidCombType,
    double% Td
)
```

```
abstract GetModalComb_1 :
    Name : string *
```

```
MyType : int byref *  
F1 : float byref *  
F2 : float byref *  
PeriodicRigidCombType : int byref *  
Td : float byref -> int
```

Parameters

Name

Type:Â [SystemString](#)

MyType

Type:Â [SystemInt32](#)

F1

Type:Â [SystemDouble](#)

F2

Type:Â [SystemDouble](#)

PeriodicRigidCombType

Type:Â [SystemInt32](#)

Td

Type:Â [SystemDouble](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cCaseResponseSpectrum Interface](#)

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cCaseResponseSpectrumAddLanguageSpecificTextSet Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetCase(
    string Name
)
```

```
Function SetCase (
    Name As String
) As Integer
```

```
Dim instance As cCaseResponseSpectrum
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.SetCase(Name)
```

```
int SetCase(
    String^ Name
)
```

```
abstract SetCase :
    Name : string -> int
```

Parameters

Name

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cCaseResponseSpectrum Interface](#)

[ETABSv1 Namespace](#)

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Introduction

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CSI API ETABS v1

cCaseResponseSpectrumAddLanguageSpecificTextSet Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetEccentricity(
    string Name,
    double Eccen
)
```

```
Function SetEccentricity (
    Name As String,
    Eccen As Double
) As Integer
```

```
Dim instance As cCaseResponseSpectrum
Dim Name As String
Dim Eccen As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetEccentricity(Name,
    Eccen)
```

```
int SetEccentricity(
    String^ Name,
    double Eccen
)
```

```
abstract SetEccentricity :
    Name : string *
    Eccen : float -> int
```

Parameters

Name

Type: SystemString

Eccen

Type: SystemDouble

Return Value

Type: Int32

▲ See Also

Reference

cCaseResponseSpectrum Interface

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CSI API ETABS v1

cCaseResponseSpectrumAddLanguageSpecificTextSet Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetLoads(
    string Name,
    int NumberLoads,
    ref string[] LoadName,
    ref string[] Func,
    ref double[] SF,
    ref string[] CSys,
    ref double[] Ang
)
```

```
Function SetLoads (
    Name As String,
    NumberLoads As Integer,
    ByRef LoadName As String(),
    ByRef Func As String(),
    ByRef SF As Double(),
    ByRef CSys As String(),
    ByRef Ang As Double()
) As Integer
```

```
Dim instance As cCaseResponseSpectrum
Dim Name As String
Dim NumberLoads As Integer
Dim LoadName As String()
Dim Func As String()
Dim SF As Double()
Dim CSys As String()
Dim Ang As Double()
Dim returnValue As Integer
```

```
returnValue = instance.SetLoads(Name,
    NumberLoads, LoadName, Func, SF, CSys,
    Ang)
```

```
int SetLoads(
    String^ Name,
    int NumberLoads,
    array<String^>^% LoadName,
    array<String^>^% Func,
    array<double>^% SF,
    array<String^>^% CSys,
    array<double>^% Ang
```

)

```

abstract SetLoads :
    Name : string *
    NumberLoads : int *
    LoadName : string[] byref *
    Func : string[] byref *
    SF : float[] byref *
    CSys : string[] byref *
    Ang : float[] byref -> int

```

Parameters

Name

Type:Â [SystemString](#)

NumberLoads

Type:Â [SystemInt32](#)

LoadName

Type:Â [SystemString](#)

Func

Type:Â [SystemString](#)

SF

Type:Â [SystemDouble](#)

CSys

Type:Â [SystemString](#)

Ang

Type:Â [SystemDouble](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cCaseResponseSpectrum Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cCaseResponseSpectrumAddLanguageSpecificTextSet Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetModalCase(
    string Name,
    string ModalCase
)
```

```
Function SetModalCase (
    Name As String,
    ModalCase As String
) As Integer
```

```
Dim instance As cCaseResponseSpectrum
Dim Name As String
Dim ModalCase As String
Dim returnValue As Integer
```

```
returnValue = instance.SetModalCase(Name,
    ModalCase)
```

```
int SetModalCase(
    String^ Name,
    String^ ModalCase
)
```

```
abstract SetModalCase :
    Name : string *
    ModalCase : string -> int
```

Parameters

Name

Type: [SystemString](#)

ModalCase

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cCaseResponseSpectrum Interface

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CSI API ETABS v1

cCaseStaticLinear Interface

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
public interface cCaseStaticLinear
```

```
Public Interface cCaseStaticLinear
```

```
Dim instance As cCaseStaticLinear
```

```
public interface class cCaseStaticLinear
```

```
type cCaseStaticLinear = interface end
```

The cCaseStaticLinear type exposes the following members.

▲ Methods

▲	Name	Description
➤	<u>GetInitialCase</u>	Retrieves the initial condition assumed for the specified load case.
➤	<u>GetLoads</u>	Retrieves the load data for the specified load case
➤	<u>SetCase</u>	Initializes a static linear load case.
➤	<u>SetInitialCase</u>	Sets the initial condition for the specified load case.
➤	<u>SetLoads</u>	Sets the load data for the specified analysis case.

Top

▲ See Also

Reference

ETABSv1 Namespace

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CSI API ETABS v1

cCaseStaticLinear Methods

The [cCaseStaticLinear](#) type exposes the following members.

Methods

⌵	Name	Description
⇒	GetInitialCase	Retrieves the initial condition assumed for the specified load case.
⇒	GetLoads	Retrieves the load data for the specified load case
⇒	SetCase	Initializes a static linear load case.
⇒	SetInitialCase	Sets the initial condition for the specified load case.
⇒	SetLoads	Sets the load data for the specified analysis case.

[Top](#)

⌵ See Also

Reference

[cCaseStaticLinear Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cCaseStaticLinearAddLanguageSpecificTextSet("LST7 Method

Retrieves the initial condition assumed for the specified load case.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetInitialCase(
    string Name,
    ref string InitialCase
)
```

```
Function GetInitialCase (
    Name As String,
    ByRef InitialCase As String
) As Integer
```

```
Dim instance As cCaseStaticLinear
Dim Name As String
Dim InitialCase As String
Dim returnValue As Integer
```

```
returnValue = instance.GetInitialCase(Name,
    InitialCase)
```

```
int GetInitialCase(
    String^ Name,
    String^% InitialCase
)
```

```
abstract GetInitialCase :
    Name : string *
    InitialCase : string byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing static linear load case.

InitialCase

Type: SystemString

This is blank, None, or the name of an existing analysis case. This item specifies if the load case starts from zero initial conditions, that is, an unstressed state, or if it starts using the stiffness that occurs at the end of a nonlinear static or nonlinear direct integration time history load case.

If the specified initial case is a nonlinear static or nonlinear direct integration time history load case, the stiffness at the end of that case is used. If the initial case is anything else, zero initial conditions are assumed.

Return Value

Type: $\hat{A}$ Int32

Returns zero if the initial condition is successfully retrieved; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim InitialCase As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'get initial condition
    ret = SapModel.LoadCases.StaticLinear.GetInitialCase("DEAD", InitialCase)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cCaseStaticLinear Interface

ETABSV1 Namespace

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Introduction

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CSI API ETABS v1

cCaseStaticLinearAddLanguageSpecificTextSet("LST4 Method

Retrieves the load data for the specified load case

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetLoads(
    string Name,
    ref int NumberLoads,
    ref string[] LoadType,
    ref string[] LoadName,
    ref double[] SF
)
```

```
Function GetLoads (
    Name As String,
    ByRef NumberLoads As Integer,
    ByRef LoadType As String(),
    ByRef LoadName As String(),
    ByRef SF As Double()
) As Integer
```

```
Dim instance As cCaseStaticLinear
Dim Name As String
Dim NumberLoads As Integer
Dim LoadType As String()
Dim LoadName As String()
Dim SF As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetLoads(Name,
    NumberLoads, LoadType, LoadName,
    SF)
```

```
int GetLoads(
    String^ Name,
    int% NumberLoads,
    array<String^>^% LoadType,
    array<String^>^% LoadName,
    array<double>^% SF
)
```

```
abstract GetLoads :
    Name : string *
    NumberLoads : int byref *
    LoadType : string[] byref *
    LoadName : string[] byref *
```

cCaseStaticLinearspan id="LST4E7196BC_0"AddLanguageSpecificTextSet("LST4E7196BC_0?676=::|nu=.

SF : float[] byref -> int

Parameters

Name

Type:Â SystemString

The name of an existing static linear load case.

NumberLoads

Type:Â SystemInt32

The number of loads assigned to the specified analysis case.

LoadType

Type:Â SystemString

This is an array that includes either Load or Accel, indicating the type of each load assigned to the load case.

LoadName

Type:Â SystemString

This is an array that includes the name of each load assigned to the load case.

If the LoadType item is Load, this item is the name of a defined load pattern.

If the LoadType item is Accel, this item is UX, UY, UZ, RX, RY or RZ, indicating the direction of the load.

SF

Type:Â SystemDouble

This is an array that includes the scale factor of each load assigned to the load case. [L/s²] for Accel UX UY and UZ; otherwise unitless

Return Value

Type:Â Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyLoadType() As String
    Dim MyLoadName() As String
    Dim MySF() As Double
    Dim NumberLoads As Integer
    Dim LoadType() As String
    Dim LoadName() As String
    Dim SF() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
```

Parameters

Introduction

```
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'add static linear load case
ret = SapModel.LoadCases.StaticLinear.SetCase("LCASE1")

'set load data
ReDim MyLoadType(1)
ReDim MyLoadName(1)
ReDim MySF(1)
MyLoadType(0) = "Load"
MyLoadName(0) = "DEAD"
MySF(0) = 0.7
MyLoadType(1) = "Accel"
MyLoadName(1) = "UZ"
MySF(1) = 1.2
ret = SapModel.LoadCases.StaticLinear.SetLoads("LCASE1", 2, MyLoadType, MyLoadName, MySF)

'get load data
ret = SapModel.LoadCases.StaticLinear.GetLoads("LCASE1", NumberLoads, LoadType, LoadName, SF)

'refresh view
ret = SapModel.View.RefreshView(0, False)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

[cCaseStaticLinear Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cCaseStaticLinearAddLanguageSpecificTextSet("LSTD Method

Initializes a static linear load case.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetCase(
    string Name
)
```

```
Function SetCase (
    Name As String
) As Integer
```

```
Dim instance As cCaseStaticLinear
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.SetCase(Name)
```

```
int SetCase(
    String^ Name
)
```

```
abstract SetCase :
    Name : string -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing or new load case. If this is an existing case, that case is modified; otherwise, a new case is added.

Return Value

Type: [Int32](#)

Returns zero if the load case is successfully initialized; otherwise it returns a nonzero value.

▲ Remarks

If this function is called for an existing load case, all items for the case are reset to their default value.

▲ Examples

cCaseStaticLinearid="LSTD683DD9E_0"AddLanguageSpecificTextSet("LSTD683DD9E_0079p=::|nu=

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'add static linear load case
    ret = SapModel.LoadCases.StaticLinear.SetCase("LCASE1")

    'refresh view
    ret = SapModel.View.RefreshView(0, False)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

 See Also
Reference[cCaseStaticLinear Interface](#)[ETABSV1 Namespace](#)

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CSI API ETABS v1

cCaseStaticLinearAddLanguageSpecificTextSet("LST4 Method

Sets the initial condition for the specified load case.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetInitialCase(
    string Name,
    string InitialCase
)
```

```
Function SetInitialCase (
    Name As String,
    InitialCase As String
) As Integer
```

```
Dim instance As cCaseStaticLinear
Dim Name As String
Dim InitialCase As String
Dim returnValue As Integer
```

```
returnValue = instance.SetInitialCase(Name,
    InitialCase)
```

```
int SetInitialCase(
    String^ Name,
    String^ InitialCase
)
```

```
abstract SetInitialCase :
    Name : string *
    InitialCase : string -> int
```

Parameters

Name

Type: SystemString

The name of an existing static linear load case.

InitialCase

Type: SystemString

This is blank, None, or the name of an existing analysis case. This item specifies if the load case starts from zero initial conditions, that is, an unstressed state, or if it starts using the stiffness that occurs at the end of a nonlinear static or nonlinear direct integration time history load case.

If the specified initial case is a nonlinear static or nonlinear direct integration time history load case, the stiffness at the end of that case is used. If the initial case is anything else, zero initial conditions are assumed.

Return Value

Type: [Int32](#)

Returns zero if the initial condition is successfully set; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'add static linear load case
    ret = SapModel.LoadCases.StaticLinear.SetCase("LCASE1")

    'set initial condition
    ret = SapModel.LoadCases.StaticLinear.SetInitialCase("LCASE1", "None")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ **See Also**

Reference

[cCaseStaticLinear Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cCaseStaticLinearAddLanguageSpecificTextSet("LST2 Method

Sets the load data for the specified analysis case.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetLoads(
    string Name,
    int NumberLoads,
    ref string[] LoadType,
    ref string[] LoadName,
    ref double[] SF
)
```

```
Function SetLoads (
    Name As String,
    NumberLoads As Integer,
    ByRef LoadType As String(),
    ByRef LoadName As String(),
    ByRef SF As Double()
) As Integer
```

```
Dim instance As cCaseStaticLinear
Dim Name As String
Dim NumberLoads As Integer
Dim LoadType As String()
Dim LoadName As String()
Dim SF As Double()
Dim returnValue As Integer
```

```
returnValue = instance.SetLoads(Name,
    NumberLoads, LoadType, LoadName,
    SF)
```

```
int SetLoads(
    String^ Name,
    int NumberLoads,
    array<String^>^% LoadType,
    array<String^>^% LoadName,
    array<double>^% SF
)
```

```
abstract SetLoads :
    Name : string *
    NumberLoads : int *
    LoadType : string[] byref *
    LoadName : string[] byref *
```

cCaseStaticLinearspan id="LST21A7A9FB_0"AddLanguageSpecificTextSet("LST21A7A9FB_0?684=::|nu=.

SF : float[] byref -> int

Parameters

Name

Type:Â SystemString

The name of an existing static linear load case.

NumberLoads

Type:Â SystemInt32

The number of loads assigned to the specified analysis case.

LoadType

Type:Â SystemString

This is an array that includes either Load or Accel, indicating the type of each load assigned to the load case.

LoadName

Type:Â SystemString

This is an array that includes the name of each load assigned to the load case.

If the LoadType item is Load, this item is the name of a defined load pattern.

If the LoadType item is Accel, this item is UX, UY, UZ, RX, RY or RZ, indicating the direction of the load.

SF

Type:Â SystemDouble

This is an array that includes the scale factor of each load assigned to the load case. [L/s²] for Accel UX UY and UZ; otherwise unitless

Return Value

Type:Â Int32

Returns zero if the data is successfully set; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyLoadType() As String
    Dim MyLoadName() As String
    Dim MySF() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()
```

Introduction

```
'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'add static linear load case
    ret = SapModel.LoadCases.StaticLinear.SetCase("LCASE1")

'set load data
    ReDim MyLoadType(1)
    ReDim MyLoadName(1)
    ReDim MySF(1)
    MyLoadType(0) = "Load"
    MyLoadName(0) = "DEAD"
    MySF(0) = 0.7
    MyLoadType(1) = "Accel"
    MyLoadName(1) = "UZ"
    MySF(1) = 1.2
    ret = SapModel.LoadCases.StaticLinear.SetLoads("LCASE1", 2, MyLoadType, MyLoadName, MySF)

'refresh view
    ret = SapModel.View.RefreshView(0, False)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

📌 See Also

Reference

[cCaseStaticLinear Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cCaseStaticNonlinear Interface

Namespace: $\hat{A}$ ETABSv1

Assembly: $\hat{A}$ ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
public interface cCaseStaticNonlinear
```

```
Public Interface cCaseStaticNonlinear
```

```
Dim instance As cCaseStaticNonlinear
```

```
public interface class cCaseStaticNonlinear
```

```
type cCaseStaticNonlinear = interface end
```

The cCaseStaticNonlinear type exposes the following members.

Methods

$\hat{A}$	Name	Description
	<u>GetGeometricNonlinearity</u>	
	<u>GetHingeUnloading</u>	
	<u>GetInitialCase</u>	
	<u>GetLoadApplication</u>	
	<u>GetLoads</u>	Retrieves the load data for the specified load case.
	<u>GetMassSource</u>	
	<u>GetModalCase</u>	
	<u>GetResultsSaved</u>	
	<u>GetSolControlParameters</u>	
	<u>GetTargetForceParameters</u>	
	<u>SetCase</u>	Initializes a static nonlinear load case.
	<u>SetGeometricNonlinearity</u>	Sets the geometric nonlinearity option for the specified load case.
	<u>SetHingeUnloading</u>	
	<u>SetInitialCase</u>	
	<u>SetLoadApplication</u>	
	<u>SetLoads</u>	Sets the load data for the specified analysis case.
	<u>SetMassSource</u>	
	<u>SetModalCase</u>	

-  [SetResultsSaved](#)
-  [SetSolControlParameters](#)
-  [SetTargetForceParameters](#)

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CSI API ETABS v1

cCaseStaticNonlinear Methods

The [cCaseStaticNonlinear](#) type exposes the following members.

Methods

⌵	Name	Description
⇒	GetGeometricNonlinearity	
⇒	GetHingeUnloading	
⇒	GetInitialCase	
⇒	GetLoadApplication	
⇒	GetLoads	Retrieves the load data for the specified load case
⇒	GetMassSource	
⇒	GetModalCase	
⇒	GetResultsSaved	
⇒	GetSolControlParameters	
⇒	GetTargetForceParameters	
⇒	SetCase	Initializes a static nonlinear load case.
⇒	SetGeometricNonlinearity	Sets the geometric nonlinearity option for the specified load case.
⇒	SetHingeUnloading	
⇒	SetInitialCase	
⇒	SetLoadApplication	
⇒	SetLoads	Sets the load data for the specified analysis case.
⇒	SetMassSource	
⇒	SetModalCase	
⇒	SetResultsSaved	
⇒	SetSolControlParameters	
⇒	SetTargetForceParameters	

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⌵ See Also

Reference

[cCaseStaticNonlinear Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cCaseStaticNonlinearAddLanguageSpecificTextSet("LST470B9AEA_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetGeometricNonlinearity(
    string Name,
    ref int NLGeomType
)
```

```
Function GetGeometricNonlinearity (
    Name As String,
    ByRef NLGeomType As Integer
) As Integer
```

```
Dim instance As cCaseStaticNonlinear
Dim Name As String
Dim NLGeomType As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetGeometricNonlinearity(Name,
    NLGeomType)
```

```
int GetGeometricNonlinearity(
    String^ Name,
    int% NLGeomType
)
```

```
abstract GetGeometricNonlinearity :
    Name : string *
    NLGeomType : int byref -> int
```

Parameters

Name

Type: [SystemString](#)

NLGeomType

Type: [SystemInt32](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cCaseStaticNonlinear Interface

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CSI API ETABS v1

cCaseStaticNonlinearAddLanguageSpecificTextSet("LST8E6240AC_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetHingeUnloading(
    string Name,
    ref int UnloadType
)
```

```
Function GetHingeUnloading (
    Name As String,
    ByRef UnloadType As Integer
) As Integer
```

```
Dim instance As cCaseStaticNonlinear
Dim Name As String
Dim UnloadType As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetHingeUnloading(Name,
    UnloadType)
```

```
int GetHingeUnloading(
    String^ Name,
    int% UnloadType
)
```

```
abstract GetHingeUnloading :
    Name : string *
    UnloadType : int byref -> int
```

Parameters

Name

Type: [SystemString](#)

UnloadType

Type: [SystemInt32](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cCaseStaticNonlinear Interface

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CSI API ETABS v1

cCaseStaticNonlinearAddLanguageSpecificTextSet("LSTBB73ADDC_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetInitialCase(
    string Name,
    ref string InitialCase
)
```

```
Function GetInitialCase (
    Name As String,
    ByRef InitialCase As String
) As Integer
```

```
Dim instance As cCaseStaticNonlinear
Dim Name As String
Dim InitialCase As String
Dim returnValue As Integer
```

```
returnValue = instance.GetInitialCase(Name,
    InitialCase)
```

```
int GetInitialCase(
    String^ Name,
    String^% InitialCase
)
```

```
abstract GetInitialCase :
    Name : string *
    InitialCase : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

InitialCase

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cCaseStaticNonlinear Interface

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CSI API ETABS v1

cCaseStaticNonlinearAddLanguageSpecificTextSet("LSTC7032C40_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetLoadApplication(
    string Name,
    ref int LoadControl,
    ref int DispType,
    ref double Displ,
    ref int Monitor,
    ref int DOF,
    ref string PointName,
    ref string GDispl
)
```

```
Function GetLoadApplication (
    Name As String,
    ByRef LoadControl As Integer,
    ByRef DispType As Integer,
    ByRef Displ As Double,
    ByRef Monitor As Integer,
    ByRef DOF As Integer,
    ByRef PointName As String,
    ByRef GDispl As String
) As Integer
```

```
Dim instance As cCaseStaticNonlinear
Dim Name As String
Dim LoadControl As Integer
Dim DispType As Integer
Dim Displ As Double
Dim Monitor As Integer
Dim DOF As Integer
Dim PointName As String
Dim GDispl As String
Dim returnValue As Integer
```

```
returnValue = instance.GetLoadApplication(Name,
    LoadControl, DispType, Displ, Monitor,
    DOF, PointName, GDispl)
```

```
int GetLoadApplication(
    String^ Name,
    int% LoadControl,
    int% DispType,
    double% Displ,
```

cCaseStaticNonlinearid="LSTC7032C40_0"AddLanguageSpecificTextSet("LSTC7032C40_0")

```

int% Monitor,
int% DOF,
String^% PointName,
String^% GDispl
)

```

```

abstract GetLoadApplication :
    Name : string *
    LoadControl : int byref *
    DispType : int byref *
    Displ : float byref *
    Monitor : int byref *
    DOF : int byref *
    PointName : string byref *
    GDispl : string byref -> int

```

Parameters

Name

Type:Â [SystemString](#)

LoadControl

Type:Â [SystemInt32](#)

DispType

Type:Â [SystemInt32](#)

Displ

Type:Â [SystemDouble](#)

Monitor

Type:Â [SystemInt32](#)

DOF

Type:Â [SystemInt32](#)

PointName

Type:Â [SystemString](#)

GDispl

Type:Â [SystemString](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cCaseStaticNonlinear Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cCaseStaticNonlinearAddLanguageSpecificTextSet("LSTE0206169_0")

Method

Retrieves the load data for the specified load case

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetLoads(
    string Name,
    ref int NumberLoads,
    ref string[] LoadType,
    ref string[] LoadName,
    ref double[] SF
)
```

```
Function GetLoads (
    Name As String,
    ByRef NumberLoads As Integer,
    ByRef LoadType As String(),
    ByRef LoadName As String(),
    ByRef SF As Double()
) As Integer
```

```
Dim instance As cCaseStaticNonlinear
Dim Name As String
Dim NumberLoads As Integer
Dim LoadType As String()
Dim LoadName As String()
Dim SF As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetLoads(Name,
    NumberLoads, LoadType, LoadName,
    SF)
```

```
int GetLoads(
    String^ Name,
    int% NumberLoads,
    array<String^>^% LoadType,
    array<String^>^% LoadName,
    array<double>^% SF
)
```

```
abstract GetLoads :
    Name : string *
    NumberLoads : int byref *
    LoadType : string[] byref *
    LoadName : string[] byref *
```

cCaseStaticNonlinearspan id="LSTE0206169_0"AddLanguageSpecificTextSet("LSTE0206169_0")

SF : float[] byref -> int

Parameters

Name

Type:Â SystemString

The name of an existing static nonlinear load case.

NumberLoads

Type:Â SystemInt32

The number of loads assigned to the specified analysis case.

LoadType

Type:Â SystemString

This is an array that includes either Load or Accel, indicating the type of each load assigned to the load case.

LoadName

Type:Â SystemString

This is an array that includes the name of each load assigned to the load case.

If the LoadType item is Load, this item is the name of a defined load pattern.

If the LoadType item is Accel, this item is UX, UY, UZ, RX, RY or RZ, indicating the direction of the load.

SF

Type:Â SystemDouble

This is an array that includes the scale factor of each load assigned to the load case. [L/s²] for Accel UX UY and UZ; otherwise unitless

Return Value

Type:Â Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyLoadType() As String
    Dim MyLoadName() As String
    Dim MySF() As Double
    Dim NumberLoads As Integer
    Dim LoadType() As String
    Dim LoadName() As String
    Dim SF() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
```

Introduction

```
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'add static nonlinear load case
ret = SapModel.LoadCases.StaticNonlinear.SetCase("LCASE1")

'set load data
ReDim MyLoadType(1)
ReDim MyLoadName(1)
ReDim MySF(1)
MyLoadType(0) = "Load"
MyLoadName(0) = "DEAD"
MySF(0) = 0.7
MyLoadType(1) = "Accel"
MyLoadName(1) = "UZ"
MySF(1) = 1.2
ret = SapModel.LoadCases.StaticNonlinear.SetLoads("LCASE1", 2, MyLoadType, MyLoadName, MySF)

'get load data
ret = SapModel.LoadCases.StaticNonlinear.GetLoads("LCASE1", NumberLoads, LoadType, LoadName,

'refresh view
ret = SapModel.View.RefreshView(0, False)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

[cCaseStaticNonlinear Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cCaseStaticNonlinearAddLanguageSpecificTextSet("LSTAE651035_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetMassSource(
    string Name,
    ref string mSource
)
```

```
Function GetMassSource (
    Name As String,
    ByRef mSource As String
) As Integer
```

```
Dim instance As cCaseStaticNonlinear
Dim Name As String
Dim mSource As String
Dim returnValue As Integer
```

```
returnValue = instance.GetMassSource(Name,
    mSource)
```

```
int GetMassSource(
    String^ Name,
    String^% mSource
)
```

```
abstract GetMassSource :
    Name : string *
    mSource : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

mSource

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cCaseStaticNonlinear Interface

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CSI API ETABS v1

cCaseStaticNonlinearAddLanguageSpecificTextSet("LST2FAFF26B_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetModalCase(
    string Name,
    ref string ModalCase
)
```

```
Function GetModalCase (
    Name As String,
    ByRef ModalCase As String
) As Integer
```

```
Dim instance As cCaseStaticNonlinear
Dim Name As String
Dim ModalCase As String
Dim returnValue As Integer
```

```
returnValue = instance.GetModalCase(Name,
    ModalCase)
```

```
int GetModalCase(
    String^ Name,
    String^% ModalCase
)
```

```
abstract GetModalCase :
    Name : string *
    ModalCase : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

ModalCase

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cCaseStaticNonlinear Interface

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CSI API ETABS v1

cCaseStaticNonlinearAddLanguageSpecificTextSet("LSTF258FB6D_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetResultsSaved(
    string Name,
    ref bool SaveMultipleSteps,
    ref int MinSavedStates,
    ref int MaxSavedStates,
    ref bool PositiveOnly
)
```

```
Function GetResultsSaved (
    Name As String,
    ByRef SaveMultipleSteps As Boolean,
    ByRef MinSavedStates As Integer,
    ByRef MaxSavedStates As Integer,
    ByRef PositiveOnly As Boolean
) As Integer
```

```
Dim instance As cCaseStaticNonlinear
Dim Name As String
Dim SaveMultipleSteps As Boolean
Dim MinSavedStates As Integer
Dim MaxSavedStates As Integer
Dim PositiveOnly As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetResultsSaved(Name,
    SaveMultipleSteps, MinSavedStates,
    MaxSavedStates, PositiveOnly)
```

```
int GetResultsSaved(
    String^ Name,
    bool% SaveMultipleSteps,
    int% MinSavedStates,
    int% MaxSavedStates,
    bool% PositiveOnly
)
```

```
abstract GetResultsSaved :
    Name : string *
    SaveMultipleSteps : bool byref *
    MinSavedStates : int byref *
    MaxSavedStates : int byref *
    PositiveOnly : bool byref -> int
```

cCaseStaticNonlinearAddLanguageSpecificTextSet("LSTF258FB6D_0")
cpp=::n

Parameters

Name

Type: SystemString

SaveMultipleSteps

Type: SystemBoolean

MinSavedStates

Type: SystemInt32

MaxSavedStates

Type: SystemInt32

PositiveOnly

Type: SystemBoolean

Return Value

Type: Int32

See Also

Reference

cCaseStaticNonlinear Interface

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CSI API ETABS v1

cCaseStaticNonlinearAddLanguageSpecificTextSet("LSTB2725095_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetSolControlParameters(
    string Name,
    ref int MaxTotalSteps,
    ref int MaxFailedSubSteps,
    ref int MaxIterCS,
    ref int MaxIterNR,
    ref double TolConvD,
    ref bool UseEventStepping,
    ref double TolEventD,
    ref int MaxLineSearchPerIter,
    ref double TolLineSearch,
    ref double LineSearchStepFact
)
```

```
Function GetSolControlParameters (
    Name As String,
    ByRef MaxTotalSteps As Integer,
    ByRef MaxFailedSubSteps As Integer,
    ByRef MaxIterCS As Integer,
    ByRef MaxIterNR As Integer,
    ByRef TolConvD As Double,
    ByRef UseEventStepping As Boolean,
    ByRef TolEventD As Double,
    ByRef MaxLineSearchPerIter As Integer,
    ByRef TolLineSearch As Double,
    ByRef LineSearchStepFact As Double
) As Integer
```

```
Dim instance As cCaseStaticNonlinear
Dim Name As String
Dim MaxTotalSteps As Integer
Dim MaxFailedSubSteps As Integer
Dim MaxIterCS As Integer
Dim MaxIterNR As Integer
Dim TolConvD As Double
Dim UseEventStepping As Boolean
Dim TolEventD As Double
Dim MaxLineSearchPerIter As Integer
Dim TolLineSearch As Double
Dim LineSearchStepFact As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetSolControlParameters(Name,
```

cCaseStaticNonlinearspan id="LSTB2725095_0"AddLanguageSpecificTextSet("LSTB2725095_0")pp=::|nu

```
MaxTotalSteps, MaxFailedSubSteps,
MaxIterCS, MaxIterNR, TolConvD, UseEventStepping,
TolEventD, MaxLineSearchPerIter,
TolLineSearch, LineSearchStepFact)
```

```
int GetSolControlParameters(
String^ Name,
int% MaxTotalSteps,
int% MaxFailedSubSteps,
int% MaxIterCS,
int% MaxIterNR,
double% TolConvD,
bool% UseEventStepping,
double% TolEventD,
int% MaxLineSearchPerIter,
double% TolLineSearch,
double% LineSearchStepFact
)
```

```
abstract GetSolControlParameters :
    Name : string *
    MaxTotalSteps : int byref *
    MaxFailedSubSteps : int byref *
    MaxIterCS : int byref *
    MaxIterNR : int byref *
    TolConvD : float byref *
    UseEventStepping : bool byref *
    TolEventD : float byref *
    MaxLineSearchPerIter : int byref *
    TolLineSearch : float byref *
    LineSearchStepFact : float byref -> int
```

Parameters

Name

Type: SystemString

MaxTotalSteps

Type: SystemInt32

MaxFailedSubSteps

Type: SystemInt32

MaxIterCS

Type: SystemInt32

MaxIterNR

Type: SystemInt32

TolConvD

Type: SystemDouble

UseEventStepping

Type: SystemBoolean

TolEventD

Type: SystemDouble

MaxLineSearchPerIter

Type: SystemInt32

TolLineSearch

Type: SystemDouble

LineSearchStepFact

Type: SystemDouble

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cCaseStaticNonlinear Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cCaseStaticNonlinearAddLanguageSpecificTextSet("LST8A829E94_0") Method

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetTargetForceParameters(
    string Name,
    ref double TolConvF,
    ref int MaxIter,
    ref double AccelFact,
    ref bool NoStop
)
```

```
Function GetTargetForceParameters (
    Name As String,
    ByRef TolConvF As Double,
    ByRef MaxIter As Integer,
    ByRef AccelFact As Double,
    ByRef NoStop As Boolean
) As Integer
```

```
Dim instance As cCaseStaticNonlinear
Dim Name As String
Dim TolConvF As Double
Dim MaxIter As Integer
Dim AccelFact As Double
Dim NoStop As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetTargetForceParameters(Name,
    TolConvF, MaxIter, AccelFact, NoStop)
```

```
int GetTargetForceParameters(
    String^ Name,
    double% TolConvF,
    int% MaxIter,
    double% AccelFact,
    bool% NoStop
)
```

```
abstract GetTargetForceParameters :
    Name : string *
    TolConvF : float byref *
    MaxIter : int byref *
    AccelFact : float byref *
    NoStop : bool byref -> int
```

Parameters

Name

Type: SystemString

TolConvF

Type: SystemDouble

MaxIter

Type: SystemInt32

AccelFact

Type: SystemDouble

NoStop

Type: SystemBoolean

Return Value

Type: Int32

See Also

Reference

cCaseStaticNonlinear Interface

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CSI API ETABS v1

cCaseStaticNonlinearAddLanguageSpecificTextSet("LSTFC716138_0") Method

Initializes a static nonlinear load case.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetCase(
    string Name
)
```

```
Function SetCase (
    Name As String
) As Integer
```

```
Dim instance As cCaseStaticNonlinear
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.SetCase(Name)
```

```
int SetCase(
    String^ Name
)
```

```
abstract SetCase :
    Name : string -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing or new load case. If this is an existing case, that case is modified; otherwise, a new case is added.

Return Value

Type: [Int32](#)

Returns zero if the load case is successfully initialized; otherwise it returns a nonzero value.

▲ Remarks

If this function is called for an existing load case, all items for the case are reset to their default value.

▲ Examples

cCaseStaticNonlinearAddLanguageSpecificTextSet("LSTFC716138_0")
pp=::|n

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'add static nonlinear load case
    ret = SapModel.LoadCases.StaticNonlinear.SetCase("LCASE1")

    'refresh view
    ret = SapModel.View.RefreshView(0, False)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

 See Also
ReferencecCaseStaticNonlinear InterfaceETABSV1 Namespace

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CSI API ETABS v1

cCaseStaticNonlinearAddLanguageSpecificTextSet("LSTBB55B511_0")

Method

Sets the geometric nonlinearity option for the specified load case.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetGeometricNonlinearity(
    string Name,
    int NLGeomType
)
```

```
Function SetGeometricNonlinearity (
    Name As String,
    NLGeomType As Integer
) As Integer
```

```
Dim instance As cCaseStaticNonlinear
Dim Name As String
Dim NLGeomType As Integer
Dim returnValue As Integer
```

```
returnValue = instance.SetGeometricNonlinearity(Name,
    NLGeomType)
```

```
int SetGeometricNonlinearity(
    String^ Name,
    int NLGeomType
)
```

```
abstract SetGeometricNonlinearity :
    Name : string *
    NLGeomType : int -> int
```

Parameters

Name

Type: SystemString

The name of an existing static nonlinear load case.

NLGeomType

Type: SystemInt32

Return ValueType: Int32

Returns zero if the option is successfully set; otherwise it returns a nonzero value.

▲ **Remarks**▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'add static nonlinear load case
    ret = SapModel.LoadCases.StaticNonlinear.SetCase("LCASE1")

    'set geometric nonlinearity option
    ret = SapModel.LoadCases.StaticNonlinear.SetGeometricNonlinearity("LCASE1", 2)

    'refresh view
    ret = SapModel.View.RefreshView(0, False)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also****Reference**cCaseStaticNonlinear InterfaceETABSV1 Namespace

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CSI API ETABS v1

cCaseStaticNonlinearAddLanguageSpecificTextSet("LST9DF57426_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetHingeUnloading(
    string Name,
    int UnloadType
)
```

```
Function SetHingeUnloading (
    Name As String,
    UnloadType As Integer
) As Integer
```

```
Dim instance As cCaseStaticNonlinear
Dim Name As String
Dim UnloadType As Integer
Dim returnValue As Integer
```

```
returnValue = instance.SetHingeUnloading(Name,
    UnloadType)
```

```
int SetHingeUnloading(
    String^ Name,
    int UnloadType
)
```

```
abstract SetHingeUnloading :
    Name : string *
    UnloadType : int -> int
```

Parameters

Name

Type: [SystemString](#)

UnloadType

Type: [SystemInt32](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cCaseStaticNonlinear Interface

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CSI API ETABS v1

cCaseStaticNonlinearAddLanguageSpecificTextSet("LST7806B191_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetInitialCase(
    string Name,
    string InitialCase
)
```

```
Function SetInitialCase (
    Name As String,
    InitialCase As String
) As Integer
```

```
Dim instance As cCaseStaticNonlinear
Dim Name As String
Dim InitialCase As String
Dim returnValue As Integer
```

```
returnValue = instance.SetInitialCase(Name,
    InitialCase)
```

```
int SetInitialCase(
    String^ Name,
    String^ InitialCase
)
```

```
abstract SetInitialCase :
    Name : string *
    InitialCase : string -> int
```

Parameters

Name

Type: [SystemString](#)

InitialCase

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cCaseStaticNonlinear Interface

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CSI API ETABS v1

cCaseStaticNonlinearAddLanguageSpecificTextSet("LST Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetLoadApplication(
    string Name,
    int LoadControl,
    int DispType,
    double Displ,
    int Monitor,
    int DOF,
    string PointName,
    string GDispl
)
```

```
Function SetLoadApplication (
    Name As String,
    LoadControl As Integer,
    DispType As Integer,
    Displ As Double,
    Monitor As Integer,
    DOF As Integer,
    PointName As String,
    GDispl As String
) As Integer
```

```
Dim instance As cCaseStaticNonlinear
Dim Name As String
Dim LoadControl As Integer
Dim DispType As Integer
Dim Displ As Double
Dim Monitor As Integer
Dim DOF As Integer
Dim PointName As String
Dim GDispl As String
Dim returnValue As Integer
```

```
returnValue = instance.SetLoadApplication(Name,
    LoadControl, DispType, Displ, Monitor,
    DOF, PointName, GDispl)
```

```
int SetLoadApplication(
    String^ Name,
    int LoadControl,
    int DispType,
    double Displ,
```

```

int Monitor,
int DOF,
String^ PointName,
String^ GDispl
)

abstract SetLoadApplication :
    Name : string *
    LoadControl : int *
    DispType : int *
    Displ : float *
    Monitor : int *
    DOF : int *
    PointName : string *
    GDispl : string -> int

```

Parameters

Name
Type:Â [SystemString](#)

LoadControl
Type:Â [SystemInt32](#)

DispType
Type:Â [SystemInt32](#)

Displ
Type:Â [SystemDouble](#)

Monitor
Type:Â [SystemInt32](#)

DOF
Type:Â [SystemInt32](#)

PointName
Type:Â [SystemString](#)

GDispl
Type:Â [SystemString](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cCaseStaticNonlinear Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cCaseStaticNonlinearAddLanguageSpecificTextSet("LST Method

Sets the load data for the specified analysis case.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetLoads(
    string Name,
    int NumberLoads,
    ref string[] LoadType,
    ref string[] LoadName,
    ref double[] SF
)
```

```
Function SetLoads (
    Name As String,
    NumberLoads As Integer,
    ByRef LoadType As String(),
    ByRef LoadName As String(),
    ByRef SF As Double()
) As Integer
```

```
Dim instance As cCaseStaticNonlinear
Dim Name As String
Dim NumberLoads As Integer
Dim LoadType As String()
Dim LoadName As String()
Dim SF As Double()
Dim returnValue As Integer
```

```
returnValue = instance.SetLoads(Name,
    NumberLoads, LoadType, LoadName,
    SF)
```

```
int SetLoads(
    String^ Name,
    int NumberLoads,
    array<String^>^% LoadType,
    array<String^>^% LoadName,
    array<double>^% SF
)
```

```
abstract SetLoads :
    Name : string *
    NumberLoads : int *
    LoadType : string[] byref *
    LoadName : string[] byref *
```

SF : float[] byref -> int

Parameters

Name

Type:Â SystemString

The name of an existing static nonlinear load case.

NumberLoads

Type:Â SystemInt32

The number of loads assigned to the specified analysis case.

LoadType

Type:Â SystemString

This is an array that includes either Load or Accel, indicating the type of each load assigned to the load case.

LoadName

Type:Â SystemString

This is an array that includes the name of each load assigned to the load case.

If the LoadType item is Load, this item is the name of a defined load pattern.

If the LoadType item is Accel, this item is UX, UY, UZ, RX, RY or RZ, indicating the direction of the load.

SF

Type:Â SystemDouble

This is an array that includes the scale factor of each load assigned to the load case. [L/s²] for Accel UX UY and UZ; otherwise unitless

Return Value

Type:Â Int32

Returns zero if the data is successfully set; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyLoadType() As String
    Dim MyLoadName() As String
    Dim MySF() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()
```

Parameters

Introduction

```
'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'add static nonlinear load case
    ret = SapModel.LoadCases.StaticNonlinear.SetCase("LCASE1")

'set load data
    ReDim MyLoadType(1)
    ReDim MyLoadName(1)
    ReDim MySF(1)
    MyLoadType(0) = "Load"
    MyLoadName(0) = "DEAD"
    MySF(0) = 0.7
    MyLoadType(1) = "Accel"
    MyLoadName(1) = "UZ"
    MySF(1) = 1.2
    ret = SapModel.LoadCases.StaticNonlinear.SetLoads("LCASE1", 2, MyLoadType, MyLoadName, MySF)

'refresh view
    ret = SapModel.View.RefreshView(0, False)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

🔗 See Also

Reference

[cCaseStaticNonlinear Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cCaseStaticNonlinearAddLanguageSpecificTextSet("LST68C77F48_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetMassSource(
    string Name,
    string mSource
)
```

```
Function SetMassSource (
    Name As String,
    mSource As String
) As Integer
```

```
Dim instance As cCaseStaticNonlinear
Dim Name As String
Dim mSource As String
Dim returnValue As Integer
```

```
returnValue = instance.SetMassSource(Name,
    mSource)
```

```
int SetMassSource(
    String^ Name,
    String^ mSource
)
```

```
abstract SetMassSource :
    Name : string *
    mSource : string -> int
```

Parameters

Name

Type: [SystemString](#)

mSource

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cCaseStaticNonlinear Interface

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CSI API ETABS v1

cCaseStaticNonlinearAddLanguageSpecificTextSet("LST Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetModalCase(
    string Name,
    string ModalCase
)
```

```
Function SetModalCase (
    Name As String,
    ModalCase As String
) As Integer
```

```
Dim instance As cCaseStaticNonlinear
Dim Name As String
Dim ModalCase As String
Dim returnValue As Integer
```

```
returnValue = instance.SetModalCase(Name,
    ModalCase)
```

```
int SetModalCase(
    String^ Name,
    String^ ModalCase
)
```

```
abstract SetModalCase :
    Name : string *
    ModalCase : string -> int
```

Parameters

Name

Type: [SystemString](#)

ModalCase

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cCaseStaticNonlinear Interface

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CSI API ETABS v1

cCaseStaticNonlinearAddLanguageSpecificTextSet("LST Method

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetResultsSaved(
    string Name,
    bool SaveMultipleSteps,
    int MinSavedStates = 10,
    int MaxSavedStates = 100,
    bool PositiveOnly = true
)
```

```
Function SetResultsSaved (
    Name As String,
    SaveMultipleSteps As Boolean,
    MinSavedStates As Integer = 10,
    MaxSavedStates As Integer = 100,
    PositiveOnly As Boolean = true
) As Integer
```

```
Dim instance As cCaseStaticNonlinear
Dim Name As String
Dim SaveMultipleSteps As Boolean
Dim MinSavedStates As Integer
Dim MaxSavedStates As Integer
Dim PositiveOnly As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetResultsSaved(Name,
    SaveMultipleSteps, MinSavedStates,
    MaxSavedStates, PositiveOnly)
```

```
int SetResultsSaved(
    String^ Name,
    bool SaveMultipleSteps,
    int MinSavedStates = 10,
    int MaxSavedStates = 100,
    bool PositiveOnly = true
)
```

```
abstract SetResultsSaved :
    Name : string *
    SaveMultipleSteps : bool *
    ?MinSavedStates : int *
    ?MaxSavedStates : int *
    ?PositiveOnly : bool
```

```
(* Defaults:
    let _MinSavedStates = defaultArg MinSavedStates 10
    let _MaxSavedStates = defaultArg MaxSavedStates 100
    let _PositiveOnly = defaultArg PositiveOnly true
*)
-> int
```

Parameters

Name

Type: SystemString

SaveMultipleSteps

Type: SystemBoolean

MinSavedStates (Optional)

Type: SystemInt32

MaxSavedStates (Optional)

Type: SystemInt32

PositiveOnly (Optional)

Type: SystemBoolean

Return Value

Type: Int32

See Also

Reference

cCaseStaticNonlinear Interface

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CSI API ETABS v1

cCaseStaticNonlinearAddLanguageSpecificTextSet("LST5063950C_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetSolControlParameters(
    string Name,
    int MaxTotalSteps,
    int MaxFailedSubSteps,
    int MaxIterCS,
    int MaxIterNR,
    double TolConvD,
    bool UseEventStepping,
    double TolEventD,
    int MaxLineSearchPerIter,
    double TolLineSearch,
    double LineSearchStepFact
)
```

```
Function SetSolControlParameters (
    Name As String,
    MaxTotalSteps As Integer,
    MaxFailedSubSteps As Integer,
    MaxIterCS As Integer,
    MaxIterNR As Integer,
    TolConvD As Double,
    UseEventStepping As Boolean,
    TolEventD As Double,
    MaxLineSearchPerIter As Integer,
    TolLineSearch As Double,
    LineSearchStepFact As Double
) As Integer
```

```
Dim instance As cCaseStaticNonlinear
Dim Name As String
Dim MaxTotalSteps As Integer
Dim MaxFailedSubSteps As Integer
Dim MaxIterCS As Integer
Dim MaxIterNR As Integer
Dim TolConvD As Double
Dim UseEventStepping As Boolean
Dim TolEventD As Double
Dim MaxLineSearchPerIter As Integer
Dim TolLineSearch As Double
Dim LineSearchStepFact As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetSolControlParameters(Name,
```

```
cCaseStaticNonlinearspan id="LST5063950C_0"AddLanguageSpecificTextSet("LST5063950C_034pp=::|nu
```

```
MaxTotalSteps, MaxFailedSubSteps,
MaxIterCS, MaxIterNR, TolConvD, UseEventStepping,
TolEventD, MaxLineSearchPerIter,
TolLineSearch, LineSearchStepFact)
```

```
int SetSolControlParameters(
String^ Name,
int MaxTotalSteps,
int MaxFailedSubSteps,
int MaxIterCS,
int MaxIterNR,
double TolConvD,
bool UseEventStepping,
double TolEventD,
int MaxLineSearchPerIter,
double TolLineSearch,
double LineSearchStepFact
)
```

```
abstract SetSolControlParameters :
    Name : string *
    MaxTotalSteps : int *
    MaxFailedSubSteps : int *
    MaxIterCS : int *
    MaxIterNR : int *
    TolConvD : float *
    UseEventStepping : bool *
    TolEventD : float *
    MaxLineSearchPerIter : int *
    TolLineSearch : float *
    LineSearchStepFact : float -> int
```

Parameters

Name

Type: SystemString

MaxTotalSteps

Type: SystemInt32

MaxFailedSubSteps

Type: SystemInt32

MaxIterCS

Type: SystemInt32

MaxIterNR

Type: SystemInt32

TolConvD

Type: SystemDouble

UseEventStepping

Type: SystemBoolean

TolEventD

Type: SystemDouble

MaxLineSearchPerIter

Type: SystemInt32

TolLineSearch

Type: SystemDouble

LineSearchStepFact

Type: SystemDouble

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cCaseStaticNonlinear Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cCaseStaticNonlinearAddLanguageSpecificTextSet("LST1AD3786C_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetTargetForceParameters(
    string Name,
    double TolConvF,
    int MaxIter,
    double AccelFact,
    bool NoStop
)
```

```
Function SetTargetForceParameters (
    Name As String,
    TolConvF As Double,
    MaxIter As Integer,
    AccelFact As Double,
    NoStop As Boolean
) As Integer
```

```
Dim instance As cCaseStaticNonlinear
Dim Name As String
Dim TolConvF As Double
Dim MaxIter As Integer
Dim AccelFact As Double
Dim NoStop As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetTargetForceParameters(Name,
    TolConvF, MaxIter, AccelFact, NoStop)
```

```
int SetTargetForceParameters(
    String^ Name,
    double TolConvF,
    int MaxIter,
    double AccelFact,
    bool NoStop
)
```

```
abstract SetTargetForceParameters :
    Name : string *
    TolConvF : float *
    MaxIter : int *
    AccelFact : float *
    NoStop : bool -> int
```

Parameters

Name

Type: SystemString

TolConvF

Type: SystemDouble

MaxIter

Type: SystemInt32

AccelFact

Type: SystemDouble

NoStop

Type: SystemBoolean

Return Value

Type: Int32

See Also

Reference

cCaseStaticNonlinear Interface

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CSI API ETABS v1

cCaseStaticNonlinearStaged Interface

Namespace: $\hat{A}$ ETABSv1

Assembly: $\hat{A}$ ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
public interface cCaseStaticNonlinearStaged
```

```
Public Interface cCaseStaticNonlinearStaged
```

```
Dim instance As cCaseStaticNonlinearStaged
```

```
public interface class cCaseStaticNonlinearStaged
```

```
type cCaseStaticNonlinearStaged = interface end
```

The cCaseStaticNonlinearStaged type exposes the following members.

▲ Methods

$\hat{A}$	Name	Description
	<u>GetGeometricNonlinearity</u>	
	<u>GetHingeUnloading</u>	
	<u>GetInitialCase</u>	
	<u>GetMassSource</u>	
	<u>GetMaterialNonlinearity</u>	
	<u>GetResultsSaved</u>	
	<u>GetSolControlParameters</u>	
	<u>GetStageData</u>	
	<u>GetStageData_1</u>	Retrieves stage data for the specified stage in the specified load case
	<u>GetStageData_2</u>	Retrieves stage data for the specified stage in the specified load case
	<u>GetStageDefinitions</u>	
	<u>GetStageDefinitions_1</u>	Retrieves the stage definition data for the specified load case
	<u>GetStageDefinitions_2</u>	Retrieves the stage definition data for the specified load case
	<u>GetTargetForceParameters</u>	
	<u>SetCase</u>	
	<u>SetGeometricNonlinearity</u>	

- ⇒ [SetHingeUnloading](#)
- ⇒ [SetInitialCase](#)
- ⇒ [SetMassSource](#)
- ⇒ [SetMaterialNonlinearity](#)
- ⇒ [SetResultsSaved](#)
- ⇒ [SetSolControlParameters](#)
- ⇒ [SetStageData](#)
- ⇒ [SetStageData_1](#)
- ⇒ [SetStageData_2](#)
- ⇒ [SetStageDefinitions](#)
- ⇒ [SetStageDefinitions_1](#)
- ⇒ [SetStageDefinitions_2](#)
- ⇒ [SetTargetForceParameters](#)

[Top](#)

▲ See Also

Reference

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CSI API ETABS v1

cCaseStaticNonlinearStaged Methods

The cCaseStaticNonlinearStaged type exposes the following members.

Methods

Á	Name	Description
⇒	<u>GetGeometricNonlinearity</u>	
⇒	<u>GetHingeUnloading</u>	
⇒	<u>GetInitialCase</u>	
⇒	<u>GetMassSource</u>	
⇒	<u>GetMaterialNonlinearity</u>	
⇒	<u>GetResultsSaved</u>	
⇒	<u>GetSolControlParameters</u>	
⇒	<u>GetStageData</u>	
⇒	<u>GetStageData_1</u>	Retrieves stage data for the specified stage in the specified load case
⇒	<u>GetStageData_2</u>	Retrieves stage data for the specified stage in the specified load case
⇒	<u>GetStageDefinitions</u>	
⇒	<u>GetStageDefinitions_1</u>	Retrieves the stage definition data for the specified load case
⇒	<u>GetStageDefinitions_2</u>	Retrieves the stage definition data for the specified load case
⇒	<u>GetTargetForceParameters</u>	
⇒	<u>SetCase</u>	
⇒	<u>SetGeometricNonlinearity</u>	
⇒	<u>SetHingeUnloading</u>	
⇒	<u>SetInitialCase</u>	
⇒	<u>SetMassSource</u>	
⇒	<u>SetMaterialNonlinearity</u>	
⇒	<u>SetResultsSaved</u>	
⇒	<u>SetSolControlParameters</u>	
⇒	<u>SetStageData</u>	
⇒	<u>SetStageData_1</u>	
⇒	<u>SetStageData_2</u>	
⇒	<u>SetStageDefinitions</u>	
⇒	<u>SetStageDefinitions_1</u>	
⇒	<u>SetStageDefinitions_2</u>	
⇒	<u>SetTargetForceParameters</u>	

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▲ See Also

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cCaseStaticNonlinearStaged Interface

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CSI API ETABS v1

cCaseStaticNonlinearStagedAddLanguageSpecificText Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetGeometricNonlinearity(
    string Name,
    ref int NLGeomType
)
```

```
Function GetGeometricNonlinearity (
    Name As String,
    ByRef NLGeomType As Integer
) As Integer
```

```
Dim instance As cCaseStaticNonlinearStaged
Dim Name As String
Dim NLGeomType As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetGeometricNonlinearity(Name,
    NLGeomType)
```

```
int GetGeometricNonlinearity(
    String^ Name,
    int% NLGeomType
)
```

```
abstract GetGeometricNonlinearity :
    Name : string *
    NLGeomType : int byref -> int
```

Parameters

Name

Type: [SystemString](#)

NLGeomType

Type: [SystemInt32](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cCaseStaticNonlinearStaged Interface

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cCaseStaticNonlinearStagedAddLanguageSpecificText Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetHingeUnloading(
    string Name,
    ref int UnloadType
)
```

```
Function GetHingeUnloading (
    Name As String,
    ByRef UnloadType As Integer
) As Integer
```

```
Dim instance As cCaseStaticNonlinearStaged
Dim Name As String
Dim UnloadType As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetHingeUnloading(Name,
    UnloadType)
```

```
int GetHingeUnloading(
    String^ Name,
    int% UnloadType
)
```

```
abstract GetHingeUnloading :
    Name : string *
    UnloadType : int byref -> int
```

Parameters

Name

Type: [SystemString](#)

UnloadType

Type: [SystemInt32](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cCaseStaticNonlinearStaged Interface

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cCaseStaticNonlinearStagedAddLanguageSpecificText Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetInitialCase(
    string Name,
    ref string InitialCase
)
```

```
Function GetInitialCase (
    Name As String,
    ByRef InitialCase As String
) As Integer
```

```
Dim instance As cCaseStaticNonlinearStaged
Dim Name As String
Dim InitialCase As String
Dim returnValue As Integer
```

```
returnValue = instance.GetInitialCase(Name,
    InitialCase)
```

```
int GetInitialCase(
    String^ Name,
    String^% InitialCase
)
```

```
abstract GetInitialCase :
    Name : string *
    InitialCase : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

InitialCase

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cCaseStaticNonlinearStaged Interface

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cCaseStaticNonlinearStagedAddLanguageSpecificText Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetMassSource(
    string Name,
    ref string mSource
)
```

```
Function GetMassSource (
    Name As String,
    ByRef mSource As String
) As Integer
```

```
Dim instance As cCaseStaticNonlinearStaged
Dim Name As String
Dim mSource As String
Dim returnValue As Integer
```

```
returnValue = instance.GetMassSource(Name,
    mSource)
```

```
int GetMassSource(
    String^ Name,
    String^% mSource
)
```

```
abstract GetMassSource :
    Name : string *
    mSource : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

mSource

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

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cCaseStaticNonlinearStagedAddLanguageSpecificText Method

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Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetMaterialNonlinearity(
    string Name,
    ref bool TimeDepMatProp
)
```

```
Function GetMaterialNonlinearity (
    Name As String,
    ByRef TimeDepMatProp As Boolean
) As Integer
```

```
Dim instance As cCaseStaticNonlinearStaged
Dim Name As String
Dim TimeDepMatProp As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetMaterialNonlinearity(Name,
    TimeDepMatProp)
```

```
int GetMaterialNonlinearity(
    String^ Name,
    bool% TimeDepMatProp
)
```

```
abstract GetMaterialNonlinearity :
    Name : string *
    TimeDepMatProp : bool byref -> int
```

Parameters

Name

Type: [SystemString](#)

TimeDepMatProp

Type: [SystemBoolean](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

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Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetResultsSaved(
    string Name,
    ref int StagedSaveOption,
    ref int StagedMinSteps,
    ref int StagedMinStepsTD
)
```

```
Function GetResultsSaved (
    Name As String,
    ByRef StagedSaveOption As Integer,
    ByRef StagedMinSteps As Integer,
    ByRef StagedMinStepsTD As Integer
) As Integer
```

```
Dim instance As cCaseStaticNonlinearStaged
Dim Name As String
Dim StagedSaveOption As Integer
Dim StagedMinSteps As Integer
Dim StagedMinStepsTD As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetResultsSaved(Name,
    StagedSaveOption, StagedMinSteps,
    StagedMinStepsTD)
```

```
int GetResultsSaved(
    String^ Name,
    int% StagedSaveOption,
    int% StagedMinSteps,
    int% StagedMinStepsTD
)
```

```
abstract GetResultsSaved :
    Name : string *
    StagedSaveOption : int byref *
    StagedMinSteps : int byref *
    StagedMinStepsTD : int byref -> int
```

Parameters

Name

Type:Â [SystemString](#)

StagedSaveOption

Type:Â [SystemInt32](#)

StagedMinSteps

Type:Â [SystemInt32](#)

StagedMinStepsTD

Type:Â [SystemInt32](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

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CSI API ETABS v1

cCaseStaticNonlinearStagedAddLanguageSpecificText Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetSolControlParameters(
    string Name,
    ref int MaxTotalSteps,
    ref int MaxFailedSubSteps,
    ref int MaxIterCS,
    ref int MaxIterNR,
    ref double TolConvD,
    ref bool UseEventStepping,
    ref double TolEventD,
    ref int MaxLineSearchPerIter,
    ref double TolLineSearch,
    ref double LineSearchStepFact
)
```

```
Function GetSolControlParameters (
    Name As String,
    ByRef MaxTotalSteps As Integer,
    ByRef MaxFailedSubSteps As Integer,
    ByRef MaxIterCS As Integer,
    ByRef MaxIterNR As Integer,
    ByRef TolConvD As Double,
    ByRef UseEventStepping As Boolean,
    ByRef TolEventD As Double,
    ByRef MaxLineSearchPerIter As Integer,
    ByRef TolLineSearch As Double,
    ByRef LineSearchStepFact As Double
) As Integer
```

```
Dim instance As cCaseStaticNonlinearStaged
Dim Name As String
Dim MaxTotalSteps As Integer
Dim MaxFailedSubSteps As Integer
Dim MaxIterCS As Integer
Dim MaxIterNR As Integer
Dim TolConvD As Double
Dim UseEventStepping As Boolean
Dim TolEventD As Double
Dim MaxLineSearchPerIter As Integer
Dim TolLineSearch As Double
Dim LineSearchStepFact As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetSolControlParameters(Name,
```

cCaseStaticNonlinearStagedspan id="LST83BF22E5_0"AddLanguageSpecificTextSet("LST83BF22E5_0?c

```
MaxTotalSteps, MaxFailedSubSteps,
MaxIterCS, MaxIterNR, TolConvD, UseEventStepping,
TolEventD, MaxLineSearchPerIter,
TolLineSearch, LineSearchStepFact)
```

```
int GetSolControlParameters(
String^ Name,
int% MaxTotalSteps,
int% MaxFailedSubSteps,
int% MaxIterCS,
int% MaxIterNR,
double% TolConvD,
bool% UseEventStepping,
double% TolEventD,
int% MaxLineSearchPerIter,
double% TolLineSearch,
double% LineSearchStepFact
)
```

```
abstract GetSolControlParameters :
    Name : string *
    MaxTotalSteps : int byref *
    MaxFailedSubSteps : int byref *
    MaxIterCS : int byref *
    MaxIterNR : int byref *
    TolConvD : float byref *
    UseEventStepping : bool byref *
    TolEventD : float byref *
    MaxLineSearchPerIter : int byref *
    TolLineSearch : float byref *
    LineSearchStepFact : float byref -> int
```

Parameters

Name

Type: SystemString

MaxTotalSteps

Type: SystemInt32

MaxFailedSubSteps

Type: SystemInt32

MaxIterCS

Type: SystemInt32

MaxIterNR

Type: SystemInt32

TolConvD

Type: SystemDouble

UseEventStepping

Type: SystemBoolean

TolEventD

Type: SystemDouble

MaxLineSearchPerIter

Type: SystemInt32

TolLineSearch

Type: SystemDouble

LineSearchStepFact

Type: SystemDouble

Return Value

Type: [Int32](#)

▲ See Also

Reference

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CSI API ETABS v1

cCaseStaticNonlinearStagedAddLanguageSpecificText Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetStageData(
    string Name,
    ref int Stage,
    ref int NumberOperations,
    ref int[] Operation,
    ref string[] GroupName,
    ref int[] Age,
    ref string[] LoadType,
    ref string[] LoadName,
    ref double[] SF
)
```

```
Function GetStageData (
    Name As String,
    ByRef Stage As Integer,
    ByRef NumberOperations As Integer,
    ByRef Operation As Integer(),
    ByRef GroupName As String(),
    ByRef Age As Integer(),
    ByRef LoadType As String(),
    ByRef LoadName As String(),
    ByRef SF As Double()
) As Integer
```

```
Dim instance As cCaseStaticNonlinearStaged
Dim Name As String
Dim Stage As Integer
Dim NumberOperations As Integer
Dim Operation As Integer()
Dim GroupName As String()
Dim Age As Integer()
Dim LoadType As String()
Dim LoadName As String()
Dim SF As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetStageData(Name,
    Stage, NumberOperations, Operation,
    GroupName, Age, LoadType, LoadName,
    SF)
```

```
int GetStageData(
```

cCaseStaticNonlinearStagedspan id="LST2CF21D52_0"AddLanguageSpecificTextSet("LST2CF21D52_0?o

```

String^ Name,
int% Stage,
int% NumberOperations,
array<int>^% Operation,
array<String^>^% GroupName,
array<int>^% Age,
array<String^>^% LoadType,
array<String^>^% LoadName,
array<double>^% SF
)

abstract GetStageData :
    Name : string *
    Stage : int byref *
    NumberOperations : int byref *
    Operation : int[] byref *
    GroupName : string[] byref *
    Age : int[] byref *
    LoadType : string[] byref *
    LoadName : string[] byref *
    SF : float[] byref -> int

```

Parameters

Name
Type:Â [SystemString](#)

Stage
Type:Â [SystemInt32](#)

NumberOperations
Type:Â [SystemInt32](#)

Operation
Type:Â [SystemInt32](#)

GroupName
Type:Â [SystemString](#)

Age
Type:Â [SystemInt32](#)

LoadType
Type:Â [SystemString](#)

LoadName
Type:Â [SystemString](#)

SF
Type:Â [SystemDouble](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cCaseStaticNonlinearStaged Interface](#)

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CSI API ETABS v1

cCaseStaticNonlinearStagedAddLanguageSpecificText Method

Retrieves stage data for the specified stage in the specified load case

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetStageData_1(
    string Name,
    ref int Stage,
    ref int NumberOperations,
    ref int[] Operation,
    ref string[] ObjectType,
    ref string[] ObjectName,
    ref int[] Age,
    ref string[] MyType,
    ref string[] MyName,
    ref double[] SF
)
```

```
Function GetStageData_1 (
    Name As String,
    ByRef Stage As Integer,
    ByRef NumberOperations As Integer,
    ByRef Operation As Integer(),
    ByRef ObjectType As String(),
    ByRef ObjectName As String(),
    ByRef Age As Integer(),
    ByRef MyType As String(),
    ByRef MyName As String(),
    ByRef SF As Double()
) As Integer
```

```
Dim instance As cCaseStaticNonlinearStaged
Dim Name As String
Dim Stage As Integer
Dim NumberOperations As Integer
Dim Operation As Integer()
Dim ObjectType As String()
Dim ObjectName As String()
Dim Age As Integer()
Dim MyType As String()
Dim MyName As String()
Dim SF As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetStageData_1(Name,
    Stage, NumberOperations, Operation,
```

ObjectType, ObjectName, Age, MyType,
MyName, SF)

```
int GetStageData_1(
String^ Name,
int% Stage,
int% NumberOperations,
array<int>^% Operation,
array<String>^% ObjectType,
array<String>^% ObjectName,
array<int>^% Age,
array<String>^% MyType,
array<String>^% MyName,
array<double>^% SF
)
```

```
abstract GetStageData_1 :
    Name : string *
    Stage : int byref *
    NumberOperations : int byref *
    Operation : int[] byref *
    ObjectType : string[] byref *
    ObjectName : string[] byref *
    Age : int[] byref *
    MyType : string[] byref *
    MyName : string[] byref *
    SF : float[] byref -> int
```

Parameters

Name

Type:Â [SystemString](#)

The name of an existing static nonlinear staged load case

Stage

Type:Â [SystemInt32](#)

The stage in the specified load case for which data is requested. Stages are numbered sequentially starting from 1

NumberOperations

Type:Â [SystemInt32](#)

The number of operations in the specified stage

Operation

Type:Â [SystemInt32](#)

This is an array that includes 1, 2, 3, 4, 5, 6, 7, or 11, indicating an operation type.

- 1 Add structure
- 2 Remove structure
- 3 Load objects if new
- 4 Load objects
- 5 Change section properties
- 6 Change section property modifiers
- 7 Change releases
- 11 Change section properties and age

ObjectType

Type: SystemString

This is an array that includes the object type associated with the specified operation. The object type may be one of the following:

- ◇ Group
- ◇ Frame
- ◇ Cable
- ◇ Tendon
- ◇ Area
- ◇ Solid
- ◇ Link
- ◇ Point

The following list shows which object types are applicable to each operation type:

Operation = 1 (Add structure)	All object types
Operation = 2 (Remove structure)	All object types
Operation = 3 (Load objects if new)	All object types
Operation = 4 (Load objects)	All object types
Operation = 5 (Change section properties)	All object types except Point
Operation = 6 (Change section property modifiers)	Group, Frame, Cable, Area
Operation = 7 (Change releases)	Group, Frame
Operation = 11 (Change section properties and age)	All object types except Point

ObjectName

Type: SystemString

This is an array that includes the name of the object associated with the specified operation. This is the name of a Group, Frame object, Cable object, Tendon object, Area object, Solid object, Link object or Point object, depending on the ObjectType item

Age

Type: SystemInt32

This is an array that includes the age of the added structure, at the time it is added, in days. This item applies only to operations with Operation = 1.

MyType

Type: SystemString

This is an array that includes a load type or an object type, depending on what is specified for the Operation item. This item applies only to operations with Operation = 3, 4, 5, 6, 7, or 11.

When Operation = 3 or 4, this is an array that includes Load or Accel, indicating the load type of an added load.

When Operation = 5 or 11, and the ObjectType item is Group, this is an array that includes Frame, Cable, Tendon, Area, Solid or Link, indicating the object type for which the section property is changed.

When Operation = 6 and the ObjectType item is Group, this is an array that includes Frame, Cable or Area, indicating the object type for which the section

property modifiers are changed.

When Operation = 7 and the ObjectType item is Group, this is an array that includes Frame, indicating the object type for which the releases are changed.

When Operation = 5, 6, 7, or 11, and the ObjectType item is not Group and not Point, this item is ignored and the type is picked up from the ObjectType item.

MyName

Type: SystemString

This is an array that includes a load assignment or an object name, depending on what is specified for the Operation item. This item applies only to operations with Operation = 3, 4, 5, 6, 7, or 11.

When Operation = 3 or 4, this is an array that includes the name of the load assigned to the operation. If the associated MyType (load type) item is Load, this item is the name of a defined load pattern. If the associated MyType (load type) item is Accel, this item is UX, UY, UZ, RX, RY or RZ, indicating the direction of the load.

When Operation = 5 or 11, this is the name of a Frame, Cable, Tendon, Area, Solid or Link object, depending on the object type specified.

When Operation = 6, this is the name of a Frame, Cable or Area object, depending on the object type specified.

When Operation = 7, this is the name of a Frame object.

SF

Type: SystemDouble

This is an array that includes the scale factor for the load assigned to the operation, if any. [L/s²] for Accel UX UY and UZ; otherwise unitless

This item applies only to operations with Operation = 3 or 4.

Return Value

Type: Int32

Returns zero if the data is successfully retrieved; otherwise, it returns a nonzero value

Remarks

This function is obsolete and has been superseded by GetStageData_2(String, Int32, Int32, Int32, String, String, Double, String, String, Double) as of version 16.0.2. This function is maintained for backwards compatibility.

See Also

Reference

cCaseStaticNonlinearStaged Interface

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CSI API ETABS v1

cCaseStaticNonlinearStagedAddLanguageSpecificTextMethod

Retrieves stage data for the specified stage in the specified load case

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetStageData_2(
    string Name,
    ref int Stage,
    ref int NumberOperations,
    ref int[] Operation,
    ref string[] ObjectType,
    ref string[] ObjectName,
    ref double[] Age,
    ref string[] MyType,
    ref string[] MyName,
    ref double[] SF
)
```

```
Function GetStageData_2 (
    Name As String,
    ByRef Stage As Integer,
    ByRef NumberOperations As Integer,
    ByRef Operation As Integer(),
    ByRef ObjectType As String(),
    ByRef ObjectName As String(),
    ByRef Age As Double(),
    ByRef MyType As String(),
    ByRef MyName As String(),
    ByRef SF As Double()
) As Integer
```

```
Dim instance As cCaseStaticNonlinearStaged
Dim Name As String
Dim Stage As Integer
Dim NumberOperations As Integer
Dim Operation As Integer()
Dim ObjectType As String()
Dim ObjectName As String()
Dim Age As Double()
Dim MyType As String()
Dim MyName As String()
Dim SF As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetStageData_2(Name,
    Stage, NumberOperations, Operation,
```

ObjectType, ObjectName, Age, MyType,
MyName, SF)

```
int GetStageData_2(
String^ Name,
int% Stage,
int% NumberOperations,
array<int>^% Operation,
array<String^>^% ObjectType,
array<String^>^% ObjectName,
array<double>^% Age,
array<String^>^% MyType,
array<String^>^% MyName,
array<double>^% SF
)
```

```
abstract GetStageData_2 :
    Name : string *
    Stage : int byref *
    NumberOperations : int byref *
    Operation : int[] byref *
    ObjectType : string[] byref *
    ObjectName : string[] byref *
    Age : float[] byref *
    MyType : string[] byref *
    MyName : string[] byref *
    SF : float[] byref -> int
```

Parameters

Name

Type:Â [SystemString](#)

The name of an existing static nonlinear staged load case

Stage

Type:Â [SystemInt32](#)

The stage in the specified load case for which data is requested. Stages are numbered sequentially starting from 1

NumberOperations

Type:Â [SystemInt32](#)

The number of operations in the specified stage

Operation

Type:Â [SystemInt32](#)

This is an array that includes 1, 2, 3, 4, 5, 6, 7, or 11, indicating an operation type.

- 1 Add structure
- 2 Remove structure
- 3 Load objects if new
- 4 Load objects
- 5 Change section properties
- 6 Change section property modifiers
- 7 Change releases
- 11 Change section properties and age

ObjectType

Type: SystemString

This is an array that includes the object type associated with the specified operation. The object type may be one of the following:

- ◇ Group
- ◇ Frame
- ◇ Cable
- ◇ Tendon
- ◇ Area
- ◇ Solid
- ◇ Link
- ◇ Point

The following list shows which object types are applicable to each operation type:

Operation = 1 (Add structure)	All object types
Operation = 2 (Remove structure)	All object types
Operation = 3 (Load objects if new)	All object types
Operation = 4 (Load objects)	All object types
Operation = 5 (Change section properties)	All object types except Point
Operation = 6 (Change section property modifiers)	Group, Frame, Cable, Area
Operation = 7 (Change releases)	Group, Frame
Operation = 11 (Change section properties and age)	All object types except Point

ObjectName

Type: SystemString

This is an array that includes the name of the object associated with the specified operation. This is the name of a Group, Frame object, Cable object, Tendon object, Area object, Solid object, Link object or Point object, depending on the ObjectType item

Age

Type: SystemDouble

This is an array that includes the age of the added structure, at the time it is added, in days. This item applies only to operations with Operation = 1.

MyType

Type: SystemString

This is an array that includes a load type or an object type, depending on what is specified for the Operation item. This item applies only to operations with Operation = 3, 4, 5, 6, 7, or 11.

When Operation = 3 or 4, this is an array that includes Load or Accel, indicating the load type of an added load.

When Operation = 5 or 11, and the ObjectType item is Group, this is an array that includes Frame, Cable, Tendon, Area, Solid or Link, indicating the object type for which the section property is changed.

When Operation = 6 and the ObjectType item is Group, this is an array that includes Frame, Cable or Area, indicating the object type for which the section

property modifiers are changed.

When Operation = 7 and the ObjectType item is Group, this is an array that includes Frame, indicating the object type for which the releases are changed.

When Operation = 5, 6, 7, or 11, and the ObjectType item is not Group and not Point, this item is ignored and the type is picked up from the ObjectType item.

MyName

Type: SystemString

This is an array that includes a load assignment or an object name, depending on what is specified for the Operation item. This item applies only to operations with Operation = 3, 4, 5, 6, 7, or 11.

When Operation = 3 or 4, this is an array that includes the name of the load assigned to the operation. If the associated MyType (load type) item is Load, this item is the name of a defined load pattern. If the associated MyType (load type) item is Accel, this item is UX, UY, UZ, RX, RY or RZ, indicating the direction of the load.

When Operation = 5 or 11, this is the name of a Frame, Cable, Tendon, Area, Solid or Link object, depending on the object type specified.

When Operation = 6, this is the name of a Frame, Cable or Area object, depending on the object type specified.

When Operation = 7, this is the name of a Frame object.

SF

Type: SystemDouble

This is an array that includes the scale factor for the load assigned to the operation, if any. [L/s²] for Accel UX UY and UZ; otherwise unitless

This item applies only to operations with Operation = 3 or 4.

Return Value

Type: Int32

Returns zero if the data is successfully retrieved; otherwise, it returns a nonzero value

▲ Remarks

▲ See Also

Reference

cCaseStaticNonlinearStaged Interface

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CSI API ETABS v1

cCaseStaticNonlinearStagedAddLanguageSpecificText Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetStageDefinitions(
    string Name,
    ref int NumberStages,
    ref int[] Duration,
    ref string[] Comment
)
```

```
Function GetStageDefinitions (
    Name As String,
    ByRef NumberStages As Integer,
    ByRef Duration As Integer(),
    ByRef Comment As String()
) As Integer
```

```
Dim instance As cCaseStaticNonlinearStaged
Dim Name As String
Dim NumberStages As Integer
Dim Duration As Integer()
Dim Comment As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetStageDefinitions(Name,
    NumberStages, Duration, Comment)
```

```
int GetStageDefinitions(
    String^ Name,
    int% NumberStages,
    array<int>^% Duration,
    array<String^>^% Comment
)
```

```
abstract GetStageDefinitions :
    Name : string *
    NumberStages : int byref *
    Duration : int[] byref *
    Comment : string[] byref -> int
```

Parameters

Name

Type: SystemString

cCaseStaticNonlinearStagedspan id="LST47EB00F5_0"AddLanguageSpecificTextSet("LST47EB00F5_0?c

NumberStages

Type: SystemInt32

Duration

Type: SystemInt32

Comment

Type: SystemString

Return Value

Type: Int32

See Also

Reference

cCaseStaticNonlinearStaged Interface

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CSI API ETABS v1

cCaseStaticNonlinearStagedAddLanguageSpecificText Method

Retrieves the stage definition data for the specified load case

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetStageDefinitions_1(
    string Name,
    ref int NumberStages,
    ref int[] Duration,
    ref bool[] Output,
    ref string[] OutputName,
    ref string[] Comment
)
```

```
Function GetStageDefinitions_1 (
    Name As String,
    ByRef NumberStages As Integer,
    ByRef Duration As Integer(),
    ByRef Output As Boolean(),
    ByRef OutputName As String(),
    ByRef Comment As String()
) As Integer
```

```
Dim instance As cCaseStaticNonlinearStaged
Dim Name As String
Dim NumberStages As Integer
Dim Duration As Integer()
Dim Output As Boolean()
Dim OutputName As String()
Dim Comment As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetStageDefinitions_1(Name,
    NumberStages, Duration, Output, OutputName,
    Comment)
```

```
int GetStageDefinitions_1(
    String^ Name,
    int% NumberStages,
    array<int>^% Duration,
    array<bool>^% Output,
    array<String>^% OutputName,
    array<String>^% Comment
)
```

```
abstract GetStageDefinitions_1 :
```

cCaseStaticNonlinearStagedspan id="LST5829E959_0"AddLanguageSpecificTextSet("LST5829E959_0?cp

```

Name : string *
NumberStages : int byref *
Duration : int[] byref *
Output : bool[] byref *
OutputName : string[] byref *
Comment : string[] byref -> int

```

Parameters

Name

Type:Â SystemString

The name of an existing static nonlinear staged load case

NumberStages

Type:Â SystemInt32

The number of stages defined for the specified load case

Duration

Type:Â SystemInt32

This is an array that includes the duration in days for each stage

Output

Type:Â SystemBoolean

This is an array that includes True or False, indicating if analysis output is to be saved for each stage

OutputName

Type:Â SystemString

This is an array that includes a user-specified output name for each stage

Comment

Type:Â SystemString

This is an array that includes a comment for each stage. The comment may be a blank string.

Return Value

Type:Â Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value

Remarks

This function is obsolete and has been superseded by GetStageDefinitions_2(String, Int32, Double, Boolean, String, String) as of version 16.0.2. This function is maintained for backwards compatibility.

See Also

Reference

cCaseStaticNonlinearStaged Interface

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CSI API ETABS v1

cCaseStaticNonlinearStagedAddLanguageSpecificTextMethod

Retrieves the stage definition data for the specified load case

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetStageDefinitions_2(
    string Name,
    ref int NumberStages,
    ref double[] Duration,
    ref bool[] Output,
    ref string[] OutputName,
    ref string[] Comment
)
```

```
Function GetStageDefinitions_2 (
    Name As String,
    ByRef NumberStages As Integer,
    ByRef Duration As Double(),
    ByRef Output As Boolean(),
    ByRef OutputName As String(),
    ByRef Comment As String()
) As Integer
```

```
Dim instance As cCaseStaticNonlinearStaged
Dim Name As String
Dim NumberStages As Integer
Dim Duration As Double()
Dim Output As Boolean()
Dim OutputName As String()
Dim Comment As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetStageDefinitions_2(Name,
    NumberStages, Duration, Output, OutputName,
    Comment)
```

```
int GetStageDefinitions_2(
    String^ Name,
    int% NumberStages,
    array<double>^% Duration,
    array<bool>^% Output,
    array<String^>^% OutputName,
    array<String^>^% Comment
)
```

```
abstract GetStageDefinitions_2 :
```

cCaseStaticNonlinearStagedspan id="LSTF2FBA534_0"AddLanguageSpecificTextSet("LSTF2FBA534_0?c

```

Name : string *
NumberStages : int byref *
Duration : float[] byref *
Output : bool[] byref *
OutputName : string[] byref *
Comment : string[] byref -> int

```

Parameters

Name

Type: SystemString

The name of an existing static nonlinear staged load case

NumberStages

Type: SystemInt32

The number of stages defined for the specified load case

Duration

Type: SystemDouble

This is an array that includes the duration in days for each stage

Output

Type: SystemBoolean

This is an array that includes True or False, indicating if analysis output is to be saved for each stage

OutputName

Type: SystemString

This is an array that includes a user-specified output name for each stage

Comment

Type: SystemString

This is an array that includes a comment for each stage. The comment may be a blank string.

Return Value

Type: Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ See Also

Reference

cCaseStaticNonlinearStaged Interface

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CSI API ETABS v1

cCaseStaticNonlinearStagedAddLanguageSpecificText Method

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetTargetForceParameters(
    string Name,
    ref double TolConvF,
    ref int MaxIter,
    ref double AccelFact,
    ref bool NoStop
)
```

```
Function GetTargetForceParameters (
    Name As String,
    ByRef TolConvF As Double,
    ByRef MaxIter As Integer,
    ByRef AccelFact As Double,
    ByRef NoStop As Boolean
) As Integer
```

```
Dim instance As cCaseStaticNonlinearStaged
Dim Name As String
Dim TolConvF As Double
Dim MaxIter As Integer
Dim AccelFact As Double
Dim NoStop As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetTargetForceParameters(Name,
    TolConvF, MaxIter, AccelFact, NoStop)
```

```
int GetTargetForceParameters(
    String^ Name,
    double% TolConvF,
    int% MaxIter,
    double% AccelFact,
    bool% NoStop
)
```

```
abstract GetTargetForceParameters :
    Name : string *
    TolConvF : float byref *
    MaxIter : int byref *
    AccelFact : float byref *
    NoStop : bool byref -> int
```

Parameters

Name

Type: SystemString

TolConvF

Type: SystemDouble

MaxIter

Type: SystemInt32

AccelFact

Type: SystemDouble

NoStop

Type: SystemBoolean

Return Value

Type: Int32

See Also

Reference

cCaseStaticNonlinearStaged Interface

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CSI API ETABS v1

cCaseStaticNonlinearStagedAddLanguageSpecificText Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetCase(
    string Name
)
```

```
Function SetCase (
    Name As String
) As Integer
```

```
Dim instance As cCaseStaticNonlinearStaged
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.SetCase(Name)
```

```
int SetCase(
    String^ Name
)
```

```
abstract SetCase :
    Name : string -> int
```

Parameters

Name

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cCaseStaticNonlinearStaged Interface](#)

[ETABSv1 Namespace](#)

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cCaseStaticNonlinearStagedspan id="LST2A1EC84E_0"AddLanguageSpecificTextSet("LST2A1EC84E_0?")

Introduction

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CSI API ETABS v1

cCaseStaticNonlinearStagedAddLanguageSpecificText Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetGeometricNonlinearity(
    string Name,
    int NLGeomType
)
```

```
Function SetGeometricNonlinearity (
    Name As String,
    NLGeomType As Integer
) As Integer
```

```
Dim instance As cCaseStaticNonlinearStaged
Dim Name As String
Dim NLGeomType As Integer
Dim returnValue As Integer
```

```
returnValue = instance.SetGeometricNonlinearity(Name,
NLGeomType)
```

```
int SetGeometricNonlinearity(
    String^ Name,
    int NLGeomType
)
```

```
abstract SetGeometricNonlinearity :
    Name : string *
    NLGeomType : int -> int
```

Parameters

Name

Type: SystemString

NLGeomType

Type: SystemInt32

Return Value

Type: Int32

▲ See Also

Reference

cCaseStaticNonlinearStaged Interface

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CSI API ETABS v1

cCaseStaticNonlinearStagedAddLanguageSpecificText Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetHingeUnloading(
    string Name,
    int UnloadType
)
```

```
Function SetHingeUnloading (
    Name As String,
    UnloadType As Integer
) As Integer
```

```
Dim instance As cCaseStaticNonlinearStaged
Dim Name As String
Dim UnloadType As Integer
Dim returnValue As Integer
```

```
returnValue = instance.SetHingeUnloading(Name,
    UnloadType)
```

```
int SetHingeUnloading(
    String^ Name,
    int UnloadType
)
```

```
abstract SetHingeUnloading :
    Name : string *
    UnloadType : int -> int
```

Parameters

Name

Type: [SystemString](#)

UnloadType

Type: [SystemInt32](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cCaseStaticNonlinearStaged Interface

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CSI API ETABS v1

cCaseStaticNonlinearStagedAddLanguageSpecificTextMethod

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetInitialCase(
    string Name,
    string InitialCase
)
```

```
Function SetInitialCase (
    Name As String,
    InitialCase As String
) As Integer
```

```
Dim instance As cCaseStaticNonlinearStaged
Dim Name As String
Dim InitialCase As String
Dim returnValue As Integer
```

```
returnValue = instance.SetInitialCase(Name,
    InitialCase)
```

```
int SetInitialCase(
    String^ Name,
    String^ InitialCase
)
```

```
abstract SetInitialCase :
    Name : string *
    InitialCase : string -> int
```

Parameters

Name

Type: [SystemString](#)

InitialCase

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cCaseStaticNonlinearStaged Interface

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CSI API ETABS v1

cCaseStaticNonlinearStagedAddLanguageSpecificText Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetMassSource(
    string Name,
    string mSource
)
```

```
Function SetMassSource (
    Name As String,
    mSource As String
) As Integer
```

```
Dim instance As cCaseStaticNonlinearStaged
Dim Name As String
Dim mSource As String
Dim returnValue As Integer
```

```
returnValue = instance.SetMassSource(Name,
mSource)
```

```
int SetMassSource(
    String^ Name,
    String^ mSource
)
```

```
abstract SetMassSource :
    Name : string *
    mSource : string -> int
```

Parameters

Name

Type: [SystemString](#)

mSource

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cCaseStaticNonlinearStaged Interface

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CSI API ETABS v1

cCaseStaticNonlinearStagedAddLanguageSpecificTextMethod

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetMaterialNonlinearity(
    string Name,
    bool TimeDepMatProp
)
```

```
Function SetMaterialNonlinearity (
    Name As String,
    TimeDepMatProp As Boolean
) As Integer
```

```
Dim instance As cCaseStaticNonlinearStaged
Dim Name As String
Dim TimeDepMatProp As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetMaterialNonlinearity(Name,
    TimeDepMatProp)
```

```
int SetMaterialNonlinearity(
    String^ Name,
    bool TimeDepMatProp
)
```

```
abstract SetMaterialNonlinearity :
    Name : string *
    TimeDepMatProp : bool -> int
```

Parameters

Name

Type: [SystemString](#)

TimeDepMatProp

Type: [SystemBoolean](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cCaseStaticNonlinearStaged Interface

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CSI API ETABS v1

cCaseStaticNonlinearStagedAddLanguageSpecificText Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetResultsSaved(
    string Name,
    int StagedSaveOption,
    int StagedMinSteps = 1,
    int StagedMinStepsTD = 1
)
```

```
Function SetResultsSaved (
    Name As String,
    StagedSaveOption As Integer,
    StagedMinSteps As Integer = 1,
    StagedMinStepsTD As Integer = 1
) As Integer
```

```
Dim instance As cCaseStaticNonlinearStaged
Dim Name As String
Dim StagedSaveOption As Integer
Dim StagedMinSteps As Integer
Dim StagedMinStepsTD As Integer
Dim returnValue As Integer
```

```
returnValue = instance.SetResultsSaved(Name,
    StagedSaveOption, StagedMinSteps,
    StagedMinStepsTD)
```

```
int SetResultsSaved(
    String^ Name,
    int StagedSaveOption,
    int StagedMinSteps = 1,
    int StagedMinStepsTD = 1
)
```

```
abstract SetResultsSaved :
    Name : string *
    StagedSaveOption : int *
    ?StagedMinSteps : int *
    ?StagedMinStepsTD : int
(* Defaults:
    let _StagedMinSteps = defaultArg StagedMinSteps 1
    let _StagedMinStepsTD = defaultArg StagedMinStepsTD 1
*)
-> int
```

cCaseStaticNonlinearStagedspan id="LSTC0D21F3D_0"AddLanguageSpecificTextSet("LSTC0D21F3D_0?")

Parameters

Name

Type: SystemString

StagedSaveOption

Type: SystemInt32

StagedMinSteps (Optional)

Type: SystemInt32

StagedMinStepsTD (Optional)

Type: SystemInt32**Return Value**Type: Int32

▲ See Also

ReferencecCaseStaticNonlinearStaged InterfaceETABSv1 Namespace

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CSI API ETABS v1

cCaseStaticNonlinearStagedAddLanguageSpecificTextMethod

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetSolControlParameters(
    string Name,
    int MaxTotalSteps,
    int MaxFailedSubSteps,
    int MaxIterCS,
    int MaxIterNR,
    double TolConvD,
    bool UseEventStepping,
    double TolEventD,
    int MaxLineSearchPerIter,
    double TolLineSearch,
    double LineSearchStepFact
)
```

```
Function SetSolControlParameters (
    Name As String,
    MaxTotalSteps As Integer,
    MaxFailedSubSteps As Integer,
    MaxIterCS As Integer,
    MaxIterNR As Integer,
    TolConvD As Double,
    UseEventStepping As Boolean,
    TolEventD As Double,
    MaxLineSearchPerIter As Integer,
    TolLineSearch As Double,
    LineSearchStepFact As Double
) As Integer
```

```
Dim instance As cCaseStaticNonlinearStaged
Dim Name As String
Dim MaxTotalSteps As Integer
Dim MaxFailedSubSteps As Integer
Dim MaxIterCS As Integer
Dim MaxIterNR As Integer
Dim TolConvD As Double
Dim UseEventStepping As Boolean
Dim TolEventD As Double
Dim MaxLineSearchPerIter As Integer
Dim TolLineSearch As Double
Dim LineSearchStepFact As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetSolControlParameters(Name,
```

cCaseStaticNonlinearStagedspan id="LST41C4BBA_0"AddLanguageSpecificTextSet("LST41C4BBA_0?cp

```
MaxTotalSteps, MaxFailedSubSteps,
MaxIterCS, MaxIterNR, TolConvD, UseEventStepping,
TolEventD, MaxLineSearchPerIter,
TolLineSearch, LineSearchStepFact)
```

```
int SetSolControlParameters(
String^ Name,
int MaxTotalSteps,
int MaxFailedSubSteps,
int MaxIterCS,
int MaxIterNR,
double TolConvD,
bool UseEventStepping,
double TolEventD,
int MaxLineSearchPerIter,
double TolLineSearch,
double LineSearchStepFact
)
```

```
abstract SetSolControlParameters :
    Name : string *
    MaxTotalSteps : int *
    MaxFailedSubSteps : int *
    MaxIterCS : int *
    MaxIterNR : int *
    TolConvD : float *
    UseEventStepping : bool *
    TolEventD : float *
    MaxLineSearchPerIter : int *
    TolLineSearch : float *
    LineSearchStepFact : float -> int
```

Parameters

Name

Type: SystemString

MaxTotalSteps

Type: SystemInt32

MaxFailedSubSteps

Type: SystemInt32

MaxIterCS

Type: SystemInt32

MaxIterNR

Type: SystemInt32

TolConvD

Type: SystemDouble

UseEventStepping

Type: SystemBoolean

TolEventD

Type: SystemDouble

MaxLineSearchPerIter

Type: SystemInt32

TolLineSearch

Type: SystemDouble

LineSearchStepFact

Type: SystemDouble

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cCaseStaticNonlinearStaged Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cCaseStaticNonlinearStagedAddLanguageSpecificText Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetStageData(
    string Name,
    int Stage,
    int NumberOperations,
    ref int[] Operation,
    ref string[] GroupName,
    ref int[] Age,
    ref string[] LoadType,
    ref string[] LoadName,
    ref double[] SF
)
```

```
Function SetStageData (
    Name As String,
    Stage As Integer,
    NumberOperations As Integer,
    ByRef Operation As Integer(),
    ByRef GroupName As String(),
    ByRef Age As Integer(),
    ByRef LoadType As String(),
    ByRef LoadName As String(),
    ByRef SF As Double()
) As Integer
```

```
Dim instance As cCaseStaticNonlinearStaged
Dim Name As String
Dim Stage As Integer
Dim NumberOperations As Integer
Dim Operation As Integer()
Dim GroupName As String()
Dim Age As Integer()
Dim LoadType As String()
Dim LoadName As String()
Dim SF As Double()
Dim returnValue As Integer
```

```
returnValue = instance.SetStageData(Name,
    Stage, NumberOperations, Operation,
    GroupName, Age, LoadType, LoadName,
    SF)
```

```
int SetStageData(
```

cCaseStaticNonlinearStagedspan id="LST9B6F70B1_0"AddLanguageSpecificTextSet("LST9B6F70B1_0?c

```

String^ Name,
int Stage,
int NumberOperations,
array<int>^% Operation,
array<String^>^% GroupName,
array<int>^% Age,
array<String^>^% LoadType,
array<String^>^% LoadName,
array<double>^% SF
)

abstract SetStageData :
    Name : string *
    Stage : int *
    NumberOperations : int *
    Operation : int[] byref *
    GroupName : string[] byref *
    Age : int[] byref *
    LoadType : string[] byref *
    LoadName : string[] byref *
    SF : float[] byref -> int

```

Parameters

Name
Type:Â [SystemString](#)

Stage
Type:Â [SystemInt32](#)

NumberOperations
Type:Â [SystemInt32](#)

Operation
Type:Â [SystemInt32](#)

GroupName
Type:Â [SystemString](#)

Age
Type:Â [SystemInt32](#)

LoadType
Type:Â [SystemString](#)

LoadName
Type:Â [SystemString](#)

SF
Type:Â [SystemDouble](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cCaseStaticNonlinearStaged Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cCaseStaticNonlinearStagedAddLanguageSpecificText Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetStageData_1(
    string Name,
    int Stage,
    int NumberOperations,
    ref int[] Operation,
    ref string[] ObjectType,
    ref string[] ObjectName,
    ref int[] Age,
    ref string[] MyType,
    ref string[] MyName,
    ref double[] SF
)
```

```
Function SetStageData_1 (
    Name As String,
    Stage As Integer,
    NumberOperations As Integer,
    ByRef Operation As Integer(),
    ByRef ObjectType As String(),
    ByRef ObjectName As String(),
    ByRef Age As Integer(),
    ByRef MyType As String(),
    ByRef MyName As String(),
    ByRef SF As Double()
) As Integer
```

```
Dim instance As cCaseStaticNonlinearStaged
Dim Name As String
Dim Stage As Integer
Dim NumberOperations As Integer
Dim Operation As Integer()
Dim ObjectType As String()
Dim ObjectName As String()
Dim Age As Integer()
Dim MyType As String()
Dim MyName As String()
Dim SF As Double()
Dim returnValue As Integer
```

```
returnValue = instance.SetStageData_1(Name,
    Stage, NumberOperations, Operation,
    ObjectType, ObjectName, Age, MyType,
    MyName, SF)
```

cCaseStaticNonlinearStagedspan id="LST54C503AE_0"AddLanguageSpecificTextSet("LST54C503AE_0?c

```

int SetStageData_1(
String^ Name,
int Stage,
int NumberOperations,
array<int>^% Operation,
array<String^>^% ObjectType,
array<String^>^% ObjectName,
array<int>^% Age,
array<String^>^% MyType,
array<String^>^% MyName,
array<double>^% SF
)

```

```

abstract SetStageData_1 :
    Name : string *
    Stage : int *
    NumberOperations : int *
    Operation : int[] byref *
    ObjectType : string[] byref *
    ObjectName : string[] byref *
    Age : int[] byref *
    MyType : string[] byref *
    MyName : string[] byref *
    SF : float[] byref -> int

```

Parameters

Name

Type:Â [SystemString](#)

Stage

Type:Â [SystemInt32](#)

NumberOperations

Type:Â [SystemInt32](#)

Operation

Type:Â [SystemInt32](#)

ObjectType

Type:Â [SystemString](#)

ObjectName

Type:Â [SystemString](#)

Age

Type:Â [SystemInt32](#)

MyType

Type:Â [SystemString](#)

MyName

Type:Â [SystemString](#)

SF

Type:Â [SystemDouble](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

cCaseStaticNonlinearStaged Interface

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CSI API ETABS v1

cCaseStaticNonlinearStagedAddLanguageSpecificText Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetStageData_2(
    string Name,
    int Stage,
    int NumberOperations,
    ref int[] Operation,
    ref string[] ObjectType,
    ref string[] ObjectName,
    ref double[] Age,
    ref string[] MyType,
    ref string[] MyName,
    ref double[] SF
)
```

```
Function SetStageData_2 (
    Name As String,
    Stage As Integer,
    NumberOperations As Integer,
    ByRef Operation As Integer(),
    ByRef ObjectType As String(),
    ByRef ObjectName As String(),
    ByRef Age As Double(),
    ByRef MyType As String(),
    ByRef MyName As String(),
    ByRef SF As Double()
) As Integer
```

```
Dim instance As cCaseStaticNonlinearStaged
Dim Name As String
Dim Stage As Integer
Dim NumberOperations As Integer
Dim Operation As Integer()
Dim ObjectType As String()
Dim ObjectName As String()
Dim Age As Double()
Dim MyType As String()
Dim MyName As String()
Dim SF As Double()
Dim returnValue As Integer
```

```
returnValue = instance.SetStageData_2(Name,
    Stage, NumberOperations, Operation,
    ObjectType, ObjectName, Age, MyType,
    MyName, SF)
```

cCaseStaticNonlinearStagedspan id="LST40579E74_0"AddLanguageSpecificTextSet("LST40579E74_0?cp

```

int SetStageData_2(
String^ Name,
int Stage,
int NumberOperations,
array<int>^% Operation,
array<String^>^% ObjectType,
array<String^>^% ObjectName,
array<double>^% Age,
array<String^>^% MyType,
array<String^>^% MyName,
array<double>^% SF
)

```

```

abstract SetStageData_2 :
    Name : string *
    Stage : int *
    NumberOperations : int *
    Operation : int[] byref *
    ObjectType : string[] byref *
    ObjectName : string[] byref *
    Age : float[] byref *
    MyType : string[] byref *
    MyName : string[] byref *
    SF : float[] byref -> int

```

Parameters

Name

Type:Â [SystemString](#)

Stage

Type:Â [SystemInt32](#)

NumberOperations

Type:Â [SystemInt32](#)

Operation

Type:Â [SystemInt32](#)

ObjectType

Type:Â [SystemString](#)

ObjectName

Type:Â [SystemString](#)

Age

Type:Â [SystemDouble](#)

MyType

Type:Â [SystemString](#)

MyName

Type:Â [SystemString](#)

SF

Type:Â [SystemDouble](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

cCaseStaticNonlinearStaged Interface

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CSI API ETABS v1

cCaseStaticNonlinearStagedAddLanguageSpecificText Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetStageDefinitions(
    string Name,
    int NumberStages,
    ref int[] Duration,
    ref string[] Comment
)
```

```
Function SetStageDefinitions (
    Name As String,
    NumberStages As Integer,
    ByRef Duration As Integer(),
    ByRef Comment As String()
) As Integer
```

```
Dim instance As cCaseStaticNonlinearStaged
Dim Name As String
Dim NumberStages As Integer
Dim Duration As Integer()
Dim Comment As String()
Dim returnValue As Integer
```

```
returnValue = instance.SetStageDefinitions(Name,
    NumberStages, Duration, Comment)
```

```
int SetStageDefinitions(
    String^ Name,
    int NumberStages,
    array<int>^% Duration,
    array<String^>^% Comment
)
```

```
abstract SetStageDefinitions :
    Name : string *
    NumberStages : int *
    Duration : int[] byref *
    Comment : string[] byref -> int
```

Parameters

Name

Type: SystemString

cCaseStaticNonlinearStagedspan id="LSTEBBB2BEC_0"AddLanguageSpecificTextSet("LSTEBBB2BEC_0

NumberStages

Type: SystemInt32

Duration

Type: SystemInt32

Comment

Type: SystemString

Return Value

Type: Int32

▲ See Also

Reference

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CSI API ETABS v1

cCaseStaticNonlinearStagedAddLanguageSpecificText Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetStageDefinitions_1(
    string Name,
    int NumberStages,
    ref int[] Duration,
    ref bool[] Output,
    ref string[] OutputName,
    ref string[] Comment
)
```

```
Function SetStageDefinitions_1 (
    Name As String,
    NumberStages As Integer,
    ByRef Duration As Integer(),
    ByRef Output As Boolean(),
    ByRef OutputName As String(),
    ByRef Comment As String()
) As Integer
```

```
Dim instance As cCaseStaticNonlinearStaged
Dim Name As String
Dim NumberStages As Integer
Dim Duration As Integer()
Dim Output As Boolean()
Dim OutputName As String()
Dim Comment As String()
Dim returnValue As Integer
```

```
returnValue = instance.SetStageDefinitions_1(Name,
    NumberStages, Duration, Output, OutputName,
    Comment)
```

```
int SetStageDefinitions_1(
    String^ Name,
    int NumberStages,
    array<int>^% Duration,
    array<bool>^% Output,
    array<String^>^% OutputName,
    array<String^>^% Comment
)
```

```
abstract SetStageDefinitions_1 :
    Name : string *
```

```
NumberStages : int *  
Duration : int[] byref *  
Output : bool[] byref *  
OutputName : string[] byref *  
Comment : string[] byref -> int
```

Parameters

Name

Type:Â [SystemString](#)

NumberStages

Type:Â [SystemInt32](#)

Duration

Type:Â [SystemInt32](#)

Output

Type:Â [SystemBoolean](#)

OutputName

Type:Â [SystemString](#)

Comment

Type:Â [SystemString](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cCaseStaticNonlinearStaged Interface](#)

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CSI API ETABS v1

cCaseStaticNonlinearStagedAddLanguageSpecificTextMethod

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetStageDefinitions_2(
    string Name,
    int NumberStages,
    ref double[] Duration,
    ref bool[] Output,
    ref string[] OutputName,
    ref string[] Comment
)
```

```
Function SetStageDefinitions_2 (
    Name As String,
    NumberStages As Integer,
    ByRef Duration As Double(),
    ByRef Output As Boolean(),
    ByRef OutputName As String(),
    ByRef Comment As String()
) As Integer
```

```
Dim instance As cCaseStaticNonlinearStaged
Dim Name As String
Dim NumberStages As Integer
Dim Duration As Double()
Dim Output As Boolean()
Dim OutputName As String()
Dim Comment As String()
Dim returnValue As Integer
```

```
returnValue = instance.SetStageDefinitions_2(Name,
    NumberStages, Duration, Output, OutputName,
    Comment)
```

```
int SetStageDefinitions_2(
    String^ Name,
    int NumberStages,
    array<double>^% Duration,
    array<bool>^% Output,
    array<String^>^% OutputName,
    array<String^>^% Comment
)
```

```
abstract SetStageDefinitions_2 :
    Name : string *
```

```
NumberStages : int *  
Duration : float[] byref *  
Output : bool[] byref *  
OutputName : string[] byref *  
Comment : string[] byref -> int
```

Parameters

Name

Type:Â [SystemString](#)

NumberStages

Type:Â [SystemInt32](#)

Duration

Type:Â [SystemDouble](#)

Output

Type:Â [SystemBoolean](#)

OutputName

Type:Â [SystemString](#)

Comment

Type:Â [SystemString](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cCaseStaticNonlinearStaged Interface](#)

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CSI API ETABS v1

cCaseStaticNonlinearStagedAddLanguageSpecificText Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetTargetForceParameters(
    string Name,
    double TolConvF,
    int MaxIter,
    double AccelFact,
    bool NoStop
)
```

```
Function SetTargetForceParameters (
    Name As String,
    TolConvF As Double,
    MaxIter As Integer,
    AccelFact As Double,
    NoStop As Boolean
) As Integer
```

```
Dim instance As cCaseStaticNonlinearStaged
Dim Name As String
Dim TolConvF As Double
Dim MaxIter As Integer
Dim AccelFact As Double
Dim NoStop As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetTargetForceParameters(Name,
    TolConvF, MaxIter, AccelFact, NoStop)
```

```
int SetTargetForceParameters(
    String^ Name,
    double TolConvF,
    int MaxIter,
    double AccelFact,
    bool NoStop
)
```

```
abstract SetTargetForceParameters :
    Name : string *
    TolConvF : float *
    MaxIter : int *
    AccelFact : float *
    NoStop : bool -> int
```

Parameters

Name

Type: SystemString

TolConvF

Type: SystemDouble

MaxIter

Type: SystemInt32

AccelFact

Type: SystemDouble

NoStop

Type: SystemBoolean

Return Value

Type: Int32

See Also

Reference

cCaseStaticNonlinearStaged Interface

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CSI API ETABS v1

cCombo Interface

Namespace: $\hat{A}$ ETABSv1

Assembly: $\hat{A}$ ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▀ Syntax

C#

VB

C++

F#

Copy

```
public interface cCombo
```

```
Public Interface cCombo
```

```
Dim instance As cCombo
```

```
public interface class cCombo
```

```
type cCombo = interface end
```

The cCombo type exposes the following members.

▀ Methods

$\hat{A}$	Name	Description
	<u>Add</u>	Adds a new load combination.
	<u>AddDesignDefaultCombos</u>	
	<u>Delete</u>	Deletes the specified load combination.
	<u>DeleteCase</u>	Deletes one load case or load combination from the list of cases included in the specified load combination.
	<u>GetCaseList</u>	Retrieves all load cases and response combinations included in the load combination specified by the Name item.
	<u>GetCaseList_1</u>	Retrieves all load cases and response combinations included in the load combination specified by the Name item.
	<u>GetNameList</u>	Retrieves the names of all defined response combinations.
	<u>GetTypeCombo</u>	Retrieves the combination type for specified load combination.
	<u>GetTypeOAPI</u>	
	<u>SetCaseList</u>	Adds or modifies one load case or response combination in the list of cases included in the load combination specified by the Name item.
	<u>SetCaseList_1</u>	Adds or modifies one load case or response combination in the list of cases included in the load

combination specified by the Name item.

[Top](#)

▲ [See Also](#)

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cCombo Methods

The cCombo type exposes the following members.

Methods

Á	Name	Description
⇒	<u>Add</u>	Adds a new load combination.
⇒	<u>AddDesignDefaultCombos</u>	
⇒	<u>Delete</u>	Deletes the specified load combination.
⇒	<u>DeleteCase</u>	Deletes one load case or load combination from the list of cases included in the specified load combination.
⇒	<u>GetCaseList</u>	Retrieves all load cases and response combinations included in the load combination specified by the Name item.
⇒	<u>GetCaseList_1</u>	Retrieves all load cases and response combinations included in the load combination specified by the Name item.
⇒	<u>GetNameList</u>	Retrieves the names of all defined response combinations.
⇒	<u>GetTypeCombo</u>	Retrieves the combination type for specified load combination.
⇒	<u>GetTypeOAPI</u>	
⇒	<u>SetCaseList</u>	Adds or modifies one load case or response combination in the list of cases included in the load combination specified by the Name item.
⇒	<u>SetCaseList_1</u>	Adds or modifies one load case or response combination in the list of cases included in the load combination specified by the Name item.

Top

See Also

Reference

cCombo Interface

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CSI API ETABS v1

cComboAddLanguageSpecificTextSet("LSTA7C71E05_0")

Method

Adds a new load combination.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int Add(
    string Name,
    int ComboType
)
```

```
Function Add (
    Name As String,
    ComboType As Integer
) As Integer
```

```
Dim instance As cCombo
Dim Name As String
Dim ComboType As Integer
Dim returnValue As Integer
```

```
returnValue = instance.Add(Name, ComboType)
```

```
int Add(
    String^ Name,
    int ComboType
)
```

```
abstract Add :
    Name : string *
    ComboType : int -> int
```

Parameters

Name

Type: SystemString

The name of a new load combination.

ComboType

Type: SystemInt32

This is 0, 1, 2, 3 or 4 indicating the load combination type.

◊ 0 = Linear Additive

◊ 1 = Envelope

◊ 2 = Absolute Additive

◊ 3 = SRSS

◇ 4 = Range Additive

Return Value

Type: Int32

Returns zero if the load combination is successfully added, otherwise it returns a nonzero value.

Remarks

The new load combination must have a different name from all other load combinations and all load cases. If the name is not unique, an error will be returned.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add combo
    ret = SapModel.RespCombo.Add("COMB1", 1)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

cCombo Interface

ETABSv1 Namespace

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CSI API ETABS v1

cComboAddLanguageSpecificTextSet("LSTB44EBEF3_0") Method

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int AddDesignDefaultCombos(
    bool DesignSteel,
    bool DesignConcrete,
    bool DesignAluminum,
    bool DesignColdFormed
)
```

```
Function AddDesignDefaultCombos (
    DesignSteel As Boolean,
    DesignConcrete As Boolean,
    DesignAluminum As Boolean,
    DesignColdFormed As Boolean
) As Integer
```

```
Dim instance As cCombo
Dim DesignSteel As Boolean
Dim DesignConcrete As Boolean
Dim DesignAluminum As Boolean
Dim DesignColdFormed As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.AddDesignDefaultCombos(DesignSteel,
    DesignConcrete, DesignAluminum,
    DesignColdFormed)
```

```
int AddDesignDefaultCombos(
    bool DesignSteel,
    bool DesignConcrete,
    bool DesignAluminum,
    bool DesignColdFormed
)
```

```
abstract AddDesignDefaultCombos :
    DesignSteel : bool *
    DesignConcrete : bool *
    DesignAluminum : bool *
    DesignColdFormed : bool -> int
```

Parameters

DesignSteel

Type: SystemBoolean

DesignConcrete

Type: SystemBoolean

DesignAluminum

Type: SystemBoolean

DesignColdFormed

Type: SystemBoolean

Return Value

Type: Int32

▲ See Also

Reference

cCombo Interface

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CSI API ETABS v1

cComboAddLanguageSpecificTextSet("LST75CB30FE_0") Delete Method

Deletes the specified load combination.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int Delete(
    string Name
)
```

```
Function Delete (
    Name As String
) As Integer
```

```
Dim instance As cCombo
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.Delete(Name)
```

```
int Delete(
    String^ Name
)
```

```
abstract Delete :
    Name : string -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing load combination.

Return Value

Type: [Int32](#)

Returns zero if the combination is successfully deleted, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

[Copy](#)

```
Public Sub Example()
```

Introduction

```
Dim SapModel As cSapModel
Dim EtabsObject As cOAPI
Dim ret As Integer = -1

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'add combo
ret = SapModel.RespCombo.Add("COMB1", 0)

'delete combo
ret = SapModel.RespCombo.Delete("COMB1")

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cCombo Interface](#)

[ETABSv1 Namespace](#)

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»

CSI API ETABS v1

cComboAddLanguageSpecificTextSet("LST261C687B_0") DeleteC

Method

Deletes one load case or load combination from the list of cases included in the specified load combination.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int DeleteCase(
    string Name,
    eCNameType CNameType,
    string CName
)
```

```
Function DeleteCase (
    Name As String,
    CNameType As eCNameType,
    CName As String
) As Integer
```

```
Dim instance As cCombo
Dim Name As String
Dim CNameType As eCNameType
Dim CName As String
Dim returnValue As Integer
```

```
returnValue = instance.DeleteCase(Name,
    CNameType, CName)
```

```
int DeleteCase(
    String^ Name,
    eCNameType CNameType,
    String^ CName
)
```

```
abstract DeleteCase :
    Name : string *
    CNameType : eCNameType *
    CName : string -> int
```

Parameters

Name

Type: SystemString

The name of an existing load combination.

CNameType

Type: ETABSv1eCNameType

This is one of the following items in the eCNameType enumeration:

◊ LoadCase = 0

◊ LoadCombo = 1

This item indicates whether the CName item is an analysis case (LoadCase) or a load combination (LoadCombo).

CName

Type: SystemString

The name of the load case or load combination to be deleted from the specified combination.

Return Value

Type: Int32

Returns zero if the item is successfully deleted, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add combo
    ret = SapModel.RespCombo.Add("COMB1", 0)

    'add load case to combo
    ret = SapModel.RespCombo.SetCaseList("COMB1", LoadCase, "DEAD", 1.4)

    'delete load case from combo
    ret = SapModel.RespCombo.DeleteCase("COMB1", LoadCase, "DEAD")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cCombo Interface

ETABSv1 Namespace

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»

CSI API ETABS v1

cComboAddLanguageSpecificTextSet("LST11E0ACC2_0")

Method

Retrieves all load cases and response combinations included in the load combination specified by the Name item.

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetCaseList(
    string Name,
    ref int NumberItems,
    ref eCNameType[] CNameType,
    ref string[] CName,
    ref double[] SF
)
```

```
Function GetCaseList (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef CNameType As eCNameType(),
    ByRef CName As String(),
    ByRef SF As Double()
) As Integer
```

```
Dim instance As cCombo
Dim Name As String
Dim NumberItems As Integer
Dim CNameType As eCNameType()
Dim CName As String()
Dim SF As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetCaseList(Name,
    NumberItems, CNameType, CName, SF)
```

```
int GetCaseList(
    String^ Name,
    int% NumberItems,
    array<eCNameType>^% CNameType,
    array<String^>^% CName,
    array<double>^% SF
)
```

```
abstract GetCaseList :
    Name : string *
    NumberItems : int byref *
    CNameType : eCNameType[] byref *
    CName : string[] byref *
```

cCombospan id="LST11E0ACC2_0"AddLanguageSpecificTextSet("LST11E0ACC2_0?cpp=::|nu82");GetCas

SF : float[] byref -> int

Parameters

Name

Type:Â SystemString

The name of an existing load combination.

NumberItems

Type:Â SystemInt32

The total number of load cases and load combinations included in the load combination specified by the Name item.

CNameType

Type:Â ETABSv1eCNameType

This is one of the following items in the eCNameType enumeration:

◊ LoadCase = 0

◊ LoadCombo = 1

This item indicates whether the CName item is an analysis case (LoadCase) or a load combination (LoadCombo).

CName

Type:Â SystemString

This is an array of the names of the load cases or load combinations included in the load combination specified by the Name item.

SF

Type:Â SystemDouble

The scale factor multiplying the case or combination indicated by the CName item.

Return Value

Type:Â Int32

Returns zero if the data is successfully retrieved, otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberItems As Integer
    Dim CNameType() As eCNameType
    Dim CName() As String
    Dim SF() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
```

Introduction

```
ret = SapModel.InitializeNewModel()  
  
'create steel deck template model  
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)  
  
'add combo  
ret = SapModel.RespCombo.Add("COMB1", 0)  
  
'add load case to combo  
ret = SapModel.RespCombo.SetCaseList("COMB1", LoadCase, "DEAD", 1.4)  
  
'get all cases and combos included in combo COMB1  
ret = SapModel.RespCombo.GetCaseList("COMB1", NumberItems, CNameType, CName, SF)  
  
'close ETABS  
EtabsObject.ApplicationExit(False)  
  
'clean up variables  
SapModel = Nothing  
EtabsObject = Nothing  
End Sub
```

▲ See Also

Reference

cCombo Interface

ETABSv1 Namespace

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CSI API ETABS v1

cComboAddLanguageSpecificTextSet("LST3F0EF5BE_0") Method

Retrieves all load cases and response combinations included in the load combination specified by the Name item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetCaseList_1(
    string Name,
    ref int NumberItems,
    ref eCNameType[] CNameType,
    ref string[] CName,
    ref int[] ModeNumber,
    ref double[] SF
)
```

```
Function GetCaseList_1 (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef CNameType As eCNameType(),
    ByRef CName As String(),
    ByRef ModeNumber As Integer(),
    ByRef SF As Double()
) As Integer
```

```
Dim instance As cCombo
Dim Name As String
Dim NumberItems As Integer
Dim CNameType As eCNameType()
Dim CName As String()
Dim ModeNumber As Integer()
Dim SF As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetCaseList_1(Name,
    NumberItems, CNameType, CName, ModeNumber,
    SF)
```

```
int GetCaseList_1(
    String^ Name,
    int% NumberItems,
    array<eCNameType>^% CNameType,
    array<String^>^% CName,
    array<int>^% ModeNumber,
    array<double>^% SF
)
```

```

abstract GetCaseList_1 :
    Name : string *
    NumberItems : int byref *
    CNameType : eCNameType[] byref *
    CName : string[] byref *
    ModeNumber : int[] byref *
    SF : float[] byref -> int

```

Parameters

Name

Type:Â [SystemString](#)

The name of an existing load combination.

NumberItems

Type:Â [SystemInt32](#)

The total number of load cases and load combinations included in the load combination specified by the Name item.

CNameType

Type:Â [ETABSv1eCNameType](#)

This is one of the following items in the [eCNameType](#) enumeration:

◊ LoadCase = 0

◊ LoadCombo = 1

This item indicates whether the CName item is an analysis case (LoadCase) or a load combination (LoadCombo).

CName

Type:Â [SystemString](#)

This is an array of the names of the load cases or load combinations included in the load combination specified by the Name item.

ModeNumber

Type:Â [SystemInt32](#)

The mode number for the case indicated by the CName item. This item applies when by the CNameType item is LoadCase and the type of load case specified by the CName item is either Modal or Buckling. Any other case type or combo returns a zero for this item.

SF

Type:Â [SystemDouble](#)

The scale factor multiplying the case or combination indicated by the CName item.

Return Value

Type:Â [Int32](#)

Returns zero if the data is successfully retrieved, otherwise it returns a nonzero value.

▲ Remarks

This function supercedes [GetCaseList\(String, Int32, eCNameType, String, Double\)](#)

▲ Examples

VB

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberItems As Integer

```

Introduction

```
Dim CNameType() As eCNameType
Dim CName() As String
Dim ModeNumber() As Integer
Dim SF() As Double

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'add combo
ret = SapModel.RespCombo.Add("COMB1", 0)

'add load case to combo
ret = SapModel.RespCombo.SetCaseList("COMB1", LoadCase, "DEAD", 1.4)

'get all cases and combos included in combo COMB1
ret = SapModel.RespCombo.GetCaseList("COMB1", NumberItems, CNameType, CName, ModeNumber, S

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cCombo Interface

ETABSv1 Namespace

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CSI API ETABS v1

cComboAddLanguageSpecificTextSet("LSTC3CDC584_0") Method

Retrieves the names of all defined response combinations.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetNameList(
    ref int NumberNames,
    ref string[] MyName
)
```

```
Function GetNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cCombo
Dim NumberNames As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetNameList(NumberNames,
    MyName)
```

```
int GetNameList(
    int% NumberNames,
    array<String^>^% MyName
)
```

```
abstract GetNameList :
    NumberNames : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberNames

Type: System.Int32

The number of load combination names retrieved by the program.

MyName

Type: System.String

This is a one-dimensional array of load combination names. The MyName array is created as a dynamic, zero-based, array by the API user:

```
Dim MyName() as String
```

The array is dimensioned to (NumberNames + 1) inside the ETABS program, filled with the names, and returned to the API user.

Return Value

Type: Int32

Returns zero if the names are successfully retrieved, otherwise it returns nonzero.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberNames As Integer
    Dim MyName() As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add combos
    ret = SapModel.RespCombo.Add("COMB1", 0)
    ret = SapModel.RespCombo.Add("COMB2", 2)

    'get combo names
    ret = SapModel.RespCombo.GetNameList(NumberNames, MyName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cCombo Interface

ETABSv1 Namespace

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»

CSI API ETABS v1

cComboAddLanguageSpecificTextSet("LST5214D883_0")

Method

Retrieves the combination type for specified load combination.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetTypeCombo(
    string Name,
    ref int ComboType
)
```

```
Function GetTypeCombo (
    Name As String,
    ByRef ComboType As Integer
) As Integer
```

```
Dim instance As cCombo
Dim Name As String
Dim ComboType As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetTypeCombo(Name,
    ComboType)
```

```
int GetTypeCombo(
    String^ Name,
    int% ComboType
)
```

```
abstract GetTypeCombo :
    Name : string *
    ComboType : int byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing load combination.

ComboType

Type: [SystemInt32](#)

This is 0, 1, 2, 3 or 4 indicating the load combination type.

◊ 0 = Linear Additive

◊ 1 = Envelope

◊ 2 = Absolute Additive

- ◇ 3 = SRSS
- ◇ 4 = Range Additive

Return Value

Type: Int32

Returns zero if the type is successfully retrieved, otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim ComboType As Integer

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add combo
    ret = SapModel.RespCombo.Add("COMB1", 2)

    'get combo type
    ret = SapModel.RespCombo.GetTypeCombo("COMB1", ComboType )

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cCombo Interface

ETABSv1 Namespace

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Introduction

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CSI API ETABS v1

cComboAddLanguageSpecificTextSet("LST9DE5BBD5_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetType0API(
    string name,
    ref int ComboType
)
```

```
Function GetType0API (
    name As String,
    ByRef ComboType As Integer
) As Integer
```

```
Dim instance As cCombo
Dim name As String
Dim ComboType As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetType0API(name,
    ComboType)
```

```
int GetType0API(
    String^ name,
    int% ComboType
)
```

```
abstract GetType0API :
    name : string *
    ComboType : int byref -> int
```

Parameters

name

Type: [SystemString](#)

ComboType

Type: [SystemInt32](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cCombo Interface

ETABSv1 Namespace

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»

CSI API ETABS v1

cComboAddLanguageSpecificTextSet("LST3DD12DC5_0")

Method

Adds or modifies one load case or response combination in the list of cases included in the load combination specified by the Name item.

Namespace: `ETABSv1`

Assembly: `ETABSv1 (in ETABSv1.dll)` Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetCaseList(
    string Name,
    ref eCNameType CNameType,
    string CName,
    double SF
)
```

```
Function SetCaseList (
    Name As String,
    ByRef CNameType As eCNameType,
    CName As String,
    SF As Double
) As Integer
```

```
Dim instance As cCombo
Dim Name As String
Dim CNameType As eCNameType
Dim CName As String
Dim SF As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetCaseList(Name,
    CNameType, CName, SF)
```

```
int SetCaseList(
    String^ Name,
    eCNameType% CNameType,
    String^ CName,
    double SF
)
```

```
abstract SetCaseList :
    Name : string *
    CNameType : eCNameType byref *
    CName : string *
    SF : float -> int
```

Parameters

Name

Type: SystemString

The name of an existing load combination.

CNameType

Type: ETABSv1eCNameTypeThis is one of the following items in the eCNameType enumeration:

◊ LoadCase = 0

◊ LoadCombo = 1

This item indicates whether the CName item is an analysis case (LoadCase) or a load combination (LoadCombo).

CName

Type: SystemString

The name of the load case or load combination to be added to or modified in the combination specified by the Name item. If the load case or combination already exists in the combination specified by the Name item, the scale factor is modified as indicated by the SF item for that load case or combination. If the analysis case or combination does not exist in the combination specified by the Name item, it is added.

SF

Type: SystemDouble

The scale factor multiplying the case or combination indicated by the CName item.

Return ValueType: Int32

Returns zero if the item is successfully added or modified, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)
```


Introduction

```
'add combo
    ret = SapModel.RespCombo.Add("COMB1", 0)

'add load case to combo
    ret = SapModel.RespCombo.SetCaseList("COMB1", eCNameType.LoadCase, "DEAD", 1.4)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cCombo Interface

ETABSv1 Namespace

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»

CSI API ETABS v1

cComboAddLanguageSpecificTextSet("LST6B7F1CE4_0")

Method

Adds or modifies one load case or response combination in the list of cases included in the load combination specified by the Name item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetCaseList_1(
    string Name,
    ref eCNameType CNameType,
    string CName,
    int ModeNumber,
    double SF
)
```

```
Function SetCaseList_1 (
    Name As String,
    ByRef CNameType As eCNameType,
    CName As String,
    ModeNumber As Integer,
    SF As Double
) As Integer
```

```
Dim instance As cCombo
Dim Name As String
Dim CNameType As eCNameType
Dim CName As String
Dim ModeNumber As Integer
Dim SF As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetCaseList_1(Name,
    CNameType, CName, ModeNumber, SF)
```

```
int SetCaseList_1(
    String^ Name,
    eCNameType% CNameType,
    String^ CName,
    int ModeNumber,
    double SF
)
```

```
abstract SetCaseList_1 :
    Name : string *
    CNameType : eCNameType byref *
    CName : string *
    ModeNumber : int *
```

cCombospan id="LST6B7F1CE4_0"AddLanguageSpecificTextSet("LST6B7F1CE4_0?cpp=::|nu838SetCas

SF : float -> int

Parameters

Name

Type: SystemString

The name of an existing load combination.

CNameType

Type: ETABSv1eCNameType

This is one of the following items in the eCNameType enumeration:

◊ LoadCase = 0

◊ LoadCombo = 1

This item indicates whether the CName item is an analysis case (LoadCase) or a load combination (LoadCombo).

CName

Type: SystemString

The name of the load case or load combination to be added to or modified in the combination specified by the Name item. If the load case or combination already exists in the combination specified by the Name item, the scale factor is modified as indicated by the SF item for that load case or combination. If the analysis case or combination does not exist in the combination specified by the Name item, it is added.

ModeNumber

Type: SystemInt32

The mode number for the case indicated by the CName item. This item applies when by the CNameType item is LoadCase and the type of load case specified by the CName item is either Modal or Buckling. Any other case type or combo should specify a zero for this item.

SF

Type: SystemDouble

The scale factor multiplying the case or combination indicated by the CName item.

Return Value

Type: Int32

Returns zero if the item is successfully added or modified, otherwise it returns a nonzero value.

Remarks

This function supercedes SetCaseList(String, eCNameType, String, Double)

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
```

Introduction

```
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'add combo
ret = SapModel.RespCombo.Add("COMB1", 0)

'add load case to combo
ret = SapModel.RespCombo.SetCaseList("COMB1", eCNameType.LoadCase, "DEAD", 0, 1.4)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

🔗 See Also

Reference

[cCombo Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cConstraint Interface

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cConstraint
```

```
Public Interface cConstraint
```

```
Dim instance As cConstraint
```

```
public interface class cConstraint
```

```
type cConstraint = interface end
```

The cConstraint type exposes the following members.

▲ Methods

▲	Name	Description
🔗	Delete	
🔗	GetDiaphragm	
🔗	GetNameList	
🔗	SetDiaphragm	DEPRECATED. Defines a Diaphragm constraint.

[Top](#)

▲ See Also

Reference

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cConstraint Methods

The [cConstraint](#) type exposes the following members.

Methods

⌵	Name	Description
⇒	Delete	
⇒	GetDiaphragm	
⇒	GetNameList	
⇒	SetDiaphragm	DEPRECATED. Defines a Diaphragm constraint.

[Top](#)

⌵ See Also

Reference

[cConstraint Interface](#)

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CSI API ETABS v1

cConstraintAddLanguageSpecificTextSet("LST1D78732B_0?cpp=::|843");Delete

Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int Delete(
    string Name
)
```

```
Function Delete (
    Name As String
) As Integer
```

```
Dim instance As cConstraint
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.Delete(Name)
```

```
int Delete(
    String^ Name
)
```

```
abstract Delete :
    Name : string -> int
```

Parameters

Name

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cConstraint Interface](#)

[ETABSv1 Namespace](#)

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Introduction

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CSI API ETABS v1

cConstraintAddLanguageSpecificTextSet("LST5F511511_0?cpp=::|n845");GetDi

Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetDiaphragm(
    string Name,
    ref eConstraintAxis Axis,
    ref string CSys
)
```

```
Function GetDiaphragm (
    Name As String,
    ByRef Axis As eConstraintAxis,
    ByRef CSys As String
) As Integer
```

```
Dim instance As cConstraint
Dim Name As String
Dim Axis As eConstraintAxis
Dim CSys As String
Dim returnValue As Integer
```

```
returnValue = instance.GetDiaphragm(Name,
    Axis, CSys)
```

```
int GetDiaphragm(
    String^ Name,
    eConstraintAxis% Axis,
    String^% CSys
)
```

```
abstract GetDiaphragm :
    Name : string *
    Axis : eConstraintAxis byref *
    CSys : string byref -> int
```

Parameters

Name

Type: SystemString

Axis

Type: ETABSv1eConstraintAxis

CSys

Type: SystemString

cConstraintsAddLanguageSpecificTextSet("LST5F511511_0?cpp=::|n845");GetDi

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cConstraint Interface](#)

[ETABSv1 Namespace](#)

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Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

C#

C++

F#

Copy

```
abstract GetNameList :
    NumberNames : int byref *
    MyName : string[] byref -> int
```

NumberNames

MyName

Type:Â SystemStringType: Int32

See Also

Reference

cConstraint Interface

ETABSv1 Namespace

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CSI API ETABS v1

cConstraintAddLanguageSpecificTextSet("LST1CA3F5AA_0") Method

DEPRECATED. Defines a Diaphragm constraint.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetDiaphragm(
    string Name,
    eConstraintAxis Axis = eConstraintAxis.AutoAxis,
    string CSys = "Global"
)
```

```
Function SetDiaphragm (
    Name As String,
    Axis As eConstraintAxis = eConstraintAxis.AutoAxis,
    CSys As String = "Global"
) As Integer
```

```
Dim instance As cConstraint
Dim Name As String
Dim Axis As eConstraintAxis
Dim CSys As String
Dim returnValue As Integer
```

```
returnValue = instance.SetDiaphragm(Name,
    Axis, CSys)
```

```
int SetDiaphragm(
    String^ Name,
    eConstraintAxis Axis = eConstraintAxis::AutoAxis,
    String^ CSys = L"Global"
)
```

```
abstract SetDiaphragm :
    Name : string *
    ?Axis : eConstraintAxis *
    ?CSys : string
(* Defaults:
    let _Axis = defaultArg Axis eConstraintAxis.AutoAxis
    let _CSys = defaultArg CSys "Global"
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of a constraint.

Axis (Optional)

Type:Â ETABSv1eConstraintAxis

CSys (Optional)

Type:Â SystemString

The name of the coordinate system in which the constraint is defined.

Return ValueType:Â Int32

Returns zero if the constraint data is successfully added or modified, otherwise it returns a nonzero value.

▲ **Remarks**This function is DEPRECATED. Please refer to cDiaphragm.▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define a new constraint
    ret = SapModel.ConstraintDef.SetDiaphragm("Diaph1")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cConstraint Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDatabaseTables Interface

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
public interface cDatabaseTables
```

```
Public Interface cDatabaseTables
```

```
Dim instance As cDatabaseTables
```

```
public interface class cDatabaseTables
```

```
type cDatabaseTables = interface end
```

The cDatabaseTables type exposes the following members.

▲ Methods

$\hat{A}$

Name

Description

 ApplyEditedTables

Instructs the program to interactively import all tables in the list using the SetTableForEditing... functions.

 CancelTableEditing

Clears all tables that were stored in the table list using the SetTableForEditing... functions.

 GetAllFieldsInTable

Returns the available fields in a specified table.

 GetAllTables

Returns all of the tables along with their import status that are available in the model to fill the table.

 GetAvailableTables

Returns the tables that are currently available for editing.

 GetLoadCasesSelectedForDisplay

Returns a list of load cases that are selected for display.

 GetLoadCombinationsSelectedForDisplay

Returns a list of load combinations that are selected for display.

 GetLoadPatternsSelectedForDisplay

Returns a list of load patterns that are selected for display.

 GetObsoleteTableKeyList

Returns a list of obsolete table keys for the program.

 GetOutputOptionsForDisplay

Returns the database table output options for load cases.

 GetTableForDisplayArray

Returns data for a single table in a single array. If the table then no data is returned.

 GetTableForDisplayCSVFile

Returns data for a single table in a CSV file. If the table then no data is returned.

 GetTableForDisplayCSVString

Returns data for a single table in a CSV string. If the table then no data is returned.

 GetTableForDisplayXMLString

Returns data for a single table as XML in a single string. If the table then no data is returned.

⇒ <u>GetTableForEditingArray</u>	Returns a single table in a string array for inter
⇒ <u>GetTableForEditingCSVFile</u>	Returns a single table in a CSV file for interacti
⇒ <u>GetTableForEditingCSVString</u>	Returns a single table in a CSV-formatted string
⇒ <u>SetLoadCasesSelectedForDisplay</u>	Sets the load cases that are selected for table d
⇒ <u>SetLoadCombinationsSelectedForDisplay</u>	Sets the load combinations that are selected for
⇒ <u>SetLoadPatternsSelectedForDisplay</u>	Sets the load patterns that are selected for tabl
⇒ <u>SetOutputOptionsForDisplay</u>	Sets the database table output options for load
⇒ <u>SetTableForEditingArray</u>	Reads a table from a string array and adds it to <u>ApplyEditedTables(Boolean, Int32, Int32, Int32)</u> <u>CancelTableEditing</u> command is issued.
⇒ <u>SetTableForEditingCSVFile</u>	Reads a table from a CSV file and adds it to a s <u>ApplyEditedTables(Boolean, Int32, Int32, Int32)</u> <u>CancelTableEditing</u> command is issued.
⇒ <u>SetTableForEditingCSVString</u>	Reads a table from a CSV-formatted string and either the <u>ApplyEditedTables(Boolean, Int32, In</u> <u>CancelTableEditing</u> command is issued.
⇒ <u>ShowTablesInExcel</u>	Exports the specified tables to Excel.

[Top](#)

🔍 See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDatabaseTables Methods

The cDatabaseTables type exposes the following members.

Methods

Á	Name	Description
⇒	<u>ApplyEditedTables</u>	Instructs the program to interactively import all tables in the table list using the SetTableForEditing... functions.
⇒	<u>CancelTableEditing</u>	Clears all tables that were stored in the table list using the SetTableForEditing... functions.
⇒	<u>GetAllFieldsInTable</u>	Returns the available fields in a specified table.
⇒	<u>GetAllTables</u>	Returns all of the tables along with their import status that are available in the model to fill the table.
⇒	<u>GetAvailableTables</u>	Returns the tables that are currently available for use in the model.
⇒	<u>GetLoadCasesSelectedForDisplay</u>	Returns a list of load cases that are selected for display in the table.
⇒	<u>GetLoadCombinationsSelectedForDisplay</u>	Returns a list of load combinations that are selected for display in the table.
⇒	<u>GetLoadPatternsSelectedForDisplay</u>	Returns a list of load patterns that are selected for display in the table.
⇒	<u>GetObsoleteTableKeyList</u>	Returns a list of obsolete table keys for the project.
⇒	<u>GetOutputOptionsForDisplay</u>	Returns the database table output options for load cases.
⇒	<u>GetTableForDisplayArray</u>	Returns data for a single table in a single array. If the table then no data is returned.
⇒	<u>GetTableForDisplayCSVFile</u>	Returns data for a single table in a CSV file. If the table then no data is returned.
⇒	<u>GetTableForDisplayCSVString</u>	Returns data for a single table in a CSV string. If the table then no data is returned.
⇒	<u>GetTableForDisplayXMLString</u>	Returns data for a single table as XML in a single string. If the table then no data is returned.
⇒	<u>GetTableForEditingArray</u>	Returns a single table in a string array for interactive editing.
⇒	<u>GetTableForEditingCSVFile</u>	Returns a single table in a CSV file for interactive editing.
⇒	<u>GetTableForEditingCSVString</u>	Returns a single table in a CSV-formatted string for interactive editing.
⇒	<u>SetLoadCasesSelectedForDisplay</u>	Sets the load cases that are selected for table display.
⇒	<u>SetLoadCombinationsSelectedForDisplay</u>	Sets the load combinations that are selected for table display.
⇒	<u>SetLoadPatternsSelectedForDisplay</u>	Sets the load patterns that are selected for table display.
⇒	<u>SetOutputOptionsForDisplay</u>	Sets the database table output options for load cases.
⇒	<u>SetTableForEditingArray</u>	Reads a table from a string array and adds it to the table list. <u>ApplyEditedTables(Boolean, Int32, Int32, Int32)</u> command is issued.
⇒	<u>SetTableForEditingCSVFile</u>	Reads a table from a CSV file and adds it to a string array. <u>ApplyEditedTables(Boolean, Int32, Int32, Int32)</u> command is issued.
⇒	<u>SetTableForEditingCSVString</u>	Reads a table from a CSV-formatted string and adds it to a string array. <u>ApplyEditedTables(Boolean, Int32, Int32, Int32)</u> command is issued.

 [ShowTablesInExcel](#)

Exports the specified tables to Excel.

[Top](#)

 [See Also](#)

Reference

[cDatabaseTables Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDatabaseTablesAddLanguageSpecificTextSet("LST93 Method

Instructs the program to interactively import all of the tables stored in the table list using the SetTableForEditing... functions.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int ApplyEditedTables(
bool FillImportLog,
ref int NumFatalErrors,
ref int NumErrorMsgs,
ref int NumWarnMsgs,
ref int NumInfoMsgs,
ref string ImportLog
)
```

```
Function ApplyEditedTables (
FillImportLog As Boolean,
ByRef NumFatalErrors As Integer,
ByRef NumErrorMsgs As Integer,
ByRef NumWarnMsgs As Integer,
ByRef NumInfoMsgs As Integer,
ByRef ImportLog As String
) As Integer
```

```
Dim instance As cDatabaseTables
Dim FillImportLog As Boolean
Dim NumFatalErrors As Integer
Dim NumErrorMsgs As Integer
Dim NumWarnMsgs As Integer
Dim NumInfoMsgs As Integer
Dim ImportLog As String
Dim returnValue As Integer
```

```
returnValue = instance.ApplyEditedTables(FillImportLog,
NumFatalErrors, NumErrorMsgs, NumWarnMsgs,
NumInfoMsgs, ImportLog)
```

```
int ApplyEditedTables(
bool FillImportLog,
int% NumFatalErrors,
int% NumErrorMsgs,
int% NumWarnMsgs,
int% NumInfoMsgs,
String^% ImportLog
)
```

```

abstract ApplyEditedTables :
    FillImportLog : bool *
    NumFatalErrors : int byref *
    NumErrorMsgs : int byref *
    NumWarnMsgs : int byref *
    NumInfoMsgs : int byref *
    ImportLog : string byref -> int

```

Parameters

FillImportLog

Type: [SystemBoolean](#)

Input Item: Whether the ImportLog string should be filled. Please note that the import log may be very large.

NumFatalErrors

Type: [SystemInt32](#)

Returned Item: The number of fatal errors that occurred during the import

NumErrorMsgs

Type: [SystemInt32](#)

Returned Item: The number of error messages logged during the import

NumWarnMsgs

Type: [SystemInt32](#)

Returned Item: The number of warning messages logged during the import

NumInfoMsgs

Type: [SystemInt32](#)

Returned Item: The number of informational messages logged during the import

ImportLog

Type: [SystemString](#)

Returned Item: A string containing all messages logged during the import

Return Value

Type: [Int32](#)

Returns 0 if the function executes correctly, otherwise returns nonzero

Remarks

If the model is locked at the time this command is called then only tables that can be interactively imported when the model is locked will be imported.

IMPORTANT NOTE: Please save your model before calling this function. If a fatal error occurs, or this function returns nonzero, the model may be in a corrupted state and it is recommended that the user close the model without saving and reopen.

See Also

Reference

[cDatabaseTables Interface](#)

[ETABSv1 Namespace](#)

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Introduction

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CSI API ETABS v1

cDatabaseTablesAddLanguageSpecificTextSet("LSTEF Method

Clears all tables that were stored in the table list using one of the SetTableForEditing... functions.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int CancelTableEditing()
```

```
Function CancelTableEditing As Integer
```

```
Dim instance As cDatabaseTables
```

```
Dim returnValue As Integer
```

```
returnValue = instance.CancelTableEditing()
```

```
int CancelTableEditing()
```

```
abstract CancelTableEditing : unit -> int
```

Return Value

Type: Int32

Returns 0 if the function executes correctly, otherwise returns nonzero

▲ See Also

Reference

cDatabaseTables Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDatabaseTablesAddLanguageSpecificTextSet("LST67 Method

Returns the available fields in a specified table.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetAllFieldsInTable(
    string TableKey,
    ref int TableVersion,
    ref int NumberFields,
    ref string[] FieldKey,
    ref string[] FieldName,
    ref string[] Description,
    ref string[] UnitsString,
    ref bool[] IsImportable
)
```

```
Function GetAllFieldsInTable (
    TableKey As String,
    ByRef TableVersion As Integer,
    ByRef NumberFields As Integer,
    ByRef FieldKey As String(),
    ByRef FieldName As String(),
    ByRef Description As String(),
    ByRef UnitsString As String(),
    ByRef IsImportable As Boolean()
) As Integer
```

```
Dim instance As cDatabaseTables
Dim TableKey As String
Dim TableVersion As Integer
Dim NumberFields As Integer
Dim FieldKey As String()
Dim FieldName As String()
Dim Description As String()
Dim UnitsString As String()
Dim IsImportable As Boolean()
Dim returnValue As Integer
```

```
returnValue = instance.GetAllFieldsInTable(TableKey,
    TableVersion, NumberFields, FieldKey,
    FieldName, Description, UnitsString,
    IsImportable)
```

```
int GetAllFieldsInTable(
    String^ TableKey,
    int% TableVersion,
```

cDatabaseTablesAddLanguageSpecificTextSet("LST679E0ACB_0?860=::|nu=.


```

int% NumberFields,
array<String^>^% FieldKey,
array<String^>^% FieldName,
array<String^>^% Description,
array<String^>^% UnitsString,
array<bool>^% IsImportable
)

abstract GetAllFieldsInTable :
    TableKey : string *
    TableVersion : int byref *
    NumberFields : int byref *
    FieldKey : string[] byref *
    FieldName : string[] byref *
    Description : string[] byref *
    UnitsString : string[] byref *
    IsImportable : bool[] byref -> int

```

Parameters

TableKey

Type:Â [SystemString](#)

Input Item: The table key for the table for which the fields will be returned.

TableVersion

Type:Â [SystemInt32](#)

Returned Item: The version number of the specified table.

NumberFields

Type:Â [SystemInt32](#)

Returned Item: The number of available fields in the specified table.

FieldKey

Type:Â [SystemString](#)

Returned Item: A zero-based array of the field keys for the specified table.

FieldName

Type:Â [SystemString](#)

Returned Item: A zero-based array of the field names for the specified table.

Description

Type:Â [SystemString](#)

Returned Item: A zero-based array of the field descriptions for the specified table.

UnitsString

Type:Â [SystemString](#)

Returned Item: A zero-based array of the field units for the specified table.

IsImportable

Type:Â [SystemBoolean](#)

Returned Item: A zero-based array of whether the field is importable for the specified table.

Return Value

Type:Â [Int32](#)

Returns 0 if the function executes correctly, otherwise returns nonzero

▲ See Also

Reference

cDatabaseTables Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDatabaseTablesAddLanguageSpecificTextSet("LSTB2 Method

Returns all of the tables along with their import type and indicates if any data is available in the model to fill the table.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetAllTables(
    ref int NumberTables,
    ref string[] TableKey,
    ref string[] TableName,
    ref int[] ImportType,
    ref bool[] IsEmpty
)
```

```
Function GetAllTables (
    ByRef NumberTables As Integer,
    ByRef TableKey As String(),
    ByRef TableName As String(),
    ByRef ImportType As Integer(),
    ByRef IsEmpty As Boolean()
) As Integer
```

```
Dim instance As cDatabaseTables
Dim NumberTables As Integer
Dim TableKey As String()
Dim TableName As String()
Dim ImportType As Integer()
Dim IsEmpty As Boolean()
Dim returnValue As Integer
```

```
returnValue = instance.GetAllTables(NumberTables,
    TableKey, TableName, ImportType,
    IsEmpty)
```

```
int GetAllTables(
    int% NumberTables,
    array<String^>^% TableKey,
    array<String^>^% TableName,
    array<int>^% ImportType,
    array<bool>^% IsEmpty
)
```

```
abstract GetAllTables :
    NumberTables : int byref *
    TableKey : string[] byref *
    TableName : string[] byref *
```

cDatabaseTablesAddLanguageSpecificTextSet("LSTB2073819_0?cp=3;|nu=.")

```

ImportType : int[] byref *
IsEmpty : bool[] byref -> int

```

Parameters

NumberTables

Type: SystemInt32

Returned Item: The number of tables that are currently available for display.

TableKey

Type: SystemString

Returned Item: A zero-based array of the table keys for the available tables.

TableName

Type: SystemString

Returned Item: A zero-based array of the table names for the available tables.

ImportType

Type: SystemInt32

Returned Item: This is either 0, 1, 2 or 3 indicating the import type for the table.

0. not importable

1. importable but not interactively importable

2. importable and interactively importable when the model is unlocked

3. importable and interactively importable when the model is unlocked and locked

IsEmpty

Type: SystemBoolean

Returned Item: False means data is available in the model to fill the table. True means there is no data in the model to fill the table.

Return Value

Type: Int32

▲ Remarks

▲ See Also

Reference

cDatabaseTables Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDatabaseTablesAddLanguageSpecificTextSet("LST5597C642_0") Method

Returns the tables that are currently available for display.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetAvailableTables(
    ref int NumberTables,
    ref string[] TableKey,
    ref string[] TableName,
    ref int[] ImportType
)
```

```
Function GetAvailableTables (
    ByRef NumberTables As Integer,
    ByRef TableKey As String(),
    ByRef TableName As String(),
    ByRef ImportType As Integer()
) As Integer
```

```
Dim instance As cDatabaseTables
Dim NumberTables As Integer
Dim TableKey As String()
Dim TableName As String()
Dim ImportType As Integer()
Dim returnValue As Integer
```

```
returnValue = instance.GetAvailableTables(NumberTables,
    TableKey, TableName, ImportType)
```

```
int GetAvailableTables(
    int% NumberTables,
    array<String^>^% TableKey,
    array<String^>^% TableName,
    array<int>^% ImportType
)
```

```
abstract GetAvailableTables :
    NumberTables : int byref *
    TableKey : string[] byref *
    TableName : string[] byref *
    ImportType : int[] byref -> int
```

Parameters

NumberTables

Type: SystemInt32

Returned Item: The number of tables that are currently available for display.

TableKey

Type: SystemString

Returned Item: A zero-based array of the table keys for the available tables.

TableName

Type: SystemString

Returned Item: A zero-based array of the table names for the available tables.

ImportType

Type: SystemInt32

Returned Item: This is either 0, 1, 2 or 3 indicating the import type for the table.

- 0. not importable
- 1. importable but not interactively importable
- 2. importable and interactively importable when the model is unlocked
- 3. importable and interactively importable when the model is unlocked and locked

Return Value

Type: Int32

Returns 0 if the function executes correctly, otherwise returns nonzero

▲ See Also

Reference

cDatabaseTables Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDatabaseTablesAddLanguageSpecificTextSet("LST27 Method

Returns a list of load cases that are selected for table display.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetLoadCasesSelectedForDisplay(
    ref int NumberSelectedLoadCases,
    ref string[] LoadCaseList
)
```

```
Function GetLoadCasesSelectedForDisplay (
    ByRef NumberSelectedLoadCases As Integer,
    ByRef LoadCaseList As String()
) As Integer
```

```
Dim instance As cDatabaseTables
Dim NumberSelectedLoadCases As Integer
Dim LoadCaseList As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetLoadCasesSelectedForDisplay(NumberSelectedLoadCases,
    LoadCaseList)
```

```
int GetLoadCasesSelectedForDisplay(
    int% NumberSelectedLoadCases,
    array<String^>% LoadCaseList
)
```

```
abstract GetLoadCasesSelectedForDisplay :
    NumberSelectedLoadCases : int byref *
    LoadCaseList : string[] byref -> int
```

Parameters

NumberSelectedLoadCases

Type: SystemInt32

Returned Item: The number of load cases selected for table display.

LoadCaseList

Type: SystemString

Returned Item: The zero-based list of load cases selected for table display.

Return Value

Type: Int32

Returns 0 if the function executes correctly, otherwise returns nonzero

Remarks

This list sets the load cases that are included when displaying analysis results.

See Also

Reference

cDatabaseTables Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDatabaseTablesAddLanguageSpecificTextSet("LSTC2192561_0") Method

Returns a list of load combinations that are selected for table display.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetLoadCombinationsSelectedForDisplay(
    ref int NumberSelectedLoadCombinations,
    ref string[] LoadCombinationList
)
```

```
Function GetLoadCombinationsSelectedForDisplay (
    ByRef NumberSelectedLoadCombinations As Integer,
    ByRef LoadCombinationList As String()
) As Integer
```

```
Dim instance As cDatabaseTables
Dim NumberSelectedLoadCombinations As Integer
Dim LoadCombinationList As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetLoadCombinationsSelectedForDisplay(NumberSelectedLoadCombinations,
    LoadCombinationList)
```

```
int GetLoadCombinationsSelectedForDisplay(
    int% NumberSelectedLoadCombinations,
    array<String^>% LoadCombinationList
)
```

```
abstract GetLoadCombinationsSelectedForDisplay :
    NumberSelectedLoadCombinations : int byref *
    LoadCombinationList : string[] byref -> int
```

Parameters

NumberSelectedLoadCombinations

Type: [System.Int32](#)

Returned Item: The number of load combinations selected for table display.

LoadCombinationList

Type: [System.String](#)

Returned Item: The zero-based list of load combinations selected for table display.

Return Value

Type: Int32

Returns 0 if the function executes correctly, otherwise returns nonzero

Remarks

This list sets the load combinations that are included when displaying analysis results.

See Also

Reference

cDatabaseTables Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDatabaseTablesAddLanguageSpecificTextSet("LSTD0 Method

Returns a list of load patterns that are selected for table display.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetLoadPatternsSelectedForDisplay(
    ref int NumberSelectedLoadPatterns,
    ref string[] LoadPatternList
)
```

```
Function GetLoadPatternsSelectedForDisplay (
    ByRef NumberSelectedLoadPatterns As Integer,
    ByRef LoadPatternList As String()
) As Integer
```

```
Dim instance As cDatabaseTables
Dim NumberSelectedLoadPatterns As Integer
Dim LoadPatternList As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetLoadPatternsSelectedForDisplay(NumberSelectedLoadPatterns,
    LoadPatternList)
```

```
int GetLoadPatternsSelectedForDisplay(
    int% NumberSelectedLoadPatterns,
    array<String^>^% LoadPatternList
)
```

```
abstract GetLoadPatternsSelectedForDisplay :
    NumberSelectedLoadPatterns : int byref *
    LoadPatternList : string[] byref -> int
```

Parameters

NumberSelectedLoadPatterns

Type: [System.Int32](#)

Returned Item: The number of load patterns selected for table display.

LoadPatternList

Type: [System.String](#)

Returned Item: The zero-based list of load patterns selected for table display.

Return Value

Type: Int32

Returns 0 if the function executes correctly, otherwise returns nonzero

Remarks

This list sets the load patterns that are included when displaying load assignments on the model.

See Also

Reference

cDatabaseTables Interface

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CSI API ETABS v1

cDatabaseTablesAddLanguageSpecificTextSet("LST52 Method

Returns a list of obsolete table keys for the program.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetObsoleteTableKeyList(
    ref int NumberTableKeys,
    ref string[] TableKeyList,
    ref string[] NotesList
)
```

```
Function GetObsoleteTableKeyList (
    ByRef NumberTableKeys As Integer,
    ByRef TableKeyList As String(),
    ByRef NotesList As String()
) As Integer
```

```
Dim instance As cDatabaseTables
Dim NumberTableKeys As Integer
Dim TableKeyList As String()
Dim NotesList As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetObsoleteTableKeyList(NumberTableKeys,
    TableKeyList, NotesList)
```

```
int GetObsoleteTableKeyList(
    int% NumberTableKeys,
    array<String^>% TableKeyList,
    array<String^>% NotesList
)
```

```
abstract GetObsoleteTableKeyList :
    NumberTableKeys : int byref *
    TableKeyList : string[] byref *
    NotesList : string[] byref -> int
```

Parameters

NumberTableKeys

Type: System.Int32

Returned Item: The number of obsolete table keys.

TableKeyList

Type: SystemString

Returned Item: A zero-based list of obsolete table keys.

NotesList

Type: SystemString

Returned Item: A zero-based list of notes, one for each table key.

Return Value

Type: Int32

Returns 0 if the function executes correctly, otherwise returns nonzero

▀ Remarks

▀ See Also

Reference

cDatabaseTables Interface

ETABSv1 Namespace

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Namespace: ETABSV1

▲ Syntax

C#

VBC++

F#

Copy

```
Function GetOutputOptionsForDisplay (
ByRef IsUserBaseReactionLocation As Boolean,
ByRef UserBaseReactionX As Double,
ByRef UserBaseReactionY As Double,
ByRef UserBaseReactionZ As Double,
ByRef IsAllModes As Boolean,
ByRef StartMode As Integer,
ByRef EndMode As Integer,
ByRef IsAllBucklingModes As Boolean,
ByRef StartBucklingMode As Integer,
ByRef EndBucklingMode As Integer,
ByRef MultistepStatic As Integer,
ByRef NonlinearStatic As Integer,
ByRef ModalHistory As Integer,
ByRef DirectHistory As Integer,
ByRef Combo As Integer
) As Integer
```

cDatabaseTable

Introduction

```
Dim IsAllModes As Boolean
Dim StartMode As Integer
Dim EndMode As Integer
Dim IsAllBucklingModes As Boolean
Dim StartBucklingMode As Integer
Dim EndBucklingMode As Integer
Dim MultistepStatic As Integer
Dim NonlinearStatic As Integer
Dim ModalHistory As Integer
Dim DirectHistory As Integer
Dim Combo As Integer
Dim returnValue As Integer

returnValue = instance.GetOutputOptionsForDisplay(IsUserBaseReactionLocation,
UserBaseReactionX, UserBaseReactionY,
UserBaseReactionZ, IsAllModes, StartMode,
EndMode, IsAllBucklingModes, StartBucklingMode,
EndBucklingMode, MultistepStatic,
NonlinearStatic, ModalHistory, DirectHistory,
Combo)

int GetOutputOptionsForDisplay(
bool% IsUserBaseReactionLocation,
double% UserBaseReactionX,
double% UserBaseReactionY,
double% UserBaseReactionZ,
bool% IsAllModes,
int% StartMode,
int% EndMode,
bool% IsAllBucklingModes,
int% StartBucklingMode,
int% EndBucklingMode,
int% MultistepStatic,
int% NonlinearStatic,
int% ModalHistory,
int% DirectHistory,
int% Combo
)

abstract GetOutputOptionsForDisplay :
    IsUserBaseReactionLocation : bool byref *
    UserBaseReactionX : float byref *
    UserBaseReactionY : float byref *
    UserBaseReactionZ : float byref *
    IsAllModes : bool byref *
    StartMode : int byref *
    EndMode : int byref *
    IsAllBucklingModes : bool byref *
    StartBucklingMode : int byref *
    EndBucklingMode : int byref *
    MultistepStatic : int byref *
    NonlinearStatic : int byref *
    ModalHistory : int byref *
    DirectHistory : int byref *
    Combo : int byref -> int
```

Parameters

IsUserBaseReactionLocation
Type: System.Boolean

Returned Item: Indicates if the base reaction location is user specified instead of program determined.

UserBaseReactionX

Type:Â SystemDouble

Returned Item: The global X coordinate of the user specified base reaction location. This item only applies when the IsUserBaseReactionLocation item is True.

UserBaseReactionY

Type:Â SystemDouble

Returned Item: The global Y coordinate of the user specified base reaction location. This item only applies when the IsUserBaseReactionLocation item is True.

UserBaseReactionZ

Type:Â SystemDouble

Returned Item: The global Z coordinate of the user specified base reaction location. This item only applies when the IsUserBaseReactionLocation item is True.

IsAllModes

Type:Â SystemBoolean

Returned Item: Indicates if results are to be displayed for all modes of modal load cases.

StartMode

Type:Â SystemInt32

Returned Item: The first mode for which results are shown for modal load cases. This item only applies when the IsAllModes item is False.

EndMode

Type:Â SystemInt32

Returned Item: The last mode for which results are shown for modal load cases. This item only applies when the IsAllModes item is False.

IsAllBucklingModes

Type:Â SystemBoolean

Returned Item: Indicates if results are to be displayed for all modes of buckling load cases.

StartBucklingMode

Type:Â SystemInt32

Returned Item: The first mode for which results are shown for buckling load cases. This item only applies when the IsAllBucklingModes item is False.

EndBucklingMode

Type:Â SystemInt32

Returned Item: The last mode for which results are shown for buckling load cases. This item only applies when the IsAllBucklingModes item is False.

MultistepStatic

Type:Â SystemInt32

Returned Item: Indicates how multistep static load case results are displayed: 1=Envelopes, 2=Step-by-step, 3=Last Step.

NonlinearStatic

Type:Â SystemInt32

Returned Item: Indicates how nonlinear static load case results are displayed: 1=Envelopes, 2=Step-by-step, 3=Last Step.

ModalHistory

Type: SystemInt32

Returned Item: Indicates how multistep modal time history load case results are displayed: 1=Envelopes, 2=Step-by-step, 3=Last Step.

DirectHistory

Type: SystemInt32

Returned Item: Indicates how direct integration time load case results are displayed: 1=Envelopes, 2=Step-by-step, 3=Last Step.

Combo

Type: SystemInt32

Returned Item: Indicates how load combination results are displayed:
1=Envelopes, 2=Multiple Values If Possible.

Return Value

Type: Int32

Returns 0 if the function executes correctly, otherwise returns nonzero

▲ See Also

Reference

cDatabaseTables Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDatabaseTablesAddLanguageSpecificTextSet("LST59 Method

Returns data for a single table in a single array. If there is nothing to be shown in the table then no data is returned.

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetTableForDisplayArray(
    string TableKey,
    ref string[] FieldKeyList,
    string GroupName,
    ref int TableVersion,
    ref string[] FieldsKeysIncluded,
    ref int NumberRecords,
    ref string[] TableData
)
```

```
Function GetTableForDisplayArray (
    TableKey As String,
    ByRef FieldKeyList As String(),
    GroupName As String,
    ByRef TableVersion As Integer,
    ByRef FieldsKeysIncluded As String(),
    ByRef NumberRecords As Integer,
    ByRef TableData As String()
) As Integer
```

```
Dim instance As cDatabaseTables
Dim TableKey As String
Dim FieldKeyList As String()
Dim GroupName As String
Dim TableVersion As Integer
Dim FieldsKeysIncluded As String()
Dim NumberRecords As Integer
Dim TableData As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetTableForDisplayArray(TableKey,
    FieldKeyList, GroupName, TableVersion,
    FieldsKeysIncluded, NumberRecords,
    TableData)
```

```
int GetTableForDisplayArray(
    String^ TableKey,
    array<String^>^% FieldKeyList,
    String^ GroupName,
    int% TableVersion,
```

cDatabaseTablesid="LST5939D311_0"AddLanguageSpecificTextSet("LST5939D311_0?cp=879;|nu=.")

```
array<String>^% FieldsKeysIncluded,
int% NumberRecords,
array<String>^% TableData
)
```

```
abstract GetTableForDisplayArray :
    TableKey : string *
    FieldKeyList : string[] byref *
    GroupName : string *
    TableVersion : int byref *
    FieldsKeysIncluded : string[] byref *
    NumberRecords : int byref *
    TableData : string[] byref -> int
```

Parameters

TableKey

Type:Â SystemString

Input Item: The table key of the table for which data is requested.

FieldKeyList

Type:Â SystemString

Input Item: A zero-based array listing the field keys associated with the specified table for which data is requested.

GroupName

Type:Â SystemString

Input Item: The name of a group for which data will be returned.

TableVersion

Type:Â SystemInt32

Returned Item: The version number of the specified table.

FieldsKeysIncluded

Type:Â SystemString

Returned Item: A zero-based array listing the field keys associated with the specified table for which data is reported in the order it is reported in the TableData array. These are essentially the column headers of the data returned in TableData.

NumberRecords

Type:Â SystemInt32

Returned Item: The number of records of data returned for each field. This is essentially the number of rows of data.

TableData

Type:Â SystemString

Returned Item: A zero-based, one-dimensional array of the table data, excluding headers, returned row by row. The format of the data is explained below.

As an example, suppose there are three fields in the FieldsKeysIncluded array and the NumberRecords is five.

EXAMPLE Material Name	Type	Density
A992Fy50	Steel	490
4000Psi	Concrete	150
A615Gr60	Rebar	480
A416Gr270	Tendon	470

6061T6

Aluminum 170

The data is returned row by row, and items 0 thru 2 in this array would be the first row, items 3 thru 5 the second row, etc, and items 12 thru 14 the fifth row. Using the example of the table above, the returned array would look like this:
 {A992Fy50, Steel, 490, 4000Psi, Concrete, 150, A615Gr60, Rebar, 480, A416Gr270, Tendon, 470, 6061T6, Aluminum, 170}

Return Value

Type: `Int32`

Returns 0 if the function executes correctly, otherwise returns nonzero

Remarks

If the FieldKeyList array contains a single blank string the data will be provided for all fields. If the GroupName is All, or a blank string, then data will be returned for all applicable objects in the model.

See Also

Reference

[cDatabaseTables Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDatabaseTablesAddLanguageSpecificTextSet("LST3B325081_0?cp=2;nu=.") Method

Returns data for a single table in a CSV file. If there is nothing to be shown in the table then no data is returned.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetTableForDisplayCSVFile(
    string TableKey,
    ref string[] FieldKeyList,
    string GroupName,
    ref int TableVersion,
    string csvFilePath,
    string sepChar = ",")
)
```

```
Function GetTableForDisplayCSVFile (
    TableKey As String,
    ByRef FieldKeyList As String(),
    GroupName As String,
    ByRef TableVersion As Integer,
    csvFilePath As String,
    Optional sepChar As String = ",",
    Optional returnVal As Integer
) As Integer
```

```
Dim instance As cDatabaseTables
Dim TableKey As String
Dim FieldKeyList As String()
Dim GroupName As String
Dim TableVersion As Integer
Dim csvFilePath As String
Dim sepChar As String
Dim returnValue As Integer
```

```
returnValue = instance.GetTableForDisplayCSVFile(TableKey,
    FieldKeyList, GroupName, TableVersion,
    csvFilePath, sepChar)
```

```
int GetTableForDisplayCSVFile(
    String^ TableKey,
    array<String^>^% FieldKeyList,
    String^ GroupName,
    int% TableVersion,
    String^ csvFilePath,
    String^ sepChar = L",")
)
```

```

abstract GetTableForDisplayCSVFile :
    TableKey : string *
    FieldKeyList : string[] byref *
    GroupName : string *
    TableVersion : int byref *
    csvFilePath : string *
    ?sepChar : string
(* Defaults:
    let _sepChar = defaultArg sepChar ", "
*)
-> int

```

Parameters

TableKey

Type: SystemString

Input Item: The table key of the table for which data is requested.

FieldKeyList

Type: SystemString

Input Item: A zero-based array listing the field keys associated with the specified table for which data is requested.

GroupName

Type: SystemString

Input Item: The name of a group for which data will be returned.

TableVersion

Type: SystemInt32

Returned Item: The version number of the specified table.

csvFilePath

Type: SystemString

Input Item: The fully-qualified path for the CSV file containing the table data.

sepChar (Optional)

Type: SystemString

Optional Input Item: The delimiter between data items, by default ","

Return Value

Type: Int32

Returns 0 if the function executes correctly, otherwise returns nonzero

Remarks

If the FieldKeyList array contains a single blank string the data will be provided for all fields. If the GroupName is All, or a blank string, then data will be returned for all applicable objects in the model.

See Also

Reference

cDatabaseTables Interface

ETABSv1 Namespace

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Namespace: ETABSv1

▲ Syntax

C#

VB

C++

F#

Copy

```
Function GetTableForDisplayCSVString (
    TableKey As String,
    ByRef FieldKeyList As String(),
    GroupName As String,
    ByRef TableVersion As Integer,
    ByRef csvString As String,
    ByRef what As String = ",",
    ) As Integer
```

```
returnValue = instance.GetTableForDisplayCSVString(TableKey,
FieldKeyList, GroupName, TableVersion,
csvString, sepChar)
```

cDatabaseTablespan id="LST95A358BF_0"AddLanguageSpecificTextSet("LST95A358BF_0?c884::lnu="


```

abstract GetTableForDisplayCSVString :
    TableKey : string *
    FieldKeyList : string[] byref *
    GroupName : string *
    TableVersion : int byref *
    csvString : string byref *
    ?sepChar : string
(* Defaults:
    let _sepChar = defaultArg sepChar ", "
*)
-> int

```

Parameters

TableKey

Type: [SystemString](#)

Input Item: The table key of the table for which data is requested.

FieldKeyList

Type: [SystemString](#)

Input Item: A zero-based array listing the field keys associated with the specified table for which data is requested.

GroupName

Type: [SystemString](#)

Input Item: The name of a group for which data will be returned.

TableVersion

Type: [SystemInt32](#)

Returned Item: The version number of the specified table.

csvString

Type: [SystemString](#)

Returned Item: A CSV-formatted string containing all the table data.

sepChar (Optional)

Type: [SystemString](#)

Optional Input Item: The delimiter between data items, by default ","

Return Value

Type: [Int32](#)

Returns 0 if the function executes correctly, otherwise returns nonzero

Remarks

If the FieldKeyList array contains a single blank string the data will be provided for all fields. If the GroupName is All, or a blank string, then data will be returned for all applicable objects in the model.

See Also

Reference

[cDatabaseTables Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDatabaseTablesAddLanguageSpecificTextSet("LSTDC23F0AA_0") Method

Returns data for a single table as XML in a single string. If there is nothing to be shown in the table then no data is returned.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetTableForDisplayXMLString(
    string TableKey,
    ref string[] FieldKeyList,
    string GroupName,
    bool IncludeSchema,
    ref int TableVersion,
    ref string XMLTableData
)
```

```
Function GetTableForDisplayXMLString (
    TableKey As String,
    ByRef FieldKeyList As String(),
    GroupName As String,
    IncludeSchema As Boolean,
    ByRef TableVersion As Integer,
    ByRef XMLTableData As String
) As Integer
```

```
Dim instance As cDatabaseTables
Dim TableKey As String
Dim FieldKeyList As String()
Dim GroupName As String
Dim IncludeSchema As Boolean
Dim TableVersion As Integer
Dim XMLTableData As String
Dim returnValue As Integer
```

```
returnValue = instance.GetTableForDisplayXMLString(TableKey,
    FieldKeyList, GroupName, IncludeSchema,
    TableVersion, XMLTableData)
```

```
int GetTableForDisplayXMLString(
    String^ TableKey,
    array<String^>^% FieldKeyList,
    String^ GroupName,
    bool IncludeSchema,
    int% TableVersion,
    String^% XMLTableData
)
```

```

abstract GetTableForDisplayXMLString :
    TableKey : string *
    FieldKeyList : string[] byref *
    GroupName : string *
    IncludeSchema : bool *
    TableVersion : int byref *
    XMLTableData : string byref -> int

```

Parameters

TableKey

Type: [SystemString](#)

Input Item: The table key of the table for which data is requested.

FieldKeyList

Type: [SystemString](#)

Input Item: A zero-based array listing the field keys associated with the specified table for which data is requested.

GroupName

Type: [SystemString](#)

Input Item: The name of a group for which data will be returned.

IncludeSchema

Type: [SystemBoolean](#)

Input Item: Flag indicating if the schema should be included with the table data.

TableVersion

Type: [SystemInt32](#)

Returned Item: The version number of the specified table.

XMLTableData

Type: [SystemString](#)

Returned Item: A string containing the XML data for the table

Return Value

Type: [Int32](#)

Returns 0 if the function executes correctly, otherwise returns nonzero

Remarks

If the FieldKeyList array contains a single blank string the data will be provided for all fields. If the GroupName is All, or a blank string, then data will be returned for all applicable objects in the model.

See Also

Reference

[cDatabaseTables Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDatabaseTablesAddLanguageSpecificTextSet("LST84 Method

Returns a single table in a string array for interactive editing.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetTableForEditingArray(
    string TableKey,
    string GroupName,
    ref int TableVersion,
    ref string[] FieldsKeysIncluded,
    ref int NumberRecords,
    ref string[] TableData
)
```

```
Function GetTableForEditingArray (
    TableKey As String,
    GroupName As String,
    ByRef TableVersion As Integer,
    ByRef FieldsKeysIncluded As String(),
    ByRef NumberRecords As Integer,
    ByRef TableData As String()
) As Integer
```

```
Dim instance As cDatabaseTables
Dim TableKey As String
Dim GroupName As String
Dim TableVersion As Integer
Dim FieldsKeysIncluded As String()
Dim NumberRecords As Integer
Dim TableData As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetTableForEditingArray(TableKey,
    GroupName, TableVersion, FieldsKeysIncluded,
    NumberRecords, TableData)
```

```
int GetTableForEditingArray(
    String^ TableKey,
    String^ GroupName,
    int% TableVersion,
    array<String^>^% FieldsKeysIncluded,
    int% NumberRecords,
    array<String^>^% TableData
)
```

```
abstract GetTableForEditingArray :
```

cDatabaseTablesAddLanguageSpecificTextSet("LST849041B_0?cpp=888u=.");G

```

TableKey : string *
GroupName : string *
TableVersion : int byref *
FieldsKeysIncluded : string[] byref *
NumberRecords : int byref *
TableData : string[] byref -> int

```

Parameters

TableKey

Type: SystemString

Input Item: The table key for a table which will be interactively edited. The table must be one that can be interactively edited.

GroupName

Type: SystemString

Input Item: IMPORTANT NOTE: This parameter is not active in this release.

The name of a group for which data will be returned.

TableVersion

Type: SystemInt32

Returned Item: The version number of the specified table.

FieldsKeysIncluded

Type: SystemString

Returned Item: A zero-based array listing the field keys associated with the specified table for which data is reported in the order it is reported in the TableData array. These are essentially the column headers of the data returned in TableData.

NumberRecords

Type: SystemInt32

Returned Item: The number of records of data returned for each field. This is essentially the number of rows of data.

TableData

Type: SystemString

Returned Item: A zero-based, one-dimensional array of the table data, excluding headers, returned row by row. The format of the data is explained below.

As an example, suppose there are three fields in the FieldsKeysIncluded array and the NumberRecords is five.

EXAMPLE Material Name	Type	Density
A992Fy50	Steel	490
4000Psi	Concrete	150
A615Gr60	Rebar	480
A416Gr270	Tendon	470
6061T6	Aluminum	170

The data is returned row by row, and items 0 thru 2 in this array would be the first row, items 3 thru 5 the second row, etc, and items 12 thru 14 the fifth row. Using the example of the table above, the returned array would look like this: {A992Fy50, Steel, 490, 4000Psi, Concrete, 150, A615Gr60, Rebar, 480, A416Gr270, Tendon, 470, 6061T6, Aluminum, 170}

Return Value

Type: Int32

Returns 0 if the function executes correctly, otherwise returns nonzero

▲ See Also

Reference

cDatabaseTables Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDatabaseTablesAddLanguageSpecificTextSet("LSTE1 Method

Returns a single table in a CSV file for interactive editing.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetTableForEditingCSVFile(
    string TableKey,
    string GroupName,
    ref int TableVersion,
    string csvFilePath,
    string sepChar = ", "
)
```

```
Function GetTableForEditingCSVFile (
    TableKey As String,
    GroupName As String,
    ByRef TableVersion As Integer,
    csvFilePath As String,
    Optional sepChar As String = ", "
) As Integer
```

```
Dim instance As cDatabaseTables
Dim TableKey As String
Dim GroupName As String
Dim TableVersion As Integer
Dim csvFilePath As String
Dim sepChar As String
Dim returnValue As Integer
```

```
returnValue = instance.GetTableForEditingCSVFile(TableKey,
    GroupName, TableVersion, csvFilePath,
    sepChar)
```

```
int GetTableForEditingCSVFile(
    String^ TableKey,
    String^ GroupName,
    int% TableVersion,
    String^ csvFilePath,
    String^ sepChar = L", "
)
```

```
abstract GetTableForEditingCSVFile :
    TableKey : string *
    GroupName : string *
    TableVersion : int byref *
    csvFilePath : string *
```

cDatabaseTablesid="LSTE1829168_0"AddLanguageSpecificTextSet("LSTE1829168_0?cp=1;nu=.")

```

        ?sepChar : string
(* Defaults:
    let _sepChar = defaultArg sepChar ", "
*)
-> int

```

Parameters

TableKey

Type: SystemString

Input Item: The table key for a table which will be interactively edited. The table must be one that can be interactively edited.

GroupName

Type: SystemString

Input Item: IMPORTANT NOTE: This parameter is not active in this release.

The name of a group for which data will be returned.

TableVersion

Type: SystemInt32

Returned Item: The version number of the specified table.

csvFilePath

Type: SystemString

Input Item: The fully-qualified path for the CSV file containing the table data.

Only the importable fields are included in the table.

sepChar (Optional)

Type: SystemString

Optional Input Item: The delimiter between data items, by default ",",

Return Value

Type: Int32

Returns 0 if the function executes correctly, otherwise returns nonzero

See Also

Reference

cDatabaseTables Interface

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CSI API ETABS v1

cDatabaseTablesAddLanguageSpecificTextSet("LST4573F2DC_0?app=::|nu=.") Method

Returns a single table in a CSV-formatted string for interactive editing.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetTableForEditingCSVString(
    string TableKey,
    string GroupName,
    ref int TableVersion,
    ref string csvString,
    string sepChar = ",")
)
```

```
Function GetTableForEditingCSVString (
    TableKey As String,
    GroupName As String,
    ByRef TableVersion As Integer,
    ByRef csvString As String,
    Optional sepChar As String = ","
) As Integer
```

```
Dim instance As cDatabaseTables
Dim TableKey As String
Dim GroupName As String
Dim TableVersion As Integer
Dim csvString As String
Dim sepChar As String
Dim returnValue As Integer
```

```
returnValue = instance.GetTableForEditingCSVString(TableKey,
    GroupName, TableVersion, csvString,
    sepChar)
```

```
int GetTableForEditingCSVString(
    String^ TableKey,
    String^ GroupName,
    int% TableVersion,
    String^% csvString,
    String^ sepChar = L",")
)
```

```
abstract GetTableForEditingCSVString :
    TableKey : string *
    GroupName : string *
    TableVersion : int byref *
    csvString : string byref *
```

cDatabaseTablesid="LST4573F2DC_0"AddLanguageSpecificTextSet("LST4573F2DC_0?app=::|nu=.")

```

        ?sepChar : string
(* Defaults:
    let _sepChar = defaultArg sepChar ", "
*)
-> int

```

Parameters

TableKey

Type: SystemString

Input Item: The table key for a table which will be interactively edited. The table must be one that can be interactively edited.

GroupName

Type: SystemString

Input Item: IMPORTANT NOTE: This parameter is not active in this release.

The name of a group for which data will be returned.

TableVersion

Type: SystemInt32

Returned Item: The version number of the specified table.

csvString

Type: SystemString

Returned Item: A CSV-formatted string containing all the table data.

sepChar (Optional)

Type: SystemString

Optional Input Item: The delimiter between data items, by default ",",

Return Value

Type: Int32

Returns 0 if the function executes correctly, otherwise returns nonzero

🔗 See Also

Reference

cDatabaseTables Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDatabaseTablesAddLanguageSpecificTextSet("LSTC52E5A4F_0") Method

Sets the load cases that are selected for table display.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetLoadCasesSelectedForDisplay(
    ref string[] LoadCaseList
)
```

```
Function SetLoadCasesSelectedForDisplay (
    ByRef LoadCaseList As String()
) As Integer
```

```
Dim instance As cDatabaseTables
Dim LoadCaseList As String()
Dim returnValue As Integer
```

```
returnValue = instance.SetLoadCasesSelectedForDisplay(LoadCaseList)
```

```
int SetLoadCasesSelectedForDisplay(
    array<String^>^% LoadCaseList
)
```

```
abstract SetLoadCasesSelectedForDisplay :
    LoadCaseList : string[] byref -> int
```

Parameters

LoadCaseList

Type: [SystemString](#)

Input Item: The zero-based list of load cases selected for table display.

Return Value

Type: [Int32](#)

Returns 0 if the function executes correctly, otherwise returns nonzero

▲ Remarks

This list sets the load cases that are included when displaying analysis results. If no load cases are to be selected then the LoadCaseList item should include a single blank string.

▲ See Also

Reference

cDatabaseTables Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDatabaseTablesAddLanguageSpecificTextSet("LSTA64AD453_0?op=::|nu="

Method

Sets the load combinations that are selected for table display.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetLoadCombinationsSelectedForDisplay(
    ref string[] LoadCombinationList
)
```

```
Function SetLoadCombinationsSelectedForDisplay (
    ByRef LoadCombinationList As String()
) As Integer
```

```
Dim instance As cDatabaseTables
Dim LoadCombinationList As String()
Dim returnValue As Integer
```

```
returnValue = instance.SetLoadCombinationsSelectedForDisplay(LoadCombinationList)
```

```
int SetLoadCombinationsSelectedForDisplay(
    array<String^>^% LoadCombinationList
)
```

```
abstract SetLoadCombinationsSelectedForDisplay :
    LoadCombinationList : string[] byref -> int
```

Parameters

LoadCombinationList

Type: [SystemString](#)

Input Item: The zero-based list of load combinations selected for table display.

Return Value

Type: [Int32](#)

Returns 0 if the function executes correctly, otherwise returns nonzero

Remarks

This list sets the load combinations that are included when displaying analysis results. If no load combinations are to be selected then the LoadCombinationList item should include a single blank string.

See Also

Reference

cDatabaseTables Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDatabaseTablesAddLanguageSpecificTextSet("LST46 Method

Sets the load patterns that are selected for table display.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetLoadPatternsSelectedForDisplay(
    ref string[] LoadPatternList
)
```

```
Function SetLoadPatternsSelectedForDisplay (
    ByRef LoadPatternList As String()
) As Integer
```

```
Dim instance As cDatabaseTables
Dim LoadPatternList As String()
Dim returnValue As Integer
```

```
returnValue = instance.SetLoadPatternsSelectedForDisplay(LoadPatternList)
```

```
int SetLoadPatternsSelectedForDisplay(
    array<String^>^% LoadPatternList
)
```

```
abstract SetLoadPatternsSelectedForDisplay :
    LoadPatternList : string[] byref -> int
```

Parameters

LoadPatternList

Type: [SystemString](#)

Input Item: The zero-based list of load patterns selected for table display.

Return Value

Type: [Int32](#)

Returns 0 if the function executes correctly, otherwise returns nonzero

▲ Remarks

This list sets the load patterns that are included when displaying load assignments on the model. If no load patterns are to be selected then the LoadPatternList item should include a single blank string.

▲ See Also

Reference

cDatabaseTables Interface

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CSI API ETABS v1

cDatabaseTablesAddLanguageSpecificTextSet("LSTD224FE3B_0?app=::|nu="

Method

Sets the database table output options for load case results.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetOutputOptionsForDisplay(
bool IsUserBaseReactionLocation,
double UserBaseReactionX,
double UserBaseReactionY,
double UserBaseReactionZ,
bool IsAllModes,
int StartMode,
int EndMode,
bool IsAllBucklingModes,
int StartBucklingMode,
int EndBucklingMode,
int MultistepStatic,
int NonlinearStatic,
int ModalHistory,
int DirectHistory,
int Combo
)
```

```
Function SetOutputOptionsForDisplay (
IsUserBaseReactionLocation As Boolean,
UserBaseReactionX As Double,
UserBaseReactionY As Double,
UserBaseReactionZ As Double,
IsAllModes As Boolean,
StartMode As Integer,
EndMode As Integer,
IsAllBucklingModes As Boolean,
StartBucklingMode As Integer,
EndBucklingMode As Integer,
MultistepStatic As Integer,
NonlinearStatic As Integer,
ModalHistory As Integer,
DirectHistory As Integer,
Combo As Integer
) As Integer
```

```
Dim instance As cDatabaseTables
Dim IsUserBaseReactionLocation As Boolean
Dim UserBaseReactionX As Double
Dim UserBaseReactionY As Double
Dim UserBaseReactionZ As Double
```

cDatabaseTablesAddLanguageSpecificTextSet("LSTD224FE3B_0?app=::|nu="

Introduction

```
Dim IsAllModes As Boolean
Dim StartMode As Integer
Dim EndMode As Integer
Dim IsAllBucklingModes As Boolean
Dim StartBucklingMode As Integer
Dim EndBucklingMode As Integer
Dim MultistepStatic As Integer
Dim NonlinearStatic As Integer
Dim ModalHistory As Integer
Dim DirectHistory As Integer
Dim Combo As Integer
Dim returnValue As Integer

returnValue = instance.SetOutputOptionsForDisplay(IsUserBaseReactionLocation,
UserBaseReactionX, UserBaseReactionY,
UserBaseReactionZ, IsAllModes, StartMode,
EndMode, IsAllBucklingModes, StartBucklingMode,
EndBucklingMode, MultistepStatic,
NonlinearStatic, ModalHistory, DirectHistory,
Combo)

int SetOutputOptionsForDisplay(
bool IsUserBaseReactionLocation,
double UserBaseReactionX,
double UserBaseReactionY,
double UserBaseReactionZ,
bool IsAllModes,
int StartMode,
int EndMode,
bool IsAllBucklingModes,
int StartBucklingMode,
int EndBucklingMode,
int MultistepStatic,
int NonlinearStatic,
int ModalHistory,
int DirectHistory,
int Combo
)

abstract SetOutputOptionsForDisplay :
    IsUserBaseReactionLocation : bool *
    UserBaseReactionX : float *
    UserBaseReactionY : float *
    UserBaseReactionZ : float *
    IsAllModes : bool *
    StartMode : int *
    EndMode : int *
    IsAllBucklingModes : bool *
    StartBucklingMode : int *
    EndBucklingMode : int *
    MultistepStatic : int *
    NonlinearStatic : int *
    ModalHistory : int *
    DirectHistory : int *
    Combo : int -> int
```

Parameters

IsUserBaseReactionLocation
Type: SystemBoolean

Input Item: Indicates if the base reaction location is user specified instead of program determined.

UserBaseReactionX

Type:Â SystemDouble

Input Item: The global X coordinate of the user specified base reaction location. This item only applies when the IsUserBaseReactionLocation item is True.

UserBaseReactionY

Type:Â SystemDouble

Input Item: The global Y coordinate of the user specified base reaction location. This item only applies when the IsUserBaseReactionLocation item is True.

UserBaseReactionZ

Type:Â SystemDouble

Input Item: The global Z coordinate of the user specified base reaction location. This item only applies when the IsUserBaseReactionLocation item is True.

IsAllModes

Type:Â SystemBoolean

Input Item: Indicates if results are to be displayed for all modes of modal load cases.

StartMode

Type:Â SystemInt32

Input Item: The first mode for which results are shown for modal load cases. This item only applies when the IsAllModes item is False.

EndMode

Type:Â SystemInt32

Input Item: The last mode for which results are shown for modal load cases. This item only applies when the IsAllModes item is False.

IsAllBucklingModes

Type:Â SystemBoolean

Input Item: Indicates if results are to be displayed for all modes of buckling load cases.

StartBucklingMode

Type:Â SystemInt32

Input Item: The first mode for which results are shown for buckling load cases. This item only applies when the IsAllBucklingModes item is False.

EndBucklingMode

Type:Â SystemInt32

Input Item: The last mode for which results are shown for buckling load cases. This item only applies when the IsAllBucklingModes item is False.

MultistepStatic

Type:Â SystemInt32

Input Item: Indicates how multistep static load case results are displayed: 1=Envelopes, 2=Step-by-step, 3=Last Step.

NonlinearStatic

Type:Â SystemInt32

Input Item: Indicates how nonlinear static load case results are displayed: 1=Envelopes, 2=Step-by-step, 3=Last Step.

ModalHistory

Type:Â SystemInt32

Input Item: Indicates how multistep modal time history load case results are displayed: 1=Envelopes, 2=Step-by-step, 3=Last Step.

DirectHistory

Type: SystemInt32

Input Item: Indicates how direct integration time load case results are displayed: 1=Envelopes, 2=Step-by-step, 3=Last Step.

Combo

Type: SystemInt32

Returned Item: Indicates how load combination results are displayed: 1=Envelopes, 2=Multiple Values If Possible.

Return Value

Type: Int32

Returns 0 if the function executes correctly, otherwise returns nonzero

■ See Also

Reference

cDatabaseTables Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDatabaseTablesAddLanguageSpecificTextSet("LST2F99A871_0?cp=905;|nu=.") Method

Reads a table from a string array and adds it to a stored table list until either the ApplyEditedTables(Boolean, Int32, Int32, Int32, Int32, String) or CancelTableEditing command is issued.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetTableForEditingArray(
    string TableKey,
    ref int TableVersion,
    ref string[] FieldsKeysIncluded,
    int NumberRecords,
    ref string[] TableData
)
```

```
Function SetTableForEditingArray (
    TableKey As String,
    ByRef TableVersion As Integer,
    ByRef FieldsKeysIncluded As String(),
    NumberRecords As Integer,
    ByRef TableData As String()
) As Integer
```

```
Dim instance As cDatabaseTables
Dim TableKey As String
Dim TableVersion As Integer
Dim FieldsKeysIncluded As String()
Dim NumberRecords As Integer
Dim TableData As String()
Dim returnValue As Integer
```

```
returnValue = instance.SetTableForEditingArray(TableKey,
    TableVersion, FieldsKeysIncluded,
    NumberRecords, TableData)
```

```
int SetTableForEditingArray(
    String^ TableKey,
    int% TableVersion,
    array<String^>^% FieldsKeysIncluded,
    int NumberRecords,
    array<String^>^% TableData
)
```

```
abstract SetTableForEditingArray :
    TableKey : string *
    TableVersion : int byref *
```

Introduction

```
FieldsKeysIncluded : string[] byref *  
NumberRecords : int *  
TableData : string[] byref -> int
```

Parameters

TableKey

Type: SystemString

Input Item: The table key for a table which has been interactively edited. The table must be one that can be interactively edited.

TableVersion

Type: SystemInt32

Returned Item: The version number of the specified table.

FieldsKeysIncluded

Type: SystemString

Input Item: A zero-based array listing the field keys associated with the specified table for which data is reported in the order it is reported in the TableData array. These are essentially the column headers of the data reported in TableData.

NumberRecords

Type: SystemInt32

Input Item: The number of records of data for each field. This is essentially the number of rows of data.

TableData

Type: SystemString

Input Item: A zero-based, one-dimensional array of the table data, excluding headers, reported row by row. See GetTableForDisplayArray(String,String,String,Int32,String,Int32,String) for a more detailed explanation of the data format.

Return Value

Type: Int32

Returns 0 if the function executes correctly, otherwise returns nonzero

Remarks

Here is an example of how to set the inputs:

```
TableKey = "EXAMPLE Material Table"
```

```
FieldsKeysIncluded = {"Material Name", "Material Type", "Density"}
```

```
NumberRecords = 5
```

```
TableData = {A992Fy50, Steel, 490, 4000Psi, Concrete, 150, A615Gr60, Rebar, 480,  
A416Gr270, Tendon, 470, 6061T6, Aluminum, 170}
```

Executing the function with these inputs would be equivalent to setting the following table interactively:

EXAMPLE Material Name	Type	Density
A992Fy50	Steel	490

Introduction

4000Psi	Concrete	150
A615Gr60	Rebar	480
A416Gr270	Tendon	470
6061T6	Aluminum	170

▲ See Also

Reference

cDatabaseTables Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDatabaseTablesAddLanguageSpecificTextSet("LST7C Method

Reads a table from a CSV file and adds it to a stored table list until either the ApplyEditedTables(Boolean, Int32, Int32, Int32, Int32, String) or CancelTableEditing command is issued.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetTableForEditingCSVFile(
    string TableKey,
    ref int TableVersion,
    string csvFilePath,
    string sepChar = ", "
)
```

```
Function SetTableForEditingCSVFile (
    TableKey As String,
    ByRef TableVersion As Integer,
    csvFilePath As String,
    Optional sepChar As String = ", "
) As Integer
```

```
Dim instance As cDatabaseTables
Dim TableKey As String
Dim TableVersion As Integer
Dim csvFilePath As String
Dim sepChar As String
Dim returnValue As Integer
```

```
returnValue = instance.SetTableForEditingCSVFile(TableKey,
    TableVersion, csvFilePath, sepChar)
```

```
int SetTableForEditingCSVFile(
    String^ TableKey,
    int% TableVersion,
    String^ csvFilePath,
    String^ sepChar = L", "
)
```

```
abstract SetTableForEditingCSVFile :
    TableKey : string *
    TableVersion : int byref *
    csvFilePath : string *
    ?sepChar : string
(* Defaults:
    let _sepChar = defaultArg sepChar ", "
*)
```

cDatabaseTablesid="LST7CAD4A89_0"AddLanguageSpecificTextSet("LST7CAD4A89_0?006=::|nu=.

-> int

Parameters

TableKey

Type: SystemString

Input Item: The table key for a table which has been interactively edited. The table must be one that can be interactively edited.

TableVersion

Type: SystemInt32

Returned Item: The version number of the specified table.

csvFilePath

Type: SystemString

Input Item: The fully-qualified path for the CSV file containing the table data.

sepChar (Optional)

Type: SystemString

Optional Input Item: The delimiter between data items, by default "," . However, please note that this delimiter must match what is used in the csvFilePath file.

Return Value

Type: Int32

Returns 0 if the function executes correctly, otherwise returns nonzero

▲ See Also

Reference

cDatabaseTables Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDatabaseTablesAddLanguageSpecificTextSet("LSTE0395D5_0") Method

Reads a table from a CSV-formatted string and adds it to a stored table list until either the ApplyEditedTables(Boolean, Int32, Int32, Int32, Int32, String) or CancelTableEditing command is issued.

Namespace: Â ETABSV1

Assembly: Â ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetTableForEditingCSVString(
    string TableKey,
    ref int TableVersion,
    ref string csvString,
    string sepChar = ",",
)
```

```
Function SetTableForEditingCSVString (
    TableKey As String,
    ByRef TableVersion As Integer,
    ByRef csvString As String,
    Optional sepChar As String = ",",
) As Integer
```

```
Dim instance As cDatabaseTables
Dim TableKey As String
Dim TableVersion As Integer
Dim csvString As String
Dim sepChar As String
Dim returnValue As Integer
```

```
returnValue = instance.SetTableForEditingCSVString(TableKey,
    TableVersion, csvString, sepChar)
```

```
int SetTableForEditingCSVString(
    String^ TableKey,
    int% TableVersion,
    String^% csvString,
    String^ sepChar = L",",
)
```

```
abstract SetTableForEditingCSVString :
    TableKey : string *
    TableVersion : int byref *
    csvString : string byref *
    ?sepChar : string
(* Defaults:
    let _sepChar = defaultArg sepChar " , "
*)
```

cDatabaseTablesspan id="LSTE0395D5_0"AddLanguageSpecificTextSet("LSTE0395D5_0?cpp9110u=.");S

-> int

Parameters

TableKey

Type: SystemString

Input Item: The table key for a table which has been interactively edited. The table must be one that can be interactively edited.

TableVersion

Type: SystemInt32

Returned Item: The version number of the specified table.

csvString

Type: SystemString

Input Item: A CSV-formatted string containing all the table data. Newline characters should be used to demarcate rows.

sepChar (Optional)

Type: SystemString

Optional Input Item: The delimiter between data items in the same row, by default ",". This delimiter must match what is used in the csvString.

Return Value

Type: Int32

Returns 0 if the function executes correctly, otherwise returns nonzero

▲ See Also

Reference

cDatabaseTables Interface

ETABSV1 Namespace

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CSI API ETABS v1

cDatabaseTablesAddLanguageSpecificTextSet("LST8B34CB36_0?012=::|nu=.") Method

Exports the specified tables to Excel.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int ShowTablesInExcel(
    ref string[] TableKeyList,
    int WindowHandle
)
```

```
Function ShowTablesInExcel (
    ByRef TableKeyList As String(),
    WindowHandle As Integer
) As Integer
```

```
Dim instance As cDatabaseTables
Dim TableKeyList As String()
Dim WindowHandle As Integer
Dim returnValue As Integer
```

```
returnValue = instance.ShowTablesInExcel(TableKeyList,
    WindowHandle)
```

```
int ShowTablesInExcel(
    array<String^>^% TableKeyList,
    int WindowHandle
)
```

```
abstract ShowTablesInExcel :
    TableKeyList : string[] byref *
    WindowHandle : int -> int
```

Parameters

TableKeyList

Type: [SystemString](#)

A zero-based array of the table keys for the tables to be included in the output.

WindowHandle

Type: [SystemInt32](#)

If a valid window handle is entered here Excel will be located in front of that window.

Return Value

Type: Int32

Returns 0 if the function executes correctly, otherwise returns nonzero

Remarks

Excel must be present on the computer for this function to work. If there is nothing to be shown in the table then no data is returned and Excel is not displayed.

See Also

Reference

cDatabaseTables Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDCoACI318_08_IBC2009 Interface

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
public interface cDCoACI318_08_IBC2009
```

```
Public Interface cDCoACI318_08_IBC2009
```

```
Dim instance As cDCoACI318_08_IBC2009
```

```
public interface class cDCoACI318_08_IBC2009
```

```
type cDCoACI318_08_IBC2009 = interface end
```

The cDCoACI318_08_IBC2009 type exposes the following members.

▲ Methods

▲	Name	Description
	<u>GetOverwrite</u>	Retrieves the value of a concrete design overwrite item.
	<u>GetPreference</u>	Retrieves the value of a concrete design preference item.
	<u>SetOverwrite</u>	Sets the value of a concrete design overwrite item.
	<u>SetPreference</u>	Sets the value of a concrete design preference item.

Top

▲ See Also

Reference

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CSI API ETABS v1

cDCoACI318_08_IBC2009 Methods

The cDCoACI318_08_IBC2009 type exposes the following members.

Methods

	Name	Description
	<u>GetOverwrite</u>	Retrieves the value of a concrete design overwrite item.
	<u>GetPreference</u>	Retrieves the value of a concrete design preference item.
	<u>SetOverwrite</u>	Sets the value of a concrete design overwrite item.
	<u>SetPreference</u>	Sets the value of a concrete design preference item.

Top

See Also

Reference

cDCoACI318_08_IBC2009 Interface

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CSI API ETABS v1

cDCoACI318_08_IBC2009AddLanguageSpecificTextSet Method

Retrieves the value of a concrete design overwrite item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDCoACI318_08_IBC2009
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```


Parameters

Name

Type: SystemString

The name of a frame object with a concrete frame design procedure.

Item

Type: SystemInt32

This is an integer between 1 and 12, inclusive, indicating the overwrite item considered.

1. Framing type
2. Live load reduction factor
3. Unbraced length ratio, Major
4. Unbraced length ratio, Minor
5. Effective length factor, K Major
6. Effective length factor, K Minor
7. Moment coefficient, Cm Major
8. Moment coefficient, Cm Minor
9. Nonsway moment factor, Db Major
10. Nonsway moment factor, Db Minor
11. Sway moment factor, Ds Major
12. Sway moment factor, Ds Minor

Value

Type: SystemDouble

The value of the considered overwrite item.

1. Framing type
 - 0 = Program Default
 - 1 = Sway special
 - 2 = Sway Intermediate
 - 3 = Sway Ordinary
 - 4 = Non-sway
2. Live load reduction factor

Value ≥ 0 ; 0 means use program determined value
3. Unbraced length ratio, Major

Value ≥ 0 ; 0 means use program determined value
4. Unbraced length ratio, Minor

Value ≥ 0 ; 0 means use program determined value
5. Effective length factor, K Major

Value ≥ 0 ; 0 means use program determined value
6. Effective length factor, K Minor

Value ≥ 0 ; 0 means use program determined value
7. Moment coefficient, Cm Major

Value ≥ 0 ; 0 means use program determined value
8. Moment coefficient, Cm Minor

Value ≥ 0 ; 0 means use program determined value

9. Nonsway moment factor, Db Major

Value ≥ 0 ; 0 means use program determined value

10. Nonsway moment factor, Db Minor

Value ≥ 0 ; 0 means use program determined value

11. Sway moment factor, Ds Major

Value ≥ 0 ; 0 means use program determined value

12. Sway moment factor, Ds Minor

Value ≥ 0 ; 0 means use program determined value

ProgDet

Type: SystemBoolean

If this item is True, the specified value is program determined.

Return ValueType: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double
    Dim ProgDet As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'create new concrete frame section property
    ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

    'set frame section property
    ret = SapModel.FrameObj.SetSection("8", "R1")

    'set concrete design code
    ret = SapModel.DesignConcrete.SetCode("ACI 318-08/IBC 2009")

    'run analysis
```

Return Value

Introduction

```
System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
ret = SapModel.Analyze.RunAnalysis

'start concrete design
ret = SapModel.DesignConcrete.StartDesign()

'get overwrite item
ret = SapModel.DesignConcrete.ACI318_08_IBC2009.GetOverwrite("8", 1, Value, ProgDet)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

[cDCoACI318_08_IBC2009 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDCoACI318_08_IBC2009AddLanguageSpecificTextSet Method

Retrieves the value of a concrete design preference item.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDCoACI318_08_IBC2009
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: SystemInt32

This is an integer between 1 and 13, inclusive, indicating the preference item considered.

1. Number of interaction curves
2. Number of interaction points
3. Consider minimum eccentricity
4. Seismic design category
5. Phi tension controlled

6. Phi compression controlled tied
7. Phi compression controlled spiral
8. Phi shear and/or torsion
9. Phi shear seismic
10. Phi joint shear
11. Pattern live load factor
12. Utilization factor limit
13. Multi-response case design

Value

Type: SystemDouble

The value of the considered preference item.

1. Number of interaction curves

Value ≥ 4 and divisible by 4

2. Number of interaction points

Value ≥ 5 and odd

3. Consider minimum eccentricity

0 = No

Any other value = Yes

4. Seismic design category

. 1 = A

. 2 = B

. 3 = C

. 4 = D

. 5 = E

. 6 = F

5. Phi tension controlled

Value > 0

6. Phi compression controlled tied

Value > 0

7. Phi compression controlled spiral

Value > 0

8. Phi shear and/or torsion

Value > 0

9. Phi shear seismic

Value > 0

10. Phi joint shear

Value > 0

11. Pattern live load factor

Value ≥ 0

12. Utilization factor limit

Value > 0

13. Multi-response case design

- 1 = Envelopes
- 2 = Step-by step
- 3 = Last step
- 4 = Envelopes - All
- 5 = Step-by step - All

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'create new concrete frame section property
    ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

    'set frame section property
    ret = SapModel.FrameObj.SetSection("8", "R1")

    'set concrete design code
    ret = SapModel.DesignConcrete.SetCode("ACI 318-08/IBC 2009")

    'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis

    'start concrete design
    ret = SapModel.DesignConcrete.StartDesign()

    'get preference item
    ret = SapModel.DesignConcrete.ACI318_08_IBC2009.GetPreference(2, Value)

    'close ETABS
```

Return Value

Introduction

```
EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cDCoACI318_08_IBC2009 Interface

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CSI API ETABS v1

cDCoACI318_08_IBC2009AddLanguageSpecificTextSet Method

Sets the value of a concrete design overwrite item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDCoACI318_08_IBC2009
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```


Parameters**Name**

Type:Â SystemString

The name of an existing frame object or group, depending on the value of the ItemType item.

Item

Type:Â SystemInt32

This is an integer between 1 and 12, inclusive, indicating the overwrite item considered.

1. Framing type
2. Live load reduction factor
3. Unbraced length ratio, Major
4. Unbraced length ratio, Minor
5. Effective length factor, K Major
6. Effective length factor, K Minor
7. Moment coefficient, Cm Major
8. Moment coefficient, Cm Minor
9. Nonsway moment factor, Db Major
10. Nonsway moment factor, Db Minor
11. Sway moment factor, Ds Major
12. Sway moment factor, Ds Minor

Value

Type:Â SystemDouble

The value of the considered overwrite item.

1. Framing type
 - 0 = Program Default
 - 1 = Sway special
 - 2 = Sway Intermediate
 - 3 = Sway Ordinary
 - 4 = Non-sway
2. Live load reduction factor

Value >= 0; 0 means use program determined value

3. Unbraced length ratio, Major

Value >= 0; 0 means use program determined value

4. Unbraced length ratio, Minor

Value >= 0; 0 means use program determined value

5. Effective length factor, K Major

Value >= 0; 0 means use program determined value

6. Effective length factor, K Minor

Value >= 0; 0 means use program determined value

7. Moment coefficient, Cm Major

Value >= 0; 0 means use program determined value

8. Moment coefficient, Cm Minor

Introduction

Value ≥ 0 ; 0 means use program determined value

9. Nonsway moment factor, Db Major

Value ≥ 0 ; 0 means use program determined value

10. Nonsway moment factor, Db Minor

Value ≥ 0 ; 0 means use program determined value

11. Sway moment factor, Ds Major

Value ≥ 0 ; 0 means use program determined value

12. Sway moment factor, Ds Minor

Value ≥ 0 ; 0 means use program determined value

ItemType (Optional)

Type: $\hat{A}$ ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the assignment is made to the frame object specified by the Name item.

If this item is Group, the assignment is made to all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the assignment is made to all selected frame objects and the Name item is ignored.

Return Value

Type: $\hat{A}$ Int32

Returns zero if the item is successfully set; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()
```

Introduction

```
'create steel deck template model
  ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'create new concrete frame section property
  ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

'set frame section property
  ret = SapModel.FrameObj.SetSection("8", "R1")

'set concrete design code
  ret = SapModel.DesignConcrete.SetCode("ACI 318-08/IBC 2009")

'set overwrite item
  ret = SapModel.DesignConcrete.ACI318_08_IBC2009.SetOverwrite("8", 1, 2)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cDCoACI318_08_IBC2009 Interface

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CSI API ETABS v1

cDCoACI318_08_IBC2009AddLanguageSpecificTextSet Method

Sets the value of a concrete design preference item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDCoACI318_08_IBC2009
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: SystemInt32

This is an integer between 1 and 13, inclusive, indicating the preference item considered.

1. Number of interaction curves
2. Number of interaction points
3. Consider minimum eccentricity
4. Seismic design category
5. Phi tension controlled

6. Phi compression controlled tied
7. Phi compression controlled spiral
8. Phi shear and/or torsion
9. Phi shear seismic
10. Phi joint shear
11. Pattern live load factor
12. Utilization factor limit
13. Multi-response case design

Value

Type: SystemDouble

The value of the considered preference item.

1. Number of interaction curves

Value ≥ 4 and divisible by 4

2. Number of interaction points

Value ≥ 5 and odd

3. Consider minimum eccentricity

0 = No

Any other value = Yes

4. Seismic design category

- . 1 = A
- . 2 = B
- . 3 = C
- . 4 = D
- . 5 = E
- . 6 = F

5. Phi tension controlled

Value > 0

6. Phi compression controlled tied

Value > 0

7. Phi compression controlled spiral

Value > 0

8. Phi shear and/or torsion

Value > 0

9. Phi shear seismic

Value > 0

10. Phi joint shear

Value > 0

11. Pattern live load factor

Value ≥ 0

12. Utilization factor limit

Value > 0

13. Multi-response case design

- 1 = Envelopes
- 2 = Step-by step
- 3 = Last step
- 4 = Envelopes - All
- 5 = Step-by step - All

Return Value

Type: Int32

Returns zero if the item is successfully set; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'create new concrete frame section property
    ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

    'set frame section property
    ret = SapModel.FrameObj.SetSection("8", "R1")

    'set concrete design code
    ret = SapModel.DesignConcrete.SetCode("ACI 318-08/IBC 2009")

    'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis

    'start concrete design
    ret = SapModel.DesignConcrete.StartDesign()

    'set preference item
    ret = SapModel.DesignConcrete.ACI318_08_IBC2009.SetPreference(2, 9)

    'close ETABS
    EtabsObject.ApplicationExit(False)
```

Return Value

```
'clean up variables  
  SapModel = Nothing  
  EtabsObject = Nothing
```

End Sub

▲ See Also

Reference

cDCoACI318_08_IBC2009 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDCoACI318_11 Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDCoACI318_11
```

```
Public Interface cDCoACI318_11
```

```
Dim instance As cDCoACI318_11
```

```
public interface class cDCoACI318_11
```

```
type cDCoACI318_11 = interface end
```

The cDCoACI318_11 type exposes the following members.

▲ Methods

▲	Name	Description
---	------	-------------

➦	GetOverwrite	
---	------------------------------	--

➦	GetPreference	
---	-------------------------------	--

➦	SetOverwrite	
---	------------------------------	--

➦	SetPreference	
---	-------------------------------	--

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDCoACI318_11 Methods

The cDCoACI318_11 type exposes the following members.

Methods

	Name	Description
--	------	-------------

	<u>GetOverwrite</u>	
--	---------------------	--

	<u>GetPreference</u>	
--	----------------------	--

	<u>SetOverwrite</u>	
--	---------------------	--

	<u>SetPreference</u>	
--	----------------------	--

Top

See Also

Reference

cDCoACI318_11 Interface

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Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

C#

C++

F#

Copy

```
abstract GetOverwrite :
  Name : string *
  Item : int *
  Value : float byref *
  ProgDet : bool byref -> int
```

Name

cDCoACI318_11span id="LST8D2C945A_0"AddLanguageSpecificTextSet("LST8D2C945A_0?c004::|nu="

Item

Type:Â [SystemInt32](#)

Value

Type:Â [SystemDouble](#)

ProgDet

Type:Â [SystemBoolean](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cDCoACI318_11 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDCoACI318_11AddLanguageSpecificTextSet("LST92A Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDCoACI318_11
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: [SystemInt32](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDCoACI318 11 Interface

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CSI API ETABS v1

cDCoACI318_11AddLanguageSpecificTextSet("LST6D3 Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDCoACI318_11
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

Item

Type: SystemInt32

Value

Type: SystemDouble

ItemType (Optional)

Type: ETABSv1eItemType

Return Value

Type: Int32

▲ See Also

Reference

cDCoACI318_11 Interface

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CSI API ETABS v1

cDCoACI318_11AddLanguageSpecificTextSet("LSTB5F477ED_0?c010::|nu=.") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDCoACI318_11
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: [SystemInt32](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDCoACI318 11 Interface

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CSI API ETABS v1

cDCoACI318_14 Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDCoACI318_14
```

```
Public Interface cDCoACI318_14
```

```
Dim instance As cDCoACI318_14
```

```
public interface class cDCoACI318_14
```

```
type cDCoACI318_14 = interface end
```

The cDCoACI318_14 type exposes the following members.

▲ Methods

▲	Name	Description
	GetOverwrite	Retrieves the value of a concrete design overwrite item.
	GetPreference	Retrieves the value of a concrete design preference item.
	SetOverwrite	Sets the value of a concrete design overwrite item.
	SetPreference	Sets the value of a concrete design preference item.

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDCoACI318_14 Methods

The [cDCoACI318_14](#) type exposes the following members.

Methods

	Name	Description
	GetOverwrite	Retrieves the value of a concrete design overwrite item.
	GetPreference	Retrieves the value of a concrete design preference item.
	SetOverwrite	Sets the value of a concrete design overwrite item.
	SetPreference	Sets the value of a concrete design preference item.

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 See Also

Reference

[cDCoACI318_14 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDCoACI318_14AddLanguageSpecificTextSet("LST861 Method

Retrieves the value of a concrete design overwrite item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDCoACI318_14
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```

Parameters

Name

Type: SystemString

The name of a frame object with a concrete frame design procedure.

Item

Type: SystemInt32

This is an integer between 1 and 13, inclusive, indicating the overwrite item considered.

1. Framing type
2. Live load reduction factor
3. Unbraced length ratio, Major
4. Unbraced length ratio, Minor
5. Effective length factor, K Major
6. Effective length factor, K Minor
7. Moment coefficient, Cm Major
8. Moment coefficient, Cm Minor
9. Nonsway moment factor, Db Major
10. Nonsway moment factor, Db Minor
11. Sway moment factor, Ds Major
12. Sway moment factor, Ds Minor
13. Consider Minimum Eccentricity

Value

Type: SystemDouble

The value of the considered overwrite item.

1. Framing type
 - 0 = Program Default
 - 1 = Sway special
 - 2 = Sway Intermediate
 - 3 = Sway Ordinary
 - 4 = Non-sway
2. Live load reduction factor

Value ≥ 0 ; 0 means use program determined value

3. Unbraced length ratio, Major

Value ≥ 0 ; 0 means use program determined value

4. Unbraced length ratio, Minor

Value ≥ 0 ; 0 means use program determined value

5. Effective length factor, K Major

Value ≥ 0 ; 0 means use program determined value

6. Effective length factor, K Minor

Value ≥ 0 ; 0 means use program determined value

7. Moment coefficient, Cm Major

Value ≥ 0 ; 0 means use program determined value

8. Moment coefficient, Cm Minor

Introduction

Value ≥ 0 ; 0 means use program determined value

9. Nonsway moment factor, Db Major

Value ≥ 0 ; 0 means use program determined value

10. Nonsway moment factor, Db Minor

Value ≥ 0 ; 0 means use program determined value

11. Sway moment factor, Ds Major

Value ≥ 0 ; 0 means use program determined value

12. Sway moment factor, Ds Minor

Value ≥ 0 ; 0 means use program determined value

13. Consider Minimum Eccentricity

0 = No

Any other value = Yes

ProgDet

Type: SystemBoolean

If this item is True, the specified value is program determined.

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim Value As Double  
    Dim ProgDet As Boolean  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)  
  
    'create new concrete frame section property  
    ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)
```

Return Value

Introduction

```
'set frame section property
    ret = SapModel.FrameObj.SetSection("8", "R1")

'set concrete design code
    ret = SapModel.DesignConcrete.SetCode("ACI 318-14")

'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis

'start concrete design
    ret = SapModel.DesignConcrete.StartDesign()

'get overwrite item
    ret = SapModel.DesignConcrete.ACI318_14.GetOverwrite("8", 1, Value, ProgDet)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

📌 See Also

Reference

[cDCoACI318_14 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDcOACI318_14AddLanguageSpecificTextSet("LSTB3B90B08_0?cp=948:|nu=.")

Method

Retrieves the value of a concrete design preference item.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDcOACI318_14
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: [SystemInt32](#)

This is an integer between 1 and 18, inclusive, indicating the preference item considered.

1. Number of interaction curves
2. Number of interaction points
3. Consider minimum eccentricity
4. Design for B/C Capacity Ratio?
5. Seismic design category

6. Design System Omega0
7. Design System Rho
8. Design System Sds
9. Consider ICC_ESR 2017
10. Phi tension controlled
11. Phi compression controlled tied
12. Phi compression controlled spiral
13. Phi shear and/or torsion
14. Phi shear seismic
15. Phi joint shear
16. Pattern live load factor
17. Utilization factor limit
18. Multi-response case design

Value

Type: SystemDouble

The value of the considered preference item.

1. Number of interaction curves

Value ≥ 4 and divisible by 4

2. Number of interaction points

Value ≥ 5 and odd

3. Consider minimum eccentricity

0 = No

Any other value = Yes

4. Design for B/C Capacity Ratio?

1 = No

2 = Yes

5. Seismic design category

. 1 = A

. 2 = B

. 3 = C

. 4 = D

. 5 = E

. 6 = F

6. Design System Omega0

Value > 0

7. Design System Rho

Value > 0

8. Design System Sds

Value ≥ 0

9. Consider ICC_ESR 2017

1 = No

2 = Yes

10. Phi tension controlled

Value > 0

11. Phi compression controlled tied

Value > 0

12. Phi compression controlled spiral

Value > 0

13. Phi shear and/or torsion

Value > 0

14. Phi shear seismic

Value > 0

15. Phi joint shear

Value > 0

16. Pattern live load factor

Value >= 0

17. Utilization factor limit

Value > 0

18. Multi-response case design

- 1 = Envelopes
- 2 = Step-by step
- 3 = Last step
- 4 = Envelopes - All
- 5 = Step-by step - All

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()
```

Introduction

```
'create SapModel object
  SapModel = EtabsObject.SapModel

'initialize model
  ret = SapModel.InitializeNewModel()

'create steel deck template model
  ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'create new concrete frame section property
  ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

'set frame section property
  ret = SapModel.FrameObj.SetSection("8", "R1")

'set concrete design code
  ret = SapModel.DesignConcrete.SetCode("ACI 318-14")

'run analysis
  System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
  ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
  ret = SapModel.Analyze.RunAnalysis

'start concrete design
  ret = SapModel.DesignConcrete.StartDesign()

'get preference item
  ret = SapModel.DesignConcrete.ACI318_14.GetPreference(2, Value)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cDCoACI318_14 Interface

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CSI API ETABS v1

cDCoACI318_14AddLanguageSpecificTextSet("LST4F2506FD_0") Method

Sets the value of a concrete design overwrite item.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDCoACI318_14
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters**Name**Type: SystemString

The name of an existing frame object or group, depending on the value of the ItemType

ItemType: SystemInt32

This is an integer between 1 and 13, inclusive, indicating the overwrite item considered.

1. Framing type
2. Live load reduction factor
3. Unbraced length ratio, Major
4. Unbraced length ratio, Minor
5. Effective length factor, K Major
6. Effective length factor, K Minor
7. Moment coefficient, Cm Major
8. Moment coefficient, Cm Minor
9. Nonsway moment factor, Db Major
10. Nonsway moment factor, Db Minor
11. Sway moment factor, Ds Major
12. Sway moment factor, Ds Minor
13. Consider Minimum Eccentricity

ValueType: SystemDouble

The value of the considered overwrite item.

1. Framing type
 - 0 = Program Default
 - 1 = Sway special
 - 2 = Sway Intermediate
 - 3 = Sway Ordinary
 - 4 = Non-sway
2. Live load reduction factor

Value ≥ 0 ; 0 means use program determined value

3. Unbraced length ratio, Major

Value ≥ 0 ; 0 means use program determined value

4. Unbraced length ratio, Minor

Value ≥ 0 ; 0 means use program determined value

5. Effective length factor, K Major

Value ≥ 0 ; 0 means use program determined value

6. Effective length factor, K Minor

Value ≥ 0 ; 0 means use program determined value

7. Moment coefficient, Cm Major

Value ≥ 0 ; 0 means use program determined value

8. Moment coefficient, Cm Minor

Introduction

Value ≥ 0 ; 0 means use program determined value

9. Nonsway moment factor, Db Major

Value ≥ 0 ; 0 means use program determined value

10. Nonsway moment factor, Db Minor

Value ≥ 0 ; 0 means use program determined value

11. Sway moment factor, Ds Major

Value ≥ 0 ; 0 means use program determined value

12. Sway moment factor, Ds Minor

Value ≥ 0 ; 0 means use program determined value

13. Consider Minimum Eccentricity

0 = No

Any other value = Yes

ItemType (Optional)

Type: $\hat{A}$ ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◇ Object = 0

◇ Group = 1

◇ SelectedObjects = 2

If this item is Objects, the overwrite is set for the frame object specified by the Name item.

If this item is Group, the overwrite is set for all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the overwrite is set for all selected frame objects and the Name item is ignored.

Return Value

Type: $\hat{A}$ Int32

Returns zero if the item is successfully set; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim Value As Double  
    Dim ProgDet As Boolean  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application
```

Return Value

Introduction

```
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'create new concrete frame section property
ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

'set frame section property
ret = SapModel.FrameObj.SetSection("8", "R1")

'set concrete design code
ret = SapModel.DesignConcrete.SetCode("ACI 318-14")

'run analysis
System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
ret = SapModel.Analyze.RunAnalysis

'start concrete design
ret = SapModel.DesignConcrete.StartDesign()

'set overwrite item
ret = SapModel.DesignConcrete.ACI318_14.SetOverwrite("8", 1, 1)

'get overwrite item
ret = SapModel.DesignConcrete.ACI318_14.GetOverwrite("8", 1, Value, ProgDet)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

[cDCoACI318_14 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDcOACI318_14AddLanguageSpecificTextSet("LSTD7087255_0?cp=56|nu=.") Method

Sets the value of a concrete design preference item.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDcOACI318_14
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: [SystemInt32](#)

This is an integer between 1 and 18, inclusive, indicating the preference item considered.

1. Number of interaction curves
2. Number of interaction points
3. Consider minimum eccentricity
4. Design for B/C Capacity Ratio?
5. Seismic design category

6. Design System Omega0
7. Design System Rho
8. Design System Sds
9. Consider ICC_ESR 2017
10. Phi tension controlled
11. Phi compression controlled tied
12. Phi compression controlled spiral
13. Phi shear and/or torsion
14. Phi shear seismic
15. Phi joint shear
16. Pattern live load factor
17. Utilization factor limit
18. Multi-response case design

Value

Type: SystemDouble

The value of the considered preference item.

1. Number of interaction curves

Value ≥ 4 and divisible by 4

2. Number of interaction points

Value ≥ 5 and odd

3. Consider minimum eccentricity

0 = No

Any other value = Yes

4. Design for B/C Capacity Ratio?

1 = No

2 = Yes

5. Seismic design category

. 1 = A

. 2 = B

. 3 = C

. 4 = D

. 5 = E

. 6 = F

6. Design System Omega0

Value > 0

7. Design System Rho

Value > 0

8. Design System Sds

Value ≥ 0

9. Consider ICC_ESR 2017

1 = No

2 = Yes

10. Phi tension controlled

Value > 0

11. Phi compression controlled tied

Value > 0

12. Phi compression controlled spiral

Value > 0

13. Phi shear and/or torsion

Value > 0

14. Phi shear seismic

Value > 0

15. Phi joint shear

Value > 0

16. Pattern live load factor

Value >= 0

17. Utilization factor limit

Value > 0

18. Multi-response case design

- 1 = Envelopes
- 2 = Step-by step
- 3 = Last step
- 4 = Envelopes - All
- 5 = Step-by step - All

Return Value

Type: Int32

Returns zero if the item is successfully set; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()
```

Introduction

```
'create SapModel object
  SapModel = EtabsObject.SapModel

'initialize model
  ret = SapModel.InitializeNewModel()

'create steel deck template model
  ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'create new concrete frame section property
  ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

'set frame section property
  ret = SapModel.FrameObj.SetSection("8", "R1")

'set concrete design code
  ret = SapModel.DesignConcrete.SetCode("ACI 318-14")

'run analysis
  System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
  ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
  ret = SapModel.Analyze.RunAnalysis

'start concrete design
  ret = SapModel.DesignConcrete.StartDesign()

'set preference item
  ret = SapModel.DesignConcrete.ACI318_14.SetPreference(2, 7)

'get preference item
  ret = SapModel.DesignConcrete.ACI318_14.GetPreference(2, Value)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

See Also

Reference

cDCoACI318_14 Interface

ETABSV1 Namespace

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CSI API ETABS v1

cDCoACI318_19 Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDCoACI318_19
```

```
Public Interface cDCoACI318_19
```

```
Dim instance As cDCoACI318_19
```

```
public interface class cDCoACI318_19
```

```
type cDCoACI318_19 = interface end
```

The cDCoACI318_19 type exposes the following members.

▲ Methods

	Name	Description
	GetOverwrite	Retrieves the value of a concrete design overwrite item.
	GetPreference	Retrieves the value of a concrete design preference item.
	SetOverwrite	Sets the value of a concrete design overwrite item.
	SetPreference	Sets the value of a concrete design preference item.

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDCoACI318_19 Methods

The [cDCoACI318_19](#) type exposes the following members.

Methods

	Name	Description
	GetOverwrite	Retrieves the value of a concrete design overwrite item.
	GetPreference	Retrieves the value of a concrete design preference item.
	SetOverwrite	Sets the value of a concrete design overwrite item.
	SetPreference	Sets the value of a concrete design preference item.

[Top](#)

 See Also

Reference

[cDCoACI318_19 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDCoACI318_19AddLanguageSpecificTextSet("LST390 Method

Retrieves the value of a concrete design overwrite item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDCoACI318_19
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```

Parameters

Name

Type:Â SystemString

The name of a frame object with a concrete frame design procedure.

Item

Type:Â SystemInt32

This is an integer between 1 and 13, inclusive, indicating the overwrite item considered.

1. Framing type
2. Live load reduction factor
3. Unbraced length ratio, Major
4. Unbraced length ratio, Minor
5. Effective length factor, K Major
6. Effective length factor, K Minor
7. Moment coefficient, Cm Major
8. Moment coefficient, Cm Minor
9. Nonsway moment factor, Db Major
10. Nonsway moment factor, Db Minor
11. Sway moment factor, Ds Major
12. Sway moment factor, Ds Minor
13. Consider Minimum Eccentricity

Value

Type:Â SystemDouble

The value of the considered overwrite item.

1. Framing type
 - 0 = Program Default
 - 1 = Sway special
 - 2 = Sway Intermediate
 - 3 = Sway Ordinary
 - 4 = Non-sway
2. Live load reduction factor

Value ≥ 0 ; 0 means use program determined value

3. Unbraced length ratio, Major

Value ≥ 0 ; 0 means use program determined value

4. Unbraced length ratio, Minor

Value ≥ 0 ; 0 means use program determined value

5. Effective length factor, K Major

Value ≥ 0 ; 0 means use program determined value

6. Effective length factor, K Minor

Value ≥ 0 ; 0 means use program determined value

7. Moment coefficient, Cm Major

Value ≥ 0 ; 0 means use program determined value

8. Moment coefficient, Cm Minor

Introduction

Value ≥ 0 ; 0 means use program determined value

9. Nonsway moment factor, Db Major

Value ≥ 0 ; 0 means use program determined value

10. Nonsway moment factor, Db Minor

Value ≥ 0 ; 0 means use program determined value

11. Sway moment factor, Ds Major

Value ≥ 0 ; 0 means use program determined value

12. Sway moment factor, Ds Minor

Value ≥ 0 ; 0 means use program determined value

13. Consider Minimum Eccentricity

0 = No

Any other value = Yes

ProgDet

Type: SystemBoolean

If this item is True, the specified value is program determined.

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim Value As Double  
    Dim ProgDet As Boolean  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)  
  
    'create new concrete frame section property  
    ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)
```

Return Value

Introduction

```
'set frame section property
    ret = SapModel.FrameObj.SetSection("8", "R1")

'set concrete design code
    ret = SapModel.DesignConcrete.SetCode("ACI 318-19")

'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis

'start concrete design
    ret = SapModel.DesignConcrete.StartDesign()

'get overwrite item
    ret = SapModel.DesignConcrete.ACI318_19.GetOverwrite("8", 1, Value, ProgDet)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

📌 See Also

Reference

[cDCoACI318_19 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDCoACI318_19AddLanguageSpecificTextSet("LST8ED13BB6_0?cpg=::|nu=.") Method

Retrieves the value of a concrete design preference item.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDCoACI318_19
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: [SystemInt32](#)

This is an integer between 1 and 18, inclusive, indicating the preference item considered.

1. Number of interaction curves
2. Number of interaction points
3. Consider minimum eccentricity
4. Design for B/C Capacity Ratio?
5. Seismic design category

6. Design System Omega0
7. Design System Rho
8. Design System Sds
9. Consider ICC_ESR 2017
10. Phi tension controlled
11. Phi compression controlled tied
12. Phi compression controlled spiral
13. Phi shear and/or torsion
14. Phi shear seismic
15. Phi joint shear
16. Pattern live load factor
17. Utilization factor limit
18. Multi-response case design

Value

Type: SystemDouble

The value of the considered preference item.

1. Number of interaction curves

Value ≥ 4 and divisible by 4

2. Number of interaction points

Value ≥ 5 and odd

3. Consider minimum eccentricity

0 = No

Any other value = Yes

4. Design for B/C Capacity Ratio?

1 = No

2 = Yes

5. Seismic design category

. 1 = A

. 2 = B

. 3 = C

. 4 = D

. 5 = E

. 6 = F

6. Design System Omega0

Value > 0

7. Design System Rho

Value > 0

8. Design System Sds

Value ≥ 0

9. Consider ICC_ESR 2017

1 = No

2 = Yes

10. Phi tension controlled

Value > 0

11. Phi compression controlled tied

Value > 0

12. Phi compression controlled spiral

Value > 0

13. Phi shear and/or torsion

Value > 0

14. Phi shear seismic

Value > 0

15. Phi joint shear

Value > 0

16. Pattern live load factor

Value >= 0

17. Utilization factor limit

Value > 0

18. Multi-response case design

- 1 = Envelopes
- 2 = Step-by step
- 3 = Last step
- 4 = Envelopes - All
- 5 = Step-by step - All

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()
```

Introduction

```
'create SapModel object
  SapModel = EtabsObject.SapModel

'initialize model
  ret = SapModel.InitializeNewModel()

'create steel deck template model
  ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'create new concrete frame section property
  ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

'set frame section property
  ret = SapModel.FrameObj.SetSection("8", "R1")

'set concrete design code
  ret = SapModel.DesignConcrete.SetCode("ACI 318-19")

'run analysis
  System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
  ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
  ret = SapModel.Analyze.RunAnalysis

'start concrete design
  ret = SapModel.DesignConcrete.StartDesign()

'get preference item
  ret = SapModel.DesignConcrete.ACI318_19.GetPreference(2, Value)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cDCoACI318_19 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDCoACI318_19AddLanguageSpecificTextSet("LST39A Method

Sets the value of a concrete design overwrite item.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDCoACI318_19
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters**Name**Type: SystemString

The name of an existing frame object or group, depending on the value of the ItemType

ItemType: SystemInt32

This is an integer between 1 and 13, inclusive, indicating the overwrite item considered.

1. Framing type
2. Live load reduction factor
3. Unbraced length ratio, Major
4. Unbraced length ratio, Minor
5. Effective length factor, K Major
6. Effective length factor, K Minor
7. Moment coefficient, Cm Major
8. Moment coefficient, Cm Minor
9. Nonsway moment factor, Db Major
10. Nonsway moment factor, Db Minor
11. Sway moment factor, Ds Major
12. Sway moment factor, Ds Minor
13. Consider Minimum Eccentricity

ValueType: SystemDouble

The value of the considered overwrite item.

1. Framing type
 - 0 = Program Default
 - 1 = Sway special
 - 2 = Sway Intermediate
 - 3 = Sway Ordinary
 - 4 = Non-sway
2. Live load reduction factor

Value ≥ 0 ; 0 means use program determined value

3. Unbraced length ratio, Major

Value ≥ 0 ; 0 means use program determined value

4. Unbraced length ratio, Minor

Value ≥ 0 ; 0 means use program determined value

5. Effective length factor, K Major

Value ≥ 0 ; 0 means use program determined value

6. Effective length factor, K Minor

Value ≥ 0 ; 0 means use program determined value

7. Moment coefficient, Cm Major

Value ≥ 0 ; 0 means use program determined value

8. Moment coefficient, Cm Minor

Introduction

Value ≥ 0 ; 0 means use program determined value

9. Nonsway moment factor, Db Major

Value ≥ 0 ; 0 means use program determined value

10. Nonsway moment factor, Db Minor

Value ≥ 0 ; 0 means use program determined value

11. Sway moment factor, Ds Major

Value ≥ 0 ; 0 means use program determined value

12. Sway moment factor, Ds Minor

Value ≥ 0 ; 0 means use program determined value

13. Consider Minimum Eccentricity

0 = No

Any other value = Yes

ItemType (Optional)

Type: $\hat{A}$ ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◇ Object = 0

◇ Group = 1

◇ SelectedObjects = 2

If this item is Objects, the overwrite is set for the frame object specified by the Name item.

If this item is Group, the overwrite is set for all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the overwrite is set for all selected frame objects and the Name item is ignored.

Return Value

Type: $\hat{A}$ Int32

Returns zero if the item is successfully set; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim Value As Double  
    Dim ProgDet As Boolean  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application
```

Return Value

Introduction

```
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'create new concrete frame section property
ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

'set frame section property
ret = SapModel.FrameObj.SetSection("8", "R1")

'set concrete design code
ret = SapModel.DesignConcrete.SetCode("ACI 318-19")

'run analysis
System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
ret = SapModel.Analyze.RunAnalysis

'start concrete design
ret = SapModel.DesignConcrete.StartDesign()

'set overwrite item
ret = SapModel.DesignConcrete.ACI318_19.SetOverwrite("8", 1, 1)

'get overwrite item
ret = SapModel.DesignConcrete.ACI318_19.GetOverwrite("8", 1, Value, ProgDet)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cDCoACI318_19 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDcOACI318_19AddLanguageSpecificTextSet("LSTB220A303_0") Method

Sets the value of a concrete design preference item.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDcOACI318_19
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: [System.Int32](#)

This is an integer between 1 and 18, inclusive, indicating the preference item considered.

1. Number of interaction curves
2. Number of interaction points
3. Consider minimum eccentricity
4. Design for B/C Capacity Ratio?
5. Seismic design category

6. Design System Omega0
7. Design System Rho
8. Design System Sds
9. Consider ICC_ESR 2017
10. Phi tension controlled
11. Phi compression controlled tied
12. Phi compression controlled spiral
13. Phi shear and/or torsion
14. Phi shear seismic
15. Phi joint shear
16. Pattern live load factor
17. Utilization factor limit
18. Multi-response case design

Value

Type: SystemDouble

The value of the considered preference item.

1. Number of interaction curves

Value ≥ 4 and divisible by 4

2. Number of interaction points

Value ≥ 5 and odd

3. Consider minimum eccentricity

0 = No

Any other value = Yes

4. Design for B/C Capacity Ratio?

1 = No

2 = Yes

5. Seismic design category

. 1 = A

. 2 = B

. 3 = C

. 4 = D

. 5 = E

. 6 = F

6. Design System Omega0

Value > 0

7. Design System Rho

Value > 0

8. Design System Sds

Value ≥ 0

9. Consider ICC_ESR 2017

1 = No

2 = Yes

10. Phi tension controlled

Value > 0

11. Phi compression controlled tied

Value > 0

12. Phi compression controlled spiral

Value > 0

13. Phi shear and/or torsion

Value > 0

14. Phi shear seismic

Value > 0

15. Phi joint shear

Value > 0

16. Pattern live load factor

Value >= 0

17. Utilization factor limit

Value > 0

18. Multi-response case design

- 1 = Envelopes
- 2 = Step-by step
- 3 = Last step
- 4 = Envelopes - All
- 5 = Step-by step - All

Return Value

Type: Int32

Returns zero if the item is successfully set; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()
```

Introduction

```
'create SapModel object
  SapModel = EtabsObject.SapModel

'initialize model
  ret = SapModel.InitializeNewModel()

'create steel deck template model
  ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'create new concrete frame section property
  ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

'set frame section property
  ret = SapModel.FrameObj.SetSection("8", "R1")

'set concrete design code
  ret = SapModel.DesignConcrete.SetCode("ACI 318-19")

'run analysis
  System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
  ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
  ret = SapModel.Analyze.RunAnalysis

'start concrete design
  ret = SapModel.DesignConcrete.StartDesign()

'set preference item
  ret = SapModel.DesignConcrete.ACI318_19.SetPreference(2, 7)

'get preference item
  ret = SapModel.DesignConcrete.ACI318_19.GetPreference(2, Value)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

🔗 See Also

Reference

[cDCoACI318_19 Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cDCoAS_3600_09 Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDCoAS_3600_09
```

```
Public Interface cDCoAS_3600_09
```

```
Dim instance As cDCoAS_3600_09
```

```
public interface class cDCoAS_3600_09
```

```
type cDCoAS_3600_09 = interface end
```

The cDCoAS_3600_09 type exposes the following members.

▲ Methods

▲	Name	Description
---	------	-------------

➤	GetOverwrite	
---	------------------------------	--

➤	GetPreference	
---	-------------------------------	--

➤	SetOverwrite	
---	------------------------------	--

➤	SetPreference	
---	-------------------------------	--

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▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDCoAS_3600_09 Methods

The cDCoAS_3600_09 type exposes the following members.

Methods

	Name	Description
	<u>GetOverwrite</u>	
	<u>GetPreference</u>	
	<u>SetOverwrite</u>	
	<u>SetPreference</u>	

Top

See Also

Reference

cDCoAS_3600_09 Interface

ETABSV1 Namespace

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CSI API ETABS v1

cDCoAS_3600_09AddLanguageSpecificTextSet("LSTB Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDCoAS_3600_09
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```

Parameters

Name

Type: SystemString

cDCoAS_3600_09280AddLanguageSpecificTextSet("LSTBB7D6BC6_0980p=::|nu

Item

Type:Â [SystemInt32](#)

Value

Type:Â [SystemDouble](#)

ProgDet

Type:Â [SystemBoolean](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cDCoAS_3600_09 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDCoAS_3600_09AddLanguageSpecificTextSet("LSTA Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDCoAS_3600_09
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: [SystemInt32](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDCoAS_3600_09 Interface

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CSI API ETABS v1

cDCoAS_3600_09AddLanguageSpecificTextSet("LST81B97A21_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDCoAS_3600_09
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

Item

Type: SystemInt32

Value

Type: SystemDouble

ItemType (Optional)

Type: ETABSv1eItemType

Return Value

Type: Int32

▲ See Also

Reference

cDCoAS_3600_09 Interface

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CSI API ETABS v1

cDCoAS_3600_09AddLanguageSpecificTextSet("LST7A Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDCoAS_3600_09
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: SystemInt32

Value

Type: SystemDouble

Return Value

Type: Int32

▲ See Also

Reference

cDCoAS_3600_09 Interface

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CSI API ETABS v1

cDCoAS_3600_2018 Interface

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
public interface cDCoAS_3600_2018
```

```
Public Interface cDCoAS_3600_2018
```

```
Dim instance As cDCoAS_3600_2018
```

```
public interface class cDCoAS_3600_2018
```

```
type cDCoAS_3600_2018 = interface end
```

The cDCoAS_3600_2018 type exposes the following members.

▲ Methods

▲	Name	Description
	<u>GetOverwrite</u>	Retrieves the value of a concrete design overwrite item.
	<u>GetPreference</u>	Retrieves the value of a concrete design preference item.
	<u>SetOverwrite</u>	Sets the value of a concrete design overwrite item.
	<u>SetPreference</u>	Sets the value of a concrete design preference item.

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▲ See Also

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CSI API ETABS v1

cDCoAS_3600_2018 Methods

The cDCoAS_3600_2018 type exposes the following members.

Methods

	Name	Description
	<u>GetOverwrite</u>	Retrieves the value of a concrete design overwrite item.
	<u>GetPreference</u>	Retrieves the value of a concrete design preference item.
	<u>SetOverwrite</u>	Sets the value of a concrete design overwrite item.
	<u>SetPreference</u>	Sets the value of a concrete design preference item.

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See Also

Reference

cDCoAS_3600_2018 Interface

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CSI API ETABS v1

cDCoAS_3600_2018AddLanguageSpecificTextSet("LST22447370_0") Method

Retrieves the value of a concrete design overwrite item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDCoAS_3600_2018
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```

Parameters

Name

Type: SystemString

The name of a frame object with a concrete frame design procedure.

Item

Type: SystemInt32

This is an integer between 1 and 12, inclusive, indicating the overwrite item considered.

1. Framing type
2. Live load reduction factor
3. Unbraced length ratio, Major
4. Unbraced length ratio, Minor
5. Effective length factor, K Major
6. Effective length factor, K Minor
7. Moment coefficient, Cm Major
8. Moment coefficient, Cm Minor
9. Nonsway moment factor, Db Major
10. Nonsway moment factor, Db Minor
11. Sway moment factor, Ds Major
12. Sway moment factor, Ds Minor

Value

Type: SystemDouble

The value of the considered overwrite item.

1. Framing type
 - 0 = Program Default
 - 1 = Sway Intermediate
 - 2 = Sway Ordinary
 - 3 = Non-sway
2. Live load reduction factor

Value ≥ 0 ; 0 means use program determined value

3. Unbraced length ratio, Major

Value ≥ 0 ; 0 means use program determined value

4. Unbraced length ratio, Minor

Value ≥ 0 ; 0 means use program determined value

5. Effective length factor, K Major

Value ≥ 0 ; 0 means use program determined value

6. Effective length factor, K Minor

Value ≥ 0 ; 0 means use program determined value

7. Moment coefficient, Cm Major

Value ≥ 0 ; 0 means use program determined value

8. Moment coefficient, Cm Minor

Value ≥ 0 ; 0 means use program determined value

9. Nonsway moment factor, Db Major

Introduction

Value ≥ 0 ; 0 means use program determined value

10. Nonsway moment factor, Db Minor

Value ≥ 0 ; 0 means use program determined value

11. Sway moment factor, Ds Major

Value ≥ 0 ; 0 means use program determined value

12. Sway moment factor, Ds Minor

Value ≥ 0 ; 0 means use program determined value

ProgDet

Type: SystemBoolean

If this item is True, the specified value is program determined.

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ See Also

Reference

cDCoAS_3600_2018 Interface

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CSI API ETABS v1

cDCoAS_3600_2018AddLanguageSpecificTextSet("LST541EB62C_0") Method

Retrieves the value of a concrete design preference item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDCoAS_3600_2018
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: SystemInt32

This is an integer between 1 and 11, inclusive, indicating the preference item considered.

1. Number of interaction curves
2. Number of interaction points
3. Consider minimum eccentricity
4. Phi tension controlled
5. Phi compression controlled tied

6. Phi shear and/or torsion
7. Phi shear seismic
8. Phi joint shear
9. Pattern live load factor
10. Utilization factor limit
11. Multi-response case design

Value

Type: SystemDouble

The value of the considered preference item.

1. Number of interaction curves

Value ≥ 4 and divisible by 4

2. Number of interaction points

Value ≥ 5 and odd

3. Consider minimum eccentricity

0 = No

Any other value = Yes

4. Phi tension controlled

Value > 0

5. Phi compression controlled tied

Value > 0

6. Phi shear and/or torsion

Value > 0

7. Phi shear seismic

Value > 0

8. Phi joint shear

Value > 0

9. Pattern live load factor

Value ≥ 0

10. Utilization factor limit

Value > 0

11. Multi-response case design

- 1 = Envelopes
- 2 = Step-by step
- 3 = Last step
- 4 = Envelopes - All
- 5 = Step-by step - All

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ See Also

Reference

cDCoAS_3600_2018 Interface

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CSI API ETABS v1

cDCoAS_3600_2018AddLanguageSpecificTextSet("LST14181EAD_0") Method

Sets the value of a concrete design overwrite item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDCoAS_3600_2018
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```


Parameters**Name**Type: SystemString

The name of an existing frame object or group, depending on the value of the ItemType

ItemType: SystemInt32

This is an integer between 1 and 12, inclusive, indicating the overwrite item considered.

1. Framing type
2. Live load reduction factor
3. Unbraced length ratio, Major
4. Unbraced length ratio, Minor
5. Effective length factor, K Major
6. Effective length factor, K Minor
7. Moment coefficient, Cm Major
8. Moment coefficient, Cm Minor
9. Nonsway moment factor, Db Major
10. Nonsway moment factor, Db Minor
11. Sway moment factor, Ds Major
12. Sway moment factor, Ds Minor

ValueType: SystemDouble

The value of the considered overwrite item.

1. Framing type
 - 0 = Program Default
 - 1 = Sway Intermediate
 - 2 = Sway Ordinary
 - 3 = Non-sway
2. Live load reduction factor

Value ≥ 0 ; 0 means use program determined value

3. Unbraced length ratio, Major

Value ≥ 0 ; 0 means use program determined value

4. Unbraced length ratio, Minor

Value ≥ 0 ; 0 means use program determined value

5. Effective length factor, K Major

Value ≥ 0 ; 0 means use program determined value

6. Effective length factor, K Minor

Value ≥ 0 ; 0 means use program determined value

7. Moment coefficient, Cm Major

Value ≥ 0 ; 0 means use program determined value

8. Moment coefficient, Cm Minor

Value ≥ 0 ; 0 means use program determined value

9. Nonsway moment factor, Db Major

Value ≥ 0 ; 0 means use program determined value

10. Nonsway moment factor, Db Minor

Value ≥ 0 ; 0 means use program determined value

11. Sway moment factor, Ds Major

Value ≥ 0 ; 0 means use program determined value

12. Sway moment factor, Ds Minor

Value ≥ 0 ; 0 means use program determined value

ItemType (Optional)

Type: $\hat{A}$ ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the overwrite is set for the frame object specified by the Name item.

If this item is Group, the overwrite is set for all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the overwrite is set for all selected frame objects and the Name item is ignored.

Return Value

Type: $\hat{A}$ Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ See Also

Reference

cDCoAS_3600_2018 Interface

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CSI API ETABS v1

cDCoAS_3600_2018AddLanguageSpecificTextSet("LST697E2F00_0") Method

Sets the value of a concrete design preference item.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDCoAS_3600_2018
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: [SystemInt32](#)

This is an integer between 1 and 11, inclusive, indicating the preference item considered.

1. Number of interaction curves
2. Number of interaction points
3. Consider minimum eccentricity
4. Phi tension controlled
5. Phi compression controlled tied

6. Phi shear and/or torsion
7. Phi shear seismic
8. Phi joint shear
9. Pattern live load factor
10. Utilization factor limit
11. Multi-response case design

Value

Type: SystemDouble

The value of the considered preference item.

1. Number of interaction curves

Value ≥ 4 and divisible by 4

2. Number of interaction points

Value ≥ 5 and odd

3. Consider minimum eccentricity

0 = No

Any other value = Yes

4. Phi tension controlled

Value > 0

5. Phi compression controlled tied

Value > 0

6. Phi shear and/or torsion

Value > 0

7. Phi shear seismic

Value > 0

8. Phi joint shear

Value > 0

9. Pattern live load factor

Value ≥ 0

10. Utilization factor limit

Value > 0

11. Multi-response case design

- 1 = Envelopes
- 2 = Step-by step
- 3 = Last step
- 4 = Envelopes - All
- 5 = Step-by step - All

Return Value

Type: Int32

Returns zero if the item is successfully set; otherwise it returns a nonzero value.

▲ See Also

Reference

cDCoAS_3600_2018 Interface

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CSI API ETABS v1

cDCoBS8110_97 Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDCoBS8110_97
```

```
Public Interface cDCoBS8110_97
```

```
Dim instance As cDCoBS8110_97
```

```
public interface class cDCoBS8110_97
```

```
type cDCoBS8110_97 = interface end
```

The cDCoBS8110_97 type exposes the following members.

▲ Methods

▲	Name	Description
---	------	-------------

➦	GetOverwrite	
---	------------------------------	--

➦	GetPreference	
---	-------------------------------	--

➦	SetOverwrite	
---	------------------------------	--

➦	SetPreference	
---	-------------------------------	--

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▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDCoBS8110_97 Methods

The [cDCoBS8110_97](#) type exposes the following members.

Methods

	Name	Description
	GetOverwrite	
	GetPreference	
	SetOverwrite	
	SetPreference	

[Top](#)

See Also

Reference

[cDCoBS8110_97 Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cDcBS8110_97AddLanguageSpecificTextSet("LSTD304D6D7_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDcBS8110_97
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```

Parameters

Name

Type: SystemString

cDcBS8110_97AddLanguageSpecificTextSet("LSTD304D6D7_0")

Item

Type:Â [SystemInt32](#)

Value

Type:Â [SystemDouble](#)

ProgDet

Type:Â [SystemBoolean](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cDCoBS8110_97 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDcBS8110_97AddLanguageSpecificTextSet("LSTBD Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDcBS8110_97
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: [SystemInt32](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDCoBS8110_97 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDCoBS8110_97AddLanguageSpecificTextSet("LST7397DDFE_01008") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDCoBS8110_97
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

Item

Type: SystemInt32

Value

Type: SystemDouble

ItemType (Optional)

Type: ETABSv1eItemType

Return Value

Type: Int32

▲ See Also

Reference

cDCoBS8110_97 Interface

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CSI API ETABS v1

cDcBS8110_97AddLanguageSpecificTextSet("LST5E74A0ED_01010") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDcBS8110_97
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: [SystemInt32](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDCoBS8110_97 Interface

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CSI API ETABS v1

cDCoChinese_2010 Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDCoChinese_2010
```

```
Public Interface cDCoChinese_2010
```

```
Dim instance As cDCoChinese_2010
```

```
public interface class cDCoChinese_2010
```

```
type cDCoChinese_2010 = interface end
```

The cDCoChinese_2010 type exposes the following members.

▲ Methods

▲	Name	Description
---	------	-------------

⇒	GetOverwrite	
---	------------------------------	--

⇒	GetPreference	
---	-------------------------------	--

⇒	SetOverwrite	
---	------------------------------	--

⇒	SetPreference	
---	-------------------------------	--

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDCoChinese_2010 Methods

The cDCoChinese_2010 type exposes the following members.

Methods

	Name	Description
⇒	<u>GetOverwrite</u>	
⇒	<u>GetPreference</u>	
⇒	<u>SetOverwrite</u>	
⇒	<u>SetPreference</u>	

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See Also

Reference

cDCoChinese_2010 Interface

ETABSV1 Namespace

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Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

C#

C++

F#

Copy

```
abstract GetOverwrite :
  Name : string *
  Item : int *
  Value : float byref *
  ProgDet : bool byref -> int
```

Name

cDcCoChinese_2010span id="LST6BE00B02_0"AddLanguageSpecificTextSet("LST6BE00B02_004p::nu

Item

Type:Â [SystemInt32](#)

Value

Type:Â [SystemDouble](#)

ProgDet

Type:Â [SystemBoolean](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cDCoChinese_2010 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDCoChinese_2010AddLanguageSpecificTextSet("LST Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDCoChinese_2010
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: [SystemInt32](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDcoChinese_2010 Interface

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CSI API ETABS v1

cDCoChinese_2010AddLanguageSpecificTextSet("LST Method

Namespace: Â ETABSv1

Assembly: Â ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDCoChinese_2010
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

Item

Type: SystemInt32

Value

Type: SystemDouble

ItemType (Optional)

Type: ETABSv1eItemType

Return Value

Type: Int32

▲ See Also

Reference

cDCoChinese_2010 Interface

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CSI API ETABS v1

cDCoChinese_2010AddLanguageSpecificTextSet("LST Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDCoChinese_2010
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: [SystemInt32](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDcoChinese_2010 Interface

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CSI API ETABS v1

cDCoEurocode_2_2004 Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDCoEurocode_2_2004
```

```
Public Interface cDCoEurocode_2_2004
```

```
Dim instance As cDCoEurocode_2_2004
```

```
public interface class cDCoEurocode_2_2004
```

```
type cDCoEurocode_2_2004 = interface end
```

The cDCoEurocode_2_2004 type exposes the following members.

▲ Methods

	Name	Description
	GetOverwrite	Retrieves the value of a concrete design overwrite item.
	GetPreference	Retrieves the value of a concrete design preference item.
	SetOverwrite	Sets the value of a concrete design overwrite item.
	SetPreference	Sets the value of a concrete design preference item.

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▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDCoEurocode_2_2004 Methods

The cDCoEurocode_2_2004 type exposes the following members.

Methods

	Name	Description
	<u>GetOverwrite</u>	Retrieves the value of a concrete design overwrite item.
	<u>GetPreference</u>	Retrieves the value of a concrete design preference item.
	<u>SetOverwrite</u>	Sets the value of a concrete design overwrite item.
	<u>SetPreference</u>	Sets the value of a concrete design preference item.

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See Also

Reference

cDCoEurocode_2_2004 Interface

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CSI API ETABS v1

cDCoEurocode_2_2004AddLanguageSpecificTextSet("Method

Retrieves the value of a concrete design overwrite item.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDCoEurocode_2_2004
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```

Parameters

Name

Type:Â SystemString

The name of a frame object with a concrete frame design procedure.

Item

Type:Â SystemInt32

This is an integer between 1 and 27, inclusive, indicating the overwrite item considered.

1. Framing type
2. Live load reduction factor
3. Unbraced length ratio, Major
4. Unbraced length ratio, Minor
5. Effective length factor, Beta Major (columns only)
6. Effective length factor, Beat Minor (columns only)
7. Moment coefficient, Cm Major (not used)
8. Moment coefficient, Cm Minor (not used)
9. Nonsway moment factor, Db Major (not used)
10. Nonsway moment factor, Db Minor (not used)
11. Sway moment factor, Ds Major (not used)
12. Sway moment factor, Ds Minor (not used)
13. Correction factor depending on axial load in Nominal Curvature method, Kr Major (columns only)
14. Factor accounting for creep in Nominal Curvature method, KPhi Major (columns only)
16. Correction factor depending on axial load in Nominal Curvature method, Kr Minor (columns only)
17. Factor accounting for creep in Nominal Curvature method, KPhi Minor (columns only)
18. Coefficient depending on the distribution of first-order moment in both Nominal Stiffness and Nominal Curvature methods, c Major (columns only)
19. Coefficient depending on the distribution of first-order moment in both Nominal Stiffness and Nominal Curvature methods, c Minor (columns only)
20. Factor for contribution of reinforcement, Ks Major (columns only)
21. Ignore the benefit of Pu for beam design? (beams only)
22. Factor for contribution of reinforcement, Ks Minor (columns only)
23. Factor for effects of cracking, creep etc, Kc Major (columns only)
24. Factor for effects of cracking, creep etc, Kc Minor (columns only)
25. Effective creep coefficient, Phief (columns only)
26. Consider torsion (beams only)
27. Consider minimum eccentricity (columns only)

Value

Type:Â SystemDouble

The value of the considered overwrite item.

1. Framing type
 - 0 = Program Determined
 - 1 = DC High
 - 2 = DC Medium
 - 3 = DC Low

- 4 = Secondary
- 2. Live load reduction factor
 - Value ≥ 0 ; 0 means use program determined value
- 3. Unbraced length ratio, Major
 - Value ≥ 0 ; 0 means use program determined value
- 4. Unbraced length ratio, Minor
 - Value ≥ 0 ; 0 means use program determined value
- 5. Effective length factor, Beta Major
 - Value ≥ 0 ; 0 means use program determined value
- 6. Effective length factor, Beta Minor
 - Value ≥ 0 ; 0 means use program determined value
- 7. Moment coefficient, Cm Major
 - Value ≥ 0 ; 0 means use program determined value
- 8. Moment coefficient, Cm Minor
 - Value ≥ 0 ; 0 means use program determined value
- 9. Nonsway moment factor, Db Major
 - Value ≥ 0 ; 0 means use program determined value
- 10. Nonsway moment factor, Db Minor
 - Value ≥ 0 ; 0 means use program determined value
- 11. Sway moment factor, Ds Major
 - Value ≥ 0 ; 0 means use program determined value
- 12. Sway moment factor, Ds Minor
 - Value ≥ 0 ; 0 means use program determined value
- 13. Correction factor depending on axial load in Nominal Curvature method, Kr Major
 - Value ≥ 0 ; 0 means use program determined value
- 14. Factor accounting for creep in Nominal Curvature method, KPhi Major
 - Value ≥ 0 ; 0 means use program determined value
- 16. Correction factor depending on axial load in Nominal Curvature method, Kr Minor
 - Value ≥ 0 ; 0 means use program determined value
- 17. Factor accounting for creep in Nominal Curvature method, KPhi Minor
 - Value ≥ 0 ; 0 means use program determined value
- 18. Coefficient depending on the distribution of first-order moment in both Nominal Stiffness and Nominal Curvature methods, c Major
 - Value ≥ 0 ; 0 means use program determined value

Introduction

Value ≥ 0 ; 0 means use program determined value

19. Coefficient depending on the distribution of first-order moment in both Nominal Stiffness and Nominal Curvature methods, c Minor

Value ≥ 0 ; 0 means use program determined value

20. Factor for contribution of reinforcement, Ks Major

Value ≥ 0 ; 0 means use program determined value

21. Ignore the benefit of Pu for beam design?

· 0 = No

· Any other value = Yes

22. Factor for contribution of reinforcement, Ks Minor

Value ≥ 0 ; 0 means use program determined value

23. Factor for effects of cracking, creep etc, Kc Major

Value ≥ 0 ; 0 means use program determined value

24. Factor for effects of cracking, creep etc, Kc Minor

Value ≥ 0 ; 0 means use program determined value

25. Effective creep coefficient, Phief

Value ≥ 0 ; 0 means use program determined value

26. Consider torsion

· 0 = No

· Any other value = Yes

27. Consider minimum eccentricity

· 0 = No

· Any other value = Yes

ProgDet

Type: System.Boolean

If this item is True, the specified value is program determined.

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim Value As Double  
    Dim ProgDet As Boolean  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application
```

Return Value

Introduction

```
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'create new concrete frame section property
ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

'set frame section property
ret = SapModel.FrameObj.SetSection("8", "R1")

'set concrete design code
ret = SapModel.DesignConcrete.SetCode("Eurocode 2-2004")

'run analysis
System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
ret = SapModel.Analyze.RunAnalysis

'start concrete design
ret = SapModel.DesignConcrete.StartDesign()

'get overwrite item
ret = SapModel.DesignConcrete.Eurocode_2_2004.GetOverwrite("8", 1, Value, ProgDet)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

📌 See Also

Reference

[cDCEurocode 2 2004 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDCoEurocode_2_2004AddLanguageSpecificTextSet("Method

Retrieves the value of a concrete design preference item.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDCoEurocode_2_2004
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: [SystemInt32](#)

This is an integer between 1 and 31, inclusive, indicating the preference item considered.

1. Country
2. Combos equation
3. Second order method
4. Number of interaction curves
5. Number of interaction points

6. Consider minimum eccentricity
7. Theta0
8. Gamma steel
9. Gamma concrete
10. AlphaCC
11. AlphaCT
12. AlphaLCC
13. AlphaLCT
14. Pattern live load factor
15. Utilization factor limit
16. Multi-response case design
17. Reliability class
18. Gamma concrete modulus
19. Consider torsion
20. Design for B/C capacity ratio
21. Ignore Beneficial Pu for Beam Design?
22. User defined allowable PT stresses
23. Concrete Strength Ratio at Transfer $f_{ck}(t) / f_{ck}$
24. Transfer: Top Fiber Tensile Stress / $f_{ctm}(t)$
25. Transfer: Bottom Fiber Tensile Stress / $f_{ctm}(t)$
26. Transfer: Extreme Fiber Compressive Stress / $f_{ck}(t)$
27. Final: Top Fiber Tensile Stress / f_{ctm}
28. Final: Bottom Fiber Tensile Stress / f_{ctm}
29. Final: Extreme Fiber Compressive Stress / f_{ck}
30. Sustained: Extreme Fiber Compressive Stress / f_{ck}
31. Sustained: Fraction of Live Load Considered

Value

Type: SystemDouble

The value of the considered preference item.

1. Country
 - . 1 = CEN Default
 - . 2 = Denmark
 - . 3 = Finland
 - . 4 = Germany
 - . 5 = Ireland
 - . 6 = Norway
 - . 7 = Poland
 - . 8 = Portugal
 - . 9 = Singapore
 - . 10 = Slovenia
 - . 11 = Swedent
 - . 12 = United Kingdom
2. Combos equation
 - . 1 = Eq. 6.10
 - . 2 = Max of Eqs. 6.10a and 6.10b
3. Second order method
 - . 1 = Nominal stiffness
 - . 2 = Nominal curvature
 - . 3 = None
4. Number of interaction curves

Value ≥ 4 and divisible by 4
5. Number of interaction points

Value ≥ 5 and odd
6. Consider minimum eccentricity

0 = No

Any other value = Yes
7. Theta0

Value > 0
8. Gamma steel

Value > 0
9. Gamma concrete

Value > 0
10. AlphaCC

Value > 0
11. AlphaCT

Value > 0
12. AlphaLCC

Value > 0
13. AlphaLCT

Value > 0
14. Pattern live load factor

Value ≥ 0
15. Utilization factor limit

Value > 0
16. Multi-response case design
· 1 = Envelopes
· 2 = Step-by step
· 3 = Last step
· 4 = Envelopes - All
· 5 = Step-by step - All

17. Reliability class
· 1 = Class 1
· 2 = Class 2
· 3 = Class 3

18. Gamma concrete modulus

Value > 0
19. Consider torsion
· 0 = No

Introduction

- Any other value = Yes
- 20. Design for B/C capacity ratio
 - 0 = No
 - Any other value = Yes
- 21. Ignore the benefit of Pu for beam design?
 - 0 = No
 - Any other value = Yes
- 22. User defined allowable PT stresses
 - 0 = No
 - Any other value = Yes
- 23. Concrete Strength Ratio at Transfer $f_{ck}(t) / f_{ck}$

Value > 0

- 24. Transfer: Top Fiber Tensile Stress / $f_{ctm}(t)$

Value > 0

- 25. Transfer: Bottom Fiber Tensile Stress / $f_{ctm}(t)$

Value > 0

- 26. Transfer: Extreme Fiber Compressive Stress / $f_{ck}(t)$

Value > 0

- 27. Final: Top Fiber Tensile Stress / f_{ctm}

Value > 0

- 28. Final: Bottom Fiber Tensile Stress / f_{ctm}

Value > 0

- 29. Final: Extreme Fiber Compressive Stress / f_{ck}

Value > 0

- 30. Sustained: Extreme Fiber Compressive Stress / f_{ck}

Value > 0

- 31. Sustained: Fraction of Live Load Considered

Value > 0

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim Value As Double
```

Introduction

```
'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
    ret = EtabsObject.ApplicationStart()

'create SapModel object
    SapModel = EtabsObject.SapModel

'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'create new concrete frame section property
    ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

'set frame section property
    ret = SapModel.FrameObj.SetSection("8", "R1")

'set concrete design code
    ret = SapModel.DesignConcrete.SetCode("Eurocode 2-2004")

'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis

'start concrete design
    ret = SapModel.DesignConcrete.StartDesign()

'get preference item
    ret = SapModel.DesignConcrete.Eurocode_2_2004.GetPreference(3, Value)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

📌 See Also

Reference

[cDCEurocode 2 2004 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDCoEurocode_2_2004AddLanguageSpecificTextSet("Method

Sets the value of a concrete design overwrite item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDCoEurocode_2_2004
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

cDCoEurocode_2_2004AddLanguageSpecificTextSet("LSTBA958138_0?cpp=::

Parameters**Name**Type: SystemString

The name of an existing frame object or group, depending on the value of the ItemType

ItemType: SystemInt32

This is an integer between 1 and 27, inclusive, indicating the overwrite item considered.

1. Framing type
2. Live load reduction factor
3. Unbraced length ratio, Major
4. Unbraced length ratio, Minor
5. Effective length factor, Beta Major (columns only)
6. Effective length factor, Beta Minor (columns only)
7. Moment coefficient, Cm Major (not used)
8. Moment coefficient, Cm Minor (not used)
9. Nonsway moment factor, Db Major (not used)
10. Nonsway moment factor, Db Minor (not used)
11. Sway moment factor, Ds Major (not used)
12. Sway moment factor, Ds Minor (not used)
13. Correction factor depending on axial load in Nominal Curvature method, Kr Major (columns only)
14. Factor accounting for creep in Nominal Curvature method, KPhi Major (columns only)
16. Correction factor depending on axial load in Nominal Curvature method, Kr Minor (columns only)
17. Factor accounting for creep in Nominal Curvature method, KPhi Minor (columns only)
18. Coefficient depending on the distribution of first-order moment in both Nominal Stiffness and Nominal Curvature methods, c Major (columns only)
19. Coefficient depending on the distribution of first-order moment in both Nominal Stiffness and Nominal Curvature methods, c Minor (columns only)
20. Factor for contribution of reinforcement, Ks Major (columns only)
21. Ignore the benefit of Pu for beam design? (beams only)
22. Factor for contribution of reinforcement, Ks Minor (columns only)
23. Factor for effects of cracking, creep etc, Kc Major (columns only)
24. Factor for effects of cracking, creep etc, Kc Minor (columns only)
25. Effective creep coefficient, Phief (columns only)
26. Consider torsion (beams only)
27. Consider minimum eccentricity (columns only)

ValueType: SystemDouble

The value of the considered overwrite item.

1. Framing type
 - 0 = Program Determined
 - 1 = DC High
 - 2 = DC Medium

Introduction

- 3 = DC Low
- 4 = Secondary
- 2. Live load reduction factor

Value ≥ 0 ; 0 means use program determined value
- 3. Unbraced length ratio, Major

Value ≥ 0 ; 0 means use program determined value
- 4. Unbraced length ratio, Minor

Value ≥ 0 ; 0 means use program determined value
- 5. Effective length factor, Beta Major

Value ≥ 0 ; 0 means use program determined value
- 6. Effective length factor, Beta Minor

Value ≥ 0 ; 0 means use program determined value
- 7. Moment coefficient, Cm Major

Value ≥ 0 ; 0 means use program determined value
- 8. Moment coefficient, Cm Minor

Value ≥ 0 ; 0 means use program determined value
- 9. Nonsway moment factor, Db Major

Value ≥ 0 ; 0 means use program determined value
- 10. Nonsway moment factor, Db Minor

Value ≥ 0 ; 0 means use program determined value
- 11. Sway moment factor, Ds Major

Value ≥ 0 ; 0 means use program determined value
- 12. Sway moment factor, Ds Minor

Value ≥ 0 ; 0 means use program determined value
- 13. Correction factor depending on axial load in Nominal Curvature method, Kr Major

Value ≥ 0 ; 0 means use program determined value
- 14. Factor accounting for creep in Nominal Curvature method, KPhi Major

Value ≥ 0 ; 0 means use program determined value
- 15. Correction factor depending on axial load in Nominal Curvature method, Kr Minor

Value ≥ 0 ; 0 means use program determined value
- 16. Factor accounting for creep in Nominal Curvature method, KPhi Minor

Value ≥ 0 ; 0 means use program determined value
- 17. Coefficient depending on the distribution of first-order moment in both Nominal Stiffness and Nominal Curvature methods, c Major

Introduction

Value ≥ 0 ; 0 means use program determined value

18. Coefficient depending on the distribution of first-order moment in both Nominal Stiffness and Nominal Curvature methods, c Minor

Value ≥ 0 ; 0 means use program determined value

19. Factor for contribution of reinforcement, Ks Major

Value ≥ 0 ; 0 means use program determined value

20. Ignore the benefit of Pu for beam design?

· 0 = No

· Any other value = Yes

21. Factor for contribution of reinforcement, Ks Minor

Value ≥ 0 ; 0 means use program determined value

22. Factor for effects of cracking, creep etc, Kc Major

Value ≥ 0 ; 0 means use program determined value

23. Factor for effects of cracking, creep etc, Kc Minor

Value ≥ 0 ; 0 means use program determined value

24. Effective creep coefficient, Phief

Value ≥ 0 ; 0 means use program determined value

25. Consider torsion

· 0 = No

· Any other value = Yes

26. Consider minimum eccentricity

· 0 = No

· Any other value = Yes

ItemType (Optional)

Type: $\hat{A}$ ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◇ Object = 0

◇ Group = 1

◇ SelectedObjects = 2

If this item is Objects, the overwrite is set for the frame object specified by the Name item.

If this item is Group, the overwrite is set for all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the overwrite is set for all selected frame objects and the Name item is ignored.

Return Value

Type: $\hat{A}$ Int32

Returns zero if the item is successfully set; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Return Value

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double
    Dim ProgDet As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'create new concrete frame section property
    ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

    'set frame section property
    ret = SapModel.FrameObj.SetSection("8", "R1")

    'set concrete design code
    ret = SapModel.DesignConcrete.SetCode("Eurocode 2-2004")

    'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis

    'start concrete design
    ret = SapModel.DesignConcrete.StartDesign()

    'set overwrite item
    ret = SapModel.DesignConcrete.Eurocode_2_2004.SetOverwrite("8", 1, 1)

    'get overwrite item
    ret = SapModel.DesignConcrete.Eurocode_2_2004.GetOverwrite("8", 1, Value, ProgDet)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub

```

▲ See Also

Reference

cDCoEurocode 2 2004 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDCoEurocode_2_2004AddLanguageSpecificTextSet("Method

Sets the value of a concrete design preference item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDCoEurocode_2_2004
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: SystemInt32

This is an integer between 1 and 31, inclusive, indicating the preference item considered.

1. Country
2. Combos equation
3. Second order method
4. Number of interaction curves
5. Number of interaction points

6. Consider minimum eccentricity
7. Theta0
8. Gamma steel
9. Gamma concrete
10. AlphaCC
11. AlphaCT
12. AlphaLCC
13. AlphaLCT
14. Pattern live load factor
15. Utilization factor limit
16. Multi-response case design
17. Reliability class
18. Gamma concrete modulus
19. Consider torsion
20. Design for B/C capacity ratio
21. Ignore Beneficial Pu for Beam Design?
22. User defined allowable PT stresses
23. Concrete Strength Ratio at Transfer $f_{ck}(t) / f_{ck}$
24. Transfer: Top Fiber Tensile Stress / $f_{ctm}(t)$
25. Transfer: Bottom Fiber Tensile Stress / $f_{ctm}(t)$
26. Transfer: Extreme Fiber Compressive Stress / $f_{ck}(t)$
27. Final: Top Fiber Tensile Stress / f_{ctm}
28. Final: Bottom Fiber Tensile Stress / f_{ctm}
29. Final: Extreme Fiber Compressive Stress / f_{ck}
30. Sustained: Extreme Fiber Compressive Stress / f_{ck}
31. Sustained: Fraction of Live Load Considered

Value

Type: SystemDouble

The value of the considered preference item.

1. Country
 - . 1 = CEN Default
 - . 2 = Denmark
 - . 3 = Finland
 - . 4 = Germany
 - . 5 = Ireland
 - . 6 = Norway
 - . 7 = Poland
 - . 8 = Portugal
 - . 9 = Singapore
 - . 10 = Slovenia
 - . 11 = Sweden
 - . 12 = United Kingdom
2. Combos equation
 - . 1 = Eq. 6.10
 - . 2 = Max of Eqs. 6.10a and 6.10b
3. Second order method
 - . 1 = Nominal stiffness
 - . 2 = Nominal curvature
 - . 3 = None
4. Number of interaction curves

Value ≥ 4 and divisible by 4
5. Number of interaction points

Value ≥ 5 and odd
6. Consider minimum eccentricity

0 = No

Any other value = Yes
7. Theta0

Value > 0
8. Gamma steel

Value > 0
9. Gamma concrete

Value > 0
10. AlphaCC

Value > 0
11. AlphaCT

Value > 0
12. AlphaLCC

Value > 0
13. AlphaLCT

Value > 0
14. Pattern live load factor

Value ≥ 0
15. Utilization factor limit

Value > 0
16. Multi-response case design
· 1 = Envelopes
· 2 = Step-by step
· 3 = Last step
· 4 = Envelopes - All
· 5 = Step-by step - All

17. Reliability class
· 1 = Class 1
· 2 = Class 2
· 3 = Class 3

18. Gamma concrete modulus

Value > 0
19. Consider torsion
· 0 = No

Introduction

- Any other value = Yes
- 20. Design for B/C capacity ratio
 - 0 = No
 - Any other value = Yes
- 21. Ignore the benefit of Pu for beam design?
 - 0 = No
 - Any other value = Yes
- 22. User defined allowable PT stresses
 - 0 = No
 - Any other value = Yes
- 23. Concrete Strength Ratio at Transfer $f_{ck}(t) / f_{ck}$

Value > 0

- 24. Transfer: Top Fiber Tensile Stress / $f_{ctm}(t)$

Value > 0

- 25. Transfer: Bottom Fiber Tensile Stress / $f_{ctm}(t)$

Value > 0

- 26. Transfer: Extreme Fiber Compressive Stress / $f_{ck}(t)$

Value > 0

- 27. Final: Top Fiber Tensile Stress / f_{ctm}

Value > 0

- 28. Final: Bottom Fiber Tensile Stress / f_{ctm}

Value > 0

- 29. Final: Extreme Fiber Compressive Stress / f_{ck}

Value > 0

- 30. Sustained: Extreme Fiber Compressive Stress / f_{ck}

Value > 0

- 31. Sustained: Fraction of Live Load Considered

Value > 0

Return Value

Type: $\hat{A}$ Int32

Returns zero if the item is successfully set; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim Value As Double
```

Introduction

```
'create ETABS object
  Dim myHelper as cHelper = New Helper
  EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
  ret = EtabsObject.ApplicationStart()

'create SapModel object
  SapModel = EtabsObject.SapModel

'initialize model
  ret = SapModel.InitializeNewModel()

'create steel deck template model
  ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'create new concrete frame section property
  ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

'set frame section property
  ret = SapModel.FrameObj.SetSection("8", "R1")

'set concrete design code
  ret = SapModel.DesignConcrete.SetCode("Eurocode 2-2004")

'run analysis
  System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
  ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
  ret = SapModel.Analyze.RunAnalysis

'start concrete design
  ret = SapModel.DesignConcrete.StartDesign()

'set preference item
  ret = SapModel.DesignConcrete.Eurocode_2_2004.SetPreference(2, 7)

'get preference item
  ret = SapModel.DesignConcrete.Eurocode_2_2004.GetPreference(2, Value)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cDCoEurocode 2 2004 Interface

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CSI API ETABS v1

cDCoHong_Kong_CP_2013 Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDCoHong_Kong_CP_2013
```

```
Public Interface cDCoHong_Kong_CP_2013
```

```
Dim instance As cDCoHong_Kong_CP_2013
```

```
public interface class cDCoHong_Kong_CP_2013
```

```
type cDCoHong_Kong_CP_2013 = interface end
```

The cDCoHong_Kong_CP_2013 type exposes the following members.

▲ Methods

▲	Name	Description
---	------	-------------

◆	GetOverwrite	
---	------------------------------	--

◆	GetPreference	
---	-------------------------------	--

◆	SetOverwrite	
---	------------------------------	--

◆	SetPreference	
---	-------------------------------	--

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDCoHong_Kong_CP_2013 Methods

The [cDCoHong_Kong_CP_2013](#) type exposes the following members.

Methods

	Name	Description
	GetOverwrite	
	GetPreference	
	SetOverwrite	
	SetPreference	

[Top](#)

See Also

Reference

[cDCoHong_Kong_CP_2013 Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cDCoHong_Kong_CP_2013AddLanguageSpecificTextSet Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDCoHong_Kong_CP_2013
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```

Parameters

Name

Type: SystemString

Item

Type:Â [SystemInt32](#)

Value

Type:Â [SystemDouble](#)

ProgDet

Type:Â [SystemBoolean](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cDcOHong_Kong_CP_2013 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDCoHong_Kong_CP_2013AddLanguageSpecificTextSet Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDCoHong_Kong_CP_2013
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: [SystemInt32](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDCoHong_Kong_CP_2013 Interface

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ï»¿

CSI API ETABS v1

cDCoHong_Kong_CP_2013AddLanguageSpecificTextSet Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDCoHong_Kong_CP_2013
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

Item

Type: SystemInt32

Value

Type: SystemDouble

ItemType (Optional)

Type: ETABSv1eItemType

Return Value

Type: Int32

▲ See Also

Reference

cDcOHong_Kong_CP_2013 Interface

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CSI API ETABS v1

cDCoHong_Kong_CP_2013AddLanguageSpecificTextSet Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDCoHong_Kong_CP_2013
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: [SystemInt32](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDCoHong_Kong_CP_2013 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDColIndian_IS_456_2000 Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDColIndian_IS_456_2000
```

```
Public Interface cDColIndian_IS_456_2000
```

```
Dim instance As cDColIndian_IS_456_2000
```

```
public interface class cDColIndian_IS_456_2000
```

```
type cDColIndian_IS_456_2000 = interface end
```

The cDColIndian_IS_456_2000 type exposes the following members.

▲ Methods

▲	Name	Description
	GetOverwrite	Retrieves the value of a concrete design overwrite item.
	GetPreference	Retrieves the value of a concrete design preference item.
	SetOverwrite	Sets the value of a concrete design overwrite item.
	SetPreference	Sets the value of a concrete design preference item.

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDCoIndian_IS_456_2000 Methods

The cDCoIndian_IS_456_2000 type exposes the following members.

Methods

	Name	Description
	<u>GetOverwrite</u>	Retrieves the value of a concrete design overwrite item.
	<u>GetPreference</u>	Retrieves the value of a concrete design preference item.
	<u>SetOverwrite</u>	Sets the value of a concrete design overwrite item.
	<u>SetPreference</u>	Sets the value of a concrete design preference item.

Top

See Also

Reference

cDCoIndian_IS_456_2000 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDColIndian_IS_456_2000AddLanguageSpecificTextSet Method

Retrieves the value of a concrete design overwrite item.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDColIndian_IS_456_2000
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```

Parameters

Name

Type:Â SystemString

The name of a frame object with a concrete frame design procedure.

Item

Type:Â SystemInt32

This is an integer between 1 and 21, inclusive, indicating the overwrite item considered.

1. Framing type
2. Live load reduction factor
3. Unbraced length ratio, Major
4. Unbraced length ratio, Minor
5. Effective length factor sway, K Major
6. Effective length factor sway, K Minor
7. Moment coefficient, Cm Major
8. Moment coefficient, Cm Minor
9. Nonsway moment factor, Db Major
10. Nonsway moment factor, Db Minor
11. Sway moment factor, Ds Major
12. Sway moment factor, Ds Minor
13. Top rebar area of a beam at the left end (I-end)
14. Bottom rebar area of a beam at the left end (I-end)
15. Top rebar area of a beam at the right end (J-end)
16. Bottom rebar area of a beam at the right end (J-end)
17. Effective length factor braced, K Major
18. Effective length factor braced, K Minor
19. Q Factor in Global X Dir
20. Q Factor in Global Y Dir
21. Consider Minimum Eccentricity

Value

Type:Â SystemDouble

The value of the considered overwrite item.

1. Framing type
 - 0 = Program Default
 - 1 = Ductile
 - 2 = Ordinary
 - 3 = Non-sway
2. Live load reduction factor

Value >= 0; 0 means use program determined value

3. Unbraced length ratio, Major

Value >= 0; 0 means use program determined value

4. Unbraced length ratio, Minor

Value >= 0; 0 means use program determined value

5. Effective length factor sway, K Major

Value >= 0; 0 means use program determined value

6. Effective length factor sway, K Minor

Introduction

Value ≥ 0 ; 0 means use program determined value

7. Moment coefficient, Cm Major

Value ≥ 0 ; 0 means use program determined value

8. Moment coefficient, Cm Minor

Value ≥ 0 ; 0 means use program determined value

9. Nonsway moment factor, Db Major

Value ≥ 0 ; 0 means use program determined value

10. Nonsway moment factor, Db Minor

Value ≥ 0 ; 0 means use program determined value

11. Sway moment factor, Ds Major

Value ≥ 0 ; 0 means use program determined value

12. Sway moment factor, Ds Minor

Value ≥ 0 ; 0 means use program determined value

13. Top rebar area of a beam at the left end (I-end)

Value ≥ 0 ; 0 means use program determined value

14. Bottom rebar area of a beam at the left end (I-end)

Value ≥ 0 ; 0 means use program determined value

15. Top rebar area of a beam at the right end (J-end)

Value ≥ 0 ; 0 means use program determined value

16. Bottom rebar area of a beam at the right end (J-end)

Value ≥ 0 ; 0 means use program determined value

17. Effective length factor braced, K Major

Value ≥ 0 ; 0 means use program determined value

18. Effective length factor braced, K Minor

Value ≥ 0 ; 0 means use program determined value

19. Q Factor in Global X Dir

Value ≥ 0 ; 0 means use program determined value

20. Q Factor in Global Y Dir

Value ≥ 0 ; 0 means use program determined value

21. Consider Minimum Eccentricity

0 = No

Any other value = Yes

ProgDet

Type: System.Boolean

If this item is True, the specified value is program determined.

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ See Also

Reference

cDcoIndian IS 456 2000 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDColIndian_IS_456_2000AddLanguageSpecificTextSet Method

Retrieves the value of a concrete design preference item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDColIndian_IS_456_2000
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: SystemInt32

This is an integer between 1 and 8, inclusive, indicating the preference item considered.

1. Number of interaction curves
2. Number of interaction points
3. Consider minimum eccentricity
4. Gamma steel
5. Gamma concrete

- 6. Pattern live load factor
- 7. Utilization factor limit
- 8. Multi-response case design

Value

Type: SystemDouble

The value of the considered preference item.

- 1. Number of interaction curves

Value ≥ 4 and divisible by 4

- 2. Number of interaction points

Value ≥ 5 and odd

- 3. Consider minimum eccentricity

0 = No

Any other value = Yes

- 4. Gamma steel

Value > 0

- 5. Gamma concrete

Value > 0

- 6. Pattern live load factor

Value ≥ 0

- 7. Utilization factor limit

Value > 0

- 8. Multi-response case design
 - 1 = Envelopes
 - 2 = Step-by step
 - 3 = Last step
 - 4 = Envelopes - All
 - 5 = Step-by step - All

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ See Also

Reference

cDcoIndian IS 456 2000 Interface

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CSI API ETABS v1

cDColIndian_IS_456_2000AddLanguageSpecificTextSet Method

Sets the value of a concrete design overwrite item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDColIndian_IS_456_2000
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters**Name**Type:Â SystemString

The name of an existing frame object or group, depending on the value of the ItemType

ItemType:Â SystemInt32

This is an integer between 1 and 21, inclusive, indicating the overwrite item considered.

1. Framing type
2. Live load reduction factor
3. Unbraced length ratio, Major
4. Unbraced length ratio, Minor
5. Effective length factor sway, K Major
6. Effective length factor sway, K Minor
7. Moment coefficient, Cm Major
8. Moment coefficient, Cm Minor
9. Nonsway moment factor, Db Major
10. Nonsway moment factor, Db Minor
11. Sway moment factor, Ds Major
12. Sway moment factor, Ds Minor
13. Top rebar area of a beam at the left end (I-end)
14. Bottom rebar area of a beam at the left end (I-end)
15. Top rebar area of a beam at the right end (J-end)
16. Bottom rebar area of a beam at the right end (J-end)
17. Effective length factor braced, K Major
18. Effective length factor braced, K Minor
19. Q Factor in Global X Dir
20. Q Factor in Global Y Dir
21. Consider Minimum Eccentricity

ValueType:Â SystemDouble

The value of the considered overwrite item.

1. Framing type
 - 0 = Program Default
 - 1 = Ductile
 - 2 = Ordinary
 - 3 = Non-sway
2. Live load reduction factor

Value >= 0; 0 means use program determined value

3. Unbraced length ratio, Major

Value >= 0; 0 means use program determined value

4. Unbraced length ratio, Minor

Value >= 0; 0 means use program determined value

5. Effective length factor sway, K Major

Value >= 0; 0 means use program determined value

Introduction

6. Effective length factor sway, K Minor

Value ≥ 0 ; 0 means use program determined value

7. Moment coefficient, Cm Major

Value ≥ 0 ; 0 means use program determined value

8. Moment coefficient, Cm Minor

Value ≥ 0 ; 0 means use program determined value

9. Nonsway moment factor, Db Major

Value ≥ 0 ; 0 means use program determined value

10. Nonsway moment factor, Db Minor

Value ≥ 0 ; 0 means use program determined value

11. Sway moment factor, Ds Major

Value ≥ 0 ; 0 means use program determined value

12. Sway moment factor, Ds Minor

Value ≥ 0 ; 0 means use program determined value

13. Top rebar area of a beam at the left end (I-end)

Value ≥ 0 ; 0 means use program determined value

14. Bottom rebar area of a beam at the left end (I-end)

Value ≥ 0 ; 0 means use program determined value

15. Top rebar area of a beam at the right end (J-end)

Value ≥ 0 ; 0 means use program determined value

16. Bottom rebar area of a beam at the right end (J-end)

Value ≥ 0 ; 0 means use program determined value

17. Effective length factor braced, K Major

Value ≥ 0 ; 0 means use program determined value

18. Effective length factor braced, K Minor

Value ≥ 0 ; 0 means use program determined value

19. Q Factor in Global X Dir

Value ≥ 0 ; 0 means use program determined value

20. Q Factor in Global Y Dir

Value ≥ 0 ; 0 means use program determined value

21. Consider Minimum Eccentricity

0 = No

Any other value = Yes

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

- ◊ Object = 0
- ◊ Group = 1
- ◊ SelectedObjects = 2

If this item is Objects, the overwrite is set for the frame object specified by the Name item.

If this item is Group, the overwrite is set for all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the overwrite is set for all selected frame objects and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the item is successfully set; otherwise it returns a nonzero value.

▲ See Also

Reference

cDcoIndian IS 456 2000 Interface

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CSI API ETABS v1

cDColIndian_IS_456_2000AddLanguageSpecificTextSet Method

Sets the value of a concrete design preference item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDColIndian_IS_456_2000
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: SystemInt32

This is an integer between 1 and 8, inclusive, indicating the preference item considered.

1. Number of interaction curves
2. Number of interaction points
3. Consider minimum eccentricity
4. Gamma steel
5. Gamma concrete

- 6. Pattern live load factor
- 7. Utilization factor limit
- 8. Multi-response case design

Value

Type: SystemDouble

The value of the considered preference item.

- 1. Number of interaction curves

Value ≥ 4 and divisible by 4

- 2. Number of interaction points

Value ≥ 5 and odd

- 3. Consider minimum eccentricity

0 = No

Any other value = Yes

- 4. Gamma steel

Value > 0

- 5. Gamma concrete

Value > 0

- 6. Pattern live load factor

Value ≥ 0

- 7. Utilization factor limit

Value > 0

- 8. Multi-response case design
 - 1 = Envelopes
 - 2 = Step-by step
 - 3 = Last step
 - 4 = Envelopes - All
 - 5 = Step-by step - All

Return Value

Type: Int32

Returns zero if the item is successfully set; otherwise it returns a nonzero value.

▲ See Also

Reference

cDcoIndian IS 456 2000 Interface

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CSI API ETABS v1

cDCoItalianNTC2008C Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDCoItalianNTC2008C
```

```
Public Interface cDCoItalianNTC2008C
```

```
Dim instance As cDCoItalianNTC2008C
```

```
public interface class cDCoItalianNTC2008C
```

```
type cDCoItalianNTC2008C = interface end
```

The cDCoItalianNTC2008C type exposes the following members.

▲ Methods

▲	Name	Description
	GetOverwrite	Retrieves the value of a concrete design overwrite item.
	GetPreference	Retrieves the value of a concrete design preference item.
	SetOverwrite	Sets the value of a concrete design overwrite item.
	SetPreference	Sets the value of a concrete design preference item.

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDColtalianNTC2008C Methods

The [cDColtalianNTC2008C](#) type exposes the following members.

Methods

	Name	Description
	<u>GetOverwrite</u>	Retrieves the value of a concrete design overwrite item.
	<u>GetPreference</u>	Retrieves the value of a concrete design preference item.
	<u>SetOverwrite</u>	Sets the value of a concrete design overwrite item.
	<u>SetPreference</u>	Sets the value of a concrete design preference item.

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See Also

Reference

[cDColtalianNTC2008C Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDColtalianNTC2008CAddLanguageSpecificTextSet("L Method

Retrieves the value of a concrete design overwrite item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDColtalianNTC2008C
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```

Parameters

Name

Type:Â SystemString

The name of a frame object with a concrete frame design procedure.

Item

Type:Â SystemInt32

This is an integer between 1 and 27, inclusive, indicating the overwrite item considered.

1. Framing type
2. Live load reduction factor
3. Unbraced length ratio, Major
4. Unbraced length ratio, Minor
5. Effective length factor, Beta Major (columns only)
6. Effective length factor, Beat Minor (columns only)
7. Moment coefficient, Cm Major (not used)
8. Moment coefficient, Cm Minor (not used)
9. Nonsway moment factor, Db Major (not used)
10. Nonsway moment factor, Db Minor (not used)
11. Sway moment factor, Ds Major (not used)
12. Sway moment factor, Ds Minor (not used)
13. Correction factor depending on axial load in Nominal Curvature method, Kr Major (columns only)
14. Factor accounting for creep in Nominal Curvature method, KPhi Major (columns only)
16. Correction factor depending on axial load in Nominal Curvature method, Kr Minor (columns only)
17. Factor accounting for creep in Nominal Curvature method, KPhi Minor (columns only)
18. Coefficient depending on the distribution of first-order moment in both Nominal Stiffness and Nominal Curvature methods, c Major (columns only)
19. Coefficient depending on the distribution of first-order moment in both Nominal Stiffness and Nominal Curvature methods, c Minor (columns only)
20. Factor for contribution of reinforcement, Ks Major (columns only)
21. Factor for contribution of reinforcement, Ks Minor (columns only)
22. Factor for effects of cracking, creep etc, Kc Major (columns only)
23. Factor for effects of cracking, creep etc, Kc Minor (columns only)
24. Effective creep coefficient, Phief (columns only)
25. Ignore the benefit of Pu for beam design? (beams only)
26. Consider torsion (beams only)
27. Consider minimum eccentricity (columns only)

Value

Type:Â SystemDouble

The value of the considered overwrite item.

1. Framing type
 - 0 = Program Determined
 - 1 = DC High
 - 2 = DC Low
 - 3 = Secondary

Introduction

2. Live load reduction factor

Value ≥ 0 ; 0 means use program determined value

3. Unbraced length ratio, Major

Value ≥ 0 ; 0 means use program determined value

4. Unbraced length ratio, Minor

Value ≥ 0 ; 0 means use program determined value

5. Effective length factor, Beta Major

Value ≥ 0 ; 0 means use program determined value

6. Effective length factor, Beta Minor

Value ≥ 0 ; 0 means use program determined value

7. Moment coefficient, Cm Major

Value ≥ 0 ; 0 means use program determined value

8. Moment coefficient, Cm Minor

Value ≥ 0 ; 0 means use program determined value

9. Nonsway moment factor, Db Major

Value ≥ 0 ; 0 means use program determined value

10. Nonsway moment factor, Db Minor

Value ≥ 0 ; 0 means use program determined value

11. Sway moment factor, Ds Major

Value ≥ 0 ; 0 means use program determined value

12. Sway moment factor, Ds Minor

Value ≥ 0 ; 0 means use program determined value

13. Correction factor depending on axial load in Nominal Curvature method, Kr Major

Value ≥ 0 ; 0 means use program determined value

14. Factor accounting for creep in Nominal Curvature method, KPhi Major

Value ≥ 0 ; 0 means use program determined value

15. Correction factor depending on axial load in Nominal Curvature method, Kr Minor

Value ≥ 0 ; 0 means use program determined value

16. Factor accounting for creep in Nominal Curvature method, KPhi Minor

Value ≥ 0 ; 0 means use program determined value

17. Coefficient depending on the distribution of first-order moment in both Nominal Stiffness and Nominal Curvature methods, c Major

Value ≥ 0 ; 0 means use program determined value

Introduction

18. Coefficient depending on the distribution of first-order moment in both Nominal Stiffness and Nominal Curvature methods, c Minor

Value ≥ 0 ; 0 means use program determined value

19. Factor for contribution of reinforcement, Ks Major

Value ≥ 0 ; 0 means use program determined value

20. Factor for contribution of reinforcement, Ks Minor

Value ≥ 0 ; 0 means use program determined value

21. Factor for effects of cracking, creep etc, Kc Major

Value ≥ 0 ; 0 means use program determined value

22. Factor for effects of cracking, creep etc, Kc Minor

Value ≥ 0 ; 0 means use program determined value

23. Effective creep coefficient, Phief

Value ≥ 0 ; 0 means use program determined value

24. Ignore the benefit of Pu for beam design?

· 0 = No

· Any other value = Yes

25. Consider torsion

· 0 = No

· Any other value = Yes

26. Consider minimum eccentricity

· 0 = No

· Any other value = Yes

ProgDet

Type: System.Boolean

If this item is True, the specified value is program determined.

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim Value As Double  
    Dim ProgDet As Boolean  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()
```

Return Value

Introduction

```
'create SapModel object
  SapModel = EtabsObject.SapModel

'initialize model
  ret = SapModel.InitializeNewModel()

'create steel deck template model
  ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'create new concrete frame section property
  ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

'set frame section property
  ret = SapModel.FrameObj.SetSection("8", "R1")

'set concrete design code
  ret = SapModel.DesignConcrete.SetCode("Italian NTC 2008")

'run analysis
  System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
  ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
  ret = SapModel.Analyze.RunAnalysis

'start concrete design
  ret = SapModel.DesignConcrete.StartDesign()

'get overwrite item
  ret = SapModel.DesignConcrete.ItalianNTC2008C.GetOverwrite("8", 1, Value, ProgDet)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

[cDcoItalianNTC2008C Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDColtalianNTC2008CAddLanguageSpecificTextSet("L Method

Retrieves the value of a concrete design preference item.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDColtalianNTC2008C
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: [SystemInt32](#)

This is an integer between 3 and 31, inclusive, indicating the preference item considered.

3. Second order method
4. Number of interaction curves
5. Number of interaction points
6. Consider minimum eccentricity
7. Theta0

8. Gamma steel
9. Gamma concrete
10. AlphaCC
11. AlphaCT
12. AlphaLCC
13. AlphaLCT
14. Pattern live load factor
15. Utilization factor limit
16. Multi-response case design
18. Gamma concrete modulus
19. Consider torsion
20. Design for B/C capacity ratio
21. Ignore Beneficial Pu for Beam Design?
22. User defined allowable PT stresses
23. Concrete Strength Ratio at Transfer $f_{ck}(t) / f_{ck}$
24. Transfer: Top Fiber Tensile Stress / $f_{ctm}(t)$
25. Transfer: Bottom Fiber Tensile Stress / $f_{ctm}(t)$
26. Transfer: Extreme Fiber Compressive Stress / $f_{ck}(t)$
27. Final: Top Fiber Tensile Stress / f_{ctm}
28. Final: Bottom Fiber Tensile Stress / f_{ctm}
29. Final: Extreme Fiber Compressive Stress / f_{ck}
30. Sustained: Extreme Fiber Compressive Stress / f_{ck}
31. Sustained: Fraction of Live Load Considered

Value

Type: SystemDouble

The value of the considered preference item.

3. Second order method
 - . 1 = Nominal stiffness
 - . 2 = Nominal curvature
 - . 3 = None
4. Number of interaction curves

Value ≥ 4 and divisible by 4
5. Number of interaction points

Value ≥ 5 and odd
6. Consider minimum eccentricity

0 = No

Any other value = Yes
7. Theta0

Value > 0
8. Gamma steel

Value > 0
9. Gamma concrete

Value > 0
10. AlphaCC

Value > 0

- Value > 0
- 11. AlphaCT
- Value > 0
- 12. AlphaLCC
- Value > 0
- 13. AlphaLCT
- Value > 0
- 14. Pattern live load factor
- Value >= 0
- 15. Utilization factor limit
- Value > 0
- 16. Multi-response case design
 - 1 = Envelopes
 - 2 = Step-by step
 - 3 = Last step
 - 4 = Envelopes - All
 - 5 = Step-by step - All
- 18. Gamma concrete modulus
- Value > 0
- 19. Consider torsion
 - 0 = No
 - Any other value = Yes
- 20. Design for B/C capacity ratio
 - 0 = No
 - Any other value = Yes
- 21. Ignore the benefit of Pu for beam design?
 - 0 = No
 - Any other value = Yes
- 22. User defined allowable PT stresses
 - 0 = No
 - Any other value = Yes
- 23. Concrete Strength Ratio at Transfer $f_{ck}(t) / f_{ck}$
- Value > 0
- 24. Transfer: Top Fiber Tensile Stress / $f_{ctm}(t)$
- Value > 0
- 25. Transfer: Bottom Fiber Tensile Stress / $f_{ctm}(t)$
- Value > 0
- 26. Transfer: Extreme Fiber Compressive Stress / $f_{ck}(t)$
- Value > 0
- 27. Final: Top Fiber Tensile Stress / f_{ctm}

Value > 0

28. Final: Bottom Fiber Tensile Stress / fctm

Value > 0

29. Final: Extreme Fiber Compressive Stress / fck

Value > 0

30. Sustained: Extreme Fiber Compressive Stress / fck

Value > 0

31. Sustained: Fraction of Live Load Considered

Value > 0

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'create new concrete frame section property
    ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

    'set frame section property
    ret = SapModel.FrameObj.SetSection("8", "R1")

    'set concrete design code
    ret = SapModel.DesignConcrete.SetCode("Italian NTC 2008")

    'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis
```

Introduction

```
'start concrete design
    ret = SapModel.DesignConcrete.StartDesign()

'get preference item
    ret = SapModel.DesignConcrete.ItalianNTC2008C.GetPreference(3, Value)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

📌 See Also

Reference

[cDcoItalianNTC2008C Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDColtalianNTC2008CAddLanguageSpecificTextSet("L Method

Sets the value of a concrete design overwrite item.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDCoItalianNTC2008C
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters**Name**Type: SystemString

The name of an existing frame object or group, depending on the value of the ItemType

ItemType: SystemInt32

This is an integer between 1 and 27, inclusive, indicating the overwrite item considered.

1. Framing type
2. Live load reduction factor
3. Unbraced length ratio, Major
4. Unbraced length ratio, Minor
5. Effective length factor, Beta Major (columns only)
6. Effective length factor, Beta Minor (columns only)
7. Moment coefficient, Cm Major (not used)
8. Moment coefficient, Cm Minor (not used)
9. Nonsway moment factor, Db Major (not used)
10. Nonsway moment factor, Db Minor (not used)
11. Sway moment factor, Ds Major (not used)
12. Sway moment factor, Ds Minor (not used)
13. Correction factor depending on axial load in Nominal Curvature method, Kr Major (columns only)
14. Factor accounting for creep in Nominal Curvature method, KPhi Major (columns only)
16. Correction factor depending on axial load in Nominal Curvature method, Kr Minor (columns only)
17. Factor accounting for creep in Nominal Curvature method, KPhi Minor (columns only)
18. Coefficient depending on the distribution of first-order moment in both Nominal Stiffness and Nominal Curvature methods, c Major (columns only)
19. Coefficient depending on the distribution of first-order moment in both Nominal Stiffness and Nominal Curvature methods, c Minor (columns only)
20. Factor for contribution of reinforcement, Ks Major (columns only)
21. Factor for contribution of reinforcement, Ks Minor (columns only)
22. Factor for effects of cracking, creep etc, Kc Major (columns only)
23. Factor for effects of cracking, creep etc, Kc Minor (columns only)
24. Effective creep coefficient, Phief (columns only)
25. Ignore the benefit of Pu for beam design? (beams only)
26. Consider torsion (beams only)
27. Consider minimum eccentricity (columns only)

ValueType: SystemDouble

The value of the considered overwrite item.

1. Framing type
 - 0 = Program Determined
 - 1 = DC High
 - 2 = DC Low

- 3 = Secondary
- 2. Live load reduction factor
 - Value ≥ 0 ; 0 means use program determined value
- 3. Unbraced length ratio, Major
 - Value ≥ 0 ; 0 means use program determined value
- 4. Unbraced length ratio, Minor
 - Value ≥ 0 ; 0 means use program determined value
- 5. Effective length factor, Beta Major
 - Value ≥ 0 ; 0 means use program determined value
- 6. Effective length factor, Beta Minor
 - Value ≥ 0 ; 0 means use program determined value
- 7. Moment coefficient, Cm Major
 - Value ≥ 0 ; 0 means use program determined value
- 8. Moment coefficient, Cm Minor
 - Value ≥ 0 ; 0 means use program determined value
- 9. Nonsway moment factor, Db Major
 - Value ≥ 0 ; 0 means use program determined value
- 10. Nonsway moment factor, Db Minor
 - Value ≥ 0 ; 0 means use program determined value
- 11. Sway moment factor, Ds Major
 - Value ≥ 0 ; 0 means use program determined value
- 12. Sway moment factor, Ds Minor
 - Value ≥ 0 ; 0 means use program determined value
- 13. Correction factor depending on axial load in Nominal Curvature method, Kr Major
 - Value ≥ 0 ; 0 means use program determined value
- 14. Factor accounting for creep in Nominal Curvature method, KPhi Major
 - Value ≥ 0 ; 0 means use program determined value
- 15. Correction factor depending on axial load in Nominal Curvature method, Kr Minor
 - Value ≥ 0 ; 0 means use program determined value
- 16. Factor accounting for creep in Nominal Curvature method, KPhi Minor
 - Value ≥ 0 ; 0 means use program determined value
- 17. Coefficient depending on the distribution of first-order moment in both Nominal Stiffness and Nominal Curvature methods, c Major
 - Value ≥ 0 ; 0 means use program determined value

Introduction

Value ≥ 0 ; 0 means use program determined value

18. Coefficient depending on the distribution of first-order moment in both Nominal Stiffness and Nominal Curvature methods, c Minor

Value ≥ 0 ; 0 means use program determined value

19. Factor for contribution of reinforcement, Ks Major

Value ≥ 0 ; 0 means use program determined value

20. Factor for contribution of reinforcement, Ks Minor

Value ≥ 0 ; 0 means use program determined value

21. Factor for effects of cracking, creep etc, Kc Major

Value ≥ 0 ; 0 means use program determined value

22. Factor for effects of cracking, creep etc, Kc Minor

Value ≥ 0 ; 0 means use program determined value

23. Effective creep coefficient, Phief

Value ≥ 0 ; 0 means use program determined value

24. Ignore the benefit of Pu for beam design?

- 0 = No
- Any other value = Yes

25. Consider torsion

- 0 = No
- Any other value = Yes

26. Consider minimum eccentricity

- 0 = No
- Any other value = Yes

ItemType (Optional)

Type: $\hat{A}$ ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

- ◊ Object = 0
- ◊ Group = 1
- ◊ SelectedObjects = 2

If this item is Objects, the overwrite is set for the frame object specified by the Name item.

If this item is Group, the overwrite is set for all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the overwrite is set for all selected frame objects and the Name item is ignored.

Return Value

Type: $\hat{A}$ Int32

Returns zero if the item is successfully set; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Return Value

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double
    Dim ProgDet As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'create new concrete frame section property
    ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

    'set frame section property
    ret = SapModel.FrameObj.SetSection("8", "R1")

    'set concrete design code
    ret = SapModel.DesignConcrete.SetCode("Italian NTC 2008")

    'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis

    'start concrete design
    ret = SapModel.DesignConcrete.StartDesign()

    'set overwrite item
    ret = SapModel.DesignConcrete.ItalianNTC2008C.SetOverwrite("8", 1, 1)

    'get overwrite item
    ret = SapModel.DesignConcrete.ItalianNTC2008C.GetOverwrite("8", 1, Value, ProgDet)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub

```

▲ See Also

Reference

cDcoItalianNTC2008C Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDColtalianNTC2008CAddLanguageSpecificTextSet("L Method

Sets the value of a concrete design preference item.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDColtalianNTC2008C
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: [SystemInt32](#)

This is an integer between 3 and 31, inclusive, indicating the preference item considered.

3. Second order method
4. Number of interaction curves
5. Number of interaction points
6. Consider minimum eccentricity
7. Theta0

8. Gamma steel
9. Gamma concrete
10. AlphaCC
11. AlphaCT
12. AlphaLCC
13. AlphaLCT
14. Pattern live load factor
15. Utilization factor limit
16. Multi-response case design
17. Gamma concrete modulus
18. Gamma concrete modulus
19. Consider torsion
20. Design for B/C capacity ratio
21. Ignore Beneficial Pu for Beam Design?
22. User defined allowable PT stresses
23. Concrete Strength Ratio at Transfer $f_{ck}(t) / f_{ck}$
24. Transfer: Top Fiber Tensile Stress / $f_{ctm}(t)$
25. Transfer: Bottom Fiber Tensile Stress / $f_{ctm}(t)$
26. Transfer: Extreme Fiber Compressive Stress / $f_{ck}(t)$
27. Final: Top Fiber Tensile Stress / f_{ctm}
28. Final: Bottom Fiber Tensile Stress / f_{ctm}
29. Final: Extreme Fiber Compressive Stress / f_{ck}
30. Sustained: Extreme Fiber Compressive Stress / f_{ck}
31. Sustained: Fraction of Live Load Considered

Value

Type: SystemDouble

The value of the considered preference item.

3. Second order method
 - . 1 = Nominal stiffness
 - . 2 = Nominal curvature
 - . 3 = None
4. Number of interaction curves

Value ≥ 4 and divisible by 4

5. Number of interaction points

Value ≥ 5 and odd

6. Consider minimum eccentricity

0 = No

Any other value = Yes

7. Theta0

Value > 0

8. Gamma steel

Value > 0

9. Gamma concrete

Value > 0

10. AlphaCC

Value > 0

11. AlphaCT

Value > 0

12. AlphaLCC

Value > 0

13. AlphaLCT

Value > 0

14. Pattern live load factor

Value >= 0

15. Utilization factor limit

Value > 0

16. Multi-response case design

- 1 = Envelopes
- 2 = Step-by step
- 3 = Last step
- 4 = Envelopes - All
- 5 = Step-by step - All

18. Gamma concrete modulus

Value > 0

19. Consider torsion

- 0 = No
- Any other value = Yes

20. Design for B/C capacity ratio

- 0 = No
- Any other value = Yes

21. Ignore the benefit of Pu for beam design?

- 0 = No
- Any other value = Yes

22. User defined allowable PT stresses

- 0 = No
- Any other value = Yes

23. Concrete Strength Ratio at Transfer $f_{ck}(t) / f_{ck}$

Value > 0

24. Transfer: Top Fiber Tensile Stress / $f_{ctm}(t)$

Value > 0

25. Transfer: Bottom Fiber Tensile Stress / $f_{ctm}(t)$

Value > 0

26. Transfer: Extreme Fiber Compressive Stress / $f_{ck}(t)$

Value > 0

27. Final: Top Fiber Tensile Stress / fctm

Value > 0

28. Final: Bottom Fiber Tensile Stress / fctm

Value > 0

29. Final: Extreme Fiber Compressive Stress / fck

Value > 0

30. Sustained: Extreme Fiber Compressive Stress / fck

Value > 0

31. Sustained: Fraction of Live Load Considered

Value > 0

Return Value

Type: Int32

Returns zero if the item is successfully set; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'create new concrete frame section property
    ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

    'set frame section property
    ret = SapModel.FrameObj.SetSection("8", "R1")

    'set concrete design code
    ret = SapModel.DesignConcrete.SetCode("Italian NTC 2008")

    'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
```

Return Value

Introduction

```
ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
ret = SapModel.Analyze.RunAnalysis

'start concrete design
ret = SapModel.DesignConcrete.StartDesign()

'set preference item
ret = SapModel.DesignConcrete.ItalianNTC2008C.SetPreference(2, 7)

'get preference item
ret = SapModel.DesignConcrete.ItalianNTC2008C.GetPreference(2, Value)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cDcoItalianNTC2008C Interface

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CSI API ETABS v1

cDCoMexican_RCDF_2004 Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDCoMexican_RCDF_2004
```

```
Public Interface cDCoMexican_RCDF_2004
```

```
Dim instance As cDCoMexican_RCDF_2004
```

```
public interface class cDCoMexican_RCDF_2004
```

```
type cDCoMexican_RCDF_2004 = interface end
```

The cDCoMexican_RCDF_2004 type exposes the following members.

▲ Methods

▲	Name	Description
---	------	-------------

◆	GetOverwrite	
---	------------------------------	--

◆	GetPreference	
---	-------------------------------	--

◆	SetOverwrite	
---	------------------------------	--

◆	SetPreference	
---	-------------------------------	--

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDCoMexican_RCDF_2004 Methods

The cDCoMexican_RCDF_2004 type exposes the following members.

Methods

	Name	Description
	<u>GetOverwrite</u>	
	<u>GetPreference</u>	
	<u>SetOverwrite</u>	
	<u>SetPreference</u>	

Top

See Also

Reference

cDCoMexican_RCDF_2004 Interface

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CSI API ETABS v1

cDCoMexican_RCDF_2004AddLanguageSpecificTextSet Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDCoMexican_RCDF_2004
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```

Parameters

Name

Type: SystemString

cDCoMexican_RCDF_2004AddLanguageSpecificTextSet("LSTC3E563D9_0?c

Item

Type:Â SystemInt32

Value

Type:Â SystemDouble

ProgDet

Type:Â SystemBoolean

Return Value

Type:Â Int32

▲ See Also

Reference

cDCoMexican_RCDF_2004 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDCoMexican_RCDF_2004AddLanguageSpecificTextSet Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDCoMexican_RCDF_2004
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: [SystemInt32](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDCoMexican_RCDF_2004 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDCoMexican_RCDF_2004AddLanguageSpecificTextSet Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDCoMexican_RCDF_2004
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

Item

Type: SystemInt32

Value

Type: SystemDouble

ItemType (Optional)

Type: ETABSv1eItemType

Return Value

Type: Int32

▲ See Also

Reference

cDCoMexican_RCDF_2004 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDCoMexican_RCDF_2004AddLanguageSpecificTextSet Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDCoMexican_RCDF_2004
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: [SystemInt32](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDCoMexican_RCDF_2004 Interface

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CSI API ETABS v1

cDCoMexican_RCDF_2017 Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDCoMexican_RCDF_2017
```

```
Public Interface cDCoMexican_RCDF_2017
```

```
Dim instance As cDCoMexican_RCDF_2017
```

```
public interface class cDCoMexican_RCDF_2017
```

```
type cDCoMexican_RCDF_2017 = interface end
```

The cDCoMexican_RCDF_2017 type exposes the following members.

▲ Methods

▲	Name	Description
	GetOverwrite	Retrieves the value of a concrete design overwrite item.
	GetPreference	Retrieves the value of a concrete design preference item.
	SetOverwrite	Sets the value of a concrete design overwrite item.
	SetPreference	Sets the value of a concrete design preference item.

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDCoMexican_RCDF_2017 Methods

The cDCoMexican_RCDF_2017 type exposes the following members.

Methods

	Name	Description
	<u>GetOverwrite</u>	Retrieves the value of a concrete design overwrite item.
	<u>GetPreference</u>	Retrieves the value of a concrete design preference item.
	<u>SetOverwrite</u>	Sets the value of a concrete design overwrite item.
	<u>SetPreference</u>	Sets the value of a concrete design preference item.

Top

See Also

Reference

cDCoMexican_RCDF_2017 Interface

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CSI API ETABS v1

cDCoMexican_RCDF_2017AddLanguageSpecificTextSet

Method

Retrieves the value of a concrete design overwrite item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDCoMexican_RCDF_2017
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```

Parameters

Name

Type:Â SystemString

The name of a frame object with a concrete frame design procedure.

Item

Type:Â SystemInt32

This is an integer between 1 and 13, inclusive, indicating the overwrite item considered.

1. Framing type
2. Live load reduction factor
3. Unbraced length ratio, Major
4. Unbraced length ratio, Minor
5. Effective length factor, K Major
6. Effective length factor, K Minor
7. Moment coefficient, Cm Major
8. Moment coefficient, Cm Minor
9. Nonsway moment factor, Db Major
10. Nonsway moment factor, Db Minor
11. Sway moment factor, Ds Major
12. Sway moment factor, Ds Minor
13. Consider Minimum Eccentricity

Value

Type:Â SystemDouble

The value of the considered overwrite item.

1. Framing type
 - 0 = Program Default
 - 1 = Sway special
 - 2 = Sway Intermediate
 - 3 = Sway Ordinary
 - 4 = Non-sway
2. Live load reduction factor

Value >= 0; 0 means use program determined value

3. Unbraced length ratio, Major

Value >= 0; 0 means use program determined value

4. Unbraced length ratio, Minor

Value >= 0; 0 means use program determined value

5. Effective length factor, K Major

Value >= 0; 0 means use program determined value

6. Effective length factor, K Minor

Value >= 0; 0 means use program determined value

7. Moment coefficient, Cm Major

Value >= 0; 0 means use program determined value

8. Moment coefficient, Cm Minor

Introduction

Value ≥ 0 ; 0 means use program determined value

9. Nonsway moment factor, Db Major

Value ≥ 0 ; 0 means use program determined value

10. Nonsway moment factor, Db Minor

Value ≥ 0 ; 0 means use program determined value

11. Sway moment factor, Ds Major

Value ≥ 0 ; 0 means use program determined value

12. Sway moment factor, Ds Minor

Value ≥ 0 ; 0 means use program determined value

13. Consider Minimum Eccentricity

0 = No

Any other value = Yes

ProgDet

Type: SystemBoolean

If this item is True, the specified value is program determined.

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ See Also

Reference

cDCoMexican_RCDF_2017 Interface

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CSI API ETABS v1

cDCoMexican_RCDF_2017AddLanguageSpecificTextSet

Method

Retrieves the value of a concrete design preference item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDCoMexican_RCDF_2017
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: SystemInt32

This is an integer between 1 and 12, inclusive, indicating the preference item considered.

1. Number of interaction curves
2. Number of interaction points
3. Consider minimum eccentricity
4. Design for B/C Capacity Ratio?
5. Phi bending

6. Phi tension
7. Phi compression controlled tied
8. Phi compression controlled spiral
9. Phi shear and/or torsion
10. Pattern live load factor
11. Utilization factor limit
12. Multi-response case design

Value

Type: SystemDouble

The value of the considered preference item.

1. Number of interaction curves

Value ≥ 4 and divisible by 4

2. Number of interaction points

Value ≥ 5 and odd

3. Consider minimum eccentricity

0 = No

Any other value = Yes

4. Design for B/C Capacity Ratio?

1 = No

2 = Yes

5. Phi bending

Value > 0

6. Phi tension

Value > 0

7. Phi compression controlled tied

Value > 0

8. Phi compression controlled spiral

Value > 0

9. Phi shear and/or torsion

Value > 0

10. Pattern live load factor

Value ≥ 0

11. Utilization factor limit

Value > 0

12. Multi-response case design

- 1 = Envelopes
- 2 = Step-by step
- 3 = Last step

- 4 = Envelopes - All
- 5 = Step-by step - All

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ See Also

Reference

cDCoMexican RCDF 2017 Interface

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CSI API ETABS v1

cDCoMexican_RCDF_2017AddLanguageSpecificTextSet

Method

Sets the value of a concrete design overwrite item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDCoMexican_RCDF_2017
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters**Name**Type: SystemString

The name of an existing frame object or group, depending on the value of the ItemType

ItemType: SystemInt32

This is an integer between 1 and 13, inclusive, indicating the overwrite item considered.

1. Framing type
2. Live load reduction factor
3. Unbraced length ratio, Major
4. Unbraced length ratio, Minor
5. Effective length factor, K Major
6. Effective length factor, K Minor
7. Moment coefficient, Cm Major
8. Moment coefficient, Cm Minor
9. Nonsway moment factor, Db Major
10. Nonsway moment factor, Db Minor
11. Sway moment factor, Ds Major
12. Sway moment factor, Ds Minor
13. Consider Minimum Eccentricity

ValueType: SystemDouble

The value of the considered overwrite item.

1. Framing type
 - 0 = Program Default
 - 1 = Sway special
 - 2 = Sway Intermediate
 - 3 = Sway Ordinary
 - 4 = Non-sway
2. Live load reduction factor

Value ≥ 0 ; 0 means use program determined value

3. Unbraced length ratio, Major

Value ≥ 0 ; 0 means use program determined value

4. Unbraced length ratio, Minor

Value ≥ 0 ; 0 means use program determined value

5. Effective length factor, K Major

Value ≥ 0 ; 0 means use program determined value

6. Effective length factor, K Minor

Value ≥ 0 ; 0 means use program determined value

7. Moment coefficient, Cm Major

Value ≥ 0 ; 0 means use program determined value

8. Moment coefficient, Cm Minor

Value ≥ 0 ; 0 means use program determined value

9. Nonsway moment factor, Db Major

Value ≥ 0 ; 0 means use program determined value

10. Nonsway moment factor, Db Minor

Value ≥ 0 ; 0 means use program determined value

11. Sway moment factor, Ds Major

Value ≥ 0 ; 0 means use program determined value

12. Sway moment factor, Ds Minor

Value ≥ 0 ; 0 means use program determined value

13. Consider Minimum Eccentricity

0 = No

Any other value = Yes

ItemType (Optional)

Type: $\hat{A}$ ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the overwrite is set for the frame object specified by the Name item.

If this item is Group, the overwrite is set for all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the overwrite is set for all selected frame objects and the Name item is ignored.

Return Value

Type: $\hat{A}$ Int32

Returns zero if the item is successfully set; otherwise it returns a nonzero value.

▲ See Also

Reference

cDCoMexican RCDF 2017 Interface

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CSI API ETABS v1

cDCoMexican_RCDF_2017AddLanguageSpecificTextSet

Method

Sets the value of a concrete design preference item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDCoMexican_RCDF_2017
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: SystemInt32

This is an integer between 1 and 12, inclusive, indicating the preference item considered.

1. Number of interaction curves
2. Number of interaction points
3. Consider minimum eccentricity
4. Design for B/C Capacity Ratio?
5. Phi bending

6. Phi tension
7. Phi compression controlled tied
8. Phi compression controlled spiral
9. Phi shear and/or torsion
10. Pattern live load factor
11. Utilization factor limit
12. Multi-response case design

Value

Type: SystemDouble

The value of the considered preference item.

1. Number of interaction curves

Value ≥ 4 and divisible by 4

2. Number of interaction points

Value ≥ 5 and odd

3. Consider minimum eccentricity

0 = No

Any other value = Yes

4. Design for B/C Capacity Ratio?

1 = No

2 = Yes

5. Phi bending

Value > 0

6. Phi tension

Value > 0

7. Phi compression controlled tied

Value > 0

8. Phi compression controlled spiral

Value > 0

9. Phi shear and/or torsion

Value > 0

10. Pattern live load factor

Value ≥ 0

11. Utilization factor limit

Value > 0

12. Multi-response case design

- 1 = Envelopes
- 2 = Step-by step
- 3 = Last step

- 4 = Envelopes - All
- 5 = Step-by step - All

Return Value

Type: Int32

Returns zero if the item is successfully set; otherwise it returns a nonzero value.

▲ See Also

Reference

cDCoMexican RCDF 2017 Interface

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CSI API ETABS v1

cDConcShellEurocode_2_2004 Interface

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDConcShellEurocode_2_2004
```

```
Public Interface cDConcShellEurocode_2_2004
```

```
Dim instance As cDConcShellEurocode_2_2004
```

```
public interface class cDConcShellEurocode_2_2004
```

```
type cDConcShellEurocode_2_2004 = interface end
```

▲ See Also

Reference

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cDConcSlabACI318_14 Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDConcSlabACI318_14
```

```
Public Interface cDConcSlabACI318_14
```

```
Dim instance As cDConcSlabACI318_14
```

```
public interface class cDConcSlabACI318_14
```

```
type cDConcSlabACI318_14 = interface end
```

The cDConcSlabACI318_14 type exposes the following members.

▲ Methods

	Name	Description
	GetPreference	Retrieves the value of a concrete slab design preference item

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDConcSlabACI318_14 Methods

The [cDConcSlabACI318_14](#) type exposes the following members.

Methods

	Name	Description
	GetPreference	Retrieves the value of a concrete slab design preference item

[Top](#)

 See Also

Reference

[cDConcSlabACI318_14 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDConcSlabACI318_14AddLanguageSpecificTextSet("I Method

Retrieves the value of a concrete slab design preference item

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetPreference(
    int Item,
    ref string textValue,
    ref double numericValue
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef textValue As String,
    ByRef numericValue As Double
) As Integer
```

```
Dim instance As cDConcSlabACI318_14
Dim Item As Integer
Dim textValue As String
Dim numericValue As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    textValue, numericValue)
```

```
int GetPreference(
    int Item,
    String^% textValue,
    double% numericValue
)
```

```
abstract GetPreference :
    Item : int *
    textValue : string byref *
    numericValue : float byref -> int
```

Parameters

Item

Type: SystemInt32

This is an integer between 1 and 21, inclusive, indicating the preference item considered

1. The resistance factor Phi (tension controlled)

cDConcSlabACI318_14id="LST2E27A186_0"AddLanguageSpecificTextSet("LST2E27A186_19?cpp=::

Introduction

2. The resistance factor Φ (compression controlled)
3. The resistance factor Φ (shear)
4. The clear cover for the top non-prestressed reinforcement [L]
5. The clear cover for the bottom non-prestressed reinforcement [L]
6. The name of the reinforcement bar that is the preferred rebar for non-prestressed reinforcement
7. This is either A or B indicating the design strip layer that represents the inner slab rebar layer
8. The top CGS of the tendon [L]
9. The bottom CGS of the tendon in exterior bays [L]
10. The bottom CGS of the tendon in interior bays [L]
11. This is either One Way or Two Way indicating the slab type assumed when calculating the required minimum reinforcing for the slab
12. Indicates if the allowable PT stresses are user defined
13. The concrete strength ratio at transfer used when calculating initial stresses
14. The top fiber tensile stress ratio used when calculating initial stresses
15. The bottom fiber tensile stress ratio used when calculating initial stresses
16. The extreme fiber compressive stress ratio used when calculating initial stresses
17. The top fiber tensile stress ratio used when calculating final stresses
18. The bottom fiber tensile stress ratio used when calculating final stresses
19. The extreme fiber compressive stress ratio used when calculating final stresses
20. The extreme fiber compressive stress ratio used when calculating sustained stresses
21. The fraction of live load considered used when calculating sustained stresses

textValue

Type: SystemString

If the indicated Item is a text value, this argument will be filled, and numericValue should be ignored

numericValue

Type: SystemDouble

If the indicated Item is a number, this argument will be filled, and textValue will be blank

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value

▲ See Also

Reference

cDConcSlabACI318_14 Interface

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Introduction

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CSI API ETABS v1

cDConcSlabACI318_19 Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDConcSlabACI318_19
```

```
Public Interface cDConcSlabACI318_19
```

```
Dim instance As cDConcSlabACI318_19
```

```
public interface class cDConcSlabACI318_19
```

```
type cDConcSlabACI318_19 = interface end
```

The cDConcSlabACI318_19 type exposes the following members.

▲ Methods

	Name	Description
	GetPreference	Retrieves the value of a concrete slab design preference item

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDConcSlabACI318_19 Methods

The [cDConcSlabACI318_19](#) type exposes the following members.

Methods

	Name	Description
	GetPreference	Retrieves the value of a concrete slab design preference item

[Top](#)

 See Also

Reference

[cDConcSlabACI318_19 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDConcSlabACI318_19AddLanguageSpecificTextSet("Method

Retrieves the value of a concrete slab design preference item

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPreference(
    int Item,
    ref string textValue,
    ref double numericValue
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef textValue As String,
    ByRef numericValue As Double
) As Integer
```

```
Dim instance As cDConcSlabACI318_19
Dim Item As Integer
Dim textValue As String
Dim numericValue As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    textValue, numericValue)
```

```
int GetPreference(
    int Item,
    String^% textValue,
    double% numericValue
)
```

```
abstract GetPreference :
    Item : int *
    textValue : string byref *
    numericValue : float byref -> int
```

Parameters

Item

Type: [System.Int32](#)

This is an integer between 1 and 21, inclusive, indicating the preference item considered

1. The resistance factor Phi (tension controlled)

Introduction

2. The resistance factor Φ (compression controlled)
3. The resistance factor Φ (shear)
4. The clear cover for the top non-prestressed reinforcement [L]
5. The clear cover for the bottom non-prestressed reinforcement [L]
6. The name of the reinforcement bar that is the preferred rebar for non-prestressed reinforcement
7. This is either A or B indicating the design strip layer that represents the inner slab rebar layer
8. The top CGS of the tendon [L]
9. The bottom CGS of the tendon in exterior bays [L]
10. The bottom CGS of the tendon in interior bays [L]
11. This is either One Way or Two Way indicating the slab type assumed when calculating the required minimum reinforcing for the slab
12. Indicates if the allowable PT stresses are user defined
13. The concrete strength ratio at transfer used when calculating initial stresses
14. The top fiber tensile stress ratio used when calculating initial stresses
15. The bottom fiber tensile stress ratio used when calculating initial stresses
16. The extreme fiber compressive stress ratio used when calculating initial stresses
17. The top fiber tensile stress ratio used when calculating final stresses
18. The bottom fiber tensile stress ratio used when calculating final stresses
19. The extreme fiber compressive stress ratio used when calculating final stresses
20. The extreme fiber compressive stress ratio used when calculating sustained stresses
21. The fraction of live load considered used when calculating sustained stresses

textValue

Type: SystemString

If the indicated Item is a text value, this argument will be filled, and numericValue should be ignored

numericValue

Type: SystemDouble

If the indicated Item is a number, this argument will be filled, and textValue will be blank

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value

▲ See Also

Reference

cDConcSlabACI318_19 Interface

ETABSv1 Namespace

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Introduction

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CSI API ETABS v1

cDCoNZS_3101_2006 Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDCoNZS_3101_2006
```

```
Public Interface cDCoNZS_3101_2006
```

```
Dim instance As cDCoNZS_3101_2006
```

```
public interface class cDCoNZS_3101_2006
```

```
type cDCoNZS_3101_2006 = interface end
```

The cDCoNZS_3101_2006 type exposes the following members.

▲ Methods

▲	Name	Description
---	------	-------------

◆	GetOverwrite	
---	------------------------------	--

◆	GetPreference	
---	-------------------------------	--

◆	SetOverwrite	
---	------------------------------	--

◆	SetPreference	
---	-------------------------------	--

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDCoNZS_3101_2006 Methods

The cDCoNZS_3101_2006 type exposes the following members.

Methods

	Name	Description
⇒	<u>GetOverwrite</u>	
⇒	<u>GetPreference</u>	
⇒	<u>SetOverwrite</u>	
⇒	<u>SetPreference</u>	

Top

See Also

Reference

cDCoNZS_3101_2006 Interface

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CSI API ETABS v1

cDConZS_3101_2006AddLanguageSpecificTextSet("LST2A33AEFF_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDConZS_3101_2006
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```

Parameters

Name

Type: SystemString

Item

Type:Â [SystemInt32](#)

Value

Type:Â [SystemDouble](#)

ProgDet

Type:Â [SystemBoolean](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cDCoNZS_3101_2006 Interface](#)

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CSI API ETABS v1

cDCoNZS_3101_2006AddLanguageSpecificTextSet("LST Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDCoNZS_3101_2006
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: [SystemInt32](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDConZS_3101_2006 Interface

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CSI API ETABS v1

cDCoNZS_3101_2006AddLanguageSpecificTextSet("LST6E1F5C2C_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDCoNZS_3101_2006
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

Item

Type: SystemInt32

Value

Type: SystemDouble

ItemType (Optional)

Type: ETABSv1eItemType

Return Value

Type: Int32

▲ See Also

Reference

cDCoNZS_3101_2006 Interface

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CSI API ETABS v1

cDConZS_3101_2006AddLanguageSpecificTextSet("LST8FB24733_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDConZS_3101_2006
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: [SystemInt32](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDConZS_3101_2006 Interface

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CSI API ETABS v1

cDCoSP63133302011 Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDCoSP63133302011
```

```
Public Interface cDCoSP63133302011
```

```
Dim instance As cDCoSP63133302011
```

```
public interface class cDCoSP63133302011
```

```
type cDCoSP63133302011 = interface end
```

The cDCoSP63133302011 type exposes the following members.

▲ Methods

▲	Name	Description
	GetOverwrite	Retrieves the value of a concrete design overwrite item.
	GetPreference	Retrieves the value of a concrete design preference item.
	SetOverwrite	Sets the value of a concrete design overwrite item.
	SetPreference	Sets the value of a concrete design preference item.

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▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDCoSP63133302011 Methods

The [cDCoSP63133302011](#) type exposes the following members.

Methods

	Name	Description
	GetOverwrite	Retrieves the value of a concrete design overwrite item.
	GetPreference	Retrieves the value of a concrete design preference item.
	SetOverwrite	Sets the value of a concrete design overwrite item.
	SetPreference	Sets the value of a concrete design preference item.

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See Also

Reference

[cDCoSP63133302011 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDCoSP63133302011AddLanguageSpecificTextSet("LS Method

Retrieves the value of a concrete design overwrite item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDCoSP63133302011
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```

Parameters

Name

Type: SystemString

The name of a frame object with a concrete frame design procedure.

Item

Type: SystemInt32

This is an integer between 1 and 22, inclusive, indicating the overwrite item considered.

1. Framing type (beams and columns)
2. Live load reduction factor (beams and columns)
3. Unbraced length ratio, Major (beams and columns)
4. Unbraced length ratio, Minor (beams and columns)
5. Effective length factor, K Major (columns only)
6. Effective length factor, K Minor (columns only)
7. Moment amplification factor, Eta Major (columns only)
8. Moment amplification factor, Eta Minor (columns only)
9. Gammab3 for column (columns only)
10. Gammab3 for beam (beams only)
11. Consider torsion (beams only)
12. $(q_{sw}, 1 * Z_1) / (R_s * A_s, 1)$ (beams only)
13. Corner rebar fraction top (beams only)
14. Corner rebar fraction bottom (beams only)
15. Consider minimum eccentricity (columns only)
16. Consider crack analysis (beams only)
17. Crack width limit full load, $a_{crc,ult}$ [mm] (beams only)
18. Crack width limit long term, $a_{crc,l,ult}$ [mm] (beams only)
19. Longitudinal rebar size top (beams only)
20. Longitudinal rebar size bottom (beams only)
21. Ignore beneficial P_u for beam design (beams only)
22. Is longitudinal rebar ribbed (beams only)

Value

Type: SystemDouble

The value of the considered overwrite item.

1. Framing type
 - 0 = Program Default
 - 1 = Sway
 - 2 = NonSway
2. Live load reduction factor
 - Value ≥ 0 ; 0 means use program determined value
3. Unbraced length ratio, Major
 - Value ≥ 0 ; 0 means use program determined value
4. Unbraced length ratio, Minor
 - Value ≥ 0 ; 0 means use program determined value
5. Effective length factor, K Major
 - Value ≥ 0 ; 0 means use program determined value
6. Effective length factor, K Minor
 - Value ≥ 0 ; 0 means use program determined value

Introduction

Value ≥ 0 ; 0 means use program determined value

7. Moment amplification factor, Eta Major

Value ≥ 0 ; 0 means use program determined value

8. Moment amplification factor, Eta Minor

Value ≥ 0 ; 0 means use program determined value

9. Gammab3 for column

Value ≥ 0 ; 0 means use program determined value

10. Gammab3 for beam

Value ≥ 0 ; 0 means use program determined value

11. Consider torsion

· 0 = No

· Any other value = Yes

12. $(q_{sw,1} * Z_1) / (R_s * A_{s,1})$

Value ≥ 0 ; 0 means use program determined value

13. Corner rebar fraction top

Value ≥ 0 ; 0 means use program determined value

14. Corner rebar fraction bottom

Value ≥ 0 ; 0 means use program determined value

15. Consider minimum eccentricity

· 0 = No

· Any other value = Yes

16. Consider crack analysis

· 0 = No

· Any other value = Yes

17. Crack Width Limit Full Load, $a_{crc,ult}$ [mm]

Value ≥ 0 ; 0 means use program determined value

18. Crack Width Limit Long Term, $a_{crc,l,ult}$ [mm]

Value ≥ 0 ; 0 means use program determined value

19. Longitudinal Rebar Size Top

Value is the index in the list of the rebar defined in the Reinforcing Bar Sizes form

20. Longitudinal Rebar Size Bottom

Value is the index in the list of the rebar defined in the Reinforcing Bar Sizes form

21. Ignore Beneficial Pu for Beam Design

· 0 = No

· Any other value = Yes

22. Is Longitudinal Rebar Ribbed

· 0 = No

· Any other value = Yes

ProgDet

Type: SystemBoolean

If this item is True, the specified value is program determined.

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double
    Dim ProgDet As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'create new concrete frame section property
    ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

    'set frame section property
    ret = SapModel.FrameObj.SetSection("8", "R1")

    'set concrete design code
    ret = SapModel.DesignConcrete.SetCode("SP 63.13330.2012")

    'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis

    'start concrete design
    ret = SapModel.DesignConcrete.StartDesign()

    'get overwrite item
    ret = SapModel.DesignConcrete.SP63133302011.GetOverwrite("8", 1, Value, ProgDet)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
```

Return Value

```
EtabsObject = Nothing
```

```
End Sub
```

▲ See Also

Reference

cDCoSP63133302011 Interface

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CSI API ETABS v1

cDCoSP63133302011AddLanguageSpecificTextSet("LS Method

Retrieves the value of a concrete design preference item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDCoSP63133302011
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: SystemInt32

This is an integer between 1 and 47, inclusive, indicating the preference item considered.

1. Multi-response case design
2. Number of interaction curves
3. Number of interaction points
4. Consider minimum eccentricity
5. Consider torsion

6. $(q_{sw,1} \cdot Z_1) / (R_s \cdot A_s,1)$
7. Corner rebar fraction top
8. Corner rebar fraction bottom
9. Relative Humidity (%)
10. Moisture Content (fraction)
11. Γ_b
12. Γ_{bt}
13. Γ_{b1} Short Term
14. Γ_{b1} Long Term
15. Γ_{b2}
16. Γ_{b3} Beams
17. Γ_{b3} Columns
18. Γ_{b4}
19. Γ_{b5}
20. Γ_S
21. Γ_{S1}
22. Pattern Live Load Factor
23. Utilization Factor Limit
24. Live Load Duration Factor
25. Snow Load Duration Factor
26. Reliability Factor, Γ_n
27. Seismic Factor, $m_{tr,strength}$
28. Seismic Factor, $m_{tr,shear}$
29. Site Seismicity
31. Ignore Beneficial P_u for Beam Design
32. Consider Crack Analysis
33. Crack Width Limit Full Load, $a_{crc,ult}$ (mm)
34. Crack Width Limit Long Term, $a_{crc,l,ult}$ (mm)
35. Longitudinal Rebar Size Top
36. Longitudinal Rebar Size Bottom
37. Is Longitudinal Rebar Ribbed
38. User Defined Allowable PT Stresses
39. Concrete Strength Ratio at Transfer f'_{ci} / f'_c
40. Transfer: Top Fiber Tensile Stress / $f'_{ci}^{(1/2)}$
41. Transfer: Bottom Fiber Tensile Stress / $f'_{ci}^{(1/2)}$
42. Transfer: Extreme Fiber Compressive Stress / f'_{ci}
43. Final: Top Fiber Tensile Stress / $f'_c^{(1/2)}$
44. Final: Bottom Fiber Tensile Stress / $f'_c^{(1/2)}$
45. Final: Extreme Fiber Compressive Stress / f'_c
46. Sustained: Extreme Fiber Compressive Stress / f'_c
47. Sustained: Fraction of Live Load Considered

Value

Type: SystemDouble

The value of the considered preference item.

1. Multi-response case design
 - 1 = Envelopes
 - 2 = Step-by-Step
 - 3 = Last Step
 - 4 = Envelope -- All
 - 5 = Step-by-Step -- All
2. Number of interaction curves

Value ≥ 4 and divisible by 4
 3. Number of interaction points

Value ≥ 5 and odd
 4. Consider minimum eccentricity
 · 0 = No
 · Any other value = Yes
 5. Consider torsion
 · 0 = No
 · Any other value = Yes
 6. $(q_{sw,1} \cdot Z_1) / (R_s \cdot A_{s,1})$

Value > 0
 7. Corner rebar fraction top

Value > 0
 8. Corner rebar fraction bottom

Value > 0
 9. Relative Humidity (%)

Value > 0
 10. Moisture Content (fraction)

Value > 0
 11. Γ_b

Value > 0
 12. Γ_{bt}

Value > 0
 13. Γ_{b1} Short Term

Value > 0
 14. Γ_{b1} Long Term

Value > 0
 15. Γ_{b2}

Value > 0
 16. Γ_{b3} Beams

Value > 0
 17. Γ_{b3} Columns

Value > 0
 18. Γ_{b4}

Value > 0
 19. Γ_{b5}

- Value > 0
20. Gamma_b4
- Value > 0
21. Gamma_S
- Value > 0
22. Gamma_S1
- Value > 0
23. Pattern Live Load Factor
- Value > 0
24. Utilization Factor Limit
- Value > 0
25. Live Load Duration Factor
- Value > 0
26. Snow Load Duration Factor
- Value > 0
27. Reliability Factor, Gamma_n
- Value > 0
28. Seismic Factor, m_tr,strength
- Value > 0
29. Seismic Factor, m_tr,shear
- Value > 0
30. Site Seismicity
 · 1 = Site Seismicity 9
 · 2 = Site Seismicity 8
 · 3 = Site Seismicity 7
 · 4 = Non-seismic
31. Ignore Beneficial Pu for Beam Design
 · 0 = No
 · Any other value = Yes
32. Consider Crack Analysis
 · 0 = No
 · Any other value = Yes
33. Crack Width Limit Full Load, a_crc,ult (mm)
- Value > 0
34. Crack Width Limit Long Term, a_crc,l,ult (mm)
- Value > 0
35. Longitudinal Rebar Size Top

Introduction

Value is the index in the list of the rebar defined in the Reinforcing Bar Sizes form

36. Longitudinal Rebar Size Bottom

Value is the index in the list of the rebar defined in the Reinforcing Bar Sizes form

37. Is Longitudinal Rebar Ribbed

- 0 = No
- Any other value = Yes

38. User Defined Allowable PT Stresses

- 0 = No
- Any other value = Yes

39. Concrete Strength Ratio at Transfer f'_{ci} / f'_c

Value > 0

40. Transfer: Top Fiber Tensile Stress / $f'_{ci}^{(1/2)}$

Value > 0

41. Transfer: Bottom Fiber Tensile Stress / $f'_{ci}^{(1/2)}$

Value > 0

42. Transfer: Extreme Fiber Compressive Stress / f'_{ci}

Value > 0

43. Final: Top Fiber Tensile Stress / $f'_c^{(1/2)}$

Value > 0

44. Final: Bottom Fiber Tensile Stress / $f'_c^{(1/2)}$

Value > 0

45. Final: Extreme Fiber Compressive Stress / f'_c

Value > 0

46. Sustained: Extreme Fiber Compressive Stress / f'_c

Value > 0

47. Sustained: Fraction of Live Load Considered

Value > 0

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1
```


Introduction

Dim Value As Double

```
'create ETABS object
  Dim myHelper as cHelper = New Helper
  EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
  ret = EtabsObject.ApplicationStart()

'create SapModel object
  SapModel = EtabsObject.SapModel

'initialize model
  ret = SapModel.InitializeNewModel()

'create steel deck template model
  ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'create new concrete frame section property
  ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

'set frame section property
  ret = SapModel.FrameObj.SetSection("8", "R1")

'set concrete design code
  ret = SapModel.DesignConcrete.SetCode("SP 63.13330.2012")

'run analysis
  System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
  ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
  ret = SapModel.Analyze.RunAnalysis

'start concrete design
  ret = SapModel.DesignConcrete.StartDesign()

'get preference item
  ret = SapModel.DesignConcrete.SP63133302011.GetPreference(35, Value)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
```

End Sub

▲ See Also

Reference

cDCoSP63133302011 Interface

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CSI API ETABS v1

cDCoSP63133302011AddLanguageSpecificTextSet("LS Method

Sets the value of a concrete design overwrite item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDCoSP63133302011
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters**Name**

Type: SystemString

The name of an existing frame object or group, depending on the value of the ItemType item.

Item

Type: SystemInt32

This is an integer between 1 and 22, inclusive, indicating the overwrite item considered.

1. Framing type (beams and columns)
2. Live load reduction factor (beams and columns)
3. Unbraced length ratio, Major (beams and columns)
4. Unbraced length ratio, Minor (beams and columns)
5. Effective length factor, K Major (columns only)
6. Effective length factor, K Minor (columns only)
7. Moment amplification factor, Eta Major (columns only)
8. Moment amplification factor, Eta Minor (columns only)
9. Gammab3 for column (columns only)
10. Gammab3 for beam (beams only)
11. Consider torsion (beams only)
12. $(q_{sw}, 1 * Z_1) / (R_s * A_s, 1)$ (beams only)
13. Corner rebar fraction top (beams only)
14. Corner rebar fraction bottom (beams only)
15. Consider minimum eccentricity (columns only)
16. Consider crack analysis (beams only)
17. Crack width limit full load, $a_{crc,ult}$ [mm] (beams only)
18. Crack width limit long term, $a_{crc,l,ult}$ [mm] (beams only)
19. Longitudinal rebar size top (beams only)
20. Longitudinal rebar size bottom (beams only)
21. Ignore beneficial P_u for beam design (beams only)
22. Is longitudinal rebar ribbed (beams only)

Value

Type: SystemDouble

The value of the considered overwrite item.

1. Framing type
 - 0 = Program Default
 - 1 = Sway
 - 2 = NonSway
2. Live load reduction factor

Value ≥ 0 ; 0 means use program determined value
3. Unbraced length ratio, Major

Value ≥ 0 ; 0 means use program determined value
4. Unbraced length ratio, Minor

Value ≥ 0 ; 0 means use program determined value
5. Effective length factor, K Major

Value ≥ 0 ; 0 means use program determined value

6. Effective length factor, K Minor

Value ≥ 0 ; 0 means use program determined value

7. Moment amplification factor, Eta Major

Value ≥ 0 ; 0 means use program determined value

8. Moment amplification factor, Eta Minor

Value ≥ 0 ; 0 means use program determined value

9. Gammab3 for column

Value ≥ 0 ; 0 means use program determined value

10. Gammab3 for beam

Value ≥ 0 ; 0 means use program determined value

11. Consider torsion

· 0 = No

· Any other value = Yes

12. $(q_{sw,1} \cdot Z_1) / (R_s \cdot A_s, 1)$

Value ≥ 0 ; 0 means use program determined value

13. Corner rebar fraction top

Value ≥ 0 ; 0 means use program determined value

14. Corner rebar fraction bottom

Value ≥ 0 ; 0 means use program determined value

15. Consider minimum eccentricity

· 0 = No

· Any other value = Yes

16. Consider crack analysis

· 0 = No

· Any other value = Yes

17. Crack Width Limit Full Load, $a_{crc,ult}$ [mm]

Value ≥ 0 ; 0 means use program determined value

18. Crack Width Limit Long Term, $a_{crc,l,ult}$ [mm]

Value ≥ 0 ; 0 means use program determined value

19. Longitudinal Rebar Size Top

Value is the index in the list of the rebar defined in the Reinforcing Bar Sizes form

20. Longitudinal Rebar Size Bottom

Value is the index in the list of the rebar defined in the Reinforcing Bar Sizes form

21. Ignore Beneficial Pu for Beam Design

· 0 = No

· Any other value = Yes

22. Is Longitudinal Rebar Ribbed

- 0 = No
- Any other value = Yes

ItemType (Optional)

Type: $\hat{A}$ ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

- ◊ Object = 0
- ◊ Group = 1
- ◊ SelectedObjects = 2

If this item is Objects, the assignment is made to the frame object specified by the Name item.

If this item is Group, the assignment is made to all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the assignment is made to all selected frame objects and the Name item is ignored.

Return Value

Type: $\hat{A}$ Int32

Returns zero if the item is successfully set; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'create new concrete frame section property
    ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

    'set frame section property
    ret = SapModel.FrameObj.SetSection("8", "R1")

    'set concrete design code
    ret = SapModel.DesignConcrete.SetCode("SP 63.13330.2012")

    'set overwrite item
    ret = SapModel.DesignConcrete.SP63133302011.SetOverwrite("8", 1, 2)
```

Return Value

```
'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

📌 See Also

Reference

cDCoSP63133302011 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDCoSP63133302011AddLanguageSpecificTextSet("LS Method

Sets the value of a concrete design preference item.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDCoSP63133302011
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: [SystemInt32](#)

This is an integer between 1 and 47, inclusive, indicating the preference item considered.

1. Multi-response case design
2. Number of interaction curves
3. Number of interaction points
4. Consider minimum eccentricity
5. Consider torsion

6. $(q_{sw,1} \cdot Z_1) / (R_s \cdot A_s,1)$
7. Corner rebar fraction top
8. Corner rebar fraction bottom
9. Relative Humidity (%)
10. Moisture Content (fraction)
11. Γ_b
12. Γ_{bt}
13. Γ_{b1} Short Term
14. Γ_{b1} Long Term
15. Γ_{b2}
16. Γ_{b3} Beams
17. Γ_{b3} Columns
18. Γ_{b4}
19. Γ_{b5}
20. Γ_S
21. Γ_{S1}
22. Pattern Live Load Factor
23. Utilization Factor Limit
24. Live Load Duration Factor
25. Snow Load Duration Factor
26. Reliability Factor, Γ_n
27. Seismic Factor, $m_{tr,strength}$
28. Seismic Factor, $m_{tr,shear}$
29. Site Seismicity
31. Ignore Beneficial P_u for Beam Design
32. Consider Crack Analysis
33. Crack Width Limit Full Load, $a_{crc,ult}$ (mm)
34. Crack Width Limit Long Term, $a_{crc,l,ult}$ (mm)
35. Longitudinal Rebar Size Top
36. Longitudinal Rebar Size Bottom
37. Is Longitudinal Rebar Ribbed
38. User Defined Allowable PT Stresses
39. Concrete Strength Ratio at Transfer f'_{ci} / f'_c
40. Transfer: Top Fiber Tensile Stress / $f'_{ci}^{(1/2)}$
41. Transfer: Bottom Fiber Tensile Stress / $f'_{ci}^{(1/2)}$
42. Transfer: Extreme Fiber Compressive Stress / f'_{ci}
43. Final: Top Fiber Tensile Stress / $f'_c^{(1/2)}$
44. Final: Bottom Fiber Tensile Stress / $f'_c^{(1/2)}$
45. Final: Extreme Fiber Compressive Stress / f'_c
46. Sustained: Extreme Fiber Compressive Stress / f'_c
47. Sustained: Fraction of Live Load Considered

Value

Type: SystemDouble

The value of the considered preference item.

1. Multi-response case design
 - 1 = Envelopes
 - 2 = Step-by-Step
 - 3 = Last Step
 - 4 = Envelopes -- All
 - 5 = Step-by-Step -- All
2. Number of interaction curves

Value ≥ 4 and divisible by 4
 3. Number of interaction points

Value ≥ 5 and odd
 4. Consider minimum eccentricity
 · 0 = No
 · Any other value = Yes
 5. Consider torsion
 · 0 = No
 · Any other value = Yes
 6. $(q_{sw,1} \cdot Z_1) / (R_s \cdot A_{s,1})$

Value > 0
 7. Corner rebar fraction top

Value > 0
 8. Corner rebar fraction bottom

Value > 0
 9. Relative Humidity (%)

Value > 0
 10. Moisture Content (fraction)

Value > 0
 11. Γ_b

Value > 0
 12. Γ_{bt}

Value > 0
 13. Γ_{b1} Short Term

Value > 0
 14. Γ_{b1} Long Term

Value > 0
 15. Γ_{b2}

Value > 0
 16. Γ_{b3} Beams

Value > 0
 17. Γ_{b3} Columns

Value > 0
 18. Γ_{b4}

Value > 0
 19. Γ_{b5}

- Value > 0
20. Gamma_b4
- Value > 0
21. Gamma_S
- Value > 0
22. Gamma_S1
- Value > 0
23. Pattern Live Load Factor
- Value > 0
24. Utilization Factor Limit
- Value > 0
25. Live Load Duration Factor
- Value > 0
26. Snow Load Duration Factor
- Value > 0
27. Reliability Factor, Gamma_n
- Value > 0
28. Seismic Factor, m_tr,strength
- Value > 0
29. Seismic Factor, m_tr,shear
- Value > 0
30. Site Seismicity
 · 1 = Site Seismicity 9
 · 2 = Site Seismicity 8
 · 3 = Site Seismicity 7
 · 4 = Non-seismic
31. Ignore Beneficial Pu for Beam Design
 · 0 = No
 · Any other value = Yes
32. Consider Crack Analysis
 · 0 = No
 · Any other value = Yes
33. Crack Width Limit Full Load, a_crc,ult (mm)
- Value > 0
34. Crack Width Limit Long Term, a_crc,l,ult (mm)
- Value > 0
35. Longitudinal Rebar Size Top

Introduction

Value is the index in the list of the rebar defined in the Reinforcing Bar Sizes form

36. Longitudinal Rebar Size Bottom

Value is the index in the list of the rebar defined in the Reinforcing Bar Sizes form

37. Is Longitudinal Rebar Ribbed

- 0 = No
- Any other value = Yes

38. User Defined Allowable PT Stresses

- 0 = No
- Any other value = Yes

39. Concrete Strength Ratio at Transfer f'_{ci} / f'_c

Value > 0

40. Transfer: Top Fiber Tensile Stress / $f'_{ci}^{(1/2)}$

Value > 0

41. Transfer: Bottom Fiber Tensile Stress / $f'_{ci}^{(1/2)}$

Value > 0

42. Transfer: Extreme Fiber Compressive Stress / f'_{ci}

Value > 0

43. Final: Top Fiber Tensile Stress / $f'_c^{(1/2)}$

Value > 0

44. Final: Bottom Fiber Tensile Stress / $f'_c^{(1/2)}$

Value > 0

45. Final: Extreme Fiber Compressive Stress / f'_c

Value > 0

46. Sustained: Extreme Fiber Compressive Stress / f'_c

Value > 0

47. Sustained: Fraction of Live Load Considered

Value > 0

Return Value

Type: Int32

Returns zero if the item is successfully set; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1
```

Introduction

```
'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
    ret = EtabsObject.ApplicationStart()

'create SapModel object
    SapModel = EtabsObject.SapModel

'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'create new concrete frame section property
    ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

'set frame section property
    ret = SapModel.FrameObj.SetSection("8", "R1")

'set concrete design code
    ret = SapModel.DesignConcrete.SetCode("SP 63.13330.2012")

'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis

'start concrete design
    ret = SapModel.DesignConcrete.StartDesign()

'set preference item
    ret = SapModel.DesignConcrete.SP63133302011.SetPreference(35, 9)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

[cDCoSP63133302011 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDCoTS_500_2000 Interface

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDCoTS_500_2000
```

```
Public Interface cDCoTS_500_2000
```

```
Dim instance As cDCoTS_500_2000
```

```
public interface class cDCoTS_500_2000
```

```
type cDCoTS_500_2000 = interface end
```

The cDCoTS_500_2000 type exposes the following members.

▲ Methods

▲	Name	Description
---	------	-------------

◆	GetOverwrite	
---	------------------------------	--

◆	GetPreference	
---	-------------------------------	--

◆	SetOverwrite	
---	------------------------------	--

◆	SetPreference	
---	-------------------------------	--

[Top](#)

▲ See Also

Reference

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cDCoTS_500_2000 Methods

The cDCoTS_500_2000 type exposes the following members.

Methods

	Name	Description
⇒	<u>GetOverwrite</u>	
⇒	<u>GetPreference</u>	
⇒	<u>SetOverwrite</u>	
⇒	<u>SetPreference</u>	

Top

See Also

Reference

cDCoTS_500_2000 Interface

ETABSV1 Namespace

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CSI API ETABS v1

cDCoTS_500_2000AddLanguageSpecificTextSet("LSTC Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDCoTS_500_2000
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```

Parameters

Name

Type: SystemString

cDCoTS_500_2000AddLanguageSpecificTextSet("LSTC4A26CBE1063pp=::|n

Item

Type:Â [SystemInt32](#)

Value

Type:Â [SystemDouble](#)

ProgDet

Type:Â [SystemBoolean](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cDCoTS_500_2000 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDCoTS_500_2000AddLanguageSpecificTextSet("LST5 Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDCoTS_500_2000
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: [SystemInt32](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDCoTS_500_2000 Interface

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CSI API ETABS v1

cDCoTS_500_2000AddLanguageSpecificTextSet("LST500_2000") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDCoTS_500_2000
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

Item

Type: SystemInt32

Value

Type: SystemDouble

ItemType (Optional)

Type: ETABSv1eItemType

Return Value

Type: Int32

▲ See Also

Reference

cDCoTS_500_2000 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDCoTS_500_2000AddLanguageSpecificTextSet("LSTC Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDCoTS_500_2000
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: [SystemInt32](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDCoTS_500_2000 Interface

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CSI API ETABS v1

cDCoTS_500_2000_R2018 Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDCoTS_500_2000_R2018
```

```
Public Interface cDCoTS_500_2000_R2018
```

```
Dim instance As cDCoTS_500_2000_R2018
```

```
public interface class cDCoTS_500_2000_R2018
```

```
type cDCoTS_500_2000_R2018 = interface end
```

The cDCoTS_500_2000_R2018 type exposes the following members.

▲ Methods

▲	Name	Description
	<u>GetOverwrite</u>	Retrieves the value of a concrete design overwrite item.
	<u>GetPreference</u>	Retrieves the value of a concrete design preference item.
	<u>SetOverwrite</u>	Sets the value of a concrete design overwrite item.
	<u>SetPreference</u>	Sets the value of a concrete design preference item.

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDCoTS_500_2000_R2018 Methods

The [cDCoTS_500_2000_R2018](#) type exposes the following members.

Methods

	Name	Description
	<u>GetOverwrite</u>	Retrieves the value of a concrete design overwrite item.
	<u>GetPreference</u>	Retrieves the value of a concrete design preference item.
	<u>SetOverwrite</u>	Sets the value of a concrete design overwrite item.
	<u>SetPreference</u>	Sets the value of a concrete design preference item.

[Top](#)

 See Also

Reference

[cDCoTS_500_2000_R2018 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDCoTS_500_2000_R2018AddLanguageSpecificTextSet Method

Retrieves the value of a concrete design overwrite item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDCoTS_500_2000_R2018
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```

Parameters

Name

Type: SystemString

The name of a frame object with a concrete frame design procedure.

Item

Type: SystemInt32

This is an integer between 1 and 13, inclusive, indicating the overwrite item considered.

1. Framing type
2. Live load reduction factor
3. Unbraced length ratio, Major
4. Unbraced length ratio, Minor
5. Effective length factor, K Major
6. Effective length factor, K Minor
7. Moment coefficient, Cm Major
8. Moment coefficient, Cm Minor
9. Nonsway moment factor, Db Major
10. Nonsway moment factor, Db Minor
11. Sway moment factor, Ds Major
12. Sway moment factor, Ds Minor
13. Consider Minimum Eccentricity

Value

Type: SystemDouble

The value of the considered overwrite item.

1. Framing type
 - 0 = Program Default
 - 1 = High Ductile
 - 2 = Nominal Ductile
 - 3 = Ordinary
 - 4 = Non-sway
2. Live load reduction factor

Value >= 0; 0 means use program determined value
3. Unbraced length ratio, Major

Value >= 0; 0 means use program determined value
4. Unbraced length ratio, Minor

Value >= 0; 0 means use program determined value
5. Effective length factor, K Major

Value >= 0; 0 means use program determined value
6. Effective length factor, K Minor

Value >= 0; 0 means use program determined value
7. Moment coefficient, Cm Major

Value >= 0; 0 means use program determined value
8. Moment coefficient, Cm Minor

Value >= 0; 0 means use program determined value

Introduction

Value ≥ 0 ; 0 means use program determined value

9. Nonsway moment factor, Db Major

Value ≥ 0 ; 0 means use program determined value

10. Nonsway moment factor, Db Minor

Value ≥ 0 ; 0 means use program determined value

11. Sway moment factor, Ds Major

Value ≥ 0 ; 0 means use program determined value

12. Sway moment factor, Ds Minor

Value ≥ 0 ; 0 means use program determined value

13. Consider Minimum Eccentricity

0 = No

Any other value = Yes

ProgDet

Type: SystemBoolean

If this item is True, the specified value is program determined.

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ See Also

Reference

cDCoTS_500_2000_R2018 Interface

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CSI API ETABS v1

cDCoTS_500_2000_R2018AddLanguageSpecificTextSet

Method

Retrieves the value of a concrete design preference item.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDCoTS_500_2000_R2018
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: SystemInt32

This is an integer between 1 and 9, inclusive, indicating the preference item considered.

1. Number of interaction curves
2. Number of interaction points
3. Consider minimum eccentricity
4. Design for B/C Capacity Ratio?
5. Gamma steel

- 6. Gamma concrete
- 7. Pattern live load factor
- 8. Utilization factor limit
- 9. Multi-response case design

Value

Type: SystemDouble

The value of the considered preference item.

- 1. Number of interaction curves

Value ≥ 4 and divisible by 4

- 2. Number of interaction points

Value ≥ 5 and odd

- 3. Consider minimum eccentricity

0 = No

Any other value = Yes

- 4. Design for B/C Capacity Ratio?

1 = No

2 = Yes

- 5. Gamma steel

Value > 0

- 6. Gamma concrete

Value > 0

- 7. Pattern live load factor

Value ≥ 0

- 8. Utilization factor limit

Value > 0

- 9. Multi-response case design

- 1 = Envelopes
- 2 = Step-by step
- 3 = Last step
- 4 = Envelopes - All
- 5 = Step-by step - All

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ See Also

Reference

cDCoTS_500_2000_R2018 Interface

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CSI API ETABS v1

cDCoTS_500_2000_R2018AddLanguageSpecificTextSet Method

Sets the value of a concrete design overwrite item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDCoTS_500_2000_R2018
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters**Name**Type:Â SystemString

The name of an existing frame object or group, depending on the value of the ItemType

ItemType:Â SystemInt32

This is an integer between 1 and 13, inclusive, indicating the overwrite item considered.

1. Framing type
2. Live load reduction factor
3. Unbraced length ratio, Major
4. Unbraced length ratio, Minor
5. Effective length factor, K Major
6. Effective length factor, K Minor
7. Moment coefficient, Cm Major
8. Moment coefficient, Cm Minor
9. Nonsway moment factor, Db Major
10. Nonsway moment factor, Db Minor
11. Sway moment factor, Ds Major
12. Sway moment factor, Ds Minor
13. Consider Minimum Eccentricity

ValueType:Â SystemDouble

The value of the considered overwrite item.

1. Framing type
 - 0 = Program Default
 - 1 = High Ductile
 - 2 = Nominal Ductile
 - 3 = Ordinary
 - 4 = Non-sway
2. Live load reduction factor

Value >= 0; 0 means use program determined value
3. Unbraced length ratio, Major

Value >= 0; 0 means use program determined value
4. Unbraced length ratio, Minor

Value >= 0; 0 means use program determined value
5. Effective length factor, K Major

Value >= 0; 0 means use program determined value
6. Effective length factor, K Minor

Value >= 0; 0 means use program determined value
7. Moment coefficient, Cm Major

Value >= 0; 0 means use program determined value
8. Moment coefficient, Cm Minor

Value >= 0; 0 means use program determined value

Introduction

Value ≥ 0 ; 0 means use program determined value

9. Nonsway moment factor, Db Major

Value ≥ 0 ; 0 means use program determined value

10. Nonsway moment factor, Db Minor

Value ≥ 0 ; 0 means use program determined value

11. Sway moment factor, Ds Major

Value ≥ 0 ; 0 means use program determined value

12. Sway moment factor, Ds Minor

Value ≥ 0 ; 0 means use program determined value

13. Consider Minimum Eccentricity

0 = No

Any other value = Yes

ItemType (Optional)

Type: $\hat{A}$ ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the overwrite is set for the frame object specified by the Name item.

If this item is Group, the overwrite is set for all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the overwrite is set for all selected frame objects and the Name item is ignored.

Return Value

Type: $\hat{A}$ Int32

Returns zero if the item is successfully set; otherwise it returns a nonzero value.

▲ See Also

Reference

cDCoTS_500_2000_R2018 Interface

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CSI API ETABS v1

cDCoTS_500_2000_R2018AddLanguageSpecificTextSet Method

Sets the value of a concrete design preference item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDCoTS_500_2000_R2018
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: SystemInt32

This is an integer between 1 and 9, inclusive, indicating the preference item considered.

1. Number of interaction curves
2. Number of interaction points
3. Consider minimum eccentricity
4. Design for B/C Capacity Ratio?
5. Gamma steel

- 6. Gamma concrete
- 7. Pattern live load factor
- 8. Utilization factor limit
- 9. Multi-response case design

Value

Type: SystemDouble

The value of the considered preference item.

- 1. Number of interaction curves

Value ≥ 4 and divisible by 4

- 2. Number of interaction points

Value ≥ 5 and odd

- 3. Consider minimum eccentricity

0 = No

Any other value = Yes

- 4. Design for B/C Capacity Ratio?

1 = No

2 = Yes

- 5. Gamma steel

Value > 0

- 6. Gamma concrete

Value > 0

- 7. Pattern live load factor

Value ≥ 0

- 8. Utilization factor limit

Value > 0

- 9. Multi-response case design

- 1 = Envelopes
- 2 = Step-by step
- 3 = Last step
- 4 = Envelopes - All
- 5 = Step-by step - All

Return Value

Type: Int32

Returns zero if the item is successfully set; otherwise it returns a nonzero value.

▲ See Also

Reference

cDCoTS_500_2000_R2018 Interface

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CSI API ETABS v1

cDesignCompositeBeam Interface

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
public interface cDesignCompositeBeam
```

```
Public Interface cDesignCompositeBeam
```

```
Dim instance As cDesignCompositeBeam
```

```
public interface class cDesignCompositeBeam
```

```
type cDesignCompositeBeam = interface end
```

The cDesignCompositeBeam type exposes the following members.

▲ Methods

$\hat{A}$	Name	Description
	<u>DeleteResults</u>	
	<u>GetCode</u>	Retrieves the composite beam design code.
	<u>GetComboDeflection</u>	
	<u>GetComboStrength</u>	
	<u>GetDesignSection</u>	
	<u>GetGroup</u>	
	<u>GetResultsAvailable</u>	This function determines if the composite beam design results are available.
	<u>GetSummaryResults</u>	
	<u>GetTargetDispl</u>	
	<u>GetTargetPeriod</u>	
	<u>ResetOverwrites</u>	
	<u>SetAutoSelectNull</u>	
	<u>SetCode</u>	Sets the composite beam design code.
	<u>SetComboDeflection</u>	
	<u>SetComboStrength</u>	
	<u>SetDesignSection</u>	
	<u>SetGroup</u>	
	<u>SetTargetDispl</u>	

⇒ [SetTargetPeriod](#)

⇒ [StartDesign](#) Starts the composite beam design.

⇒ [VerifyPassed](#)

⇒ [VerifySections](#)

[Top](#)

🔍 See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignCompositeBeam Methods

The [cDesignCompositeBeam](#) type exposes the following members.

Methods

Á	Name	Description
⇒	DeleteResults	
⇒	GetCode	Retrieves the composite beam design code.
⇒	GetComboDeflection	
⇒	GetComboStrength	
⇒	GetDesignSection	
⇒	GetGroup	
⇒	GetResultsAvailable	This function determines if the composite beam design results are available.
⇒	GetSummaryResults	
⇒	GetTargetDispl	
⇒	GetTargetPeriod	
⇒	ResetOverwrites	
⇒	SetAutoSelectNull	
⇒	SetCode	Sets the composite beam design code.
⇒	SetComboDeflection	
⇒	SetComboStrength	
⇒	SetDesignSection	
⇒	SetGroup	
⇒	SetTargetDispl	
⇒	SetTargetPeriod	
⇒	StartDesign	Starts the composite beam design.
⇒	VerifyPassed	
⇒	VerifySections	

[Top](#)

See Also

Reference

[cDesignCompositeBeam Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignCompositeBeamAddLanguageSpecificTextSet Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

int DeleteResults()

Function DeleteResults As Integer

Dim instance As cDesignCompositeBeam

Dim returnValue As Integer

returnValue = instance.DeleteResults()

int DeleteResults()

abstract DeleteResults : unit -> int

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cDesignCompositeBeam Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignCompositeBeamAddLanguageSpecificTextSet(

Method

Retrieves the composite beam design code.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetCode(
    ref string CodeName
)
```

```
Function GetCode (
    ByRef CodeName As String
) As Integer
```

```
Dim instance As cDesignCompositeBeam
Dim CodeName As String
Dim returnValue As Integer
```

```
returnValue = instance.GetCode(CodeName)
```

```
int GetCode(
    String^% CodeName
)
```

```
abstract GetCode :
    CodeName : string byref -> int
```

Parameters

CodeName

Type: SystemString

This is one of the following composite beam design code names.

- ◇ AISC 360-10
- ◇ AISC 360-05
- ◇ BS 5950-1990
- ◇ Chinese 2010
- ◇ CSA S16-14
- ◇ CSA S16-09
- ◇ Eurocode 4-2004

Return Value

Type: Int32

Returns zero if the code is successfully retrieved; otherwise it returns a nonzero value

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim CodeName As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'get composite beam design code
    ret = SapModel.DesignCompositeBeam.GetCode(CodeName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ **See Also**

Reference

cDesignCompositeBeam Interface

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CSI API ETABS v1

cDesignCompositeBeamAddLanguageSpecificTextSet Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetComboDeflection(
    ref int NumberItems,
    ref string[] MyName
)
```

```
Function GetComboDeflection (
    ByRef NumberItems As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cDesignCompositeBeam
Dim NumberItems As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetComboDeflection(NumberItems,
MyName)
```

```
int GetComboDeflection(
    int% NumberItems,
    array<String^>^% MyName
)
```

```
abstract GetComboDeflection :
    NumberItems : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberItems

Type: [System.Int32](#)

MyName

Type: [System.String](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDesignCompositeBeam Interface

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CSI API ETABS v1

cDesignCompositeBeamAddLanguageSpecificTextSet Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetComboStrength(
    ref int NumberItems,
    ref string[] MyName
)
```

```
Function GetComboStrength (
    ByRef NumberItems As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cDesignCompositeBeam
Dim NumberItems As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetComboStrength(NumberItems,
MyName)
```

```
int GetComboStrength(
    int% NumberItems,
    array<String^>^% MyName
)
```

```
abstract GetComboStrength :
    NumberItems : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberItems

Type: [System.Int32](#)

MyName

Type: [System.String](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDesignCompositeBeam Interface

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CSI API ETABS v1

cDesignCompositeBeamAddLanguageSpecificTextSet Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetDesignSection(
    string Name,
    ref string PropName
)
```

```
Function GetDesignSection (
    Name As String,
    ByRef PropName As String
) As Integer
```

```
Dim instance As cDesignCompositeBeam
Dim Name As String
Dim PropName As String
Dim returnValue As Integer
```

```
returnValue = instance.GetDesignSection(Name,
    PropName)
```

```
int GetDesignSection(
    String^ Name,
    String^% PropName
)
```

```
abstract GetDesignSection :
    Name : string *
    PropName : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

PropName

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDesignCompositeBeam Interface

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CSI API ETABS v1

cDesignCompositeBeamAddLanguageSpecificTextSet Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetGroup(
    ref int NumberItems,
    ref string[] MyName
)
```

```
Function GetGroup (
    ByRef NumberItems As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cDesignCompositeBeam
Dim NumberItems As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetGroup(NumberItems,
    MyName)
```

```
int GetGroup(
    int% NumberItems,
    array<String^>^% MyName
)
```

```
abstract GetGroup :
    NumberItems : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberItems

Type: [System.Int32](#)

MyName

Type: [System.String](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDesignCompositeBeam Interface

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CSI API ETABS v1

cDesignCompositeBeamAddLanguageSpecificTextSet Method

This function determines if the composite beam design results are available.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
bool GetResultsAvailable()
```

```
Function GetResultsAvailable As Boolean
```

```
Dim instance As cDesignCompositeBeam
```

```
Dim returnValue As Boolean
```

```
returnValue = instance.GetResultsAvailable()
```

```
bool GetResultsAvailable()
```

```
abstract GetResultsAvailable : unit -> bool
```

Return Value

Type: Boolean

Returns True if the composite beam design results are available, otherwise it returns False.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim ResultsAvailable as Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
```

cDesignCompositeBeamsAddLanguageSpecificTextSet("LST80EEE392_0?cpp=

Introduction

```
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'run analysis
System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
ret = SapModel.Analyze.RunAnalysis

'start steel design
ret = SapModel.DesignCompositeBeam.StartDesign()

'check if design results are available
ResultsAvailable = SapModel.DesignCompositeBeam.GetResultsAvailable

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cDesignCompositeBeam Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignCompositeBeamAddLanguageSpecificTextSet Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetSummaryResults(
    string Name,
    ref int NumberItems,
    ref string[] DesignSect,
    ref double[] BeamFy,
    ref double[] StudDia,
    ref string[] StudLayout,
    ref bool[] BeamShored,
    ref double[] BeamCamber,
    ref string[] PassFail,
    ref double[] ReacLeft,
    ref double[] ReacRt,
    ref double[] MMaxNeg,
    ref double[] MMaxPos,
    ref double[] PCC,
    ref double[] OverallRatio,
    ref double[] StudRatio,
    ref double[] StrPMRat,
    ref double[] ConstPMRat,
    ref double[] StrShrRat,
    ref double[] ConShrRat,
    ref double[] PCDLDfRat,
    ref double[] SDLDfRat,
    ref double[] LLDfRat,
    ref double[] TotCamDfRat,
    ref double[] FreqRat,
    ref double[] MDampRat,
    itemType itemType = itemType.Objects
)
```

```
Function GetSummaryResults (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef DesignSect As String(),
    ByRef BeamFy As Double(),
    ByRef StudDia As Double(),
    ByRef StudLayout As String(),
    ByRef BeamShored As Boolean(),
    ByRef BeamCamber As Double(),
    ByRef PassFail As String(),
    ByRef ReacLeft As Double(),
    ByRef ReacRt As Double(),
    ByRef MMaxNeg As Double(),
```

Introduction

```
ByRef MMaxPos As Double(),  
ByRef PCC As Double(),  
ByRef OverallRatio As Double(),  
ByRef StudRatio As Double(),  
ByRef StrPMRat As Double(),  
ByRef ConstPMRat As Double(),  
ByRef StrShrRat As Double(),  
ByRef ConShrRat As Double(),  
ByRef PCDLDfRat As Double(),  
ByRef SDLDFRat As Double(),  
ByRef LLDfRat As Double(),  
ByRef TotCamDfRat As Double(),  
ByRef FreqRat As Double(),  
ByRef MDampRat As Double(),  
ByRef ItemType As eItemType = eItemType.Objects  
) As Integer
```

```
Dim instance As cDesignCompositeBeam  
Dim Name As String  
Dim NumberItems As Integer  
Dim DesignSect As String()  
Dim BeamFy As Double()  
Dim StudDia As Double()  
Dim StudLayout As String()  
Dim BeamShored As Boolean()  
Dim BeamCamber As Double()  
Dim PassFail As String()  
Dim ReacLeft As Double()  
Dim ReacRt As Double()  
Dim MMaxNeg As Double()  
Dim MMaxPos As Double()  
Dim PCC As Double()  
Dim OverallRatio As Double()  
Dim StudRatio As Double()  
Dim StrPMRat As Double()  
Dim ConstPMRat As Double()  
Dim StrShrRat As Double()  
Dim ConShrRat As Double()  
Dim PCDLDfRat As Double()  
Dim SDLDFRat As Double()  
Dim LLDfRat As Double()  
Dim TotCamDfRat As Double()  
Dim FreqRat As Double()  
Dim MDampRat As Double()  
Dim ItemType As eItemType  
Dim returnValue As Integer  
  
returnValue = instance.GetSummaryResults(Name,  
NumberItems, DesignSect, BeamFy,  
StudDia, StudLayout, BeamShored,  
BeamCamber, PassFail, ReacLeft, ReacRt,  
MMaxNeg, MMaxPos, PCC, OverallRatio,  
StudRatio, StrPMRat, ConstPMRat,  
StrShrRat, ConShrRat, PCDLDfRat,  
SDLDFRat, LLDfRat, TotCamDfRat, FreqRat,  
MDampRat, ItemType)
```

```
int GetSummaryResults(  
String^ Name,  
int% NumberItems,  
array<String^>^% DesignSect,  
array<double>^% BeamFy,
```

Introduction

```
array<double>^% StudDia,  
array<String^>^% StudLayout,  
array<bool>^% BeamShored,  
array<double>^% BeamCamber,  
array<String^>^% PassFail,  
array<double>^% ReacLeft,  
array<double>^% ReacRt,  
array<double>^% MMaxNeg,  
array<double>^% MMaxPos,  
array<double>^% PCC,  
array<double>^% OverallRatio,  
array<double>^% StudRatio,  
array<double>^% StrPMRat,  
array<double>^% ConstPMRat,  
array<double>^% StrShrRat,  
array<double>^% ConShrRat,  
array<double>^% PCDLDfRat,  
array<double>^% SDLDFRat,  
array<double>^% LLDfRat,  
array<double>^% TotCamDfRat,  
array<double>^% FreqRat,  
array<double>^% MDampRat,  
eItemType ItemType = eItemType::Objects  
)
```

```
abstract GetSummaryResults :  
    Name : string *  
    NumberItems : int byref *  
    DesignSect : string[] byref *  
    BeamFy : float[] byref *  
    StudDia : float[] byref *  
    StudLayout : string[] byref *  
    BeamShored : bool[] byref *  
    BeamCamber : float[] byref *  
    PassFail : string[] byref *  
    ReacLeft : float[] byref *  
    ReacRt : float[] byref *  
    MMaxNeg : float[] byref *  
    MMaxPos : float[] byref *  
    PCC : float[] byref *  
    OverallRatio : float[] byref *  
    StudRatio : float[] byref *  
    StrPMRat : float[] byref *  
    ConstPMRat : float[] byref *  
    StrShrRat : float[] byref *  
    ConShrRat : float[] byref *  
    PCDLDfRat : float[] byref *  
    SDLDFRat : float[] byref *  
    LLDfRat : float[] byref *  
    TotCamDfRat : float[] byref *  
    FreqRat : float[] byref *  
    MDampRat : float[] byref *  
    ?ItemType : eItemType  
(* Defaults:  
    let _ItemType = defaultArg ItemType eItemType.Objects  
*)  
-> int
```

Parameters

Name

Type:Â SystemString

NumberItems

Type:Â SystemInt32

DesignSect

Type:Â SystemString

BeamFy

Type:Â SystemDouble

StudDia

Type:Â SystemDouble

StudLayout

Type:Â SystemString

BeamShored

Type:Â SystemBoolean

BeamCamber

Type:Â SystemDouble

PassFail

Type:Â SystemString

ReacLeft

Type:Â SystemDouble

ReacRt

Type:Â SystemDouble

MMaxNeg

Type:Â SystemDouble

MMaxPos

Type:Â SystemDouble

PCC

Type:Â SystemDouble

OverallRatio

Type:Â SystemDouble

StudRatio

Type:Â SystemDouble

StrPMRat

Type:Â SystemDouble

ConstPMRat

Type:Â SystemDouble

StrShrRat

Type:Â SystemDouble

ConShrRat

Type:Â SystemDouble

PCDLDFRat

Type:Â SystemDouble

SDLDFRat

Type:Â SystemDouble

LLDFRat

Type:Â SystemDouble

TotCamDFRat

Type:Â SystemDouble

FreqRat

Parameters

Type: SystemDouble
MDampRat
Type: SystemDouble
ItemType (Optional)
Type: ETABSv1eItemType

Return Value

Type: Int32

▲ See Also

Reference

cDesignCompositeBeam Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDesignCompositeBeamAddLanguageSpecificTextSet Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetTargetDispl(
    ref int NumberItems,
    ref string[] LoadCase,
    ref string[] Point,
    ref double[] Displ,
    ref bool Active
)
```

```
Function GetTargetDispl (
    ByRef NumberItems As Integer,
    ByRef LoadCase As String(),
    ByRef Point As String(),
    ByRef Displ As Double(),
    ByRef Active As Boolean
) As Integer
```

```
Dim instance As cDesignCompositeBeam
Dim NumberItems As Integer
Dim LoadCase As String()
Dim Point As String()
Dim Displ As Double()
Dim Active As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetTargetDispl(NumberItems,
    LoadCase, Point, Displ, Active)
```

```
int GetTargetDispl(
    int% NumberItems,
    array<String^>^% LoadCase,
    array<String^>^% Point,
    array<double>^% Displ,
    bool% Active
)
```

```
abstract GetTargetDispl :
    NumberItems : int byref *
    LoadCase : string[] byref *
    Point : string[] byref *
    Displ : float[] byref *
    Active : bool byref -> int
```

Parameters

NumberItems

Type: SystemInt32

LoadCase

Type: SystemString

Point

Type: SystemString

Displ

Type: SystemDouble

Active

Type: SystemBoolean

Return Value

Type: Int32

See Also

Reference

cDesignCompositeBeam Interface

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CSI API ETABS v1

cDesignCompositeBeamAddLanguageSpecificTextSet Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetTargetPeriod(
    ref int NumberItems,
    ref string ModalCase,
    ref int[] Mode,
    ref double[] Period,
    ref bool Active
)
```

```
Function GetTargetPeriod (
    ByRef NumberItems As Integer,
    ByRef ModalCase As String,
    ByRef Mode As Integer(),
    ByRef Period As Double(),
    ByRef Active As Boolean
) As Integer
```

```
Dim instance As cDesignCompositeBeam
Dim NumberItems As Integer
Dim ModalCase As String
Dim Mode As Integer()
Dim Period As Double()
Dim Active As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetTargetPeriod(NumberItems,
    ModalCase, Mode, Period, Active)
```

```
int GetTargetPeriod(
    int% NumberItems,
    String^% ModalCase,
    array<int>^% Mode,
    array<double>^% Period,
    bool% Active
)
```

```
abstract GetTargetPeriod :
    NumberItems : int byref *
    ModalCase : string byref *
    Mode : int[] byref *
    Period : float[] byref *
    Active : bool byref -> int
```

Parameters

NumberItems

Type: SystemInt32

ModalCase

Type: SystemString

Mode

Type: SystemInt32

Period

Type: SystemDouble

Active

Type: SystemBoolean

Return Value

Type: Int32

See Also

Reference

cDesignCompositeBeam Interface

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CSI API ETABS v1

cDesignCompositeBeamAddLanguageSpecificTextSet Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

int ResetOverwrites()

Function ResetOverwrites As Integer

Dim instance As cDesignCompositeBeam

Dim returnValue As Integer

returnValue = instance.ResetOverwrites()

int ResetOverwrites()

abstract ResetOverwrites : unit -> int

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cDesignCompositeBeam Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignCompositeBeamAddLanguageSpecificTextSet Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetAutoSelectNull(
    string Name,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetAutoSelectNull (
    Name As String,
    Optional ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDesignCompositeBeam
Dim Name As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetAutoSelectNull(Name,
    ItemType)
```

```
int SetAutoSelectNull(
    String^ Name,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetAutoSelectNull :
    Name : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

ItemType (Optional)

Type: ETABSv1eItemType

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cDesignCompositeBeam Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignCompositeBeamAddLanguageSpecificTextSet Method

Sets the composite beam design code.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetCode(
    string CodeName
)
```

```
Function SetCode (
    CodeName As String
) As Integer
```

```
Dim instance As cDesignCompositeBeam
Dim CodeName As String
Dim returnValue As Integer
```

```
returnValue = instance.SetCode(CodeName)
```

```
int SetCode(
    String^ CodeName
)
```

```
abstract SetCode :
    CodeName : string -> int
```

Parameters

CodeName

Type: SystemString

This is one of the following composite beam design code names.

- ◇ AISC 360-10
- ◇ AISC 360-05
- ◇ BS 5950-1990
- ◇ Chinese 2010
- ◇ CSA S16-14
- ◇ CSA S16-09
- ◇ Eurocode 4-2004

Return Value

Type: Int32

Returns zero if the code is successfully set; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim CodeName As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set composite beam design code
    ret = SapModel.DesignCompositeBeam.SetCode("Eurocode 3-2005")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ **See Also**

Reference

cDesignCompositeBeam Interface

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CSI API ETABS v1

cDesignCompositeBeamAddLanguageSpecificTextSet Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetComboDeflection(
    string Name,
    bool Selected
)
```

```
Function SetComboDeflection (
    Name As String,
    Selected As Boolean
) As Integer
```

```
Dim instance As cDesignCompositeBeam
Dim Name As String
Dim Selected As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetComboDeflection(Name,
    Selected)
```

```
int SetComboDeflection(
    String^ Name,
    bool Selected
)
```

```
abstract SetComboDeflection :
    Name : string *
    Selected : bool -> int
```

Parameters

Name

Type: [SystemString](#)

Selected

Type: [SystemBoolean](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

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CSI API ETABS v1

cDesignCompositeBeamAddLanguageSpecificTextSet Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetComboStrength(
    string Name,
    bool Selected
)
```

```
Function SetComboStrength (
    Name As String,
    Selected As Boolean
) As Integer
```

```
Dim instance As cDesignCompositeBeam
Dim Name As String
Dim Selected As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetComboStrength(Name,
    Selected)
```

```
int SetComboStrength(
    String^ Name,
    bool Selected
)
```

```
abstract SetComboStrength :
    Name : string *
    Selected : bool -> int
```

Parameters

Name

Type: [SystemString](#)

Selected

Type: [SystemBoolean](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

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CSI API ETABS v1

cDesignCompositeBeamAddLanguageSpecificTextSet Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetDesignSection(
    string Name,
    string PropName,
    bool LastAnalysis,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetDesignSection (
    Name As String,
    PropName As String,
    LastAnalysis As Boolean,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDesignCompositeBeam
Dim Name As String
Dim PropName As String
Dim LastAnalysis As Boolean
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetDesignSection(Name,
    PropName, LastAnalysis, ItemType)
```

```
int SetDesignSection(
    String^ Name,
    String^ PropName,
    bool LastAnalysis,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetDesignSection :
    Name : string *
    PropName : string *
    LastAnalysis : bool *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

PropName

Type: SystemString

LastAnalysis

Type: SystemBoolean

ItemType (Optional)

Type: ETABSv1eItemType

Return Value

Type: Int32

▲ See Also

Reference

cDesignCompositeBeam Interface

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CSI API ETABS v1

cDesignCompositeBeamAddLanguageSpecificTextSet Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetGroup(
    string Name,
    bool Selected
)
```

```
Function SetGroup (
    Name As String,
    Selected As Boolean
) As Integer
```

```
Dim instance As cDesignCompositeBeam
Dim Name As String
Dim Selected As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetGroup(Name,
    Selected)
```

```
int SetGroup(
    String^ Name,
    bool Selected
)
```

```
abstract SetGroup :
    Name : string *
    Selected : bool -> int
```

Parameters

Name

Type: [SystemString](#)

Selected

Type: [SystemBoolean](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDesignCompositeBeam Interface

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CSI API ETABS v1

cDesignCompositeBeamAddLanguageSpecificTextSet Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetTargetDispl(
int NumberItems,
ref string[] LoadCase,
ref string[] Point,
ref double[] Displ,
bool Active = true
)
```

```
Function SetTargetDispl (
NumberItems As Integer,
ByRef LoadCase As String(),
ByRef Point As String(),
ByRef Displ As Double(),
Optional Active As Boolean = true
) As Integer
```

```
Dim instance As cDesignCompositeBeam
Dim NumberItems As Integer
Dim LoadCase As String()
Dim Point As String()
Dim Displ As Double()
Dim Active As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetTargetDispl(NumberItems,
LoadCase, Point, Displ, Active)
```

```
int SetTargetDispl(
int NumberItems,
array<String^>^% LoadCase,
array<String^>^% Point,
array<double>^% Displ,
bool Active = true
)
```

```
abstract SetTargetDispl :
    NumberItems : int *
    LoadCase : string[] byref *
    Point : string[] byref *
    Displ : float[] byref *
    ?Active : bool
(* Defaults:
```

```
        let _Active = defaultArg Active true
*)
-> int
```

Parameters

NumberItems

Type:Â [SystemInt32](#)

LoadCase

Type:Â [SystemString](#)

Point

Type:Â [SystemString](#)

Displ

Type:Â [SystemDouble](#)

Active (Optional)

Type:Â [SystemBoolean](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cDesignCompositeBeam Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignCompositeBeamAddLanguageSpecificTextSet Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetTargetPeriod(
    int NumberItems,
    string ModalCase,
    ref int[] Mode,
    ref double[] Period,
    bool Active = true
)
```

```
Function SetTargetPeriod (
    NumberItems As Integer,
    ModalCase As String,
    ByRef Mode As Integer(),
    ByRef Period As Double(),
    Optional Active As Boolean = true
) As Integer
```

```
Dim instance As cDesignCompositeBeam
Dim NumberItems As Integer
Dim ModalCase As String
Dim Mode As Integer()
Dim Period As Double()
Dim Active As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetTargetPeriod(NumberItems,
    ModalCase, Mode, Period, Active)
```

```
int SetTargetPeriod(
    int NumberItems,
    String^ ModalCase,
    array<int>^% Mode,
    array<double>^% Period,
    bool Active = true
)
```

```
abstract SetTargetPeriod :
    NumberItems : int *
    ModalCase : string *
    Mode : int[] byref *
    Period : float[] byref *
    ?Active : bool
(* Defaults:
```

```
        let _Active = defaultArg Active true
*)
-> int
```

Parameters

NumberItems

Type:Â SystemInt32

ModalCase

Type:Â SystemString

Mode

Type:Â SystemInt32

Period

Type:Â SystemDouble

Active (Optional)

Type:Â SystemBoolean

Return Value

Type:Â Int32

▲ See Also

Reference

cDesignCompositeBeam Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDesignCompositeBeamAddLanguageSpecificTextSet Method

Starts the composite beam design.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int StartDesign()
```

```
Function StartDesign As Integer
```

```
Dim instance As cDesignCompositeBeam
```

```
Dim returnValue As Integer
```

```
returnValue = instance.StartDesign()
```

```
int StartDesign()
```

```
abstract StartDesign : unit -> int
```

Return Value

Type: [Int32](#)

Returns zero if the composite beam design is successfully started; otherwise it returns a nonzero value.

▲ Remarks

The function will fail if no composite beam frame objects are present. It will also fail if analysis results are not available.

▲ Examples

[VB](#)

[Copy](#)

```
Public Sub Example()
```

```
    Dim SapModel As cSapModel
```

```
    Dim EtabsObject As cOAPI
```

```
    Dim ret As Integer = -1
```

```
'create ETABS object
```

```
    Dim myHelper as cHelper = New Helper
```

```
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

```
'start ETABS application
```

```
    ret = EtabsObject.ApplicationStart()
```

```
'create SapModel object
```

```
    SapModel = EtabsObject.SapModel
```

cDesignCompositeBeams id="LSTCA01F4A3_0"AddLanguageSpecificTextSet("LSTCA01F4A3_0?cpp=

Introduction

```
'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'create new concrete frame section property
    ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

'set frame section property
    ret = SapModel.FrameObj.SetSection("8", "R1")

'set concrete design code
    ret = SapModel.DesignConcrete.SetCode("ACI 318-08/IBC 2009")

'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis

'start composite beam design
    ret = SapModel.Designcomposite.StartDesign()

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

[cDesignCompositeBeam Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignCompositeBeamAddLanguageSpecificTextSet Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int VerifyPassed(
    ref int NumberItems,
    ref int N1,
    ref int N2,
    ref string[] MyName
)
```

```
Function VerifyPassed (
    ByRef NumberItems As Integer,
    ByRef N1 As Integer,
    ByRef N2 As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cDesignCompositeBeam
Dim NumberItems As Integer
Dim N1 As Integer
Dim N2 As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.VerifyPassed(NumberItems,
    N1, N2, MyName)
```

```
int VerifyPassed(
    int% NumberItems,
    int% N1,
    int% N2,
    array<String^>^% MyName
)
```

```
abstract VerifyPassed :
    NumberItems : int byref *
    N1 : int byref *
    N2 : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberItems

Type: System.Int32

N1

Type:Â [SystemInt32](#)

N2

Type:Â [SystemInt32](#)

MyName

Type:Â [SystemString](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cDesignCompositeBeam Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignCompositeBeamAddLanguageSpecificTextSet Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int VerifySections(
    ref int NumberItems,
    ref string[] MyName
)
```

```
Function VerifySections (
    ByRef NumberItems As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cDesignCompositeBeam
Dim NumberItems As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.VerifySections(NumberItems,
MyName)
```

```
int VerifySections(
    int% NumberItems,
    array<String^>^% MyName
)
```

```
abstract VerifySections :
    NumberItems : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberItems

Type: [System.Int32](#)

MyName

Type: [System.String](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDesignCompositeBeam Interface

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CSI API ETABS v1

cDesignConcrete Interface

Namespace: Â ETABSv1

Assembly: Â ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
public interface cDesignConcrete
```

```
Public Interface cDesignConcrete
```

```
Dim instance As cDesignConcrete
```

```
public interface class cDesignConcrete
```

```
type cDesignConcrete = interface end
```

The cDesignConcrete type exposes the following members.

▲ Properties

Â	Name	Description
	<u>ACI318_08_IBC2009</u>	
	<u>ACI318_14</u>	
	<u>ACI318_19</u>	
	<u>AS_3600_09</u>	
	<u>AS_3600_2018</u>	
	<u>BS8110_97</u>	
	<u>Chinese_2010</u>	
	<u>Eurocode_2_2004</u>	
	<u>Indian_IS_456_2000</u>	
	<u>Mexican_RCDF_2017</u>	
	<u>SP63_13330_2012</u>	
	<u>TS_500_2000_R2018</u>	

Top

▲ Methods

Â	Name	Description
	<u>GetCode</u>	Retrieves the concrete design code.
	<u>GetDesignSection</u>	Retrieves the design section for a specified concrete frame object
	<u>GetRebarPrefsBeam</u>	Retrieves data from the Rebar Selection Rules for Beams

Introduction

- ⇒ [GetRebarPrefsColumn](#) Retrieves data from the Rebar Selection Rules for Columns
- ⇒ [GetResultsAvailable](#) This function determines if the concrete frame design results are available.
- ⇒ [GetSeismicFramingType](#)
- ⇒ [GetSummaryResultsBeam](#) Retrieves beam summary results for concrete design.
- ⇒ [GetSummaryResultsBeam_2](#)
- ⇒ [GetSummaryResultsColumn](#) Retrieves column summary results for concrete design
- ⇒ [GetSummaryResultsJoint](#)
- ⇒ [SetCode](#) Sets the concrete design code.
- ⇒ [SetDesignSection](#) Modifies the design section for all specified frame objects that have a concrete frame design procedure
- ⇒ [StartDesign](#) Starts the concrete frame design.

[Top](#)

▲ [See Also](#)

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignConcrete Properties

The cDesignConcrete type exposes the following members.

Properties

Ä	Name	Description
	<u>ACI318_08_IBC2009</u>	
	<u>ACI318_14</u>	
	<u>ACI318_19</u>	
	<u>AS_3600_09</u>	
	<u>AS_3600_2018</u>	
	<u>BS8110_97</u>	
	<u>Chinese_2010</u>	
	<u>Eurocode_2_2004</u>	
	<u>Indian_IS_456_2000</u>	
	<u>Mexican_RCDF_2017</u>	
	<u>SP63_13330_2012</u>	
	<u>TS_500_2000_R2018</u>	

Top

Ä See Also

Reference

cDesignConcrete Interface

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CSI API ETABS v1

cDesignConcreteAddLanguageSpecificTextSet("LST70 Property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
cDCoACI318_08_IBC2009 ACI318_08_IBC2009 { get; }
```

```
ReadOnly Property ACI318_08_IBC2009 As cDCoACI318_08_IBC2009  
Get
```

```
Dim instance As cDesignConcrete  
Dim value As cDCoACI318_08_IBC2009
```

```
value = instance.ACI318_08_IBC2009
```

```
property cDCoACI318_08_IBC2009^ ACI318_08_IBC2009 {  
cDCoACI318_08_IBC2009^ get ();  
}
```

```
abstract ACI318_08_IBC2009 : cDCoACI318_08_IBC2009 with get
```

Property Value

Type: cDCoACI318_08_IBC2009

▲ See Also

Reference

cDesignConcrete Interface

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CSI API ETABS v1

cDesignConcreteAddLanguageSpecificTextSet("LST1C Property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
cDCoACI318_14 ACI318_14 { get; }
```

```
ReadOnly Property ACI318_14 As cDCoACI318_14  
Get
```

```
Dim instance As cDesignConcrete  
Dim value As cDCoACI318_14
```

```
value = instance.ACI318_14
```

```
property cDCoACI318_14^ ACI318_14 {  
cDCoACI318_14^ get ();  
}
```

```
abstract ACI318_14 : cDCoACI318_14 with get
```

Property Value

Type: cDCoACI318_14

▲ See Also

Reference

cDesignConcrete Interface

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CSI API ETABS v1

cDesignConcreteAddLanguageSpecificTextSet("LST1A Property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
cDcCoACI318_19 ACI318_19 { get; }
```

```
ReadOnly Property ACI318_19 As cDcCoACI318_19
Get
```

```
Dim instance As cDesignConcrete
Dim value As cDcCoACI318_19
```

```
value = instance.ACI318_19
```

```
property cDcCoACI318_19^ ACI318_19 {
cDcCoACI318_19^ get ();
}
```

```
abstract ACI318_19 : cDcCoACI318_19 with get
```

Property Value

Type: cDcCoACI318_19

▲ See Also

Reference

cDesignConcrete Interface

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CSI API ETABS v1

cDesignConcreteAddLanguageSpecificTextSet("LSTE5 Property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
cDCoAS_3600_09 AS_3600_09 { get; }
```

```
ReadOnly Property AS_3600_09 As cDCoAS_3600_09
Get
```

```
Dim instance As cDesignConcrete
Dim value As cDCoAS_3600_09
```

```
value = instance.AS_3600_09
```

```
property cDCoAS_3600_09^ AS_3600_09 {
cDCoAS_3600_09^ get ();
}
```

```
abstract AS_3600_09 : cDCoAS_3600_09 with get
```

Property Value

Type: cDCoAS_3600_09

▲ See Also

Reference

cDesignConcrete Interface

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CSI API ETABS v1

cDesignConcreteAddLanguageSpecificTextSet("LST53 Property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
cDCoAS_3600_2018 AS_3600_2018 { get; }
```

```
ReadOnly Property AS_3600_2018 As cDCoAS_3600_2018
Get
```

```
Dim instance As cDesignConcrete
Dim value As cDCoAS_3600_2018
```

```
value = instance.AS_3600_2018
```

```
property cDCoAS 3600_2018^ AS_3600_2018 {
cDCoAS_3600_2018^ get ();
}
```

```
abstract AS_3600_2018 : cDCoAS_3600_2018 with get
```

Property Value

Type: cDCoAS_3600_2018

▲ See Also

Reference

cDesignConcrete Interface

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CSI API ETABS v1

cDesignConcreteAddLanguageSpecificTextSet("LST6D Property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
cDcBS8110_97 BS8110_97 { get; }
```

```
ReadOnly Property BS8110_97 As cDcBS8110_97  
Get
```

```
Dim instance As cDesignConcrete  
Dim value As cDcBS8110_97
```

```
value = instance.BS8110_97
```

```
property cDcBS8110_97^ BS8110_97 {  
cDcBS8110_97^ get ();  
}
```

```
abstract BS8110_97 : cDcBS8110_97 with get
```

Property Value

Type: cDcBS8110_97

▲ See Also

Reference

cDesignConcrete Interface

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CSI API ETABS v1

cDesignConcreteAddLanguageSpecificTextSet("LSTDA Property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
cDCoChinese_2010 Chinese_2010 { get; }
```

```
ReadOnly Property Chinese_2010 As cDCoChinese_2010
Get
```

```
Dim instance As cDesignConcrete
Dim value As cDCoChinese_2010
```

```
value = instance.Chinese_2010
```

```
property cDCoChinese_2010^ Chinese_2010 {
cDCoChinese_2010^ get ();
}
```

```
abstract Chinese_2010 : cDCoChinese_2010 with get
```

Property Value

Type: cDCoChinese_2010

▲ See Also

Reference

cDesignConcrete Interface

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CSI API ETABS v1

cDesignConcreteAddLanguageSpecificTextSet("LSTFF Property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
cDCoEurocode_2_2004 Eurocode_2_2004 { get; }
```

```
ReadOnly Property Eurocode_2_2004 As cDCoEurocode_2_2004  
Get
```

```
Dim instance As cDesignConcrete  
Dim value As cDCoEurocode_2_2004
```

```
value = instance.Eurocode_2_2004
```

```
property cDCoEurocode_2_2004^ Eurocode_2_2004 {  
cDCoEurocode_2_2004^ get ();  
}
```

```
abstract Eurocode_2_2004 : cDCoEurocode_2_2004 with get
```

Property Value

Type: cDCoEurocode_2_2004

▲ See Also

Reference

cDesignConcrete Interface

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CSI API ETABS v1

cDesignConcreteAddLanguageSpecificTextSet("LSTCDBBA2D1_0") Property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
cDcoIndian_IS_456_2000 Indian_IS_456_2000 { get; }
```

```
ReadOnly Property Indian_IS_456_2000 As cDcoIndian_IS_456_2000
Get
```

```
Dim instance As cDesignConcrete
Dim value As cDcoIndian_IS_456_2000
```

```
value = instance.Indian_IS_456_2000
```

```
property cDcoIndian_IS_456_2000^ Indian_IS_456_2000 {
cDcoIndian_IS_456_2000^ get ();
}
```

```
abstract Indian_IS_456_2000 : cDcoIndian_IS_456_2000 with get
```

Property Value

Type: cDcoIndian_IS_456_2000

▲ See Also

Reference

cDesignConcrete Interface

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CSI API ETABS v1

cDesignConcreteAddLanguageSpecificTextSet("LST74AD03C8_0?1245::|nu=.") Property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
cDcoMexican_RCDF_2017 Mexican_RCDF_2017 { get; }
```

```
ReadOnly Property Mexican_RCDF_2017 As cDcoMexican_RCDF_2017
Get
```

```
Dim instance As cDesignConcrete
Dim value As cDcoMexican_RCDF_2017
```

```
value = instance.Mexican_RCDF_2017
```

```
property cDcoMexican_RCDF_2017^ Mexican_RCDF_2017 {
cDcoMexican_RCDF_2017^ get ();
}
```

```
abstract Mexican_RCDF_2017 : cDcoMexican_RCDF_2017 with get
```

Property Value

Type: cDcoMexican_RCDF_2017

▲ See Also

Reference

cDesignConcrete Interface

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CSI API ETABS v1

cDesignConcreteAddLanguageSpecificTextSet("LSTD8 Property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
cDCoSP63133302011 SP63_13330_2012 { get; }
```

```
ReadOnly Property SP63_13330_2012 As cDCoSP63133302011  
Get
```

```
Dim instance As cDesignConcrete  
Dim value As cDCoSP63133302011
```

```
value = instance.SP63_13330_2012
```

```
property cDCoSP63133302011^ SP63_13330_2012 {  
cDCoSP63133302011^ get ();  
}
```

```
abstract SP63_13330_2012 : cDCoSP63133302011 with get
```

Property Value

Type: cDCoSP63133302011

▲ See Also

Reference

cDesignConcrete Interface

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CSI API ETABS v1

cDesignConcreteAddLanguageSpecificTextSet("LST768C7129_0") Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cDcCoTS_500_2000_R2018 TS_500_2000_R2018 { get; }
```

```
ReadOnly Property TS_500_2000_R2018 As cDcCoTS_500_2000_R2018
Get
```

```
Dim instance As cDesignConcrete
Dim value As cDcCoTS_500_2000_R2018
```

```
value = instance.TS_500_2000_R2018
```

```
property cDcCoTS_500_2000_R2018^ TS_500_2000_R2018 {
cDcCoTS_500_2000_R2018^ get ();
}
```

```
abstract TS_500_2000_R2018 : cDcCoTS_500_2000_R2018 with get
```

Property Value

Type: [cDcCoTS_500_2000_R2018](#)

▲ See Also

Reference

[cDesignConcrete Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignConcrete Methods

The cDesignConcrete type exposes the following members.

Methods

	Name	Description
◆	<u>GetCode</u>	Retrieves the concrete design code.
◆	<u>GetDesignSection</u>	Retrieves the design section for a specified concrete frame object
◆	<u>GetRebarPrefsBeam</u>	Retrieves data from the Rebar Selection Rules for Beams
◆	<u>GetRebarPrefsColumn</u>	Retrieves data from the Rebar Selection Rules for Columns
◆	<u>GetResultsAvailable</u>	This function determines if the concrete frame design results are available.
◆	<u>GetSeismicFramingType</u>	
◆	<u>GetSummaryResultsBeam</u>	Retrieves beam summary results for concrete design.
◆	<u>GetSummaryResultsBeam_2</u>	
◆	<u>GetSummaryResultsColumn</u>	Retrieves column summary results for concrete design
◆	<u>GetSummaryResultsJoint</u>	
◆	<u>SetCode</u>	Sets the concrete design code.
◆	<u>SetDesignSection</u>	Modifies the design section for all specified frame objects that have a concrete frame design procedure
◆	<u>StartDesign</u>	Starts the concrete frame design.

Top

◆ See Also

Reference

cDesignConcrete Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDesignConcreteAddLanguageSpecificTextSet("LST87 Method

Retrieves the concrete design code.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetCode(
    ref string CodeName
)
```

```
Function GetCode (
    ByRef CodeName As String
) As Integer
```

```
Dim instance As cDesignConcrete
Dim CodeName As String
Dim returnValue As Integer
```

```
returnValue = instance.GetCode(CodeName)
```

```
int GetCode(
    String^% CodeName
)
```

```
abstract GetCode :
    CodeName : string byref -> int
```

Parameters

CodeName

Type: SystemString

This is one of the following concrete design code names.

- ◇ ACI 318-14
- ◇ ACI 318-11
- ◇ ACI 318-08
- ◇ AS 3600-2018
- ◇ AS 3600-09
- ◇ BS 8110-97
- ◇ Chinese 2010
- ◇ CSA A23.3-14
- ◇ CSA A23.3-04
- ◇ Eurocode 2-2004
- ◇ Hong Kong CP 2013

◇ IS 456:2000
 ◇ Italian NTC 2008
 ◇ KBC 2016
 ◇ KBC 2009
 ◇ Mexican RCDF 2017
 ◇ Mexican RCDF 2004
 ◇ NZS 3101:2006
 ◇ Singapore CP 65:99
 ◇ SP 63.13330.2012
 ◇ TS 500-2000(R2018)
 ◇ TS 500-2000
 ◇ TCVN 5574:2012

Return Value

Type: Int32

Returns zero if the code is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim CodeName As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'get concrete design code
    ret = SapModel.DesignConcrete.GetCode(CodeName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cDesignConcrete Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDesignConcreteAddLanguageSpecificTextSet("LST43037CB_0") Method

Retrieves the design section for a specified concrete frame object

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetDesignSection(
    string Name,
    ref string PropName
)
```

```
Function GetDesignSection (
    Name As String,
    ByRef PropName As String
) As Integer
```

```
Dim instance As cDesignConcrete
Dim Name As String
Dim PropName As String
Dim returnValue As Integer
```

```
returnValue = instance.GetDesignSection(Name,
    PropName)
```

```
int GetDesignSection(
    String^ Name,
    String^% PropName
)
```

```
abstract GetDesignSection :
    Name : string *
    PropName : string byref -> int
```

Parameters

Name

Type: SystemString

The name of a frame object with a concrete frame design procedure

PropName

Type: SystemString

The name of the design section for the specified frame object

Return ValueType: Int32

Returns zero if the section is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    ret = SapModel.File.NewGridOnly(1, 10, 10, 2, 2, 10, 10)

    'create new concrete frame section properties
    ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)
    ret = SapModel.PropFrame.SetRectangle("R2", "4000Psi", 20, 16)

    'create a concrete column
    Dim FrameName1 As String
    ret = SapModel.FrameObj.AddByCoord(0, 0, 0, 0, 0, 10, FrameName1, "R1")

    'run analysis
    ret = SapModel.File.Save("C:\ETABSAPI\example.edb")
    ret = SapModel.Analyze.RunAnalysis

    'start concrete design
    ret = SapModel.DesignConcrete.StartDesign

    'get design section
    Dim PropName As String
    ret = SapModel.DesignConcrete.GetDesignSection(FrameName1, PropName)

    'set design section
    ret = SapModel.DesignConcrete.SetDesignSection(FrameName1, "R2", False)

    ret = SapModel.DesignConcrete.GetDesignSection(FrameName1, PropName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
```

End Sub

▲ See Also

Reference

cDesignConcrete Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDesignConcreteAddLanguageSpecificTextSet("LSTE6E89D23_0") Method

Retrieves data from the Rebar Selection Rules for Beams

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetRebarPrefsBeam(
    int Item,
    ref string Value
)
```

```
Function GetRebarPrefsBeam (
    Item As Integer,
    ByRef Value As String
) As Integer
```

```
Dim instance As cDesignConcrete
Dim Item As Integer
Dim Value As String
Dim returnValue As Integer
```

```
returnValue = instance.GetRebarPrefsBeam(Item,
    Value)
```

```
int GetRebarPrefsBeam(
    int Item,
    String^% Value
)
```

```
abstract GetRebarPrefsBeam :
    Item : int *
    Value : string byref -> int
```

Parameters

Item

Type: [SystemInt32](#)

An integer value corresponding to the data to be retrieved, from the following list

1. Non-Seismic Smallest Longitudinal Bar Size Top
2. Non-Seismic Largest Longitudinal Bar Size Top
3. Non-Seismic Preferred Longitudinal Bar Size Top
4. Non-Seismic Min Number of Bars Top
5. Non-Seismic Smallest Longitudinal Bar Size Bottom

Introduction

6. Non-Seismic Largest Longitudinal Bar Size Bottom
7. Non-Seismic Preferred Longitudinal Bar Size Bottom
8. Non-Seismic Min Number of Bars Bottom
9. Non-Seismic Smallest Stirrup Bar Size
10. Non-Seismic Largest Stirrup Bar Size
11. Non-Seismic Stirrup Min Spacing
12. Non-Seismic Stirrup Max Spacing
13. Seismic Smallest Longitudinal Bar Size Top
14. Seismic Largest Longitudinal Bar Size Top
15. Seismic Preferred Longitudinal Bar Size Top
16. Seismic Min Number of Bars Top
17. Seismic Smallest Longitudinal Bar Size Bottom
18. Seismic Largest Longitudinal Bar Size Bottom
19. Seismic Preferred Longitudinal Bar Size Bottom
20. Seismic Min Number of Bars Bottom
21. Seismic Smallest Stirrup Bar Size
22. Seismic Largest Stirrup Bar Size
23. Seismic Stirrup Min Spacing
24. Seismic Stirrup Max Spacing

Value

Type: SystemString

The retrieved data; note that numerical values are converted to String

Return Value

Type: Int32

Returns zero if the data is successfully retrieved; otherwise returns non-zero

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim Value as String  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)  
  
    'retrieve rebar data  
    ret = SapModel.File.GetRebarPrefsBeam(2, Value)
```

```
'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cDesignConcrete Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDesignConcreteAddLanguageSpecificTextSet("LST161 Method

Retrieves data from the Rebar Selection Rules for Columns

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetRebarPrefsColumn(
    int Item,
    ref string Value
)
```

```
Function GetRebarPrefsColumn (
    Item As Integer,
    ByRef Value As String
) As Integer
```

```
Dim instance As cDesignConcrete
Dim Item As Integer
Dim Value As String
Dim returnValue As Integer
```

```
returnValue = instance.GetRebarPrefsColumn(Item,
    Value)
```

```
int GetRebarPrefsColumn(
    int Item,
    String^% Value
)
```

```
abstract GetRebarPrefsColumn :
    Item : int *
    Value : string byref -> int
```

Parameters

Item

Type: SystemInt32

An integer value corresponding to the data to be retrieved, from the following list

1. Non-Seismic Smallest Longitudinal Bar Size
2. Non-Seismic Largest Longitudinal Bar Size
3. Non-Seismic Smallest Tie Bar Size
4. Non-Seismic Largest Tie Bar Size
5. Non-Seismic Min Spacing

6. Non-Seismic Max Spacing
7. Seismic Smallest Longitudinal Bar Size
8. Seismic Largest Longitudinal Bar Size
9. Seismic Smallest Tie Bar Size
10. Seismic Largest Tie Bar Size
11. Seismic Min Spacing
12. Seismic Max Spacing

Value

Type: SystemString

The retrieved data; note that numerical values are converted to String

Return Value

Type: Int32

Returns zero if the data is successfully retrieved; otherwise returns non-zero

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value as String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'retrieve rebar data
    ret = SapModel.File.GetRebarPrefsColumn(3, Value)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cDesignConcrete Interface

ETABSv1 Namespace

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Namespace: ETABSv1

▲ Syntax

C#

VBC++

F#

Copy

Function GetResultsAvailable As Boolean

```
Dim instance As cDesignConcrete
```

```
returnValue = instance.GetResultsAvailable()
```

```
bool GetResultsAvailable()
```

```
abstract GetResultsAvailable : unit -> bool
```

Type: Boolean

Remarks

Examples

VB

Copy

```
Public Sub Example()
```

```
Dim SapModel As cSapModel
```

```
Dim EtabsObject As c0API
```

```
Dim ret As Integer = -1
```

Dim ResultsAvailable as Boolean

```
'create ETABS object
```

```
Dim myHelper as cHelper = New Helper
```

```
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

```
'start ETABS application
```

```
ret = EtabsObject.ApplicationStart()
```

```
'create SapModel object
```

```
SapModel = Etabsobject.SapModel
```

```
'initialize model
```

Introduction

```
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'create new concrete frame section property
ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

'set frame section property
ret = SapModel.FrameObj.SetSection("8", "R1")

'run analysis
System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
ret = SapModel.Analyze.RunAnalysis

'start concrete design
ret = SapModel.DesignConcrete.StartDesign()

'check if design results are available
ResultsAvailable = SapModel.DesignConcrete.GetResultsAvailable

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cDesignConcrete Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignConcreteAddLanguageSpecificTextSet("LST8B8FDA81_0?1263::|nu=." Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetSeismicFramingType(
    string Name,
    ref int NumberItems,
    ref string[] FrameName,
    ref int[] FramingType,
    eItemType ItemType = eItemType.Objects
)
```

```
Function GetSeismicFramingType (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef FrameName As String(),
    ByRef FramingType As Integer(),
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDesignConcrete
Dim Name As String
Dim NumberItems As Integer
Dim FrameName As String()
Dim FramingType As Integer()
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.GetSeismicFramingType(Name,
    NumberItems, FrameName, FramingType,
    ItemType)
```

```
int GetSeismicFramingType(
    String^ Name,
    int% NumberItems,
    array<String^>^% FrameName,
    array<int>^% FramingType,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract GetSeismicFramingType :
    Name : string *
    NumberItems : int byref *
    FrameName : string[] byref *
    FramingType : int[] byref *
    ?ItemType : eItemType
```

```
(* Defaults:  
    let _ItemType = defaultArg ItemType eItemType.Objects  
*)  
-> int
```

Parameters

Name

Type: SystemString

NumberItems

Type: SystemInt32

FrameName

Type: SystemString

FramingType

Type: SystemInt32

ItemType (Optional)

Type: ETABSv1eItemType

Return Value

Type: Int32

See Also

Reference

cDesignConcrete Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDesignConcreteAddLanguageSpecificTextSet("LST175FFFE5_0") Method

Retrieves beam summary results for concrete design.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetSummaryResultsBeam(
    string Name,
    ref int NumberItems,
    ref string[] FrameName,
    ref double[] Location,
    ref string[] TopCombo,
    ref double[] TopArea,
    ref string[] BotCombo,
    ref double[] BotArea,
    ref string[] VMajorCombo,
    ref double[] VMajorArea,
    ref string[] TLCombo,
    ref double[] TLArea,
    ref string[] TTCombo,
    ref double[] TTArea,
    ref string[] ErrorSummary,
    ref string[] WarningSummary,
    eItemType ItemType = eItemType.Objects
)
```

```
Function GetSummaryResultsBeam (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef FrameName As String(),
    ByRef Location As Double(),
    ByRef TopCombo As String(),
    ByRef TopArea As Double(),
    ByRef BotCombo As String(),
    ByRef BotArea As Double(),
    ByRef VMajorCombo As String(),
    ByRef VMajorArea As Double(),
    ByRef TLCombo As String(),
    ByRef TLArea As Double(),
    ByRef TTCombo As String(),
    ByRef TTArea As Double(),
    ByRef ErrorSummary As String(),
    ByRef WarningSummary As String(),
    Optional ItemType As eItemType = eItemType.Objects
) As Integer
```

Dim instance As cDesignConcrete

cDesignConcretespan id="LST175FFFE5_0"AddLanguageSpecificTextSet("LST175FFFE5_0?1265::|nu=.")

Introduction

```
Dim Name As String
Dim NumberItems As Integer
Dim FrameName As String()
Dim Location As Double()
Dim TopCombo As String()
Dim TopArea As Double()
Dim BotCombo As String()
Dim BotArea As Double()
Dim VMajorCombo As String()
Dim VMajorArea As Double()
Dim TLCombo As String()
Dim TLArea As Double()
Dim TTCombo As String()
Dim TTArea As Double()
Dim ErrorSummary As String()
Dim WarningSummary As String()
Dim ItemType As eItemType
Dim returnValue As Integer

returnValue = instance.GetSummaryResultsBeam(Name,
NumberItems, FrameName, Location,
TopCombo, TopArea, BotCombo, BotArea,
VMajorCombo, VMajorArea, TLCombo,
TLArea, TTCombo, TTArea, ErrorSummary,
WarningSummary, ItemType)
```

```
int GetSummaryResultsBeam(
String^ Name,
int% NumberItems,
array<String^>^% FrameName,
array<double>^% Location,
array<String^>^% TopCombo,
array<double>^% TopArea,
array<String^>^% BotCombo,
array<double>^% BotArea,
array<String^>^% VMajorCombo,
array<double>^% VMajorArea,
array<String^>^% TLCombo,
array<double>^% TLArea,
array<String^>^% TTCombo,
array<double>^% TTArea,
array<String^>^% ErrorSummary,
array<String^>^% WarningSummary,
eItemType ItemType = eItemType::Objects
)
```

```
abstract GetSummaryResultsBeam :
    Name : string *
    NumberItems : int byref *
    FrameName : string[] byref *
    Location : float[] byref *
    TopCombo : string[] byref *
    TopArea : float[] byref *
    BotCombo : string[] byref *
    BotArea : float[] byref *
    VMajorCombo : string[] byref *
    VMajorArea : float[] byref *
    TLCombo : string[] byref *
    TLArea : float[] byref *
    TTCombo : string[] byref *
    TTArea : float[] byref *
    ErrorSummary : string[] byref *
```

Introduction

```
WarningSummary : string[] byref *  
?ItemType : eItemType  
(* Defaults:  
    let _ItemType = defaultArg ItemType eItemType.Objects  
*)  
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame object or group, depending on the value of the ItemType item.

NumberItems

Type:Â SystemInt32

The number of frame objects for which results are obtained.

FrameName

Type:Â SystemString

This is an array that includes each frame object name for which results are obtained.

Location

Type:Â SystemDouble

This is an array that includes the distance from the I-end of the frame object to the location where the results are reported. [L]

TopCombo

Type:Â SystemString

This is an array that includes the name of the design combination for which the controlling top longitudinal rebar area for flexure occurs. A combination name followed by (Sp) indicates that the design loads were obtained by applying special, code-specific multipliers to all or part of the specified design load combination, or that the design was based on the capacity of other objects (or other design locations for the same object).

TopArea

Type:Â SystemDouble

This is an array that includes the total top longitudinal rebar area required for the flexure at the specified location. It does not include the area of steel required for torsion. [L2]

BotCombo

Type:Â SystemString

This is an array that includes the name of the design combination for which the controlling bottom longitudinal rebar area for flexure occurs. A combination name followed by (Sp) indicates that the design loads were obtained by applying special, code-specific, multipliers to all or part of the specified design load combination, or that the design was based on the capacity of other objects (or other design locations for the same object).

BotArea

Type:Â SystemDouble

This is an array that includes the total bottom longitudinal rebar area required for the flexure at the specified location. It does not include the area of steel required for torsion. [L2]

VMajorCombo

Type:Â SystemString

VMajorArea

Type: SystemDouble

TLCombo

Type: SystemString

This is an array that includes the name of the design combination for which the controlling longitudinal rebar area for torsion occurs. A combination name followed by (Sp) indicates that the design loads were obtained by applying special, code-specific, multipliers to all or part of the specified design load combination, or that the design was based on the capacity of other objects (or other design locations for the same object).

TlArea

Type: SystemDouble

This is an array that includes the total longitudinal rebar area required for torsion. [L2]

TTCombo

Type: SystemString

This is an array that includes the name of the design combination for which the controlling transverse reinforcing for torsion occurs. A combination name followed by (Sp) indicates that the design loads were obtained by applying special, code-specific, multipliers to all or part of the specified design load combination, or that the design was based on the capacity of other objects (or other design locations for the same object).

TTArea

Type: SystemDouble

This is an array that includes the required area of transverse torsional shear reinforcing per unit length along the frame object for torsion at the specified location. [L2/L]

ErrorSummary

Type: SystemString

This is an array that includes the design error messages for the frame object, if any.

WarningSummary

Type: SystemString

This is an array that includes the design warning messages for the frame object, if any.

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

- ◊ Object = 0
- ◊ Group = 1
- ◊ SelectedObjects = 2

If this item is Objects, the design results are retrieved for the frame object specified by the Name item.

If this item is Group, the design results are retrieved for all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the design results are retrieved for all selected frame objects, and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the results are successfully retrieved; otherwise it returns a nonzero value.

▲ **Remarks**

Note that torsional design is only included for some codes.

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim NumberItems As Integer
    Dim FrameName() As String
    Dim Location() As Double
    Dim TopCombo() As String
    Dim TopArea() As Double
    Dim BotCombo() As String
    Dim BotArea() As Double
    Dim VmajorCombo() As String
    Dim VmajorArea() As Double
    Dim TLCombo() As String
    Dim TLArea() As Double
    Dim TTCombo() As String
    Dim TTArea() As Double
    Dim ErrorSummary() As String
    Dim WarningSummary() As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'add ASTM A706 rebar material
    ret = SapModel.PropMaterial.AddMaterial(Name, eMatType.Rebar, "United States", "ASTM A706", "C")

    'create new concrete frame section properties
    ret = SapModel.PropFrame.SetRectangle("COL", "4000Psi", 20, 20)
    ret = SapModel.PropFrame.SetRectangle("BEAM", "4000Psi", 20, 12)
    ret = SapModel.PropFrame.SetRebarBeam("BEAM", Name, Name, 2, 2, 2, 2, 2, 2)

    'assign concrete beam
    ret = SapModel.FrameObj.SetSection("74", "BEAM")

    'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
```

Introduction

```
ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
ret = SapModel.Analyze.RunAnalysis()

'start concrete design
ret = SapModel.DesignConcrete.StartDesign()

'get summary result data
ret = SapModel.DesignConcrete.GetSummaryResultsBeam("74", NumberItems, FrameName, Location, To

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

[cDesignConcrete Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignConcreteAddLanguageSpecificTextSet("LSTCC Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetSummaryResultsBeam_2(
    string Name,
    ref int NumberItems,
    ref string[] FrameName,
    ref double[] Location,
    ref string[] TopCombo,
    ref double[] TopArea,
    ref double[] TopAreaReq,
    ref double[] TopAreaMin,
    ref double[] TopAreaProvided,
    ref string[] BotCombo,
    ref double[] BotArea,
    ref double[] BotAreaReq,
    ref double[] BotAreaMin,
    ref double[] BotAreaProvided,
    ref string[] VmajorCombo,
    ref double[] VmajorArea,
    ref double[] VmajorAreaReq,
    ref double[] VmajorAreaMin,
    ref double[] VmajorAreaProvided,
    ref string[] TLCombo,
    ref double[] TLArea,
    ref string[] TTCombo,
    ref double[] TTArea,
    ref string[] ErrorSummary,
    ref string[] WarningSummary,
    eItemType ItemType = eItemType.Objects
)
```

```
Function GetSummaryResultsBeam_2 (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef FrameName As String(),
    ByRef Location As Double(),
    ByRef TopCombo As String(),
    ByRef TopArea As Double(),
    ByRef TopAreaReq As Double(),
    ByRef TopAreaMin As Double(),
    ByRef TopAreaProvided As Double(),
    ByRef BotCombo As String(),
    ByRef BotArea As Double(),
    ByRef BotAreaReq As Double(),
    ByRef BotAreaMin As Double(),
```

cDesignConcreteAddLanguageSpecificTextSet("LSTCC21A60F_0?app=::|nu="

Introduction

```
ByRef BotAreaProvided As Double(),
ByRef VmajorCombo As String(),
ByRef VmajorArea As Double(),
ByRef VmajorAreaReq As Double(),
ByRef VmajorAreaMin As Double(),
ByRef VmajorAreaProvided As Double(),
ByRef TLCombo As String(),
ByRef TLArea As Double(),
ByRef TTCombo As String(),
ByRef TTArea As Double(),
ByRef ErrorSummary As String(),
ByRef WarningSummary As String(),
    itemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDesignConcrete
Dim Name As String
Dim NumberItems As Integer
Dim FrameName As String()
Dim Location As Double()
Dim TopCombo As String()
Dim TopArea As Double()
Dim TopAreaReq As Double()
Dim TopAreaMin As Double()
Dim TopAreaProvided As Double()
Dim BotCombo As String()
Dim BotArea As Double()
Dim BotAreaReq As Double()
Dim BotAreaMin As Double()
Dim BotAreaProvided As Double()
Dim VmajorCombo As String()
Dim VmajorArea As Double()
Dim VmajorAreaReq As Double()
Dim VmajorAreaMin As Double()
Dim VmajorAreaProvided As Double()
Dim TLCombo As String()
Dim TLArea As Double()
Dim TTCombo As String()
Dim TTArea As Double()
Dim ErrorSummary As String()
Dim WarningSummary As String()
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.GetSummaryResultsBeam_2(Name,
NumberItems, FrameName, Location,
TopCombo, TopArea, TopAreaReq, TopAreaMin,
TopAreaProvided, BotCombo, BotArea,
BotAreaReq, BotAreaMin, BotAreaProvided,
VmajorCombo, VmajorArea, VmajorAreaReq,
VmajorAreaMin, VmajorAreaProvided,
TLCombo, TLArea, TTCombo, TTArea,
ErrorSummary, WarningSummary, ItemType)
```

```
int GetSummaryResultsBeam_2(
String^ Name,
int% NumberItems,
array<String^>^% FrameName,
array<double>^% Location,
array<String^>^% TopCombo,
array<double>^% TopArea,
array<double>^% TopAreaReq,
```

cDesignConcretespan id="LSTCC21A60F_0"AddLanguageSpecificTextSet("LSTCC21A60F_01272::|nu="

```

array<double>^% TopAreaMin,
array<double>^% TopAreaProvided,
array<String^>^% BotCombo,
array<double>^% BotArea,
array<double>^% BotAreaReq,
array<double>^% BotAreaMin,
array<double>^% BotAreaProvided,
array<String^>^% VmajorCombo,
array<double>^% VmajorArea,
array<double>^% VmajorAreaReq,
array<double>^% VmajorAreaMin,
array<double>^% VmajorAreaProvided,
array<String^>^% TLCombo,
array<double>^% TLArea,
array<String^>^% TTCombo,
array<double>^% TTArea,
array<String^>^% ErrorSummary,
array<String^>^% WarningSummary,
eItemType ItemType = eItemType::Objects
)

abstract GetSummaryResultsBeam_2 :
    Name : string *
    NumberItems : int byref *
    FrameName : string[] byref *
    Location : float[] byref *
    TopCombo : string[] byref *
    TopArea : float[] byref *
    TopAreaReq : float[] byref *
    TopAreaMin : float[] byref *
    TopAreaProvided : float[] byref *
    BotCombo : string[] byref *
    BotArea : float[] byref *
    BotAreaReq : float[] byref *
    BotAreaMin : float[] byref *
    BotAreaProvided : float[] byref *
    VmajorCombo : string[] byref *
    VmajorArea : float[] byref *
    VmajorAreaReq : float[] byref *
    VmajorAreaMin : float[] byref *
    VmajorAreaProvided : float[] byref *
    TLCombo : string[] byref *
    TLArea : float[] byref *
    TTCombo : string[] byref *
    TTArea : float[] byref *
    ErrorSummary : string[] byref *
    WarningSummary : string[] byref *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int

```

Parameters

Name

Type:Â SystemString

NumberItems

Type:Â SystemInt32

FrameName

Parameters

Type:Â SystemString
 Location
 Type:Â SystemDouble
 TopCombo
 Type:Â SystemString
 TopArea
 Type:Â SystemDouble
 TopAreaReq
 Type:Â SystemDouble
 TopAreaMin
 Type:Â SystemDouble
 TopAreaProvided
 Type:Â SystemDouble
 BotCombo
 Type:Â SystemString
 BotArea
 Type:Â SystemDouble
 BotAreaReq
 Type:Â SystemDouble
 BotAreaMin
 Type:Â SystemDouble
 BotAreaProvided
 Type:Â SystemDouble
 VmajorCombo
 Type:Â SystemString
 VmajorArea
 Type:Â SystemDouble
 VmajorAreaReq
 Type:Â SystemDouble
 VmajorAreaMin
 Type:Â SystemDouble
 VmajorAreaProvided
 Type:Â SystemDouble
 TLCombo
 Type:Â SystemString
 TLArea
 Type:Â SystemDouble
 TTCombo
 Type:Â SystemString
 TTArea
 Type:Â SystemDouble
 ErrorSummary
 Type:Â SystemString
 WarningSummary
 Type:Â SystemString
 ItemType (Optional)
 Type:Â ETABSv1eItemType

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cDesignConcrete Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignConcreteAddLanguageSpecificTextSet("LST7E Method

Retrieves column summary results for concrete design

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetSummaryResultsColumn(
    string Name,
    ref int NumberItems,
    ref string[] FrameName,
    ref int[] MyOption,
    ref double[] Location,
    ref string[] PMMCombo,
    ref double[] PMMArea,
    ref double[] PMMRatio,
    ref string[] VMajorCombo,
    ref double[] AVMajor,
    ref string[] VMinorCombo,
    ref double[] AVMinor,
    ref string[] ErrorSummary,
    ref string[] WarningSummary,
    eItemType ItemType = eItemType.Objects
)
```

```
Function GetSummaryResultsColumn (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef FrameName As String(),
    ByRef MyOption As Integer(),
    ByRef Location As Double(),
    ByRef PMMCombo As String(),
    ByRef PMMArea As Double(),
    ByRef PMMRatio As Double(),
    ByRef VMajorCombo As String(),
    ByRef AVMajor As Double(),
    ByRef VMinorCombo As String(),
    ByRef AVMinor As Double(),
    ByRef ErrorSummary As String(),
    ByRef WarningSummary As String(),
    Optional ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDesignConcrete
Dim Name As String
Dim NumberItems As Integer
Dim FrameName As String()
Dim MyOption As Integer()
```

cDesignConcreteAddLanguageSpecificTextSet("LST7EC1A668_0?c=1076::|nu=."

Introduction

```
Dim Location As Double()
Dim PMMCombo As String()
Dim PMMArea As Double()
Dim PMMRatio As Double()
Dim VMajorCombo As String()
Dim AVMajor As Double()
Dim VMinorCombo As String()
Dim AVMinor As Double()
Dim ErrorSummary As String()
Dim WarningSummary As String()
Dim ItemType As eItemType
Dim returnValue As Integer

returnValue = instance.GetSummaryResultsColumn(Name,
NumberItems, FrameName, MyOption,
Location, PMMCombo, PMMArea, PMMRatio,
VMajorCombo, AVMajor, VMinorCombo,
AVMinor, ErrorSummary, WarningSummary,
ItemType)

int GetSummaryResultsColumn(
String^ Name,
int% NumberItems,
array<String^>^% FrameName,
array<int>^% MyOption,
array<double>^% Location,
array<String^>^% PMMCombo,
array<double>^% PMMArea,
array<double>^% PMMRatio,
array<String^>^% VMajorCombo,
array<double>^% AVMajor,
array<String^>^% VMinorCombo,
array<double>^% AVMinor,
array<String^>^% ErrorSummary,
array<String^>^% WarningSummary,
eItemType ItemType = eItemType::Objects
)

abstract GetSummaryResultsColumn :
    Name : string *
    NumberItems : int byref *
    FrameName : string[] byref *
    MyOption : int[] byref *
    Location : float[] byref *
    PMMCombo : string[] byref *
    PMMArea : float[] byref *
    PMMRatio : float[] byref *
    VMajorCombo : string[] byref *
    AVMajor : float[] byref *
    VMinorCombo : string[] byref *
    AVMinor : float[] byref *
    ErrorSummary : string[] byref *
    WarningSummary : string[] byref *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters**Name**

Type:Â SystemString

The name of an existing frame object or group, depending on the value of the ItemType item.

NumberItems

Type:Â SystemInt32

The number of frame objects for which results are obtained.

FrameName

Type:Â SystemString

This is an array that includes each frame object name for which results are obtained.

MyOption

Type:Â SystemInt32

This is an array that includes 1 or 2, indicating the design option for each frame object

1. Check

2. Design

Location

Type:Â SystemDouble

This is an array that includes the distance from the I-end of the frame object to the location where the results are reported. [L]

PMMCombo

Type:Â SystemString

This is an array that includes the name of the design combination for which the controlling PMM ratio or rebar area occurs. A combination name followed by (Sp) indicates that the design loads were obtained by applying special, code-specific multipliers to all or part of the specified design load combination, or that the design was based on the capacity of other objects (or other design locations for the same object).

PMMArea

Type:Â SystemDouble

This is an array that includes the total longitudinal rebar area required for the axial force plus biaxial moment (PMM) design at the specified location. [L2]

This item applies only when MyOption = 2 (design).

PMMRatio

Type:Â SystemDouble

This is an array that includes the axial force plus biaxial moment (PMM) stress ratio at the specified location

This item applies only when MyOption = 1 (check).

VMajorCombo

Type:Â SystemString

This is an array that includes the name of the design combination for which the controlling major shear occurs.

AVMajor

Type:Â SystemDouble

This is an array that includes the required area of transverse shear reinforcing per unit length along the frame object for major shear at the specified location.

[L2/L]

VMinorCombo

Type:Â SystemString

This is an array that includes the name of the design combination for which the controlling minor shear occurs.

AVMinor

Type:Â SystemDouble

This is an array that includes the required area of transverse shear reinforcing per unit length along the frame object for minor shear at the specified location.

[L2/L]

ErrorSummary

Type:Â SystemString

This is an array that includes the design error messages for the frame object, if any.

WarningSummary

Type:Â SystemString

This is an array that includes the design warning messages for the frame object, if any.

ItemType (Optional)

Type:Â ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

- ◊ Object = 0

- ◊ Group = 1

- ◊ SelectedObjects = 2

If this item is Objects, the design results are retrieved for the frame object specified by the Name item.

If this item is Group, the design results are retrieved for all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the design results are retrieved for all selected frame objects, and the Name item is ignored.

Return Value

Type:Â Int32

Returns zero if the results are successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim NumberItems As Integer
    Dim FrameName() As String
    Dim MyOption() As Integer
    Dim Location() As Double
    Dim PMMCombo() As String
    Dim PMMArea() As Double
```

Introduction

```
Dim PMMRatio() As Double
Dim VmajorCombo() As String
Dim AVmajor() As Double
Dim VminorCombo() As String
Dim AVminor() As Double
Dim ErrorSummary() As String
Dim WarningSummary() As String

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'add ASTM A706 rebar material
ret = SapModel.PropMaterial.AddMaterial(Name, eMatType.Rebar, "United States", "ASTM A706", "C")

'create new concrete frame section properties
ret = SapModel.PropFrame.SetRectangle("COL", "4000Psi", 20, 20)
ret = SapModel.PropFrame.SetRectangle("BEAM", "4000Psi", 20, 12)
ret = SapModel.PropFrame.SetRebarBeam("BEAM", Name, Name, 2, 2, 2, 2, 2, 2)

'assign concrete column
ret = SapModel.FrameObj.SetSection("4", "COL")

'run analysis
System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
ret = SapModel.Analyze.RunAnalysis()

'start concrete design
ret = SapModel.DesignConcrete.StartDesign()

'get summary result data
ret = SapModel.DesignConcrete.GetSummaryResultsColumn("4", NumberItems, FrameName, MyOption, L)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

📌 See Also

Reference

[cDesignConcrete Interface](#)
[ETABSv1 Namespace](#)

Introduction

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CSI API ETABS v1

cDesignConcreteAddLanguageSpecificTextSet("LSTF4 Method

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetSummaryResultsJoint(
    string Name,
    ref int NumberItems,
    ref string[] FrameName,
    ref string[] LCJSRatioMajor,
    ref double[] JSRatioMajor,
    ref string[] LCJSRatioMinor,
    ref double[] JSRatioMinor,
    ref string[] LCBCCRatioMajor,
    ref double[] BCCRatioMajor,
    ref string[] LCBCCRatioMinor,
    ref double[] BCCRatioMinor,
    ref string[] ErrorSummary,
    ref string[] WarningSummary,
    eItemType ItemType = eItemType.Objects
)
```

```
Function GetSummaryResultsJoint (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef FrameName As String(),
    ByRef LCJSRatioMajor As String(),
    ByRef JSRatioMajor As Double(),
    ByRef LCJSRatioMinor As String(),
    ByRef JSRatioMinor As Double(),
    ByRef LCBCCRatioMajor As String(),
    ByRef BCCRatioMajor As Double(),
    ByRef LCBCCRatioMinor As String(),
    ByRef BCCRatioMinor As Double(),
    ByRef ErrorSummary As String(),
    ByRef WarningSummary As String(),
    ByRef ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDesignConcrete
Dim Name As String
Dim NumberItems As Integer
Dim FrameName As String()
Dim LCJSRatioMajor As String()
Dim JSRatioMajor As Double()
Dim LCJSRatioMinor As String()
Dim JSRatioMinor As Double()
Dim LCBCCRatioMajor As String()
```

cDesignConcreteAddLanguageSpecificTextSet("LSTF41B5101_0?cp=2;nu=.")

Introduction

```
Dim BCCRatioMajor As Double()
Dim LCBCCRatioMinor As String()
Dim BCCRatioMinor As Double()
Dim ErrorSummary As String()
Dim WarningSummary As String()
Dim ItemType As eItemType
Dim returnValue As Integer

returnValue = instance.GetSummaryResultsJoint(Name,
NumberItems, FrameName, LCJSRatioMajor,
JSRatioMajor, LCJSRatioMinor, JSRatioMinor,
LCBCCRatioMajor, BCCRatioMajor,
LCBCCRatioMinor, BCCRatioMinor,
ErrorSummary, WarningSummary, ItemType)

int GetSummaryResultsJoint(
String^ Name,
int% NumberItems,
array<String^>^% FrameName,
array<String^>^% LCJSRatioMajor,
array<double>^% JSRatioMajor,
array<String^>^% LCJSRatioMinor,
array<double>^% JSRatioMinor,
array<String^>^% LCBCCRatioMajor,
array<double>^% BCCRatioMajor,
array<String^>^% LCBCCRatioMinor,
array<double>^% BCCRatioMinor,
array<String^>^% ErrorSummary,
array<String^>^% WarningSummary,
eItemType ItemType = eItemType::Objects
)

abstract GetSummaryResultsJoint :
    Name : string *
    NumberItems : int byref *
    FrameName : string[] byref *
    LCJSRatioMajor : string[] byref *
    JSRatioMajor : float[] byref *
    LCJSRatioMinor : string[] byref *
    JSRatioMinor : float[] byref *
    LCBCCRatioMajor : string[] byref *
    BCCRatioMajor : float[] byref *
    LCBCCRatioMinor : string[] byref *
    BCCRatioMinor : float[] byref *
    ErrorSummary : string[] byref *
    WarningSummary : string[] byref *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name
Type:Â SystemString

NumberItems
Type:Â SystemInt32

FrameName
Type:Â SystemString

Parameters

LCJSRatioMajor
 Type:Â [SystemString](#)
 JSRatioMajor
 Type:Â [SystemDouble](#)
 LCJSRatioMinor
 Type:Â [SystemString](#)
 JSRatioMinor
 Type:Â [SystemDouble](#)
 LCBCCRatioMajor
 Type:Â [SystemString](#)
 BCCRatioMajor
 Type:Â [SystemDouble](#)
 LCBCCRatioMinor
 Type:Â [SystemString](#)
 BCCRatioMinor
 Type:Â [SystemDouble](#)
 ErrorSummary
 Type:Â [SystemString](#)
 WarningSummary
 Type:Â [SystemString](#)
 ItemType (Optional)
 Type:Â [ETABSv1eItemType](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cDesignConcrete Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignConcreteAddLanguageSpecificTextSet("LST69D05DBD_0") Method

Sets the concrete design code.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetCode(
    string CodeName
)
```

```
Function SetCode (
    CodeName As String
) As Integer
```

```
Dim instance As cDesignConcrete
Dim CodeName As String
Dim returnValue As Integer
```

```
returnValue = instance.SetCode(CodeName)
```

```
int SetCode(
    String^ CodeName
)
```

```
abstract SetCode :
    CodeName : string -> int
```

Parameters

CodeName

Type: SystemString

This is one of the following concrete design code names.

- ◇ ACI 318-14
- ◇ ACI 318-11
- ◇ ACI 318-08
- ◇ AS 3600-2018
- ◇ AS 3600-09
- ◇ BS 8110-97
- ◇ Chinese 2010
- ◇ CSA A23.3-14
- ◇ CSA A23.3-04
- ◇ Eurocode 2-2004
- ◇ Hong Kong CP 2013

- ◇ IS 456:2000
- ◇ Italian NTC 2008
- ◇ KBC 2016
- ◇ KBC 2009
- ◇ Mexican RCDF 2017
- ◇ Mexican RCDF 2004
- ◇ NZS 3101:2006
- ◇ Singapore CP 65:99
- ◇ SP 63.13330.2012
- ◇ TS 500-2000(R2018)
- ◇ TS 500-2000
- ◇ TCVN 5574:2012

Return Value

Type: Int32

Returns zero if the code is successfully set; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'create new concrete frame section property
    ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

    'set frame section property
    ret = SapModel.FrameObj.SetSection("8", "R1")

    'set concrete design code
    ret = SapModel.DesignConcrete.SetCode("ACI 318-08/IBC 2009")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
```

End Sub

▲ See Also

Reference

cDesignConcrete Interface

ETABSv1 Namespace

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i>ç

CSI API ETABS v1

cDesignConcreteAddLanguageSpecificTextSet("LST4E6DFF58_0?cpe::|nu=")

Method

Modifies the design section for all specified frame objects that have a concrete frame design procedure

Namespace: Â ETABSv1

Assembly: Â ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetDesignSection(
    string Name,
    string PropName,
    bool LastAnalysis,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetDesignSection (
    Name As String,
    PropName As String,
    LastAnalysis As Boolean,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDesignConcrete
Dim Name As String
Dim PropName As String
Dim LastAnalysis As Boolean
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetDesignSection(Name,
    PropName, LastAnalysis, ItemType)
```

```
int SetDesignSection(
    String^ Name,
    String^ PropName,
    bool LastAnalysis,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetDesignSection :
    Name : string *
    PropName : string *
    LastAnalysis : bool *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

cDesignConcreteAddLanguageSpecificTextSet("LST4E6DFF58_0?cpe::|nu=")

Parameters**Name**Type: SystemString

The name of an existing frame object or group, depending on the value of the ItemType item

PropNameType: SystemString

The name of an existing frame section property to be used as the design section for the specified frame objects. This item applies only when LastAnalysis = False

LastAnalysisType: SystemBoolean

If this item is True, the design section for the specified frame objects is reset to the last analysis section for the frame object. If it is False, the design section is set to that specified by PropName

ItemType (Optional)Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the assignment is made to the frame object specified by the Name item.

If this item is Group, the assignment is made to all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the assignment is made to all selected frame objects and the Name item is ignored.

Return ValueType: Int32

Returns zero if the design section is successfully modified; otherwise it returns a nonzero value

Remarks**Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
```

Introduction

```
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

ret = SapModel.File.NewGridOnly(1, 10, 10, 2, 2, 10, 10)

'create new concrete frame section properties
ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)
ret = SapModel.PropFrame.SetRectangle("R2", "4000Psi", 20, 16)

'create a concrete column
Dim FrameName1 As String
ret = SapModel.FrameObj.AddByCoord(0, 0, 0, 0, 0, 10, FrameName1, "R1")

'run analysis
ret = SapModel.File.Save("C:\ETABSAPI\example.edb")
ret = SapModel.Analyze.RunAnalysis

'start concrete design
ret = SapModel.DesignConcrete.StartDesign

'get design section
Dim PropName As String
ret = SapModel.DesignConcrete.GetDesignSection(FrameName1, PropName)

'set design section
ret = SapModel.DesignConcrete.SetDesignSection(FrameName1, "R2", False)

ret = SapModel.DesignConcrete.GetDesignSection(FrameName1, PropName)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

📌 See Also

Reference

[cDesignConcrete Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignConcreteAddLanguageSpecificTextSet("LSTA56FCC24_0") Method

Starts the concrete frame design.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int StartDesign()
```

```
Function StartDesign As Integer
```

```
Dim instance As cDesignConcrete
```

```
Dim returnValue As Integer
```

```
returnValue = instance.StartDesign()
```

```
int StartDesign()
```

```
abstract StartDesign : unit -> int
```

Return Value

Type: [Int32](#)

Returns zero if the concrete frame design is successfully started; otherwise it returns a nonzero value.

▲ Remarks

The function will fail if no concrete frame objects are present. It also will fail if analysis results are not available.

▲ Examples

VB

[Copy](#)

```
Public Sub Example()
```

```
    Dim SapModel As cSapModel
```

```
    Dim EtabsObject As cOAPI
```

```
    Dim ret As Integer = -1
```

```
'create ETABS object
```

```
    Dim myHelper as cHelper = New Helper
```

```
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

```
'start ETABS application
```

```
    ret = EtabsObject.ApplicationStart()
```

```
'create SapModel object
```

```
    SapModel = EtabsObject.SapModel
```

cDesignConcreteAddLanguageSpecificTextSet("LSTA56FCC24_0")

Introduction

```
'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'create new concrete frame section property
    ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

'set frame section property
    ret = SapModel.FrameObj.SetSection("8", "R1")

'set concrete design code
    ret = SapModel.DesignConcrete.SetCode("ACI 318-08/IBC 2009")

'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis

'start concrete design
    ret = SapModel.DesignConcrete.StartDesign()

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cDesignConcrete Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDesignConcreteShell Interface

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDesignConcreteShell
```

```
Public Interface cDesignConcreteShell
```

```
Dim instance As cDesignConcreteShell
```

```
public interface class cDesignConcreteShell
```

```
type cDesignConcreteShell = interface end
```

▲ See Also

Reference

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cDesignConcreteSlab Interface

Namespace: $\hat{A}$ [ETABSv1](#)

Assembly: $\hat{A}$ ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▀ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDesignConcreteSlab
```

```
Public Interface cDesignConcreteSlab
```

```
Dim instance As cDesignConcreteSlab
```

```
public interface class cDesignConcreteSlab
```

```
type cDesignConcreteSlab = interface end
```

The cDesignConcreteSlab type exposes the following members.

▀ Properties

$\hat{A}$	Name	Description
-----------	------	-------------

	ACI318_14	
---	---------------------------	--

	DesignStrip	
---	-----------------------------	--

[Top](#)

▀ Methods

$\hat{A}$	Name	Description
-----------	------	-------------

	GetFlexureAndShear	
---	------------------------------------	--

	GetSummaryResultsFlexureAndShear	
---	--	--

	GetSummaryResultsSpanDefinition	
---	---	--

	StartSlabDesign	Starts the concrete slab design.
---	---------------------------------	----------------------------------

[Top](#)

▀ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignConcreteSlab Properties

The [cDesignConcreteSlab](#) type exposes the following members.

Properties

	Name	Description
--	------	-------------

	ACI318_14	
---	---------------------------	--

	DesignStrip	
---	-----------------------------	--

[Top](#)

 See Also

Reference

[cDesignConcreteSlab Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignConcreteSlabAddLanguageSpecificTextSet("LST Property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
cDConcSlabACI318_14 ACI318_14 { get; }
```

```
ReadOnly Property ACI318_14 As cDConcSlabACI318_14  
Get
```

```
Dim instance As cDesignConcreteSlab  
Dim value As cDConcSlabACI318_14
```

```
value = instance.ACI318_14
```

```
property cDConcSlabACI318_14^ ACI318_14 {  
cDConcSlabACI318_14^ get ();  
}
```

```
abstract ACI318_14 : cDConcSlabACI318_14 with get
```

Property Value

Type: cDConcSlabACI318_14

▲ See Also

Reference

cDesignConcreteSlab Interface

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CSI API ETABS v1

cDesignConcreteSlabAddLanguageSpecificTextSet("LST5E6F06DC_0") Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cDesignStrip DesignStrip { get; }
```

```
ReadOnly Property DesignStrip As cDesignStrip
Get
```

```
Dim instance As cDesignConcreteSlab
Dim value As cDesignStrip
```

```
value = instance.DesignStrip
```

```
property cDesignStrip^ DesignStrip {
cDesignStrip^ get ();
}
```

```
abstract DesignStrip : cDesignStrip with get
```

Property Value

Type: [cDesignStrip](#)

▲ See Also

Reference

[cDesignConcreteSlab Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignConcreteSlab Methods

The [cDesignConcreteSlab](#) type exposes the following members.

Methods

	Name	Description
➤	GetFlexureAndShear	
➤	GetSummaryResultsFlexureAndShear	
➤	GetSummaryResultsSpanDefinition	
➤	StartSlabDesign	Starts the concrete slab design.

[Top](#)

➤ See Also

Reference

[cDesignConcreteSlab Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cDesignConcreteSlabAddLanguageSpecificTextSet("LST64D8521C_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetFlexureAndShear(
    ref string[] StoryName,
    ref string[] DesignStripName,
    ref double[] Station,
    ref double[] ConcWidth,
    ref string[] FTopCombo,
    ref double[] FTopMoment,
    ref double[] FTopArea,
    ref double[] FTopAMin,
    ref string[] FBotCombo,
    ref double[] FBotMoment,
    ref double[] FBotArea,
    ref double[] FBotAMin,
    ref double[] AxialForce,
    ref string[] VCombo,
    ref double[] VForce,
    ref double[] VArea,
    ref string[] Status,
    ref double[] GlobalX,
    ref double[] GlobalY,
    ref string[] Layer
)
```

```
Function GetFlexureAndShear (
    ByRef StoryName As String(),
    ByRef DesignStripName As String(),
    ByRef Station As Double(),
    ByRef ConcWidth As Double(),
    ByRef FTopCombo As String(),
    ByRef FTopMoment As Double(),
    ByRef FTopArea As Double(),
    ByRef FTopAMin As Double(),
    ByRef FBotCombo As String(),
    ByRef FBotMoment As Double(),
    ByRef FBotArea As Double(),
    ByRef FBotAMin As Double(),
    ByRef AxialForce As Double(),
    ByRef VCombo As String(),
    ByRef VForce As Double(),
    ByRef VArea As Double(),
    ByRef Status As String(),
    ByRef GlobalX As Double(),
    ByRef GlobalY As Double(),
```

cDesignConcreteSlabsid="LST64D8521C_0"AddLanguageSpecificTextSet("LST64D8521C_0")
cpp=::in

Introduction

```
ByRef Layer As String()  
) As Integer
```

```
Dim instance As cDesignConcreteSlab  
Dim StoryName As String()  
Dim DesignStripName As String()  
Dim Station As Double()  
Dim ConcWidth As Double()  
Dim FTopCombo As String()  
Dim FTopMoment As Double()  
Dim FTopArea As Double()  
Dim FTopAMin As Double()  
Dim FBotCombo As String()  
Dim FBotMoment As Double()  
Dim FBotArea As Double()  
Dim FBotAMin As Double()  
Dim AxialForce As Double()  
Dim VCombo As String()  
Dim VForce As Double()  
Dim VArea As Double()  
Dim Status As String()  
Dim GlobalX As Double()  
Dim GlobalY As Double()  
Dim Layer As String()  
Dim returnValue As Integer
```

```
returnValue = instance.GetFlexureAndShear(StoryName,  
DesignStripName, Station, ConcWidth,  
FTopCombo, FTopMoment, FTopArea,  
FTopAMin, FBotCombo, FBotMoment,  
FBotArea, FBotAMin, AxialForce, VCombo,  
VForce, VArea, Status, GlobalX, GlobalY,  
Layer)
```

```
int GetFlexureAndShear(  
array<String>^% StoryName,  
array<String>^% DesignStripName,  
array<double>^% Station,  
array<double>^% ConcWidth,  
array<String>^% FTopCombo,  
array<double>^% FTopMoment,  
array<double>^% FTopArea,  
array<double>^% FTopAMin,  
array<String>^% FBotCombo,  
array<double>^% FBotMoment,  
array<double>^% FBotArea,  
array<double>^% FBotAMin,  
array<double>^% AxialForce,  
array<String>^% VCombo,  
array<double>^% VForce,  
array<double>^% VArea,  
array<String>^% Status,  
array<double>^% GlobalX,  
array<double>^% GlobalY,  
array<String>^% Layer  
)
```

```
abstract GetFlexureAndShear :  
    StoryName : string[] byref *  
    DesignStripName : string[] byref *  
    Station : float[] byref *  
    ConcWidth : float[] byref *
```



```

FTopCombo : string[] byref *
FTopMoment : float[] byref *
FTopArea : float[] byref *
FTopAMin : float[] byref *
FBotCombo : string[] byref *
FBotMoment : float[] byref *
FBotArea : float[] byref *
FBotAMin : float[] byref *
AxialForce : float[] byref *
VCombo : string[] byref *
VForce : float[] byref *
VArea : float[] byref *
Status : string[] byref *
GlobalX : float[] byref *
GlobalY : float[] byref *
Layer : string[] byref -> int

```

Parameters

```

StoryName
    Type:Â SystemString
DesignStripName
    Type:Â SystemString
Station
    Type:Â SystemDouble
ConcWidth
    Type:Â SystemDouble
FTopCombo
    Type:Â SystemString
FTopMoment
    Type:Â SystemDouble
FTopArea
    Type:Â SystemDouble
FTopAMin
    Type:Â SystemDouble
FBotCombo
    Type:Â SystemString
FBotMoment
    Type:Â SystemDouble
FBotArea
    Type:Â SystemDouble
FBotAMin
    Type:Â SystemDouble
AxialForce
    Type:Â SystemDouble
VCombo
    Type:Â SystemString
VForce
    Type:Â SystemDouble
VArea
    Type:Â SystemDouble
Status
    Type:Â SystemString
GlobalX

```

Type:Â SystemDouble
GlobalY
Type:Â SystemDouble
Layer
Type:Â SystemString

Return Value

Type:Â Int32

▲ See Also

Reference

cDesignConcreteSlab Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDesignConcreteSlabAddLanguageSpecificTextSet("LST83E301F0_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetSummaryResultsFlexureAndShear(
    ref string[] StoryName,
    ref string[] DesignStripName,
    ref string[] SpanID,
    ref string[] Location,
    ref string[] FTopCombo,
    ref double[] FTopMoment,
    ref double[] FTopArea,
    ref string[] FBotCombo,
    ref double[] FBotMoment,
    ref double[] FBotArea,
    ref string[] VCombo,
    ref double[] VForce,
    ref double[] VArea,
    ref string[] Status,
    ref string[] Layer
)
```

```
Function GetSummaryResultsFlexureAndShear (
    ByRef StoryName As String(),
    ByRef DesignStripName As String(),
    ByRef SpanID As String(),
    ByRef Location As String(),
    ByRef FTopCombo As String(),
    ByRef FTopMoment As Double(),
    ByRef FTopArea As Double(),
    ByRef FBotCombo As String(),
    ByRef FBotMoment As Double(),
    ByRef FBotArea As Double(),
    ByRef VCombo As String(),
    ByRef VForce As Double(),
    ByRef VArea As Double(),
    ByRef Status As String(),
    ByRef Layer As String()
) As Integer
```

```
Dim instance As cDesignConcreteSlab
Dim StoryName As String()
Dim DesignStripName As String()
Dim SpanID As String()
Dim Location As String()
Dim FTopCombo As String()
Dim FTopMoment As Double()
```

cDesignConcreteSlabspan id="LST83E301F0_0"AddLanguageSpecificTextSet("LST83E301F01000pp=::|nu

Introduction

```
Dim FTopArea As Double()
Dim FBotCombo As String()
Dim FBotMoment As Double()
Dim FBotArea As Double()
Dim VCombo As String()
Dim VForce As Double()
Dim VArea As Double()
Dim Status As String()
Dim Layer As String()
Dim returnValue As Integer

returnValue = instance.GetSummaryResultsFlexureAndShear(StoryName,
DesignStripName, SpanID, Location,
FTopCombo, FTopMoment, FTopArea,
FBotCombo, FBotMoment, FBotArea,
VCombo, VForce, VArea, Status, Layer)

int GetSummaryResultsFlexureAndShear(
array<String^>^% StoryName,
array<String^>^% DesignStripName,
array<String^>^% SpanID,
array<String^>^% Location,
array<String^>^% FTopCombo,
array<double>^% FTopMoment,
array<double>^% FTopArea,
array<String^>^% FBotCombo,
array<double>^% FBotMoment,
array<double>^% FBotArea,
array<String^>^% VCombo,
array<double>^% VForce,
array<double>^% VArea,
array<String^>^% Status,
array<String^>^% Layer
)

abstract GetSummaryResultsFlexureAndShear :
    StoryName : string[] byref *
    DesignStripName : string[] byref *
    SpanID : string[] byref *
    Location : string[] byref *
    FTopCombo : string[] byref *
    FTopMoment : float[] byref *
    FTopArea : float[] byref *
    FBotCombo : string[] byref *
    FBotMoment : float[] byref *
    FBotArea : float[] byref *
    VCombo : string[] byref *
    VForce : float[] byref *
    VArea : float[] byref *
    Status : string[] byref *
    Layer : string[] byref -> int
```

Parameters

StoryName
Type:Â SystemString

DesignStripName
Type:Â SystemString

SpanID
Type:Â SystemString

Location
 Type:Â SystemString
 FTopCombo
 Type:Â SystemString
 FTopMoment
 Type:Â SystemDouble
 FTopArea
 Type:Â SystemDouble
 FBotCombo
 Type:Â SystemString
 FBotMoment
 Type:Â SystemDouble
 FBotArea
 Type:Â SystemDouble
 VCombo
 Type:Â SystemString
 VForce
 Type:Â SystemDouble
 VArea
 Type:Â SystemDouble
 Status
 Type:Â SystemString
 Layer
 Type:Â SystemString

Return Value

Type:Â Int32

▲ See Also

Reference

cDesignConcreteSlab Interface

ETABSv1 Namespace

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Assembly: Â ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VBC++F#

Copy

```
Function GetSummaryResultsSpanDefinition (
ByRef StoryName As String(),
ByRef DesignStripName As String(),
ByRef SpanID As String(),
ByRef SpanLength As Double(),
ByRef StartDist As Double(),
ByRef EndDist As Double(),
ByRef GlobalX1 As Double(),
ByRef GlobalY1 As Double(),
ByRef GlobalX2 As Double(),
ByRef GlobalY2 As Double()
) As Integer
```

```
returnValue = instance.GetSummaryResultsSpanDefinition(StoryName,
DesignStripName, SpanID, SpanLength,
StartDist, EndDist, GlobalX1, GlobalY1,
GlobalX2, GlobalY2)
```

cDesignConcreteSlabs span id="LSTD581DB36 0"AddLanguageSpecificTextSet("LSTD581DB36306cpp=::|n

```

int GetSummaryResultsSpanDefinition(
array<String>^% StoryName,
array<String>^% DesignStripName,
array<String>^% SpanID,
array<double>^% SpanLength,
array<double>^% StartDist,
array<double>^% EndDist,
array<double>^% GlobalX1,
array<double>^% GlobalY1,
array<double>^% GlobalX2,
array<double>^% GlobalY2
)

```

```

abstract GetSummaryResultsSpanDefinition :
    StoryName : string[] byref *
    DesignStripName : string[] byref *
    SpanID : string[] byref *
    SpanLength : float[] byref *
    StartDist : float[] byref *
    EndDist : float[] byref *
    GlobalX1 : float[] byref *
    GlobalY1 : float[] byref *
    GlobalX2 : float[] byref *
    GlobalY2 : float[] byref -> int

```

Parameters

StoryName

Type: [SystemString](#)

DesignStripName

Type: [SystemString](#)

SpanID

Type: [SystemString](#)

SpanLength

Type: [SystemDouble](#)

StartDist

Type: [SystemDouble](#)

EndDist

Type: [SystemDouble](#)

GlobalX1

Type: [SystemDouble](#)

GlobalY1

Type: [SystemDouble](#)

GlobalX2

Type: [SystemDouble](#)

GlobalY2

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDesignConcreteSlab Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDesignConcreteSlabAddLanguageSpecificTextSet("LST Method

Starts the concrete slab design.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int StartSlabDesign()
```

```
Function StartSlabDesign As Integer
```

```
Dim instance As cDesignConcreteSlab
```

```
Dim returnValue As Integer
```

```
returnValue = instance.StartSlabDesign()
```

```
int StartSlabDesign()
```

```
abstract StartSlabDesign : unit -> int
```

Return Value

Type: Int32

Returns zero if the concrete slab design is successfully started; otherwise it returns a nonzero value.

▲ Remarks

This function designs slab strips, individual finite elements, and checks for punching shear, if concrete slabs are present. The function will fail if analysis results are not available.

▲ See Also

Reference

cDesignConcreteSlab Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDesignForces Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in [ETABSv1.dll](#)) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDesignForces
```

```
Public Interface cDesignForces
```

```
Dim instance As cDesignForces
```

```
public interface class cDesignForces
```

```
type cDesignForces = interface end
```

The cDesignForces type exposes the following members.

▲ Methods

▲	Name	Description
➤	BeamDesignForces	Retrieves design forces for an existing designed beam
➤	BraceDesignForces	Retrieves design forces for an existing designed brace
➤	ColumnDesignForces	Retrieves design forces for an existing designed column
➤	PierDesignForces	Retrieves design forces for an existing designed pier
➤	SpandrelDesignForces	Retrieves design forces for an existing designed spandrel

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignForces Methods

The [cDesignForces](#) type exposes the following members.

Methods

	Name	Description
⇒	BeamDesignForces	Retrieves design forces for an existing designed beam
⇒	BraceDesignForces	Retrieves design forces for an existing designed brace
⇒	ColumnDesignForces	Retrieves design forces for an existing designed column
⇒	PierDesignForces	Retrieves design forces for an existing designed pier
⇒	SpandrelDesignForces	Retrieves design forces for an existing designed spandrel

[Top](#)

See Also

Reference

[cDesignForces Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignForcesAddLanguageSpecificTextSet("LST3831 Method

Retrieves design forces for an existing designed beam

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int BeamDesignForces(
    string Name,
    ref int NumberResults,
    ref string[] FrameName,
    ref string[] ComboName,
    ref double[] Station,
    ref double[] P,
    ref double[] V2,
    ref double[] V3,
    ref double[] T,
    ref double[] M2,
    ref double[] M3,
    eItemType ItemType = eItemType.Objects
)
```

```
Function BeamDesignForces (
    Name As String,
    ByRef NumberResults As Integer,
    ByRef FrameName As String(),
    ByRef ComboName As String(),
    ByRef Station As Double(),
    ByRef P As Double(),
    ByRef V2 As Double(),
    ByRef V3 As Double(),
    ByRef T As Double(),
    ByRef M2 As Double(),
    ByRef M3 As Double(),
    Optional ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDesignForces
Dim Name As String
Dim NumberResults As Integer
Dim FrameName As String()
Dim ComboName As String()
Dim Station As Double()
Dim P As Double()
Dim V2 As Double()
Dim V3 As Double()
Dim T As Double()
Dim M2 As Double()
```

cDesignForcesid="LST383173DC_0"AddLanguageSpecificTextSet("LST383173DC_0?cp1312nu=");B

Introduction

```
Dim M3 As Double()
Dim ItemType As eItemType
Dim returnValue As Integer

returnValue = instance.BeamDesignForces(Name,
NumberResults, FrameName, ComboName,
Station, P, V2, V3, T, M2, M3, ItemType)

int BeamDesignForces(
String^ Name,
int% NumberResults,
array<String^>^% FrameName,
array<String^>^% ComboName,
array<double>^% Station,
array<double>^% P,
array<double>^% V2,
array<double>^% V3,
array<double>^% T,
array<double>^% M2,
array<double>^% M3,
eItemType ItemType = eItemType::Objects
)

abstract BeamDesignForces :
    Name : string *
    NumberResults : int byref *
    FrameName : string[] byref *
    ComboName : string[] byref *
    Station : float[] byref *
    P : float[] byref *
    V2 : float[] byref *
    V3 : float[] byref *
    T : float[] byref *
    M2 : float[] byref *
    M3 : float[] byref *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame object or group, depending on the value of the ItemType item.

NumberResults

Type:Â SystemInt32

The number of results, this is the length of all output arrays

FrameName

Type:Â SystemString

The frame name associated with each result

ComboName

Type:Â SystemString

The load combination for which the results are reported

Station

- Type: SystemDouble
The location, measured from the I-end of the frame object, where the results are reported [L]
- P
Type: SystemDouble
The design axial force in the frame object local 1-axis direction [F]
- V2
Type: SystemDouble
The design shear force in the frame object local 2-axis direction [F]
- V3
Type: SystemDouble
The design shear force in the frame object local 3-axis direction [F]
- T
Type: SystemDouble
The design torsional moment about the frame object local 1-axis [FL]
- M2
Type: SystemDouble
The design bending moment about the frame object local 2-axis [FL]
- M3
Type: SystemDouble
The design bending moment about the frame object local 3-axis [FL]
- ItemType (Optional)
Type: ETABSv1eItemType
This is one of the following items in the eItemType enumeration:
 ◊ Object = 0
 ◊ Group = 1
 ◊ SelectedObjects = 2
 If this item is Objects, results are retrieved for the frame object specified by the Name item.

 If this item is Group, results are retrieved for all frame objects in the group specified by the Name item.

 If this item is SelectedObjects, results are retrieved for all selected frame objects and the Name item is ignored.

Return Value

Type: Int32
Returns zero if the results are successfully retrieved, otherwise it returns a nonzero value.

▲ See Also

Reference

cDesignForces Interface

ETABSv1 Namespace

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Introduction

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CSI API ETABS v1

cDesignForcesAddLanguageSpecificTextSet("LST61B8C9EA_0?cp=1616&nu=.");

Method

Retrieves design forces for an existing designed brace

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int BraceDesignForces(
    string Name,
    ref int NumberResults,
    ref string[] FrameName,
    ref string[] ComboName,
    ref double[] Station,
    ref double[] P,
    ref double[] V2,
    ref double[] V3,
    ref double[] T,
    ref double[] M2,
    ref double[] M3,
    eItemType ItemType = eItemType.Objects
)
```

```
Function BraceDesignForces (
    Name As String,
    ByRef NumberResults As Integer,
    ByRef FrameName As String(),
    ByRef ComboName As String(),
    ByRef Station As Double(),
    ByRef P As Double(),
    ByRef V2 As Double(),
    ByRef V3 As Double(),
    ByRef T As Double(),
    ByRef M2 As Double(),
    ByRef M3 As Double(),
    Optional ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDesignForces
Dim Name As String
Dim NumberResults As Integer
Dim FrameName As String()
Dim ComboName As String()
Dim Station As Double()
Dim P As Double()
Dim V2 As Double()
Dim V3 As Double()
Dim T As Double()
Dim M2 As Double()
```

cDesignForcesid="LST61B8C9EA_0"AddLanguageSpecificTextSet("LST61B8C9EA_0?cp=1616&nu=.");

Introduction

```
Dim M3 As Double()
Dim ItemType As eItemType
Dim returnValue As Integer

returnValue = instance.BraceDesignForces(Name,
NumberResults, FrameName, ComboName,
Station, P, V2, V3, T, M2, M3, ItemType)

int BraceDesignForces(
String^ Name,
int% NumberResults,
array<String^>^% FrameName,
array<String^>^% ComboName,
array<double>^% Station,
array<double>^% P,
array<double>^% V2,
array<double>^% V3,
array<double>^% T,
array<double>^% M2,
array<double>^% M3,
eItemType ItemType = eItemType::Objects
)

abstract BraceDesignForces :
    Name : string *
    NumberResults : int byref *
    FrameName : string[] byref *
    ComboName : string[] byref *
    Station : float[] byref *
    P : float[] byref *
    V2 : float[] byref *
    V3 : float[] byref *
    T : float[] byref *
    M2 : float[] byref *
    M3 : float[] byref *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame object or group, depending on the value of the ItemType item.

NumberResults

Type:Â SystemInt32

The number of results, this is the length of all output arrays

FrameName

Type:Â SystemString

The frame name associated with each result

ComboName

Type:Â SystemString

The load combination for which the results are reported

Station

- Type: SystemDouble
The location, measured from the I-end of the frame object, where the results are reported [L]
- P
Type: SystemDouble
The design axial force in the frame object local 1-axis direction [F]
- V2
Type: SystemDouble
The design shear force in the frame object local 2-axis direction [F]
- V3
Type: SystemDouble
The design shear force in the frame object local 3-axis direction [F]
- T
Type: SystemDouble
The design torsional moment about the frame object local 1-axis [FL]
- M2
Type: SystemDouble
The design bending moment about the frame object local 2-axis [FL]
- M3
Type: SystemDouble
The design bending moment about the frame object local 3-axis [FL]
- ItemType (Optional)
Type: ETABSv1eItemType
This is one of the following items in the eItemType enumeration:
 ◊ Object = 0
 ◊ Group = 1
 ◊ SelectedObjects = 2
 If this item is Objects, results are retrieved for the frame object specified by the Name item.

 If this item is Group, results are retrieved for all frame objects in the group specified by the Name item.

 If this item is SelectedObjects, results are retrieved for all selected frame objects and the Name item is ignored.

Return Value

Type: Int32
Returns zero if the results are successfully retrieved, otherwise it returns a nonzero value.

▲ See Also

Reference

cDesignForces Interface

ETABSv1 Namespace

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Introduction

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CSI API ETABS v1

cDesignForcesAddLanguageSpecificTextSet("LST28DB2405_0?cpu=."); Method

Retrieves design forces for an existing designed column

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int ColumnDesignForces(
    string Name,
    ref int NumberResults,
    ref string[] FrameName,
    ref string[] ComboName,
    ref double[] Station,
    ref double[] P,
    ref double[] V2,
    ref double[] V3,
    ref double[] T,
    ref double[] M2,
    ref double[] M3,
    eItemType ItemType = eItemType.Objects
)
```

```
Function ColumnDesignForces (
    Name As String,
    ByRef NumberResults As Integer,
    ByRef FrameName As String(),
    ByRef ComboName As String(),
    ByRef Station As Double(),
    ByRef P As Double(),
    ByRef V2 As Double(),
    ByRef V3 As Double(),
    ByRef T As Double(),
    ByRef M2 As Double(),
    ByRef M3 As Double(),
    Optional ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDesignForces
Dim Name As String
Dim NumberResults As Integer
Dim FrameName As String()
Dim ComboName As String()
Dim Station As Double()
Dim P As Double()
Dim V2 As Double()
Dim V3 As Double()
Dim T As Double()
Dim M2 As Double()
```

cDesignForcesid="LST28DB2405_0"AddLanguageSpecificTextSet("LST28DB2405_0?cpu=.");

Introduction

```
Dim M3 As Double()
Dim ItemType As eItemType
Dim returnValue As Integer

returnValue = instance.ColumnDesignForces(Name,
NumberResults, FrameName, ComboName,
Station, P, V2, V3, T, M2, M3, ItemType)

int ColumnDesignForces(
String^ Name,
int% NumberResults,
array<String^>^% FrameName,
array<String^>^% ComboName,
array<double>^% Station,
array<double>^% P,
array<double>^% V2,
array<double>^% V3,
array<double>^% T,
array<double>^% M2,
array<double>^% M3,
eItemType ItemType = eItemType::Objects
)

abstract ColumnDesignForces :
    Name : string *
    NumberResults : int byref *
    FrameName : string[] byref *
    ComboName : string[] byref *
    Station : float[] byref *
    P : float[] byref *
    V2 : float[] byref *
    V3 : float[] byref *
    T : float[] byref *
    M2 : float[] byref *
    M3 : float[] byref *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame object or group, depending on the value of the ItemType item.

NumberResults

Type:Â SystemInt32

The number of results, this is the length of all output arrays

FrameName

Type:Â SystemString

The frame name associated with each result

ComboName

Type:Â SystemString

The load combination for which the results are reported

Station

- Type: SystemDouble
The location, measured from the I-end of the frame object, where the results are reported [L]
- P
Type: SystemDouble
The design axial force in the frame object local 1-axis direction [F]
- V2
Type: SystemDouble
The design shear force in the frame object local 2-axis direction [F]
- V3
Type: SystemDouble
The design shear force in the frame object local 3-axis direction [F]
- T
Type: SystemDouble
The design torsional moment about the frame object local 1-axis [FL]
- M2
Type: SystemDouble
The design bending moment about the frame object local 2-axis [FL]
- M3
Type: SystemDouble
The design bending moment about the frame object local 3-axis [FL]
- ItemType (Optional)
Type: ETABSv1eItemType
This is one of the following items in the eItemType enumeration:
 ◊ Object = 0
 ◊ Group = 1
 ◊ SelectedObjects = 2
 If this item is Objects, results are retrieved for the frame object specified by the Name item.

 If this item is Group, results are retrieved for all frame objects in the group specified by the Name item.

 If this item is SelectedObjects, results are retrieved for all selected frame objects and the Name item is ignored.

Return Value

Type: Int32
Returns zero if the results are successfully retrieved, otherwise it returns a nonzero value.

▲ See Also

Reference

cDesignForces Interface

ETABSv1 Namespace

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Introduction

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CSI API ETABS v1

cDesignForcesAddLanguageSpecificTextSet("LST2472C445_0?cpp1324u=.");Pi

Method

Retrieves design forces for an existing designed pier

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int PierDesignForces(
    string InputPierLabel,
    string InputStoryName,
    ref int NumberResults,
    ref string[] Story,
    ref string[] PierLabel,
    ref string[] ComboName,
    ref string[] Location,
    ref double[] P,
    ref double[] V2,
    ref double[] V3,
    ref double[] T,
    ref double[] M2,
    ref double[] M3
)
```

```
Function PierDesignForces (
    InputPierLabel As String,
    InputStoryName As String,
    ByRef NumberResults As Integer,
    ByRef Story As String(),
    ByRef PierLabel As String(),
    ByRef ComboName As String(),
    ByRef Location As String(),
    ByRef P As Double(),
    ByRef V2 As Double(),
    ByRef V3 As Double(),
    ByRef T As Double(),
    ByRef M2 As Double(),
    ByRef M3 As Double()
) As Integer
```

```
Dim instance As cDesignForces
Dim InputPierLabel As String
Dim InputStoryName As String
Dim NumberResults As Integer
Dim Story As String()
Dim PierLabel As String()
Dim ComboName As String()
Dim Location As String()
Dim P As Double()
```

cDesignForcesspan id="LST2472C445_0"AddLanguageSpecificTextSet("LST2472C445_0?cpp1324u=.");Pi

Introduction

```
Dim V2 As Double()
Dim V3 As Double()
Dim T As Double()
Dim M2 As Double()
Dim M3 As Double()
Dim returnValue As Integer

returnValue = instance.PierDesignForces(InputPierLabel,
InputStoryName, NumberResults, Story,
PierLabel, ComboName, Location, P,
V2, V3, T, M2, M3)

int PierDesignForces(
String^ InputPierLabel,
String^ InputStoryName,
int% NumberResults,
array<String^>^% Story,
array<String^>^% PierLabel,
array<String^>^% ComboName,
array<String^>^% Location,
array<double>^% P,
array<double>^% V2,
array<double>^% V3,
array<double>^% T,
array<double>^% M2,
array<double>^% M3
)

abstract PierDesignForces :
    InputPierLabel : string *
    InputStoryName : string *
    NumberResults : int byref *
    Story : string[] byref *
    PierLabel : string[] byref *
    ComboName : string[] byref *
    Location : string[] byref *
    P : float[] byref *
    V2 : float[] byref *
    V3 : float[] byref *
    T : float[] byref *
    M2 : float[] byref *
    M3 : float[] byref -> int
```

Parameters

InputPierLabel

Type:Â SystemString

The name of an existing pier label on the story specified by InputStoryName. If this is left blank, results will be retrieved for all pier labels on the story.

InputStoryName

Type:Â SystemString

The name of an existing story for which to retrieve pier design forces. If this is left blank, results will be retrieved for all stories.

NumberResults

Type:Â SystemInt32

The number of results, this is the length of all output arrays

Story

	Type:Â <u>SystemString</u>
	The story at which the pier exists
PierLabel	
	Type:Â <u>SystemString</u>
	The label of the pier
ComboName	
	Type:Â <u>SystemString</u>
	The load combination for which the results are reported
Location	
	Type:Â <u>SystemString</u>
	The location where the results are reported
P	
	Type:Â <u>SystemDouble</u>
	The design axial force in the pier object local 1-axis direction [F]
V2	
	Type:Â <u>SystemDouble</u>
	The design shear force in the pier object local 2-axis direction [F]
V3	
	Type:Â <u>SystemDouble</u>
	The design shear force in the pier object local 3-axis direction [F]
T	
	Type:Â <u>SystemDouble</u>
	The design torsional moment about the pier object local 1-axis [FL]
M2	
	Type:Â <u>SystemDouble</u>
	The design bending moment about the pier object local 2-axis [FL]
M3	
	Type:Â <u>SystemDouble</u>
	The design bending moment about the pier object local 3-axis [FL]

Return ValueType:Â Int32

Returns zero if the results are successfully retrieved, otherwise it returns a nonzero value.

▲ **Remarks**

To retrieve results for all pier labels on all stories, leave both InputPierLabel and InputStoryName blank.

▲ **See Also****Reference**cDesignForces InterfaceETABSv1 Namespace

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CSI API ETABS v1

cDesignForcesAddLanguageSpecificTextSet("LSTC9E Method

Retrieves design forces for an existing designed spandrel

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SpandrelDesignForces(
    string InputSpandrelLabel,
    string InputStoryName,
    ref int NumberResults,
    ref string[] Story,
    ref string[] SpandrelLabel,
    ref string[] ComboName,
    ref string[] Location,
    ref double[] P,
    ref double[] V2,
    ref double[] V3,
    ref double[] T,
    ref double[] M2,
    ref double[] M3
)
```

```
Function SpandrelDesignForces (
    InputSpandrelLabel As String,
    InputStoryName As String,
    ByRef NumberResults As Integer,
    ByRef Story As String(),
    ByRef SpandrelLabel As String(),
    ByRef ComboName As String(),
    ByRef Location As String(),
    ByRef P As Double(),
    ByRef V2 As Double(),
    ByRef V3 As Double(),
    ByRef T As Double(),
    ByRef M2 As Double(),
    ByRef M3 As Double()
) As Integer
```

```
Dim instance As cDesignForces
Dim InputSpandrelLabel As String
Dim InputStoryName As String
Dim NumberResults As Integer
Dim Story As String()
Dim SpandrelLabel As String()
Dim ComboName As String()
Dim Location As String()
Dim P As Double()
```

cDesignForcesspan id="LSTC9EBD682_0"AddLanguageSpecificTextSet("LSTC9EBD682_0?cp1627|nu=.");

Introduction

```
Dim V2 As Double()
Dim V3 As Double()
Dim T As Double()
Dim M2 As Double()
Dim M3 As Double()
Dim returnValue As Integer

returnValue = instance.SpandrelDesignForces(InputSpandrelLabel,
InputStoryName, NumberResults, Story,
SpandrelLabel, ComboName, Location,
P, V2, V3, T, M2, M3)
```

```
int SpandrelDesignForces(
String^ InputSpandrelLabel,
String^ InputStoryName,
int% NumberResults,
array<String^>^% Story,
array<String^>^% SpandrelLabel,
array<String^>^% ComboName,
array<String^>^% Location,
array<double>^% P,
array<double>^% V2,
array<double>^% V3,
array<double>^% T,
array<double>^% M2,
array<double>^% M3
)
```

```
abstract SpandrelDesignForces :
    InputSpandrelLabel : string *
    InputStoryName : string *
    NumberResults : int byref *
    Story : string[] byref *
    SpandrelLabel : string[] byref *
    ComboName : string[] byref *
    Location : string[] byref *
    P : float[] byref *
    V2 : float[] byref *
    V3 : float[] byref *
    T : float[] byref *
    M2 : float[] byref *
    M3 : float[] byref -> int
```

Parameters

InputSpandrelLabel

Type: [SystemString](#)

The name of an existing spandrel label on the story specified by InputStoryName. If this is left blank, results will be retrieved for all spandrel labels on the story.

InputStoryName

Type: [SystemString](#)

The name of an existing story for which to retrieve spandrel design forces. If this is left blank, results will be retrieved for all stories.

NumberResults

Type: [SystemInt32](#)

The number of results, this is the length of all output arrays

Story

	Type:Â <u>SystemString</u>
	The story at which the spandrel exists
SpandrelLabel	Type:Â <u>SystemString</u>
	The label of the spandrel
ComboName	Type:Â <u>SystemString</u>
	The load combination for which the results are reported
Location	Type:Â <u>SystemString</u>
	The location where the results are reported
P	Type:Â <u>SystemDouble</u>
	The design axial force in the spandrel object local 1-axis direction [F]
V2	Type:Â <u>SystemDouble</u>
	The design shear force in the spandrel object local 2-axis direction [F]
V3	Type:Â <u>SystemDouble</u>
	The design shear force in the spandrel object local 3-axis direction [F]
T	Type:Â <u>SystemDouble</u>
	The design torsional moment about the spandrel object local 1-axis [FL]
M2	Type:Â <u>SystemDouble</u>
	The design bending moment about the spandrel object local 2-axis [FL]
M3	Type:Â <u>SystemDouble</u>
	The design bending moment about the spandrel object local 3-axis [FL]

Return ValueType:Â Int32

Returns zero if the results are successfully retrieved, otherwise it returns a nonzero value.

▲ **Remarks**

To retrieve results for all spandrel labels on all stories, leave both InputSpandrelLabel and InputStoryName blank.

▲ **See Also****Reference**cDesignForces InterfaceETABSv1 Namespace

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CSI API ETABS v1

cDesignResults Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDesignResults
```

```
Public Interface cDesignResults
```

```
Dim instance As cDesignResults
```

```
public interface class cDesignResults
```

```
type cDesignResults = interface end
```

The cDesignResults type exposes the following members.

▲ Properties

	Name	Description
--	------	-------------

	DesignForces	
---	------------------------------	--

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignResults Properties

The cDesignResults type exposes the following members.

Properties

Name	Description
------	-------------

 <u>DesignForces</u>	
---	--

Top

See Also

Reference

cDesignResults Interface

ETABSV1 Namespace

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CSI API ETABS v1

cDesignResultsAddLanguageSpecificTextSet("LST87C Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cDesignForces DesignForces { get; }
```

```
ReadOnly Property DesignForces As cDesignForces
Get
```

```
Dim instance As cDesignResults
Dim value As cDesignForces
```

```
value = instance.DesignForces
```

```
property cDesignForces^ DesignForces {
cDesignForces^ get ();
}
```

```
abstract DesignForces : cDesignForces with get
```

Property Value

Type: [cDesignForces](#)

▲ See Also

Reference

[cDesignResults Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignShearWall Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDesignShearWall
```

```
Public Interface cDesignShearWall
```

```
Dim instance As cDesignShearWall
```

```
public interface class cDesignShearWall
```

```
type cDesignShearWall = interface end
```

The cDesignShearWall type exposes the following members.

▲ Methods

▲	Name	Description
➤	GetPierSummaryResults	
➤	GetRebar	
➤	GetRebarPrefsPier	Retrieves data from the Rebar Selection Rules for Wall Piers
➤	GetRebarPrefsSpandrel	Retrieves data from the Rebar Selection Rules for Wall Spandrels
➤	GetSpandrelSummaryResults	

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▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignShearWall Methods

The [cDesignShearWall](#) type exposes the following members.

Methods

	Name	Description
	GetPierSummaryResults	
	GetRebar	
	GetRebarPrefsPier	Retrieves data from the Rebar Selection Rules for Wall Piers
	GetRebarPrefsSpandrel	Retrieves data from the Rebar Selection Rules for Wall Spandrels
	GetSpandrelSummaryResults	

[Top](#)

See Also

Reference

[cDesignShearWall Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cDesignShearWallAddLanguageSpecificTextSet("LST7 Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPierSummaryResults(
    ref string[] Story,
    ref string[] PierLabel,
    ref string[] Station,
    ref string[] DesignType,
    ref string[] PierSecType,
    ref string[] EdgeBar,
    ref string[] EndBar,
    ref double[] BarSpacing,
    ref double[] ReinfPercent,
    ref double[] CurrPercent,
    ref double[] DCRatio,
    ref string[] PierLeg,
    ref double[] LegX1,
    ref double[] LegY1,
    ref double[] LegX2,
    ref double[] LegY2,
    ref double[] EdgeLeft,
    ref double[] EdgeRight,
    ref double[] AsLeft,
    ref double[] AsRight,
    ref double[] ShearAv,
    ref double[] StressCompLeft,
    ref double[] StressCompRight,
    ref double[] StressLimitLeft,
    ref double[] StressLimitRight,
    ref double[] CDepthLeft,
    ref double[] CLimitLeft,
    ref double[] CDepthRight,
    ref double[] CLimitRight,
    ref double[] InelasticRotDemand,
    ref double[] InelasticRotCapacity,
    ref double[] NormCompStress,
    ref double[] NormCompStressLimit,
    ref double[] CDepth,
    ref double[] BZoneL,
    ref double[] BZoneR,
    ref double[] BZoneLength,
    ref string[] WarnMsg,
    ref string[] ErrMsg
)
```

Function GetPierSummaryResults (

cDesignShearWallspan id="LST7684374A_0"AddLanguageSpecificTextSet("LST7684374A_0?1885::|nu=".

Introduction

```
ByRef Story As String(),
ByRef PierLabel As String(),
ByRef Station As String(),
ByRef DesignType As String(),
ByRef PierSecType As String(),
ByRef EdgeBar As String(),
ByRef EndBar As String(),
ByRef BarSpacing As Double(),
ByRef ReinfPercent As Double(),
ByRef CurrPercent As Double(),
ByRef DCRatio As Double(),
ByRef PierLeg As String(),
ByRef LegX1 As Double(),
ByRef LegY1 As Double(),
ByRef LegX2 As Double(),
ByRef LegY2 As Double(),
ByRef EdgeLeft As Double(),
ByRef EdgeRight As Double(),
ByRef AsLeft As Double(),
ByRef AsRight As Double(),
ByRef ShearAv As Double(),
ByRef StressCompLeft As Double(),
ByRef StressCompRight As Double(),
ByRef StressLimitLeft As Double(),
ByRef StressLimitRight As Double(),
ByRef CDepthLeft As Double(),
ByRef CLimitLeft As Double(),
ByRef CDepthRight As Double(),
ByRef CLimitRight As Double(),
ByRef InelasticRotDemand As Double(),
ByRef InelasticRotCapacity As Double(),
ByRef NormCompStress As Double(),
ByRef NormCompStressLimit As Double(),
ByRef CDepth As Double(),
ByRef BZoneL As Double(),
ByRef BZoneR As Double(),
ByRef BZoneLength As Double(),
ByRef WarnMsg As String(),
ByRef ErrMsg As String()
) As Integer
```

```
Dim instance As cDesignShearWall
Dim Story As String()
Dim PierLabel As String()
Dim Station As String()
Dim DesignType As String()
Dim PierSecType As String()
Dim EdgeBar As String()
Dim EndBar As String()
Dim BarSpacing As Double()
Dim ReinfPercent As Double()
Dim CurrPercent As Double()
Dim DCRatio As Double()
Dim PierLeg As String()
Dim LegX1 As Double()
Dim LegY1 As Double()
Dim LegX2 As Double()
Dim LegY2 As Double()
Dim EdgeLeft As Double()
Dim EdgeRight As Double()
Dim AsLeft As Double()
Dim AsRight As Double()
```

Introduction

```
Dim ShearAv As Double()
Dim StressCompLeft As Double()
Dim StressCompRight As Double()
Dim StressLimitLeft As Double()
Dim StressLimitRight As Double()
Dim CDepthLeft As Double()
Dim CLimitLeft As Double()
Dim CDepthRight As Double()
Dim CLimitRight As Double()
Dim InelasticRotDemand As Double()
Dim InelasticRotCapacity As Double()
Dim NormCompStress As Double()
Dim NormCompStressLimit As Double()
Dim CDepth As Double()
Dim BZoneL As Double()
Dim BZoneR As Double()
Dim BZoneLength As Double()
Dim WarnMsg As String()
Dim ErrMsg As String()
Dim returnValue As Integer

returnValue = instance.GetPierSummaryResults(Story,
PierLabel, Station, DesignType, PierSecType,
EdgeBar, EndBar, BarSpacing, ReinfPercent,
CurrPercent, DCRatio, PierLeg, LegX1,
LegY1, LegX2, LegY2, EdgeLeft, EdgeRight,
AsLeft, AsRight, ShearAv, StressCompLeft,
StressCompRight, StressLimitLeft,
StressLimitRight, CDepthLeft, CLimitLeft,
CDepthRight, CLimitRight, InelasticRotDemand,
InelasticRotCapacity, NormCompStress,
NormCompStressLimit, CDepth, BZoneL,
BZoneR, BZoneLength, WarnMsg, ErrMsg)

int GetPierSummaryResults(
array<String>^% Story,
array<String>^% PierLabel,
array<String>^% Station,
array<String>^% DesignType,
array<String>^% PierSecType,
array<String>^% EdgeBar,
array<String>^% EndBar,
array<double>^% BarSpacing,
array<double>^% ReinfPercent,
array<double>^% CurrPercent,
array<double>^% DCRatio,
array<String>^% PierLeg,
array<double>^% LegX1,
array<double>^% LegY1,
array<double>^% LegX2,
array<double>^% LegY2,
array<double>^% EdgeLeft,
array<double>^% EdgeRight,
array<double>^% AsLeft,
array<double>^% AsRight,
array<double>^% ShearAv,
array<double>^% StressCompLeft,
array<double>^% StressCompRight,
array<double>^% StressLimitLeft,
array<double>^% StressLimitRight,
array<double>^% CDepthLeft,
array<double>^% CLimitLeft,
```

```

array<double>^% CDepthRight,
array<double>^% CLimitRight,
array<double>^% InelasticRotDemand,
array<double>^% InelasticRotCapacity,
array<double>^% NormCompStress,
array<double>^% NormCompStressLimit,
array<double>^% CDepth,
array<double>^% BZoneL,
array<double>^% BZoneR,
array<double>^% BZoneLength,
array<String^>^% WarnMsg,
array<String^>^% ErrMsg
)

```

```

abstract GetPierSummaryResults :
    Story : string[] byref *
    PierLabel : string[] byref *
    Station : string[] byref *
    DesignType : string[] byref *
    PierSecType : string[] byref *
    EdgeBar : string[] byref *
    EndBar : string[] byref *
    BarSpacing : float[] byref *
    ReinfPercent : float[] byref *
    CurrPercent : float[] byref *
    DCRatio : float[] byref *
    PierLeg : string[] byref *
    LegX1 : float[] byref *
    LegY1 : float[] byref *
    LegX2 : float[] byref *
    LegY2 : float[] byref *
    EdgeLeft : float[] byref *
    EdgeRight : float[] byref *
    AsLeft : float[] byref *
    AsRight : float[] byref *
    ShearAv : float[] byref *
    StressCompLeft : float[] byref *
    StressCompRight : float[] byref *
    StressLimitLeft : float[] byref *
    StressLimitRight : float[] byref *
    CDepthLeft : float[] byref *
    CLimitLeft : float[] byref *
    CDepthRight : float[] byref *
    CLimitRight : float[] byref *
    InelasticRotDemand : float[] byref *
    InelasticRotCapacity : float[] byref *
    NormCompStress : float[] byref *
    NormCompStressLimit : float[] byref *
    CDepth : float[] byref *
    BZoneL : float[] byref *
    BZoneR : float[] byref *
    BZoneLength : float[] byref *
    WarnMsg : string[] byref *
    ErrMsg : string[] byref -> int

```

Parameters

Story

Type: SystemString

PierLabel

Type:Â SystemString
 Station
 Type:Â SystemString
 DesignType
 Type:Â SystemString
 PierSecType
 Type:Â SystemString
 EdgeBar
 Type:Â SystemString
 EndBar
 Type:Â SystemString
 BarSpacing
 Type:Â SystemDouble
 ReinfPercent
 Type:Â SystemDouble
 CurrPercent
 Type:Â SystemDouble
 DCRatio
 Type:Â SystemDouble
 PierLeg
 Type:Â SystemString
 LegX1
 Type:Â SystemDouble
 LegY1
 Type:Â SystemDouble
 LegX2
 Type:Â SystemDouble
 LegY2
 Type:Â SystemDouble
 EdgeLeft
 Type:Â SystemDouble
 EdgeRight
 Type:Â SystemDouble
 AsLeft
 Type:Â SystemDouble
 AsRight
 Type:Â SystemDouble
 ShearAv
 Type:Â SystemDouble
 StressCompLeft
 Type:Â SystemDouble
 StressCompRight
 Type:Â SystemDouble
 StressLimitLeft
 Type:Â SystemDouble
 StressLimitRight
 Type:Â SystemDouble
 CDepthLeft
 Type:Â SystemDouble
 CLimitLeft
 Type:Â SystemDouble

CDepthRight
 Type:Â [SystemDouble](#)
 CLimitRight
 Type:Â [SystemDouble](#)
 InelasticRotDemand
 Type:Â [SystemDouble](#)
 InelasticRotCapacity
 Type:Â [SystemDouble](#)
 NormCompStress
 Type:Â [SystemDouble](#)
 NormCompStressLimit
 Type:Â [SystemDouble](#)
 CDepth
 Type:Â [SystemDouble](#)
 BZoneL
 Type:Â [SystemDouble](#)
 BZoneR
 Type:Â [SystemDouble](#)
 BZoneLength
 Type:Â [SystemDouble](#)
 WarnMsg
 Type:Â [SystemString](#)
 ErrMsg
 Type:Â [SystemString](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cDesignShearWall Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignShearWallAddLanguageSpecificTextSet("LST7 Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetRebar(
    ref string[] AreaObjName,
    ref string[] StoryName,
    ref string[] PierLabel,
    ref string[] StationLocation,
    ref string[] LegID,
    ref double[] LeftX1,
    ref double[] LeftY1,
    ref double[] RightX2,
    ref double[] RightY2,
    ref double[] Length,
    ref double[] Thickness,
    ref double[] fc,
    ref double[] fy,
    ref double[] fys,
    ref string[] Flexural,
    ref string[] ShearAndConfinement
)
```

```
Function GetRebar (
    ByRef AreaObjName As String(),
    ByRef StoryName As String(),
    ByRef PierLabel As String(),
    ByRef StationLocation As String(),
    ByRef LegID As String(),
    ByRef LeftX1 As Double(),
    ByRef LeftY1 As Double(),
    ByRef RightX2 As Double(),
    ByRef RightY2 As Double(),
    ByRef Length As Double(),
    ByRef Thickness As Double(),
    ByRef fc As Double(),
    ByRef fy As Double(),
    ByRef fys As Double(),
    ByRef Flexural As String(),
    ByRef ShearAndConfinement As String()
) As Integer
```

```
Dim instance As cDesignShearWall
Dim AreaObjName As String()
Dim StoryName As String()
Dim PierLabel As String()
Dim StationLocation As String()
```

cDesignShearWallspan id="LST7026F33D_0"AddLanguageSpecificTextSet("LST7026F33D_0?nu="

```

Dim LegID As String()
Dim LeftX1 As Double()
Dim LeftY1 As Double()
Dim RightX2 As Double()
Dim RightY2 As Double()
Dim Length As Double()
Dim Thickness As Double()
Dim fc As Double()
Dim fy As Double()
Dim fys As Double()
Dim Flexural As String()
Dim ShearAndConfinement As String()
Dim returnValue As Integer

returnValue = instance.GetRebar(AreaObjName,
StoryName, PierLabel, StationLocation,
LegID, LeftX1, LeftY1, RightX2, RightY2,
Length, Thickness, fc, fy, fys, Flexural,
ShearAndConfinement)

int GetRebar(
array<String^>^% AreaObjName,
array<String^>^% StoryName,
array<String^>^% PierLabel,
array<String^>^% StationLocation,
array<String^>^% LegID,
array<double>^% LeftX1,
array<double>^% LeftY1,
array<double>^% RightX2,
array<double>^% RightY2,
array<double>^% Length,
array<double>^% Thickness,
array<double>^% fc,
array<double>^% fy,
array<double>^% fys,
array<String^>^% Flexural,
array<String^>^% ShearAndConfinement
)

abstract GetRebar :
    AreaObjName : string[] byref *
    StoryName : string[] byref *
    PierLabel : string[] byref *
    StationLocation : string[] byref *
    LegID : string[] byref *
    LeftX1 : float[] byref *
    LeftY1 : float[] byref *
    RightX2 : float[] byref *
    RightY2 : float[] byref *
    Length : float[] byref *
    Thickness : float[] byref *
    fc : float[] byref *
    fy : float[] byref *
    fys : float[] byref *
    Flexural : string[] byref *
    ShearAndConfinement : string[] byref -> int

```

Parameters

AreaObjName
Type:Â SystemString

Parameters

StoryName
 Type:Â SystemString
 PierLabel
 Type:Â SystemString
 StationLocation
 Type:Â SystemString
 LegID
 Type:Â SystemString
 LeftX1
 Type:Â SystemDouble
 LeftY1
 Type:Â SystemDouble
 RightX2
 Type:Â SystemDouble
 RightY2
 Type:Â SystemDouble
 Length
 Type:Â SystemDouble
 Thickness
 Type:Â SystemDouble
 fc
 Type:Â SystemDouble
 fy
 Type:Â SystemDouble
 fys
 Type:Â SystemDouble
 Flexural
 Type:Â SystemString
 ShearAndConfinement
 Type:Â SystemString

Return Value

Type:Â Int32

▲ See Also

Reference

cDesignShearWall Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDesignShearWallAddLanguageSpecificTextSet("LST35AD2166_0") Method

Retrieves data from the Rebar Selection Rules for Wall Piers

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetRebarPrefsPier(
    int Item,
    ref string Value
)
```

```
Function GetRebarPrefsPier (
    Item As Integer,
    ByRef Value As String
) As Integer
```

```
Dim instance As cDesignShearWall
Dim Item As Integer
Dim Value As String
Dim returnValue As Integer
```

```
returnValue = instance.GetRebarPrefsPier(Item,
    Value)
```

```
int GetRebarPrefsPier(
    int Item,
    String^% Value
)
```

```
abstract GetRebarPrefsPier :
    Item : int *
    Value : string byref -> int
```

Parameters

Item

Type: [System.Int32](#)

An integer value corresponding to the data to be retrieved, from the following list

1. Vertical Bars - Unconfined Zones - Smallest Bar Size
2. Vertical Bars - Unconfined Zones - Largest Bar Size
3. Vertical Bars - Unconfined Zones - Preferred Corner/Joint Bar Size
4. Vertical Bars - Unconfined Zones - Preferred Middle Bar Size
5. Vertical Bars - Unconfined Zones - Minimum Spacing of Bars

cDesignShearWallspan id="LST35AD2166_0"AddLanguageSpecificTextSet("LST35AD2166_0")

Introduction

6. Vertical Bars - Unconfined Zones - Maximum Spacing of Bars
7. Vertical Bars - Confined Zones - Smallest Bar Size
8. Vertical Bars - Confined Zones - Largest Bar Size
9. Vertical Bars - Confined Zones - Preferred Corner/Joint Bar Size
10. Vertical Bars - Confined Zones - Preferred Middle Bar Size
11. Vertical Bars - Confined Zones - Minimum Spacing of Bars
12. Vertical Bars - Confined Zones - Maximum Spacing of Bars
13. Transverse Bars - Unconfined Zones - Smallest Bar Size
14. Transverse Bars - Unconfined Zones - Largest Bar Size
15. Transverse Bars - Unconfined Zones - Preferred Bar Size
16. Transverse Bars - Unconfined Zones - Minimum Spacing of Bars
17. Transverse Bars - Unconfined Zones - Maximum Spacing of Bars
18. Transverse Bars - Unconfined Zones - Preferred Bar Arrangement
19. Transverse Bars - Confined Zones - Smallest Bar Size
20. Transverse Bars - Confined Zones - Largest Bar Size
21. Transverse Bars - Confined Zones - Preferred Bar Size
22. Transverse Bars - Confined Zones - Minimum Spacing of Bars
23. Transverse Bars - Confined Zones - Maximum Spacing of Bars
24. Transverse Bars - Confined Zones - Preferred Bar Arrangement

Value

Type: SystemString

The retrieved data; note that numerical values are converted to String

If the Item is one of the Preferred Bar Arrangement options, then this value will be an integer from the following list:

1. Straight
2. UAndStraight
3. Hoops

Return Value

Type: Int32

Returns zero if the data is successfully retrieved; otherwise returns non-zero

▲ See Also

Reference

cDesignShearWall Interface

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CSI API ETABS v1

cDesignShearWallAddLanguageSpecificTextSet("LST5 Method

Retrieves data from the Rebar Selection Rules for Wall Spandrels

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetRebarPrefsSpandrel(
    int Item,
    ref string Value
)
```

```
Function GetRebarPrefsSpandrel (
    Item As Integer,
    ByRef Value As String
) As Integer
```

```
Dim instance As cDesignShearWall
Dim Item As Integer
Dim Value As String
Dim returnValue As Integer
```

```
returnValue = instance.GetRebarPrefsSpandrel(Item,
    Value)
```

```
int GetRebarPrefsSpandrel(
    int Item,
    String^% Value
)
```

```
abstract GetRebarPrefsSpandrel :
    Item : int *
    Value : string byref -> int
```

Parameters

Item

Type: SystemInt32

An integer value corresponding to the data to be retrieved, from the following list

1. Horizontal Top/Bottom Bars - Smallest bar size
2. Horizontal Top/Bottom Bars - Largest bar size
3. Horizontal Top/Bottom Bars - Preferred bar size
4. Horizontal Web Bars - Bar size
5. Horizontal Web Bars - Minimum Spacing of Bars

Introduction

6. Horizontal Web Bars - Maximum Spacing of Bars
7. Vertical Bars/Ties - Stirrup Bar size
8. Vertical Bars/Ties - Minimum Spacing of Bars
9. Vertical Bars/Ties - Maximum Spacing of Bars
10. Diagonal Bars - Smallest bar size
11. Diagonal Bars - Largest bar size
12. Diagonal Bars - Preferred bar size
13. Diagonal Bar Ties - Stirrup bar size
14. Diagonal Bar Ties - Minimum Spacing of ties
15. Diagonal Bar Ties - Maximum Spacing of ties

Value

Type: SystemString

The retrieved data; note that numerical values are converted to String

Return Value

Type: Int32

Returns zero if the data is successfully retrieved; otherwise returns non-zero

▲ See Also

Reference

cDesignShearWall Interface

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CSI API ETABS v1

cDesignShearWallAddLanguageSpecificTextSet("LST5 Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetSpandrelSummaryResults(
    ref string[] Story,
    ref string[] Spandrel,
    ref string[] Station,
    ref double[] TopRebar,
    ref double[] TopRebarRatio,
    ref string[] TopRebarCombo,
    ref double[] MuTop,
    ref double[] BotRebar,
    ref double[] BotRebarRatio,
    ref string[] BotRebarCombo,
    ref double[] MuBot,
    ref double[] AVert,
    ref double[] AHorz,
    ref string[] ShearCombo,
    ref double[] Vu,
    ref double[] ADiag,
    ref string[] ShearDiagCombo,
    ref double[] VuDiag,
    ref string[] WarnMsg,
    ref string[] ErrMsg
)
```

```
Function GetSpandrelSummaryResults (
    ByRef Story As String(),
    ByRef Spandrel As String(),
    ByRef Station As String(),
    ByRef TopRebar As Double(),
    ByRef TopRebarRatio As Double(),
    ByRef TopRebarCombo As String(),
    ByRef MuTop As Double(),
    ByRef BotRebar As Double(),
    ByRef BotRebarRatio As Double(),
    ByRef BotRebarCombo As String(),
    ByRef MuBot As Double(),
    ByRef AVert As Double(),
    ByRef AHorz As Double(),
    ByRef ShearCombo As String(),
    ByRef Vu As Double(),
    ByRef ADiag As Double(),
    ByRef ShearDiagCombo As String(),
    ByRef VuDiag As Double(),
    ByRef WarnMsg As String(),
```

cDesignShearWallspan id="LST59A6771C_0"AddLanguageSpecificTextSet("LST59A6771C_01048=::|nu=.

Introduction

```
ByRef ErrMsg As String()  
) As Integer
```

```
Dim instance As cDesignShearWall  
Dim Story As String()  
Dim Spandrel As String()  
Dim Station As String()  
Dim TopRebar As Double()  
Dim TopRebarRatio As Double()  
Dim TopRebarCombo As String()  
Dim MuTop As Double()  
Dim BotRebar As Double()  
Dim BotRebarRatio As Double()  
Dim BotRebarCombo As String()  
Dim MuBot As Double()  
Dim AVert As Double()  
Dim AHorz As Double()  
Dim ShearCombo As String()  
Dim Vu As Double()  
Dim ADiag As Double()  
Dim ShearDiagCombo As String()  
Dim VuDiag As Double()  
Dim WarnMsg As String()  
Dim ErrMsg As String()  
Dim returnValue As Integer
```

```
returnValue = instance.GetSpandrelSummaryResults(Story,  
Spandrel, Station, TopRebar, TopRebarRatio,  
TopRebarCombo, MuTop, BotRebar, BotRebarRatio,  
BotRebarCombo, MuBot, AVert, AHorz,  
ShearCombo, Vu, ADiag, ShearDiagCombo,  
VuDiag, WarnMsg, ErrMsg)
```

```
int GetSpandrelSummaryResults(  
array<String^>^% Story,  
array<String^>^% Spandrel,  
array<String^>^% Station,  
array<double>^% TopRebar,  
array<double>^% TopRebarRatio,  
array<String^>^% TopRebarCombo,  
array<double>^% MuTop,  
array<double>^% BotRebar,  
array<double>^% BotRebarRatio,  
array<String^>^% BotRebarCombo,  
array<double>^% MuBot,  
array<double>^% AVert,  
array<double>^% AHorz,  
array<String^>^% ShearCombo,  
array<double>^% Vu,  
array<double>^% ADiag,  
array<String^>^% ShearDiagCombo,  
array<double>^% VuDiag,  
array<String^>^% WarnMsg,  
array<String^>^% ErrMsg  
)
```

```
abstract GetSpandrelSummaryResults :  
    Story : string[] byref *  
    Spandrel : string[] byref *  
    Station : string[] byref *  
    TopRebar : float[] byref *  
    TopRebarRatio : float[] byref *
```

cDesignShearWallspan id="LST59A6771C_0"AddLanguageSpecificTextSet("LST59A6771C_01049::|nu=.

```

TopRebarCombo : string[] byref *
MuTop : float[] byref *
BotRebar : float[] byref *
BotRebarRatio : float[] byref *
BotRebarCombo : string[] byref *
MuBot : float[] byref *
AVert : float[] byref *
AHorz : float[] byref *
ShearCombo : string[] byref *
Vu : float[] byref *
ADiag : float[] byref *
ShearDiagCombo : string[] byref *
VuDiag : float[] byref *
WarnMsg : string[] byref *
ErrMsg : string[] byref -> int

```

Parameters

Story

Type:Â SystemString

Spandrel

Type:Â SystemString

Station

Type:Â SystemString

TopRebar

Type:Â SystemDouble

TopRebarRatio

Type:Â SystemDouble

TopRebarCombo

Type:Â SystemString

MuTop

Type:Â SystemDouble

BotRebar

Type:Â SystemDouble

BotRebarRatio

Type:Â SystemDouble

BotRebarCombo

Type:Â SystemString

MuBot

Type:Â SystemDouble

AVert

Type:Â SystemDouble

AHorz

Type:Â SystemDouble

ShearCombo

Type:Â SystemString

Vu

Type:Â SystemDouble

ADiag

Type:Â SystemDouble

ShearDiagCombo

Type:Â SystemString

VuDiag

Type:Â SystemDouble
WarnMsg
Type:Â SystemString
ErrMsg
Type:Â SystemString

Return Value

Type:Â Int32

▲ See Also

Reference

cDesignShearWall Interface

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CSI API ETABS v1

cDesignSteel Interface

Namespace: $\hat{A}$ ETABSv1

Assembly: $\hat{A}$ ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
public interface cDesignSteel
```

```
Public Interface cDesignSteel
```

```
Dim instance As cDesignSteel
```

```
public interface class cDesignSteel
```

```
type cDesignSteel = interface end
```

The cDesignSteel type exposes the following members.

▲ Properties

$\hat{A}$	Name	Description
	<u>AISC_LRFD93</u>	
	<u>AISC360_05_IBC2006</u>	
	<u>Australian_AS4100_98</u>	
	<u>BS5950_2000</u>	
	<u>Canadian_S16_09</u>	
	<u>Canadian_S16_14</u>	
	<u>Canadian_S16_19</u>	
	<u>Chinese_2010</u>	
	<u>Chinese_2018</u>	
	<u>Eurocode_3_2005</u>	
	<u>Indian_IS_800_2007</u>	
	<u>Italian_NTC_2008</u>	
	<u>Italian_NTC_2018</u>	
	<u>NewZealand_NZS3404_97</u>	
	<u>SP16_13330_2011</u>	

Top

▲ Methods

$\hat{A}$	Name	Description
	<u>DeleteResults</u>	
	<u>GetCode</u>	Retrieves the steel design code.

- ⇒ [GetComboDeflection](#)
- ⇒ [GetComboStrength](#)
- ⇒ [GetDesignSection](#)
- ⇒ [GetGroup](#)
- ⇒ [GetResultsAvailable](#) This function determines if the steel frame design results are available.
- ⇒ [GetSummaryResults](#)
- ⇒ [GetSummaryResults_2](#) DEPRECATED ; refer to [GetSummaryResults_3\(String, Int32, String, eFrameDesignOrientation, String, String, String, Double, eItemType\)](#)
- ⇒ [GetSummaryResults_3](#) Retrieves steel frame design summary results
- ⇒ [GetTargetDispl](#)
- ⇒ [GetTargetPeriod](#)
- ⇒ [ResetOverwrites](#)
- ⇒ [SetAutoSelectNull](#)
- ⇒ [SetCode](#) Sets the steel design code.
- ⇒ [SetComboDeflection](#)
- ⇒ [SetComboStrength](#)
- ⇒ [SetDesignSection](#)
- ⇒ [SetGroup](#)
- ⇒ [SetTargetDispl](#)
- ⇒ [SetTargetPeriod](#)
- ⇒ [StartDesign](#) Starts the steel frame design.
- ⇒ [VerifyPassed](#)
- ⇒ [VerifySections](#)

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignSteel Properties

The cDesignSteel type exposes the following members.

Properties

Â	Name	Description
	<u>AISC_LRFD93</u>	
	<u>AISC360_05_IBC2006</u>	
	<u>Australian_AS4100_98</u>	
	<u>BS5950_2000</u>	
	<u>Canadian_S16_09</u>	
	<u>Canadian_S16_14</u>	
	<u>Canadian_S16_19</u>	
	<u>Chinese_2010</u>	
	<u>Chinese_2018</u>	
	<u>Eurocode_3_2005</u>	
	<u>Indian_IS_800_2007</u>	
	<u>Italian_NTC_2008</u>	
	<u>Italian_NTC_2018</u>	
	<u>NewZealand_NZS3404_97</u>	
	<u>SP16_13330_2011</u>	

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See Also

Reference

cDesignSteel Interface

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LST17C57 Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cDStAISC_LRFD93 AISC_LRFD93 { get; }
```

```
ReadOnly Property AISC_LRFD93 As cDStAISC_LRFD93
Get
```

```
Dim instance As cDesignSteel
Dim value As cDStAISC_LRFD93
```

```
value = instance.AISC_LRFD93
```

```
property cDStAISC_LRFD93^ AISC_LRFD93 {
cDStAISC_LRFD93^ get ();
}
```

```
abstract AISC_LRFD93 : cDStAISC_LRFD93 with get
```

Property Value

Type: [cDStAISC_LRFD93](#)

▲ See Also

Reference

[cDesignSteel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LST5F422D07_0?cpp=1356=.") Property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
cDStAISC360_05_IBC2006 AISC360_05_IBC2006 { get; }
```

```
ReadOnly Property AISC360_05_IBC2006 As cDStAISC360_05_IBC2006  
Get
```

```
Dim instance As cDesignSteel  
Dim value As cDStAISC360_05_IBC2006
```

```
value = instance.AISC360_05_IBC2006
```

```
property cDStAISC360_05_IBC2006^ AISC360_05_IBC2006 {  
cDStAISC360_05_IBC2006^ get ();  
}
```

```
abstract AISC360_05_IBC2006 : cDStAISC360_05_IBC2006 with get
```

Property Value

Type: cDStAISC360_05_IBC2006

▲ See Also

Reference

cDesignSteel Interface

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LSTB6823227_0?cpp=1357-");Aus Property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
cDStAustralian_AS4100_98 Australian_AS4100_98 { get; }
```

```
ReadOnly Property Australian_AS4100_98 As cDStAustralian_AS4100_98  
Get
```

```
Dim instance As cDesignSteel  
Dim value As cDStAustralian_AS4100_98
```

```
value = instance.Australian_AS4100_98
```

```
property cDStAustralian_AS4100_98^ Australian_AS4100_98 {  
cDStAustralian_AS4100_98^ get ();  
}
```

```
abstract Australian_AS4100_98 : cDStAustralian_AS4100_98 with get
```

Property Value

Type: cDStAustralian_AS4100_98

▲ See Also

Reference

cDesignSteel Interface

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LSTB32B6AD2_0?cpp1358u=.");BS Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cDStBS5950_2000 BS5950_2000 { get; }
```

```
ReadOnly Property BS5950_2000 As cDStBS5950_2000
Get
```

```
Dim instance As cDesignSteel
Dim value As cDStBS5950_2000
```

```
value = instance.BS5950_2000
```

```
property cDStBS5950_2000^ BS5950_2000 {
cDStBS5950_2000^ get ();
}
```

```
abstract BS5950_2000 : cDStBS5950_2000 with get
```

Property Value

Type: [cDStBS5950_2000](#)

▲ See Also

Reference

[cDesignSteel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LST206D0 Property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
cDStCanadian_S16_09 Canadian_S16_09 { get; }
```

```
ReadOnly Property Canadian_S16_09 As cDStCanadian_S16_09  
Get
```

```
Dim instance As cDesignSteel  
Dim value As cDStCanadian_S16_09
```

```
value = instance.Canadian_S16_09
```

```
property cDStCanadian_S16_09^ Canadian_S16_09 {  
cDStCanadian_S16_09^ get ();  
}
```

```
abstract Canadian_S16_09 : cDStCanadian_S16_09 with get
```

Property Value

Type: cDStCanadian_S16_09

▲ See Also

Reference

cDesignSteel Interface

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LST207208BF_0?cpp=1360-");Can Property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
cDStCanadian_S16_14 Canadian_S16_14 { get; }
```

```
ReadOnly Property Canadian_S16_14 As cDStCanadian_S16_14
Get
```

```
Dim instance As cDesignSteel
Dim value As cDStCanadian_S16_14
```

```
value = instance.Canadian_S16_14
```

```
property cDStCanadian_S16_14^ Canadian_S16_14 {
cDStCanadian_S16_14^ get ();
}
```

```
abstract Canadian_S16_14 : cDStCanadian_S16_14 with get
```

Property Value

Type: cDStCanadian_S16_14

▲ See Also

Reference

cDesignSteel Interface

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LST206D0 Property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
cDStCanadian_S16_19 Canadian_S16_19 { get; }
```

```
ReadOnly Property Canadian_S16_19 As cDStCanadian_S16_19
Get
```

```
Dim instance As cDesignSteel
Dim value As cDStCanadian_S16_19
```

```
value = instance.Canadian_S16_19
```

```
property cDStCanadian_S16_19^ Canadian_S16_19 {
cDStCanadian_S16_19^ get ();
}
```

```
abstract Canadian_S16_19 : cDStCanadian_S16_19 with get
```

Property Value

Type: cDStCanadian_S16_19

▲ See Also

Reference

cDesignSteel Interface

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LSTBEE04287_0") Property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
cDStChinese_2010 Chinese_2010 { get; }
```

```
ReadOnly Property Chinese_2010 As cDStChinese_2010
Get
```

```
Dim instance As cDesignSteel
Dim value As cDStChinese_2010
```

```
value = instance.Chinese_2010
```

```
property cDStChinese_2010^ Chinese_2010 {
cDStChinese_2010^ get ();
}
```

```
abstract Chinese_2010 : cDStChinese_2010 with get
```

Property Value

Type: cDStChinese_2010

▲ See Also

Reference

cDesignSteel Interface

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LSTD41BE75F_0?cpp1360=.");Ch Property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
cDStChinese_2018 Chinese_2018 { get; }
```

```
ReadOnly Property Chinese_2018 As cDStChinese_2018  
Get
```

```
Dim instance As cDesignSteel  
Dim value As cDStChinese_2018
```

```
value = instance.Chinese_2018
```

```
property cDStChinese_2018^ Chinese_2018 {  
cDStChinese_2018^ get ();  
}
```

```
abstract Chinese_2018 : cDStChinese_2018 with get
```

Property Value

Type: cDStChinese_2018

▲ See Also

Reference

cDesignSteel Interface

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LSTFFAF43C5_0?cpp1304=.") Property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
cDStEurocode_3_2005 Eurocode_3_2005 { get; }
```

```
ReadOnly Property Eurocode_3_2005 As cDStEurocode_3_2005
Get
```

```
Dim instance As cDesignSteel
Dim value As cDStEurocode_3_2005
```

```
value = instance.Eurocode_3_2005
```

```
property cDStEurocode_3_2005^ Eurocode_3_2005 {
cDStEurocode_3_2005^ get ();
}
```

```
abstract Eurocode_3_2005 : cDStEurocode_3_2005 with get
```

Property Value

Type: cDStEurocode_3_2005

▲ See Also

Reference

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LST1DAD4E30_0?cpp1365u=.");ln Property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
cDStIndian_IS_800_2007 Indian_IS_800_2007 { get; }
```

```
ReadOnly Property Indian_IS_800_2007 As cDStIndian_IS_800_2007  
Get
```

```
Dim instance As cDesignSteel  
Dim value As cDStIndian_IS_800_2007
```

```
value = instance.Indian_IS_800_2007
```

```
property cDStIndian_IS_800_2007^ Indian_IS_800_2007 {  
cDStIndian_IS_800_2007^ get ();  
}
```

```
abstract Indian_IS_800_2007 : cDStIndian_IS_800_2007 with get
```

Property Value

Type: cDStIndian_IS_800_2007

▲ See Also

Reference

cDesignSteel Interface

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LST5AAEB6F2_0?cpp136u=.");Ita Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cDStItalianNTC2008S Italian_NTC_2008 { get; }
```

```
ReadOnly Property Italian_NTC_2008 As cDStItalianNTC2008S
Get
```

```
Dim instance As cDesignSteel
Dim value As cDStItalianNTC2008S
```

```
value = instance.Italian_NTC_2008
```

```
property cDStItalianNTC2008S^ Italian_NTC_2008 {
cDStItalianNTC2008S^ get ();
}
```

```
abstract Italian_NTC_2008 : cDStItalianNTC2008S with get
```

Property Value

Type: [cDStItalianNTC2008S](#)

▲ See Also

Reference

[cDesignSteel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LST5AAFE Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cDStItalianNTC2018S Italian_NTC_2018 { get; }
```

```
ReadOnly Property Italian_NTC_2018 As cDStItalianNTC2018S
Get
```

```
Dim instance As cDesignSteel
Dim value As cDStItalianNTC2018S
```

```
value = instance.Italian_NTC_2018
```

```
property cDStItalianNTC2018S^ Italian_NTC_2018 {
cDStItalianNTC2018S^ get ();
}
```

```
abstract Italian_NTC_2018 : cDStItalianNTC2018S with get
```

Property Value

Type: [cDStItalianNTC2018S](#)

▲ See Also

Reference

[cDesignSteel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LST96C5C Property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
cDStNewZealand_NZS3404_97 NewZealand_NZS3404_97 { get; }
```

```
ReadOnly Property NewZealand_NZS3404_97 As cDStNewZealand_NZS3404_97  
Get
```

```
Dim instance As cDesignSteel  
Dim value As cDStNewZealand_NZS3404_97
```

```
value = instance.NewZealand_NZS3404_97
```

```
property cDStNewZealand_NZS3404_97^ NewZealand_NZS3404_97 {  
cDStNewZealand_NZS3404_97^ get ();  
}
```

```
abstract NewZealand_NZS3404_97 : cDStNewZealand_NZS3404_97 with get
```

Property Value

Type: cDStNewZealand_NZS3404_97

▲ See Also

Reference

cDesignSteel Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LSTE2FE5 Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cDStSP16_13330_2011 SP16_13330_2011 { get; }
```

```
ReadOnly Property SP16_13330_2011 As cDStSP16_13330_2011
Get
```

```
Dim instance As cDesignSteel
Dim value As cDStSP16_13330_2011
```

```
value = instance.SP16_13330_2011
```

```
property cDStSP16_13330_2011^ SP16_13330_2011 {
cDStSP16_13330_2011^ get ();
}
```

```
abstract SP16_13330_2011 : cDStSP16_13330_2011 with get
```

Property Value

Type: [cDStSP16_13330_2011](#)

▲ See Also

Reference

[cDesignSteel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignSteel Methods

The cDesignSteel type exposes the following members.

Methods

Â	Name	Description
⇒	<u>DeleteResults</u>	
⇒	<u>GetCode</u>	Retrieves the steel design code.
⇒	<u>GetComboDeflection</u>	
⇒	<u>GetComboStrength</u>	
⇒	<u>GetDesignSection</u>	
⇒	<u>GetGroup</u>	
⇒	<u>GetResultsAvailable</u>	This function determines if the steel frame design results are available.
⇒	<u>GetSummaryResults</u>	
⇒	<u>GetSummaryResults_2</u>	DEPRECATED ; refer to <u>GetSummaryResults_3(String, Int32, String, eFrameDesignOrientation, String, String, String, Double, eItemType)</u>
⇒	<u>GetSummaryResults_3</u>	Retrieves steel frame design summary results
⇒	<u>GetTargetDispl</u>	
⇒	<u>GetTargetPeriod</u>	
⇒	<u>ResetOverwrites</u>	
⇒	<u>SetAutoSelectNull</u>	
⇒	<u>SetCode</u>	Sets the steel design code.
⇒	<u>SetComboDeflection</u>	
⇒	<u>SetComboStrength</u>	
⇒	<u>SetDesignSection</u>	
⇒	<u>SetGroup</u>	
⇒	<u>SetTargetDispl</u>	
⇒	<u>SetTargetPeriod</u>	
⇒	<u>StartDesign</u>	Starts the steel frame design.
⇒	<u>VerifyPassed</u>	
⇒	<u>VerifySections</u>	

Top

See Also

Reference

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Introduction

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LSTCA6CB5A8_0?cpu=.") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int DeleteResults()
```

```
Function DeleteResults As Integer
```

```
Dim instance As cDesignSteel
```

```
Dim returnValue As Integer
```

```
returnValue = instance.DeleteResults()
```

```
int DeleteResults()
```

```
abstract DeleteResults : unit -> int
```

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cDesignSteel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LST9D423C97_0?cpp=1370=.");Get

Method

Retrieves the steel design code.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetCode(
    ref string CodeName
)
```

```
Function GetCode (
    ByRef CodeName As String
) As Integer
```

```
Dim instance As cDesignSteel
Dim CodeName As String
Dim returnValue As Integer
```

```
returnValue = instance.GetCode(CodeName)
```

```
int GetCode(
    String^% CodeName
)
```

```
abstract GetCode :
    CodeName : string byref -> int
```

Parameters

CodeName

Type: SystemString

For ETABS, this is one of the following steel design code names.

- ◇ AISC 360-10
- ◇ AISC 360-05
- ◇ AISC LRFD 93
- ◇ AISC ASD 89
- ◇ AS 4100-1998
- ◇ BS 5950-2000
- ◇ Chinese 2010
- ◇ Chinese 2018
- ◇ CSA S16-14
- ◇ CSA S16-09
- ◇ Eurocode 3-2005

- ◇ IS 800:2007
- ◇ Italian NTC 2018
- ◇ Italian NTC 2008
- ◇ KBC 2009
- ◇ NZS 3404:1997
- ◇ SP 16.13330.2017

Return Value

Type: Int32

Returns zero if the code is successfully retrieved; otherwise it returns a nonzero value

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim CodeName As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'get steel design code
    ret = SapModel.DesignSteel.GetCode(CodeName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ **See Also**

Reference

cDesignSteel Interface

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LSTCEE5D84C_0?cpu=.") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetComboDeflection(
    ref int NumberItems,
    ref string[] MyName
)
```

```
Function GetComboDeflection (
    ByRef NumberItems As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cDesignSteel
Dim NumberItems As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetComboDeflection(NumberItems,
MyName)
```

```
int GetComboDeflection(
    int% NumberItems,
    array<String^>^% MyName
)
```

```
abstract GetComboDeflection :
    NumberItems : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberItems

Type: System.Int32

MyName

Type: System.String

Return Value

Type: Int32

▲ See Also

Reference

cDesignSteel Interface

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LST63F7912D_0?cpp=1378=.");Get Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetComboStrength(
    ref int NumberItems,
    ref string[] MyName
)
```

```
Function GetComboStrength (
    ByRef NumberItems As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cDesignSteel
Dim NumberItems As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetComboStrength(NumberItems,
MyName)
```

```
int GetComboStrength(
    int% NumberItems,
    array<String^>^% MyName
)
```

```
abstract GetComboStrength :
    NumberItems : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberItems

Type: SystemInt32

MyName

Type: SystemString

Return Value

Type: Int32

▲ See Also

Reference

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LSTD9F64172_0?cpp=1380=.");GetDesignSteelMethod

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetDesignSection(
    string Name,
    ref string PropName
)
```

```
Function GetDesignSection (
    Name As String,
    ByRef PropName As String
) As Integer
```

```
Dim instance As cDesignSteel
Dim Name As String
Dim PropName As String
Dim returnValue As Integer
```

```
returnValue = instance.GetDesignSection(Name,
    PropName)
```

```
int GetDesignSection(
    String^ Name,
    String^% PropName
)
```

```
abstract GetDesignSection :
    Name : string *
    PropName : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

PropName

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDesignSteel Interface

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LSTB3709 Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetGroup(
    ref int NumberItems,
    ref string[] MyName
)
```

```
Function GetGroup (
    ByRef NumberItems As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cDesignSteel
Dim NumberItems As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetGroup(NumberItems,
    MyName)
```

```
int GetGroup(
    int% NumberItems,
    array<String^>^% MyName
)
```

```
abstract GetGroup :
    NumberItems : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberItems

Type: SystemInt32

MyName

Type: SystemString

Return Value

Type: Int32

▲ See Also

Reference

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LST2446B779_0?cpp=1384-");Get

Method

This function determines if the steel frame design results are available.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
bool GetResultsAvailable()
```

```
Function GetResultsAvailable As Boolean
```

```
Dim instance As cDesignSteel
```

```
Dim returnValue As Boolean
```

```
returnValue = instance.GetResultsAvailable()
```

```
bool GetResultsAvailable()
```

```
abstract GetResultsAvailable : unit -> bool
```

Return Value

Type: Boolean

Returns True if the steel frame design results are available, otherwise it returns False.

■ Remarks

■ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim ResultsAvailable as Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()
```

cDesignSteelspan id="LST2446B779_0"AddLanguageSpecificTextSet("LST2446B779_0?cpp=1384-");Get

Introduction

```
'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis

'start steel design
    ret = SapModel.DesignSteel.StartDesign()

'check if design results are available
    ResultsAvailable = SapModel.DesignSteel.GetResultsAvailable

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cDesignSteel Interface

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LSTDFE954AE_0?cpp1386u=.");G Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetSummaryResults(
    string Name,
    ref int NumberItems,
    ref string[] FrameName,
    ref double[] Ratio,
    ref int[] RatioType,
    ref double[] Location,
    ref string[] ComboName,
    ref string[] ErrorSummary,
    ref string[] WarningSummary,
    eItemType ItemType = eItemType.Objects
)
```

```
Function GetSummaryResults (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef FrameName As String(),
    ByRef Ratio As Double(),
    ByRef RatioType As Integer(),
    ByRef Location As Double(),
    ByRef ComboName As String(),
    ByRef ErrorSummary As String(),
    ByRef WarningSummary As String(),
    ByRef ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDesignSteel
Dim Name As String
Dim NumberItems As Integer
Dim FrameName As String()
Dim Ratio As Double()
Dim RatioType As Integer()
Dim Location As Double()
Dim ComboName As String()
Dim ErrorSummary As String()
Dim WarningSummary As String()
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.GetSummaryResults(Name,
    NumberItems, FrameName, Ratio, RatioType,
    Location, ComboName, ErrorSummary,
    WarningSummary, ItemType)
```

cDesignSteelspan id="LSTDFE954AE_0"AddLanguageSpecificTextSet("LSTDFE954AE_0?cpp1386u=.");G

```

int GetSummaryResults(
String^ Name,
int% NumberItems,
array<String^>^% FrameName,
array<double>^% Ratio,
array<int>^% RatioType,
array<double>^% Location,
array<String^>^% ComboName,
array<String^>^% ErrorSummary,
array<String^>^% WarningSummary,
eItemType ItemType = eItemType::Objects
)

```

```

abstract GetSummaryResults :
    Name : string *
    NumberItems : int byref *
    FrameName : string[] byref *
    Ratio : float[] byref *
    RatioType : int[] byref *
    Location : float[] byref *
    ComboName : string[] byref *
    ErrorSummary : string[] byref *
    WarningSummary : string[] byref *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int

```

Parameters

Name
Type:Â [SystemString](#)

NumberItems
Type:Â [SystemInt32](#)

FrameName
Type:Â [SystemString](#)

Ratio
Type:Â [SystemDouble](#)

RatioType
Type:Â [SystemInt32](#)

Location
Type:Â [SystemDouble](#)

ComboName
Type:Â [SystemString](#)

ErrorSummary
Type:Â [SystemString](#)

WarningSummary
Type:Â [SystemString](#)

ItemType (Optional)
Type:Â [ETABSv1eItemType](#)

Return Value

Type:Â [Int32](#)
 See Also

Reference

cDesignSteel Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LSTA10BA Method

DEPRECATED ; refer to [GetSummaryResults_3\(String,Int32,String,eFrameDesignOrientation,String,String,String,Double,Double,Double,Double,String,eItemType\)](#)

Namespace: Â [ETABSv1](#)

Assembly: Â ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetSummaryResults_2(
string Name,
ref int NumberItems,
ref string[] FrameType,
ref string[] DesignSect,
ref string[] Status,
ref string[] PMMCombo,
ref double[] PMMRatio,
ref double[] PRatio,
ref double[] MMajRatio,
ref double[] MMinRatio,
ref string[] VMajCombo,
ref double[] VMajRatio,
ref string[] VMinCombo,
ref double[] VMinRatio,
eItemType ItemType = eItemType.Objects
)
```

```
Function GetSummaryResults_2 (
Name As String,
ByRef NumberItems As Integer,
ByRef FrameType As String(),
ByRef DesignSect As String(),
ByRef Status As String(),
ByRef PMMCombo As String(),
ByRef PMMRatio As Double(),
ByRef PRatio As Double(),
ByRef MMajRatio As Double(),
ByRef MMinRatio As Double(),
ByRef VMajCombo As String(),
ByRef VMajRatio As Double(),
ByRef VMinCombo As String(),
ByRef VMinRatio As Double(),
Optional ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDesignSteel
Dim Name As String
Dim NumberItems As Integer
```

cDesignSteelspan id="LSTA10BABF4_0"AddLanguageSpecificTextSet("LSTA10BABF4_0?cpp1380u=");G

Introduction

```
Dim FrameType As String()
Dim DesignSect As String()
Dim Status As String()
Dim PMMCombo As String()
Dim PMMRatio As Double()
Dim PRatio As Double()
Dim MMajRatio As Double()
Dim MMinRatio As Double()
Dim VMajCombo As String()
Dim VMajRatio As Double()
Dim VMinCombo As String()
Dim VMinRatio As Double()
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.GetSummaryResults_2(Name,
NumberItems, FrameType, DesignSect,
Status, PMMCombo, PMMRatio, PRatio,
MMajRatio, MMinRatio, VMajCombo,
VMajRatio, VMinCombo, VMinRatio,
ItemType)
```

```
int GetSummaryResults_2(
String^ Name,
int% NumberItems,
array<String^>^% FrameType,
array<String^>^% DesignSect,
array<String^>^% Status,
array<String^>^% PMMCombo,
array<double>^% PMMRatio,
array<double>^% PRatio,
array<double>^% MMajRatio,
array<double>^% MMinRatio,
array<String^>^% VMajCombo,
array<double>^% VMajRatio,
array<String^>^% VMinCombo,
array<double>^% VMinRatio,
eItemType ItemType = eItemType::Objects
)
```

```
abstract GetSummaryResults_2 :
    Name : string *
    NumberItems : int byref *
    FrameType : string[] byref *
    DesignSect : string[] byref *
    Status : string[] byref *
    PMMCombo : string[] byref *
    PMMRatio : float[] byref *
    PRatio : float[] byref *
    MMajRatio : float[] byref *
    MMinRatio : float[] byref *
    VMajCombo : string[] byref *
    VMajRatio : float[] byref *
    VMinCombo : string[] byref *
    VMinRatio : float[] byref *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name
Type:Â [SystemString](#)

NumberItems
Type:Â [SystemInt32](#)

FrameType
Type:Â [SystemString](#)

DesignSect
Type:Â [SystemString](#)

Status
Type:Â [SystemString](#)

PMMCombo
Type:Â [SystemString](#)

PMMRatio
Type:Â [SystemDouble](#)

PRatio
Type:Â [SystemDouble](#)

MMajRatio
Type:Â [SystemDouble](#)

MMinRatio
Type:Â [SystemDouble](#)

VMajCombo
Type:Â [SystemString](#)

VMajRatio
Type:Â [SystemDouble](#)

VMinCombo
Type:Â [SystemString](#)

VMinRatio
Type:Â [SystemDouble](#)

ItemType (Optional)
Type:Â [ETABSv1eItemType](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cDesignSteel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LSTFB55C679_0") Method

Retrieves steel frame design summary results

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetSummaryResults_3(
    string Name,
    ref int NumberItems,
    ref string[] FrameName,
    ref eFrameDesignOrientation[] FrameType,
    ref string[] DesignSect,
    ref string[] Status,
    ref string[] PMMCombo,
    ref double[] PMMRatio,
    ref double[] PRatio,
    ref double[] MMajRatio,
    ref double[] MMinRatio,
    ref string[] VMajCombo,
    ref double[] VMajRatio,
    ref string[] VMinCombo,
    ref double[] VMinRatio,
    eItemType ItemType = eItemType.Objects
)
```

```
Function GetSummaryResults_3 (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef FrameName As String(),
    ByRef FrameType As eFrameDesignOrientation(),
    ByRef DesignSect As String(),
    ByRef Status As String(),
    ByRef PMMCombo As String(),
    ByRef PMMRatio As Double(),
    ByRef PRatio As Double(),
    ByRef MMajRatio As Double(),
    ByRef MMinRatio As Double(),
    ByRef VMajCombo As String(),
    ByRef VMajRatio As Double(),
    ByRef VMinCombo As String(),
    ByRef VMinRatio As Double(),
    ByRef ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDesignSteel
Dim Name As String
Dim NumberItems As Integer
```

cDesignSteelspan id="LSTFB55C679_0"AddLanguageSpecificTextSet("LSTFB55C679_0?cpp=1392=");Ge

Introduction

```
Dim FrameName As String()
Dim FrameType As eFrameDesignOrientation()
Dim DesignSect As String()
Dim Status As String()
Dim PMMCombo As String()
Dim PMMRatio As Double()
Dim PRatio As Double()
Dim MMajRatio As Double()
Dim MMinRatio As Double()
Dim VMajCombo As String()
Dim VMajRatio As Double()
Dim VMinCombo As String()
Dim VMinRatio As Double()
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.GetSummaryResults_3(Name,
NumberItems, FrameName, FrameType,
DesignSect, Status, PMMCombo, PMMRatio,
PRatio, MMajRatio, MMinRatio, VMajCombo,
VMajRatio, VMinCombo, VMinRatio,
ItemType)
```

```
int GetSummaryResults_3(
String^ Name,
int% NumberItems,
array<String^>^% FrameName,
array<eFrameDesignOrientation>^% FrameType,
array<String^>^% DesignSect,
array<String^>^% Status,
array<String^>^% PMMCombo,
array<double>^% PMMRatio,
array<double>^% PRatio,
array<double>^% MMajRatio,
array<double>^% MMinRatio,
array<String^>^% VMajCombo,
array<double>^% VMajRatio,
array<String^>^% VMinCombo,
array<double>^% VMinRatio,
eItemType ItemType = eItemType::Objects
)
```

```
abstract GetSummaryResults_3 :
    Name : string *
    NumberItems : int byref *
    FrameName : string[] byref *
    FrameType : eFrameDesignOrientation[] byref *
    DesignSect : string[] byref *
    Status : string[] byref *
    PMMCombo : string[] byref *
    PMMRatio : float[] byref *
    PRatio : float[] byref *
    MMajRatio : float[] byref *
    MMinRatio : float[] byref *
    VMajCombo : string[] byref *
    VMajRatio : float[] byref *
    VMinCombo : string[] byref *
    VMinRatio : float[] byref *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
```

-> int

Parameters

Name

Type:Â SystemString

The name of an existing frame object or group, depending on the value of the ItemType argument.

NumberItems

Type:Â SystemInt32

The number of results, this is the length of all output arrays

FrameName

Type:Â SystemString

The frame object unique name associated with each result

FrameType

Type:Â ETABSv1eFrameDesignOrientation

The type of each frame object, a value from eFrameDesignOrientation enumeration, either Column , Beam , Brace , Other , or Null

DesignSect

Type:Â SystemString

The design section of each frame object

Status

Type:Â SystemString

The design status

PMMCombo

Type:Â SystemString

The load combination for which the PMM ratio is reported

PMMRatio

Type:Â SystemDouble

The controlling PMM ratio for the frame object

PRatio

Type:Â SystemDouble

The axial component of the controlling PMM ratio

MMajRatio

Type:Â SystemDouble

The major axis bending component of the controlling PMM ratio

MMinRatio

Type:Â SystemDouble

The minor axis bending component of the controlling PMM ratio

VMajCombo

Type:Â SystemString

The load combination for which the major direction shear ratio is reported

VMajRatio

Type:Â SystemDouble

The controlling major direction shear ratio for the frame object

VMinCombo

Type:Â SystemString

The load combination for which the minor direction shear ratio is reported

VMinRatio

Type:Â SystemDouble

The controlling minor direction shear ratio for the frame object

Item Type (Optional)

Type: $\hat{A}$ ETABSv1eItemTypeThis is one of the following items in the eItemType enumeration:

- 0. Object
- 1. Group
- 2. SelectedObjects

If this item is Objects, results are retrieved for the frame object specified by the Name item.

If this item is Group, results are retrieved for all frame objects in the group specified by the Name item.

If this item is SelectedObjects, results are retrieved for all selected frame objects and the Name item is ignored.

Return ValueType: $\hat{A}$ Int32

Returns zero if the results are successfully retrieved, otherwise it returns a nonzero value.

▲ Remarks

The function will fail if no steel frame objects are present. It will also fail if analysis results are not available.

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set steel design code
    ret = SapModel.DesignSteel.SetCode("AISC 360-10")

    'run analysis
    System.IO.Directory.CreateDirectory("C:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis()

    'start steel design
```

Introduction

```
ret = SapModel.DesignSteel.StartDesign()

Dim NumberItems As Integer
Dim FrameName As String()
Dim FrameType As eFrameDesignOrientation()
Dim DesignSect As String()
Dim Status As String()
Dim PMMCombo As String()
Dim PMMRatio As Double()
Dim PRatio As Double()
Dim MMajRatio As Double()
Dim MMinRatio As Double()
Dim VMajCombo As String()
Dim VMajRatio As Double()
Dim VMinCombo As String()
Dim VMinRatio As Double()

ret = mySapModel.DesignSteel.GetSummaryResults_3("All", NumberItems, FrameName, FrameType, Des

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cDesignSteel Interface

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LST45CA8260_0?cpp=1;307=.");Get

Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetTargetDispl(
    ref int NumberItems,
    ref string[] LoadCase,
    ref string[] Point,
    ref double[] Displ,
    ref bool Active
)
```

```
Function GetTargetDispl (
    ByRef NumberItems As Integer,
    ByRef LoadCase As String(),
    ByRef Point As String(),
    ByRef Displ As Double(),
    ByRef Active As Boolean
) As Integer
```

```
Dim instance As cDesignSteel
Dim NumberItems As Integer
Dim LoadCase As String()
Dim Point As String()
Dim Displ As Double()
Dim Active As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetTargetDispl(NumberItems,
    LoadCase, Point, Displ, Active)
```

```
int GetTargetDispl(
    int% NumberItems,
    array<String^>^% LoadCase,
    array<String^>^% Point,
    array<double>^% Displ,
    bool% Active
)
```

```
abstract GetTargetDispl :
    NumberItems : int byref *
    LoadCase : string[] byref *
    Point : string[] byref *
    Displ : float[] byref *
    Active : bool byref -> int
```

cDesignSteelspan id="LST45CA8260_0"AddLanguageSpecificTextSet("LST45CA8260_0?cpp=1;307=.");Get

Parameters

NumberItems

Type:Â SystemInt32

LoadCase

Type:Â SystemString

Point

Type:Â SystemString

Displ

Type:Â SystemDouble

Active

Type:Â SystemBoolean

Return Value

Type:Â Int32

▲ See Also

Reference

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LST3764A377_0?cpp=1.300-");GetTargetPeriod

Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetTargetPeriod(
    ref int NumberItems,
    ref string ModalCase,
    ref int[] Mode,
    ref double[] Period,
    ref bool Active
)
```

```
Function GetTargetPeriod (
    ByRef NumberItems As Integer,
    ByRef ModalCase As String,
    ByRef Mode As Integer(),
    ByRef Period As Double(),
    ByRef Active As Boolean
) As Integer
```

```
Dim instance As cDesignSteel
Dim NumberItems As Integer
Dim ModalCase As String
Dim Mode As Integer()
Dim Period As Double()
Dim Active As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetTargetPeriod(NumberItems,
    ModalCase, Mode, Period, Active)
```

```
int GetTargetPeriod(
    int% NumberItems,
    String^% ModalCase,
    array<int>^% Mode,
    array<double>^% Period,
    bool% Active
)
```

```
abstract GetTargetPeriod :
    NumberItems : int byref *
    ModalCase : string byref *
    Mode : int[] byref *
    Period : float[] byref *
    Active : bool byref -> int
```

cDesignSteelspan id="LST3764A377_0"AddLanguageSpecificTextSet("LST3764A377_0?cpp=1.300-");GetTargetPeriod

Parameters

NumberItems

Type: SystemInt32

ModalCase

Type: SystemString

Mode

Type: SystemInt32

Period

Type: SystemDouble

Active

Type: SystemBoolean

Return Value

Type: Int32

See Also

Reference

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LSTBC686B2D_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

int ResetOverwrites()

Function ResetOverwrites As Integer

Dim instance As cDesignSteel

Dim returnValue As Integer

returnValue = instance.ResetOverwrites()

int ResetOverwrites()

abstract ResetOverwrites : unit -> int

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cDesignSteel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LST3C578458_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetAutoSelectNull(
    string Name,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetAutoSelectNull (
    Name As String,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDesignSteel
Dim Name As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetAutoSelectNull(Name,
    ItemType)
```

```
int SetAutoSelectNull(
    String^ Name,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetAutoSelectNull :
    Name : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

ItemType (Optional)

Type: ETABSv1eItemType

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cDesignSteel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LST827C419F_0") Method

Sets the steel design code.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetCode(
    string CodeName
)
```

```
Function SetCode (
    CodeName As String
) As Integer
```

```
Dim instance As cDesignSteel
Dim CodeName As String
Dim returnValue As Integer
```

```
returnValue = instance.SetCode(CodeName)
```

```
int SetCode(
    String^ CodeName
)
```

```
abstract SetCode :
    CodeName : string -> int
```

Parameters

CodeName

Type: SystemString

For ETABS, this is one of the following steel design code names.

- ◇ AISC 360-10
- ◇ AISC 360-05
- ◇ AISC LRFD 93
- ◇ AISC ASD 89
- ◇ AS 4100-1998
- ◇ BS 5950-2000
- ◇ Chinese 2010
- ◇ Chinese 2018
- ◇ CSA S16-14
- ◇ CSA S16-09
- ◇ Eurocode 3-2005

- ◇ IS 800:2007
- ◇ Italian NTC 2018
- ◇ Italian NTC 2008
- ◇ KBC 2009
- ◇ NZS 3404:1997
- ◇ SP 16.13330.2017

Return Value

Type: Int32

Returns zero if the code is successfully set; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim CodeName As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set steel design code
    ret = SapModel.DesignSteel.SetCode("Eurocode 3-2005")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ **See Also**

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LST79A5016_0?cpp=1#07.")

Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetComboDeflection(
    string Name,
    bool Selected
)
```

```
Function SetComboDeflection (
    Name As String,
    Selected As Boolean
) As Integer
```

```
Dim instance As cDesignSteel
Dim Name As String
Dim Selected As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetComboDeflection(Name,
    Selected)
```

```
int SetComboDeflection(
    String^ Name,
    bool Selected
)
```

```
abstract SetComboDeflection :
    Name : string *
    Selected : bool -> int
```

Parameters

Name

Type: [SystemString](#)

Selected

Type: [SystemBoolean](#)

Return Value

Type: [Int32](#)

▲ See Also

cDesignSteelspan id="LST79A50516_0"AddLanguageSpecificTextSet("LST79A50516_0?cpp=1#07.");SetC

Reference

cDesignSteel Interface

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LST83B2A955_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetComboStrength(
    string Name,
    bool Selected
)
```

```
Function SetComboStrength (
    Name As String,
    Selected As Boolean
) As Integer
```

```
Dim instance As cDesignSteel
Dim Name As String
Dim Selected As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetComboStrength(Name,
    Selected)
```

```
int SetComboStrength(
    String^ Name,
    bool Selected
)
```

```
abstract SetComboStrength :
    Name : string *
    Selected : bool -> int
```

Parameters

Name

Type: [SystemString](#)

Selected

Type: [SystemBoolean](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDesignSteel Interface

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LST3EAA26AE_0?cpu=.") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetDesignSection(
    string Name,
    string PropName,
    bool LastAnalysis,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetDesignSection (
    Name As String,
    PropName As String,
    LastAnalysis As Boolean,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDesignSteel
Dim Name As String
Dim PropName As String
Dim LastAnalysis As Boolean
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetDesignSection(Name,
    PropName, LastAnalysis, ItemType)
```

```
int SetDesignSection(
    String^ Name,
    String^ PropName,
    bool LastAnalysis,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetDesignSection :
    Name : string *
    PropName : string *
    LastAnalysis : bool *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

cDesignSteelspan id="LST3EAA26AE_0"AddLanguageSpecificTextSet("LST3EAA26AE_0?cpu=.")

1411

Parameters

Name

Type: SystemString

PropName

Type: SystemString

LastAnalysis

Type: SystemBoolean

ItemType (Optional)

Type: ETABSv1eItemType

Return Value

Type: Int32

▲ See Also

Reference

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LSTCE4DD2EE_0?cpu=.") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetGroup(
    string Name,
    bool Selected
)
```

```
Function SetGroup (
    Name As String,
    Selected As Boolean
) As Integer
```

```
Dim instance As cDesignSteel
Dim Name As String
Dim Selected As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetGroup(Name,
    Selected)
```

```
int SetGroup(
    String^ Name,
    bool Selected
)
```

```
abstract SetGroup :
    Name : string *
    Selected : bool -> int
```

Parameters

Name

Type: [SystemString](#)

Selected

Type: [SystemBoolean](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LSTBDEA8849_0?cpp1415u=".");SetMethod

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetTargetDispl(
int NumberItems,
ref string[] LoadCase,
ref string[] Point,
ref double[] Displ,
bool Active = true
)
```

```
Function SetTargetDispl (
NumberItems As Integer,
ByRef LoadCase As String(),
ByRef Point As String(),
ByRef Displ As Double(),
Optional Active As Boolean = true
) As Integer
```

```
Dim instance As cDesignSteel
Dim NumberItems As Integer
Dim LoadCase As String()
Dim Point As String()
Dim Displ As Double()
Dim Active As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetTargetDispl(NumberItems,
LoadCase, Point, Displ, Active)
```

```
int SetTargetDispl(
int NumberItems,
array<String^>^% LoadCase,
array<String^>^% Point,
array<double>^% Displ,
bool Active = true
)
```

```
abstract SetTargetDispl :
    NumberItems : int *
    LoadCase : string[] byref *
    Point : string[] byref *
    Displ : float[] byref *
    ?Active : bool
(* Defaults:
```

cDesignSteelspan id="LSTBDEA8849_0"AddLanguageSpecificTextSet("LSTBDEA8849_0?cpp1415u=".");Set

```
let _Active = defaultArg Active true
*)
-> int
```

Parameters

NumberItems

Type:Â [SystemInt32](#)

LoadCase

Type:Â [SystemString](#)

Point

Type:Â [SystemString](#)

Displ

Type:Â [SystemDouble](#)

Active (Optional)

Type:Â [SystemBoolean](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cDesignSteel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LST50B44583_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetTargetPeriod(
    int NumberItems,
    string ModalCase,
    ref int[] Mode,
    ref double[] Period,
    bool Active = true
)
```

```
Function SetTargetPeriod (
    NumberItems As Integer,
    ModalCase As String,
    ByRef Mode As Integer(),
    ByRef Period As Double(),
    Optional Active As Boolean = true
) As Integer
```

```
Dim instance As cDesignSteel
Dim NumberItems As Integer
Dim ModalCase As String
Dim Mode As Integer()
Dim Period As Double()
Dim Active As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetTargetPeriod(NumberItems,
    ModalCase, Mode, Period, Active)
```

```
int SetTargetPeriod(
    int NumberItems,
    String^ ModalCase,
    array<int>^% Mode,
    array<double>^% Period,
    bool Active = true
)
```

```
abstract SetTargetPeriod :
    NumberItems : int *
    ModalCase : string *
    Mode : int[] byref *
    Period : float[] byref *
    ?Active : bool
(* Defaults:
```

```
cDesignSteelspan id="LST50B44583_0"AddLanguageSpecificTextSet("LST50B44583_0?cpp=1#17.");Set7
```

```
let _Active = defaultArg Active true
*)
-> int
```

Parameters

NumberItems

Type:Â [SystemInt32](#)

ModalCase

Type:Â [SystemString](#)

Mode

Type:Â [SystemInt32](#)

Period

Type:Â [SystemDouble](#)

Active (Optional)

Type:Â [SystemBoolean](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cDesignSteel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LST9D0D7C38_0?cpp1410u=.");St Method

Starts the steel frame design.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int StartDesign()
```

```
Function StartDesign As Integer
```

```
Dim instance As cDesignSteel
```

```
Dim returnValue As Integer
```

```
returnValue = instance.StartDesign()
```

```
int StartDesign()
```

```
abstract StartDesign : unit -> int
```

Return Value

Type: [Int32](#)

Returns zero if the steel frame design is successfully started; otherwise it returns a nonzero value.

▲ Remarks

The function will fail if no steel frame objects are present. It will also fail if analysis results are not available.

▲ Examples

[VB](#)

[Copy](#)

```
Public Sub Example()
```

```
    Dim SapModel As cSapModel
```

```
    Dim EtabsObject As cOAPI
```

```
    Dim ret As Integer = -1
```

```
'create ETABS object
```

```
    Dim myHelper as cHelper = New Helper
```

```
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

```
'start ETABS application
```

```
    ret = EtabsObject.ApplicationStart()
```

```
'create SapModel object
```

```
    SapModel = EtabsObject.SapModel
```

```
cDesignSteelspan id="LST9D0D7C38_0"AddLanguageSpecificTextSet("LST9D0D7C38_0?cpp1410u=.");St
```

Introduction

```
'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'set steel design code
    ret = SapModel.DesignSteel.SetCode("AISC 360-10")

'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis

'start steel design
    ret = SapModel.DesignSteel.StartDesign()

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

[cDesignSteel Interface](#)

[ETABSv1 Namespace](#)

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»

CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LST268BEB22_0?cpp1421u=");Ver

Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int VerifyPassed(
    ref int NumberItems,
    ref int N1,
    ref int N2,
    ref string[] MyName
)
```

```
Function VerifyPassed (
    ByRef NumberItems As Integer,
    ByRef N1 As Integer,
    ByRef N2 As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cDesignSteel
Dim NumberItems As Integer
Dim N1 As Integer
Dim N2 As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.VerifyPassed(NumberItems,
    N1, N2, MyName)
```

```
int VerifyPassed(
    int% NumberItems,
    int% N1,
    int% N2,
    array<String^>^% MyName
)
```

```
abstract VerifyPassed :
    NumberItems : int byref *
    N1 : int byref *
    N2 : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberItems

Type: System.Int32

cDesignSteelspan id="LST268BEB22_0"AddLanguageSpecificTextSet("LST268BEB22_0?cpp1421u=");Ver

N1

Type:Â SystemInt32

N2

Type:Â SystemInt32

MyName

Type:Â SystemString

Return Value

Type:Â Int32

▲ See Also

Reference

cDesignSteel Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDesignSteelAddLanguageSpecificTextSet("LSTFDCBEA7_0?cpp=1.423-");VerifySections Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int VerifySections(
    ref int NumberItems,
    ref string[] MyName
)
```

```
Function VerifySections (
    ByRef NumberItems As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cDesignSteel
Dim NumberItems As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.VerifySections(NumberItems,
MyName)
```

```
int VerifySections(
    int% NumberItems,
    array<String^>^% MyName
)
```

```
abstract VerifySections :
    NumberItems : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberItems

Type: [System.Int32](#)

MyName

Type: [System.String](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDesignSteel Interface

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CSI API ETABS v1

cDesignStrip Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDesignStrip
```

```
Public Interface cDesignStrip
```

```
Dim instance As cDesignStrip
```

```
public interface class cDesignStrip
```

```
type cDesignStrip = interface end
```

The cDesignStrip type exposes the following members.

▲ Methods

▲	Name	Description
🔗	ChangeName	Changes the name of a defined Design Strip
🔗	Delete	Deletes the specified Design Strip
🔗	GetDesignStrip	Retrieves the specified Design Strip
🔗	GetDesignStrip_1	Retrieves the specified Design Strip
🔗	GetGUID	Retrieves the GUID of an existing strip
🔗	GetNameList	Retrieves the names of all defined Design Strips
🔗	SetGUID	Sets the GUID of an existing strip

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDesignStrip Methods

The cDesignStrip type exposes the following members.

Methods

⌚	Name	Description
⇒	<u>ChangeName</u>	Changes the name of a defined Design Strip
⇒	<u>Delete</u>	Deletes the specified Design Strip
⇒	<u>GetDesignStrip</u>	Retrieves the specified Design Strip
⇒	<u>GetDesignStrip_1</u>	Retrieves the specified Design Strip
⇒	<u>GetGUID</u>	Retrieves the GUID of an existing strip
⇒	<u>GetNameList</u>	Retrieves the names of all defined Design Strips
⇒	<u>SetGUID</u>	Sets the GUID of an existing strip

Top

See Also

Reference

cDesignStrip Interface

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CSI API ETABS v1

cDesignStripAddLanguageSpecificTextSet("LST707DD Method

Changes the name of a defined Design Strip

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int ChangeName(
    string Name,
    string NewName
)
```

```
Function ChangeName (
    Name As String,
    NewName As String
) As Integer
```

```
Dim instance As cDesignStrip
Dim Name As String
Dim NewName As String
Dim returnValue As Integer
```

```
returnValue = instance.ChangeName(Name,
    NewName)
```

```
int ChangeName(
    String^ Name,
    String^ NewName
)
```

```
abstract ChangeName :
    Name : string *
    NewName : string -> int
```

Parameters

Name

Type: SystemString

The existing name of a defined Design Strip

NewName

Type: SystemString

The new name for the Design Strip

Return ValueType: Int32

Returns zero if the new name is successfully applied, otherwise it returns a nonzero value.

▀ **Remarks**

▀ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new Design Strip with name "MyDesignStrip1A" ; currently not available in API

    'change DesignStrip name
    ret = SapModel.DesignConcreteSlab.DesignStrip.ChangeName("MyDesignStrip1A", "MyDesStrp2B")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▀ **See Also**

Reference

cDesignStrip Interface

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CSI API ETABS v1

cDesignStripAddLanguageSpecificTextSet("LSTC90E0 Method

Deletes the specified Design Strip

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int Delete(
string Name
)
```

```
Function Delete (
Name As String
) As Integer
```

```
Dim instance As cDesignStrip
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.Delete(Name)
```

```
int Delete(
String^ Name
)
```

```
abstract Delete :
    Name : string -> int
```

Parameters

Name

Type: SystemString

The name of an existing Design Strip

Return Value

Type: Int32

Returns zero if the Design Strip is successfully deleted, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
```

cDesignStripAddLanguageSpecificTextSet("LSTC90E0DE5_0?cpp1429=.");De

Introduction

```
Dim SapModel As cSapModel
Dim EtabsObject As cOAPI
Dim ret As Integer = -1

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'define new Design Strip with name "MyDesignStrip1A" ; currently not available in API

'change DesignStrip name
ret = SapModel.DesignStrip.Delete("MyDesignStrip1A")

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

[cDesignStrip Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cDesignStripAddLanguageSpecificTextSet("LSTACFB2E75_0?cpp1431u=.");GetDesignStrip Method

Retrieves the specified Design Strip

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetDesignStrip(
    string Name,
    ref string[] Point,
    ref double[] GlobalX,
    ref double[] GlobalY,
    ref double[] GlobalZ,
    ref double[] WLeft,
    ref double[] WRight,
    ref double[] WLeft,
    ref double[] WRight,
    ref bool[] AutoWiden
)
```

```
Function GetDesignStrip (
    Name As String,
    ByRef Point As String(),
    ByRef GlobalX As Double(),
    ByRef GlobalY As Double(),
    ByRef GlobalZ As Double(),
    ByRef WLeft As Double(),
    ByRef WRight As Double(),
    ByRef WLeft As Double(),
    ByRef WRight As Double(),
    ByRef AutoWiden As Boolean()
) As Integer
```

```
Dim instance As cDesignStrip
Dim Name As String
Dim Point As String()
Dim GlobalX As Double()
Dim GlobalY As Double()
Dim GlobalZ As Double()
Dim WLeft As Double()
Dim WRight As Double()
Dim WLeft As Double()
Dim WRight As Double()
Dim AutoWiden As Boolean()
Dim returnValue As Integer
```

```
returnValue = instance.GetDesignStrip(Name,
    Point, GlobalX, GlobalY, GlobalZ,
```

WLeft, WRight, WLeft, WRight,
AutoWiden)

```
int GetDesignStrip(
String^ Name,
array<String^>^% Point,
array<double>^% GlobalX,
array<double>^% GlobalY,
array<double>^% GlobalZ,
array<double>^% WLeft,
array<double>^% WRight,
array<double>^% WLeft,
array<double>^% WRight,
array<bool>^% AutoWiden
)
```

```
abstract GetDesignStrip :
    Name : string *
    Point : string[] byref *
    GlobalX : float[] byref *
    GlobalY : float[] byref *
    GlobalZ : float[] byref *
    WLeft : float[] byref *
    WRight : float[] byref *
    WLeft : float[] byref *
    WRight : float[] byref *
    AutoWiden : bool[] byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing Design Strip

Point

Type:Â SystemString

The unique name of a point object along the polyline that defines the strip.

GlobalX

Type:Â SystemDouble

The global X coordinate of the specified point object.

GlobalY

Type:Â SystemDouble

The global Y coordinate of the specified point object.

GlobalZ

Type:Â SystemDouble

The global Z coordinate of the specified point object.

WLeft

Type:Â SystemDouble

The strip width left just before the specified point.

WRight

Type:Â SystemDouble

The strip width right just before the specified point.

WLeft

Type:Â SystemDouble

The strip width left just after the specified point.

WRight

Type: SystemDouble

The strip width right just after the specified point.

AutoWiden

Type: SystemBoolean

This is boolean indicating whether the Strip Width Option is set to Auto Widen Entire Strip (True), or User Specified Strip Segment Width (False)

Return Value

Type: Int32

Returns zero if the Design Strip is successfully retrieved, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Point as String()
    Dim GlobalX as Double()
    Dim GlobalY as Double()
    Dim GlobalZ as Double()
    Dim WLeft as Double()
    Dim WRight as Double()
    Dim WLeft as Double()
    Dim WRight as Double()
    Dim AutoWiden as Boolean()

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new Design Strip with name "MyDesignStrip1A" ; currently not available in API

    'get DesignStrip
    ret = SapModel.DesignConcreteSlab.DesignStrip.GetDesignStrip("MyDesignStrip1A", Point, GlobalX, GlobalY, GlobalZ, WLeft, WRight)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

Return Value

▲ See Also

Reference

cDesignStrip Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDesignStripAddLanguageSpecificTextSet("LST3F68D793_0") Method

Retrieves the specified Design Strip

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetDesignStrip_1(
    string Name,
    ref int DesignType,
    ref string[] Point,
    ref double[] GlobalX,
    ref double[] GlobalY,
    ref double[] GlobalZ,
    ref double[] WLeft,
    ref double[] WRight,
    ref double[] WLeft,
    ref double[] WRight,
    ref bool[] AutoWiden
)
```

```
Function GetDesignStrip_1 (
    Name As String,
    ByRef DesignType As Integer,
    ByRef Point As String(),
    ByRef GlobalX As Double(),
    ByRef GlobalY As Double(),
    ByRef GlobalZ As Double(),
    ByRef WLeft As Double(),
    ByRef WRight As Double(),
    ByRef WLeft As Double(),
    ByRef WRight As Double(),
    ByRef AutoWiden As Boolean()
) As Integer
```

```
Dim instance As cDesignStrip
Dim Name As String
Dim DesignType As Integer
Dim Point As String()
Dim GlobalX As Double()
Dim GlobalY As Double()
Dim GlobalZ As Double()
Dim WLeft As Double()
Dim WRight As Double()
Dim WLeft As Double()
Dim WRight As Double()
Dim AutoWiden As Boolean()
Dim returnValue As Integer
```

cDesignStripspan id="LST3F68D793_0"AddLanguageSpecificTextSet("LST3F68D793_0?cpp=1#35-");Get

```
returnValue = instance.GetDesignStrip_1(Name,
DesignType, Point, GlobalX, GlobalY,
GlobalZ, WLeft, WRight, WLeft,
WRight, AutoWiden)
```

```
int GetDesignStrip_1(
String^ Name,
int% DesignType,
array<String^>^% Point,
array<double>^% GlobalX,
array<double>^% GlobalY,
array<double>^% GlobalZ,
array<double>^% WLeft,
array<double>^% WRight,
array<double>^% WLeft,
array<double>^% WRight,
array<bool>^% AutoWiden
)
```

```
abstract GetDesignStrip_1 :
    Name : string *
    DesignType : int byref *
    Point : string[] byref *
    GlobalX : float[] byref *
    GlobalY : float[] byref *
    GlobalZ : float[] byref *
    WLeft : float[] byref *
    WRight : float[] byref *
    WLeft : float[] byref *
    WRight : float[] byref *
    AutoWiden : bool[] byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing Design Strip

DesignType

Type: SystemInt32

This is an integer indicating Strip Design Type

◇ None = 0

◇ Column Strip = 1

◇ Middle Strip = 2

Point

Type: SystemString

The unique name of a point object along the polyline that defines the strip.

GlobalX

Type: SystemDouble

The global X coordinate of the specified point object.

GlobalY

Type: SystemDouble

The global Y coordinate of the specified point object.

GlobalZ

Type: SystemDouble

The global Z coordinate of the specified point object.

WBLeft

Type: SystemDouble

The strip width left just before the specified point.

WBRight

Type: SystemDouble

The strip width right just before the specified point.

WALeft

Type: SystemDouble

The strip width left just after the specified point.

WARight

Type: SystemDouble

The strip width right just after the specified point.

AutoWiden

Type: SystemBoolean

This is boolean indicating whether the Strip Width Option is set to Auto Widen Entire Strip (True), or User Specified Strip Segment Width (False)

Return Value

Type: Int32

Returns zero if the Design Strip is successfully retrieved, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim DesignType as Integer
    Dim Point as String()
    Dim GlobalX as Double()
    Dim GlobalY as Double()
    Dim GlobalZ as Double()
    Dim WBLeft as Double()
    Dim WBRight as Double()
    Dim WALeft as Double()
    Dim WARight as Double()
    Dim AutoWiden as Boolean()

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)
```

Return Value

Introduction

```
'define new Design Strip with name "MyDesignStrip1A" ; currently not available in API

'get DesignStrip
    ret = SapModel.DesignConcreteSlab.DesignStrip.GetDesignStrip_1("MyDesignStrip1A", DesignType,
'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

📌 See Also

Reference

cDesignStrip Interface

ETABSv1 Namespace

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Namespace: ETABSv1

▲ Syntax

VBF#

Copy

```
Function GetGUID (
  Name As String,
  ByRef GUID As String
) As Integer
```

```
returnValue = instance.GetGUID(Name, GUID)
```

```
abstract GetGUID :
  Name : string *
  GUID : string byref -> int
```

Name

GUID

Type: SystemString
The GUID of the strip

Return Value

Type: Int32

Returns zero if the function executes correctly, otherwise it returns a nonzero value.

▲ See Also

Reference

cDesignStrip Interface

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CSI API ETABS v1

cDesignStripAddLanguageSpecificTextSet("LST4E4392E1_0")

Method

Retrieves the names of all defined Design Strips

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetNameList(
    ref int NumberNames,
    ref string[] MyName
)
```

```
Function GetNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cDesignStrip
Dim NumberNames As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetNameList(NumberNames,
    MyName)
```

```
int GetNameList(
    int% NumberNames,
    array<String^>^% MyName
)
```

```
abstract GetNameList :
    NumberNames : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberNames

Type: System.Int32

The number of Design Strip names defined in the program

MyName

Type: System.String

This is a one-dimensional array of names. The MyName array is created as a dynamic, zero-based, array by the API user:

```
Dim MyName() as String
```

cDesignStripspan id="LST4E4392E1_0"AddLanguageSpecificTextSet("LST4E4392E1_0?cpp=1#1=.");GetL

Introduction

The array is dimensioned to (NumberNames + 1) inside the ETABS program, filled with the names, and returned to the API user.

Return Value

Type: Int32

Returns zero if the names are successfully retrieved, otherwise it returns nonzero.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim NumberNames As Integer  
    Dim MyName As String()  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)  
  
    'define new Design Strip with name "MyDesignStrip1A" ; currently not available in API  
  
    'get names  
    ret = SapModel.DesignStrip.GetNameList(NumberNames, MyName)  
  
    'close ETABS  
    EtabsObject.ApplicationExit(False)  
  
    'clean up variables  
    SapModel = Nothing  
    EtabsObject = Nothing  
  
End Sub
```

▲ **See Also**

Reference

cDesignStrip Interface

ETABSv1 Namespace

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Introduction

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CSI API ETABS v1

cDesignStripAddLanguageSpecificTextSet("LST455B7D1C_0")

Method

Sets the GUID of an existing strip

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetGUID(
    string Name,
    string GUID = ""
)
```

```
Function SetGUID (
    Name As String,
    GUID As String = ""
) As Integer
```

```
Dim instance As cDesignStrip
Dim Name As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.SetGUID(Name, GUID)
```

```
int SetGUID(
    String^ Name,
    String^ GUID = L""
)
```

```
abstract SetGUID :
    Name : string *
    ?GUID : string
(* Defaults:
    let _GUID = defaultArg GUID ""
*)
-> int
```

Parameters

Name

Type: SystemString

The name of the strip

GUID (Optional)

Type: SystemString

The desired GUID for the strip. If this argument is left blank, the program will assign a new randomly-created GUID

cDesignStripspan id="LST455B7D1C_0"AddLanguageSpecificTextSet("LST455B7D1C_0?cpp1444=");Set

Return Value

Type: Int32

Returns zero if the function executes correctly, otherwise it returns a nonzero value.

▲ See Also

Reference

cDesignStrip Interface

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CSI API ETABS v1

cDetailing Interface

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
public interface cDetailing
```

```
Public Interface cDetailing
```

```
Dim instance As cDetailing
```

```
public interface class cDetailing
```

```
type cDetailing = interface end
```

The cDetailing type exposes the following members.

▲ Methods

▲

Name

⇒

ClearDetailing

⇒

GetBeamLongRebarData

⇒

GetBeamTieRebarData

⇒

GetColumnLongRebarData

⇒

GetColumnTieRebarData

⇒

GetDetailed_OneWallStack

⇒

GetDetailedBeamLineData

⇒

GetDetailedBeamLineData_1

⇒

GetDetailedBeamLineGuidData

⇒

GetDetailedBeamLines

⇒

GetDetailedColumnStackData

- ⇒ [GetDetailedColumnStackData_1](#)
- ⇒ [GetDetailedColumnStackData_2](#)
- ⇒ [GetDetailedColumnStackGuidData](#)
- ⇒ [GetDetailedColumnStacks](#)
- ⇒ [GetDetailedSlab_OneDetailingOutputInfo](#)
- ⇒ [GetDetailedSlabBotBarData](#)
- ⇒ [GetDetailedSlabBotBarData_1](#)
- ⇒ [GetDetailedSlabs](#)
- ⇒ [GetDetailedSlabTopBarData](#)
- ⇒ [GetDetailedSlabTopBarData_1](#)
- ⇒ [GetDetailedWall_OnePier_OneDesignLeg_OneTieBar_OneTiePline_OnePoint](#)
- ⇒ [GetDetailedWall_OneWallStack_OnePier_OneDesignLeg_OneTieBar_OneTiePlineInfo](#)
- ⇒ [GetDetailedWall_OneWallStack_OnePier_OneDesignLeg_OneTieBarInfo](#)
- ⇒ [GetDetailedWall_OneWallStack_OnePier_OneDesignLeg_OneVerticalBarInfo](#)
- ⇒ [GetDetailedWall_OneWallStack_OnePier_OneDesignLegOutputInfo](#)
- ⇒ [GetDetailedWall_OneWallStack_OnePierOutputInfo](#)
- ⇒ [GetDetailedWall_OneWallStack_OneSpandrel_OneLongBarInfo](#)
- ⇒ [GetDetailedWall_OneWallStack_OneSpandrel_OneStirrupsInfo](#)
- ⇒ [GetDetailedWall_OneWallStack_OneSpandrelOutputInfo](#)

- ⇒ [GetDetailingAvailable](#)
- ⇒ [GetNumberDetailedSlabs](#)
- ⇒ [GetNumberDetailedWallStacks](#)
- ⇒ [GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneBottomRebar](#)
- ⇒ [GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneBottomRebar](#)
- ⇒ [GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneBottomRebar](#)
- ⇒ [GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneTopRebar_B](#)
- ⇒ [GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneTopRebar_B](#)
- ⇒ [GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneTopRebarInt](#)
- ⇒ [GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegionInfo](#)
- ⇒ [GetOneDetailedSlab_OneDetailingOutput_StripGUID](#)
- ⇒ [GetOneDetailedSlab_OneDetailingOutput_StripInfo](#)
- ⇒ [GetSimilarBeamLines](#)
- ⇒ [GetSimilarColumnStacks](#)
- ⇒ [GetSimilarSlabs](#)
- ⇒ [StartDetailing](#)

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[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDetailing Methods

The cDetailing type exposes the following members.

Methods

Â

Name

- ⇒ ClearDetailing
- ⇒ GetBeamLongRebarData
- ⇒ GetBeamTieRebarData
- ⇒ GetColumnLongRebarData
- ⇒ GetColumnTieRebarData
- ⇒ GetDetailed_OneWallStack
- ⇒ GetDetailedBeamLineData
- ⇒ GetDetailedBeamLineData_1
- ⇒ GetDetailedBeamLineGuidData
- ⇒ GetDetailedBeamLines
- ⇒ GetDetailedColumnStackData
- ⇒ GetDetailedColumnStackData_1
- ⇒ GetDetailedColumnStackData_2
- ⇒ GetDetailedColumnStackGuidData
- ⇒ GetDetailedColumnStacks
- ⇒ GetDetailedSlab_OneDetailingOutputInfo
- ⇒ GetDetailedSlabBotBarData
- ⇒ GetDetailedSlabBotBarData_1
- ⇒ GetDetailedSlabs
- ⇒ GetDetailedSlabTopBarData
- ⇒ GetDetailedSlabTopBarData_1
- ⇒ GetDetailedWall_OnePier_OneDesignLeg_OneTieBar_OneTiePline_OnePoint
- ⇒ GetDetailedWall_OneWallStack_OnePier_OneDesignLeg_OneTieBar_OneTiePlineInfo

⇒ GetDetailedWall_OneWallStack_OnePier_OneDesignLeg_OneTieBarInfo

⇒ GetDetailedWall_OneWallStack_OnePier_OneDesignLeg_OneVerticalBarInfo

⇒ GetDetailedWall_OneWallStack_OnePier_OneDesignLegOutputInfo

⇒ GetDetailedWall_OneWallStack_OnePierOutputInfo

⇒ GetDetailedWall_OneWallStack_OneSpandrel_OneLongBarInfo

⇒ GetDetailedWall_OneWallStack_OneSpandrel_OneStirrupsInfo

⇒ GetDetailedWall_OneWallStack_OneSpandrelOutputInfo

⇒ GetDetailingAvailable

⇒ GetNumberDetailedSlabs

⇒ GetNumberDetailedWallStacks

⇒ GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneBottomRebar

⇒ GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneBottomRebar

⇒ GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneBottomRebar

⇒ GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneTopRebar_B

⇒ GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneTopRebar_B

⇒ [GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneTopRebarInfo](#)

⇒ [GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegionInfo](#)

⇒ [GetOneDetailedSlab_OneDetailingOutput_StripGUID](#)

⇒ [GetOneDetailedSlab_OneDetailingOutput_StripInfo](#)

⇒ [GetSimilarBeamLines](#)

⇒ [GetSimilarColumnStacks](#)

⇒ [GetSimilarSlabs](#)

⇒ [StartDetailing](#)

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LSTE367FDF_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int ClearDetailing()
```

Function ClearDetailing As Integer

```
Dim instance As cDetailing
```

```
Dim returnValue As Integer
```

```
returnValue = instance.ClearDetailing()
```

```
int ClearDetailing()
```

```
abstract ClearDetailing : unit -> int
```

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cDetailing Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LST37E05DCA_0?cpp=::1453");GetB

Method

Retrieves longitudinal rebar data for a beam frame object

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetBeamLongRebarData(
    string Name,
    ref int NumberRebarSets,
    ref string[] BarSizeName,
    ref double[] BarArea,
    ref int[] NumberBars,
    ref string[] Location,
    ref double[] ClearCover,
    ref double[] StartCoord1,
    ref double[] BarLength,
    ref double[] BendingAngleStart,
    ref double[] BendingAngleEnd,
    ref string[] RebarSetGUID
)
```

```
Function GetBeamLongRebarData (
    Name As String,
    ByRef NumberRebarSets As Integer,
    ByRef BarSizeName As String(),
    ByRef BarArea As Double(),
    ByRef NumberBars As Integer(),
    ByRef Location As String(),
    ByRef ClearCover As Double(),
    ByRef StartCoord1 As Double(),
    ByRef BarLength As Double(),
    ByRef BendingAngleStart As Double(),
    ByRef BendingAngleEnd As Double(),
    ByRef RebarSetGUID As String()
) As Integer
```

```
Dim instance As cDetailing
Dim Name As String
Dim NumberRebarSets As Integer
Dim BarSizeName As String()
Dim BarArea As Double()
Dim NumberBars As Integer()
Dim Location As String()
Dim ClearCover As Double()
Dim StartCoord1 As Double()
Dim BarLength As Double()
Dim BendingAngleStart As Double()
```

cDetailingspan id="LST37E05DCA_0"AddLanguageSpecificTextSet("LST37E05DCA_0?cpp=::1453");GetB

Introduction

```
Dim BendingAngleEnd As Double()
Dim RebarSetGUID As String()
Dim returnValue As Integer

returnValue = instance.GetBeamLongRebarData(Name,
NumberRebarSets, BarSizeName, BarArea,
NumberBars, Location, ClearCover,
StartCoord1, BarLength, BendingAngleStart,
BendingAngleEnd, RebarSetGUID)

int GetBeamLongRebarData(
String^ Name,
int% NumberRebarSets,
array<String^>^% BarSizeName,
array<double>^% BarArea,
array<int>^% NumberBars,
array<String^>^% Location,
array<double>^% ClearCover,
array<double>^% StartCoord1,
array<double>^% BarLength,
array<double>^% BendingAngleStart,
array<double>^% BendingAngleEnd,
array<String^>^% RebarSetGUID
)

abstract GetBeamLongRebarData :
    Name : string *
    NumberRebarSets : int byref *
    BarSizeName : string[] byref *
    BarArea : float[] byref *
    NumberBars : int[] byref *
    Location : string[] byref *
    ClearCover : float[] byref *
    StartCoord1 : float[] byref *
    BarLength : float[] byref *
    BendingAngleStart : float[] byref *
    BendingAngleEnd : float[] byref *
    RebarSetGUID : string[] byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing beam frame object

NumberRebarSets

Type:Â SystemInt32

Number of longitudinal rebar sets for this beam object

BarSizeName

Type:Â SystemString

Rebar designation

BarArea

Type:Â SystemDouble

Individual rebar area

NumberBars

Type:Â SystemInt32

Number of rebars in this set

Location

Type:Â SystemString

Location of this set of rebars - A, B, C, etc. (See picture for seismic/ non-seismic detailing)

ClearCover

Type:Â SystemDouble

Clear cover from top or bottom beam face to longitudinal rebar being specified

StartCoord1

Type:Â SystemDouble

Start of this set of rebars from start/end (depends on location) of beam object

BarLength

Type:Â SystemDouble

Length of bars from start point

BendingAngleStart

Type:Â SystemDouble

Bend angle if any at start of rebars

BendingAngleEnd

Type:Â SystemDouble

Bend angle if any at end of rebars

RebarSetGUID

Type:Â SystemString

The GUIDs of each of the rebar sets (could be duplicated for adjacent beam if rebars are continuous)

Return Value

Type:Â Int32

Returns zero if the rebar data is successfully retrieved; otherwise it returns a nonzero value.

▲ See Also

Reference

cDetailing Interface

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LSTFA1F212D_0?cpp=::|1456);GetBeamTieRebarData Method

Retrieves tie (confinement/shear) rebar data for a beam frame object

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetBeamTieRebarData(
    string Name,
    ref int NumberRebarSets,
    ref string[] BarSizeName,
    ref double[] BarArea,
    ref double[] NumberLegs,
    ref string[] Location,
    ref double[] ClearCover,
    ref double[] StartCoord1,
    ref double[] Spacing,
    ref double[] Lengths,
    ref string[] RebarSetGUID
)
```

```
Function GetBeamTieRebarData (
    Name As String,
    ByRef NumberRebarSets As Integer,
    ByRef BarSizeName As String(),
    ByRef BarArea As Double(),
    ByRef NumberLegs As Double(),
    ByRef Location As String(),
    ByRef ClearCover As Double(),
    ByRef StartCoord1 As Double(),
    ByRef Spacing As Double(),
    ByRef Lengths As Double(),
    ByRef RebarSetGUID As String()
) As Integer
```

```
Dim instance As cDetailing
Dim Name As String
Dim NumberRebarSets As Integer
Dim BarSizeName As String()
Dim BarArea As Double()
Dim NumberLegs As Double()
Dim Location As String()
Dim ClearCover As Double()
Dim StartCoord1 As Double()
Dim Spacing As Double()
Dim Lengths As Double()
Dim RebarSetGUID As String()
Dim returnValue As Integer
```

cDetailingspan id="LSTFA1F212D_0"AddLanguageSpecificTextSet("LSTFA1F212D_0?cpp=::|1456);GetBe

```
returnValue = instance.GetBeamTieRebarData(Name,
NumberRebarSets, BarSizeName, BarArea,
NumberLegs, Location, ClearCover,
StartCoord1, Spacing, Lengths, RebarSetGUID)
```

```
int GetBeamTieRebarData(
String^ Name,
int% NumberRebarSets,
array<String^>^% BarSizeName,
array<double>^% BarArea,
array<double>^% NumberLegs,
array<String^>^% Location,
array<double>^% ClearCover,
array<double>^% StartCoord1,
array<double>^% Spacing,
array<double>^% Lengths,
array<String^>^% RebarSetGUID
)
```

```
abstract GetBeamTieRebarData :
    Name : string *
    NumberRebarSets : int byref *
    BarSizeName : string[] byref *
    BarArea : float[] byref *
    NumberLegs : float[] byref *
    Location : string[] byref *
    ClearCover : float[] byref *
    StartCoord1 : float[] byref *
    Spacing : float[] byref *
    Lengths : float[] byref *
    RebarSetGUID : string[] byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing beam frame object

NumberRebarSets

Type:Â SystemInt32

Number of tie rebar sets for this beam object

BarSizeName

Type:Â SystemString

Rebar designation

BarArea

Type:Â SystemDouble

Individual rebar area

NumberLegs

Type:Â SystemDouble

Number of legs for ties in this set (legs are in the local 2 direction)

Location

Type:Â SystemString

Zone of this set of rebars - A, B, C, etc. (See picture for seismic/ non-seismic detailing)

ClearCover

Type: SystemDouble

Clear cover from beam face to ties being specified

StartCoord1

Type: SystemDouble

Start of this set of tie rebars from start/end (depends on location) of beam object

Spacing

Type: SystemDouble

Spacing of ties

Lengths

Type: SystemDouble

Length over which these tie bars are provided from start point

RebarSetGUID

Type: SystemString

The GUIDs of each of the rebar sets (could be duplicated for adjacent beam if rebars are continuous)

Return Value

Type: Int32

Returns zero if the rebar data is successfully retrieved; otherwise it returns a nonzero value.

See Also

Reference

cDetailing Interface

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LST9ADEFAF5_0?cpp=::1459");GetC Method

Retrieves longitudinal rebar data for a column frame object

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetColumnLongRebarData(
    string Name,
    ref int NumberRebarSets,
    ref string[] BarSizeName,
    ref double[] BarArea,
    ref int[] NumberCBars,
    ref int[] NumberR3Bars,
    ref int[] NumberR2Bars,
    ref string[] Location,
    ref double[] ClearCover,
    ref string[] RebarSetGUID
)
```

```
Function GetColumnLongRebarData (
    Name As String,
    ByRef NumberRebarSets As Integer,
    ByRef BarSizeName As String(),
    ByRef BarArea As Double(),
    ByRef NumberCBars As Integer(),
    ByRef NumberR3Bars As Integer(),
    ByRef NumberR2Bars As Integer(),
    ByRef Location As String(),
    ByRef ClearCover As Double(),
    ByRef RebarSetGUID As String()
) As Integer
```

```
Dim instance As cDetailing
Dim Name As String
Dim NumberRebarSets As Integer
Dim BarSizeName As String()
Dim BarArea As Double()
Dim NumberCBars As Integer()
Dim NumberR3Bars As Integer()
Dim NumberR2Bars As Integer()
Dim Location As String()
Dim ClearCover As Double()
Dim RebarSetGUID As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetColumnLongRebarData(Name,
    NumberRebarSets, BarSizeName, BarArea,
```

Introduction

NumberCBars, NumberR3Bars, NumberR2Bars,
Location, ClearCover, RebarSetGUID)

```
int GetColumnLongRebarData(  
String^ Name,  
int% NumberRebarSets,  
array<String^>^% BarSizeName,  
array<double>^% BarArea,  
array<int>^% NumberCBars,  
array<int>^% NumberR3Bars,  
array<int>^% NumberR2Bars,  
array<String^>^% Location,  
array<double>^% ClearCover,  
array<String^>^% RebarSetGUID  
)
```

```
abstract GetColumnLongRebarData :  
    Name : string *  
    NumberRebarSets : int byref *  
    BarSizeName : string[] byref *  
    BarArea : float[] byref *  
    NumberCBars : int[] byref *  
    NumberR3Bars : int[] byref *  
    NumberR2Bars : int[] byref *  
    Location : string[] byref *  
    ClearCover : float[] byref *  
    RebarSetGUID : string[] byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing beam frame object

NumberRebarSets

Type:Â SystemInt32

Number of longitudinal rebar sets for this beam object

BarSizeName

Type:Â SystemString

Rebar designation

BarArea

Type:Â SystemDouble

Individual rebar area

NumberCBars

Type:Â SystemInt32

Total number of rebars in this set (for both circular or rectangular pattern)

NumberR3Bars

Type:Â SystemInt32

Number of rectangular pattern bars along 3 direction (per face including corner bars); zero for circular pattern

NumberR2Bars

Type:Â SystemInt32

Number of rectangular pattern bars along 2 direction (per face including corner bars); zero for circular pattern

Location

Type: SystemString

Location of this set of rebars - not currently used - left initialized to blank ""

ClearCover

Type: SystemDouble

Clear cover from face to longitudinal rebar being specified

RebarSetGUID

Type: SystemString

The GUIDs of each of the rebar sets

Return Value

Type: Int32

Returns zero if the rebar data is successfully retrieved; otherwise it returns a nonzero value.

▲ See Also

Reference

cDetailing Interface

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Namespace: ETABSv1

▲ Syntax

C#

VBC++

F#

Copy

```
Function GetColumnTieRebarData (
Name As String,
ByRef NumberRebarSets As Integer,
ByRef BarSizeName As String(),
ByRef BarArea As Double(),
ByRef Pattern As Integer(),
ByRef ConfineType As Integer(),
ByRef NumberLegs2Dir As Integer(),
ByRef NumberLegs3Dir As Integer(),
ByRef Location As String(),
ByRef ClearCover As Double(),
ByRef StartCoord1 As Double(),
ByRef Spacing As Double(),
ByRef Heights As Double(),
ByRef RebarSetGUID As String()
) As Integer
```

cDetailingspan id="LSTCC606546_0"AddLanguageSpecificTextSet("LSTCC606546_0?cpp=::1462);GetCo

Introduction

```
Dim NumberLegs2Dir As Integer()
Dim NumberLegs3Dir As Integer()
Dim Location As String()
Dim ClearCover As Double()
Dim StartCoord1 As Double()
Dim Spacing As Double()
Dim Heights As Double()
Dim RebarSetGUID As String()
Dim returnValue As Integer

returnValue = instance.GetColumnTieRebarData(Name,
NumberRebarSets, BarSizeName, BarArea,
Pattern, ConfineType, NumberLegs2Dir,
NumberLegs3Dir, Location, ClearCover,
StartCoord1, Spacing, Heights, RebarSetGUID)
```

```
int GetColumnTieRebarData(
String^ Name,
int% NumberRebarSets,
array<String^>^% BarSizeName,
array<double>^% BarArea,
array<int>^% Pattern,
array<int>^% ConfineType,
array<int>^% NumberLegs2Dir,
array<int>^% NumberLegs3Dir,
array<String^>^% Location,
array<double>^% ClearCover,
array<double>^% StartCoord1,
array<double>^% Spacing,
array<double>^% Heights,
array<String^>^% RebarSetGUID
)
```

```
abstract GetColumnTieRebarData :
    Name : string *
    NumberRebarSets : int byref *
    BarSizeName : string[] byref *
    BarArea : float[] byref *
    Pattern : int[] byref *
    ConfineType : int[] byref *
    NumberLegs2Dir : int[] byref *
    NumberLegs3Dir : int[] byref *
    Location : string[] byref *
    ClearCover : float[] byref *
    StartCoord1 : float[] byref *
    Spacing : float[] byref *
    Heights : float[] byref *
    RebarSetGUID : string[] byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing beam frame object

NumberRebarSets

Type:Â SystemInt32

Number of tie rebar sets for this beam object

BarSizeName

Type:Â SystemString
Rebar designation

BarArea
Type:Â SystemDouble
Individual rebar area

Pattern
Type:Â SystemInt32
Rebar pattern - 1 - Rectangular, 2 - Circular

ConfineType
Type:Â SystemInt32
Confinement type - 1 - Ties, 2 - Spiral (Spiral only possible when Pattern is Circular)

NumberLegs2Dir
Type:Â SystemInt32
Number of legs for ties in this set along the local 2 direction)

NumberLegs3Dir
Type:Â SystemInt32
Number of legs for ties in this set along the local 3 direction)

Location
Type:Â SystemString
Zone of this set of rebars - A, B, C, etc. (See picture for seismic/ non-seismic detailing)

ClearCover
Type:Â SystemDouble
Clear cover from face to ties being specified

StartCoord1
Type:Â SystemDouble
Start of this set of tie rebars from start/end (depends on location) of column object

Spacing
Type:Â SystemDouble
Spacing of ties

Heights
Type:Â SystemDouble
Height over which these tie bars are provided from start point

RebarSetGUID
Type:Â SystemString
The GUIDs of each of the rebar sets (could be duplicated for adjacent beam if rebars are continuous)

Return Value

Type:Â Int32

Returns zero if the rebar data is successfully retrieved; otherwise it returns a nonzero value.

▲ See Also

Reference

cDetailing Interface
ETABSv1 Namespace

Introduction

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LST63B28EF2_0?cpp=::int466;GetDe Method

Detailed output for Wall

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetDetailed_OneWallStack(
int WallStackIndex,
ref string GUID,
ref int TowerID,
ref int NumberPiers,
ref int NumberSpandrels
)
```

```
Function GetDetailed_OneWallStack (
WallStackIndex As Integer,
ByRef GUID As String,
ByRef TowerID As Integer,
ByRef NumberPiers As Integer,
ByRef NumberSpandrels As Integer
) As Integer
```

```
Dim instance As cDetailing
Dim WallStackIndex As Integer
Dim GUID As String
Dim TowerID As Integer
Dim NumberPiers As Integer
Dim NumberSpandrels As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetDetailed_OneWallStack(WallStackIndex,
GUID, TowerID, NumberPiers, NumberSpandrels)
```

```
int GetDetailed_OneWallStack(
int WallStackIndex,
String^% GUID,
int% TowerID,
int% NumberPiers,
int% NumberSpandrels
)
```

```
abstract GetDetailed_OneWallStack :
    WallStackIndex : int *
    GUID : string byref *
    TowerID : int byref *
    NumberPiers : int byref *
    NumberSpandrels : int byref -> int
```

cDetailingspan id="LST63B28EF2_0"AddLanguageSpecificTextSet("LST63B28EF2_0?cpp=::int466;GetDe

Parameters

WallStackIndex

Type: SystemInt32

Wall Stack Index

GUID

Type: SystemString

Unique GUID assigned to the Wall Stack

TowerID

Type: SystemInt32

The index of Tower, in which this Stack belongs to. Taken from ETABS

NumberPiers

Type: SystemInt32

All Piers in the wall stack, on all floors

NumberSpandrels

Type: SystemInt32

Return Value

Type: Int32

Returns zero if the function is successful; otherwise it returns a nonzero value.

▲ See Also

Reference

cDetailing Interface

ETABSV1 Namespace

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LSTC0EC4DA Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetDetailedBeamLineData(
    string BeamLineID,
    ref string[] ObjectUniqueNames,
    ref int NumberSpans,
    ref double[] SpanLength,
    ref int[] NumLongBars,
    ref double[] LongBarDiameter,
    ref string[] LongBarNotation,
    ref double[] LongBarStartDist,
    ref int[] LongBarStartBend,
    ref int[] LongBarEndBend,
    ref double[] LongBarLength,
    ref int[] LongBarNumLayers,
    ref int[] NumTieBars,
    ref int[] NumTieVertLegs,
    ref double[] TieBarDiameter,
    ref string[] TieBarNotation,
    ref double[] TieBarStartDist,
    ref double[] TieBarSpacing,
    ref int[] TieBarType
)
```

```
Function GetDetailedBeamLineData (
    BeamLineID As String,
    ByRef ObjectUniqueNames As String(),
    ByRef NumberSpans As Integer,
    ByRef SpanLength As Double(),
    ByRef NumLongBars As Integer(),
    ByRef LongBarDiameter As Double(),
    ByRef LongBarNotation As String(),
    ByRef LongBarStartDist As Double(),
    ByRef LongBarStartBend As Integer(),
    ByRef LongBarEndBend As Integer(),
    ByRef LongBarLength As Double(),
    ByRef LongBarNumLayers As Integer(),
    ByRef NumTieBars As Integer(),
    ByRef NumTieVertLegs As Integer(),
    ByRef TieBarDiameter As Double(),
    ByRef TieBarNotation As String(),
    ByRef TieBarStartDist As Double(),
    ByRef TieBarSpacing As Double(),
    ByRef TieBarType As Integer()
) As Integer
```

cDetailingspan id="LSTC0EC4DAF_0"AddLanguageSpecificTextSet("LSTC0EC4DAF_0?cpp=:1468.");GetD

Introduction

```
Dim instance As cDetailing
Dim BeamLineID As String
Dim ObjectUniqueNames As String()
Dim NumberSpans As Integer
Dim SpanLength As Double()
Dim NumLongBars As Integer()
Dim LongBarDiameter As Double()
Dim LongBarNotation As String()
Dim LongBarStartDist As Double()
Dim LongBarStartBend As Integer()
Dim LongBarEndBend As Integer()
Dim LongBarLength As Double()
Dim LongBarNumLayers As Integer()
Dim NumTieBars As Integer()
Dim NumTieVertLegs As Integer()
Dim TieBarDiameter As Double()
Dim TieBarNotation As String()
Dim TieBarStartDist As Double()
Dim TieBarSpacing As Double()
Dim TieBarType As Integer()
Dim returnValue As Integer
```

```
returnValue = instance.GetDetailedBeamLineData(BeamLineID,
ObjectUniqueNames, NumberSpans,
SpanLength, NumLongBars, LongBarDiameter,
LongBarNotation, LongBarStartDist,
LongBarStartBend, LongBarEndBend,
LongBarLength, LongBarNumLayers,
NumTieBars, NumTieVertLegs, TieBarDiameter,
TieBarNotation, TieBarStartDist,
TieBarSpacing, TieBarType)
```

```
int GetDetailedBeamLineData(
String^ BeamLineID,
array<String^>^% ObjectUniqueNames,
int% NumberSpans,
array<double>^% SpanLength,
array<int>^% NumLongBars,
array<double>^% LongBarDiameter,
array<String^>^% LongBarNotation,
array<double>^% LongBarStartDist,
array<int>^% LongBarStartBend,
array<int>^% LongBarEndBend,
array<double>^% LongBarLength,
array<int>^% LongBarNumLayers,
array<int>^% NumTieBars,
array<int>^% NumTieVertLegs,
array<double>^% TieBarDiameter,
array<String^>^% TieBarNotation,
array<double>^% TieBarStartDist,
array<double>^% TieBarSpacing,
array<int>^% TieBarType
)
```

```
abstract GetDetailedBeamLineData :
    BeamLineID : string *
    ObjectUniqueNames : string[] byref *
    NumberSpans : int byref *
    SpanLength : float[] byref *
    NumLongBars : int[] byref *
    LongBarDiameter : float[] byref *
```

cDetailingspan id="LSTC0EC4DAF_0"AddLanguageSpecificTextSet("LSTC0EC4DAF_0?cpp=:1469.");GetD

```

LongBarNotation : string[] byref *
LongBarStartDist : float[] byref *
LongBarStartBend : int[] byref *
LongBarEndBend : int[] byref *
LongBarLength : float[] byref *
LongBarNumLayers : int[] byref *
NumTieBars : int[] byref *
NumTieVertLegs : int[] byref *
TieBarDiameter : float[] byref *
TieBarNotation : string[] byref *
TieBarStartDist : float[] byref *
TieBarSpacing : float[] byref *
TieBarType : int[] byref -> int

```

Parameters

```

BeamLineID
    Type:Â SystemString
ObjectUniqueNames
    Type:Â SystemString
NumberSpans
    Type:Â SystemInt32
SpanLength
    Type:Â SystemDouble
NumLongBars
    Type:Â SystemInt32
LongBarDiameter
    Type:Â SystemDouble
LongBarNotation
    Type:Â SystemString
LongBarStartDist
    Type:Â SystemDouble
LongBarStartBend
    Type:Â SystemInt32
LongBarEndBend
    Type:Â SystemInt32
LongBarLength
    Type:Â SystemDouble
LongBarNumLayers
    Type:Â SystemInt32
NumTieBars
    Type:Â SystemInt32
NumTieVertLegs
    Type:Â SystemInt32
TieBarDiameter
    Type:Â SystemDouble
TieBarNotation
    Type:Â SystemString
TieBarStartDist
    Type:Â SystemDouble
TieBarSpacing
    Type:Â SystemDouble
TieBarType

```

Type: SystemInt32

Return Value

Type: Int32

▲ See Also

Reference

cDetailing Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LST19DB13C2_0?cpp=::|1472");GetDe Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetDetailedBeamLineData_1(
    string BeamLineID,
    ref string[] ObjectUniqueNames,
    ref int NumberSpans,
    ref double[] SpanLength,
    ref int[] NumLongBars,
    ref string[] LongBarGUID,
    ref double[] LongBarDiameter,
    ref string[] LongBarNotation,
    ref double[] LongBarStartDist,
    ref int[] LongBarStartBend,
    ref int[] LongBarEndBend,
    ref double[] LongBarLength,
    ref int[] LongBarNumLayers,
    ref int[] NumTieBars,
    ref int[] NumTieVertLegs,
    ref string[] TieBarGUID,
    ref double[] TieBarDiameter,
    ref string[] TieBarNotation,
    ref double[] TieBarStartDist,
    ref double[] TieBarSpacing,
    ref int[] TieBarType
)
```

```
Function GetDetailedBeamLineData_1 (
    BeamLineID As String,
    ByRef ObjectUniqueNames As String(),
    ByRef NumberSpans As Integer,
    ByRef SpanLength As Double(),
    ByRef NumLongBars As Integer(),
    ByRef LongBarGUID As String(),
    ByRef LongBarDiameter As Double(),
    ByRef LongBarNotation As String(),
    ByRef LongBarStartDist As Double(),
    ByRef LongBarStartBend As Integer(),
    ByRef LongBarEndBend As Integer(),
    ByRef LongBarLength As Double(),
    ByRef LongBarNumLayers As Integer(),
    ByRef NumTieBars As Integer(),
    ByRef NumTieVertLegs As Integer(),
    ByRef TieBarGUID As String(),
    ByRef TieBarDiameter As Double(),
    ByRef TieBarNotation As String(),
```

cDetailingspan id="LST19DB13C2_0"AddLanguageSpecificTextSet("LST19DB13C2_0?cpp=::|1472");GetDe

Introduction

```
ByRef TieBarStartDist As Double(),  
ByRef TieBarSpacing As Double(),  
ByRef TieBarType As Integer()  
) As Integer
```

```
Dim instance As cDetailing  
Dim BeamLineID As String  
Dim ObjectUniqueNames As String()  
Dim NumberSpans As Integer  
Dim SpanLength As Double()  
Dim NumLongBars As Integer()  
Dim LongBarGUID As String()  
Dim LongBarDiameter As Double()  
Dim LongBarNotation As String()  
Dim LongBarStartDist As Double()  
Dim LongBarStartBend As Integer()  
Dim LongBarEndBend As Integer()  
Dim LongBarLength As Double()  
Dim LongBarNumLayers As Integer()  
Dim NumTieBars As Integer()  
Dim NumTieVertLegs As Integer()  
Dim TieBarGUID As String()  
Dim TieBarDiameter As Double()  
Dim TieBarNotation As String()  
Dim TieBarStartDist As Double()  
Dim TieBarSpacing As Double()  
Dim TieBarType As Integer()  
Dim returnValue As Integer
```

```
returnValue = instance.GetDetailedBeamLineData_1(BeamLineID,  
ObjectUniqueNames, NumberSpans,  
SpanLength, NumLongBars, LongBarGUID,  
LongBarDiameter, LongBarNotation,  
LongBarStartDist, LongBarStartBend,  
LongBarEndBend, LongBarLength, LongBarNumLayers,  
NumTieBars, NumTieVertLegs, TieBarGUID,  
TieBarDiameter, TieBarNotation,  
TieBarStartDist, TieBarSpacing,  
TieBarType)
```

```
int GetDetailedBeamLineData_1(  
String^ BeamLineID,  
array<String^>^% ObjectUniqueNames,  
int% NumberSpans,  
array<double>^% SpanLength,  
array<int>^% NumLongBars,  
array<String^>^% LongBarGUID,  
array<double>^% LongBarDiameter,  
array<String^>^% LongBarNotation,  
array<double>^% LongBarStartDist,  
array<int>^% LongBarStartBend,  
array<int>^% LongBarEndBend,  
array<double>^% LongBarLength,  
array<int>^% LongBarNumLayers,  
array<int>^% NumTieBars,  
array<int>^% NumTieVertLegs,  
array<String^>^% TieBarGUID,  
array<double>^% TieBarDiameter,  
array<String^>^% TieBarNotation,  
array<double>^% TieBarStartDist,  
array<double>^% TieBarSpacing,  
array<int>^% TieBarType
```

cDetailingspan id="LST19DB13C2_0"AddLanguageSpecificTextSet("LST19DB13C2_0?cpp=::|1473);GetDe

)

```

abstract GetDetailedBeamLineData_1 :
    BeamLineID : string *
    ObjectUniqueNames : string[] byref *
    NumberSpans : int byref *
    SpanLength : float[] byref *
    NumLongBars : int[] byref *
    LongBarGUID : string[] byref *
    LongBarDiameter : float[] byref *
    LongBarNotation : string[] byref *
    LongBarStartDist : float[] byref *
    LongBarStartBend : int[] byref *
    LongBarEndBend : int[] byref *
    LongBarLength : float[] byref *
    LongBarNumLayers : int[] byref *
    NumTieBars : int[] byref *
    NumTieVertLegs : int[] byref *
    TieBarGUID : string[] byref *
    TieBarDiameter : float[] byref *
    TieBarNotation : string[] byref *
    TieBarStartDist : float[] byref *
    TieBarSpacing : float[] byref *
    TieBarType : int[] byref -> int

```

Parameters

BeamLineID
 Type:Â SystemString
 ObjectUniqueNames
 Type:Â SystemString
 NumberSpans
 Type:Â SystemInt32
 SpanLength
 Type:Â SystemDouble
 NumLongBars
 Type:Â SystemInt32
 LongBarGUID
 Type:Â SystemString
 LongBarDiameter
 Type:Â SystemDouble
 LongBarNotation
 Type:Â SystemString
 LongBarStartDist
 Type:Â SystemDouble
 LongBarStartBend
 Type:Â SystemInt32
 LongBarEndBend
 Type:Â SystemInt32
 LongBarLength
 Type:Â SystemDouble
 LongBarNumLayers
 Type:Â SystemInt32
 NumTieBars
 Type:Â SystemInt32

NumTieVertLegs
Type:Â [SystemInt32](#)
TieBarGUID
Type:Â [SystemString](#)
TieBarDiameter
Type:Â [SystemDouble](#)
TieBarNotation
Type:Â [SystemString](#)
TieBarStartDist
Type:Â [SystemDouble](#)
TieBarSpacing
Type:Â [SystemDouble](#)
TieBarType
Type:Â [SystemInt32](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cDetailing Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LSTD406094D Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetDetailedBeamLineGuidData(
    string BeamLineID,
    string SimilarFirstBeamUniqueID,
    ref string[] LongitudinalABars,
    ref string[] LongitudinalBBars,
    ref string[] LongitudinalCBars,
    ref string[] LongitudinalDBars,
    ref string[] LongitudinalEBars,
    ref string[] LongitudinalFBars,
    ref string[] LongitudinalGBars,
    ref string[] LongitudinalHBars,
    ref string[] ZoneATies,
    ref string[] ZoneBTies,
    ref string[] ZoneCTies
)
```

```
Function GetDetailedBeamLineGuidData (
    BeamLineID As String,
    SimilarFirstBeamUniqueID As String,
    ByRef LongitudinalABars As String(),
    ByRef LongitudinalBBars As String(),
    ByRef LongitudinalCBars As String(),
    ByRef LongitudinalDBars As String(),
    ByRef LongitudinalEBars As String(),
    ByRef LongitudinalFBars As String(),
    ByRef LongitudinalGBars As String(),
    ByRef LongitudinalHBars As String(),
    ByRef ZoneATies As String(),
    ByRef ZoneBTies As String(),
    ByRef ZoneCTies As String()
) As Integer
```

```
Dim instance As cDetailing
Dim BeamLineID As String
Dim SimilarFirstBeamUniqueID As String
Dim LongitudinalABars As String()
Dim LongitudinalBBars As String()
Dim LongitudinalCBars As String()
Dim LongitudinalDBars As String()
Dim LongitudinalEBars As String()
Dim LongitudinalFBars As String()
Dim LongitudinalGBars As String()
Dim LongitudinalHBars As String()
```

cDetailingspan id="LSTD406094D_0"AddLanguageSpecificTextSet("LSTD406094D_0?cpp=::|n1476);GetDe

Introduction

```
Dim ZoneATies As String()
Dim ZoneBTies As String()
Dim ZoneCTies As String()
Dim returnValue As Integer

returnValue = instance.GetDetailedBeamLineGuidData(BeamLineID,
SimilarFirstBeamUniqueID, LongitudinalABars,
LongitudinalBBars, LongitudinalCBars,
LongitudinalDBars, LongitudinalEBars,
LongitudinalFBars, LongitudinalGBars,
LongitudinalHBars, ZoneATies, ZoneBTies,
ZoneCTies)

int GetDetailedBeamLineGuidData(
String^ BeamLineID,
String^ SimilarFirstBeamUniqueID,
array<String^>^% LongitudinalABars,
array<String^>^% LongitudinalBBars,
array<String^>^% LongitudinalCBars,
array<String^>^% LongitudinalDBars,
array<String^>^% LongitudinalEBars,
array<String^>^% LongitudinalFBars,
array<String^>^% LongitudinalGBars,
array<String^>^% LongitudinalHBars,
array<String^>^% ZoneATies,
array<String^>^% ZoneBTies,
array<String^>^% ZoneCTies
)

abstract GetDetailedBeamLineGuidData :
    BeamLineID : string *
    SimilarFirstBeamUniqueID : string *
    LongitudinalABars : string[] byref *
    LongitudinalBBars : string[] byref *
    LongitudinalCBars : string[] byref *
    LongitudinalDBars : string[] byref *
    LongitudinalEBars : string[] byref *
    LongitudinalFBars : string[] byref *
    LongitudinalGBars : string[] byref *
    LongitudinalHBars : string[] byref *
    ZoneATies : string[] byref *
    ZoneBTies : string[] byref *
    ZoneCTies : string[] byref -> int
```

Parameters

BeamLineID
Type:Â SystemString

SimilarFirstBeamUniqueID
Type:Â SystemString

LongitudinalABars
Type:Â SystemString

LongitudinalBBars
Type:Â SystemString

LongitudinalCBars
Type:Â SystemString

LongitudinalDBars
Type:Â SystemString

LongitudinalEBars
Type:Â [SystemString](#)
LongitudinalFBars
Type:Â [SystemString](#)
LongitudinalGBars
Type:Â [SystemString](#)
LongitudinalHBars
Type:Â [SystemString](#)
ZoneATies
Type:Â [SystemString](#)
ZoneBTies
Type:Â [SystemString](#)
ZoneCTies
Type:Â [SystemString](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cDetailing Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LSTD9CAEB77_0?cpp=::1479.");GetD Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetDetailedBeamLines(
    ref int NumberItems,
    ref string[] BeamLineIDs
)
```

```
Function GetDetailedBeamLines (
    ByRef NumberItems As Integer,
    ByRef BeamLineIDs As String()
) As Integer
```

```
Dim instance As cDetailing
Dim NumberItems As Integer
Dim BeamLineIDs As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetDetailedBeamLines(NumberItems,
    BeamLineIDs)
```

```
int GetDetailedBeamLines(
    int% NumberItems,
    array<String^>^% BeamLineIDs
)
```

```
abstract GetDetailedBeamLines :
    NumberItems : int byref *
    BeamLineIDs : string[] byref -> int
```

Parameters

NumberItems

Type: [System.Int32](#)

BeamLineIDs

Type: [System.String](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDetailing Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LSTEF764B8E0?cpp=::|148.1");GetDe Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetDetailedColumnStackData(
    string ColumnStackID,
    ref string[] ObjectUniqueNames,
    ref int NumLongBarSets,
    ref int[] NumLongBars,
    ref double[] LongBarDiameter,
    ref string[] LongBarNotation,
    ref double[] LongBarStartDist,
    ref int[] LongBarStartBend,
    ref int[] LongBarEndBend,
    ref double[] LongBarLength,
    ref int[] LongBarNumLayers,
    ref int NumTieZones,
    ref string[] TieBarZones,
    ref int[] NumTieBars,
    ref int[] NumTieVertLegs,
    ref double[] TieBarDiameter,
    ref string[] TieBarNotation,
    ref double[] TieBarStartDist,
    ref double[] TieBarSpacing,
    ref int[] TieBarType
)
```

```
Function GetDetailedColumnStackData (
    ColumnStackID As String,
    ByRef ObjectUniqueNames As String(),
    ByRef NumLongBarSets As Integer,
    ByRef NumLongBars As Integer(),
    ByRef LongBarDiameter As Double(),
    ByRef LongBarNotation As String(),
    ByRef LongBarStartDist As Double(),
    ByRef LongBarStartBend As Integer(),
    ByRef LongBarEndBend As Integer(),
    ByRef LongBarLength As Double(),
    ByRef LongBarNumLayers As Integer(),
    ByRef NumTieZones As Integer,
    ByRef TieBarZones As String(),
    ByRef NumTieBars As Integer(),
    ByRef NumTieVertLegs As Integer(),
    ByRef TieBarDiameter As Double(),
    ByRef TieBarNotation As String(),
    ByRef TieBarStartDist As Double(),
    ByRef TieBarSpacing As Double(),
```

cDetailingspan id="LSTEF764B8E_0"AddLanguageSpecificTextSet("LSTEF764B8E_0?cpp=::|148.1");GetDe

Introduction

```
ByRef TieBarType As Integer()  
) As Integer
```

```
Dim instance As cDetailing  
Dim ColumnStackID As String  
Dim ObjectUniqueNames As String()  
Dim NumLongBarSets As Integer  
Dim NumLongBars As Integer()  
Dim LongBarDiameter As Double()  
Dim LongBarNotation As String()  
Dim LongBarStartDist As Double()  
Dim LongBarStartBend As Integer()  
Dim LongBarEndBend As Integer()  
Dim LongBarLength As Double()  
Dim LongBarNumLayers As Integer()  
Dim NumTieZones As Integer  
Dim TieBarZones As String()  
Dim NumTieBars As Integer()  
Dim NumTieVertLegs As Integer()  
Dim TieBarDiameter As Double()  
Dim TieBarNotation As String()  
Dim TieBarStartDist As Double()  
Dim TieBarSpacing As Double()  
Dim TieBarType As Integer()  
Dim returnValue As Integer
```

```
returnValue = instance.GetDetailedColumnStackData(ColumnStackID,  
ObjectUniqueNames, NumLongBarSets,  
NumLongBars, LongBarDiameter, LongBarNotation,  
LongBarStartDist, LongBarStartBend,  
LongBarEndBend, LongBarLength, LongBarNumLayers,  
NumTieZones, TieBarZones, NumTieBars,  
NumTieVertLegs, TieBarDiameter,  
TieBarNotation, TieBarStartDist,  
TieBarSpacing, TieBarType)
```

```
int GetDetailedColumnStackData(  
String^ ColumnStackID,  
array<String^>^% ObjectUniqueNames,  
int% NumLongBarSets,  
array<int>^% NumLongBars,  
array<double>^% LongBarDiameter,  
array<String^>^% LongBarNotation,  
array<double>^% LongBarStartDist,  
array<int>^% LongBarStartBend,  
array<int>^% LongBarEndBend,  
array<double>^% LongBarLength,  
array<int>^% LongBarNumLayers,  
int% NumTieZones,  
array<String^>^% TieBarZones,  
array<int>^% NumTieBars,  
array<int>^% NumTieVertLegs,  
array<double>^% TieBarDiameter,  
array<String^>^% TieBarNotation,  
array<double>^% TieBarStartDist,  
array<double>^% TieBarSpacing,  
array<int>^% TieBarType  
)
```

```
abstract GetDetailedColumnStackData :  
    ColumnStackID : string *  
    ObjectUniqueNames : string[] byref *
```

cDetailingspan id="LSTEF764B8E_0"AddLanguageSpecificTextSet("LSTEF764B8E_0?cpp=::|1482);GetDe

```

NumLongBarSets : int byref *
NumLongBars : int[] byref *
LongBarDiameter : float[] byref *
LongBarNotation : string[] byref *
LongBarStartDist : float[] byref *
LongBarStartBend : int[] byref *
LongBarEndBend : int[] byref *
LongBarLength : float[] byref *
LongBarNumLayers : int[] byref *
NumTieZones : int byref *
TieBarZones : string[] byref *
NumTieBars : int[] byref *
NumTieVertLegs : int[] byref *
TieBarDiameter : float[] byref *
TieBarNotation : string[] byref *
TieBarStartDist : float[] byref *
TieBarSpacing : float[] byref *
TieBarType : int[] byref -> int

```

Parameters

```

ColumnStackID
    Type:Â SystemString
ObjectUniqueNames
    Type:Â SystemString
NumLongBarSets
    Type:Â SystemInt32
NumLongBars
    Type:Â SystemInt32
LongBarDiameter
    Type:Â SystemDouble
LongBarNotation
    Type:Â SystemString
LongBarStartDist
    Type:Â SystemDouble
LongBarStartBend
    Type:Â SystemInt32
LongBarEndBend
    Type:Â SystemInt32
LongBarLength
    Type:Â SystemDouble
LongBarNumLayers
    Type:Â SystemInt32
NumTieZones
    Type:Â SystemInt32
TieBarZones
    Type:Â SystemString
NumTieBars
    Type:Â SystemInt32
NumTieVertLegs
    Type:Â SystemInt32
TieBarDiameter
    Type:Â SystemDouble
TieBarNotation

```

Type:Â SystemString
TieBarStartDist
Type:Â SystemDouble
TieBarSpacing
Type:Â SystemDouble
TieBarType
Type:Â SystemInt32

Return Value

Type:Â Int32
▲ See Also

Reference

cDetailing Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LST2ED23505_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetDetailedColumnStackData_1(
    string ColumnStackID,
    ref string[] ObjectUniqueNames,
    ref int NumLongBarSets,
    ref int[] NumLongBars,
    ref string[] LongBarGUID,
    ref double[] LongBarDiameter,
    ref string[] LongBarNotation,
    ref double[] LongBarStartDist,
    ref int[] LongBarStartBend,
    ref int[] LongBarEndBend,
    ref double[] LongBarLength,
    ref int[] LongBarNumLayers,
    ref int NumTieZones,
    ref string[] TieBarZones,
    ref int[] NumTieBars,
    ref int[] NumTieHorLegs,
    ref int[] NumTieVertLegs,
    ref string[] TieBarGUID,
    ref double[] TieBarDiameter,
    ref string[] TieBarNotation,
    ref double[] TieBarStartDist,
    ref double[] TieBarSpacing,
    ref int[] TieBarType
)
```

```
Function GetDetailedColumnStackData_1 (
    ColumnStackID As String,
    ByRef ObjectUniqueNames As String(),
    ByRef NumLongBarSets As Integer,
    ByRef NumLongBars As Integer(),
    ByRef LongBarGUID As String(),
    ByRef LongBarDiameter As Double(),
    ByRef LongBarNotation As String(),
    ByRef LongBarStartDist As Double(),
    ByRef LongBarStartBend As Integer(),
    ByRef LongBarEndBend As Integer(),
    ByRef LongBarLength As Double(),
    ByRef LongBarNumLayers As Integer(),
    ByRef NumTieZones As Integer,
    ByRef TieBarZones As String(),
    ByRef NumTieBars As Integer(),
    ByRef NumTieHorLegs As Integer(),
```

cDetailingspan id="LST2ED23505_0"AddLanguageSpecificTextSet("LST2ED23505_0?cpp=::int485;GetDe

Introduction

```
ByRef NumTieVertLegs As Integer(),  
ByRef TieBarGUID As String(),  
ByRef TieBarDiameter As Double(),  
ByRef TieBarNotation As String(),  
ByRef TieBarStartDist As Double(),  
ByRef TieBarSpacing As Double(),  
ByRef TieBarType As Integer()  
) As Integer
```

```
Dim instance As cDetailing  
Dim ColumnStackID As String  
Dim ObjectUniqueNames As String()  
Dim NumLongBarSets As Integer  
Dim NumLongBars As Integer()  
Dim LongBarGUID As String()  
Dim LongBarDiameter As Double()  
Dim LongBarNotation As String()  
Dim LongBarStartDist As Double()  
Dim LongBarStartBend As Integer()  
Dim LongBarEndBend As Integer()  
Dim LongBarLength As Double()  
Dim LongBarNumLayers As Integer()  
Dim NumTieZones As Integer  
Dim TieBarZones As String()  
Dim NumTieBars As Integer()  
Dim NumTieHorLegs As Integer()  
Dim NumTieVertLegs As Integer()  
Dim TieBarGUID As String()  
Dim TieBarDiameter As Double()  
Dim TieBarNotation As String()  
Dim TieBarStartDist As Double()  
Dim TieBarSpacing As Double()  
Dim TieBarType As Integer()  
Dim returnValue As Integer
```

```
returnValue = instance.GetDetailedColumnStackData_1(ColumnStackID,  
ObjectUniqueNames, NumLongBarSets,  
NumLongBars, LongBarGUID, LongBarDiameter,  
LongBarNotation, LongBarStartDist,  
LongBarStartBend, LongBarEndBend,  
LongBarLength, LongBarNumLayers,  
NumTieZones, TieBarZones, NumTieBars,  
NumTieHorLegs, NumTieVertLegs, TieBarGUID,  
TieBarDiameter, TieBarNotation,  
TieBarStartDist, TieBarSpacing,  
TieBarType)
```

```
int GetDetailedColumnStackData_1(  
String^ ColumnStackID,  
array<String^>^% ObjectUniqueNames,  
int% NumLongBarSets,  
array<int>^% NumLongBars,  
array<String^>^% LongBarGUID,  
array<double>^% LongBarDiameter,  
array<String^>^% LongBarNotation,  
array<double>^% LongBarStartDist,  
array<int>^% LongBarStartBend,  
array<int>^% LongBarEndBend,  
array<double>^% LongBarLength,  
array<int>^% LongBarNumLayers,  
int% NumTieZones,  
array<String^>^% TieBarZones,
```

cDetailingspan id="LST2ED23505_0"AddLanguageSpecificTextSet("LST2ED23505_0?cpp=::|nu486;GetDe

```

array<int>^% NumTieBars,
array<int>^% NumTieHorLegs,
array<int>^% NumTieVertLegs,
array<String^>^% TieBarGUID,
array<double>^% TieBarDiameter,
array<String^>^% TieBarNotation,
array<double>^% TieBarStartDist,
array<double>^% TieBarSpacing,
array<int>^% TieBarType
)

abstract GetDetailedColumnStackData_1 :
    ColumnStackID : string *
    ObjectUniqueNames : string[] byref *
    NumLongBarSets : int byref *
    NumLongBars : int[] byref *
    LongBarGUID : string[] byref *
    LongBarDiameter : float[] byref *
    LongBarNotation : string[] byref *
    LongBarStartDist : float[] byref *
    LongBarStartBend : int[] byref *
    LongBarEndBend : int[] byref *
    LongBarLength : float[] byref *
    LongBarNumLayers : int[] byref *
    NumTieZones : int byref *
    TieBarZones : string[] byref *
    NumTieBars : int[] byref *
    NumTieHorLegs : int[] byref *
    NumTieVertLegs : int[] byref *
    TieBarGUID : string[] byref *
    TieBarDiameter : float[] byref *
    TieBarNotation : string[] byref *
    TieBarStartDist : float[] byref *
    TieBarSpacing : float[] byref *
    TieBarType : int[] byref -> int

```

Parameters

ColumnStackID
Type:Â [SystemString](#)

ObjectUniqueNames
Type:Â [SystemString](#)

NumLongBarSets
Type:Â [SystemInt32](#)

NumLongBars
Type:Â [SystemInt32](#)

LongBarGUID
Type:Â [SystemString](#)

LongBarDiameter
Type:Â [SystemDouble](#)

LongBarNotation
Type:Â [SystemString](#)

LongBarStartDist
Type:Â [SystemDouble](#)

LongBarStartBend
Type:Â [SystemInt32](#)

LongBarEndBend

Type:Â SystemInt32
 LongBarLength
 Type:Â SystemDouble
 LongBarNumLayers
 Type:Â SystemInt32
 NumTieZones
 Type:Â SystemInt32
 TieBarZones
 Type:Â SystemString
 NumTieBars
 Type:Â SystemInt32
 NumTieHorLegs
 Type:Â SystemInt32
 NumTieVertLegs
 Type:Â SystemInt32
 TieBarGUID
 Type:Â SystemString
 TieBarDiameter
 Type:Â SystemDouble
 TieBarNotation
 Type:Â SystemString
 TieBarStartDist
 Type:Â SystemDouble
 TieBarSpacing
 Type:Â SystemDouble
 TieBarType
 Type:Â SystemInt32

Return Value

Type:Â Int32

▲ See Also

Reference

cDetailing Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LSTD778CC50_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetDetailedColumnStackData_2(
    string ColumnStackID,
    ref string[] ObjectUniqueNames,
    ref int NumLongBarSets,
    ref int[] NumLongBars,
    ref int[] NumLongR2Bars,
    ref int[] NumLongR3Bars,
    ref string[] LongBarGUID,
    ref double[] LongBarDiameter,
    ref string[] LongBarNotation,
    ref double[] LongBarStartDist,
    ref int[] LongBarStartBend,
    ref int[] LongBarEndBend,
    ref double[] LongBarLength,
    ref int[] LongBarNumLayers,
    ref int NumTieZones,
    ref string[] TieBarZone,
    ref int[] NumTieBars,
    ref int[] NumTieR2Legs,
    ref int[] NumTieR3Legs,
    ref string[] TieBarGUID,
    ref double[] TieBarDiameter,
    ref string[] TieBarNotation,
    ref double[] TieBarStartDist,
    ref double[] TieBarSpacing,
    ref int[] TieBarType
)
```

```
Function GetDetailedColumnStackData_2 (
    ColumnStackID As String,
    ByRef ObjectUniqueNames As String(),
    ByRef NumLongBarSets As Integer,
    ByRef NumLongBars As Integer(),
    ByRef NumLongR2Bars As Integer(),
    ByRef NumLongR3Bars As Integer(),
    ByRef LongBarGUID As String(),
    ByRef LongBarDiameter As Double(),
    ByRef LongBarNotation As String(),
    ByRef LongBarStartDist As Double(),
    ByRef LongBarStartBend As Integer(),
    ByRef LongBarEndBend As Integer(),
    ByRef LongBarLength As Double(),
    ByRef LongBarNumLayers As Integer(),
```

cDetailingspan id="LSTD778CC50_0"AddLanguageSpecificTextSet("LSTD778CC50_0?cpp=::1489");GetD

Introduction

```
ByRef NumTieZones As Integer,  
ByRef TieBarZone As String(),  
ByRef NumTieBars As Integer(),  
ByRef NumTieR2Legs As Integer(),  
ByRef NumTieR3Legs As Integer(),  
ByRef TieBarGUID As String(),  
ByRef TieBarDiameter As Double(),  
ByRef TieBarNotation As String(),  
ByRef TieBarStartDist As Double(),  
ByRef TieBarSpacing As Double(),  
ByRef TieBarType As Integer()  
) As Integer
```

```
Dim instance As cDetailing  
Dim ColumnStackID As String  
Dim ObjectUniqueNames As String()  
Dim NumLongBarSets As Integer  
Dim NumLongBars As Integer()  
Dim NumLongR2Bars As Integer()  
Dim NumLongR3Bars As Integer()  
Dim LongBarGUID As String()  
Dim LongBarDiameter As Double()  
Dim LongBarNotation As String()  
Dim LongBarStartDist As Double()  
Dim LongBarStartBend As Integer()  
Dim LongBarEndBend As Integer()  
Dim LongBarLength As Double()  
Dim LongBarNumLayers As Integer()  
Dim NumTieZones As Integer  
Dim TieBarZone As String()  
Dim NumTieBars As Integer()  
Dim NumTieR2Legs As Integer()  
Dim NumTieR3Legs As Integer()  
Dim TieBarGUID As String()  
Dim TieBarDiameter As Double()  
Dim TieBarNotation As String()  
Dim TieBarStartDist As Double()  
Dim TieBarSpacing As Double()  
Dim TieBarType As Integer()  
Dim returnValue As Integer
```

```
returnValue = instance.GetDetailedColumnStackData_2(ColumnStackID,  
ObjectUniqueNames, NumLongBarSets,  
NumLongBars, NumLongR2Bars, NumLongR3Bars,  
LongBarGUID, LongBarDiameter, LongBarNotation,  
LongBarStartDist, LongBarStartBend,  
LongBarEndBend, LongBarLength, LongBarNumLayers,  
NumTieZones, TieBarZone, NumTieBars,  
NumTieR2Legs, NumTieR3Legs, TieBarGUID,  
TieBarDiameter, TieBarNotation,  
TieBarStartDist, TieBarSpacing,  
TieBarType)
```

```
int GetDetailedColumnStackData_2(  
String^ ColumnStackID,  
array<String^>^% ObjectUniqueNames,  
int% NumLongBarSets,  
array<int>^% NumLongBars,  
array<int>^% NumLongR2Bars,  
array<int>^% NumLongR3Bars,  
array<String^>^% LongBarGUID,  
array<double>^% LongBarDiameter,
```

cDetailingspan id="LSTD778CC50_0"AddLanguageSpecificTextSet("LSTD778CC50_0?cpp=::1490");GetD

```

array<String^>^% LongBarNotation,
array<double>^% LongBarStartDist,
array<int>^% LongBarStartBend,
array<int>^% LongBarEndBend,
array<double>^% LongBarLength,
array<int>^% LongBarNumLayers,
int% NumTieZones,
array<String^>^% TieBarZone,
array<int>^% NumTieBars,
array<int>^% NumTieR2Legs,
array<int>^% NumTieR3Legs,
array<String^>^% TieBarGUID,
array<double>^% TieBarDiameter,
array<String^>^% TieBarNotation,
array<double>^% TieBarStartDist,
array<double>^% TieBarSpacing,
array<int>^% TieBarType
)

```

```

abstract GetDetailedColumnStackData_2 :
    ColumnStackID : string *
    ObjectUniqueNames : string[] byref *
    NumLongBarSets : int byref *
    NumLongBars : int[] byref *
    NumLongR2Bars : int[] byref *
    NumLongR3Bars : int[] byref *
    LongBarGUID : string[] byref *
    LongBarDiameter : float[] byref *
    LongBarNotation : string[] byref *
    LongBarStartDist : float[] byref *
    LongBarStartBend : int[] byref *
    LongBarEndBend : int[] byref *
    LongBarLength : float[] byref *
    LongBarNumLayers : int[] byref *
    NumTieZones : int byref *
    TieBarZone : string[] byref *
    NumTieBars : int[] byref *
    NumTieR2Legs : int[] byref *
    NumTieR3Legs : int[] byref *
    TieBarGUID : string[] byref *
    TieBarDiameter : float[] byref *
    TieBarNotation : string[] byref *
    TieBarStartDist : float[] byref *
    TieBarSpacing : float[] byref *
    TieBarType : int[] byref -> int

```

Parameters

ColumnStackID

Type:Â [SystemString](#)

ObjectUniqueNames

Type:Â [SystemString](#)

NumLongBarSets

Type:Â [SystemInt32](#)

NumLongBars

Type:Â [SystemInt32](#)

NumLongR2Bars

Type:Â [SystemInt32](#)

NumLongR3Bars

Type:Â SystemInt32
 LongBarGUID
 Type:Â SystemString
 LongBarDiameter
 Type:Â SystemDouble
 LongBarNotation
 Type:Â SystemString
 LongBarStartDist
 Type:Â SystemDouble
 LongBarStartBend
 Type:Â SystemInt32
 LongBarEndBend
 Type:Â SystemInt32
 LongBarLength
 Type:Â SystemDouble
 LongBarNumLayers
 Type:Â SystemInt32
 NumTieZones
 Type:Â SystemInt32
 TieBarZone
 Type:Â SystemString
 NumTieBars
 Type:Â SystemInt32
 NumTieR2Legs
 Type:Â SystemInt32
 NumTieR3Legs
 Type:Â SystemInt32
 TieBarGUID
 Type:Â SystemString
 TieBarDiameter
 Type:Â SystemDouble
 TieBarNotation
 Type:Â SystemString
 TieBarStartDist
 Type:Â SystemDouble
 TieBarSpacing
 Type:Â SystemDouble
 TieBarType
 Type:Â SystemInt32

Return Value

Type:Â Int32

▲ See Also

Reference

cDetailing Interface

ETABSv1 Namespace

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Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

C#

C++

F#

Copy

```
Function GetDetailedColumnStackGuidData (
ColumnStackID As String,
SimilarFirstColumnUniqueID As String,
ByRef LongitudinalBars As String(),
ByRef TiesA As String(),
ByRef TiesB As String(),
ByRef TiesC As String()
) As Integer
```

```
Dim instance As cDetailing
Dim ColumnStackID As String
Dim SimilarFirstColumnUniqueID As String
Dim LongitudinalBars As String()
Dim TiesA As String()
Dim TiesB As String()
Dim TiesC As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetDetailedColumnStackGuidData(ColumnStackID,
SimilarFirstColumnUniqueID, LongitudinalBars,
TiesA, TiesB, TiesC)
```

```
int GetDetailedColumnStackGuidData(
String^ ColumnStackID,
String^ SimilarFirstColumnUniqueID,
array<String^>^% LongitudinalBars,
array<String^>^% TiesA,
array<String^>^% TiesB,
array<String^>^% TiesC
)
```

```
abstract GetDetailedColumnStackGuidData :  
    ColumnStackID : string *
```

Introduction

```
SimilarFirstColumnUniqueID : string *  
LongitudinalBars : string[] byref *  
TiesA : string[] byref *  
TiesB : string[] byref *  
TiesC : string[] byref -> int
```

Parameters

ColumnStackID
Type:Â [SystemString](#)
SimilarFirstColumnUniqueID
Type:Â [SystemString](#)
LongitudinalBars
Type:Â [SystemString](#)
TiesA
Type:Â [SystemString](#)
TiesB
Type:Â [SystemString](#)
TiesC
Type:Â [SystemString](#)

Return Value

Type:Â [Int32](#)
▲ See Also

Reference

[cDetailing Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LSTD90E9E83_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetDetailedColumnStacks(
    ref int NumberItems,
    ref string[] ColumnStackIDs
)
```

```
Function GetDetailedColumnStacks (
    ByRef NumberItems As Integer,
    ByRef ColumnStackIDs As String()
) As Integer
```

```
Dim instance As cDetailing
Dim NumberItems As Integer
Dim ColumnStackIDs As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetDetailedColumnStacks(NumberItems,
    ColumnStackIDs)
```

```
int GetDetailedColumnStacks(
    int% NumberItems,
    array<String^>% ColumnStackIDs
)
```

```
abstract GetDetailedColumnStacks :
    NumberItems : int byref *
    ColumnStackIDs : string[] byref -> int
```

Parameters

NumberItems

Type: [System.Int32](#)

ColumnStackIDs

Type: [System.String](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDetailing Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LSTFE601F20_0") Method

Detailing output of one slab on a particular floor

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetDetailedSlab_OneDetailingOutputInfo(
int DetailingOutputIndex,
ref string Guid_ETABS,
ref string Floor,
ref string StoryNameETABS,
ref double LevelZ,
ref int NumberStrips
)
```

```
Function GetDetailedSlab_OneDetailingOutputInfo (
DetailingOutputIndex As Integer,
ByRef Guid_ETABS As String,
ByRef Floor As String,
ByRef StoryNameETABS As String,
ByRef LevelZ As Double,
ByRef NumberStrips As Integer
) As Integer
```

```
Dim instance As cDetailing
Dim DetailingOutputIndex As Integer
Dim Guid_ETABS As String
Dim Floor As String
Dim StoryNameETABS As String
Dim LevelZ As Double
Dim NumberStrips As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetDetailedSlab_OneDetailingOutputInfo(DetailingOutputIndex,
Guid_ETABS, Floor, StoryNameETABS,
LevelZ, NumberStrips)
```

```
int GetDetailedSlab_OneDetailingOutputInfo(
int DetailingOutputIndex,
String% Guid_ETABS,
String% Floor,
String% StoryNameETABS,
double% LevelZ,
int% NumberStrips
)
```

abstract GetDetailedSlab_OneDetailingOutputInfo :

cDetailingspan id="LSTFE601F20_0"AddLanguageSpecificTextSet("LSTFE601F20_0?cpp=::int498;GetDet

```
DetailingOutputIndex : int *  
Guid_ETABS : string byref *  
Floor : string byref *  
StoryNameETABS : string byref *  
LevelZ : float byref *  
NumberStrips : int byref -> int
```

Parameters

DetailingOutputIndex

Type:Â [SystemInt32](#)

Detailing Output Index

Guid_ETABS

Type:Â [SystemString](#)

ETABS GUID

Floor

Type:Â [SystemString](#)

Floor ID in CSiDetail, based on ETABS story Name

StoryNameETABS

Type:Â [SystemString](#)

Story Name from ETABS

LevelZ

Type:Â [SystemDouble](#)

The reference Level of the slab where Design Strips lie

NumberStrips

Type:Â [SystemInt32](#)

Number of Strips present in current detailing output

Return Value

Type:Â [Int32](#)

Returns zero if the function is successful; otherwise it returns a nonzero value.

▲ See Also

Reference

[cDetailing Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LST7D424A21_0?cpp=::|n1500;GetDe Method

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetDetailedSlabBotBarData(
    string SlabName,
    ref int NumData,
    ref string[] Names,
    ref int[] NumBars,
    ref double[] BarDiameter,
    ref string[] BarNotation,
    ref string[] BarMaterial,
    ref double[] StartX,
    ref double[] StartY,
    ref double[] StartZ,
    ref double[] EndX,
    ref double[] EndY,
    ref double[] EndZ,
    ref double[] WidthLeft,
    ref double[] WidthRight,
    ref double[] OffsetFromTop,
    ref double[] OffsetFromBot,
    ref int[] StartBarBend,
    ref int[] EndBarBend,
    ref string[] GUIDs
)
```

```
Function GetDetailedSlabBotBarData (
    SlabName As String,
    ByRef NumData As Integer,
    ByRef Names As String(),
    ByRef NumBars As Integer(),
    ByRef BarDiameter As Double(),
    ByRef BarNotation As String(),
    ByRef BarMaterial As String(),
    ByRef StartX As Double(),
    ByRef StartY As Double(),
    ByRef StartZ As Double(),
    ByRef EndX As Double(),
    ByRef EndY As Double(),
    ByRef EndZ As Double(),
    ByRef WidthLeft As Double(),
    ByRef WidthRight As Double(),
    ByRef OffsetFromTop As Double(),
    ByRef OffsetFromBot As Double(),
    ByRef StartBarBend As Integer(),
    ByRef EndBarBend As Integer(),
```

cDetailingspan id="LST7D424A21_0"AddLanguageSpecificTextSet("LST7D424A21_0?cpp=::|n1500;GetDe

Introduction

```
ByRef GUIDs As String()  
) As Integer
```

```
Dim instance As cDetailing  
Dim SlabName As String  
Dim NumData As Integer  
Dim Names As String()  
Dim NumBars As Integer()  
Dim BarDiameter As Double()  
Dim BarNotation As String()  
Dim BarMaterial As String()  
Dim StartX As Double()  
Dim StartY As Double()  
Dim StartZ As Double()  
Dim EndX As Double()  
Dim EndY As Double()  
Dim EndZ As Double()  
Dim WidthLeft As Double()  
Dim WidthRight As Double()  
Dim OffsetFromTop As Double()  
Dim OffsetFromBot As Double()  
Dim StartBarBend As Integer()  
Dim EndBarBend As Integer()  
Dim GUIDs As String()  
Dim returnValue As Integer
```

```
returnValue = instance.GetDetailedSlabBotBarData(SlabName,  
NumData, Names, NumBars, BarDiameter,  
BarNotation, BarMaterial, StartX,  
StartY, StartZ, EndX, EndY, EndZ, WidthLeft,  
WidthRight, OffsetFromTop, OffsetFromBot,  
StartBarBend, EndBarBend, GUIDs)
```

```
int GetDetailedSlabBotBarData(  
String^ SlabName,  
int% NumData,  
array<String^>^% Names,  
array<int>^% NumBars,  
array<double>^% BarDiameter,  
array<String^>^% BarNotation,  
array<String^>^% BarMaterial,  
array<double>^% StartX,  
array<double>^% StartY,  
array<double>^% StartZ,  
array<double>^% EndX,  
array<double>^% EndY,  
array<double>^% EndZ,  
array<double>^% WidthLeft,  
array<double>^% WidthRight,  
array<double>^% OffsetFromTop,  
array<double>^% OffsetFromBot,  
array<int>^% StartBarBend,  
array<int>^% EndBarBend,  
array<String^>^% GUIDs  
)
```

```
abstract GetDetailedSlabBotBarData :  
    SlabName : string *  
    NumData : int byref *  
    Names : string[] byref *  
    NumBars : int[] byref *  
    BarDiameter : float[] byref *
```

cDetailingspan id="LST7D424A21_0"AddLanguageSpecificTextSet("LST7D424A21_0?cpp=::|n150");GetDe

```

BarNotation : string[] byref *
BarMaterial : string[] byref *
StartX : float[] byref *
StartY : float[] byref *
StartZ : float[] byref *
EndX : float[] byref *
EndY : float[] byref *
EndZ : float[] byref *
WidthLeft : float[] byref *
WidthRight : float[] byref *
OffsetFromTop : float[] byref *
OffsetFromBot : float[] byref *
StartBarBend : int[] byref *
EndBarBend : int[] byref *
GUIDs : string[] byref -> int

```

Parameters

SlabName
Type:Â SystemString

NumData
Type:Â SystemInt32

Names
Type:Â SystemString

NumBars
Type:Â SystemInt32

BarDiameter
Type:Â SystemDouble

BarNotation
Type:Â SystemString

BarMaterial
Type:Â SystemString

StartX
Type:Â SystemDouble

StartY
Type:Â SystemDouble

StartZ
Type:Â SystemDouble

EndX
Type:Â SystemDouble

EndY
Type:Â SystemDouble

EndZ
Type:Â SystemDouble

WidthLeft
Type:Â SystemDouble

WidthRight
Type:Â SystemDouble

OffsetFromTop
Type:Â SystemDouble

OffsetFromBot
Type:Â SystemDouble

StartBarBend

Type:Â SystemInt32
EndBarBend
Type:Â SystemInt32
GUIDs
Type:Â SystemString

Return Value

Type:Â Int32
▲ See Also

Reference

cDetailing Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LSTE2068AE5_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetDetailedSlabBotBarData_1(
    string SlabName,
    ref int NumData,
    ref string[] Names,
    ref int[] NumBars,
    ref double[] BarDiameter,
    ref string[] BarNotation,
    ref string[] BarMaterial,
    ref double[] StartX,
    ref double[] StartY,
    ref double[] StartZ,
    ref double[] EndX,
    ref double[] EndY,
    ref double[] EndZ,
    ref double[] WidthLeft,
    ref double[] WidthRight,
    ref double[] OffsetFromTop,
    ref double[] OffsetFromBot,
    ref int[] StartBarBend,
    ref int[] EndBarBend,
    ref string[] GUIDs,
    ref string[] StripNames,
    ref int[] SpanNos
)
```

```
Function GetDetailedSlabBotBarData_1 (
    SlabName As String,
    ByRef NumData As Integer,
    ByRef Names As String(),
    ByRef NumBars As Integer(),
    ByRef BarDiameter As Double(),
    ByRef BarNotation As String(),
    ByRef BarMaterial As String(),
    ByRef StartX As Double(),
    ByRef StartY As Double(),
    ByRef StartZ As Double(),
    ByRef EndX As Double(),
    ByRef EndY As Double(),
    ByRef EndZ As Double(),
    ByRef WidthLeft As Double(),
    ByRef WidthRight As Double(),
    ByRef OffsetFromTop As Double(),
    ByRef OffsetFromBot As Double(),

```

cDetailingspan id="LSTE2068AE5_0"AddLanguageSpecificTextSet("LSTE2068AE5_0?cpp=::|1504);GetDe

Introduction

```
ByRef StartBarBend As Integer(),  
ByRef EndBarBend As Integer(),  
ByRef GUIDs As String(),  
ByRef StripNames As String(),  
ByRef SpanNos As Integer()  
) As Integer
```

```
Dim instance As cDetailing  
Dim SlabName As String  
Dim NumData As Integer  
Dim Names As String()  
Dim NumBars As Integer()  
Dim BarDiameter As Double()  
Dim BarNotation As String()  
Dim BarMaterial As String()  
Dim StartX As Double()  
Dim StartY As Double()  
Dim StartZ As Double()  
Dim EndX As Double()  
Dim EndY As Double()  
Dim EndZ As Double()  
Dim WidthLeft As Double()  
Dim WidthRight As Double()  
Dim OffsetFromTop As Double()  
Dim OffsetFromBot As Double()  
Dim StartBarBend As Integer()  
Dim EndBarBend As Integer()  
Dim GUIDs As String()  
Dim StripNames As String()  
Dim SpanNos As Integer()  
Dim returnValue As Integer
```

```
returnValue = instance.GetDetailedSlabBotBarData_1(SlabName,  
NumData, Names, NumBars, BarDiameter,  
BarNotation, BarMaterial, StartX,  
StartY, StartZ, EndX, EndY, EndZ, WidthLeft,  
WidthRight, OffsetFromTop, OffsetFromBot,  
StartBarBend, EndBarBend, GUIDs,  
StripNames, SpanNos)
```

```
int GetDetailedSlabBotBarData_1(  
String^ SlabName,  
int% NumData,  
array<String^>^% Names,  
array<int>^% NumBars,  
array<double>^% BarDiameter,  
array<String^>^% BarNotation,  
array<String^>^% BarMaterial,  
array<double>^% StartX,  
array<double>^% StartY,  
array<double>^% StartZ,  
array<double>^% EndX,  
array<double>^% EndY,  
array<double>^% EndZ,  
array<double>^% WidthLeft,  
array<double>^% WidthRight,  
array<double>^% OffsetFromTop,  
array<double>^% OffsetFromBot,  
array<int>^% StartBarBend,  
array<int>^% EndBarBend,  
array<String^>^% GUIDs,  
array<String^>^% StripNames,
```

cDetailingspan id="LSTE2068AE5_0"AddLanguageSpecificTextSet("LSTE2068AE5_0?cpp=::|1505);GetDe

```
array<int>^% SpanNos
)
```

```
abstract GetDetailedSlabBotBarData_1 :
    SlabName : string *
    NumData : int byref *
    Names : string[] byref *
    NumBars : int[] byref *
    BarDiameter : float[] byref *
    BarNotation : string[] byref *
    BarMaterial : string[] byref *
    StartX : float[] byref *
    StartY : float[] byref *
    StartZ : float[] byref *
    EndX : float[] byref *
    EndY : float[] byref *
    EndZ : float[] byref *
    WidthLeft : float[] byref *
    WidthRight : float[] byref *
    OffsetFromTop : float[] byref *
    OffsetFromBot : float[] byref *
    StartBarBend : int[] byref *
    EndBarBend : int[] byref *
    GUIDs : string[] byref *
    StripNames : string[] byref *
    SpanNos : int[] byref -> int
```

Parameters

SlabName
Type:Â SystemString

NumData
Type:Â SystemInt32

Names
Type:Â SystemString

NumBars
Type:Â SystemInt32

BarDiameter
Type:Â SystemDouble

BarNotation
Type:Â SystemString

BarMaterial
Type:Â SystemString

StartX
Type:Â SystemDouble

StartY
Type:Â SystemDouble

StartZ
Type:Â SystemDouble

EndX
Type:Â SystemDouble

EndY
Type:Â SystemDouble

EndZ
Type:Â SystemDouble

WidthLeft
Type:Â [SystemDouble](#)
WidthRight
Type:Â [SystemDouble](#)
OffsetFromTop
Type:Â [SystemDouble](#)
OffsetFromBot
Type:Â [SystemDouble](#)
StartBarBend
Type:Â [SystemInt32](#)
EndBarBend
Type:Â [SystemInt32](#)
GUIDs
Type:Â [SystemString](#)
StripNames
Type:Â [SystemString](#)
SpanNos
Type:Â [SystemInt32](#)

Return Value

Type:Â [Int32](#)

🔍 See Also

Reference

[cDetailing Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LST9C5BF1B8_0?cpp=::|1508");GetDe Method

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetDetailedSlabs(
    ref int NumberItems,
    ref string[] Names,
    ref double[] SlabElevations,
    ref string[] GUIDs
)
```

```
Function GetDetailedSlabs (
    ByRef NumberItems As Integer,
    ByRef Names As String(),
    ByRef SlabElevations As Double(),
    ByRef GUIDs As String()
) As Integer
```

```
Dim instance As cDetailing
Dim NumberItems As Integer
Dim Names As String()
Dim SlabElevations As Double()
Dim GUIDs As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetDetailedSlabs(NumberItems,
    Names, SlabElevations, GUIDs)
```

```
int GetDetailedSlabs(
    int% NumberItems,
    array<String^>% Names,
    array<double>% SlabElevations,
    array<String^>% GUIDs
)
```

```
abstract GetDetailedSlabs :
    NumberItems : int byref *
    Names : string[] byref *
    SlabElevations : float[] byref *
    GUIDs : string[] byref -> int
```

Parameters

NumberItems

Type: [System.Int32](#)

cDetailingspan id="LST9C5BF1B8_0"AddLanguageSpecificTextSet("LST9C5BF1B8_0?cpp=::|1508");GetDe

Names

Type:Â [SystemString](#)

SlabElevations

Type:Â [SystemDouble](#)

GUIDs

Type:Â [SystemString](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cDetailing Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LST2521E30B_0?cpp=::|n1510";GetDet

Method

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetDetailedSlabTopBarData(
    string SlabName,
    ref int NumData,
    ref string[] Names,
    ref int[] NumBars,
    ref double[] BarDiameter,
    ref string[] BarNotation,
    ref string[] BarMaterial,
    ref double[] StartX,
    ref double[] StartY,
    ref double[] StartZ,
    ref double[] EndX,
    ref double[] EndY,
    ref double[] EndZ,
    ref double[] WidthLeft,
    ref double[] WidthRight,
    ref double[] OffsetFromTop,
    ref double[] OffsetFromBot,
    ref int[] StartBarBend,
    ref int[] EndBarBend,
    ref string[] GUIDs
)
```

```
Function GetDetailedSlabTopBarData (
    SlabName As String,
    ByRef NumData As Integer,
    ByRef Names As String(),
    ByRef NumBars As Integer(),
    ByRef BarDiameter As Double(),
    ByRef BarNotation As String(),
    ByRef BarMaterial As String(),
    ByRef StartX As Double(),
    ByRef StartY As Double(),
    ByRef StartZ As Double(),
    ByRef EndX As Double(),
    ByRef EndY As Double(),
    ByRef EndZ As Double(),
    ByRef WidthLeft As Double(),
    ByRef WidthRight As Double(),
    ByRef OffsetFromTop As Double(),
    ByRef OffsetFromBot As Double(),
    ByRef StartBarBend As Integer(),
    ByRef EndBarBend As Integer(),
```

cDetailingspan id="LST2521E30B_0"AddLanguageSpecificTextSet("LST2521E30B_0?cpp=::|n1510";GetDet

Introduction

```
ByRef GUIDs As String()  
) As Integer
```

```
Dim instance As cDetailing  
Dim SlabName As String  
Dim NumData As Integer  
Dim Names As String()  
Dim NumBars As Integer()  
Dim BarDiameter As Double()  
Dim BarNotation As String()  
Dim BarMaterial As String()  
Dim StartX As Double()  
Dim StartY As Double()  
Dim StartZ As Double()  
Dim EndX As Double()  
Dim EndY As Double()  
Dim EndZ As Double()  
Dim WidthLeft As Double()  
Dim WidthRight As Double()  
Dim OffsetFromTop As Double()  
Dim OffsetFromBot As Double()  
Dim StartBarBend As Integer()  
Dim EndBarBend As Integer()  
Dim GUIDs As String()  
Dim returnValue As Integer
```

```
returnValue = instance.GetDetailedSlabTopBarData(SlabName,  
NumData, Names, NumBars, BarDiameter,  
BarNotation, BarMaterial, StartX,  
StartY, StartZ, EndX, EndY, EndZ, WidthLeft,  
WidthRight, OffsetFromTop, OffsetFromBot,  
StartBarBend, EndBarBend, GUIDs)
```

```
int GetDetailedSlabTopBarData(  
String^ SlabName,  
int% NumData,  
array<String^>^% Names,  
array<int>^% NumBars,  
array<double>^% BarDiameter,  
array<String^>^% BarNotation,  
array<String^>^% BarMaterial,  
array<double>^% StartX,  
array<double>^% StartY,  
array<double>^% StartZ,  
array<double>^% EndX,  
array<double>^% EndY,  
array<double>^% EndZ,  
array<double>^% WidthLeft,  
array<double>^% WidthRight,  
array<double>^% OffsetFromTop,  
array<double>^% OffsetFromBot,  
array<int>^% StartBarBend,  
array<int>^% EndBarBend,  
array<String^>^% GUIDs  
)
```

```
abstract GetDetailedSlabTopBarData :  
    SlabName : string *  
    NumData : int byref *  
    Names : string[] byref *  
    NumBars : int[] byref *  
    BarDiameter : float[] byref *
```

cDetailingspan id="LST2521E30B_0"AddLanguageSpecificTextSet("LST2521E30B_0?cpp=::|n151");GetDet

```

BarNotation : string[] byref *
BarMaterial : string[] byref *
StartX : float[] byref *
StartY : float[] byref *
StartZ : float[] byref *
EndX : float[] byref *
EndY : float[] byref *
EndZ : float[] byref *
WidthLeft : float[] byref *
WidthRight : float[] byref *
OffsetFromTop : float[] byref *
OffsetFromBot : float[] byref *
StartBarBend : int[] byref *
EndBarBend : int[] byref *
GUIDs : string[] byref -> int

```

Parameters

SlabName
Type:Â SystemString

NumData
Type:Â SystemInt32

Names
Type:Â SystemString

NumBars
Type:Â SystemInt32

BarDiameter
Type:Â SystemDouble

BarNotation
Type:Â SystemString

BarMaterial
Type:Â SystemString

StartX
Type:Â SystemDouble

StartY
Type:Â SystemDouble

StartZ
Type:Â SystemDouble

EndX
Type:Â SystemDouble

EndY
Type:Â SystemDouble

EndZ
Type:Â SystemDouble

WidthLeft
Type:Â SystemDouble

WidthRight
Type:Â SystemDouble

OffsetFromTop
Type:Â SystemDouble

OffsetFromBot
Type:Â SystemDouble

StartBarBend

Type:Â SystemInt32
EndBarBend
Type:Â SystemInt32
GUIDs
Type:Â SystemString

Return Value

Type:Â Int32

▲ See Also

Reference

cDetailing Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LSTE03C3353 Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetDetailedSlabTopBarData_1(
    string SlabName,
    ref int NumData,
    ref string[] Names,
    ref int[] NumBars,
    ref double[] BarDiameter,
    ref string[] BarNotation,
    ref string[] BarMaterial,
    ref double[] StartX,
    ref double[] StartY,
    ref double[] StartZ,
    ref double[] EndX,
    ref double[] EndY,
    ref double[] EndZ,
    ref double[] WidthLeft,
    ref double[] WidthRight,
    ref double[] OffsetFromTop,
    ref double[] OffsetFromBot,
    ref int[] StartBarBend,
    ref int[] EndBarBend,
    ref string[] GUIDs,
    ref string[] StripNames,
    ref int[] SpanNos
)
```

```
Function GetDetailedSlabTopBarData_1 (
    SlabName As String,
    ByRef NumData As Integer,
    ByRef Names As String(),
    ByRef NumBars As Integer(),
    ByRef BarDiameter As Double(),
    ByRef BarNotation As String(),
    ByRef BarMaterial As String(),
    ByRef StartX As Double(),
    ByRef StartY As Double(),
    ByRef StartZ As Double(),
    ByRef EndX As Double(),
    ByRef EndY As Double(),
    ByRef EndZ As Double(),
    ByRef WidthLeft As Double(),
    ByRef WidthRight As Double(),
    ByRef OffsetFromTop As Double(),
    ByRef OffsetFromBot As Double(),
```

cDetailingspan id="LSTE03C3353_0"AddLanguageSpecificTextSet("LSTE03C3353_0?cpp=::int1514;GetDe

Introduction

```
ByRef StartBarBend As Integer(),  
ByRef EndBarBend As Integer(),  
ByRef GUIDs As String(),  
ByRef StripNames As String(),  
ByRef SpanNos As Integer()  
 ) As Integer
```

```
Dim instance As cDetailing  
Dim SlabName As String  
Dim NumData As Integer  
Dim Names As String()  
Dim NumBars As Integer()  
Dim BarDiameter As Double()  
Dim BarNotation As String()  
Dim BarMaterial As String()  
Dim StartX As Double()  
Dim StartY As Double()  
Dim StartZ As Double()  
Dim EndX As Double()  
Dim EndY As Double()  
Dim EndZ As Double()  
Dim WidthLeft As Double()  
Dim WidthRight As Double()  
Dim OffsetFromTop As Double()  
Dim OffsetFromBot As Double()  
Dim StartBarBend As Integer()  
Dim EndBarBend As Integer()  
Dim GUIDs As String()  
Dim StripNames As String()  
Dim SpanNos As Integer()  
Dim returnValue As Integer
```

```
returnValue = instance.GetDetailedSlabTopBarData_1(SlabName,  
NumData, Names, NumBars, BarDiameter,  
BarNotation, BarMaterial, StartX,  
StartY, StartZ, EndX, EndY, EndZ, WidthLeft,  
WidthRight, OffsetFromTop, OffsetFromBot,  
StartBarBend, EndBarBend, GUIDs,  
StripNames, SpanNos)
```

```
int GetDetailedSlabTopBarData_1(  
String^ SlabName,  
int% NumData,  
array<String^>^% Names,  
array<int>^% NumBars,  
array<double>^% BarDiameter,  
array<String^>^% BarNotation,  
array<String^>^% BarMaterial,  
array<double>^% StartX,  
array<double>^% StartY,  
array<double>^% StartZ,  
array<double>^% EndX,  
array<double>^% EndY,  
array<double>^% EndZ,  
array<double>^% WidthLeft,  
array<double>^% WidthRight,  
array<double>^% OffsetFromTop,  
array<double>^% OffsetFromBot,  
array<int>^% StartBarBend,  
array<int>^% EndBarBend,  
array<String^>^% GUIDs,  
array<String^>^% StripNames,
```

cDetailingspan id="LSTE03C3353_0"AddLanguageSpecificTextSet("LSTE03C3353_0?cpp=::int15;GetDe

```
array<int>^% SpanNos
)
```

```
abstract GetDetailedSlabTopBarData_1 :
    SlabName : string *
    NumData : int byref *
    Names : string[] byref *
    NumBars : int[] byref *
    BarDiameter : float[] byref *
    BarNotation : string[] byref *
    BarMaterial : string[] byref *
    StartX : float[] byref *
    StartY : float[] byref *
    StartZ : float[] byref *
    EndX : float[] byref *
    EndY : float[] byref *
    EndZ : float[] byref *
    WidthLeft : float[] byref *
    WidthRight : float[] byref *
    OffsetFromTop : float[] byref *
    OffsetFromBot : float[] byref *
    StartBarBend : int[] byref *
    EndBarBend : int[] byref *
    GUIDs : string[] byref *
    StripNames : string[] byref *
    SpanNos : int[] byref -> int
```

Parameters

SlabName
Type:Â SystemString

NumData
Type:Â SystemInt32

Names
Type:Â SystemString

NumBars
Type:Â SystemInt32

BarDiameter
Type:Â SystemDouble

BarNotation
Type:Â SystemString

BarMaterial
Type:Â SystemString

StartX
Type:Â SystemDouble

StartY
Type:Â SystemDouble

StartZ
Type:Â SystemDouble

EndX
Type:Â SystemDouble

EndY
Type:Â SystemDouble

EndZ
Type:Â SystemDouble

WidthLeft
Type:Â [SystemDouble](#)
WidthRight
Type:Â [SystemDouble](#)
OffsetFromTop
Type:Â [SystemDouble](#)
OffsetFromBot
Type:Â [SystemDouble](#)
StartBarBend
Type:Â [SystemInt32](#)
EndBarBend
Type:Â [SystemInt32](#)
GUIDs
Type:Â [SystemString](#)
StripNames
Type:Â [SystemString](#)
SpanNos
Type:Â [SystemInt32](#)

Return Value

Type:Â [Int32](#)

🔍 See Also

Reference

[cDetailing Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LSTB96D3B2D_0?cpp=::1518") Method

Polyline(s) of the Bar shapes, in Local Design Leg Coordinates

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetDetailedWall_OnePier_OneDesignLeg_OneTieBar_OneTiePline_OnePoint(
int WallStackIndex,
int PierIndex,
int DesignLegIndex,
int TieBarIndex,
int TiePLineIndex,
int TPLinePointIndex,
ref double X,
ref double Y,
ref double Z
)
```

```
Function GetDetailedWall_OnePier_OneDesignLeg_OneTieBar_OneTiePline_OnePoint (
WallStackIndex As Integer,
PierIndex As Integer,
DesignLegIndex As Integer,
TieBarIndex As Integer,
TiePLineIndex As Integer,
TPLinePointIndex As Integer,
ByRef X As Double,
ByRef Y As Double,
ByRef Z As Double
) As Integer
```

```
Dim instance As cDetailing
Dim WallStackIndex As Integer
Dim PierIndex As Integer
Dim DesignLegIndex As Integer
Dim TieBarIndex As Integer
Dim TiePLineIndex As Integer
Dim TPLinePointIndex As Integer
Dim X As Double
Dim Y As Double
Dim Z As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetDetailedWall_OnePier_OneDesignLeg_OneTieBar_OneTiePline_OnePoint(WallStackIndex,
PierIndex, DesignLegIndex, TieBarIndex,
TiePLineIndex, TPLinePointIndex,
X, Y, Z)
```

Introduction

```
int GetDetailedWall_OnePier_OneDesignLeg_OneTieBar_OneTiePline_OnePoint(  
int WallStackIndex,  
int PierIndex,  
int DesignLegIndex,  
int TieBarIndex,  
int TiePLineIndex,  
int TPLinePointIndex,  
double% X,  
double% Y,  
double% Z  
)
```

```
abstract GetDetailedWall_OnePier_OneDesignLeg_OneTieBar_OneTiePline_OnePoint :  
    WallStackIndex : int *  
    PierIndex : int *  
    DesignLegIndex : int *  
    TieBarIndex : int *  
    TiePLineIndex : int *  
    TPLinePointIndex : int *  
    X : float byref *  
    Y : float byref *  
    Z : float byref -> int
```

Parameters

WallStackIndex

Type:Â SystemInt32
Wall Stack Index

PierIndex

Type:Â SystemInt32
Pier Output Index

DesignLegIndex

Type:Â SystemInt32
Design leg Index

TieBarIndex

Type:Â SystemInt32
Tie bar Index

TiePLineIndex

Type:Â SystemInt32
Tie Polyline Index

TPLinePointIndex

Type:Â SystemInt32
Tie Polyline Point Index

X

Type:Â SystemDouble
X coordinates

Y

Type:Â SystemDouble
Y coordinates

Z

Type:Â SystemDouble
Z coordinates

Return Value

Type: Int32

Returns zero if the function is successful; otherwise it returns a nonzero value.

▲ See Also

Reference

cDetailing Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LST805C29D_0") Method

Polyline(s) of the Bar shapes, in Local Design Leg Coordinates

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetDetailedWall_OneWallStack_OnePier_OneDesignLeg_OneTieBar_OneTiePlineInfo(
int WallStackIndex,
int PierIndex,
int DesignLegIndex,
int TieBarIndex,
int TiePlineIndex,
ref double Dia,
ref int NumberPoints,
ref double ZoneLength,
ref int LocationCode
)
```

```
Function GetDetailedWall_OneWallStack_OnePier_OneDesignLeg_OneTieBar_OneTiePlineInfo (
WallStackIndex As Integer,
PierIndex As Integer,
DesignLegIndex As Integer,
TieBarIndex As Integer,
TiePlineIndex As Integer,
ByRef Dia As Double,
ByRef NumberPoints As Integer,
ByRef ZoneLength As Double,
ByRef LocationCode As Integer
) As Integer
```

```
Dim instance As cDetailing
Dim WallStackIndex As Integer
Dim PierIndex As Integer
Dim DesignLegIndex As Integer
Dim TieBarIndex As Integer
Dim TiePlineIndex As Integer
Dim Dia As Double
Dim NumberPoints As Integer
Dim ZoneLength As Double
Dim LocationCode As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetDetailedWall_OneWallStack_OnePier_OneDesignLeg_OneTieBar_OneTiePlineInfo(
PierIndex, DesignLegIndex, TieBarIndex,
TiePlineIndex, Dia, NumberPoints,
ZoneLength, LocationCode)
```

Introduction

```
int GetDetailedWall_OneWallStack_OnePier_OneDesignLeg_OneTieBar_OneTiePlineInfo(  
int WallStackIndex,  
int PierIndex,  
int DesignLegIndex,  
int TieBarIndex,  
int TiePlineIndex,  
double% Dia,  
int% NumberPoints,  
double% ZoneLength,  
int% LocationCode  
)
```

```
abstract GetDetailedWall_OneWallStack_OnePier_OneDesignLeg_OneTieBar_OneTiePlineInfo :  
    WallStackIndex : int *  
    PierIndex : int *  
    DesignLegIndex : int *  
    TieBarIndex : int *  
    TiePlineIndex : int *  
    Dia : float byref *  
    NumberPoints : int byref *  
    ZoneLength : float byref *  
    LocationCode : int byref -> int
```

Parameters

WallStackIndex

Type:Â SystemInt32
Wall Stack Index

PierIndex

Type:Â SystemInt32
Pier Output Index

DesignLegIndex

Type:Â SystemInt32
Design leg Index

TieBarIndex

Type:Â SystemInt32
Tie bar Index

TiePlineIndex

Type:Â SystemInt32
Tie Polyline Index

Dia

Type:Â SystemDouble
Information about the size of Rebar Dia

NumberPoints

Type:Â SystemInt32
Number of 3D points defining the polyline

ZoneLength

Type:Â SystemDouble
Confinement Zone Length

LocationCode

Type:Â SystemInt32
0 - horizontal distributed, 1 - horizontal confinement zone I, 2 - horizontal
confinement zone J

Return Value

Type: Int32

Returns zero if the function is successful; otherwise it returns a nonzero value.

▲ See Also

Reference

cDetailing Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LSTDD987978_0")

Method

Represents one Rebar Tie/Stirrup set in a Design Leg, in local axis

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetDetailedWall_OneWallStack_OnePier_OneDesignLeg_OneTieBarInfo(
int WallStackIndex,
int PierIndex,
int DesignLegIndex,
int TieBarIndex,
ref string GUID,
ref double BarSize_Dia,
ref double BarSize_Area,
ref double BarSize_Fy,
ref string BarSize_Notation,
ref double Spacing,
ref double StartZ,
ref double EndZ,
ref int TieShape,
ref int NumberOfTiePlines
)
```

```
Function GetDetailedWall_OneWallStack_OnePier_OneDesignLeg_OneTieBarInfo (
WallStackIndex As Integer,
PierIndex As Integer,
DesignLegIndex As Integer,
TieBarIndex As Integer,
ByRef GUID As String,
ByRef BarSize_Dia As Double,
ByRef BarSize_Area As Double,
ByRef BarSize_Fy As Double,
ByRef BarSize_Notation As String,
ByRef Spacing As Double,
ByRef StartZ As Double,
ByRef EndZ As Double,
ByRef TieShape As Integer,
ByRef NumberOfTiePlines As Integer
) As Integer
```

```
Dim instance As cDetailing
Dim WallStackIndex As Integer
Dim PierIndex As Integer
Dim DesignLegIndex As Integer
Dim TieBarIndex As Integer
Dim GUID As String
Dim BarSize_Dia As Double
```

cDetailingspan id="LSTDD987978_0"AddLanguageSpecificTextSet("LSTDD987978_0?cpp=::|1524);GetDe

Introduction

```
Dim BarSize_Area As Double
Dim BarSize_Fy As Double
Dim BarSize_Notation As String
Dim Spacing As Double
Dim StartZ As Double
Dim EndZ As Double
Dim TieShape As Integer
Dim NumberOfTiePlines As Integer
Dim returnValue As Integer

returnValue = instance.GetDetailedWall_OneWallStack_OnePier_OneDesignLeg_OneTieBarInfo(WallStackIndex,
PierIndex, DesignLegIndex, TieBarIndex,
GUID, BarSize_Dia, BarSize_Area,
BarSize_Fy, BarSize_Notation, Spacing,
StartZ, EndZ, TieShape, NumberOfTiePlines)

int GetDetailedWall_OneWallStack_OnePier_OneDesignLeg_OneTieBarInfo(
int WallStackIndex,
int PierIndex,
int DesignLegIndex,
int TieBarIndex,
String^% GUID,
double% BarSize_Dia,
double% BarSize_Area,
double% BarSize_Fy,
String^% BarSize_Notation,
double% Spacing,
double% StartZ,
double% EndZ,
int% TieShape,
int% NumberOfTiePlines
)

abstract GetDetailedWall_OneWallStack_OnePier_OneDesignLeg_OneTieBarInfo :
    WallStackIndex : int *
    PierIndex : int *
    DesignLegIndex : int *
    TieBarIndex : int *
    GUID : string byref *
    BarSize_Dia : float byref *
    BarSize_Area : float byref *
    BarSize_Fy : float byref *
    BarSize_Notation : string byref *
    Spacing : float byref *
    StartZ : float byref *
    EndZ : float byref *
    TieShape : int byref *
    NumberOfTiePlines : int byref -> int
```

Parameters

WallStackIndex
Type: System.Int32
Wall Stack Index

PierIndex
Type: System.Int32
Pier Output Index

DesignLegIndex

Type:Â SystemInt32
 Design leg Index

TieBarIndex
 Type:Â SystemInt32
 Tie bar Index

GUID
 Type:Â SystemString
 GUID

BarSize_Dia
 Type:Â SystemDouble
 Information about the size of Rebar Dia

BarSize_Area
 Type:Â SystemDouble
 Information about the size of Rebar Area

BarSize_Fy
 Type:Â SystemDouble
 Information about the size of Rebar Fy

BarSize_Notation
 Type:Â SystemString
 Information about the size of Rebar Notation

Spacing
 Type:Â SystemDouble
 Spacing of Tie, along the height

StartZ
 Type:Â SystemDouble
 Starting offset for placing the ties, along height, with respect to Design Leg Z Level

EndZ
 Type:Â SystemDouble
 Starting offset for placing the ties, along height, with respect to Design Leg Z Level

TieShape
 Type:Â SystemInt32
 Shape of the Tie Straight, JoggleSingle, JoggleDouble, Rectangular, Diamond, UEqual, UunEqual, Circle, Spiral, RectangularLapped 'Incident #16730, Rectangle90, Rectangle135, RectLap90 'Incident #16730, RectLap135 'Incident #16730, U90, U135, UandTie90, UandTie135, LTieWithLip, Circular

NumberOfTiePlines
 Type:Â SystemInt32
 Number of Polyline(s) of the Bar shapes, in Local Design Leg Coordinates

Return Value

Type:Â Int32

Returns zero if the function is successful; otherwise it returns a nonzero value.

▲ See Also

Reference

cDetailing Interface
ETABSv1 Namespace

Introduction

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LST43C821A_0") Method

Defines one set of Vertical bars, along local X axis of the Design Leg. Each data set represents one group of bars, placed along a line, with reference to Design Leg centerline

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetDetailedWall_OneWallStack_OnePier_OneDesignLeg_OneVerticalBarInfo(
int WallStackIndex,
int PierIndex,
int DesignLegIndex,
int VerticalBarIndex,
ref string GUID,
ref double BarSizeFirst_Dia,
ref double BarSizeFirst_Area,
ref double BarSizeFirst_Fy,
ref string BarSizeFirst_Notation,
ref double BarSizeLast_Dia,
ref double BarSizeLast_Area,
ref double BarSizeLast_Fy,
ref string BarSizeLast_Notation,
ref double BarSizeOthers_Dia,
ref double BarSizeOthers_Area,
ref double BarSizeOthers_Fy,
ref string BarSizeOthers_Notation,
ref int Number,
ref double StartX,
ref double StartY,
ref double EndX,
ref double EndY,
ref int StartBarBend,
ref int EndBarBend,
ref double OffsetZ,
ref double BarLength,
ref int LocationCode
)
```

```
Function GetDetailedWall_OneWallStack_OnePier_OneDesignLeg_OneVerticalBarInfo (
WallStackIndex As Integer,
PierIndex As Integer,
DesignLegIndex As Integer,
VerticalBarIndex As Integer,
ByRef GUID As String,
ByRef BarSizeFirst_Dia As Double,
ByRef BarSizeFirst_Area As Double,
ByRef BarSizeFirst_Fy As Double,
```

Introduction

```
ByRef BarSizeFirst_Notation As String,  
ByRef BarSizeLast_Dia As Double,  
ByRef BarSizeLast_Area As Double,  
ByRef BarSizeLast_Fy As Double,  
ByRef BarSizeLast_Notation As String,  
ByRef BarSizeOthers_Dia As Double,  
ByRef BarSizeOthers_Area As Double,  
ByRef BarSizeOthers_Fy As Double,  
ByRef BarSizeOthers_Notation As String,  
ByRef Number As Integer,  
ByRef StartX As Double,  
ByRef StartY As Double,  
ByRef EndX As Double,  
ByRef EndY As Double,  
ByRef StartBarBend As Integer,  
ByRef EndBarBend As Integer,  
ByRef OffsetZ As Double,  
ByRef BarLength As Double,  
ByRef LocationCode As Integer  
) As Integer
```

```
Dim instance As cDetailing  
Dim WallStackIndex As Integer  
Dim PierIndex As Integer  
Dim DesignLegIndex As Integer  
Dim VerticalBarIndex As Integer  
Dim GUID As String  
Dim BarSizeFirst_Dia As Double  
Dim BarSizeFirst_Area As Double  
Dim BarSizeFirst_Fy As Double  
Dim BarSizeFirst_Notation As String  
Dim BarSizeLast_Dia As Double  
Dim BarSizeLast_Area As Double  
Dim BarSizeLast_Fy As Double  
Dim BarSizeLast_Notation As String  
Dim BarSizeOthers_Dia As Double  
Dim BarSizeOthers_Area As Double  
Dim BarSizeOthers_Fy As Double  
Dim BarSizeOthers_Notation As String  
Dim Number As Integer  
Dim StartX As Double  
Dim StartY As Double  
Dim EndX As Double  
Dim EndY As Double  
Dim StartBarBend As Integer  
Dim EndBarBend As Integer  
Dim OffsetZ As Double  
Dim BarLength As Double  
Dim LocationCode As Integer  
Dim returnValue As Integer
```

```
returnValue = instance.GetDetailedWall_OneWallStack_OnePier_OneDesignLeg_OneVerticalBarInfo(WallStackIndex,  
PierIndex, DesignLegIndex, VerticalBarIndex,  
GUID, BarSizeFirst_Dia, BarSizeFirst_Area,  
BarSizeFirst_Fy, BarSizeFirst_Notation,  
BarSizeLast_Dia, BarSizeLast_Area,  
BarSizeLast_Fy, BarSizeLast_Notation,  
BarSizeOthers_Dia, BarSizeOthers_Area,  
BarSizeOthers_Fy, BarSizeOthers_Notation,  
Number, StartX, StartY, EndX, EndY,  
StartBarBend, EndBarBend, OffsetZ,  
BarLength, LocationCode)
```

cDetailingAddLanguageSpecificTextSet("LST43C821A_0?cpp=::|nu=1529GetDetail

Introduction

```
int GetDetailedWall_OneWallStack_OnePier_OneDesignLeg_OneVerticalBarInfo(  
int WallStackIndex,  
int PierIndex,  
int DesignLegIndex,  
int VerticalBarIndex,  
String^% GUID,  
double% BarSizeFirst_Dia,  
double% BarSizeFirst_Area,  
double% BarSizeFirst_Fy,  
String^% BarSizeFirst_Notation,  
double% BarSizeLast_Dia,  
double% BarSizeLast_Area,  
double% BarSizeLast_Fy,  
String^% BarSizeLast_Notation,  
double% BarSizeOthers_Dia,  
double% BarSizeOthers_Area,  
double% BarSizeOthers_Fy,  
String^% BarSizeOthers_Notation,  
int% Number,  
double% StartX,  
double% StartY,  
double% EndX,  
double% EndY,  
int% StartBarBend,  
int% EndBarBend,  
double% OffsetZ,  
double% BarLength,  
int% LocationCode  
)
```

```
abstract GetDetailedWall_OneWallStack_OnePier_OneDesignLeg_OneVerticalBarInfo :  
    WallStackIndex : int *  
    PierIndex : int *  
    DesignLegIndex : int *  
    VerticalBarIndex : int *  
    GUID : string byref *  
    BarSizeFirst_Dia : float byref *  
    BarSizeFirst_Area : float byref *  
    BarSizeFirst_Fy : float byref *  
    BarSizeFirst_Notation : string byref *  
    BarSizeLast_Dia : float byref *  
    BarSizeLast_Area : float byref *  
    BarSizeLast_Fy : float byref *  
    BarSizeLast_Notation : string byref *  
    BarSizeOthers_Dia : float byref *  
    BarSizeOthers_Area : float byref *  
    BarSizeOthers_Fy : float byref *  
    BarSizeOthers_Notation : string byref *  
    Number : int byref *  
    StartX : float byref *  
    StartY : float byref *  
    EndX : float byref *  
    EndY : float byref *  
    StartBarBend : int byref *  
    EndBarBend : int byref *  
    OffsetZ : float byref *  
    BarLength : float byref *  
    LocationCode : int byref -> int
```

Parameters

WallStackIndex

Type: SystemInt32

Wall Stack Index

PierIndex

Type: SystemInt32

Pier Output Index

DesignLegIndex

Type: SystemInt32

Design leg Index

VerticalBarIndex

Type: SystemInt32

Vertical Index

GUID

Type: SystemString

GUID

BarSizeFirst_Dia

Type: SystemDouble

Information about the first bar in bar group Dia

BarSizeFirst_Area

Type: SystemDouble

Information about the first bar in bar group Area

BarSizeFirst_Fy

Type: SystemDouble

Information about the first bar in bar group Fy

BarSizeFirst_Notation

Type: SystemString

Information about the first bar in bar group notation

BarSizeLast_Dia

Type: SystemDouble

Information about the last bar in bar group Dia

BarSizeLast_Area

Type: SystemDouble

Information about the last bar in bar group Area

BarSizeLast_Fy

Type: SystemDouble

Information about the last bar in bar group Fy

BarSizeLast_Notation

Type: SystemString

Information about the last bar in bar group notation

BarSizeOthers_Dia

Type: SystemDouble

Information about the other bar in bar group Dia

BarSizeOthers_Area

Type: SystemDouble

Information about the other bar in bar group Area

BarSizeOthers_Fy

Type: SystemDouble

Information about the other bar in bar group Fy

BarSizeOthers_Notation

Type:Â SystemString
 Information about the other bar in bar group notation

Number
 Type:Â SystemInt32
 Total number of bars in the group, including First and Last

StartX
 Type:Â SystemDouble
 Start distance of bar zone, with respect to first point of Design Leg centerline

StartY
 Type:Â SystemDouble
 Offset Y from the Design Leg centerline at first bar

EndX
 Type:Â SystemDouble
 End distance of bar zone, with respect to first point of Design Leg centerline

EndY
 Type:Â SystemDouble
 Offset Y from the Design Leg centerline at last bar

StartBarBend
 Type:Â SystemInt32
 The bar bend that applies at the start of the bars, at lower end. None, Cut, Curtail, Bend45, Bend90, Bend135, Bend180, Joggle

EndBarBend
 Type:Â SystemInt32
 The bar bend that applies at the end of the bars, at upper end.

OffsetZ
 Type:Â SystemDouble
 Vertical starting offset from Design Leg Z level

BarLength
 Type:Â SystemDouble
 the length of these bars, along height of the pier, above Offset Z

LocationCode
 Type:Â SystemInt32
 : 0 - vertical distributed, 1 - vertical confinement zone I, 2 - vertical confinement zone J,

Return Value

Type:Â Int32
 Returns zero if the function is successful; otherwise it returns a nonzero value.

▲ See Also

Reference

cDetailing Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LSTC3D935E9_0?cpp=::|1533");GetDe

Method

Represents Rebars in one Design Leg, on one story as defined in ETABS

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetDetailedWall_OneWallStack_OnePier_OneDesignLegOutputInfo(
int WallStackIndex,
int PierIndex,
int DesignLegIndex,
ref string GUID,
ref string PierLabel,
ref double x1,
ref double y1,
ref double z1,
ref double x2,
ref double y2,
ref double z2,
ref double ZLevel,
ref int NumberVerticalBars,
ref int NumberHorizontalBars,
ref int TotalAreaObjects,
ref string[] AreaObjectNames
)
```

```
Function GetDetailedWall_OneWallStack_OnePier_OneDesignLegOutputInfo (
WallStackIndex As Integer,
PierIndex As Integer,
DesignLegIndex As Integer,
ByRef GUID As String,
ByRef PierLabel As String,
ByRef x1 As Double,
ByRef y1 As Double,
ByRef z1 As Double,
ByRef x2 As Double,
ByRef y2 As Double,
ByRef z2 As Double,
ByRef ZLevel As Double,
ByRef NumberVerticalBars As Integer,
ByRef NumberHorizontalBars As Integer,
ByRef TotalAreaObjects As Integer,
ByRef AreaObjectNames As String()
) As Integer
```

```
Dim instance As cDetailing
Dim WallStackIndex As Integer
Dim PierIndex As Integer
```

cDetailingspan id="LSTC3D935E9_0"AddLanguageSpecificTextSet("LSTC3D935E9_0?cpp=::|1533");GetDe

Introduction

```
Dim DesignLegIndex As Integer
Dim GUID As String
Dim PierLabel As String
Dim x1 As Double
Dim y1 As Double
Dim z1 As Double
Dim x2 As Double
Dim y2 As Double
Dim z2 As Double
Dim ZLevel As Double
Dim NumberVerticalBars As Integer
Dim NumberHorizontalBars As Integer
Dim TotalAreaObjects As Integer
Dim AreaObjectNames As String()
Dim returnValue As Integer

returnValue = instance.GetDetailedWall_OneWallStack_OnePier_OneDesignLegOutputInfo(WallStackIndex,
PierIndex, DesignLegIndex, GUID,
PierLabel, x1, y1, z1, x2, y2, z2, ZLevel,
NumberVerticalBars, NumberHorizontalBars,
TotalAreaObjects, AreaObjectNames)

int GetDetailedWall_OneWallStack_OnePier_OneDesignLegOutputInfo(
int WallStackIndex,
int PierIndex,
int DesignLegIndex,
String^% GUID,
String^% PierLabel,
double% x1,
double% y1,
double% z1,
double% x2,
double% y2,
double% z2,
double% ZLevel,
int% NumberVerticalBars,
int% NumberHorizontalBars,
int% TotalAreaObjects,
array<String^>^% AreaObjectNames
)

abstract GetDetailedWall_OneWallStack_OnePier_OneDesignLegOutputInfo :
    WallStackIndex : int *
    PierIndex : int *
    DesignLegIndex : int *
    GUID : string byref *
    PierLabel : string byref *
    x1 : float byref *
    y1 : float byref *
    z1 : float byref *
    x2 : float byref *
    y2 : float byref *
    z2 : float byref *
    ZLevel : float byref *
    NumberVerticalBars : int byref *
    NumberHorizontalBars : int byref *
    TotalAreaObjects : int byref *
    AreaObjectNames : string[] byref -> int
```

Parameters

WallStackIndex

Type:Â SystemInt32

Wall Stack Index

PierIndex

Type:Â SystemInt32

Pier Output Index

DesignLegIndex

Type:Â SystemInt32

Design leg Index

GUID

Type:Â SystemString

GUID

PierLabel

Type:Â SystemString

Label of the Piersection in ETABS from which the Design Leg is taken

x1

Type:Â SystemDouble

Global Centerline of the Design Leg. Matching with ETABS Design leg data X1

y1

Type:Â SystemDouble

Global Centerline of the Design Leg. Matching with ETABS Design leg data Y1

z1

Type:Â SystemDouble

Global Centerline of the Design Leg. Matching with ETABS Design leg data Z1

x2

Type:Â SystemDouble

Global Centerline of the Design Leg. Matching with ETABS Design leg data X2

y2

Type:Â SystemDouble

Global Centerline of the Design Leg. Matching with ETABS Design leg data Y2

z2

Type:Â SystemDouble

Global Centerline of the Design Leg. Matching with ETABS Design leg data Z2

ZLevel

Type:Â SystemDouble

The reference Z level of Design Leg, in global coordinates

NumberVerticalBars

Type:Â SystemInt32

Number of Vertical Bars in Local Coordinates of Desig Leg

NumberHorizontalBars

Type:Â SystemInt32

Number of Horizontal Bars, in local coordinates of Design Leg

TotalAreaObjects

Type:Â SystemInt32

Total number of ETABS Area Objects

AreaObjectNames

Type:Â SystemString

ETABS Area Object Name

Return Value

Type: Int32

Returns zero if the function is successful; otherwise it returns a nonzero value.

▲ See Also

Reference

cDetailing Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LSTD940F4260")

Method

Represents a Pier in the wall stack, for one story. Contains the Design Legs from one or more ETABS PierSections/WallSections

Namespace: `ETABSV1`

Assembly: `ETABSV1 (in ETABSV1.dll)` Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetDetailedWall_OneWallStack_OnePierOutputInfo(
int WallStackIndex,
int PierIndex,
ref int StoryID,
ref string ETABSStoryName,
ref int NumberDesignLegs
)
```

```
Function GetDetailedWall_OneWallStack_OnePierOutputInfo (
WallStackIndex As Integer,
PierIndex As Integer,
ByRef StoryID As Integer,
ByRef ETABSStoryName As String,
ByRef NumberDesignLegs As Integer
) As Integer
```

```
Dim instance As cDetailing
Dim WallStackIndex As Integer
Dim PierIndex As Integer
Dim StoryID As Integer
Dim ETABSStoryName As String
Dim NumberDesignLegs As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetDetailedWall_OneWallStack_OnePierOutputInfo(WallStackIndex,
PierIndex, StoryID, ETABSStoryName,
NumberDesignLegs)
```

```
int GetDetailedWall_OneWallStack_OnePierOutputInfo(
int WallStackIndex,
int PierIndex,
int% StoryID,
String^% ETABSStoryName,
int% NumberDesignLegs
)
```

```
abstract GetDetailedWall_OneWallStack_OnePierOutputInfo :
    WallStackIndex : int *
    PierIndex : int *
    StoryID : int byref *
```

cDetailingAddLanguageSpecificTextSet("LSTD940F426_0?cpp=::|nu537;GetDet

```
ETABSStoryName : string byref *  
NumberDesignLegs : int byref -> int
```

Parameters

WallStackIndex

Type: SystemInt32

Wall Stack Index

PierIndex

Type: SystemInt32

Pier Output Index

StoryID

Type: SystemInt32

CSiDetail Index of Story on which the pier lies, in the corresponding Tower

ETABSStoryName

Type: SystemString

The Story Name used in ETABS on which the pier lies, in the corresponding Tower

NumberDesignLegs

Type: SystemInt32

Number of Design Legs

Return Value

Type: Int32

Returns zero if the function is successful; otherwise it returns a nonzero value.

▲ See Also

Reference

cDetailing Interface

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LST156B1F22_0")

Method

Represents Rebars in one Spandrel on a particular story , as defined in ETABS

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetDetailedWall_OneWallStack_OneSpandrel_OneLongBarInfo(
int WallStackIndex,
int SpandrelIndex,
int LongBarIndex,
ref double BarSize_Dia,
ref double BarSize_Area,
ref double BarSize_Fy,
ref string BarSize_Notation,
ref int NumberPoints,
ref double[] X,
ref double[] Y,
ref double[] Z,
ref int StartType,
ref int EndType
)
```

```
Function GetDetailedWall_OneWallStack_OneSpandrel_OneLongBarInfo (
WallStackIndex As Integer,
SpandrelIndex As Integer,
LongBarIndex As Integer,
ByRef BarSize_Dia As Double,
ByRef BarSize_Area As Double,
ByRef BarSize_Fy As Double,
ByRef BarSize_Notation As String,
ByRef NumberPoints As Integer,
ByRef X As Double(),
ByRef Y As Double(),
ByRef Z As Double(),
ByRef StartType As Integer,
ByRef EndType As Integer
) As Integer
```

```
Dim instance As cDetailing
Dim WallStackIndex As Integer
Dim SpandrelIndex As Integer
Dim LongBarIndex As Integer
Dim BarSize_Dia As Double
Dim BarSize_Area As Double
Dim BarSize_Fy As Double
Dim BarSize_Notation As String
Dim NumberPoints As Integer
```

cDetailingspan id="LST156B1F22_0"AddLanguageSpecificTextSet("LST156B1F22_0?cpp=::|n1539;GetDet

Introduction

```
Dim X As Double()
Dim Y As Double()
Dim Z As Double()
Dim StartType As Integer
Dim EndType As Integer
Dim returnValue As Integer

returnValue = instance.GetDetailedWall_OneWallStack_OneSpandrel_OneLongBarInfo(WallStackIndex,
SpandrelIndex, LongBarIndex, BarSize_Dia,
BarSize_Area, BarSize_Fy, BarSize_Notation,
NumberPoints, X, Y, Z, StartType, EndType)

int GetDetailedWall_OneWallStack_OneSpandrel_OneLongBarInfo(
int WallStackIndex,
int SpandrelIndex,
int LongBarIndex,
double% BarSize_Dia,
double% BarSize_Area,
double% BarSize_Fy,
String^% BarSize_Notation,
int% NumberPoints,
array<double>^% X,
array<double>^% Y,
array<double>^% Z,
int% StartType,
int% EndType
)

abstract GetDetailedWall_OneWallStack_OneSpandrel_OneLongBarInfo :
    WallStackIndex : int *
    SpandrelIndex : int *
    LongBarIndex : int *
    BarSize_Dia : float byref *
    BarSize_Area : float byref *
    BarSize_Fy : float byref *
    BarSize_Notation : string byref *
    NumberPoints : int byref *
    X : float[] byref *
    Y : float[] byref *
    Z : float[] byref *
    StartType : int byref *
    EndType : int byref -> int
```

Parameters

WallStackIndex

Type: SystemInt32

Wall Stack Index

SpandrelIndex

Type: SystemInt32

Spandrel Output Index

LongBarIndex

Type: SystemInt32

Long bar index

BarSize_Dia

Type: SystemDouble

Information about the bar In bar group Dia

BarSize_Area

Type:Â SystemDouble
Information about the bar In bar group Area

BarSize_Fy
Type:Â SystemDouble
Information about the bar In bar group Fy

BarSize_Notation
Type:Â SystemString
Information about the bar In bar group Notation

NumberPoints
Type:Â SystemInt32
Number of Points

X
Type:Â SystemDouble
Coordinates of bar in local beam axis as defined in ETABS (X is along 1, Y is along 2, Z is along 3)

Y
Type:Â SystemDouble
Coordinates of bar in local beam axis as defined in ETABS (X is along 1, Y is along 2, Z is along 3)

Z
Type:Â SystemDouble
Coordinates of bar in local beam axis as defined in ETABS (X is along 1, Y is along 2, Z is along 3)

StartType
Type:Â SystemInt32
Represents Bar End Types, Cut, Ben90, Bend180, Bend135

EndType
Type:Â SystemInt32
Represents Bar End Types, Cut, Ben90, Bend180, Bend135

Return Value

Type:Â Int32
Returns zero if the function is successful; otherwise it returns a nonzero value.
▲ See Also

Reference

cDetailing Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LST8BAD85CF_0") Method

Represents Rebars in one Spandrel on a particular story , as defined in ETABS

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetDetailedWall_OneWallStack_OneSpandrel_OneStirrupsInfo(
int WallStackIndex,
int SpandrelIndex,
int StirrupsIndex,
ref double BarSize_Dia,
ref double BarSize_Area,
ref double BarSize_Fy,
ref string BarSize_Notation,
ref double X1,
ref double X2,
ref double Spacing,
ref int NumberLegs
)
```

```
Function GetDetailedWall_OneWallStack_OneSpandrel_OneStirrupsInfo (
WallStackIndex As Integer,
SpandrelIndex As Integer,
StirrupsIndex As Integer,
ByRef BarSize_Dia As Double,
ByRef BarSize_Area As Double,
ByRef BarSize_Fy As Double,
ByRef BarSize_Notation As String,
ByRef X1 As Double,
ByRef X2 As Double,
ByRef Spacing As Double,
ByRef NumberLegs As Integer
) As Integer
```

```
Dim instance As cDetailing
Dim WallStackIndex As Integer
Dim SpandrelIndex As Integer
Dim StirrupsIndex As Integer
Dim BarSize_Dia As Double
Dim BarSize_Area As Double
Dim BarSize_Fy As Double
Dim BarSize_Notation As String
Dim X1 As Double
Dim X2 As Double
Dim Spacing As Double
Dim NumberLegs As Integer
Dim returnValue As Integer
```

cDetailingspan id="LST8BAD85CF_0"AddLanguageSpecificTextSet("LST8BAD85CF_0?cpp=::1542");GetD

Introduction

```
returnValue = instance.GetDetailedWall_OneWallStack_OneSpandrel_OneStirrupsInfo(WallStackIndex,  
SpandrelIndex, StirrupsIndex, BarSize_Dia,  
BarSize_Area, BarSize_Fy, BarSize_Notation,  
X1, X2, Spacing, NumberLegs)
```

```
int GetDetailedWall_OneWallStack_OneSpandrel_OneStirrupsInfo(  
int WallStackIndex,  
int SpandrelIndex,  
int StirrupsIndex,  
double% BarSize_Dia,  
double% BarSize_Area,  
double% BarSize_Fy,  
String^% BarSize_Notation,  
double% X1,  
double% X2,  
double% Spacing,  
int% NumberLegs  
)
```

```
abstract GetDetailedWall_OneWallStack_OneSpandrel_OneStirrupsInfo :  
    WallStackIndex : int *  
    SpandrelIndex : int *  
    StirrupsIndex : int *  
    BarSize_Dia : float byref *  
    BarSize_Area : float byref *  
    BarSize_Fy : float byref *  
    BarSize_Notation : string byref *  
    X1 : float byref *  
    X2 : float byref *  
    Spacing : float byref *  
    NumberLegs : int byref -> int
```

Parameters

WallStackIndex

Type: SystemInt32
Wall Stack Index

SpandrelIndex

Type: SystemInt32
Spandrel Output Index

StirrupsIndex

Type: SystemInt32
Long bar index

BarSize_Dia

Type: SystemDouble
Information about the bar In bar group Dia

BarSize_Area

Type: SystemDouble
Information about the bar In bar group Area

BarSize_Fy

Type: SystemDouble
Information about the bar In bar group Fy

BarSize_Notation

Type: SystemString
Information about the bar In bar group Notation

X1

Type: SystemDouble

Coordinates of bar start and End, in local beam axis as defined in ETABS

X2

Type: SystemDouble

Coordinates of bar start and End, in local beam axis as defined in ETABS

Spacing

Type: SystemDouble

Spacing

NumberLegs

Type: SystemInt32

Number of legs

Return Value

Type: Int32

Returns zero if the function is successful; otherwise it returns a nonzero value.

■ See Also

Reference

cDetailing Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LST9F308BBD_0?cpp=::|1545");GetDe

Method

Represents Rebars in one Spandrel on a particular story , as defined in ETABS

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetDetailedWall_OneWallStack_OneSpandrelOutputInfo(
int WallStackIndex,
int SpandrelIndex,
ref string GUID,
ref string Name,
ref double Height,
ref double Width,
ref double Thickness,
ref double CoverLongBar,
ref double CoverStirrups,
ref double x1,
ref double y1,
ref double z1,
ref double x2,
ref double y2,
ref double z2,
ref int NumberLongBars,
ref int NumberStirrups,
ref int TotalAreaObjects,
ref string[] AreaObjectNames
)
```

```
Function GetDetailedWall_OneWallStack_OneSpandrelOutputInfo (
WallStackIndex As Integer,
SpandrelIndex As Integer,
ByRef GUID As String,
ByRef Name As String,
ByRef Height As Double,
ByRef Width As Double,
ByRef Thickness As Double,
ByRef CoverLongBar As Double,
ByRef CoverStirrups As Double,
ByRef x1 As Double,
ByRef y1 As Double,
ByRef z1 As Double,
ByRef x2 As Double,
ByRef y2 As Double,
ByRef z2 As Double,
ByRef NumberLongBars As Integer,
ByRef NumberStirrups As Integer,
ByRef TotalAreaObjects As Integer,
```

cDetailingspan id="LST9F308BBD_0"AddLanguageSpecificTextSet("LST9F308BBD_0?cpp=::|1545");GetDe

Introduction

```
ByRef AreaObjectNames As String()  
) As Integer
```

```
Dim instance As cDetailing  
Dim WallStackIndex As Integer  
Dim SpandrelIndex As Integer  
Dim GUID As String  
Dim Name As String  
Dim Height As Double  
Dim Width As Double  
Dim Thickness As Double  
Dim CoverLongBar As Double  
Dim CoverStirrups As Double  
Dim x1 As Double  
Dim y1 As Double  
Dim z1 As Double  
Dim x2 As Double  
Dim y2 As Double  
Dim z2 As Double  
Dim NumberLongBars As Integer  
Dim NumberStirrups As Integer  
Dim TotalAreaObjects As Integer  
Dim AreaObjectNames As String()  
Dim returnValue As Integer
```

```
returnValue = instance.GetDetailedWall_OneWallStack_OneSpandrelOutputInfo(WallStackIndex,  
SpandrelIndex, GUID, Name, Height,  
Width, Thickness, CoverLongBar, CoverStirrups,  
x1, y1, z1, x2, y2, z2, NumberLongBars,  
NumberStirrups, TotalAreaObjects,  
AreaObjectNames)
```

```
int GetDetailedWall_OneWallStack_OneSpandrelOutputInfo(  
int WallStackIndex,  
int SpandrelIndex,  
String^% GUID,  
String^% Name,  
double% Height,  
double% Width,  
double% Thickness,  
double% CoverLongBar,  
double% CoverStirrups,  
double% x1,  
double% y1,  
double% z1,  
double% x2,  
double% y2,  
double% z2,  
int% NumberLongBars,  
int% NumberStirrups,  
int% TotalAreaObjects,  
array<String^>^% AreaObjectNames  
)
```

```
abstract GetDetailedWall_OneWallStack_OneSpandrelOutputInfo :  
    WallStackIndex : int *  
    SpandrelIndex : int *  
    GUID : string byref *  
    Name : string byref *  
    Height : float byref *  
    Width : float byref *  
    Thickness : float byref *
```

cDetailingspan id="LST9F308BBD_0"AddLanguageSpecificTextSet("LST9F308BBD_0?cpp=::|1546");GetDe

```

CoverLongBar : float byref *
CoverStirrups : float byref *
x1 : float byref *
y1 : float byref *
z1 : float byref *
x2 : float byref *
y2 : float byref *
z2 : float byref *
NumberLongBars : int byref *
NumberStirrups : int byref *
TotalAreaObjects : int byref *
AreaObjectName : string[] byref -> int

```

Parameters

WallStackIndex

Type: System.Int32

Wall Stack Index

SpandrelIndex

Type: System.Int32

Spandrel Output Index

GUID

Type: System.String

GUID

Name

Type: System.String

Name of Spandrel

Height

Type: System.Double

Spandrel Height

Width

Type: System.Double

Spandrel Width

Thickness

Type: System.Double

Spandrel Thickness

CoverLongBar

Type: System.Double

Cover for LongBar

CoverStirrups

Type: System.Double

Cover for Stirrup

x1

Type: System.Double

Global Centerline of Spandrel. Matching with ETABS Design leg data X1

y1

Type: System.Double

Global Centerline of Spandrel. Matching with ETABS Design leg data Y1

z1

Type: System.Double

Global Centerline of Spandrel. Matching with ETABS Design leg data Z1

x2

Type:Â SystemDouble
Global Centerline of Spandrel. Matching with ETABS Design leg data X2

y2
Type:Â SystemDouble
Global Centerline of Spandrel. Matching with ETABS Design leg data Y2

z2
Type:Â SystemDouble
Global Centerline of Spandrel. Matching with ETABS Design leg data Z2

NumberLongBars
Type:Â SystemInt32
Contains all Longitudinal Rebars in a beam

NumberStirrups
Type:Â SystemInt32
Represents one Zone of Stirrups with same size and spacing

TotalAreaObjects
Type:Â SystemInt32
Total number of ETABS Area Objects

AreaObjectNames
Type:Â SystemString
ETABS Area Object Name

Return Value

Type:Â Int32
Returns zero if the function is successful; otherwise it returns a nonzero value.
▲ See Also

Reference

cDetailing Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LSTA8050D9F_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

bool GetDetailingAvailable()

Function GetDetailingAvailable As Boolean

Dim instance As cDetailing

Dim returnValue As Boolean

returnValue = instance.GetDetailingAvailable()

bool GetDetailingAvailable()

abstract GetDetailingAvailable : unit -> bool

Return Value

Type: [Boolean](#)

▲ See Also

Reference

[cDetailing Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LSTC0C5E8C Method

Retrieves number of Detailed outputs for Slab

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNumberDetailedSlabs(
    ref int NumberDetailingOutput
)
```

```
Function GetNumberDetailedSlabs (
    ByRef NumberDetailingOutput As Integer
) As Integer
```

```
Dim instance As cDetailing
Dim NumberDetailingOutput As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetNumberDetailedSlabs(NumberDetailingOutput)
```

```
int GetNumberDetailedSlabs(
    int% NumberDetailingOutput
)
```

```
abstract GetNumberDetailedSlabs :
    NumberDetailingOutput : int byref -> int
```

Parameters

NumberDetailingOutput

Type: [System.Int32](#)

Number of detailed outputs present in model. One output for each story.

Return Value

Type: [Int32](#)

Returns zero if the function is successful; otherwise it returns a nonzero value.

▲ See Also

Reference

[cDetailing Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LST502AD5E8_0?cpp=::|1552);GetNu Method

Retrieves number of Detailed outputs for Wall

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNumberDetailedWallStacks(  
    ref int NumberWallStacks  
)
```

```
Function GetNumberDetailedWallStacks (  
    ByRef NumberWallStacks As Integer  
) As Integer
```

```
Dim instance As cDetailing  
Dim NumberWallStacks As Integer  
Dim returnValue As Integer
```

```
returnValue = instance.GetNumberDetailedWallStacks(NumberWallStacks)
```

```
int GetNumberDetailedWallStacks(  
    int% NumberWallStacks  
)
```

```
abstract GetNumberDetailedWallStacks :  
    NumberWallStacks : int byref -> int
```

Parameters

NumberWallStacks

Type: [System.Int32](#)

Number of wall stacks in model.

Return Value

Type: [Int32](#)

Returns zero if the function is successful; otherwise it returns a nonzero value.

▲ See Also

Reference

[cDetailing Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LSTFC6E9F18_0?cpp=::|1554);GetOn

Method

Represents one Bar in extent - BOTTOM BAR1

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneBottomRebar_Bar1Info(
int DetailingOutputIndex,
int StripIndex,
int DetailingRegionIndex,
int BottomRebarIndex,
ref string GUID,
ref string PlaceCode,
ref double Dia,
ref string Size,
ref int Number,
ref double StartDist,
ref double EndDist,
ref int StartBend,
ref int EndBend,
ref string Material
)
```

```
Function GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneBottomRebar_Bar1Info(
DetailingOutputIndex As Integer,
StripIndex As Integer,
DetailingRegionIndex As Integer,
BottomRebarIndex As Integer,
ByRef GUID As String,
ByRef PlaceCode As String,
ByRef Dia As Double,
ByRef Size As String,
ByRef Number As Integer,
ByRef StartDist As Double,
ByRef EndDist As Double,
ByRef StartBend As Integer,
ByRef EndBend As Integer,
ByRef Material As String
) As Integer
```

```
Dim instance As cDetailing
Dim DetailingOutputIndex As Integer
Dim StripIndex As Integer
Dim DetailingRegionIndex As Integer
Dim BottomRebarIndex As Integer
Dim GUID As String
Dim PlaceCode As String
```

cDetailingspan id="LSTFC6E9F18_0"AddLanguageSpecificTextSet("LSTFC6E9F18_0?cpp=::|1554);GetOn

Introduction

```
Dim Dia As Double
Dim Size As String
Dim Number As Integer
Dim StartDist As Double
Dim EndDist As Double
Dim StartBend As Integer
Dim EndBend As Integer
Dim Material As String
Dim returnValue As Integer
```

```
returnValue = instance.GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneBottomRebar_Bar1Info(
StripIndex, DetailingRegionIndex,
BottomRebarIndex, GUID, PlaceCode,
Dia, Size, Number, StartDist, EndDist,
StartBend, EndBend, Material)
```

```
int GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneBottomRebar_Bar1Info(
int DetailingOutputIndex,
int StripIndex,
int DetailingRegionIndex,
int BottomRebarIndex,
String^% GUID,
String^% PlaceCode,
double% Dia,
String^% Size,
int% Number,
double% StartDist,
double% EndDist,
int% StartBend,
int% EndBend,
String^% Material
)
```

```
abstract GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneBottomRebar_Bar1Info
    DetailingOutputIndex : int *
    StripIndex : int *
    DetailingRegionIndex : int *
    BottomRebarIndex : int *
    GUID : string byref *
    PlaceCode : string byref *
    Dia : float byref *
    Size : string byref *
    Number : int byref *
    StartDist : float byref *
    EndDist : float byref *
    StartBend : int byref *
    EndBend : int byref *
    Material : string byref -> int
```

Parameters

DetailingOutputIndex
Type: System.Int32
Detailing Output Index

StripIndex
Type: System.Int32
Strip Index

DetailingRegionIndex

Type: SystemInt32
 Detailing region index

BottomRebarIndex
 Type: SystemInt32
 Bottom rebar index

GUID
 Type: SystemString
 GUID assigned by CSiDetail

PlaceCode
 Type: SystemString
 Place Code. A = Top Long Bar in Exterior Span. B = Top Short Bar in Exterior Span. C = Top Long Bar in Interior Span. D = Top Short Bar in Interior Span. E = Bottom Long Bar in Interior/Exterior Span. F = Bottom Short Bar in Interior/Exterior Span

Dia
 Type: SystemDouble
 Diameter of Bar

Size
 Type: SystemString
 Bar Designation, based on selected Rebar Set

Number
 Type: SystemInt32
 Total number of Bars of this type

StartDist
 Type: SystemDouble
 Start Location of the Bar, along X in local axis of region

EndDist
 Type: SystemDouble
 End Location of the Bar, along X in local axis of region

StartBend
 Type: SystemInt32
 Bend type at start of bar

EndBend
 Type: SystemInt32
 Bend type at end of of bar

Material
 Type: SystemString
 Material assigned to Slab

Return Value

Type: Int32
 Returns zero if the function is successful; otherwise it returns a nonzero value.

▲ See Also

Reference

cDetailing Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LST85ADB4D1_0?cpp=::1558");GetO

Method

Represents one Bar in extent - BOTTOM BAR2

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneBottomRebar_Bar2Info(
int DetailingOutputIndex,
int StripIndex,
int DetailingRegionIndex,
int BottomRebarIndex,
ref string GUID,
ref string PlaceCode,
ref double Dia,
ref string Size,
ref int Number,
ref double StartDist,
ref double EndDist,
ref int StartBend,
ref int EndBend,
ref string Material
)
```

```
Function GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneBottomRebar_Bar2Info(
DetailingOutputIndex As Integer,
StripIndex As Integer,
DetailingRegionIndex As Integer,
BottomRebarIndex As Integer,
ByRef GUID As String,
ByRef PlaceCode As String,
ByRef Dia As Double,
ByRef Size As String,
ByRef Number As Integer,
ByRef StartDist As Double,
ByRef EndDist As Double,
ByRef StartBend As Integer,
ByRef EndBend As Integer,
ByRef Material As String
) As Integer
```

```
Dim instance As cDetailing
Dim DetailingOutputIndex As Integer
Dim StripIndex As Integer
Dim DetailingRegionIndex As Integer
Dim BottomRebarIndex As Integer
Dim GUID As String
Dim PlaceCode As String
```

cDetailingspan id="LST85ADB4D1_0"AddLanguageSpecificTextSet("LST85ADB4D1_0?cpp=::1558");GetO

Introduction

```
Dim Dia As Double
Dim Size As String
Dim Number As Integer
Dim StartDist As Double
Dim EndDist As Double
Dim StartBend As Integer
Dim EndBend As Integer
Dim Material As String
Dim returnValue As Integer
```

```
returnValue = instance.GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneBottomRebar_Bar2Info(
StripIndex, DetailingRegionIndex,
BottomRebarIndex, GUID, PlaceCode,
Dia, Size, Number, StartDist, EndDist,
StartBend, EndBend, Material)
```

```
int GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneBottomRebar_Bar2Info(
int DetailingOutputIndex,
int StripIndex,
int DetailingRegionIndex,
int BottomRebarIndex,
String^% GUID,
String^% PlaceCode,
double% Dia,
String^% Size,
int% Number,
double% StartDist,
double% EndDist,
int% StartBend,
int% EndBend,
String^% Material
)
```

```
abstract GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneBottomRebar_Bar2Info
    DetailingOutputIndex : int *
    StripIndex : int *
    DetailingRegionIndex : int *
    BottomRebarIndex : int *
    GUID : string byref *
    PlaceCode : string byref *
    Dia : float byref *
    Size : string byref *
    Number : int byref *
    StartDist : float byref *
    EndDist : float byref *
    StartBend : int byref *
    EndBend : int byref *
    Material : string byref -> int
```

Parameters

DetailingOutputIndex
Type: System.Int32
Detailing Output Index

StripIndex
Type: System.Int32
Strip Index

DetailingRegionIndex

Type:Â SystemInt32
 Detailing region index
 BottomRebarIndex
 Type:Â SystemInt32
 Bottom rebar index
 GUID
 Type:Â SystemString
 GUID assigned by CSiDetail
 PlaceCode
 Type:Â SystemString
 Place Code. A = Top Long Bar in Exterior Span. B = Top Short Bar in Exterior Span. C = Top Long Bar in Interior Span. D = Top Short Bar in Interior Span. E = Bottom Long Bar in Interior/Exterior Span. F = Bottom Short Bar in Interior/Exterior Span
 Dia
 Type:Â SystemDouble
 Diameter of Bar
 Size
 Type:Â SystemString
 Bar Designation, based on selected Rebar Set
 Number
 Type:Â SystemInt32
 Total number of Bars of this type
 StartDist
 Type:Â SystemDouble
 Start Location of the Bar, along X in local axis of region
 EndDist
 Type:Â SystemDouble
 End Location of the Bar, along X in local axis of region
 StartBend
 Type:Â SystemInt32
 Bend type at start of bar
 EndBend
 Type:Â SystemInt32
 Bend type at end of of bar
 Material
 Type:Â SystemString
 Material assigned to Slab

Return Value

Type:Â Int32
 Returns zero if the function is successful; otherwise it returns a nonzero value.

▲ See Also

Reference

cDetailing Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LST3CC36B8D_0?cpp=::1562");GetO

Method

Represents one Bar in extent - BOTTOM BARS

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneBottomRebarInfo(
int DetailingOutputIndex,
int StripIndex,
int DetailingRegionIndex,
int BottomRebarIndex,
ref double X1,
ref double Y1,
ref double X2,
ref double Y2,
ref double WidthRight,
ref double WidthLeft,
ref double Z,
ref double ReqAst,
ref double ProvAst
)
```

```
Function GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneBottomRebarInfo (
DetailingOutputIndex As Integer,
StripIndex As Integer,
DetailingRegionIndex As Integer,
BottomRebarIndex As Integer,
ByRef X1 As Double,
ByRef Y1 As Double,
ByRef X2 As Double,
ByRef Y2 As Double,
ByRef WidthRight As Double,
ByRef WidthLeft As Double,
ByRef Z As Double,
ByRef ReqAst As Double,
ByRef ProvAst As Double
) As Integer
```

```
Dim instance As cDetailing
Dim DetailingOutputIndex As Integer
Dim StripIndex As Integer
Dim DetailingRegionIndex As Integer
Dim BottomRebarIndex As Integer
Dim X1 As Double
Dim Y1 As Double
Dim X2 As Double
Dim Y2 As Double
```

cDetailingspan id="LST3CC36B8D_0"AddLanguageSpecificTextSet("LST3CC36B8D_0?cpp=::1562");GetO

Introduction

```
Dim WidthRight As Double
Dim WidthLeft As Double
Dim Z As Double
Dim ReqAst As Double
Dim ProvAst As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneBottomRebarInfo(
StripIndex, DetailingRegionIndex,
BottomRebarIndex, X1, Y1, X2, Y2, WidthRight,
WidthLeft, Z, ReqAst, ProvAst)
```

```
int GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneBottomRebarInfo(
int DetailingOutputIndex,
int StripIndex,
int DetailingRegionIndex,
int BottomRebarIndex,
double% X1,
double% Y1,
double% X2,
double% Y2,
double% WidthRight,
double% WidthLeft,
double% Z,
double% ReqAst,
double% ProvAst
)
```

```
abstract GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneBottomRebarInfo :
    DetailingOutputIndex : int *
    StripIndex : int *
    DetailingRegionIndex : int *
    BottomRebarIndex : int *
    X1 : float byref *
    Y1 : float byref *
    X2 : float byref *
    Y2 : float byref *
    WidthRight : float byref *
    WidthLeft : float byref *
    Z : float byref *
    ReqAst : float byref *
    ProvAst : float byref -> int
```

Parameters

DetailingOutputIndex

Type: System.Int32

Detailing Output Index

StripIndex

Type: System.Int32

Strip Index

DetailingRegionIndex

Type: System.Int32

Detailing region index

BottomRebarIndex

Type: System.Int32

Top rebar index

X1

Type: SystemDouble
Start point X of Extent CLine, Local wrt to Region

Y1
Type: SystemDouble
Start point y of Extent CLine, Local wrt to Region

X2
Type: SystemDouble
End point X of Extent CLine, Local wrt to Region

Y2
Type: SystemDouble
End point Y of Extent CLine, Local wrt to Region

WidthRight
Type: SystemDouble
Bar distribution Width, on right side of the extent centerline

WidthLeft
Type: SystemDouble
Bar distribution Width, on left side of the extent centerline

Z
Type: SystemDouble
Z value in Global Coordinates, of centerline of all rebar in the extent

ReqAst
Type: SystemDouble
Required area of reinforcement, as determined from ETABS

ProvAst
Type: SystemDouble
Provided area of reinforcement, as detailed

Return Value

Type: Int32
Returns zero if the function is successful; otherwise it returns a nonzero value.

See Also

Reference

cDetailing Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LST87F46454_0") Method

Represents one Bar in extent - TOP BAR1

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneTopRebar_Bar1Info(
int DetailingOutputIndex,
int StripIndex,
int DetailingRegionIndex,
int TopRebarIndex,
ref string GUID,
ref string PlaceCode,
ref double Dia,
ref string Size,
ref int Number,
ref double StartDist,
ref double EndDist,
ref int StartBend,
ref int EndBend,
ref string Material
)
```

```
Function GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneTopRebar_Bar1Info (
DetailingOutputIndex As Integer,
StripIndex As Integer,
DetailingRegionIndex As Integer,
TopRebarIndex As Integer,
ByRef GUID As String,
ByRef PlaceCode As String,
ByRef Dia As Double,
ByRef Size As String,
ByRef Number As Integer,
ByRef StartDist As Double,
ByRef EndDist As Double,
ByRef StartBend As Integer,
ByRef EndBend As Integer,
ByRef Material As String
) As Integer
```

```
Dim instance As cDetailing
Dim DetailingOutputIndex As Integer
Dim StripIndex As Integer
Dim DetailingRegionIndex As Integer
Dim TopRebarIndex As Integer
Dim GUID As String
Dim PlaceCode As String
```

cDetailingspan id="LST87F46454_0"AddLanguageSpecificTextSet("LST87F46454_0?cpp=::|nu1565GetOne

Introduction

```
Dim Dia As Double
Dim Size As String
Dim Number As Integer
Dim StartDist As Double
Dim EndDist As Double
Dim StartBend As Integer
Dim EndBend As Integer
Dim Material As String
Dim returnValue As Integer
```

```
returnValue = instance.GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneTopRebar_
StripIndex, DetailingRegionIndex,
TopRebarIndex, GUID, PlaceCode, Dia,
Size, Number, StartDist, EndDist,
StartBend, EndBend, Material)
```

```
int GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneTopRebar_Bar1Info(
int DetailingOutputIndex,
int StripIndex,
int DetailingRegionIndex,
int TopRebarIndex,
String^% GUID,
String^% PlaceCode,
double% Dia,
String^% Size,
int% Number,
double% StartDist,
double% EndDist,
int% StartBend,
int% EndBend,
String^% Material
)
```

```
abstract GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneTopRebar_Bar1Info :
    DetailingOutputIndex : int *
    StripIndex : int *
    DetailingRegionIndex : int *
    TopRebarIndex : int *
    GUID : string byref *
    PlaceCode : string byref *
    Dia : float byref *
    Size : string byref *
    Number : int byref *
    StartDist : float byref *
    EndDist : float byref *
    StartBend : int byref *
    EndBend : int byref *
    Material : string byref -> int
```

Parameters

DetailingOutputIndex
Type:Â SystemInt32
Detailing Output Index

StripIndex
Type:Â SystemInt32
Strip Index

DetailingRegionIndex

Type: SystemInt32
 Detailing region index
 TopRebarIndex
 Type: SystemInt32
 Top rebar index
 GUID
 Type: SystemString
 GUID assigned by CSiDetail
 PlaceCode
 Type: SystemString
 Place Code. A = Top Long Bar in Exterior Span. B = Top Short Bar in Exterior Span. C = Top Long Bar in Interior Span. D = Top Short Bar in Interior Span. E = Bottom Long Bar in Interior/Exterior Span. F = Bottom Short Bar in Interior/Exterior Span
 Dia
 Type: SystemDouble
 Diameter of Bar
 Size
 Type: SystemString
 Bar Designation, based on selected Rebar Set
 Number
 Type: SystemInt32
 Total number of Bars of this type
 StartDist
 Type: SystemDouble
 Start Location of the Bar, along X in local axis of region
 EndDist
 Type: SystemDouble
 End Location of the Bar, along X in local axis of region
 StartBend
 Type: SystemInt32
 Bend type at start of bar
 EndBend
 Type: SystemInt32
 Bend type at end of of bar
 Material
 Type: SystemString
 Material assigned to Slab

Return Value

Type: Int32
 Returns zero if the function is successful; otherwise it returns a nonzero value.

▲ See Also

Reference

cDetailing Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LST1C59DDC Method

Represents one Bar in extent - TOP BAR2

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneTopRebar_Bar2Info(
int DetailingOutputIndex,
int StripIndex,
int DetailingRegionIndex,
int TopRebarIndex,
ref string GUID,
ref string PlaceCode,
ref double Dia,
ref string Size,
ref int Number,
ref double StartDist,
ref double EndDist,
ref int StartBend,
ref int EndBend,
ref string Material
)
```

```
Function GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneTopRebar_Bar2Info (
DetailingOutputIndex As Integer,
StripIndex As Integer,
DetailingRegionIndex As Integer,
TopRebarIndex As Integer,
ByRef GUID As String,
ByRef PlaceCode As String,
ByRef Dia As Double,
ByRef Size As String,
ByRef Number As Integer,
ByRef StartDist As Double,
ByRef EndDist As Double,
ByRef StartBend As Integer,
ByRef EndBend As Integer,
ByRef Material As String
) As Integer
```

```
Dim instance As cDetailing
Dim DetailingOutputIndex As Integer
Dim StripIndex As Integer
Dim DetailingRegionIndex As Integer
Dim TopRebarIndex As Integer
Dim GUID As String
Dim PlaceCode As String
```

cDetailingspan id="LST1C59DDCE_0"AddLanguageSpecificTextSet("LST1C59DDCE_0?cpp=:1569.");GetC

Introduction

```
Dim Dia As Double
Dim Size As String
Dim Number As Integer
Dim StartDist As Double
Dim EndDist As Double
Dim StartBend As Integer
Dim EndBend As Integer
Dim Material As String
Dim returnValue As Integer
```

```
returnValue = instance.GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneTopRebar_Bar2Info(
StripIndex, DetailingRegionIndex,
TopRebarIndex, GUID, PlaceCode, Dia,
Size, Number, StartDist, EndDist,
StartBend, EndBend, Material)
```

```
int GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneTopRebar_Bar2Info(
int DetailingOutputIndex,
int StripIndex,
int DetailingRegionIndex,
int TopRebarIndex,
String^% GUID,
String^% PlaceCode,
double% Dia,
String^% Size,
int% Number,
double% StartDist,
double% EndDist,
int% StartBend,
int% EndBend,
String^% Material
)
```

```
abstract GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneTopRebar_Bar2Info :
    DetailingOutputIndex : int *
    StripIndex : int *
    DetailingRegionIndex : int *
    TopRebarIndex : int *
    GUID : string byref *
    PlaceCode : string byref *
    Dia : float byref *
    Size : string byref *
    Number : int byref *
    StartDist : float byref *
    EndDist : float byref *
    StartBend : int byref *
    EndBend : int byref *
    Material : string byref -> int
```

Parameters

DetailingOutputIndex
Type:Â SystemInt32
Detailing Output Index

StripIndex
Type:Â SystemInt32
Strip Index

DetailingRegionIndex

Type:Â SystemInt32
 Detailing region index
 TopRebarIndex
 Type:Â SystemInt32
 Top rebar index
 GUID
 Type:Â SystemString
 GUID assigned by CSiDetail
 PlaceCode
 Type:Â SystemString
 Place Code. A = Top Long Bar in Exterior Span. B = Top Short Bar in Exterior Span. C = Top Long Bar in Interior Span. D = Top Short Bar in Interior Span. E = Bottom Long Bar in Interior/Exterior Span. F = Bottom Short Bar in Interior/Exterior Span
 Dia
 Type:Â SystemDouble
 Diameter of Bar
 Size
 Type:Â SystemString
 Bar Designation, based on selected Rebar Set
 Number
 Type:Â SystemInt32
 Total number of Bars of this type
 StartDist
 Type:Â SystemDouble
 Start Location of the Bar, along X in local axis of region
 EndDist
 Type:Â SystemDouble
 End Location of the Bar, along X in local axis of region
 StartBend
 Type:Â SystemInt32
 Bend type at start of bar
 EndBend
 Type:Â SystemInt32
 Bend type at end of of bar
 Material
 Type:Â SystemString
 Material assigned to Slab

Return Value

Type:Â Int32
 Returns zero if the function is successful; otherwise it returns a nonzero value.

▲ See Also

Reference

cDetailing Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LSTA94619B6_0")

Method

Represents one Bar in extent - TOP BARS

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneTopRebarInfo(
int DetailingOutputIndex,
int StripIndex,
int DetailingRegionIndex,
int TopRebarIndex,
ref double X1,
ref double Y1,
ref double X2,
ref double Y2,
ref double WidthRight,
ref double WidthLeft,
ref double Z,
ref double ReqAst,
ref double ProvAst
)
```

```
Function GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneTopRebarInfo (
DetailingOutputIndex As Integer,
StripIndex As Integer,
DetailingRegionIndex As Integer,
TopRebarIndex As Integer,
ByRef X1 As Double,
ByRef Y1 As Double,
ByRef X2 As Double,
ByRef Y2 As Double,
ByRef WidthRight As Double,
ByRef WidthLeft As Double,
ByRef Z As Double,
ByRef ReqAst As Double,
ByRef ProvAst As Double
) As Integer
```

```
Dim instance As cDetailing
Dim DetailingOutputIndex As Integer
Dim StripIndex As Integer
Dim DetailingRegionIndex As Integer
Dim TopRebarIndex As Integer
Dim X1 As Double
Dim Y1 As Double
Dim X2 As Double
Dim Y2 As Double
```

cDetailingspan id="LSTA94619B6_0"AddLanguageSpecificTextSet("LSTA94619B6_0?cpp=::|nu1573;GetOn

Introduction

```
Dim WidthRight As Double
Dim WidthLeft As Double
Dim Z As Double
Dim ReqAst As Double
Dim ProvAst As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneTopRebarInfo(
StripIndex, DetailingRegionIndex,
TopRebarIndex, X1, Y1, X2, Y2, WidthRight,
WidthLeft, Z, ReqAst, ProvAst)
```

```
int GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneTopRebarInfo(
int DetailingOutputIndex,
int StripIndex,
int DetailingRegionIndex,
int TopRebarIndex,
double% X1,
double% Y1,
double% X2,
double% Y2,
double% WidthRight,
double% WidthLeft,
double% Z,
double% ReqAst,
double% ProvAst
)
```

```
abstract GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegion_OneTopRebarInfo :
    DetailingOutputIndex : int *
    StripIndex : int *
    DetailingRegionIndex : int *
    TopRebarIndex : int *
    X1 : float byref *
    Y1 : float byref *
    X2 : float byref *
    Y2 : float byref *
    WidthRight : float byref *
    WidthLeft : float byref *
    Z : float byref *
    ReqAst : float byref *
    ProvAst : float byref -> int
```

Parameters

DetailingOutputIndex

Type:Â SystemInt32

Detailing Output Index

StripIndex

Type:Â SystemInt32

Strip Index

DetailingRegionIndex

Type:Â SystemInt32

Detailing region index

TopRebarIndex

Type:Â SystemInt32

Top rebar index

X1

Type: SystemDouble
Start point X of Extent CLine, Local wrt to Region

Y1
Type: SystemDouble
Start point y of Extent CLine, Local wrt to Region

X2
Type: SystemDouble
End point X of Extent CLine, Local wrt to Region

Y2
Type: SystemDouble
End point Y of Extent CLine, Local wrt to Region

WidthRight
Type: SystemDouble
Bar distribution Width, on right side of the extent centerline

WidthLeft
Type: SystemDouble
Bar distribution Width, on left side of the extent centerline

Z
Type: SystemDouble
Z value in Global Coordinates, of centerline of all rebar in the extent

ReqAst
Type: SystemDouble
Required area of reinforcement, as determined from ETABS

ProvAst
Type: SystemDouble
Provided area of reinforcement, as detailed

Return Value

Type: Int32
Returns zero if the function is successful; otherwise it returns a nonzero value.

🔍 See Also

Reference

cDetailing Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LST16D21C1A_0?cpp=::|1576");GetOn

Method

Represents one region of Design Strip, based on strip segment and slab geometry

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegionInfo(
int DetailingOutputIndex,
int StripIndex,
int DetailingRegionIndex,
ref double X1,
ref double Y1,
ref double X2,
ref double Y2,
ref int NumberTopBars,
ref int NumberBottomBars
)
```

```
Function GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegionInfo (
DetailingOutputIndex As Integer,
StripIndex As Integer,
DetailingRegionIndex As Integer,
ByRef X1 As Double,
ByRef Y1 As Double,
ByRef X2 As Double,
ByRef Y2 As Double,
ByRef NumberTopBars As Integer,
ByRef NumberBottomBars As Integer
) As Integer
```

```
Dim instance As cDetailing
Dim DetailingOutputIndex As Integer
Dim StripIndex As Integer
Dim DetailingRegionIndex As Integer
Dim X1 As Double
Dim Y1 As Double
Dim X2 As Double
Dim Y2 As Double
Dim NumberTopBars As Integer
Dim NumberBottomBars As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegionInfo(Deta
StripIndex, DetailingRegionIndex,
X1, Y1, X2, Y2, NumberTopBars, NumberBottomBars)
```

```
int GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegionInfo(
```

```
cDetailingspan id="LST16D21C1A_0"AddLanguageSpecificTextSet("LST16D21C1A_0?cpp=::|1576");GetOn
```

```

int DetailingOutputIndex,
int StripIndex,
int DetailingRegionIndex,
double% X1,
double% Y1,
double% X2,
double% Y2,
int% NumberTopBars,
int% NumberBottomBars
)

```

```

abstract GetOneDetailedSlab_OneDetailingOutput_OneStrip_OneDetailingRegionInfo :
    DetailingOutputIndex : int *
    StripIndex : int *
    DetailingRegionIndex : int *
    X1 : float byref *
    Y1 : float byref *
    X2 : float byref *
    Y2 : float byref *
    NumberTopBars : int byref *
    NumberBottomBars : int byref -> int

```

Parameters

DetailingOutputIndex

Type:Â SystemInt32

Detailing Output Index

StripIndex

Type:Â SystemInt32

Strip Index

DetailingRegionIndex

Type:Â SystemInt32

Detailing region index

X1

Type:Â SystemDouble

X1 for Global location Centerline of the region

Y1

Type:Â SystemDouble

Y1 for Global location Centerline of the region

X2

Type:Â SystemDouble

X2 for Global location Centerline of the region

Y2

Type:Â SystemDouble

Y2 for Global location Centerline of the region

NumberTopBars

Type:Â SystemInt32

Top Rebar Extents in local centerline of the region

NumberBottomBars

Type:Â SystemInt32

Bottom Rebar Extents in local centerline of the region

Return Value

Type: Int32

Returns zero if the function is successful; otherwise it returns a nonzero value.

▲ See Also

Reference

cDetailing Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LST20BD842C_0?cpp=::|1579");GetOn

Method

Represents the detailed rebars in one rebar strip of the slab, as defined in ETABS

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetOneDetailedSlab_OneDetailingOutput_StripGUID(
int DetailingOutputIndex,
int StripIndex,
ref string GUID_ETABS
)
```

```
Function GetOneDetailedSlab_OneDetailingOutput_StripGUID (
DetailingOutputIndex As Integer,
StripIndex As Integer,
ByRef GUID_ETABS As String
) As Integer
```

```
Dim instance As cDetailing
Dim DetailingOutputIndex As Integer
Dim StripIndex As Integer
Dim GUID_ETABS As String
Dim returnValue As Integer
```

```
returnValue = instance.GetOneDetailedSlab_OneDetailingOutput_StripGUID(DetailingOutputIndex,
StripIndex, GUID_ETABS)
```

```
int GetOneDetailedSlab_OneDetailingOutput_StripGUID(
int DetailingOutputIndex,
int StripIndex,
String^% GUID_ETABS
)
```

```
abstract GetOneDetailedSlab_OneDetailingOutput_StripGUID :
    DetailingOutputIndex : int *
    StripIndex : int *
    GUID_ETABS : string byref -> int
```

Parameters

DetailingOutputIndex

Type: [System.Int32](#)

Detailing Output Index

StripIndex

Type:Â SystemInt32
Strip Index
GUID_ETABS
Type:Â SystemString
ETABS GUID

Return Value

Type:Â Int32

Returns zero if the function is successful; otherwise it returns a nonzero value.

▲ See Also

Reference

cDetailing Interface

ETABSv1 Namespace

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»

CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LST795D92D7_0") Method

Represents the detailed rebars in one rebar strip of the slab, as defined in ETABS

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetOneDetailedSlab_OneDetailingOutput_StripInfo(
int DetailingOutputIndex,
int StripIndex,
ref string Name,
ref string LayerName,
ref string StripType,
ref int NumberRegions
)
```

```
Function GetOneDetailedSlab_OneDetailingOutput_StripInfo (
DetailingOutputIndex As Integer,
StripIndex As Integer,
ByRef Name As String,
ByRef LayerName As String,
ByRef StripType As String,
ByRef NumberRegions As Integer
) As Integer
```

```
Dim instance As cDetailing
Dim DetailingOutputIndex As Integer
Dim StripIndex As Integer
Dim Name As String
Dim LayerName As String
Dim StripType As String
Dim NumberRegions As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetOneDetailedSlab_OneDetailingOutput_StripInfo(DetailingOutputIndex,
StripIndex, Name, LayerName, StripType,
NumberRegions)
```

```
int GetOneDetailedSlab_OneDetailingOutput_StripInfo(
int DetailingOutputIndex,
int StripIndex,
String^% Name,
String^% LayerName,
String^% StripType,
int% NumberRegions
)
```

```
abstract GetOneDetailedSlab_OneDetailingOutput_StripInfo :
```

cDetailingspan id="LST795D92D7_0"AddLanguageSpecificTextSet("LST795D92D7_0?cpp=::|nls=1");GetOn

```
DetailingOutputIndex : int *  
StripIndex : int *  
Name : string byref *  
LayerName : string byref *  
StripType : string byref *  
NumberRegions : int byref -> int
```

Parameters

DetailingOutputIndex

Type: SystemInt32

Detailing Output Index

StripIndex

Type: SystemInt32

Strip Index

Name

Type: SystemString

Strip Name, as defined in ETABS

LayerName

Type: SystemString

Layer Name, as defined in ETABS

StripType

Type: SystemString

Type of Design Strip, as defined in ETABS - Column, Middle, None

NumberRegions

Type: SystemInt32

Regions of Rebars in the Design Strip. Generated by CSiDetail based on strip geometry and slab geometry

Return Value

Type: Int32

Returns zero if the function is successful; otherwise it returns a nonzero value.

▲ See Also

Reference

cDetailing Interface

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LSTD0FA445E_0?cpp=::|1583");GetSi Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetSimilarBeamLines(
    string BeamLineID,
    ref int NumberSimilarBeams,
    ref int[] NumberUniqueObjects,
    ref string[] ObjectUniqueNames
)
```

```
Function GetSimilarBeamLines (
    BeamLineID As String,
    ByRef NumberSimilarBeams As Integer,
    ByRef NumberUniqueObjects As Integer(),
    ByRef ObjectUniqueNames As String()
) As Integer
```

```
Dim instance As cDetailing
Dim BeamLineID As String
Dim NumberSimilarBeams As Integer
Dim NumberUniqueObjects As Integer()
Dim ObjectUniqueNames As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetSimilarBeamLines(BeamLineID,
    NumberSimilarBeams, NumberUniqueObjects,
    ObjectUniqueNames)
```

```
int GetSimilarBeamLines(
    String^ BeamLineID,
    int% NumberSimilarBeams,
    array<int>^% NumberUniqueObjects,
    array<String>^% ObjectUniqueNames
)
```

```
abstract GetSimilarBeamLines :
    BeamLineID : string *
    NumberSimilarBeams : int byref *
    NumberUniqueObjects : int[] byref *
    ObjectUniqueNames : string[] byref -> int
```

Parameters

BeamLineID

Type: SystemString

NumberSimilarBeams

Type: SystemInt32

NumberUniqueObjects

Type: SystemInt32

ObjectUniqueNames

Type: SystemString

Return Value

Type: Int32

▲ See Also

Reference

cDetailing Interface

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LST982AAD9A_0?cpp=::1585");GetSi

Method

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetSimilarColumnStacks(
    string ColumnStackID,
    ref int NumberSimilarColumns,
    ref int[] NumberUniqueObjects,
    ref string[] ObjectUniqueNames
)
```

```
Function GetSimilarColumnStacks (
    ColumnStackID As String,
    ByRef NumberSimilarColumns As Integer,
    ByRef NumberUniqueObjects As Integer(),
    ByRef ObjectUniqueNames As String()
) As Integer
```

```
Dim instance As cDetailing
Dim ColumnStackID As String
Dim NumberSimilarColumns As Integer
Dim NumberUniqueObjects As Integer()
Dim ObjectUniqueNames As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetSimilarColumnStacks(ColumnStackID,
    NumberSimilarColumns, NumberUniqueObjects,
    ObjectUniqueNames)
```

```
int GetSimilarColumnStacks(
    String^ ColumnStackID,
    int% NumberSimilarColumns,
    array<int>^% NumberUniqueObjects,
    array<String>^% ObjectUniqueNames
)
```

```
abstract GetSimilarColumnStacks :
    ColumnStackID : string *
    NumberSimilarColumns : int byref *
    NumberUniqueObjects : int[] byref *
    ObjectUniqueNames : string[] byref -> int
```

Parameters

ColumnStackID

Type: SystemString

NumberSimilarColumns

Type: SystemInt32

NumberUniqueObjects

Type: SystemInt32

ObjectUniqueNames

Type: SystemString

Return Value

Type: Int32

▲ See Also

Reference

cDetailing Interface

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LSTF61A23E7_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetSimilarSlabs(
    string SlabName,
    ref int NumberSimilarSlabs,
    ref string[] Names
)
```

```
Function GetSimilarSlabs (
    SlabName As String,
    ByRef NumberSimilarSlabs As Integer,
    ByRef Names As String()
) As Integer
```

```
Dim instance As cDetailing
Dim SlabName As String
Dim NumberSimilarSlabs As Integer
Dim Names As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetSimilarSlabs(SlabName,
    NumberSimilarSlabs, Names)
```

```
int GetSimilarSlabs(
    String^ SlabName,
    int% NumberSimilarSlabs,
    array<String^>^% Names
)
```

```
abstract GetSimilarSlabs :
    SlabName : string *
    NumberSimilarSlabs : int byref *
    Names : string[] byref -> int
```

Parameters

SlabName

Type: SystemString

NumberSimilarSlabs

Type: SystemInt32

Names

Type: SystemString

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cDetailing Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDetailingAddLanguageSpecificTextSet("LST89CD2325_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int StartDetailing(
    bool OverwriteExisting
)
```

```
Function StartDetailing (
    OverwriteExisting As Boolean
) As Integer
```

```
Dim instance As cDetailing
Dim OverwriteExisting As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.StartDetailing(OverwriteExisting)
```

```
int StartDetailing(
    bool OverwriteExisting
)
```

```
abstract StartDetailing :
    OverwriteExisting : bool -> int
```

Parameters

OverwriteExisting
Type: [System.Boolean](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cDetailing Interface](#)

[ETABSv1 Namespace](#)

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Introduction

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CSI API ETABS v1

cDiaphragm Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDiaphragm
```

```
Public Interface cDiaphragm
```

```
Dim instance As cDiaphragm
```

```
public interface class cDiaphragm
```

```
type cDiaphragm = interface end
```

The cDiaphragm type exposes the following members.

▲ Methods

▲	Name	Description
➤	ChangeName	Changes the name of a defined diaphragm
➤	Delete	Deletes the specified diaphragm
➤	GetDiaphragm	Retrieves the specified diaphragm
➤	GetNameList	Retrieves the names of all defined diaphragms
➤	SetDiaphragm	Adds a new diaphragm, or modifies an existing diaphragm

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDiaphragm Methods

The cDiaphragm type exposes the following members.

Methods

⌕	Name	Description
⇒	<u>ChangeName</u>	Changes the name of a defined diaphragm
⇒	<u>Delete</u>	Deletes the specified diaphragm
⇒	<u>GetDiaphragm</u>	Retrieves the specified diaphragm
⇒	<u>GetNameList</u>	Retrieves the names of all defined diaphragms
⇒	<u>SetDiaphragm</u>	Adds a new diaphragm, or modifies an existing diaphragm

Top

See Also

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cDiaphragm Interface

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CSI API ETABS v1

cDiaphragmAddLanguageSpecificTextSet("LSTBA408D Method

Changes the name of a defined diaphragm

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int ChangeName(
    string Name,
    string NewName
)
```

```
Function ChangeName (
    Name As String,
    NewName As String
) As Integer
```

```
Dim instance As cDiaphragm
Dim Name As String
Dim NewName As String
Dim returnValue As Integer
```

```
returnValue = instance.ChangeName(Name,
    NewName)
```

```
int ChangeName(
    String^ Name,
    String^ NewName
)
```

```
abstract ChangeName :
    Name : string *
    NewName : string -> int
```

Parameters

Name

Type: SystemString

The existing name of a defined diaphragm

NewName

Type: SystemString

The new name for the diaphragm

Return Value

Type: Int32

Returns zero if the new name is successfully applied, otherwise it returns a nonzero value.

▀ **Remarks**

▀ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new diaphragm
    ret = SapModel.Diaphragm.SetDiaphragm("MyDiaphragm1A", SemiRigid:=True)

    'change diaphragm name
    ret = SapModel.Diaphragm.ChangeName("MyDiaphragm1A", "MyDiaph2B")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▀ **See Also**

Reference

cDiaphragm Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDiaphragmAddLanguageSpecificTextSet("LSTEF1690B2_0?cpp=:1505.");Delete Method

Deletes the specified diaphragm

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int Delete(
    string Name
)
```

```
Function Delete (
    Name As String
) As Integer
```

```
Dim instance As cDiaphragm
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.Delete(Name)
```

```
int Delete(
    String^ Name
)
```

```
abstract Delete :
    Name : string -> int
```

Parameters

Name

Type: SystemString

The name of an existing diaphragm

Return Value

Type: Int32

Returns zero if the diaphragm is successfully deleted, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
```

cDiaphragmid="LSTEF1690B2_0"AddLanguageSpecificTextSet("LSTEF1690B2_0?cpp=:1505.");Delete

Introduction

```
Dim SapModel As cSapModel
Dim EtabsObject As cOAPI
Dim ret As Integer = -1

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'define new diaphragm
ret = SapModel.Diaphragm.SetDiaphragm("MyDiaphragm1A", SemiRigid:=True)

'change diaphragm name
ret = SapModel.Diaphragm.Delete("MyDiaphragm1A")

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

🔗 See Also

Reference

[cDiaphragm Interface](#)

[ETABSv1 Namespace](#)

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Namespace: ETABSv1

Syntax

VBF#

```
int GetDiaphragm(
string Name,
ref bool SemiRigid
)
```

```
abstract GetDiaphragm :
  Name : string *
  SemiRigid : bool byref -> int
```

Type:Â SystemString

SemiRigid

Type: SystemBoolean

cDiaphragmspan id="LST58E31ED9_0"AddLanguageSpecificTextSet("LST58E31ED9_0?cpp=1597=");Get

Return ValueType: Int32

Returns zero if the diaphragm is successfully retrieved, otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim SemiRigid As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 4, 4, 24, 24)

    'define new diaphragm
    ret = SapModel.Diaphragm.SetDiaphragm("MyDiaphragm1A", SemiRigid:=True)

    'get diaphragm
    ret = SapModel.Diaphragm.GetDiaphragm("MyDiaphragm1A", SemiRigid)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ **See Also**

Reference

cDiaphragm Interface

ETABSv1 Namespace

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»

CSI API ETABS v1

cDiaphragmAddLanguageSpecificTextSet("LSTE8E16E Method

Retrieves the names of all defined diaphragms

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetNameList(
    ref int NumberNames,
    ref string[] MyName
)
```

```
Function GetNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cDiaphragm
Dim NumberNames As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetNameList(NumberNames,
    MyName)
```

```
int GetNameList(
    int% NumberNames,
    array<String^>^% MyName
)
```

```
abstract GetNameList :
    NumberNames : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberNames

Type: SystemInt32

The number of diaphragm names defined in the program

MyName

Type: SystemString

This is a one-dimensional array of names. The MyName array is created as a dynamic, zero-based, array by the API user:

```
Dim MyName() as String
```

Introduction

The array is dimensioned to (NumberNames + 1) inside the ETABS program, filled with the names, and returned to the API user.

Return Value

Type: Int32

Returns zero if the names are successfully retrieved, otherwise it returns nonzero.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim NumberNames As Integer  
    Dim MyName As String()  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)  
  
    'define new diaphragm  
    ret = SapModel.Diaphragm.SetDiaphragm("MyDiaphragm1A", SemiRigid:=True)  
  
    'get names  
    ret = SapModel.Diaphragm.GetNameList(NumberNames, MyName)  
  
    'close ETABS  
    EtabsObject.ApplicationExit(False)  
  
    'clean up variables  
    SapModel = Nothing  
    EtabsObject = Nothing  
  
End Sub
```

▲ **See Also**

Reference

cDiaphragm Interface

ETABSv1 Namespace

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»

CSI API ETABS v1

cDiaphragmAddLanguageSpecificTextSet("LST7B14B877_0?cpp=:1602.");SetD

Method

Adds a new diaphragm, or modifies an existing diaphragm

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetDiaphragm(
    string Name,
    bool SemiRigid
)
```

```
Function SetDiaphragm (
    Name As String,
    SemiRigid As Boolean
) As Integer
```

```
Dim instance As cDiaphragm
Dim Name As String
Dim SemiRigid As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetDiaphragm(Name,
    SemiRigid)
```

```
int SetDiaphragm(
    String^ Name,
    bool SemiRigid
)
```

```
abstract SetDiaphragm :
    Name : string *
    SemiRigid : bool -> int
```

Parameters

Name

Type: [SystemString](#)

This is the name of a diaphragm. If this is the name of an existing diaphragm, that diaphragm is modified, otherwise a new diaphragm is added.

SemiRigid

Type: [SystemBoolean](#)

This item is True if the diaphragm is to be semi-rigid, and False otherwise

Return Value

Type: Int32

Returns zero if the diaphragm is successfully added or modified, otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberNames As Integer
    Dim MyName As String()

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new diaphragm
    ret = SapModel.Diaphragm.SetDiaphragm("MyDiaphragm1A", SemiRigid:=True)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ **See Also**

Reference

cDiaphragm Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDStAISC_ASD89 Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDStAISC_ASD89
```

```
Public Interface cDStAISC_ASD89
```

```
Dim instance As cDStAISC_ASD89
```

```
public interface class cDStAISC_ASD89
```

```
type cDStAISC_ASD89 = interface end
```

The cDStAISC_ASD89 type exposes the following members.

▲ Methods

▲	Name	Description
---	------	-------------

➤	GetOverwrite	
---	------------------------------	--

➤	GetPreference	
---	-------------------------------	--

➤	SetOverwrite	
---	------------------------------	--

➤	SetPreference	
---	-------------------------------	--

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDStAISC_ASD89 Methods

The cDStAISC_ASD89 type exposes the following members.

Methods

	Name	Description
⇒	<u>GetOverwrite</u>	
⇒	<u>GetPreference</u>	
⇒	<u>SetOverwrite</u>	
⇒	<u>SetPreference</u>	

Top

See Also

Reference

cDStAISC_ASD89 Interface

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CSI API ETABS v1

cDStAISC_ASD89AddLanguageSpecificTextSet("LST89 Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDStAISC_ASD89
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```

Parameters

Name

Type: SystemString

cDStAISC_ASD89AddLanguageSpecificTextSet("LST89E66247_0?1006::|nu=."

Item

Type: [SystemInt32](#)

Value

Type: [SystemDouble](#)

ProgDet

Type: [SystemBoolean](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cDStAISC_ASD89 Interface](#)

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CSI API ETABS v1

cDStAISC_ASD89AddLanguageSpecificTextSet("LSTA Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDStAISC_ASD89
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: [SystemInt32](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDStAISC ASD89 Interface

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CSI API ETABS v1

cDStAISC_ASD89AddLanguageSpecificTextSet("LST80C1812B_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDStAISC_ASD89
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

Item

Type: SystemInt32

Value

Type: SystemDouble

ItemType (Optional)

Type: ETABSv1eItemType

Return Value

Type: Int32

▲ See Also

Reference

cDStAISC ASD89 Interface

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CSI API ETABS v1

cDStAISC_ASD89AddLanguageSpecificTextSet("LSTB Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDStAISC_ASD89
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: [SystemInt32](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDStAISC ASD89 Interface

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CSI API ETABS v1

cDStAISC_LRFD93 Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDStAISC_LRFD93
```

```
Public Interface cDStAISC_LRFD93
```

```
Dim instance As cDStAISC_LRFD93
```

```
public interface class cDStAISC_LRFD93
```

```
type cDStAISC_LRFD93 = interface end
```

The cDStAISC_LRFD93 type exposes the following members.

▲ Methods

▲	Name	Description
---	------	-------------

➤	GetOverwrite	
---	------------------------------	--

➤	GetPreference	
---	-------------------------------	--

➤	SetOverwrite	
---	------------------------------	--

➤	SetPreference	
---	-------------------------------	--

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▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDStAISC_LRFD93 Methods

The cDStAISC_LRFD93 type exposes the following members.

Methods

	Name	Description
⇒	<u>GetOverwrite</u>	
⇒	<u>GetPreference</u>	
⇒	<u>SetOverwrite</u>	
⇒	<u>SetPreference</u>	

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See Also

Reference

cDStAISC_LRFD93 Interface

ETABSV1 Namespace

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CSI API ETABS v1

cDStAISC_LRFD93AddLanguageSpecificTextSet("LST3 Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDStAISC_LRFD93
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```

Parameters

Name

Type: SystemString

cDStAISC_LRFD93AddLanguageSpecificTextSet("LST3D917A31_016p=::|nu=

Item

Type: [SystemInt32](#)

Value

Type: [SystemDouble](#)

ProgDet

Type: [SystemBoolean](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cDStAISC_LRFD93 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDStAISC_LRFD93AddLanguageSpecificTextSet("LST4 Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDStAISC_LRFD93
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: [SystemInt32](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDStAISC LRFD93 Interface

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CSI API ETABS v1

cDStAISC_LRFD93AddLanguageSpecificTextSet("LST1 Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDStAISC_LRFD93
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```


Parameters

Name

Type: SystemString

Item

Type: SystemInt32

Value

Type: SystemDouble

ItemType (Optional)

Type: ETABSv1eItemType

Return Value

Type: Int32

▲ See Also

Reference

cDStAISC_LRFD93 Interface

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Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

C#

C++

F#

Copy

```
Function SetPreference (
Item As Integer,
Value As Double
) As Integer
```

```
returnValue = instance.SetPreference(Item,
Value)
```

```
int SetPreference(
int Item,
double Value
)
```

```
abstract SetPreference :  
  Item : int *  
  Value : float -> int
```

Item

Type: SystemInt32

Value

Type: SystemDoubleType:Â Int32

▲ See Also

Reference

cDStAISC LRFD93 Interface

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CSI API ETABS v1

cDStAISC360_05_IBC2006 Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDStAISC360_05_IBC2006
```

```
Public Interface cDStAISC360_05_IBC2006
```

```
Dim instance As cDStAISC360_05_IBC2006
```

```
public interface class cDStAISC360_05_IBC2006
```

```
type cDStAISC360_05_IBC2006 = interface end
```

The cDStAISC360_05_IBC2006 type exposes the following members.

▲ Methods

▲	Name	Description
⇒	GetOverwrite	Retrieves the value of a steel design overwrite item.
⇒	GetPreference	Retrieves the value of a steel design preference item.
⇒	SetOverwrite	Sets the value of a steel design overwrite item.
⇒	SetPreference	Sets the value of a steel design preference item.

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDStAISC360_05_IBC2006 Methods

The cDStAISC360_05_IBC2006 type exposes the following members.

Methods

⌕	Name	Description
⇒	<u>GetOverwrite</u>	Retrieves the value of a steel design overwrite item.
⇒	<u>GetPreference</u>	Retrieves the value of a steel design preference item.
⇒	<u>SetOverwrite</u>	Sets the value of a steel design overwrite item.
⇒	<u>SetPreference</u>	Sets the value of a steel design preference item.

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See Also

Reference

cDStAISC360_05_IBC2006 Interface

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CSI API ETABS v1

cDStAISC360_05_IBC2006AddLanguageSpecificTextSet Method

Retrieves the value of a steel design overwrite item.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDStAISC360_05_IBC2006
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```

Parameters

Name

Type: SystemString

The name of a frame object with a steel frame design procedure.

Item

Type: SystemInt32

This is an integer between 1 and 43, inclusive, indicating the overwrite item considered.

1. Framing type
2. Omega0
3. Consider deflection
4. Deflection check type
5. DL deflection limit, L/Value
6. SDL + LL deflection limit, L/Value
7. LL deflection limit, L/Value
8. Total load deflection limit, L/Value
9. Total camber limit, L/Value
10. DL deflection limit, absolute
11. SDL + LL deflection limit, absolute
12. LL deflection limit, absolute
13. Total load deflection limit, absolute
14. Total camber limit, absolute
15. Specified camber
16. Net area to total area ratio
17. Live load reduction factor
18. Unbraced length ratio, Major
19. Unbraced length ratio, Minor
20. Unbraced length ratio, Lateral Torsional Buckling
21. Effective length factor, K1 Major
22. Effective length factor, K1 Minor
23. Effective length factor, K2 Major
24. Effective length factor, K2 Minor
25. Effective length factor, K Lateral Torsional Buckling
26. Moment coefficient, Cm Major
27. Moment coefficient, Cm Minor
28. Bending coefficient, Cb
29. Nonsway moment factor, B1 Major
30. Nonsway moment factor, B1 Minor
31. Sway moment factor, B2 Major
32. Sway moment factor, B2 Minor
33. Reduce HSS thickness
34. HSS welding type
35. Yield stress, Fy
36. Expected to specified Fy ratio, Ry
37. Compressive capacity, Pnc
38. Tensile capacity, Pnt
39. Major bending capacity, Mn3
40. Minor bending capacity, Mn2
41. Major shear capacity, Vn2
42. Minor shear capacity, Vn3

43. Demand/capacity ratio limit

Value

Type: SystemDouble

The value of the considered overwrite item.

1. Framing type

- . 0 = Program Default
- . 1 = SMF
- . 2 = IMF
- . 3 = OMF
- . 4 = SCBF
- . 5 = OCBF
- . 6 = OCBFI
- . 7 = EBF

2. Omega0

Value ≥ 0 ; 0 means use a program determined value.

3. Consider deflection

- . 0 = Program Default
- . 1 = No
- . 2 = Yes

4. Deflection check type

- . 0 = Program Default
- . 1 = Ratio
- . 2 = Absolute
- . 3 = Both

5. DL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item.

6. SDL + LL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item.

7. LL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item.

8. Total load deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item.

9. Total camber limit, L/Value

Value ≥ 0 ; 0 means no check for this item.

10. DL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

11. SDL + LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

12. LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

13. Total load deflection limit, absolute

Introduction

Value ≥ 0 ; 0 means no check for this item. [L]

14. Total camber limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

15. Specified camber

Value ≥ 0 . [L]

16. Net area to total area ratio

Value ≥ 0 ; 0 means use program default value.

17. Live load reduction factor

Value ≥ 0 ; 0 means use program determined value.

18. Unbraced length ratio, Major

Value ≥ 0 ; 0 means use program determined value.

19. Unbraced length ratio, Minor

Value ≥ 0 ; 0 means use program determined value.

20. Unbraced length ratio, Lateral Torsional Buckling

Value ≥ 0 ; 0 means use program determined value.

21. Effective length factor, K1 Major

Value ≥ 0 ; 0 means use program determined value.

22. Effective length factor, K1 Minor

Value ≥ 0 ; 0 means use program determined value.

23. Effective length factor, K2 Major

Value ≥ 0 ; 0 means use program determined value.

24. Effective length factor, K2 Minor

Value ≥ 0 ; 0 means use program determined value.

25. Effective length factor, K Lateral Torsional Buckling

Value ≥ 0 ; 0 means use program determined value.

26. Moment coefficient, Cm Major

Value ≥ 0 ; 0 means use program determined value.

27. Moment coefficient, Cm Minor

Value ≥ 0 ; 0 means use program determined value.

28. Bending coefficient, Cb

Value ≥ 0 ; 0 means use program determined value.

29. Nonsway moment factor, B1 Major

Value ≥ 0 ; 0 means use program determined value.

30. Nonsway moment factor, B1 Minor

Introduction

Value ≥ 0 ; 0 means use program determined value.

31. Sway moment factor, B2 Major

Value ≥ 0 ; 0 means use program determined value.

32. Sway moment factor, B2 Minor

Value ≥ 0 ; 0 means use program determined value.

33. Reduce HSS thickness

· 0 = Program Default

· 1 = No

· 2 = Yes

34. HSS welding type

· 0 = Program Default

· 1 = ERW

· 2 = SAW

35. Yield stress, Fy

Value ≥ 0 ; 0 means use program determined value. $[F/L^2]$

36. Expected to specified Fy ratio, Ry

Value ≥ 0 ; 0 means use program determined value.

37. Compressive capacity, Pnc

Value ≥ 0 ; 0 means use program determined value. $[F]$

38. Tensile capacity, Pnt

Value ≥ 0 ; 0 means use program determined value. $[F]$

39. Major bending capacity, Mn3

Value ≥ 0 ; 0 means use program determined value. $[FL]$

40. Minor bending capacity, Mn2

Value ≥ 0 ; 0 means use program determined value. $[FL]$

41. Major shear capacity, Vn2

Value ≥ 0 ; 0 means use program determined value. $[F]$

42. Minor shear capacity, Vn3

Value ≥ 0 ; 0 means use program determined value. $[F]$

43. Demand/capacity ratio limit

Value ≥ 0 ; 0 means use program determined value.

ProgDet

Type: SystemBoolean

If this item is True, the specified value is program determined.

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

Return Value

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double
    Dim ProgDet As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set steel design code
    ret = SapModel.DesignSteel.SetCode("AISC360-05/IBC2006")

    'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis

    'start steel design
    ret = SapModel.DesignSteel.StartDesign

    'get overwrite item
    ret = SapModel.DesignSteel.AISC360_05_IBC2006.GetOverwrite("8", 1, Value, ProgDet)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

See Also

Reference

[cDStAISC360_05_IBC2006 Interface](#)

[ETABSV1 Namespace](#)

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Introduction

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CSI API ETABS v1

cDStAISC360_05_IBC2006AddLanguageSpecificTextSet

Method

Retrieves the value of a steel design preference item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDStAISC360_05_IBC2006
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: SystemInt32

This is an integer between 1 and 29, inclusive, indicating the preference item considered.

1. Framing type
2. Seismic design category
3. Design provision
4. Analysis method (Obsolete, replaced by 27, 28, and 29)
5. Notional load coefficient

6. Phi bending
7. Phi compression
8. Phi tension yielding
9. Phi tension fracture
10. Phi shear
11. Phi shear sort webbed rolled I
12. Phi torsion
13. Ignore seismic code
14. Ignore special seismic load
15. Doubler plate is plug welded
16. HSS welding type
17. Reduce HSS thickness
18. Consider deflection
19. DL deflection limit, L/Value
20. SDL + LL deflection limit, L/Value
21. LL deflection limit, L/Value
22. Total load deflection limit, L/Value
23. Total camber limit, L/Value
24. Pattern live load factor
25. Demand/capacity ratio limit
26. Multi-response case design
27. Analysis Method
28. Second Order Method
29. Stiffness Reduction Method

Value

Type: SystemDouble

The value of the considered preference item.

1. Framing type
 1. SMF
 2. IMF
 3. OMF
 4. SCBF
 5. OCBF
 6. OCBFI
 7. EBF
2. Seismic design category
 1. A
 2. B
 3. C
 4. D
 5. E
 6. F
3. Design provision
 1. LRFD
 2. ASD
4. Analysis method (Obsolete, replaced by 27, 28, and 29)
 1. Gen 2nd Order Elastic
 2. 2nd Order By Amp 1st Order
 3. Limited 1st Order Elastic
 4. DAM Gen 2nd Order Taub Variable
 5. DAM Gen 2nd Order Taub Fixed

Introduction

- 6. DAM Amp 1st Order Taub Variable
- 7. DAM Amp 1st Order Taub Fixed
- 5. Notional load coefficient

Value > 0

- 6. Phi bending

Value > 0

- 7. Phi compression

Value > 0

- 8. Phi tension yielding

Value > 0

- 9. Phi tension fracture

Value > 0

- 10. Phi shear

Value > 0

- 11. Phi shear sort webbed rolled I

Value > 0

- 12. Phi torsion

Value > 0

- 13. Ignore seismic code

0 = No

Any other value = Yes

- 14. Ignore special seismic load

0 = No

Any other value = Yes

- 15. Doubler plate is plug welded

0 = No

Any other value = Yes

- 16. HSS welding type

1. ERW

2. SAW

- 17. Reduce HSS thickness

0 = No

Any other value = Yes

- 18. Consider deflection

0 = No

Any other value = Yes

19. DL deflection limit, L/Value

Value > 0

20. SDL + LL deflection limit, L/Value

Value > 0

21. LL deflection limit, L/Value

Value > 0

22. Total load deflection limit, L/Value

Value > 0

23. Total camber limit, L/Value

Value > 0

24. Pattern live load factor

Value > 0

25. Demand/capacity ratio limit

Value > 0

26. Multi-response case design

1. Envelopes
2. Step-by-step
3. Last step
4. Envelopes -- All
5. Step-by-step -- All

27. Analysis Method

1. Direct Analysis
2. Effective Length
3. Limited 1st Order

28. Second Order Method

1. General 2nd Order
2. Amplified 1st Order

29. Stiffness Reduction Method

1. Tau-b variable
2. Tau-b Fixed
3. No Modification

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ See Also

Reference

cDStAISC360_05_IBC2006 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDStAISC360_05_IBC2006AddLanguageSpecificTextSet Method

Sets the value of a steel design overwrite item.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDStAISC360_05_IBC2006
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters**Name**

Type: SystemString

The name of an existing frame object or group, depending on the value of the ItemType item.

Item

Type: SystemInt32

This is an integer between 1 and 43, inclusive, indicating the overwrite item considered.

1. Framing type
2. Omega0
3. Consider deflection
4. Deflection check type
5. DL deflection limit, L/Value
6. SDL + LL deflection limit, L/Value
7. LL deflection limit, L/Value
8. Total load deflection limit, L/Value
9. Total camber limit, L/Value
10. DL deflection limit, absolute
11. SDL + LL deflection limit, absolute
12. LL deflection limit, absolute
13. Total load deflection limit, absolute
14. Total camber limit, absolute
15. Specified camber
16. Net area to total area ratio
17. Live load reduction factor
18. Unbraced length ratio, Major
19. Unbraced length ratio, Minor
20. Unbraced length ratio, Lateral Torsional Buckling
21. Effective length factor, K1 Major
22. Effective length factor, K1 Minor
23. Effective length factor, K2 Major
24. Effective length factor, K2 Minor
25. Effective length factor, K Lateral Torsional Buckling
26. Moment coefficient, Cm Major
27. Moment coefficient, Cm Minor
28. Bending coefficient, Cb
29. Nonsway moment factor, B1 Major
30. Nonsway moment factor, B1 Minor
31. Sway moment factor, B2 Major
32. Sway moment factor, B2 Minor
33. Reduce HSS thickness
34. HSS welding type
35. Yield stress, Fy
36. Expected to specified Fy ratio, Ry
37. Compressive capacity, Pnc
38. Tensile capacity, Pnt
39. Major bending capacity, Mn3
40. Minor bending capacity, Mn2
41. Major shear capacity, Vn2

- 42. Minor shear capacity, V_n3
- 43. Demand/capacity ratio limit

Value

Type: SystemDouble

The value of the considered overwrite item.

1. Framing type
 - . 0 = Program Default
 - . 1 = SMF
 - . 2 = IMF
 - . 3 = OMF
 - . 4 = SCBF
 - . 5 = OCBF
 - . 6 = OCBFI
 - . 7 = EBF
2. Omega0

Value ≥ 0 ; 0 means use a program determined value.

3. Consider deflection
 - . 0 = Program Default
 - . 1 = No
 - . 2 = Yes
4. Deflection check type
 - . 0 = Program Default
 - . 1 = Ratio
 - . 2 = Absolute
 - . 3 = Both
5. DL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item.

6. SDL + LL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item.

7. LL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item.

8. Total load deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item.

9. Total camber limit, L/Value

Value ≥ 0 ; 0 means no check for this item.

10. DL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

11. SDL + LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

12. LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

13. Total load deflection limit, absolute

Introduction

Value ≥ 0 ; 0 means no check for this item. [L]

14. Total camber limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

15. Specified camber

Value ≥ 0 . [L]

16. Net area to total area ratio

Value ≥ 0 ; 0 means use program default value.

17. Live load reduction factor

Value ≥ 0 ; 0 means use program determined value.

18. Unbraced length ratio, Major

Value ≥ 0 ; 0 means use program determined value.

19. Unbraced length ratio, Minor

Value ≥ 0 ; 0 means use program determined value.

20. Unbraced length ratio, Lateral Torsional Buckling

Value ≥ 0 ; 0 means use program determined value.

21. Effective length factor, K1 Major

Value ≥ 0 ; 0 means use program determined value.

22. Effective length factor, K1 Minor

Value ≥ 0 ; 0 means use program determined value.

23. Effective length factor, K2 Major

Value ≥ 0 ; 0 means use program determined value.

24. Effective length factor, K2 Minor

Value ≥ 0 ; 0 means use program determined value.

25. Effective length factor, K Lateral Torsional Buckling

Value ≥ 0 ; 0 means use program determined value.

26. Moment coefficient, Cm Major

Value ≥ 0 ; 0 means use program determined value.

27. Moment coefficient, Cm Minor

Value ≥ 0 ; 0 means use program determined value.

28. Bending coefficient, Cb

Value ≥ 0 ; 0 means use program determined value.

29. Nonsway moment factor, B1 Major

Value ≥ 0 ; 0 means use program determined value.

30. Nonsway moment factor, B1 Minor

Introduction

Value ≥ 0 ; 0 means use program determined value.

31. Sway moment factor, B2 Major

Value ≥ 0 ; 0 means use program determined value.

32. Sway moment factor, B2 Minor

Value ≥ 0 ; 0 means use program determined value.

33. Reduce HSS thickness

· 0 = Program Default

· 1 = No

· 2 = Yes

34. HSS welding type

· 0 = Program Default

· 1 = ERW

· 2 = SAW

35. Yield stress, F_y

Value ≥ 0 ; 0 means use program determined value. $[F/L^2]$

36. Expected to specified F_y ratio, R_y

Value ≥ 0 ; 0 means use program determined value.

37. Compressive capacity, P_{nc}

Value ≥ 0 ; 0 means use program determined value. $[F]$

38. Tensile capacity, P_{nt}

Value ≥ 0 ; 0 means use program determined value. $[F]$

39. Major bending capacity, M_{n3}

Value ≥ 0 ; 0 means use program determined value. $[FL]$

40. Minor bending capacity, M_{n2}

Value ≥ 0 ; 0 means use program determined value. $[FL]$

41. Major shear capacity, V_{n2}

Value ≥ 0 ; 0 means use program determined value. $[F]$

42. Minor shear capacity, V_{n3}

Value ≥ 0 ; 0 means use program determined value. $[F]$

43. Demand/capacity ratio limit

Value ≥ 0 ; 0 means use program determined value.

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◇ Object = 0

◇ Group = 1

◇ SelectedObjects = 2

If this item is Objects, the assignment is made to the frame object specified by the Name item.

If this item is Group, the assignment is made to all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the assignment is made to all selected frame objects and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the item is successfully set; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double
    Dim ProgDet As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set steel design code
    ret = SapModel.DesignSteel.SetCode("AISC360-05/IBC2006")

    'set overwrite item
    ret = SapModel.DesignSteel.AISC360_05_IBC2006.SetOverwrite("8", 1, 7)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cDStAISC360_05_IBC2006 Interface

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CSI API ETABS v1

cDStAISC360_05_IBC2006AddLanguageSpecificTextSet Method

Sets the value of a steel design preference item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDStAISC360_05_IBC2006
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: SystemInt32

This is an integer between 1 and 29, inclusive, indicating the preference item considered.

1. Framing type
2. Seismic design category
3. Design provision
4. Analysis method (Obsolete, replaced by 27, 28, and 29)
5. Notional load coefficient

6. Phi bending
7. Phi compression
8. Phi tension yielding
9. Phi tension fracture
10. Phi shear
11. Phi shear sort webbed rolled I
12. Phi torsion
13. Ignore seismic code
14. Ignore special seismic load
15. Doubler plate is plug welded
16. HSS welding type
17. Reduce HSS thickness
18. Consider deflection
19. DL deflection limit, L/Value
20. SDL + LL deflection limit, L/Value
21. LL deflection limit, L/Value
22. Total load deflection limit, L/Value
23. Total camber limit, L/Value
24. Pattern live load factor
25. Demand/capacity ratio limit
26. Multi-response case design
27. Analysis Method
28. Second Order Method
29. Stiffness Reduction Method

Value

Type: SystemDouble

The value of the considered preference item.

1. Framing type
 1. SMF
 2. IMF
 3. OMF
 4. SCBF
 5. OCBF
 6. OCBFI
 7. EBF
2. Seismic design category
 1. A
 2. B
 3. C
 4. D
 5. E
 6. F
3. Design provision
 1. LRFD
 2. ASD
4. Analysis method (Obsolete, replaced by 27, 28, and 29)
 1. Gen 2nd Order Elastic
 2. 2nd Order By Amp 1st Order
 3. Limited 1st Order Elastic
 4. DAM Gen 2nd Order Taub Variable
 5. DAM Gen 2nd Order Taub Fixed

Introduction

- 6. DAM Amp 1st Order Taub Variable
- 7. DAM Amp 1st Order Taub Fixed
- 5. Notional load coefficient

Value > 0

- 6. Phi bending

Value > 0

- 7. Phi compression

Value > 0

- 8. Phi tension yielding

Value > 0

- 9. Phi tension fracture

Value > 0

- 10. Phi shear

Value > 0

- 11. Phi shear sort webbed rolled I

Value > 0

- 12. Phi torsion

Value > 0

- 13. Ignore seismic code

0 = No

Any other value = Yes

- 14. Ignore special seismic load

0 = No

Any other value = Yes

- 15. Doubler plate is plug welded

0 = No

Any other value = Yes

- 16. HSS welding type

1. ERW

2. SAW

- 17. Reduce HSS thickness

0 = No

Any other value = Yes

- 18. Consider deflection

0 = No

Any other value = Yes

19. DL deflection limit, L/Value

Value > 0

20. SDL + LL deflection limit, L/Value

Value > 0

21. LL deflection limit, L/Value

Value > 0

22. Total load deflection limit, L/Value

Value > 0

23. Total camber limit, L/Value

Value > 0

24. Pattern live load factor

Value > 0

25. Demand/capacity ratio limit

Value > 0

26. Multi-response case design

1. Envelopes
2. Step-by-step
3. Last step
4. Envelopes -- All
5. Step-by-step -- All

27. Analysis Method

1. Direct Analysis
2. Effective Length
3. Limited 1st Order

28. Second Order Method

1. General 2nd Order
2. Amplified 1st Order

29. Stiffness Reduction Method

1. Tau-b variable
2. Tau-b Fixed
3. No Modification

Return Value

Type: Int32

Returns zero if the item is successfully set; otherwise it returns a nonzero value.

▲ Remarks

▲ See Also

Reference

cDStAISC360_05_IBC2006 Interface

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CSI API ETABS v1

cDStAISC360_10 Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDStAISC360_10
```

```
Public Interface cDStAISC360_10
```

```
Dim instance As cDStAISC360_10
```

```
public interface class cDStAISC360_10
```

```
type cDStAISC360_10 = interface end
```

The cDStAISC360_10 type exposes the following members.

▲ Methods

▲	Name	Description
---	------	-------------

◆	GetOverwrite	
---	------------------------------	--

◆	GetPreference	
---	-------------------------------	--

◆	SetOverwrite	
---	------------------------------	--

◆	SetPreference	
---	-------------------------------	--

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDStAISC360_10 Methods

The [cDStAISC360_10](#) type exposes the following members.

Methods

	Name	Description
	GetOverwrite	
	GetPreference	
	SetOverwrite	
	SetPreference	

[Top](#)

See Also

Reference

[cDStAISC360_10 Interface](#)

[ETABSV1 Namespace](#)

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Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

C#

C++

F#

Copy

```
abstract GetOverwrite :
  Name : string *
  Item : int *
  Value : float byref *
  ProgDet : bool byref -> int
```

Name

cDStAlSC360_10span id="LST4A52D9CE_0"AddLanguageSpecificTextSet("LST4A52D9CE_01652::|nu=

Item

Type:Â [SystemInt32](#)

Value

Type:Â [SystemDouble](#)

ProgDet

Type:Â [SystemBoolean](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cDStAISC360_10 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDStAISC360_10AddLanguageSpecificTextSet("LSTE0 Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDStAISC360_10
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: [SystemInt32](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDStAISC360_10 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDStAISC360_10AddLanguageSpecificTextSet("LSTC5 Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDStAISC360_10
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

Item

Type: SystemInt32

Value

Type: SystemDouble

ItemType (Optional)

Type: ETABSv1eItemType

Return Value

Type: Int32

▲ See Also

Reference

cDStAISC360_10 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDStAISC360_10AddLanguageSpecificTextSet("LST23D62388_0?nu=.") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDStAISC360_10
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: [SystemInt32](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDStAISC360_10 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDStAustralian_AS4100_98 Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDStAustralian_AS4100_98
```

```
Public Interface cDStAustralian_AS4100_98
```

```
Dim instance As cDStAustralian_AS4100_98
```

```
public interface class cDStAustralian_AS4100_98
```

```
type cDStAustralian_AS4100_98 = interface end
```

The cDStAustralian_AS4100_98 type exposes the following members.

▲ Methods

▲	Name	Description
---	------	-------------

⇒	GetOverwrite	
---	------------------------------	--

⇒	GetPreference	
---	-------------------------------	--

⇒	SetOverwrite	
---	------------------------------	--

⇒	SetPreference	
---	-------------------------------	--

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDStAustralian_AS4100_98 Methods

The cDStAustralian_AS4100_98 type exposes the following members.

Methods

	Name	Description
⇒	<u>GetOverwrite</u>	
⇒	<u>GetPreference</u>	
⇒	<u>SetOverwrite</u>	
⇒	<u>SetPreference</u>	

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See Also

Reference

cDStAustralian_AS4100_98 Interface

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CSI API ETABS v1

cDStAustralian_AS4100_98AddLanguageSpecificTextSet Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDStAustralian_AS4100_98
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```

Parameters

Name

Type: SystemString

cDStAustralian_AS4100_98span id="LST6DB37A09_0"AddLanguageSpecificTextSet("LST6DB37A09_0?cp

Item

Type:Â [SystemInt32](#)

Value

Type:Â [SystemDouble](#)

ProgDet

Type:Â [SystemBoolean](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cDStAustralian_AS4100_98 Interface](#)

[ETABSv1 Namespace](#)

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Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

C#

C++

F#

Copy

```
Function GetPreference (
Item As Integer,
ByRef Value As Double
) As Integer
```

```
returnValue = instance.GetPreference(Item,
Value)
```

```
int GetPreference(
int Item,
double% Value
)
```

```
abstract GetPreference :
  Item : int *
  Value : float byref -> int
```

Item

Type: SystemInt32

Value

Type: SystemDoubleType: Int32

See Also

Reference

cDStAustralian_AS4100_98 Interface

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Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

C#

C++

F#

Copy

```

abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int

```

Parameters

Name

Type: SystemString

Item

Type: SystemInt32

Value

Type: SystemDouble

ItemType (Optional)

Type: ETABSv1eItemType

Return Value

Type: Int32

▲ See Also

Reference

cDStAustralian_AS4100_98 Interface

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CSI API ETABS v1

cDStAustralian_AS4100_98AddLanguageSpecificTextSet Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDStAustralian_AS4100_98
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: [SystemInt32](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDStAustralian_AS4100_98 Interface

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CSI API ETABS v1

cDStBS5950_2000 Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDStBS5950_2000
```

```
Public Interface cDStBS5950_2000
```

```
Dim instance As cDStBS5950_2000
```

```
public interface class cDStBS5950_2000
```

```
type cDStBS5950_2000 = interface end
```

The cDStBS5950_2000 type exposes the following members.

▲ Methods

▲	Name	Description
---	------	-------------

➦	GetOverwrite	
---	------------------------------	--

➦	GetPreference	
---	-------------------------------	--

➦	SetOverwrite	
---	------------------------------	--

➦	SetPreference	
---	-------------------------------	--

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▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDStBS5950_2000 Methods

The cDStBS5950_2000 type exposes the following members.

Methods

	Name	Description
	<u>GetOverwrite</u>	
	<u>GetPreference</u>	
	<u>SetOverwrite</u>	
	<u>SetPreference</u>	

Top

See Also

Reference

cDStBS5950_2000 Interface

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CSI API ETABS v1

cDStBS5950_2000AddLanguageSpecificTextSet("LSTB Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDStBS5950_2000
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```

Parameters

Name

Type: SystemString

cDStBS5950_2000AddLanguageSpecificTextSet("LSTB1DFBC0E1672app=::|nu

Item

Type:Â [SystemInt32](#)

Value

Type:Â [SystemDouble](#)

ProgDet

Type:Â [SystemBoolean](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cDStBS5950_2000 Interface](#)

[ETABSv1 Namespace](#)

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Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

C#

C++F#

Copy

```
int GetPreference(
int Item,
ref double Value
)
```

```
Function GetPreference (
Item As Integer,
ByRef Value As Double
) As Integer
```

```
Dim instance As cDStBS5950_2000
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
Value)
```

```
int GetPreference(
int Item,
double% Value
)
```

```
abstract GetPreference :  
  Item : int *  
  Value : float byref -> int
```

Item

Type: SystemInt32

Value

Type: SystemDoubleType: Int32

See Also

Reference

cDStBS5950_2000 Interface

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CSI API ETABS v1

cDStBS5950_2000AddLanguageSpecificTextSet("LSTE Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDStBS5950_2000
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```


Parameters

Name

Type:Â SystemString

Item

Type:Â SystemInt32

Value

Type:Â SystemDouble

ItemType (Optional)

Type:Â ETABSv1eItemType

Return Value

Type:Â Int32

▲ See Also

Reference

cDStBS5950_2000 Interface

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CSI API ETABS v1

cDStBS5950_2000AddLanguageSpecificTextSet("LST7 Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDStBS5950_2000
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: [SystemInt32](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDStBS5950_2000 Interface

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CSI API ETABS v1

cDStCanadian_S16_09 Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDStCanadian_S16_09
```

```
Public Interface cDStCanadian_S16_09
```

```
Dim instance As cDStCanadian_S16_09
```

```
public interface class cDStCanadian_S16_09
```

```
type cDStCanadian_S16_09 = interface end
```

The cDStCanadian_S16_09 type exposes the following members.

▲ Methods

▲	Name	Description
➤	GetOverwrite	Retrieves the value of a steel design overwrite item
➤	GetPreference	Retrieves the value of a steel design preference item
➤	SetOverwrite	Sets the value of a steel design overwrite item
➤	SetPreference	Sets the value of a steel design preference item

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▲ See Also

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CSI API ETABS v1

cDStCanadian_S16_09 Methods

The cDStCanadian_S16_09 type exposes the following members.

Methods

⚡	Name	Description
⇒	<u>GetOverwrite</u>	Retrieves the value of a steel design overwrite item
⇒	<u>GetPreference</u>	Retrieves the value of a steel design preference item
⇒	<u>SetOverwrite</u>	Sets the value of a steel design overwrite item
⇒	<u>SetPreference</u>	Sets the value of a steel design preference item

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⚡ See Also

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cDStCanadian_S16_09 Interface

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CSI API ETABS v1

cDStCanadian_S16_09AddLanguageSpecificTextSet("L Method

Retrieves the value of a steel design overwrite item

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDStCanadian_S16_09
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing frame object with a steel frame design procedure

Item

Type: SystemInt32

This is an integer between 1 and 39, inclusive, indicating the overwrite item considered

1. Framing type
2. Consider deflection
3. Deflection check type
4. DL deflection limit, L/Value
5. SDL + LL deflection limit, L/Value
6. LL deflection limit, L/Value
7. Total load deflection limit, L/Value
8. Total-camber limit, L/Value
9. DL deflection limit, absolute
10. SDL + LL deflection limit, absolute
11. LL deflection limit, absolute
12. Total load deflection limit, absolute
13. Total-camber limit, absolute
14. Specified camber
15. Net area to total area ratio
16. Live load reduction factor
17. Unbraced length ratio, Major
18. Unbraced length ratio, Minor LTB
19. Unbraced length ratio, Lateral Torsional Buckling
20. Effective length factor, K Major
21. Effective length factor, K Minor
22. Effective length factor, K LTB
23. Moment coefficient, Omega1 Major
24. Moment coefficient, Omega1 Minor
25. Bending coefficient, Omega2
26. Nonsway moment factor, U1 Major
27. Nonsway moment factor, U1 Minor
28. Sway moment factor, U2 Major
29. Sway moment factor, U2 Minor
30. Parameter for compressive resistance, n
31. Yield stress, Fy
32. Expected to specified Fy ratio, Ry
33. Compressive resistance, Cr
34. Tensile resistance, Tr
35. Major bending resistance, Mr3
36. Minor bending resistance, Mr2
37. Major shear resistance, Vr2
38. Minor shear resistance, Vr3
39. Demand/capacity ratio limit

Value

Type: SystemDouble

The value of the considered overwrite item

1. Framing type
 - **0** - As specified in preferences
 - **1** - Type LD MRF
 - **2** - Type MD MRF
 - **3** - Type D MRF
 - **4** - Type LD CBF(V)
 - **5** - Type LD CBF(TC)
 - **6** - Type LD CBF(TO)
 - **7** - Type LD CBF(OT)
 - **8** - Type MD CBF(V)
 - **9** - Type MD CBF(TC)
 - **10** - Type MD CBF(TO)
 - **11** - Type MD CBF(OT)
 - **12** - EBF
 - **13** - Cantilever Column
 - **14** - Conventional MF
 - **15** - Conventional BF

2. Consider deflection?
 - 0 = No
 - Any other value = Yes

3. Deflection check type
 - **0** - Program Determined
 - **1** - Ratio
 - **2** - Absolute
 - **3** - Both

4. DL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

5. SDL + LL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

6. LL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

7. Total load deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

8. Total-camber limit, L/Value

Value ≥ 0 ; 0 means no check for this item

9. DL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

10. SDL + LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

11. LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

12. Total load deflection limit, absolute

Introduction

Value ≥ 0 ; 0 means no check for this item. [L]

13. Total-camber limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

14. Specified camber

Value ≥ 0 . [L]

15. Net area to total area ratio

Value ≥ 0 ; 0 means use program default value

16. Live load reduction factor

Value ≥ 0 ; 0 means use program determined value

17. Unbraced length ratio, Major

Value ≥ 0 ; 0 means use program determined value

18. Unbraced length ratio, Minor

Value ≥ 0 ; 0 means use program determined value

19. Unbraced length ratio, Lateral Torsional Buckling

Value ≥ 0 ; 0 means use program determined value

20. Effective length factor, K Major

Value ≥ 0 ; 0 means use program determined value

21. Effective length factor, K Minor

Value ≥ 0 ; 0 means use program determined value

22. Effective length factor, K LTB

Value ≥ 0 ; 0 means use program determined value

23. Moment coefficient, Omega1 Major

Value ≥ 0 ; 0 means use program determined value

24. Moment coefficient, Omega1 Minor

Value ≥ 0 ; 0 means use program determined value

25. Moment coefficient, Omega2

Value ≥ 0 ; 0 means use program determined value

26. Nonsway moment factor, U1 Major

Value ≥ 0 ; 0 means use program determined value

27. Nonsway moment factor, U1 Minor

Value ≥ 0 ; 0 means use program determined value

28. Sway moment factor, U2 Major

Value ≥ 0 ; 0 means use program determined value

29. Sway moment factor, U2 Minor

Introduction

Value ≥ 0 ; 0 means use program determined value
30. Parameter for compressive resistance, n

Value ≥ 0 ; 0 means use program determined value
31. Yield stress, F_y

Value ≥ 0 ; 0 means use program determined value [F/L2]
32. Expected to specified F_y ratio, R_y

Value ≥ 0 ; 0 means use program determined value [F/L2]=
33. Compressive resistance, C_r

Value ≥ 0 ; 0 means use program determined value [F]
34. Tensile resistance, T_r

Value ≥ 0 ; 0 means use program determined value [F]
35. Major bending resistance, M_{r3}

Value ≥ 0 ; 0 means use program determined value [FL]
36. Minor bending resistance, M_{r2}

Value ≥ 0 ; 0 means use program determined value [FL]
37. Major shear resistance, V_{r2}

Value ≥ 0 ; 0 means use program determined value [F]
38. Minor shear resistance, V_{r3}

Value ≥ 0 ; 0 means use program determined value [F]
39. Demand/capacity ratio limit

Value ≥ 0 ; 0 means use program determined value

ProgDet

Type: System.Boolean

If this item is True, the specified value is program determined

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim Value As Double  
    Dim ProgDet As Boolean
```

'create ETABS object

```
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

Return Value

```
'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 4, 4, 24, 24)

'set steel design code
ret = SapModel.DesignSteel.SetCode("CSA-S16-09")

'get overwrite item
ret = SapModel.DesignSteel.Canadian_S16_09.GetOverwrite("8", 1, Value, ProgDet)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cDStCanadian_S16_09 Interface

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CSI API ETABS v1

cDStCanadian_S16_09AddLanguageSpecificTextSet("L Method

Retrieves the value of a steel design preference item

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDStCanadian_S16_09
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: SystemInt32

This is an integer between 1 and 21, inclusive, indicating the preference item considered

1. Multi-response case design
2. Framing type
3. Spectral Acceleration Ratio, $I_e \cdot F_a \cdot S_a(0.2)$
4. Ductility related modification factor, R_d
5. Overstrength related modification factor, R_o

6. Capacity factor, Phi bending
7. Capacity factor, Phi compression
8. Capacity factor, Phi tension
9. Capacity factor, Phi shear
10. Slender section modification
11. Ignore seismic code
12. Ignore special seismic load
13. Doubler plate is plug welded
14. Consider deflection
15. DL deflection limit, L/Value
16. SDL + LL deflection limit, L/Value
17. LL deflection limit, L/Value
18. Total load deflection limit, L/Value
19. Total camber limit, L/Value
20. Pattern live load factor
21. Demand/capacity ratio limit

Value

Type: SystemDouble

The value of the considered preference item

1. Multi-response case design
 - **1** - Envelopes
 - **2** - Step-by-step
 - **3** - Last step
 - **4** - Envelopes -- All
 - **5** - Step-by-step -- All
2. Framing type
 - **1** - Type LD MRF
 - **2** - Type MD MRF
 - **3** - Type D MRF
 - **4** - Type LD CBF(V)
 - **5** - Type LD CBF(TC)
 - **6** - Type LD CBF(TO)
 - **7** - Type LD CBF(OT)
 - **8** - Type MD CBF(V)
 - **9** - Type MD CBF(TC)
 - **10** - Type MD CBF(TO)
 - **11** - Type MD CBF(OT)
 - **12** - EBF
 - **13** - Cantilever Column
 - **14** - Conventional MF
 - **15** - Conventional BF
3. Spectral Acceleration Ratio, $I_e \cdot F_a \cdot S_a(0.2)$

Value > 0

4. Ductility related modification factor, R_d

Value > 0

5. Overstrength related modification factor, R_o

Value > 0

6. Capacity factor, Phi bending

Value > 0

7. Capacity factor, Phi compression

Value > 0

8. Capacity factor, Phi tension

Value > 0

9. Capacity factor, Phi shear

Value > 0

10. Slender section modification

· **1** - modified geometry

· **2** - modified Fy

11. Ignore seismic code

· 0 = No

· Any other value = Yes

12. Ignore special seismic load

· 0 = No

· Any other value = Yes

13. Doubler plate is plug welded

· 0 = No

· Any other value = Yes

14. Consider deflection

· 0 = No

· Any other value = Yes

15. DL deflection limit, L/Value

Value > 0

16. SDL + LL deflection limit, L/Value

Value > 0

17. LL deflection limit, L/Value

Value > 0

18. Total deflection limit, L/Value

Value > 0

19. Total camber limit, L/Value

Value > 0

20. Pattern live load factor

Value >= 0

21. Demand/capacity ratio limit

Value > 0

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise, it returns a nonzero value

Remarks

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set steel design code
    ret = SapModel.DesignSteel.SetCode("CSA-S16-09")

    'get preference item
    ret = SapModel.DesignSteel.Canadian_S16_09.GetPreference(4, Value)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

See Also

Reference

cDStCanadian_S16_09 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDStCanadian_S16_09AddLanguageSpecificTextSet("L Method

Sets the value of a steel design overwrite item

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDStCanadian_S16_09
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```


Parameters

Name

Type: SystemString

The name of an existing frame object with a steel frame design procedure

Item

Type: SystemInt32

This is an integer between 1 and 39, inclusive, indicating the overwrite item considered

1. Framing type
2. Consider deflection
3. Deflection check type
4. DL deflection limit, L/Value
5. SDL + LL deflection limit, L/Value
6. LL deflection limit, L/Value
7. Total load deflection limit, L/Value
8. Total-camber limit, L/Value
9. DL deflection limit, absolute
10. SDL + LL deflection limit, absolute
11. LL deflection limit, absolute
12. Total load deflection limit, absolute
13. Total-camber limit, absolute
14. Specified camber
15. Net area to total area ratio
16. Live load reduction factor
17. Unbraced length ratio, Major
18. Unbraced length ratio, Minor LTB
19. Unbraced length ratio, Lateral Torsional Buckling
20. Effective length factor, K Major
21. Effective length factor, K Minor
22. Effective length factor, K LTB
23. Moment coefficient, Omega1 Major
24. Moment coefficient, Omega1 Minor
25. Bending coefficient, Omega2
26. Nonsway moment factor, U1 Major
27. Nonsway moment factor, U1 Minor
28. Sway moment factor, U2 Major
29. Sway moment factor, U2 Minor
30. Parameter for compressive resistance, n
31. Yield stress, Fy
32. Expected to specified Fy ratio, Ry
33. Compressive resistance, Cr
34. Tensile resistance, Tr
35. Major bending resistance, Mr3
36. Minor bending resistance, Mr2
37. Major shear resistance, Vr2
38. Minor shear resistance, Vr3
39. Demand/capacity ratio limit

Value

Type: SystemDouble

The value of the considered overwrite item

1. Framing type
 - **0** - As specified in preferences
 - **1** - Type LD MRF
 - **2** - Type MD MRF
 - **3** - Type D MRF
 - **4** - Type LD CBF(V)
 - **5** - Type LD CBF(TC)
 - **6** - Type LD CBF(TO)
 - **7** - Type LD CBF(OT)
 - **8** - Type MD CBF(V)
 - **9** - Type MD CBF(TC)
 - **10** - Type MD CBF(TO)
 - **11** - Type MD CBF(OT)
 - **12** - EBF
 - **13** - Cantilever Column
 - **14** - Conventional MF
 - **15** - Conventional BF

2. Consider deflection?
 - 0 = No
 - Any other value = Yes

3. Deflection check type
 - **0** - Program Determined
 - **1** - Ratio
 - **2** - Absolute
 - **3** - Both

4. DL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

5. SDL + LL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

6. LL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

7. Total load deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

8. Total-camber limit, L/Value

Value ≥ 0 ; 0 means no check for this item

9. DL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

10. SDL + LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

11. LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

12. Total load deflection limit, absolute

Introduction

Value ≥ 0 ; 0 means no check for this item. [L]

13. Total-camber limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

14. Specified camber

Value ≥ 0 . [L]

15. Net area to total area ratio

Value ≥ 0 ; 0 means use program default value

16. Live load reduction factor

Value ≥ 0 ; 0 means use program determined value

17. Unbraced length ratio, Major

Value ≥ 0 ; 0 means use program determined value

18. Unbraced length ratio, Minor

Value ≥ 0 ; 0 means use program determined value

19. Unbraced length ratio, Lateral Torsional Buckling

Value ≥ 0 ; 0 means use program determined value

20. Effective length factor, K Major

Value ≥ 0 ; 0 means use program determined value

21. Effective length factor, K Minor

Value ≥ 0 ; 0 means use program determined value

22. Effective length factor, K LTB

Value ≥ 0 ; 0 means use program determined value

23. Moment coefficient, Omega1 Major

Value ≥ 0 ; 0 means use program determined value

24. Moment coefficient, Omega1 Minor

Value ≥ 0 ; 0 means use program determined value

25. Moment coefficient, Omega2

Value ≥ 0 ; 0 means use program determined value

26. Nonsway moment factor, U1 Major

Value ≥ 0 ; 0 means use program determined value

27. Nonsway moment factor, U1 Minor

Value ≥ 0 ; 0 means use program determined value

28. Sway moment factor, U2 Major

Value ≥ 0 ; 0 means use program determined value

29. Sway moment factor, U2 Minor

Introduction

Value ≥ 0 ; 0 means use program determined value

30. Parameter for compressive resistance, n

Value ≥ 0 ; 0 means use program determined value

31. Yield stress, F_y

Value ≥ 0 ; 0 means use program determined value [F/L2]

32. Expected to specified F_y ratio, R_y

Value ≥ 0 ; 0 means use program determined value [F/L2]=

33. Compressive resistance, C_r

Value ≥ 0 ; 0 means use program determined value [F]

34. Tensile resistance, T_r

Value ≥ 0 ; 0 means use program determined value [F]

35. Major bending resistance, M_{r3}

Value ≥ 0 ; 0 means use program determined value [FL]

36. Minor bending resistance, M_{r2}

Value ≥ 0 ; 0 means use program determined value [FL]

37. Major shear resistance, V_{r2}

Value ≥ 0 ; 0 means use program determined value [F]

38. Minor shear resistance, V_{r3}

Value ≥ 0 ; 0 means use program determined value [F]

39. Demand/capacity ratio limit

Value ≥ 0 ; 0 means use program determined value

ItemType (Optional)

Type: $\hat{A}$ ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◇ Object = 0

◇ Group = 1

◇ SelectedObjects = 2

If this item is Objects, the assignment is made to the frame object specified by the Name item.

If this item is Group, the assignment is made to all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the assignment is made to all selected frame objects and the Name item is ignored.

Return Value

Type: $\hat{A}$ Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set steel design code
    ret = SapModel.DesignSteel.SetCode("CSA-S16-09")

    'set overwrite item
    ret = SapModel.DesignSteel.Canadian_S16_09.SetOverwrite("8", 1, 7)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

[cDStCanadian_S16_09 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDStCanadian_S16_09AddLanguageSpecificTextSet("L Method

Sets the value of a steel design preference item

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDStCanadian_S16_09
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: SystemInt32

This is an integer between 1 and 21, inclusive, indicating the preference item considered

1. Multi-response case design
2. Framing type
3. Spectral Acceleration Ratio, $I_e \cdot F_a \cdot S_a(0.2)$
4. Ductility related modification factor, R_d
5. Overstrength related modification factor, R_o

6. Capacity factor, Phi bending
7. Capacity factor, Phi compression
8. Capacity factor, Phi tension
9. Capacity factor, Phi shear
10. Slender section modification
11. Ignore seismic code
12. Ignore special seismic load
13. Doubler plate is plug welded
14. Consider deflection
15. DL deflection limit, L/Value
16. SDL + LL deflection limit, L/Value
17. LL deflection limit, L/Value
18. Total load deflection limit, L/Value
19. Total camber limit, L/Value
20. Pattern live load factor
21. Demand/capacity ratio limit

Value

Type: SystemDouble

The value of the considered preference item

1. Multi-response case design
 - **1** - Envelopes
 - **2** - Step-by-step
 - **3** - Last step
 - **4** - Envelopes -- All
 - **5** - Step-by-step -- All
2. Framing type
 - **1** - Type LD MRF
 - **2** - Type MD MRF
 - **3** - Type D MRF
 - **4** - Type LD CBF(V)
 - **5** - Type LD CBF(TC)
 - **6** - Type LD CBF(TO)
 - **7** - Type LD CBF(OT)
 - **8** - Type MD CBF(V)
 - **9** - Type MD CBF(TC)
 - **10** - Type MD CBF(TO)
 - **11** - Type MD CBF(OT)
 - **12** - EBF
 - **13** - Cantilever Column
 - **14** - Conventional MF
 - **15** - Conventional BF
3. Spectral Acceleration Ratio, $I_e \cdot F_a \cdot S_a(0.2)$

Value > 0

4. Ductility related modification factor, R_d

Value > 0

5. Overstrength related modification factor, R_o

Value > 0

6. Capacity factor, Phi bending

Value > 0

7. Capacity factor, Phi compression

Value > 0

8. Capacity factor, Phi tension

Value > 0

9. Capacity factor, Phi shear

Value > 0

10. Slender section modification

· **1** - modified geometry

· **2** - modified Fy

11. Ignore seismic code

· 0 = No

· Any other value = Yes

12. Ignore special seismic load

· 0 = No

· Any other value = Yes

13. Doubler plate is plug welded

· 0 = No

· Any other value = Yes

14. Consider deflection

· 0 = No

· Any other value = Yes

15. DL deflection limit, L/Value

Value > 0

16. SDL + LL deflection limit, L/Value

Value > 0

17. LL deflection limit, L/Value

Value > 0

18. Total deflection limit, L/Value

Value > 0

19. Total camber limit, L/Value

Value > 0

20. Pattern live load factor

Value >= 0

21. Demand/capacity ratio limit

Value > 0

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise, it returns a nonzero value

Remarks

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set steel design code
    ret = SapModel.DesignSteel.SetCode("CSA-S16-09")

    'set preference item
    ret = SapModel.DesignSteel.Canadian_S16_09.SetPreference(2, 6)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

See Also

Reference

cDStCanadian_S16_09 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDStCanadian_S16_14 Interface

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDStCanadian_S16_14
```

```
Public Interface cDStCanadian_S16_14
```

```
Dim instance As cDStCanadian_S16_14
```

```
public interface class cDStCanadian_S16_14
```

```
type cDStCanadian_S16_14 = interface end
```

The cDStCanadian_S16_14 type exposes the following members.

▲ Methods

▲	Name	Description
➤	GetOverwrite	Retrieves the value of a steel design overwrite item
➤	GetPreference	Retrieves the value of a steel design preference item
➤	SetOverwrite	Sets the value of a steel design overwrite item
➤	SetPreference	Sets the value of a steel design preference item

[Top](#)

▲ See Also

Reference

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cDStCanadian_S16_14 Methods

The cDStCanadian_S16_14 type exposes the following members.

Methods

⌕	Name	Description
⇒	<u>GetOverwrite</u>	Retrieves the value of a steel design overwrite item
⇒	<u>GetPreference</u>	Retrieves the value of a steel design preference item
⇒	<u>SetOverwrite</u>	Sets the value of a steel design overwrite item
⇒	<u>SetPreference</u>	Sets the value of a steel design preference item

Top

See Also

Reference

cDStCanadian_S16_14 Interface

ETABSV1 Namespace

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Namespace: ETABSv1

▲ Syntax

C#

VBC++F#

Copy

```
Function GetOverwrite (
Name As String,
Item As Integer,
ByRef Value As Double,
ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDStCanadian_S16_14
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
Item, Value, ProgDet)
```

```
int GetOverwrite(
String^ Name,
int Item,
double% Value,
bool% ProgDet
)
```

```
abstract GetOverwrite :
  Name : string *
  Item : int *
  Value : float byref *
  ProgDet : bool byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing frame object with a steel frame design procedure

Item

Type: SystemInt32

This is an integer between 1 and 39, inclusive, indicating the overwrite item considered

1. Framing type
2. Consider deflection
3. Deflection check type
4. DL deflection limit, L/Value
5. SDL + LL deflection limit, L/Value
6. LL deflection limit, L/Value
7. Total load deflection limit, L/Value
8. Total-camber limit, L/Value
9. DL deflection limit, absolute
10. SDL + LL deflection limit, absolute
11. LL deflection limit, absolute
12. Total load deflection limit, absolute
13. Total-camber limit, absolute
14. Specified camber
15. Net area to total area ratio
16. Live load reduction factor
17. Unbraced length ratio, Major
18. Unbraced length ratio, Minor LTB
19. Unbraced length ratio, Lateral Torsional Buckling
20. Effective length factor, K Major
21. Effective length factor, K Minor
22. Effective length factor, K LTB
23. Moment coefficient, Omega1 Major
24. Moment coefficient, Omega1 Minor
25. Bending coefficient, Omega2
26. Nonsway moment factor, U1 Major
27. Nonsway moment factor, U1 Minor
28. Sway moment factor, U2 Major
29. Sway moment factor, U2 Minor
30. Parameter for compressive resistance, n
31. Yield stress, Fy
32. Expected to specified Fy ratio, Ry
33. Compressive resistance, Cr
34. Tensile resistance, Tr
35. Major bending resistance, Mr3
36. Minor bending resistance, Mr2
37. Major shear resistance, Vr2
38. Minor shear resistance, Vr3
39. Demand/capacity ratio limit

Value

Type: SystemDouble

The value of the considered overwrite item

1. Framing type
 - **0** - As specified in preferences
 - **1** - Type LD MRF
 - **2** - Type MD MRF
 - **3** - Type D MRF
 - **4** - Type LD CBF(V)
 - **5** - Type LD CBF(TC)
 - **6** - Type LD CBF(TO)
 - **7** - Type LD CBF(OT)
 - **8** - Type MD CBF(V)
 - **9** - Type MD CBF(TC)
 - **10** - Type MD CBF(TO)
 - **11** - Type MD CBF(OT)
 - **12** - EBF
 - **13** - Cantilever Column
 - **14** - Conventional MF
 - **15** - Conventional BF

2. Consider deflection?
 - 0 = No
 - Any other value = Yes

3. Deflection check type
 - **0** - Program Determined
 - **1** - Ratio
 - **2** - Absolute
 - **3** - Both

4. DL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

5. SDL + LL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

6. LL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

7. Total load deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

8. Total-camber limit, L/Value

Value ≥ 0 ; 0 means no check for this item

9. DL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

10. SDL + LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

11. LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

12. Total load deflection limit, absolute

Introduction

Value ≥ 0 ; 0 means no check for this item. [L]

13. Total-camber limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

14. Specified camber

Value ≥ 0 . [L]

15. Net area to total area ratio

Value ≥ 0 ; 0 means use program default value

16. Live load reduction factor

Value ≥ 0 ; 0 means use program determined value

17. Unbraced length ratio, Major

Value ≥ 0 ; 0 means use program determined value

18. Unbraced length ratio, Minor

Value ≥ 0 ; 0 means use program determined value

19. Unbraced length ratio, Lateral Torsional Buckling

Value ≥ 0 ; 0 means use program determined value

20. Effective length factor, K Major

Value ≥ 0 ; 0 means use program determined value

21. Effective length factor, K Minor

Value ≥ 0 ; 0 means use program determined value

22. Effective length factor, K LTB

Value ≥ 0 ; 0 means use program determined value

23. Moment coefficient, Omega1 Major

Value ≥ 0 ; 0 means use program determined value

24. Moment coefficient, Omega1 Minor

Value ≥ 0 ; 0 means use program determined value

25. Moment coefficient, Omega2

Value ≥ 0 ; 0 means use program determined value

26. Nonsway moment factor, U1 Major

Value ≥ 0 ; 0 means use program determined value

27. Nonsway moment factor, U1 Minor

Value ≥ 0 ; 0 means use program determined value

28. Sway moment factor, U2 Major

Value ≥ 0 ; 0 means use program determined value

29. Sway moment factor, U2 Minor

Introduction

- Value ≥ 0 ; 0 means use program determined value
30. Parameter for compressive resistance, n
- Value ≥ 0 ; 0 means use program determined value
31. Yield stress, F_y
- Value ≥ 0 ; 0 means use program determined value [F/L2]
32. Expected to specified F_y ratio, R_y
- Value ≥ 0 ; 0 means use program determined value [F/L2]=
33. Compressive resistance, C_r
- Value ≥ 0 ; 0 means use program determined value [F]
34. Tensile resistance, T_r
- Value ≥ 0 ; 0 means use program determined value [F]
35. Major bending resistance, M_{r3}
- Value ≥ 0 ; 0 means use program determined value [FL]
36. Minor bending resistance, M_{r2}
- Value ≥ 0 ; 0 means use program determined value [FL]
37. Major shear resistance, V_{r2}
- Value ≥ 0 ; 0 means use program determined value [F]
38. Minor shear resistance, V_{r3}
- Value ≥ 0 ; 0 means use program determined value [F]
39. Demand/capacity ratio limit
- Value ≥ 0 ; 0 means use program determined value

ProgDet

Type: System.Boolean

If this item is True, the specified value is program determined

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim Value As Double  
    Dim ProgDet As Boolean
```

'create ETABS object

```
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

Return Value

Introduction

```
'start ETABS application
  ret = EtabsObject.ApplicationStart()

'create SapModel object
  SapModel = EtabsObject.SapModel

'initialize model
  ret = SapModel.InitializeNewModel()

'create steel deck template model
  ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'set steel design code
  ret = SapModel.DesignSteel.SetCode("CSA S16-14")

'get overwrite item
  ret = SapModel.DesignSteel.Canadian_S16_14.GetOverwrite("8", 1, Value, ProgDet)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

[cDStCanadian_S16_14 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDStCanadian_S16_14AddLanguageSpecificTextSet("L Method

Retrieves the value of a steel design preference item

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDStCanadian_S16_14
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: SystemInt32

This is an integer between 1 and 21, inclusive, indicating the preference item considered

1. Multi-response case design
2. Framing type
3. Spectral Acceleration Ratio, $I_e \cdot F_a \cdot S_a(0.2)$
4. Ductility related modification factor, R_d
5. Overstrength related modification factor, R_o

6. Capacity factor, Phi bending
7. Capacity factor, Phi compression
8. Capacity factor, Phi tension
9. Capacity factor, Phi shear
10. Slender section modification
11. Ignore seismic code
12. Ignore special seismic load
13. Doubler plate is plug welded
14. Consider deflection
15. DL deflection limit, L/Value
16. SDL + LL deflection limit, L/Value
17. LL deflection limit, L/Value
18. Total load deflection limit, L/Value
19. Total camber limit, L/Value
20. Pattern live load factor
21. Demand/capacity ratio limit

Value

Type: SystemDouble

The value of the considered preference item

1. Multi-response case design
 - **1** - Envelopes
 - **2** - Step-by-step
 - **3** - Last step
 - **4** - Envelopes -- All
 - **5** - Step-by-step -- All
2. Framing type
 - **1** - Type LD MRF
 - **2** - Type MD MRF
 - **3** - Type D MRF
 - **4** - Type LD CBF(V)
 - **5** - Type LD CBF(TC)
 - **6** - Type LD CBF(TO)
 - **7** - Type LD CBF(OT)
 - **8** - Type MD CBF(V)
 - **9** - Type MD CBF(TC)
 - **10** - Type MD CBF(TO)
 - **11** - Type MD CBF(OT)
 - **12** - EBF
 - **13** - Cantilever Column
 - **14** - Conventional MF
 - **15** - Conventional BF
3. Spectral Acceleration Ratio, $I_e \cdot F_a \cdot S_a(0.2)$

Value > 0

4. Ductility related modification factor, R_d

Value > 0

5. Overstrength related modification factor, R_o

Value > 0

6. Capacity factor, Phi bending

Value > 0

7. Capacity factor, Phi compression

Value > 0

8. Capacity factor, Phi tension

Value > 0

9. Capacity factor, Phi shear

Value > 0

10. Slender section modification

· **1** - modified geometry

· **2** - modified Fy

11. Ignore seismic code

· 0 = No

· Any other value = Yes

12. Ignore special seismic load

· 0 = No

· Any other value = Yes

13. Doubler plate is plug welded

· 0 = No

· Any other value = Yes

14. Consider deflection

· 0 = No

· Any other value = Yes

15. DL deflection limit, L/Value

Value > 0

16. SDL + LL deflection limit, L/Value

Value > 0

17. LL deflection limit, L/Value

Value > 0

18. Total deflection limit, L/Value

Value > 0

19. Total camber limit, L/Value

Value > 0

20. Pattern live load factor

Value >= 0

21. Demand/capacity ratio limit

Value > 0

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise, it returns a nonzero value

Remarks

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set steel design code
    ret = SapModel.DesignSteel.SetCode("CSA S16-14")

    'get preference item
    ret = SapModel.DesignSteel.Canadian_S16_14.GetPreference(4, Value)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

See Also

Reference

cDStCanadian_S16_14 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDStCanadian_S16_14AddLanguageSpecificTextSet("L Method

Sets the value of a steel design overwrite item

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDStCanadian_S16_14
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing frame object with a steel frame design procedure

Item

Type: SystemInt32

This is an integer between 1 and 39, inclusive, indicating the overwrite item considered

1. Framing type
2. Consider deflection
3. Deflection check type
4. DL deflection limit, L/Value
5. SDL + LL deflection limit, L/Value
6. LL deflection limit, L/Value
7. Total load deflection limit, L/Value
8. Total-camber limit, L/Value
9. DL deflection limit, absolute
10. SDL + LL deflection limit, absolute
11. LL deflection limit, absolute
12. Total load deflection limit, absolute
13. Total-camber limit, absolute
14. Specified camber
15. Net area to total area ratio
16. Live load reduction factor
17. Unbraced length ratio, Major
18. Unbraced length ratio, Minor LTB
19. Unbraced length ratio, Lateral Torsional Buckling
20. Effective length factor, K Major
21. Effective length factor, K Minor
22. Effective length factor, K LTB
23. Moment coefficient, Omega1 Major
24. Moment coefficient, Omega1 Minor
25. Bending coefficient, Omega2
26. Nonsway moment factor, U1 Major
27. Nonsway moment factor, U1 Minor
28. Sway moment factor, U2 Major
29. Sway moment factor, U2 Minor
30. Parameter for compressive resistance, n
31. Yield stress, Fy
32. Expected to specified Fy ratio, Ry
33. Compressive resistance, Cr
34. Tensile resistance, Tr
35. Major bending resistance, Mr3
36. Minor bending resistance, Mr2
37. Major shear resistance, Vr2
38. Minor shear resistance, Vr3
39. Demand/capacity ratio limit

Value

Type: SystemDouble

The value of the considered overwrite item

1. Framing type
 - **0** - As specified in preferences
 - **1** - Type LD MRF
 - **2** - Type MD MRF
 - **3** - Type D MRF
 - **4** - Type LD CBF(V)
 - **5** - Type LD CBF(TC)
 - **6** - Type LD CBF(TO)
 - **7** - Type LD CBF(OT)
 - **8** - Type MD CBF(V)
 - **9** - Type MD CBF(TC)
 - **10** - Type MD CBF(TO)
 - **11** - Type MD CBF(OT)
 - **12** - EBF
 - **13** - Cantilever Column
 - **14** - Conventional MF
 - **15** - Conventional BF

2. Consider deflection?
 - 0 = No
 - Any other value = Yes

3. Deflection check type
 - **0** - Program Determined
 - **1** - Ratio
 - **2** - Absolute
 - **3** - Both

4. DL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

5. SDL + LL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

6. LL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

7. Total load deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

8. Total-camber limit, L/Value

Value ≥ 0 ; 0 means no check for this item

9. DL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

10. SDL + LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

11. LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

12. Total load deflection limit, absolute

Introduction

Value ≥ 0 ; 0 means no check for this item. [L]

13. Total-camber limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

14. Specified camber

Value ≥ 0 . [L]

15. Net area to total area ratio

Value ≥ 0 ; 0 means use program default value

16. Live load reduction factor

Value ≥ 0 ; 0 means use program determined value

17. Unbraced length ratio, Major

Value ≥ 0 ; 0 means use program determined value

18. Unbraced length ratio, Minor

Value ≥ 0 ; 0 means use program determined value

19. Unbraced length ratio, Lateral Torsional Buckling

Value ≥ 0 ; 0 means use program determined value

20. Effective length factor, K Major

Value ≥ 0 ; 0 means use program determined value

21. Effective length factor, K Minor

Value ≥ 0 ; 0 means use program determined value

22. Effective length factor, K LTB

Value ≥ 0 ; 0 means use program determined value

23. Moment coefficient, Omega1 Major

Value ≥ 0 ; 0 means use program determined value

24. Moment coefficient, Omega1 Minor

Value ≥ 0 ; 0 means use program determined value

25. Moment coefficient, Omega2

Value ≥ 0 ; 0 means use program determined value

26. Nonsway moment factor, U1 Major

Value ≥ 0 ; 0 means use program determined value

27. Nonsway moment factor, U1 Minor

Value ≥ 0 ; 0 means use program determined value

28. Sway moment factor, U2 Major

Value ≥ 0 ; 0 means use program determined value

29. Sway moment factor, U2 Minor

Introduction

Value ≥ 0 ; 0 means use program determined value

30. Parameter for compressive resistance, n

Value ≥ 0 ; 0 means use program determined value

31. Yield stress, F_y

Value ≥ 0 ; 0 means use program determined value [F/L2]

32. Expected to specified F_y ratio, R_y

Value ≥ 0 ; 0 means use program determined value [F/L2]=

33. Compressive resistance, C_r

Value ≥ 0 ; 0 means use program determined value [F]

34. Tensile resistance, T_r

Value ≥ 0 ; 0 means use program determined value [F]

35. Major bending resistance, M_{r3}

Value ≥ 0 ; 0 means use program determined value [FL]

36. Minor bending resistance, M_{r2}

Value ≥ 0 ; 0 means use program determined value [FL]

37. Major shear resistance, V_{r2}

Value ≥ 0 ; 0 means use program determined value [F]

38. Minor shear resistance, V_{r3}

Value ≥ 0 ; 0 means use program determined value [F]

39. Demand/capacity ratio limit

Value ≥ 0 ; 0 means use program determined value

ItemType (Optional)

Type: $\hat{A}$ ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◇ Object = 0

◇ Group = 1

◇ SelectedObjects = 2

If this item is Objects, the assignment is made to the frame object specified by the Name item.

If this item is Group, the assignment is made to all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the assignment is made to all selected frame objects and the Name item is ignored.

Return Value

Type: $\hat{A}$ Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set steel design code
    ret = SapModel.DesignSteel.SetCode("CSA S16-14")

    'set overwrite item
    ret = SapModel.DesignSteel.Canadian_S16_14.SetOverwrite("8", 1, 7)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

[cDStCanadian_S16_14 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDStCanadian_S16_14AddLanguageSpecificTextSet("L Method

Sets the value of a steel design preference item

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDStCanadian_S16_14
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: SystemInt32

This is an integer between 1 and 21, inclusive, indicating the preference item considered

1. Multi-response case design
2. Framing type
3. Spectral Acceleration Ratio, $I_e \cdot F_a \cdot S_a(0.2)$
4. Ductility related modification factor, R_d
5. Overstrength related modification factor, R_o

6. Capacity factor, Phi bending
7. Capacity factor, Phi compression
8. Capacity factor, Phi tension
9. Capacity factor, Phi shear
10. Slender section modification
11. Ignore seismic code
12. Ignore special seismic load
13. Doubler plate is plug welded
14. Consider deflection
15. DL deflection limit, L/Value
16. SDL + LL deflection limit, L/Value
17. LL deflection limit, L/Value
18. Total load deflection limit, L/Value
19. Total camber limit, L/Value
20. Pattern live load factor
21. Demand/capacity ratio limit

Value

Type: SystemDouble

The value of the considered preference item

1. Multi-response case design
 - **1** - Envelopes
 - **2** - Step-by-step
 - **3** - Last step
 - **4** - Envelopes -- All
 - **5** - Step-by-step -- All
2. Framing type
 - **1** - Type LD MRF
 - **2** - Type MD MRF
 - **3** - Type D MRF
 - **4** - Type LD CBF(V)
 - **5** - Type LD CBF(TC)
 - **6** - Type LD CBF(TO)
 - **7** - Type LD CBF(OT)
 - **8** - Type MD CBF(V)
 - **9** - Type MD CBF(TC)
 - **10** - Type MD CBF(TO)
 - **11** - Type MD CBF(OT)
 - **12** - EBF
 - **13** - Cantilever Column
 - **14** - Conventional MF
 - **15** - Conventional BF
3. Spectral Acceleration Ratio, $I_e \cdot F_a \cdot S_a(0.2)$

Value > 0

4. Ductility related modification factor, R_d

Value > 0

5. Overstrength related modification factor, R_o

Value > 0

6. Capacity factor, Phi bending

Value > 0

7. Capacity factor, Phi compression

Value > 0

8. Capacity factor, Phi tension

Value > 0

9. Capacity factor, Phi shear

Value > 0

10. Slender section modification

· **1** - modified geometry

· **2** - modified Fy

11. Ignore seismic code

· 0 = No

· Any other value = Yes

12. Ignore special seismic load

· 0 = No

· Any other value = Yes

13. Doubler plate is plug welded

· 0 = No

· Any other value = Yes

14. Consider deflection

· 0 = No

· Any other value = Yes

15. DL deflection limit, L/Value

Value > 0

16. SDL + LL deflection limit, L/Value

Value > 0

17. LL deflection limit, L/Value

Value > 0

18. Total deflection limit, L/Value

Value > 0

19. Total camber limit, L/Value

Value > 0

20. Pattern live load factor

Value >= 0

21. Demand/capacity ratio limit

Value > 0

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise, it returns a nonzero value

Remarks

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set steel design code
    ret = SapModel.DesignSteel.SetCode("CSA S16-14")

    'set preference item
    ret = SapModel.DesignSteel.Canadian_S16_14.SetPreference(2, 6)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

See Also

Reference

cDStCanadian_S16_14 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDStCanadian_S16_19 Interface

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDStCanadian_S16_19
```

```
Public Interface cDStCanadian_S16_19
```

```
Dim instance As cDStCanadian_S16_19
```

```
public interface class cDStCanadian_S16_19
```

```
type cDStCanadian_S16_19 = interface end
```

The cDStCanadian_S16_19 type exposes the following members.

▲ Methods

▲	Name	Description
⇒	GetOverwrite	Retrieves the value of a steel design overwrite item
⇒	GetPreference	Retrieves the value of a steel design preference item
⇒	SetOverwrite	Sets the value of a steel design overwrite item
⇒	SetPreference	Sets the value of a steel design preference item

[Top](#)

▲ See Also

Reference

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cDStCanadian_S16_19 Methods

The cDStCanadian_S16_19 type exposes the following members.

Methods

⚡	Name	Description
⇒	<u>GetOverwrite</u>	Retrieves the value of a steel design overwrite item
⇒	<u>GetPreference</u>	Retrieves the value of a steel design preference item
⇒	<u>SetOverwrite</u>	Sets the value of a steel design overwrite item
⇒	<u>SetPreference</u>	Sets the value of a steel design preference item

Top

⚡ See Also

Reference

cDStCanadian_S16_19 Interface

ETABSV1 Namespace

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CSI API ETABS v1

cDStCanadian_S16_19AddLanguageSpecificTextSet("L Method

Retrieves the value of a steel design overwrite item

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDStCanadian_S16_19
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing frame object with a steel frame design procedure

Item

Type: SystemInt32

This is an integer between 1 and 39, inclusive, indicating the overwrite item considered

1. Framing type
2. Consider deflection
3. Deflection check type
4. DL deflection limit, L/Value
5. SDL + LL deflection limit, L/Value
6. LL deflection limit, L/Value
7. Total load deflection limit, L/Value
8. Total-camber limit, L/Value
9. DL deflection limit, absolute
10. SDL + LL deflection limit, absolute
11. LL deflection limit, absolute
12. Total load deflection limit, absolute
13. Total-camber limit, absolute
14. Specified camber
15. Net area to total area ratio
16. Live load reduction factor
17. Unbraced length ratio, Major
18. Unbraced length ratio, Minor LTB
19. Unbraced length ratio, Lateral Torsional Buckling
20. Effective length factor, K Major
21. Effective length factor, K Minor
22. Effective length factor, K LTB
23. Moment coefficient, Omega1 Major
24. Moment coefficient, Omega1 Minor
25. Bending coefficient, Omega2
26. Nonsway moment factor, U1 Major
27. Nonsway moment factor, U1 Minor
28. Sway moment factor, U2 Major
29. Sway moment factor, U2 Minor
30. Parameter for compressive resistance, n
31. Yield stress, Fy
32. Expected to specified Fy ratio, Ry
33. Compressive resistance, Cr
34. Tensile resistance, Tr
35. Major bending resistance, Mr3
36. Minor bending resistance, Mr2
37. Major shear resistance, Vr2
38. Minor shear resistance, Vr3
39. Demand/capacity ratio limit

Value

Type: SystemDouble

The value of the considered overwrite item

1. Framing type
 - **0** - As specified in preferences
 - **1** - Type LD MRF
 - **2** - Type MD MRF
 - **3** - Type D MRF
 - **4** - Type LD CBF(V)
 - **5** - Type LD CBF(TC)
 - **6** - Type LD CBF(TO)
 - **7** - Type LD CBF(OT)
 - **8** - Type MD CBF(V)
 - **9** - Type MD CBF(TC)
 - **10** - Type MD CBF(TO)
 - **11** - Type MD CBF(OT)
 - **12** - EBF
 - **13** - Cantilever Column
 - **14** - Conventional MF
 - **15** - Conventional BF

2. Consider deflection?
 - 0 = No
 - Any other value = Yes

3. Deflection check type
 - **0** - Program Determined
 - **1** - Ratio
 - **2** - Absolute
 - **3** - Both

4. DL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

5. SDL + LL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

6. LL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

7. Total load deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

8. Total-camber limit, L/Value

Value ≥ 0 ; 0 means no check for this item

9. DL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

10. SDL + LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

11. LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

12. Total load deflection limit, absolute

Introduction

Value ≥ 0 ; 0 means no check for this item. [L]

13. Total-camber limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

14. Specified camber

Value ≥ 0 . [L]

15. Net area to total area ratio

Value ≥ 0 ; 0 means use program default value

16. Live load reduction factor

Value ≥ 0 ; 0 means use program determined value

17. Unbraced length ratio, Major

Value ≥ 0 ; 0 means use program determined value

18. Unbraced length ratio, Minor

Value ≥ 0 ; 0 means use program determined value

19. Unbraced length ratio, Lateral Torsional Buckling

Value ≥ 0 ; 0 means use program determined value

20. Effective length factor, K Major

Value ≥ 0 ; 0 means use program determined value

21. Effective length factor, K Minor

Value ≥ 0 ; 0 means use program determined value

22. Effective length factor, K LTB

Value ≥ 0 ; 0 means use program determined value

23. Moment coefficient, Omega1 Major

Value ≥ 0 ; 0 means use program determined value

24. Moment coefficient, Omega1 Minor

Value ≥ 0 ; 0 means use program determined value

25. Moment coefficient, Omega2

Value ≥ 0 ; 0 means use program determined value

26. Nonsway moment factor, U1 Major

Value ≥ 0 ; 0 means use program determined value

27. Nonsway moment factor, U1 Minor

Value ≥ 0 ; 0 means use program determined value

28. Sway moment factor, U2 Major

Value ≥ 0 ; 0 means use program determined value

29. Sway moment factor, U2 Minor

Introduction

Value ≥ 0 ; 0 means use program determined value
30. Parameter for compressive resistance, n

Value ≥ 0 ; 0 means use program determined value
31. Yield stress, F_y

Value ≥ 0 ; 0 means use program determined value [F/L2]
32. Expected to specified F_y ratio, R_y

Value ≥ 0 ; 0 means use program determined value [F/L2]=
33. Compressive resistance, C_r

Value ≥ 0 ; 0 means use program determined value [F]
34. Tensile resistance, T_r

Value ≥ 0 ; 0 means use program determined value [F]
35. Major bending resistance, M_{r3}

Value ≥ 0 ; 0 means use program determined value [FL]
36. Minor bending resistance, M_{r2}

Value ≥ 0 ; 0 means use program determined value [FL]
37. Major shear resistance, V_{r2}

Value ≥ 0 ; 0 means use program determined value [F]
38. Minor shear resistance, V_{r3}

Value ≥ 0 ; 0 means use program determined value [F]
39. Demand/capacity ratio limit

Value ≥ 0 ; 0 means use program determined value

ProgDet

Type: System.Boolean

If this item is True, the specified value is program determined

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim Value As Double  
    Dim ProgDet As Boolean
```

'create ETABS object

```
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

Return Value

```
'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'set steel design code
ret = SapModel.DesignSteel.SetCode("CSA S16-19")

'get overwrite item
ret = SapModel.DesignSteel.Canadian_S16_19.GetOverwrite("8", 1, Value, ProgDet)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

[cDStCanadian_S16_19 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDStCanadian_S16_19AddLanguageSpecificTextSet("L Method

Retrieves the value of a steel design preference item

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDStCanadian_S16_19
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: SystemInt32

This is an integer between 1 and 22, inclusive, indicating the preference item considered

1. Multi-response case design
2. Framing type
3. Spectral Acceleration Ratio, $I_e \cdot F_a \cdot S_a(0.2)$
4. Ductility related modification factor, R_d
5. Overstrength related modification factor, R_o

6. Capacity factor, Phi bending
7. Capacity factor, Phi compression
8. Capacity factor, Phi tension
9. Capacity factor, Phi shear
10. Slender section modification
11. Ignore seismic code
12. Ignore special seismic load
13. Doubler plate is plug welded
14. Consider deflection
15. DL deflection limit, L/Value
16. SDL + LL deflection limit, L/Value
17. LL deflection limit, L/Value
18. Total load deflection limit, L/Value
19. Total camber limit, L/Value
20. Pattern live load factor
21. Demand/capacity ratio limit
22. Seismic Design Category

Value

Type: SystemDouble

The value of the considered preference item

1. Multi-response case design
 - **1** - Envelopes
 - **2** - Step-by-step
 - **3** - Last step
 - **4** - Envelopes -- All
 - **5** - Step-by-step -- All
2. Framing type
 - **1** - Type LD MRF
 - **2** - Type MD MRF
 - **3** - Type D MRF
 - **4** - Type LD CBF(V)
 - **5** - Type LD CBF(TC)
 - **6** - Type LD CBF(TO)
 - **7** - Type LD CBF(OT)
 - **8** - Type MD CBF(V)
 - **9** - Type MD CBF(TC)
 - **10** - Type MD CBF(TO)
 - **11** - Type MD CBF(OT)
 - **12** - EBF
 - **13** - Cantilever Column
 - **14** - Conventional MF
 - **15** - Conventional BF
3. Spectral Acceleration Ratio, $I_e \cdot F_a \cdot S_a(0.2)$

Value > 0

4. Ductility related modification factor, R_d

Value > 0

5. Overstrength related modification factor, R_o

Value > 0

6. Capacity factor, Phi bending

Value > 0

7. Capacity factor, Phi compression

Value > 0

8. Capacity factor, Phi tension

Value > 0

9. Capacity factor, Phi shear

Value > 0

10. Slender section modification

- **1** - modified geometry
- **2** - modified Fy

11. Ignore seismic code

- 0 = No
- Any other value = Yes

12. Ignore special seismic load

- 0 = No
- Any other value = Yes

13. Doubler plate is plug welded

- 0 = No
- Any other value = Yes

14. Consider deflection

- 0 = No
- Any other value = Yes

15. DL deflection limit, L/Value

Value > 0

16. SDL + LL deflection limit, L/Value

Value > 0

17. LL deflection limit, L/Value

Value > 0

18. Total deflection limit, L/Value

Value > 0

19. Total camber limit, L/Value

Value > 0

20. Pattern live load factor

Value >= 0

21. Demand/capacity ratio limit

Value > 0

22. Seismic Design Category

- **1** - SC0
- **2** - SC1

- 3 - SC2
- 4 - SC3
- 5 - SC4

Return Value

Type: [Int32](#)

Returns zero if the item is successfully retrieved; otherwise, it returns a nonzero value

▲ Remarks

▲ Examples

VB

[Copy](#)

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set steel design code
    ret = SapModel.DesignSteel.SetCode("CSA S16-19")

    'get preference item
    ret = SapModel.DesignSteel.Canadian_S16_19.GetPreference(4, Value)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

[cDStCanadian_S16_19 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDStCanadian_S16_19AddLanguageSpecificTextSet("L Method

Sets the value of a steel design overwrite item

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDStCanadian_S16_19
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

cDStCanadian_S16_19span id="LSTBA15809E_0"AddLanguageSpecificTextSet("LSTBA15809E_30?cpp=::

Parameters

Name

Type: SystemString

The name of an existing frame object with a steel frame design procedure

Item

Type: SystemInt32

This is an integer between 1 and 39, inclusive, indicating the overwrite item considered

1. Framing type
2. Consider deflection
3. Deflection check type
4. DL deflection limit, L/Value
5. SDL + LL deflection limit, L/Value
6. LL deflection limit, L/Value
7. Total load deflection limit, L/Value
8. Total-camber limit, L/Value
9. DL deflection limit, absolute
10. SDL + LL deflection limit, absolute
11. LL deflection limit, absolute
12. Total load deflection limit, absolute
13. Total-camber limit, absolute
14. Specified camber
15. Net area to total area ratio
16. Live load reduction factor
17. Unbraced length ratio, Major
18. Unbraced length ratio, Minor LTB
19. Unbraced length ratio, Lateral Torsional Buckling
20. Effective length factor, K Major
21. Effective length factor, K Minor
22. Effective length factor, K LTB
23. Moment coefficient, Omega1 Major
24. Moment coefficient, Omega1 Minor
25. Bending coefficient, Omega2
26. Nonsway moment factor, U1 Major
27. Nonsway moment factor, U1 Minor
28. Sway moment factor, U2 Major
29. Sway moment factor, U2 Minor
30. Parameter for compressive resistance, n
31. Yield stress, Fy
32. Expected to specified Fy ratio, Ry
33. Compressive resistance, Cr
34. Tensile resistance, Tr
35. Major bending resistance, Mr3
36. Minor bending resistance, Mr2
37. Major shear resistance, Vr2
38. Minor shear resistance, Vr3
39. Demand/capacity ratio limit

Value

Type: SystemDouble

The value of the considered overwrite item

1. Framing type
 - **0** - As specified in preferences
 - **1** - Type LD MRF
 - **2** - Type MD MRF
 - **3** - Type D MRF
 - **4** - Type LD CBF(V)
 - **5** - Type LD CBF(TC)
 - **6** - Type LD CBF(TO)
 - **7** - Type LD CBF(OT)
 - **8** - Type MD CBF(V)
 - **9** - Type MD CBF(TC)
 - **10** - Type MD CBF(TO)
 - **11** - Type MD CBF(OT)
 - **12** - EBF
 - **13** - Cantilever Column
 - **14** - Conventional MF
 - **15** - Conventional BF

2. Consider deflection?
 - 0 = No
 - Any other value = Yes

3. Deflection check type
 - **0** - Program Determined
 - **1** - Ratio
 - **2** - Absolute
 - **3** - Both

4. DL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

5. SDL + LL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

6. LL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

7. Total load deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

8. Total-camber limit, L/Value

Value ≥ 0 ; 0 means no check for this item

9. DL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

10. SDL + LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

11. LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

12. Total load deflection limit, absolute

Introduction

Value ≥ 0 ; 0 means no check for this item. [L]

13. Total-camber limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

14. Specified camber

Value ≥ 0 . [L]

15. Net area to total area ratio

Value ≥ 0 ; 0 means use program default value

16. Live load reduction factor

Value ≥ 0 ; 0 means use program determined value

17. Unbraced length ratio, Major

Value ≥ 0 ; 0 means use program determined value

18. Unbraced length ratio, Minor

Value ≥ 0 ; 0 means use program determined value

19. Unbraced length ratio, Lateral Torsional Buckling

Value ≥ 0 ; 0 means use program determined value

20. Effective length factor, K Major

Value ≥ 0 ; 0 means use program determined value

21. Effective length factor, K Minor

Value ≥ 0 ; 0 means use program determined value

22. Effective length factor, K LTB

Value ≥ 0 ; 0 means use program determined value

23. Moment coefficient, Omega1 Major

Value ≥ 0 ; 0 means use program determined value

24. Moment coefficient, Omega1 Minor

Value ≥ 0 ; 0 means use program determined value

25. Moment coefficient, Omega2

Value ≥ 0 ; 0 means use program determined value

26. Nonsway moment factor, U1 Major

Value ≥ 0 ; 0 means use program determined value

27. Nonsway moment factor, U1 Minor

Value ≥ 0 ; 0 means use program determined value

28. Sway moment factor, U2 Major

Value ≥ 0 ; 0 means use program determined value

29. Sway moment factor, U2 Minor

Introduction

Value ≥ 0 ; 0 means use program determined value

30. Parameter for compressive resistance, n

Value ≥ 0 ; 0 means use program determined value

31. Yield stress, F_y

Value ≥ 0 ; 0 means use program determined value [F/L2]

32. Expected to specified F_y ratio, R_y

Value ≥ 0 ; 0 means use program determined value [F/L2]=

33. Compressive resistance, C_r

Value ≥ 0 ; 0 means use program determined value [F]

34. Tensile resistance, T_r

Value ≥ 0 ; 0 means use program determined value [F]

35. Major bending resistance, M_{r3}

Value ≥ 0 ; 0 means use program determined value [FL]

36. Minor bending resistance, M_{r2}

Value ≥ 0 ; 0 means use program determined value [FL]

37. Major shear resistance, V_{r2}

Value ≥ 0 ; 0 means use program determined value [F]

38. Minor shear resistance, V_{r3}

Value ≥ 0 ; 0 means use program determined value [F]

39. Demand/capacity ratio limit

Value ≥ 0 ; 0 means use program determined value

ItemType (Optional)

Type: $\hat{A}$ ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◇ Object = 0

◇ Group = 1

◇ SelectedObjects = 2

If this item is Objects, the assignment is made to the frame object specified by the Name item.

If this item is Group, the assignment is made to all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the assignment is made to all selected frame objects and the Name item is ignored.

Return Value

Type: $\hat{A}$ Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set steel design code
    ret = SapModel.DesignSteel.SetCode("CSA S16-19")

    'set overwrite item
    ret = SapModel.DesignSteel.Canadian_S16_19.SetOverwrite("8", 1, 7)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

[cDStCanadian_S16_19 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDStCanadian_S16_19AddLanguageSpecificTextSet("L Method

Sets the value of a steel design preference item

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDStCanadian_S16_19
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: [SystemInt32](#)

This is an integer between 1 and 22, inclusive, indicating the preference item considered

1. Multi-response case design
2. Framing type
3. Spectral Acceleration Ratio, $I_e \cdot F_a \cdot S_a(0.2)$
4. Ductility related modification factor, R_d
5. Overstrength related modification factor, R_o

6. Capacity factor, Phi bending
7. Capacity factor, Phi compression
8. Capacity factor, Phi tension
9. Capacity factor, Phi shear
10. Slender section modification
11. Ignore seismic code
12. Ignore special seismic load
13. Doubler plate is plug welded
14. Consider deflection
15. DL deflection limit, L/Value
16. SDL + LL deflection limit, L/Value
17. LL deflection limit, L/Value
18. Total load deflection limit, L/Value
19. Total camber limit, L/Value
20. Pattern live load factor
21. Demand/capacity ratio limit
22. Seismic Design Category

Value

Type: SystemDouble

The value of the considered preference item

1. Multi-response case design
 - **1** - Envelopes
 - **2** - Step-by-step
 - **3** - Last step
 - **4** - Envelopes -- All
 - **5** - Step-by-step -- All
2. Framing type
 - **1** - Type LD MRF
 - **2** - Type MD MRF
 - **3** - Type D MRF
 - **4** - Type LD CBF(V)
 - **5** - Type LD CBF(TC)
 - **6** - Type LD CBF(TO)
 - **7** - Type LD CBF(OT)
 - **8** - Type MD CBF(V)
 - **9** - Type MD CBF(TC)
 - **10** - Type MD CBF(TO)
 - **11** - Type MD CBF(OT)
 - **12** - EBF
 - **13** - Cantilever Column
 - **14** - Conventional MF
 - **15** - Conventional BF
3. Spectral Acceleration Ratio, $I_e \cdot F_a \cdot S_a(0.2)$

Value > 0

4. Ductility related modification factor, R_d

Value > 0

5. Overstrength related modification factor, R_o

Value > 0

6. Capacity factor, Phi bending

Value > 0

7. Capacity factor, Phi compression

Value > 0

8. Capacity factor, Phi tension

Value > 0

9. Capacity factor, Phi shear

Value > 0

10. Slender section modification

- **1** - modified geometry
- **2** - modified Fy

11. Ignore seismic code

- 0 = No
- Any other value = Yes

12. Ignore special seismic load

- 0 = No
- Any other value = Yes

13. Doubler plate is plug welded

- 0 = No
- Any other value = Yes

14. Consider deflection

- 0 = No
- Any other value = Yes

15. DL deflection limit, L/Value

Value > 0

16. SDL + LL deflection limit, L/Value

Value > 0

17. LL deflection limit, L/Value

Value > 0

18. Total deflection limit, L/Value

Value > 0

19. Total camber limit, L/Value

Value > 0

20. Pattern live load factor

Value >= 0

21. Demand/capacity ratio limit

Value > 0

22. Seismic Design Category

- **1** - SC0
- **2** - SC1

- 3 - SC2
- 4 - SC3
- 5 - SC4

Return Value

Type: [Int32](#)

Returns zero if the item is successfully retrieved; otherwise, it returns a nonzero value

▲ Remarks

▲ Examples

VB

[Copy](#)

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set steel design code
    ret = SapModel.DesignSteel.SetCode("CSA S16-19")

    'set preference item
    ret = SapModel.DesignSteel.Canadian_S16_19.SetPreference(2, 6)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

[cDStCanadian_S16_19 Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cDStChinese_2010 Interface

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDStChinese_2010
```

```
Public Interface cDStChinese_2010
```

```
Dim instance As cDStChinese_2010
```

```
public interface class cDStChinese_2010
```

```
type cDStChinese_2010 = interface end
```

The cDStChinese_2010 type exposes the following members.

▲ Methods

▲	Name	Description
---	------	-------------

⇒	GetOverwrite	
---	------------------------------	--

⇒	GetPreference	
---	-------------------------------	--

⇒	SetOverwrite	
---	------------------------------	--

⇒	SetPreference	
---	-------------------------------	--

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▲ See Also

Reference

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cDStChinese_2010 Methods

The cDStChinese_2010 type exposes the following members.

Methods

	Name	Description
⇒	<u>GetOverwrite</u>	
⇒	<u>GetPreference</u>	
⇒	<u>SetOverwrite</u>	
⇒	<u>SetPreference</u>	

Top

See Also

Reference

cDStChinese_2010 Interface

ETABSV1 Namespace

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CSI API ETABS v1

cDStChinese_2010AddLanguageSpecificTextSet("LST94E5D18_0?cp=1750|nu=.") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDStChinese_2010
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```

Parameters

Name

Type: SystemString

cDStChinese_2010AddLanguageSpecificTextSet("LST94E5D18_0?cp=1750|nu=.")

Item

Type:Â [SystemInt32](#)

Value

Type:Â [SystemDouble](#)

ProgDet

Type:Â [SystemBoolean](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cDStChinese_2010 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDStChinese_2010AddLanguageSpecificTextSet("LST5 Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDStChinese_2010
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: SystemInt32

Value

Type: SystemDouble

Return Value

Type: Int32

▲ See Also

Reference

cDStChinese_2010 Interface

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CSI API ETABS v1

cDStChinese_2010AddLanguageSpecificTextSet("LST20D96EFF_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDStChinese_2010
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

Item

Type: SystemInt32

Value

Type: SystemDouble

ItemType (Optional)

Type: ETABSv1eItemType

Return Value

Type: Int32

▲ See Also

Reference

cDStChinese_2010 Interface

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CSI API ETABS v1

cDStChinese_2010AddLanguageSpecificTextSet("LST9 Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDStChinese_2010
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: SystemInt32

Value

Type: SystemDouble

Return Value

Type: Int32

▲ See Also

Reference

cDStChinese_2010 Interface

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CSI API ETABS v1

cDStChinese_2018 Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDStChinese_2018
```

```
Public Interface cDStChinese_2018
```

```
Dim instance As cDStChinese_2018
```

```
public interface class cDStChinese_2018
```

```
type cDStChinese_2018 = interface end
```

The cDStChinese_2018 type exposes the following members.

▲ Methods

▲	Name	Description
⇒	GetOverwrite	Sets the value of a steel design overwrite item.
⇒	GetPreference	Sets the value of a steel design preference item.
⇒	SetOverwrite	Sets the value of a steel design overwrite item.
⇒	SetPreference	Sets the value of a steel design preference item.

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDStChinese_2018 Methods

The cDStChinese_2018 type exposes the following members.

Methods

⌵	Name	Description
⇒	<u>GetOverwrite</u>	Sets the value of a steel design overwrite item.
⇒	<u>GetPreference</u>	Sets the value of a steel design preference item.
⇒	<u>SetOverwrite</u>	Sets the value of a steel design overwrite item.
⇒	<u>SetPreference</u>	Sets the value of a steel design preference item.

Top

⌵ See Also

Reference

cDStChinese_2018 Interface

ETABSV1 Namespace

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CSI API ETABS v1

cDStChinese_2018AddLanguageSpecificTextSet("LSTD37E262F_0") Method

Sets the value of a steel design overwrite item.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDStChinese_2018
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```

Parameters

Name

Type: SystemString

Item

Type: SystemInt32

This is an integer between 1 and 45, inclusive, indicating the overwrite item considered.

1. Framing Type
2. Element Type
3. Seismic Design Grade
4. Energy Dissipation Member?
5. Is Transfer Member?
6. Seismic Magnification Factor
7. Dual System Magnification Factor
8. Ignore B/T Check?
9. Classify Beam as Flexo-Compression Member?
10. Is Rolled Section?
11. Is Flange Edge Cut by Gas?
12. Is Both End Pinned?
13. Ignore Beam Φ_B ?
14. Is Beam Top Loaded?
15. Consider Deflection?
16. Live Load Deflection Limit, L/Value
17. Total Load Deflection Limit, L/Value
18. Total-Camber Deflection Limit, L/Value
19. Specified Camber
20. Net Area to Total Area Ratio
21. Live Load Reduction Factor
22. Unbraced Length Ratio (Major)
23. Unbraced Length Ratio (Minor), Lateral Torsional Buckling
24. Effective Length Factor (μ_e Major)
25. Effective Length Factor (μ_e Minor)
26. Sway M Amplification Factor (α_{II} Major)
27. Sway M Amplification Factor (α_{II} Minor)
28. Axial Stability Coefficient (Φ Major)
29. Axial Stability Coefficient (Φ Minor)
30. Flexural Stability Coefficient (Φ_b Major)
31. Flexural Stability Coefficient (Φ_b Minor)
32. Moment Coefficient (β_m Major)
33. Moment Coefficient (β_m Minor)
34. Moment Coefficient (β_t Major)
35. Moment Coefficient (β_t Minor)
36. Plasticity Factor (γ Major)
37. Plasticity Factor (γ Minor)
38. Section Influence Coefficient (η)
39. Beam/Column Capacity Factor (η)
40. Slenderness Limit in Compression (l/r)
41. Slenderness Limit in Tension (l/r)
42. Yield stress, F_y
43. Allowable Normal Stress, f

- 44. Allowable Shear Stress, f_v
- 45. Consider Fictitious shear?

Value

Type: SystemDouble

The value of the considered overwrite item.

- 1. Framing Type
 - 1. Sway Moment Frame, SMF
 - 2. Centrically Braced Frame, CBF
 - 3. Eccentrically Braced Frame, EBF
 - 4. NonSway Moment Frame, NMF
- 2. Element Type
 - 1. Column
 - 2. Beam
 - 3. Brace
- 3. Seismic Design Grade
 - 1. Grade I
 - 2. Grade II
 - 3. Grade III
 - 4. Grade IV
 - 5. Non Seismic
- 4. Energy Dissipation Member?

0 = No

Any other value = Yes

- 5. Is Transfer Member?

0 = No

Any other value = Yes

- 6. Seismic Magnification Factor

Value > 0

- 7. Dual System Magnification Factor

Value > 0

- 8. Ignore B/T Check?

0 = No

Any other value = Yes

- 9. "Classify Beam as Flexo-Compression Member?"

0 = No

Any other value = Yes

- 10. "Is Rolled Section?"

0 = No

Any other value = Yes

11. "Is Flange Edge Cut by Gas?"

0 = No

Any other value = Yes

12. "Is Both End Pinned?"

0 = No

Any other value = Yes

13. "Ignore Beam PhiB?"

0 = No

Any other value = Yes

14. "Is Beam Top Loaded?"

0 = No

Any other value = Yes

15. Consider deflection

0 = No

Any other value = Yes

16. "Live Load Deflection Limit, L/Value"

Value > 0

17. "Total Load Deflection Limit, L/Value"

Value > 0

18. "Total-Camber Deflection Limit, L/Value"

Value > 0

19. "Specified Camber"

Value > 0

20. "Net Area to Total Area Ratio"

Value > 0

21. "Live Load Reduction Factor"

Value > 0

22. Unbraced Length Ratio (Major)

Value > 0

23. Unbraced Length Ratio (Minor), Lateral Torsional Buckling

Value > 0

24. Effective Length Factor (Mue Major)

Value > 0

25. Effective Length Factor (μ_e Minor)

Value > 0

26. Sway M Amplification Factor (α_{II} Major)

Value > 0

27. Sway M Amplification Factor (α_{II} Minor)

Value > 0

28. Axial Stability Coefficient (ϕ Major)

Value > 0

29. Axial Stability Coefficient (ϕ Minor)

Value > 0

30. Flexural Stability Coefficient (ϕ_b Major)

Value > 0

31. Flexural Stability Coefficient (ϕ_b Minor)

Value > 0

32. Moment Coefficient (β_m Major)

Value > 0

33. Moment Coefficient (β_m Minor)

Value > 0

34. Moment Coefficient (β_t Major)

Value > 0

35. Moment Coefficient (β_t Minor)

Value > 0

36. Plasticity Factor (γ Major)

Value > 0

37. Plasticity Factor (γ Minor)

Value > 0

38. Section Influence Coefficient (η)

Value > 0

39. Beam/Column Capacity Factor (η)

Value > 0

40. Slenderness Limit in Compression (l/r)

Value > 0

41. Slenderness Limit in Tension (l/r)

Value > 0

42. Yield stress, F_y

Value > 0

43. Allowable Normal Stress, f

Value > 0

44. Allowable Shear Stress, f_v

Value > 0

45. Consider Fictitious shear?

0 = No

Any other value = Yes

ProgDet

Type: SystemBoolean

If this item is True, the specified value is program determined.

Return Value

Type: Int32

Returns zero if the item is successfully set; otherwise it returns a nonzero value.

▲ Remarks

▲ See Also

Reference

cDStChinese_2018 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDStChinese_2018AddLanguageSpecificTextSet("LST246189CA_0") Method

Sets the value of a steel design preference item.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDStChinese_2018
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: [SystemInt32](#)

This is an integer between 1 and 29, inclusive, indicating the preference item considered.

1. Multi-Response Case Designn
2. Framing Type
3. Seismic Design Grade
4. Ignore b/t Check?
5. Classify Beam as Flexo-Compression Member?

6. Analysis Method
7. Stability Factor
8. Ignore Beam PhiB?
9. Reserved for Future Use
10. Reserved for Future Use
11. Consider Deflection?
12. LL deflection limit, L/Value
13. Total Load Deflection Limit, L/Value
14. Total - Camber Limit, L/Value
15. Reserved for Future Use
16. Reserved for Future Use
17. Reserved for Future Use
18. Reserved for Future Use
19. Reserved for Future Use
20. Reserved for Future Use
21. Demand/Capacity Ratio Limit
22. Maximum Number of Auto Iterations

Value

Type: SystemDouble

The value of the considered preference item.

1. Multi-Response Case Designn
 1. Envelopes
 2. Step-by-step
 3. Last step
 4. Envelopes -- All
 5. Step-by-step -- All
2. Framing Type
 1. Sway Moment Frame, SMF
 2. Centrally Braced Frame, CBF
 3. Eccentrically Braced Frame, EBF
 4. NonSway Moment Frame, NMF
3. Seismic Design Grade
 1. Grade I
 2. Grade II
 3. Grade III
 4. Grade IV
 5. Non Seismic
4. Ignore b/t Check?

0 = No

Any other value = Yes
5. Classify Beam as Flexo-Compression Member?

0 = No

Any other value = Yes
6. Analysis Method
 1. Direct Analysis
 2. Second Order
 3. Amplified 1st Order

- 4. Limited 1st Order
- 7. Stability Factor

Value > 0

- 8. Ignore Beam PhiB?

0 = No

Any other value = Yes

- 9. Reserved for Future Use
- 10. Reserved for Future Use
- 11. Consider deflection

0 = No

Any other value = Yes

- 12. LL deflection limit, L/Value

Value > 0

- 13. Total Load Deflection Limit, L/Value

Value > 0

- 14. Total - Camber Limit, L/Value

Value > 0

- 15. Reserved for Future Use
- 16. Reserved for Future Use
- 17. Reserved for Future Use
- 18. Reserved for Future Use
- 19. Reserved for Future Use
- 20. Reserved for Future Use
- 21. Demand/Capacity Ratio Limit

Value > 0

- 22. Maximum Number of Auto Iterations

Value >= 1

Return Value

Type: Int32

Returns zero if the item is successfully set; otherwise it returns a nonzero value.

▲ Remarks

▲ See Also

Reference

cDStChinese 2018 Interface

ETABSv1 Namespace

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Namespace: ETABSv1

▲ Syntax

VB

F#

Copy

```
Function SetOverwrite (
Name As String,
Item As Integer,
Value As Double,
Optional OptType As eItemType = eItemType.Objects
) As Integer
```

```
returnValue = instance.SetOverwrite(Name,
Item, Value, ItemType)
```

```
int SetOverwrite(
String^ Name,
int Item,
double Value,
eItemType ItemType = eItemType::Objects
)
```

cDStChinese_2018span id="LST5F63F36F_0"AddLanguageSpecificTextSet("LST5F63F36F_01770::|nu=

Parameters

Name

Type: SystemString

Item

Type: SystemInt32

This is an integer between 1 and 45, inclusive, indicating the overwrite item considered.

1. Framing Type
2. Element Type
3. Seismic Design Grade
4. Energy Dissipation Member?
5. Is Transfer Member?
6. Seismic Magnification Factor
7. Dual System Magnification Factor
8. Ignore B/T Check?
9. Classify Beam as Flexo-Compression Member?
10. Is Rolled Section?
11. Is Flange Edge Cut by Gas?
12. Is Both End Pinned?
13. Ignore Beam Φ_B ?
14. Is Beam Top Loaded?
15. Consider Deflection?
16. Live Load Deflection Limit, L/Value
17. Total Load Deflection Limit, L/Value
18. Total-Camber Deflection Limit, L/Value
19. Specified Camber
20. Net Area to Total Area Ratio
21. Live Load Reduction Factor
22. Unbraced Length Ratio (Major)
23. Unbraced Length Ratio (Minor), Lateral Torsional Buckling
24. Effective Length Factor (μ_e Major)
25. Effective Length Factor (μ_e Minor)
26. Sway M Amplification Factor (α_{II} Major)
27. Sway M Amplification Factor (α_{II} Minor)
28. Axial Stability Coefficient (Φ Major)
29. Axial Stability Coefficient (Φ Minor)
30. Flexural Stability Coefficient (Φ_b Major)
31. Flexural Stability Coefficient (Φ_b Minor)
32. Moment Coefficient (β_m Major)
33. Moment Coefficient (β_m Minor)
34. Moment Coefficient (β_t Major)
35. Moment Coefficient (β_t Minor)
36. Plasticity Factor (γ Major)
37. Plasticity Factor (γ Minor)
38. Section Influence Coefficient (η)
39. Beam/Column Capacity Factor (η)
40. Slenderness Limit in Compression (l/r)
41. Slenderness Limit in Tension (l/r)
42. Yield stress, F_y
43. Allowable Normal Stress, f

- 44. Allowable Shear Stress, f_v
- 45. Consider Fictitious shear?

Value

Type: SystemDouble

The value of the considered overwrite item.

- 1. Framing Type
 - 1. Sway Moment Frame, SMF
 - 2. Centrically Braced Frame, CBF
 - 3. Eccentrically Braced Frame, EBF
 - 4. NonSway Moment Frame, NMF
- 2. Element Type
 - 1. Column
 - 2. Beam
 - 3. Brace
- 3. Seismic Design Grade
 - 1. Grade I
 - 2. Grade II
 - 3. Grade III
 - 4. Grade IV
 - 5. Non Seismic
- 4. Energy Dissipation Member?

0 = No

Any other value = Yes

- 5. Is Transfer Member?

0 = No

Any other value = Yes

- 6. Seismic Magnification Factor

Value > 0

- 7. Dual System Magnification Factor

Value > 0

- 8. Ignore B/T Check?

0 = No

Any other value = Yes

- 9. "Classify Beam as Flexo-Compression Member?"

0 = No

Any other value = Yes

- 10. "Is Rolled Section?"

0 = No

Any other value = Yes

11. "Is Flange Edge Cut by Gas?"

0 = No

Any other value = Yes

12. "Is Both End Pinned?"

0 = No

Any other value = Yes

13. "Ignore Beam PhiB?"

0 = No

Any other value = Yes

14. "Is Beam Top Loaded?"

0 = No

Any other value = Yes

15. Consider deflection

0 = No

Any other value = Yes

16. "Live Load Deflection Limit, L/Value"

Value > 0

17. "Total Load Deflection Limit, L/Value"

Value > 0

18. "Total-Camber Deflection Limit, L/Value"

Value > 0

19. "Specified Camber"

Value > 0

20. "Net Area to Total Area Ratio"

Value > 0

21. "Live Load Reduction Factor"

Value > 0

22. Unbraced Length Ratio (Major)

Value > 0

23. Unbraced Length Ratio (Minor), Lateral Torsional Buckling

Value > 0

24. Effective Length Factor (Mue Major)

Value > 0

25. Effective Length Factor (μ_e Minor)

Value > 0

26. Sway M Amplification Factor (α_{II} Major)

Value > 0

27. Sway M Amplification Factor (α_{II} Minor)

Value > 0

28. Axial Stability Coefficient (ϕ Major)

Value > 0

29. Axial Stability Coefficient (ϕ Minor)

Value > 0

30. Flexural Stability Coefficient (ϕ_b Major)

Value > 0

31. Flexural Stability Coefficient (ϕ_b Minor)

Value > 0

32. Moment Coefficient (β_m Major)

Value > 0

33. Moment Coefficient (β_m Minor)

Value > 0

34. Moment Coefficient (β_t Major)

Value > 0

35. Moment Coefficient (β_t Minor)

Value > 0

36. Plasticity Factor (γ Major)

Value > 0

37. Plasticity Factor (γ Minor)

Value > 0

38. Section Influence Coefficient (η)

Value > 0

39. Beam/Column Capacity Factor (η)

Value > 0

40. Slenderness Limit in Compression (l/r)

Value > 0

41. Slenderness Limit in Tension (l/r)

Value > 0

42. Yield stress, F_y

Value > 0

43. Allowable Normal Stress, f

Value > 0

44. Allowable Shear Stress, f_v

Value > 0

45. Consider Fictitious shear?

0 = No

Any other value = Yes

ItemType (Optional)

Type: ETABSv1eItemType

Return Value

Type: Int32

Returns zero if the item is successfully set; otherwise it returns a nonzero value.

▀ Remarks

▀ See Also

Reference

cDStChinese_2018 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDStChinese_2018AddLanguageSpecificTextSet("LST8FB5C44F_0") Method

Sets the value of a steel design preference item.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDStChinese_2018
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: [System.Int32](#)

This is an integer between 1 and 29, inclusive, indicating the preference item considered.

1. Multi-Response Case Design
2. Framing Type
3. Seismic Design Grade
4. Ignore b/t Check?
5. Classify Beam as Flexo-Compression Member?

6. Analysis Method
7. Stability Factor
8. Ignore Beam PhiB?
9. Reserved for Future Use
10. Reserved for Future Use
11. Consider Deflection?
12. LL deflection limit, L/Value
13. Total Load Deflection Limit, L/Value
14. Total - Camber Limit, L/Value
15. Reserved for Future Use
16. Reserved for Future Use
17. Reserved for Future Use
18. Reserved for Future Use
19. Reserved for Future Use
20. Reserved for Future Use
21. Demand/Capacity Ratio Limit
22. Maximum Number of Auto Iterations

Value

Type: SystemDouble

The value of the considered preference item.

1. Multi-Response Case Designn
 1. Envelopes
 2. Step-by-step
 3. Last step
 4. Envelopes -- All
 5. Step-by-step -- All
2. Framing Type
 1. Sway Moment Frame, SMF
 2. Concentrically Braced Frame, CBF
 3. Eccentrically Braced Frame, EBF
 4. NonSway Moment Frame, NMF
3. Seismic Design Grade
 1. Grade I
 2. Grade II
 3. Grade III
 4. Grade IV
 5. Non Seismic
4. Ignore b/t Check?

0 = No

Any other value = Yes
5. Classify Beam as Flexo-Compression Member?

0 = No

Any other value = Yes
6. Analysis Method
 1. Direct Analysis
 2. Second Order
 3. Amplified 1st Order

- 4. Limited 1st Order
- 7. Stability Factor

Value > 0

- 8. Ignore Beam PhiB?

0 = No

Any other value = Yes

- 9. Reserved for Future Use
- 10. Reserved for Future Use
- 11. Consider deflection

0 = No

Any other value = Yes

- 12. LL deflection limit, L/Value

Value > 0

- 13. Total Load Deflection Limit, L/Value

Value > 0

- 14. Total - Camber Limit, L/Value

Value > 0

- 15. Reserved for Future Use
- 16. Reserved for Future Use
- 17. Reserved for Future Use
- 18. Reserved for Future Use
- 19. Reserved for Future Use
- 20. Reserved for Future Use
- 21. Demand/Capacity Ratio Limit

Value > 0

- 22. Maximum Number of Auto Iterations

Value >= 1

Return Value

Type: Int32

Returns zero if the item is successfully set; otherwise it returns a nonzero value.

▲ Remarks

▲ See Also

Reference

cDStChinese 2018 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDStEurocode_3_2005 Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDStEurocode_3_2005
```

```
Public Interface cDStEurocode_3_2005
```

```
Dim instance As cDStEurocode_3_2005
```

```
public interface class cDStEurocode_3_2005
```

```
type cDStEurocode_3_2005 = interface end
```

The cDStEurocode_3_2005 type exposes the following members.

▲ Methods

▲	Name	Description
⇒	GetOverwrite	Retrieves the value of a steel design overwrite item
⇒	GetPreference	Retrieves the value of a steel design preference item.
⇒	SetOverwrite	Sets the value of a steel design overwrite item
⇒	SetPreference	Sets the value of a steel design preference item.

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDStEurocode_3_2005 Methods

The [cDStEurocode_3_2005](#) type exposes the following members.

Methods

	Name	Description
⇒	GetOverwrite	Retrieves the value of a steel design overwrite item
⇒	GetPreference	Retrieves the value of a steel design preference item.
⇒	SetOverwrite	Sets the value of a steel design overwrite item
⇒	SetPreference	Sets the value of a steel design preference item.

[Top](#)

See Also

Reference

[cDStEurocode_3_2005 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDStEurocode_3_2005AddLanguageSpecificTextSet("L Method

Retrieves the value of a steel design overwrite item

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDStEurocode_3_2005
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```

Parameters

Name

Type: SystemString

The name of a frame object with a steel frame design procedure

Item

Type: SystemInt32

This is an integer between 1 and 59, inclusive, indicating the overwrite item considered.

1. Framing Type
2. Consider Deflection
3. Deflection check type
4. DL deflection limit, L/Value
5. SDL + LL deflection limits, L/Value
6. LL deflection limit, L/Value
7. Total load deflection limit, L/Value
8. Total camber limit, L/Value
9. DL deflection limit, absolute
10. SDL + LL deflection limit, absolute
11. LL deflection limit, absolute
12. Total load deflection limit, absolute
13. Total camber limit, absolute
14. Specified camber
15. Net area to total area ratio
16. Live load reduction factor
17. Unbraced length ratio, L_y
18. Unbraced length ratio, L_z
19. Effective length factor, K_{2y}
20. Effective length factor, K_{2z}
21. Moment coefficient, k_{yy}
22. Moment coefficient, k_{zz}
23. Bending coefficient, C_1
24. Effective length factor, K_{1y}
25. Effective length factor, K_{1z}
26. Yield stress, F_y
27. Compressive capacity, N_c, R_d
28. Tensile capacity, N_t, R_d
29. Bending capacity Y-Y, M_{cy}, R_d
30. Bending capacity Z-Z, M_{cz}, R_d
31. Buckling resistance moment, M_b, R_d
32. Shear capacity Z-Z, V_z, R_d
33. Shear capacity Y-Y, V_y, R_d
34. Demand/capacity ratio limit
35. Section class
36. Column buckling curve, y-y
37. Column buckling curve, z-z
38. Buckling curve for LTB
39. System overstrength factor, Ω
40. Is rolled section
41. Unbraced length ratio, LLTB
44. Effective length factor, KLTB

- 45. Material overstrength, GammaOv
- 49. Bending coefficient, C2
- 50. Bending coefficient, C3
- 51. Warping constant, Iw
- 52. Elastic torsional buckling force, Ncr,T
- 53. Elastic torsional-flexural buckling force, Ncr,TF
- 54. Elastic critical moment for lateral-torsional buckling, Mcr
- 55. Moment coefficient, kzy
- 56. Moment coefficient, kyz
- 57. Warping coefficient, kw
- 58. Coordinate of load application, za
- 59. Coordinate of shear center, zs

Value

Type: SystemDouble

The numeric value of the considered overwrite item.

1. Design Section
2. Framing Type
 - . 0 = Program Default
 - . 1 = DCH-MRF (Ductility Class High - Moment Frame)
 - . 2 = DCM-MRF (Ductility Class Medium - Moment Frame)
 - . 3 = DCL-MRF (Ductility Class Low - Moment Frame)
 - . 4 = DCH-CBF (Ductility Class High - Concentrically Braced Frame)
 - . 5 = DCM-CBF (Ductility Class Medium - Concentrically Braced Frame)
 - . 6 = DCL-CBF (Ductility Class Low - Concentrically Braced Frame)
 - . 7 = DCH-EBF (Ductility Class High - Eccentrically Braced Frame)
 - . 8 = DCM-EBF (Ductility Class Medium - Eccentrically Braced Frame)
 - . 9 = DCL-EBF (Ductility Class Low - Eccentrically Braced Frame)
 - . 10 = Inverted Pendulum Structure
 - . 11 = Secondary
 - . 12 = Non Seismic
3. Consider Deflection
 - . 0 = Program Default
 - . 1 = No
 - . 2 = Yes
4. Deflection check type
 - . 0 = Program Default
 - . 1 = Ratio
 - . 2 = Absolute
 - . 3 = Both
5. DL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item.
6. SDL + LL deflection limits, L/Value

Value ≥ 0 ; 0 means no check for this item.
7. Value ≥ 0 ; 0 means no check for this item.

LL deflection limit, L/Value
8. Total load deflection limit, L/Value

Introduction

Value ≥ 0 ; 0 means no check for this item.

9. Total camber limit, $L/Value$

Value ≥ 0 ; 0 means no check for this item.

10. DL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

11. SDL + LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

12. LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

13. Total load deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

14. Total camber limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

15. Specified camber

Value ≥ 0 ; 0 means no check for this item. [L]

16. Net area to total area ratio

Value ≥ 0 ; 0 means use program determined value.

17. Live load reduction factor

Value ≥ 0 ; 0 means use program determined value.

18. Unbraced length ratio, L_y

Value ≥ 0 ; 0 means use program determined value.

19. Unbraced length ratio, L_z

Value ≥ 0 ; 0 means use program determined value.

20. Effective length factor, K_{2y}

Value ≥ 0 ; 0 means use program determined value.

21. Effective length factor, K_{2z}

Value ≥ 0 ; 0 means use program determined value.

22. Moment coefficient, k_{yy}

Value ≥ 0 ; 0 means use program determined value.

23. Moment coefficient, k_{zz}

Value ≥ 0 ; 0 means use program determined value.

24. Bending coefficient, C_1

Value ≥ 0 ; 0 means use program determined value.

25. Effective length factor, K_{1y}

Value ≥ 0 ; 0 means use program determined value.

26. Effective length factor, $K1z$

Value ≥ 0 ; 0 means use program determined value.

27. Yield stress, F_y

Value ≥ 0 ; 0 means use program determined value. $[F/L^2]$

28. Compressive capacity, N_c, R_d

Value ≥ 0 ; 0 means use program determined value. $[F]$

29. Tensile capacity, N_t, R_d

Value ≥ 0 ; 0 means use program determined value. $[F]$

30. Bending capacity Y-Y, M_{cy}, R_d

Value ≥ 0 ; 0 means use program determined value. $[F.L]$

31. Bending capacity Z-Z, M_{cz}, R_d

Value ≥ 0 ; 0 means use program determined value. $[F.L]$

32. Buckling resistance moment, M_b, R_d

Value ≥ 0 ; 0 means use program determined value. $[F.L]$

33. Shear capacity Z-Z, V_z, R_d

Value ≥ 0 ; 0 means use program determined value. $[F]$

34. Shear capacity Y-Y, V_y, R_d

Value ≥ 0 ; 0 means use program determined value. $[F]$

35. Demand/capacity ratio limit

Value ≥ 0 ; 0 means use program determined value.

36. Section Class

- 0 = Program Default
- 1 = Class 1
- 2 = Class 2
- 3 = Class 3
- 4 = Class 4

37. Column buckling curve (y-y)

- 0 = Program Default
- 1 = a0
- 2 = a
- 3 = b
- 4 = c
- 5 = d

38. Column buckling curve (z-z)

- 0 = Program Default
- 1 = a0
- 2 = a
- 3 = b
- 4 = c
- 5 = d

39. Buckling curve for LTB

- 0 = Program Default
- 1 = a0
- 2 = a
- 3 = b
- 4 = c
- 5 = d

40. System overstrength factor, Omega

Value ≥ 0 ; 0 means use a program determined value.

41. Is rolled section

- 0 = Program Default
- 1 = No
- 2 = Yes

42. Unbraced length ratio, LLTB

Value ≥ 0 ; 0 means use program determined value.

44. Effective length factor, KLTB

Value ≥ 0 ; 0 means use program determined value.

45. Material overstrength, GammaOv

Value ≥ 0 ; 0 means use program determined value.

49. Bending coefficient, C2

Value ≥ 0 ; 0 means use program determined value.

50. Bending coefficient, C3

Value ≥ 0 ; 0 means use program determined value.

51. Warping constant, Iw

Value ≥ 0 ; 0 means use program determined value.

52. Elastic torsional buckling force, Ncr,T

Value ≥ 0 ; 0 means use program determined value.

53. Elastic torsional-flexural buckling force, Ncr,TF

Value ≥ 0 ; 0 means use program determined value.

54. Elastic critical moment for lateral-torsional buckling, Mcr

Value ≥ 0 ; 0 means use program determined value.

55. Moment coefficient, kzy

Value ≥ 0 ; 0 means use program determined value.

56. Moment coefficient, kyz

Value ≥ 0 ; 0 means use program determined value.

57. Warping coefficient, kw

Value ≥ 0 ; 0 means use program determined value.

58. Coordinate of load application, za

59. Coordinate of shear center, zs

ProgDet

Type: System.Boolean

If this item is True, the specified value is program determined.

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double
    Dim ProgDet As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set steel design code
    ret = SapModel.DesignSteel.SetCode("Eurocode 3-2005")

    'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis

    'start steel design
    ret = SapModel.DesignSteel.StartDesign

    'get overwrite item
    ret = SapModel.DesignSteel.Eurocode_3_2005.GetOverwrite("8", 2, Value, ProgDet)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

Return Value

▲ See Also

Reference

cDStEurocode_3_2005 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDStEurocode_3_2005AddLanguageSpecificTextSet("L Method

Retrieves the value of a steel design preference item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDStEurocode_3_2005
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: SystemInt32

This is an integer between 1 and 24, inclusive, indicating the preference item considered.

1. Country
2. Combos equation
3. K factor method
4. Framing type
5. Partial safety factor, GammaM0

6. Partial safety factor, GammaM1
7. Partial safety factor, GammaM2
8. Consider deflection
9. DL deflection limit, L/Value
10. SDL + LL deflection limit, L/Value
11. LL deflection limit, L/Value
12. Total load deflection limit, L/Value
13. Total camber limit, L/Value
14. Pattern live load factor
15. Demand/capacity ratio limit
16. Multi-response case design
17. Reliability class
18. Behavior factor, q
19. System overstrength, Omega
20. Consider P-delta
21. Consider torsion
22. Ignore seismic code
23. Ignore special seismic load
24. Are Doubler plates plug-welded

Value

Type: SystemDouble

The numeric value of the considered preference item.

1. Country
 - . 1 = CEN Default
 - . 2 = United Kingdom
 - . 3 = Slovenia
 - . 4 = Bulgaria
 - . 5 = Norway
 - . 6 = Singapore
 - . 7 = Sweden
 - . 8 = Finland
 - . 9 = Denmark
 - . 10 = Portugal
 - . 11 = Germany
 - . 12 = Poland
 - . 13 = Ireland
2. Combos equation
 - . 1 = Eq. 6.10
 - . 2 = Max of Eqs. 6.10a and 6.10b
3. K factor method
 - . 1 = Method A
 - . 2 = Method B
 - . 3 = Both
4. Framing type
 - . 1 = DCH-MRF (Ductility Class High - Moment Frame)
 - . 2 = DCM-MRF (Ductility Class Medium - Moment Frame)
 - . 3 = DCL-MRF (Ductility Class Low - Moment Frame)
 - . 4 = DCH-CBF (Ductility Class High - Concentrically Braced Frame)
 - . 5 = DCM-CBF (Ductility Class Medium - Concentrically Braced Frame)
 - . 6 = DCL-CBF (Ductility Class Low - Concentrically Braced Frame)

Introduction

- 7 = DCH-EBF (Ductility Class High - Eccentrically Braced Frame)
- 8 = DCM-EBF (Ductility Class Medium - Eccentrically Braced Frame)
- 9 = DCL-EBF (Ductility Class Low - Eccentrically Braced Frame)
- 10 = Inverted Pendulum Structure
- 11 = Secondary
- 12 = Non Seismic

5. Partial safety factor, GammaM0

Value > 0

6. Partial safety factor, GammaM1

Value > 0

7. Partial safety factor, GammaM2

Value > 0

8. Consider deflection

0 = No

Any other value = Yes

9. DL deflection limit, L/Value

Value > 0

10. SDL + LL deflection limit, L/Value

Value > 0

11. LL deflection limit, L/Value

Value > 0

12. Total load deflection limit, L/Value

Value > 0

13. Total camber limit limit, L/Value

Value > 0

14. Pattern live load factor

Value > 0

15. Demand/capacity ratio limit

Value > 0

16. Multi-response case design

- 1 = Envelopes
- 2 = Step-by-Step
- 3 = Last Step
- 4 = Envelope -- All
- 5 = Step-by-Step -- All

17. Reliability class

- 1 = Class 1
- 2 = Class 2

- 3 = Class 3
18. Behavior factor, q
- Value > 0
19. System overstrength, Ω
- Value > 0
20. Consider P-delta
- 0 = No
- Any other value = Yes
21. Consider Torsion
- 0 = No
- Any other value = Yes
22. Ignore seismic code
- 0 = No
- Any other value = Yes
23. Ignore special seismic load
- 0 = No
- Any other value = Yes
24. Are Doubler plates plug-welded
- 0 = No
- Any other value = Yes

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double
    Dim ProgDet As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
```

Return Value

Introduction

```
ret = EtabsObject.ApplicationStart()  
  
'create SapModel object  
  SapModel = EtabsObject.SapModel  
  
'initialize model  
  ret = SapModel.InitializeNewModel()  
  
'create steel deck template model  
  ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)  
  
'set steel design code  
  ret = SapModel.DesignSteel.SetCode("Eurocode 3-2005")  
  
'set overwrite item  
  ret = SapModel.DesignSteel.Eurocode_3_2005.GetPreference(2, Value)  
  
'close ETABS  
  EtabsObject.ApplicationExit(False)  
  
'clean up variables  
  SapModel = Nothing  
  EtabsObject = Nothing  
  
End Sub
```

📌 See Also

Reference

cDStEurocode_3_2005 Interface

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CSI API ETABS v1

cDStEurocode_3_2005AddLanguageSpecificTextSet("L Method

Sets the value of a steel design overwrite item

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDStEurocode_3_2005
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters**Name**

Type: SystemString

The name of an existing frame object or group, depending on the value of the ItemType item.

Item

Type: SystemInt32

This is an integer between 1 and 59, inclusive, indicating the overwrite item considered.

1. Framing Type
2. Consider Deflection
3. Deflection check type
4. DL deflection limit, L/Value
5. SDL + LL deflection limits, L/Value
6. LL deflection limit, L/Value
7. Total load deflection limit, L/Value
8. Total camber limit, L/Value
9. DL deflection limit, absolute
10. SDL + LL deflection limit, absolute
11. LL deflection limit, absolute
12. Total load deflection limit, absolute
13. Total camber limit, absolute
14. Specified camber
15. Net area to total area ratio
16. Live load reduction factor
17. Unbraced length ratio, Ly
18. Unbraced length ratio, Lz
19. Effective length factor, K2y
20. Effective length factor, K2z
21. Moment coefficient, kyy
22. Moment coefficient, kzz
23. Bending coefficient, C1
24. Effective length factor, K1y
25. Effective length factor, K1z
26. Yield stress, Fy
27. Compressive capacity, Nc,Rd
28. Tensile capacity Nt,Rd
29. Bending capacity Y-Y, Mcy,Rd
30. Bending capacity Z-Z, Mcz,Rd
31. Buckling resistance moment, Mb,Rd
32. Shear capacity Z-Z, Vz,Rd
33. Shear capacity Y-Y, Vy,Rd
34. Demand/capacity ratio limit
35. Section class
36. Column buckling curve, y-y
37. Column buckling curve, z-z
38. Buckling curve for LTB
39. System overstrength factor, Omega
40. Is rolled section
41. Unbraced length ratio, LTB

- 44. Effective length factor, K LTB
- 45. Material overstrength, GammaOv
- 49. Bending coefficient, C2
- 50. Bending coefficient, C3
- 51. Warping constant, Iw
- 52. Elastic torsional buckling force, Ncr,T
- 53. Elastic torsional-flexural buckling force, Ncr,TF
- 54. Elastic critical moment for lateral-torsional buckling, Mcr
- 55. Moment coefficient, kzy
- 56. Moment coefficient, kyz
- 57. Warping coefficient, kw
- 58. Coordinate of load application, za
- 59. Coordinate of shear center, zs

Value

Type: SystemDouble

The numeric value of the considered overwrite item.

1. Design Section
2. Framing Type
 - 0 = Program Default
 - 1 = DCH-MRF (Ductility Class High - Moment Frame)
 - 2 = DCM-MRF (Ductility Class Medium - Moment Frame)
 - 3 = DCL-MRF (Ductility Class Low - Moment Frame)
 - 4 = DCH-CBF (Ductility Class High - Concentrically Braced Frame)
 - 5 = DCM-CBF (Ductility Class Medium - Concentrically Braced Frame)
 - 6 = DCL-CBF (Ductility Class Low - Concentrically Braced Frame)
 - 7 = DCH-EBF (Ductility Class High - Eccentrically Braced Frame)
 - 8 = DCM-EBF (Ductility Class Medium - Eccentrically Braced Frame)
 - 9 = DCL-EBF (Ductility Class Low - Eccentrically Braced Frame)
 - 10 = Inverted Pendulum Structure
 - 11 = Secondary
 - 12 = Non Seismic
3. Consider Deflection
 - 0 = Program Default
 - 1 = No
 - 2 = Yes
4. Deflection check type
 - 0 = Program Default
 - 1 = Ratio
 - 2 = Absolute
 - 3 = Both
5. DL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item.

6. SDL + LL deflection limits, L/Value

Value ≥ 0 ; 0 means no check for this item.

7. Value ≥ 0 ; 0 means no check for this item.

LL deflection limit, L/Value

Introduction

8. Total load deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item.

9. Total camber limit, L/Value

Value ≥ 0 ; 0 means no check for this item.

10. DL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

11. SDL + LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

12. LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

13. Total load deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

14. Total camber limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

15. Specified camber

Value ≥ 0 ; 0 means no check for this item. [L]

16. Net area to total area ratio

Value ≥ 0 ; 0 means use program determined value.

17. Live load reduction factor

Value ≥ 0 ; 0 means use program determined value.

18. Unbraced length ratio, L_y

Value ≥ 0 ; 0 means use program determined value.

19. Unbraced length ratio, L_z

Value ≥ 0 ; 0 means use program determined value.

20. Effective length factor, K_{2y}

Value ≥ 0 ; 0 means use program determined value.

21. Effective length factor, K_{2z}

Value ≥ 0 ; 0 means use program determined value.

22. Moment coefficient, k_{yy}

Value ≥ 0 ; 0 means use program determined value.

23. Moment coefficient, k_{zz}

Value ≥ 0 ; 0 means use program determined value.

24. Bending coefficient, C_1

Value ≥ 0 ; 0 means use program determined value.

25. Effective length factor, K_{1y}

Value ≥ 0 ; 0 means use program determined value.

26. Effective length factor, K_{1z}

Value ≥ 0 ; 0 means use program determined value.

27. Yield stress, F_y

Value ≥ 0 ; 0 means use program determined value. $[F/L^2]$

28. Compressive capacity, N_c, R_d

Value ≥ 0 ; 0 means use program determined value. $[F]$

29. Tensile capacity N_t, R_d

Value ≥ 0 ; 0 means use program determined value. $[F]$

30. Bending capacity Y-Y, M_{cy}, R_d

Value ≥ 0 ; 0 means use program determined value. $[F.L]$

31. Bending capacity Z-Z, M_{cz}, R_d

Value ≥ 0 ; 0 means use program determined value. $[F.L]$

32. Buckling resistance moment, M_b, R_d

Value ≥ 0 ; 0 means use program determined value. $[F.L]$

33. Shear capacity Z-Z, V_z, R_d

Value ≥ 0 ; 0 means use program determined value. $[F]$

34. Shear capacity Y-Y, V_y, R_d

Value ≥ 0 ; 0 means use program determined value. $[F]$

35. Demand/capacity ratio limit

Value ≥ 0 ; 0 means use program determined value.

36. Section Class

- . 0 = Program Default
- . 1 = Class 1
- . 2 = Class 2
- . 3 = Class 3
- . 4 = Class 4

37. Column buckling curve (y-y)

- . 0 = Program Default
- . 1 = a0
- . 2 = a
- . 3 = b
- . 4 = c
- . 5 = d

38. Column buckling curve (z-z)

- . 0 = Program Default
- . 1 = a0
- . 2 = a
- . 3 = b

- 4 = c
- 5 = d
- 39. Buckling curve for LTB
 - 0 = Program Default
 - 1 = a0
 - 2 = a
 - 3 = b
 - 4 = c
 - 5 = d
- 40. System overstrength factor, Omega

Value ≥ 0 ; 0 means use a program determined value.
- 41. Is rolled section
 - 0 = Program Default
 - 1 = No
 - 2 = Yes
- 42. Unbraced length ratio, LTB

Value ≥ 0 ; 0 means use program determined value.
- 44. Effective length factor, KLTB

Value ≥ 0 ; 0 means use program determined value.
- 45. Material overstrength, GammaOv

Value ≥ 0 ; 0 means use program determined value.
- 49. Bending coefficient, C2

Value ≥ 0 ; 0 means use program determined value.
- 50. Bending coefficient, C3

Value ≥ 0 ; 0 means use program determined value.
- 51. Warping constant, Iw

Value ≥ 0 ; 0 means use program determined value.
- 52. Elastic torsional buckling force, Ncr,T

Value ≥ 0 ; 0 means use program determined value.
- 53. Elastic torsional-flexural buckling force, Ncr,TF

Value ≥ 0 ; 0 means use program determined value.
- 54. Elastic critical moment for lateral-torsional buckling, Mcr

Value ≥ 0 ; 0 means use program determined value.
- 55. Moment coefficient, kzy

Value ≥ 0 ; 0 means use program determined value.
- 56. Moment coefficient, kyz

Value ≥ 0 ; 0 means use program determined value.
- 57. Warping coefficient, kw

Value ≥ 0 ; 0 means use program determined value.

Value ≥ 0 ; 0 means use program determined value.

58. Coordinate of load application, za

59. Coordinate of shear center, zs

ItemType (Optional)

Type: $\hat{A}$ ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the assignment is made to the frame object specified by the Name item.

If this item is Group, the assignment is made to all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the assignment is made to all selected frame objects and the Name item is ignored.

Return Value

Type: $\hat{A}$ Int32

Returns zero if the item is successfully set; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double
    Dim ProgDet As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set steel design code
    ret = SapModel.DesignSteel.SetCode("Eurocode 3-2005")

    'set overwrite item
    ret = SapModel.DesignSteel.Eurocode_3_2005.SetOverwrite("8", 2, 1)

    'close ETABS
```

Introduction

```
EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

[cDStEurocode_3_2005 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDStEurocode_3_2005AddLanguageSpecificTextSet("L Method

Sets the value of a steel design preference item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDStEurocode_3_2005
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: SystemInt32

This is an integer between 1 and 24, inclusive, indicating the preference item considered.

1. Country
2. Combos equation
3. K factor method
4. Framing type
5. Partial safety factor, GammaM0

6. Partial safety factor, Γ_{M1}
7. Partial safety factor, Γ_{M2}
8. Consider deflection
9. DL deflection limit, $L/Value$
10. SDL + LL deflection limit, $L/Value$
11. LL deflection limit, $L/Value$
12. Total load deflection limit, $L/Value$
13. Total camber limit, $L/Value$
14. Pattern live load factor
15. Demand/capacity ratio limit
16. Multi-response case design
17. Reliability class
18. Behavior factor, q
19. System overstrength, Ω
20. Consider P-delta
21. Consider torsion
22. Ignore seismic code
23. Ignore special seismic load
24. Are Doubler plates plug-welded

Value

Type: SystemDouble

The numeric value of the considered preference item.

1. Country
 - 1 = CEN Default
 - 2 = United Kingdom
 - 3 = Slovenia
 - 4 = Bulgaria
 - 5 = Norway
 - 6 = Singapore
 - 7 = Sweden
 - 8 = Finland
 - 9 = Denmark
 - 10 = Portugal
 - 11 = Germany
 - 12 = Poland
 - 13 = Ireland
2. Combos equation
 - 1 = Eq. 6.10
 - 2 = Max of Eqs. 6.10a and 6.10b
3. K factor method
 - 1 = Method A
 - 2 = Method B
 - 3 = Both
4. Framing type
 - 1 = DCH-MRF (Ductility Class High - Moment Frame)
 - 2 = DCM-MRF (Ductility Class Medium - Moment Frame)
 - 3 = DCL-MRF (Ductility Class Low - Moment Frame)
 - 4 = DCH-CBF (Ductility Class High - Concentrically Braced Frame)
 - 5 = DCM-CBF (Ductility Class Medium - Concentrically Braced Frame)
 - 6 = DCL-CBF (Ductility Class Low - Concentrically Braced Frame)

Introduction

- 7 = DCH-EBF (Ductility Class High - Eccentrically Braced Frame)
- 8 = DCM-EBF (Ductility Class Medium - Eccentrically Braced Frame)
- 9 = DCL-EBF (Ductility Class Low - Eccentrically Braced Frame)
- 10 = Inverted Pendulum Structure
- 11 = Secondary
- 12 = Non Seismic

5. Partial safety factor, Γ_{M0}

Value > 0

6. Partial safety factor, Γ_{M1}

Value > 0

7. Partial safety factor, Γ_{M2}

Value > 0

8. Consider deflection

0 = No

Any other value = Yes

9. DL deflection limit, $L/Value$

Value > 0

10. SDL + LL deflection limit, $L/Value$

Value > 0

11. LL deflection limit, $L/Value$

Value > 0

12. Total load deflection limit, $L/Value$

Value > 0

13. Total camber limit limit, $L/Value$

Value > 0

14. Pattern live load factor

Value > 0

15. Demand/capacity ratio limit

Value > 0

16. Multi-response case design

- 1 = Envelopes
- 2 = Step-by-Step
- 3 = Last Step
- 4 = Envelope -- All
- 5 = Step-by-Step -- All

17. Reliability class

- 1 = Class 1
- 2 = Class 2

- 3 = Class 3
18. Behavior factor, q
- Value > 0
19. System overstrength, Ω
- Value > 0
20. Consider P-delta
- 0 = No
- Any other value = Yes
21. Consider Torsion
- 0 = No
- Any other value = Yes
22. Ignore seismic code
- 0 = No
- Any other value = Yes
23. Ignore special seismic load
- 0 = No
- Any other value = Yes
24. Are Doubler plates plug-welded
- 0 = No
- Any other value = Yes

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double
    Dim ProgDet As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
```

Return Value

Introduction

```
ret = EtabsObject.ApplicationStart()  
  
'create SapModel object  
  SapModel = EtabsObject.SapModel  
  
'initialize model  
  ret = SapModel.InitializeNewModel()  
  
'create steel deck template model  
  ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)  
  
'set steel design code  
  ret = SapModel.DesignSteel.SetCode("Eurocode 3-2005")  
  
'set overwrite item  
  ret = SapModel.DesignSteel.Eurocode_3_2005.SetPreference(2, 1)  
  
'close ETABS  
  EtabsObject.ApplicationExit(False)  
  
'clean up variables  
  SapModel = Nothing  
  EtabsObject = Nothing  
  
End Sub
```

🔗 See Also

Reference

[cDStEurocode_3_2005 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDStIndian_IS_800_2007 Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDStIndian_IS_800_2007
```

```
Public Interface cDStIndian_IS_800_2007
```

```
Dim instance As cDStIndian_IS_800_2007
```

```
public interface class cDStIndian_IS_800_2007
```

```
type cDStIndian_IS_800_2007 = interface end
```

The cDStIndian_IS_800_2007 type exposes the following members.

▲ Methods

▲	Name	Description
---	------	-------------

◆	GetOverwrite	
---	------------------------------	--

◆	GetPreference	
---	-------------------------------	--

◆	SetOverwrite	
---	------------------------------	--

◆	SetPreference	
---	-------------------------------	--

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDStIndian_IS_800_2007 Methods

The cDStIndian_IS_800_2007 type exposes the following members.

Methods

	Name	Description
--	------	-------------

	<u>GetOverwrite</u>	
--	---------------------	--

	<u>GetPreference</u>	
--	----------------------	--

	<u>SetOverwrite</u>	
--	---------------------	--

	<u>SetPreference</u>	
--	----------------------	--

Top

See Also

Reference

cDStIndian_IS_800_2007 Interface

ETABSV1 Namespace

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CSI API ETABS v1

cDStIndian_IS_800_2007AddLanguageSpecificTextSet(

Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDStIndian_IS_800_2007
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```

Parameters

Name

Type: SystemString

cDStIndian_IS_800_2007AddLanguageSpecificTextSet("LSTCF42C029_0?cpp

Item

Type:Â SystemInt32

Value

Type:Â SystemDouble

ProgDet

Type:Â SystemBoolean

Return Value

Type:Â Int32

▲ See Also

Reference

cDStIndian_IS_800_2007 Interface

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CSI API ETABS v1

cDStIndian_IS_800_2007AddLanguageSpecificTextSet(

Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDStIndian_IS_800_2007
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: [SystemInt32](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDStIndian_IS_800_2007 Interface

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CSI API ETABS v1

cDStIndian_IS_800_2007AddLanguageSpecificTextSet(

Method

Namespace: Â ETABSv1

Assembly: Â ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDStIndian_IS_800_2007
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type:Â SystemString

Item

Type:Â SystemInt32

Value

Type:Â SystemDouble

ItemType (Optional)

Type:Â ETABSv1eItemType

Return Value

Type:Â Int32

▲ See Also

Reference

cDStIndian IS_800_2007 Interface

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CSI API ETABS v1

cDStIndian_IS_800_2007AddLanguageSpecificTextSet(

Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDStIndian_IS_800_2007
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: [SystemInt32](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDStIndian_IS_800_2007 Interface

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CSI API ETABS v1

cDStItalianNTC2008S Interface

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDStItalianNTC2008S
```

```
Public Interface cDStItalianNTC2008S
```

```
Dim instance As cDStItalianNTC2008S
```

```
public interface class cDStItalianNTC2008S
```

```
type cDStItalianNTC2008S = interface end
```

The cDStItalianNTC2008S type exposes the following members.

▲ Methods

▲	Name	Description
⇒	GetOverwrite	Retrieves the value of a steel design overwrite item
⇒	GetPreference	Retrieves the value of a steel design preference item.
⇒	SetOverwrite	Sets the value of a steel design overwrite item
⇒	SetPreference	Sets the value of a steel design preference item.

[Top](#)

▲ See Also

Reference

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cDStItalianNTC2008S Methods

The [cDStItalianNTC2008S](#) type exposes the following members.

Methods

⚡	Name	Description
⇒	GetOverwrite	Retrieves the value of a steel design overwrite item
⇒	GetPreference	Retrieves the value of a steel design preference item.
⇒	SetOverwrite	Sets the value of a steel design overwrite item
⇒	SetPreference	Sets the value of a steel design preference item.

[Top](#)

⚡ See Also

Reference

[cDStItalianNTC2008S Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cDStItalianNTC2008SAddLanguageSpecificTextSet("LST Method

Retrieves the value of a steel design overwrite item

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDStItalianNTC2008S
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```


Parameters

Name

Type: SystemString

The name of a frame object with a steel frame design procedure

Item

Type: SystemInt32

This is an integer between 1 and 57, inclusive, indicating the overwrite item considered.

1. Design Section
2. Framing Type
3. Section Class
4. Column buckling curve (y-y)
5. Column buckling curve (z-z)
6. Buckling curve for LTB
7. System Overstrength
8. Is Shape Rolled
9. Consider Deflection
10. Deflection check type
11. DL deflection limit, L/Value
12. SDL + LL deflection limits, L/Value
13. LL deflection limit, L/Value
14. Total load deflection limit, L/Value
15. Total camber limit, L/Value
16. DL deflection limit, absolute
17. SDL + LL deflection limit, absolute
18. LL deflection limit, absolute
19. Total load deflection limit, absolute
20. Total camber limit, absolute
21. Specified camber
22. Net area to total area ratio
23. Live load reduction factor
24. Unbraced length ratio, L_y
25. Unbraced length ratio, L_z
26. Unbraced length ratio, LLTB
27. Effective length factor, K_{1y}
28. Effective length factor, K_{1z}
29. Effective length factor, K_{2y}
30. Effective length factor, K_{2z}
31. Effective length factor, KLTB
32. Bending coefficient, C_1
33. Moment coefficient, k_{yy}
34. Moment coefficient, k_{zz}
35. Moment coefficient, k_{zy}
36. Moment coefficient, k_{yz}
37. Yield stress, F_y
38. Material Overstrength, Γ_{Ov}
39. Compressive capacity, N_{cRd}
40. Tensile capacity, N_{tRd}
41. Bending capacity Y-Y, $M_{cy,Rd}$
42. Bending capacity Z-Z, $M_{cz,Rd}$

- 43. Buckling resistance moment, M_b, R_d
- 44. Shear capacity Z-Z, V_z, R_d
- 45. Shear capacity Y-Y, V_y, R_d
- 46. Demand/capacity ratio limit
- 49. Bending coefficient, C_2
- 50. Bending coefficient, C_3
- 51. Warping constant, I_w
- 52. Elastic torsional buckling force, $N_{cr, T}$
- 53. Elastic torsional-flexural buckling force, $N_{cr, TF}$
- 54. Elastic critical moment for lateral-torsional buckling, M_{cr}
- 55. Warping coefficient, k_w
- 56. Coordinate of load application, z_a
- 57. Coordinate of shear center, z_s

Value

Type: SystemDouble

The numeric value of the considered overwrite item.

1. Design Section
2. Framing Type
 - . 0 = Program Default
 - . 1 = DCH-MRF
 - . 2 = DCL-MRF
 - . 3 = DCH-CBF
 - . 4 = DCL-CBF
 - . 5 = DCH-EBF
 - . 6 = DCL-EBF
 - . 7 = InvPendulum
 - . 8 = Non Dissipative
3. Section Class
 - . 0 = Program Default
 - . 1 = Class 1
 - . 2 = Class 2
 - . 3 = Class 3
 - . 4 = Class 4
4. Column buckling curve (y-y)
 - . 0 = Program Default
 - . 1 = a0
 - . 2 = a
 - . 3 = b
 - . 4 = c
 - . 5 = d
5. Column buckling curve (z-z)
 - . 0 = Program Default
 - . 1 = a0
 - . 2 = a
 - . 3 = b
 - . 4 = c
 - . 5 = d
6. Buckling curve for LTB
 - . 0 = Program Default
 - . 1 = a0
 - . 2 = a

- 3 = b
- 4 = c
- 5 = d

7. System Overstrength

Value ≥ 0 ; 0 means use a program determined value.

8. Is Shape Rolled

- 0 = Program Default
- 1 = No
- 2 = Yes

9. Consider Deflection

- 0 = Program Default
- 1 = No
- 2 = Yes

10. Deflection check type

- 0 = Program Default
- 1 = Ratio
- 2 = Absolute
- 3 = Both

11. DL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item.

12. SDL + LL deflection limits, L/Value

Value ≥ 0 ; 0 means no check for this item.

13. Value ≥ 0 ; 0 means no check for this item.

LL deflection limit, L/Value

14. Total load deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item.

15. Total camber limit, L/Value

Value ≥ 0 ; 0 means no check for this item.

16. DL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

17. SDL + LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

18. LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

19. Total load deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

20. Total camber limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

21. Specified camber

Introduction

Value ≥ 0 ; 0 means no check for this item. [L]

22. Net area to total area ratio

Value ≥ 0 ; 0 means use program determined value.

23. Live load reduction factor

Value ≥ 0 ; 0 means use program determined value.

24. Unbraced length ratio, L_y

Value ≥ 0 ; 0 means use program determined value.

25. Unbraced length ratio, L_z

Value ≥ 0 ; 0 means use program determined value.

26. Unbraced length ratio, LLTB

Value ≥ 0 ; 0 means use program determined value.

27. Effective length factor, K_{1y}

Value ≥ 0 ; 0 means use program determined value.

28. Effective length factor, K_{1z}

Value ≥ 0 ; 0 means use program determined value.

29. Effective length factor, K_{2y}

Value ≥ 0 ; 0 means use program determined value.

30. Effective length factor, K_{2z}

Value ≥ 0 ; 0 means use program determined value.

31. Effective length factor, KLTB

Value ≥ 0 ; 0 means use program determined value.

32. Bending coefficient, C_1

Value ≥ 0 ; 0 means use program determined value.

33. Bending coefficient, k_{yy}

Value ≥ 0 ; 0 means use program determined value.

34. Moment coefficient, k_{zz}

Value ≥ 0 ; 0 means use program determined value.

35. Moment coefficient, k_{zy}

Value ≥ 0 ; 0 means use program determined value.

36. Moment coefficient, k_{yz}

Value ≥ 0 ; 0 means use program determined value.

37. Yield stress, F_y

Value ≥ 0 ; 0 means use program determined value. [F/L^2]

38. Material Overstrength, Γ_{Ov}

Introduction

Value ≥ 0 ; 0 means use program determined value.

39. Compressive capacity, N_c, R_d

Value ≥ 0 ; 0 means use program determined value. [F]

40. Tensile capacity, N_t, R_d

Value ≥ 0 ; 0 means use program determined value. [F]

41. Bending capacity Y-Y, M_{cy}, R_d

Value ≥ 0 ; 0 means use program determined value. [F.L]

42. Bending capacity Z-Z, M_{cz}, R_d

Value ≥ 0 ; 0 means use program determined value. [F.L]

43. Buckling resistance moment, M_b, R_d

Value ≥ 0 ; 0 means use program determined value. [F.L]

44. Shear capacity Z-Z, V_z, R_d

Value ≥ 0 ; 0 means use program determined value. [F]

45. Shear capacity Y-Y, V_y, R_d

Value ≥ 0 ; 0 means use program determined value. [F]

46. Demand/capacity ratio limit

Value ≥ 0 ; 0 means use program determined value.

49. Bending coefficient, C_2

Value ≥ 0 ; 0 means use program determined value.

50. Bending coefficient, C_3

Value ≥ 0 ; 0 means use program determined value.

51. Warping constant, I_w

Value ≥ 0 ; 0 means use program determined value.

52. Elastic torsional buckling force, N_{crT}

Value ≥ 0 ; 0 means use program determined value.

53. Elastic torsional-flexural buckling force, N_{crTF}

Value ≥ 0 ; 0 means use program determined value.

54. Elastic critical moment for lateral-torsional buckling, M_{cr}

Value ≥ 0 ; 0 means use program determined value.

55. Warping coefficient, k_w

Value ≥ 0 ; 0 means use program determined value.

56. Coordinate of load application, z_a

57. Coordinate of shear center, z_s

ProgDet

Type: SystemBoolean

If this item is True, the specified value is program determined.

Return ValueType: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double
    Dim ProgDet As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set steel design code
    ret = SapModel.DesignSteel.SetCode("Italian NTC 2008")

    'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis

    'start steel design
    ret = SapModel.DesignSteel.StartDesign

    'get overwrite item
    ret = SapModel.DesignSteel.Italian_NTC_2008.GetOverwrite("8", 2, Value, ProgDet)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cDStItalianNTC2008S Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDStItalianNTC2008SAddLanguageSpecificTextSet("LSTCF2DBFC8_0")

Method

Retrieves the value of a steel design preference item.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDStItalianNTC2008S
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: [SystemInt32](#)

This is an integer between 3 and 23, inclusive, indicating the preference item considered.

3. Method used for Buckling in P-M-M
4. Framing type
5. Partial safety factor, GammaM0
6. Partial safety factor, GammaM1
7. Partial safety factor, GammaM2

8. Consider deflection
9. DL deflection limit, L/Value
10. SDL + LL deflection limit, L/Value
11. LL deflection limit, L/Value
12. Total load deflection limit, L/Value
13. Total camber limit, L/Value
14. Pattern live load factor
15. Demand/capacity ratio limit
16. Multi-response case design
17. Behavior factor, q
18. System overstrength, Ω
19. Consider P-delta
20. Consider torsion
21. Ignore seismic code
22. Ignore special seismic load
23. Are Doubler plates plug-welded

Value

Type: SystemDouble

The numeric value of the considered preference item.

3. Method used for Buckling in P-M-M
 - . 1 = Method A
 - . 2 = Method B
 - . 3 = Both
4. Framing type
 - . 1 = DCH-MRF (Ductility Class High - Moment Frame)
 - . 2 = DCL-MRF (Ductility Class Low - Moment Frame)
 - . 3 = DCH-CBF (Ductility Class High - Concentrically Braced Frame)
 - . 4 = DCL-CBF (Ductility Class Low - Concentrically Braced Frame)
 - . 5 = DCH-EBF (Ductility Class High - Eccentrically Braced Frame)
 - . 6 = DCL-EBF (Ductility Class Low - Eccentrically Braced Frame)
 - . 7 = Inverted Pendulum Structure
 - . 8 = Non Dissipative
5. Partial safety factor, Γ_{M0}

Value > 0

6. Partial safety factor, Γ_{M1}

Value > 0

7. Partial safety factor, Γ_{M2}

Value > 0

8. Consider deflection

1 or "No" - Default

2 or "Yes"

9. DL deflection limit, L/Value

Value > 0

10. SDL + LL deflection limit, L/Value

- Value > 0
11. LL deflection limit, L/Value
- Value > 0
12. Total load deflection limit, L/Value
- Value > 0
13. Total camber limit limit, L/Value
- Value > 0
14. Pattern live load factor
- Value > 0
15. Demand/capacity ratio limit
- Value > 0
16. Multi-response case design
 · 1 = Envelopes
 · 2 = Step-by-Step
 · 3 = Last Step
 · 4 = Envelope -- All
 · 5 = Step-by-Step -- All
 17. Behavior factor, q
- Value > 0
18. System overstrength, Omega
- Value > 0
19. Consider P-delta
- 1 or "No" - Default
2 or "Yes"
20. Consider Torsion
- 1 or "No" - Default
2 or "Yes"
21. Ignore seismic code
- 1 or "No" - Default
2 or "Yes"
22. Ignore special seismic load
- 1 or "No" - Default
2 or "Yes"
23. Are Doubler plates plug-welded
- 1 or "No" - Default

2 or "Yes"

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double
    Dim ProgDet As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set steel design code
    ret = SapModel.DesignSteel.SetCode("Italian NTC 2008")

    'set overwrite item
    ret = SapModel.DesignSteel.Italian_NTC_2008.GetPreference(2, Value)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ **See Also**

Reference

cDStItalianNTC2008S Interface

ETABSv1 Namespace

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Introduction

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CSI API ETABS v1

cDStItalianNTC2008SAddLanguageSpecificTextSet("LS

Method

Sets the value of a steel design overwrite item

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDStItalianNTC2008S
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters**Name**

Type: SystemString

The name of an existing frame object or group, depending on the value of the ItemType item.

Item

Type: SystemInt32

This is an integer between 1 and 57, inclusive, indicating the overwrite item considered.

1. Design Section
2. Framing Type
3. Section Class
4. Column buckling curve (y-y)
5. Column buckling curve (z-z)
6. Buckling curve for LTB
7. System Overstrength
8. Is Shape Rolled
9. Consider Deflection
10. Deflection check type
11. DL deflection limit, L/Value
12. SDL + LL deflection limits, L/Value
13. LL deflection limit, L/Value
14. Total load deflection limit, L/Value
15. Total camber limit, L/Value
16. DL deflection limit, absolute
17. SDL + LL deflection limit, absolute
18. LL deflection limit, absolute
19. Total load deflection limit, absolute
20. Total camber limit, absolute
21. Specified camber
22. Net area to total area ratio
23. Live load reduction factor
24. Unbraced length ratio, Ly
25. Unbraced length ratio, Lz
26. Unbraced length ratio, LLTB
27. Effective length factor, K1y
28. Effective length factor, K1z
29. Effective length factor, K2y
30. Effective length factor, K2z
31. Effective length factor, KLTB
32. Bending coefficient, C1
33. Moment coefficient, kyy
34. Moment coefficient, kzz
35. Moment coefficient, kzy
36. Moment coefficient, kyz
37. Yield stress, Fy
38. Material Overstrength, GammaOv
39. Compressive capacity, Nc,Rd
40. Tensile capacity, Nt,Rd
41. Bending capacity Y-Y, Mcy,Rd

- 42. Bending capacity Z-Z, M_{cz}, R_d
- 43. Buckling resistance moment, M_b, R_d
- 44. Shear capacity Z-Z, V_z, R_d
- 45. Shear capacity Y-Y, V_y, R_d
- 46. Demand/capacity ratio limit
- 49. Bending coefficient, C2
- 50. Bending coefficient, C3
- 51. Warping constant, I_w
- 52. Elastic torsional buckling force, $N_{cr, T}$
- 53. Elastic torsional-flexural buckling force, $N_{cr, TF}$
- 54. Elastic critical moment for lateral-torsional buckling, M_{cr}
- 55. Warping coefficient, k_w
- 56. Coordinate of load application, z_a
- 57. Coordinate of shear center, z_s

Value

Type: SystemDouble

The numeric value of the considered overwrite item.

- 1. Design Section
- 2. Framing Type
 - . 0 = Program Default
 - . 1 = DCH-MRF
 - . 2 = DCL-MRF
 - . 3 = DCH-CBF
 - . 4 = DCL-CBF
 - . 5 = DCH-EBF
 - . 6 = DCL-EBF
 - . 7 = InvPendulum
 - . 8 = Non Dissipative
- 3. Section Class
 - . 0 = Program Default
 - . 1 = Class 1
 - . 2 = Class 2
 - . 3 = Class 3
 - . 4 = Class 4
- 4. Column buckling curve (y-y)
 - . 0 = Program Default
 - . 1 = a0
 - . 2 = a
 - . 3 = b
 - . 4 = c
 - . 5 = d
- 5. Column buckling curve (z-z)
 - . 0 = Program Default
 - . 1 = a0
 - . 2 = a
 - . 3 = b
 - . 4 = c
 - . 5 = d
- 6. Buckling curve for LTB
 - . 0 = Program Default
 - . 1 = a0

- . 2 = a
- . 3 = b
- . 4 = c
- . 5 = d

7. System Overstrength

Value ≥ 0 ; 0 means use a program determined value.

8. Is Shape Rolled

- . 0 = Program Default
- . 1 = No
- . 2 = Yes

9. Consider Deflection

- . 0 = Program Default
- . 1 = No
- . 2 = Yes

10. Deflection check type

- . 0 = Program Default
- . 1 = Ratio
- . 2 = Absolute
- . 3 = Both

11. DL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item.

12. SDL + LL deflection limits, L/Value

Value ≥ 0 ; 0 means no check for this item.

13. Value ≥ 0 ; 0 means no check for this item.

LL deflection limit, L/Value

14. Total load deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item.

15. Total camber limit, L/Value

Value ≥ 0 ; 0 means no check for this item.

16. DL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

17. SDL + LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

18. LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

19. Total load deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

20. Total camber limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

21. Specified camber

Introduction

Value ≥ 0 ; 0 means no check for this item. [L]

22. Net area to total area ratio

Value ≥ 0 ; 0 means use program determined value.

23. Live load reduction factor

Value ≥ 0 ; 0 means use program determined value.

24. Unbraced length ratio, L_y

Value ≥ 0 ; 0 means use program determined value.

25. Unbraced length ratio, L_z

Value ≥ 0 ; 0 means use program determined value.

26. Unbraced length ratio, LLTB

Value ≥ 0 ; 0 means use program determined value.

27. Effective length factor, K_{1y}

Value ≥ 0 ; 0 means use program determined value.

28. Effective length factor, K_{1z}

Value ≥ 0 ; 0 means use program determined value.

29. Effective length factor, K_{2y}

Value ≥ 0 ; 0 means use program determined value.

30. Effective length factor, K_{2z}

Value ≥ 0 ; 0 means use program determined value.

31. Effective length factor, KLTB

Value ≥ 0 ; 0 means use program determined value.

32. Bending coefficient, C_1

Value ≥ 0 ; 0 means use program determined value.

33. Moment coefficient, k_{yy}

Value ≥ 0 ; 0 means use program determined value.

34. Moment coefficient, k_{zz}

Value ≥ 0 ; 0 means use program determined value.

35. Moment coefficient, k_{zy}

Value ≥ 0 ; 0 means use program determined value.

36. Moment coefficient, k_{yz}

Value ≥ 0 ; 0 means use program determined value.

37. Yield stress, F_y

Value ≥ 0 ; 0 means use program determined value. [F/L^2]

38. Material Overstrength, Γ_{Ov}

Introduction

Value ≥ 0 ; 0 means use program determined value.

39. Compressive capacity, N_c, R_d

Value ≥ 0 ; 0 means use program determined value. [F]

40. Tensile capacity, N_t, R_d

Value ≥ 0 ; 0 means use program determined value. [F]

41. Bending capacity Y-Y, M_{cy}, R_d

Value ≥ 0 ; 0 means use program determined value. [F.L]

42. Bending capacity Z-Z, M_{cz}, R_d

Value ≥ 0 ; 0 means use program determined value. [F.L]

43. Buckling resistance moment, M_b, R_d

Value ≥ 0 ; 0 means use program determined value. [F.L]

44. Shear capacity Z-Z, V_z, R_d

Value ≥ 0 ; 0 means use program determined value. [F]

45. Shear capacity Y-Y, V_y, R_d

Value ≥ 0 ; 0 means use program determined value. [F]

46. Demand/capacity ratio limit

Value ≥ 0 ; 0 means use program determined value.

49. Bending coefficient, C_2

Value ≥ 0 ; 0 means use program determined value.

50. Bending coefficient, C_3

Value ≥ 0 ; 0 means use program determined value.

51. Warping constant, I_w

Value ≥ 0 ; 0 means use program determined value.

52. Elastic torsional buckling force, N_{cr}, T

Value ≥ 0 ; 0 means use program determined value.

53. Elastic torsional-flexural buckling force, N_{cr}, T_F

Value ≥ 0 ; 0 means use program determined value.

54. Elastic critical moment for lateral-torsional buckling, M_{cr}

Value ≥ 0 ; 0 means use program determined value.

55. Warping coefficient, k_w

Value ≥ 0 ; 0 means use program determined value.

56. Coordinate of load application, z_a

57. Coordinate of shear center, z_s

ItemType (Optional)

Type: $\hat{A}$ ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

- ◇ Object = 0
- ◇ Group = 1
- ◇ SelectedObjects = 2

If this item is Objects, the assignment is made to the frame object specified by the Name item.

If this item is Group, the assignment is made to all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the assignment is made to all selected frame objects and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the item is successfully set; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double
    Dim ProgDet As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set steel design code
    ret = SapModel.DesignSteel.SetCode("Italian NTC 2008")

    'set overwrite item
    ret = SapModel.DesignSteel.Italian_NTC_2008.SetOverwrite("8", 2, 1)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cDStItalianNTC2008S Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDStItalianNTC2008SAddLanguageSpecificTextSet("LST19D83C28_0")

Method

Sets the value of a steel design preference item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDStItalianNTC2008S
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: SystemInt32

This is an integer between 3 and 23, inclusive, indicating the preference item considered.

3. Method used for Buckling in P-M-M
4. Framing type
5. Partial safety factor, GammaM0
6. Partial safety factor, GammaM1
7. Partial safety factor, GammaM2

8. Consider deflection
9. DL deflection limit, L/Value
10. SDL + LL deflection limit, L/Value
11. LL deflection limit, L/Value
12. Total load deflection limit, L/Value
13. Total camber limit, L/Value
14. Pattern live load factor
15. Demand/capacity ratio limit
16. Multi-response case design
17. Behavior factor, q
18. System overstrength, Ω
19. Consider P-delta
20. Consider torsion
21. Ignore seismic code
22. Ignore special seismic load
23. Are Doubler plates plug-welded

Value

Type: SystemDouble

The numeric value of the considered preference item.

3. Method used for Buckling in P-M-M

- 1 = Method A
- 2 = Method B
- 3 = Both

4. Framing type

- 1 = DCH-MRF
- 2 = DCL-MRF
- 3 = DCH-CBF
- 4 = DCL-CBF
- 5 = DCH-EBF
- 6 = DCL-EBF
- 7 = InvPendulum
- 8 = Non Dissipative

5. Partial safety factor, Γ_{M0}

Value > 0

6. Partial safety factor, Γ_{M1}

Value > 0

7. Partial safety factor, Γ_{M2}

Value > 0

8. Consider deflection

1 or "No" - Default

2 or "Yes"

9. DL deflection limit, L/Value

Value > 0

10. SDL + LL deflection limit, L/Value

- Value > 0
11. LL deflection limit, L/Value
- Value > 0
12. Total load deflection limit, L/Value
- Value > 0
13. Total camber limit limit, L/Value
- Value > 0
14. Pattern live load factor
- Value > 0
15. Demand/capacity ratio limit
- Value > 0
16. Multi-response case design
 · 1 = Envelopes
 · 2 = Step-by-Step
 · 3 = Last Step
 · 4 = Envelope -- All
 · 5 = Step-by-Step -- All
 17. Behavior factor, q
- Value > 0
18. System overstrength, Omega
- Value > 0
19. Consider P-delta
- 1 or "No" - Default
2 or "Yes"
20. Consider Torsion
- 1 or "No" - Default
2 or "Yes"
21. Ignore seismic code
- 1 or "No" - Default
2 or "Yes"
22. Ignore special seismic load
- 1 or "No" - Default
2 or "Yes"
23. Are Doubler plates plug-welded
- 1 or "No" - Default

2 or "Yes"

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double
    Dim ProgDet As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set steel design code
    ret = SapModel.DesignSteel.SetCode("Italian NTC 2008")

    'set overwrite item
    ret = SapModel.DesignSteel.Italian_NTC_2008.SetPreference(2, 1)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ **See Also**

Reference

cDStItalianNTC2008S Interface

ETABSv1 Namespace

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Introduction

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CSI API ETABS v1

cDStItalianNTC2018S Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDStItalianNTC2018S
```

```
Public Interface cDStItalianNTC2018S
```

```
Dim instance As cDStItalianNTC2018S
```

```
public interface class cDStItalianNTC2018S
```

```
type cDStItalianNTC2018S = interface end
```

The cDStItalianNTC2018S type exposes the following members.

▲ Methods

▲	Name	Description
⇒	GetOverwrite	Retrieves the value of a steel design overwrite item
⇒	GetPreference	Retrieves the value of a steel design preference item.
⇒	SetOverwrite	Sets the value of a steel design overwrite item
⇒	SetPreference	Sets the value of a steel design preference item.

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDStItalianNTC2018S Methods

The [cDStItalianNTC2018S](#) type exposes the following members.

Methods

⚡	Name	Description
⇒	GetOverwrite	Retrieves the value of a steel design overwrite item
⇒	GetPreference	Retrieves the value of a steel design preference item.
⇒	SetOverwrite	Sets the value of a steel design overwrite item
⇒	SetPreference	Sets the value of a steel design preference item.

[Top](#)

⚡ See Also

Reference

[cDStItalianNTC2018S Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cDStItalianNTC2018SAddLanguageSpecificTextSet("LST5356496E1048")

Method

Retrieves the value of a steel design overwrite item

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetOverwrite(
    string Name,
    int Item,
    ref string textValue,
    ref double numericValue,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef textValue As String,
    ByRef numericValue As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDStItalianNTC2018S
Dim Name As String
Dim Item As Integer
Dim textValue As String
Dim numericValue As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, textValue, numericValue, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    String^% textValue,
    double% numericValue,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    textValue : string byref *
    numericValue : float byref *
    ProgDet : bool byref -> int
```

Parameters

Name

Type: SystemString

The name of a frame object with a steel frame design procedure

Item

Type: SystemInt32

This is an integer between 1 and 57, inclusive, indicating the overwrite item considered.

1. Design Section
2. Framing Type
3. Section Class
4. Column buckling curve (y-y)
5. Column buckling curve (z-z)
6. Buckling curve for LTB
7. System Overstrength
8. Is Shape Rolled
9. Consider Deflection
10. Deflection check type
11. DL deflection limit, L/Value
12. SDL + LL deflection limits, L/Value
13. LL deflection limit, L/Value
14. Total load deflection limit, L/Value
15. Total camber limit, L/Value
16. DL deflection limit, absolute
17. SDL + LL deflection limit, absolute
18. LL deflection limit, absolute
19. Total load deflection limit, absolute
20. Total camber limit, absolute
21. Specified camber
22. Net area to total area ratio
23. Live load reduction factor
24. Unbraced length ratio, L_y
25. Unbraced length ratio, L_z
26. Unbraced length ratio, LLTB
27. Effective length factor, K_{1y}
28. Effective length factor, K_{1z}
29. Effective length factor, K_{2y}
30. Effective length factor, K_{2z}
31. Effective length factor, KLTB
32. Bending coefficient, C_1
33. Moment coefficient, k_{yy}
34. Moment coefficient, k_{zz}
35. Moment coefficient, k_{zy}
36. Moment coefficient, k_{yz}
37. Yield stress, F_y
38. Material Overstrength, Γ_{Ov}
39. Compressive capacity, N_c, R_d
40. Tensile capacity N_t, R_d
41. Bending capacity Y-Y, M_{cy}, R_d
42. Bending capacity Z-Z, M_{cz}, R_d

- 43. Buckling resistance moment, Mb,Rd
- 44. Shear capacity Z-Z, Vz,Rd
- 45. Shear capacity Y-Y, Vy,Rd
- 46. Demand/capacity ratio limit
- 49. Bending coefficient, C2
- 50. Bending coefficient, C3
- 51. Warping constant, Iw
- 52. Elastic torsional buckling force, Ncr,T
- 53. Elastic torsional-flexural buckling force, Ncr,TF
- 54. Elastic critical moment for lateral-torsional buckling, Mcr
- 55. Warping coefficient, kw
- 56. Coordinate of load application, za
- 57. Coordinate of shear center, zs

textValue

Type: SystemString

The string value of the considered overwrite item.

numericValue

Type: SystemDouble

The numeric value of the considered overwrite item.

- 1. Design Section
- 2. Framing Type
 - . 0 = Program Default
 - . 1 = DCH-MRF
 - . 2 = DCL-MRF
 - . 3 = DCH-CBF
 - . 4 = DCL-CBF
 - . 5 = DCH-EBF
 - . 6 = DCL-EBF
 - . 7 = InvPendulum
 - . 8 = Non Dissipative
- 3. Section Class
 - . 0 = Program Default
 - . 1 = Class 1
 - . 2 = Class 2
 - . 3 = Class 3
 - . 4 = Class 4
- 4. Column buckling curve (y-y)
 - . 0 = Program Default
 - . 1 = a0
 - . 2 = a
 - . 3 = b
 - . 4 = c
 - . 5 = d
- 5. Column buckling curve (z-z)
 - . 0 = Program Default
 - . 1 = a0
 - . 2 = a
 - . 3 = b
 - . 4 = c
 - . 5 = d
- 6. Buckling curve for LTB

Introduction

- . 0 = Program Default
- . 1 = a0
- . 2 = a
- . 3 = b
- . 4 = c
- . 5 = d

7. System Overstrength

Value ≥ 0 ; 0 means use a program determined value.

8. Is Shape Rolled

- . 0 = Program Default
- . 1 = No
- . 2 = Yes

9. Consider Deflection

- . 0 = Program Default
- . 1 = No
- . 2 = Yes

10. Deflection check type

- . 0 = Program Default
- . 1 = Ratio
- . 2 = Absolute
- . 3 = Both

11. DL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item.

12. SDL + LL deflection limits, L/Value

Value ≥ 0 ; 0 means no check for this item.

13. Value ≥ 0 ; 0 means no check for this item.

LL deflection limit, L/Value

14. Total load deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item.

15. Total camber limit, L/Value

Value ≥ 0 ; 0 means no check for this item.

16. DL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

17. SDL + LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

18. LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

19. Total load deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

20. Total camber limit, absolute

Introduction

Value ≥ 0 ; 0 means no check for this item. [L]

21. Specified camber

Value ≥ 0 ; 0 means no check for this item. [L]

22. Net area to total area ratio

Value ≥ 0 ; 0 means use program determined value.

23. Live load reduction factor

Value ≥ 0 ; 0 means use program determined value.

24. Unbraced length ratio, L_y

Value ≥ 0 ; 0 means use program determined value.

25. Unbraced length ratio, L_z

Value ≥ 0 ; 0 means use program determined value.

26. Unbraced length ratio, LLTB

Value ≥ 0 ; 0 means use program determined value.

27. Effective length factor, K_{1y}

Value ≥ 0 ; 0 means use program determined value.

28. Effective length factor, K_{1z}

Value ≥ 0 ; 0 means use program determined value.

29. Effective length factor, K_{2y}

Value ≥ 0 ; 0 means use program determined value.

30. Effective length factor, K_{2z}

Value ≥ 0 ; 0 means use program determined value.

31. Effective length factor, KLTB

Value ≥ 0 ; 0 means use program determined value.

32. Bending coefficient, C_1

Value ≥ 0 ; 0 means use program determined value.

33. Moment coefficient, k_{yy}

Value ≥ 0 ; 0 means use program determined value.

34. Moment coefficient, k_{zz}

Value ≥ 0 ; 0 means use program determined value.

35. Moment coefficient, k_{zy}

Value ≥ 0 ; 0 means use program determined value.

36. Moment coefficient, k_{yz}

Value ≥ 0 ; 0 means use program determined value.

37. Yield stress, F_y

Introduction

- Value ≥ 0 ; 0 means use program determined value. [F/L²]
38. Material Overstrength, Γ_{Ov}
- Value ≥ 0 ; 0 means use program determined value.
39. Compressive capacity, $N_{c,Rd}$
- Value ≥ 0 ; 0 means use program determined value. [F]
40. Tensile capacity, $N_{t,Rd}$
- Value ≥ 0 ; 0 means use program determined value. [F]
41. Bending capacity Y-Y, $M_{cy,Rd}$
- Value ≥ 0 ; 0 means use program determined value. [F.L]
42. Bending capacity Z-Z, $M_{cz,Rd}$
- Value ≥ 0 ; 0 means use program determined value. [F.L]
43. Buckling resistance moment, $M_{b,Rd}$
- Value ≥ 0 ; 0 means use program determined value. [F.L]
44. Shear capacity Z-Z, $V_{z,Rd}$
- Value ≥ 0 ; 0 means use program determined value. [F]
45. Shear capacity Y-Y, $V_{y,Rd}$
- Value ≥ 0 ; 0 means use program determined value. [F]
46. Demand/capacity ratio limit
- Value ≥ 0 ; 0 means use program determined value.
49. Bending coefficient, C_2
- Value ≥ 0 ; 0 means use program determined value.
50. Bending coefficient, C_3
- Value ≥ 0 ; 0 means use program determined value.
51. Warping constant, I_w
- Value ≥ 0 ; 0 means use program determined value.
52. Elastic torsional buckling force, $N_{cr,T}$
- Value ≥ 0 ; 0 means use program determined value.
53. Elastic torsional-flexural buckling force, $N_{cr,TF}$
- Value ≥ 0 ; 0 means use program determined value.
54. Elastic critical moment for lateral-torsional buckling, M_{cr}
- Value ≥ 0 ; 0 means use program determined value.
55. Warping coefficient, k_w
- Value ≥ 0 ; 0 means use program determined value.
56. Coordinate of load application, z_a
57. Coordinate of shear center, z_s

ProgDet

Type: System.Boolean

If this item is True, the specified value is program determined.

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim textValue As String
    Dim Value As Double
    Dim ProgDet As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set steel design code
    ret = SapModel.DesignSteel.SetCode("Italian NTC 2018")

    'run analysis
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")
    ret = SapModel.Analyze.RunAnalysis

    'start steel design
    ret = SapModel.DesignSteel.StartDesign

    'get overwrite item
    ret = SapModel.DesignSteel.Italian_NTC_2018.GetOverwrite("8", 1, textValue, Value, ProgDet)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

See Also

Reference

cDStItalianNTC2018S Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDStItalianNTC2018SAddLanguageSpecificTextSet("LST7BF205B3_0") Method

Retrieves the value of a steel design preference item.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetPreference(
    int Item,
    ref string textValue,
    ref double numericValue
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef textValue As String,
    ByRef numericValue As Double
) As Integer
```

```
Dim instance As cDStItalianNTC2018S
Dim Item As Integer
Dim textValue As String
Dim numericValue As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    textValue, numericValue)
```

```
int GetPreference(
    int Item,
    String^% textValue,
    double% numericValue
)
```

```
abstract GetPreference :
    Item : int *
    textValue : string byref *
    numericValue : float byref -> int
```

Parameters

Item

Type: System.Int32

This is an integer between 3 and 23, inclusive, indicating the preference item considered.

3. Method used for Buckling in P-M-M

4. Framing type
5. Partial safety factor, GammaM0
6. Partial safety factor, GammaM1
7. Partial safety factor, GammaM2
8. Consider deflection
9. DL deflection limit, L/Value
10. SDL + LL deflection limit, L/Value
11. LL deflection limit, L/Value
12. Total load deflection limit, L/Value
13. Total camber limit, L/Value
14. Pattern live load factor
15. Demand/capacity ratio limit
16. Multi-response case design
17. Behavior factor, q
18. System overstrength, Omega
19. Consider P-delta
20. Consider torsion
21. Ignore seismic code
22. Ignore special seismic load
23. Are Doubler plates plug-welded

textValue

Type: SystemString

The string value of the considered preference item.

numericValue

Type: SystemDouble

The numeric value of the considered preference item.

3. Method used for Buckling in P-M-M
 - 1 = Method A
 - 2 = Method B
 - 3 = Both
4. Framing type
 - 1 = DCH-MRF (Ductility Class High - Moment Frame)
 - 2 = DCL-MRF (Ductility Class Low - Moment Frame)
 - 3 = DCH-CBF (Ductility Class High - Concentrically Braced Frame)
 - 4 = DCL-CBF (Ductility Class Low - Concentrically Braced Frame)
 - 5 = DCH-EBF (Ductility Class High - Eccentrically Braced Frame)
 - 6 = DCL-EBF (Ductility Class Low - Eccentrically Braced Frame)
 - 7 = Inverted Pendulum Structure
 - 8 = Non Dissipative
5. Partial safety factor, GammaM0

Value > 0
6. Partial safety factor, GammaM1

Value > 0
7. Partial safety factor, GammaM2

Value > 0
8. Consider deflection

1 or "No" - Default

2 or "Yes"

9. DL deflection limit, L/Value

Value > 0

10. SDL + LL deflection limit, L/Value

Value > 0

11. LL deflection limit, L/Value

Value > 0

12. Total load deflection limit, L/Value

Value > 0

13. Total camber limit limit, L/Value

Value > 0

14. Pattern live load factor

Value > 0

15. Demand/capacity ratio limit

Value > 0

16. Multi-response case design

- 1 = Envelopes
- 2 = Step-by-Step
- 3 = Last Step
- 4 = Envelope -- All
- 5 = Step-by-Step -- All

17. Behavior factor, q

Value > 0

18. System overstrength, Omega

Value > 0

19. Consider P-delta

1 or "No" - Default

2 or "Yes"

20. Consider Torsion

1 or "No" - Default

2 or "Yes"

21. Ignore seismic code

1 or "No" - Default

2 or "Yes"

22. Ignore special seismic load

1 or "No" - Default

2 or "Yes"

23. Are Doubler plates plug-welded

1 or "No" - Default

2 or "Yes"

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim textValue As String
    Dim Value As Double
    Dim ProgDet As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set steel design code
    ret = SapModel.DesignSteel.SetCode("Italian NTC 2018")

    'set overwrite item
    ret = SapModel.DesignSteel.Italian_NTC_2018.GetPreference(2, textValue, Value)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cDStItalianNTC2018S Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDStItalianNTC2018SAddLanguageSpecificTextSet("LST89262CD3_0")

Method

Sets the value of a steel design overwrite item

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetOverwrite(
    string Name,
    int Item,
    string textValue,
    double numericValue,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    textValue As String,
    numericValue As Double,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDStItalianNTC2018S
Dim Name As String
Dim Item As Integer
Dim textValue As String
Dim numericValue As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, textValue, numericValue, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    String^ textValue,
    double numericValue,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    textValue : string *
    numericValue : float *
    ?ItemType : eItemType
```

```
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing frame object or group, depending on the value of the ItemType item.

Item

Type: SystemInt32

This is an integer between 1 and 57, inclusive, indicating the overwrite item considered.

1. Design Section
2. Framing Type
3. Section Class
4. Column buckling curve (y-y)
5. Column buckling curve (z-z)
6. Buckling curve for LTB
7. System Overstrength
8. Is Shape Rolled
9. Consider Deflection
10. Deflection check type
11. DL deflection limit, L/Value
12. SDL + LL deflection limits, L/Value
13. LL deflection limit, L/Value
14. Total load deflection limit, L/Value
15. Total camber limit, L/Value
16. DL deflection limit, absolute
17. SDL + LL deflection limit, absolute
18. LL deflection limit, absolute
19. Total load deflection limit, absolute
20. Total camber limit, absolute
21. Specified camber
22. Net area to total area ratio
23. Live load reduction factor
24. Unbraced length ratio, Ly
25. Unbraced length ratio, Lz
26. Unbraced length ratio, LLTB
27. Effective length factor, K1y
28. Effective length factor, K1z
29. Effective length factor, K2y
30. Effective length factor, K2z
31. Effective length factor KLTB
32. Bending coefficient, C1
33. Bending coefficient, kyy
34. Bending coefficient, kzz
35. Bending coefficient, kzy
36. Bending coefficient, kyz

37. Yield stress, F_y
38. Material Overstrength, Γ_{ov}
39. Compressive capacity, $N_{c,Rd}$
40. Tensile capacity $N_{t,Rd}$
41. Bending capacity Y-Y, $M_{cy,Rd}$
42. Bending capacity Z-Z, $M_{cz,Rd}$
43. Buckling resistance moment, $M_{b,Rd}$
44. Shear capacity Z-Z, $V_{z,Rd}$
45. Shear capacity Y-Y, $V_{y,Rd}$
46. Demand/capacity ratio limit
49. Bending coefficient, C_2
50. Bending coefficient, C_3
51. Warping constant, I_w
52. Elastic torsional buckling force, $N_{cr,T}$
53. Elastic torsional-flexural buckling force, $N_{cr,TF}$
54. Elastic critical moment for lateral-torsional buckling, M_{cr}
55. Warping coefficient, k_w
56. Coordinate of load application, z_a
57. Coordinate of shear center, z_s

textValue

Type: SystemString

The string value of the considered overwrite item.

numericValue

Type: SystemDouble

The numeric value of the considered overwrite item.

1. Design Section
2. Framing Type
 - 0 = Program Default
 - 1 = DCH-MRF
 - 2 = DCL-MRF
 - 3 = DCH-CBF
 - 4 = DCL-CBF
 - 5 = DCH-EBF
 - 6 = DCL-EBF
 - 7 = InvPendulum
 - 8 = Non Dissipative
3. Section Class
 - 0 = Program Default
 - 1 = Class 1
 - 2 = Class 2
 - 3 = Class 3
 - 4 = Class 4
4. Column buckling curve (y-y)
 - 0 = Program Default
 - 1 = a0
 - 2 = a
 - 3 = b
 - 4 = c
 - 5 = d
5. Column buckling curve (z-z)
 - 0 = Program Default

Introduction

- . 1 = a0
 - . 2 = a
 - . 3 = b
 - . 4 = c
 - . 5 = d
6. Buckling curve for LTB
- . 0 = Program Default
 - . 1 = a0
 - . 2 = a
 - . 3 = b
 - . 4 = c
 - . 5 = d
7. System Overstrength
- Value ≥ 0 ; 0 means use a program determined value.
8. Is Shape Rolled
- . 0 = Program Default
 - . 1 = No
 - . 2 = Yes
9. Consider Deflection
- . 0 = Program Default
 - . 1 = No
 - . 2 = Yes
10. Deflection check type
- . 0 = Program Default
 - . 1 = Ratio
 - . 2 = Absolute
 - . 3 = Both
11. DL deflection limit, L/Value
- Value ≥ 0 ; 0 means no check for this item.
12. SDL + LL deflection limits, L/Value
- Value ≥ 0 ; 0 means no check for this item.
13. Value ≥ 0 ; 0 means no check for this item.
- LL deflection limit, L/Value
14. Total load deflection limit, L/Value
- Value ≥ 0 ; 0 means no check for this item.
15. Total camber limit, L/Value
- Value ≥ 0 ; 0 means no check for this item.
16. DL deflection limit, absolute
- Value ≥ 0 ; 0 means no check for this item. [L]
17. SDL + LL deflection limit, absolute
- Value ≥ 0 ; 0 means no check for this item. [L]
18. LL deflection limit, absolute

Introduction

Value ≥ 0 ; 0 means no check for this item. [L]

19. Total load deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

20. Total camber limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

21. Specified camber

Value ≥ 0 ; 0 means no check for this item. [L]

22. Net area to total area ratio

Value ≥ 0 ; 0 means use program determined value.

23. Live load reduction factor

Value ≥ 0 ; 0 means use program determined value.

24. Unbraced length ratio, L_y

Value ≥ 0 ; 0 means use program determined value.

25. Unbraced length ratio, L_z

Value ≥ 0 ; 0 means use program determined value.

26. Unbraced length ratio, LLTB

Value ≥ 0 ; 0 means use program determined value.

27. Effective length factor, K_{1y}

Value ≥ 0 ; 0 means use program determined value.

28. Effective length factor, K_{1z}

Value ≥ 0 ; 0 means use program determined value.

29. Effective length factor, K_{2y}

Value ≥ 0 ; 0 means use program determined value.

30. Effective length factor, K_{2z}

Value ≥ 0 ; 0 means use program determined value.

31. Effective length factor, KLTB

Value ≥ 0 ; 0 means use program determined value.

32. Bending coefficient, C_1

Value ≥ 0 ; 0 means use program determined value.

33. Moment coefficient, k_{yy}

Value ≥ 0 ; 0 means use program determined value.

34. Moment coefficient, k_{zz}

Value ≥ 0 ; 0 means use program determined value.

35. Moment coefficient, k_{zy}

Introduction

- Value ≥ 0 ; 0 means use program determined value.
36. Moment coefficient, k_{yz}
- Value ≥ 0 ; 0 means use program determined value.
37. Yield stress, F_y
- Value ≥ 0 ; 0 means use program determined value. $[F/L^2]$
38. Material Overstrength, $\Gamma_{M,Ov}$
- Value ≥ 0 ; 0 means use program determined value.
39. Compressive capacity, $N_{c,Rd}$
- Value ≥ 0 ; 0 means use program determined value. $[F]$
40. Tensile capacity, $N_{t,Rd}$
- Value ≥ 0 ; 0 means use program determined value. $[F]$
41. Bending capacity Y-Y, $M_{cy,Rd}$
- Value ≥ 0 ; 0 means use program determined value. $[F.L]$
42. Bending capacity Z-Z, $M_{cz,Rd}$
- Value ≥ 0 ; 0 means use program determined value. $[F.L]$
43. Buckling resistance moment, $M_{b,Rd}$
- Value ≥ 0 ; 0 means use program determined value. $[F.L]$
44. Shear capacity Z-Z, $V_{z,Rd}$
- Value ≥ 0 ; 0 means use program determined value. $[F]$
45. Shear capacity Y-Y, $V_{y,Rd}$
- Value ≥ 0 ; 0 means use program determined value. $[F]$
46. Demand/capacity ratio limit
- Value ≥ 0 ; 0 means use program determined value.
49. Bending coefficient, C_2
- Value ≥ 0 ; 0 means use program determined value.
50. Bending coefficient, C_3
- Value ≥ 0 ; 0 means use program determined value.
51. Warping constant, I_w
- Value ≥ 0 ; 0 means use program determined value.
52. Elastic torsional buckling force, $N_{cr,T}$
- Value ≥ 0 ; 0 means use program determined value.
53. Elastic torsional-flexural buckling force, $N_{cr,TF}$
- Value ≥ 0 ; 0 means use program determined value.
54. Elastic critical moment for lateral-torsional buckling, M_{cr}

Value ≥ 0 ; 0 means use program determined value.

55. Warping coefficient, kw

Value ≥ 0 ; 0 means use program determined value.

56. Coordinate of load application, za

57. Coordinate of shear center, zs

ItemType (Optional)

Type: $\hat{A}$ ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the assignment is made to the frame object specified by the Name item.

If this item is Group, the assignment is made to all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the assignment is made to all selected frame objects and the Name item is ignored.

Return Value

Type: $\hat{A}$ Int32

Returns zero if the item is successfully set; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double
    Dim ProgDet As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set steel design code
    ret = SapModel.DesignSteel.SetCode("Italian NTC 2018")

    'set overwrite item
```

Return Value

Introduction

```
ret = SapModel.DesignSteel.Italian_NTC_2018.SetOverwrite("8", 2, "DCH-MRF", 1)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

📌 See Also

Reference

cDStItalianNTC2018S Interface

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CSI API ETABS v1

cDStItalianNTC2018SAddLanguageSpecificTextSet("LST3F59757E1860") Method

Sets the value of a steel design preference item.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetPreference(
    int Item,
    string textValue,
    double numericValue
)
```

```
Function SetPreference (
    Item As Integer,
    textValue As String,
    numericValue As Double
) As Integer
```

```
Dim instance As cDStItalianNTC2018S
Dim Item As Integer
Dim textValue As String
Dim numericValue As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    textValue, numericValue)
```

```
int SetPreference(
    int Item,
    String^ textValue,
    double numericValue
)
```

```
abstract SetPreference :
    Item : int *
    textValue : string *
    numericValue : float -> int
```

Parameters

Item

Type: [System.Int32](#)

This is an integer between 3 and 23, inclusive, indicating the preference item considered.

3. Method used for Buckling in P-M-M

4. Framing type
5. Partial safety factor, GammaM0
6. Partial safety factor, GammaM1
7. Partial safety factor, GammaM2
8. Consider deflection
9. DL deflection limit, L/Value
10. SDL + LL deflection limit, L/Value
11. LL deflection limit, L/Value
12. Total load deflection limit, L/Value
13. Total camber limit, L/Value
14. Pattern live load factor
15. Demand/capacity ratio limit
16. Multi-response case design
17. Behavior factor, q
18. System overstrength, Omega
19. Consider P-delta
20. Consider torsion
21. Ignore seismic code
22. Ignore special seismic load
23. Are Doubler plates plug-welded

textValue

Type:Â SystemString

The string value of the considered preference item.

numericValue

Type:Â SystemDouble

The numeric value of the considered preference item.

3. Method used for Buckling in P-M-M

- 1 = Method A
- 2 = Method B
- 3 = Both

4. Framing type

- 1 = DCH-MRF
- 2 = DCL-MRF
- 3 = DCH-CBF
- 4 = DCL-CBF
- 5 = DCH-EBF
- 6 = DCL-EBF
- 7 = InvPendulum
- 8 = Non Dissipative

5. Partial safety factor, GammaM0

Value > 0

6. Partial safety factor, GammaM1

Value > 0

7. Partial safety factor, GammaM2

Value > 0

8. Consider deflection

1 or "No" - Default

2 or "Yes"

9. DL deflection limit, L/Value

Value > 0

10. SDL + LL deflection limit, L/Value

Value > 0

11. LL deflection limit, L/Value

Value > 0

12. Total load deflection limit, L/Value

Value > 0

13. Total camber limit limit, L/Value

Value > 0

14. Pattern live load factor

Value > 0

15. Demand/capacity ratio limit

Value > 0

16. Multi-response case design

- 1 = Envelopes
- 2 = Step-by-Step
- 3 = Last Step
- 4 = Envelope -- All
- 5 = Step-by-Step -- All

17. Behavior factor, q

Value > 0

18. System overstrength, Omega

Value > 0

19. Consider P-delta

1 or "No" - Default

2 or "Yes"

20. Consider Torsion

1 or "No" - Default

2 or "Yes"

21. Ignore seismic code

1 or "No" - Default

2 or "Yes"

22. Ignore special seismic load

1 or "No" - Default

2 or "Yes"

23. Are Doubler plates plug-welded

1 or "No" - Default

2 or "Yes"

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim textValue As String
    Dim Value As Double
    Dim ProgDet As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set steel design code
    ret = SapModel.DesignSteel.SetCode("Italian NTC 2018")

    'set overwrite item
    ret = SapModel.DesignSteel.Italian_NTC_2018.SetPreference(2, "DCH-MRF", 1)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cDStItalianNTC2018S Interface

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CSI API ETABS v1

cDStNewZealand_NZS3404_97 Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDStNewZealand_NZS3404_97
```

```
Public Interface cDStNewZealand_NZS3404_97
```

```
Dim instance As cDStNewZealand_NZS3404_97
```

```
public interface class cDStNewZealand_NZS3404_97
```

```
type cDStNewZealand_NZS3404_97 = interface end
```

The cDStNewZealand_NZS3404_97 type exposes the following members.

▲ Methods

▲	Name	Description
---	------	-------------

⇒	GetOverwrite	
---	------------------------------	--

⇒	GetPreference	
---	-------------------------------	--

⇒	SetOverwrite	
---	------------------------------	--

⇒	SetPreference	
---	-------------------------------	--

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDStNewZealand_NZS3404_97 Methods

The cDStNewZealand_NZS3404_97 type exposes the following members.

Methods

	Name	Description
	<u>GetOverwrite</u>	
	<u>GetPreference</u>	
	<u>SetOverwrite</u>	
	<u>SetPreference</u>	

Top

See Also

Reference

cDStNewZealand_NZS3404_97 Interface

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CSI API ETABS v1

cDStNewZealand_NZS3404_97AddLanguageSpecificTextSet Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDStNewZealand_NZS3404_97
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```

Parameters

Name

Type: SystemString

Item

Type:Â [SystemInt32](#)

Value

Type:Â [SystemDouble](#)

ProgDet

Type:Â [SystemBoolean](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cDStNewZealand_NZS3404_97 Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDStNewZealand_NZS3404_97AddLanguageSpecificTextSet Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDStNewZealand_NZS3404_97
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: [SystemInt32](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDStNewZealand_NZS3404_97 Interface

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CSI API ETABS v1

cDStNewZealand_NZS3404_97AddLanguageSpecificTextSet Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDStNewZealand_NZS3404_97
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

Item

Type: SystemInt32

Value

Type: SystemDouble

ItemType (Optional)

Type: ETABSv1eItemType

Return Value

Type: Int32

▲ See Also

Reference

cDStNewZealand_NZS3404_97 Interface

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CSI API ETABS v1

cDStNewZealand_NZS3404_97AddLanguageSpecificTextSet Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDStNewZealand_NZS3404_97
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: [SystemInt32](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cDStNewZealand_NZS3404_97 Interface

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CSI API ETABS v1

cDStSP16_13330_2011 Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDStSP16_13330_2011
```

```
Public Interface cDStSP16_13330_2011
```

```
Dim instance As cDStSP16_13330_2011
```

```
public interface class cDStSP16_13330_2011
```

```
type cDStSP16_13330_2011 = interface end
```

The cDStSP16_13330_2011 type exposes the following members.

▲ Methods

▲	Name	Description
➤	GetOverwrite	Retrieves the value of a steel design overwrite item
➤	GetPreference	Retrieves the value of a steel design preference item
➤	SetOverwrite	Sets the value of a steel design overwrite item
➤	SetPreference	Sets the value of a steel design preference item

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDStSP16_13330_2011 Methods

The cDStSP16_13330_2011 type exposes the following members.

Methods

⚡	Name	Description
⇒	<u>GetOverwrite</u>	Retrieves the value of a steel design overwrite item
⇒	<u>GetPreference</u>	Retrieves the value of a steel design preference item
⇒	<u>SetOverwrite</u>	Sets the value of a steel design overwrite item
⇒	<u>SetPreference</u>	Sets the value of a steel design preference item

Top

⚡ See Also

Reference

cDStSP16_13330_2011 Interface

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CSI API ETABS v1

cDStSP16_13330_2011AddLanguageSpecificTextSet("L Method

Retrieves the value of a steel design overwrite item

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDStSP16_13330_2011
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing frame object with a steel frame design procedure

Item

Type: SystemInt32

This is an integer between 1 and 40, inclusive, indicating the overwrite item considered

1. Framing type (Not used anymore)
2. Section class (Not used anymore)
3. Service factor, GammaC
4. Safety factor, GammaU
5. Factor for L-Shapes, GammaC1
6. Column buckling curve major
7. Column buckling curve minor
8. Is rolled section?
9. Is beam top loaded?
10. Consider deflection
11. Deflection check type
12. DL deflection limit, L/Value
13. SDL + LL deflection limit, L/Value
14. LL deflection limit, L/Value (Not used anymore)
15. Total load deflection limit, L/Value
16. Total-camber limit, L/Value
17. DL deflection limit, absolute
18. SDL + LL deflection limit, absolute
19. LL deflection limit, absolute (Not used anymore)
20. Total load deflection limit, absolute
21. Total-camber limit, absolute
22. Specified camber
23. Net area to total area ratio
24. Live load reduction factor
25. Unbraced length ratio, Major
26. Unbraced length ratio, Minor
27. Unbraced length ratio, LTB
28. Effective length factor, K1 Major (Not used anymore)
29. Effective length factor, K1 Minor (Not used anymore)
30. Effective length factor, K Major
31. Effective length factor, K Minor
32. Effective length factor, K LTB
33. Characteristic strength, Ryn
34. Design yield strength, Ry
35. Design fracture strength, Ru
36. Design shear strength, Rs
37. Demand/capacity ratio limit
38. Limit slenderness compression, A
39. Limit slenderness compression, B
40. Limit slenderness tension

Value

Type: SystemDouble

The value of the considered overwrite item

1. Framing type (Not used anymore)
2. Section class (Not used anymore)
3. Service factor, GammaC

Value ≥ 0 ; 0 means use program determined value

4. Safety factor, GammaU

Value ≥ 0 ; 0 means use program determined value

5. Factor for L-Shapes, GammaC1

Value ≥ 0 ; 0 means use program determined value

6. Column buckling curve major
 - . 0 = Program Determined
 - . 1 = a
 - . 2 = b
 - . 3 = c

7. Column buckling curve minor
 - . 0 = Program Determined
 - . 1 = a
 - . 2 = b
 - . 3 = c

8. Is rolled section?
 - . 0 = Program Determined
 - . 1 = No
 - . 2 = Yes

9. Is beam top loaded?
 - . 0 = Program Determined
 - . 1 = No
 - . 2 = Yes

10. Consider deflection?
 - . 0 = Program Determined
 - . 1 = No
 - . 2 = Yes

11. Deflection check type
 - . 0 = Program Determined
 - . 1 = Ratio
 - . 2 = Absolute
 - . 3 = Both

12. DL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

13. SDL + LL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

14. LL deflection limit, L/Value (Not used anymore)
15. Total load deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

16. Total-camber limit, L/Value

Introduction

Value ≥ 0 ; 0 means no check for this item

17. DL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

18. SDL + LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

19. LL deflection limit, absolute (Not used anymore)

20. Total load deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

21. Total-camber limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

22. Specified camber

Value ≥ 0 . [L]

23. Net area to total area ratio

Value ≥ 0 ; 0 means use program default value

24. Live load reduction factor

Value ≥ 0 ; 0 means use program determined value

25. Unbraced length ratio, Major

Value ≥ 0 ; 0 means use program determined value

26. Unbraced length ratio, Minor

Value ≥ 0 ; 0 means use program determined value

27. Unbraced length ratio, LTB

Value ≥ 0 ; 0 means use program determined value

28. Effective length factor, K1 Major (Not used anymore)

29. Effective length factor, K1 Minor (Not used anymore)

30. Effective length factor, K Major

Value ≥ 0 ; 0 means use program determined value

31. Effective length factor, K Minor

Value ≥ 0 ; 0 means use program determined value

32. Effective length factor, K LTB

Value ≥ 0 ; 0 means use program determined value

33. Characteristic strength, R_y

Value ≥ 0 ; 0 means use program determined value [F/L2]

34. Design yield strength, R_y

Value ≥ 0 ; 0 means use program determined value [F/L2]

35. Design fracture strength, R_u

Introduction

Value ≥ 0 ; 0 means use program determined value [F/L2]

36. Design shear strength, R_s

Value ≥ 0 ; 0 means use program determined value [F/L2]

37. Demand/capacity ratio limit

Value ≥ 0 ; 0 means use program determined value

38. Limit slenderness compression, A

Value ≥ 0 ; 0 means use program determined value

39. Limit slenderness compression, B

Value ≥ 0 ; 0 means use program determined value

40. Limit slenderness tension

Value ≥ 0 ; 0 means use program determined value

ProgDet

Type: System.Boolean

If this item is True, the specified value is program determined

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim Value As Double  
    Dim ProgDet As Boolean  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)  
  
    'set steel design code  
    ret = SapModel.DesignSteel.SetCode("SP 16.13330.2011")  
  
    'get overwrite item  
    ret = SapModel.DesignSteel.SP16_13330_2011.GetOverwrite("8", 1, Value, ProgDet)
```

Return Value

```
'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

📌 See Also

Reference

cDStSP16_13330_2011 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDStSP16_13330_2011AddLanguageSpecificTextSet("L Method

Retrieves the value of a steel design preference item

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDStSP16_13330_2011
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: [SystemInt32](#)

This is an integer between 1 and 16, inclusive, indicating the preference item considered

1. Multi-response case design
2. Framing type (Not used anymore)
3. Section class (Not used anymore)
4. GammaM
5. GammaC

- 6. GammeU
- 7. GammaC1
- 8. Consider deflection
- 9. DL deflection limit, L/Value
- 10. SDL + LL deflection limit, L/Value
- 11. LL deflection limit, L/Value (Not used anymore)
- 12. Total load deflection limit, L/Value
- 13. Total - Camber limit, L/Value
- 14. Pattern live load factor
- 15. Demand/capacity ratio limit
- 16. Reliability factor
- 17. Max Number of Auto Iterations

Value

Type: SystemDouble

The value of the considered preference item

- 1. Multi-response case design
 - 1 = Envelopes
 - 2 = Step-by-step
 - 3 = Last step
 - 4 = Envelopes -- All
 - 5 = Step-by-step -- All
- 2. Framing type (Not used anymore)
- 3. Section class (Not used anymore)
- 4. GammaM

Value > 0

- 5. GammaC

Value > 0

- 6. GammaU

Value > 0

- 7. GammaC1

Value > 0

- 8. Consider deflection
 - 0 = No
 - Any other value = Yes
- 9. DL deflection limit, L/Value

Value > 0

- 10. SDL + LL deflection limit, L/Value

Value > 0

- 11. LL deflection limit, L/Value (Not used anymore)
- 12. Total deflection limit, L/Value

Value > 0

- 13. Total camber limit, L/Value

Value > 0

14. Pattern live load factor

Value ≥ 0

15. Demand/capacity ratio limit

Value > 0

16. Reliability factor

Value > 0

17. Max Number of Auto Iterations

Value > 0 **Return Value**Type: $\hat{A}$ Int32

Returns zero if the item is successfully retrieved; otherwise, it returns a nonzero value

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set steel design code
    ret = SapModel.DesignSteel.SetCode("SP 16.13330.2011")

    'get preference item
    ret = SapModel.DesignSteel.SP16_13330_2011.GetPreference(4, Value)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cDStSP16_13330_2011 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDStSP16_13330_2011AddLanguageSpecificTextSet("L Method

Sets the value of a steel design overwrite item

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDStSP16_13330_2011
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing frame object with a steel frame design procedure

Item

Type: SystemInt32

This is an integer between 1 and 40, inclusive, indicating the overwrite item considered

1. Framing type (Not used anymore)
2. Section class (Not used anymore)
3. Service factor, GammaC
4. Safety factor, GammaU
5. Factor for L-Shapes, GammaC1
6. Column buckling curve major
7. Column buckling curve minor
8. Is rolled section?
9. Is beam top loaded?
10. Consider deflection
11. Deflection check type
12. DL deflection limit, L/Value
13. SDL + LL deflection limit, L/Value
14. LL deflection limit, L/Value (Not used anymore)
15. Total load deflection limit, L/Value
16. Total-camber limit, L/Value
17. DL deflection limit, absolute
18. SDL + LL deflection limit, absolute
19. LL deflection limit, absolute (Not used anymore)
20. Total load deflection limit, absolute
21. Total-camber limit, absolute
22. Specified camber
23. Net area to total area ratio
24. Live load reduction factor
25. Unbraced length ratio, Major
26. Unbraced length ratio, Minor
27. Unbraced length ratio, LTB
28. Effective length factor, K1 Major (Not used anymore)
29. Effective length factor, K1 Minor (Not used anymore)
30. Effective length factor, K Major
31. Effective length factor, K Minor
32. Effective length factor, K LTB
33. Characteristic strength, Ryn
34. Design yield strength, Ry
35. Design fracture strength, Ru
36. Design shear strength, Rs
37. Demand/capacity ratio limit
38. Limit slenderness compression, A
39. Limit slenderness compression, B
40. Limit slenderness tension

Value

Type: SystemDouble

The value of the considered overwrite item

1. Framing type (Not used anymore)
2. Section class (Not used anymore)
3. Service factor, GammaC

Value ≥ 0 ; 0 means use program determined value

4. Safety factor, GammaU

Value ≥ 0 ; 0 means use program determined value

5. Factor for L-Shapes, GammaC1

Value ≥ 0 ; 0 means use program determined value

6. Column buckling curve major
 - 0 = Program Determined
 - 1 = a
 - 2 = b
 - 3 = c

7. Column buckling curve minor
 - 0 = Program Determined
 - 1 = a
 - 2 = b
 - 3 = c

8. Is rolled section?
 - 0 = Program Determined
 - 1 = No
 - 2 = Yes

9. Is beam top loaded?
 - 0 = Program Determined
 - 1 = No
 - 2 = Yes

10. Consider deflection?
 - 0 = Program Determined
 - 1 = No
 - 2 = Yes

11. Deflection check type
 - 0 = Program Determined
 - 1 = Ratio
 - 2 = Absolute
 - 3 = Both

12. DL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

13. SDL + LL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

14. LL deflection limit, L/Value (Not used anymore)

15. Total load deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

16. Total-camber limit, L/Value

Introduction

Value ≥ 0 ; 0 means no check for this item

17. DL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

18. SDL + LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

19. LL deflection limit, absolute (Not used anymore)

20. Total load deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

21. Total-camber limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

22. Specified camber

Value ≥ 0 . [L]

23. Net area to total area ratio

Value ≥ 0 ; 0 means use program default value

24. Live load reduction factor

Value ≥ 0 ; 0 means use program determined value

25. Unbraced length ratio, Major

Value ≥ 0 ; 0 means use program determined value

26. Unbraced length ratio, Minor

Value ≥ 0 ; 0 means use program determined value

27. Unbraced length ratio, LTB

Value ≥ 0 ; 0 means use program determined value

28. Effective length factor, K1 Major (Not used anymore)

29. Effective length factor, K1 Minor (Not used anymore)

30. Effective length factor, K Major

Value ≥ 0 ; 0 means use program determined value

31. Effective length factor, K Minor

Value ≥ 0 ; 0 means use program determined value

32. Effective length factor, K LTB

Value ≥ 0 ; 0 means use program determined value

33. Characteristic strength, R_{yn}

Value ≥ 0 ; 0 means use program determined value [F/L2]

34. Design yield strength, R_y

Value ≥ 0 ; 0 means use program determined value [F/L2]

35. Design fracture strength, R_u

Introduction

Value ≥ 0 ; 0 means use program determined value [F/L2]

36. Design shear strength, R_s

Value ≥ 0 ; 0 means use program determined value [F/L2]

37. Demand/capacity ratio limit

Value ≥ 0 ; 0 means use program determined value

38. Limit slenderness compression, A

Value ≥ 0 ; 0 means use program determined value

39. Limit slenderness compression, B

Value ≥ 0 ; 0 means use program determined value

40. Limit slenderness tension

Value ≥ 0 ; 0 means use program determined value

ItemType (Optional)

Type: $\hat{A}$ ETABSv1eItemType

Return Value

Type: $\hat{A}$ Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim Value As Double  
    Dim ProgDet As Boolean  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)  
  
    'set steel design code  
    ret = SapModel.DesignSteel.SetCode("SP 16.13330.2011")  
  
    'get overwrite item  
    ret = SapModel.DesignSteel.SP16_13330_2011.GetOverwrite("8", 1, Value, ProgDet)
```


Introduction

```
'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cDStSP16_13330_2011 Interface

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CSI API ETABS v1

cDStSP16_13330_2011AddLanguageSpecificTextSet("L Method

Sets the value of a steel design preference item

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDStSP16_13330_2011
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: SystemInt32

This is an integer between 1 and 16, inclusive, indicating the preference item considered

1. Multi-response case design
2. Framing type (Not used anymore)
3. Section class (Not used anymore)
4. GammaM
5. GammaC

6. GammeU
7. GammaC1
8. Consider deflection
9. DL deflection limit, L/Value
10. SDL + LL deflection limit, L/Value
11. LL deflection limit, L/Value (Not used anymore)
12. Total load deflection limit, L/Value
13. Total - Camber limit, L/Value
14. Pattern live load factor
15. Demand/capacity ratio limit
16. Reliability factor
17. Max Number of Auto Iterations

Value

Type: SystemDouble

The value of the considered preference item

1. Multi-response case design
 - 1 = Envelopes
 - 2 = Step-by-step
 - 3 = Last step
 - 4 = Envelopes -- All
 - 5 = Step-by-step -- All
2. Framing type (Not used anymore)
3. Section class (Not used anymore)
4. GammaM

Value > 0

5. GammaC

Value > 0

6. GammaU

Value > 0

7. GammaC1

Value > 0

8. Consider deflection
 - 0 = No
 - Any other value = Yes
9. DL deflection limit, L/Value

Value > 0

10. SDL + LL deflection limit, L/Value

Value > 0

11. LL deflection limit, L/Value (Not used anymore)
12. Total deflection limit, L/Value

Value > 0

13. Total camber limit, L/Value

Value > 0

14. Pattern live load factor

Value ≥ 0

15. Demand/capacity ratio limit

Value > 0

16. Reliability factor

Value > 0

17. Max Number of Auto Iterations

Value > 0 **Return Value**Type: $\hat{A}$ Int32

Returns zero if the item is successfully retrieved; otherwise, it returns a nonzero value

▲ Remarks

▲ Examples

VB

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set steel design code
    ret = SapModel.DesignSteel.SetCode("SP 16.13330.2011")

    'get preference item
    ret = SapModel.DesignSteel.SP16_13330_2011.GetPreference(4, Value)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub

```

▲ See Also

Reference

cDStSP16_13330_2011 Interface

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CSI API ETABS v1

cDStSP16_13330_2017 Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cDStSP16_13330_2017
```

```
Public Interface cDStSP16_13330_2017
```

```
Dim instance As cDStSP16_13330_2017
```

```
public interface class cDStSP16_13330_2017
```

```
type cDStSP16_13330_2017 = interface end
```

The cDStSP16_13330_2017 type exposes the following members.

▲ Methods

▲	Name	Description
➤	GetOverwrite	Retrieves the value of a steel design overwrite item
➤	GetPreference	Retrieves the value of a steel design preference item
➤	SetOverwrite	Sets the value of a steel design overwrite item
➤	SetPreference	Sets the value of a steel design preference item

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cDStSP16_13330_2017 Methods

The cDStSP16_13330_2017 type exposes the following members.

Methods

⚡	Name	Description
⇒	<u>GetOverwrite</u>	Retrieves the value of a steel design overwrite item
⇒	<u>GetPreference</u>	Retrieves the value of a steel design preference item
⇒	<u>SetOverwrite</u>	Sets the value of a steel design overwrite item
⇒	<u>SetPreference</u>	Sets the value of a steel design preference item

Top

⚡ See Also

Reference

cDStSP16_13330_2017 Interface

ETABSV1 Namespace

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CSI API ETABS v1

cDStSP16_13330_2017AddLanguageSpecificTextSet("L Method

Retrieves the value of a steel design overwrite item

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetOverwrite(
    string Name,
    int Item,
    ref double Value,
    ref bool ProgDet
)
```

```
Function GetOverwrite (
    Name As String,
    Item As Integer,
    ByRef Value As Double,
    ByRef ProgDet As Boolean
) As Integer
```

```
Dim instance As cDStSP16_13330_2017
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ProgDet As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOverwrite(Name,
    Item, Value, ProgDet)
```

```
int GetOverwrite(
    String^ Name,
    int Item,
    double% Value,
    bool% ProgDet
)
```

```
abstract GetOverwrite :
    Name : string *
    Item : int *
    Value : float byref *
    ProgDet : bool byref -> int
```


Parameters

Name

Type: SystemString

The name of an existing frame object with a steel frame design procedure

Item

Type: SystemInt32

This is an integer between 1 and 40, inclusive, indicating the overwrite item considered

1. Framing type (Not used anymore)
2. Section class (Not used anymore)
3. Service factor, GammaC
4. Safety factor, GammaU
5. Factor for L-Shapes, GammaC1
6. Column buckling curve major
7. Column buckling curve minor
8. Is rolled section?
9. Is beam top loaded?
10. Consider deflection
11. Deflection check type
12. DL deflection limit, L/Value
13. SDL + LL deflection limit, L/Value
14. LL deflection limit, L/Value (Not used anymore)
15. Total load deflection limit, L/Value
16. Total-camber limit, L/Value
17. DL deflection limit, absolute
18. SDL + LL deflection limit, absolute
19. LL deflection limit, absolute (Not used anymore)
20. Total load deflection limit, absolute
21. Total-camber limit, absolute
22. Specified camber
23. Net area to total area ratio
24. Live load reduction factor
25. Unbraced length ratio, Major
26. Unbraced length ratio, Minor
27. Unbraced length ratio, LTB
28. Effective length factor, K1 Major (Not used anymore)
29. Effective length factor, K1 Minor (Not used anymore)
30. Effective length factor, K Major
31. Effective length factor, K Minor
32. Effective length factor, K LTB
33. Characteristic strength, Ryn
34. Design yield strength, Ry
35. Design fracture strength, Ru
36. Design shear strength, Rs
37. Demand/capacity ratio limit
38. Limit slenderness compression, A
39. Limit slenderness compression, B
40. Limit slenderness tension

Value

Type: SystemDouble

The value of the considered overwrite item

1. Framing type (Not used anymore)
2. Section class (Not used anymore)
3. Service factor, GammaC

Value ≥ 0 ; 0 means use program determined value

4. Safety factor, GammaU

Value ≥ 0 ; 0 means use program determined value

5. Factor for L-Shapes, GammaC1

Value ≥ 0 ; 0 means use program determined value

6. Column buckling curve major
 - . 0 = Program Determined
 - . 1 = a
 - . 2 = b
 - . 3 = c

7. Column buckling curve minor
 - . 0 = Program Determined
 - . 1 = a
 - . 2 = b
 - . 3 = c

8. Is rolled section?
 - . 0 = Program Determined
 - . 1 = No
 - . 2 = Yes

9. Is beam top loaded?
 - . 0 = Program Determined
 - . 1 = No
 - . 2 = Yes

10. Consider deflection?
 - . 0 = Program Determined
 - . 1 = No
 - . 2 = Yes

11. Deflection check type
 - . 0 = Program Determined
 - . 1 = Ratio
 - . 2 = Absolute
 - . 3 = Both

12. DL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

13. SDL + LL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

14. LL deflection limit, L/Value (Not used anymore)

15. Total load deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

16. Total-camber limit, L/Value

Introduction

Value ≥ 0 ; 0 means no check for this item

17. DL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

18. SDL + LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

19. LL deflection limit, absolute (Not used anymore)

20. Total load deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

21. Total-camber limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

22. Specified camber

Value ≥ 0 . [L]

23. Net area to total area ratio

Value ≥ 0 ; 0 means use program default value

24. Live load reduction factor

Value ≥ 0 ; 0 means use program determined value

25. Unbraced length ratio, Major

Value ≥ 0 ; 0 means use program determined value

26. Unbraced length ratio, Minor

Value ≥ 0 ; 0 means use program determined value

27. Unbraced length ratio, LTB

Value ≥ 0 ; 0 means use program determined value

28. Effective length factor, K1 Major (Not used anymore)

29. Effective length factor, K1 Minor (Not used anymore)

30. Effective length factor, K Major

Value ≥ 0 ; 0 means use program determined value

31. Effective length factor, K Minor

Value ≥ 0 ; 0 means use program determined value

32. Effective length factor, K LTB

Value ≥ 0 ; 0 means use program determined value

33. Characteristic strength, R_y

Value ≥ 0 ; 0 means use program determined value [F/L2]

34. Design yield strength, R_y

Value ≥ 0 ; 0 means use program determined value [F/L2]

35. Design fracture strength, R_u

Introduction

Value ≥ 0 ; 0 means use program determined value [F/L2]

36. Design shear strength, R_s

Value ≥ 0 ; 0 means use program determined value [F/L2]

37. Demand/capacity ratio limit

Value ≥ 0 ; 0 means use program determined value

38. Limit slenderness compression, A

Value ≥ 0 ; 0 means use program determined value

39. Limit slenderness compression, B

Value ≥ 0 ; 0 means use program determined value

40. Limit slenderness tension

Value ≥ 0 ; 0 means use program determined value

ProgDet

Type: System.Boolean

If this item is True, the specified value is program determined

Return Value

Type: Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim Value As Double  
    Dim ProgDet As Boolean  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)  
  
    'set steel design code  
    ret = SapModel.DesignSteel.SetCode("SP 16.13330.2017")  
  
    'get overwrite item  
    ret = SapModel.DesignSteel.SP16_13330_2017.GetOverwrite("8", 1, Value, ProgDet)
```

Return Value

```
'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

📌 See Also

Reference

cDStSP16_13330_2017 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDStSP16_13330_2017AddLanguageSpecificTextSet("L Method

Retrieves the value of a steel design preference item

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPreference(
    int Item,
    ref double Value
)
```

```
Function GetPreference (
    Item As Integer,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cDStSP16_13330_2017
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPreference(Item,
    Value)
```

```
int GetPreference(
    int Item,
    double% Value
)
```

```
abstract GetPreference :
    Item : int *
    Value : float byref -> int
```

Parameters

Item

Type: [SystemInt32](#)

This is an integer between 1 and 16, inclusive, indicating the preference item considered

1. Multi-response case design
2. Framing type (Not used anymore)
3. Section class (Not used anymore)
4. GammaM
5. GammaC

6. GammeU
7. GammaC1
8. Consider deflection
9. DL deflection limit, L/Value
10. SDL + LL deflection limit, L/Value
11. LL deflection limit, L/Value (Not used anymore)
12. Total load deflection limit, L/Value
13. Total - Camber limit, L/Value
14. Pattern live load factor
15. Demand/capacity ratio limit
16. Reliability factor
17. Max Number of Auto Iterations

Value

Type: SystemDouble

The value of the considered preference item

1. Multi-response case design
 - 1 = Envelopes
 - 2 = Step-by-step
 - 3 = Last step
 - 4 = Envelopes -- All
 - 5 = Step-by-step -- All
2. Framing type (Not used anymore)
3. Section class (Not used anymore)
4. GammaM

Value > 0

5. GammaC

Value > 0

6. GammaU

Value > 0

7. GammaC1

Value > 0

8. Consider deflection
 - 0 = No
 - Any other value = Yes
9. DL deflection limit, L/Value

Value > 0

10. SDL + LL deflection limit, L/Value

Value > 0

11. LL deflection limit, L/Value (Not used anymore)
12. Total deflection limit, L/Value

Value > 0

13. Total camber limit, L/Value

Value > 0

14. Pattern live load factor

Value ≥ 0

15. Demand/capacity ratio limit

Value > 0

16. Reliability factor

Value > 0

17. Max Number of Auto Iterations

Value > 0 **Return Value**Type: $\hat{A}$ Int32

Returns zero if the item is successfully retrieved; otherwise, it returns a nonzero value

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set steel design code
    ret = SapModel.DesignSteel.SetCode("SP 16.13330.2017")

    'get preference item
    ret = SapModel.DesignSteel.SP16_13330_2017.GetPreference(4, Value)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```


▲ See Also

Reference

cDStSP16_13330_2017 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDStSP16_13330_2017AddLanguageSpecificTextSet("L Method

Sets the value of a steel design overwrite item

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetOverwrite(
    string Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOverwrite (
    Name As String,
    Item As Integer,
    Value As Double,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cDStSP16_13330_2017
Dim Name As String
Dim Item As Integer
Dim Value As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOverwrite(Name,
    Item, Value, ItemType)
```

```
int SetOverwrite(
    String^ Name,
    int Item,
    double Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetOverwrite :
    Name : string *
    Item : int *
    Value : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing frame object with a steel frame design procedure

Item

Type: SystemInt32

This is an integer between 1 and 40, inclusive, indicating the overwrite item considered

1. Framing type (Not used anymore)
2. Section class (Not used anymore)
3. Service factor, GammaC
4. Safety factor, GammaU
5. Factor for L-Shapes, GammaC1
6. Column buckling curve major
7. Column buckling curve minor
8. Is rolled section?
9. Is beam top loaded?
10. Consider deflection
11. Deflection check type
12. DL deflection limit, L/Value
13. SDL + LL deflection limit, L/Value
14. LL deflection limit, L/Value (Not used anymore)
15. Total load deflection limit, L/Value
16. Total-camber limit, L/Value
17. DL deflection limit, absolute
18. SDL + LL deflection limit, absolute
19. LL deflection limit, absolute (Not used anymore)
20. Total load deflection limit, absolute
21. Total-camber limit, absolute
22. Specified camber
23. Net area to total area ratio
24. Live load reduction factor
25. Unbraced length ratio, Major
26. Unbraced length ratio, Minor
27. Unbraced length ratio, LTB
28. Effective length factor, K1 Major (Not used anymore)
29. Effective length factor, K1 Minor (Not used anymore)
30. Effective length factor, K Major
31. Effective length factor, K Minor
32. Effective length factor, K LTB
33. Characteristic strength, Ryn
34. Design yield strength, Ry
35. Design fracture strength, Ru
36. Design shear strength, Rs
37. Demand/capacity ratio limit
38. Limit slenderness compression, A
39. Limit slenderness compression, B
40. Limit slenderness tension

Value

Type: SystemDouble

The value of the considered overwrite item

1. Framing type (Not used anymore)
2. Section class (Not used anymore)
3. Service factor, GammaC

Value ≥ 0 ; 0 means use program determined value

4. Safety factor, GammaU

Value ≥ 0 ; 0 means use program determined value

5. Factor for L-Shapes, GammaC1

Value ≥ 0 ; 0 means use program determined value

6. Column buckling curve major
 - . 0 = Program Determined
 - . 1 = a
 - . 2 = b
 - . 3 = c

7. Column buckling curve minor
 - . 0 = Program Determined
 - . 1 = a
 - . 2 = b
 - . 3 = c

8. Is rolled section?
 - . 0 = Program Determined
 - . 1 = No
 - . 2 = Yes

9. Is beam top loaded?
 - . 0 = Program Determined
 - . 1 = No
 - . 2 = Yes

10. Consider deflection?
 - . 0 = Program Determined
 - . 1 = No
 - . 2 = Yes

11. Deflection check type
 - . 0 = Program Determined
 - . 1 = Ratio
 - . 2 = Absolute
 - . 3 = Both

12. DL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

13. SDL + LL deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

14. LL deflection limit, L/Value (Not used anymore)

15. Total load deflection limit, L/Value

Value ≥ 0 ; 0 means no check for this item

16. Total-camber limit, L/Value

Introduction

Value ≥ 0 ; 0 means no check for this item

17. DL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

18. SDL + LL deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

19. LL deflection limit, absolute (Not used anymore)

20. Total load deflection limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

21. Total-camber limit, absolute

Value ≥ 0 ; 0 means no check for this item. [L]

22. Specified camber

Value ≥ 0 . [L]

23. Net area to total area ratio

Value ≥ 0 ; 0 means use program default value

24. Live load reduction factor

Value ≥ 0 ; 0 means use program determined value

25. Unbraced length ratio, Major

Value ≥ 0 ; 0 means use program determined value

26. Unbraced length ratio, Minor

Value ≥ 0 ; 0 means use program determined value

27. Unbraced length ratio, LTB

Value ≥ 0 ; 0 means use program determined value

28. Effective length factor, K1 Major (Not used anymore)

29. Effective length factor, K1 Minor (Not used anymore)

30. Effective length factor, K Major

Value ≥ 0 ; 0 means use program determined value

31. Effective length factor, K Minor

Value ≥ 0 ; 0 means use program determined value

32. Effective length factor, K LTB

Value ≥ 0 ; 0 means use program determined value

33. Characteristic strength, R_y

Value ≥ 0 ; 0 means use program determined value [F/L2]

34. Design yield strength, R_y

Value ≥ 0 ; 0 means use program determined value [F/L2]

35. Design fracture strength, R_u

Introduction

Value ≥ 0 ; 0 means use program determined value [F/L2]

36. Design shear strength, R_s

Value ≥ 0 ; 0 means use program determined value [F/L2]

37. Demand/capacity ratio limit

Value ≥ 0 ; 0 means use program determined value

38. Limit slenderness compression, A

Value ≥ 0 ; 0 means use program determined value

39. Limit slenderness compression, B

Value ≥ 0 ; 0 means use program determined value

40. Limit slenderness tension

Value ≥ 0 ; 0 means use program determined value

ItemType (Optional)

Type: $\hat{A}$ ETABSv1eItemType

Return Value

Type: $\hat{A}$ Int32

Returns zero if the item is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim Value As Double  
    Dim ProgDet As Boolean  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)  
  
    'set steel design code  
    ret = SapModel.DesignSteel.SetCode("SP 16.13330.2017")  
  
    'get overwrite item  
    ret = SapModel.DesignSteel.SP16_13330_2017.GetOverwrite("8", 1, Value, ProgDet)
```

Introduction

```
'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cDStSP16_13330_2017 Interface

ETABSv1 Namespace

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CSI API ETABS v1

cDStSP16_13330_2017AddLanguageSpecificTextSet("L Method

Sets the value of a steel design preference item

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetPreference(
    int Item,
    double Value
)
```

```
Function SetPreference (
    Item As Integer,
    Value As Double
) As Integer
```

```
Dim instance As cDStSP16_13330_2017
Dim Item As Integer
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetPreference(Item,
    Value)
```

```
int SetPreference(
    int Item,
    double Value
)
```

```
abstract SetPreference :
    Item : int *
    Value : float -> int
```

Parameters

Item

Type: SystemInt32

This is an integer between 1 and 16, inclusive, indicating the preference item considered

1. Multi-response case design
2. Framing type (Not used anymore)
3. Section class (Not used anymore)
4. GammaM
5. GammaC

- 6. GammeU
- 7. GammaC1
- 8. Consider deflection
- 9. DL deflection limit, L/Value
- 10. SDL + LL deflection limit, L/Value
- 11. LL deflection limit, L/Value (Not used anymore)
- 12. Total load deflection limit, L/Value
- 13. Total - Camber limit, L/Value
- 14. Pattern live load factor
- 15. Demand/capacity ratio limit
- 16. Reliability factor
- 17. Max Number of Auto Iterations

Value

Type: SystemDouble

The value of the considered preference item

- 1. Multi-response case design
 - 1 = Envelopes
 - 2 = Step-by-step
 - 3 = Last step
 - 4 = Envelopes -- All
 - 5 = Step-by-step -- All
- 2. Framing type (Not used anymore)
- 3. Section class (Not used anymore)
- 4. GammaM

Value > 0

- 5. GammaC

Value > 0

- 6. GammaU

Value > 0

- 7. GammaC1

Value > 0

- 8. Consider deflection
 - 0 = No
 - Any other value = Yes
- 9. DL deflection limit, L/Value

Value > 0

- 10. SDL + LL deflection limit, L/Value

Value > 0

- 11. LL deflection limit, L/Value (Not used anymore)
- 12. Total deflection limit, L/Value

Value > 0

- 13. Total camber limit, L/Value

Value > 0

14. Pattern live load factor

Value ≥ 0

15. Demand/capacity ratio limit

Value > 0

16. Reliability factor

Value > 0

17. Max Number of Auto Iterations

Value > 0 **Return Value**Type: $\hat{A}$ Int32

Returns zero if the item is successfully retrieved; otherwise, it returns a nonzero value

▲ Remarks

▲ Examples

VB

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set steel design code
    ret = SapModel.DesignSteel.SetCode("SP 16.13330.2017")

    'get preference item
    ret = SapModel.DesignSteel.SP16_13330_2017.GetPreference(4, Value)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub

```

See Also

Reference

cDStSP16_13330_2017 Interface

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CSI API ETABS v1

cEditFrame Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cEditFrame
```

```
Public Interface cEditFrame
```

```
Dim instance As cEditFrame
```

```
public interface class cEditFrame
```

```
type cEditFrame = interface end
```

The cEditFrame type exposes the following members.

▲ Methods

▲

Name

Description

▲

[ChangeConnectivity](#) Modifies the connectivity of a frame object

[Top](#)

▲

See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cEditFrame Methods

The [cEditFrame](#) type exposes the following members.

Methods

	Name	Description
	ChangeConnectivity	Modifies the connectivity of a frame object

[Top](#)

See Also

Reference

[cEditFrame Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cEditFrameAddLanguageSpecificTextSet("LSTC27CDE Method

Modifies the connectivity of a frame object

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int ChangeConnectivity(
    string Name,
    string Point1,
    string Point2
)
```

```
Function ChangeConnectivity (
    Name As String,
    Point1 As String,
    Point2 As String
) As Integer
```

```
Dim instance As cEditFrame
Dim Name As String
Dim Point1 As String
Dim Point2 As String
Dim returnValue As Integer
```

```
returnValue = instance.ChangeConnectivity(Name,
    Point1, Point2)
```

```
int ChangeConnectivity(
    String^ Name,
    String^ Point1,
    String^ Point2
)
```

```
abstract ChangeConnectivity :
    Name : string *
    Point1 : string *
    Point2 : string -> int
```

Parameters

Name

Type: SystemString

The name of an existing frame object

Point1

Type: SystemString

The name of the point object to set as the I-End of the frame object

Point2

Type: SystemString

The name of the point object to set as the J-End of the frame object

Return Value

Type: Int32

Returns zero if the frame object connectivity is successfully modified; otherwise it returns a nonzero value

Remarks

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'modify connectivity
    ret = SapModel.EditFrame.ChangeConnectivity("8", "3", "5")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

cEditFrame Interface

ETABSv1 Namespace

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Introduction

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CSI API ETABS v1

cEditGeneral Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cEditGeneral
```

```
Public Interface cEditGeneral
```

```
Dim instance As cEditGeneral
```

```
public interface class cEditGeneral
```

```
type cEditGeneral = interface end
```

The cEditGeneral type exposes the following members.

▲ Methods

Â Name

Description

 [Move](#) Moves selected point, frame, cable, tendon, area, solid and link objects.

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cEditGeneral Methods

The [cEditGeneral](#) type exposes the following members.

Methods

Name

Description

 [Move](#) Moves selected point, frame, cable, tendon, area, solid and link objects.

[Top](#)

 See Also

Reference

[cEditGeneral Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cEditGeneralAddLanguageSpecificTextSet("LST8ED49 Method

Moves selected point, frame, cable, tendon, area, solid and link objects.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int Move(
double DX,
double DY,
double DZ
)
```

```
Function Move (
DX As Double,
DY As Double,
DZ As Double
) As Integer
```

```
Dim instance As cEditGeneral
Dim DX As Double
Dim DY As Double
Dim DZ As Double
Dim returnValue As Integer
```

```
returnValue = instance.Move(DX, DY, DZ)
```

```
int Move(
double DX,
double DY,
double DZ
)
```

```
abstract Move :
    DX : float *
    DY : float *
    DZ : float -> int
```

Parameters

DX

Type: [System.Double](#)

This is the x offset used, in the present coordinate system, for moving the selected objects.

DY

Type: SystemDouble

This is the y offset used, in the present coordinate system, for moving the selected objects.

DZ

Type: SystemDouble

This is the z offset used, in the present coordinate system, for moving the selected objects.

Return Value

Type: Int32

Returns zero if the move is successful; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'move objects
    ret = SapModel.SelectObj.ClearSelection
    ret = SapModel.PointObj.SetSelected("8", True)
    ret = SapModel.PointObj.SetSelected("16", True)
    ret = SapModel.FrameObj.SetSelected("74", True)
    ret = SapModel.PointObj.SetSelected("92", True)
    ret = SapModel.PointObj.SetSelected("91", True)
    ret = SapModel.FrameObj.SetSelected("166", True)
    ret = SapModel.EditGeneral.Move(0, 0, -36)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cEditGeneral Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFile Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cFile
```

Public Interface cFile

```
Dim instance As cFile
```

```
public interface class cFile
```

```
type cFile = interface end
```

The cFile type exposes the following members.

▲ Methods

▲	Name	Description
⇒	GetFilePath	
⇒	NewBlank	Creates a new, blank model.
⇒	NewGridOnly	Creates a new grid-only model from template.
⇒	NewSteelDeck	Creates a new steel deck model from template.
⇒	OpenFile	Opens an existing file.
⇒	Save	Saves the present model.

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cFile Methods

The [cFile](#) type exposes the following members.

Methods

⌕	Name	Description
⌕	GetFilePath	
⌕	NewBlank	Creates a new, blank model.
⌕	NewGridOnly	Creates a new grid-only model from template.
⌕	NewSteelDeck	Creates a new steel deck model from template.
⌕	OpenFile	Opens an existing file.
⌕	Save	Saves the present model.

[Top](#)

⌕ See Also

Reference

[cFile Interface](#)

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CSI API ETABS v1

cFileAddLanguageSpecificTextSet("LSTF4062F67_0?cpp=::|nu=."); Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetFilePath(
    ref string FilePath
)
```

```
Function GetFilePath (
    ByRef FilePath As String
) As Integer
```

```
Dim instance As cFile
Dim FilePath As String
Dim returnValue As Integer
```

```
returnValue = instance.GetFilePath(FilePath)
```

```
int GetFilePath(
    String^% FilePath
)
```

```
abstract GetFilePath :
    FilePath : string byref -> int
```

Parameters

FilePath
Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cFile Interface](#)

[ETABSv1 Namespace](#)

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Introduction

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CSI API ETABS v1

cFileAddLanguageSpecificTextSet("LST997383AB_0?c Method

Creates a new, blank model.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int NewBlank()
```

```
Function NewBlank As Integer
```

```
Dim instance As cFile
```

```
Dim returnValue As Integer
```

```
returnValue = instance.NewBlank()
```

```
int NewBlank()
```

```
abstract NewBlank : unit -> int
```

Return Value

Type: Int32

Returns zero if the new blank model is successfully created, otherwise it returns a nonzero value.

▲ Remarks

Do not use this function to add to an existing model. This function should be used only for creating a new model and typically would be preceded by calls to ApplicationStart or InitializeNewModel(eUnits).

▲ Examples

VB

Copy

```
Public Sub Example()
```

```
    Dim SapModel As cSapModel
```

```
    Dim EtabsObject As cOAPI
```

```
    Dim ret As Integer = -1
```

```
    'create ETABS object
```

```
    Dim myHelper as cHelper = New Helper
```

```
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

```
    'start ETABS application
```

```
    ret = EtabsObject.ApplicationStart()
```

```
    'create SapModel object
```

Introduction

```
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create a blank model from template
ret = SapModel.File.NewBlank()

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

📌 See Also

Reference

cFile Interface

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CSI API ETABS v1

cFileAddLanguageSpecificTextSet("LSTC398FFDB_0?c

Method

Creates a new grid-only model from template.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int NewGridOnly(
    int NumberStorys,
    double TypicalStoryHeight,
    double BottomStoryHeight,
    int NumberLinesX,
    int NumberLinesY,
    double SpacingX,
    double SpacingY
)
```

```
Function NewGridOnly (
    NumberStorys As Integer,
    TypicalStoryHeight As Double,
    BottomStoryHeight As Double,
    NumberLinesX As Integer,
    NumberLinesY As Integer,
    SpacingX As Double,
    SpacingY As Double
) As Integer
```

```
Dim instance As cFile
Dim NumberStorys As Integer
Dim TypicalStoryHeight As Double
Dim BottomStoryHeight As Double
Dim NumberLinesX As Integer
Dim NumberLinesY As Integer
Dim SpacingX As Double
Dim SpacingY As Double
Dim returnValue As Integer
```

```
returnValue = instance.NewGridOnly(NumberStorys,
    TypicalStoryHeight, BottomStoryHeight,
    NumberLinesX, NumberLinesY, SpacingX,
    SpacingY)
```

```
int NewGridOnly(
    int NumberStorys,
    double TypicalStoryHeight,
    double BottomStoryHeight,
    int NumberLinesX,
    int NumberLinesY,
```

```
double SpacingX,
double SpacingY
)
```

```
abstract NewGridOnly :
    NumberStorys : int *
    TypicalStoryHeight : float *
    BottomStoryHeight : float *
    NumberLinesX : int *
    NumberLinesY : int *
    SpacingX : float *
    SpacingY : float -> int
```

Parameters

NumberStorys

Type: SystemInt32

The number of stories in the model

TypicalStoryHeight

Type: SystemDouble

The story height that will be used for all stories in the model, except the bottom story. [L]

BottomStoryHeight

Type: SystemDouble

The story height will be used for the bottom story. [L]

NumberLinesX

Type: SystemInt32

The number of grid lines in the X direction.

NumberLinesY

Type: SystemInt32

The number of grid lines in the Y direction.

SpacingX

Type: SystemDouble

The uniform spacing for grid lines in the X direction. [L]

SpacingY

Type: SystemDouble

The uniform spacing for grid lines in the Y direction. [L]

Return Value

Type: Int32

Returns zero if the new grid only model is successfully created; otherwise it returns a nonzero value.

Remarks

Do not use this function to add to an existing model. This function should be used only for creating a new model and typically would be preceded by a call to InitializeNewModel(eUnits).

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
```

Introduction

```
'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
    ret = EtabsObject.ApplicationStart()

'create SapModel object
    SapModel = EtabsObject.SapModel

'initialize model
    ret = SapModel.InitializeNewModel()

'create grid-only template model
    ret = SapModel.File.NewGridOnly(4,12,12,4,4,24,24)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

[cFile Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cFileAddLanguageSpecificTextSet("LSTD1362B8D_0?c Method

Creates a new steel deck model from template.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int NewSteelDeck(
    int NumberStorys,
    double TypicalStoryHeight,
    double BottomStoryHeight,
    int NumberLinesX,
    int NumberLinesY,
    double SpacingX,
    double SpacingY
)
```

```
Function NewSteelDeck (
    NumberStorys As Integer,
    TypicalStoryHeight As Double,
    BottomStoryHeight As Double,
    NumberLinesX As Integer,
    NumberLinesY As Integer,
    SpacingX As Double,
    SpacingY As Double
) As Integer
```

```
Dim instance As cFile
Dim NumberStorys As Integer
Dim TypicalStoryHeight As Double
Dim BottomStoryHeight As Double
Dim NumberLinesX As Integer
Dim NumberLinesY As Integer
Dim SpacingX As Double
Dim SpacingY As Double
Dim returnValue As Integer
```

```
returnValue = instance.NewSteelDeck(NumberStorys,
    TypicalStoryHeight, BottomStoryHeight,
    NumberLinesX, NumberLinesY, SpacingX,
    SpacingY)
```

```
int NewSteelDeck(
    int NumberStorys,
    double TypicalStoryHeight,
    double BottomStoryHeight,
    int NumberLinesX,
    int NumberLinesY,
```

cFilesid="LSTD1362B8D_0"AddLanguageSpecificTextSet("LSTD1362B8D_0?cpp=::|nu=.);NewSteelD

```
double SpacingX,
double SpacingY
)
```

```
abstract NewSteelDeck :
    NumberStorys : int *
    TypicalStoryHeight : float *
    BottomStoryHeight : float *
    NumberLinesX : int *
    NumberLinesY : int *
    SpacingX : float *
    SpacingY : float -> int
```

Parameters

NumberStorys

Type: SystemInt32

The number of stories in the model

TypicalStoryHeight

Type: SystemDouble

The story height that will be used for all stories in the model, except the bottom story. [L]

BottomStoryHeight

Type: SystemDouble

The story height that will be used for the bottom story. [L]

NumberLinesX

Type: SystemInt32

The number of grid lines in the X direction.

NumberLinesY

Type: SystemInt32

The number of grid lines in the Y direction.

SpacingX

Type: SystemDouble

The uniform spacing for grid lines in the X direction. [L]

SpacingY

Type: SystemDouble

The uniform spacing for grid lines in the Y direction. [L]

Return Value

Type: Int32

Returns zero if the new steel deck model is successfully created; otherwise it returns a nonzero value.

Remarks

Do not use this function to add to an existing model. This function should be used only for creating a new model and typically would be preceded by a call to InitializeNewModel(eUnits).

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
```


Introduction

```
'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
    ret = EtabsObject.ApplicationStart()

'create SapModel object
    SapModel = EtabsObject.SapModel

'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

🔗 See Also

Reference

[cFile Interface](#)

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CSI API ETABS v1

cFileAddLanguageSpecificTextSet("LST2382CBF2_0?c Method

Opens an existing file.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int OpenFile(
    string FileName
)
```

```
Function OpenFile (
    FileName As String
) As Integer
```

```
Dim instance As cFile
Dim FileName As String
Dim returnValue As Integer
```

```
returnValue = instance.OpenFile(FileName)
```

```
int OpenFile(
    String^ FileName
)
```

```
abstract OpenFile :
    FileName : string -> int
```

Parameters

FileName

Type: SystemString

The full path of a model file to be opened in the application.

Return Value

Type: Int32

Returns zero if the file is successfully opened and nonzero if it is not opened.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As c0API
```

cFilesid="LST2382CBF2_0"AddLanguageSpecificTextSet("LST2382CBF2_0?cpp=::|nu=.")1050nFile M

Introduction

```
Dim FileName as String
Dim ret As Integer = -1

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'open an existing file - If no file exists, run the Save example first.
FileName = "c:\CSI_API_temp\example.edb"
ret = SapModel.File.OpenFile(FileName)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cFile Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cFileAddLanguageSpecificTextSet("LST9B53E4F1_0?c Method

Saves the present model.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int Save(
    string FileName = ""
)
```

```
Function Save (
    0FileName As String = ""
) As Integer
```

```
Dim instance As cFile
Dim FileName As String
Dim returnValue As Integer
```

```
returnValue = instance.Save(FileName)
```

```
int Save(
    String^ FileName = L""
)
```

```
abstract Save :
    ?FileName : string
(* Defaults:
    let _FileName = defaultArg FileName ""
*)
-> int
```

Parameters

FileName (Optional)

Type: SystemString

The full path to which the model file is saved.

Return Value

Type: Int32

Returns zero if the file is successfully saved and nonzero if it is not saved.

▲ Remarks

If no file name is specified, the file is saved using its current name. If there is no current name for the file (the file has not been saved previously) and this function is

called with no file name specified, an error will be returned.

Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)  
  
    'save file  
    System.IO.Directory.CreateDirectory("c:\CSI_API_temp")  
    ret = SapModel.File.Save("C:\CSI_API_temp\example.edb")  
  
    'close ETABS  
    EtabsObject.ApplicationExit(False)  
  
    'clean up variables  
    SapModel = Nothing  
    EtabsObject = Nothing  
End Sub
```

See Also

Reference

cFile Interface

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CSI API ETABS v1

cFrameObj Interface

Namespace: Â ETABSv1

Assembly: Â ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
public interface cFrameObj
```

```
Public Interface cFrameObj
```

```
Dim instance As cFrameObj
```

```
public interface class cFrameObj
```

```
type cFrameObj = interface end
```

The cFrameObj type exposes the following members.

▲ Methods

Â	Name	Description
⇒	<u>AddByCoord</u>	Adds a new frame object whose end points are at the specified coordinates.
⇒	<u>AddByPoint</u>	Adds a new frame object whose end points are specified by name.
⇒	<u>ChangeName</u>	
⇒	<u>Count</u>	
⇒	<u>Delete</u>	Deletes frame objects.
⇒	<u>DeleteLateralBracing</u>	
⇒	<u>DeleteLoadDistributed</u>	Deletes the distributed load assignments to the specified frame objects for the specified load pattern.
⇒	<u>DeleteLoadPoint</u>	Deletes the point load assignments to the specified frame objects for the specified load pattern.
⇒	<u>DeleteLoadTemperature</u>	
⇒	<u>DeleteMass</u>	
⇒	<u>DeleteModifiers</u>	
⇒	<u>DeleteSpring</u>	
⇒	<u>GetAllFrames</u>	Retrieves select data for all frame objects in the model
⇒	<u>GetColumnSpliceOverwrite</u>	Retrieves the frame object column splice overwrite assignment.
⇒	<u>GetCurved_2</u>	Retrieves frame object curve data

Introduction

⇒ <u>GetDesignOrientation</u>	Retrieves the design orientation for a given frame object
⇒ <u>GetDesignProcedure</u>	Retrieves the design procedure for a frame object
⇒ <u>GetElm</u>	
⇒ <u>GetEndLengthOffset</u>	Retrieves the frame object end offsets along the 1-axis of the object.
⇒ <u>GetGroupAssign</u>	Retrieves the groups to which a frame object is assigned.
⇒ <u>GetGUID</u>	Retrieves the GUID for the specified frame object.
⇒ <u>GetHingeAssigns</u>	DEPRECATED Retrieves frame object hinge assignments.
⇒ <u>GetHingeAssigns_1</u>	Retrieves frame object hinge assignments. The assignments include the hinge property, type, and location
⇒ <u>GetInsertionPoint</u>	Retrieves frame object insertion point assignments. The assignments include the cardinal point and end joint offsets.
⇒ <u>GetLabelFromName</u>	Retrieves the label and story for a unique frame object name
⇒ <u>GetLabelNameList</u>	Retrieves the names and labels of all defined frame objects.
⇒ <u>GetLateralBracing</u>	
⇒ <u>GetLoadDistributed</u>	Retrieves the distributed load assignments to frame objects.
⇒ <u>GetLoadPoint</u>	Retrieves the point load assignments to frame objects.
⇒ <u>GetLoadTemperature</u>	Retrieves the temperature load assignments to frame objects.
⇒ <u>GetLocalAxes</u>	Retrieves the frame local axis angle assignment for frame objects.
⇒ <u>GetMass</u>	
⇒ <u>GetMaterialOverwrite</u>	
⇒ <u>GetModifiers</u>	Retrieves the frame modifier assignment for frame objects. The default value for all modifiers is one.
⇒ <u>GetNameFromLabel</u>	Retrieves the unique name of a frame object, given the label and story level
⇒ <u>GetNameList</u>	Retrieves the names of all defined frame objects.
⇒ <u>GetNameListOnStory</u>	Retrieves the names of the defined frame objects on a given story.
⇒ <u>GetOutputStations</u>	Retrieves frame object output station data
⇒ <u>GetPier</u>	Retrieves the pier label assignments of a frame object
⇒ <u>GetPoints</u>	Retrieves the names of the point objects at each end of a specified frame object.
⇒ <u>GetReleases</u>	
⇒ <u>GetSection</u>	Retrieves the frame section property assigned to a frame object.

⇒ <u>GetSectionNonPrismatic</u>	
⇒ <u>GetSelected</u>	
⇒ <u>GetSpandrel</u>	Retrieves the Spandrel label assignments of a frame object
⇒ <u>GetSpringAssignment</u>	Retrieves the named line spring property assignment for a frame object
⇒ <u>GetSupports</u>	Retrieves support data for a given frame beam object
⇒ <u>GetTCLimits</u>	
⇒ <u>GetTransformationMatrix</u>	
⇒ <u>GetTypeOAPI</u>	Retrieves the type of frame object (straight or curved)
⇒ <u>SetColumnSpliceOverwrite</u>	Sets the frame object column splice overwrite assignment.
⇒ <u>SetDesignProcedure</u>	Sets the design procedure for frame objects.
⇒ <u>SetEndLengthOffset</u>	Assigns frame object end offsets along the 1-axis of the object.
⇒ <u>SetGroupAssign</u>	Adds or removes frame objects from a specified group.
⇒ <u>SetGUID</u>	Sets the GUID for the specified frame object. If the GUID is passed in as a blank string, the program automatically creates a GUID for the object.
⇒ <u>SetInsertionPoint</u>	Assigns frame object insertion point data. The assignments include the cardinal point and end joint offsets.
⇒ <u>SetLateralBracing</u>	
⇒ <u>SetLoadDistributed</u>	Assigns distributed loads to frame objects.
⇒ <u>SetLoadPoint</u>	Assigns point loads to frame objects.
⇒ <u>SetLoadTemperature</u>	
⇒ <u>SetLocalAxes</u>	Assigns a local axis angle to frame objects.
⇒ <u>SetMass</u>	
⇒ <u>SetMaterialOverwrite</u>	
⇒ <u>SetModifiers</u>	Sets the frame modifier assignment for frame objects. The default value for all modifiers is one
⇒ <u>SetOutputStations</u>	Assigns frame object output station data.
⇒ <u>SetPier</u>	Sets the pier label assignment of one or more frame objects
⇒ <u>SetReleases</u>	Makes end release and partial fixity assignments to frame objects.
⇒ <u>SetSection</u>	Assigns a frame section property to a frame object.
⇒ <u>SetSelected</u>	Sets the selected status for a frame object.
⇒ <u>SetSpandrel</u>	Sets the Spandrel label assignment of one or more frame objects
⇒ <u>SetSpringAssignment</u>	Assigns an existing named line spring property to frame objects
⇒ <u>SetTCLimits</u>	Makes tension/compression force limit assignments to frame objects.

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CSI API ETABS v1

cFrameObj Methods

The cFrameObj type exposes the following members.

Methods

⌘	Name	Description
⇒	<u>AddByCoord</u>	Adds a new frame object whose end points are at the specified coordinates.
⇒	<u>AddByPoint</u>	Adds a new frame object whose end points are specified by name.
⇒	<u>ChangeName</u>	
⇒	<u>Count</u>	
⇒	<u>Delete</u>	Deletes frame objects.
⇒	<u>DeleteLateralBracing</u>	
⇒	<u>DeleteLoadDistributed</u>	Deletes the distributed load assignments to the specified frame objects for the specified load pattern.
⇒	<u>DeleteLoadPoint</u>	Deletes the point load assignments to the specified frame objects for the specified load pattern.
⇒	<u>DeleteLoadTemperature</u>	
⇒	<u>DeleteMass</u>	
⇒	<u>DeleteModifiers</u>	
⇒	<u>DeleteSpring</u>	
⇒	<u>GetAllFrames</u>	Retrieves select data for all frame objects in the model
⇒	<u>GetColumnSpliceOverwrite</u>	Retrieves the frame object column splice overwrite assignment.
⇒	<u>GetCurved_2</u>	Retrieves frame object curve data
⇒	<u>GetDesignOrientation</u>	Retrieves the design orientation for a given frame object
⇒	<u>GetDesignProcedure</u>	Retrieves the design procedure for a frame object
⇒	<u>GetElm</u>	
⇒	<u>GetEndLengthOffset</u>	Retrieves the frame object end offsets along the 1-axis of the object.
⇒	<u>GetGroupAssign</u>	Retrieves the groups to which a frame object is assigned.
⇒	<u>GetGUID</u>	Retrieves the GUID for the specified frame object.
⇒	<u>GetHingeAssigns</u>	DEPRECATED Retrieves frame object hinge assignments.
⇒	<u>GetHingeAssigns_1</u>	Retrieves frame object hinge assignments. The assignments include the hinge property, type, and location
⇒	<u>GetInsertionPoint</u>	Retrieves frame object insertion point assignments. The assignments include the cardinal point and end joint offsets.

Introduction

⇒ <u>GetLabelFromName</u>	Retrieves the label and story for a unique frame object name
⇒ <u>GetLabelNameList</u>	Retrieves the names and labels of all defined frame objects.
⇒ <u>GetLateralBracing</u>	
⇒ <u>GetLoadDistributed</u>	Retrieves the distributed load assignments to frame objects.
⇒ <u>GetLoadPoint</u>	Retrieves the point load assignments to frame objects.
⇒ <u>GetLoadTemperature</u>	Retrieves the temperature load assignments to frame objects.
⇒ <u>GetLocalAxes</u>	Retrieves the frame local axis angle assignment for frame objects.
⇒ <u>GetMass</u>	
⇒ <u>GetMaterialOverwrite</u>	
⇒ <u>GetModifiers</u>	Retrieves the frame modifier assignment for frame objects. The default value for all modifiers is one.
⇒ <u>GetNameFromLabel</u>	Retrieves the unique name of a frame object, given the label and story level
⇒ <u>GetNameList</u>	Retrieves the names of all defined frame objects.
⇒ <u>GetNameListOnStory</u>	Retrieves the names of the defined frame objects on a given story.
⇒ <u>GetOutputStations</u>	Retrieves frame object output station data
⇒ <u>GetPier</u>	Retrieves the pier label assignments of a frame object
⇒ <u>GetPoints</u>	Retrieves the names of the point objects at each end of a specified frame object.
⇒ <u>GetReleases</u>	
⇒ <u>GetSection</u>	Retrieves the frame section property assigned to a frame object.
⇒ <u>GetSectionNonPrismatic</u>	
⇒ <u>GetSelected</u>	
⇒ <u>GetSpandrel</u>	Retrieves the Spandrel label assignments of a frame object
⇒ <u>GetSpringAssignment</u>	Retrieves the named line spring property assignment for a frame object
⇒ <u>GetSupports</u>	Retrieves support data for a given frame beam object
⇒ <u>GetTCLimits</u>	
⇒ <u>GetTransformationMatrix</u>	
⇒ <u>GetTypeOAPI</u>	Retrieves the type of frame object (straight or curved)
⇒ <u>SetColumnSpliceOverwrite</u>	Sets the frame object column splice overwrite assignment.
⇒ <u>SetDesignProcedure</u>	Sets the design procedure for frame objects.
⇒ <u>SetEndLengthOffset</u>	Assigns frame object end offsets along the 1-axis of the object.
⇒ <u>SetGroupAssign</u>	Adds or removes frame objects from a specified group.

 <u>SetGUID</u>	Sets the GUID for the specified frame object. If the GUID is passed in as a blank string, the program automatically creates a GUID for the object.
 <u>SetInsertionPoint</u>	Assigns frame object insertion point data. The assignments include the cardinal point and end joint offsets.
 <u>SetLateralBracing</u>	
 <u>SetLoadDistributed</u>	Assigns distributed loads to frame objects.
 <u>SetLoadPoint</u>	Assigns point loads to frame objects.
 <u>SetLoadTemperature</u>	
 <u>SetLocalAxes</u>	Assigns a local axis angle to frame objects.
 <u>SetMass</u>	
 <u>SetMaterialOverwrite</u>	
 <u>SetModifiers</u>	Sets the frame modifier assignment for frame objects. The default value for all modifiers is one
 <u>SetOutputStations</u>	Assigns frame object output station data.
 <u>SetPier</u>	Sets the pier label assignment of one or more frame objects
 <u>SetReleases</u>	Makes end release and partial fixity assignments to frame objects.
 <u>SetSection</u>	Assigns a frame section property to a frame object.
 <u>SetSelected</u>	Sets the selected status for a frame object.
 <u>SetSpandrel</u>	Sets the Spandrel label assignment of one or more frame objects
 <u>SetSpringAssignment</u>	Assigns an existing named line spring property to frame objects
 <u>SetTCLimits</u>	Makes tension/compression force limit assignments to frame objects.

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LSTEC8CD2A9_0?cpp=1961=.") Method

Adds a new frame object whose end points are at the specified coordinates.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int AddByCoord(
    double XI,
    double YI,
    double ZI,
    double XJ,
    double YJ,
    double ZJ,
    ref string Name,
    string PropName = "Default",
    string UserName = "",
    string CSys = "Global"
)
```

```
Function AddByCoord (
    XI As Double,
    YI As Double,
    ZI As Double,
    XJ As Double,
    YJ As Double,
    ZJ As Double,
    ByRef Name As String,
    PropName As String = "Default",
    UserName As String = "",
    CSys As String = "Global"
) As Integer
```

```
Dim instance As cFrameObj
Dim XI As Double
Dim YI As Double
Dim ZI As Double
Dim XJ As Double
Dim YJ As Double
Dim ZJ As Double
Dim Name As String
Dim PropName As String
Dim UserName As String
Dim CSys As String
Dim returnValue As Integer
```

```
returnValue = instance.AddByCoord(XI,
    YI, ZI, XJ, YJ, ZJ, Name, PropName, UserName,
```

cFrameObjid="LSTEC8CD2A9_0"AddLanguageSpecificTextSet("LSTEC8CD2A9_0?cpp=1961=.");Add

CSys)

```
int AddByCoord(
double XI,
double YI,
double ZI,
double XJ,
double YJ,
double ZJ,
String^% Name,
String^ PropName = L"Default",
String^ UserName = L"",
String^ CSys = L"Global"
)
```

```
abstract AddByCoord :
    XI : float *
    YI : float *
    ZI : float *
    XJ : float *
    YJ : float *
    ZJ : float *
    Name : string byref *
    ?PropName : string *
    ?UserName : string *
    ?CSys : string
(* Defaults:
    let _PropName = defaultArg PropName "Default"
    let _UserName = defaultArg UserName ""
    let _CSys = defaultArg CSys "Global"
*)
-> int
```

Parameters

XI

Type:Â SystemDouble

The coordinates of the I-End of the added frame object. The coordinates are in the coordinate system defined by the CSys item.

YI

Type:Â SystemDouble

The coordinates of the I-End of the added frame object. The coordinates are in the coordinate system defined by the CSys item.

ZI

Type:Â SystemDouble

The coordinates of the I-End of the added frame object. The coordinates are in the coordinate system defined by the CSys item.

XJ

Type:Â SystemDouble

The coordinates of the J-End of the added frame object. The coordinates are in the coordinate system defined by the CSys item.

YJ

Type:Â SystemDouble

The coordinates of the J-End of the added frame object. The coordinates are in the coordinate system defined by the CSys item.

ZJ

Type:Â SystemDouble

The coordinates of the J-End of the added frame object. The coordinates are in the coordinate system defined by the CSys item.

Name

Type:Â SystemString

This is the name that the program ultimately assigns for the frame object. If no userName is specified, the program assigns a default name to the frame object. If a userName is specified and that name is not used for another frame, cable or tendon object, the userName is assigned to the frame object, otherwise a default name is assigned to the frame object.

PropName (Optional)

Type:Â SystemString

This is Default, None, or the name of a defined frame section property.

If it is Default, the program assigns a default section property to the frame object. If it is None, no section property is assigned to the frame object. If it is the name of a defined frame section property, that property is assigned to the frame object.

UserName (Optional)

Type:Â SystemString

This is an optional user specified name for the frame object. If a UserName is specified and that name is already used for another frame object, the program ignores the UserName.

CSys (Optional)

Type:Â SystemString

The name of the coordinate system in which the frame object end point coordinates are defined.

Return Value

Type:Â Int32

Returns zero if the frame object is successfully added, otherwise it returns a nonzero value.

▲Remarks

Note that the End I and End J joints of the added frame object are determined based on the orientation of the object and may not be the same order as specified in the parameters of this API functions. Please use GetPoints(String, String, String) to check the position of the End I and End J joints.

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()
```

Return Value

Introduction

```
'create SapModel object
  SapModel = EtabsObject.SapModel

'initialize model
  ret = SapModel.InitializeNewModel(eUnits.lb_ft_F)

'create steel deck template model
  ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'add frame object by coordinates
  ret = SapModel.FrameObj.AddByCoord(0, 72, 48, 0, 48, 36, Name)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cFrameObj Interface

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST1A5655F0_0?cpp=::|1965");AddByMethod

Adds a new frame object whose end points are specified by name.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int AddByPoint(
    string Point1,
    string Point2,
    ref string Name,
    string PropName = "Default",
    string UserName = ""
)
```

```
Function AddByPoint (
    Point1 As String,
    Point2 As String,
    ByRef Name As String,
    PropName As String = "Default",
    UserName As String = ""
) As Integer
```

```
Dim instance As cFrameObj
Dim Point1 As String
Dim Point2 As String
Dim Name As String
Dim PropName As String
Dim UserName As String
Dim returnValue As Integer
```

```
returnValue = instance.AddByPoint(Point1,
    Point2, Name, PropName, UserName)
```

```
int AddByPoint(
    String^ Point1,
    String^ Point2,
    String^% Name,
    String^ PropName = L"Default",
    String^ UserName = L""
)
```

```
abstract AddByPoint :
    Point1 : string *
    Point2 : string *
    Name : string byref *
    ?PropName : string *
    ?UserName : string
```

cFrameObjid="LST1A5655F0_0"AddLanguageSpecificTextSet("LST1A5655F0_0?cpp=::|1965");AddBy

```
(* Defaults:
    let _PropName = defaultArg PropName "Default"
    let _UserName = defaultArg UserName ""
*)
-> int
```

Parameters

Point1

Type: SystemString

The name of a defined point object at the I-End of the added frame object.

Point2

Type: SystemString

The name of a defined point object at the J-End of the added frame object.

Name

Type: SystemString

This is the name that the program ultimately assigns for the frame object. If no UserName is specified, the program assigns a default name to the frame object. If a UserName is specified and that name is not used for another frame, cable or tendon object, the UserName is assigned to the frame object, otherwise a default name is assigned to the frame object.

PropName (Optional)

Type: SystemString

This is Default, None, or the name of a defined frame section property.

If it is Default, the program assigns a default section property to the frame object. If it is None, no section property is assigned to the frame object. If it is the name of a defined frame section property, that property is assigned to the frame object.

UserName (Optional)

Type: SystemString

This is an optional user specified name for the frame object. If a UserName is specified and that name is already used for another frame object, the program ignores the UserName.

Return Value

Type: Int32

Returns zero if the frame object is successfully added, otherwise it returns a nonzero value.

Remarks

Note that the End I and End J joints of the added frame object are determined based on the orientation of the object and may not be the same order as specified in the parameters of this API functions. Please use GetPoints(String, String, String) to check the position of the End I and End J joints.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
```

Introduction

```
'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
    ret = EtabsObject.ApplicationStart()

'create SapModel object
    SapModel = EtabsObject.SapModel

'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'add frame object by points
    ret = SapModel.FrameObj.AddByPoint("1", "6", Name)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST923ECC37_0?cpp=:1068.");Char

Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int ChangeName(
    string Name,
    string NewName
)
```

```
Function ChangeName (
    Name As String,
    NewName As String
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim NewName As String
Dim returnValue As Integer
```

```
returnValue = instance.ChangeName(Name,
    NewName)
```

```
int ChangeName(
    String^ Name,
    String^ NewName
)
```

```
abstract ChangeName :
    Name : string *
    NewName : string -> int
```

Parameters

Name

Type: [SystemString](#)

NewName

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST26CC3295_0?cpp=::1070");Count

Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int Count(
    string MyType = "All"
)
```

```
Function Count (
    MyType As String = "All"
) As Integer
```

```
Dim instance As cFrameObj
Dim MyType As String
Dim returnValue As Integer
```

```
returnValue = instance.Count(MyType)
```

```
int Count(
    String^ MyType = L"All"
)
```

```
abstract Count :
    ?MyType : string
(* Defaults:
    let _MyType = defaultArg MyType "All"
*)
-> int
```

Parameters

MyType (Optional)
Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST49780FA1_0?cpp=::|1072);DeleteMethod

Deletes frame objects.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int Delete(
    string Name,
    eItemType ItemType = eItemType.Objects
)
```

```
Function Delete (
    Name As String,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.Delete(Name, ItemType)
```

```
int Delete(
    String^ Name,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract Delete :
    Name : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing frame object or group depending on the value of the ItemType item.

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

cFrameObjid="LST49780FA1_0"AddLanguageSpecificTextSet("LST49780FA1_0?cpp=::|1072);Delete

- ◇ Object = 0
- ◇ Group = 1
- ◇ SelectedObjects = 2

If this item is Objects, the frame object specified by the Name item is deleted.

If this item is Group, all of the frame objects in the group specified by the Name item are deleted.

If this item is SelectedObjects, all selected frame objects are deleted, and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the frame object is successfully deleted, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'delete frame object
    ret = SapModel.FrameObj.Delete("1")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cFrameObj Interface

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LSTE3B552F8_0?cpp=::1075");Delete

Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int DeleteLateralBracing(
    string Name,
    int MyType = 3,
    eItemType ItemType = eItemType.Objects
)
```

```
Function DeleteLateralBracing (
    Name As String,
    MyType As Integer = 3,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim MyType As Integer
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.DeleteLateralBracing(Name,
MyType, ItemType)
```

```
int DeleteLateralBracing(
    String^ Name,
    int MyType = 3,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract DeleteLateralBracing :
    Name : string *
    ?MyType : int *
    ?ItemType : eItemType
(* Defaults:
    let _MyType = defaultArg MyType 3
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: [SystemString](#)

cFrameObjspan id="LSTE3B552F8_0"AddLanguageSpecificTextSet("LSTE3B552F8_0?cpp=::1075");Delete

MyType (Optional)

Type:Â SystemInt32

ItemType (Optional)

Type:Â ETABSv1eItemType

Return Value

Type:Â Int32

▲ See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LSTA7133175_0") Method

Deletes the distributed load assignments to the specified frame objects for the specified load pattern.

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int DeleteLoadDistributed(
    string Name,
    string LoadPat,
    eItemType ItemType = eItemType.Objects
)
```

```
Function DeleteLoadDistributed (
    Name As String,
    LoadPat As String,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim LoadPat As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.DeleteLoadDistributed(Name,
    LoadPat, ItemType)
```

```
int DeleteLoadDistributed(
    String^ Name,
    String^ LoadPat,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract DeleteLoadDistributed :
    Name : string *
    LoadPat : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame object or group depending on the value of the ItemType item.

LoadPat

Type:Â SystemString

The name of a defined load pattern.

ItemType (Optional)

Type:Â ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the load assignments are deleted for the frame object specified by the Name item.

If this item is Group, the load assignments are deleted for all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the load assignments are deleted for all selected frame objects, and the Name item is ignored.

Return ValueType:Â Int32

Returns zero if the load assignments are successfully deleted, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'assign frame distributed loads
```

Introduction

```
ret = SapModel.FrameObj.SetLoadDistributed("14", "DEAD", 1, 10, 0, 1, 0.08, 0.08)
ret = SapModel.FrameObj.SetLoadDistributed("15", "DEAD", 1, 10, 0, 1, 0.08, 0.08)

'delete frame distributed load
ret = SapModel.FrameObj.DeleteLoadDistributed("14", "DEAD")

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

📌 See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST1C2013F1_0?cpp=::1080");DeleteLoadPoint Method

Deletes the point load assignments to the specified frame objects for the specified load pattern.

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int DeleteLoadPoint(
    string Name,
    string LoadPat,
    eItemType ItemType = eItemType.Objects
)
```

```
Function DeleteLoadPoint (
    Name As String,
    LoadPat As String,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim LoadPat As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.DeleteLoadPoint(Name,
    LoadPat, ItemType)
```

```
int DeleteLoadPoint(
    String^ Name,
    String^ LoadPat,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract DeleteLoadPoint :
    Name : string *
    LoadPat : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```


Parameters**Name**

Type: SystemString

The name of an existing frame object or group depending on the value of the ItemType item.

LoadPat

Type: SystemString

The name of a defined load pattern.

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the load assignments are deleted for the frame object specified by the Name item.

If this item is Group, the load assignments are deleted for all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the load assignments are deleted for all selected frame objects, and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the load assignments are successfully deleted, otherwise it returns a nonzero value.

Remarks**Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'assign frame point loads
```

Introduction

```
ret = SapModel.FrameObj.SetLoadPoint("14", "DEAD", 1, 10, .5, 20)
ret = SapModel.FrameObj.SetLoadPoint("15", "DEAD", 1, 10, .5, 20)

'delete frame point load
ret = SapModel.FrameObj.DeleteLoadPoint("14", "DEAD")

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

📌 See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LSTC2D11653_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int DeleteLoadTemperature(
    string Name,
    string LoadPat,
    eItemType ItemType = eItemType.Objects
)
```

```
Function DeleteLoadTemperature (
    Name As String,
    LoadPat As String,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim LoadPat As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.DeleteLoadTemperature(Name,
    LoadPat, ItemType)
```

```
int DeleteLoadTemperature(
    String^ Name,
    String^ LoadPat,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract DeleteLoadTemperature :
    Name : string *
    LoadPat : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: [SystemString](#)

LoadPat

cFrameObjid="LSTC2D11653_0"AddLanguageSpecificTextSet("LSTC2D11653_0?cpp=::11653");Delete

Type: SystemString
ItemType (Optional)
Type: ETABSv1eItemType

Return Value

Type: Int32

■ See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST4FE61D45_0?cpp=::1085");DeleteMethod

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int DeleteMass(
    string Name,
    eItemType ItemType = eItemType.Objects
)
```

```
Function DeleteMass (
    Name As String,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.DeleteMass(Name,
    ItemType)
```

```
int DeleteMass(
    String^ Name,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract DeleteMass :
    Name : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

ItemType (Optional)

Type: ETABSv1eItemType

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cFrameObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST77E36816_0?cpp=::|1987);DeleteMethod

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int DeleteModifiers(
    string Name,
    eItemType ItemType = eItemType.Objects
)
```

```
Function DeleteModifiers (
    Name As String,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.DeleteModifiers(Name,
    ItemType)
```

```
int DeleteModifiers(
    String^ Name,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract DeleteModifiers :
    Name : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

ItemType (Optional)

Type: ETABSv1eItemType

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cFrameObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST94A68D94_0?cpp=::1089");Delete

Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int DeleteSpring(
    string Name,
    eItemType ItemType = eItemType.Objects
)
```

```
Function DeleteSpring (
    Name As String,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.DeleteSpring(Name,
    ItemType)
```

```
int DeleteSpring(
    String^ Name,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract DeleteSpring :
    Name : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

ItemType (Optional)

Type: ETABSv1eItemType

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cFrameObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST10E182FD_0") Method

Retrieves select data for all frame objects in the model

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetAllFrames(
    ref int NumberNames,
    ref string[] MyName,
    ref string[] PropName,
    ref string[] StoryName,
    ref string[] PointName1,
    ref string[] PointName2,
    ref double[] Point1X,
    ref double[] Point1Y,
    ref double[] Point1Z,
    ref double[] Point2X,
    ref double[] Point2Y,
    ref double[] Point2Z,
    ref double[] Angle,
    ref double[] Offset1X,
    ref double[] Offset2X,
    ref double[] Offset1Y,
    ref double[] Offset2Y,
    ref double[] Offset1Z,
    ref double[] Offset2Z,
    ref int[] CardinalPoint,
    string csys = "Global"
)
```

```
Function GetAllFrames (
    ByRef NumberNames As Integer,
    ByRef MyName As String(),
    ByRef PropName As String(),
    ByRef StoryName As String(),
    ByRef PointName1 As String(),
    ByRef PointName2 As String(),
    ByRef Point1X As Double(),
    ByRef Point1Y As Double(),
    ByRef Point1Z As Double(),
    ByRef Point2X As Double(),
    ByRef Point2Y As Double(),
    ByRef Point2Z As Double(),
    ByRef Angle As Double(),
    ByRef Offset1X As Double(),
    ByRef Offset2X As Double(),
    ByRef Offset1Y As Double(),
    ByRef Offset2Y As Double(),
    ByRef Offset1Z As Double(),
    ByRef Offset2Z As Double(),
    ByRef CardinalPoint As Integer(),
    csys As String = "Global"
)
```

cFrameObjAddLanguageSpecificTextSet("LST10E182FD_0?cpp=::1001.");GetA

Introduction

```
ByRef Offset2Y As Double(),  
ByRef Offset1Z As Double(),  
ByRef Offset2Z As Double(),  
ByRef CardinalPoint As Integer(),  
    Option As String = "Global"  
) As Integer
```

```
Dim instance As cFrameObj  
Dim NumberNames As Integer  
Dim MyName As String()  
Dim PropName As String()  
Dim StoryName As String()  
Dim PointName1 As String()  
Dim PointName2 As String()  
Dim Point1X As Double()  
Dim Point1Y As Double()  
Dim Point1Z As Double()  
Dim Point2X As Double()  
Dim Point2Y As Double()  
Dim Point2Z As Double()  
Dim Angle As Double()  
Dim Offset1X As Double()  
Dim Offset2X As Double()  
Dim Offset1Y As Double()  
Dim Offset2Y As Double()  
Dim Offset1Z As Double()  
Dim Offset2Z As Double()  
Dim CardinalPoint As Integer()  
Dim csys As String  
Dim returnValue As Integer
```

```
returnValue = instance.GetAllFrames(NumberNames,  
MyName, PropName, StoryName, PointName1,  
PointName2, Point1X, Point1Y, Point1Z,  
Point2X, Point2Y, Point2Z, Angle,  
Offset1X, Offset2X, Offset1Y, Offset2Y,  
Offset1Z, Offset2Z, CardinalPoint,  
csys)
```

```
int GetAllFrames(  
int% NumberNames,  
array<String^>^% MyName,  
array<String^>^% PropName,  
array<String^>^% StoryName,  
array<String^>^% PointName1,  
array<String^>^% PointName2,  
array<double>^% Point1X,  
array<double>^% Point1Y,  
array<double>^% Point1Z,  
array<double>^% Point2X,  
array<double>^% Point2Y,  
array<double>^% Point2Z,  
array<double>^% Angle,  
array<double>^% Offset1X,  
array<double>^% Offset2X,  
array<double>^% Offset1Y,  
array<double>^% Offset2Y,  
array<double>^% Offset1Z,  
array<double>^% Offset2Z,  
array<int>^% CardinalPoint,  
String^ csys = L"Global"  
)
```

cFrameObjspan id="LST10E182FD_0"AddLanguageSpecificTextSet("LST10E182FD_0?cpp=::1002");GetA

```

abstract GetAllFrames :
    NumberNames : int byref *
    MyName : string[] byref *
    PropName : string[] byref *
    StoryName : string[] byref *
    PointName1 : string[] byref *
    PointName2 : string[] byref *
    Point1X : float[] byref *
    Point1Y : float[] byref *
    Point1Z : float[] byref *
    Point2X : float[] byref *
    Point2Y : float[] byref *
    Point2Z : float[] byref *
    Angle : float[] byref *
    Offset1X : float[] byref *
    Offset2X : float[] byref *
    Offset1Y : float[] byref *
    Offset2Y : float[] byref *
    Offset1Z : float[] byref *
    Offset2Z : float[] byref *
    CardinalPoint : int[] byref *
    ?csys : string
(* Defaults:
    let _csys = defaultArg csys "Global"
*)
-> int

```

Parameters

NumberNames
 Type:Â SystemInt32
 MyName
 Type:Â SystemString
 PropName
 Type:Â SystemString
 StoryName
 Type:Â SystemString
 PointName1
 Type:Â SystemString
 PointName2
 Type:Â SystemString
 Point1X
 Type:Â SystemDouble
 Point1Y
 Type:Â SystemDouble
 Point1Z
 Type:Â SystemDouble
 Point2X
 Type:Â SystemDouble
 Point2Y
 Type:Â SystemDouble
 Point2Z
 Type:Â SystemDouble
 Angle
 Type:Â SystemDouble

Offset1X
Type:Â SystemDouble
Offset2X
Type:Â SystemDouble
Offset1Y
Type:Â SystemDouble
Offset2Y
Type:Â SystemDouble
Offset1Z
Type:Â SystemDouble
Offset2Z
Type:Â SystemDouble
CardinalPoint
Type:Â SystemInt32
csys (Optional)
Type:Â SystemString

Return Value

Type:Â Int32

▲ Remarks

▲ See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LSTEEA6CC24_0?cpp=1995.") Method

Retrieves the frame object column splice overwrite assignment.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetColumnSpliceOverwrite(
    string Name,
    ref int SpliceOption,
    ref double Height
)
```

```
Function GetColumnSpliceOverwrite (
    Name As String,
    ByRef SpliceOption As Integer,
    ByRef Height As Double
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim SpliceOption As Integer
Dim Height As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetColumnSpliceOverwrite(Name,
    SpliceOption, Height)
```

```
int GetColumnSpliceOverwrite(
    String^ Name,
    int% SpliceOption,
    double% Height
)
```

```
abstract GetColumnSpliceOverwrite :
    Name : string *
    SpliceOption : int byref *
    Height : float byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing frame object.

SpliceOption

Type: SystemInt32

This is a numeric value from 1 to 3 that specifies the option used for defining the splice overwrite.

1. from story data (default)
2. no splice
3. splice at height above story at bottom of the column object

Height

Type: SystemDouble

If the SpliceOption=3, this specifies the height of the splice above the story at the bottom of the column object.

Return Value

Type: Int32

Returns zero if the column splice overwrite data is successfully retrieved, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'assign column splice overwrite
    ret = SapModel.FrameObj.SetColumnSpliceOverwrite("15", 2, 0)

    'get column splice overwrite
    Dim SpliceOption As Integer
    Dim Height As Double
    ret = SapModel.FrameObj.GetColumnSpliceOverwrite("15", SpliceOption, Height)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```


▲ See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST4A2E0073_0") Method

Retrieves frame object curve data

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetCurved_2(
    string Name,
    ref int CurveType,
    ref double Tension,
    ref int NumPnts,
    ref double[] gx,
    ref double[] gy,
    ref double[] gz
)
```

```
Function GetCurved_2 (
    Name As String,
    ByRef CurveType As Integer,
    ByRef Tension As Double,
    ByRef NumPnts As Integer,
    ByRef gx As Double(),
    ByRef gy As Double(),
    ByRef gz As Double()
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim CurveType As Integer
Dim Tension As Double
Dim NumPnts As Integer
Dim gx As Double()
Dim gy As Double()
Dim gz As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetCurved_2(Name,
    CurveType, Tension, NumPnts, gx, gy,
    gz)
```

```
int GetCurved_2(
    String^ Name,
    int% CurveType,
    double% Tension,
    int% NumPnts,
    array<double>^% gx,
    array<double>^% gy,
```

cFrameObjAddLanguageSpecificTextSet("LST4A2E0073_0?cpp=::1098");GetC

```
array<double>^% gz
)
```

```
abstract GetCurved 2 :
    Name : string *
    CurveType : int byref *
    Tension : float byref *
    NumPnts : int byref *
    gx : float[] byref *
    gy : float[] byref *
    gz : float[] byref -> int
```

Parameters

Name

Type:Â [SystemString](#)

The name of an existing curved frame object

CurveType

Type:Â [SystemInt32](#)

The type of curve, this is an integer from the following list:

- ◊ Straight = 0
- ◊ Circular Curve = 1
- ◊ Multilinear Curve = 2
- ◊ Bezier Curve = 3
- ◊ Spline Curve = 4

Tension

Type:Â [SystemDouble](#)

Used for spline, a value that specifies the amount that the curve bends between control points. Values greater than 1 produce unpredictable results

NumPnts

Type:Â [SystemInt32](#)

The number of points along the curve

gx

Type:Â [SystemDouble](#)

An array of size NumPnts, the global X coordinates

gy

Type:Â [SystemDouble](#)

An array of size NumPnts, the global Y coordinates

gz

Type:Â [SystemDouble](#)

An array of size NumPnts, the global Z coordinates

Return Value

Type:Â [Int32](#)

Returns zero if the data is successfully retrieved, otherwise it returns a nonzero value

▲ See Also

Reference

[cFrameObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LSTFC4759F7_0?cpp=::2001");GetDesignOrientation

Method

Retrieves the design orientation for a given frame object

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetDesignOrientation(
    string Name,
    ref eFrameDesignOrientation DesignOrientation
)
```

```
Function GetDesignOrientation (
    Name As String,
    ByRef DesignOrientation As eFrameDesignOrientation
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim DesignOrientation As eFrameDesignOrientation
Dim returnValue As Integer
```

```
returnValue = instance.GetDesignOrientation(Name,
    DesignOrientation)
```

```
int GetDesignOrientation(
    String^ Name,
    eFrameDesignOrientation% DesignOrientation
)
```

```
abstract GetDesignOrientation :
    Name : string *
    DesignOrientation : eFrameDesignOrientation byref -> int
```

Parameters

Name

Type: SystemString

The name of the frame object

DesignOrientation

Type: ETABSv1eFrameDesignOrientation

A value from the eFrameDesignOrientation enumeration

Return Value

Type: Int32

▲ Remarks

▲ See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST7A3C1C5_0")

Method

Retrieves the design procedure for a frame object

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetDesignProcedure(
    string Name,
    ref int MyType
)
```

```
Function GetDesignProcedure (
    Name As String,
    ByRef MyType As Integer
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim MyType As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetDesignProcedure(Name,
    MyType)
```

```
int GetDesignProcedure(
    String^ Name,
    int% MyType
)
```

```
abstract GetDesignProcedure :
    Name : string *
    MyType : int byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing frame object

MyType

Type: SystemInt32

This is an integer from the list below, indicating the design procedure type of the specified frame object.

◇ Program determined = 0

◇ Steel Frame Design = 1

- ◇ Concrete Frame Design = 2
- ◇ Composite Beam Design = 3
- ◇ Steel Joist Design = 4
- ◇ No Design = 7
- ◇ Composite Column Design = 13

Return Value

Type: Int32

Returns zero if the design procedure is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim MyType as Integer  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)  
  
    'set design procedure  
    ret = SapModel.FrameObj.GetDesignProcedure("8", MyType)  
  
    'close ETABS  
    EtabsObject.ApplicationExit(False)  
  
    'clean up variables  
    SapModel = Nothing  
    EtabsObject = Nothing  
  
End Sub
```

▲ See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST18744A29_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetElm(
    string Name,
    ref int NElm,
    ref string[] Elm,
    ref double[] RDI,
    ref double[] RDJ
)
```

```
Function GetElm (
    Name As String,
    ByRef NElm As Integer,
    ByRef Elm As String(),
    ByRef RDI As Double(),
    ByRef RDJ As Double()
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim NElm As Integer
Dim Elm As String()
Dim RDI As Double()
Dim RDJ As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetElm(Name, NElm,
    Elm, RDI, RDJ)
```

```
int GetElm(
    String^ Name,
    int% NElm,
    array<String^>^% Elm,
    array<double>^% RDI,
    array<double>^% RDJ
)
```

```
abstract GetElm :
    Name : string *
    NElm : int byref *
    Elm : string[] byref *
    RDI : float[] byref *
    RDJ : float[] byref -> int
```

Parameters

Name

Type: SystemString

NElm

Type: SystemInt32

Elm

Type: SystemString

RDI

Type: SystemDouble

RDJ

Type: SystemDouble

Return Value

Type: Int32

See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST11D12F23_0?cpp=::2008");GetEndLengthOffset Method

Retrieves the frame object end offsets along the 1-axis of the object.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetEndLengthOffset(
    string Name,
    ref bool AutoOffset,
    ref double Length1,
    ref double Length2,
    ref double RZ
)
```

```
Function GetEndLengthOffset (
    Name As String,
    ByRef AutoOffset As Boolean,
    ByRef Length1 As Double,
    ByRef Length2 As Double,
    ByRef RZ As Double
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim AutoOffset As Boolean
Dim Length1 As Double
Dim Length2 As Double
Dim RZ As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetEndLengthOffset(Name,
    AutoOffset, Length1, Length2, RZ)
```

```
int GetEndLengthOffset(
    String^ Name,
    bool% AutoOffset,
    double% Length1,
    double% Length2,
    double% RZ
)
```

```
abstract GetEndLengthOffset :
    Name : string *
    AutoOffset : bool byref *
    Length1 : float byref *
    Length2 : float byref *
    RZ : float byref -> int
```

cFrameObjid="LST11D12F23_0"AddLanguageSpecificTextSet("LST11D12F23_0?cpp=::2008");GetEndLengthOffset

Parameters

Name

Type:Â [SystemString](#)

The name of an existing frame object.

AutoOffset

Type:Â [SystemBoolean](#)

If this item is True, the end length offsets are automatically determined by the program from object connectivity.

Length1

Type:Â [SystemDouble](#)

The offset length along the 1-axis of the frame object at the I-End of the frame object. [L]

Length2

Type:Â [SystemDouble](#)

The offset length along the 1-axis of the frame object at the J-End of the frame object. [L]

RZ

Type:Â [SystemDouble](#)

The rigid zone factor. This is the fraction of the end offset length assumed to be rigid for bending and shear deformations.

Return Value

Type:Â [Int32](#)

Returns zero if the offsets are successfully retrieved, otherwise it returns a nonzero value.

▲ Remarks

▲ See Also

Reference

[cFrameObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST731D63E8_0?cpp=::2014") Method

Retrieves the groups to which a frame object is assigned.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetGroupAssign(
    string Name,
    ref int NumberGroups,
    ref string[] Groups
)
```

```
Function GetGroupAssign (
    Name As String,
    ByRef NumberGroups As Integer,
    ByRef Groups As String()
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim NumberGroups As Integer
Dim Groups As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetGroupAssign(Name,
    NumberGroups, Groups)
```

```
int GetGroupAssign(
    String^ Name,
    int% NumberGroups,
    array<String^>^% Groups
)
```

```
abstract GetGroupAssign :
    Name : string *
    NumberGroups : int byref *
    Groups : string[] byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing frame object.

NumberGroups

Type:Â SystemInt32

The number of group names retrieved.

Groups

Type:Â SystemString

This is an array of the names of the groups to which the frame object is assigned.

Return Value

Type:Â Int32

Returns zero if the group assignments are successfully retrieved, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberGroups As Integer
    Dim Groups() as String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new groups
    ret = SapModel.GroupDef.SetGroup("Group1")
    ret = SapModel.GroupDef.SetGroup("Group2")

    'add frame object to groups
    ret = SapModel.FrameObj.SetGroupAssign("3", "Group1")
    ret = SapModel.FrameObj.SetGroupAssign("3", "Group2")

    'get frame object groups
    ret = SapModel.FrameObj.GetGroupAssign("3", NumberGroups, Groups)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST64E8BD82_0") Method

Retrieves the GUID for the specified frame object.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetGUID(
    string Name,
    ref string GUID
)
```

```
Function GetGUID (
    Name As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.GetGUID(Name, GUID)
```

```
int GetGUID(
    String^ Name,
    String^% GUID
)
```

```
abstract GetGUID :
    Name : string *
    GUID : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing frame object.

GUID

Type: [SystemString](#)

The GUID (Global Unique ID) for the specified frame object.

Return Value

Type: Int32

Returns zero if the frame object GUID is successfully retrieved; otherwise it returns nonzero

▀ **Remarks**

▀ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim GUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'get GUID
    ret = SapModel.FrameObj.GetGUID("1", GUID)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▀ **See Also**

Reference

cFrameObj Interface

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST50965F91") Method

DEPRECATED Retrieves frame object hinge assignments.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetHingeAssigns(
    string Name,
    ref int NumberHinges,
    ref int[] HingeNum,
    ref string[] Prop,
    ref int[] MyType,
    ref int[] Behavior,
    ref string[] Source,
    ref double[] RD
)
```

```
Function GetHingeAssigns (
    Name As String,
    ByRef NumberHinges As Integer,
    ByRef HingeNum As Integer(),
    ByRef Prop As String(),
    ByRef MyType As Integer(),
    ByRef Behavior As Integer(),
    ByRef Source As String(),
    ByRef RD As Double()
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim NumberHinges As Integer
Dim HingeNum As Integer()
Dim Prop As String()
Dim MyType As Integer()
Dim Behavior As Integer()
Dim Source As String()
Dim RD As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetHingeAssigns(Name,
    NumberHinges, HingeNum, Prop, MyType,
    Behavior, Source, RD)
```

```
int GetHingeAssigns(
    String^ Name,
    int% NumberHinges,
    array<int>^% HingeNum,
```

cFrameObjid="LST50965F91_0"AddLanguageSpecificTextSet("LST50965F91_0?cpp=::|2015);GetHir

```
array<String>^% Prop,
array<int>^% MyType,
array<int>^% Behavior,
array<String>^% Source,
array<double>^% RD
)

abstract GetHingeAssigns :
    Name : string *
    NumberHinges : int byref *
    HingeNum : int[] byref *
    Prop : string[] byref *
    MyType : int[] byref *
    Behavior : int[] byref *
    Source : string[] byref *
    RD : float[] byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing frame object.

NumberHinges

Type: SystemInt32

The number of frame hinges assigned to the frame object.

HingeNum

Type: SystemInt32

An array of size NumberHinges. This is the hinge identification number.

Prop

Type: SystemString

An array of size NumberHinges. This is the name of the property of the hinge.

MyType

Type: SystemInt32

An array of size NumberHinges. This is a numeric value from 1 to 14 that specifies the type of the hinge.

1. P
2. V2
3. V3
4. T
5. M2
6. M3
7. Interacting P-M2
8. Interacting P-M3
9. Interacting M2-M3
10. Interacting P-M2-M3
11. Fiber P-M2-M3
12. Fiber P-M3
13. Parametric Concrete P-M2-M3
14. Parametric Steel P-M2-M3

Behavior

Type: SystemInt32

An array of size NumberHinges. This is a numeric value from 1 to 2 that specifies behavior of the hinge.

1. Force-controlled
2. Deformation-controlled

Source

Type: SystemString

An array of size NumberHinges. This is the name of the assigned hinge property, or "Auto" if an auto-generated hinge.

RD

Type: SystemDouble

An array of size NumberHinges. This is the distance of the hinge from the end of the i-end offset, as a ratio to the clear length of the frame object.

Return Value

Type: Int32

Returns zero if the hinge data is successfully retrieved, otherwise it returns a nonzero value.

▲ Remarks

This function is DEPRECATED. Please refer to GetHingeAssigns_1(String, Int32, Int32, String, Int32, Int32, String, eHingeLocationType, Double, Double)

▲ See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST9A2D9DDB_0?cpp=2018=.");Get

Method

Retrieves frame object hinge assignments. The assignments include the hinge property, type, and location

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetHingeAssigns_1(
    string Name,
    ref int NumberHinges,
    ref int[] HingeNum,
    ref string[] Prop,
    ref int[] MyType,
    ref int[] Behavior,
    ref string[] Source,
    ref eHingeLocationType[] LocType,
    ref double[] RD,
    ref double[] AD
)
```

```
Function GetHingeAssigns_1 (
    Name As String,
    ByRef NumberHinges As Integer,
    ByRef HingeNum As Integer(),
    ByRef Prop As String(),
    ByRef MyType As Integer(),
    ByRef Behavior As Integer(),
    ByRef Source As String(),
    ByRef LocType As eHingeLocationType(),
    ByRef RD As Double(),
    ByRef AD As Double()
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim NumberHinges As Integer
Dim HingeNum As Integer()
Dim Prop As String()
Dim MyType As Integer()
Dim Behavior As Integer()
Dim Source As String()
Dim LocType As eHingeLocationType()
Dim RD As Double()
Dim AD As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetHingeAssigns_1(Name,
```

cFrameObjid="LST9A2D9DDB_0"AddLanguageSpecificTextSet("LST9A2D9DDB_0?cpp=2018=.");Get

Introduction

NumberHinges, HingeNum, Prop, MyType,
Behavior, Source, LocType, RD, AD)

```
int GetHingeAssigns_1(  
String^ Name,  
int% NumberHinges,  
array<int>^% HingeNum,  
array<String^>^% Prop,  
array<int>^% MyType,  
array<int>^% Behavior,  
array<String^>^% Source,  
array<eHingeLocationType>^% LocType,  
array<double>^% RD,  
array<double>^% AD  
)
```

```
abstract GetHingeAssigns_1 :  
    Name : string *  
    NumberHinges : int byref *  
    HingeNum : int[] byref *  
    Prop : string[] byref *  
    MyType : int[] byref *  
    Behavior : int[] byref *  
    Source : string[] byref *  
    LocType : eHingeLocationType[] byref *  
    RD : float[] byref *  
    AD : float[] byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame object.

NumberHinges

Type:Â SystemInt32

The number of frame hinges assigned to the frame object.

HingeNum

Type:Â SystemInt32

An array of size NumberHinges. This is the hinge identification number.

Prop

Type:Â SystemString

An array of size NumberHinges. This is the name of the property of the hinge.

MyType

Type:Â SystemInt32

An array of size NumberHinges. This is a numeric value from 1 to 14 that specifies the type of the hinge.

1. P
2. V2
3. V3
4. T
5. M2
6. M3
7. Interacting P-M2
8. Interacting P-M3
9. Interacting M2-M3

- 10. Interacting P-M2-M3
- 11. Fiber P-M2-M3
- 12. Fiber P-M3
- 13. Parametric Concrete P-M2-M3
- 14. Parametric Steel P-M2-M3

Behavior

Type: SystemInt32

An array of size NumberHinges. This is a numeric value from 1 to 2 that specifies behavior of the hinge.

- 1. Force-controlled
- 2. Deformation-controlled

Source

Type: SystemString

An array of size NumberHinges. This is the name of the assigned hinge property, or "Auto" if an auto-generated hinge.

LocType

Type: ETABSv1eHingeLocationType

An array of size NumberHinges. This is a value from the eHingeLocationType enumeration, specifying the type used to define the location of the hinge.

RD

Type: SystemDouble

An array of size NumberHinges. If LocType = RelativeDistance, this is the distance of the hinge from the end of the i-end offset, as a ratio to the clear length of the frame object.

AD

Type: SystemDouble

An array of size NumberHinges. If LocType = OffsetFromIEnd, this is the absolute distance of the hinge from the end of the i-end offset. If LocType = OffsetFromJEnd, this is the absolute distance of the hinge from the end of the j-end offset.

Return Value

Type: Int32

Returns zero if the hinge data is successfully retrieved, otherwise it returns a nonzero value.

▲ See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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i»j

CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LSTCC3EBF86_0?cpp=2021-")

Method

Retrieves frame object insertion point assignments. The assignments include the cardinal point and end joint offsets.

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetInsertionPoint(
    string Name,
    ref int CardinalPoint,
    ref bool Mirror2,
    ref bool StiffTransform,
    ref double[] Offset1,
    ref double[] Offset2,
    ref string CSys
)
```

```
Function GetInsertionPoint (
    Name As String,
    ByRef CardinalPoint As Integer,
    ByRef Mirror2 As Boolean,
    ByRef StiffTransform As Boolean,
    ByRef Offset1 As Double(),
    ByRef Offset2 As Double(),
    ByRef CSys As String
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim CardinalPoint As Integer
Dim Mirror2 As Boolean
Dim StiffTransform As Boolean
Dim Offset1 As Double()
Dim Offset2 As Double()
Dim CSys As String
Dim returnValue As Integer
```

```
returnValue = instance.GetInsertionPoint(Name,
    CardinalPoint, Mirror2, StiffTransform,
    Offset1, Offset2, CSys)
```

```
int GetInsertionPoint(
    String^ Name,
    int% CardinalPoint,
    bool% Mirror2,
    bool% StiffTransform,
    array<double>^% Offset1,
```

cFrameObjspan id="LSTCC3EBF86_0"AddLanguageSpecificTextSet("LSTCC3EBF86_0?cpp=2021-");Get

```
array<double>^% Offset2,
String^% CSys
)
```

```
abstract GetInsertionPoint :
    Name : string *
    CardinalPoint : int byref *
    Mirror2 : bool byref *
    StiffTransform : bool byref *
    Offset1 : float[] byref *
    Offset2 : float[] byref *
    CSys : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame object.

CardinalPoint

Type:Â SystemInt32

This is a numeric value from 1 to 11 that specifies the cardinal point for the frame object. The cardinal point specifies the relative position of the frame section on the line representing the frame object.

1. bottom left
2. bottom center
3. bottom right
4. middle left
5. middle center
6. middle right
7. top left
8. top center
9. top right
10. centroid
11. shear center

Mirror2

Type:Â SystemBoolean

If this item is True, the frame object section is assumed to be mirrored (flipped) about its local 2-axis.

StiffTransform

Type:Â SystemBoolean

If this item is True, the frame object stiffness is transformed for cardinal point and joint offsets from the frame section centroid.

Offset1

Type:Â SystemDouble

This is an array of three joint offset distances, in the coordinate directions specified by CSys, at the I-End of the frame object. [L]

- ◊ Offset1(0) = Offset in the 1-axis or X-axis direction
- ◊ Offset1(1) = Offset in the 2-axis or Y-axis direction
- ◊ Offset1(2) = Offset in the 3-axis or Z-axis direction

Offset2

Type:Â SystemDouble

This is an array of three joint offset distances, in the coordinate directions specified by CSys, at the J-End of the frame object. [L]

Introduction

- ◇ Offset2(0) = Offset in the 1-axis or X-axis direction
- ◇ Offset2(1) = Offset in the 2-axis or Y-axis direction
- ◇ Offset2(2) = Offset in the 3-axis or Z-axis direction

CSys

Type: SystemString

This is Local or the name of a defined coordinate system. It is the coordinate system in which the Offset1 and Offset2 items are specified.

Return Value

Type: Int32

Returns zero if the insertion point data is successfully retrieved, otherwise it returns a nonzero value.

Remarks

Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)  
  
    'assign frame insertion point  
    ReDim Offset1(2)  
    ReDim Offset2(2)  
    For i=0 To 2  
        Offset1(i)=10 + i  
        Offset2(i)=20 + i  
    Next i  
    ret = SapModel.FrameObj.SetInsertionPoint("15", 7, False, True, Offset1, Offset2)  
  
    'get frame insertion point  
    ReDim Offset1(2)  
    ReDim Offset2(2)  
    ret = SapModel.FrameObj.GetInsertionPoint("15", CardinalPoint, Mirror2, StiffTransform, Offset1, Offset2)  
  
    'close ETABS  
    EtabsObject.ApplicationExit(False)  
  
    'clean up variables  
    SapModel = Nothing  
    EtabsObject = Nothing
```

Return Value

End Sub

▲ See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST68D89F6A_0?cpp=::2025");GetL

Method

Retrieves the label and story for a unique frame object name

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetLabelFromName(
    string Name,
    ref string Label,
    ref string Story
)
```

```
Function GetLabelFromName (
    Name As String,
    ByRef Label As String,
    ByRef Story As String
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim Label As String
Dim Story As String
Dim returnValue As Integer
```

```
returnValue = instance.GetLabelFromName(Name,
    Label, Story)
```

```
int GetLabelFromName(
    String^ Name,
    String^% Label,
    String^% Story
)
```

```
abstract GetLabelFromName :
    Name : string *
    Label : string byref *
    Story : string byref -> int
```

Parameters

Name

Type: SystemString

The name of the frame object

Label

Type:Â SystemString
 The frame object label

Story

Type:Â SystemString
 The frame object story level

Return Value

Type:Â Int32
 Returns zero if the data is successfully retrieved, otherwise it returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberNames As Integer
    Dim Label As String
    Dim Story As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'get area object data
    ret = SapModel.FrameObj.GetLabelFromName("12", Label, Story)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST2996B2C7_0") Method

Retrieves the names and labels of all defined frame objects.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetLabelNameList(
    ref int NumberNames,
    ref string[] MyName,
    ref string[] MyLabel,
    ref string[] MyStory
)
```

```
Function GetLabelNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String(),
    ByRef MyLabel As String(),
    ByRef MyStory As String()
) As Integer
```

```
Dim instance As cFrameObj
Dim NumberNames As Integer
Dim MyName As String()
Dim MyLabel As String()
Dim MyStory As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetLabelNameList(NumberNames,
    MyName, MyLabel, MyStory)
```

```
int GetLabelNameList(
    int% NumberNames,
    array<String^>^% MyName,
    array<String^>^% MyLabel,
    array<String^>^% MyStory
)
```

```
abstract GetLabelNameList :
    NumberNames : int byref *
    MyName : string[] byref *
    MyLabel : string[] byref *
    MyStory : string[] byref -> int
```


Parameters**NumberNames**

Type: SystemInt32

The number of frame object names retrieved by the program.

MyName

Type: SystemString

This is a one-dimensional array of frame object names. The MyName array is created as a dynamic, zero-based, array by the API user:

```
Dim MyName() as String
```

The array is dimensioned to (NumberNames + 1) inside the program, filled with the names, and returned to the API user.

MyLabel

Type: SystemString

This is a one-dimensional array of frame object labels. The MyLabel array is created as a dynamic, zero-based, array by the API user:

```
Dim MyLabel() as String
```

The array is dimensioned to (NumberNames + 1) inside the program, filled with the labels, and returned to the API user.

MyStory

Type: SystemString

This is a one-dimensional array of the story levels of the frame objects. The MyStory array is created as a dynamic, zero-based, array by the API user:

```
Dim MyStory() as String
```

The array is dimensioned to (NumberNames + 1) inside the program, filled with the story levels, and returned to the API user.

Return Value

Type: Int32

Returns zero if the data is successfully retrieved, otherwise it returns nonzero.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberNames As Integer
    Dim MyName() As String
    Dim MyLabel() As String
    Dim MyStory() As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

```
'start ETABS application
  ret = EtabsObject.ApplicationStart()

'create SapModel object
  SapModel = EtabsObject.SapModel

'initialize model
  ret = SapModel.InitializeNewModel()

'create steel deck template model
  ret = SapModel.File.NewSteelDeck(4, 12, 4, 4, 24, 24)

'get area object data
  ret = SapModel.FrameObj.GetLabelNameList(NumberNames, MyName, MyLabel, MyStory)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

🔗 See Also

Reference

[cFrameObj Interface](#)

[ETABSv1 Namespace](#)

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»

CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST3DB32C22_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetLateralBracing(
    string Name,
    ref int NumberItems,
    ref string[] FrameName,
    ref int[] MyType,
    ref int[] Loc,
    ref double[] RD1,
    ref double[] RD2,
    ref double[] Dist1,
    ref double[] Dist2
)
```

```
Function GetLateralBracing (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef FrameName As String(),
    ByRef MyType As Integer(),
    ByRef Loc As Integer(),
    ByRef RD1 As Double(),
    ByRef RD2 As Double(),
    ByRef Dist1 As Double(),
    ByRef Dist2 As Double()
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim NumberItems As Integer
Dim FrameName As String()
Dim MyType As Integer()
Dim Loc As Integer()
Dim RD1 As Double()
Dim RD2 As Double()
Dim Dist1 As Double()
Dim Dist2 As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetLateralBracing(Name,
    NumberItems, FrameName, MyType, Loc,
    RD1, RD2, Dist1, Dist2)
```

```
int GetLateralBracing(
    String^ Name,
```

cFrameObjid="LST3DB32C22_0"AddLanguageSpecificTextSet("LST3DB32C22_0?cpp=:2004.");GetL

```

int% NumberItems,
array<String>^% FrameName,
array<int>^% MyType,
array<int>^% Loc,
array<double>^% RD1,
array<double>^% RD2,
array<double>^% Dist1,
array<double>^% Dist2
)

abstract GetLateralBracing :
    Name : string *
    NumberItems : int byref *
    FrameName : string[] byref *
    MyType : int[] byref *
    Loc : int[] byref *
    RD1 : float[] byref *
    RD2 : float[] byref *
    Dist1 : float[] byref *
    Dist2 : float[] byref -> int

```

Parameters

Name
Type: [SystemString](#)

NumberItems
Type: [SystemInt32](#)

FrameName
Type: [SystemString](#)

MyType
Type: [SystemInt32](#)

Loc
Type: [SystemInt32](#)

RD1
Type: [SystemDouble](#)

RD2
Type: [SystemDouble](#)

Dist1
Type: [SystemDouble](#)

Dist2
Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

See Also

Reference

[cFrameObj Interface](#)

[ETABSv1 Namespace](#)

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Introduction

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»

CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST84283CE1_0?cpp=::2034");GetL

Method

Retrieves the distributed load assignments to frame objects.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetLoadDistributed(
    string Name,
    ref int NumberItems,
    ref string[] FrameName,
    ref string[] LoadPat,
    ref int[] MyType,
    ref string[] CSys,
    ref int[] Dir,
    ref double[] RD1,
    ref double[] RD2,
    ref double[] Dist1,
    ref double[] Dist2,
    ref double[] Val1,
    ref double[] Val2,
    eItemType ItemType = eItemType.Objects
)
```

```
Function GetLoadDistributed (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef FrameName As String(),
    ByRef LoadPat As String(),
    ByRef MyType As Integer(),
    ByRef CSys As String(),
    ByRef Dir As Integer(),
    ByRef RD1 As Double(),
    ByRef RD2 As Double(),
    ByRef Dist1 As Double(),
    ByRef Dist2 As Double(),
    ByRef Val1 As Double(),
    ByRef Val2 As Double(),
    ByRef ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim NumberItems As Integer
Dim FrameName As String()
Dim LoadPat As String()
Dim MyType As Integer()
Dim CSys As String()
```

cFrameObjAddLanguageSpecificTextSet("LST84283CE1_0?cpp=::2034");GetL

Introduction

```
Dim Dir As Integer()
Dim RD1 As Double()
Dim RD2 As Double()
Dim Dist1 As Double()
Dim Dist2 As Double()
Dim Val1 As Double()
Dim Val2 As Double()
Dim ItemType As eItemType
Dim returnValue As Integer

returnValue = instance.GetLoadDistributed(Name,
NumberItems, FrameName, LoadPat,
MyType, CSys, Dir, RD1, RD2, Dist1,
Dist2, Val1, Val2, ItemType)

int GetLoadDistributed(
String^ Name,
int% NumberItems,
array<String^>^% FrameName,
array<String^>^% LoadPat,
array<int>^% MyType,
array<String^>^% CSys,
array<int>^% Dir,
array<double>^% RD1,
array<double>^% RD2,
array<double>^% Dist1,
array<double>^% Dist2,
array<double>^% Val1,
array<double>^% Val2,
eItemType ItemType = eItemType::Objects
)

abstract GetLoadDistributed :
    Name : string *
    NumberItems : int byref *
    FrameName : string[] byref *
    LoadPat : string[] byref *
    MyType : int[] byref *
    CSys : string[] byref *
    Dir : int[] byref *
    RD1 : float[] byref *
    RD2 : float[] byref *
    Dist1 : float[] byref *
    Dist2 : float[] byref *
    Val1 : float[] byref *
    Val2 : float[] byref *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing frame object or group, depending on the value of the ItemType item.

NumberItems

	Type:Â <u>SystemInt32</u> The total number of distributed loads retrieved for the specified frame objects.
FrameName	Type:Â <u>SystemString</u> This is an array that includes the name of the frame object associated with each distributed load.
LoadPat	Type:Â <u>SystemString</u> This is an array that includes the name of the load pattern associated with each distributed load.
MyType	Type:Â <u>SystemInt32</u> This is an array that includes 1 or 2, indicating the type of distributed load. 1. Force per unit length 2. Moment per unit length
CSys	Type:Â <u>SystemString</u> This is an array that includes the name of the coordinate system in which each distributed load is defined. It may be Local or the name of a defined coordinate system.
Dir	Type:Â <u>SystemInt32</u> This is an integer between 1 and 11, indicating the direction of the load. 1. Local 1 axis (only applies when CSys is Local) 2. Local 2 axis (only applies when CSys is Local) 3. Local 3 axis (only applies when CSys is Local) 4. X direction (does not apply when CSys is Local) 5. Y direction (does not apply when CSys is Local) 6. Z direction (does not apply when CSys is Local) 7. Projected X direction (does not apply when CSys is Local) 8. Projected Y direction (does not apply when CSys is Local) 9. Projected Z direction (does not apply when CSys is Local) 10. Gravity direction (only applies when CSys is Global) 11. Projected Gravity direction (only applies when CSys is Global) The positive gravity direction (see Dir = 10 and 11) is in the negative Global Z direction.
RD1	Type:Â <u>SystemDouble</u>
RD2	Type:Â <u>SystemDouble</u>
Dist1	Type:Â <u>SystemDouble</u> This is an array that includes the actual distance from the I-End of the frame object to the start of the distributed load. [L]
Dist2	Type:Â <u>SystemDouble</u> This is an array that includes the actual distance from the I-End of the frame object to the end of the distributed load. [L]
Val1	Type:Â <u>SystemDouble</u> This is an array that includes the load value at the start of the distributed load.

[F/L] when MyType is 1 and [FL/L] when MyType is 2

Val2

Type: SystemDouble

This is an array that includes the load value at the end of the distributed load.

[F/L] when MyType is 1 and [FL/L] when MyType is 2

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the assignments are retrieved for the frame object specified by the Name item.

If this item is Group, the assignments are retrieved for all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the assignments are retrieved for all selected frame objects and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the load assignments are successfully retrieved, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberItems As Integer
    Dim FrameName() As String
    Dim LoadPat() As String
    Dim MyType() As Integer
    Dim CSys() As String
    Dim Dir() As Integer
    Dim RD1() As Double
    Dim RD2() As Double
    Dim Dist1() As Double
    Dim Dist2() As Double
    Dim Val1() As Double
    Dim Val2() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel
```

Introduction

```
'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'assign frame distributed loads
    ret = SapModel.FrameObj.SetLoadDistributed("14", "DEAD", 1, 10, 0, 1, 0.08, 0.08)
    ret = SapModel.FrameObj.SetLoadDistributed("15", "DEAD", 1, 10, 0, 1, 0.08, 0.08)

'get frame distributed loads
    ret = SapModel.FrameObj.GetLoadDistributed("ALL", NumberItems, FrameName, LoadPat, MyType, CSy

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST5A235C90_0") Method

Retrieves the point load assignments to frame objects.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetLoadPoint(
    string Name,
    ref int NumberItems,
    ref string[] FrameName,
    ref string[] LoadPat,
    ref int[] MyType,
    ref string[] CSys,
    ref int[] Dir,
    ref double[] RelDist,
    ref double[] Dist,
    ref double[] Val,
    eItemType ItemType = eItemType.Objects
)
```

```
Function GetLoadPoint (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef FrameName As String(),
    ByRef LoadPat As String(),
    ByRef MyType As Integer(),
    ByRef CSys As String(),
    ByRef Dir As Integer(),
    ByRef RelDist As Double(),
    ByRef Dist As Double(),
    ByRef Val As Double(),
    ByRef ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim NumberItems As Integer
Dim FrameName As String()
Dim LoadPat As String()
Dim MyType As Integer()
Dim CSys As String()
Dim Dir As Integer()
Dim RelDist As Double()
Dim Dist As Double()
Dim Val As Double()
Dim ItemType As eItemType
Dim returnValue As Integer
```

cFrameObjAddLanguageSpecificTextSet("LST5A235C90_0?cpp=::2009");GetL

```
returnValue = instance.GetLoadPoint(Name,
NumberItems, FrameName, LoadPat,
MyType, CSys, Dir, RelDist, Dist, Val,
ItemType)
```

```
int GetLoadPoint(
String^ Name,
int% NumberItems,
array<String^>^% FrameName,
array<String^>^% LoadPat,
array<int>^% MyType,
array<String^>^% CSys,
array<int>^% Dir,
array<double>^% RelDist,
array<double>^% Dist,
array<double>^% Val,
eItemType ItemType = eItemType::Objects
)
```

```
abstract GetLoadPoint :
    Name : string *
    NumberItems : int byref *
    FrameName : string[] byref *
    LoadPat : string[] byref *
    MyType : int[] byref *
    CSys : string[] byref *
    Dir : int[] byref *
    RelDist : float[] byref *
    Dist : float[] byref *
    Val : float[] byref *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame object or group, depending on the value of the ItemType item.

NumberItems

Type:Â SystemInt32

The total number of point loads retrieved for the specified frame objects.

FrameName

Type:Â SystemString

This is an array that includes the name of the frame object associated with each point load.

LoadPat

Type:Â SystemString

This is an array that includes the name of the load pattern associated with each point load.

MyType

Type:Â SystemInt32

This is an array that includes 1 or 2, indicating the type of point load.

1. Force
2. Moment

CSys

Type:Â SystemString

This is an array that includes the name of the coordinate system in which each point load is defined. It may be Local or the name of a defined coordinate system.

Dir

Type:Â SystemInt32

This is an integer between 1 and 11, indicating the direction of the load.

1. Local 1 axis (only applies when CSys is Local)
2. Local 2 axis (only applies when CSys is Local)
3. Local 3 axis (only applies when CSys is Local)
4. X direction (does not apply when CSys is Local)
5. Y direction (does not apply when CSys is Local)
6. Z direction (does not apply when CSys is Local)
7. Projected X direction (does not apply when CSys is Local)
8. Projected Y direction (does not apply when CSys is Local)
9. Projected Z direction (does not apply when CSys is Local)
10. Gravity direction (only applies when CSys is Global)
11. Projected Gravity direction (only applies when CSys is Global)

The positive gravity direction (see Dir = 10 and 11) is in the negative Global Z direction.

RelDist

Type:Â SystemDouble

This is an array that includes the relative distance from the I-End of the frame object to the location where the point load is applied.

Dist

Type:Â SystemDouble

Val

Type:Â SystemDouble

ItemType (Optional)

Type:Â ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

- ◇ Object = 0
- ◇ Group = 1
- ◇ SelectedObjects = 2

If this item is Objects, the assignments are retrieved for the frame object specified by the Name item.

If this item is Group, the assignments are retrieved for all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the assignments are retrieved for all selected frame objects and the Name item is ignored.

Return Value

Type:Â Int32

Returns zero if the load assignments are successfully retrieved, otherwise it returns a nonzero value.

Return Value

Remarks

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberItems As Integer
    Dim FrameName() As String
    Dim LoadPat() As String
    Dim MyType() As Integer
    Dim CSys() As String
    Dim Dir() As Integer
    Dim RelDist() As Double
    Dim Dist() As Double
    Dim Val() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'assign frame point loads
    ret = SapModel.FrameObj.SetLoadPoint("14", "DEAD", 1, 10, .5, 20)
    ret = SapModel.FrameObj.SetLoadPoint("15", "DEAD", 1, 10, .5, 20)

    'get frame point loads
    ret = SapModel.FrameObj.GetLoadPoint("ALL", NumberItems, FrameName, LoadPat, MyType, CSys, Dir, RelDist, Dist, Val)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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Introduction

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LSTD297AF90_0?cpp=::2014") Method

Retrieves the temperature load assignments to frame objects.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetLoadTemperature(
    string Name,
    ref int NumberItems,
    ref string[] FrameName,
    ref string[] LoadPat,
    ref int[] MyType,
    ref double[] Val,
    ref string[] PatternName,
    eItemType ItemType = eItemType.Objects
)
```

```
Function GetLoadTemperature (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef FrameName As String(),
    ByRef LoadPat As String(),
    ByRef MyType As Integer(),
    ByRef Val As Double(),
    ByRef PatternName As String(),
    ByRef ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim NumberItems As Integer
Dim FrameName As String()
Dim LoadPat As String()
Dim MyType As Integer()
Dim Val As Double()
Dim PatternName As String()
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.GetLoadTemperature(Name,
    NumberItems, FrameName, LoadPat,
    MyType, Val, PatternName, ItemType)
```

```
int GetLoadTemperature(
    String^ Name,
    int% NumberItems,
    array<String^>% FrameName,
```

cFrameObjid="LSTD297AF90_0"AddLanguageSpecificTextSet("LSTD297AF90_0?cpp=::2014");GetL


```

array<String>^% LoadPat,
array<int>^% MyType,
array<double>^% Val,
array<String>^% PatternName,
eItemType ItemType = eItemType::Objects
)

abstract GetLoadTemperature :
    Name : string *
    NumberItems : int byref *
    FrameName : string[] byref *
    LoadPat : string[] byref *
    MyType : int[] byref *
    Val : float[] byref *
    PatternName : string[] byref *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of an existing frame object or group, depending on the value of the ItemType item.

NumberItems

Type:Â SystemInt32

The total number of temperature loads retrieved for the specified frame objects.

FrameName

Type:Â SystemString

This is an array that includes the name of the frame object associated with each temperature load.

LoadPat

Type:Â SystemString

This is an array that includes the name of the load pattern associated with each temperature load.

MyType

Type:Â SystemInt32

This is an array that includes 1, 2 or 3, indicating the type of temperature load.

1. Temperature

2. Temperature gradient along local 2 axis

3. Temperature gradient along local 3 axis

Val

Type:Â SystemDouble

PatternName

Type:Â SystemString

This is an array that includes the joint pattern name, if any, used to specify the temperature load.

ItemType (Optional)

Type:Â ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◇ Group = 1

◇ SelectedObjects = 2

If this item is Objects, the assignments are retrieved for the frame object specified by the Name item.

If this item is Group, the assignments are retrieved for all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the assignments are retrieved for all selected frame objects and the Name item is ignored.

Return Value

Type: $\hat{A}$ Int32

Returns zero if the load assignments are successfully retrieved, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberItems As Integer
    Dim FrameName() As String
    Dim LoadPat() As String
    Dim MyType() As Integer
    Dim Val() As Double
    Dim PatternName() As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'assign frame temperature load
    ret = SapModel.FrameObj.SetLoadTemperature("All", "DEAD", 1, 50, , , eItemType.Group)

    'get frame temperature load
    ret = SapModel.FrameObj.GetLoadTemperature("ALL", NumberItems, FrameName, LoadPat, MyType, Val)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
```

```
EtabsObject = Nothing
```

```
End Sub
```

▲ See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST1E0A2CC5_0") Method

Retrieves the frame local axis angle assignment for frame objects.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetLocalAxes(
    string Name,
    ref double Ang,
    ref bool Advanced
)
```

```
Function GetLocalAxes (
    Name As String,
    ByRef Ang As Double,
    ByRef Advanced As Boolean
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim Ang As Double
Dim Advanced As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetLocalAxes(Name,
    Ang, Advanced)
```

```
int GetLocalAxes(
    String^ Name,
    double% Ang,
    bool% Advanced
)
```

```
abstract GetLocalAxes :
    Name : string *
    Ang : float byref *
    Advanced : bool byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing frame object.

Ang

Type: SystemDouble

This is the angle that the local 2 and 3 axes are rotated about the positive local 1 axis, from the default orientation or, if the Advanced item is True, from the orientation determined by the plane reference vector. The rotation for a positive angle appears counter clockwise when the local +1 axis is pointing toward you.
[deg]

Advanced

Type: SystemBoolean

This item is True if the line object local axes orientation was obtained using advanced local axes parameters.

Return Value

Type: Int32

Returns zero if the assignment is successfully retrieved, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'get frame local axis angle
    ret = SapModel.FrameObj.GetLocalAxes("3", Ang, Advanced)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST4141C470_0?cpp=::|205.1");GetMass

Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetMass(
    string Name,
    ref double MassOverL
)
```

```
Function GetMass (
    Name As String,
    ByRef MassOverL As Double
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim MassOverL As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetMass(Name, MassOverL)
```

```
int GetMass(
    String^ Name,
    double% MassOverL
)
```

```
abstract GetMass :
    Name : string *
    MassOverL : float byref -> int
```

Parameters

Name

Type: [SystemString](#)

MassOverL

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST9991EC2B_0?cpp=:2053.");GetMaterialOverwrite

Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetMaterialOverwrite(
    string Name,
    ref string PropName
)
```

```
Function GetMaterialOverwrite (
    Name As String,
    ByRef PropName As String
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim PropName As String
Dim returnValue As Integer
```

```
returnValue = instance.GetMaterialOverwrite(Name,
    PropName)
```

```
int GetMaterialOverwrite(
    String^ Name,
    String^% PropName
)
```

```
abstract GetMaterialOverwrite :
    Name : string *
    PropName : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

PropName

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST53B2C_0")

Method

Retrieves the frame modifier assignment for frame objects. The default value for all modifiers is one.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetModifiers(
    string Name,
    ref double[] Value
)
```

```
Function GetModifiers (
    Name As String,
    ByRef Value As Double()
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim Value As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetModifiers(Name,
    Value)
```

```
int GetModifiers(
    String^ Name,
    array<double>^% Value
)
```

```
abstract GetModifiers :
    Name : string *
    Value : float[] byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing frame object

Value

Type: [SystemDouble](#)

This is an array of eight unitless modifiers:

Value

Modifier

Value(0) Cross sectional area modifier

cFrameObjid="LST53B2C_0"AddLanguageSpecificTextSet("LST53B2C_0?cpp=::|nu=.");**GetModifiers**

Introduction

Value(1) Shear area in local 2 direction modifier
Value(2) Shear area in local 3 direction modifier
Value(3) Torsional constant modifier
Value(4) Moment of inertia about local 2 axis modifier
Value(5) Moment of inertia about local 3 axis modifier
Value(6) Mass modifier
Value(7) Weight modifier

Return Value

Type: Int32

Returns zero if the modifier assignments are successfully retrieved, otherwise it returns a nonzero value

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim Value() As Double  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)  
  
    'assign modifiers  
    ReDim Value(7)  
    For i = 0 To 7  
        Value(i) = 1  
    Next i  
    Value(5) = 100  
    ret = SapModel.FrameObj.SetModifiers("3", Value, eItemType.Objects)  
  
    'get modifiers  
    Dim myValue As Double()  
    ret = SapModel.FrameObj.GetModifiers("3", myValue)  
  
    'close ETABS  
    EtabsObject.ApplicationExit(False)  
  
    'clean up variables  
    SapModel = Nothing  
    EtabsObject = Nothing
```

Parameters

End Sub

▲ See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LSTAAAE7F Method

Retrieves the unique name of a frame object, given the label and story level

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNameFromLabel(
    string Label,
    string Story,
    ref string Name
)
```

```
Function GetNameFromLabel (
    Label As String,
    Story As String,
    ByRef Name As String
) As Integer
```

```
Dim instance As cFrameObj
Dim Label As String
Dim Story As String
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.GetNameFromLabel(Label,
    Story, Name)
```

```
int GetNameFromLabel(
    String^ Label,
    String^ Story,
    String^% Name
)
```

```
abstract GetNameFromLabel :
    Label : string *
    Story : string *
    Name : string byref -> int
```

Parameters

Label

Type: [SystemString](#)

The frame object label

Story

Type: SystemString

The frame object story level

Name

Type: SystemString

The unique name of the frame object

Return Value

Type: Int32

Returns zero if the data is successfully retrieved, otherwise it returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberNames As Integer
    Dim Name As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'get area object data
    ret = SapModel.FrameObj.GetNameFromLabel("B3", "Story2", Name)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cFrameObj Interface

ETABSV1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LSTD221F6BB_0")

Method

Retrieves the names of all defined frame objects.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetNameList(
    ref int NumberNames,
    ref string[] MyName
)
```

```
Function GetNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cFrameObj
Dim NumberNames As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetNameList(NumberNames,
    MyName)
```

```
int GetNameList(
    int% NumberNames,
    array<String^>^% MyName
)
```

```
abstract GetNameList :
    NumberNames : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberNames

Type: System.Int32

The number of frame object names retrieved by the program.

MyName

Type: System.String

This is a one-dimensional array of frame object names. The MyName array is created as a dynamic, zero-based, array by the API user:

```
Dim MyName() as String
```

cFrameObjspan id="LSTD221F6BB_0"AddLanguageSpecificTextSet("LSTD221F6BB_0?cpp=:2064.");GetN

The array is dimensioned to (NumberNames + 1) inside the ETABS program, filled with the names, and returned to the API user.

Return Value

Type: Int32

Returns zero if the names are successfully retrieved, otherwise it returns nonzero.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberNames As Integer
    Dim MyName() As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'get frame object names
    ret = SapModel.FrameObj.GetNameList(NumberNames, MyName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ **See Also**

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST6A90DE7_0")

Method

Retrieves the names of the defined frame objects on a given story.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNameListOnStory(
    string StoryName,
    ref int NumberNames,
    ref string[] MyName
)
```

```
Function GetNameListOnStory (
    StoryName As String,
    ByRef NumberNames As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cFrameObj
Dim StoryName As String
Dim NumberNames As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetNameListOnStory(StoryName,
    NumberNames, MyName)
```

```
int GetNameListOnStory(
    String^ StoryName,
    int% NumberNames,
    array<String^>^% MyName
)
```

```
abstract GetNameListOnStory :
    StoryName : string *
    NumberNames : int byref *
    MyName : string[] byref -> int
```

Parameters

StoryName

Type: [SystemString](#)

The name of an existing story.

NumberNames

Type: SystemInt32

The number of frame object names retrieved by the program.

MyName

Type: SystemString

This is a one-dimensional array of frame object names. The MyName array is created as a dynamic, zero-based, array by the API user:

```
Dim MyName() as String
```

The array is dimensioned to (NumberNames + 1) inside the ETABS program, filled with the names, and returned to the API user.

Return Value

Type: Int32

Returns zero if the names are successfully retrieved, otherwise it returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberNames As Integer
    Dim MyName() As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'get frame object names
    ret = SapModel.FrameObj.GetNameListOnStory("Story2", NumberNames, MyName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST4723D8B Method

Retrieves frame object output station data

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetOutputStations(
    string Name,
    ref int MyType,
    ref double MaxSegSize,
    ref int MinSections,
    ref bool NoOutPutAndDesignAtElementEnds,
    ref bool NoOutPutAndDesignAtPointLoads
)
```

```
Function GetOutputStations (
    Name As String,
    ByRef MyType As Integer,
    ByRef MaxSegSize As Double,
    ByRef MinSections As Integer,
    ByRef NoOutPutAndDesignAtElementEnds As Boolean,
    ByRef NoOutPutAndDesignAtPointLoads As Boolean
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim MyType As Integer
Dim MaxSegSize As Double
Dim MinSections As Integer
Dim NoOutPutAndDesignAtElementEnds As Boolean
Dim NoOutPutAndDesignAtPointLoads As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetOutputStations(Name,
    MyType, MaxSegSize, MinSections,
    NoOutPutAndDesignAtElementEnds,
    NoOutPutAndDesignAtPointLoads)
```

```
int GetOutputStations(
    String^ Name,
    int% MyType,
    double% MaxSegSize,
    int% MinSections,
    bool% NoOutPutAndDesignAtElementEnds,
    bool% NoOutPutAndDesignAtPointLoads
)
```

cFrameObjAddLanguageSpecificTextSet("LST4723D8BD_0?cpp=:2066.");GetC

Introduction

```
abstract GetOutputStations :  
    Name : string *  
    MyType : int byref *  
    MaxSegSize : float byref *  
    MinSections : int byref *  
    NoOutPutAndDesignAtElementEnds : bool byref *  
    NoOutPutAndDesignAtPointLoads : bool byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame object

MyType

Type:Â SystemInt32

This is either 1 or 2 indicating how the output stations are specified

1. maximum segment size, that is, maximum station spacing
2. minimum number of stations

MaxSegSize

Type:Â SystemDouble

The maximum segment size, that is, the maximum station spacing. This item applies only when MyType = 1. [L]

MinSections

Type:Â SystemInt32

The minimum number of stations. This item applies only when MyType = 2.

NoOutPutAndDesignAtElementEnds

Type:Â SystemBoolean

If this item is True, no additional output stations are added at the ends of line elements when the frame object is internally meshed. In ETABS, this item will always be False.

NoOutPutAndDesignAtPointLoads

Type:Â SystemBoolean

If this item is True, no additional output stations are added at point load locations. In ETABS, this item will always be False.

Return Value

Type:Â Int32

Returns zero if the data is successfully retrieved, otherwise it returns a nonzero value

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim MyType As Integer  
    Dim MaxSegSize As Double  
    Dim MinSections As Integer  
    Dim NoOutPutAndDesignAtElementEnds As Boolean  
    Dim NoOutPutAndDesignAtPointLoads As Boolean
```

'create ETABS object

Introduction

```
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'get frame output station data
ret = SapModel.FrameObj.GetOutputStations("15", MyType, MaxSegSize, MinSections, NoOutPutAndDe

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

[cFrameObj Interface](#)

[ETABSv1 Namespace](#)

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»

CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST94138DF7_0?cpp=::2069");GetPi

Method

Retrieves the pier label assignments of a frame object

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetPier(
    string Name,
    ref string PierName
)
```

```
Function GetPier (
    Name As String,
    ByRef PierName As String
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim PierName As String
Dim returnValue As Integer
```

```
returnValue = instance.GetPier(Name, PierName)
```

```
int GetPier(
    String^ Name,
    String^% PierName
)
```

```
abstract GetPier :
    Name : string *
    PierName : string byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing frame object

PierName

Type: SystemString

The name of the pier assignment, if any, or "None"

Return Value

Type: Int32

Returns zero if the assignment is successfully retrieved; otherwise it returns a nonzero value.

▀ **Remarks**

▀ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim PierName as String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new pier label
    ret = SapModel.PierLabel.SetPier("MyPier")

    'set assignment
    ret = SapModel.FrameObj.SetPier("3", "MyPier")

    'get assignment
    ret = SapModel.FrameObj.GetPier("3", PierName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▀ **See Also**

Reference

cFrameObj Interface

ETABSV1 Namespace

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Introduction

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LSTB4F68C08_0")

Method

Retrieves the names of the point objects at each end of a specified frame object.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPoints(
    string Name,
    ref string Point1,
    ref string Point2
)
```

```
Function GetPoints (
    Name As String,
    ByRef Point1 As String,
    ByRef Point2 As String
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim Point1 As String
Dim Point2 As String
Dim returnValue As Integer
```

```
returnValue = instance.GetPoints(Name,
    Point1, Point2)
```

```
int GetPoints(
    String^ Name,
    String^% Point1,
    String^% Point2
)
```

```
abstract GetPoints :
    Name : string *
    Point1 : string byref *
    Point2 : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of a defined frame object.

Point1

Type: SystemString

The name of the point object at the I-End of the specified frame object.

Point2

Type: SystemString

The name of the point object at the J-End of the specified frame object.

Return Value

Type: Int32

Returns zero if the point names are successfully retrieved, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Point1 As String
    Dim Point2 As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'get names of points
    ret = SapModel.FrameObj.GetPoints("3", Point1, Point2)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cFrameObj Interface

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST28D20105_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetReleases(
    string Name,
    ref bool[] II,
    ref bool[] JJ,
    ref double[] StartValue,
    ref double[] EndValue
)
```

```
Function GetReleases (
    Name As String,
    ByRef II As Boolean(),
    ByRef JJ As Boolean(),
    ByRef StartValue As Double(),
    ByRef EndValue As Double()
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim II As Boolean()
Dim JJ As Boolean()
Dim StartValue As Double()
Dim EndValue As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetReleases(Name,
    II, JJ, StartValue, EndValue)
```

```
int GetReleases(
    String^ Name,
    array<bool>^% II,
    array<bool>^% JJ,
    array<double>^% StartValue,
    array<double>^% EndValue
)
```

```
abstract GetReleases :
    Name : string *
    II : bool[] byref *
    JJ : bool[] byref *
    StartValue : float[] byref *
    EndValue : float[] byref -> int
```

Parameters

Name

Type: SystemString

II

Type: SystemBoolean

JJ

Type: SystemBoolean

StartValue

Type: SystemDouble

EndValue

Type: SystemDouble

Return Value

Type: Int32

See Also

Reference

cFrameObj Interface

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LSTC40B0890_0?cpp=::2017");GetS

Method

Retrieves the frame section property assigned to a frame object.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetSection(
    string Name,
    ref string PropName,
    ref string SAuto
)
```

```
Function GetSection (
    Name As String,
    ByRef PropName As String,
    ByRef SAuto As String
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim PropName As String
Dim SAuto As String
Dim returnValue As Integer
```

```
returnValue = instance.GetSection(Name,
    PropName, SAuto)
```

```
int GetSection(
    String^ Name,
    String^% PropName,
    String^% SAuto
)
```

```
abstract GetSection :
    Name : string *
    PropName : string byref *
    SAuto : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of a defined frame object.

PropName

Type: SystemString

If no auto select list is assigned to the frame object, this is the name of the frame section property assigned to the frame object. If an auto select list is assigned to the frame object, this is the name of the frame section property, within the auto select list, which is currently being used as the analysis property for the frame object. If this item is None, no frame section property is assigned to the frame object.

SAuto

Type: SystemString

This is the name of the auto select list assigned to the frame object, if any. If this item is returned as a blank string, no auto select list is assigned to the frame object.

Return Value

Type: Int32

Returns zero if the frame object property is successfully retrieved, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim PropName As String
    Dim SAuto As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'get frame section property
    ret = SapModel.FrameObj.GetSection("3", PropName, SAuto)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

[cFrameObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST732EC4E5_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetSectionNonPrismatic(
    string Name,
    ref string PropName,
    ref double SVarTotalLength,
    ref double SVarRelStartLoc
)
```

```
Function GetSectionNonPrismatic (
    Name As String,
    ByRef PropName As String,
    ByRef SVarTotalLength As Double,
    ByRef SVarRelStartLoc As Double
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim PropName As String
Dim SVarTotalLength As Double
Dim SVarRelStartLoc As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetSectionNonPrismatic(Name,
    PropName, SVarTotalLength, SVarRelStartLoc)
```

```
int GetSectionNonPrismatic(
    String^ Name,
    String^% PropName,
    double% SVarTotalLength,
    double% SVarRelStartLoc
)
```

```
abstract GetSectionNonPrismatic :
    Name : string *
    PropName : string byref *
    SVarTotalLength : float byref *
    SVarRelStartLoc : float byref -> int
```

Parameters

Name

Type: SystemString

cFrameObjAddLanguageSpecificTextSet("LST732EC4E5_0?cpp=:2089.");GetS

PropName

Type:Â [SystemString](#)

SVarTotalLength

Type:Â [SystemDouble](#)

SVarRelStartLoc

Type:Â [SystemDouble](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cFrameObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LSTBCDA7C0A_0?cpp2082=.");Ge Method

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetSelected(
    string Name,
    ref bool Selected
)
```

```
Function GetSelected (
    Name As String,
    ByRef Selected As Boolean
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim Selected As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetSelected(Name,
    Selected)
```

```
int GetSelected(
    String^ Name,
    bool% Selected
)
```

```
abstract GetSelected :
    Name : string *
    Selected : bool byref -> int
```

Parameters

Name

Type: [SystemString](#)

Selected

Type: [SystemBoolean](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST138BC5B7_0") Method

Retrieves the Spandrel label assignments of a frame object

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetSpandrel(
    string Name,
    ref string SpandrelName
)
```

```
Function GetSpandrel (
    Name As String,
    ByRef SpandrelName As String
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim SpandrelName As String
Dim returnValue As Integer
```

```
returnValue = instance.GetSpandrel(Name,
    SpandrelName)
```

```
int GetSpandrel(
    String^ Name,
    String^% SpandrelName
)
```

```
abstract GetSpandrel :
    Name : string *
    SpandrelName : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing frame object

SpandrelName

Type: [SystemString](#)

The name of the Spandrel assignment, if any, or "None"

Return ValueType: Int32

Returns zero if the assignment is successfully retrieved; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim SpandrelName As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new Spandrel label
    ret = SapModel.SpandrelLabel.SetSpandrel("MySpandrel")

    'set assignment
    ret = SapModel.FrameObj.SetSpandrel("3", "MySpandrel")

    'get assignment
    ret = SapModel.FrameObj.GetSpandrel("3", SpandrelName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cFrameObj Interface

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Introduction

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST3F88A4A_0?cpp=::|nu2087GetSpringAssignment")

Method

Retrieves the named line spring property assignment for a frame object

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetSpringAssignment(
    string Name,
    ref string SpringProp
)
```

```
Function GetSpringAssignment (
    Name As String,
    ByRef SpringProp As String
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim SpringProp As String
Dim returnValue As Integer
```

```
returnValue = instance.GetSpringAssignment(Name,
    SpringProp)
```

```
int GetSpringAssignment(
    String^ Name,
    String^% SpringProp
)
```

```
abstract GetSpringAssignment :
    Name : string *
    SpringProp : string byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing frame object

SpringProp

Type: SystemString

The name of an existing line spring property

Return Value

Type: Int32

Returns zero if the assignment is successfully retrieved, otherwise it returns a nonzero value

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim SpringProp As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 4, 4, 24, 24)

    'define new spring property
    ret = SapModel.PropLineSpring.SetLineSpringProp("mySpringProp1", 0, 10, 20, 0, 0, 1)

    'assign property
    ret = SapModel.FrameObj.SetSpringAssignment("1", "mySpringProp1")

    'get assigned property
    Dim prop as String
    ret = SapModel.FrameObj.GetSpringAssignment("1", prop)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cFrameObj Interface

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST2E5F35E3_0?cpp=::2090");GetS

Method

Retrieves support data for a given frame beam object

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetSupports(
    string Name,
    ref string SupportName1,
    ref eObjType SupportType1,
    ref string SupportName2,
    ref eObjType SupportType2
)
```

```
Function GetSupports (
    Name As String,
    ByRef SupportName1 As String,
    ByRef SupportType1 As eObjType,
    ByRef SupportName2 As String,
    ByRef SupportType2 As eObjType
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim SupportName1 As String
Dim SupportType1 As eObjType
Dim SupportName2 As String
Dim SupportType2 As eObjType
Dim returnValue As Integer
```

```
returnValue = instance.GetSupports(Name,
    SupportName1, SupportType1, SupportName2,
    SupportType2)
```

```
int GetSupports(
    String^ Name,
    String^% SupportName1,
    eObjType% SupportType1,
    String^% SupportName2,
    eObjType% SupportType2
)
```

```
abstract GetSupports :
    Name : string *
    SupportName1 : string byref *
    SupportType1 : eObjType byref *
    SupportName2 : string byref *
```

cFrameObjid="LST2E5F35E3_0"AddLanguageSpecificTextSet("LST2E5F35E3_0?cpp=::2090");GetS

SupportType2 : eObjType byref -> int

Parameters

Name

Type: SystemString

The name of an existing frame beam object

SupportName1

Type: SystemString

The name of the column frame object, beam frame object or wall area object which supports the beam at its start node

SupportType1

Type: ETABSv1eObjType

A value from the eObjType enumeration

SupportName2

Type: SystemString

The name of the column frame object, beam frame object or wall area object which supports the beam at its end node

SupportType2

Type: ETABSv1eObjType

A value from the eObjType enumeration

Return Value

Type: Int32

▲ Remarks

▲ See Also

Reference

cFrameObj Interface

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST4B5D7E88_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetTCLimits(
    string Name,
    ref bool LimitCompressionExists,
    ref double LimitCompression,
    ref bool LimitTensionExists,
    ref double LimitTension
)
```

```
Function GetTCLimits (
    Name As String,
    ByRef LimitCompressionExists As Boolean,
    ByRef LimitCompression As Double,
    ByRef LimitTensionExists As Boolean,
    ByRef LimitTension As Double
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim LimitCompressionExists As Boolean
Dim LimitCompression As Double
Dim LimitTensionExists As Boolean
Dim LimitTension As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetTCLimits(Name,
    LimitCompressionExists, LimitCompression,
    LimitTensionExists, LimitTension)
```

```
int GetTCLimits(
    String^ Name,
    bool% LimitCompressionExists,
    double% LimitCompression,
    bool% LimitTensionExists,
    double% LimitTension
)
```

```
abstract GetTCLimits :
    Name : string *
    LimitCompressionExists : bool byref *
    LimitCompression : float byref *
    LimitTensionExists : bool byref *
    LimitTension : float byref -> int
```

cFrameObjAddLanguageSpecificTextSet("LST4B5D7E88_0?cpp=:2002");GetT

Parameters

Name

Type: SystemString

LimitCompressionExists

Type: SystemBoolean

LimitCompression

Type: SystemDouble

LimitTensionExists

Type: SystemBoolean

LimitTension

Type: SystemDouble

Return Value

Type: Int32

See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST585042E0_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetTransformationMatrix(
    string Name,
    ref double[] Value,
    bool IsGlobal = true
)
```

```
Function GetTransformationMatrix (
    Name As String,
    ByRef Value As Double(),
    Optional IsGlobal As Boolean = true
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim Value As Double()
Dim IsGlobal As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetTransformationMatrix(Name,
    Value, IsGlobal)
```

```
int GetTransformationMatrix(
    String^ Name,
    array<double>^% Value,
    bool IsGlobal = true
)
```

```
abstract GetTransformationMatrix :
    Name : string *
    Value : float[] byref *
    ?IsGlobal : bool
(* Defaults:
    let _IsGlobal = defaultArg IsGlobal true
*)
-> int
```

Parameters

Name

Type: SystemString

Value

cFrameObjid="LST585042E0_0"AddLanguageSpecificTextSet("LST585042E0_0?cpp=::|2094);GetTr

Type: SystemDouble
IsGlobal (Optional)
Type: SystemBoolean

Return Value

Type: Int32

See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LSTFAA67885_0")

Method

Retrieves the type of frame object (straight or curved)

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetType0API(
    string Name,
    ref string MyType
)
```

```
Function GetType0API (
    Name As String,
    ByRef MyType As String
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim MyType As String
Dim returnValue As Integer
```

```
returnValue = instance.GetType0API(Name,
    MyType)
```

```
int GetType0API(
    String^ Name,
    String^% MyType
)
```

```
abstract GetType0API :
    Name : string *
    MyType : string byref -> int
```

Parameters

Name

Type: SystemString

The name of a defined frame object

MyType

Type: SystemString

This is "Straight" or "Curved", indicating the type of frame object

Return Value

Type: Int32

Returns zero if the frame object type is successfully retrieved, otherwise it returns a nonzero value

▀ **Remarks**

▀ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyType As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'get frame type
    ret = SapModel.FrameObj.GetTypeOAPI("3", MyType)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▀ **See Also**

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST2AE1F7C2_0") Method

Sets the frame object column splice overwrite assignment.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetColumnSpliceOverwrite(
    string Name,
    int SpliceOption,
    double Height,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetColumnSpliceOverwrite (
    Name As String,
    SpliceOption As Integer,
    Height As Double,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim SpliceOption As Integer
Dim Height As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetColumnSpliceOverwrite(Name,
    SpliceOption, Height, ItemType)
```

```
int SetColumnSpliceOverwrite(
    String^ Name,
    int SpliceOption,
    double Height,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetColumnSpliceOverwrite :
    Name : string *
    SpliceOption : int *
    Height : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

cFrameObjid="LST2AE1F7C2_0"AddLanguageSpecificTextSet("LST2AE1F7C2_0?cpp=:2008.");SetC

Parameters

Name

Type:Â SystemString

The name of an existing frame object.

SpliceOption

Type:Â SystemInt32

This is a numeric value from 1 to 3 that specifies the option used for defining the splice overwrite.

1. from story data (default)

2. no splice

3. splice at height above story at bottom of the column object

Height

Type:Â SystemDouble

If the SpliceOption=3, this specifies the height of the splice above the story at the bottom of the column object.

ItemType (Optional)

Type:Â ETABSv1eItemType**Return Value**Type:Â Int32

Returns zero if the column splice overwrite assignment is successful, otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'assign column splice overwrite
    ret = SapModel.FrameObj.SetColumnSpliceOverwrite("15", 2, 0)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
```

Introduction

```
SapModel = Nothing  
EtabsObject = Nothing
```

End Sub

▲ See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST5A8CAC76_0?cpp=21011.") Method

Sets the design procedure for frame objects.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetDesignProcedure(
    string Name,
    int MyType,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetDesignProcedure (
    Name As String,
    MyType As Integer,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim MyType As Integer
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetDesignProcedure(Name,
    MyType, ItemType)
```

```
int SetDesignProcedure(
    String^ Name,
    int MyType,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetDesignProcedure :
    Name : string *
    MyType : int *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing frame object or group, depending on the value of the ItemType item.

MyType

Type: SystemInt32

This is an integer from the list below, indicating the design procedure type desired for the specified frame object.

- ◊ Program determined = 0
- ◊ Steel Frame Design = 1
- ◊ Concrete Frame Design = 2
- ◊ Composite Beam Design = 3
- ◊ Steel Joist Design = 4
- ◊ No Design = 7
- ◊ Composite Column Design = 13

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

- ◊ Object = 0
- ◊ Group = 1
- ◊ SelectedObjects = 2

If this item is Objects, the assignment is made to the frame object specified by the Name item.

If this item is Group, the assignment is made to all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the assignment is made to all selected frame objects and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the design procedure is successfully set; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
```

Parameters

Introduction

```
ret = SapModel.InitializeNewModel()  
  
'create steel deck template model  
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)  
  
'set design procedure  
ret = SapModel.FrameObj.SetDesignProcedure("8", 3)  
  
'close ETABS  
EtabsObject.ApplicationExit(False)  
  
'clean up variables  
SapModel = Nothing  
EtabsObject = Nothing  
  
End Sub
```

🔍 See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LSTA3AF17A_0?cpp=::|n24.04#SetEndLengthOffset")

Method

Assigns frame object end offsets along the 1-axis of the object.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetEndLengthOffset(
    string Name,
    bool AutoOffset,
    double Length1,
    double Length2,
    double RZ,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetEndLengthOffset (
    Name As String,
    AutoOffset As Boolean,
    Length1 As Double,
    Length2 As Double,
    RZ As Double,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim AutoOffset As Boolean
Dim Length1 As Double
Dim Length2 As Double
Dim RZ As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetEndLengthOffset(Name,
    AutoOffset, Length1, Length2, RZ,
    ItemType)
```

```
int SetEndLengthOffset(
    String^ Name,
    bool AutoOffset,
    double Length1,
    double Length2,
    double RZ,
    eItemType ItemType = eItemType::Objects
)
```

abstract SetEndLengthOffset :

cFrameObjid="LSTA3AF17A_0"AddLanguageSpecificTextSet("LSTA3AF17A_0?cpp=::|n24.04#SetEndLengthOffset")

Introduction

```
Name : string *
AutoOffset : bool *
Length1 : float *
Length2 : float *
RZ : float *
?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame object or group, depending on the value of the ItemType item.

AutoOffset

Type:Â SystemBoolean

If this item is True, the end length offsets are automatically determined by the program from object connectivity, and the Length1, Length2 and RZ items are ignored.

Length1

Type:Â SystemDouble

The offset length along the 1-axis of the frame object at the I-End of the frame object. [L]

Length2

Type:Â SystemDouble

The offset length along the 1-axis of the frame object at the J-End of the frame object. [L]

RZ

Type:Â SystemDouble

The rigid zone factor. This is the fraction of the end offset length assumed to be rigid for bending and shear deformations.

ItemType (Optional)

Type:Â ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the assignment is made to the frame object specified by the Name item.

If this item is Group, the assignment is made to all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the assignment is made to all selected frame objects and the Name item is ignored.

Return Value

Type:Â Int32

Returns zero if the offsets are successfully assigned, otherwise it returns a nonzero

value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'assign offsets
    ret = SapModel.FrameObj.SetEndLengthOffset("15", False, 12, 12, 0.5)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST2E9C2A66_0") Method

Adds or removes frame objects from a specified group.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetGroupAssign(
    string Name,
    string GroupName,
    bool Remove = false,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetGroupAssign (
    Name As String,
    GroupName As String,
    Remove As Boolean = false,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim GroupName As String
Dim Remove As Boolean
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetGroupAssign(Name,
    GroupName, Remove, ItemType)
```

```
int SetGroupAssign(
    String^ Name,
    String^ GroupName,
    bool Remove = false,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetGroupAssign :
    Name : string *
    GroupName : string *
    ?Remove : bool *
    ?ItemType : eItemType
(* Defaults:
    let _Remove = defaultArg Remove false
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

cFrameObjid="LST2E9C2A66_0"AddLanguageSpecificTextSet("LST2E9C2A66_0?cpp=:2107.");SetG

Parameters

Name

Type:Â SystemString

The name of an existing point object or group depending on the value of the ItemType item.

GroupName

Type:Â SystemString

The name of an existing group to which the assignment is made.

Remove (Optional)

Type:Â SystemBoolean

ItemType (Optional)

Type:Â ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the frame object specified by the Name item is added or removed from the group specified by the GroupName item.

If this item is Group, all frame objects in the group specified by the Name item are added or removed from the group specified by the GroupName item.

If this item is SelectedObjects, all selected frame objects are added or removed from the group specified by the GroupName item and the Name item is ignored.

Return ValueType:Â Int32

Returns zero if the group assignment is successful, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
```


Introduction

```
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'define new group
ret = SapModel.GroupDef.SetGroup("Group1")

'add frame objects to group
ret = SapModel.FrameObj.SetGroupAssign("8", "Group1")
ret = SapModel.FrameObj.SetGroupAssign("10", "Group1")

'select objects in group
ret = SapModel.SelectObj.Group("Group1")

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

See Also

Reference

cFrameObj Interface

ETABSV1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST8D1CF84C_0")

Method

Sets the GUID for the specified frame object. If the GUID is passed in as a blank string, the program automatically creates a GUID for the object.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetGUID(
    string Name,
    string GUID = ""
)
```

```
Function SetGUID (
    Name As String,
    GUID As String = ""
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.SetGUID(Name, GUID)
```

```
int SetGUID(
    String^ Name,
    String^ GUID = L""
)
```

```
abstract SetGUID :
    Name : string *
    ?GUID : string
(* Defaults:
    let _GUID = defaultArg GUID ""
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing frame object.

GUID (Optional)

Type: SystemString

The GUID (Global Unique ID) for the specified frame object.

cFrameObjspan id="LST8D1CF84C_0"AddLanguageSpecificTextSet("LST8D1CF84C_0?cpp=2110.");SetC

Return Value

Type: Int32

Returns zero if the frame object GUID is successfully set; otherwise it returns nonzero.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set program created GUID
    ret = SapModel.FrameObj.SetGUID("1")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ **See Also**

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST75B3566E_0?cpp=::20142");SetIn

Method

Assigns frame object insertion point data. The assignments include the cardinal point and end joint offsets.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetInsertionPoint(
    string Name,
    int CardinalPoint,
    bool Mirror2,
    bool StiffTransform,
    ref double[] Offset1,
    ref double[] Offset2,
    string CSys = "Local",
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetInsertionPoint (
    Name As String,
    CardinalPoint As Integer,
    Mirror2 As Boolean,
    StiffTransform As Boolean,
    ByRef Offset1 As Double(),
    ByRef Offset2 As Double(),
    CSys As String = "Local",
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim CardinalPoint As Integer
Dim Mirror2 As Boolean
Dim StiffTransform As Boolean
Dim Offset1 As Double()
Dim Offset2 As Double()
Dim CSys As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetInsertionPoint(Name,
    CardinalPoint, Mirror2, StiffTransform,
    Offset1, Offset2, CSys, ItemType)
```

```
int SetInsertionPoint(
    String^ Name,
    int CardinalPoint,
```

cFrameObjid="LST75B3566E_0"AddLanguageSpecificTextSet("LST75B3566E_0?cpp=::20142");SetIn

```

bool Mirror2,
bool StiffTransform,
array<double>^% Offset1,
array<double>^% Offset2,
String^ CSys = L"Local",
eItemType ItemType = eItemType::Objects
)

abstract SetInsertionPoint :
    Name : string *
    CardinalPoint : int *
    Mirror2 : bool *
    StiffTransform : bool *
    Offset1 : float[] byref *
    Offset2 : float[] byref *
    ?CSys : string *
    ?ItemType : eItemType
(* Defaults:
    let _CSys = defaultArg CSys "Local"
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of an existing frame object or group, depending on the value of the ItemType item.

CardinalPoint

Type:Â SystemInt32

This is a numeric value from 1 to 11 that specifies the cardinal point for the frame object. The cardinal point specifies the relative position of the frame section on the line representing the frame object.

1. bottom left
2. bottom center
3. bottom right
4. middle left
5. middle center
6. middle right
7. top left
8. top center
9. top right
10. centroid
11. shear center

Mirror2

Type:Â SystemBoolean

If this item is True, the frame object section is assumed to be mirrored (flipped) about its local 2-axis.

StiffTransform

Type:Â SystemBoolean

If this item is True, the frame object stiffness is transformed for cardinal point and joint offsets from the frame section centroid.

Offset1

Type: SystemDouble

This is an array of three joint offset distances, in the coordinate directions specified by CSys, at the I-End of the frame object. [L]

- ◊ Offset1(0) = Offset in the 1-axis or X-axis direction
- ◊ Offset1(1) = Offset in the 2-axis or Y-axis direction
- ◊ Offset1(2) = Offset in the 3-axis or Z-axis direction

Offset2

Type: SystemDouble

This is an array of three joint offset distances, in the coordinate directions specified by CSys, at the J-End of the frame object. [L]

- ◊ Offset2(0) = Offset in the 1-axis or X-axis direction
- ◊ Offset2(1) = Offset in the 2-axis or Y-axis direction
- ◊ Offset2(2) = Offset in the 3-axis or Z-axis direction

CSys (Optional)

Type: SystemString

This is Local or the name of a defined coordinate system. It is the coordinate system in which the Offset1 and Offset2 items are specified.

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

- ◊ Object = 0
- ◊ Group = 1
- ◊ SelectedObjects = 2

If this item is Objects, the assignment is made to the frame object specified by the Name item.

If this item is Group, the assignment is made to all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the assignment is made to all selected frame objects and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the insertion point data is successfully assigned, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As c0API
    Dim ret As Integer = -1
    Dim i As Integer
    Dim Offset1() As Double
    Dim Offset2() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

Introduction

```
'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'assign frame insertion point
ReDim Offset1(2)
ReDim Offset2(2)
For i=0 To 2
    Offset1(i)=10 + i
    Offset2(i)=20 + i
Next i
ret = SapModel.FrameObj.SetInsertionPoint("15", 7, False, True, Offset1, Offset2)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

[cFrameObj Interface](#)

[ETABSV1 Namespace](#)

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»

CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST612C99B9_0?cpp=::2116");SetLa

Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetLateralBracing(
    string Name,
    int MyType,
    int Loc,
    double MyDist1,
    double MyDist2,
    bool RelDist = true,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetLateralBracing (
    Name As String,
    MyType As Integer,
    Loc As Integer,
    MyDist1 As Double,
    MyDist2 As Double,
    RelDist As Boolean = true,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim MyType As Integer
Dim Loc As Integer
Dim MyDist1 As Double
Dim MyDist2 As Double
Dim RelDist As Boolean
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetLateralBracing(Name,
    MyType, Loc, MyDist1, MyDist2, RelDist,
    ItemType)
```

```
int SetLateralBracing(
    String^ Name,
    int MyType,
    int Loc,
    double MyDist1,
    double MyDist2,
    bool RelDist = true,
    eItemType ItemType = eItemType::Objects
```

cFrameObjAddLanguageSpecificTextSet("LST612C99B9_0?cpp=::2116");SetLa


```

)

abstract SetLateralBracing :
    Name : string *
    MyType : int *
    Loc : int *
    MyDist1 : float *
    MyDist2 : float *
    ?RelDist : bool *
    ?ItemType : eItemType
(* Defaults:
    let _RelDist = defaultArg RelDist true
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int

```

Parameters

Name
Type:Â [SystemString](#)

MyType
Type:Â [SystemInt32](#)

Loc
Type:Â [SystemInt32](#)

MyDist1
Type:Â [SystemDouble](#)

MyDist2
Type:Â [SystemDouble](#)

RelDist (Optional)
Type:Â [SystemBoolean](#)

ItemType (Optional)
Type:Â [ETABSv1eItemType](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cFrameObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST228DF8E Method

Assigns distributed loads to frame objects.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetLoadDistributed(
    string Name,
    string LoadPat,
    int MyType,
    int Dir,
    double Dist1,
    double Dist2,
    double Val1,
    double Val2,
    string CSys = "Global",
    bool RelDist = true,
    bool Replace = true,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetLoadDistributed (
    Name As String,
    LoadPat As String,
    MyType As Integer,
    Dir As Integer,
    Dist1 As Double,
    Dist2 As Double,
    Val1 As Double,
    Val2 As Double,
    CSys As String = "Global",
    RelDist As Boolean = true,
    Replace As Boolean = true,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim LoadPat As String
Dim MyType As Integer
Dim Dir As Integer
Dim Dist1 As Double
Dim Dist2 As Double
Dim Val1 As Double
Dim Val2 As Double
Dim CSys As String
Dim RelDist As Boolean
```

cFrameObjAddLanguageSpecificTextSet("LST228DF8EF_0?cpp=:2118");SetL

Introduction

```
Dim Replace As Boolean
Dim ItemType As eItemType
Dim returnValue As Integer

returnValue = instance.SetLoadDistributed(Name,
LoadPat, MyType, Dir, Dist1, Dist2,
Val1, Val2, CSys, RelDist, Replace,
ItemType)

int SetLoadDistributed(
String^ Name,
String^ LoadPat,
int MyType,
int Dir,
double Dist1,
double Dist2,
double Val1,
double Val2,
String^ CSys = L"Global",
bool RelDist = true,
bool Replace = true,
eItemType ItemType = eItemType::Objects
)

abstract SetLoadDistributed :
    Name : string *
    LoadPat : string *
    MyType : int *
    Dir : int *
    Dist1 : float *
    Dist2 : float *
    Val1 : float *
    Val2 : float *
    ?CSys : string *
    ?RelDist : bool *
    ?Replace : bool *
    ?ItemType : eItemType
(* Defaults:
    let _CSys = defaultArg CSys "Global"
    let _RelDist = defaultArg RelDist true
    let _Replace = defaultArg Replace true
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame object or group, depending on the value of the ItemType item.

LoadPat

Type:Â SystemString

The name of a defined load pattern.

MyType

Type:Â SystemInt32

This is 1 or 2, indicating the type of distributed load.

1. Force per unit length

2. Moment per unit length

Dir

Type: SystemInt32

This is an integer between 1 and 11, indicating the direction of the load.

1. Local 1 axis (only applies when CSys is Local)
2. Local 2 axis (only applies when CSys is Local)
3. Local 3 axis (only applies when CSys is Local)
4. X direction (does not apply when CSys is Local)
5. Y direction (does not apply when CSys is Local)
6. Z direction (does not apply when CSys is Local)
7. Projected X direction (does not apply when CSys is Local)
8. Projected Y direction (does not apply when CSys is Local)
9. Projected Z direction (does not apply when CSys is Local)
10. Gravity direction (only applies when CSys is Global)
11. Projected Gravity direction (only applies when CSys is Global)

The positive gravity direction (see Dir = 10 and 11) is in the negative Global Z direction.

Dist1

Type: SystemDouble

This is the distance from the I-End of the frame object to the start of the distributed load. This may be a relative distance ($0 \leq \text{Dist1} \leq 1$) or an actual distance, depending on the value of the RelDist item. [L] when RelDist is False

Dist2

Type: SystemDouble

This is the distance from the I-End of the frame object to the end of the distributed load. This may be a relative distance ($0 \leq \text{Dist2} \leq 1$) or an actual distance, depending on the value of the RelDist item. [L] when RelDist is False

Val1

Type: SystemDouble

This is the load value at the start of the distributed load. [F/L] when MyType is 1 and [FL/L] when MyType is 2

Val2

Type: SystemDouble

This is the load value at the end of the distributed load. [F/L] when MyType is 1 and [FL/L] when MyType is 2

CSys (Optional)

Type: SystemString

This is Local or the name of a defined coordinate system. It is the coordinate system in which the loads are specified.

RelDist (Optional)

Type: SystemBoolean

If this item is True, the specified Dist item is a relative distance, otherwise it is an actual distance.

Replace (Optional)

Type: SystemBoolean

If this item is True, all previous loads, if any, assigned to the specified frame object(s), in the specified load pattern, are deleted before making the new assignment.

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

- ◇ Object = 0
- ◇ Group = 1
- ◇ SelectedObjects = 2

If this item is Objects, the assignment is made to the frame object specified by the Name item.

If this item is Group, the assignment is made to all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the assignment is made to all selected frame objects and the Name item is ignored.

Return Value

Type: $\hat{A}$ Int32

Returns zero if the loads are successfully assigned, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'assign frame distributed loads
    ret = SapModel.FrameObj.SetLoadDistributed("15", "DEAD", 1, 10, 0, 1, 0.08, 0.08)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LSTE372B4C Method

Assigns point loads to frame objects.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetLoadPoint(
    string Name,
    string LoadPat,
    int MyType,
    int Dir,
    double Dist,
    double Val,
    string CSys = "Global",
    bool RelDist = true,
    bool Replace = true,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetLoadPoint (
    Name As String,
    LoadPat As String,
    MyType As Integer,
    Dir As Integer,
    Dist As Double,
    Val As Double,
    CSys As String = "Global",
    RelDist As Boolean = true,
    Replace As Boolean = true,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim LoadPat As String
Dim MyType As Integer
Dim Dir As Integer
Dim Dist As Double
Dim Val As Double
Dim CSys As String
Dim RelDist As Boolean
Dim Replace As Boolean
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetLoadPoint(Name,
    LoadPat, MyType, Dir, Dist, Val, CSys,
```

cFrameObjspan id="LSTE372B4CF_0"AddLanguageSpecificTextSet("LSTE372B4CF_0?cpp=:2103.");SetL

RelDist, Replace, ItemType)

```
int SetLoadPoint(
String^ Name,
String^ LoadPat,
int MyType,
int Dir,
double Dist,
double Val,
String^ CSys = L"Global",
bool RelDist = true,
bool Replace = true,
eItemType ItemType = eItemType::Objects
)
```

```
abstract SetLoadPoint :
    Name : string *
    LoadPat : string *
    MyType : int *
    Dir : int *
    Dist : float *
    Val : float *
    ?CSys : string *
    ?RelDist : bool *
    ?Replace : bool *
    ?ItemType : eItemType
(* Defaults:
    let _CSys = defaultArg CSys "Global"
    let _RelDist = defaultArg RelDist true
    let _Replace = defaultArg Replace true
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame object or group, depending on the value of the ItemType item.

LoadPat

Type:Â SystemString

The name of a defined load pattern.

MyType

Type:Â SystemInt32

This is 1 or 2, indicating the type of point load.

1. Force
2. Moment

Dir

Type:Â SystemInt32

This is an integer between 1 and 11, indicating the direction of the load.

1. Local 1 axis (only applies when CSys is Local)
2. Local 2 axis (only applies when CSys is Local)
3. Local 3 axis (only applies when CSys is Local)
4. X direction (does not apply when CSys is Local)
5. Y direction (does not apply when CSys is Local)

6. Z direction (does not apply when CSys is Local)
7. Projected X direction (does not apply when CSys is Local)
8. Projected Y direction (does not apply when CSys is Local)
9. Projected Z direction (does not apply when CSys is Local)
10. Gravity direction (only applies when CSys is Global)
11. Projected Gravity direction (only applies when CSys is Global)

The positive gravity direction (see Dir = 10 and 11) is in the negative Global Z direction.

Dist

Type: SystemDouble

Val

Type: SystemDouble

CSys (Optional)

Type: SystemString

This is Local or the name of a defined coordinate system. It is the coordinate system in which the loads are specified.

RelDist (Optional)

Type: SystemBoolean

If this item is True, the specified dist item is a relative distance, otherwise it is an actual distance.

Replace (Optional)

Type: SystemBoolean

If this item is True, all previous loads, if any, assigned to the specified frame object(s), in the specified load pattern, are deleted before making the new assignment.

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

- ◊ Object = 0
- ◊ Group = 1
- ◊ SelectedObjects = 2

If this item is Objects, the assignment is made to the frame object specified by the Name item.

If this item is Group, the assignment is made to all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the assignment is made to all selected frame objects and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the loads are successfully assigned, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
```

```
    Dim SapModel As cSapModel
```

Introduction

```
Dim EtabsObject As cOAPI
Dim ret As Integer = -1

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'assign frame point loads
ret = SapModel.FrameObj.SetLoadPoint("15", "DEAD", 1, 10, .5, 20)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

📌 See Also

Reference

[cFrameObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST19934A57_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetLoadTemperature(
    string Name,
    string LoadPat,
    int MyType,
    double Val,
    string PatternName = "",
    bool Replace = true,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetLoadTemperature (
    Name As String,
    LoadPat As String,
    MyType As Integer,
    Val As Double,
    PatternName As String = "",
    Replace As Boolean = true,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim LoadPat As String
Dim MyType As Integer
Dim Val As Double
Dim PatternName As String
Dim Replace As Boolean
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetLoadTemperature(Name,
    LoadPat, MyType, Val, PatternName,
    Replace, ItemType)
```

```
int SetLoadTemperature(
    String^ Name,
    String^ LoadPat,
    int MyType,
    double Val,
    String^ PatternName = L"",
    bool Replace = true,
    eItemType ItemType = eItemType::Objects
)
```

cFrameObjAddLanguageSpecificTextSet("LST19934A57_0?cpp=::|2127);SetLo

```

)

abstract SetLoadTemperature :
    Name : string *
    LoadPat : string *
    MyType : int *
    Val : float *
    ?PatternName : string *
    ?Replace : bool *
    ?ItemType : eItemType
(* Defaults:
    let _PatternName = defaultArg PatternName ""
    let _Replace = defaultArg Replace true
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int

```

Parameters

Name
Type:Â [SystemString](#)

LoadPat
Type:Â [SystemString](#)

MyType
Type:Â [SystemInt32](#)

Val
Type:Â [SystemDouble](#)

PatternName (Optional)
Type:Â [SystemString](#)

Replace (Optional)
Type:Â [SystemBoolean](#)

ItemType (Optional)
Type:Â [ETABSv1eItemType](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cFrameObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST7FBE6435_0?cpp=::2129");SetLocal

Method

Assigns a local axis angle to frame objects.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetLocalAxes(
    string Name,
    double Ang,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetLocalAxes (
    Name As String,
    Ang As Double,
    Optional ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim Ang As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetLocalAxes(Name,
    Ang, ItemType)
```

```
int SetLocalAxes(
    String^ Name,
    double Ang,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetLocalAxes :
    Name : string *
    Ang : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing frame object or group, depending on the value of the ItemType item.

Ang

Type: SystemDouble

This is the angle that the local 2 and 3 axes are rotated about the positive local 1 axis, from the default orientation or, if the Advanced item is True, from the orientation determined by the plane reference vector. The rotation for a positive angle appears counter clockwise when the local +1 axis is pointing toward you.
[deg]

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

- ◊ Object = 0
- ◊ Group = 1
- ◊ SelectedObjects = 2

If this item is Objects, the local axes assignment is made to the frame object specified by the Name item.

If this item is Group, the local axes assignment is made to all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the local axes assignment is made to all selected frame objects and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the local axis angle is successfully assigned, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)
```

Introduction

```
'assign frame local axis angle
  ret = SapModel.FrameObj.SetLocalAxes("3", 30)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

🔍 See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST1185A2C6_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetMass(
    string Name,
    double MassOverL,
    bool Replace = false,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetMass (
    Name As String,
    MassOverL As Double,
    Replace As Boolean = false,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim MassOverL As Double
Dim Replace As Boolean
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetMass(Name, MassOverL,
    Replace, ItemType)
```

```
int SetMass(
    String^ Name,
    double MassOverL,
    bool Replace = false,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetMass :
    Name : string *
    MassOverL : float *
    ?Replace : bool *
    ?ItemType : eItemType
(* Defaults:
    let _Replace = defaultArg Replace false
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

cFrameObjAddLanguageSpecificTextSet("LST1185A2C6_0?cpp=::2162");SetM

Parameters

Name

Type: SystemString

MassOverL

Type: SystemDouble

Replace (Optional)

Type: SystemBoolean

ItemType (Optional)

Type: ETABSv1eItemType

Return Value

Type: Int32

▲ See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST4C689F40_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetMaterialOverwrite(
    string Name,
    string PropName,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetMaterialOverwrite (
    Name As String,
    PropName As String,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim PropName As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetMaterialOverwrite(Name,
    PropName, ItemType)
```

```
int SetMaterialOverwrite(
    String^ Name,
    String^ PropName,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetMaterialOverwrite :
    Name : string *
    PropName : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: [SystemString](#)

PropName

cFrameObjspan id="LST4C689F40_0"AddLanguageSpecificTextSet("LST4C689F40_0?cpp=::1134");SetMaterialOverwrite(Name, PropName, eItemType.Objects);

Type: SystemString
ItemType (Optional)
Type: ETABSv1eItemType

Return Value

Type: Int32

See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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»

CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST963E12A

Method

Sets the frame modifier assignment for frame objects. The default value for all modifiers is one

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetModifiers(
    string Name,
    ref double[] Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetModifiers (
    Name As String,
    ByRef Value As Double(),
    Optional ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim Value As Double()
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetModifiers(Name,
    Value, ItemType)
```

```
int SetModifiers(
    String^ Name,
    array<double>^% Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetModifiers :
    Name : string *
    Value : float[] byref *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters**Name**Type:Â SystemStringThe name of an existing frame object or group, depending on the value of the ItemType item.**Value**Type:Â SystemDouble

This is an array of eight unitless modifiers:

Value**Modifier**

Value(0) Cross sectional area modifier

Value(1) Shear area in local 2 direction modifier

Value(2) Shear area in local 3 direction modifier

Value(3) Torsional constant modifier

Value(4) Moment of inertia about local 2 axis modifier

Value(5) Moment of inertia about local 3 axis modifier

Value(6) Mass modifier

Value(7) Weight modifier

ItemType (Optional)Type:Â ETABSv1eItemTypeThis is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the assignment is made to the frame object specified by the Name item.If this item is Group, the assignment is made to all frame objects in the group specified by the Name item.If this item is SelectedObjects, the assignment is made to all selected frame objects and the Name item is ignored.**Return Value**Type:Â Int32

Returns zero if the modifier assignments are successfully assigned, otherwise it returns a nonzero value

▲ **Remarks**▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value() As Double
```

'create ETABS object

Dim myHelper as cHelper = New Helper

EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

Parameters

```
'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 4, 4, 24, 24)

'assign modifiers
ReDim Value(7)
For i = 0 To 7
    Value(i) = 1
Next i
Value(5) = 100
ret = SapModel.FrameObj.SetModifiers("3", Value, eItemType.Objects)

'get modifiers
Dim myValue As Double()
ret = SapModel.FrameObj.GetModifiers("3", myValue)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

[cFrameObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LSTB80C8CFD_0") Method

Assigns frame object output station data.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetOutputStations(
    string Name,
    int MyType,
    double MaxSegSize,
    int MinSections,
    bool NoOutPutAndDesignAtElementEnds = false,
    bool NoOutPutAndDesignAtPointLoads = false,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetOutputStations (
    Name As String,
    MyType As Integer,
    MaxSegSize As Double,
    MinSections As Integer,
    NoOutPutAndDesignAtElementEnds As Boolean = false,
    NoOutPutAndDesignAtPointLoads As Boolean = false,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim MyType As Integer
Dim MaxSegSize As Double
Dim MinSections As Integer
Dim NoOutPutAndDesignAtElementEnds As Boolean
Dim NoOutPutAndDesignAtPointLoads As Boolean
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetOutputStations(Name,
    MyType, MaxSegSize, MinSections,
    NoOutPutAndDesignAtElementEnds,
    NoOutPutAndDesignAtPointLoads,
    ItemType)
```

```
int SetOutputStations(
    String^ Name,
    int MyType,
    double MaxSegSize,
    int MinSections,
```

cFrameObjAddLanguageSpecificTextSet("LSTB80C8CFD_0?cpp=2109.");Set

Introduction

```
bool NoOutPutAndDesignAtElementEnds = false,
bool NoOutPutAndDesignAtPointLoads = false,
eItemType ItemType = eItemType::Objects
)

abstract SetOutputStations :
    Name : string *
    MyType : int *
    MaxSegSize : float *
    MinSections : int *
    ?NoOutPutAndDesignAtElementEnds : bool *
    ?NoOutPutAndDesignAtPointLoads : bool *
    ?ItemType : eItemType
(* Defaults:
    let _NoOutPutAndDesignAtElementEnds = defaultArg NoOutPutAndDesignAtElementEnds false
    let _NoOutPutAndDesignAtPointLoads = defaultArg NoOutPutAndDesignAtPointLoads false
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing frame object or group, depending on the value of the ItemType item.

MyType

Type: SystemInt32

This is 1 or 2, indicating how the output stations are specified.

1. maximum segment size, that is, maximum station spacing
2. minimum number of stations

MaxSegSize

Type: SystemDouble

The maximum segment size, that is, the maximum station spacing. This item applies only when MyType = 1. [L]

MinSections

Type: SystemInt32

The minimum number of stations. This item applies only when MyType = 2.

NoOutPutAndDesignAtElementEnds (Optional)

Type: SystemBoolean

If this item is True, no additional output stations are added at the ends of line elements when the frame object is internally meshed. In ETABS, this item will always be False

NoOutPutAndDesignAtPointLoads (Optional)

Type: SystemBoolean

If this item is True, no additional output stations are added at point load locations. In ETABS, this item will always be False

ItemType (Optional)

Type: ETABSV1eItemType

This is one of the following items in the eItemType enumeration:

- ◇ Object = 0
- ◇ Group = 1
- ◇ SelectedObjects = 2

Introduction

If this item is Objects, the assignment is made to the frame object specified by the Name item.

If this item is Group, the assignment is made to all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the assignment is made to all selected frame objects and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the data is successfully assigned, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)  
  
    'assign frame output station data  
    ret = SapModel.FrameObj.SetOutputStations("15", 1, 18, 0)  
  
    'close ETABS  
    EtabsObject.ApplicationExit(False)  
  
    'clean up variables  
    SapModel = Nothing  
    EtabsObject = Nothing  
  
End Sub
```

▲ See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST226F5BA_0") Method

Sets the pier label assignment of one or more frame objects

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetPier(
    string Name,
    string PierName,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetPier (
    Name As String,
    PierName As String,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim PierName As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetPier(Name, PierName,
    ItemType)
```

```
int SetPier(
    String^ Name,
    String^ PierName,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetPier :
    Name : string *
    PierName : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing frame object or group, depending on the value of the ItemType item.

PierName

Type: SystemString

The name of the pier assignment, or "None", to unset any assignment

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the assignment is made to the frame object specified by the Name item.

If this item is Group, the assignment is made to all frame objects in the group specified by the Name item.

If this item is SelectedObjects, assignment is made to all selected frame objects, and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the assignment is successful; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim PierName as String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 4, 4, 24, 24)

    'define new pier label
    ret = SapModel.PierLabel.SetPier("MyPier")

    'set assignment
    ret = SapModel.FrameObj.SetPier("3", "MyPier")
```

Introduction

```
'get assignment
    ret = SapModel.FrameObj.GetPier("3", PierName)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST10F9BDCC_0") Method

Makes end release and partial fixity assignments to frame objects.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetReleases(
    string Name,
    ref bool[] II,
    ref bool[] JJ,
    ref double[] StartValue,
    ref double[] EndValue,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetReleases (
    Name As String,
    ByRef II As Boolean(),
    ByRef JJ As Boolean(),
    ByRef StartValue As Double(),
    ByRef EndValue As Double(),
    ByRef ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim II As Boolean()
Dim JJ As Boolean()
Dim StartValue As Double()
Dim EndValue As Double()
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetReleases(Name,
    II, JJ, StartValue, EndValue, ItemType)
```

```
int SetReleases(
    String^ Name,
    array<bool>^% II,
    array<bool>^% JJ,
    array<double>^% StartValue,
    array<double>^% EndValue,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetReleases :
    Name : string *
```

cFrameObjAddLanguageSpecificTextSet("LST10F9BDCC_0?cpp=2146-");Set

```

II : bool[] byref *
JJ : bool[] byref *
StartValue : float[] byref *
EndValue : float[] byref *
?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of an existing frame object or group, depending on the value of the ItemType item.

II

Type:Â SystemBoolean

JJ

Type:Â SystemBoolean

These is an array of six booleans indicating the J-End releases for the frame object.

- ◊ ii(0) and jj(0) = U1 release
- ◊ ii(1) and jj(1) = U2 release
- ◊ ii(2) and jj(2) = U3 release
- ◊ ii(3) and jj(3) = R1 release
- ◊ ii(4) and jj(4) = R2 release
- ◊ ii(5) and jj(5) = R3 release

StartValue

Type:Â SystemDouble

These is an array of six values indicating the I-End partial fixity springs for the frame object.

- ◊ StartValue(0) and EndValue(0) = U1 partial fixity [F/L]
- ◊ StartValue(1) and EndValue(1) = U2 partial fixity [F/L]
- ◊ StartValue(2) and EndValue(2) = U3 partial fixity [F/L]
- ◊ StartValue(3) and EndValue(3) = R1 partial fixity [FL/rad]
- ◊ StartValue(4) and EndValue(4) = R2 partial fixity [FL/rad]
- ◊ StartValue(5) and EndValue(5) = R3 partial fixity [FL/rad]

EndValue

Type:Â SystemDouble

These is an array of six values indicating the J-End partial fixity springs for the frame object.

- ◊ StartValue(0) and EndValue(0) = U1 partial fixity [F/L]
- ◊ StartValue(1) and EndValue(1) = U2 partial fixity [F/L]
- ◊ StartValue(2) and EndValue(2) = U3 partial fixity [F/L]
- ◊ StartValue(3) and EndValue(3) = R1 partial fixity [FL/rad]
- ◊ StartValue(4) and EndValue(4) = R2 partial fixity [FL/rad]
- ◊ StartValue(5) and EndValue(5) = R3 partial fixity [FL/rad]

ItemType (Optional)

Type:Â ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

- ◊ Object = 0

◇ Group = 1

◇ SelectedObjects = 2

If this item is Objects, the assignment is made to the frame object specified by the Name item.

If this item is Group, the assignment is made to all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the assignment is made to all selected frame objects and the Name item is ignored.

Return Value

Type: $\hat{A}$ Int32

Returns zero if the assignments are successfully retrieved, otherwise it returns a nonzero value.

Remarks

Partial fixity assignments are made to degrees of freedom that have been released only.

Some release assignments would cause instability in the model. An error is returned if this type of assignment is made. Unstable release assignments include the following:

- U1 released at both ends
- U2 released at both ends
- U3 released at both ends
- R1 released at both ends
- R2 released at both ends and U3 at either end
- R3 released at both ends and U2 at either end

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim ii() As Boolean
    Dim jj() As Boolean
    Dim StartValue() As Double
    Dim EndValue() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()
```


Introduction

```
'create steel deck template model
  ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'assign end releases
  ReDim ii(5)
  ReDim jj(5)
  ReDim StartValue(5)
  ReDim EndValue(5)
  ii(5) = True
  jj(5) = True
  ret = SapModel.FrameObj.SetReleases("13", ii, jj, StartValue, EndValue)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

[cFrameObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST2EAF6E3D_0?cpp=2150.");SetS

Method

Assigns a frame section property to a frame object.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetSection(
    string Name,
    string PropName,
    eItemType ItemType = eItemType.Objects,
    double SVarRelStartLoc = 0,
    double SVarTotalLength = 0
)
```

```
Function SetSection (
    Name As String,
    PropName As String,
    ?ItemType As eItemType = eItemType.Objects,
    ?SVarRelStartLoc As Double = 0,
    ?SVarTotalLength As Double = 0
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim PropName As String
Dim ItemType As eItemType
Dim SVarRelStartLoc As Double
Dim SVarTotalLength As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetSection(Name,
    PropName, ItemType, SVarRelStartLoc,
    SVarTotalLength)
```

```
int SetSection(
    String^ Name,
    String^ PropName,
    eItemType ItemType = eItemType::Objects,
    double SVarRelStartLoc = 0,
    double SVarTotalLength = 0
)
```

```
abstract SetSection :
    Name : string *
    PropName : string *
    ?ItemType : eItemType *
    ?SVarRelStartLoc : float *
```

cFrameObjid="LST2EAF6E3D_0"AddLanguageSpecificTextSet("LST2EAF6E3D_0?cpp=2150.");SetS

```

        ?SVarTotalLength : float
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
    let _SVarRelStartLoc = defaultArg SVarRelStartLoc 0
    let _SVarTotalLength = defaultArg SVarTotalLength 0
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of an existing frame object or group, depending on the value of the ItemType item.

PropName

Type:Â SystemString

This is None or the name of a frame section property to be assigned to the specified frame object(s).

ItemType (Optional)

Type:Â ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the assignment is made to the frame object specified by the Name item.

If this item is Group, the assignment is made to all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the assignment is made to all selected frame objects and the Name item is ignored.

SVarRelStartLoc (Optional)

Type:Â SystemDouble

SVarTotalLength (Optional)

Type:Â SystemDouble

Return Value

Type:Â Int32

Returns zero if the frame section property data is successfully assigned, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

```

'create ETABS object

```

    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

```

Introduction

```
'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 4, 4, 24, 24)

'create new concrete frame section property
ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

'set frame section property
ret = SapModel.FrameObj.SetSection("8", "R1")

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
```

End Sub

▲ See Also

Reference

[cFrameObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST9593BE44_0") Method

Sets the selected status for a frame object.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetSelected(
    string Name,
    bool Selected,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetSelected (
    Name As String,
    Selected As Boolean,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim Selected As Boolean
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetSelected(Name,
    Selected, ItemType)
```

```
int SetSelected(
    String^ Name,
    bool Selected,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetSelected :
    Name : string *
    Selected : bool *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing frame object or group depending on the value of the ItemType item.

Selected

Type: SystemBoolean

This item is True if the specified frame object is selected, otherwise it is False.

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the selected status is set for the frame object specified by the Name item.

If this item is Group, the selected status is set for all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the selected status is set for all selected frame objects and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the selected status is successfully set, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set point selected
    ret = SapModel.FrameObj.SetSelected("8", True)

    'close ETABS
    EtabsObject.ApplicationExit(False)
```

```
'clean up variables  
  SapModel = Nothing  
  EtabsObject = Nothing  
End Sub
```

📌 See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LST32B63482_0") Method

Sets the Spandrel label assignment of one or more frame objects

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetSpandrel(
    string Name,
    string SpandrelName,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetSpandrel (
    Name As String,
    SpandrelName As String,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim SpandrelName As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetSpandrel(Name,
    SpandrelName, ItemType)
```

```
int SetSpandrel(
    String^ Name,
    String^ SpandrelName,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetSpandrel :
    Name : string *
    SpandrelName : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing frame object or group, depending on the value of the ItemType item.

SpandrelName

Type: SystemString

The name of the Spandrel assignment, or "None", to unset any assignment

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the assignment is made to the frame object specified by the Name item.

If this item is Group, the assignment is made to all frame objects in the group specified by the Name item.

If this item is SelectedObjects, assignment is made to all selected frame objects, and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the assignment is successful; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim SpandrelName as String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 4, 4, 24, 24)

    'define new Spandrel label
    ret = SapModel.SpandrelLabel.SetSpandrel("MySpandrel")

    'set assignment
    ret = SapModel.FrameObj.SetSpandrel("3", "MySpandrel")
```

Parameters

Introduction

```
'get assignment
    ret = SapModel.FrameObj.GetSpandrel("3", SpandrelName)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LSTE04E73B9_0") Method

Assigns an existing named line spring property to frame objects

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetSpringAssignment(
    string Name,
    string SpringProp,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetSpringAssignment (
    Name As String,
    SpringProp As String,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim SpringProp As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetSpringAssignment(Name,
    SpringProp, ItemType)
```

```
int SetSpringAssignment(
    String^ Name,
    String^ SpringProp,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetSpringAssignment :
    Name : string *
    SpringProp : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing frame object or group depending on the value of the ItemType item.

SpringProp

Type: SystemString

The name of an existing line spring property

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the property assignment is made to the frame object specified by the Name item.

If this item is Group, the property assignment is made to all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the property assignment is made to all selected frame objects and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the property is successfully assigned, otherwise it returns a nonzero value

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim SpringProp As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new spring property
    ret = SapModel.PropLineSpring.SetLineSpringProp("mySpringProp1", 0, 10, 20, 0, 0, 1)

    'assign property
```

Parameters

Introduction

```
ret = SapModel.FrameObj.SetSpringAssignment("1", "mySpringProp1")

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

🔍 See Also

Reference

cFrameObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFrameObjAddLanguageSpecificTextSet("LSTB982EC93_0")

Method

Makes tension/compression force limit assignments to frame objects.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetTCLimits(
    string Name,
    bool LimitCompressionExists,
    double LimitCompression,
    bool LimitTensionExists,
    double LimitTension,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetTCLimits (
    Name As String,
    LimitCompressionExists As Boolean,
    LimitCompression As Double,
    LimitTensionExists As Boolean,
    LimitTension As Double,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cFrameObj
Dim Name As String
Dim LimitCompressionExists As Boolean
Dim LimitCompression As Double
Dim LimitTensionExists As Boolean
Dim LimitTension As Double
Dim ItemType As eItemType
Dim returnValue As Integer

returnValue = instance.SetTCLimits(Name,
    LimitCompressionExists, LimitCompression,
    LimitTensionExists, LimitTension,
    ItemType)
```

```
int SetTCLimits(
    String^ Name,
    bool LimitCompressionExists,
    double LimitCompression,
    bool LimitTensionExists,
    double LimitTension,
    eItemType ItemType = eItemType::Objects
)
```

cFrameObjAddLanguageSpecificTextSet("LSTB982EC93_0?cpp=:2162");SetT

```

abstract SetTCLimits :
  Name : string *
  LimitCompressionExists : bool *
  LimitCompression : float *
  LimitTensionExists : bool *
  LimitTension : float *
  ?ItemType : eItemType
(* Defaults:
   let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of an existing frame object or group depending on the value of the ItemType item.

LimitCompressionExists

Type:Â SystemBoolean

This item is True if a compression force limit exists for the frame object.

LimitCompression

Type:Â SystemDouble

The compression force limit for the frame object. [F]

LimitTensionExists

Type:Â SystemBoolean

This item is True if a tension force limit exists for the frame object.

LimitTension

Type:Â SystemDouble

The tension force limit for the frame object. [F]

ItemType (Optional)

Type:Â ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the assignment is made to the frame object specified by the Name item.

If this item is Group, the assignment is made to all frame objects in the group specified by the Name item.

If this item is SelectedObjects, the assignment is made to all selected frame objects and the Name item is ignored.

Return Value

Type:Â Int32

Returns zero if the assignments are successfully applied, otherwise it returns a nonzero value.

▲ Remarks

Note that the tension and compression limits are only used in nonlinear analyses.

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'assign tension/compression limits
    ret = SapModel.FrameObj.SetTCLimits("1", False, 0, True, 100)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

 See Also
ReferencecFrameObj InterfaceETABSv1 Namespace

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CSI API ETABS v1

cFunction Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cFunction
```

```
Public Interface cFunction
```

```
Dim instance As cFunction
```

```
public interface class cFunction
```

```
type cFunction = interface end
```

The cFunction type exposes the following members.

Properties

Name	Description
------	-------------

 FuncRS	
--	--

 FuncTH	
--	--

[Top](#)

Methods

Name	Description
------	-------------

 ChangeName	
--	--

 ConvertToUser	
---	--

 Count	
---	--

 Delete	
--	--

 GetNameList	
---	--

 GetTypeOAPI	Retrieves the function type for the specified function
---	--

 GetValues	
---	--

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[ETABSv1 Namespace](#)

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Introduction

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CSI API ETABS v1

cFunction Properties

The cFunction type exposes the following members.

Properties

	Name	Description
--	------	-------------

	<u>FuncRS</u>	
---	---------------	--

	<u>FuncTH</u>	
---	---------------	--

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 See Also

Reference

cFunction Interface

ETABSV1 Namespace

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CSI API ETABS v1

cFunctionAddLanguageSpecificTextSet("LSTD214E044_0") Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cFunctionRS FuncRS { get; }
```

```
ReadOnly Property FuncRS As cFunctionRS
Get
```

```
Dim instance As cFunction
Dim value As cFunctionRS
```

```
value = instance.FuncRS
```

```
property cFunctionRS^ FuncRS {
cFunctionRS^ get ();
}
```

```
abstract FuncRS : cFunctionRS with get
```

Property Value

Type: [cFunctionRS](#)

▲ See Also

Reference

[cFunction Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cFunctionAddLanguageSpecificTextSet("LSTD1FFE042_0") Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cFunctionTH FuncTH { get; }
```

```
ReadOnly Property FuncTH As cFunctionTH
Get
```

```
Dim instance As cFunction
Dim value As cFunctionTH
```

```
value = instance.FuncTH
```

```
property cFunctionTH^ FuncTH {
cFunctionTH^ get ();
}
```

```
abstract FuncTH : cFunctionTH with get
```

Property Value

Type: [cFunctionTH](#)

▲ See Also

Reference

[cFunction Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cFunction Methods

The [cFunction](#) type exposes the following members.

Methods

⌕	Name	Description
⇒	ChangeName	
⇒	ConvertToUser	
⇒	Count	
⇒	Delete	
⇒	GetNameList	
⇒	GetTypeOAPI	Retrieves the function type for the specified function
⇒	GetValues	

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⌕ See Also

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CSI API ETABS v1

cFunctionAddLanguageSpecificTextSet("LSTC9BD3DC0_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int ChangeName(
    string Name,
    string NewName
)
```

```
Function ChangeName (
    Name As String,
    NewName As String
) As Integer
```

```
Dim instance As cFunction
Dim Name As String
Dim NewName As String
Dim returnValue As Integer
```

```
returnValue = instance.ChangeName(Name,
    NewName)
```

```
int ChangeName(
    String^ Name,
    String^ NewName
)
```

```
abstract ChangeName :
    Name : string *
    NewName : string -> int
```

Parameters

Name

Type: [SystemString](#)

NewName

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cFunction Interface

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CSI API ETABS v1

cFunctionAddLanguageSpecificTextSet("LST9263C684_0?cpp=::|n2473;Conver Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int ConvertToUser(
    string Name
)
```

```
Function ConvertToUser (
    Name As String
) As Integer
```

```
Dim instance As cFunction
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.ConvertToUser(Name)
```

```
int ConvertToUser(
    String^ Name
)
```

```
abstract ConvertToUser :
    Name : string -> int
```

Parameters

Name

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cFunction Interface](#)

[ETABSv1 Namespace](#)

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Introduction

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CSI API ETABS v1

cFunctionAddLanguageSpecificTextSet("LSTD3FF7219_0?cpp=::|n2175;Count

Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int Count(
    int FuncType = 0
)
```

```
Function Count (
    Optional FuncType As Integer = 0
) As Integer
```

```
Dim instance As cFunction
Dim FuncType As Integer
Dim returnValue As Integer
```

```
returnValue = instance.Count(FuncType)
```

```
int Count(
    int FuncType = 0
)
```

```
abstract Count :
    ?FuncType : int
(* Defaults:
    let _FuncType = defaultArg FuncType 0
*)
-> int
```

Parameters

FuncType (Optional)
Type: [SystemInt32](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cFunction Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cFunctionAddLanguageSpecificTextSet("LSTD573F0BC0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int Delete(
    string Name
)
```

```
Function Delete (
    Name As String
) As Integer
```

```
Dim instance As cFunction
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.Delete(Name)
```

```
int Delete(
    String^ Name
)
```

```
abstract Delete :
    Name : string -> int
```

Parameters

Name

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cFunction Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cFunctionAddLanguageSpecificTextSet("LSTFC36A4C1_0?cpp=::|0179");GetN

Method

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNameList(
    ref int NumberNames,
    ref string[] MyName,
    int FuncType = 0
)
```

```
Function GetNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String(),
    Optional FuncType As Integer = 0
) As Integer
```

```
Dim instance As cFunction
Dim NumberNames As Integer
Dim MyName As String()
Dim FuncType As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetNameList(NumberNames,
    MyName, FuncType)
```

```
int GetNameList(
    int% NumberNames,
    array<String^>^% MyName,
    int FuncType = 0
)
```

```
abstract GetNameList :
    NumberNames : int byref *
    MyName : string[] byref *
    ?FuncType : int
(* Defaults:
    let _FuncType = defaultArg FuncType 0
*)
-> int
```

Parameters

NumberNames

Type: [System.Int32](#)

MyName

cFunctionsAddLanguageSpecificTextSet("LSTFC36A4C1_0?cpp=::|0179");GetN

Type:Â SystemString
FuncType (Optional)
Type:Â SystemInt32

Return Value

Type:Â Int32

▀ See Also

Reference

cFunction Interface

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CSI API ETABS v1

cFunctionAddLanguageSpecificTextSet("LST8857D8D9_0?cpp=::|r2181");GetTy

Method

Retrieves the function type for the specified function

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetType0API(
    string Name,
    ref int FuncType,
    ref int AddType
)
```

```
Function GetType0API (
    Name As String,
    ByRef FuncType As Integer,
    ByRef AddType As Integer
) As Integer
```

```
Dim instance As cFunction
Dim Name As String
Dim FuncType As Integer
Dim AddType As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetType0API(Name,
    FuncType, AddType)
```

```
int GetType0API(
    String^ Name,
    int% FuncType,
    int% AddType
)
```

```
abstract GetType0API :
    Name : string *
    FuncType : int byref *
    AddType : int byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing function

FuncType

Type: SystemInt32

This is one of the following numbers, indicating the type of function

1. Response Spectrum
2. Time History
3. Power spectral density
4. Steady state
5. Heat Transfer

AddType

Type: SystemInt32

The is one of the following items, indicating the function subtype

Response Spectrum Functions

- ◇ 0 = From file
- ◇ 1 = User
- ◇ 2 = UBC94
- ◇ 3 = UBC97
- ◇ 4 = BOCA96
- ◇ 5 = NBCC95
- ◇ 6 = ASCE7-02
- ◇ 7 = NEHRP97
- ◇ 8 = EUROCODE8
- ◇ 9 = NZS4203
- ◇ 10 = Chinese2010
- ◇ 11 = Italian3274
- ◇ 12 = IS1893:2002
- ◇ 13 = ASCE7-05
- ◇ 14 = NBCC2005
- ◇ 15 = EUROCODE8-2004
- ◇ 16 = AS1170-2007
- ◇ 17 = NZS1170-2004
- ◇ 18 = ASCE7-10
- ◇ 19 = NBCC2010
- ◇ 20 = Italian NTC2008
- ◇ 21 = TSC-2007
- ◇ 22 = Uniform
- ◇ 23 = SI 413
- ◇ 24 = Argentina INPRES-CIRSOC 103
- ◇ 25 = Chile Norma NCh433+DS61
- ◇ 26 = Chile Norma NCh2369-2003
- ◇ 27 = Colombia NSR-10
- ◇ 28 = Ecuador NEC-11 Capitulo 2
- ◇ 29 = Guatemala AGIES NSE 2-10
- ◇ 30 = Mexico NTC-2004
- ◇ 31 = Peru Norma E.030
- ◇ 32 = Dominican Republic R-001
- ◇ 33 = Venezuela COVENIN 1756-2:2001
- ◇ 34 = Korean KBC 2009
- ◇ 35 = Mexico CFE-93
- ◇ 36 = Peru NTE E.030 2014
- ◇ 37 = Mexico CFE-2008

Introduction

- ◇ 38 = Ecuador Norma NEC-SE-DS 2015
- ◇ 39 = Costa Rica Seismic Code 2010
- ◇ 40 = NBCC2015
- ◇ 41 = SP 14.13330.2014
- ◇ 42 = TCVN 9386:2012
- ◇ 43 = IS 1893:2016
- ◇ 44 = ASCE7-16
- ◇ 45 = Korean KBC 2016

Time History Functions

- ◇ 0 = From file
- ◇ 1 = User
- ◇ 2 = Sine
- ◇ 3 = Cosine
- ◇ 4 = Ramp
- ◇ 5 = Sawtooth
- ◇ 6 = Triangular
- ◇ 7 = User periodic
- ◇ 8 N/A
- ◇ 9 = Matched to Response Spectrum

Power Spectral Density Functions

- ◇ 0 = From file
- ◇ 1 = User

Steady State Functions

- ◇ 0 = From file
- ◇ 1 = User

Return Value

Type: Int32

Returns zero if the type is successfully retrieved; otherwise it returns nonzero

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
```

```
    'dimension variables
    Dim SapObject as c0API
    Dim SapModel As cSapModel
    Dim ret As Integer
    Dim FuncType As Integer
    Dim AddType As Integer

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
```

Return Value

Introduction

```
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'get function type
ret = SapModel.Func.GetType0API("UNIFRS", FuncType, AddType)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cFunction Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cFunctionAddLanguageSpecificTextSet("LST96D20CD1_0?cpp=::|20185");GetVa Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetValues(
    string Name,
    ref int NumberItems,
    ref double[] MyTime,
    ref double[] Value
)
```

```
Function GetValues (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef MyTime As Double(),
    ByRef Value As Double()
) As Integer
```

```
Dim instance As cFunction
Dim Name As String
Dim NumberItems As Integer
Dim MyTime As Double()
Dim Value As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetValues(Name,
    NumberItems, MyTime, Value)
```

```
int GetValues(
    String^ Name,
    int% NumberItems,
    array<double>^% MyTime,
    array<double>^% Value
)
```

```
abstract GetValues :
    Name : string *
    NumberItems : int byref *
    MyTime : float[] byref *
    Value : float[] byref -> int
```

Parameters

Name

Type: [SystemString](#)

cFunctionsAddLanguageSpecificTextSet("LST96D20CD1_0?cpp=::|20185");GetVa

NumberItems

Type: SystemInt32

MyTime

Type: SystemDouble

Value

Type: SystemDouble

Return Value

Type: Int32

▲ See Also

Reference

cFunction Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFunctionRS Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cFunctionRS
```

```
Public Interface cFunctionRS
```

```
Dim instance As cFunctionRS
```

```
public interface class cFunctionRS
```

```
type cFunctionRS = interface end
```

The cFunctionRS type exposes the following members.

▲ Methods

▲	Name	Description
⇒	GetNTC2008	Retrieves the definition of a NTC2008 response spectrum function
⇒	GetNTC2018	Retrieves the definition of a NTC 2018 response spectrum function
⇒	SetNTC2008	Defines a NTC2008 response spectrum function
⇒	SetNTC2018	Defines a NTC 2018 response spectrum function

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cFunctionRS Methods

The [cFunctionRS](#) type exposes the following members.

Methods

⌕	Name	Description
⇒	GetNTC2008	Retrieves the definition of a NTC2008 response spectrum function
⇒	GetNTC2018	Retrieves the definition of a NTC 2018 response spectrum function
⇒	SetNTC2008	Defines a NTC2008 response spectrum function
⇒	SetNTC2018	Defines a NTC 2018 response spectrum function

[Top](#)

⌕ See Also

Reference

[cFunctionRS Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cFunctionRSAddLanguageSpecificTextSet("LST2E2F5F21_0?cpp=2189-");Get

Method

Retrieves the definition of a NTC2008 response spectrum function

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNTC2008(
    string Name,
    ref int ParamsOption,
    ref double Latitude,
    ref double Longitude,
    ref int Island,
    ref int LimitState,
    ref int UsageClass,
    ref double NomLife,
    ref double PeakAccel,
    ref double F0,
    ref double Tcs,
    ref int SpecType,
    ref int SoilType,
    ref int Topography,
    ref double hRatio,
    ref double Damping,
    ref double q
)
```

```
Function GetNTC2008 (
    Name As String,
    ByRef ParamsOption As Integer,
    ByRef Latitude As Double,
    ByRef Longitude As Double,
    ByRef Island As Integer,
    ByRef LimitState As Integer,
    ByRef UsageClass As Integer,
    ByRef NomLife As Double,
    ByRef PeakAccel As Double,
    ByRef F0 As Double,
    ByRef Tcs As Double,
    ByRef SpecType As Integer,
    ByRef SoilType As Integer,
    ByRef Topography As Integer,
    ByRef hRatio As Double,
    ByRef Damping As Double,
    ByRef q As Double
) As Integer
```

```
Dim instance As cFunctionRS
```

cFunctionRSspan id="LST2E2F5F21_0"AddLanguageSpecificTextSet("LST2E2F5F21_0?cpp=2189-");Get

Introduction

```
Dim Name As String
Dim ParamsOption As Integer
Dim Latitude As Double
Dim Longitude As Double
Dim Island As Integer
Dim LimitState As Integer
Dim UsageClass As Integer
Dim NomLife As Double
Dim PeakAccel As Double
Dim F0 As Double
Dim Tcs As Double
Dim SpecType As Integer
Dim SoilType As Integer
Dim Topography As Integer
Dim hRatio As Double
Dim Damping As Double
Dim q As Double
Dim returnValue As Integer

returnValue = instance.GetNTC2008(Name,
ParamsOption, Latitude, Longitude,
Island, LimitState, UsageClass, NomLife,
PeakAccel, F0, Tcs, SpecType, SoilType,
Topography, hRatio, Damping, q)
```

```
int GetNTC2008(
String^ Name,
int% ParamsOption,
double% Latitude,
double% Longitude,
int% Island,
int% LimitState,
int% UsageClass,
double% NomLife,
double% PeakAccel,
double% F0,
double% Tcs,
int% SpecType,
int% SoilType,
int% Topography,
double% hRatio,
double% Damping,
double% q
)
```

```
abstract GetNTC2008 :
    Name : string *
    ParamsOption : int byref *
    Latitude : float byref *
    Longitude : float byref *
    Island : int byref *
    LimitState : int byref *
    UsageClass : int byref *
    NomLife : float byref *
    PeakAccel : float byref *
    F0 : float byref *
    Tcs : float byref *
    SpecType : int byref *
    SoilType : int byref *
    Topography : int byref *
    hRatio : float byref *
    Damping : float byref *
```

q : float byref -> int

Parameters

Name

Type:Â SystemString

The name of a NTC2008 response spectrum function

ParamsOption

Type:Â SystemInt32

This is 1, 2, or 3, indicating the option for defining the parameters

1. by latitude and longitude
2. by island
3. user specified

Latitude

Type:Â SystemDouble

The latitude for which the seismic coefficients are obtained. This item is meaningful only when ParamsOption = 1

Longitude

Type:Â SystemDouble

The longitude for which the seismic coefficients are obtained. This item is meaningful only when ParamsOption = 1

Island

Type:Â SystemInt32

This is one of the following values. This item is used only when ParamsOption = 2

1. Alicudi
2. Arcipelago Toscano
3. Filcudi
4. Isole Egadi
5. Lampedusa
6. Linosa
7. Lipari
8. Palmarola
9. Panarea
10. Pantelleria
11. Ponza
12. Salina
13. Santo Stefano
14. Sardegna
15. Stromboli
16. Tremiti
17. Ustica
18. Ventotene
19. Vulcano
20. Zannone

LimitState

Type:Â SystemInt32

This is 1, 2, 3, or 4, indicating the limit state

1. SLO
2. SLD
3. SLV

4. SLC

UsageClass

Type: SystemInt32

This is 1, 2, 3, or 4, indicating the usage class

1. I
2. II
3. III
4. IV

NomLife

Type: SystemDouble

The nominal life to be considered

PeakAccel

Type: SystemDouble

The peak ground acceleration, ag/g

F0

Type: SystemDouble

The magnitude factor

Tcs

Type: SystemDouble

The reference period, Tc*[s]

SpecType

Type: SystemInt32

This is 1, 2, 3, or 4, indicating the type of spectrum to consider

1. Elastic horizontal
2. Elastic vertical
3. Design horizontal
4. Design vertical

SoilType

Type: SystemInt32

This is 1, 2, 3, 4, or 5, indicating the subsoil type

1. A
2. B
3. C
4. D
5. E

Topography

Type: SystemInt32

This is 1, 2, 3, or 4, indicating the topography type

1. T1
2. T2
3. T3
4. T4

hRatio

Type: SystemDouble

The ratio for the site altitude at the base of the hill to the height of the hill

Damping

Type: SystemDouble

The damping, in percent. This is only applicable for SpecType 1 and 2

q

Type: SystemDouble

The behavior correction factor. This is only applicable for SpecType 3 and 4

Return ValueType: Int32

returns zero if the function is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
```

```
    'dimension variables
    Dim SapObject as cOAPI
    Dim SapModel As cSapModel
    Dim ret As Integer
    Dim ParamsOption As Integer
    Dim Latitude As Double
    Dim Longitude As Double
    Dim Island As Integer
    Dim LimitState As Integer
    Dim UsageClass As Integer
    Dim NomLife As Double
    Dim PeakAccel As Double
    Dim F0 As Double
    Dim Tcs As Double
    Dim SpecType As Integer
    Dim SoilType As Integer
    Dim Topography As Integer
    Dim hRatio As Double
    Dim Damping As Double
    Dim q As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add NTC2008 RS function
    ret = SapModel.Func.FuncRS.SetNTC2008("RS-1", 1, 45.9, 12.6, 1, 3, 2, 50, 0.2, 2.4, 0.3, 3

    'get NTC2008 RS function
    ret = SapModel.Func.FuncRS.GetNTC2008("RS-1", ParamsOption, Latitude, Longitude, Island, L

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
```

End Sub

▲ See Also
Reference

cFunctionRS Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFunctionRSAddLanguageSpecificTextSet("LSTA620F9E9_0?cpp=2105.");GetN

Method

Retrieves the definition of a NTC 2018 response spectrum function

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNTC2018(
    string Name,
    ref int ParamsOption,
    ref double Latitude,
    ref double Longitude,
    ref int Island,
    ref int LimitState,
    ref int UsageClass,
    ref double NomLife,
    ref double PeakAccel,
    ref double F0,
    ref double Tcs,
    ref int SpecType,
    ref int SoilType,
    ref int Topography,
    ref double hRatio,
    ref double Damping,
    ref double q
)
```

```
Function GetNTC2018 (
    Name As String,
    ByRef ParamsOption As Integer,
    ByRef Latitude As Double,
    ByRef Longitude As Double,
    ByRef Island As Integer,
    ByRef LimitState As Integer,
    ByRef UsageClass As Integer,
    ByRef NomLife As Double,
    ByRef PeakAccel As Double,
    ByRef F0 As Double,
    ByRef Tcs As Double,
    ByRef SpecType As Integer,
    ByRef SoilType As Integer,
    ByRef Topography As Integer,
    ByRef hRatio As Double,
    ByRef Damping As Double,
    ByRef q As Double
) As Integer
```

```
Dim instance As cFunctionRS
```

cFunctionRSspan id="LSTA620F9E9_0"AddLanguageSpecificTextSet("LSTA620F9E9_0?cpp=2105.");GetN

Introduction

```
Dim Name As String
Dim ParamsOption As Integer
Dim Latitude As Double
Dim Longitude As Double
Dim Island As Integer
Dim LimitState As Integer
Dim UsageClass As Integer
Dim NomLife As Double
Dim PeakAccel As Double
Dim F0 As Double
Dim Tcs As Double
Dim SpecType As Integer
Dim SoilType As Integer
Dim Topography As Integer
Dim hRatio As Double
Dim Damping As Double
Dim q As Double
Dim returnValue As Integer

returnValue = instance.GetNTC2018(Name,
ParamsOption, Latitude, Longitude,
Island, LimitState, UsageClass, NomLife,
PeakAccel, F0, Tcs, SpecType, SoilType,
Topography, hRatio, Damping, q)
```

```
int GetNTC2018(
String^ Name,
int% ParamsOption,
double% Latitude,
double% Longitude,
int% Island,
int% LimitState,
int% UsageClass,
double% NomLife,
double% PeakAccel,
double% F0,
double% Tcs,
int% SpecType,
int% SoilType,
int% Topography,
double% hRatio,
double% Damping,
double% q
)
```

```
abstract GetNTC2018 :
    Name : string *
    ParamsOption : int byref *
    Latitude : float byref *
    Longitude : float byref *
    Island : int byref *
    LimitState : int byref *
    UsageClass : int byref *
    NomLife : float byref *
    PeakAccel : float byref *
    F0 : float byref *
    Tcs : float byref *
    SpecType : int byref *
    SoilType : int byref *
    Topography : int byref *
    hRatio : float byref *
    Damping : float byref *
```


q : float byref -> int

Parameters

Name

Type:Â SystemString

The name of a NTC 2018 response spectrum function

ParamsOption

Type:Â SystemInt32

This is 1, 2, or 3, indicating the option for defining the parameters

1. by latitude and longitude
2. by island
3. user specified

Latitude

Type:Â SystemDouble

The latitude for which the seismic coefficients are obtained. This item is meaningful only when ParamsOption = 1

Longitude

Type:Â SystemDouble

The longitude for which the seismic coefficients are obtained. This item is meaningful only when ParamsOption = 1

Island

Type:Â SystemInt32

This is one of the following values. This item is used only when ParamsOption = 2

1. Alicudi
2. Arcipelago Toscano
3. Filcudi
4. Isole Egadi
5. Lampedusa
6. Linosa
7. Lipari
8. Palmarola
9. Panarea
10. Pantelleria
11. Ponza
12. Salina
13. Santo Stefano
14. Sardegna
15. Stromboli
16. Tremiti
17. Ustica
18. Ventotene
19. Vulcano
20. Zannone

LimitState

Type:Â SystemInt32

This is 1, 2, 3, or 4, indicating the limit state

1. SLO
2. SLD
3. SLV

4. SLC

UsageClass

Type: SystemInt32

This is 1, 2, 3, or 4, indicating the usage class

1. I
2. II
3. III
4. IV

NomLife

Type: SystemDouble

The nominal life to be considered

PeakAccel

Type: SystemDouble

The peak ground acceleration, ag/g

F0

Type: SystemDouble

The magnitude factor

Tcs

Type: SystemDouble

The reference period, Tc*[s]

SpecType

Type: SystemInt32

This is 1, 2, 3, or 4, indicating the type of spectrum to consider

1. Elastic horizontal
2. Elastic vertical
3. Design horizontal
4. Design vertical

SoilType

Type: SystemInt32

This is 1, 2, 3, 4, or 5, indicating the subsoil type

1. A
2. B
3. C
4. D
5. E

Topography

Type: SystemInt32

This is 1, 2, 3, or 4, indicating the topography type

1. T1
2. T2
3. T3
4. T4

hRatio

Type: SystemDouble

The ratio for the site altitude at the base of the hill to the height of the hill

Damping

Type: SystemDouble

The damping, in percent. This is only applicable for SpecType 1 and 2

q

Type: SystemDouble

The behavior correction factor. This is only applicable for SpecType 3 and 4

Return ValueType: Int32

returns zero if the function is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
```

```
    'dimension variables
    Dim SapObject as cOAPI
    Dim SapModel As cSapModel
    Dim ret As Integer
    Dim ParamsOption As Integer
    Dim Latitude As Double
    Dim Longitude As Double
    Dim Island As Integer
    Dim LimitState As Integer
    Dim UsageClass As Integer
    Dim NomLife As Double
    Dim PeakAccel As Double
    Dim F0 As Double
    Dim Tcs As Double
    Dim SpecType As Integer
    Dim SoilType As Integer
    Dim Topography As Integer
    Dim hRatio As Double
    Dim Damping As Double
    Dim q As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add NTC 2018 RS function
    ret = SapModel.Func.FuncRS.SetNTC2018("RS-1", 1, 45.9, 12.6, 1, 3, 2, 50, 0.2, 2.4, 0.3, 3

    'get NTC 2018 RS function
    ret = SapModel.Func.FuncRS.GetNTC2018("RS-1", ParamsOption, Latitude, Longitude, Island, L

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
```

End Sub

▲ See Also
Reference

cFunctionRS Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFunctionRSAddLanguageSpecificTextSet("LSTD39E0D60_0") Method

Defines a NTC2008 response spectrum function

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetNTC2008(
    string Name,
    int ParamsOption,
    double Latitude,
    double Longitude,
    int Island,
    int LimitState,
    int UsageClass,
    double NomLife,
    double PeakAccel,
    double F0,
    double Tcs,
    int SpecType,
    int SoilType,
    int Topography,
    double hRatio,
    double Damping,
    double q
)
```

```
Function SetNTC2008 (
    Name As String,
    ParamsOption As Integer,
    Latitude As Double,
    Longitude As Double,
    Island As Integer,
    LimitState As Integer,
    UsageClass As Integer,
    NomLife As Double,
    PeakAccel As Double,
    F0 As Double,
    Tcs As Double,
    SpecType As Integer,
    SoilType As Integer,
    Topography As Integer,
    hRatio As Double,
    Damping As Double,
    q As Double
) As Integer
```

```
Dim instance As cFunctionRS
```

cFunctionRSspan id="LSTD39E0D60_0"AddLanguageSpecificTextSet("LSTD39E0D60_0?cpp2201u=.");Se

Introduction

```
Dim Name As String
Dim ParamsOption As Integer
Dim Latitude As Double
Dim Longitude As Double
Dim Island As Integer
Dim LimitState As Integer
Dim UsageClass As Integer
Dim NomLife As Double
Dim PeakAccel As Double
Dim F0 As Double
Dim Tcs As Double
Dim SpecType As Integer
Dim SoilType As Integer
Dim Topography As Integer
Dim hRatio As Double
Dim Damping As Double
Dim q As Double
Dim returnValue As Integer

returnValue = instance.SetNTC2008(Name,
ParamsOption, Latitude, Longitude,
Island, LimitState, UsageClass, NomLife,
PeakAccel, F0, Tcs, SpecType, SoilType,
Topography, hRatio, Damping, q)
```

```
int SetNTC2008(
String^ Name,
int ParamsOption,
double Latitude,
double Longitude,
int Island,
int LimitState,
int UsageClass,
double NomLife,
double PeakAccel,
double F0,
double Tcs,
int SpecType,
int SoilType,
int Topography,
double hRatio,
double Damping,
double q
)
```

```
abstract SetNTC2008 :
    Name : string *
    ParamsOption : int *
    Latitude : float *
    Longitude : float *
    Island : int *
    LimitState : int *
    UsageClass : int *
    NomLife : float *
    PeakAccel : float *
    F0 : float *
    Tcs : float *
    SpecType : int *
    SoilType : int *
    Topography : int *
    hRatio : float *
    Damping : float *
```

q : float -> int

Parameters

Name

Type:Â SystemString

The name of an existing or new function. If this is an existing function, that function is modified; otherwise, a new function is added

ParamsOption

Type:Â SystemInt32

This is 1, 2, or 3, indicating the option for defining the parameters

1. by latitude and longitude
2. by island
3. user specified

Latitude

Type:Â SystemDouble

The latitude for which the seismic coefficients are obtained. This item is meaningful only when ParamsOption = 1

Longitude

Type:Â SystemDouble

The longitude for which the seismic coefficients are obtained. This item is meaningful only when ParamsOption = 1

Island

Type:Â SystemInt32

This is one of the following values. This item is used only when ParamsOption = 2

1. Alicudi
2. Arcipelago Toscano
3. Filcudi
4. Isole Egadi
5. Lampedusa
6. Linosa
7. Lipari
8. Palmarola
9. Panarea
10. Pantelleria
11. Ponza
12. Salina
13. Santo Stefano
14. Sardegna
15. Stromboli
16. Tremiti
17. Ustica
18. Ventotene
19. Vulcano
20. Zannone

LimitState

Type:Â SystemInt32

This is 1, 2, 3, or 4, indicating the limit state

1. SLO
2. SLD

3. SLV
4. SLC

UsageClass

Type:Â SystemInt32

This is 1, 2, 3, or 4, indicating the usage class

1. I
2. II
3. III
4. IV

NomLife

Type:Â SystemDouble

The nominal life to be considered

PeakAccel

Type:Â SystemDouble

The peak ground acceleration, ag/g

F0

Type:Â SystemDouble

The magnitude factor

Tcs

Type:Â SystemDouble

The reference period, Tc*[s]

SpecType

Type:Â SystemInt32

This is 1, 2, 3, or 4, indicating the type of spectrum to consider

1. Elastic horizontal
2. Elastic vertical
3. Design horizontal
4. Design vertical

SoilType

Type:Â SystemInt32

This is 1, 2, 3, 4, or 5, indicating the subsoil type

1. A
2. B
3. C
4. D
5. E

Topography

Type:Â SystemInt32

This is 1, 2, 3, or 4, indicating the topography type

1. T1
2. T2
3. T3
4. T4

hRatio

Type:Â SystemDouble

The ratio for the site altitude at the base of the hill to the height of the hill

Damping

Type:Â SystemDouble

The damping, in percent. This is only applicable for SpecType 1 and 2

q

Type: SystemDouble

The behavior correction factor. This is only applicable for SpecType 3 and 4

Return Value

Type: Int32

Returns zero if the function is successfully defined; otherwise it returns a nonzero value

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
```

```
    'dimension variables
    Dim SapObject as cOAPI
    Dim SapModel As cSapModel
    Dim ret As Integer
    Dim ParamsOption As Integer
    Dim Latitude As Double
    Dim Longitude As Double
    Dim Island As Integer
    Dim LimitState As Integer
    Dim UsageClass As Integer
    Dim NomLife As Double
    Dim PeakAccel As Double
    Dim F0 As Double
    Dim Tcs As Double
    Dim SpecType As Integer
    Dim SoilType As Integer
    Dim Topography As Integer
    Dim hRatio As Double
    Dim Damping As Double
    Dim q As Double
```

```
    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

```
    'start ETABS application
    ret = EtabsObject.ApplicationStart()
```

```
    'create SapModel object
    SapModel = EtabsObject.SapModel
```

```
    'initialize model
    ret = SapModel.InitializeNewModel()
```

```
    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)
```

```
    'add NTC2008 RS function
    ret = SapModel.Func.FuncRS.SetNTC2008("RS-1", 1, 45.9, 12.6, 1, 3, 2, 50, 0.2, 2.4, 0.3, 3
```

```
    'get NTC2008 RS function
    ret = SapModel.Func.FuncRS.GetNTC2008("RS-1", ParamsOption, Latitude, Longitude, Island, L
```

```
    'close ETABS
    EtabsObject.ApplicationExit(False)
```

Introduction

```
'clean up variables  
  SapModel = Nothing  
  EtabsObject = Nothing  
End Sub
```

🔍 See Also

Reference

cFunctionRS Interface

ETABSv1 Namespace

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CSI API ETABS v1

cFunctionRSAddLanguageSpecificTextSet("LSTA228C Method

Defines a NTC 2018 response spectrum function

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetNTC2018(
    string Name,
    int ParamsOption,
    double Latitude,
    double Longitude,
    int Island,
    int LimitState,
    int UsageClass,
    double NomLife,
    double PeakAccel,
    double F0,
    double Tcs,
    int SpecType,
    int SoilType,
    int Topography,
    double hRatio,
    double Damping,
    double q
)
```

```
Function SetNTC2018 (
    Name As String,
    ParamsOption As Integer,
    Latitude As Double,
    Longitude As Double,
    Island As Integer,
    LimitState As Integer,
    UsageClass As Integer,
    NomLife As Double,
    PeakAccel As Double,
    F0 As Double,
    Tcs As Double,
    SpecType As Integer,
    SoilType As Integer,
    Topography As Integer,
    hRatio As Double,
    Damping As Double,
    q As Double
) As Integer
```

```
Dim instance As cFunctionRS
```

cFunctionRSspan id="LSTA228CA48_0"AddLanguageSpecificTextSet("LSTA228CA48_0?cpp=2207=.");Set

Introduction

```
Dim Name As String
Dim ParamsOption As Integer
Dim Latitude As Double
Dim Longitude As Double
Dim Island As Integer
Dim LimitState As Integer
Dim UsageClass As Integer
Dim NomLife As Double
Dim PeakAccel As Double
Dim F0 As Double
Dim Tcs As Double
Dim SpecType As Integer
Dim SoilType As Integer
Dim Topography As Integer
Dim hRatio As Double
Dim Damping As Double
Dim q As Double
Dim returnValue As Integer

returnValue = instance.SetNTC2018(Name,
ParamsOption, Latitude, Longitude,
Island, LimitState, UsageClass, NomLife,
PeakAccel, F0, Tcs, SpecType, SoilType,
Topography, hRatio, Damping, q)
```

```
int SetNTC2018(
String^ Name,
int ParamsOption,
double Latitude,
double Longitude,
int Island,
int LimitState,
int UsageClass,
double NomLife,
double PeakAccel,
double F0,
double Tcs,
int SpecType,
int SoilType,
int Topography,
double hRatio,
double Damping,
double q
)
```

```
abstract SetNTC2018 :
    Name : string *
    ParamsOption : int *
    Latitude : float *
    Longitude : float *
    Island : int *
    LimitState : int *
    UsageClass : int *
    NomLife : float *
    PeakAccel : float *
    F0 : float *
    Tcs : float *
    SpecType : int *
    SoilType : int *
    Topography : int *
    hRatio : float *
    Damping : float *
```

q : float -> int

Parameters

Name

Type:Â SystemString

The name of an existing or new function. If this is an existing function, that function is modified; otherwise, a new function is added

ParamsOption

Type:Â SystemInt32

This is 1, 2, or 3, indicating the option for defining the parameters

1. by latitude and longitude
2. by island
3. user specified

Latitude

Type:Â SystemDouble

The latitude for which the seismic coefficients are obtained. This item is meaningful only when ParamsOption = 1

Longitude

Type:Â SystemDouble

The longitude for which the seismic coefficients are obtained. This item is meaningful only when ParamsOption = 1

Island

Type:Â SystemInt32

This is one of the following values. This item is used only when ParamsOption = 2

1. Alicudi
2. Arcipelago Toscano
3. Filcudi
4. Isole Egadi
5. Lampedusa
6. Linosa
7. Lipari
8. Palmarola
9. Panarea
10. Pantelleria
11. Ponza
12. Salina
13. Santo Stefano
14. Sardegna
15. Stromboli
16. Tremiti
17. Ustica
18. Ventotene
19. Vulcano
20. Zannone

LimitState

Type:Â SystemInt32

This is 1, 2, 3, or 4, indicating the limit state

1. SLO
2. SLD

3. SLV
4. SLC

UsageClass

Type:Â SystemInt32

This is 1, 2, 3, or 4, indicating the usage class

1. I
2. II
3. III
4. IV

NomLife

Type:Â SystemDouble

The nominal life to be considered

PeakAccel

Type:Â SystemDouble

The peak ground acceleration, ag/g

F0

Type:Â SystemDouble

The magnitude factor

Tcs

Type:Â SystemDouble

The reference period, Tc*[s]

SpecType

Type:Â SystemInt32

This is 1, 2, 3, or 4, indicating the type of spectrum to consider

1. Elastic horizontal
2. Elastic vertical
3. Design horizontal
4. Design vertical

SoilType

Type:Â SystemInt32

This is 1, 2, 3, 4, or 5, indicating the subsoil type

1. A
2. B
3. C
4. D
5. E

Topography

Type:Â SystemInt32

This is 1, 2, 3, or 4, indicating the topography type

1. T1
2. T2
3. T3
4. T4

hRatio

Type:Â SystemDouble

The ratio for the site altitude at the base of the hill to the height of the hill

Damping

Type:Â SystemDouble

The damping, in percent. This is only applicable for SpecType 1 and 2

q

Type: SystemDouble

The behavior correction factor. This is only applicable for SpecType 3 and 4

Return Value

Type: Int32

Returns zero if the function is successfully defined; otherwise it returns a nonzero value

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
```

```
    'dimension variables
    Dim SapObject as cOAPI
    Dim SapModel As cSapModel
    Dim ret As Integer
    Dim ParamsOption As Integer
    Dim Latitude As Double
    Dim Longitude As Double
    Dim Island As Integer
    Dim LimitState As Integer
    Dim UsageClass As Integer
    Dim NomLife As Double
    Dim PeakAccel As Double
    Dim F0 As Double
    Dim Tcs As Double
    Dim SpecType As Integer
    Dim SoilType As Integer
    Dim Topography As Integer
    Dim hRatio As Double
    Dim Damping As Double
    Dim q As Double
```

```
    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

```
    'start ETABS application
    ret = EtabsObject.ApplicationStart()
```

```
    'create SapModel object
    SapModel = EtabsObject.SapModel
```

```
    'initialize model
    ret = SapModel.InitializeNewModel()
```

```
    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)
```

```
    'add NTC 2018 RS function
    ret = SapModel.Func.FuncRS.SetNTC2018("RS-1", 1, 45.9, 12.6, 1, 3, 2, 50, 0.2, 2.4, 0.3, 3)
```

```
    'get NTC 2018 RS function
    ret = SapModel.Func.FuncRS.GetNTC2018("RS-1", ParamsOption, Latitude, Longitude, Island, L
```

```
    'close ETABS
    EtabsObject.ApplicationExit(False)
```

Introduction

```
'clean up variables  
  SapModel = Nothing  
  EtabsObject = Nothing  
End Sub
```

📌 See Also

Reference

cFunctionRS Interface

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CSI API ETABS v1

cGenDispl Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cGenDispl
```

```
Public Interface cGenDispl
```

```
Dim instance As cGenDispl
```

```
public interface class cGenDispl
```

```
type cGenDispl = interface end
```

The cGenDispl type exposes the following members.

▲ Methods

▲	Name	Description
---	------	-------------

⇒	Add	
---	---------------------	--

⇒	ChangeName	
---	----------------------------	--

⇒	Count	
---	-----------------------	--

⇒	CountPoint	
---	----------------------------	--

⇒	Delete	
---	------------------------	--

⇒	DeletePoint	
---	-----------------------------	--

⇒	GetNameList	
---	-----------------------------	--

⇒	GetPoint	
---	--------------------------	--

⇒	GetTypeGenDispl	
---	---------------------------------	--

⇒	GetTypeOAPI	
---	-----------------------------	--

⇒	SetPoint	
---	--------------------------	--

⇒	SetType	
---	-------------------------	--

⇒	SetTypeOAPI	
---	-----------------------------	--

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cGenDispl Methods

The [cGenDispl](#) type exposes the following members.

Methods

	Name	Description
⇒	Add	
⇒	ChangeName	
⇒	Count	
⇒	CountPoint	
⇒	Delete	
⇒	DeletePoint	
⇒	GetNameList	
⇒	GetPoint	
⇒	GetTypeGenDispl	
⇒	GetTypeOAPI	
⇒	SetPoint	
⇒	SetType	
⇒	SetTypeOAPI	

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[cGenDispl Interface](#)

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CSI API ETABS v1

cGenDisplAddLanguageSpecificTextSet("LSTA69DD547_0?cpp=::2216");Add M

Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int Add(
    string Name,
    int MyType
)
```

```
Function Add (
    Name As String,
    MyType As Integer
) As Integer
```

```
Dim instance As cGenDispl
Dim Name As String
Dim MyType As Integer
Dim returnValue As Integer
```

```
returnValue = instance.Add(Name, MyType)
```

```
int Add(
    String^ Name,
    int MyType
)
```

```
abstract Add :
    Name : string *
    MyType : int -> int
```

Parameters

Name

Type: [SystemString](#)

MyType

Type: [SystemInt32](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cGenDispl Interface

ETABSv1 Namespace

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CSI API ETABS v1

cGenDisplAddLanguageSpecificTextSet("LSTC0148E58_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int ChangeName(
    string Name,
    string NewName
)
```

```
Function ChangeName (
    Name As String,
    NewName As String
) As Integer
```

```
Dim instance As cGenDispl
Dim Name As String
Dim NewName As String
Dim returnValue As Integer
```

```
returnValue = instance.ChangeName(Name,
    NewName)
```

```
int ChangeName(
    String^ Name,
    String^ NewName
)
```

```
abstract ChangeName :
    Name : string *
    NewName : string -> int
```

Parameters

Name

Type: [SystemString](#)

NewName

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cGenDispl Interface

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CSI API ETABS v1

cGenDisplAddLanguageSpecificTextSet("LST16864168 Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

int Count()

Function Count As Integer

```
Dim instance As cGenDispl
Dim returnValue As Integer
```

```
returnValue = instance.Count()
```

```
int Count()
```

```
abstract Count : unit -> int
```

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cGenDispl Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cGenDisplAddLanguageSpecificTextSet("LST86A10D8C_0?cpp=::221");Count

Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int CountPoint(
    string Name,
    ref int Count
)
```

```
Function CountPoint (
    Name As String,
    ByRef Count As Integer
) As Integer
```

```
Dim instance As cGenDispl
Dim Name As String
Dim Count As Integer
Dim returnValue As Integer
```

```
returnValue = instance.CountPoint(Name,
    Count)
```

```
int CountPoint(
    String^ Name,
    int% Count
)
```

```
abstract CountPoint :
    Name : string *
    Count : int byref -> int
```

Parameters

Name

Type: [SystemString](#)

Count

Type: [SystemInt32](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cGenDispl Interface

ETABSv1 Namespace

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CSI API ETABS v1

cGenDisplAddLanguageSpecificTextSet("LST8BA88054_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int Delete(
    string Name
)
```

```
Function Delete (
    Name As String
) As Integer
```

```
Dim instance As cGenDispl
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.Delete(Name)
```

```
int Delete(
    String^ Name
)
```

```
abstract Delete :
    Name : string -> int
```

Parameters

Name

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cGenDispl Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cGenDisplAddLanguageSpecificTextSet("LST25FCAAA Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int DeletePoint(
    string Name,
    string PointName
)
```

```
Function DeletePoint (
    Name As String,
    PointName As String
) As Integer
```

```
Dim instance As cGenDispl
Dim Name As String
Dim PointName As String
Dim returnValue As Integer
```

```
returnValue = instance.DeletePoint(Name,
    PointName)
```

```
int DeletePoint(
    String^ Name,
    String^ PointName
)
```

```
abstract DeletePoint :
    Name : string *
    PointName : string -> int
```

Parameters

Name

Type: [SystemString](#)

PointName

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cGenDispl Interface

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CSI API ETABS v1

cGenDisplAddLanguageSpecificTextSet("LST75EE3CD7_0?cpp=:2027.");GetNMethod

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetNameList(
    ref int NumberNames,
    ref string[] MyName
)
```

```
Function GetNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cGenDispl
Dim NumberNames As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetNameList(NumberNames,
MyName)
```

```
int GetNameList(
    int% NumberNames,
    array<String^>^% MyName
)
```

```
abstract GetNameList :
    NumberNames : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberNames

Type: SystemInt32

MyName

Type: SystemString

Return Value

Type: Int32

▲ See Also

Reference

cGenDispl Interface

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CSI API ETABS v1

cGenDisplAddLanguageSpecificTextSet("LSTFFDBDAI Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetPoint(
    string Name,
    ref int NumberItems,
    ref string[] PointName,
    ref double[] U1,
    ref double[] U2,
    ref double[] U3,
    ref double[] R1,
    ref double[] R2,
    ref double[] R3
)
```

```
Function GetPoint (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef PointName As String(),
    ByRef U1 As Double(),
    ByRef U2 As Double(),
    ByRef U3 As Double(),
    ByRef R1 As Double(),
    ByRef R2 As Double(),
    ByRef R3 As Double()
) As Integer
```

```
Dim instance As cGenDispl
Dim Name As String
Dim NumberItems As Integer
Dim PointName As String()
Dim U1 As Double()
Dim U2 As Double()
Dim U3 As Double()
Dim R1 As Double()
Dim R2 As Double()
Dim R3 As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetPoint(Name,
    NumberItems, PointName, U1, U2, U3,
    R1, R2, R3)
```

```
int GetPoint(
    String^ Name,
```

cGenDisplAddLanguageSpecificTextSet("LSTFFDBDAD3_0?cpp=2129-");Get

```

int% NumberItems,
array<String>^% PointName,
array<double>^% U1,
array<double>^% U2,
array<double>^% U3,
array<double>^% R1,
array<double>^% R2,
array<double>^% R3
)

abstract GetPoint :
    Name : string *
    NumberItems : int byref *
    PointName : string[] byref *
    U1 : float[] byref *
    U2 : float[] byref *
    U3 : float[] byref *
    R1 : float[] byref *
    R2 : float[] byref *
    R3 : float[] byref -> int

```

Parameters

Name
Type: [SystemString](#)

NumberItems
Type: [SystemInt32](#)

PointName
Type: [SystemString](#)

U1
Type: [SystemDouble](#)

U2
Type: [SystemDouble](#)

U3
Type: [SystemDouble](#)

R1
Type: [SystemDouble](#)

R2
Type: [SystemDouble](#)

R3
Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

See Also

Reference

[cGenDispl Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cGenDisplAddLanguageSpecificTextSet("LST6F9D145D_0?cpp=::2032");GetTy Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetTypeGenDispl(
    string Name,
    ref int MyType
)
```

```
Function GetTypeGenDispl (
    Name As String,
    ByRef MyType As Integer
) As Integer
```

```
Dim instance As cGenDispl
Dim Name As String
Dim MyType As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetTypeGenDispl(Name,
    MyType)
```

```
int GetTypeGenDispl(
    String^ Name,
    int% MyType
)
```

```
abstract GetTypeGenDispl :
    Name : string *
    MyType : int byref -> int
```

Parameters

Name

Type: [SystemString](#)

MyType

Type: [SystemInt32](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cGenDispl Interface

ETABSv1 Namespace

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CSI API ETABS v1

cGenDisplAddLanguageSpecificTextSet("LST46300F5E Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetType0API(
    string name,
    ref int MyType
)
```

```
Function GetType0API (
    name As String,
    ByRef MyType As Integer
) As Integer
```

```
Dim instance As cGenDispl
Dim name As String
Dim MyType As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetType0API(name,
    MyType)
```

```
int GetType0API(
    String^ name,
    int% MyType
)
```

```
abstract GetType0API :
    name : string *
    MyType : int byref -> int
```

Parameters

name

Type: [SystemString](#)

MyType

Type: [SystemInt32](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cGenDispl Interface

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CSI API ETABS v1

cGenDisplAddLanguageSpecificTextSet("LST8C776F12_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetPoint(
    string Name,
    string PointName,
    ref double[] SF
)
```

```
Function SetPoint (
    Name As String,
    PointName As String,
    ByRef SF As Double()
) As Integer
```

```
Dim instance As cGenDispl
Dim Name As String
Dim PointName As String
Dim SF As Double()
Dim returnValue As Integer
```

```
returnValue = instance.SetPoint(Name,
    PointName, SF)
```

```
int SetPoint(
    String^ Name,
    String^ PointName,
    array<double>^% SF
)
```

```
abstract SetPoint :
    Name : string *
    PointName : string *
    SF : float[] byref -> int
```

Parameters

Name

Type: SystemString

PointName

Type: SystemString

SF

Type: SystemDouble

cGenDisplAddLanguageSpecificTextSet("LST8C776F12_0?cpp=::|2236);SetPo

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cGenDispl Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cGenDisplAddLanguageSpecificTextSet("LSTE60F8E11_0?cpp=::|2238);SetType

Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetType(
    string Name,
    int MyType
)
```

```
Function SetType (
    Name As String,
    MyType As Integer
) As Integer
```

```
Dim instance As cGenDispl
Dim Name As String
Dim MyType As Integer
Dim returnValue As Integer
```

```
returnValue = instance.SetType(Name, MyType)
```

```
int SetType(
    String^ Name,
    int MyType
)
```

```
abstract SetType :
    Name : string *
    MyType : int -> int
```

Parameters

Name

Type: [SystemString](#)

MyType

Type: [SystemInt32](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cGenDispl Interface

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CSI API ETABS v1

cGenDisplAddLanguageSpecificTextSet("LST5206D2F5") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetType0API(
    string Name,
    int MyType
)
```

```
Function SetType0API (
    Name As String,
    MyType As Integer
) As Integer
```

```
Dim instance As cGenDispl
Dim Name As String
Dim MyType As Integer
Dim returnValue As Integer
```

```
returnValue = instance.SetType0API(Name,
    MyType)
```

```
int SetType0API(
    String^ Name,
    int MyType
)
```

```
abstract SetType0API :
    Name : string *
    MyType : int -> int
```

Parameters

Name

Type: [SystemString](#)

MyType

Type: [SystemInt32](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cGenDispl Interface

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CSI API ETABS v1

cGridSys Interface

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cGridSys
```

```
Public Interface cGridSys
```

```
Dim instance As cGridSys
```

```
public interface class cGridSys
```

```
type cGridSys = interface end
```

The cGridSys type exposes the following members.

▲ Methods

▲	Name	Description
➤	ChangeName	Changes the name of a defined grid system.
➤	Count	Retrieves the number of defined grid systems.
➤	Delete	Deletes the specified grid system.
➤	GetGridSys	Retrieves the grid system specified by the Name item.
➤	GetGridSys_2	Retrieves the grid system specified by the Name item.
➤	GetGridSysCartesian	
➤	GetGridSysCylindrical	
➤	GetGridSysType	
➤	GetNameList	Retrieves the names of all defined grid systems.
➤	GetNameTypeList	
➤	GetTransformationMatrix	Retrieves the grid system transformation matrix.
➤	SetGridSys	Adds a new grid system, or modifies an existing grid system.

[Top](#)

▲ See Also

Reference

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cGridSys Methods

The cGridSys type exposes the following members.

Methods

⌕	Name	Description
⇒	<u>ChangeName</u>	Changes the name of a defined grid system.
⇒	<u>Count</u>	Retrieves the number of defined grid systems.
⇒	<u>Delete</u>	Deletes the specified grid system.
⇒	<u>GetGridSys</u>	Retrieves the grid system specified by the Name item.
⇒	<u>GetGridSys_2</u>	Retrieves the grid system specified by the Name item.
⇒	<u>GetGridSysCartesian</u>	
⇒	<u>GetGridSysCylindrical</u>	
⇒	<u>GetGridSysType</u>	
⇒	<u>GetNameList</u>	Retrieves the names of all defined grid systems.
⇒	<u>GetNameTypeList</u>	
⇒	<u>GetTransformationMatrix</u>	Retrieves the grid system transformation matrix.
⇒	<u>SetGridSys</u>	Adds a new grid system, or modifies an existing grid system.

Top

▲ See Also

Reference

cGridSys Interface

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CSI API ETABS v1

cGridSysAddLanguageSpecificTextSet("LSTAB8645EC Method

Changes the name of a defined grid system.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int ChangeName(
    string Name,
    string NewName
)
```

```
Function ChangeName (
    Name As String,
    NewName As String
) As Integer
```

```
Dim instance As cGridSys
Dim Name As String
Dim NewName As String
Dim returnValue As Integer
```

```
returnValue = instance.ChangeName(Name,
    NewName)
```

```
int ChangeName(
    String^ Name,
    String^ NewName
)
```

```
abstract ChangeName :
    Name : string *
    NewName : string -> int
```

Parameters

Name

Type: SystemString

The existing name of a defined grid system.

NewName

Type: SystemString

The new name for the grid system.

Return Value

Type: Int32

Returns zero if the new name is successfully applied, otherwise it returns a nonzero value.

▀ **Remarks**

Changing the name of the Global grid system will fail and return an error.

▀ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new grid system
    ret = SapModel.GridSys.SetGridSys("GridSysA", 1000, 1000, 0, 0, 0, 0)

    'change name of new grid system
    ret = SapModel.GridSys.ChangeName("GridSysA", "MyChangedCSysA")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▀ **See Also**

Reference

cGridSys Interface

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CSI API ETABS v1

cGridSysAddLanguageSpecificTextSet("LSTD8335F13_0") Method

Retrieves the number of defined grid systems.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int Count()
```

```
Function Count As Integer
```

```
Dim instance As cGridSys
Dim returnValue As Integer
```

```
returnValue = instance.Count()
```

```
int Count()
```

```
abstract Count : unit -> int
```

Return Value

Type: [Int32](#)

Returns the number of defined grid systems.

▲ Remarks

▲ Examples

VB

[Copy](#)

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()
```

Introduction

```
'create steel deck template model
  ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'define new grid system
  ret = SapModel.GridSys.SetGridSys("GridSysA", 1000, 1000, 0, 0, 0, 0)

'return number of defined grid systems
  ret = SapModel.GridSys.Count

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

[cGridSys Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cGridSysAddLanguageSpecificTextSet("LSTA517F079_0") Method

Deletes the specified grid system.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int Delete(
    string Name
)
```

```
Function Delete (
    Name As String
) As Integer
```

```
Dim instance As cGridSys
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.Delete(Name)
```

```
int Delete(
    String^ Name
)
```

```
abstract Delete :
    Name : string -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing grid system.

Return Value

Type: [Int32](#)

Returns zero if the grid system is successfully deleted, otherwise it returns a nonzero value.

▲ Remarks

Attempting to delete the Global grid system will fail and return an error.

▲ Examples

[VB](#)

[Copy](#)

Introduction

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)  
  
    'define new grid system  
    ret = SapModel.GridSys.SetGridSys("GridSysA", 1000, 1000, 0, 0, 0, 0)  
  
    'delete grid system  
    ret = SapModel.GridSys.Delete("GridSysA")  
  
    'close ETABS  
    EtabsObject.ApplicationExit(False)  
  
    'clean up variables  
    SapModel = Nothing  
    EtabsObject = Nothing  
  
End Sub
```

🔗 See Also

Reference

[cGridSys Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cGridSysAddLanguageSpecificTextSet("LSTB219EDA7 Method

Retrieves the grid system specified by the Name item.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetGridSys(
    string Name,
    ref double x,
    ref double y,
    ref double RZ
)
```

```
Function GetGridSys (
    Name As String,
    ByRef x As Double,
    ByRef y As Double,
    ByRef RZ As Double
) As Integer
```

```
Dim instance As cGridSys
Dim Name As String
Dim x As Double
Dim y As Double
Dim RZ As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetGridSys(Name,
    x, y, RZ)
```

```
int GetGridSys(
    String^ Name,
    double% x,
    double% y,
    double% RZ
)
```

```
abstract GetGridSys :
    Name : string *
    x : float byref *
    y : float byref *
    RZ : float byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing grid system.

x

Type:Â SystemDouble

The global X grid of the origin of the grid system. [L]

y

Type:Â SystemDouble

The global Y grid of the origin of the grid system. [L]

RZ

Type:Â SystemDouble

The rotation of an axis of the new grid system relative to the global grid system is defined as follows: (1) Rotate the grid system about the positive global Z-axis as defined by the RZ item. [deg]

Return ValueType:Â Int32

Returns zero if the grid system data is successfully retrieved, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new grid system
    ret = SapModel.GridSys.SetGridSys("GridSysA", 1000, 1000, 0, 0, 0, 0)

    'get new grid system data
    ret = SapModel.GridSys.GetGridSys("GridSysA", x, y, z, RZ)

    'close ETABS
    EtabsObject.ApplicationExit(False)
```



```
'clean up variables  
  SapModel = Nothing  
  EtabsObject = Nothing
```

End Sub

📌 See Also

Reference

cGridSys Interface

ETABSv1 Namespace

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CSI API ETABS v1

cGridSysAddLanguageSpecificTextSet("LST333D4B3_0") Method

Retrieves the grid system specified by the Name item.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetGridSys_2(
    string Name,
    ref double Xo,
    ref double Yo,
    ref double RZ,
    ref string GridSysType,
    ref int NumXLines,
    ref int NumYLines,
    ref string[] GridLineIDX,
    ref string[] GridLineIDY,
    ref double[] OrdinateX,
    ref double[] OrdinateY,
    ref bool[] VisibleX,
    ref bool[] VisibleY,
    ref string[] BubbleLocX,
    ref string[] BubbleLocY
)
```

```
Function GetGridSys_2 (
    Name As String,
    ByRef Xo As Double,
    ByRef Yo As Double,
    ByRef RZ As Double,
    ByRef GridSysType As String,
    ByRef NumXLines As Integer,
    ByRef NumYLines As Integer,
    ByRef GridLineIDX As String(),
    ByRef GridLineIDY As String(),
    ByRef OrdinateX As Double(),
    ByRef OrdinateY As Double(),
    ByRef VisibleX As Boolean(),
    ByRef VisibleY As Boolean(),
    ByRef BubbleLocX As String(),
    ByRef BubbleLocY As String()
) As Integer
```

```
Dim instance As cGridSys
Dim Name As String
Dim Xo As Double
Dim Yo As Double
Dim RZ As Double
```

```

Dim GridSysType As String
Dim NumXLines As Integer
Dim NumYLines As Integer
Dim GridLineIDX As String()
Dim GridLineIDY As String()
Dim OrdinateX As Double()
Dim OrdinateY As Double()
Dim VisibleX As Boolean()
Dim VisibleY As Boolean()
Dim BubbleLocX As String()
Dim BubbleLocY As String()
Dim returnValue As Integer

returnValue = instance.GetGridSys_2(Name,
Xo, Yo, RZ, GridSysType, NumXLines,
NumYLines, GridLineIDX, GridLineIDY,
OrdinateX, OrdinateY, VisibleX, VisibleY,
BubbleLocX, BubbleLocY)

int GetGridSys_2(
String^ Name,
double% Xo,
double% Yo,
double% RZ,
String^% GridSysType,
int% NumXLines,
int% NumYLines,
array<String^>^% GridLineIDX,
array<String^>^% GridLineIDY,
array<double>^% OrdinateX,
array<double>^% OrdinateY,
array<bool>^% VisibleX,
array<bool>^% VisibleY,
array<String^>^% BubbleLocX,
array<String^>^% BubbleLocY
)

abstract GetGridSys_2 :
    Name : string *
    Xo : float byref *
    Yo : float byref *
    RZ : float byref *
    GridSysType : string byref *
    NumXLines : int byref *
    NumYLines : int byref *
    GridLineIDX : string[] byref *
    GridLineIDY : string[] byref *
    OrdinateX : float[] byref *
    OrdinateY : float[] byref *
    VisibleX : bool[] byref *
    VisibleY : bool[] byref *
    BubbleLocX : string[] byref *
    BubbleLocY : string[] byref -> int

```

Parameters

Name

Type: SystemString

The name of an existing grid system.

Xo

Type:Â SystemDouble
 The global X grid of the origin of the grid system. [L]
 Yo
 Type:Â SystemDouble
 The global Y grid of the origin of the grid system. [L]
 RZ
 Type:Â SystemDouble
 The rotation of an axis of the new grid system relative to the global grid system is defined as follows: (1) Rotate the grid system about the positive global Z-axis as defined by the RZ item. [deg]
 GridSysType
 Type:Â SystemString
 NumXLines
 Type:Â SystemInt32
 NumYLines
 Type:Â SystemInt32
 GridLineIDX
 Type:Â SystemString
 GridLineIDY
 Type:Â SystemString
 OrdinateX
 Type:Â SystemDouble
 OrdinateY
 Type:Â SystemDouble
 VisibleX
 Type:Â SystemBoolean
 VisibleY
 Type:Â SystemBoolean
 BubbleLocX
 Type:Â SystemString
 BubbleLocY
 Type:Â SystemString

Return Value

Type:Â Int32
 Returns zero if the grid system data is successfully retrieved, otherwise it returns a nonzero value.

- ▲ Remarks
- ▲ Examples
- ▲ See Also

Reference

cGridSys Interface

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CSI API ETABS v1

cGridSysAddLanguageSpecificTextSet("LST17A4CBAE Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetGridSysCartesian(
    string Name,
    ref double Xo,
    ref double Yo,
    ref double RZ,
    ref bool StoryRangeIsDefault,
    ref string TopStory,
    ref string BottomStory,
    ref double BubbleSize,
    ref int GridColor,
    ref int NumXLines,
    ref string[] GridLineIDX,
    ref double[] OrdinateX,
    ref bool[] VisibleX,
    ref string[] BubbleLocX,
    ref int NumYLines,
    ref string[] GridLineIDY,
    ref double[] OrdinateY,
    ref bool[] VisibleY,
    ref string[] BubbleLocY,
    ref int NumGenLines,
    ref string[] GridLineIDGen,
    ref double[] GenOrdX1,
    ref double[] GenOrdY1,
    ref double[] GenOrdX2,
    ref double[] GenOrdY2,
    ref bool[] VisibleGen,
    ref string[] BubbleLocGen
)
```

```
Function GetGridSysCartesian (
    Name As String,
    ByRef Xo As Double,
    ByRef Yo As Double,
    ByRef RZ As Double,
    ByRef StoryRangeIsDefault As Boolean,
    ByRef TopStory As String,
    ByRef BottomStory As String,
    ByRef BubbleSize As Double,
    ByRef GridColor As Integer,
    ByRef NumXLines As Integer,
    ByRef GridLineIDX As String(),
    ByRef OrdinateX As Double(),
```

cGridSysid="LST17A4CBAE_0"AddLanguageSpecificTextSet("LST17A4CBAE_0?cpp=::|2257");GetGr

Introduction

```
ByRef VisibleX As Boolean(),  
ByRef BubbleLocX As String(),  
ByRef NumYLines As Integer,  
ByRef GridLineIDY As String(),  
ByRef OrdinateY As Double(),  
ByRef VisibleY As Boolean(),  
ByRef BubbleLocY As String(),  
ByRef NumGenLines As Integer,  
ByRef GridLineIDGen As String(),  
ByRef GenOrdX1 As Double(),  
ByRef GenOrdY1 As Double(),  
ByRef GenOrdX2 As Double(),  
ByRef GenOrdY2 As Double(),  
ByRef VisibleGen As Boolean(),  
ByRef BubbleLocGen As String()  
) As Integer
```

```
Dim instance As cGridSys  
Dim Name As String  
Dim Xo As Double  
Dim Yo As Double  
Dim RZ As Double  
Dim StoryRangeIsDefault As Boolean  
Dim TopStory As String  
Dim BottomStory As String  
Dim BubbleSize As Double  
Dim GridColor As Integer  
Dim NumXLines As Integer  
Dim GridLineIDX As String()  
Dim OrdinateX As Double()  
Dim VisibleX As Boolean()  
Dim BubbleLocX As String()  
Dim NumYLines As Integer  
Dim GridLineIDY As String()  
Dim OrdinateY As Double()  
Dim VisibleY As Boolean()  
Dim BubbleLocY As String()  
Dim NumGenLines As Integer  
Dim GridLineIDGen As String()  
Dim GenOrdX1 As Double()  
Dim GenOrdY1 As Double()  
Dim GenOrdX2 As Double()  
Dim GenOrdY2 As Double()  
Dim VisibleGen As Boolean()  
Dim BubbleLocGen As String()  
Dim returnValue As Integer
```

```
returnValue = instance.GetGridSysCartesian(Name,  
Xo, Yo, RZ, StoryRangeIsDefault, TopStory,  
BottomStory, BubbleSize, GridColor,  
NumXLines, GridLineIDX, OrdinateX,  
VisibleX, BubbleLocX, NumYLines,  
GridLineIDY, OrdinateY, VisibleY,  
BubbleLocY, NumGenLines, GridLineIDGen,  
GenOrdX1, GenOrdY1, GenOrdX2, GenOrdY2,  
VisibleGen, BubbleLocGen)
```

```
int GetGridSysCartesian(  
String^ Name,  
double% Xo,  
double% Yo,  
double% RZ,
```

cGridSysspan id="LST17A4CBAE_0"AddLanguageSpecificTextSet("LST17A4CBAE_0?cpp=::|2258");GetGr

```

bool% StoryRangeIsDefault,
String^% TopStory,
String^% BottomStory,
double% BubbleSize,
int% GridColor,
int% NumXLines,
array<String^>^% GridLineIDX,
array<double>^% OrdinateX,
array<bool>^% VisibleX,
array<String^>^% BubbleLocX,
int% NumYLines,
array<String^>^% GridLineIDY,
array<double>^% OrdinateY,
array<bool>^% VisibleY,
array<String^>^% BubbleLocY,
int% NumGenLines,
array<String^>^% GridLineIDGen,
array<double>^% GenOrdX1,
array<double>^% GenOrdY1,
array<double>^% GenOrdX2,
array<double>^% GenOrdY2,
array<bool>^% VisibleGen,
array<String^>^% BubbleLocGen
)

abstract GetGridSysCartesian :
    Name : string *
    Xo : float byref *
    Yo : float byref *
    RZ : float byref *
    StoryRangeIsDefault : bool byref *
    TopStory : string byref *
    BottomStory : string byref *
    BubbleSize : float byref *
    GridColor : int byref *
    NumXLines : int byref *
    GridLineIDX : string[] byref *
    OrdinateX : float[] byref *
    VisibleX : bool[] byref *
    BubbleLocX : string[] byref *
    NumYLines : int byref *
    GridLineIDY : string[] byref *
    OrdinateY : float[] byref *
    VisibleY : bool[] byref *
    BubbleLocY : string[] byref *
    NumGenLines : int byref *
    GridLineIDGen : string[] byref *
    GenOrdX1 : float[] byref *
    GenOrdY1 : float[] byref *
    GenOrdX2 : float[] byref *
    GenOrdY2 : float[] byref *
    VisibleGen : bool[] byref *
    BubbleLocGen : string[] byref -> int

```

Parameters

Name

Type:Â SystemString

Xo

Type:Â SystemDouble

Yo
 Type:Â SystemDouble
 RZ
 Type:Â SystemDouble
 StoryRangeIsDefault
 Type:Â SystemBoolean
 TopStory
 Type:Â SystemString
 BottomStory
 Type:Â SystemString
 BubbleSize
 Type:Â SystemDouble
 GridColor
 Type:Â SystemInt32
 NumXLines
 Type:Â SystemInt32
 GridLineIDX
 Type:Â SystemString
 OrdinateX
 Type:Â SystemDouble
 VisibleX
 Type:Â SystemBoolean
 BubbleLocX
 Type:Â SystemString
 NumYLines
 Type:Â SystemInt32
 GridLineIDY
 Type:Â SystemString
 OrdinateY
 Type:Â SystemDouble
 VisibleY
 Type:Â SystemBoolean
 BubbleLocY
 Type:Â SystemString
 NumGenLines
 Type:Â SystemInt32
 GridLineIDGen
 Type:Â SystemString
 GenOrdX1
 Type:Â SystemDouble
 GenOrdY1
 Type:Â SystemDouble
 GenOrdX2
 Type:Â SystemDouble
 GenOrdY2
 Type:Â SystemDouble
 VisibleGen
 Type:Â SystemBoolean
 BubbleLocGen
 Type:Â SystemString

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cGridSys Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cGridSysAddLanguageSpecificTextSet("LST5E7B1A5D Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetGridSysCylindrical(
    string Name,
    ref double Xo,
    ref double Yo,
    ref double RZ,
    ref bool StoryRangeIsDefault,
    ref string TopStory,
    ref string BottomStory,
    ref double BubbleSize,
    ref int GridColor,
    ref int NumRLines,
    ref string[] GridLineIDR,
    ref double[] OrdinateR,
    ref bool[] VisibleR,
    ref string[] BubbleLocR,
    ref int NumTLines,
    ref string[] GridLineIDT,
    ref double[] OrdinateT,
    ref bool[] VisibleT,
    ref string[] BubbleLocT
)
```

```
Function GetGridSysCylindrical (
    Name As String,
    ByRef Xo As Double,
    ByRef Yo As Double,
    ByRef RZ As Double,
    ByRef StoryRangeIsDefault As Boolean,
    ByRef TopStory As String,
    ByRef BottomStory As String,
    ByRef BubbleSize As Double,
    ByRef GridColor As Integer,
    ByRef NumRLines As Integer,
    ByRef GridLineIDR As String(),
    ByRef OrdinateR As Double(),
    ByRef VisibleR As Boolean(),
    ByRef BubbleLocR As String(),
    ByRef NumTLines As Integer,
    ByRef GridLineIDT As String(),
    ByRef OrdinateT As Double(),
    ByRef VisibleT As Boolean(),
    ByRef BubbleLocT As String()
) As Integer
```

cGridSysspan id="LST5E7B1A5D_0"AddLanguageSpecificTextSet("LST5E7B1A5D_0?cpp=::|2262);GetGrid

Introduction

```
Dim instance As cGridSys
Dim Name As String
Dim Xo As Double
Dim Yo As Double
Dim RZ As Double
Dim StoryRangeIsDefault As Boolean
Dim TopStory As String
Dim BottomStory As String
Dim BubbleSize As Double
Dim GridColor As Integer
Dim NumRLines As Integer
Dim GridLineIDR As String()
Dim OrdinateR As Double()
Dim VisibleR As Boolean()
Dim BubbleLocR As String()
Dim NumTLines As Integer
Dim GridLineIDT As String()
Dim OrdinateT As Double()
Dim VisibleT As Boolean()
Dim BubbleLocT As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetGridSysCylindrical(Name,
Xo, Yo, RZ, StoryRangeIsDefault, TopStory,
BottomStory, BubbleSize, GridColor,
NumRLines, GridLineIDR, OrdinateR,
VisibleR, BubbleLocR, NumTLines,
GridLineIDT, OrdinateT, VisibleT,
BubbleLocT)
```

```
int GetGridSysCylindrical(
String^ Name,
double% Xo,
double% Yo,
double% RZ,
bool% StoryRangeIsDefault,
String^% TopStory,
String^% BottomStory,
double% BubbleSize,
int% GridColor,
int% NumRLines,
array<String^>^% GridLineIDR,
array<double>^% OrdinateR,
array<bool>^% VisibleR,
array<String^>^% BubbleLocR,
int% NumTLines,
array<String^>^% GridLineIDT,
array<double>^% OrdinateT,
array<bool>^% VisibleT,
array<String^>^% BubbleLocT
)
```

```
abstract GetGridSysCylindrical :
    Name : string *
    Xo : float byref *
    Yo : float byref *
    RZ : float byref *
    StoryRangeIsDefault : bool byref *
    TopStory : string byref *
    BottomStory : string byref *
    BubbleSize : float byref *
```

```

GridColor : int byref *
NumRLines : int byref *
GridLineIDR : string[] byref *
OrdinateR : float[] byref *
VisibleR : bool[] byref *
BubbleLocR : string[] byref *
NumTLines : int byref *
GridLineIDT : string[] byref *
OrdinateT : float[] byref *
VisibleT : bool[] byref *
BubbleLocT : string[] byref -> int

```

Parameters

Name

Type:Â SystemString

Xo

Type:Â SystemDouble

Yo

Type:Â SystemDouble

RZ

Type:Â SystemDouble

StoryRangeIsDefault

Type:Â SystemBoolean

TopStory

Type:Â SystemString

BottomStory

Type:Â SystemString

BubbleSize

Type:Â SystemDouble

GridColor

Type:Â SystemInt32

NumRLines

Type:Â SystemInt32

GridLineIDR

Type:Â SystemString

OrdinateR

Type:Â SystemDouble

VisibleR

Type:Â SystemBoolean

BubbleLocR

Type:Â SystemString

NumTLines

Type:Â SystemInt32

GridLineIDT

Type:Â SystemString

OrdinateT

Type:Â SystemDouble

VisibleT

Type:Â SystemBoolean

BubbleLocT

Type:Â SystemString

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cGridSys Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cGridSysAddLanguageSpecificTextSet("LST82D3A92A Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetGridSysType(
    string Name,
    ref string GridSysType
)
```

```
Function GetGridSysType (
    Name As String,
    ByRef GridSysType As String
) As Integer
```

```
Dim instance As cGridSys
Dim Name As String
Dim GridSysType As String
Dim returnValue As Integer
```

```
returnValue = instance.GetGridSysType(Name,
    GridSysType)
```

```
int GetGridSysType(
    String^ Name,
    String^% GridSysType
)
```

```
abstract GetGridSysType :
    Name : string *
    GridSysType : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

GridSysType

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cGridSys Interface

ETABSv1 Namespace

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CSI API ETABS v1

cGridSysAddLanguageSpecificTextSet("LST9B0F690D_0") Method

Retrieves the names of all defined grid systems.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNameList(
    ref int NumberNames,
    ref string[] MyName
)
```

```
Function GetNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cGridSys
Dim NumberNames As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetNameList(NumberNames,
    MyName)
```

```
int GetNameList(
    int% NumberNames,
    array<String^>^% MyName
)
```

```
abstract GetNameList :
    NumberNames : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberNames

Type: [System.Int32](#)

The number of grid system names retrieved by the program.

MyName

Type: [System.String](#)

This is a one-dimensional array of grid system names. The MyName array is created as a dynamic, zero-based, array by the API user:

```
Dim MyName() as String
```

cGridSysspan id="LST9B0F690D_0">AddLanguageSpecificTextSet("LST9B0F690D_0?cpp=::|n2268;GetNar

Introduction

The array is dimensioned to (NumberNames + 1) inside the ETABS program, filled with the names, and returned to the API user.

Return Value

Type: Int32

Returns zero if the names are successfully retrieved, otherwise it returns nonzero.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)  
  
    'define new grid system  
    ret = SapModel.GridSys.SetGridSys("GridSysA", 1000, 1000, 0, 0, 0, 0)  
  
    'get grid system names  
    ret = SapModel.GridSys.GetNameList(NumberNames, MyName)  
  
    'close ETABS  
    EtabsObject.ApplicationExit(False)  
  
    'clean up variables  
    SapModel = Nothing  
    EtabsObject = Nothing  
  
End Sub
```

▲ **See Also**

Reference

cGridSys Interface

ETABSv1 Namespace

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Introduction

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CSI API ETABS v1

cGridSysAddLanguageSpecificTextSet("LST9C70F2EF_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetNameTypeList(
    ref int NumberNames,
    ref string[] GridSysName,
    ref string[] GridSysType
)
```

```
Function GetNameTypeList (
    ByRef NumberNames As Integer,
    ByRef GridSysName As String(),
    ByRef GridSysType As String()
) As Integer
```

```
Dim instance As cGridSys
Dim NumberNames As Integer
Dim GridSysName As String()
Dim GridSysType As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetNameTypeList(NumberNames,
    GridSysName, GridSysType)
```

```
int GetNameTypeList(
    int% NumberNames,
    array<String^>^% GridSysName,
    array<String^>^% GridSysType
)
```

```
abstract GetNameTypeList :
    NumberNames : int byref *
    GridSysName : string[] byref *
    GridSysType : string[] byref -> int
```

Parameters

NumberNames

Type: SystemInt32

GridSysName

Type: SystemString

GridSysType

Type: SystemString

cGridSysspan id="LST9C70F2EF_0"AddLanguageSpecificTextSet("LST9C70F2EF_0?cpp=::|n227");GetNa

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cGridSys Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cGridSysAddLanguageSpecificTextSet("LST7CCD30CC0")

Method

Retrieves the grid system transformation matrix.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetTransformationMatrix(
    string Name,
    ref double[] Value
)
```

```
Function GetTransformationMatrix (
    Name As String,
    ByRef Value As Double()
) As Integer
```

```
Dim instance As cGridSys
Dim Name As String
Dim Value As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetTransformationMatrix(Name,
    Value)
```

```
int GetTransformationMatrix(
    String^ Name,
    array<double>^% Value
)
```

```
abstract GetTransformationMatrix :
    Name : string *
    Value : float[] byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing grid system.

Value

Type: [SystemDouble](#)

Value is an array of nine direction cosines that define the transformation matrix from the specified grid system to the global grid system.

Return Value

Type: Int32

Returns zero if the grid system transformation matrix is successfully returned, otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new grid system
    ret = SapModel.GridSys.SetGridSys("GridSysA", 1000, 1000, 0, 0, 0, 0)

    'get grid system transformation matrix
    ReDim Value(8)
    ret = SapModel.GridSys.GetTransformationMatrix("GridSysA", Value)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ **See Also**

Reference

cGridSys Interface

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CSI API ETABS v1

cGridSysAddLanguageSpecificTextSet("LST94730656_0") Method

Adds a new grid system, or modifies an existing grid system.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetGridSys(
    string Name,
    double x,
    double y,
    double RZ
)
```

```
Function SetGridSys (
    Name As String,
    x As Double,
    y As Double,
    RZ As Double
) As Integer
```

```
Dim instance As cGridSys
Dim Name As String
Dim x As Double
Dim y As Double
Dim RZ As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetGridSys(Name,
    x, y, RZ)
```

```
int SetGridSys(
    String^ Name,
    double x,
    double y,
    double RZ
)
```

```
abstract SetGridSys :
    Name : string *
    x : float *
    y : float *
    RZ : float -> int
```

Parameters

Name

Type:Â SystemString

This is the name of a grid system. If this is the name of an existing grid system, that grid system is modified, otherwise a new grid system is added.

x

Type:Â SystemDouble

The global X grid of the origin of the grid system. [L]

y

Type:Â SystemDouble

The global Y grid of the origin of the grid system. [L]

RZ

Type:Â SystemDouble

The rotation of an axis of the new grid system relative to the global grid system is defined as follows: (1) Rotate the grid system about the positive global Z-axis as defined by the RZ item. [deg]

Return ValueType:Â Int32

Returns zero if the grid system is successfully added or modified, otherwise it returns a nonzero value.

▲Remarks

Modifying the Global grid system will fail and return an error.

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new grid system
    ret = SapModel.GridSys.SetGridSys("GridSysA", 1000, 1000, 0)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
```


Introduction

```
SapModel = Nothing  
EtabsObject = Nothing
```

End Sub

▲ See Also

Reference

cGridSys Interface

ETABSv1 Namespace

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CSI API ETABS v1

cGroup Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cGroup
```

```
Public Interface cGroup
```

```
Dim instance As cGroup
```

```
public interface class cGroup
```

```
type cGroup = interface end
```

The cGroup type exposes the following members.

▲ Methods

▲	Name	Description
⇒	Count	Retrieves the number of defined groups.
⇒	Delete	Deletes the specified group.
⇒	GetAssignments	Retrieves the assignments to a specified group
⇒	GetGroup	Retrieves the group data.
⇒	GetGroup_1	Retrieves the group data. Primarily for ETABS
⇒	GetNameList	
⇒	SetGroup	Sets the group data.
⇒	SetGroup_1	Sets the group data. Primarily for ETABS

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▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cGroup Methods

The cGroup type exposes the following members.

Methods

⌵	Name	Description
⇒	<u>Count</u>	Retrieves the number of defined groups.
⇒	<u>Delete</u>	Deletes the specified group.
⇒	<u>GetAssignments</u>	Retrieves the assignments to a specified group
⇒	<u>GetGroup</u>	Retrieves the group data.
⇒	<u>GetGroup_1</u>	Retrieves the group data. Primarily for ETABS
⇒	<u>GetNameList</u>	
⇒	<u>SetGroup</u>	Sets the group data.
⇒	<u>SetGroup_1</u>	Sets the group data. Primarily for ETABS

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⌵ See Also

Reference

cGroup Interface

ETABSV1 Namespace

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CSI API ETABS v1

cGroupAddLanguageSpecificTextSet("LST20FF6263_0 Method

Retrieves the number of defined groups.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int Count()
```

Function Count As Integer

```
Dim instance As cGroup
```

```
Dim returnValue As Integer
```

```
returnValue = instance.Count()
```

```
int Count()
```

```
abstract Count : unit -> int
```

Return Value

Type: Int32

Returns the number of defined groups.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Count as Integer

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()
```

Introduction

```
'create steel deck template model
  ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'return number of defined groups
  Count = SapModel.GroupDef.Count

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

cGroup Interface

ETABSv1 Namespace

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CSI API ETABS v1

cGroupAddLanguageSpecificTextSet("LST13790709_0") Method

Deletes the specified group.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int Delete(
    string Name
)
```

```
Function Delete (
    Name As String
) As Integer
```

```
Dim instance As cGroup
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.Delete(Name)
```

```
int Delete(
    String^ Name
)
```

```
abstract Delete :
    Name : string -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing group.

Return Value

Type: [Int32](#)

Returns zero if the group is successfully deleted, otherwise it returns a nonzero value.

▲ Remarks

An error will be returned if an attempt is made to delete the Group named "ALL".

▲ Examples

VB

[Copy](#)

```
Public Sub Example()
```

cGroupspan id="LST13790709_0"AddLanguageSpecificTextSet("LST13790709_0?cpp=::|nu=.2232Delete Me

Introduction

```
Dim SapModel As cSapModel
Dim EtabsObject As cOAPI
Dim ret As Integer = -1

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'define new group
ret = SapModel.GroupDef.SetGroup("Group1")

'delete group
ret = SapModel.GroupDef.Delete("Group1")

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cGroup Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cGroupAddLanguageSpecificTextSet("LSTED40DAEE_0") Method

Retrieves the assignments to a specified group

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetAssignments(
    string Name,
    ref int NumberItems,
    ref int[] ObjectType,
    ref string[] ObjectName
)
```

```
Function GetAssignments (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef ObjectType As Integer(),
    ByRef ObjectName As String()
) As Integer
```

```
Dim instance As cGroup
Dim Name As String
Dim NumberItems As Integer
Dim ObjectType As Integer()
Dim ObjectName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetAssignments(Name,
    NumberItems, ObjectType, ObjectName)
```

```
int GetAssignments(
    String^ Name,
    int% NumberItems,
    array<int>^% ObjectType,
    array<String>^% ObjectName
)
```

```
abstract GetAssignments :
    Name : string *
    NumberItems : int byref *
    ObjectType : int[] byref *
    ObjectName : string[] byref -> int
```


Parameters

Name

Type:Â SystemString

The name of an existing group

NumberItems

Type:Â SystemInt32

The number of assignments made to the specified group

ObjectType

Type:Â SystemInt32

This is an array that includes the object type of each item in the group.

1. Point object
2. Frame object
3. Cable object
4. Tendon object
5. Area object
6. Solid object
7. Link object

ObjectName

Type:Â SystemString

This is an array that includes the name of each item in the group

Return ValueType:Â Int32

Returns zero if the group assignment is successfully retrieved, otherwise it returns a nonzero value

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberItems As Integer
    Dim ObjectType() As Integer
    Dim ObjectName() As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)
```

Introduction

```
'define new group
    ret = SapModel.GroupDef.SetGroup("Group1")

'add point objects to group
    ret = SapModel.PointObj.SetGroupAssign("3", "Group1")
    ret = SapModel.PointObj.SetGroupAssign("6", "Group1")
    ret = SapModel.PointObj.SetGroupAssign("9", "Group1")

'add frame objects to group
    ret = SapModel.FrameObj.SetGroupAssign("8", "Group1")
    ret = SapModel.FrameObj.SetGroupAssign("10", "Group1")

'get group assignments
    ret = SapModel.GroupDef.GetAssignments("Group1", NumberItems, ObjectType, ObjectName)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cGroup Interface](#)

[ETABSv1 Namespace](#)

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»

CSI API ETABS v1

cGroupAddLanguageSpecificTextSet("LST7C35E9E4_0") Method

Retrieves the group data.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetGroup(
    string Name,
    ref int Color,
    ref bool SpecifiedForSelection,
    ref bool SpecifiedForSectionCutDefinition,
    ref bool SpecifiedForSteelDesign,
    ref bool SpecifiedForConcreteDesign,
    ref bool SpecifiedForAluminumDesign,
    ref bool SpecifiedForColdFormedDesign,
    ref bool SpecifiedForStaticNActiveStage,
    ref bool SpecifiedForBridgeResponseOutput,
    ref bool SpecifiedForAutoSeismicOutput,
    ref bool SpecifiedForAutoWindOutput,
    ref bool SpecifiedForMassAndWeight
)
```

```
Function GetGroup (
    Name As String,
    ByRef Color As Integer,
    ByRef SpecifiedForSelection As Boolean,
    ByRef SpecifiedForSectionCutDefinition As Boolean,
    ByRef SpecifiedForSteelDesign As Boolean,
    ByRef SpecifiedForConcreteDesign As Boolean,
    ByRef SpecifiedForAluminumDesign As Boolean,
    ByRef SpecifiedForColdFormedDesign As Boolean,
    ByRef SpecifiedForStaticNActiveStage As Boolean,
    ByRef SpecifiedForBridgeResponseOutput As Boolean,
    ByRef SpecifiedForAutoSeismicOutput As Boolean,
    ByRef SpecifiedForAutoWindOutput As Boolean,
    ByRef SpecifiedForMassAndWeight As Boolean
) As Integer
```

```
Dim instance As cGroup
Dim Name As String
Dim Color As Integer
Dim SpecifiedForSelection As Boolean
Dim SpecifiedForSectionCutDefinition As Boolean
Dim SpecifiedForSteelDesign As Boolean
Dim SpecifiedForConcreteDesign As Boolean
Dim SpecifiedForAluminumDesign As Boolean
Dim SpecifiedForColdFormedDesign As Boolean
```

Introduction

```
Dim SpecifiedForStaticNActiveStage As Boolean
Dim SpecifiedForBridgeResponseOutput As Boolean
Dim SpecifiedForAutoSeismicOutput As Boolean
Dim SpecifiedForAutoWindOutput As Boolean
Dim SpecifiedForMassAndWeight As Boolean
Dim returnValue As Integer

returnValue = instance.GetGroup(Name,
Color, SpecifiedForSelection, SpecifiedForSectionCutDefinition,
SpecifiedForSteelDesign, SpecifiedForConcreteDesign,
SpecifiedForAluminumDesign, SpecifiedForColdFormedDesign,
SpecifiedForStaticNActiveStage,
SpecifiedForBridgeResponseOutput,
SpecifiedForAutoSeismicOutput,
SpecifiedForAutoWindOutput, SpecifiedForMassAndWeight)
```

```
int GetGroup(
String^ Name,
int% Color,
bool% SpecifiedForSelection,
bool% SpecifiedForSectionCutDefinition,
bool% SpecifiedForSteelDesign,
bool% SpecifiedForConcreteDesign,
bool% SpecifiedForAluminumDesign,
bool% SpecifiedForColdFormedDesign,
bool% SpecifiedForStaticNActiveStage,
bool% SpecifiedForBridgeResponseOutput,
bool% SpecifiedForAutoSeismicOutput,
bool% SpecifiedForAutoWindOutput,
bool% SpecifiedForMassAndWeight
)
```

```
abstract GetGroup :
    Name : string *
    Color : int byref *
    SpecifiedForSelection : bool byref *
    SpecifiedForSectionCutDefinition : bool byref *
    SpecifiedForSteelDesign : bool byref *
    SpecifiedForConcreteDesign : bool byref *
    SpecifiedForAluminumDesign : bool byref *
    SpecifiedForColdFormedDesign : bool byref *
    SpecifiedForStaticNActiveStage : bool byref *
    SpecifiedForBridgeResponseOutput : bool byref *
    SpecifiedForAutoSeismicOutput : bool byref *
    SpecifiedForAutoWindOutput : bool byref *
    SpecifiedForMassAndWeight : bool byref -> int
```

Parameters

Name

Type:Â SystemString

This is the name of an existing group.

Color

Type:Â SystemInt32

SpecifiedForSelection

Type:Â SystemBoolean

This item is True if the group is specified to be used for selection; otherwise it is False.

SpecifiedForSectionCutDefinition

Type: SystemBoolean

This item is True if the group is specified to be used for defining section cuts; otherwise it is False.

SpecifiedForSteelDesign

Type: SystemBoolean

This item is True if the group is specified to be used for defining steel frame design groups; otherwise it is False.

SpecifiedForConcreteDesign

Type: SystemBoolean

This item is True if the group is specified to be used for defining concrete frame design groups; otherwise it is False.

SpecifiedForAluminumDesign

Type: SystemBoolean

This item is True if the group is specified to be used for defining aluminum frame design groups; otherwise it is False.

SpecifiedForColdFormedDesign

Type: SystemBoolean

This item is True if the group is specified to be used for defining cold formed frame design groups; otherwise it is False.

SpecifiedForStaticNLActiveStage

Type: SystemBoolean

This item is True if the group is specified to be used for defining stages for nonlinear static analysis; otherwise it is False.

SpecifiedForBridgeResponseOutput

Type: SystemBoolean

This item is True if the group is specified to be used for reporting bridge response output; otherwise it is False.

SpecifiedForAutoSeismicOutput

Type: SystemBoolean

This item is True if the group is specified to be used for reporting auto seismic loads; otherwise it is False.

SpecifiedForAutoWindOutput

Type: SystemBoolean

This item is True if the group is specified to be used for reporting auto wind loads; otherwise it is False.

SpecifiedForMassAndWeight

Type: SystemBoolean

This item is True if the group is specified to be used for reporting group masses and weight; otherwise it is False.

Return Value

Type: Int32

Returns zero if the group data is successfully set; otherwise it returns a nonzero value.

▲ Remarks

Items SpecifiedForAluminumDesign, SpecifiedForColdFormedDesignSpecifiedForBridgeResponseOutput are not supported in ETABS.

▲ Examples

VB

Copy

Introduction

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim Color as Integer  
    Dim SpecifiedForSelection as Boolean  
    Dim SpecifiedForSectionCutDefinition as Boolean  
    Dim SpecifiedForSteelDesign as Boolean  
    Dim SpecifiedForConcreteDesign as Boolean  
    Dim SpecifiedForAluminumDesign as Boolean  
    Dim SpecifiedForColdFormedDesign as Boolean  
    Dim SpecifiedForStaticNLActiveStage as Boolean  
    Dim SpecifiedForBridgeResponseOutput as Boolean  
    Dim SpecifiedForAutoSeismicOutput as Boolean  
    Dim SpecifiedForAutoWindOutput as Boolean  
    Dim SpecifiedForMassAndWeight as Boolean  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)  
  
    'define new group  
    ret = SapModel.GroupDef.SetGroup("Group1", -1, True)  
  
    'get group data  
    ret = SapModel.GroupDef.GetGroup("Group1", Color, SpecifiedForSelection, SpecifiedForSectionCutDefinition)  
  
    'close ETABS  
    EtabsObject.ApplicationExit(False)  
  
    'clean up variables  
    SapModel = Nothing  
    EtabsObject = Nothing  
End Sub
```

▲ See Also

Reference

cGroup Interface

ETABSv1 Namespace

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CSI API ETABS v1

cGroupAddLanguageSpecificTextSet("LSTC2E021D7_0") Method

Retrieves the group data. Primarily for ETABS

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetGroup_1(
    string Name,
    ref int color,
    ref bool SpecifiedForSelection,
    ref bool SpecifiedForSectionCutDefinition,
    ref bool SpecifiedForSteelDesign,
    ref bool SpecifiedForConcreteDesign,
    ref bool SpecifiedForAluminumDesign,
    ref bool SpecifiedForStaticNActiveStage,
    ref bool SpecifiedForAutoSeismicOutput,
    ref bool SpecifiedForAutoWindOutput,
    ref bool SpecifiedForMassAndWeight,
    ref bool SpecifiedForSteelJoistDesign,
    ref bool SpecifiedForWallDesign,
    ref bool SpecifiedForBasePlateDesign,
    ref bool SpecifiedForConnectionDesign
)
```

```
Function GetGroup_1 (
    Name As String,
    ByRef color As Integer,
    ByRef SpecifiedForSelection As Boolean,
    ByRef SpecifiedForSectionCutDefinition As Boolean,
    ByRef SpecifiedForSteelDesign As Boolean,
    ByRef SpecifiedForConcreteDesign As Boolean,
    ByRef SpecifiedForAluminumDesign As Boolean,
    ByRef SpecifiedForStaticNActiveStage As Boolean,
    ByRef SpecifiedForAutoSeismicOutput As Boolean,
    ByRef SpecifiedForAutoWindOutput As Boolean,
    ByRef SpecifiedForMassAndWeight As Boolean,
    ByRef SpecifiedForSteelJoistDesign As Boolean,
    ByRef SpecifiedForWallDesign As Boolean,
    ByRef SpecifiedForBasePlateDesign As Boolean,
    ByRef SpecifiedForConnectionDesign As Boolean
) As Integer
```

```
Dim instance As cGroup
Dim Name As String
Dim color As Integer
Dim SpecifiedForSelection As Boolean
Dim SpecifiedForSectionCutDefinition As Boolean
```

cGroupspan id="LSTC2E021D7_0"AddLanguageSpecificTextSet("LSTC2E021D7_0?cpp=::|nu229;GetGroup_1")

Introduction

```
Dim SpecifiedForSteelDesign As Boolean
Dim SpecifiedForConcreteDesign As Boolean
Dim SpecifiedForAluminumDesign As Boolean
Dim SpecifiedForStaticNActiveStage As Boolean
Dim SpecifiedForAutoSeismicOutput As Boolean
Dim SpecifiedForAutoWindOutput As Boolean
Dim SpecifiedForMassAndWeight As Boolean
Dim SpecifiedForSteelJoistDesign As Boolean
Dim SpecifiedForWallDesign As Boolean
Dim SpecifiedForBasePlateDesign As Boolean
Dim SpecifiedForConnectionDesign As Boolean
Dim returnValue As Integer

returnValue = instance.GetGroup_1(Name,
color, SpecifiedForSelection, SpecifiedForSectionCutDefinition,
SpecifiedForSteelDesign, SpecifiedForConcreteDesign,
SpecifiedForAluminumDesign, SpecifiedForStaticNActiveStage,
SpecifiedForAutoSeismicOutput,
SpecifiedForAutoWindOutput, SpecifiedForMassAndWeight,
SpecifiedForSteelJoistDesign, SpecifiedForWallDesign,
SpecifiedForBasePlateDesign, SpecifiedForConnectionDesign)

int GetGroup_1(
String^ Name,
int% color,
bool% SpecifiedForSelection,
bool% SpecifiedForSectionCutDefinition,
bool% SpecifiedForSteelDesign,
bool% SpecifiedForConcreteDesign,
bool% SpecifiedForAluminumDesign,
bool% SpecifiedForStaticNActiveStage,
bool% SpecifiedForAutoSeismicOutput,
bool% SpecifiedForAutoWindOutput,
bool% SpecifiedForMassAndWeight,
bool% SpecifiedForSteelJoistDesign,
bool% SpecifiedForWallDesign,
bool% SpecifiedForBasePlateDesign,
bool% SpecifiedForConnectionDesign
)

abstract GetGroup_1 :
    Name : string *
    color : int byref *
    SpecifiedForSelection : bool byref *
    SpecifiedForSectionCutDefinition : bool byref *
    SpecifiedForSteelDesign : bool byref *
    SpecifiedForConcreteDesign : bool byref *
    SpecifiedForAluminumDesign : bool byref *
    SpecifiedForStaticNActiveStage : bool byref *
    SpecifiedForAutoSeismicOutput : bool byref *
    SpecifiedForAutoWindOutput : bool byref *
    SpecifiedForMassAndWeight : bool byref *
    SpecifiedForSteelJoistDesign : bool byref *
    SpecifiedForWallDesign : bool byref *
    SpecifiedForBasePlateDesign : bool byref *
    SpecifiedForConnectionDesign : bool byref -> int
```

Parameters

Name

Parameters

Type:Â SystemString

This is the name of an existing group.

color

Type:Â SystemInt32

The display color for the group specified as a Integer.

SpecifiedForSelection

Type:Â SystemBoolean

This item is True if the group is specified to be used for selection; otherwise it is False.

SpecifiedForSectionCutDefinition

Type:Â SystemBoolean

This item is True if the group is specified to be used for defining section cuts; otherwise it is False.

SpecifiedForSteelDesign

Type:Â SystemBoolean

This item is True if the group is specified to be used for defining steel frame design groups; otherwise it is False.

SpecifiedForConcreteDesign

Type:Â SystemBoolean

This item is True if the group is specified to be used for defining concrete frame design groups; otherwise it is False.

SpecifiedForAluminumDesign

Type:Â SystemBoolean

This item is True if the group is specified to be used for defining aluminum frame design groups; otherwise it is False.

SpecifiedForStaticNLActiveStage

Type:Â SystemBoolean

This item is True if the group is specified to be used for defining stages for nonlinear static analysis; otherwise it is False.

SpecifiedForAutoSeismicOutput

Type:Â SystemBoolean

This item is True if the group is specified to be used for reporting auto seismic loads; otherwise it is False.

SpecifiedForAutoWindOutput

Type:Â SystemBoolean

This item is True if the group is specified to be used for reporting auto wind loads; otherwise it is False.

SpecifiedForMassAndWeight

Type:Â SystemBoolean

This item is True if the group is specified to be used for reporting group masses and weight; otherwise it is False.

SpecifiedForSteelJoistDesign

Type:Â SystemBoolean

This item is True if the group is specified to be used for defining steel joist design groups; otherwise it is False.

SpecifiedForWallDesign

Type:Â SystemBoolean

This item is True if the group is specified to be used for defining wall design groups; otherwise it is False.

SpecifiedForBasePlateDesign

Type: SystemBoolean

This item is True if the group is specified to be used for defining base plate design groups; otherwise it is False.

SpecifiedForConnectionDesign

Type: SystemBoolean

This item is True if the group is specified to be used for defining connection design groups; otherwise it is False.

Return Value

Type: Int32

Returns zero if the group data is successfully set; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Color as Integer
    Dim SpecifiedForSelection as Boolean
    Dim SpecifiedForSectionCutDefinition as Boolean
    Dim SpecifiedForSteelDesign as Boolean
    Dim SpecifiedForConcreteDesign as Boolean
    Dim SpecifiedForAluminumDesign as Boolean
    Dim SpecifiedForStaticNLActiveStage as Boolean
    Dim SpecifiedForAutoSeismicOutput as Boolean
    Dim SpecifiedForAutoWindOutput as Boolean
    Dim SpecifiedForMassAndWeight as Boolean
    Dim SpecifiedForSteelJoistDesign as Boolean
    Dim SpecifiedForWallDesign as Boolean
    Dim SpecifiedForBasePlateDesign as Boolean
    Dim SpecifiedForConnectionDesign as Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'define new group
    ret = SapModel.GroupDef.SetGroup_1("Group1", -1, True)

    'get group data
    ret = SapModel.GroupDef.GetGroup_1("Group1", Color, SpecifiedForSelection, SpecifiedForSe

    'close ETABS
    EtabsObject.ApplicationExit(False)
```

```
'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

📌 See Also

Reference

cGroup Interface

ETABSv1 Namespace

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CSI API ETABS v1

cGroupAddLanguageSpecificTextSet("LSTCA3669D9_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNameList(
    ref int NumberNames,
    ref string[] MyName
)
```

```
Function GetNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cGroup
Dim NumberNames As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetNameList(NumberNames,
    MyName)
```

```
int GetNameList(
    int% NumberNames,
    array<String^>^% MyName
)
```

```
abstract GetNameList :
    NumberNames : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberNames

Type: [System.Int32](#)

MyName

Type: [System.String](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cGroup Interface

ETABSv1 Namespace

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i»j

CSI API ETABS v1

cGroupAddLanguageSpecificTextSet("LSTA3773B03_0 Method

Sets the group data.

Namespace: ETABSV1**Assembly:** ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#VBC++F#Copy

```
int SetGroup(
    string Name,
    int Color = -1,
    bool SpecifiedForSelection = true,
    bool SpecifiedForSectionCutDefinition = true,
    bool SpecifiedForSteelDesign = true,
    bool SpecifiedForConcreteDesign = true,
    bool SpecifiedForAluminumDesign = true,
    bool SpecifiedForColdFormedDesign = true,
    bool SpecifiedForStaticNActiveStage = true,
    bool SpecifiedForBridgeResponseOutput = true,
    bool SpecifiedForAutoSeismicOutput = false,
    bool SpecifiedForAutoWindOutput = false,
    bool SpecifiedForMassAndWeight = true
)
```

```
Function SetGroup (
    Name As String,
    Optional As Integer = -1,
    SpecifiedForSelection As Boolean = true,
    SpecifiedForSectionCutDefinition As Boolean = true,
    SpecifiedForSteelDesign As Boolean = true,
    SpecifiedForConcreteDesign As Boolean = true,
    SpecifiedForAluminumDesign As Boolean = true,
    SpecifiedForColdFormedDesign As Boolean = true,
    SpecifiedForStaticNActiveStage As Boolean = true,
    SpecifiedForBridgeResponseOutput As Boolean = true,
    SpecifiedForAutoSeismicOutput As Boolean = false,
    SpecifiedForAutoWindOutput As Boolean = false,
    SpecifiedForMassAndWeight As Boolean = true
) As Integer
```

```
Dim instance As cGroup
Dim Name As String
Dim Color As Integer
Dim SpecifiedForSelection As Boolean
Dim SpecifiedForSectionCutDefinition As Boolean
Dim SpecifiedForSteelDesign As Boolean
Dim SpecifiedForConcreteDesign As Boolean
Dim SpecifiedForAluminumDesign As Boolean
Dim SpecifiedForColdFormedDesign As Boolean
```

Introduction

```
Dim SpecifiedForStaticNLAActiveStage As Boolean
Dim SpecifiedForBridgeResponseOutput As Boolean
Dim SpecifiedForAutoSeismicOutput As Boolean
Dim SpecifiedForAutoWindOutput As Boolean
Dim SpecifiedForMassAndWeight As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetGroup(Name,
Color, SpecifiedForSelection, SpecifiedForSectionCutDefinition,
SpecifiedForSteelDesign, SpecifiedForConcreteDesign,
SpecifiedForAluminumDesign, SpecifiedForColdFormedDesign,
SpecifiedForStaticNLAActiveStage,
SpecifiedForBridgeResponseOutput,
SpecifiedForAutoSeismicOutput,
SpecifiedForAutoWindOutput, SpecifiedForMassAndWeight)
```

```
int SetGroup(
String^ Name,
int Color = -1,
bool SpecifiedForSelection = true,
bool SpecifiedForSectionCutDefinition = true,
bool SpecifiedForSteelDesign = true,
bool SpecifiedForConcreteDesign = true,
bool SpecifiedForAluminumDesign = true,
bool SpecifiedForColdFormedDesign = true,
bool SpecifiedForStaticNLAActiveStage = true,
bool SpecifiedForBridgeResponseOutput = true,
bool SpecifiedForAutoSeismicOutput = false,
bool SpecifiedForAutoWindOutput = false,
bool SpecifiedForMassAndWeight = true
)
```

```
abstract SetGroup :
    Name : string *
    ?Color : int *
    ?SpecifiedForSelection : bool *
    ?SpecifiedForSectionCutDefinition : bool *
    ?SpecifiedForSteelDesign : bool *
    ?SpecifiedForConcreteDesign : bool *
    ?SpecifiedForAluminumDesign : bool *
    ?SpecifiedForColdFormedDesign : bool *
    ?SpecifiedForStaticNLAActiveStage : bool *
    ?SpecifiedForBridgeResponseOutput : bool *
    ?SpecifiedForAutoSeismicOutput : bool *
    ?SpecifiedForAutoWindOutput : bool *
    ?SpecifiedForMassAndWeight : bool
(* Defaults:
    let _Color = defaultArg Color -1
    let _SpecifiedForSelection = defaultArg SpecifiedForSelection true
    let _SpecifiedForSectionCutDefinition = defaultArg SpecifiedForSectionCutDefinition true
    let _SpecifiedForSteelDesign = defaultArg SpecifiedForSteelDesign true
    let _SpecifiedForConcreteDesign = defaultArg SpecifiedForConcreteDesign true
    let _SpecifiedForAluminumDesign = defaultArg SpecifiedForAluminumDesign true
    let _SpecifiedForColdFormedDesign = defaultArg SpecifiedForColdFormedDesign true
    let _SpecifiedForStaticNLAActiveStage = defaultArg SpecifiedForStaticNLAActiveStage true
    let _SpecifiedForBridgeResponseOutput = defaultArg SpecifiedForBridgeResponseOutput true
    let _SpecifiedForAutoSeismicOutput = defaultArg SpecifiedForAutoSeismicOutput false
    let _SpecifiedForAutoWindOutput = defaultArg SpecifiedForAutoWindOutput false
    let _SpecifiedForMassAndWeight = defaultArg SpecifiedForMassAndWeight true
*)
-> int
```

Parameters**Name**Type:Â SystemString

This is the name of a group. If this is the name of an existing group, that group is modified, otherwise a new group is added.

Color (Optional)Type:Â SystemInt32**SpecifiedForSelection (Optional)**Type:Â SystemBoolean

This item is True if the group is specified to be used for selection; otherwise it is False.

SpecifiedForSectionCutDefinition (Optional)Type:Â SystemBoolean

This item is True if the group is specified to be used for defining section cuts; otherwise it is False.

SpecifiedForSteelDesign (Optional)Type:Â SystemBoolean

This item is True if the group is specified to be used for defining steel frame design groups; otherwise it is False.

SpecifiedForConcreteDesign (Optional)Type:Â SystemBoolean

This item is True if the group is specified to be used for defining concrete frame design groups; otherwise it is False.

SpecifiedForAluminumDesign (Optional)Type:Â SystemBoolean

This item is True if the group is specified to be used for defining aluminum frame design groups; otherwise it is False.

SpecifiedForColdFormedDesign (Optional)Type:Â SystemBoolean

This item is True if the group is specified to be used for defining cold formed frame design groups; otherwise it is False.

SpecifiedForStaticNLActiveStage (Optional)Type:Â SystemBoolean

This item is True if the group is specified to be used for defining stages for nonlinear static analysis; otherwise it is False.

SpecifiedForBridgeResponseOutput (Optional)Type:Â SystemBoolean

This item is True if the group is specified to be used for reporting bridge response output; otherwise it is False.

SpecifiedForAutoSeismicOutput (Optional)Type:Â SystemBoolean

This item is True if the group is specified to be used for reporting auto seismic loads; otherwise it is False.

SpecifiedForAutoWindOutput (Optional)Type:Â SystemBoolean

This item is True if the group is specified to be used for reporting auto wind loads; otherwise it is False.

SpecifiedForMassAndWeight (Optional)Type:Â SystemBoolean

This item is True if the group is specified to be used for reporting group masses

and weight; otherwise it is False.

Return Value

Type: Int32

Returns zero if the group data is successfully set; otherwise it returns a nonzero value.

Remarks

Items SpecifiedForAluminumDesign, SpecifiedForColdFormedDesignSpecifiedForBridgeResponseOutput are not supported in ETABS.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'define new group
    ret = SapModel.GroupDef.SetGroup("Group1")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

cGroup Interface

ETABSv1 Namespace

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CSI API ETABS v1

cGroupAddLanguageSpecificTextSet("LST8FB836F9_0 Method

Sets the group data. Primarily for ETABS

Namespace: Â ETABSv1

Assembly: Â ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▀ Syntax

C#

VB

C++

F#

Copy

```
int SetGroup_1(
    string Name,
    int color = -1,
    bool SpecifiedForSelection = true,
    bool SpecifiedForSectionCutDefinition = true,
    bool SpecifiedForSteelDesign = true,
    bool SpecifiedForConcreteDesign = true,
    bool SpecifiedForAluminumDesign = true,
    bool SpecifiedForStaticNLActiveStage = true,
    bool SpecifiedForAutoSeismicOutput = false,
    bool SpecifiedForAutoWindOutput = false,
    bool SpecifiedForMassAndWeight = true,
    bool SpecifiedForSteelJoistDesign = true,
    bool SpecifiedForWallDesign = true,
    bool SpecifiedForBasePlateDesign = true,
    bool SpecifiedForConnectionDesign = true
)
```

Function SetGroup_1 (

Name As String,

Optional As Integer = -1,

SpecifiedForSelection As Boolean = true,

SpecifiedForSectionCutDefinition As Boolean = true,

SpecifiedForSteelDesign As Boolean = true,

SpecifiedForConcreteDesign As Boolean = true,

SpecifiedForAluminumDesign As Boolean = true,

SpecifiedForStaticNLActiveStage As Boolean = true,

SpecifiedForAutoSeismicOutput As Boolean = false,

SpecifiedForAutoWindOutput As Boolean = false,

SpecifiedForMassAndWeight As Boolean = true,

SpecifiedForSteelJoistDesign As Boolean = true,

SpecifiedForWallDesign As Boolean = true,

SpecifiedForBasePlateDesign As Boolean = true,

SpecifiedForConnectionDesign As Boolean = true

) As Integer

Dim instance As cGroup

Dim Name As String

Dim color As Integer

Dim SpecifiedForSelection As Boolean

Dim SpecifiedForSectionCutDefinition As Boolean

Introduction

```
Dim SpecifiedForSteelDesign As Boolean
Dim SpecifiedForConcreteDesign As Boolean
Dim SpecifiedForAluminumDesign As Boolean
Dim SpecifiedForStaticNActiveStage As Boolean
Dim SpecifiedForAutoSeismicOutput As Boolean
Dim SpecifiedForAutoWindOutput As Boolean
Dim SpecifiedForMassAndWeight As Boolean
Dim SpecifiedForSteelJoistDesign As Boolean
Dim SpecifiedForWallDesign As Boolean
Dim SpecifiedForBasePlateDesign As Boolean
Dim SpecifiedForConnectionDesign As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetGroup_1(Name,
color, SpecifiedForSelection, SpecifiedForSectionCutDefinition,
SpecifiedForSteelDesign, SpecifiedForConcreteDesign,
SpecifiedForAluminumDesign, SpecifiedForStaticNActiveStage,
SpecifiedForAutoSeismicOutput,
SpecifiedForAutoWindOutput, SpecifiedForMassAndWeight,
SpecifiedForSteelJoistDesign, SpecifiedForWallDesign,
SpecifiedForBasePlateDesign, SpecifiedForConnectionDesign)
```

```
int SetGroup_1(
String^ Name,
int color = -1,
bool SpecifiedForSelection = true,
bool SpecifiedForSectionCutDefinition = true,
bool SpecifiedForSteelDesign = true,
bool SpecifiedForConcreteDesign = true,
bool SpecifiedForAluminumDesign = true,
bool SpecifiedForStaticNActiveStage = true,
bool SpecifiedForAutoSeismicOutput = false,
bool SpecifiedForAutoWindOutput = false,
bool SpecifiedForMassAndWeight = true,
bool SpecifiedForSteelJoistDesign = true,
bool SpecifiedForWallDesign = true,
bool SpecifiedForBasePlateDesign = true,
bool SpecifiedForConnectionDesign = true
)
```

```
abstract SetGroup_1 :
    Name : string *
    ?color : int *
    ?SpecifiedForSelection : bool *
    ?SpecifiedForSectionCutDefinition : bool *
    ?SpecifiedForSteelDesign : bool *
    ?SpecifiedForConcreteDesign : bool *
    ?SpecifiedForAluminumDesign : bool *
    ?SpecifiedForStaticNActiveStage : bool *
    ?SpecifiedForAutoSeismicOutput : bool *
    ?SpecifiedForAutoWindOutput : bool *
    ?SpecifiedForMassAndWeight : bool *
    ?SpecifiedForSteelJoistDesign : bool *
    ?SpecifiedForWallDesign : bool *
    ?SpecifiedForBasePlateDesign : bool *
    ?SpecifiedForConnectionDesign : bool
(* Defaults:
    let _color = defaultArg color -1
    let _SpecifiedForSelection = defaultArg SpecifiedForSelection true
    let _SpecifiedForSectionCutDefinition = defaultArg SpecifiedForSectionCutDefinition true
    let _SpecifiedForSteelDesign = defaultArg SpecifiedForSteelDesign true
    let _SpecifiedForConcreteDesign = defaultArg SpecifiedForConcreteDesign true
```

Introduction

```
let _SpecifiedForAluminumDesign = defaultArg SpecifiedForAluminumDesign true
let _SpecifiedForStaticNLActiveStage = defaultArg SpecifiedForStaticNLActiveStage true
let _SpecifiedForAutoSeismicOutput = defaultArg SpecifiedForAutoSeismicOutput false
let _SpecifiedForAutoWindOutput = defaultArg SpecifiedForAutoWindOutput false
let _SpecifiedForMassAndWeight = defaultArg SpecifiedForMassAndWeight true
let _SpecifiedForSteelJoistDesign = defaultArg SpecifiedForSteelJoistDesign true
let _SpecifiedForWallDesign = defaultArg SpecifiedForWallDesign true
let _SpecifiedForBasePlateDesign = defaultArg SpecifiedForBasePlateDesign true
let _SpecifiedForConnectionDesign = defaultArg SpecifiedForConnectionDesign true
*)
-> int
```

Parameters

Name

Type:Â SystemString

This is the name of a group. If this is the name of an existing group, that group is modified, otherwise a new group is added.

color (Optional)

Type:Â SystemInt32

The display color for the group specified as a Integer. If this value is input as â 1, the program automatically selects a display color for the group.

SpecifiedForSelection (Optional)

Type:Â SystemBoolean

This item is True if the group is specified to be used for selection; otherwise it is False.

SpecifiedForSectionCutDefinition (Optional)

Type:Â SystemBoolean

This item is True if the group is specified to be used for defining section cuts; otherwise it is False.

SpecifiedForSteelDesign (Optional)

Type:Â SystemBoolean

This item is True if the group is specified to be used for defining steel frame design groups; otherwise it is False.

SpecifiedForConcreteDesign (Optional)

Type:Â SystemBoolean

This item is True if the group is specified to be used for defining concrete frame design groups; otherwise it is False.

SpecifiedForAluminumDesign (Optional)

Type:Â SystemBoolean

This item is True if the group is specified to be used for defining aluminum frame design groups; otherwise it is False.

SpecifiedForStaticNLActiveStage (Optional)

Type:Â SystemBoolean

This item is True if the group is specified to be used for defining stages for nonlinear static analysis; otherwise it is False.

SpecifiedForAutoSeismicOutput (Optional)

Type:Â SystemBoolean

This item is True if the group is specified to be used for reporting auto seismic loads; otherwise it is False.

SpecifiedForAutoWindOutput (Optional)

Type:Â SystemBoolean

This item is True if the group is specified to be used for reporting auto wind

loads; otherwise it is False.

SpecifiedForMassAndWeight (Optional)

Type:Â SystemBoolean

This item is True if the group is specified to be used for reporting group masses and weight; otherwise it is False.

SpecifiedForSteelJoistDesign (Optional)

Type:Â SystemBoolean

This item is True if the group is specified to be used for defining steel joist design groups; otherwise it is False.

SpecifiedForWallDesign (Optional)

Type:Â SystemBoolean

This item is True if the group is specified to be used for defining wall design groups; otherwise it is False.

SpecifiedForBasePlateDesign (Optional)

Type:Â SystemBoolean

This item is True if the group is specified to be used for defining base plate design groups; otherwise it is False.

SpecifiedForConnectionDesign (Optional)

Type:Â SystemBoolean

This item is True if the group is specified to be used for defining connection design groups; otherwise it is False.

Return Value

Type:Â Int32

Returns zero if the group data is successfully set; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'define new group
    ret = SapModel.GroupDef.SetGroup_1("Group1")

    'close ETABS
    EtabsObject.ApplicationExit(False)
```

Return Value

```
'clean up variables  
  SapModel = Nothing  
  EtabsObject = Nothing  
End Sub
```

📌 See Also

Reference

cGroup Interface

ETABSv1 Namespace

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CSI API ETABS v1

cHelper Interface

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
public interface cHelper
```

```
Public Interface cHelper
```

```
Dim instance As cHelper
```

```
public interface class cHelper
```

```
type cHelper = interface end
```

The cHelper type exposes the following members.

Methods

<u>Name</u>	<u>Description</u>
<u>CreateObject</u>	Starts the program at the given path.
<u>CreateObjectHost</u>	Starts the program at the given path on a given computer using the default TCP port.
<u>CreateObjectHostPort</u>	Starts the program at the given path on a given computer using the given TCP port.
<u>CreateObjectProgID</u>	Starts the program and returns the API object for the given program ID.
<u>CreateObjectProgIDHost</u>	Starts the program and returns the API object for the given program ID on a given computer using the default TCP port.
<u>CreateObjectProgIDHostPort</u>	Starts the program and returns the API object for the given program ID on a given computer using the given TCP port.
<u>GetOAPIVersionNumber</u>	Retrieves the API version.
<u>GetObject</u>	Attaches to the active running instance of the program.
<u>GetObjectHost</u>	Attaches to the active running instance of the program on a given computer using the default TCP port.
<u>GetObjectHostPort</u>	Attaches to the active running instance of the program on a given computer using the given TCP

port.

Attaches to the running instance of the program with the given process id.

 [GetObjectProcess](#)

[Top](#)

▲ [See Also](#)

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cHelper Methods

The cHelper type exposes the following members.

Methods

⚡	Name	Description
⇒	<u>CreateObject</u>	Starts the program at the given path.
⇒	<u>CreateObjectHost</u>	Starts the program at the given path on a given computer using the default TCP port.
⇒	<u>CreateObjectHostPort</u>	Starts the program at the given path on a given computer using the given TCP port.
⇒	<u>CreateObjectProgID</u>	Starts the program and returns the API object for the given program ID.
⇒	<u>CreateObjectProgIDHost</u>	Starts the program and returns the API object for the given program ID on a given computer using the default TCP port.
⇒	<u>CreateObjectProgIDHostPort</u>	Starts the program and returns the API object for the given program ID on a given computer using the given TCP port.
⇒	<u>GetOAPIVersionNumber</u>	Retrieves the API version.
⇒	<u>GetObject</u>	Attaches to the active running instance of the program.
⇒	<u>GetObjectHost</u>	Attaches to the active running instance of the program on a given computer using the default TCP port.
⇒	<u>GetObjectHostPort</u>	Attaches to the active running instance of the program on a given computer using the given TCP port.
⇒	<u>GetObjectProcess</u>	Attaches to the running instance of the program with the given process id.

Top

⚡ See Also

Reference

cHelper Interface

ETABSv1 Namespace

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CSI API ETABS v1

cHelperAddLanguageSpecificTextSet("LSTC7D7B990_0") Method

Starts the program at the given path.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cOAPI CreateObject(
    string fullPath
)
```

```
Function CreateObject (
    fullPath As String
) As cOAPI
```

```
Dim instance As cHelper
Dim fullPath As String
Dim returnValue As cOAPI
```

```
returnValue = instance.CreateObject(fullPath)
```

```
cOAPI^ CreateObject(
    String^ fullPath
)
```

```
abstract CreateObject :
    fullPath : string -> cOAPI
```

Parameters

fullPath

Type: [SystemString](#)

The full path to the program executable.

Return Value

Type: [cOAPI](#)

An instance of [cOAPI](#) if successful, nothing otherwise.

▲ Remarks

▲ Examples

[VB](#)

[Copy](#)

```
Public Sub Example()
    Dim myHelper As cHelper
    Dim SapModel As cSapModel
```

cHelperspan id="LSTC7D7B990_0"AddLanguageSpecificTextSet("LSTC7D7B990_0?cpp=::|nu23")CreateC

Introduction

```
Dim EtabsObject As cOAPI
Dim ret As Integer = -1

'create ETABS object
myHelper = New Helper
EtabsObject = myHelper.CreateObject("C:\Program Files\Computers and Structures\ETABS 19\ETABS

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
myHelper = Nothing
End Sub
```

🔍 See Also

Reference

[cHelper Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cHelperAddLanguageSpecificTextSet("LST1CE1B393_0") Method

Starts the program at the given path on a given computer using the default TCP port.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
c0API CreateObjectHost(
    string hostName,
    string fullPath
)
```

```
Function CreateObjectHost (
    hostName As String,
    fullPath As String
) As c0API
```

```
Dim instance As cHelper
Dim hostName As String
Dim fullPath As String
Dim returnValue As c0API
```

```
returnValue = instance.CreateObjectHost(hostName,
    fullPath)
```

```
c0API^ CreateObjectHost(
    String^ hostName,
    String^ fullPath
)
```

```
abstract CreateObjectHost :
    hostName : string *
    fullPath : string -> c0API
```

Parameters

hostName

Type: SystemString

The host computer name.

fullPath

Type: SystemString

The full path to the program executable on the host computer.

Return Value

Type: cOAPI

An instance of cOAPI if successful, nothing otherwise.

▲ **Remarks**

CSiAPIService.exe must be running on the host computer and listening to the default TCP port for the call to succeed.

The default TCP port can be overridden by setting the environment variable ETABSv1_cOAPI_DEFAULT_PORT on both the Main and the Host computers.

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim myHelper As cHelper
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    myHelper = New Helper
    EtabsObject = myHelper.CreateObjectHost("RemoteServer", "C:\Program Files\Computers and Structures\ETABS\ETABSv1\cOAPI\cOAPI.dll")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
    myHelper = Nothing
End Sub
```

▲ **See Also**

Reference

cHelper Interface

ETABSv1 Namespace

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CSI API ETABS v1

cHelperAddLanguageSpecificTextSet("LSTE4A79BCE_0") Method

Starts the program at the given path on a given computer using the given TCP port.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cOAPI CreateObjectHostPort(
    string hostName,
    int portNumber,
    string fullPath
)
```

```
Function CreateObjectHostPort (
    hostName As String,
    portNumber As Integer,
    fullPath As String
) As cOAPI
```

```
Dim instance As cHelper
Dim hostName As String
Dim portNumber As Integer
Dim fullPath As String
Dim returnValue As cOAPI
```

```
returnValue = instance.CreateObjectHostPort(hostName,
    portNumber, fullPath)
```

```
cOAPI^ CreateObjectHostPort(
    String^ hostName,
    int portNumber,
    String^ fullPath
)
```

```
abstract CreateObjectHostPort :
    hostName : string *
    portNumber : int *
    fullPath : string -> cOAPI
```

Parameters

hostName

Type: [SystemString](#)

The host computer name.

portNumber

Type: SystemInt32

The TCP port to connect to the host computer.

fullPath

Type: SystemString

The full path to the program executable on the host computer.

Return Value

Type: cOAPI

An instance of cOAPI if successful, nothing otherwise.

Remarks

CSiAPIService.exe must be running on the host computer and listening to the given TCP port for the call to succeed.

Examples

VB

Copy

```
Public Sub Example()  
    Dim myHelper As cHelper  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
  
    'create ETABS object  
    myHelper = New Helper  
    EtabsObject = myHelper.CreateObjectHostPort("RemoteServer", 8500, "C:\Program Files\Computers  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'close ETABS  
    EtabsObject.ApplicationExit(False)  
  
    'clean up variables  
    SapModel = Nothing  
    EtabsObject = Nothing  
    myHelper = Nothing  
End Sub
```

See Also

Reference

cHelper Interface

ETABSv1 Namespace

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CSI API ETABS v1

cHelperAddLanguageSpecificTextSet("LST85084D8D_0") Method

Starts the program and returns the API object for the given program ID.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
cOAPI CreateObjectProgID(
    string progID
)
```

```
Function CreateObjectProgID (
    progID As String
) As cOAPI
```

```
Dim instance As cHelper
Dim progID As String
Dim returnValue As cOAPI
```

```
returnValue = instance.CreateObjectProgID(progID)
```

```
cOAPI^ CreateObjectProgID(
    String^ progID
)
```

```
abstract CreateObjectProgID :
    progID : string -> cOAPI
```

Parameters

progID

Type: SystemString

The program ID of the API object.

Return Value

Type: cOAPI

An instance of cOAPI if successful, nothing otherwise.

■ Remarks

Searches the registry for the newest version unless overridden with an environment variable containing the full path of the API object's assembly. The overriding environment variable is found by replacing the "."s with "_" in the program id and appending "_PATH". e.g. the environment variable of "CSI.ETABS.API.ETABSObject" is "CSI_ETABS_API_ETABSObject_PATH".

cHelperspan id="LST85084D8D_0"AddLanguageSpecificTextSet("LST85084D8D_0?cpp=::|nu23);CreateO

Examples

VB

Copy

```
Public Sub Example()
    Dim myHelper As cHelper
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    myHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
    myHelper = Nothing
End Sub
```

See Also

Reference

cHelper Interface

ETABSv1 Namespace

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CSI API ETABS v1

cHelperAddLanguageSpecificTextSet("LST5FD2BFAC_0") Method

Starts the program and returns the API object for the given program ID on a given computer using the default TCP port.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cOAPI CreateObjectProgIDHost(
    string hostName,
    string progID
)
```

```
Function CreateObjectProgIDHost (
    hostName As String,
    progID As String
) As cOAPI
```

```
Dim instance As cHelper
Dim hostName As String
Dim progID As String
Dim returnValue As cOAPI
```

```
returnValue = instance.CreateObjectProgIDHost(hostName,
    progID)
```

```
cOAPI^ CreateObjectProgIDHost(
    String^ hostName,
    String^ progID
)
```

```
abstract CreateObjectProgIDHost :
    hostName : string *
    progID : string -> cOAPI
```

Parameters

hostName

Type: [SystemString](#)

The host computer name.

progID

Type: [SystemString](#)

The program ID of the API object.

Return Value

Type: cOAPI

An instance of cOAPI if successful, nothing otherwise.

▀ **Remarks**

CSiAPIService.exe must be running on the host computer and listening to the default TCP port for the call to succeed.

▀ **Examples**

VB

Copy

```
Public Sub Example()
    Dim myHelper As cHelper
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    myHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgIDHost("RemoteServer", "CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
    myHelper = Nothing
End Sub
```

▀ **See Also**

Reference

cHelper Interface

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CSI API ETABS v1

cHelperAddLanguageSpecificTextSet("LST616E402F_0 Method

Starts the program and returns the API object for the given program ID on a given computer using the given TCP port.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cOAPI CreateObjectProgIDHostPort(
    string hostName,
    int portNumber,
    string progID
)
```

```
Function CreateObjectProgIDHostPort (
    hostName As String,
    portNumber As Integer,
    progID As String
) As cOAPI
```

```
Dim instance As cHelper
Dim hostName As String
Dim portNumber As Integer
Dim progID As String
Dim returnValue As cOAPI
```

```
returnValue = instance.CreateObjectProgIDHostPort(hostName,
    portNumber, progID)
```

```
cOAPI^ CreateObjectProgIDHostPort(
    String^ hostName,
    int portNumber,
    String^ progID
)
```

```
abstract CreateObjectProgIDHostPort :
    hostName : string *
    portNumber : int *
    progID : string -> cOAPI
```

Parameters

hostName

Type: [SystemString](#)

The host computer name.

portNumber

Type: SystemInt32

The TCP port to connect to the host computer.

progID

Type: SystemString

The program ID of the API object.

Return Value

Type: cOAPI

An instance of cOAPI if successful, nothing otherwise.

Remarks

CSiAPIService.exe must be running on the host computer and listening to the given TCP port for the call to succeed.

Examples

VB

Copy

```
Public Sub Example()
    Dim myHelper As cHelper
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    myHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgIDHostPort("RemoteServer", 8500, "CSI.ETABS.API.ETABS")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
    myHelper = Nothing
End Sub
```

See Also

Reference

cHelper Interface

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CSI API ETABS v1

cHelperAddLanguageSpecificTextSet("LST309572A_0? Method

Retrieves the API version.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
double GetOAPIVersionNumber()
```

```
Function GetOAPIVersionNumber As Double
```

```
Dim instance As cHelper
```

```
Dim returnValue As Double
```

```
returnValue = instance.GetOAPIVersionNumber()
```

```
double GetOAPIVersionNumber()
```

```
abstract GetOAPIVersionNumber : unit -> float
```

Return Value

Type: Double

The API version.

▲ Remarks

The returned API version can be compared to GetOAPIVersionNumber to determine compatibility with the program.

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim myHelper As cHelper
    Dim EtabsObject As cOAPI
    Dim clientAPIVersion As Double
    Dim programAPIVersion As Double
    Dim ret As Integer = -1

    'create ETABS object
    myHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()
```

Introduction

```
'get client API version
clientAPIVersion = myHelper.GetOAPIVersionNumber()

'get program API version
programAPIVersion = EtabsObject.GetOAPIVersionNumber()

'API compatibility check
If clientAPIVersion > programAPIVersion Then
    'API client uses a newer version of the API than the program.
    'All API functionality may be not be available.
    Exit Sub
End If

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
EtabsObject = Nothing
myHelper = Nothing
End Sub
```

➤ See Also

Reference

cHelper Interface

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CSI API ETABS v1

cHelperAddLanguageSpecificTextSet("LSTA81B1602_0") Method

Attaches to the active running instance of the program.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cOAPI GetObject(
    string typeName
)
```

```
Function GetObject (
    typeName As String
) As cOAPI
```

```
Dim instance As cHelper
Dim typeName As String
Dim returnValue As cOAPI
```

```
returnValue = instance.GetObject(typeName)
```

```
cOAPI^ GetObject(
    String^ typeName
)
```

```
abstract GetObject :
    typeName : string -> cOAPI
```

Parameters

typeName

Type: [SystemString](#)

The program ID of the API object.

Return Value

Type: [cOAPI](#)

An instance of [cOAPI](#) if successful, nothing otherwise.

▲ Remarks

▲ Examples

[VB](#)

[Copy](#)

```
Public Sub Example()
    Dim myHelper As cHelper
    Dim SapModel As cSapModel
```

cHelperspan id="LSTA81B1602_0"AddLanguageSpecificTextSet("LSTA81B1602_0?cpp=::|nu=2324GetObje

Introduction

```
Dim EtabsObject As cOAPI
Dim ret As Integer = -1

'create ETABS object
myHelper = New Helper
EtabsObject = myHelper.GetObject("CSI.ETABS.API.ETABSObject")

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
myHelper = Nothing
End Sub
```

▲ See Also

Reference

cHelper Interface

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CSI API ETABS v1

cHelperAddLanguageSpecificTextSet("LST4BD09954_0") Method

Attaches to the active running instance of the program on a given computer using the default TCP port.

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
cOAPI GetObjectHost(
    string hostName,
    string progID
)
```

```
Function GetObjectHost (
    hostName As String,
    progID As String
) As cOAPI
```

```
Dim instance As cHelper
Dim hostName As String
Dim progID As String
Dim returnValue As cOAPI
```

```
returnValue = instance.GetObjectHost(hostName,
    progID)
```

```
cOAPI^ GetObjectHost(
    String^ hostName,
    String^ progID
)
```

```
abstract GetObjectHost :
    hostName : string *
    progID : string -> cOAPI
```

Parameters

hostName

Type: $\hat{A}$ SystemString

The host computer name.

progID

Type: $\hat{A}$ SystemString

The program ID of the API object.

Return Value

Type: cOAPI

An instance of cOAPI if successful, nothing otherwise.

▲ **Remarks**

CSiAPIService.exe must be running on the host computer and listening to the default TCP port for the call to succeed.

The default TCP port can be overridden by setting the environment variable ETABSv1_cOAPI_DEFAULT_PORT on both the Main and the Host computers.

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim myHelper As cHelper
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    myHelper = New Helper
    EtabsObject = myHelper.GetObjectHost("RemoteServer", "CSI.ETABS.API.ETABSObject")

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
    myHelper = Nothing
End Sub
```

▲ **See Also**

Reference

cHelper Interface

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CSI API ETABS v1

cHelperAddLanguageSpecificTextSet("LST2692A164_0") Method

Attaches to the active running instance of the program on a given computer using the given TCP port.

Namespace: `ETABSV1`

Assembly: `ETABSV1 (in ETABSV1.dll)` Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
cOAPI GetObjectHostPort(
    string hostName,
    int portNumber,
    string progID
)
```

```
Function GetObjectHostPort (
    hostName As String,
    portNumber As Integer,
    progID As String
) As cOAPI
```

```
Dim instance As cHelper
Dim hostName As String
Dim portNumber As Integer
Dim progID As String
Dim returnValue As cOAPI
```

```
returnValue = instance.GetObjectHostPort(hostName,
    portNumber, progID)
```

```
cOAPI^ GetObjectHostPort(
    String^ hostName,
    int portNumber,
    String^ progID
)
```

```
abstract GetObjectHostPort :
    hostName : string *
    portNumber : int *
    progID : string -> cOAPI
```

Parameters

hostName

Type: `System.String`

The host computer name.

portNumber

Type: SystemInt32

The TCP port to connect to the host computer.

progID

Type: SystemString

The program ID of the API object.

Return Value

Type: cOAPI

An instance of cOAPI if successful, nothing otherwise.

Remarks

CSiAPIService.exe must be running on the host computer and listening to the given TCP port for the call to succeed.

Examples

VB

Copy

```
Public Sub Example()
    Dim myHelper As cHelper
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    myHelper = New Helper
    EtabsObject = myHelper.GetObjectHostPort("RemoteServer", 8500, "CSI.ETABS.API.ETABSObject")

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
    myHelper = Nothing
End Sub
```

See Also

Reference

cHelper Interface

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CSI API ETABS v1

cHelperAddLanguageSpecificTextSet("LSTB1CB8126_0") Method

Attaches to the running instance of the program with the given process id.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
c0API GetObjectProcess(
    string typeName,
    int pid
)
```

```
Function GetObjectProcess (
    typeName As String,
    pid As Integer
) As c0API
```

```
Dim instance As cHelper
Dim typeName As String
Dim pid As Integer
Dim returnValue As c0API
```

```
returnValue = instance.GetObjectProcess(typeName,
    pid)
```

```
c0API^ GetObjectProcess(
    String^ typeName,
    int pid
)
```

```
abstract GetObjectProcess :
    typeName : string *
    pid : int -> c0API
```

Parameters

typeName

Type: [SystemString](#)

The program ID of the API object.

pid

Type: [SystemInt32](#)

The system-generated unique identifier (process ID) of the desired instance of the program as shown in Windows Task Manager and/or under [Tools](#) menu.

Return Value

Type: cOAPI

An instance of cOAPI if successful, nothing otherwise.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim myHelper As cHelper
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim pid As Integer
    Dim ret As Integer = -1

    'create ETABS object
    myHelper = New Helper
    pid = 1234 'set to process ID of the program instance to attach to
    EtabsObject = myHelper.GetObjectProcess("CSI.ETABS.API.ETABSObject", pid)

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
    myHelper = Nothing
End Sub
```

▲ See Also

Reference

cHelper Interface

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CSI API ETABS v1

cLineElm Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cLineElm
```

```
Public Interface cLineElm
```

```
Dim instance As cLineElm
```

```
public interface class cLineElm
```

```
type cLineElm = interface end
```

The cLineElm type exposes the following members.

▲ Methods

▲	Name	Description
⇒	Count	
⇒	GetEndLengthOffset	
⇒	GetInsertionPoint	
⇒	GetLoadDistributed	
⇒	GetLoadPoint	
⇒	GetLoadTemperature	
⇒	GetLocalAxes	
⇒	GetMaterialOverwrite	
⇒	GetModifiers	
⇒	GetNameList	
⇒	GetObj	
⇒	GetPoints	
⇒	GetProperty	
⇒	GetReleases	
⇒	GetTCLimits	
⇒	GetTransformationMatrix	

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▲ See Also

Reference

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CSI API ETABS v1

cLineElm Methods

The [cLineElm](#) type exposes the following members.

Methods

	Name	Description
⇒	Count	
⇒	GetEndLengthOffset	
⇒	GetInsertionPoint	
⇒	GetLoadDistributed	
⇒	GetLoadPoint	
⇒	GetLoadTemperature	
⇒	GetLocalAxes	
⇒	GetMaterialOverwrite	
⇒	GetModifiers	
⇒	GetNameList	
⇒	GetObj	
⇒	GetPoints	
⇒	GetProperty	
⇒	GetReleases	
⇒	GetTCLimits	
⇒	GetTransformationMatrix	

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See Also

Reference

[cLineElm Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cLineElmAddLanguageSpecificTextSet("LST3FDEED12_0?cpp=::|2025");Count

Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

int Count()

Function Count As Integer

Dim instance As cLineElm

Dim returnValue As Integer

returnValue = instance.Count()

int Count()

abstract Count : unit -> int

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cLineElm Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLineElmAddLanguageSpecificTextSet("LSTF557393D_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetEndLengthOffset(
    string Name,
    ref double Length1,
    ref double Length2,
    ref double RZ
)
```

```
Function GetEndLengthOffset (
    Name As String,
    ByRef Length1 As Double,
    ByRef Length2 As Double,
    ByRef RZ As Double
) As Integer
```

```
Dim instance As cLineElm
Dim Name As String
Dim Length1 As Double
Dim Length2 As Double
Dim RZ As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetEndLengthOffset(Name,
    Length1, Length2, RZ)
```

```
int GetEndLengthOffset(
    String^ Name,
    double% Length1,
    double% Length2,
    double% RZ
)
```

```
abstract GetEndLengthOffset :
    Name : string *
    Length1 : float byref *
    Length2 : float byref *
    RZ : float byref -> int
```

Parameters

Name

Type: [SystemString](#)

cLineElmspan id="LSTF557393D_0">AddLanguageSpecificTextSet("LSTF557393D_0?cpp=::|nu2336GetEnd

Length1
Type:Â [SystemDouble](#)
Length2
Type:Â [SystemDouble](#)
RZ
Type:Â [SystemDouble](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cLineElm Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLineElmAddLanguageSpecificTextSet("LST50CBBEEA/

Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetInsertionPoint(
    string Name,
    ref double[] Offset1,
    ref double[] Offset2
)
```

```
Function GetInsertionPoint (
    Name As String,
    ByRef Offset1 As Double(),
    ByRef Offset2 As Double()
) As Integer
```

```
Dim instance As cLineElm
Dim Name As String
Dim Offset1 As Double()
Dim Offset2 As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetInsertionPoint(Name,
    Offset1, Offset2)
```

```
int GetInsertionPoint(
    String^ Name,
    array<double>^% Offset1,
    array<double>^% Offset2
)
```

```
abstract GetInsertionPoint :
    Name : string *
    Offset1 : float[] byref *
    Offset2 : float[] byref -> int
```

Parameters

Name

Type: [SystemString](#)

Offset1

Type: [SystemDouble](#)

Offset2

Type: [SystemDouble](#)

cLineElm

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cLineElm Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLineElmAddLanguageSpecificTextSet("LSTB8A12ED4_0?cpp=::|2340);GetLo

Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetLoadDistributed(
    string Name,
    ref int NumberItems,
    ref string[] LineName,
    ref string[] LoadPat,
    ref int[] MyType,
    ref string[] CSys,
    ref int[] Dir,
    ref double[] RD1,
    ref double[] RD2,
    ref double[] Dist1,
    ref double[] Dist2,
    ref double[] Val1,
    ref double[] Val2,
    eItemTypeElm ItemTypeElm = eItemTypeElm.Element
)
```

```
Function GetLoadDistributed (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef LineName As String(),
    ByRef LoadPat As String(),
    ByRef MyType As Integer(),
    ByRef CSys As String(),
    ByRef Dir As Integer(),
    ByRef RD1 As Double(),
    ByRef RD2 As Double(),
    ByRef Dist1 As Double(),
    ByRef Dist2 As Double(),
    ByRef Val1 As Double(),
    ByRef Val2 As Double(),
    Optional ItemTypeElm As eItemTypeElm = eItemTypeElm.Element
) As Integer
```

```
Dim instance As cLineElm
Dim Name As String
Dim NumberItems As Integer
Dim LineName As String()
Dim LoadPat As String()
Dim MyType As Integer()
Dim CSys As String()
Dim Dir As Integer()
Dim RD1 As Double()
```

cLineElmsid="LSTB8A12ED4_0"AddLanguageSpecificTextSet("LSTB8A12ED4_0?cpp=::|2340);GetLo

Introduction

```
Dim RD2 As Double()
Dim Dist1 As Double()
Dim Dist2 As Double()
Dim Val1 As Double()
Dim Val2 As Double()
Dim ItemTypeElm As eItemTypeElm
Dim returnValue As Integer

returnValue = instance.GetLoadDistributed(Name,
NumberItems, LineName, LoadPat, MyType,
CSys, Dir, RD1, RD2, Dist1, Dist2, Val1,
Val2, ItemTypeElm)

int GetLoadDistributed(
String^ Name,
int% NumberItems,
array<String^>^% LineName,
array<String^>^% LoadPat,
array<int>^% MyType,
array<String^>^% CSys,
array<int>^% Dir,
array<double>^% RD1,
array<double>^% RD2,
array<double>^% Dist1,
array<double>^% Dist2,
array<double>^% Val1,
array<double>^% Val2,
eItemTypeElm ItemTypeElm = eItemTypeElm::Element
)

abstract GetLoadDistributed :
    Name : string *
    NumberItems : int byref *
    LineName : string[] byref *
    LoadPat : string[] byref *
    MyType : int[] byref *
    CSys : string[] byref *
    Dir : int[] byref *
    RD1 : float[] byref *
    RD2 : float[] byref *
    Dist1 : float[] byref *
    Dist2 : float[] byref *
    Val1 : float[] byref *
    Val2 : float[] byref *
    ?ItemTypeElm : eItemTypeElm
(* Defaults:
    let _ItemTypeElm = defaultArg ItemTypeElm eItemTypeElm.Element
*)
-> int
```

Parameters

Name
Type:Â SystemString

NumberItems
Type:Â SystemInt32

LineName
Type:Â SystemString

LoadPat

Type:Â SystemString
 MyType
 Type:Â SystemInt32
 CSys
 Type:Â SystemString
 Dir
 Type:Â SystemInt32
 RD1
 Type:Â SystemDouble
 RD2
 Type:Â SystemDouble
 Dist1
 Type:Â SystemDouble
 Dist2
 Type:Â SystemDouble
 Val1
 Type:Â SystemDouble
 Val2
 Type:Â SystemDouble
 ItemTypeElm (Optional)
 Type:Â ETABSv1eItemTypeElm

Return Value

Type:Â Int32

▲ See Also

Reference

cLineElm Interface

ETABSv1 Namespace

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CSI API ETABS v1

cLineElmAddLanguageSpecificTextSet("LST549239E0_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetLoadPoint(
    string Name,
    ref int NumberItems,
    ref string[] LineName,
    ref string[] LoadPat,
    ref int[] MyType,
    ref string[] CSys,
    ref int[] Dir,
    ref double[] RelDist,
    ref double[] Dist,
    ref double[] Val,
    eItemTypeElm ItemTypeElm = eItemTypeElm.Element
)
```

```
Function GetLoadPoint (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef LineName As String(),
    ByRef LoadPat As String(),
    ByRef MyType As Integer(),
    ByRef CSys As String(),
    ByRef Dir As Integer(),
    ByRef RelDist As Double(),
    ByRef Dist As Double(),
    ByRef Val As Double(),
    ByRef ItemTypeElm As eItemTypeElm = eItemTypeElm.Element
) As Integer
```

```
Dim instance As cLineElm
Dim Name As String
Dim NumberItems As Integer
Dim LineName As String()
Dim LoadPat As String()
Dim MyType As Integer()
Dim CSys As String()
Dim Dir As Integer()
Dim RelDist As Double()
Dim Dist As Double()
Dim Val As Double()
Dim ItemTypeElm As eItemTypeElm
Dim returnValue As Integer
```

```
returnValue = instance.GetLoadPoint(Name,
```

cLineElmspan id="LST549239E0_0"AddLanguageSpecificTextSet("LST549239E0_0?cpp=::|nu2343GetLoad

Introduction

NumberItems, LineName, LoadPat, MyType,
CSys, Dir, RelDist, Dist, Val, ItemTypeElm)

```
int GetLoadPoint(  
String^ Name,  
int% NumberItems,  
array<String^>^% LineName,  
array<String^>^% LoadPat,  
array<int>^% MyType,  
array<String^>^% CSys,  
array<int>^% Dir,  
array<double>^% RelDist,  
array<double>^% Dist,  
array<double>^% Val,  
eItemTypeElm ItemTypeElm = eItemTypeElm::Element  
)
```

```
abstract GetLoadPoint :  
    Name : string *  
    NumberItems : int byref *  
    LineName : string[] byref *  
    LoadPat : string[] byref *  
    MyType : int[] byref *  
    CSys : string[] byref *  
    Dir : int[] byref *  
    RelDist : float[] byref *  
    Dist : float[] byref *  
    Val : float[] byref *  
    ?ItemTypeElm : eItemTypeElm  
(* Defaults:  
    let _ItemTypeElm = defaultArg ItemTypeElm eItemTypeElm.Element  
*)  
-> int
```

Parameters

Name
Type:Â [SystemString](#)
NumberItems
Type:Â [SystemInt32](#)
LineName
Type:Â [SystemString](#)
LoadPat
Type:Â [SystemString](#)
MyType
Type:Â [SystemInt32](#)
CSys
Type:Â [SystemString](#)
Dir
Type:Â [SystemInt32](#)
RelDist
Type:Â [SystemDouble](#)
Dist
Type:Â [SystemDouble](#)
Val
Type:Â [SystemDouble](#)

ItemTypeElm (Optional)

Type: ETABSv1eItemTypeElm

Return Value

Type: Int32

See Also

Reference

cLineElm Interface

ETABSv1 Namespace

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CSI API ETABS v1

cLineElmAddLanguageSpecificTextSet("LSTA32B4292_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetLoadTemperature(
    string Name,
    ref int NumberItems,
    ref string[] LineName,
    ref string[] LoadPat,
    ref int[] MyType,
    ref double[] Val,
    ref string[] PatternName,
    eItemTypeElm ItemTypeElm = eItemTypeElm.Element
)
```

```
Function GetLoadTemperature (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef LineName As String(),
    ByRef LoadPat As String(),
    ByRef MyType As Integer(),
    ByRef Val As Double(),
    ByRef PatternName As String(),
    ByRef ItemTypeElm As eItemTypeElm = eItemTypeElm.Element
) As Integer
```

```
Dim instance As cLineElm
Dim Name As String
Dim NumberItems As Integer
Dim LineName As String()
Dim LoadPat As String()
Dim MyType As Integer()
Dim Val As Double()
Dim PatternName As String()
Dim ItemTypeElm As eItemTypeElm
Dim returnValue As Integer
```

```
returnValue = instance.GetLoadTemperature(Name,
    NumberItems, LineName, LoadPat, MyType,
    Val, PatternName, ItemTypeElm)
```

```
int GetLoadTemperature(
    String^ Name,
    int% NumberItems,
    array<String^>^% LineName,
    array<String^>^% LoadPat,
```

cLineElmspan id="LSTA32B4292_0">AddLanguageSpecificTextSet("LSTA32B4292_0?cpp=::|nu2346GetLoa

```

array<int>^% MyType,
array<double>^% Val,
array<String>^% PatternName,
eItemTypeElm ItemTypeElm = eItemTypeElm::Element
)

abstract GetLoadTemperature :
    Name : string *
    NumberItems : int byref *
    LineName : string[] byref *
    LoadPat : string[] byref *
    MyType : int[] byref *
    Val : float[] byref *
    PatternName : string[] byref *
    ?ItemTypeElm : eItemTypeElm
(* Defaults:
    let _ItemTypeElm = defaultArg ItemTypeElm eItemTypeElm.Element
*)
-> int

```

Parameters

Name
Type:Â [SystemString](#)

NumberItems
Type:Â [SystemInt32](#)

LineName
Type:Â [SystemString](#)

LoadPat
Type:Â [SystemString](#)

MyType
Type:Â [SystemInt32](#)

Val
Type:Â [SystemDouble](#)

PatternName
Type:Â [SystemString](#)

ItemTypeElm (Optional)
Type:Â [ETABSv1eItemTypeElm](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cLineElm Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLineElmAddLanguageSpecificTextSet("LST7F411CA8_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetLocalAxes(
    string Name,
    ref double Ang
)
```

```
Function GetLocalAxes (
    Name As String,
    ByRef Ang As Double
) As Integer
```

```
Dim instance As cLineElm
Dim Name As String
Dim Ang As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetLocalAxes(Name,
    Ang)
```

```
int GetLocalAxes(
    String^ Name,
    double% Ang
)
```

```
abstract GetLocalAxes :
    Name : string *
    Ang : float byref -> int
```

Parameters

Name

Type: [SystemString](#)

Ang

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cLineElm Interface

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CSI API ETABS v1

cLineElmAddLanguageSpecificTextSet("LSTF8AC7158_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetMaterialOverwrite(
    string Name,
    ref string PropName
)
```

```
Function GetMaterialOverwrite (
    Name As String,
    ByRef PropName As String
) As Integer
```

```
Dim instance As cLineElm
Dim Name As String
Dim PropName As String
Dim returnValue As Integer
```

```
returnValue = instance.GetMaterialOverwrite(Name,
    PropName)
```

```
int GetMaterialOverwrite(
    String^ Name,
    String^% PropName
)
```

```
abstract GetMaterialOverwrite :
    Name : string *
    PropName : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

PropName

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cLineElm Interface

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CSI API ETABS v1

cLineElmAddLanguageSpecificTextSet("LST2447535_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetModifiers(
    string Name,
    ref double[] Value
)
```

```
Function GetModifiers (
    Name As String,
    ByRef Value As Double()
) As Integer
```

```
Dim instance As cLineElm
Dim Name As String
Dim Value As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetModifiers(Name,
    Value)
```

```
int GetModifiers(
    String^ Name,
    array<double>^% Value
)
```

```
abstract GetModifiers :
    Name : string *
    Value : float[] byref -> int
```

Parameters

Name

Type: [SystemString](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cLineElm Interface

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CSI API ETABS v1

cLineElmAddLanguageSpecificTextSet("LSTCE5AE60F0") Method

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNameList(
    ref int NumberNames,
    ref string[] MyName
)
```

```
Function GetNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cLineElm
Dim NumberNames As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetNameList(NumberNames,
    MyName)
```

```
int GetNameList(
    int% NumberNames,
    array<String^>^% MyName
)
```

```
abstract GetNameList :
    NumberNames : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberNames

Type: [System.Int32](#)

MyName

Type: [System.String](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cLineElm Interface

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CSI API ETABS v1

cLineElmAddLanguageSpecificTextSet("LSTF7A24978_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetObj(
    string Name,
    ref string Obj,
    ref int ObjType,
    ref double RDI,
    ref double RDJ
)
```

```
Function GetObj (
    Name As String,
    ByRef Obj As String,
    ByRef ObjType As Integer,
    ByRef RDI As Double,
    ByRef RDJ As Double
) As Integer
```

```
Dim instance As cLineElm
Dim Name As String
Dim Obj As String
Dim ObjType As Integer
Dim RDI As Double
Dim RDJ As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetObj(Name, Obj,
    ObjType, RDI, RDJ)
```

```
int GetObj(
    String^ Name,
    String^% Obj,
    int% ObjType,
    double% RDI,
    double% RDJ
)
```

```
abstract GetObj :
    Name : string *
    Obj : string byref *
    ObjType : int byref *
    RDI : float byref *
    RDJ : float byref -> int
```


Parameters

Name

Type: SystemString

Obj

Type: SystemString

ObjType

Type: SystemInt32

RDI

Type: SystemDouble

RDJ

Type: SystemDouble

Return Value

Type: Int32

See Also

Reference

cLineElm Interface

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CSI API ETABS v1

cLineElmAddLanguageSpecificTextSet("LSTBD79A7F_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPoints(
    string Name,
    ref string Point1,
    ref string Point2
)
```

```
Function GetPoints (
    Name As String,
    ByRef Point1 As String,
    ByRef Point2 As String
) As Integer
```

```
Dim instance As cLineElm
Dim Name As String
Dim Point1 As String
Dim Point2 As String
Dim returnValue As Integer
```

```
returnValue = instance.GetPoints(Name,
    Point1, Point2)
```

```
int GetPoints(
    String^ Name,
    String^% Point1,
    String^% Point2
)
```

```
abstract GetPoints :
    Name : string *
    Point1 : string byref *
    Point2 : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

Point1

Type: [SystemString](#)

Point2

Type: [SystemString](#)

cLineElmAddLanguageSpecificTextSet("LSTBD79A7F_0?cpp=::|nu=2358",GetPoints

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cLineElm Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLineElmAddLanguageSpecificTextSet("LSTF11ACE95 Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetProperty(
    string Name,
    ref string PropName,
    ref int ObjType,
    ref bool Var,
    ref double SVarRelStartLoc,
    ref double SVarTotalLength
)
```

```
Function GetProperty (
    Name As String,
    ByRef PropName As String,
    ByRef ObjType As Integer,
    ByRef Var As Boolean,
    ByRef SVarRelStartLoc As Double,
    ByRef SVarTotalLength As Double
) As Integer
```

```
Dim instance As cLineElm
Dim Name As String
Dim PropName As String
Dim ObjType As Integer
Dim Var As Boolean
Dim SVarRelStartLoc As Double
Dim SVarTotalLength As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetProperty(Name,
    PropName, ObjType, Var, SVarRelStartLoc,
    SVarTotalLength)
```

```
int GetProperty(
    String^ Name,
    String^% PropName,
    int% ObjType,
    bool% Var,
    double% SVarRelStartLoc,
    double% SVarTotalLength
)
```

```
abstract GetProperty :
    Name : string *
```

```
PropName : string byref *  
ObjType : int byref *  
Var : bool byref *  
SVarRelStartLoc : float byref *  
SVarTotalLength : float byref -> int
```

Parameters

Name

Type:Â [SystemString](#)

PropName

Type:Â [SystemString](#)

ObjType

Type:Â [SystemInt32](#)

Var

Type:Â [SystemBoolean](#)

SVarRelStartLoc

Type:Â [SystemDouble](#)

SVarTotalLength

Type:Â [SystemDouble](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cLineElm Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLineElmAddLanguageSpecificTextSet("LSTE5CCF75E0?cpp=::|0062");GetRe

Method

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetReleases(
    string Name,
    ref bool[] II,
    ref bool[] JJ,
    ref double[] StartValue,
    ref double[] EndValue
)
```

```
Function GetReleases (
    Name As String,
    ByRef II As Boolean(),
    ByRef JJ As Boolean(),
    ByRef StartValue As Double(),
    ByRef EndValue As Double()
) As Integer
```

```
Dim instance As cLineElm
Dim Name As String
Dim II As Boolean()
Dim JJ As Boolean()
Dim StartValue As Double()
Dim EndValue As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetReleases(Name,
    II, JJ, StartValue, EndValue)
```

```
int GetReleases(
    String^ Name,
    array<bool>^% II,
    array<bool>^% JJ,
    array<double>^% StartValue,
    array<double>^% EndValue
)
```

```
abstract GetReleases :
    Name : string *
    II : bool[] byref *
    JJ : bool[] byref *
    StartValue : float[] byref *
    EndValue : float[] byref -> int
```

cLineElmspan id="LSTE5CCF75E_0"AddLanguageSpecificTextSet("LSTE5CCF75E_0?cpp=::|0062");GetRe

Parameters

Name

Type: SystemString

II

Type: SystemBoolean

JJ

Type: SystemBoolean

StartValue

Type: SystemDouble

EndValue

Type: SystemDouble

Return Value

Type: Int32

See Also

Reference

cLineElm Interface

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CSI API ETABS v1

cLineElmAddLanguageSpecificTextSet("LSTF2B7A21C_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetTCLimits(
    string Name,
    ref bool LimitCompressionExists,
    ref double LimitCompression,
    ref bool LimitTensionExists,
    ref double LimitTension
)
```

```
Function GetTCLimits (
    Name As String,
    ByRef LimitCompressionExists As Boolean,
    ByRef LimitCompression As Double,
    ByRef LimitTensionExists As Boolean,
    ByRef LimitTension As Double
) As Integer
```

```
Dim instance As cLineElm
Dim Name As String
Dim LimitCompressionExists As Boolean
Dim LimitCompression As Double
Dim LimitTensionExists As Boolean
Dim LimitTension As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetTCLimits(Name,
    LimitCompressionExists, LimitCompression,
    LimitTensionExists, LimitTension)
```

```
int GetTCLimits(
    String^ Name,
    bool% LimitCompressionExists,
    double% LimitCompression,
    bool% LimitTensionExists,
    double% LimitTension
)
```

```
abstract GetTCLimits :
    Name : string *
    LimitCompressionExists : bool byref *
    LimitCompression : float byref *
    LimitTensionExists : bool byref *
    LimitTension : float byref -> int
```

cLineElmid="LSTF2B7A21C_0"AddLanguageSpecificTextSet("LSTF2B7A21C_0?cpp=::|2364);GetTC

Parameters

Name

Type: SystemString

LimitCompressionExists

Type: SystemBoolean

LimitCompression

Type: SystemDouble

LimitTensionExists

Type: SystemBoolean

LimitTension

Type: SystemDouble

Return Value

Type: Int32

See Also

Reference

cLineElm Interface

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CSI API ETABS v1

cLineElmAddLanguageSpecificTextSet("LSTE5F197FD_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetTransformationMatrix(
    string Name,
    ref double[] Value
)
```

```
Function GetTransformationMatrix (
    Name As String,
    ByRef Value As Double()
) As Integer
```

```
Dim instance As cLineElm
Dim Name As String
Dim Value As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetTransformationMatrix(Name,
    Value)
```

```
int GetTransformationMatrix(
    String^ Name,
    array<double>^% Value
)
```

```
abstract GetTransformationMatrix :
    Name : string *
    Value : float[] byref -> int
```

Parameters

Name

Type: [SystemString](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cLineElm Interface

ETABSv1 Namespace

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CSI API ETABS v1

cLinkObj Interface

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
public interface cLinkObj
```

```
Public Interface cLinkObj
```

```
Dim instance As cLinkObj
```

```
public interface class cLinkObj
```

```
type cLinkObj = interface end
```

The cLinkObj type exposes the following members.

▲ Methods

<u>Â</u>	Name	Description
➤	<u>AddByCoord</u>	Adds a new link object whose end points are at the specified coordinates.
➤	<u>AddByPoint</u>	Adds a new link object whose end points are specified by name.
➤	<u>ChangeName</u>	Assigns a new name to an link object.
➤	<u>Count</u>	Retrieves the count of the link objects in the model.
➤	<u>Delete</u>	Deletes link objects.
➤	<u>GetElm</u>	Retrieves the name of the link element (analysis model link) associated with a specified link object in the object-based model.
➤	<u>GetGroupAssign</u>	Retrieves the groups to which a link object is assigned.
➤	<u>GetGUID</u>	Retrieves the GUID for the specified link object.
➤	<u>GetLocalAxes</u>	Retrieves the local axis angle assignment for link objects.
➤	<u>GetLocalAxesAdvanced</u>	Retrieves advanced local axes assignment for link objects.
➤	<u>GetNameList</u>	Retrieves the names of all defined link objects.
➤	<u>GetNameListOnStory</u>	Retrieves the names of the defined link objects on a given story.
➤	<u>GetPoints</u>	Retrieves the names of the point objects at each end of a specified link object. If names of the two point objects

are the same, the specified link object is a one-joint link object.

- ⇒ [GetProperty](#) Retrieves the link property assigned to a link object.
- ⇒ [GetSelected](#) Retrieves the selected status for a link object.
- ⇒ [GetTransformationMatrix](#) Retrieves the transformation matrix for a link object.
- ⇒ [SetGroupAssign](#) Adds/removes link objects to/from a specified group.
- ⇒ [SetGUID](#) Sets the GUID for the specified link object. If the GUID is passed in as a blank string, the program automatically creates a GUID for the object.
- ⇒ [SetLocalAxes](#) Assigns a local axis angle to link objects.
- ⇒ [SetLocalAxesAdvanced](#) Assigns advanced local axes to link objects
- ⇒ [SetProperty](#) Assigns a link property to link objects.
- ⇒ [SetSelected](#) Sets the selected status for link objects.

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLinkObj Methods

The cLinkObj type exposes the following members.

Methods

Á	Name	Description
⇒	<u>AddByCoord</u>	Adds a new link object whose end points are at the specified coordinates.
⇒	<u>AddByPoint</u>	Adds a new link object whose end points are specified by name.
⇒	<u>ChangeName</u>	Assigns a new name to an link object.
⇒	<u>Count</u>	Retrieves the count of the link objects in the model.
⇒	<u>Delete</u>	Deletes link objects.
⇒	<u>GetElm</u>	Retrieves the name of the link element (analysis model link) associated with a specified link object in the object-based model.
⇒	<u>GetGroupAssign</u>	Retrieves the groups to which a link object is assigned.
⇒	<u>GetGUID</u>	Retrieves the GUID for the specified link object.
⇒	<u>GetLocalAxes</u>	Retrieves the local axis angle assignment for link objects.
⇒	<u>GetLocalAxesAdvanced</u>	Retrieves advanced local axes assignment for link objects.
⇒	<u>GetNameList</u>	Retrieves the names of all defined link objects.
⇒	<u>GetNameListOnStory</u>	Retrieves the names of the defined link objects on a given story.
⇒	<u>GetPoints</u>	Retrieves the names of the point objects at each end of a specified link object. If names of the two point objects are the same, the specified link object is a one-joint link object.
⇒	<u>GetProperty</u>	Retrieves the link property assigned to a link object.
⇒	<u>GetSelected</u>	Retrieves the selected status for a link object.
⇒	<u>GetTransformationMatrix</u>	Retrieves the transformation matrix for a link object.
⇒	<u>SetGroupAssign</u>	Adds/removes link objects to/from a specified group.
⇒	<u>SetGUID</u>	Sets the GUID for the specified link object. If the GUID is passed in as a blank string, the program automatically creates a GUID for the object.
⇒	<u>SetLocalAxes</u>	Assigns a local axis angle to link objects.
⇒	<u>SetLocalAxesAdvanced</u>	Assigns advanced local axes to link objects
⇒	<u>SetProperty</u>	Assigns a link property to link objects.
⇒	<u>SetSelected</u>	Sets the selected status for link objects.

Top

See Also

Reference

cLinkObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cLinkObjAddLanguageSpecificTextSet("LST30D8DEA0 Method

Adds a new link object whose end points are at the specified coordinates.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int AddByCoord(
    double XI,
    double YI,
    double ZI,
    double XJ,
    double YJ,
    double ZJ,
    ref string Name,
    bool IsSingleJoint = false,
    string PropName = "Default",
    string UserName = "",
    string CSys = "Global"
)
```

```
Function AddByCoord (
    XI As Double,
    YI As Double,
    ZI As Double,
    XJ As Double,
    YJ As Double,
    ZJ As Double,
    ByRef Name As String,
    IsSingleJoint As Boolean = false,
    PropName As String = "Default",
    UserName As String = "",
    CSys As String = "Global"
) As Integer
```

```
Dim instance As cLinkObj
Dim XI As Double
Dim YI As Double
Dim ZI As Double
Dim XJ As Double
Dim YJ As Double
Dim ZJ As Double
Dim Name As String
Dim IsSingleJoint As Boolean
Dim PropName As String
Dim UserName As String
Dim CSys As String
Dim returnValue As Integer
```

cLinkObjid="LST30D8DEA0_0"AddLanguageSpecificTextSet("LST30D8DEA0_0?cpp=::|2372);AddBy


```
returnValue = instance.AddByCoord(XI,
YI, ZI, XJ, YJ, ZJ, Name, IsSingleJoint,
PropName, UserName, CSys)
```

```
int AddByCoord(
double XI,
double YI,
double ZI,
double XJ,
double YJ,
double ZJ,
String^% Name,
bool IsSingleJoint = false,
String^ PropName = L"Default",
String^ UserName = L"",
String^ CSys = L"Global"
)
```

```
abstract AddByCoord :
    XI : float *
    YI : float *
    ZI : float *
    XJ : float *
    YJ : float *
    ZJ : float *
    Name : string byref *
    ?IsSingleJoint : bool *
    ?PropName : string *
    ?UserName : string *
    ?CSys : string
(* Defaults:
    let _IsSingleJoint = defaultArg IsSingleJoint false
    let _PropName = defaultArg PropName "Default"
    let _UserName = defaultArg UserName ""
    let _CSys = defaultArg CSys "Global"
*)
-> int
```

Parameters

XI

Type:Â SystemDouble

The X coordinate of the I-End of the added link object in the coordinate system defined by the CSys item.

YI

Type:Â SystemDouble

The Y coordinate of the I-End of the added link object in the coordinate system defined by the CSys item.

ZI

Type:Â SystemDouble

The Z coordinate of the I-End of the added link object in the coordinate system defined by the CSys item.

XJ

Type:Â SystemDouble

The X coordinate of the J-End of the added link object in the coordinate system defined by the CSys item.

YJ

Type:Â SystemDouble

The Y coordinate of the J-End of the added link object in the coordinate system defined by the CSys item.

ZJ

Type:Â SystemDouble

The Z coordinate of the J-End of the added link object in the coordinate system defined by the CSys item.

Name

Type:Â SystemString

This is the name that the program ultimately assigns for the link object. If no UserName is specified, the program assigns a default name to the link object. If a UserName is specified and that name is not used for another link object, the UserName is assigned to the link object; otherwise a default name is assigned to the link object.

IsSingleJoint (Optional)

Type:Â SystemBoolean

This item is True if a one-joint link is added and False if a two-joint link is added.

PropName (Optional)

Type:Â SystemString

This is either Default or the name of a defined link property.

If it is Default the program assigns a default link property to the link object. If it is the name of a defined link property, that property is assigned to the link object

UserName (Optional)

Type:Â SystemString

CSys (Optional)

Type:Â SystemString

The name of the coordinate system in which the link object end point coordinates are defined.

Return Value

Type:Â Int32

Returns zero if the link object is successfully added; otherwise it returns a nonzero value.

▲Remarks

One-joint links are not supported in ETABS.

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name1 As String
    Dim Name2 As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

Return Value

Introduction

```
'start ETABS application
    ret = EtabsObject.ApplicationStart()

'create SapModel object
    SapModel = EtabsObject.SapModel

'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'add link object by coordinates
    ret = SapModel.LinkObj.AddByCoord(-288, 0, 288, 0, 0, 0, Name1, True)
    ret = SapModel.LinkObj.AddByCoord(-288, 0, 0, 0, 0, 144, Name2)

'refresh view
    ret = SapModel.View.RefreshView

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cLinkObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLinkObjAddLanguageSpecificTextSet("LST58F760AD_0") Method

Adds a new link object whose end points are specified by name.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int AddByPoint(
    string Point1,
    string Point2,
    ref string Name,
    bool IsSingleJoint = false,
    string PropName = "Default",
    string UserName = ""
)
```

```
Function AddByPoint (
    Point1 As String,
    Point2 As String,
    ByRef Name As String,
    IsSingleJoint As Boolean = false,
    PropName As String = "Default",
    UserName As String = ""
) As Integer
```

```
Dim instance As cLinkObj
Dim Point1 As String
Dim Point2 As String
Dim Name As String
Dim IsSingleJoint As Boolean
Dim PropName As String
Dim UserName As String
Dim returnValue As Integer
```

```
returnValue = instance.AddByPoint(Point1,
    Point2, Name, IsSingleJoint, PropName,
    UserName)
```

```
int AddByPoint(
    String^ Point1,
    String^ Point2,
    String^% Name,
    bool IsSingleJoint = false,
    String^ PropName = L"Default",
    String^ UserName = L""
)
```

abstract AddByPoint :

cLinkObjid="LST58F760AD_0"AddLanguageSpecificTextSet("LST58F760AD_0?cpp=::|n2376AddByP

```

Point1 : string *
Point2 : string *
Name : string byref *
?IsSingleJoint : bool *
?PropName : string *
?UserName : string
(* Defaults:
    let _IsSingleJoint = defaultArg IsSingleJoint false
    let _PropName = defaultArg PropName "Default"
    let _UserName = defaultArg UserName ""
*)
-> int

```

Parameters

Point1

Type: SystemString

The name of a defined point object at the I-End of the added link object.

Point2

Type: SystemString

The name of a defined point object at the J-End of the added link object.

This item is ignored if the IsSingleJoint item is True.

Name

Type: SystemString

This is the name that the program ultimately assigns for the link object. If no UserName is specified, the program assigns a default name to the link object. If a UserName is specified and that name is not used for another link object, the UserName is assigned to the link object; otherwise a default name is assigned to the link object.

IsSingleJoint (Optional)

Type: SystemBoolean

This item is True if a one-joint link is added and False if a two-joint link is added.

PropName (Optional)

Type: SystemString

This is either Default or the name of a defined link property.

If it is Default the program assigns a default link property to the link object. If it is the name of a defined link property, that property is assigned to the link object

UserName (Optional)

Type: SystemString

Return Value

Type: Int32

Returns zero if the link object is successfully added; otherwise it returns a nonzero value.

Remarks

One-joint links are not supported in ETABS.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name1 As String
    Dim Name2 As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link object by points
    ret = SapModel.LinkObj.AddByPoint("2", "4", Name1)
    ret = SapModel.LinkObj.AddByPoint("1", "5", Name2)

    'refresh view
    ret = SapModel.View.RefreshView

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cLinkObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLinkObjAddLanguageSpecificTextSet("LST2163EFFC_0") Method

Assigns a new name to an link object.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int ChangeName(
    string Name,
    string NewName
)
```

```
Function ChangeName (
    Name As String,
    NewName As String
) As Integer
```

```
Dim instance As cLinkObj
Dim Name As String
Dim NewName As String
Dim returnValue As Integer
```

```
returnValue = instance.ChangeName(Name,
    NewName)
```

```
int ChangeName(
    String^ Name,
    String^ NewName
)
```

```
abstract ChangeName :
    Name : string *
    NewName : string -> int
```

Parameters

Name

Type: SystemString

The existing name of a defined link object.

NewName

Type: SystemString

The new name for the link object.

Return Value

Type: Int32

Returns zero if the new name is successfully assigned; otherwise it returns a nonzero value.

▀ **Remarks**

▀ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link object by points
    ret = SapModel.LinkObj.AddByPoint("1", "5", Name)

    'refresh view
    ret = SapModel.View.RefreshView

    'change name
    ret = SapModel.LinkObj.ChangeName(Name, "MyLink")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▀ **See Also**

Reference

cLinkObj Interface

ETABSv1 Namespace

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Introduction

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CSI API ETABS v1

cLinkObjAddLanguageSpecificTextSet("LST80A201A_0") Method

Retrieves the count of the link objects in the model.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int Count()
```

Function Count As Integer

```
Dim instance As cLinkObj
Dim returnValue As Integer
```

```
returnValue = instance.Count()
```

```
int Count()
```

```
abstract Count : unit -> int
```

Return Value

Type: Int32

returns a count of the link objects in the model.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name1 As String
    Dim Name2 As String
    Dim Count as Integer

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel
```

Introduction

```
'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'add link object by coordinates
    ret = SapModel.LinkObj.AddByCoord(-288, 0, 288, 0, 0, 0, Name1, True)
    ret = SapModel.LinkObj.AddByCoord(-288, 0, 0, 0, 0, 144, Name2)

'refresh view
    ret = SapModel.View.RefreshView

'get number of link objects
    Count = SapModel.LinkObj.Count

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cLinkObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLinkObjAddLanguageSpecificTextSet("LSTC7CC91BE0") Method

Deletes link objects.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int Delete(
    string Name,
    eItemType ItemType = eItemType.Objects
)
```

```
Function Delete (
    Name As String,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cLinkObj
Dim Name As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.Delete(Name, ItemType)
```

```
int Delete(
    String^ Name,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract Delete :
    Name : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing link object or group, depending on the value of the ItemType item.

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the items in the eItemType enumeration.

cLinkObjspan id="LSTC7CC91BE_0"AddLanguageSpecificTextSet("LSTC7CC91BE_0?cpp=::|0004");Delete

Introduction

If this item is Objects, the link object specified by the Name item is deleted.

If this item is Group, the all link objects in the group specified by the Name item are deleted.

If this item is SelectedObjects, all selected link objects are deleted, and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the link objects are successfully deleted; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim Name1 As String  
    Dim Name2 As String  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)  
  
    'add link object by points  
    ret = SapModel.LinkObj.AddByPoint("2", "4", Name1)  
    ret = SapModel.LinkObj.AddByPoint("1", "5", Name2)  
  
    'delete link object  
    ret = SapModel.LinkObj.Delete(Name1)  
  
    'refresh view  
    ret = SapModel.View.RefreshView  
  
    'close ETABS  
    EtabsObject.ApplicationExit(False)  
  
    'clean up variables  
    SapModel = Nothing  
    EtabsObject = Nothing  
End Sub
```

▲ See Also

Reference

[cLinkObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLinkObjAddLanguageSpecificTextSet("LSTCE9FE67D0") Method

Retrieves the name of the link element (analysis model link) associated with a specified link object in the object-based model.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetElm(
    string Name,
    ref string Elm
)
```

```
Function GetElm (
    Name As String,
    ByRef Elm As String
) As Integer
```

```
Dim instance As cLinkObj
Dim Name As String
Dim Elm As String
Dim returnValue As Integer
```

```
returnValue = instance.GetElm(Name, Elm)
```

```
int GetElm(
    String^ Name,
    String^% Elm
)
```

```
abstract GetElm :
    Name : string *
    Elm : string byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing link object.

Elm

Type: SystemString

The name of the link element created from the specified link object.

Return ValueType: Int32

Returns zero if the link element name is successfully retrieved; otherwise it returns nonzero. An error occurs if the analysis model does not exist.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim Elm As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link object by points
    ret = SapModel.LinkObj.AddByPoint("1", "5", Name)

    'create the analysis model
    ret = SapModel.Analyze.CreateAnalysisModel

    'get link element name
    ret = SapModel.LinkObj.GetElm(Name, Elm)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cLinkObj Interface

ETABSv1 Namespace

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Introduction

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CSI API ETABS v1

cLinkObjAddLanguageSpecificTextSet("LST2EEE8AB50") Method

Retrieves the groups to which a link object is assigned.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetGroupAssign(
    string Name,
    ref int NumberGroups,
    ref string[] Groups
)
```

```
Function GetGroupAssign (
    Name As String,
    ByRef NumberGroups As Integer,
    ByRef Groups As String()
) As Integer
```

```
Dim instance As cLinkObj
Dim Name As String
Dim NumberGroups As Integer
Dim Groups As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetGroupAssign(Name,
    NumberGroups, Groups)
```

```
int GetGroupAssign(
    String^ Name,
    int% NumberGroups,
    array<String^>^% Groups
)
```

```
abstract GetGroupAssign :
    Name : string *
    NumberGroups : int byref *
    Groups : string[] byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing link object.

NumberGroups

Type: SystemInt32

The number of group names retrieved.

Groups

Type: SystemString

This is an array of the names of the groups to which the link object is assigned.

Return Value

Type: Int32

Returns zero if the group assignments are successfully retrieved, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim NumberGroups As Integer
    Dim Groups() as String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'add link object by points
    ret = SapModel.LinkObj.AddByPoint("1", "5", Name)

    'define new groups
    ret = SapModel.GroupDef.SetGroup("Group1")
    ret = SapModel.GroupDef.SetGroup("Group2")

    'add link object to groups
    ret = SapModel.LinkObj.SetGroupAssign(Name, "Group1")
    ret = SapModel.LinkObj.SetGroupAssign(Name, "Group2")

    'get link object groups
    ret = SapModel.LinkObj.GetGroupAssign(Name, NumberGroups, Groups)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
```

End Sub

▲ See Also

Reference

[cLinkObj Interface](#)

[ETABSv1 Namespace](#)

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»

CSI API ETABS v1

cLinkObjAddLanguageSpecificTextSet("LSTDE9CA3CE0?cpp=::2893");GetGUID Method

Retrieves the GUID for the specified link object.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetGUID(
    string Name,
    ref string GUID
)
```

```
Function GetGUID (
    Name As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cLinkObj
Dim Name As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.GetGUID(Name, GUID)
```

```
int GetGUID(
    String^ Name,
    String^% GUID
)
```

```
abstract GetGUID :
    Name : string *
    GUID : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing link object.

GUID

Type: [SystemString](#)

The GUID (Global Unique ID) for the specified link object.

Return ValueType: Int32

Returns zero if the link object GUID is successfully retrieved; otherwise it returns nonzero.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim MyGUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link object by points
    ret = SapModel.LinkObj.AddByPoint("1", "5", Name)

    'set GUID
    ret = SapModel.LinkObj.SetGUID(Name, New Guid().ToString)

    'get GUID
    ret = SapModel.LinkObj.GetGUID(Name, GUID)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cLinkObj Interface

ETABSv1 Namespace

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Introduction

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CSI API ETABS v1

cLinkObjAddLanguageSpecificTextSet("LSTBEC53FE_0") Method

Retrieves the local axis angle assignment for link objects.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetLocalAxes(
    string Name,
    ref double Ang,
    ref bool Advanced
)
```

```
Function GetLocalAxes (
    Name As String,
    ByRef Ang As Double,
    ByRef Advanced As Boolean
) As Integer
```

```
Dim instance As cLinkObj
Dim Name As String
Dim Ang As Double
Dim Advanced As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetLocalAxes(Name,
    Ang, Advanced)
```

```
int GetLocalAxes(
    String^ Name,
    double% Ang,
    bool% Advanced
)
```

```
abstract GetLocalAxes :
    Name : string *
    Ang : float byref *
    Advanced : bool byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing link object.

Ang

Type: SystemDouble

This is the angle that the local 2 and 3 axes are rotated about the positive local 1 axis, from the default orientation. The rotation for a positive angle appears counter clockwise when the local +1 axis is pointing toward you. [deg].

Advanced

Type: SystemBoolean

This item is True if the link object local axes orientation was obtained using advanced local axes parameters.

Return Value

Type: Int32

Returns zero if the assignment is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim Ang As Double
    Dim Advanced As boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link object by points
    ret = SapModel.LinkObj.AddByPoint("1", "5", Name)

    'assign link local axis angle
    ret = SapModel.LinkObj.SetLocalAxes(Name, 30)

    'get link local axis angle
    ret = SapModel.LinkObj.GetLocalAxes(Name, Ang, Advanced)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cLinkObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLinkObjAddLanguageSpecificTextSet("LST8CDDEE700") Method

Retrieves advanced local axes assignment for link objects.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetLocalAxesAdvanced(
    string Name,
    ref bool Active,
    ref int AxVectOpt,
    ref string AxCSys,
    ref int[] AxDir,
    ref string[] AxPt,
    ref double[] AxVect,
    ref int Plane2,
    ref int PlVectOpt,
    ref string PlCSys,
    ref int[] PlDir,
    ref string[] PlPt,
    ref double[] PlVect
)
```

```
Function GetLocalAxesAdvanced (
    Name As String,
    ByRef Active As Boolean,
    ByRef AxVectOpt As Integer,
    ByRef AxCSys As String,
    ByRef AxDir As Integer(),
    ByRef AxPt As String(),
    ByRef AxVect As Double(),
    ByRef Plane2 As Integer,
    ByRef PlVectOpt As Integer,
    ByRef PlCSys As String,
    ByRef PlDir As Integer(),
    ByRef PlPt As String(),
    ByRef PlVect As Double()
) As Integer
```

```
Dim instance As cLinkObj
Dim Name As String
Dim Active As Boolean
Dim AxVectOpt As Integer
Dim AxCSys As String
Dim AxDir As Integer()
Dim AxPt As String()
Dim AxVect As Double()
Dim Plane2 As Integer
```

cLinkObjid="LST8CDDEE70_0"AddLanguageSpecificTextSet("LST8CDDEE70_0?cpp=::|0099");GetLocalAxesAdvanced

```

Dim PlVectOpt As Integer
Dim PlCSys As String
Dim PlDir As Integer()
Dim PlPt As String()
Dim PlVect As Double()
Dim returnValue As Integer

returnValue = instance.GetLocalAxesAdvanced(Name,
Active, AxVectOpt, AxCSys, AxDir,
AxPt, AxVect, Plane2, PlVectOpt, PlCSys,
PlDir, PlPt, PlVect)

```

```

int GetLocalAxesAdvanced(
String^ Name,
bool% Active,
int% AxVectOpt,
String^% AxCSys,
array<int>^% AxDir,
array<String^>^% AxPt,
array<double>^% AxVect,
int% Plane2,
int% PlVectOpt,
String^% PlCSys,
array<int>^% PlDir,
array<String^>^% PlPt,
array<double>^% PlVect
)

```

```

abstract GetLocalAxesAdvanced :
    Name : string *
    Active : bool byref *
    AxVectOpt : int byref *
    AxCSys : string byref *
    AxDir : int[] byref *
    AxPt : string[] byref *
    AxVect : float[] byref *
    Plane2 : int byref *
    PlVectOpt : int byref *
    PlCSys : string byref *
    PlDir : int[] byref *
    PlPt : string[] byref *
    PlVect : float[] byref -> int

```

Parameters

Name

Type: SystemString

The name of an existing link object.

Active

Type: SystemBoolean

This is True if advanced local axes exist

AxVectOpt

Type: SystemInt32

This is 1, 2, or 3, indicating the axis reference vector option. This item applies only when the Active item is True.

Value **AxVectOpt**

1 Coordinate direction

- 2 Two joints
- 3 User vector

AxCSSys

Type:Â SystemString

The coordinate system used to define the axis reference vector coordinate directions and the axis user vector. This item applies when the Active item is True and the AxVectOpt item is 1 or 3.

AxDir

Type:Â SystemInt32

AxPt

Type:Â SystemString

This is an array dimensioned to 1 (2 strings), indicating the labels of two joints that define the axis reference vector. Either of these joints may be specified as None to indicate the center of the specified object. If both joints are specified as None, they are not used to define the axis reference vector. This item applies when the Active item is True and the AxVectOpt item is 2.

AxVect

Type:Â SystemDouble

This is an array dimensioned to 2 (3 doubles) that defines the axis reference vector. This item applies when the Active item is True and the AxVectOpt item is 3.

Plane2

Type:Â SystemInt32

This is 12 or 13, indicating that the local plane determined by the plane reference vector is the 1-2 or 1-3 plane. This item applies only when the Active item is True.

PlVectOpt

Type:Â SystemInt32

This is 1, 2, or 3, indicating the plane reference vector option. This item applies only when the Active item is True.

Value	PlVectOpt
1	Coordinate direction
2	Two joints
3	User vector

PlCSys

Type:Â SystemString

PlDir

Type:Â SystemInt32

PlPt

Type:Â SystemString

This is an array dimensioned to 1 (2 strings), indicating the labels of two joints that define the plane reference vector. Either of these joints may be specified as None to indicate the center of the specified object. If both joints are specified as None, they are not used to define the plane reference vector. This item applies when the Active item is True and the PlVectOpt item is 2.

PlVect

Type:Â SystemDouble

This is an array dimensioned to 2 (3 doubles) that defines the plane reference vector. This item applies when the Active item is True and the PlVectOpt item is

3.

Return ValueType: Int32

Returns zero if the advanced local axes assignments are retrieved successfully; otherwise, it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyName As String
    Dim MyAxDir(1) As Integer
    Dim MyAxPt(1) As String
    Dim MyAxVect(2) As Double
    Dim MyPlDir(1) As Integer
    Dim MyPlPt(1) As String
    Dim MyPlVect(2) As Double
    Dim Ang As Double
    Dim Advanced As Boolean
    Dim Active As Boolean
    Dim AxVectOpt As Integer
    Dim AxCSys As String
    Dim AxDir() As Integer
    Dim AxPt() As String
    Dim AxVect() As Double
    Dim Plane2 As Integer
    Dim PlVectOpt As Integer
    Dim PlCSys As String
    Dim PlDir() As Integer
    Dim PlPt() As String
    Dim PlVect() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link object by points
    ret = SapModel.LinkObj.AddByPoint("1", "5", MyName)

    'assign link advanced local axes
    MyAxVect(0)=0.707
    MyAxVect(1)=0.707
    MyAxVect(2)=0
```

Introduction

```
MyPlDir(0) = 2
MyPlDir(1) = 3
ret = SapModel.LinkObj.SetLocalAxesAdvanced(MyName, True, 3, "Global", MyAxDir, MyAxPt, MyAxPt)

'get link local axis angle
ret = SapModel.LinkObj.GetLocalAxes(MyName, Ang, Advanced)

'get link advanced local axes data
If Advanced Then
    ret = SapModel.LinkObj.GetLocalAxesAdvanced(MyName, Active, AxVectOpt, AxCSys, AxDir, AxPt)
End If

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

📌 See Also

Reference

[cLinkObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLinkObjAddLanguageSpecificTextSet("LST4592306C_0?cpp=::|nu240;GetNam Method

Retrieves the names of all defined link objects.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNameList(
    ref int NumberNames,
    ref string[] MyName
)
```

```
Function GetNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cLinkObj
Dim NumberNames As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetNameList(NumberNames,
    MyName)
```

```
int GetNameList(
    int% NumberNames,
    array<String^>^% MyName
)
```

```
abstract GetNameList :
    NumberNames : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberNames

Type: [System.Int32](#)

The number of link object names retrieved by the program.

MyName

Type: [System.String](#)

This is a one-dimensional array of link object names. The MyName array is created as a dynamic, zero-based, array by the API user: Dim MyName() as String

The array is dimensioned to (NumberNames - 1) inside the program, filled with the names, and returned to the API user.

Return Value

Type: Int32

Returns zero if the names are successfully retrieved; otherwise it returns nonzero.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim NumberNames As Integer
    Dim MyName() As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link object by points
    ret = SapModel.LinkObj.AddByPoint("2", "4", Name)
    ret = SapModel.LinkObj.AddByPoint("1", "5", Name)

    'get link object names
    ret = SapModel.LinkObj.GetNameList(NumberNames, MyName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cLinkObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cLinkObjAddLanguageSpecificTextSet("LSTA96CE6F5_0") Method

Retrieves the names of the defined link objects on a given story.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNameListOnStory(
    string StoryName,
    ref int NumberNames,
    ref string[] MyName
)
```

```
Function GetNameListOnStory (
    StoryName As String,
    ByRef NumberNames As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cLinkObj
Dim StoryName As String
Dim NumberNames As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetNameListOnStory(StoryName,
    NumberNames, MyName)
```

```
int GetNameListOnStory(
    String^ StoryName,
    int% NumberNames,
    array<String^>^% MyName
)
```

```
abstract GetNameListOnStory :
    StoryName : string *
    NumberNames : int byref *
    MyName : string[] byref -> int
```

Parameters

StoryName

Type: [SystemString](#)

The name of an existing story.

NumberNames

Type: SystemInt32

The number of link object names retrieved by the program.

MyName

Type: SystemString

This is a one-dimensional array of link object names. The MyName array is created as a dynamic, zero-based, array by the API user: Dim MyName() as String

The array is dimensioned to (NumberNames - 1) inside the program, filled with the names, and returned to the API user.

Return Value

Type: Int32

Returns zero if the names are successfully retrieved; otherwise it returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim NumberNames As Integer
    Dim MyName() As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link object by points
    ret = SapModel.LinkObj.AddByPoint("2", "4", Name)
    ret = SapModel.LinkObj.AddByPoint("1", "5", Name)

    'get link object names
    ret = SapModel.LinkObj.GetNameListOnStory("Story4", NumberNames, MyName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cLinkObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLinkObjAddLanguageSpecificTextSet("LST7A75A6_0") Method

Retrieves the names of the point objects at each end of a specified link object. If names of the two point objects are the same, the specified link object is a one-joint link object.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPoints(
    string Name,
    ref string Point1,
    ref string Point2
)
```

```
Function GetPoints (
    Name As String,
    ByRef Point1 As String,
    ByRef Point2 As String
) As Integer
```

```
Dim instance As cLinkObj
Dim Name As String
Dim Point1 As String
Dim Point2 As String
Dim returnValue As Integer
```

```
returnValue = instance.GetPoints(Name,
    Point1, Point2)
```

```
int GetPoints(
    String^ Name,
    String^% Point1,
    String^% Point2
)
```

```
abstract GetPoints :
    Name : string *
    Point1 : string byref *
    Point2 : string byref -> int
```

Parameters

Name

Type: [System.String](#)

The name of a defined link object.

cLinkObjid="LST7A75A6_0"AddLanguageSpecificTextSet("LST7A75A6_0?cpp=::|nu=.");[GetPoints](#) Method

Point1

Type:Â SystemString

The name of the point object at the I-End of the specified link object.

Point2

Type:Â SystemString

The name of the point object at the J-End of the specified link object.

Return Value

Type:Â Int32

Returns zero if the point names are successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim Point1 As String
    Dim Point2 As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link object by points
    ret = SapModel.LinkObj.AddByPoint("1", "5", Name)

    'get names of points
    ret = SapModel.LinkObj.GetPoints(Name, Point1, Point2)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cLinkObj Interface

ETABSv1 Namespace

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»

CSI API ETABS v1

cLinkObjAddLanguageSpecificTextSet("LSTC7DE0416_0") Method

Retrieves the link property assigned to a link object.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetProperty(
    string Name,
    ref string PropName
)
```

```
Function GetProperty (
    Name As String,
    ByRef PropName As String
) As Integer
```

```
Dim instance As cLinkObj
Dim Name As String
Dim PropName As String
Dim returnValue As Integer
```

```
returnValue = instance.GetProperty(Name,
    PropName)
```

```
int GetProperty(
    String^ Name,
    String^% PropName
)
```

```
abstract GetProperty :
    Name : string *
    PropName : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing link object.

PropName

Type: [SystemString](#)

This is the name of a link property assigned to the link object.

Return ValueType: Int32

Returns zero if the property is successfully retrieved; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim PropName As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link object by points
    ret = SapModel.LinkObj.AddByPoint("1", "5", Name)

    'set link property
    ret = SapModel.LinkObj.SetProperty(Name, "Link1")

    'get link property
    ret = SapModel.LinkObj.GetProperty(Name, PropName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cLinkObj Interface

ETABSv1 Namespace

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Introduction

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»

CSI API ETABS v1

cLinkObjAddLanguageSpecificTextSet("LST380BBA5A Method

Retrieves the selected status for a link object.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetSelected(
    string Name,
    ref bool Selected
)
```

```
Function GetSelected (
    Name As String,
    ByRef Selected As Boolean
) As Integer
```

```
Dim instance As cLinkObj
Dim Name As String
Dim Selected As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetSelected(Name,
    Selected)
```

```
int GetSelected(
    String^ Name,
    bool% Selected
)
```

```
abstract GetSelected :
    Name : string *
    Selected : bool byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing link object.

Selected

Type: SystemBoolean

This item is True if the specified link object is selected; otherwise it is False.

Return Value

Type: Int32

Returns zero if the selected status is successfully retrieved; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim Selected As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link object by points
    ret = SapModel.LinkObj.AddByPoint("1", "5", Name)

    'set link object selected
    ret = SapModel.LinkObj.SetSelected("ALL", True, eItemType.Group)

    'get link object selected status
    ret = SapModel.LinkObj.GetSelected(Name, Selected)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cLinkObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cLinkObjAddLanguageSpecificTextSet("LST8A19823D_0") Method

Retrieves the transformation matrix for a link object.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetTransformationMatrix(
    string Name,
    ref double[] Value,
    bool IsGlobal = true
)
```

```
Function GetTransformationMatrix (
    Name As String,
    ByRef Value As Double(),
    IsGlobal As Boolean = true
) As Integer
```

```
Dim instance As cLinkObj
Dim Name As String
Dim Value As Double()
Dim IsGlobal As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetTransformationMatrix(Name,
    Value, IsGlobal)
```

```
int GetTransformationMatrix(
    String^ Name,
    array<double>^% Value,
    bool IsGlobal = true
)
```

```
abstract GetTransformationMatrix :
    Name : string *
    Value : float[] byref *
    ?IsGlobal : bool
(* Defaults:
    let _IsGlobal = defaultArg IsGlobal true
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing link object.

Value

Type: SystemDouble

Value is an array of nine direction cosines that define the transformation matrix.

The following matrix equation shows how the transformation matrix is used to convert items from the link object local coordinate system to the global coordinate system.

$$[c_0 \ c_1 \ c_2] [Local1] [GlobalX]$$

$$[c_3 \ c_4 \ c_5] * [Local2] = [GlobalY]$$

$$[c_6 \ c_7 \ c_8] [Local3] [Globalz]$$

In the equation, c0 through c8 are the nine values from the transformation array, (Local1, Local2, Local3) are an item (such as a load) in the object local coordinate system, and (GlobalX, GlobalY, GlobalZ) are the same item in the global coordinate system.

The transformation from the local coordinate system to the present coordinate system is the same as that shown above for the global system if you substitute the present system for the global system.

IsGlobal (Optional)

Type: SystemBoolean

If this item is True, the transformation matrix is between the Global coordinate system and the link object local coordinate system. If this item is False, the transformation matrix is between the present coordinate system, and the link object local coordinate system.

Return Value

Type: Int32

Returns zero if the link object transformation matrix is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim Value() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()
```


Introduction

```
'create SapModel object
  SapModel = EtabsObject.SapModel

'initialize model
  ret = SapModel.InitializeNewModel()

'create steel deck template model
  ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'add link object by points
  ret = SapModel.LinkObj.AddByPoint("1", "5", Name)

'get link object transformation matrix
  ReDim Value(8)
  ret = SapModel.LinkObj.GetTransformationMatrix(Name, Value)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

🔗 See Also

Reference

[cLinkObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLinkObjAddLanguageSpecificTextSet("LST1DCE8616_0") Method

Adds/removes link objects to/from a specified group.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetGroupAssign(
    string Name,
    string GroupName,
    bool Remove = false,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetGroupAssign (
    Name As String,
    GroupName As String,
    Remove As Boolean = false,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cLinkObj
Dim Name As String
Dim GroupName As String
Dim Remove As Boolean
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetGroupAssign(Name,
    GroupName, Remove, ItemType)
```

```
int SetGroupAssign(
    String^ Name,
    String^ GroupName,
    bool Remove = false,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetGroupAssign :
    Name : string *
    GroupName : string *
    ?Remove : bool *
    ?ItemType : eItemType
(* Defaults:
    let _Remove = defaultArg Remove false
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

cLinkObjid="LST1DCE8616_0"AddLanguageSpecificTextSet("LST1DCE8616_0?cpp=::|n2422;SetGro

Parameters**Name**

Type:Â SystemString

The name of an existing link object or group, depending on the value of the ItemType item.

GroupName

Type:Â SystemString

The name of an existing group to which the assignment is made.

Remove (Optional)

Type:Â SystemBoolean

If this item is False, the specified link objects are added to the group specified by the GroupName item. If it is True, the link objects are removed from the group.

ItemType (Optional)

Type:Â ETABSv1eItemType

This is one of the items in the eItemType enumeration.

If this item is Objects, the link object specified by the Name item is added/removed to/from the group specified by the GroupName item.

If this item is Group, the link objects in the group specified by the Name item is added/removed to/from the group specified by the GroupName item.

If this item is SelectedObjects, all selected link objects are added/removed to/from the group specified by the GroupName item, and the Name item is ignored.

Return Value

Type:Â Int32

Returns zero if the group assignment is successful; otherwise it returns a nonzero value.

▲ **Remarks**▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
```

Introduction

```
ret = SapModel.InitializeNewModel()  
  
'create steel deck template model  
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)  
  
'add link object by points  
ret = SapModel.LinkObj.AddByPoint("1", "5", Name)  
  
'define new group  
ret = SapModel.GroupDef.SetGroup("Group1")  
  
'add link object to group  
ret = SapModel.LinkObj.SetGroupAssign(Name, "Group1")  
  
'close ETABS  
EtabsObject.ApplicationExit(False)  
  
'clean up variables  
SapModel = Nothing  
EtabsObject = Nothing  
End Sub
```

See Also

Reference

[cLinkObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLinkObjAddLanguageSpecificTextSet("LSTC6A30F1_0")

Method

Sets the GUID for the specified link object. If the GUID is passed in as a blank string, the program automatically creates a GUID for the object.

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetGUID(
    string Name,
    string GUID = ""
)
```

```
Function SetGUID (
    Name As String,
    GUID As String = ""
) As Integer
```

```
Dim instance As cLinkObj
Dim Name As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.SetGUID(Name, GUID)
```

```
int SetGUID(
    String^ Name,
    String^ GUID = L""
)
```

```
abstract SetGUID :
    Name : string *
    ?GUID : string
(* Defaults:
    let _GUID = defaultArg GUID ""
*)
-> int
```

Parameters

Name

Type: $\hat{A}$ SystemString

The name of an existing link object.

GUID (Optional)

Type: $\hat{A}$ SystemString

The GUID (Global Unique ID) for the specified link object.

cLinkObjid="LSTC6A30F1_0"AddLanguageSpecificTextSet("LSTC6A30F1_0?cpp=::|nu=2435")SetGUID I

Return ValueType: Int32

Returns zero if the link object GUID is successfully set; otherwise it returns nonzero.

▲ **Remarks**▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim MyGUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link object by points
    ret = SapModel.LinkObj.AddByPoint("1", "5", Name)

    'set GUID
    ret = SapModel.LinkObj.SetGUID(Name, New Guid().ToString)

    'get GUID
    ret = SapModel.LinkObj.GetGUID(Name, GUID)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also****Reference**cLinkObj InterfaceETABSv1 Namespace

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Introduction

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CSI API ETABS v1

cLinkObjAddLanguageSpecificTextSet("LST18B8F629_0") Method

Assigns a local axis angle to link objects.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetLocalAxes(
    string Name,
    double Ang,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetLocalAxes (
    Name As String,
    Ang As Double,
    Optional ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cLinkObj
Dim Name As String
Dim Ang As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetLocalAxes(Name,
    Ang, ItemType)
```

```
int SetLocalAxes(
    String^ Name,
    double Ang,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetLocalAxes :
    Name : string *
    Ang : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing link object or group, depending on the value of the ItemType item.

Ang

Type:Â SystemDouble

This is the angle that the local 2 and 3 axes are rotated about the positive local 1 axis, from the default orientation. The rotation for a positive angle appears counter clockwise when the local +1 axis is pointing toward you. [deg].

ItemType (Optional)

Type:Â ETABSV1eItemType

This is one of the items in the eItemType enumeration.

If this item is Objects, the assignment is made to the link object specified by the Name item.

If this item is Group, the assignment is made to all link objects in the group specified by the Name item.

If this item is SelectedObjects, the assignment is made to all selected link objects, and the Name item is ignored.

Return Value

Type:Â Int32

Returns zero if the local axis angle is successfully assigned; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link object by points
    ret = SapModel.LinkObj.AddByPoint("1", "5", Name)

    'assign link local axis angle
```

Introduction

```
ret = SapModel.LinkObj.SetLocalAxes(Name, 30)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

📌 See Also

Reference

[cLinkObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLinkObjAddLanguageSpecificTextSet("LSTF696B4D6_0?cpp=::|nu24.3)|SetLocalAxesAdvanced")

Method

Assigns advanced local axes to link objects

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetLocalAxesAdvanced(
    string Name,
    bool Active,
    int AxVectOpt,
    string AxCSys,
    ref int[] AxDir,
    ref string[] AxPt,
    ref double[] AxVect,
    int Plane2,
    int PlVectOpt,
    string PlCSys,
    ref int[] PlDir,
    ref string[] PlPt,
    ref double[] PlVect,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetLocalAxesAdvanced (
    Name As String,
    Active As Boolean,
    AxVectOpt As Integer,
    AxCSys As String,
    ByRef AxDir As Integer(),
    ByRef AxPt As String(),
    ByRef AxVect As Double(),
    Plane2 As Integer,
    PlVectOpt As Integer,
    PlCSys As String,
    ByRef PlDir As Integer(),
    ByRef PlPt As String(),
    ByRef PlVect As Double(),
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cLinkObj
Dim Name As String
Dim Active As Boolean
Dim AxVectOpt As Integer
Dim AxCSys As String
Dim AxDir As Integer()
Dim AxPt As String()
```

Introduction

```
Dim AxVect As Double()
Dim Plane2 As Integer
Dim PlVectOpt As Integer
Dim PlCSys As String
Dim PlDir As Integer()
Dim PlPt As String()
Dim PlVect As Double()
Dim ItemType As eItemType
Dim returnValue As Integer

returnValue = instance.SetLocalAxesAdvanced(Name,
Active, AxVectOpt, AxCSys, AxDir,
AxPt, AxVect, Plane2, PlVectOpt, PlCSys,
PlDir, PlPt, PlVect, ItemType)

int SetLocalAxesAdvanced(
String^ Name,
bool Active,
int AxVectOpt,
String^ AxCSys,
array<int>^% AxDir,
array<String^>^% AxPt,
array<double>^% AxVect,
int Plane2,
int PlVectOpt,
String^ PlCSys,
array<int>^% PlDir,
array<String^>^% PlPt,
array<double>^% PlVect,
eItemType ItemType = eItemType::Objects
)

abstract SetLocalAxesAdvanced :
    Name : string *
    Active : bool *
    AxVectOpt : int *
    AxCSys : string *
    AxDir : int[] byref *
    AxPt : string[] byref *
    AxVect : float[] byref *
    Plane2 : int *
    PlVectOpt : int *
    PlCSys : string *
    PlDir : int[] byref *
    PlPt : string[] byref *
    PlVect : float[] byref *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing link object or group depending on the value of the ItemType item.

Active

Type:Â SystemBoolean

This is True if advanced local axes exist

AxVectOpt

Type:Â SystemInt32

This is 1, 2, or 3, indicating the axis reference vector option. This item applies only when the Active item is True.

Value AxVectOpt

- 1 Coordinate direction
- 2 Two joints
- 3 User vector

AxCSSys

Type:Â SystemString

The coordinate system used to define the axis reference vector coordinate directions and the axis user vector. This item applies when the Active item is True and the AxVectOpt item is 1 or 3.

AxDir

Type:Â SystemInt32

AxPt

Type:Â SystemString

This is an array dimensioned to 1 (2 strings), indicating the labels of two joints that define the axis reference vector. Either of these joints may be specified as None to indicate the center of the specified object. If both joints are specified as None, they are not used to define the axis reference vector. This item applies when the Active item is True and the AxVectOpt item is 2.

AxVect

Type:Â SystemDouble

This is an array dimensioned to 2 (3 doubles) that defines the axis reference vector. This item applies when the Active item is True and the AxVectOpt item is 3.

Plane2

Type:Â SystemInt32

This is 12 or 13, indicating that the local plane determined by the plane reference vector is the 1-2 or 1-3 plane. This item applies only when the Active item is True.

PlVectOpt

Type:Â SystemInt32

This is 1, 2, or 3, indicating the plane reference vector option. This item applies only when the Active item is True.

Value PlVectOpt

- 1 Coordinate direction
- 2 Two joints
- 3 User vector

PlCSys

Type:Â SystemString

PlDir

Type:Â SystemInt32

PlPt

Type:Â SystemString

This is an array dimensioned to 1 (2 strings), indicating the labels of two joints

that define the plane reference vector. Either of these joints may be specified as None to indicate the center of the specified object. If both joints are specified as None, they are not used to define the plane reference vector. This item applies when the Active item is True and the PlVectOpt item is 2.

PlVect

Type:Â SystemDouble

This is an array dimensioned to 2 (3 doubles) that defines the plane reference vector. This item applies when the Active item is True and the PlVectOpt item is 3.

ItemType (Optional)

Type:Â ETABSv1eItemType

This is one of the items in the eItemType enumeration.

If this item is Objects, the assignment is made to the link object specified by the Name item.

If this item is Group, the assignment is made to the all link objects in the group specified by the Name item.

If this item is SelectedObjects, the assignment is made to all selected link objects, and the Name item is ignored.

Return Value

Type:Â Int32

Returns zero if the advanced local axes assignments are assigned successfully; otherwise, it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyName As String
    Dim MyAxDi(1) As Integer
    Dim MyAxPt(1) As String
    Dim MyAxVect(2) As Double
    Dim MyPlDi(1) As Integer
    Dim MyPlPt(1) As String
    Dim MyPlVect(2) As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
```

Introduction

```
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'add link object by points
ret = SapModel.LinkObj.AddByPoint("1", "5", MyName)

'assign link advanced local axes
MyAxVect(0)=0.707
MyAxVect(1)=0.707
MyAxVect(2)=0
MyPlDir(0) = 2
MyPlDir(1) = 3
ret = SapModel.LinkObj.SetLocalAxesAdvanced(MyName, True, 3, "Global", MyAxDir, MyAxPt, MyPlDir)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cLinkObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLinkObjAddLanguageSpecificTextSet("LST94599681_0") Method

Assigns a link property to link objects.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetProperty(
    string Name,
    string PropName,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetProperty (
    Name As String,
    PropName As String,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cLinkObj
Dim Name As String
Dim PropName As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetProperty(Name,
    PropName, ItemType)
```

```
int SetProperty(
    String^ Name,
    String^ PropName,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetProperty :
    Name : string *
    PropName : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing link object or group, depending on the value of the ItemType item.

PropName

Type: SystemString

This is the name of a link property to be assigned to the specified link object(s).

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the items in the eItemType enumeration.

If this item is Objects, the assignment is made to the link object specified by the Name item.

If this item is Group, the assignment is made to all link objects in the group specified by the Name item.

If this item is SelectedObjects, the assignment is made to all selected link objects, and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the property is successfully assigned; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link object by points
    ret = SapModel.LinkObj.AddByPoint("1", "5", Name)

    'set link property
    ret = SapModel.LinkObj.SetProperty(Name, "Link1")
```

Introduction

```
'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cLinkObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLinkObjAddLanguageSpecificTextSet("LST64549485_0") Method

Sets the selected status for link objects.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetSelected(
    string Name,
    bool Selected,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetSelected (
    Name As String,
    Selected As Boolean,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cLinkObj
Dim Name As String
Dim Selected As Boolean
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetSelected(Name,
    Selected, ItemType)
```

```
int SetSelected(
    String^ Name,
    bool Selected,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetSelected :
    Name : string *
    Selected : bool *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing link object or group, depending on the value of the ItemType item.

Selected

Type: SystemBoolean

This item is True if the specified link object is selected; otherwise it is False.

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the items in the eItemType enumeration.

If this item is Objects, the selected status is set for the link object specified by the Name item.

If this item is Group, the selected status is set for all link objects in the group specified by the Name item.

If this item is SelectedObjects, the selected status is set for all selected link objects, and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the selected status is successfully set; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link object by points
    ret = SapModel.LinkObj.AddByPoint("1", "5", Name)

    'set link object selected
    ret = SapModel.LinkObj.SetSelected("ALL", True, eItemType.Group)
```

Introduction

```
'update view
    ret = SapModel.View.RefreshView

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

[cLinkObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLoadCases Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cLoadCases
```

Public Interface cLoadCases

```
Dim instance As cLoadCases
```

```
public interface class cLoadCases
```

```
type cLoadCases = interface end
```

The cLoadCases type exposes the following members.

Properties

	Name	Description
	Buckling	
	DirHistLinear	
	DirHistNonlinear	
	HyperStatic	
	ModalEigen	
	ModalRitz	
	ModHistLinear	
	ModHistNonlinear	
	ResponseSpectrum	
	StaticLinear	
	StaticNonlinear	
	StaticNonlinearStaged	

[Top](#)

Methods

	Name	Description
	ChangeName	
	Count	
	Delete	Deletes the specified load case.
	GetNameList	Retrieves the names of all defined load cases of the specified type.
	GetTypeOAPI	

⇒ GetTypeOAPI_1 Retrieves the case type, design type, and auto flag for the specified load case

⇒ SetDesignType

Top

▲ See Also

Reference

ETABSv1 Namespace

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CSI API ETABS v1

cLoadCases Properties

The [cLoadCases](#) type exposes the following members.

Properties

	Name	Description
	Buckling	
	DirHistLinear	
	DirHistNonlinear	
	HyperStatic	
	ModalEigen	
	ModalRitz	
	ModHistLinear	
	ModHistNonlinear	
	ResponseSpectrum	
	StaticLinear	
	StaticNonlinear	
	StaticNonlinearStaged	

[Top](#)

See Also

Reference

[cLoadCases Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLoadCasesAddLanguageSpecificTextSet("LST242AA96_0?cpp=::|n2445;Buckling Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cCaseBuckling Buckling { get; }
```

```
ReadOnly Property Buckling As cCaseBuckling
Get
```

```
Dim instance As cLoadCases
Dim value As cCaseBuckling
```

```
value = instance.Buckling
```

```
property cCaseBuckling^ Buckling {
cCaseBuckling^ get ();
}
```

```
abstract Buckling : cCaseBuckling with get
```

Property Value

Type: [cCaseBuckling](#)

▲ See Also

Reference

[cLoadCases Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLoadCasesAddLanguageSpecificTextSet("LST423DC2A5_0?cpp=2446=.");DirL Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cCaseDirectHistoryLinear DirHistLinear { get; }
```

```
ReadOnly Property DirHistLinear As cCaseDirectHistoryLinear  
Get
```

```
Dim instance As cLoadCases  
Dim value As cCaseDirectHistoryLinear
```

```
value = instance.DirHistLinear
```

```
property cCaseDirectHistoryLinear^ DirHistLinear {  
    cCaseDirectHistoryLinear^ get ();  
}
```

```
abstract DirHistLinear : cCaseDirectHistoryLinear with get
```

Property Value

Type: [cCaseDirectHistoryLinear](#)

▲ See Also

Reference

[cLoadCases Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLoadCasesAddLanguageSpecificTextSet("LST434D0C1E_0?cpp=2447=");DirL Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cCaseDirectHistoryNonlinear DirHistNonlinear { get; }
```

```
ReadOnly Property DirHistNonlinear As cCaseDirectHistoryNonlinear  
Get
```

```
Dim instance As cLoadCases  
Dim value As cCaseDirectHistoryNonlinear
```

```
value = instance.DirHistNonlinear
```

```
property cCaseDirectHistoryNonlinear^ DirHistNonlinear {  
    cCaseDirectHistoryNonlinear^ get ();  
}
```

```
abstract DirHistNonlinear : cCaseDirectHistoryNonlinear with get
```

Property Value

Type: [cCaseDirectHistoryNonlinear](#)

▲ See Also

Reference

[cLoadCases Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLoadCasesAddLanguageSpecificTextSet("LST24EDFEA0_0?cpp2448=.");Hy Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cCaseHyperStatic HyperStatic { get; }
```

```
ReadOnly Property HyperStatic As cCaseHyperStatic
Get
```

```
Dim instance As cLoadCases
Dim value As cCaseHyperStatic
```

```
value = instance.HyperStatic
```

```
property cCaseHyperStatic^ HyperStatic {
cCaseHyperStatic^ get ();
}
```

```
abstract HyperStatic : cCaseHyperStatic with get
```

Property Value

Type: [cCaseHyperStatic](#)

▲ See Also

Reference

[cLoadCases Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLoadCasesAddLanguageSpecificTextSet("LST4A3B02E5_0?cpp=2449-");ModalEigen Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cCaseModalEigen ModalEigen { get; }
```

```
ReadOnly Property ModalEigen As cCaseModalEigen
Get
```

```
Dim instance As cLoadCases
Dim value As cCaseModalEigen
```

```
value = instance.ModalEigen
```

```
property cCaseModalEigen^ ModalEigen {
cCaseModalEigen^ get ();
}
```

```
abstract ModalEigen : cCaseModalEigen with get
```

Property Value

Type: [cCaseModalEigen](#)

▲ See Also

Reference

[cLoadCases Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLoadCasesAddLanguageSpecificTextSet("LST7DB441D4_0?cpp=2450=");Mo Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cCaseModalRitz ModalRitz { get; }
```

```
ReadOnly Property ModalRitz As cCaseModalRitz
Get
```

```
Dim instance As cLoadCases
Dim value As cCaseModalRitz
```

```
value = instance.ModalRitz
```

```
property cCaseModalRitz^ ModalRitz {
cCaseModalRitz^ get ();
}
```

```
abstract ModalRitz : cCaseModalRitz with get
```

Property Value

Type: [cCaseModalRitz](#)

▲ See Also

Reference

[cLoadCases Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLoadCasesAddLanguageSpecificTextSet("LST85685E Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cCaseModalHistoryLinear ModHistLinear { get; }
```

```
ReadOnly Property ModHistLinear As cCaseModalHistoryLinear  
Get
```

```
Dim instance As cLoadCases  
Dim value As cCaseModalHistoryLinear
```

```
value = instance.ModHistLinear
```

```
property cCaseModalHistoryLinear^ ModHistLinear {  
    cCaseModalHistoryLinear^ get ();  
}
```

```
abstract ModHistLinear : cCaseModalHistoryLinear with get
```

Property Value

Type: [cCaseModalHistoryLinear](#)

▲ See Also

Reference

[cLoadCases Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLoadCasesAddLanguageSpecificTextSet("LST677F881A_0?cpp=:2452.");ModProperty

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cCaseModalHistoryNonlinear ModHistNonlinear { get; }
```

```
ReadOnly Property ModHistNonlinear As cCaseModalHistoryNonlinear
Get
```

```
Dim instance As cLoadCases
Dim value As cCaseModalHistoryNonlinear
```

```
value = instance.ModHistNonlinear
```

```
property cCaseModalHistoryNonlinear^ ModHistNonlinear {
cCaseModalHistoryNonlinear^ get ();
}
```

```
abstract ModHistNonlinear : cCaseModalHistoryNonlinear with get
```

Property Value

Type: [cCaseModalHistoryNonlinear](#)

▲ See Also

Reference

[cLoadCases Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLoadCasesAddLanguageSpecificTextSet("LST281A149_0") Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cCaseResponseSpectrum ResponseSpectrum { get; }
```

```
ReadOnly Property ResponseSpectrum As cCaseResponseSpectrum
Get
```

```
Dim instance As cLoadCases
Dim value As cCaseResponseSpectrum
```

```
value = instance.ResponseSpectrum
```

```
property cCaseResponseSpectrum^ ResponseSpectrum {
cCaseResponseSpectrum^ get ();
}
```

```
abstract ResponseSpectrum : cCaseResponseSpectrum with get
```

Property Value

Type: [cCaseResponseSpectrum](#)

▲ See Also

Reference

[cLoadCases Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLoadCasesAddLanguageSpecificTextSet("LSTFEDA4C33_0?cpp2454u=.");StaticLinear Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cCaseStaticLinear StaticLinear { get; }
```

```
ReadOnly Property StaticLinear As cCaseStaticLinear
Get
```

```
Dim instance As cLoadCases
Dim value As cCaseStaticLinear
```

```
value = instance.StaticLinear
```

```
property cCaseStaticLinear^ StaticLinear {
cCaseStaticLinear^ get ();
}
```

```
abstract StaticLinear : cCaseStaticLinear with get
```

Property Value

Type: [cCaseStaticLinear](#)

▲ See Also

Reference

[cLoadCases Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLoadCasesAddLanguageSpecificTextSet("LST43933A Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cCaseStaticNonlinear StaticNonlinear { get; }
```

```
ReadOnly Property StaticNonlinear As cCaseStaticNonlinear  
Get
```

```
Dim instance As cLoadCases  
Dim value As cCaseStaticNonlinear
```

```
value = instance.StaticNonlinear
```

```
property cCaseStaticNonlinear^ StaticNonlinear {  
    cCaseStaticNonlinear^ get ();  
}
```

```
abstract StaticNonlinear : cCaseStaticNonlinear with get
```

Property Value

Type: [cCaseStaticNonlinear](#)

▲ See Also

Reference

[cLoadCases Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLoadCasesAddLanguageSpecificTextSet("LSTCF2D65B0_0?cpp=2456=.");StaticNonlinearStaged Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cCaseStaticNonlinearStaged StaticNonlinearStaged { get; }
```

```
ReadOnly Property StaticNonlinearStaged As cCaseStaticNonlinearStaged
Get
```

```
Dim instance As cLoadCases
Dim value As cCaseStaticNonlinearStaged
```

```
value = instance.StaticNonlinearStaged
```

```
property cCaseStaticNonlinearStaged^ StaticNonlinearStaged {
cCaseStaticNonlinearStaged^ get ();
}
```

```
abstract StaticNonlinearStaged : cCaseStaticNonlinearStaged with get
```

Property Value

Type: [cCaseStaticNonlinearStaged](#)

▲ See Also

Reference

[cLoadCases Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLoadCases Methods

The cLoadCases type exposes the following members.

Methods

⌕	Name	Description
⌕	<u>ChangeName</u>	
⌕	<u>Count</u>	
⌕	<u>Delete</u>	Deletes the specified load case.
⌕	<u>GetNameList</u>	Retrieves the names of all defined load cases of the specified type.
⌕	<u>GetTypeOAPI</u>	
⌕	<u>GetTypeOAPI_1</u>	Retrieves the case type, design type, and auto flag for the specified load case
⌕	<u>SetDesignType</u>	

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⌕ See Also

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cLoadCases Interface

ETABSV1 Namespace

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CSI API ETABS v1

cLoadCasesAddLanguageSpecificTextSet("LST867ECADD_0?cpp2456u=.");CMethod

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int ChangeName(
    string Name,
    string NewName
)
```

```
Function ChangeName (
    Name As String,
    NewName As String
) As Integer
```

```
Dim instance As cLoadCases
Dim Name As String
Dim NewName As String
Dim returnValue As Integer
```

```
returnValue = instance.ChangeName(Name,
    NewName)
```

```
int ChangeName(
    String^ Name,
    String^ NewName
)
```

```
abstract ChangeName :
    Name : string *
    NewName : string -> int
```

Parameters

Name

Type: [SystemString](#)

NewName

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cLoadCases Interface

ETABSv1 Namespace

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CSI API ETABS v1

cLoadCasesAddLanguageSpecificTextSet("LSTBF8AB Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int Count(
    eLoadCaseType CaseType =
)
```

```
Function Count (
    Optional CaseType As eLoadCaseType =
) As Integer
```

```
Dim instance As cLoadCases
Dim CaseType As eLoadCaseType
Dim returnValue As Integer
```

```
returnValue = instance.Count(CaseType)
```

```
int Count(
    eLoadCaseType CaseType =
)
```

```
abstract Count :
    ?CaseType : eLoadCaseType
(* Defaults:
    let _CaseType = defaultArg CaseType
*)
-> int
```

Parameters

CaseType (Optional)

Type: [ETABSv1eLoadCaseType](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cLoadCases Interface](#)

[ETABSv1 Namespace](#)

Introduction

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CSI API ETABS v1

cLoadCasesAddLanguageSpecificTextSet("LSTD2D3042_0") Method

Deletes the specified load case.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int Delete(
    string Name
)
```

```
Function Delete (
    Name As String
) As Integer
```

```
Dim instance As cLoadCases
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.Delete(Name)
```

```
int Delete(
    String^ Name
)
```

```
abstract Delete :
    Name : string -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing load case.

Return Value

Type: [Int32](#)

Returns zero if the load case is successfully deleted, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

[Copy](#)

```
Public Sub Example()
```

```
cLoadCases.AddLanguageSpecificTextSet("LSTD2D3042_0?cpp=::|2462");Delete
```

Introduction

```
Dim SapModel As cSapModel
Dim EtabsObject As cOAPI
Dim ret As Integer = -1

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'delete load case
ret = SapModel.LoadCases.Delete("DEAD")

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

[cLoadCases Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLoadCasesAddLanguageSpecificTextSet("LST52B729F_0?cpp=:2404.")

Method

Retrieves the names of all defined load cases of the specified type.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNameList(
    ref int NumberNames,
    ref string[] MyName,
    eLoadCaseType CaseType =
)
```

```
Function GetNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String(),
    Optional CaseType As eLoadCaseType =
0 As Integer
) As Integer
```

```
Dim instance As cLoadCases
Dim NumberNames As Integer
Dim MyName As String()
Dim CaseType As eLoadCaseType
Dim returnValue As Integer
```

```
returnValue = instance.GetNameList(NumberNames,
MyName, CaseType)
```

```
int GetNameList(
    int% NumberNames,
    array<String^>^% MyName,
    eLoadCaseType CaseType =
)
```

```
abstract GetNameList :
    NumberNames : int byref *
    MyName : string[] byref *
    ?CaseType : eLoadCaseType
(* Defaults:
    let _CaseType = defaultArg CaseType
*)
-> int
```

Parameters

NumberNames

Type: SystemInt32

The number of load case names retrieved by the program.

MyName

Type: SystemString

This is a one-dimensional array of area object names. The MyName array is created as a dynamic, zero-based, array by the API user:

```
Dim MyName() as String
```

The array is dimensioned to (NumberNames + 1) inside the ETABS program, filled with the names, and returned to the API user.

CaseType (Optional)

Type: ETABSv1eLoadCaseType

Return Value

Type: Int32

Returns zero if the names are successfully retrieved, otherwise it returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberNames As Integer
    Dim MyName() As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'get load case names
    ret = SapModel.LoadCases.GetNameList(NumberNames, MyName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Parameters

Reference

cLoadCases Interface

ETABSv1 Namespace

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CSI API ETABS v1

cLoadCasesAddLanguageSpecificTextSet("LST27DA3846_0?cpp=2467-");GetT Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetType0API(
    string Name,
    ref eLoadCaseType CaseType,
    ref int SubType
)
```

```
Function GetType0API (
    Name As String,
    ByRef CaseType As eLoadCaseType,
    ByRef SubType As Integer
) As Integer
```

```
Dim instance As cLoadCases
Dim Name As String
Dim CaseType As eLoadCaseType
Dim SubType As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetType0API(Name,
    CaseType, SubType)
```

```
int GetType0API(
    String^ Name,
    eLoadCaseType% CaseType,
    int% SubType
)
```

```
abstract GetType0API :
    Name : string *
    CaseType : eLoadCaseType byref *
    SubType : int byref -> int
```

Parameters

Name

Type: SystemString

CaseType

Type: ETABSv1eLoadCaseType

SubType

Type: SystemInt32

cLoadCasesAddLanguageSpecificTextSet("LST27DA3846_0?cpp=2467-");GetT

Return Value

Type: Int32

▲ Remarks

▲ See Also

Reference

cLoadCases Interface

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CSI API ETABS v1

cLoadCasesAddLanguageSpecificTextSet("LST4887C2BA_0?cpp=2469-");Get

Method

Retrieves the case type, design type, and auto flag for the specified load case

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetType0API_1(
    string Name,
    ref eLoadCaseType CaseType,
    ref int SubType,
    ref eLoadPatternType DesignType,
    ref int DesignTypeOption,
    ref int Auto
)
```

```
Function GetType0API_1 (
    Name As String,
    ByRef CaseType As eLoadCaseType,
    ByRef SubType As Integer,
    ByRef DesignType As eLoadPatternType,
    ByRef DesignTypeOption As Integer,
    ByRef Auto As Integer
) As Integer
```

```
Dim instance As cLoadCases
Dim Name As String
Dim CaseType As eLoadCaseType
Dim SubType As Integer
Dim DesignType As eLoadPatternType
Dim DesignTypeOption As Integer
Dim Auto As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetType0API_1(Name,
    CaseType, SubType, DesignType, DesignTypeOption,
    Auto)
```

```
int GetType0API_1(
    String^ Name,
    eLoadCaseType% CaseType,
    int% SubType,
    eLoadPatternType% DesignType,
    int% DesignTypeOption,
    int% Auto
)
```

```
abstract GetType0API_1 :
```

cLoadCasesAddLanguageSpecificTextSet("LST4887C2BA_0?cpp=2469-");Get

```
Name : string *
CaseType : eLoadCaseType byref *
SubType : int byref *
DesignType : eLoadPatternType byref *
DesignTypeOption : int byref *
Auto : int byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing load case

CaseType

Type:Â ETABSv1eLoadCaseType

This is one of the items in the eLoadCaseType enumeration

SubType

Type:Â SystemInt32

This is an integer representing the load case sub type. This item applies only for certain case types.

For NonlinearStatic

1. Nonlinear
2. Nonlinear staged construction

For Modal

1. Eigen
2. Ritz

For LinearHistory

1. Transient
2. Periodic

DesignType

Type:Â ETABSv1eLoadPatternType

This is one of the items in the eLoadPatternType enumeration

DesignTypeOption

Type:Â SystemInt32

This is one of the following options for the DesignType item

0 = Program determined

1 = User specified

Auto

Type:Â SystemInt32

This is one of the following values indicating if the load case has been automatically created

0 = Not automatically created

1 = Automatically created

Return Value

Type: Int32

Returns zero if the type is successfully retrieved; otherwise it returns nonzero

▲ Remarks

This function supercedes GetTypeOAPI(String, eLoadCaseType, Int32)

▲ See Also

Reference

cLoadCases Interface

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CSI API ETABS v1

cLoadCasesAddLanguageSpecificTextSet("LST6C88F4BB_0?DesignType=.") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetDesignType(
    string Name,
    int DesignTypeOption,
    eLoadPatternType DesignType = eLoadPatternType.Dead
)
```

```
Function SetDesignType (
    Name As String,
    DesignTypeOption As Integer,
    DesignType As eLoadPatternType = eLoadPatternType.Dead
) As Integer
```

```
Dim instance As cLoadCases
Dim Name As String
Dim DesignTypeOption As Integer
Dim DesignType As eLoadPatternType
Dim returnValue As Integer
```

```
returnValue = instance.SetDesignType(Name,
    DesignTypeOption, DesignType)
```

```
int SetDesignType(
    String^ Name,
    int DesignTypeOption,
    eLoadPatternType DesignType = eLoadPatternType::Dead
)
```

```
abstract SetDesignType :
    Name : string *
    DesignTypeOption : int *
    ?DesignType : eLoadPatternType
(* Defaults:
    let _DesignType = defaultArg DesignType eLoadPatternType.Dead
*)
-> int
```

Parameters

Name

Type: SystemString

DesignTypeOption

cLoadCasesAddLanguageSpecificTextSet("LST6C88F4BB_0?cpp=2470=.");Set

Type: SystemInt32
DesignType (Optional)
Type: ETABSv1eLoadPatternType

Return Value

Type: Int32

See Also

Reference

cLoadCases Interface

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CSI API ETABS v1

cLoadPatterns Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cLoadPatterns
```

Public Interface cLoadPatterns

```
Dim instance As cLoadPatterns
```

```
public interface class cLoadPatterns
```

```
type cLoadPatterns = interface end
```

The cLoadPatterns type exposes the following members.

Properties

	Name	Description
--	------	-------------

	AutoSeismic	
---	-----------------------------	--

	AutoWind	
---	--------------------------	--

[Top](#)

Methods

	Name	Description
--	------	-------------

	Add	Adds a new load pattern.
---	---------------------	--------------------------

	ChangeName	Applies a new name to a load pattern.
---	----------------------------	---------------------------------------

	Count	Retrieves the number of defined load patterns.
---	-----------------------	--

	Delete	Deletes the specified load pattern.
---	------------------------	-------------------------------------

	GetAutoSeismicCode	Retrieves the code name used for auto seismic parameters in Quake-type load patterns.
---	------------------------------------	---

	GetAutoWindCode	Retrieves the code name used for auto wind parameters in Wind-type load patterns.
---	---------------------------------	---

	GetLoadType	Retrieves the load type for a specified load pattern.
---	-----------------------------	---

	GetNameList	Retrieves the names of all defined load cases.
---	-----------------------------	--

	GetSelfWTMultiplier	Retrieves the self weight multiplier for a specified load pattern.
---	-------------------------------------	--

	SetLoadType	Assigns a load type to a load pattern.
---	-----------------------------	--

	SetSelfWTMultiplier	Assigns a self weight multiplier to a load case.
---	-------------------------------------	--

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[See Also](#)

Reference

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CSI API ETABS v1

cLoadPatterns Properties

The [cLoadPatterns](#) type exposes the following members.

Properties

	Name	Description
--	------	-------------

	AutoSeismic	
---	-----------------------------	--

	AutoWind	
---	--------------------------	--

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 See Also

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[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLoadPatternsAddLanguageSpecificTextSet("LSTF0E8 Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cAutoSeismic AutoSeismic { get; }
```

```
ReadOnly Property AutoSeismic As cAutoSeismic
Get
```

```
Dim instance As cLoadPatterns
Dim value As cAutoSeismic
```

```
value = instance.AutoSeismic
```

```
property cAutoSeismic^ AutoSeismic {
cAutoSeismic^ get ();
}
```

```
abstract AutoSeismic : cAutoSeismic with get
```

Property Value

Type: [cAutoSeismic](#)

▲ See Also

Reference

[cLoadPatterns Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLoadPatternsAddLanguageSpecificTextSet("LST688D Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cAutoWind AutoWind { get; }
```

```
ReadOnly Property AutoWind As cAutoWind
Get
```

```
Dim instance As cLoadPatterns
Dim value As cAutoWind
```

```
value = instance.AutoWind
```

```
property cAutoWind^ AutoWind {
cAutoWind^ get ();
}
```

```
abstract AutoWind : cAutoWind with get
```

Property Value

Type: [cAutoWind](#)

▲ See Also

Reference

[cLoadPatterns Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cLoadPatterns Methods

The [cLoadPatterns](#) type exposes the following members.

Methods

⌕	Name	Description
⌕	Add	Adds a new load pattern.
⌕	ChangeName	Applies a new name to a load pattern.
⌕	Count	Retrieves the number of defined load patterns.
⌕	Delete	Deletes the specified load pattern.
⌕	GetAutoSeismicCode	Retrieves the code name used for auto seismic parameters in Quake-type load patterns.
⌕	GetAutoWindCode	Retrieves the code name used for auto wind parameters in Wind-type load patterns.
⌕	GetLoadType	Retrieves the load type for a specified load pattern.
⌕	GetNameList	Retrieves the names of all defined load cases.
⌕	GetSelfWTMultiplier	Retrieves the self weight multiplier for a specified load pattern.
⌕	SetLoadType	Assigns a load type to a load pattern.
⌕	SetSelfWTMultiplier	Assigns a self weight multiplier to a load case.

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⌕ See Also

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[cLoadPatterns Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cLoadPatternsAddLanguageSpecificTextSet("LST7D2FF18D_0?cpu=.") Method

Adds a new load pattern.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int Add(
    string Name,
    eLoadPatternType MyType,
    double SelfWTMultiplier = 0,
    bool AddAnalysisCase = true
)
```

```
Function Add (
    Name As String,
    MyType As eLoadPatternType,
    SelfWTMultiplier As Double = 0,
    AddAnalysisCase As Boolean = true
) As Integer
```

```
Dim instance As cLoadPatterns
Dim Name As String
Dim MyType As eLoadPatternType
Dim SelfWTMultiplier As Double
Dim AddAnalysisCase As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.Add(Name, MyType,
    SelfWTMultiplier, AddAnalysisCase)
```

```
int Add(
    String^ Name,
    eLoadPatternType MyType,
    double SelfWTMultiplier = 0,
    bool AddAnalysisCase = true
)
```

```
abstract Add :
    Name : string *
    MyType : eLoadPatternType *
    ?SelfWTMultiplier : float *
    ?AddAnalysisCase : bool
(* Defaults:
    let _SelfWTMultiplier = defaultArg SelfWTMultiplier 0
    let _AddAnalysisCase = defaultArg AddAnalysisCase true
*)
-> int
```

cLoadPatternsid="LST7D2FF18D_0"AddLanguageSpecificTextSet("LST7D2FF18D_0?cpu=.")

Parameters

Name

Type: SystemString

The name for the new load pattern.

MyType

Type: ETABSv1eLoadPatternTypeThis is one of the items in the eLoadPatternType enumeration.

SelfWTMultiplier (Optional)

Type: SystemDouble

The self weight multiplier for the new load pattern.

AddAnalysisCase (Optional)

Type: SystemBoolean

If this item is True, a linear static load case corresponding to the new load pattern is added.

Return ValueType: Int32

Returns 0 if the load pattern is successfully added; otherwise it returns nonzero.

▲ **Remarks**

An error is returned if the Name item is already used for an existing load pattern.

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add new load pattern
    ret = SapModel.LoadPatterns.Add("LIVE1A", eLoadPatternType.Live)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cLoadPatterns Interface

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CSI API ETABS v1

cLoadPatternsAddLanguageSpecificTextSet("LST888A Method

Applies a new name to a load pattern.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int ChangeName(
    string Name,
    string NewName
)
```

```
Function ChangeName (
    Name As String,
    NewName As String
) As Integer
```

```
Dim instance As cLoadPatterns
Dim Name As String
Dim NewName As String
Dim returnValue As Integer
```

```
returnValue = instance.ChangeName(Name,
    NewName)
```

```
int ChangeName(
    String^ Name,
    String^ NewName
)
```

```
abstract ChangeName :
    Name : string *
    NewName : string -> int
```

Parameters

Name

Type: SystemString

The name of a defined load pattern.

NewName

Type: SystemString

The new name for the load pattern.

Return ValueType: Int32

Returns zero if the new name is successfully applied; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'change name
    ret = SapModel.LoadPatterns.ChangeName("DEAD", "DL")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cLoadPatterns Interface

ETABSv1 Namespace

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»

CSI API ETABS v1

cLoadPatternsAddLanguageSpecificTextSet("LST512283DD_0?cp2485u=");C

Method

Retrieves the number of defined load patterns.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int Count()
```

Function Count As Integer

```
Dim instance As cLoadPatterns
```

```
Dim returnValue As Integer
```

```
returnValue = instance.Count()
```

```
int Count()
```

```
abstract Count : unit -> int
```

Return Value

Type: [Int32](#)

Returns the number of defined load patterns.

▲ Remarks

▲ Examples

VB

[Copy](#)

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()
```

Introduction

```
'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'get number of defined load patterns
    ret = SapModel.LoadPatterns.Count

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cLoadPatterns Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cLoadPatternsAddLanguageSpecificTextSet("LSTE33D6E89_0") Method

Deletes the specified load pattern.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int Delete(
    string Name
)
```

```
Function Delete (
    Name As String
) As Integer
```

```
Dim instance As cLoadPatterns
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.Delete(Name)
```

```
int Delete(
    String^ Name
)
```

```
abstract Delete :
    Name : string -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing load pattern.

Return Value

Type: [Int32](#)

Returns zero if the load pattern is successfully deleted, otherwise it returns a nonzero value.

▲ Remarks

The load pattern is not deleted and the function returns an error if the load pattern is assigned to an load case or if it is the only defined load pattern.

▲ Examples

VB

cLoadPatternsAddLanguageSpecificTextSet("LSTE33D6E89_0?cp24877u=.");D

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'delete load case
    ret = SapModel.LoadPatterns.Delete("DEAD")

    'Note: ret in the above line returns 1, indicating an error.
    'This is because the load pattern DEAD is assigned to
    'load case DEAD.

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

🔗 See Also

Reference

cLoadPatterns Interface

ETABSv1 Namespace

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CSI API ETABS v1

cLoadPatternsAddLanguageSpecificTextSet("LST15968960_0?cpp=2480=.");GetAutoSeismicCode

Method

Retrieves the code name used for auto seismic parameters in Quake-type load patterns.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetAutoSeismicCode(
    string Name,
    ref string CodeName
)
```

```
Function GetAutoSeismicCode (
    Name As String,
    ByRef CodeName As String
) As Integer
```

```
Dim instance As cLoadPatterns
Dim Name As String
Dim CodeName As String
Dim returnValue As Integer
```

```
returnValue = instance.GetAutoSeismicCode(Name,
    CodeName)
```

```
int GetAutoSeismicCode(
    String^ Name,
    String^% CodeName
)
```

```
abstract GetAutoSeismicCode :
    Name : string *
    CodeName : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing Quake-type load pattern.

CodeName

Type: [SystemString](#)

This is either blank or the name code used for the auto seismic parameters.

Blank means no auto seismic load is specified for the Quake-type load pattern.

cLoadPatternsspan id="LST15968960_0"AddLanguageSpecificTextSet("LST15968960_0?cpp=2480=.");GetAutoSeismicCode

Return Value

Type: Int32

Returns zero if the code is successfully retrieved; otherwise it returns a nonzero value.

▲ **Remarks**

An error is returned if the specified load pattern is not a Quake-type load pattern.

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim CodeName As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add new load pattern
    ret = SapModel.LoadPatterns.Add("EQX", LTYPE_QUAKE)

    'assign BOCA96 parameters
    ret = SapModel.LoadPatterns.AutoSeismic.SetBOCA96("EQX", 1, 0.05, 1, 0.035, 0, False, 0, 0)

    'get auto seismic code
    ret = SapModel.LoadPatterns.GetAutoSeismicCode("EQX", CodeName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cLoadPatterns Interface

ETABSv1 Namespace

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Introduction

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CSI API ETABS v1

cLoadPatternsAddLanguageSpecificTextSet("LST4AD29D9E_0?cpnu=");

Method

Retrieves the code name used for auto wind parameters in Wind-type load patterns.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetAutoWindCode(
    string Name,
    ref string CodeName
)
```

```
Function GetAutoWindCode (
    Name As String,
    ByRef CodeName As String
) As Integer
```

```
Dim instance As cLoadPatterns
Dim Name As String
Dim CodeName As String
Dim returnValue As Integer
```

```
returnValue = instance.GetAutoWindCode(Name,
    CodeName)
```

```
int GetAutoWindCode(
    String^ Name,
    String^% CodeName
)
```

```
abstract GetAutoWindCode :
    Name : string *
    CodeName : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing Wind-type load pattern.

CodeName

Type: [SystemString](#)

This is either blank or the name code used for the auto wind parameters. Blank means no auto wind load is specified for the Wind-type load pattern.

Return Value

Type: Int32

Returns zero if the code is successfully retrieved; otherwise it returns a nonzero value.

▀ **Remarks**

An error is returned if the specified load pattern is not a Wind-type load pattern.

▀ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim CodeName As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add new load pattern
    ret = SapModel.LoadPatterns.Add("WIND", LTYPE_WIND)

    'get auto wind code
    ret = SapModel.LoadPatterns.GetAutoWindCode("WIND", CodeName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▀ **See Also**

Reference

cLoadPatterns Interface

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CSI API ETABS v1

cLoadPatternsAddLanguageSpecificTextSet("LSTE8EF Method

Retrieves the load type for a specified load pattern.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetLoadType(
    string Name,
    ref eLoadPatternType MyType
)
```

```
Function GetLoadType (
    Name As String,
    ByRef MyType As eLoadPatternType
) As Integer
```

```
Dim instance As cLoadPatterns
Dim Name As String
Dim MyType As eLoadPatternType
Dim returnValue As Integer
```

```
returnValue = instance.GetLoadType(Name,
    MyType)
```

```
int GetLoadType(
    String^ Name,
    eLoadPatternType% MyType
)
```

```
abstract GetLoadType :
    Name : string *
    MyType : eLoadPatternType byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing load pattern.

MyType

Type: ETABSV1eLoadPatternType

This is one of the items in the eLoadPatternType enumeration.

Return Value

Type: Int32

Returns zero if the load type is successfully retrieved; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyType As eLoadPatternType

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'get load pattern type
    ret = SapModel.LoadPatterns.GetLoadType("DEAD", MyType)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

[cLoadPatterns Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cLoadPatternsAddLanguageSpecificTextSet("LST7D848DA0_0?cpu=.") Method

Retrieves the names of all defined load cases.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetNameList(
    ref int NumberNames,
    ref string[] MyName
)
```

```
Function GetNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cLoadPatterns
Dim NumberNames As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetNameList(NumberNames,
    MyName)
```

```
int GetNameList(
    int% NumberNames,
    array<String^>^% MyName
)
```

```
abstract GetNameList :
    NumberNames : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberNames

Type: System.Int32

The number of load pattern names retrieved by the program.

MyName

Type: System.String

This is a one-dimensional array of load pattern names. The MyName array is created as a dynamic, zero-based, array by the API user:

```
Dim MyName() as String
```

Introduction

The array is dimensioned to (NumberNames + 1) inside the ETABS program, filled with the names, and returned to the API user.

Return Value

Type: Int32

Returns zero if the names are successfully retrieved, otherwise it returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim NumberNames As Integer  
    Dim MyName() As String  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)  
  
    'get load pattern names  
    ret = SapModel.LoadPatterns.GetNameList(NumberNames, MyName)  
  
    'close ETABS  
    EtabsObject.ApplicationExit(False)  
  
    'clean up variables  
    SapModel = Nothing  
    EtabsObject = Nothing  
End Sub
```

▲ See Also

Reference

cLoadPatterns Interface

ETABSv1 Namespace

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CSI API ETABS v1

cLoadPatternsAddLanguageSpecificTextSet("LSTA8BE7313_0?cpa=.")

Method

Retrieves the self weight multiplier for a specified load pattern.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetSelfWTMultiplier(
    string Name,
    ref double SelfWTMultiplier
)
```

```
Function GetSelfWTMultiplier (
    Name As String,
    ByRef SelfWTMultiplier As Double
) As Integer
```

```
Dim instance As cLoadPatterns
Dim Name As String
Dim SelfWTMultiplier As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetSelfWTMultiplier(Name,
    SelfWTMultiplier)
```

```
int GetSelfWTMultiplier(
    String^ Name,
    double% SelfWTMultiplier
)
```

```
abstract GetSelfWTMultiplier :
    Name : string *
    SelfWTMultiplier : float byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing load pattern.

SelfWTMultiplier

Type: [SystemDouble](#)

The self weight multiplier for the specified load pattern.

Return Value

Type: Int32

Returns zero if the multiplier is successfully retrieved; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim SelfWTMultiplier As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'get self weight multiplier
    ret = SapModel.LoadPatterns.GetSelfWtMultiplier("DEAD", SelfWTMultiplier)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

[cLoadPatterns Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cLoadPatternsAddLanguageSpecificTextSet("LSTD5D833F7_0?cpu=.");S

Method

Assigns a load type to a load pattern.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetLoadType(
    string Name,
    eLoadPatternType MyType
)
```

```
Function SetLoadType (
    Name As String,
    MyType As eLoadPatternType
) As Integer
```

```
Dim instance As cLoadPatterns
Dim Name As String
Dim MyType As eLoadPatternType
Dim returnValue As Integer
```

```
returnValue = instance.SetLoadType(Name,
    MyType)
```

```
int SetLoadType(
    String^ Name,
    eLoadPatternType MyType
)
```

```
abstract SetLoadType :
    Name : string *
    MyType : eLoadPatternType -> int
```

Parameters

Name

Type: SystemString

The name of an existing load pattern.

MyType

Type: ETABSv1eLoadPatternType

This is one of the items in the eLoadPatternType enumeration.

Return Value

Type: Int32

Returns zero if the load type is successfully assigned; otherwise it returns a nonzero value.

▀ **Remarks**

▀ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyType As eLoadPatternType

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'assign load type
    ret = SapModel.LoadPatterns.SetLoadType("DEAD", eLoadPatternType.SuperDead)

    'get load pattern type
    ret = SapModel.LoadPatterns.GetLoadType("DEAD", MyType)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▀ **See Also**

Reference

cLoadPatterns Interface

ETABSv1 Namespace

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CSI API ETABS v1

cLoadPatternsAddLanguageSpecificTextSet("LSTD713570_0?cpp=:2502");SetS

Method

Assigns a self weight multiplier to a load case.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetSelfWTMultiplier(
    string Name,
    double SelfWTMultiplier
)
```

```
Function SetSelfWTMultiplier (
    Name As String,
    SelfWTMultiplier As Double
) As Integer
```

```
Dim instance As cLoadPatterns
Dim Name As String
Dim SelfWTMultiplier As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetSelfWTMultiplier(Name,
    SelfWTMultiplier)
```

```
int SetSelfWTMultiplier(
    String^ Name,
    double SelfWTMultiplier
)
```

```
abstract SetSelfWTMultiplier :
    Name : string *
    SelfWTMultiplier : float -> int
```

Parameters

Name

Type: SystemString

The name of an existing load pattern.

SelfWTMultiplier

Type: SystemDouble

The self weight multiplier for the specified load pattern.

Return Value

Type: Int32

Returns zero if the multiplier is successfully retrieved; otherwise it returns a nonzero value.

▀ **Remarks**

▀ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim SelfWTMultiplier As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'assign self weight multiplier
    ret = SapModel.LoadPatterns.SetSelfWtMultiplier("DEAD", 2)

    'get self weight multiplier
    ret = SapModel.LoadPatterns.GetSelfWtMultiplier("DEAD", SelfWTMultiplier)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▀ **See Also**

Reference

cLoadPatterns Interface

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CSI API ETABS v1

cOAPI Interface

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
public interface cOAPI
```

```
Public Interface cOAPI
```

```
Dim instance As cOAPI
```

```
public interface class cOAPI
```

```
type cOAPI = interface end
```

The cOAPI type exposes the following members.

▲ Properties

$\hat{A}$	Name	Description
-----------	------	-------------

	<u>SapModel</u>	Gets a reference to <u>cSapModel</u> .
---	-----------------	--

Top

▲ Methods

$\hat{A}$	Name	Description
-----------	------	-------------

	<u>ApplicationExit</u>	Exits the application.
---	------------------------	------------------------

	<u>ApplicationStart</u>	This function starts the application.
---	-------------------------	---------------------------------------

	<u>GetOAPIVersionNumber</u>	Retrieves the API version number implemented by the GUI.
---	-----------------------------	--

	<u>Hide</u>	This function hides the application. When the application is hidden it is not visible on the screen or on the Windows task bar.
---	-------------	---

	<u>InternalExec</u>	For internal use.
---	---------------------	-------------------

	<u>SetActiveObject</u>	This function sets the active ETABSObject instance in the system Running Object Table (ROT), replacing the previous instance (if any).
---	------------------------	--

	<u>Unhide</u>	This function unhides the application, that is, it makes it visible. When the application is hidden, it is not visible on the screen or on the Windows task bar.
---	---------------	--

	<u>UnSetActiveObject</u>	This function removes the ETABSObject instance from the system Running Object Table (ROT).
---	--------------------------	--

	<u>Visible</u>	This function gets the application visibility.
---	----------------	--

[Top](#)

▀ [See Also](#)

Reference

ETABSv1 Namespace

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CSI API ETABS v1

cOAPI Properties

The cOAPI type exposes the following members.

Properties

Â	Name	Description
	<u>SapModel</u>	Gets a reference to <u>cSapModel</u> .

Top

See Also

Reference

cOAPI Interface

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CSI API ETABS v1

cOAPIAddLanguageSpecificTextSet("LST8E61ECD7_0") Property

Gets a reference to [cSapModel](#).

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cSapModel SapModel { get; }
```

```
ReadOnly Property SapModel As cSapModel
Get
```

```
Dim instance As cOAPI
Dim value As cSapModel
```

```
value = instance.SapModel
```

```
property cSapModel^ SapModel {
cSapModel^ get ();
}
```

```
abstract SapModel : cSapModel with get
```

Property Value

Type: [cSapModel](#)

A reference to [cSapModel](#)

Return Value

Type: [cSapModel](#)

A reference to [cSapModel](#)

▲ Remarks

▲ Examples

[VB](#)

[Copy](#)

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

Introduction

```
'start ETABS application
    ret = EtabsObject.ApplicationStart()

'create SapModel object
    SapModel = EtabsObject.SapModel

'initialize model
    ret = SapModel.InitializeNewModel()

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

cOAPI Interface

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CSI API ETABS v1

cOAPI Methods

The cOAPI type exposes the following members.

Methods

⌕	Name	Description
⌕	<u>ApplicationExit</u>	Exits the application.
⌕	<u>ApplicationStart</u>	This function starts the application.
⌕	<u>GetOAPIVersionNumber</u>	Retrieves the API version number implemented by the GUI.
⌕	<u>Hide</u>	This function hides the application. When the application is hidden it is not visible on the screen or on the Windows task bar.
⌕	<u>InternalExec</u>	For internal use.
⌕	<u>SetAsActiveObject</u>	This function sets the active ETABSObject instance in the system Running Object Table (ROT), replacing the previous instance (if any).
⌕	<u>Unhide</u>	This function unhides the application, that is, it makes it visible. When the application is hidden, it is not visible on the screen or on the Windows task bar.
⌕	<u>UnsetAsActiveObject</u>	This function removes the ETABSObject instance from the system Running Object Table (ROT).
⌕	<u>Visible</u>	This function gets the application visibility.

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⌕ See Also

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cOAPI Interface

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CSI API ETABS v1

cOAPIAddLanguageSpecificTextSet("LST38FC36A7_0? Method

Exits the application.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int ApplicationExit(
    bool FileSave
)
```

```
Function ApplicationExit (
    FileSave As Boolean
) As Integer
```

```
Dim instance As cOAPI
Dim FileSave As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.ApplicationExit(FileSave)
```

```
int ApplicationExit(
    bool FileSave
)
```

```
abstract ApplicationExit :
    FileSave : bool -> int
```

Parameters

FileSave

Type: SystemBoolean

If this item is True the existing model file is saved prior to closing the application.

Return Value

Type: Int32

This function returns zero if the function succeeds and nonzero if it fails.

▲ Remarks

If the model file is saved then it is saved with its current name. Any instance of cSapModel should be set to nothing after calling this function.

▲ Examples

cOAPIspan id="LST38FC36A7_0"AddLanguageSpecificTextSet("LST38FC36A7_0?cpp=::|nu=2510Application

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'close ETABS  
    EtabsObject.ApplicationExit(False)  
  
    'clean up variables  
    SapModel = Nothing  
    EtabsObject = Nothing  
End Sub
```

▲ See Also

Reference

cOAPI Interface

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CSI API ETABS v1

cOAPIAddLanguageSpecificTextSet("LST3351E757_0? Method

This function starts the application.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int ApplicationStart()
```

Function ApplicationStart As Integer

```
Dim instance As cOAPI
```

```
Dim returnValue As Integer
```

```
returnValue = instance.ApplicationStart()
```

```
int ApplicationStart()
```

```
abstract ApplicationStart : unit -> int
```

Return Value

Type: Int32

This function returns zero if the application successfully starts and nonzero if it fails.

▲ Remarks

When the model is not visible it does not appear on screen and it does not appear in the Windows task bar.

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'close ETABS
    EtabsObject.ApplicationExit(False)
```

```
'clean up variables  
  SapModel = Nothing  
  EtabsObject = Nothing  
End Sub
```

📌 See Also

Reference

cOAPI Interface

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CSI API ETABS v1

cOAPIAddLanguageSpecificTextSet("LST69634120_0? Method

Retrieves the API version number implemented by the GUI.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
double GetOAPIVersionNumber()
```

```
Function GetOAPIVersionNumber As Double
```

```
Dim instance As cOAPI
```

```
Dim returnValue As Double
```

```
returnValue = instance.GetOAPIVersionNumber()
```

```
double GetOAPIVersionNumber()
```

```
abstract GetOAPIVersionNumber : unit -> float
```

Return Value

Type: [Double](#)

Returns the API version number implemented by the GUI.

▲ Remarks

▲ See Also

Reference

[cOAPI Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cOAPIAddLanguageSpecificTextSet("LST1F19D410_0? Method

This function hides the application. When the application is hidden it is not visible on the screen or on the Windows task bar.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int Hide()
```

Function Hide As Integer

```
Dim instance As cOAPI
```

```
Dim returnValue As Integer
```

```
returnValue = instance.Hide()
```

```
int Hide()
```

```
abstract Hide : unit -> int
```

Return Value

Type: Int32

The function returns zero if the application is successfully hidden and nonzero if the function fails. If the application is already hidden, calling this function returns an error.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
```

```
    Dim SapModel As cSapModel
```

```
    Dim EtabsObject As cOAPI
```

```
    Dim ret As Integer = -1
```

```
    'create ETABS object
```

```
    Dim myHelper as cHelper = New Helper
```

```
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

```
    'start ETABS application
```

```
    ret = EtabsObject.ApplicationStart()
```

```
    'create SapModel object
```

```
    SapModel = EtabsObject.SapModel
```

Introduction

```
'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'hide application
    ret = SapObject.Hide

'unhide application
    ret = SapObject.Unhide

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

cOAPI Interface

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CSI API ETABS v1

cOAPIAddLanguageSpecificTextSet("LST74692DDD_0") Method

For internal use.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int InternalExec(
    int operation
)
```

```
Function InternalExec (
    operation As Integer
) As Integer
```

```
Dim instance As cOAPI
Dim operation As Integer
Dim returnValue As Integer
```

```
returnValue = instance.InternalExec(operation)
```

```
int InternalExec(
    int operation
)
```

```
abstract InternalExec :
    operation : int -> int
```

Parameters

operation
Type: [SystemInt32](#)

Return Value

Type: [Int32](#)
For internal use.

▲ Remarks

For internal use.

▲ See Also

Reference

cOAPI Interface

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CSI API ETABS v1

cOAPIAddLanguageSpecificTextSet("LST2633D627_0?Method

This function sets the active ETABSObject instance in the system Running Object Table (ROT), replacing the previous instance (if any).

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetAsActiveObject()
```

```
Function SetAsActiveObject As Integer
```

```
Dim instance As cOAPI
```

```
Dim returnValue As Integer
```

```
returnValue = instance.SetAsActiveObject()
```

```
int SetAsActiveObject()
```

```
abstract SetAsActiveObject : unit -> int
```

Return Value

Type: Int32

This function returns zero if the current instance is successfully added to the system ROT and nonzero if it fails.

▲ Remarks

When a new ETABSObject is created using the API, it is automatically added to the system ROT if none is already present. Creation of subsequent ETABSObject instances does not alter the ROT as long as at least one active instance is present in the ROT. GetObject(String) can be used to attach to the active ETABSObject instance registered in the ROT.

▲ Examples

VB

Copy

```
Public Sub Example()
```

```
    Dim EtabsObject As cOAPI
```

```
    Dim ret As Integer = -1
```

```
    'create ETABS object
```

```
    Dim myHelper as cHelper = New Helper
```

```
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

Introduction

```
'start ETABS application
    ret = EtabsObject.ApplicationStart()

'set as active instance
    ret = EtabsObject.SetAsActiveObject

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cOAPI Interface

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CSI API ETABS v1

cOAPIAddLanguageSpecificTextSet("LSTA921DB41_0") Method

This function unhides the application, that is, it makes it visible. When the application is hidden, it is not visible on the screen or on the Windows task bar.

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int Unhide()
```

```
Function Unhide As Integer
```

```
Dim instance As cOAPI
```

```
Dim returnValue As Integer
```

```
returnValue = instance.Unhide()
```

```
int Unhide()
```

```
abstract Unhide : unit -> int
```

Return Value

Type: $\hat{A}$ Int32

The function returns zero if the application is successfully unhidden (set visible) and nonzero if the function fails. If the application is already visible (not hidden) calling this function returns an error.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel
```

Introduction

```
'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'hide application
    ret = SapObject.Hide

'unhide application
    ret = SapObject.Unhide

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

cOAPI Interface

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CSI API ETABS v1

cOAPIAddLanguageSpecificTextSet("LST12127F0C_0? Method

This function removes the ETABSObject instance from the system Running Object Table (ROT).

Namespace: $\hat{A}$ ETABSv1

Assembly: $\hat{A}$ ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int UnsetAsActiveObject()
```

```
Function UnsetAsActiveObject As Integer
```

```
Dim instance As cOAPI
```

```
Dim returnValue As Integer
```

```
returnValue = instance.UnsetAsActiveObject()
```

```
int UnsetAsActiveObject()
```

```
abstract UnsetAsActiveObject : unit -> int
```

Return Value

Type: $\hat{A}$ Int32

This function returns zero if the current instance is successfully removed from the ROT and nonzero if it fails or if the current instance is not in the ROT.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'set as active instance
    ret = EtabsObject.UnsetAsActiveObject

    'close ETABS
    EtabsObject.ApplicationExit(False)
```

cOAPIspan id="LST12127F0C_0"AddLanguageSpecificTextSet("LST12127F0C_0?cpp=::|nu=.2520setAsA

```
'clean up variables  
  EtabsObject = Nothing  
End Sub
```

📌 See Also

Reference

cOAPI Interface

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CSI API ETABS v1

cOAPIAddLanguageSpecificTextSet("LSTA7572DA7_01") Method

This function gets the application visibility.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

bool Visible()

Function Visible As Boolean

Dim instance As cOAPI

Dim returnValue As Boolean

returnValue = instance.Visible()

bool Visible()

abstract Visible : unit -> bool

Return Value

Type: Boolean

Returns True if the application is visible on the screen, otherwise it returns False.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Visible as Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()
```

cOAPIspan id="LSTA7572DA7_0"AddLanguageSpecificTextSet("LSTA7572DA7_0?cpp=::|nu=2525"Visible M

Introduction

```
'create steel deck template model
  ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'get application visibility
  Visible = SapObject.Visible

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

📌 See Also

Reference

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CSI API ETABS v1

cPierLabel Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cPierLabel
```

```
Public Interface cPierLabel
```

```
Dim instance As cPierLabel
```

```
public interface class cPierLabel
```

```
type cPierLabel = interface end
```

The cPierLabel type exposes the following members.

▲ Methods

▲	Name	Description
➤	ChangeName	Changes the name of a defined Pier Label
➤	Delete	Deletes the specified Pier Label
➤	GetNameList	Retrieves the names of all defined Pier Labels
➤	GetPier	Checks whether the specified Pier Label exists
➤	GetSectionProperties	Retrieves the section properties for a specified pier.
➤	SetPier	Adds a new Pier Label

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▲ See Also

Reference

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CSI API ETABS v1

cPierLabel Methods

The [cPierLabel](#) type exposes the following members.

Methods

	Name	Description
	ChangeName	Changes the name of a defined Pier Label
	Delete	Deletes the specified Pier Label
	GetNameList	Retrieves the names of all defined Pier Labels
	GetPier	Checks whether the specified Pier Label exists
	GetSectionProperties	Retrieves the section properties for a specified pier.
	SetPier	Adds a new Pier Label

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CSI API ETABS v1

cPierLabelAddLanguageSpecificTextSet("LSTC4C88E6A_0?cpp=:2509.");Char

Method

Changes the name of a defined Pier Label

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int ChangeName(
    string Name,
    string NewName
)
```

```
Function ChangeName (
    Name As String,
    NewName As String
) As Integer
```

```
Dim instance As cPierLabel
Dim Name As String
Dim NewName As String
Dim returnValue As Integer
```

```
returnValue = instance.ChangeName(Name,
    NewName)
```

```
int ChangeName(
    String^ Name,
    String^ NewName
)
```

```
abstract ChangeName :
    Name : string *
    NewName : string -> int
```

Parameters

Name

Type: SystemString

The existing name of a defined Pier Label

NewName

Type: SystemString

The new name for the Pier Label

Return Value

Type: Int32

Returns zero if the new name is successfully applied, otherwise it returns a nonzero value.

▀ **Remarks**

▀ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new pier label
    ret = SapModel.PierLabel.SetPier("MyPier")

    'change pier label
    ret = SapModel.PierLabel.ChangeName("MyPier", "Pier1A")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▀ **See Also**

Reference

cPierLabel Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPierLabelAddLanguageSpecificTextSet("LST7AA804E3_0") Method

Deletes the specified Pier Label

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int Delete(
    string Name
)
```

```
Function Delete (
    Name As String
) As Integer
```

```
Dim instance As cPierLabel
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.Delete(Name)
```

```
int Delete(
    String^ Name
)
```

```
abstract Delete :
    Name : string -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing Pier Label

Return Value

Type: [Int32](#)

Returns zero if the Pier Label is successfully deleted, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

[Copy](#)

```
Public Sub Example()
```

cPierLabelid="LST7AA804E3_0"AddLanguageSpecificTextSet("LST7AA804E3_0?cpp=::2531");Delete

Introduction

```
Dim SapModel As cSapModel
Dim EtabsObject As cOAPI
Dim ret As Integer = -1

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'define new pier label
ret = SapModel.PierLabel.SetPier("MyPier")

'change pier name
ret = SapModel.PierLabel.Delete("MyPier")

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

📌 See Also

Reference

[cPierLabel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPierLabelAddLanguageSpecificTextSet("LST2D3E917F_0") Method

Retrieves the names of all defined Pier Labels

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNameList(
    ref int NumberNames,
    ref string[] MyName
)
```

```
Function GetNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cPierLabel
Dim NumberNames As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetNameList(NumberNames,
    MyName)
```

```
int GetNameList(
    int% NumberNames,
    array<String^>^% MyName
)
```

```
abstract GetNameList :
    NumberNames : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberNames

Type: [System.Int32](#)

The number of Pier Labels defined in the program

MyName

Type: [System.String](#)

This is a one-dimensional array of coordinate system names. The MyName array is created as a dynamic, zero-based, array by the API user:

```
Dim MyName() as String
```

The array is dimensioned to (NumberNames + 1) inside the ETABS program, filled with the names, and returned to the API user.

Return Value

Type: Int32

Returns zero if the names are successfully retrieved, otherwise it returns nonzero.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberNames As Integer
    Dim MyName As String()

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new pier label
    ret = SapModel.PierLabel.SetPier("MyPier")

    'get names
    ret = SapModel.PierLabel.GetNameList(NumberNames, MyName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ **See Also**

Reference

cPierLabel Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPierLabelAddLanguageSpecificTextSet("LST6699DB26_0?cpp=::|2536");GetPierLabel

Method

Checks whether the specified Pier Label exists

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPier(
    string Name
)
```

```
Function GetPier (
    Name As String
) As Integer
```

```
Dim instance As cPierLabel
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.GetPier(Name)
```

```
int GetPier(
    String^ Name
)
```

```
abstract GetPier :
    Name : string -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing Pier Label

Return Value

Type: [Int32](#)

Returns zero if the Pier Label exists, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

[VB](#)

[Copy](#)

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
```

cPierLabelsAddLanguageSpecificTextSet("LST6699DB26_0?cpp=::|2536");GetPier

Introduction

```
Dim ret As Integer = -1
Dim SemiRigid As Boolean

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'define new pier label
ret = SapModel.PierLabel.SetPier("MyPier")

'check pier label
ret = SapModel.PierLabel.GetPier("MyPier")

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

[cPierLabel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPierLabelAddLanguageSpecificTextSet("LSTE5457DD8_0") Method

Retrieves the section properties for a specified pier.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetSectionProperties(
    string Name,
    ref int NumberStories,
    ref string[] StoryName,
    ref double[] AxisAngle,
    ref int[] NumAreaObjs,
    ref int[] NumLineObjs,
    ref double[] WidthBot,
    ref double[] ThicknessBot,
    ref double[] WidthTop,
    ref double[] ThicknessTop,
    ref string[] MatProp,
    ref double[] CGBotX,
    ref double[] CGBotY,
    ref double[] CGBotZ,
    ref double[] CGTopX,
    ref double[] CGTopY,
    ref double[] CGTopZ
)
```

```
Function GetSectionProperties (
    Name As String,
    ByRef NumberStories As Integer,
    ByRef StoryName As String(),
    ByRef AxisAngle As Double(),
    ByRef NumAreaObjs As Integer(),
    ByRef NumLineObjs As Integer(),
    ByRef WidthBot As Double(),
    ByRef ThicknessBot As Double(),
    ByRef WidthTop As Double(),
    ByRef ThicknessTop As Double(),
    ByRef MatProp As String(),
    ByRef CGBotX As Double(),
    ByRef CGBotY As Double(),
    ByRef CGBotZ As Double(),
    ByRef CGTopX As Double(),
    ByRef CGTopY As Double(),
    ByRef CGTopZ As Double()
) As Integer
```

```
Dim instance As cPierLabel
```

cPierLabelspan id="LSTE5457DD8_0"AddLanguageSpecificTextSet("LSTE5457DD8_0?cpp=::2538");GetS

Introduction

```
Dim Name As String
Dim NumberStories As Integer
Dim StoryName As String()
Dim AxisAngle As Double()
Dim NumAreaObjs As Integer()
Dim NumLineObjs As Integer()
Dim WidthBot As Double()
Dim ThicknessBot As Double()
Dim WidthTop As Double()
Dim ThicknessTop As Double()
Dim MatProp As String()
Dim CGBotX As Double()
Dim CGBotY As Double()
Dim CGBotZ As Double()
Dim CGTopX As Double()
Dim CGTopY As Double()
Dim CGTopZ As Double()
Dim returnValue As Integer

returnValue = instance.GetSectionProperties(Name,
NumberStories, StoryName, AxisAngle,
NumAreaObjs, NumLineObjs, WidthBot,
ThicknessBot, WidthTop, ThicknessTop,
MatProp, CGBotX, CGBotY, CGBotZ, CGTopX,
CGTopY, CGTopZ)
```

```
int GetSectionProperties(
String^ Name,
int% NumberStories,
array<String^>^% StoryName,
array<double>^% AxisAngle,
array<int>^% NumAreaObjs,
array<int>^% NumLineObjs,
array<double>^% WidthBot,
array<double>^% ThicknessBot,
array<double>^% WidthTop,
array<double>^% ThicknessTop,
array<String^>^% MatProp,
array<double>^% CGBotX,
array<double>^% CGBotY,
array<double>^% CGBotZ,
array<double>^% CGTopX,
array<double>^% CGTopY,
array<double>^% CGTopZ
)
```

```
abstract GetSectionProperties :
    Name : string *
    NumberStories : int byref *
    StoryName : string[] byref *
    AxisAngle : float[] byref *
    NumAreaObjs : int[] byref *
    NumLineObjs : int[] byref *
    WidthBot : float[] byref *
    ThicknessBot : float[] byref *
    WidthTop : float[] byref *
    ThicknessTop : float[] byref *
    MatProp : string[] byref *
    CGBotX : float[] byref *
    CGBotY : float[] byref *
    CGBotZ : float[] byref *
    CGTopX : float[] byref *
```

```
CGTopY : float[] byref *
CGTopZ : float[] byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing pier.

NumberStories

Type:Â SystemInt32

The number of stories that the pier exists on and for which section properties are provided.

StoryName

Type:Â SystemString

This is an array that includes the story names at which the pier exists.

AxisAngle

Type:Â SystemDouble

This is an array that includes the pier local axis angle at each story, defined as the angle between the global x-axis and the pier local 2-axis.

NumAreaObjs

Type:Â SystemInt32

This is an array that includes the number of area objects in the pier at each story.

NumLineObjs

Type:Â SystemInt32

This is an array that includes the number of line objects in the pier at each story.

WidthBot

Type:Â SystemDouble

This is an array that includes the width of the pier at the bottom of each story.

ThicknessBot

Type:Â SystemDouble

This is an array that includes the thickness of the pier at the bottom of each story.

WidthTop

Type:Â SystemDouble

This is an array that includes the width of the pier at the top of each story.

ThicknessTop

Type:Â SystemDouble

This is an array that includes the thickness of the pier at the top of each story.

MatProp

Type:Â SystemString

This is an array that includes the name of the pier material property at each story.

CGBotX

Type:Â SystemDouble

This is an array that includes the x-coordinate of the pier center of gravity at the bottom of each story.

CGBotY

Type:Â SystemDouble

This is an array that includes the y-coordinate of the pier center of gravity at the

bottom of each story.

CGBotZ

Type: SystemDouble

This is an array that includes the z-coordinate of the pier center of gravity at the bottom of each story.

CGTopX

Type: SystemDouble

This is an array that includes the x-coordinate of the pier center of gravity at the top of each story.

CGTopY

Type: SystemDouble

This is an array that includes the y-coordinate of the pier center of gravity at the top of each story.

CGTopZ

Type: SystemDouble

This is an array that includes the z-coordinate of the pier center of gravity at the top of each story.

Return Value

Type: Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ See Also

Reference

cPierLabel Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPierLabelAddLanguageSpecificTextSet("LST635E4D1_0") Method

Adds a new Pier Label

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetPier(
    string Name
)
```

```
Function SetPier (
    Name As String
) As Integer
```

```
Dim instance As cPierLabel
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.SetPier(Name)
```

```
int SetPier(
    String^ Name
)
```

```
abstract SetPier :
    Name : string -> int
```

Parameters

Name

Type: [SystemString](#)

The name of a new Pier Label.

Return Value

Type: [Int32](#)

Returns zero if the Pier Label is successfully added, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

[VB](#)

[Copy](#)

```
Public Sub Example()
```

cPierLabelAddLanguageSpecificTextSet("LST635E4D1_0?cpp=::|nu2542SetPier I

Introduction

```
Dim SapModel As cSapModel
Dim EtabsObject As cOAPI
Dim ret As Integer = -1
Dim NumberNames As Integer
Dim MyName As String()

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'define new pier label
ret = SapModel.PierLabel.SetPier("MyPier")

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

[cPierLabel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPluginCallback Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cPluginCallback
```

```
Public Interface cPluginCallback
```

```
Dim instance As cPluginCallback
```

```
public interface class cPluginCallback
```

```
type cPluginCallback = interface end
```

The cPluginCallback type exposes the following members.

▲ Properties

▲	Name	Description
	ErrorFlag	Indicates whether an error occurred
	Finished	Indicates whether the plugin is finished

[Top](#)

▲ Methods

▲	Name	Description
	Finish	This method must be called when the plugin has finished.

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPluginCallback Properties

The [cPluginCallback](#) type exposes the following members.

Properties

Â	Name	Description
	ErrorFlag	Indicates whether an error occurred
	Finished	Indicates whether the plugin is finished

[Top](#)

 See Also

Reference

[cPluginCallback Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cPluginCallbackAddLanguageSpecificTextSet("LSTF954D1E_0?c2548|nu=.") Property

Indicates whether an error occurred

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int ErrorFlag { get; }
```

```
ReadOnly Property ErrorFlag As Integer
Get
```

```
Dim instance As cPluginCallback
Dim value As Integer
```

```
value = instance.ErrorFlag
```

```
property int ErrorFlag {
int get ();
}
```

```
abstract ErrorFlag : int with get
```

Return Value

Type: [Int32](#)

Zero if the plugin completed successfully; a non-zero value indicates that an error occurred

▲ See Also

Reference

[cPluginCallback Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPluginCallbackAddLanguageSpecificTextSet("LST2A436DA8_0?cpi=1547|nu=."); Property

Indicates whether the plugin is finished

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
bool Finished { get; }
```

ReadOnly Property Finished As Boolean
Get

```
Dim instance As cPluginCallback
Dim value As Boolean
```

```
value = instance.Finished
```

```
property bool Finished {
    bool get ();
}
```

```
abstract Finished : bool with get
```

Return Value

Type: [Boolean](#)

True if the plugin is finished; else False

▲ See Also

Reference

[cPluginCallback Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPluginCallback Methods

The [cPluginCallback](#) type exposes the following members.

Methods

Â Name	Description
 Finish	This method must be called when the plugin has finished.

[Top](#)

 See Also

Reference

[cPluginCallback Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cPluginCallbackAddLanguageSpecificTextSet("LST8F5D6FCE_0?2549:|nu=.") Method

This method must be called when the plugin has finished.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
void Finish(
    int iVal
)
```

```
Sub Finish (
    iVal As Integer
)
```

```
Dim instance As cPluginCallback
Dim iVal As Integer
```

```
instance.Finish(iVal)
```

```
void Finish(
    int iVal
)
```

```
abstract Finish :
    iVal : int -> unit
```

Parameters

iVal

Type: [System.Int32](#)

Sets the [ErrorFlag](#) property

▲ See Also

Reference

[cPluginCallback Interface](#)

[ETABSv1 Namespace](#)

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cPluginCallbackspan id="LST8F5D6FCE_0"AddLanguageSpecificTextSet("LST8F5D6FCE_0?2549:|nu=.")

»

CSI API ETABS v1

cPluginContract Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cPluginContract
```

```
Public Interface cPluginContract
```

```
Dim instance As cPluginContract
```

```
public interface class cPluginContract
```

```
type cPluginContract = interface end
```

The cPluginContract type exposes the following members.

▲ Methods

▲ Name	Description
Info	Provides information about the plugin
Main	The entry point for launching the plugin

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPluginContract Methods

The [cPluginContract](#) type exposes the following members.

Methods

Â	Name	Description
»	Info	Provides information about the plugin
»	Main	The entry point for launching the plugin

[Top](#)

» See Also

Reference

[cPluginContract Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cPluginContractAddLanguageSpecificTextSet("LSTD95B3204_0?cnu=.") Method

Provides information about the plugin

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int Info(
    ref string Text
)
```

```
Function Info (
    ByRef Text As String
) As Integer
```

```
Dim instance As cPluginContract
Dim Text As String
Dim returnValue As Integer
```

```
returnValue = instance.Info(Text)
```

```
int Info(
    String^% Text
)
```

```
abstract Info :
    Text : string byref -> int
```

Parameters

Text

Type: [SystemString](#)

Information about the plugin, e.g. author, version, description

Return Value

Type: [Int32](#)

Returns zero if the information is successfully retrieved; otherwise it returns a nonzero value

▲ See Also

Reference

cPluginContract Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPluginContractAddLanguageSpecificTextSet("LST997E8963_0") Method

The entry point for launching the plugin

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
void Main(
    ref cSapModel SapModel,
    ref cPluginCallback ISapPlugin
)
```

```
Sub Main (
    ByRef SapModel As cSapModel,
    ByRef ISapPlugin As cPluginCallback
)
```

```
Dim instance As cPluginContract
Dim SapModel As cSapModel
Dim ISapPlugin As cPluginCallback
```

```
instance.Main(SapModel, ISapPlugin)
```

```
void Main(
    cSapModel^% SapModel,
    cPluginCallback^% ISapPlugin
)
```

```
abstract Main :
    SapModel : cSapModel byref *
    ISapPlugin : cPluginCallback byref -> unit
```

Parameters

SapModel

Type: [ETABSv1cSapModel](#)

A reference to the SapModel object upon which all operations will be performed

ISapPlugin

Type: [ETABSv1cPluginCallback](#)

A reference to the callback object, used to retrieve the plugin status and notify the program when the plugin is closed

▲ See Also

Reference

cPluginContract Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPointElm Interface

Namespace: $\hat{A}$ ETABSv1

Assembly: $\hat{A}$ ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▀ Syntax

C#

VB

C++

F#

Copy

```
public interface cPointElm
```

```
Public Interface cPointElm
```

```
Dim instance As cPointElm
```

```
public interface class cPointElm
```

```
type cPointElm = interface end
```

The cPointElm type exposes the following members.

▀ Methods

$\hat{A}$	Name	Description
	<u>Count</u>	
	<u>CountConstraint</u>	
	<u>CountLoadDispl</u>	
	<u>CountLoadForce</u>	
	<u>CountRestraint</u>	
	<u>CountSpring</u>	
	<u>GetConnectivity</u>	
	<u>GetConstraint</u>	
	<u>GetCoordCartesian</u>	
	<u>GetLoadDispl</u>	
	<u>GetLoadForce</u>	
	<u>GetLocalAxes</u>	
	<u>GetNameList</u>	
	<u>GetObj</u>	
	<u>GetPatternValue</u>	
	<u>GetRestraint</u>	
	<u>GetSpring</u>	
	<u>GetSpringCoupled</u>	
	<u>GetTransformationMatrix</u>	

 IsSpringCoupled

Top

▀ See Also

Reference

ETABSv1 Namespace

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CSI API ETABS v1

cPointElm Methods

The cPointElm type exposes the following members.

Methods

Â	Name	Description
⇒	<u>Count</u>	
⇒	<u>CountConstraint</u>	
⇒	<u>CountLoadDispl</u>	
⇒	<u>CountLoadForce</u>	
⇒	<u>CountRestraint</u>	
⇒	<u>CountSpring</u>	
⇒	<u>GetConnectivity</u>	
⇒	<u>GetConstraint</u>	
⇒	<u>GetCoordCartesian</u>	
⇒	<u>GetLoadDispl</u>	
⇒	<u>GetLoadForce</u>	
⇒	<u>GetLocalAxes</u>	
⇒	<u>GetNameList</u>	
⇒	<u>GetObj</u>	
⇒	<u>GetPatternValue</u>	
⇒	<u>GetRestraint</u>	
⇒	<u>GetSpring</u>	
⇒	<u>GetSpringCoupled</u>	
⇒	<u>GetTransformationMatrix</u>	
⇒	<u>IsSpringCoupled</u>	

Top

See Also

Reference

cPointElm Interface

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CSI API ETABS v1

cPointElmAddLanguageSpecificTextSet("LSTBEA7BBE Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

int Count()

Function Count As Integer

Dim instance As cPointElm

Dim returnValue As Integer

returnValue = instance.Count()

int Count()

abstract Count : unit -> int

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cPointElm Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPointElmAddLanguageSpecificTextSet("LST35943B3F0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int CountConstraint(
    ref int Count,
    string Name = ""
)
```

```
Function CountConstraint (
    ByRef Count As Integer,
    Optional Name As String = ""
) As Integer
```

```
Dim instance As cPointElm
Dim Count As Integer
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.CountConstraint(Count,
    Name)
```

```
int CountConstraint(
    int% Count,
    String^ Name = L""
)
```

```
abstract CountConstraint :
    Count : int byref *
    ?Name : string
(* Defaults:
    let _Name = defaultArg Name ""
*)
-> int
```

Parameters

Count

Type: [System.Int32](#)

Name (Optional)

Type: [System.String](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cPointElm Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPointElmAddLanguageSpecificTextSet("LSTFE32EA0B_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int CountLoadDispl(
    ref int Count,
    string Name = "",
    string LoadPat = ""
)
```

```
Function CountLoadDispl (
    ByRef Count As Integer,
    Optional Name As String = "",
    Optional LoadPat As String = ""
) As Integer
```

```
Dim instance As cPointElm
Dim Count As Integer
Dim Name As String
Dim LoadPat As String
Dim returnValue As Integer
```

```
returnValue = instance.CountLoadDispl(Count,
    Name, LoadPat)
```

```
int CountLoadDispl(
    int% Count,
    String^ Name = L"",
    String^ LoadPat = L""
)
```

```
abstract CountLoadDispl :
    Count : int byref *
    ?Name : string *
    ?LoadPat : string
(* Defaults:
    let _Name = defaultArg Name ""
    let _LoadPat = defaultArg LoadPat ""
*)
-> int
```

Parameters

Count

Type: [System.Int32](#)

cPointElmspan id="LSTFE32EA0B_0"AddLanguageSpecificTextSet("LSTFE32EA0B_0?cpp=::2562");Count

Name (Optional)

Type: SystemString

LoadPat (Optional)

Type: SystemString

Return Value

Type: Int32

▲ See Also

Reference

cPointElm Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPointElmAddLanguageSpecificTextSet("LST6ABD6F81_0?cpp=::2564");Count

Method

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int CountLoadForce(
    ref int Count,
    string Name = "",
    string LoadPat = ""
)
```

```
Function CountLoadForce (
    ByRef Count As Integer,
    Optional Name As String = "",
    Optional LoadPat As String = ""
) As Integer
```

```
Dim instance As cPointElm
Dim Count As Integer
Dim Name As String
Dim LoadPat As String
Dim returnValue As Integer
```

```
returnValue = instance.CountLoadForce(Count,
    Name, LoadPat)
```

```
int CountLoadForce(
    int% Count,
    String^ Name = L"",
    String^ LoadPat = L""
)
```

```
abstract CountLoadForce :
    Count : int byref *
    ?Name : string *
    ?LoadPat : string
(* Defaults:
    let _Name = defaultArg Name ""
    let _LoadPat = defaultArg LoadPat ""
*)
-> int
```

Parameters

Count

Type: [System.Int32](#)

cPointElmspan id="LST6ABD6F81_0"AddLanguageSpecificTextSet("LST6ABD6F81_0?cpp=::2564");Count

Name (Optional)

Type: SystemString

LoadPat (Optional)

Type: SystemString

Return Value

Type: Int32

▲ See Also

Reference

cPointElm Interface

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CSI API ETABS v1

cPointElmAddLanguageSpecificTextSet("LSTA0F20C70") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

int CountRestraint()

Function CountRestraint As Integer

Dim instance As cPointElm

Dim returnValue As Integer

returnValue = instance.CountRestraint()

int CountRestraint()

abstract CountRestraint : unit -> int

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cPointElm Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPointElmAddLanguageSpecificTextSet("LSTF7B2061C_0?cpp=::|2567);Count Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int CountSpring()
```

```
Function CountSpring As Integer
```

```
Dim instance As cPointElm
```

```
Dim returnValue As Integer
```

```
returnValue = instance.CountSpring()
```

```
int CountSpring()
```

```
abstract CountSpring : unit -> int
```

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cPointElm Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPointElmAddLanguageSpecificTextSet("LST582119D_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetConnectivity(
    string Name,
    ref int NumberItems,
    ref int[] ObjectType,
    ref string[] ObjectName,
    ref int[] PointNumber
)
```

```
Function GetConnectivity (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef ObjectType As Integer(),
    ByRef ObjectName As String(),
    ByRef PointNumber As Integer()
) As Integer
```

```
Dim instance As cPointElm
Dim Name As String
Dim NumberItems As Integer
Dim ObjectType As Integer()
Dim ObjectName As String()
Dim PointNumber As Integer()
Dim returnValue As Integer
```

```
returnValue = instance.GetConnectivity(Name,
    NumberItems, ObjectType, ObjectName,
    PointNumber)
```

```
int GetConnectivity(
    String^ Name,
    int% NumberItems,
    array<int>^% ObjectType,
    array<String^>^% ObjectName,
    array<int>^% PointNumber
)
```

```
abstract GetConnectivity :
    Name : string *
    NumberItems : int byref *
    ObjectType : int[] byref *
    ObjectName : string[] byref *
    PointNumber : int[] byref -> int
```

Parameters

Name

Type:Â SystemString

NumberItems

Type:Â SystemInt32

ObjectType

Type:Â SystemInt32

ObjectName

Type:Â SystemString

PointNumber

Type:Â SystemInt32

Return Value

Type:Â Int32

▲ See Also

Reference

cPointElm Interface

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CSI API ETABS v1

cPointElmAddLanguageSpecificTextSet("LST7D70227A_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetConstraint(
    string Name,
    ref int NumberItems,
    ref string[] PointName,
    ref string[] ConstraintName,
    eItemTypeElm ItemTypeElm = eItemTypeElm.Element
)
```

```
Function GetConstraint (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef PointName As String(),
    ByRef ConstraintName As String(),
    Optional ItemTypeElm As eItemTypeElm = eItemTypeElm.Element
) As Integer
```

```
Dim instance As cPointElm
Dim Name As String
Dim NumberItems As Integer
Dim PointName As String()
Dim ConstraintName As String()
Dim ItemTypeElm As eItemTypeElm
Dim returnValue As Integer
```

```
returnValue = instance.GetConstraint(Name,
    NumberItems, PointName, ConstraintName,
    ItemTypeElm)
```

```
int GetConstraint(
    String^ Name,
    int% NumberItems,
    array<String^>^% PointName,
    array<String^>^% ConstraintName,
    eItemTypeElm ItemTypeElm = eItemTypeElm::Element
)
```

```
abstract GetConstraint :
    Name : string *
    NumberItems : int byref *
    PointName : string[] byref *
    ConstraintName : string[] byref *
    ?ItemTypeElm : eItemTypeElm
```

Introduction

```
(* Defaults:  
    let _ItemTypeElm = defaultArg ItemTypeElm eItemTypeElm.Element  
*)  
-> int
```

Parameters

Name

Type: SystemString

NumberItems

Type: SystemInt32

PointName

Type: SystemString

ConstraintName

Type: SystemString

ItemTypeElm (Optional)

Type: ETABSv1eItemTypeElm

Return Value

Type: Int32

▲ See Also

Reference

cPointElm Interface

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CSI API ETABS v1

cPointElmAddLanguageSpecificTextSet("LST74613E38_0?cpp=::|n2572;GetCoordCartesian")

Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetCoordCartesian(
    string Name,
    ref double X,
    ref double Y,
    ref double Z,
    string CSys = "Global"
)
```

```
Function GetCoordCartesian (
    Name As String,
    ByRef X As Double,
    ByRef Y As Double,
    ByRef Z As Double,
    Optional CSys As String = "Global"
) As Integer
```

```
Dim instance As cPointElm
Dim Name As String
Dim X As Double
Dim Y As Double
Dim Z As Double
Dim CSys As String
Dim returnValue As Integer
```

```
returnValue = instance.GetCoordCartesian(Name,
    X, Y, Z, CSys)
```

```
int GetCoordCartesian(
    String^ Name,
    double% X,
    double% Y,
    double% Z,
    String^ CSys = L"Global"
)
```

```
abstract GetCoordCartesian :
    Name : string *
    X : float byref *
    Y : float byref *
    Z : float byref *
    ?CSys : string
(* Defaults:
```



```

    let _CSys = defaultArg CSys "Global"
*)
-> int

```

Parameters

Name

Type:Â [SystemString](#)

X

Type:Â [SystemDouble](#)

Y

Type:Â [SystemDouble](#)

Z

Type:Â [SystemDouble](#)

CSys (Optional)

Type:Â [SystemString](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cPointElm Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPointElmAddLanguageSpecificTextSet("LST1F799EFF_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetLoadDispl(
    string Name,
    ref int NumberItems,
    ref string[] PointName,
    ref string[] LoadPat,
    ref int[] LcStep,
    ref string[] CSys,
    ref double[] U1,
    ref double[] U2,
    ref double[] U3,
    ref double[] R1,
    ref double[] R2,
    ref double[] R3,
    eItemTypeElm ItemTypeElm = eItemTypeElm.Element
)
```

```
Function GetLoadDispl (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef PointName As String(),
    ByRef LoadPat As String(),
    ByRef LcStep As Integer(),
    ByRef CSys As String(),
    ByRef U1 As Double(),
    ByRef U2 As Double(),
    ByRef U3 As Double(),
    ByRef R1 As Double(),
    ByRef R2 As Double(),
    ByRef R3 As Double(),
    ItemTypeElm As eItemTypeElm = eItemTypeElm.Element
) As Integer
```

```
Dim instance As cPointElm
Dim Name As String
Dim NumberItems As Integer
Dim PointName As String()
Dim LoadPat As String()
Dim LcStep As Integer()
Dim CSys As String()
Dim U1 As Double()
Dim U2 As Double()
Dim U3 As Double()
Dim R1 As Double()
```

cPointElmsAddLanguageSpecificTextSet("LST1F799EFF_0?cpp=::|2574);GetLo

Introduction

```
Dim R2 As Double()
Dim R3 As Double()
Dim ItemTypeElm As eItemTypeElm
Dim returnValue As Integer

returnValue = instance.GetLoadDispl(Name,
NumberItems, PointName, LoadPat,
LcStep, CSys, U1, U2, U3, R1, R2, R3,
ItemTypeElm)

int GetLoadDispl(
String^ Name,
int% NumberItems,
array<String^>^% PointName,
array<String^>^% LoadPat,
array<int>^% LcStep,
array<String^>^% CSys,
array<double>^% U1,
array<double>^% U2,
array<double>^% U3,
array<double>^% R1,
array<double>^% R2,
array<double>^% R3,
eItemTypeElm ItemTypeElm = eItemTypeElm::Element
)

abstract GetLoadDispl :
    Name : string *
    NumberItems : int byref *
    PointName : string[] byref *
    LoadPat : string[] byref *
    LcStep : int[] byref *
    CSys : string[] byref *
    U1 : float[] byref *
    U2 : float[] byref *
    U3 : float[] byref *
    R1 : float[] byref *
    R2 : float[] byref *
    R3 : float[] byref *
    ?ItemTypeElm : eItemTypeElm
(* Defaults:
    let _ItemTypeElm = defaultArg ItemTypeElm eItemTypeElm.Element
*)
-> int
```

Parameters

Name
Type:Â SystemString

NumberItems
Type:Â SystemInt32

PointName
Type:Â SystemString

LoadPat
Type:Â SystemString

LcStep
Type:Â SystemInt32

CSys

U1 Type:Â SystemString
 U2 Type:Â SystemDouble
 U3 Type:Â SystemDouble
 R1 Type:Â SystemDouble
 R2 Type:Â SystemDouble
 R3 Type:Â SystemDouble
 ItemTypeElm (Optional)
 Type:Â ETABSv1eItemTypeElm

Return Value

Type:Â Int32

▲ See Also

Reference

cPointElm Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPointElmAddLanguageSpecificTextSet("LST57B9EB1D_0?cpp=::257");GetLoadForce Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetLoadForce(
    string Name,
    ref int NumberItems,
    ref string[] PointName,
    ref string[] LoadPat,
    ref int[] LcStep,
    ref string[] CSys,
    ref double[] F1,
    ref double[] F2,
    ref double[] F3,
    ref double[] M1,
    ref double[] M2,
    ref double[] M3,
    eItemTypeElm ItemTypeElm = eItemTypeElm.Element
)
```

```
Function GetLoadForce (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef PointName As String(),
    ByRef LoadPat As String(),
    ByRef LcStep As Integer(),
    ByRef CSys As String(),
    ByRef F1 As Double(),
    ByRef F2 As Double(),
    ByRef F3 As Double(),
    ByRef M1 As Double(),
    ByRef M2 As Double(),
    ByRef M3 As Double(),
    ItemTypeElm As eItemTypeElm = eItemTypeElm.Element
) As Integer
```

```
Dim instance As cPointElm
Dim Name As String
Dim NumberItems As Integer
Dim PointName As String()
Dim LoadPat As String()
Dim LcStep As Integer()
Dim CSys As String()
Dim F1 As Double()
Dim F2 As Double()
Dim F3 As Double()
Dim M1 As Double()
```

cPointElmspan id="LST57B9EB1D_0"AddLanguageSpecificTextSet("LST57B9EB1D_0?cpp=::257");GetLoadForce

Introduction

```
Dim M2 As Double()
Dim M3 As Double()
Dim ItemTypeElm As eItemTypeElm
Dim returnValue As Integer

returnValue = instance.GetLoadForce(Name,
NumberItems, PointName, LoadPat,
LcStep, CSys, F1, F2, F3, M1, M2, M3,
ItemTypeElm)

int GetLoadForce(
String^ Name,
int% NumberItems,
array<String^>^% PointName,
array<String^>^% LoadPat,
array<int>^% LcStep,
array<String^>^% CSys,
array<double>^% F1,
array<double>^% F2,
array<double>^% F3,
array<double>^% M1,
array<double>^% M2,
array<double>^% M3,
eItemTypeElm ItemTypeElm = eItemTypeElm::Element
)

abstract GetLoadForce :
    Name : string *
    NumberItems : int byref *
    PointName : string[] byref *
    LoadPat : string[] byref *
    LcStep : int[] byref *
    CSys : string[] byref *
    F1 : float[] byref *
    F2 : float[] byref *
    F3 : float[] byref *
    M1 : float[] byref *
    M2 : float[] byref *
    M3 : float[] byref *
    ?ItemTypeElm : eItemTypeElm
(* Defaults:
    let _ItemTypeElm = defaultArg ItemTypeElm eItemTypeElm.Element
*)
-> int
```

Parameters

Name
Type:Â SystemString

NumberItems
Type:Â SystemInt32

PointName
Type:Â SystemString

LoadPat
Type:Â SystemString

LcStep
Type:Â SystemInt32

CSys

F1 Type:Â SystemString
 F2 Type:Â SystemDouble
 F3 Type:Â SystemDouble
 M1 Type:Â SystemDouble
 M2 Type:Â SystemDouble
 M3 Type:Â SystemDouble
 ItemTypeElm (Optional)
 Type:Â ETABSv1eItemTypeElm

Return Value

Type:Â Int32

▲ See Also

Reference

cPointElm Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPointElmAddLanguageSpecificTextSet("LST508A4A63_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetLocalAxes(
    string Name,
    ref double A,
    ref double B,
    ref double C
)
```

```
Function GetLocalAxes (
    Name As String,
    ByRef A As Double,
    ByRef B As Double,
    ByRef C As Double
) As Integer
```

```
Dim instance As cPointElm
Dim Name As String
Dim A As Double
Dim B As Double
Dim C As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetLocalAxes(Name,
    A, B, C)
```

```
int GetLocalAxes(
    String^ Name,
    double% A,
    double% B,
    double% C
)
```

```
abstract GetLocalAxes :
    Name : string *
    A : float byref *
    B : float byref *
    C : float byref -> int
```

Parameters

Name

Type: SystemString

cPointElmspan id="LST508A4A63_0"AddLanguageSpecificTextSet("LST508A4A63_0?cpp=::|n2580);GetLoc

- A Type:Â [SystemDouble](#)
- B Type:Â [SystemDouble](#)
- C Type:Â [SystemDouble](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cPointElm Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPointElmAddLanguageSpecificTextSet("LST4D580FB6") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNameList(
    ref int NumberNames,
    ref string[] MyName
)
```

```
Function GetNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cPointElm
Dim NumberNames As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetNameList(NumberNames,
    MyName)
```

```
int GetNameList(
    int% NumberNames,
    array<String^>^% MyName
)
```

```
abstract GetNameList :
    NumberNames : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberNames

Type: [System.Int32](#)

MyName

Type: [System.String](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cPointElm Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPointElmAddLanguageSpecificTextSet("LST95EAEC22_0?cpp=::2584");GetO Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetObj(
    string Name,
    ref string Obj,
    ref int ObjType
)
```

```
Function GetObj (
    Name As String,
    ByRef Obj As String,
    ByRef ObjType As Integer
) As Integer
```

```
Dim instance As cPointElm
Dim Name As String
Dim Obj As String
Dim ObjType As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetObj(Name, Obj,
    ObjType)
```

```
int GetObj(
    String^ Name,
    String^% Obj,
    int% ObjType
)
```

```
abstract GetObj :
    Name : string *
    Obj : string byref *
    ObjType : int byref -> int
```

Parameters

Name

Type: [SystemString](#)

Obj

Type: [SystemString](#)

ObjType

Type: [SystemInt32](#)

cPointElmsAddLanguageSpecificTextSet("LST95EAEC22_0?cpp=::2584");GetO

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cPointElm Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPointElmAddLanguageSpecificTextSet("LST56B9C85_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPatternValue(
    string Name,
    string PatternName,
    ref double Value
)
```

```
Function GetPatternValue (
    Name As String,
    PatternName As String,
    ByRef Value As Double
) As Integer
```

```
Dim instance As cPointElm
Dim Name As String
Dim PatternName As String
Dim Value As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPatternValue(Name,
    PatternName, Value)
```

```
int GetPatternValue(
    String^ Name,
    String^ PatternName,
    double% Value
)
```

```
abstract GetPatternValue :
    Name : string *
    PatternName : string *
    Value : float byref -> int
```

Parameters

Name

Type: [System.String](#)

PatternName

Type: [System.String](#)

Value

Type: [System.Double](#)

cPointElmspan id="LST56B9C85_0"AddLanguageSpecificTextSet("LST56B9C85_0?cpp=::|nu=2586GetPatter

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cPointElm Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPointElmAddLanguageSpecificTextSet("LST2CFE3D44") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetRestraint(
    string Name,
    ref bool[] Value
)
```

```
Function GetRestraint (
    Name As String,
    ByRef Value As Boolean()
) As Integer
```

```
Dim instance As cPointElm
Dim Name As String
Dim Value As Boolean()
Dim returnValue As Integer
```

```
returnValue = instance.GetRestraint(Name,
    Value)
```

```
int GetRestraint(
    String^ Name,
    array<bool>^% Value
)
```

```
abstract GetRestraint :
    Name : string *
    Value : bool[] byref -> int
```

Parameters

Name

Type: [SystemString](#)

Value

Type: [SystemBoolean](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cPointElm Interface

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CSI API ETABS v1

cPointElmAddLanguageSpecificTextSet("LST3E046FD5") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetSpring(
    string Name,
    ref double[] K
)
```

```
Function GetSpring (
    Name As String,
    ByRef K As Double()
) As Integer
```

```
Dim instance As cPointElm
Dim Name As String
Dim K As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetSpring(Name,
K)
```

```
int GetSpring(
    String^ Name,
    array<double>^% K
)
```

```
abstract GetSpring :
    Name : string *
    K : float[] byref -> int
```

Parameters

Name

Type: [SystemString](#)

K

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cPointElm Interface

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CSI API ETABS v1

cPointElmAddLanguageSpecificTextSet("LST13CD1AB6_0?cpp=::2592");GetS

Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetSpringCoupled(
    string Name,
    ref double[] K
)
```

```
Function GetSpringCoupled (
    Name As String,
    ByRef K As Double()
) As Integer
```

```
Dim instance As cPointElm
Dim Name As String
Dim K As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetSpringCoupled(Name,
    K)
```

```
int GetSpringCoupled(
    String^ Name,
    array<double>^% K
)
```

```
abstract GetSpringCoupled :
    Name : string *
    K : float[] byref -> int
```

Parameters

Name

Type: [SystemString](#)

K

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cPointElm Interface

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CSI API ETABS v1

cPointElmAddLanguageSpecificTextSet("LST1B538E10_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetTransformationMatrix(
    string Name,
    ref double[] Value
)
```

```
Function GetTransformationMatrix (
    Name As String,
    ByRef Value As Double()
) As Integer
```

```
Dim instance As cPointElm
Dim Name As String
Dim Value As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetTransformationMatrix(Name,
    Value)
```

```
int GetTransformationMatrix(
    String^ Name,
    array<double>^% Value
)
```

```
abstract GetTransformationMatrix :
    Name : string *
    Value : float[] byref -> int
```

Parameters

Name

Type: [SystemString](#)

Value

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cPointElm Interface

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CSI API ETABS v1

cPointElmAddLanguageSpecificTextSet("LST58BA2EF6_0?cpp=::|2596");IsSpringCoupled Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int IsSpringCoupled(
    string Name,
    ref bool IsCoupled
)
```

```
Function IsSpringCoupled (
    Name As String,
    ByRef IsCoupled As Boolean
) As Integer
```

```
Dim instance As cPointElm
Dim Name As String
Dim IsCoupled As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.IsSpringCoupled(Name,
    IsCoupled)
```

```
int IsSpringCoupled(
    String^ Name,
    bool% IsCoupled
)
```

```
abstract IsSpringCoupled :
    Name : string *
    IsCoupled : bool byref -> int
```

Parameters

Name

Type: [SystemString](#)

IsCoupled

Type: [SystemBoolean](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cPointElm Interface

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CSI API ETABS v1

cPointObj Interface

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cPointObj
```

```
Public Interface cPointObj
```

```
Dim instance As cPointObj
```

```
public interface class cPointObj
```

```
type cPointObj = interface end
```

The cPointObj type exposes the following members.

▲ Methods

▲	Name	Description
⇒	AddCartesian	Adds a point object to a model. The added point object will be tagged as a Special Point except if it was merged with another point object. Special points are allowed to exist in the model with no objects connected to them.
⇒	ChangeName	
⇒	Count	
⇒	CountLoadDispl	
⇒	CountLoadForce	
⇒	CountPanelZone	
⇒	CountRestraint	
⇒	CountSpring	
⇒	DeleteLoadDispl	Deletes all ground displacement load assignments, for the specified load pattern, from the specified point object(s).
⇒	DeleteLoadForce	Deletes all point force load assignments, for the specified load pattern, from the specified point object(s).
⇒	DeleteMass	
⇒	DeletePanelZone	
⇒	DeleteRestraint	Deletes all restraint assignments from the specified point object(s).

Introduction

⇒ <u>DeleteSpecialPoint</u>	Deletes special point objects that have no other objects connected to them.
⇒ <u>DeleteSpring</u>	Deletes all point spring assignments from the specified point object(s).
⇒ <u>GetAllPoints</u>	Retrieves selected data for all point objects in the model
⇒ <u>GetCommonTo</u>	
⇒ <u>GetConnectivity</u>	Retrieves a list of objects connected to a specified point object
⇒ <u>GetCoordCartesian</u>	Retrieves the x, y and z coordinates of the specified point object in the Present Units.
⇒ <u>GetCoordCylindrical</u>	
⇒ <u>GetCoordSpherical</u>	
⇒ <u>GetDiaphragm</u>	Retrieves the diaphragm for a specified point object
⇒ <u>GetElm</u>	
⇒ <u>GetGroupAssign</u>	Retrieves the groups to which a point object is assigned.
⇒ <u>GetGUID</u>	Retrieves the GUID for specified point object.
⇒ <u>GetLabelFromName</u>	Retrieves the label and story for a unique point object name
⇒ <u>GetLabelNameList</u>	Retrieves the names and labels of all defined point objects.
⇒ <u>GetLoadDispl</u>	Retrieves the ground displacement load assignments to point objects.
⇒ <u>GetLoadForce</u>	Retrieves the joint force load assignments to point objects.
⇒ <u>GetLocalAxes</u>	Retrieves the local axis angles for a point object.
⇒ <u>GetMass</u>	
⇒ <u>GetNameFromLabel</u>	Retrieves the unique name of a point object, given the label and story level
⇒ <u>GetNameList</u>	Retrieves the names of all defined point objects.
⇒ <u>GetNameListOnStory</u>	Retrieves the names of the defined point objects on a given story.
⇒ <u>GetPanelZone</u>	
⇒ <u>GetRestraint</u>	Retrieves the restraint assignments for a point object. The restraint assignments are always returned in the point local coordinate system.
⇒ <u>GetSelected</u>	
⇒ <u>GetSpecialPoint</u>	
⇒ <u>GetSpring</u>	Retrieves uncoupled spring stiffness assignments for a point object, that is, it retrieves the diagonal terms in the 6x6 spring matrix for the point object.
⇒ <u>GetSpringAssignment</u>	Retrieves the named point spring property assignment for a point object
⇒ <u>GetSpringCoupled</u>	
⇒ <u>GetTransformationMatrix</u>	

 <u>IsSpringCoupled</u>	
 <u>SetDiaphragm</u>	Assigns a diaphragm to a point object
 <u>SetGroupAssign</u>	Adds or removes point objects from a specified group.
 <u>SetGUID</u>	Sets the GUID for specified point object. If the GUID is passed in as a blank string, the program automatically creates a GUID for the object.
 <u>SetLoadDispl</u>	Makes ground displacement load assignments to point objects.
 <u>SetLoadForce</u>	Makes point force load assignments to point objects.
 <u>SetMass</u>	
 <u>SetMassByVolume</u>	
 <u>SetMassByWeight</u>	
 <u>SetPanelZone</u>	
 <u>SetRestraint</u>	Assigns the restraint assignments for a point object. The restraint assignments are always set in the point local coordinate system.
 <u>SetSelected</u>	Sets the selected status for a point object.
 <u>SetSpecialPoint</u>	Sets the special point status for a point object.
 <u>SetSpring</u>	Assigns coupled springs to a point object.
 <u>SetSpringAssignment</u>	Assigns an existing named point spring property to point objects
 <u>SetSpringCoupled</u>	

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▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPointObj Methods

The cPointObj type exposes the following members.

Methods

Ä	Name	Description
⇒	<u>AddCartesian</u>	Adds a point object to a model. The added point object will be tagged as a Special Point except if it was merged with another point object. Special points are allowed to exist in the model with no objects connected to them.
⇒	<u>ChangeName</u>	
⇒	<u>Count</u>	
⇒	<u>CountLoadDispl</u>	
⇒	<u>CountLoadForce</u>	
⇒	<u>CountPanelZone</u>	
⇒	<u>CountRestraint</u>	
⇒	<u>CountSpring</u>	
⇒	<u>DeleteLoadDispl</u>	Deletes all ground displacement load assignments, for the specified load pattern, from the specified point object(s).
⇒	<u>DeleteLoadForce</u>	Deletes all point force load assignments, for the specified load pattern, from the specified point object(s).
⇒	<u>DeleteMass</u>	
⇒	<u>DeletePanelZone</u>	
⇒	<u>DeleteRestraint</u>	Deletes all restraint assignments from the specified point object(s).
⇒	<u>DeleteSpecialPoint</u>	Deletes special point objects that have no other objects connected to them.
⇒	<u>DeleteSpring</u>	Deletes all point spring assignments from the specified point object(s).
⇒	<u>GetAllPoints</u>	Retrieves selected data for all point objects in the model
⇒	<u>GetCommonTo</u>	
⇒	<u>GetConnectivity</u>	Retrieves a list of objects connected to a specified point object
⇒	<u>GetCoordCartesian</u>	Retrieves the x, y and z coordinates of the specified point object in the Present Units.
⇒	<u>GetCoordCylindrical</u>	
⇒	<u>GetCoordSpherical</u>	
⇒	<u>GetDiaphragm</u>	Retrieves the diaphragm for a specified point object
⇒	<u>GetElm</u>	
⇒	<u>GetGroupAssign</u>	Retrieves the groups to which a point object is assigned.
⇒	<u>GetGUID</u>	Retrieves the GUID for specified point object.

Introduction

⇒ <u>GetLabelFromName</u>	Retrieves the label and story for a unique point object name
⇒ <u>GetLabelNameList</u>	Retrieves the names and labels of all defined point objects.
⇒ <u>GetLoadDispl</u>	Retrieves the ground displacement load assignments to point objects.
⇒ <u>GetLoadForce</u>	Retrieves the joint force load assignments to point objects.
⇒ <u>GetLocalAxes</u>	Retrieves the local axis angles for a point object.
⇒ <u>GetMass</u>	
⇒ <u>GetNameFromLabel</u>	Retrieves the unique name of a point object, given the label and story level
⇒ <u>GetNameList</u>	Retrieves the names of all defined point objects.
⇒ <u>GetNameListOnStory</u>	Retrieves the names of the defined point objects on a given story.
⇒ <u>GetPanelZone</u>	
⇒ <u>GetRestraint</u>	Retrieves the restraint assignments for a point object. The restraint assignments are always returned in the point local coordinate system.
⇒ <u>GetSelected</u>	
⇒ <u>GetSpecialPoint</u>	
⇒ <u>GetSpring</u>	Retrieves uncoupled spring stiffness assignments for a point object, that is, it retrieves the diagonal terms in the 6x6 spring matrix for the point object.
⇒ <u>GetSpringAssignment</u>	Retrieves the named point spring property assignment for a point object
⇒ <u>GetSpringCoupled</u>	
⇒ <u>GetTransformationMatrix</u>	
⇒ <u>IsSpringCoupled</u>	
⇒ <u>SetDiaphragm</u>	Assigns a diaphragm to a point object
⇒ <u>SetGroupAssign</u>	Adds or removes point objects from a specified group.
⇒ <u>SetGUID</u>	Sets the GUID for specified point object. If the GUID is passed in as a blank string, the program automatically creates a GUID for the object.
⇒ <u>SetLoadDispl</u>	Makes ground displacement load assignments to point objects.
⇒ <u>SetLoadForce</u>	Makes point force load assignments to point objects.
⇒ <u>SetMass</u>	
⇒ <u>SetMassByVolume</u>	
⇒ <u>SetMassByWeight</u>	
⇒ <u>SetPanelZone</u>	
⇒ <u>SetRestraint</u>	Assigns the restraint assignments for a point object. The restraint assignments are always set in the point local coordinate system.

Introduction

- ⇒ [SetSelected](#) Sets the selected status for a point object.
- ⇒ [SetSpecialPoint](#) Sets the special point status for a point object.
- ⇒ [SetSpring](#) Assigns coupled springs to a point object.
- ⇒ [SetSpringAssignment](#) Assigns an existing named point spring property to point objects
- ⇒ [SetSpringCoupled](#)

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▲ See Also

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[cPointObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST95D093BA_0")

Method

Adds a point object to a model. The added point object will be tagged as a Special Point except if it was merged with another point object. Special points are allowed to exist in the model with no objects connected to them.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int AddCartesian(
    double X,
    double Y,
    double Z,
    ref string Name,
    string UserName = "",
    string CSys = "Global",
    bool MergeOff = false,
    int MergeNumber = 0
)
```

```
Function AddCartesian (
    X As Double,
    Y As Double,
    Z As Double,
    ByRef Name As String,
    Optional Name As String = "",
    Optional CSys As String = "Global",
    Optional MergeOff As Boolean = false,
    Optional MergeNumber As Integer = 0
) As Integer
```

```
Dim instance As cPointObj
Dim X As Double
Dim Y As Double
Dim Z As Double
Dim Name As String
Dim UserName As String
Dim CSys As String
Dim MergeOff As Boolean
Dim MergeNumber As Integer
Dim returnValue As Integer
```

```
returnValue = instance.AddCartesian(X,
    Y, Z, Name, UserName, CSys, MergeOff,
    MergeNumber)
```

```
int AddCartesian(
    double X,
```

cPointObjspan id="LST95D093BA_0"AddLanguageSpecificTextSet("LST95D093BA_0?cpp=::|2694);AddCa


```

double Y,
double Z,
String^% Name,
String^ UserName = L"",
String^ CSys = L"Global",
bool MergeOff = false,
int MergeNumber = 0
)

abstract AddCartesian :
    X : float *
    Y : float *
    Z : float *
    Name : string byref *
    ?UserName : string *
    ?CSys : string *
    ?MergeOff : bool *
    ?MergeNumber : int
(* Defaults:
    let _UserName = defaultArg UserName ""
    let _CSys = defaultArg CSys "Global"
    let _MergeOff = defaultArg MergeOff false
    let _MergeNumber = defaultArg MergeNumber 0
*)
-> int

```

Parameters

X

Type: SystemDouble

The X-coordinate of the added point object in the specified coordinate system.
[L]

Y

Type: SystemDouble

The Y-coordinate of the added point object in the specified coordinate system.
[L]

Z

Type: SystemDouble

The Z-coordinate of the added point object in the specified coordinate system.
[L]

Name

Type: SystemString

This is the name that the program ultimately assigns for the point object. If no userName is specified, the program assigns a default name to the point object. If a userName is specified and that name is not used for another point, the userName is assigned to the point; otherwise a default name is assigned to the point.

If a point is merged with another point, this will be the name of the point object with which it was merged.

UserName (Optional)

Type: SystemString

This is an optional user specified name for the point object. If a UserName is specified and that name is already used for another point object, the program ignores the UserName.

CSys (Optional)

Type:Â SystemString

The name of the coordinate system in which the joint coordinates are defined.

MergeOff (Optional)

Type:Â SystemBoolean

If this item is False, a new point object that is added at the same location as an existing point object will be merged with the existing point object (assuming the two point objects have the same MergeNumber) and thus only one point object will exist at the location.

If this item is True, the points will not merge and two point objects will exist at the same location.

MergeNumber (Optional)

Type:Â SystemInt32

Two points objects in the same location will merge only if their merge number assignments are the same. By default all pointobjects have a merge number of zero.

Return Value

Type:Â Int32

Returns zero if the point object is successfully added or merged, otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim x As Double, y As Double, z As Double
    Dim Name As String
    Dim MyName As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'add point object to model
    x = 12
    y = 37
    z = 0
    MyName = "A1"
```

Introduction

```
ret = SapModel.PointObj.AddCartesian(x, y, z, Name, MyName)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

📌 See Also

Reference

cPointObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST25A232D00") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int ChangeName(
    string Name,
    string NewName
)
```

```
Function ChangeName (
    Name As String,
    NewName As String
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim NewName As String
Dim returnValue As Integer
```

```
returnValue = instance.ChangeName(Name,
    NewName)
```

```
int ChangeName(
    String^ Name,
    String^ NewName
)
```

```
abstract ChangeName :
    Name : string *
    NewName : string -> int
```

Parameters

Name

Type: [SystemString](#)

NewName

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cPointObj Interface

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST12B34AA8_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

int Count()

Function Count As Integer

```
Dim instance As cPointObj
Dim returnValue As Integer
```

```
returnValue = instance.Count()
```

```
int Count()
```

```
abstract Count : unit -> int
```

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cPointObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST64CB88FB_0?cpp=::|264.1");Count

Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int CountLoadDispl(
    ref int Count,
    string Name = "",
    string LoadPat = ""
)
```

```
Function CountLoadDispl (
    ByRef Count As Integer,
    Optional Name As String = "",
    Optional LoadPat As String = ""
) As Integer
```

```
Dim instance As cPointObj
Dim Count As Integer
Dim Name As String
Dim LoadPat As String
Dim returnValue As Integer
```

```
returnValue = instance.CountLoadDispl(Count,
    Name, LoadPat)
```

```
int CountLoadDispl(
    int% Count,
    String^ Name = L"",
    String^ LoadPat = L""
)
```

```
abstract CountLoadDispl :
    Count : int byref *
    ?Name : string *
    ?LoadPat : string
(* Defaults:
    let _Name = defaultArg Name ""
    let _LoadPat = defaultArg LoadPat ""
*)
-> int
```

Parameters

Count

Type: [System.Int32](#)

cPointObjspan id="LST64CB88FB_0"AddLanguageSpecificTextSet("LST64CB88FB_0?cpp=::|264.1");Count

Name (Optional)

Type: SystemString

LoadPat (Optional)

Type: SystemString

Return Value

Type: Int32

▲ See Also

Reference

cPointObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST3AD78681_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int CountLoadForce(
    ref int Count,
    string Name = "",
    string LoadPat = ""
)
```

```
Function CountLoadForce (
    ByRef Count As Integer,
    Optional Name As String = "",
    Optional LoadPat As String = ""
) As Integer
```

```
Dim instance As cPointObj
Dim Count As Integer
Dim Name As String
Dim LoadPat As String
Dim returnValue As Integer
```

```
returnValue = instance.CountLoadForce(Count,
    Name, LoadPat)
```

```
int CountLoadForce(
    int% Count,
    String^ Name = L"",
    String^ LoadPat = L""
)
```

```
abstract CountLoadForce :
    Count : int byref *
    ?Name : string *
    ?LoadPat : string
(* Defaults:
    let _Name = defaultArg Name ""
    let _LoadPat = defaultArg LoadPat ""
*)
-> int
```

Parameters

Count

Type: [System.Int32](#)

Name (Optional)

Type: SystemString

LoadPat (Optional)

Type: SystemString

Return Value

Type: Int32

▲ See Also

Reference

cPointObj Interface

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST8E1C4325_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int CountPanelZone()
```

Function CountPanelZone As Integer

```
Dim instance As cPointObj
```

```
Dim returnValue As Integer
```

```
returnValue = instance.CountPanelZone()
```

```
int CountPanelZone()
```

```
abstract CountPanelZone : unit -> int
```

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cPointObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST5370F575_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

int CountRestraint()

Function CountRestraint As Integer

Dim instance As cPointObj

Dim returnValue As Integer

returnValue = instance.CountRestraint()

int CountRestraint()

abstract CountRestraint : unit -> int

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cPointObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LSTA2ACE4E4_0?cpp=::2617");CountSpring() Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

int CountSpring()

Function CountSpring As Integer

Dim instance As cPointObj

Dim returnValue As Integer

returnValue = instance.CountSpring()

int CountSpring()

abstract CountSpring : unit -> int

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cPointObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LSTB119591D_0") Method

Deletes all ground displacement load assignments, for the specified load pattern, from the specified point object(s).

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int DeleteLoadDispl(
    string Name,
    string LoadPat,
    eItemType ItemType = eItemType.Objects
)
```

```
Function DeleteLoadDispl (
    Name As String,
    LoadPat As String,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim LoadPat As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.DeleteLoadDispl(Name,
    LoadPat, ItemType)
```

```
int DeleteLoadDispl(
    String^ Name,
    String^ LoadPat,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract DeleteLoadDispl :
    Name : string *
    LoadPat : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters**Name**

Type: SystemString

The name of an existing point object or group depending on the value of the ItemType item.

LoadPat

Type: SystemString

The name of the load pattern for the ground displacement load.

ItemType (Optional)

Type: ETABSV1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If Objects is selected, the Name item refers to a point object. The load assignments, for the specified load pattern, made to that point object, are removed.

If Group is selected, the Name item refers to a group. The load assignments, for the specified load pattern, made to all point objects in the group, are removed.

If SelectedObjects is selected, the Name item is ignored. The load assignments, for the specified load pattern, made to all selected point objects, are removed.

Return Value

Type: Int32

Returns zero if the load assignments are successfully deleted, otherwise it returns a nonzero value.

Remarks**Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value as Double()

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)
```

Introduction

```
'add ground displacement load
  Redim Value(5)
  Value(0) = 10
  ret = SapModel.PointObj.SetLoadDispl("1", "DEAD", Value)

'delete ground displacement load
  ret = SapModel.PointObj.DeleteLoadDispl("1", "DEAD")

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPointObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LSTEC80F889_0")

Method

Deletes all point force load assignments, for the specified load pattern, from the specified point object(s).

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int DeleteLoadForce(
    string Name,
    string LoadPat,
    eItemType ItemType = eItemType.Objects
)
```

```
Function DeleteLoadForce (
    Name As String,
    LoadPat As String,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim LoadPat As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.DeleteLoadForce(Name,
    LoadPat, ItemType)
```

```
int DeleteLoadForce(
    String^ Name,
    String^ LoadPat,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract DeleteLoadForce :
    Name : string *
    LoadPat : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters**Name**

Type: SystemString

The name of an existing point object or group depending on the value of the ItemType item.

LoadPat

Type: SystemString

The name of the load pattern for the point force load.

ItemType (Optional)

Type: ETABSV1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If Objects is selected, the Name item refers to a point object. The load assignments, for the specified load pattern, made to that point object, are removed.

If Group is selected, the Name item refers to a group. The load assignments, for the specified load pattern, made to all point objects in the group, are removed.

If SelectedObjects is selected, the Name item is ignored. The load assignments, for the specified load pattern, made to all selected point objects, are removed.

Return Value

Type: Int32

Returns zero if the load assignments are successfully deleted, otherwise it returns a nonzero value.

Remarks**Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value as Double()

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)
```

Introduction

```
'add point load
  Redim Value(5)
  Value(0) = 10
  ret = SapModel.PointObj.SetLoadForce("2", "DEAD", Value)

'delete point load
  ret = SapModel.PointObj.DeleteLoadForce("2", "DEAD")

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

See Also

Reference

cPointObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST390F7989_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int DeleteMass(
    string Name,
    eItemType ItemType = eItemType.Objects
)
```

```
Function DeleteMass (
    Name As String,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.DeleteMass(Name,
    ItemType)
```

```
int DeleteMass(
    String^ Name,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract DeleteMass :
    Name : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: [SystemString](#)

ItemType (Optional)

Type: [ETABSv1eItemType](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cPointObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST1C17A8E7_0?cpp=::|2626);DeleteMethod

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int DeletePanelZone(
    string Name,
    eItemType ItemType = eItemType.Objects
)
```

```
Function DeletePanelZone (
    Name As String,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.DeletePanelZone(Name,
    ItemType)
```

```
int DeletePanelZone(
    String^ Name,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract DeletePanelZone :
    Name : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

ItemType (Optional)

Type: ETABSv1eItemType

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cPointObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST46737A3A") Method

Deletes all restraint assignments from the specified point object(s).

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int DeleteRestraint(
    string Name,
    eItemType ItemType = eItemType.Objects
)
```

```
Function DeleteRestraint (
    Name As String,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.DeleteRestraint(Name,
    ItemType)
```

```
int DeleteRestraint(
    String^ Name,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract DeleteRestraint :
    Name : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing point object or group depending on the value of the ItemType item.

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

- ◊ Object = 0
- ◊ Group = 1
- ◊ SelectedObjects = 2

If Objects is selected, the Name item refers to a point object. The restraint assignments for that point object are removed.

If Group is selected, the Name item refers to a group. The restraint assignments for all point objects in the group are removed.

If SelectedObjects is selected, the Name item is ignored. The restraint assignments for all selected point objects are removed.

Return Value

Type: Int32

Returns zero if the restraint assignments are successfully deleted, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'delete joint restraint assignment
    ret = SapModel.PointObj.DeleteRestraint("65")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPointObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LSTC6F467E_0") Method

Deletes special point objects that have no other objects connected to them.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int DeleteSpecialPoint(
    string Name,
    eItemType ItemType = eItemType.Objects
)
```

```
Function DeleteSpecialPoint (
    Name As String,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.DeleteSpecialPoint(Name,
    ItemType)
```

```
int DeleteSpecialPoint(
    String^ Name,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract DeleteSpecialPoint :
    Name : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing point object or group depending on the value of the ItemType item.

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

- ◊ Object = 0
- ◊ Group = 1
- ◊ SelectedObjects = 2

If Objects is selected, the deletion applies to the point object specified by the Name item.

If Group is selected, the deletion applies to all point objects in the group specified by the Name item.

If SelectedObjects is selected, the deletion applies to all selected point objects, and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the function completes successfully, otherwise it returns a nonzero value.

Remarks

Point objects can be deleted only if they have no other objects (e.g., frame, cable, tendon, area, solid link) connected to them. If a point object is not specified to be a Special Point, the program automatically deletes that point object when it has no other objects connected to it. If a point object is specified to be a Special Point, to delete it, first delete all other objects connected to the point and then call this function to delete the point.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set as special point
    ret = SapModel.PointObj.SetSpecialPoint("3", True)

    'delete frame objects
    ret = SapModel.FrameObj.Delete("2")
    ret = SapModel.FrameObj.Delete("8")
end Sub
```

Introduction

```
'delete special point object
    ret = SapModel.PointObj.DeleteSpecialPoint("3")

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPointObj Interface

ETABSv1 Namespace

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»

CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST6BE565590") Method

Deletes all point spring assignments from the specified point object(s).

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int DeleteSpring(
    string Name,
    eItemType ItemType = eItemType.Objects
)
```

```
Function DeleteSpring (
    Name As String,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.DeleteSpring(Name,
    ItemType)
```

```
int DeleteSpring(
    String^ Name,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract DeleteSpring :
    Name : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing point object or group depending on the value of the ItemType item.

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

- ◊ Object = 0
- ◊ Group = 1
- ◊ SelectedObjects = 2

If Objects is selected, the Name item refers to a point object. The point spring assignments for that point object are removed.

If Group is selected, the Name item refers to a group. The point spring assignments for all point objects in the group are removed.

If SelectedObjects is selected, the Name item is ignored. The point spring assignments for all selected point objects are removed.

Return Value

Type: Int32

Returns zero if the restraint assignments are successfully deleted, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim i As Integer
    Dim k As Double()

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'add joint spring assignment
    Redim k(5)
    For i = 0 to 5
        k(i) = i + 1
    Next i
    ret = SapModel.PointObj.SetSpring("3", k)

    'delete joint spring assignment
    ret = SapModel.PointObj.DeleteSpring("3")

    'close ETABS
```

Introduction

```
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

cPointObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST5DAA9B41_0?cpp=::|2037");GetAll

Method

Retrieves selected data for all point objects in the model

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetAllPoints(
    ref int NumberNames,
    ref string[] MyName,
    ref double[] X,
    ref double[] Y,
    ref double[] Z,
    string csys = "Global"
)
```

```
Function GetAllPoints (
    ByRef NumberNames As Integer,
    ByRef MyName As String(),
    ByRef X As Double(),
    ByRef Y As Double(),
    ByRef Z As Double(),
    Optional csys As String = "Global"
) As Integer
```

```
Dim instance As cPointObj
Dim NumberNames As Integer
Dim MyName As String()
Dim X As Double()
Dim Y As Double()
Dim Z As Double()
Dim csys As String
Dim returnValue As Integer
```

```
returnValue = instance.GetAllPoints(NumberNames,
    MyName, X, Y, Z, csys)
```

```
int GetAllPoints(
    int% NumberNames,
    array<String^>^% MyName,
    array<double>^% X,
    array<double>^% Y,
    array<double>^% Z,
    String^ csys = L"Global"
)
```

```
abstract GetAllPoints :
    NumberNames : int byref *
```

cPointObjspan id="LST5DAA9B41_0"AddLanguageSpecificTextSet("LST5DAA9B41_0?cpp=::|2037");GetAll

```

    MyName : string[] byref *
    X : float[] byref *
    Y : float[] byref *
    Z : float[] byref *
    ?csys : string
(* Defaults:
    let _csys = defaultArg csys "Global"
*)
-> int

```

Parameters

NumberNames

Type:Â SystemInt32

MyName

Type:Â SystemString

X

Type:Â SystemDouble

Y

Type:Â SystemDouble

Z

Type:Â SystemDouble

csys (Optional)

Type:Â SystemString

Return Value

Type:Â Int32

▲ Remarks

▲ See Also

Reference

cPointObj Interface

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST528DE101_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetCommonTo(
    string Name,
    ref int CommonTo
)
```

```
Function GetCommonTo (
    Name As String,
    ByRef CommonTo As Integer
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim CommonTo As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetCommonTo(Name,
    CommonTo)
```

```
int GetCommonTo(
    String^ Name,
    int% CommonTo
)
```

```
abstract GetCommonTo :
    Name : string *
    CommonTo : int byref -> int
```

Parameters

Name

Type: [SystemString](#)

CommonTo

Type: [SystemInt32](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cPointObj Interface

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST7525B610_0?cpp=::|num=264)|GetCon Method

Retrieves a list of objects connected to a specified point object

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetConnectivity(
    string Name,
    ref int NumberItems,
    ref int[] ObjectType,
    ref string[] ObjectName,
    ref int[] PointNumber
)
```

```
Function GetConnectivity (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef ObjectType As Integer(),
    ByRef ObjectName As String(),
    ByRef PointNumber As Integer()
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim NumberItems As Integer
Dim ObjectType As Integer()
Dim ObjectName As String()
Dim PointNumber As Integer()
Dim returnValue As Integer
```

```
returnValue = instance.GetConnectivity(Name,
    NumberItems, ObjectType, ObjectName,
    PointNumber)
```

```
int GetConnectivity(
    String^ Name,
    int% NumberItems,
    array<int>^% ObjectType,
    array<String>^% ObjectName,
    array<int>^% PointNumber
)
```

```
abstract GetConnectivity :
    Name : string *
    NumberItems : int byref *
    ObjectType : int[] byref *
    ObjectName : string[] byref *
```

cPointObjid="LST7525B610_0"AddLanguageSpecificTextSet("LST7525B610_0?cpp=::|num=264)|GetCon

PointNumber : int[] byref -> int

Parameters

Name

Type: SystemString

The name of an existing point object

NumberItems

Type: SystemInt32

This is the total number of objects connected to the specified point object

ObjectType

Type: SystemInt32

This is an array that includes the object type of each object connected to the specified point object.

2 Frame object

3 Cable object

4 Tendon object

5 Area object

6 Solid object

7 Link object

ObjectName

Type: SystemString

This is an array that includes the object name of each object connected to the specified point object

PointNumber

Type: SystemInt32

This is an array that includes the point number within the considered object that corresponds to the specified point object

Return Value

Type: Int32

Returns zero if the list is successfully filled; otherwise it returns nonzero

▲ Remarks

▲ See Also

Reference

cPointObj Interface

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST8A4796A8_0") Method

Retrieves the x, y and z coordinates of the specified point object in the Present Units.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetCoordCartesian(
    string Name,
    ref double X,
    ref double Y,
    ref double Z,
    string CSys = "Global"
)
```

```
Function GetCoordCartesian (
    Name As String,
    ByRef X As Double,
    ByRef Y As Double,
    ByRef Z As Double,
    Optional CSys As String = "Global"
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim X As Double
Dim Y As Double
Dim Z As Double
Dim CSys As String
Dim returnValue As Integer
```

```
returnValue = instance.GetCoordCartesian(Name,
    X, Y, Z, CSys)
```

```
int GetCoordCartesian(
    String^ Name,
    double% X,
    double% Y,
    double% Z,
    String^ CSys = L"Global"
)
```

```
abstract GetCoordCartesian :
    Name : string *
    X : float byref *
    Y : float byref *
    Z : float byref *
    ?CSys : string
```

cPointObjid="LST8A4796A8_0"AddLanguageSpecificTextSet("LST8A4796A8_0?cpp=::|n2643;GetCo

```
(* Defaults:
    let _CSys = defaultArg CSys "Global"
*)
-> int
```

Parameters

Name

Type:Â SystemString
The name of a defined point object.

X

Type:Â SystemDouble
The X-coordinate of the specified point object in the specified coordinate system. [L]

Y

Type:Â SystemDouble
The Y-coordinate of the specified point object in the specified coordinate system. [L]

Z

Type:Â SystemDouble
The Z-coordinate of the specified point object in the specified coordinate system. [L]

CSys (Optional)

Type:Â SystemString
The name of a defined coordinate system. If Csys is not specified, the Global coordinate system is assumed.

Return Value

Type:Â Int32

Returns zero if the coordinates are successfully returned; otherwise it returns nonzero.

▲ Remarks

The coordinates are reported in the coordinate system specified by CSys.

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim x As Double, y As Double, z As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()
```


Introduction

```
'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'get point coordinates
    ret = SapModel.PointObj.GetCoordCartesian("1", x, y, z)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

[cPointObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST2363E47E0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetCoordCylindrical(
    string Name,
    ref double R,
    ref double Theta,
    ref double Z,
    string CSys = "Global"
)
```

```
Function GetCoordCylindrical (
    Name As String,
    ByRef R As Double,
    ByRef Theta As Double,
    ByRef Z As Double,
    Optional CSys As String = "Global"
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim R As Double
Dim Theta As Double
Dim Z As Double
Dim CSys As String
Dim returnValue As Integer
```

```
returnValue = instance.GetCoordCylindrical(Name,
    R, Theta, Z, CSys)
```

```
int GetCoordCylindrical(
    String^ Name,
    double% R,
    double% Theta,
    double% Z,
    String^ CSys = L"Global"
)
```

```
abstract GetCoordCylindrical :
    Name : string *
    R : float byref *
    Theta : float byref *
    Z : float byref *
    ?CSys : string
(* Defaults:
```

```

        let _CSys = defaultArg CSys "Global"
*)
-> int

```

Parameters

Name

Type:Â SystemString

R

Type:Â SystemDouble

Theta

Type:Â SystemDouble

Z

Type:Â SystemDouble

CSys (Optional)

Type:Â SystemString

Return Value

Type:Â Int32

▲ See Also

Reference

cPointObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST6D71813E0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetCoordSpherical(
    string Name,
    ref double R,
    ref double A,
    ref double B,
    string CSys = "Global"
)
```

```
Function GetCoordSpherical (
    Name As String,
    ByRef R As Double,
    ByRef A As Double,
    ByRef B As Double,
    Optional CSys As String = "Global"
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim R As Double
Dim A As Double
Dim B As Double
Dim CSys As String
Dim returnValue As Integer
```

```
returnValue = instance.GetCoordSpherical(Name,
    R, A, B, CSys)
```

```
int GetCoordSpherical(
    String^ Name,
    double% R,
    double% A,
    double% B,
    String^ CSys = L"Global"
)
```

```
abstract GetCoordSpherical :
    Name : string *
    R : float byref *
    A : float byref *
    B : float byref *
    ?CSys : string
(* Defaults:
```

```

        let _CSys = defaultArg CSys "Global"
*)
-> int

```

Parameters

Name

Type:Â [SystemString](#)

R

Type:Â [SystemDouble](#)

A

Type:Â [SystemDouble](#)

B

Type:Â [SystemDouble](#)

CSys (Optional)

Type:Â [SystemString](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cPointObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST2397FCC2_0")

Method

Retrieves the diaphragm for a specified point object

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetDiaphragm(
    string Name,
    ref eDiaphragmOption DiaphragmOption,
    ref string DiaphragmName
)
```

```
Function GetDiaphragm (
    Name As String,
    ByRef DiaphragmOption As eDiaphragmOption,
    ByRef DiaphragmName As String
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim DiaphragmOption As eDiaphragmOption
Dim DiaphragmName As String
Dim returnValue As Integer
```

```
returnValue = instance.GetDiaphragm(Name,
    DiaphragmOption, DiaphragmName)
```

```
int GetDiaphragm(
    String^ Name,
    eDiaphragmOption% DiaphragmOption,
    String^% DiaphragmName
)
```

```
abstract GetDiaphragm :
    Name : string *
    DiaphragmOption : eDiaphragmOption byref *
    DiaphragmName : string byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing point object

DiaphragmOption

Type: ETABSv1eDiaphragmOption

This is an item from the eDiaphragmOption enumeration

If this item is Disconnect then the point object is disconnected from any diaphragm

If this item is FromShellObject then the point object inherits the diaphragm assignment of its bounding area object.

If this item is DefinedDiaphragm then the point object is assigned to the existing diaphragm specified by DiaphragmName

DiaphragmName

Type: SystemString

The name of an existing diaphragm. This item will only be filled if DiaphragmOption is DefinedDiaphragm

Return Value

Type: Int32

Returns zero if the diaphragm assignment is successfully retrieved, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim DiaphragmOption As eDiaphragmOption
    Dim DiaphragmName As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define a new diaphragm
    ret = SapModel.Diaphragm.SetDiaphragm("MyDiaph1A", True)

    'assign diaphragm to point
    ret = SapModel.PointObj.SetDiaphragm("1", eDiaphragmOption.DefinedDiaphragm, "MyDiaph1A")

    'get point diaphragm assignment
    ret = SapModel.PointObj.GetDiaphragm("1", DiaphragmOption, DiaphragmName)
```

Introduction

```
'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPointObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST62224BCC_0?cpp=::|2653");GetElm Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetElm(
    string Name,
    ref string Elm
)
```

```
Function GetElm (
    Name As String,
    ByRef Elm As String
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim Elm As String
Dim returnValue As Integer
```

```
returnValue = instance.GetElm(Name, Elm)
```

```
int GetElm(
    String^ Name,
    String^% Elm
)
```

```
abstract GetElm :
    Name : string *
    Elm : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

Elm

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cPointObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LSTA8763D8F0")

Method

Retrieves the groups to which a point object is assigned.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetGroupAssign(
    string Name,
    ref int NumberGroups,
    ref string[] Groups
)
```

```
Function GetGroupAssign (
    Name As String,
    ByRef NumberGroups As Integer,
    ByRef Groups As String()
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim NumberGroups As Integer
Dim Groups As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetGroupAssign(Name,
    NumberGroups, Groups)
```

```
int GetGroupAssign(
    String^ Name,
    int% NumberGroups,
    array<String^>^% Groups
)
```

```
abstract GetGroupAssign :
    Name : string *
    NumberGroups : int byref *
    Groups : string[] byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing point object.

NumberGroups

Type: SystemInt32

The number of group names retrieved.

Groups

Type: SystemString

This is an array of the names of the groups to which the point object is assigned.

Return Value

Type: Int32

Returns zero if the group assignments are successfully retrieved, otherwise it returns a nonzero value.

▀ Remarks

▀ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberGroups As Integer
    Dim Groups() as String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new groups
    ret = SapModel.GroupDef.SetGroup("Group1")
    ret = SapModel.GroupDef.SetGroup("Group2")

    'add point object to groups
    ret = SapModel.PointObj.SetGroupAssign("3", "Group1")
    ret = SapModel.PointObj.SetGroupAssign("3", "Group2")

    'get point object groups
    ret = SapModel.PointObj.GetGroupAssign("3", NumberGroups, Groups)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▀ See Also

Reference

cPointObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LSTEC6350B9_0?cpp=::|2658);GetGUID")

Method

Retrieves the GUID for specified point object.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetGUID(
    string Name,
    ref string GUID
)
```

```
Function GetGUID (
    Name As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.GetGUID(Name, GUID)
```

```
int GetGUID(
    String^ Name,
    String^% GUID
)
```

```
abstract GetGUID :
    Name : string *
    GUID : string byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing point object.

GUID

Type: SystemString

The GUID (Global Unique ID) for the specified point object.

Return Value

Type: Int32

Returns zero if the point object GUID is successfully retrieved; otherwise it returns nonzero.

▀ **Remarks**

▀ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim GUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'get GUID
    ret = SapModel.PointObj.GetGUID("1", GUID)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▀ **See Also**

Reference

cPointObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST5AD85C6D_0?cpp=::2660");GetL

Method

Retrieves the label and story for a unique point object name

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetLabelFromName(
    string Name,
    ref string Label,
    ref string Story
)
```

```
Function GetLabelFromName (
    Name As String,
    ByRef Label As String,
    ByRef Story As String
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim Label As String
Dim Story As String
Dim returnValue As Integer
```

```
returnValue = instance.GetLabelFromName(Name,
    Label, Story)
```

```
int GetLabelFromName(
    String^ Name,
    String^% Label,
    String^% Story
)
```

```
abstract GetLabelFromName :
    Name : string *
    Label : string byref *
    Story : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of the point object

Label

Type:Â SystemString
 The point object label

Story

Type:Â SystemString
 The point object story level

Return Value

Type:Â Int32
 Returns zero if the data is successfully retrieved, otherwise it returns nonzero.

Remarks

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberNames As Integer
    Dim Label As String
    Dim Story As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'get area object data
    ret = SapModel.PointObj.GetLabelFromName("42", Label, Story)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

cPointObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST7A2E94E3_0") Method

Retrieves the names and labels of all defined point objects.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetLabelNameList(
    ref int NumberNames,
    ref string[] MyName,
    ref string[] MyLabel,
    ref string[] MyStory
)
```

```
Function GetLabelNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String(),
    ByRef MyLabel As String(),
    ByRef MyStory As String()
) As Integer
```

```
Dim instance As cPointObj
Dim NumberNames As Integer
Dim MyName As String()
Dim MyLabel As String()
Dim MyStory As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetLabelNameList(NumberNames,
    MyName, MyLabel, MyStory)
```

```
int GetLabelNameList(
    int% NumberNames,
    array<String^>^% MyName,
    array<String^>^% MyLabel,
    array<String^>^% MyStory
)
```

```
abstract GetLabelNameList :
    NumberNames : int byref *
    MyName : string[] byref *
    MyLabel : string[] byref *
    MyStory : string[] byref -> int
```

Parameters**NumberNames**

Type: SystemInt32

The number of point object names retrieved by the program.

MyName

Type: SystemString

This is a one-dimensional array of point object names. The MyName array is created as a dynamic, zero-based, array by the API user:

```
Dim MyName() as String
```

The array is dimensioned to (NumberNames + 1) inside the program, filled with the names, and returned to the API user.

MyLabel

Type: SystemString

This is a one-dimensional array of point object labels. The MyLabel array is created as a dynamic, zero-based, array by the API user:

```
Dim MyLabel() as String
```

The array is dimensioned to (NumberNames + 1) inside the program, filled with the labels, and returned to the API user.

MyStory

Type: SystemString

This is a one-dimensional array of the story levels of the point objects. The MyStory array is created as a dynamic, zero-based, array by the API user:

```
Dim MyStory() as String
```

The array is dimensioned to (NumberNames + 1) inside the program, filled with the story levels, and returned to the API user.

Return Value

Type: Int32

Returns zero if the data is successfully retrieved, otherwise it returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberNames As Integer
    Dim MyName() As String
    Dim MyLabel() As String
    Dim MyStory() As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

```
'start ETABS application
  ret = EtabsObject.ApplicationStart()

'create SapModel object
  SapModel = EtabsObject.SapModel

'initialize model
  ret = SapModel.InitializeNewModel()

'create steel deck template model
  ret = SapModel.File.NewSteelDeck(4, 12, 4, 4, 24, 24)

'get area object data
  ret = SapModel.PointObj.GetLabelNameList(NumberNames, MyName, MyLabel, MyStory)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

📌 See Also

Reference

[cPointObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST1AB80C27_0") Method

Retrieves the ground displacement load assignments to point objects.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetLoadDispl(
    string Name,
    ref int NumberItems,
    ref string[] PointName,
    ref string[] LoadPat,
    ref int[] LcStep,
    ref string[] CSys,
    ref double[] U1,
    ref double[] U2,
    ref double[] U3,
    ref double[] R1,
    ref double[] R2,
    ref double[] R3,
    eItemType ItemType = eItemType.Objects
)
```

```
Function GetLoadDispl (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef PointName As String(),
    ByRef LoadPat As String(),
    ByRef LcStep As Integer(),
    ByRef CSys As String(),
    ByRef U1 As Double(),
    ByRef U2 As Double(),
    ByRef U3 As Double(),
    ByRef R1 As Double(),
    ByRef R2 As Double(),
    ByRef R3 As Double(),
    Optional ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim NumberItems As Integer
Dim PointName As String()
Dim LoadPat As String()
Dim LcStep As Integer()
Dim CSys As String()
Dim U1 As Double()
Dim U2 As Double()
```

Introduction

```
Dim U3 As Double()
Dim R1 As Double()
Dim R2 As Double()
Dim R3 As Double()
Dim ItemType As eItemType
Dim returnValue As Integer

returnValue = instance.GetLoadDispl(Name,
NumberItems, PointName, LoadPat,
LcStep, CSys, U1, U2, U3, R1, R2, R3,
ItemType)

int GetLoadDispl(
String^ Name,
int% NumberItems,
array<String^>^% PointName,
array<String^>^% LoadPat,
array<int>^% LcStep,
array<String^>^% CSys,
array<double>^% U1,
array<double>^% U2,
array<double>^% U3,
array<double>^% R1,
array<double>^% R2,
array<double>^% R3,
eItemType ItemType = eItemType::Objects
)

abstract GetLoadDispl :
    Name : string *
    NumberItems : int byref *
    PointName : string[] byref *
    LoadPat : string[] byref *
    LcStep : int[] byref *
    CSys : string[] byref *
    U1 : float[] byref *
    U2 : float[] byref *
    U3 : float[] byref *
    R1 : float[] byref *
    R2 : float[] byref *
    R3 : float[] byref *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing point object or group depending on the value of the ItemType item.

NumberItems

Type:Â SystemInt32

This is the total number of joint ground displacement assignments returned.

PointName

Type:Â SystemString

This is an array that includes the name of the point object to which the specified

ground displacement assignment applies.

LoadPat

Type: SystemString

This is an array that includes the name of the load pattern for the ground displacement load.

LcStep

Type: SystemInt32

CSys

Type: SystemString

This is an array that includes the name of the coordinate system for the ground displacement load. This is Local or the name of a defined coordinate system.

U1

Type: SystemDouble

This is an array that includes the assigned translational ground displacement in the local 1-axis or coordinate system X-axis direction, depending on the specified CSys. [L]

U2

Type: SystemDouble

This is an array that includes the assigned translational ground displacement in the local 2-axis or coordinate system Y-axis direction, depending on the specified CSys. [L]

U3

Type: SystemDouble

R1

Type: SystemDouble

This is an array that includes the assigned rotational ground displacement about the local 1-axis or coordinate system X-axis, depending on the specified CSys. [rad]

R2

Type: SystemDouble

This is an array that includes the assigned rotational ground displacement about the local 2-axis or coordinate system Y-axis, depending on the specified CSys. [rad]

R3

Type: SystemDouble

This is an array that includes the assigned rotational ground displacement about the local 3-axis or coordinate system Z-axis, depending on the specified CSys. [rad]

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the load assignments are retrieved for the point object specified by the Name item.

If this item is Group, the load assignments are retrieved for all point objects in the group specified by the Name item.

If this item is SelectedObjects, the load assignments are retrieved for all selected point objects and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the load assignments are successfully retrieved, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberItems As Integer
    Dim PointName() As String
    Dim LoadPat() As String
    Dim LCStep() As Integer
    Dim CSys() As String
    Dim U1() As Double
    Dim U2() As Double
    Dim U3() As Double
    Dim R1() As Double
    Dim R2() As Double
    Dim R3() As Double
    Dim Value() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'add ground displacement load
    Redim Value(5)
    Value(0) = 10
    ret = SapModel.PointObj.SetLoadDispl("1", "DEAD", Value)

    'get ground displacement load
    ret = SapModel.PointObj.GetLoadDispl("ALL", NumberItems, PointName, LoadPat, LCStep, CSys, U1, U2, U3, R1, R2, R3)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
end Sub
```

End Sub

▲ See Also

Reference

cPointObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LSTDF2162A8_0") Method

Retrieves the joint force load assignments to point objects.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetLoadForce(
    string Name,
    ref int NumberItems,
    ref string[] PointName,
    ref string[] LoadPat,
    ref int[] LcStep,
    ref string[] CSys,
    ref double[] F1,
    ref double[] F2,
    ref double[] F3,
    ref double[] M1,
    ref double[] M2,
    ref double[] M3,
    eItemType ItemType = eItemType.Objects
)
```

```
Function GetLoadForce (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef PointName As String(),
    ByRef LoadPat As String(),
    ByRef LcStep As Integer(),
    ByRef CSys As String(),
    ByRef F1 As Double(),
    ByRef F2 As Double(),
    ByRef F3 As Double(),
    ByRef M1 As Double(),
    ByRef M2 As Double(),
    ByRef M3 As Double(),
    Optional ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim NumberItems As Integer
Dim PointName As String()
Dim LoadPat As String()
Dim LcStep As Integer()
Dim CSys As String()
Dim F1 As Double()
Dim F2 As Double()
```

Introduction

```
Dim F3 As Double()
Dim M1 As Double()
Dim M2 As Double()
Dim M3 As Double()
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.GetLoadForce(Name,
NumberItems, PointName, LoadPat,
LcStep, CSys, F1, F2, F3, M1, M2, M3,
ItemType)
```

```
int GetLoadForce(
String^ Name,
int% NumberItems,
array<String^>^% PointName,
array<String^>^% LoadPat,
array<int>^% LcStep,
array<String^>^% CSys,
array<double>^% F1,
array<double>^% F2,
array<double>^% F3,
array<double>^% M1,
array<double>^% M2,
array<double>^% M3,
eItemType ItemType = eItemType::Objects
)
```

```
abstract GetLoadForce :
    Name : string *
    NumberItems : int byref *
    PointName : string[] byref *
    LoadPat : string[] byref *
    LcStep : int[] byref *
    CSys : string[] byref *
    F1 : float[] byref *
    F2 : float[] byref *
    F3 : float[] byref *
    M1 : float[] byref *
    M2 : float[] byref *
    M3 : float[] byref *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing point object or group depending on the value of the ItemType item.

NumberItems

Type: [SystemInt32](#)

This is the total number of joint force load assignments returned.

PointName

Type: [SystemString](#)

This is an array that includes the name of the point object to which the specified

joint force load assignment applies.

LoadPat

Type:Â SystemString

This is an array that includes the name of the load pattern for the joint force load.

LcStep

Type:Â SystemInt32

CSys

Type:Â SystemString

This is an array that includes the name of the coordinate system for the joint force load. This is Local or the name of a defined coordinate system.

F1

Type:Â SystemDouble

This is an array that includes the assigned translational force in the local 1-axis or coordinate system X-axis direction, depending on the specified CSys. [L]

F2

Type:Â SystemDouble

This is an array that includes the assigned translational force in the local 2-axis or coordinate system Y-axis direction, depending on the specified CSys. [L]

F3

Type:Â SystemDouble

This is an array that includes the assigned translational force in the local 3-axis or coordinate system Z-axis direction, depending on the specified CSys. [L]

M1

Type:Â SystemDouble

This is an array that includes the assigned moment about the local 1-axis or coordinate system X-axis, depending on the specified CSys. [rad]

M2

Type:Â SystemDouble

This is an array that includes the assigned moment about the local 2-axis or coordinate system Y-axis, depending on the specified CSys. [rad]

M3

Type:Â SystemDouble

This is an array that includes the assigned moment about the local 3-axis or coordinate system Z-axis, depending on the specified CSys. [rad]

ItemType (Optional)

Type:Â ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the load assignments are retrieved for the point object specified by the Name item.

If this item is Group, the load assignments are retrieved for all point objects in the group specified by the Name item.

If this item is SelectedObjects, the load assignments are retrieved for all selected point objects and the Name item is ignored.

Return ValueType: Int32

Returns zero if the load assignments are successfully retrieved, otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberItems As Integer
    Dim PointName() As String
    Dim LoadPat() As String
    Dim LCStep() As Integer
    Dim CSys() As String
    Dim F1() As Double
    Dim F2() As Double
    Dim F3() As Double
    Dim M1() As Double
    Dim M2() As Double
    Dim M3() As Double
    Dim Value() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'add joint force load
    Redim Value(5)
    Value(0) = 10
    ret = SapModel.PointObj.SetLoadForce("1", "DEAD", Value)

    'get joint force load
    ret = SapModel.PointObj.GetLoadForce("ALL", NumberItems, PointName, LoadPat, LCStep, CSys, F1,

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ **See Also**

Return Value

Reference

cPointObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LSTFA79583E0") Method

Retrieves the local axis angles for a point object.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetLocalAxes(
    string Name,
    ref double A,
    ref double B,
    ref double C,
    ref bool Advanced
)
```

```
Function GetLocalAxes (
    Name As String,
    ByRef A As Double,
    ByRef B As Double,
    ByRef C As Double,
    ByRef Advanced As Boolean
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim A As Double
Dim B As Double
Dim C As Double
Dim Advanced As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetLocalAxes(Name,
    A, B, C, Advanced)
```

```
int GetLocalAxes(
    String^ Name,
    double% A,
    double% B,
    double% C,
    bool% Advanced
)
```

```
abstract GetLocalAxes :
    Name : string *
    A : float byref *
    B : float byref *
    C : float byref *
    Advanced : bool byref -> int
```


Parameters

Name

Type: SystemString

The name of an existing point object.

A

Type: SystemDouble

The local axes of the point are defined by first setting the positive local 1, 2 and 3 axes the same as the positive global X, Y and Z axes and then doing the following: [deg]

1. Rotate about the 3 axis by angle a.
2. Rotate about the resulting 2 axis by angle b.
3. Rotate about the resulting 1 axis by angle c.

B

Type: SystemDouble

The local axes of the point are defined by first setting the positive local 1, 2 and 3 axes the same as the positive global X, Y and Z axes and then doing the following: [deg]

1. Rotate about the 3 axis by angle a.
2. Rotate about the resulting 2 axis by angle b.
3. Rotate about the resulting 1 axis by angle c.

C

Type: SystemDouble

The local axes of the point are defined by first setting the positive local 1, 2 and 3 axes the same as the positive global X, Y and Z axes and then doing the following: [deg]

1. Rotate about the 3 axis by angle a.
2. Rotate about the resulting 2 axis by angle b.
3. Rotate about the resulting 1 axis by angle c.

Advanced

Type: SystemBoolean

This item is True if the point object local axes orientation was obtained using advanced local axes parameters.

Return ValueType: Int32

Returns zero if the local axes angles are successfully retrieved, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim GUID As String
```

```
'create ETABS object
```

```
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

Introduction

```
'start ETABS application
  ret = EtabsObject.ApplicationStart()

'create SapModel object
  SapModel = EtabsObject.SapModel

'initialize model
  ret = SapModel.InitializeNewModel()

'create steel deck template model
  ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'get local axes assignments
  ret = SapModel.PointObj.GetLocalAxes("1", a, b, c, Advanced)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cPointObj Interface

ETABSV1 Namespace

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»

CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST787BED94_0?cpp=::|2679);GetMa

Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetMass(
    string Name,
    ref double[] M
)
```

```
Function GetMass (
    Name As String,
    ByRef M As Double()
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim M As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetMass(Name, M)
```

```
int GetMass(
    String^ Name,
    array<double>^% M
)
```

```
abstract GetMass :
    Name : string *
    M : float[] byref -> int
```

Parameters

Name

Type: [SystemString](#)

M

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cPointObj Interface

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LSTC5050506 Method

Retrieves the unique name of a point object, given the label and story level

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNameFromLabel(
    string Label,
    string Story,
    ref string Name
)
```

```
Function GetNameFromLabel (
    Label As String,
    Story As String,
    ByRef Name As String
) As Integer
```

```
Dim instance As cPointObj
Dim Label As String
Dim Story As String
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.GetNameFromLabel(Label,
    Story, Name)
```

```
int GetNameFromLabel(
    String^ Label,
    String^ Story,
    String^% Name
)
```

```
abstract GetNameFromLabel :
    Label : string *
    Story : string *
    Name : string byref -> int
```

Parameters

Label

Type: [SystemString](#)

The point object label

Story

Type: SystemString
The point object story level

Name

Type: SystemString
The unique name of the point object

Return Value

Type: Int32

Returns zero if the data is successfully retrieved, otherwise it returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'get point object data
    ret = SapModel.PointObj.GetNameFromLabel("C1", "Story4", Name)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPointObj Interface

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Introduction

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST2A9D9825_0")

Method

Retrieves the names of all defined point objects.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNameList(
    ref int NumberNames,
    ref string[] MyName
)
```

```
Function GetNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cPointObj
Dim NumberNames As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetNameList(NumberNames,
    MyName)
```

```
int GetNameList(
    int% NumberNames,
    array<String^>^% MyName
)
```

```
abstract GetNameList :
    NumberNames : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberNames

Type: [System.Int32](#)

The number of point object names retrieved by the program.

MyName

Type: [System.String](#)

This is a one-dimensional array of point object names. The MyName array is created as a dynamic, zero-based, array by the API user:

```
Dim MyName() as String
```


The array is dimensioned to (NumberNames + 1) inside the ETABS program, filled with the names, and returned to the API user.

Return Value

Type: Int32

Returns zero if the names are successfully retrieved, otherwise it returns nonzero.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)  
  
    'get point object names  
    ret = SapModel.PointObj.GetNameList(NumberNames, MyName)  
  
    'close ETABS  
    EtabsObject.ApplicationExit(False)  
  
    'clean up variables  
    SapModel = Nothing  
    EtabsObject = Nothing  
  
End Sub
```

▲ **See Also**

Reference

cPointObj Interface

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LSTD20D4CB2_0?cpp=::2686");GetN Method

Retrieves the names of the defined point objects on a given story.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNameListOnStory(
    string StoryName,
    ref int NumberNames,
    ref string[] MyName
)
```

```
Function GetNameListOnStory (
    StoryName As String,
    ByRef NumberNames As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cPointObj
Dim StoryName As String
Dim NumberNames As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetNameListOnStory(StoryName,
    NumberNames, MyName)
```

```
int GetNameListOnStory(
    String^ StoryName,
    int% NumberNames,
    array<String^>^% MyName
)
```

```
abstract GetNameListOnStory :
    StoryName : string *
    NumberNames : int byref *
    MyName : string[] byref -> int
```

Parameters

StoryName

Type: [SystemString](#)

The name of an existing story.

NumberNames

Type: SystemInt32

The number of point object names retrieved by the program.

MyName

Type: SystemString

This is a one-dimensional array of point object names. The MyName array is created as a dynamic, zero-based, array by the API user:

```
Dim MyName() as String
```

The array is dimensioned to (NumberNames + 1) inside the ETABS program, filled with the names, and returned to the API user.

Return Value

Type: Int32

Returns zero if the names are successfully retrieved, otherwise it returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'get point object names
    ret = SapModel.PointObj.GetNameListOnStory("Story2", NumberNames, MyName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cPointObj Interface

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LSTD9C3DD6A_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetPanelZone(
    string Name,
    ref int PropType,
    ref double Thickness,
    ref double K1,
    ref double K2,
    ref string LinkProp,
    ref int Connectivity,
    ref int LocalAxisFrom,
    ref double LocalAxisAngle
)
```

```
Function GetPanelZone (
    Name As String,
    ByRef PropType As Integer,
    ByRef Thickness As Double,
    ByRef K1 As Double,
    ByRef K2 As Double,
    ByRef LinkProp As String,
    ByRef Connectivity As Integer,
    ByRef LocalAxisFrom As Integer,
    ByRef LocalAxisAngle As Double
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim PropType As Integer
Dim Thickness As Double
Dim K1 As Double
Dim K2 As Double
Dim LinkProp As String
Dim Connectivity As Integer
Dim LocalAxisFrom As Integer
Dim LocalAxisAngle As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetPanelZone(Name,
    PropType, Thickness, K1, K2, LinkProp,
    Connectivity, LocalAxisFrom, LocalAxisAngle)
```

```
int GetPanelZone(
    String^ Name,
```

cPointObjspan id="LSTD9C3DD6A_0"AddLanguageSpecificTextSet("LSTD9C3DD6A_0?cpp=:2689.");GetP

```

int% PropType,
double% Thickness,
double% K1,
double% K2,
String^% LinkProp,
int% Connectivity,
int% LocalAxisFrom,
double% LocalAxisAngle
)

```

```

abstract GetPanelZone :
    Name : string *
    PropType : int byref *
    Thickness : float byref *
    K1 : float byref *
    K2 : float byref *
    LinkProp : string byref *
    Connectivity : int byref *
    LocalAxisFrom : int byref *
    LocalAxisAngle : float byref -> int

```

Parameters

Name

Type: [SystemString](#)

PropType

Type: [SystemInt32](#)

Thickness

Type: [SystemDouble](#)

K1

Type: [SystemDouble](#)

K2

Type: [SystemDouble](#)

LinkProp

Type: [SystemString](#)

Connectivity

Type: [SystemInt32](#)

LocalAxisFrom

Type: [SystemInt32](#)

LocalAxisAngle

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

See Also

Reference

[cPointObj Interface](#)

[ETABSv1 Namespace](#)

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Introduction

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST3F06C192_0")

Method

Retrieves the restraint assignments for a point object. The restraint assignments are always returned in the point local coordinate system.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetRestraint(
    string Name,
    ref bool[] Value
)
```

```
Function GetRestraint (
    Name As String,
    ByRef Value As Boolean()
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim Value As Boolean()
Dim returnValue As Integer
```

```
returnValue = instance.GetRestraint(Name,
    Value)
```

```
int GetRestraint(
    String^ Name,
    array<bool>^% Value
)
```

```
abstract GetRestraint :
    Name : string *
    Value : bool[] byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing point object.

Value

Type: SystemBoolean

This is an array of six restraint values.

◊ Value(0) = U1

◊ Value(1) = U2

◇ Value(2) = U3
 ◇ Value(3) = R1
 ◇ Value(4) = R2
 ◇ Value(5) = R3

Return Value

Type: Int32

Returns zero if the restraint assignments are successfully retrieved, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'get point object restraints
    Redim Value(5)
    ret = SapModel.PointObj.GetRestraint("65", Value)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPointObj Interface

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST47F8CD0A_0?cpp=::|2095");GetSe

Method

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetSelected(
    string Name,
    ref bool Selected
)
```

```
Function GetSelected (
    Name As String,
    ByRef Selected As Boolean
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim Selected As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetSelected(Name,
    Selected)
```

```
int GetSelected(
    String^ Name,
    bool% Selected
)
```

```
abstract GetSelected :
    Name : string *
    Selected : bool byref -> int
```

Parameters

Name

Type: [SystemString](#)

Selected

Type: [SystemBoolean](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST5E115359 Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetSpecialPoint(
    string Name,
    ref bool SpecialPoint
)
```

```
Function GetSpecialPoint (
    Name As String,
    ByRef SpecialPoint As Boolean
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim SpecialPoint As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetSpecialPoint(Name,
    SpecialPoint)
```

```
int GetSpecialPoint(
    String^ Name,
    bool% SpecialPoint
)
```

```
abstract GetSpecialPoint :
    Name : string *
    SpecialPoint : bool byref -> int
```

Parameters

Name

Type: [SystemString](#)

SpecialPoint

Type: [SystemBoolean](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST3942726C Method

Retrieves uncoupled spring stiffness assignments for a point object, that is, it retrieves the diagonal terms in the 6x6 spring matrix for the point object.

Namespace: $\hat{A}$ ETABSv1

Assembly: $\hat{A}$ ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetSpring(
    string Name,
    ref double[] K
)
```

```
Function GetSpring (
    Name As String,
    ByRef K As Double()
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim K As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetSpring(Name,
    K)
```

```
int GetSpring(
    String^ Name,
    array<double>^% K
)
```

```
abstract GetSpring :
    Name : string *
    K : float[] byref -> int
```

Parameters

Name

Type: $\hat{A}$ SystemString

The name of an existing point object.

K

Type: $\hat{A}$ SystemDouble

This is an array of six spring stiffness values.

◊ Value(0) = U1 [F/L]

◊ Value(1) = U2 [F/L]

- ◇ Value(2) = U3 [F/L]
- ◇ Value(3) = R1 [FL/rad]
- ◇ Value(4) = R2 [FL/rad]
- ◇ Value(5) = R3 [FL/rad]

Return Value

Type: $\hat{A}$ Int32

Returns zero if the stiffnesses are successfully retrieved, otherwise it returns a nonzero value. If no springs exist at the point object, the function returns a nonzero value.

Remarks

The spring stiffnesses reported are the sum of all springs assigned to the point object. The spring stiffness values are reported in the point local coordinate system.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'assign spring to a point
    ReDim k(5)
    k(2) = 10
    ret = SapModel.PointObj.SetSpring("3", k)

    'get spring values
    ReDim k(5)
    ret = SapModel.PointObj.GetSpring("3", k)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST21307CA90") Method

Retrieves the named point spring property assignment for a point object

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetSpringAssignment(
    string Name,
    ref string SpringProp
)
```

```
Function GetSpringAssignment (
    Name As String,
    ByRef SpringProp As String
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim SpringProp As String
Dim returnValue As Integer
```

```
returnValue = instance.GetSpringAssignment(Name,
    SpringProp)
```

```
int GetSpringAssignment(
    String^ Name,
    String^% SpringProp
)
```

```
abstract GetSpringAssignment :
    Name : string *
    SpringProp : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing point object

SpringProp

Type: [SystemString](#)

The name of an existing point spring property

Return Value

Type: Int32

Returns zero if the assignment is successfully retrieved, otherwise it returns a nonzero value

▀ **Remarks**

▀ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim SpringProp As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 4, 4, 24, 24)

    'define new spring property
    Dim k() as Double
    ReDim k(5)
    k(2) = 1
    ret = SapModel.PropPointSpring.SetPointSpringProp("mySpringProp1", 1, k)

    'assign property
    ret = SapModel.PointObj.SetSpringAssignment("1", "mySpringProp1")

    'get assigned property
    Dim prop as String
    ret = SapModel.PointObj.GetSpringAssignment("1", prop)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▀ **See Also**

Reference

cPointObj Interface

ETABSv1 Namespace

Introduction

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST73D6C454 Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetSpringCoupled(
    string Name,
    ref double[] K
)
```

```
Function GetSpringCoupled (
    Name As String,
    ByRef K As Double()
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim K As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetSpringCoupled(Name,
    K)
```

```
int GetSpringCoupled(
    String^ Name,
    array<double>^% K
)
```

```
abstract GetSpringCoupled :
    Name : string *
    K : float[] byref -> int
```

Parameters

Name

Type: [SystemString](#)

K

Type: [SystemDouble](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cPointObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST5DB58E7_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetTransformationMatrix(
    string Name,
    ref double[] Value,
    bool IsGlobal = true
)
```

```
Function GetTransformationMatrix (
    Name As String,
    ByRef Value As Double(),
    Optional IsGlobal As Boolean = true
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim Value As Double()
Dim IsGlobal As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetTransformationMatrix(Name,
    Value, IsGlobal)
```

```
int GetTransformationMatrix(
    String^ Name,
    array<double>^% Value,
    bool IsGlobal = true
)
```

```
abstract GetTransformationMatrix :
    Name : string *
    Value : float[] byref *
    ?IsGlobal : bool
(* Defaults:
    let _IsGlobal = defaultArg IsGlobal true
*)
-> int
```

Parameters

Name

Type: [SystemString](#)

Value

cPointObjspan id="LST5DB58E7_0">AddLanguageSpecificTextSet("LST5DB58E7_0?cpp=::|nu=2700")GetTrans

Type: SystemDouble
IsGlobal (Optional)
Type: SystemBoolean

Return Value

Type: Int32

See Also

Reference

cPointObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LSTEE3510EF Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int IsSpringCoupled(
    string Name,
    ref bool IsCoupled
)
```

```
Function IsSpringCoupled (
    Name As String,
    ByRef IsCoupled As Boolean
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim IsCoupled As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.IsSpringCoupled(Name,
    IsCoupled)
```

```
int IsSpringCoupled(
    String^ Name,
    bool% IsCoupled
)
```

```
abstract IsSpringCoupled :
    Name : string *
    IsCoupled : bool byref -> int
```

Parameters

Name

Type: [SystemString](#)

IsCoupled

Type: [SystemBoolean](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cPointObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LSTD3BB0553_0") Method

Assigns a diaphragm to a point object

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetDiaphragm(
    string Name,
    eDiaphragmOption DiaphragmOption,
    string DiaphragmName = ""
)
```

```
Function SetDiaphragm (
    Name As String,
    DiaphragmOption As eDiaphragmOption,
    DiaphragmName As String = ""
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim DiaphragmOption As eDiaphragmOption
Dim DiaphragmName As String
Dim returnValue As Integer
```

```
returnValue = instance.SetDiaphragm(Name,
    DiaphragmOption, DiaphragmName)
```

```
int SetDiaphragm(
    String^ Name,
    eDiaphragmOption DiaphragmOption,
    String^ DiaphragmName = L""
)
```

```
abstract SetDiaphragm :
    Name : string *
    DiaphragmOption : eDiaphragmOption *
    ?DiaphragmName : string
(* Defaults:
    let _DiaphragmName = defaultArg DiaphragmName ""
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing point object

DiaphragmOption

Type:Â ETABSv1eDiaphragmOption

This is an item from the eDiaphragmOption enumeration

If this item is Disconnect then the point object will be disconnected from any diaphragm

If this item is FromShellObject then the point object will inherit the diaphragm assignment of its bounding area object.

If this item is DefinedDiaphragm then the point object will be assigned the existing diaphragm specified by DiaphragmName

DiaphragmName (Optional)

Type:Â SystemString

The name of an existing diaphragm

Return Value

Type:Â Int32

Returns zero if the diaphragm assignment is successfully made, otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim DiaphragmOption As eDiaphragmOption
    Dim DiaphragmName As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define a new diaphragm
    ret = SapModel.Diaphragm.SetDiaphragm("MyDiaph1A", True)

    'assign diaphragm to point
    ret = SapModel.PointObj.SetDiaphragm("1", eDiaphragmOption.DefinedDiaphragm, "MyDiaph1A")
```

Introduction

```
'get point diaphragm assignment
  ret = SapModel.PointObj.GetDiaphragm("1", DiaphragmOption, DiaphragmName)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

cPointObj Interface

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LSTCBC2E4F Method

Adds or removes point objects from a specified group.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetGroupAssign(
    string Name,
    string GroupName,
    bool Remove = false,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetGroupAssign (
    Name As String,
    GroupName As String,
    Remove As Boolean = false,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim GroupName As String
Dim Remove As Boolean
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetGroupAssign(Name,
    GroupName, Remove, ItemType)
```

```
int SetGroupAssign(
    String^ Name,
    String^ GroupName,
    bool Remove = false,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetGroupAssign :
    Name : string *
    GroupName : string *
    ?Remove : bool *
    ?ItemType : eItemType
(* Defaults:
    let _Remove = defaultArg Remove false
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

cPointObjid="LSTCBC2E4FE_0"AddLanguageSpecificTextSet("LSTCBC2E4FE_0?cpp=:2714.");SetG

Parameters

Name

Type:Â SystemString

The name of an existing point object or group depending on the value of the ItemType item.

GroupName

Type:Â SystemString

The name of an existing group to which the assignment is made.

Remove (Optional)

Type:Â SystemBoolean

ItemType (Optional)

Type:Â ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the point object specified by the Name item is added or removed from the group specified by the GroupName item.

If this item is Group, all point objects in the group specified by the Name item are added or removed from the group specified by the GroupName item.

If this item is SelectedObjects, all selected point objects are added or removed from the group specified by the GroupName item and the Name item is ignored.

Return ValueType:Â Int32

Returns zero if the group assignment is successful, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
```

Introduction

```
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'define new group
ret = SapModel.GroupDef.SetGroup("Group1")

'add point objects to group
ret = SapModel.PointObj.SetGroupAssign("3", "Group1")
ret = SapModel.PointObj.SetGroupAssign("6", "Group1")
ret = SapModel.PointObj.SetGroupAssign("9", "Group1")

'select objects in group
ret = SapModel.SelectObj.Group("Group1")

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

[cPointObj Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST9E75B03C_0")

Method

Sets the GUID for specified point object. If the GUID is passed in as a blank string, the program automatically creates a GUID for the object.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetGUID(
    string Name,
    string GUID = ""
)
```

```
Function SetGUID (
    Name As String,
    GUID As String = ""
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.SetGUID(Name, GUID)
```

```
int SetGUID(
    String^ Name,
    String^ GUID = L""
)
```

```
abstract SetGUID :
    Name : string *
    ?GUID : string
(* Defaults:
    let _GUID = defaultArg GUID ""
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing point object.

GUID (Optional)

Type: SystemString

The GUID (Global Unique ID) for the specified point object.

cPointObjid="LST9E75B03C_0"AddLanguageSpecificTextSet("LST9E75B03C_0?cpp=::|274.7);SetGL

Return Value

Type: Int32

Returns zero if the point object GUID is successfully set; otherwise it returns nonzero.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim GUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set program created GUID
    ret = SapModel.PointObj.SetGUID("1")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ **See Also**

Reference

cPointObj Interface

ETABSV1 Namespace

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST9B98AC7C_0?cpp=::2719");SetLoadPat

Method

Makes ground displacement load assignments to point objects.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetLoadDispl(
    string Name,
    string LoadPat,
    ref double[] Value,
    bool Replace = false,
    string CSys = "Local",
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetLoadDispl (
    Name As String,
    LoadPat As String,
    ByRef Value As Double(),
    Replace As Boolean = false,
    CSys As String = "Local",
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim LoadPat As String
Dim Value As Double()
Dim Replace As Boolean
Dim CSys As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetLoadDispl(Name,
    LoadPat, Value, Replace, CSys, ItemType)
```

```
int SetLoadDispl(
    String^ Name,
    String^ LoadPat,
    array<double>^% Value,
    bool Replace = false,
    String^ CSys = L"Local",
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetLoadDispl :
    Name : string *
```

cPointObjspan id="LST9B98AC7C_0"AddLanguageSpecificTextSet("LST9B98AC7C_0?cpp=::2719");SetLoad

```

LoadPat : string *
Value : float[] byref *
?Replace : bool *
?CSys : string *
?ItemType : eItemType
(* Defaults:
    let _Replace = defaultArg Replace false
    let _CSys = defaultArg CSys "Local"
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int

```

Parameters

Name

Type: SystemString

The name of an existing point object or group depending on the value of the ItemType item.

LoadPat

Type: SystemString

The name of the load pattern for the ground displacement load.

Value

Type: SystemDouble

This is an array of six point load values.

- ◊ Value(0) = U1 [L]
- ◊ Value(1) = U2 [L]
- ◊ Value(2) = U3 [L]
- ◊ Value(3) = R1 [rad]
- ◊ Value(4) = R2 [rad]
- ◊ Value(5) = R3 [rad]

Replace (Optional)

Type: SystemBoolean

If this item is True, all previous ground displacement loads, if any, assigned to the specified point object(s) in the specified load pattern are deleted before making the new assignment.

CSys (Optional)

Type: SystemString

The name of the coordinate system for the considered ground displacement load. This is Local or the name of a defined coordinate system.

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

- ◊ Object = 0
- ◊ Group = 1
- ◊ SelectedObjects = 2

If this item is Objects, the load assignment is made to the point object specified by the Name item.

If this item is Group, the load assignment is made to all point objects in the group specified by the Name item.

If this item is SelectedObjects, the load assignment is made to all selected point objects and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the load assignments are successfully made, otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value as Double()

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'add ground displacement load
    Redim Value(5)
    Value(0) = 10
    ret = SapModel.PointObj.SetLoadDispl("1", "DEAD", Value)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cPointObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LSTE3F26083 Method

Makes point force load assignments to point objects.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetLoadForce(
    string Name,
    string LoadPat,
    ref double[] Value,
    bool Replace = false,
    string CSys = "Global",
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetLoadForce (
    Name As String,
    LoadPat As String,
    ByRef Value As Double(),
    Replace As Boolean = false,
    CSys As String = "Global",
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim LoadPat As String
Dim Value As Double()
Dim Replace As Boolean
Dim CSys As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetLoadForce(Name,
    LoadPat, Value, Replace, CSys, ItemType)
```

```
int SetLoadForce(
    String^ Name,
    String^ LoadPat,
    array<double>^% Value,
    bool Replace = false,
    String^ CSys = L"Global",
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetLoadForce :
    Name : string *
```

cPointObjspan id="LSTE3F26083_0"AddLanguageSpecificTextSet("LSTE3F26083_0?cpp=::|n2722SetLoa

```

LoadPat : string *
Value : float[] byref *
?Replace : bool *
?CSys : string *
?ItemType : eItemType
(* Defaults:
    let _Replace = defaultArg Replace false
    let _CSys = defaultArg CSys "Global"
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int

```

Parameters

Name

Type: SystemString

The name of an existing point object or group depending on the value of the ItemType item.

LoadPat

Type: SystemString

The name of the load pattern for the point force load.

Value

Type: SystemDouble

This is an array of six point load values.

◊ Value(0) = F1 [F]

◊ Value(1) = F2 [F]

◊ Value(2) = F3 [F]

◊ Value(3) = M1 [FL]

◊ Value(4) = M2 [FL]

◊ Value(5) = M3 [FL]

Replace (Optional)

Type: SystemBoolean

If this item is True, all previous point force loads, if any, assigned to the specified point object(s) in the specified load pattern are deleted before making the new assignment.

CSys (Optional)

Type: SystemString

The name of the coordinate system for the considered point force load. This is Local or the name of a defined coordinate system.

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the load assignment is made to the point object specified by the Name item.

If this item is Group, the load assignment is made to all point objects in the group specified by the Name item.

If this item is SelectedObjects, the load assignment is made to all selected point objects and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the load assignments are successfully made, otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value as Double()

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'add point load
    Redim Value(5)
    Value(0) = 10
    ret = SapModel.PointObj.SetLoadForce("2", "DEAD", Value)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cPointObj Interface

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LSTF93EC3B8_0?cpp=::|2725);SetMa Method

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetMass(
    string Name,
    ref double[] M,
    eItemType ItemType = eItemType.Objects,
    bool IsLocalCSys = true,
    bool Replace = false
)
```

```
Function SetMass (
    Name As String,
    ByRef M As Double(),
    Optional ItemType As eItemType = eItemType.Objects,
    Optional IsLocalCSys As Boolean = true,
    Optional Replace As Boolean = false
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim M As Double()
Dim ItemType As eItemType
Dim IsLocalCSys As Boolean
Dim Replace As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetMass(Name, M,
    ItemType, IsLocalCSys, Replace)
```

```
int SetMass(
    String^ Name,
    array<double>^% M,
    eItemType ItemType = eItemType::Objects,
    bool IsLocalCSys = true,
    bool Replace = false
)
```

```
abstract SetMass :
    Name : string *
    M : float[] byref *
    ?ItemType : eItemType *
    ?IsLocalCSys : bool *
    ?Replace : bool
(* Defaults:
```

cPointObjspan id="LSTF93EC3B8_0"AddLanguageSpecificTextSet("LSTF93EC3B8_0?cpp=::|2725);SetMa

Introduction

```
let _ItemType = defaultArg ItemType eItemType.Objects
let _IsLocalCSys = defaultArg IsLocalCSys true
let _Replace = defaultArg Replace false
*)
-> int
```

Parameters

Name

Type: SystemString

M

Type: SystemDouble

ItemType (Optional)

Type: ETABSv1eItemType

IsLocalCSys (Optional)

Type: SystemBoolean

Replace (Optional)

Type: SystemBoolean

Return Value

Type: Int32

See Also

Reference

cPointObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LSTF3CBFC52_0?cpp=::2727");SetM

Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetMassByVolume(
    string Name,
    string MatProp,
    ref double[] M,
    eItemType ItemType = eItemType.Objects,
    bool IsLocalCSys = true,
    bool Replace = false
)
```

```
Function SetMassByVolume (
    Name As String,
    MatProp As String,
    ByRef M As Double(),
    Optional ItemType As eItemType = eItemType.Objects,
    Optional IsLocalCSys As Boolean = true,
    Optional Replace As Boolean = false
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim MatProp As String
Dim M As Double()
Dim ItemType As eItemType
Dim IsLocalCSys As Boolean
Dim Replace As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetMassByVolume(Name,
    MatProp, M, ItemType, IsLocalCSys,
    Replace)
```

```
int SetMassByVolume(
    String^ Name,
    String^ MatProp,
    array<double>^% M,
    eItemType ItemType = eItemType::Objects,
    bool IsLocalCSys = true,
    bool Replace = false
)
```

```
abstract SetMassByVolume :
    Name : string *
```

cPointObjid="LSTF3CBFC52_0"AddLanguageSpecificTextSet("LSTF3CBFC52_0?cpp=::2727");SetM

Introduction

```
MatProp : string *
M : float[] byref *
?ItemType : eItemType *
?IsLocalCSys : bool *
?Replace : bool
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
    let _IsLocalCSys = defaultArg IsLocalCSys true
    let _Replace = defaultArg Replace false
*)
-> int
```

Parameters

Name

Type:Â SystemString

MatProp

Type:Â SystemString

M

Type:Â SystemDouble

ItemType (Optional)

Type:Â ETABSv1eItemType

IsLocalCSys (Optional)

Type:Â SystemBoolean

Replace (Optional)

Type:Â SystemBoolean

Return Value

Type:Â Int32

▲ See Also

Reference

cPointObj Interface

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LSTB7937270_0") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetMassByWeight(
    string Name,
    ref double[] M,
    eItemType ItemType = eItemType.Objects,
    bool IsLocalCSys = true,
    bool Replace = false
)
```

```
Function SetMassByWeight (
    Name As String,
    ByRef M As Double(),
    Optional ItemType As eItemType = eItemType.Objects,
    Optional IsLocalCSys As Boolean = true,
    Optional Replace As Boolean = false
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim M As Double()
Dim ItemType As eItemType
Dim IsLocalCSys As Boolean
Dim Replace As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetMassByWeight(Name,
M, ItemType, IsLocalCSys, Replace)
```

```
int SetMassByWeight(
    String^ Name,
    array<double>^% M,
    eItemType ItemType = eItemType::Objects,
    bool IsLocalCSys = true,
    bool Replace = false
)
```

```
abstract SetMassByWeight :
    Name : string *
    M : float[] byref *
    ?ItemType : eItemType *
    ?IsLocalCSys : bool *
    ?Replace : bool
(* Defaults:
```

Introduction

```
let _ItemType = defaultArg ItemType eItemType.Objects
let _IsLocalCSys = defaultArg IsLocalCSys true
let _Replace = defaultArg Replace false
*)
-> int
```

Parameters

Name

Type:Â SystemString

M

Type:Â SystemDouble

ItemType (Optional)

Type:Â ETABSv1eItemType

IsLocalCSys (Optional)

Type:Â SystemBoolean

Replace (Optional)

Type:Â SystemBoolean

Return Value

Type:Â Int32

🔗 See Also

Reference

cPointObj Interface

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST1BECC6C Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetPanelZone(
    string Name,
    int PropType,
    double Thickness,
    double K1,
    double K2,
    string LinkProp,
    int Connectivity,
    int LocalAxisFrom,
    double LocalAxisAngle,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetPanelZone (
    Name As String,
    PropType As Integer,
    Thickness As Double,
    K1 As Double,
    K2 As Double,
    LinkProp As String,
    Connectivity As Integer,
    LocalAxisFrom As Integer,
    LocalAxisAngle As Double,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim PropType As Integer
Dim Thickness As Double
Dim K1 As Double
Dim K2 As Double
Dim LinkProp As String
Dim Connectivity As Integer
Dim LocalAxisFrom As Integer
Dim LocalAxisAngle As Double
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetPanelZone(Name,
    PropType, Thickness, K1, K2, LinkProp,
    Connectivity, LocalAxisFrom, LocalAxisAngle,
    ItemType)
```

cPointObjspan id="LST1BECC6CE_0"AddLanguageSpecificTextSet("LST1BECC6CE_0?cpp=2701.");SetP

```

int SetPanelZone(
String^ Name,
int PropType,
double Thickness,
double K1,
double K2,
String^ LinkProp,
int Connectivity,
int LocalAxisFrom,
double LocalAxisAngle,
eItemType ItemType = eItemType::Objects
)

```

```

abstract SetPanelZone :
    Name : string *
    PropType : int *
    Thickness : float *
    K1 : float *
    K2 : float *
    LinkProp : string *
    Connectivity : int *
    LocalAxisFrom : int *
    LocalAxisAngle : float *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int

```

Parameters

Name
Type:Â [SystemString](#)

PropType
Type:Â [SystemInt32](#)

Thickness
Type:Â [SystemDouble](#)

K1
Type:Â [SystemDouble](#)

K2
Type:Â [SystemDouble](#)

LinkProp
Type:Â [SystemString](#)

Connectivity
Type:Â [SystemInt32](#)

LocalAxisFrom
Type:Â [SystemInt32](#)

LocalAxisAngle
Type:Â [SystemDouble](#)

ItemType (Optional)
Type:Â [ETABSv1eItemType](#)

Return Value

Type:Â [Int32](#)
 See Also

Reference

cPointObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LSTEBDD5271_0?cpp=::2734");SetRe

Method

Assigns the restraint assignments for a point object. The restraint assignments are always set in the point local coordinate system.

Namespace: Â ETABSv1

Assembly: Â ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetRestraint(
    string Name,
    ref bool[] Value,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetRestraint (
    Name As String,
    ByRef Value As Boolean(),
    Optional ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim Value As Boolean()
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetRestraint(Name,
    Value, ItemType)
```

```
int SetRestraint(
    String^ Name,
    array<bool>^% Value,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetRestraint :
    Name : string *
    Value : bool[] byref *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters**Name**

Type:Â SystemString

The name of an existing point object or group depending on the value of the ItemType item.

Value

Type:Â SystemBoolean

This is an array of six restraint values.

◇ Value(0) = U1

◇ Value(1) = U2

◇ Value(2) = U3

◇ Value(3) = R1

◇ Value(4) = R2

◇ Value(5) = R3

ItemType (Optional)

Type:Â ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◇ Object = 0

◇ Group = 1

◇ SelectedObjects = 2

If this item is Objects, the restraint assignment is made to the point object specified by the Name item.

If this item is Group, the restraint assignment is made to all point objects in the group specified by the Name item.

If this item is SelectedObjects, the restraint assignment is made to all selected point objects and the Name item is ignored.

Return Value

Type:Â Int32

Returns zero if the restraint assignments are successfully assigned, otherwise it returns a nonzero value.

▲ **Remarks**▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value as Boolean()

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel
```

Introduction

```
'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'assign point object restraints
    ReDim Value(5)
    For i = 0 To 5
        Value(i) = True
    Next i
    ret = SapModel.PointObj.SetRestraint("1", Value)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPointObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST229B9AE9_0") Method

Sets the selected status for a point object.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetSelected(
    string Name,
    bool Selected,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetSelected (
    Name As String,
    Selected As Boolean,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim Selected As Boolean
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetSelected(Name,
    Selected, ItemType)
```

```
int SetSelected(
    String^ Name,
    bool Selected,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetSelected :
    Name : string *
    Selected : bool *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing point object or group depending on the value of the ItemType item.

Selected

Type: SystemBoolean

This item is True if the specified point object is selected, otherwise it is False.

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the selected status is set for the point object specified by the Name item.

If this item is Group, the selected status is set for all point objects in the group specified by the Name item.

If this item is SelectedObjects, the selected status is set for all selected point objects and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the selected status is successfully set, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Value As Boolean()

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'set point selected
    ret = SapModel.PointObj.SetSelected("3", True)

    'close ETABS
```

Introduction

```
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

cPointObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LST631133A0 Method

Sets the special point status for a point object.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetSpecialPoint(
    string Name,
    bool SpecialPoint,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetSpecialPoint (
    Name As String,
    SpecialPoint As Boolean,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim SpecialPoint As Boolean
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetSpecialPoint(Name,
    SpecialPoint, ItemType)
```

```
int SetSpecialPoint(
    String^ Name,
    bool SpecialPoint,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetSpecialPoint :
    Name : string *
    SpecialPoint : bool *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing point object or group depending on the value of the ItemType item.

SpecialPoint

Type:Â SystemBoolean

This item is True if the point object is specified as a special point, otherwise it is False.

ItemType (Optional)

Type:Â ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the special point status is set for the point object specified by the Name item.

If this item is Group, the special point status is set for all point objects in the group specified by the Name item.

If this item is SelectedObjects, the special point status is set for all selected point objects and the Name item is ignored.

Return Value

Type:Â Int32

Returns zero if the special point status is successfully set, otherwise it returns a nonzero value.

Remarks

Special points are allowed to exist in the model even if no objects (line, area, solid, link) are connected to them. Points that are not special are automatically deleted if no objects connect to them.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)
```

Introduction

```
'set as special point
    ret = SapModel.PointObj.SetSpecialPoint("3", True)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

cPointObj Interface

ETABSv1 Namespace

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Namespace: ETABSv1

▲ Syntax

C#

VBC++

F#

Copy

```
Function SetSpring (
Name As String,
ByRef K As Double(),
Optional eItemType As eItemType = eItemType.Objects,
Optional CSys As Boolean = true,
Optional RepNode As Boolean = false
) As Integer
```

```
returnValue = instance.SetSpring(Name,
K, ItemType, IsLocalCSys, Replace)
```

```
int SetSpring(
String^ Name,
array<double>% K,
eItemType ItemType = eItemType::Objects,
bool IsLocalCSys = true,
bool Replace = false
)
```

cPointObj

```
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
    let _IsLocalCSys = defaultArg IsLocalCSys true
    let _Replace = defaultArg Replace false
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing point object or group depending on the value of the ItemType item.

K

Type:Â SystemDouble

This is an array of six spring stiffness values.

◇ Value(0) = U1 [F/L]

◇ Value(1) = U2 [F/L]

◇ Value(2) = U3 [F/L]

◇ Value(3) = R1 [FL/rad]

◇ Value(4) = R2 [FL/rad]

◇ Value(5) = R3 [FL/rad]

ItemType (Optional)

Type:Â ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◇ Object = 0

◇ Group = 1

◇ SelectedObjects = 2

If this item is Object, the spring assignment is made to the point object specified by the Name item.

If this item is Group, the spring assignment is made to all point objects in the group specified by the Name item.

If this item is SelectedObjects, the spring assignment is made to all selected point objects and the Name item is ignored.

IsLocalCSys (Optional)

Type:Â SystemBoolean

If this item is True, the specified spring assignments are in the point object local coordinate system. If it is False, the assignments are in the Global coordinate system.

Replace (Optional)

Type:Â SystemBoolean

If this item is True, all existing point spring assignments to the specified point object(s) are deleted prior to making the assignment. If it is False, the spring assignments are added to any existing assignments.

Return Value

Type:Â Int32

Returns zero if the stiffnesses are successfully assigned, otherwise it returns a nonzero value.

Remarks

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'assign spring to a point
    ReDim k(5)
    k(2) = 10
    ret = SapModel.PointObj.SetSpring("3", k)

    'get spring values
    ReDim k(5)
    ret = SapModel.PointObj.GetSpring("3", k)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

cPointObj Interface

ETABSV1 Namespace

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LSTFD83B008_0") Method

Assigns an existing named point spring property to point objects

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetSpringAssignment(
    string Name,
    string SpringProp,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetSpringAssignment (
    Name As String,
    SpringProp As String,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim SpringProp As String
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetSpringAssignment(Name,
    SpringProp, ItemType)
```

```
int SetSpringAssignment(
    String^ Name,
    String^ SpringProp,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetSpringAssignment :
    Name : string *
    SpringProp : string *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing point object or group depending on the value of the ItemType item.

SpringProp

Type: SystemString

The name of an existing point spring property

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, the property assignment is made to the point object specified by the Name item.

If this item is Group, the property assignment is made to all point objects in the group specified by the Name item.

If this item is SelectedObjects, the property assignment is made to all selected point objects and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the property is successfully assigned, otherwise it returns a nonzero value

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim SpringProp As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new spring property
    Dim k() as Double
    ReDim k(5)
    k(2) = 1
```

Parameters

Introduction

```
ret = SapModel.PropPointSpring.SetPointSpringProp("mySpringProp1", 1, k)

'assign property
ret = SapModel.PointObj.SetSpringAssignment("1", "mySpringProp1")

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cPointObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPointObjAddLanguageSpecificTextSet("LSTC3C88A58A58_0?cpp=::|2749);SetSp Method

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetSpringCoupled(
    string Name,
    ref double[] K,
    eItemType ItemType = eItemType.Objects,
    bool IsLocalCSys = false,
    bool Replace = false
)
```

```
Function SetSpringCoupled (
    Name As String,
    ByRef K As Double(),
    Optional ItemType As eItemType = eItemType.Objects,
    Optional IsLocalCSys As Boolean = false,
    Optional Replace As Boolean = false
) As Integer
```

```
Dim instance As cPointObj
Dim Name As String
Dim K As Double()
Dim ItemType As eItemType
Dim IsLocalCSys As Boolean
Dim Replace As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetSpringCoupled(Name,
    K, ItemType, IsLocalCSys, Replace)
```

```
int SetSpringCoupled(
    String^ Name,
    array<double>^% K,
    eItemType ItemType = eItemType::Objects,
    bool IsLocalCSys = false,
    bool Replace = false
)
```

```
abstract SetSpringCoupled :
    Name : string *
    K : float[] byref *
    ?ItemType : eItemType *
    ?IsLocalCSys : bool *
    ?Replace : bool
(* Defaults:
```

cPointObjspan id="LSTC3C88A58_0"AddLanguageSpecificTextSet("LSTC3C88A58_0?cpp=::|2749);SetSp

Introduction

```
let _ItemType = defaultArg ItemType eItemType.Objects
let _IsLocalCSys = defaultArg IsLocalCSys false
let _Replace = defaultArg Replace false
*)
-> int
```

Parameters

Name

Type:Â SystemString

K

Type:Â SystemDouble

ItemType (Optional)

Type:Â ETABSv1eItemType

IsLocalCSys (Optional)

Type:Â SystemBoolean

Replace (Optional)

Type:Â SystemBoolean

Return Value

Type:Â Int32

🔗 See Also

Reference

cPointObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropArea Interface

Namespace: Â ETABSV1

Assembly: Â ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
public interface cPropArea
```

```
Public Interface cPropArea
```

```
Dim instance As cPropArea
```

```
public interface class cPropArea
```

```
type cPropArea = interface end
```

The cPropArea type exposes the following members.

▲ Methods

Â	Name	Description
⇒	<u>ChangeName</u>	Changes the name of an existing area property.
⇒	<u>Count</u>	Returns the total number of defined area properties in the model. If desired, counts can be returned for all area properties of a specified type in the model.
⇒	<u>Delete</u>	Deletes a specified area property.
⇒	<u>GetDeck</u>	Retrieves property data for a deck section.
⇒	<u>GetDeckFilled</u>	Retrieves property data for a filled deck section.
⇒	<u>GetDeckSolidSlab</u>	Retrieves property data for a solid-slab deck section.
⇒	<u>GetDeckUnfilled</u>	Retrieves property data for an unfilled deck section.
⇒	<u>GetModifiers</u>	Retrieves the modifier assignments for an area property.
⇒	<u>GetNameList</u>	Retrieves the names of all defined area properties of the specified type.
⇒	<u>GetShellDesign</u>	Retrieves area property design parameters for a shell-type area section.
⇒	<u>GetShellLayer</u>	Retrieves area property layer parameters for a shell-type area section.
⇒	<u>GetShellLayer_1</u>	Retrieves area property layer parameters for a shell-type area section.
⇒	<u>GetShellLayer_2</u>	Retrieves area property layer parameters for a shell-type area section.
⇒	<u>GetSlab</u>	Retrieves property data for a slab section.

Introduction

⇒ <u>GetSlabRibbed</u>	Retrieves property data for a ribbed slab section.
⇒ <u>GetSlabWaffle</u>	Retrieves property data for a waffle slab section.
⇒ <u>GetTypeOAPI</u>	Retrieves the property type for the specified area property.
⇒ <u>GetWall</u>	Retrieves property data for a wall section.
⇒ <u>GetWallAutoSelectList</u>	Retrieves property data for an auto select list wall section.
⇒ <u>SetDeck</u>	Initializes a deck property.
⇒ <u>SetDeckFilled</u>	Initializes a filled deck property
⇒ <u>SetDeckSolidSlab</u>	Initializes a solid-slab deck property
⇒ <u>SetDeckUnfilled</u>	Initializes an unfilled deck property
⇒ <u>SetModifiers</u>	Assigns property modifiers to an area property.
⇒ <u>SetShellDesign</u>	Assigns the design parameters for shell-type area properties.
⇒ <u>SetShellLayer</u>	Assigns the layer parameters for shell-type area properties
⇒ <u>SetShellLayer_1</u>	Assigns the layer parameters for shell-type area properties
⇒ <u>SetShellLayer_2</u>	Assigns the layer parameters for shell-type area properties
⇒ <u>SetSlab</u>	Initializes a slab property.
⇒ <u>SetSlabRibbed</u>	Initializes a ribbed slab section.
⇒ <u>SetSlabWaffle</u>	Initializes a waffle slab section.
⇒ <u>SetWall</u>	Initializes property data for a wall section.
⇒ <u>SetWallAutoSelectList</u>	Initializes property data for an auto select list wall section.

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🔍 See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropArea Methods

The cPropArea type exposes the following members.

Methods

Â	Name	Description
⇒	<u>ChangeName</u>	Changes the name of an existing area property.
⇒	<u>Count</u>	Returns the total number of defined area properties in the model. If desired, counts can be returned for all area properties of a specified type in the model.
⇒	<u>Delete</u>	Deletes a specified area property.
⇒	<u>GetDeck</u>	Retrieves property data for a deck section.
⇒	<u>GetDeckFilled</u>	Retrieves property data for a filled deck section.
⇒	<u>GetDeckSolidSlab</u>	Retrieves property data for a solid-slab deck section.
⇒	<u>GetDeckUnfilled</u>	Retrieves property data for an unfilled deck section.
⇒	<u>GetModifiers</u>	Retrieves the modifier assignments for an area property.
⇒	<u>GetNameList</u>	Retrieves the names of all defined area properties of the specified type.
⇒	<u>GetShellDesign</u>	Retrieves area property design parameters for a shell-type area section.
⇒	<u>GetShellLayer</u>	Retrieves area property layer parameters for a shell-type area section.
⇒	<u>GetShellLayer_1</u>	Retrieves area property layer parameters for a shell-type area section.
⇒	<u>GetShellLayer_2</u>	Retrieves area property layer parameters for a shell-type area section.
⇒	<u>GetSlab</u>	Retrieves property data for a slab section.
⇒	<u>GetSlabRibbed</u>	Retrieves property data for a ribbed slab section.
⇒	<u>GetSlabWaffle</u>	Retrieves property data for a waffle slab section.
⇒	<u>GetTypeOAPI</u>	Retrieves the property type for the specified area property.
⇒	<u>GetWall</u>	Retrieves property data for a wall section.
⇒	<u>GetWallAutoSelectList</u>	Retrieves property data for an auto select list wall section.
⇒	<u>SetDeck</u>	Initializes a deck property.
⇒	<u>SetDeckFilled</u>	Initializes a filled deck property
⇒	<u>SetDeckSolidSlab</u>	Initializes a solid-slab deck property
⇒	<u>SetDeckUnfilled</u>	Initializes an unfilled deck property
⇒	<u>SetModifiers</u>	Assigns property modifiers to an area property.
⇒	<u>SetShellDesign</u>	Assigns the design parameters for shell-type area properties.
⇒	<u>SetShellLayer</u>	Assigns the layer parameters for shell-type area properties
⇒	<u>SetShellLayer_1</u>	Assigns the layer parameters for shell-type area properties
⇒	<u>SetShellLayer_2</u>	Assigns the layer parameters for shell-type area properties

Introduction

- ⇒ [SetSlab](#) Initializes a slab property.
- ⇒ [SetSlabRibbed](#) Initializes a ribbed slab section.
- ⇒ [SetSlabWaffle](#) Initializes a waffle slab section.
- ⇒ [SetWall](#) Initializes property data for a wall section.
- ⇒ [SetWallAutoSelectList](#) Initializes property data for an auto select list wall section.

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▀ See Also

Reference

[cPropArea Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LST1AE65486_0?cpp=::|2755);ChangeName(Method

Changes the name of an existing area property.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int ChangeName(  
    string Name,  
    string NewName  
)
```

```
Function ChangeName (  
    Name As String,  
    NewName As String  
) As Integer
```

```
Dim instance As cPropArea  
Dim Name As String  
Dim NewName As String  
Dim returnValue As Integer
```

```
returnValue = instance.ChangeName(Name,  
    NewName)
```

```
int ChangeName(  
    String^ Name,  
    String^ NewName  
)
```

```
abstract ChangeName :  
    Name : string *  
    NewName : string -> int
```

Parameters

Name

Type: SystemString

The existing name of a defined area property.

NewName

Type: SystemString

The new name for the area property.

Return Value

Type: Int32

Returns zero if the new name is successfully applied; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'change name of area property
    ret = SapModel.PropArea.ChangeName("SLAB1", "MyArea")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cPropArea Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LSTFD578D3A_0") Method

Returns the total number of defined area properties in the model. If desired, counts can be returned for all area properties of a specified type in the model.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int Count(
    int PropType = 0
)
```

```
Function Count (
    Optional PropType As Integer = 0
) As Integer
```

```
Dim instance As cPropArea
Dim PropType As Integer
Dim returnValue As Integer
```

```
returnValue = instance.Count(PropType)
```

```
int Count(
    int PropType = 0
)
```

```
abstract Count :
    ?PropType : int
(* Defaults:
    let _PropType = defaultArg PropType 0
*)
-> int
```

Parameters

PropType (Optional)

Type: System.Int32

This optional value is 0, 1, 2 or 3, indicating the type of area properties included in the count.

Value Type

0	All
1	Shell
2	Plane
3	Asolid

cPropAreaid="LSTFD578D3A_0"AddLanguageSpecificTextSet("LSTFD578D3A_0?cpp=:2757.");Cour

Return Value

Type: Int32

The total number of defined area properties in the model.

▲ **Remarks**

Plane and Asolid area properties are not supported in ETABS.

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Count as Integer

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'return number of defined area properties of all types
    Count = SapModel.PropArea.Count

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cPropArea Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LSTFD232EF0_0") Method

Deletes a specified area property.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int Delete(
    string Name
)
```

```
Function Delete (
    Name As String
) As Integer
```

```
Dim instance As cPropArea
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.Delete(Name)
```

```
int Delete(
    String^ Name
)
```

```
abstract Delete :
    Name : string -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing area property.

Return Value

Type: [Int32](#)

Returns zero if the area property is successfully deleted; otherwise it returns a nonzero value

▲ Remarks

Returns an error if the specified area property can not be deleted, for example, if it is being used by an area object.

▲ Examples

VB

cPropAreaspan id="LSTFD232EF0_0"AddLanguageSpecificTextSet("LSTFD232EF0_0?cpp=::2759");Delete

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'delete area property
    ret = SapModel.PropArea.Delete("SLAB1")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

cPropArea Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LST16651BC6_0?cpp=::|276.1");GetDeck

Method

Retrieves property data for a deck section.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetDeck(
    string Name,
    ref eDeckType DeckType,
    ref eShellType ShellType,
    ref string MatProp,
    ref double Thickness,
    ref int color,
    ref string notes,
    ref string GUID
)
```

```
Function GetDeck (
    Name As String,
    ByRef DeckType As eDeckType,
    ByRef ShellType As eShellType,
    ByRef MatProp As String,
    ByRef Thickness As Double,
    ByRef color As Integer,
    ByRef notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropArea
Dim Name As String
Dim DeckType As eDeckType
Dim ShellType As eShellType
Dim MatProp As String
Dim Thickness As Double
Dim color As Integer
Dim notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.GetDeck(Name, DeckType,
    ShellType, MatProp, Thickness, color,
    notes, GUID)
```

```
int GetDeck(
    String^ Name,
    eDeckType% DeckType,
    eShellType% ShellType,
```

cPropAreaAddLanguageSpecificTextSet("LST16651BC6_0?cpp=::|276.1");GetDeck

Parameters

Type: SystemInt32
The display color assigned to the property.

notes

Type:Â SystemString

The notes, if any, assigned to the property.

GUID

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the property.

Return Value

Type:Â Int32

The function returns zero if the property data is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

Imports [Namespace]

```
Public Sub Example()
    Dim SapModel As [Namespace].cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim DeckType As eDeckType
    Dim ShellType As eShellType
    Dim MatProp As String
    Dim Thickness As Double
    Dim Color As Integer
    Dim Notes As String
    Dim GUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new area property
    ret = SapModel.PropArea.SetDeck("MyShellProp1A", eDeckType.Filled, eShellType.ShellThin, '

    'get area property data
    ret = SapModel.PropArea.GetDeck("MyShellProp1A", DeckType, ShellType, MatProp, Thickness,

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

Return Value

▀ See Also

Reference

cPropArea Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LSTEF7AB6B Method

Retrieves property data for a filled deck section.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetDeckFilled(
    string Name,
    ref double SlabDepth,
    ref double RibDepth,
    ref double RibWidthTop,
    ref double RibWidthBot,
    ref double RibSpacing,
    ref double ShearThickness,
    ref double UnitWeight,
    ref double ShearStudDia,
    ref double ShearStudHt,
    ref double ShearStudFu
)
```

```
Function GetDeckFilled (
    Name As String,
    ByRef SlabDepth As Double,
    ByRef RibDepth As Double,
    ByRef RibWidthTop As Double,
    ByRef RibWidthBot As Double,
    ByRef RibSpacing As Double,
    ByRef ShearThickness As Double,
    ByRef UnitWeight As Double,
    ByRef ShearStudDia As Double,
    ByRef ShearStudHt As Double,
    ByRef ShearStudFu As Double
) As Integer
```

```
Dim instance As cPropArea
Dim Name As String
Dim SlabDepth As Double
Dim RibDepth As Double
Dim RibWidthTop As Double
Dim RibWidthBot As Double
Dim RibSpacing As Double
Dim ShearThickness As Double
Dim UnitWeight As Double
Dim ShearStudDia As Double
Dim ShearStudHt As Double
Dim ShearStudFu As Double
Dim returnValue As Integer
```

cPropAreaspan id="LSTEF7AB6BB_0"AddLanguageSpecificTextSet("LSTEF7AB6BB_0?cpp=2765.");GetD

```
returnValue = instance.GetDeckFilled(Name,
SlabDepth, RibDepth, RibWidthTop,
RibWidthBot, RibSpacing, ShearThickness,
UnitWeight, ShearStudDia, ShearStudHt,
ShearStudFu)
```

```
int GetDeckFilled(
String^ Name,
double% SlabDepth,
double% RibDepth,
double% RibWidthTop,
double% RibWidthBot,
double% RibSpacing,
double% ShearThickness,
double% UnitWeight,
double% ShearStudDia,
double% ShearStudHt,
double% ShearStudFu
)
```

```
abstract GetDeckFilled :
    Name : string *
    SlabDepth : float byref *
    RibDepth : float byref *
    RibWidthTop : float byref *
    RibWidthBot : float byref *
    RibSpacing : float byref *
    ShearThickness : float byref *
    UnitWeight : float byref *
    ShearStudDia : float byref *
    ShearStudHt : float byref *
    ShearStudFu : float byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing filled deck property.

SlabDepth

Type:Â SystemDouble

Slab Depth, tc

RibDepth

Type:Â SystemDouble

Rib Depth, hr

RibWidthTop

Type:Â SystemDouble

Rib Width Top, wrt

RibWidthBot

Type:Â SystemDouble

Rib Width Bottom, wrb

RibSpacing

Type:Â SystemDouble

Rib Spacing, sr

ShearThickness

Type:Â SystemDouble

Deck Shear Thickness
UnitWeight
Type:Â SystemDouble
Deck Unit Weight
ShearStudDia
Type:Â SystemDouble
Shear Stud Diameter
ShearStudHt
Type:Â SystemDouble
Shear Stud Height, hs
ShearStudFu
Type:Â SystemDouble
Shear Stud Tensile Strength, Fu

Return Value

Type:Â Int32

The function returns zero if the property data is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim DeckType As eDeckType
    Dim ShellType As eShellType
    Dim MatProp As String
    Dim Thickness As Double
    Dim Color As Integer
    Dim Notes As String
    Dim GUID As String
    Dim SlabDepth As Double
    Dim RibDepth As Double
    Dim RibWidthTop As Double
    Dim RibWidthBot As Double
    Dim RibSpacing As Double
    Dim DeckShearThickness As Double
    Dim DeckUnitWeight As Double
    Dim ShearStudDia As Double
    Dim ShearStudHt As Double
    Dim ShearStudFu As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()
```

Introduction

```
'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new area property
    ret = SapModel.PropArea.SetDeck("MyShellProp1A", eDeckType.Filled, eShellType.ShellThin, '

'set new area property
    ret = SapModel.PropArea.SetDeckFilled(Name:="MyShellProp1A", SlabDepth:=15.6, RibDepth:=1.
        RibSpacing:=7, ShearThickness:=5, UnitWeight:=3.1, S

'get area property data
    ret = SapModel.PropArea.GetDeck("MyShellProp1A", DeckType, ShellType, MatProp, Thickness,

'get area property data
    ret = SapModel.PropArea.GetDeckFilled("MyShellProp1A", SlabDepth, RibDepth, RibWidthTop, F
        RibSpacing, DeckShearThickness, DeckUnitWeight, Shear

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

🔗 See Also

Reference

[cPropArea Interface](#)

[ETABSV1 Namespace](#)

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»

CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LSTB2D53F9C_0?cpp=:2769.");GetD

Method

Retrieves property data for a solid-slab deck section.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetDeckSolidSlab(
    string Name,
    ref double SlabDepth,
    ref double ShearStudDia,
    ref double ShearStudHt,
    ref double ShearStudFu
)
```

```
Function GetDeckSolidSlab (
    Name As String,
    ByRef SlabDepth As Double,
    ByRef ShearStudDia As Double,
    ByRef ShearStudHt As Double,
    ByRef ShearStudFu As Double
) As Integer
```

```
Dim instance As cPropArea
Dim Name As String
Dim SlabDepth As Double
Dim ShearStudDia As Double
Dim ShearStudHt As Double
Dim ShearStudFu As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetDeckSolidSlab(Name,
    SlabDepth, ShearStudDia, ShearStudHt,
    ShearStudFu)
```

```
int GetDeckSolidSlab(
    String^ Name,
    double% SlabDepth,
    double% ShearStudDia,
    double% ShearStudHt,
    double% ShearStudFu
)
```

```
abstract GetDeckSolidSlab :
    Name : string *
    SlabDepth : float byref *
    ShearStudDia : float byref *
    ShearStudHt : float byref *
```

cPropAreaspan id="LSTB2D53F9C_0"AddLanguageSpecificTextSet("LSTB2D53F9C_0?cpp=:2769.");GetD

ShearStudFu : float byref -> int

Parameters

Name

Type:Â SystemString

The name of an existing solid-slab deck property.

SlabDepth

Type:Â SystemDouble

Slab Depth, tc

ShearStudDia

Type:Â SystemDouble

Shear Stud Diameter

ShearStudHt

Type:Â SystemDouble

Shear Stud Height, hs

ShearStudFu

Type:Â SystemDouble

Shear Stud Tensile Strength, Fu

Return Value

Type:Â Int32

The function returns zero if the property data is successfully retrieved; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim DeckType As eDeckType
    Dim ShellType As eShellType
    Dim MatProp As String
    Dim Thickness As Double
    Dim Color As Integer
    Dim Notes As String
    Dim GUID As String
    Dim SlabDepth As Double
    Dim ShearStudDia As Double
    Dim ShearStudHt As Double
    Dim ShearStudFu As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
```

Introduction

```
ret = SapModel.InitializeNewModel()  
  
'create steel deck template model  
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)  
  
'set new area property  
ret = SapModel.PropArea.SetDeck("MyShellProp1A", eDeckType.SolidSlab, eShellType.ShellThin  
  
'set new area property  
ret = SapModel.PropArea.SetDeckSolidSlab(Name:="MyShellProp1A", SlabDepth:=3.5, ShearStudD  
  
'get area property data  
ret = SapModel.PropArea.GetDeck("MyShellProp1A", DeckType, ShellType, MatProp, Thickness,  
  
'get area property data  
ret = SapModel.PropArea.GetDeckSolidSlab("MyShellProp1A", SlabDepth, ShearStudDia, ShearSt  
  
'close ETABS  
EtabsObject.ApplicationExit(False)  
  
'clean up variables  
SapModel = Nothing  
EtabsObject = Nothing  
End Sub
```

🔗 See Also

Reference

[cPropArea Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LST6F8975A7") Method

Retrieves property data for an unfilled deck section.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetDeckUnfilled(
    string Name,
    ref double RibDepth,
    ref double RibWidthTop,
    ref double RibWidthBot,
    ref double RibSpacing,
    ref double ShearThickness,
    ref double UnitWeight
)
```

```
Function GetDeckUnfilled (
    Name As String,
    ByRef RibDepth As Double,
    ByRef RibWidthTop As Double,
    ByRef RibWidthBot As Double,
    ByRef RibSpacing As Double,
    ByRef ShearThickness As Double,
    ByRef UnitWeight As Double
) As Integer
```

```
Dim instance As cPropArea
Dim Name As String
Dim RibDepth As Double
Dim RibWidthTop As Double
Dim RibWidthBot As Double
Dim RibSpacing As Double
Dim ShearThickness As Double
Dim UnitWeight As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetDeckUnfilled(Name,
    RibDepth, RibWidthTop, RibWidthBot,
    RibSpacing, ShearThickness, UnitWeight)
```

```
int GetDeckUnfilled(
    String^ Name,
    double% RibDepth,
    double% RibWidthTop,
    double% RibWidthBot,
    double% RibSpacing,
    double% ShearThickness,
```

cPropAreaspan id="LST6F8975A7_0"AddLanguageSpecificTextSet("LST6F8975A7_0?cpp=::|277.2);GetDe


```
double% UnitWeight
)

abstract GetDeckUnfilled :
    Name : string *
    RibDepth : float byref *
    RibWidthTop : float byref *
    RibWidthBot : float byref *
    RibSpacing : float byref *
    ShearThickness : float byref *
    UnitWeight : float byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing unfilled deck property.

RibDepth

Type:Â SystemDouble

Rib Depth, hr

RibWidthTop

Type:Â SystemDouble

Rib Width Top, wrt

RibWidthBot

Type:Â SystemDouble

Rib Width Bottom, wrb

RibSpacing

Type:Â SystemDouble

Rib Spacing, sr

ShearThickness

Type:Â SystemDouble

Deck Shear Thickness

UnitWeight

Type:Â SystemDouble

Deck Unit Weight

Return Value

Type:Â Int32

The function returns zero if the property data is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim DeckType As eDeckType
    Dim ShellType As eShellType
    Dim MatProp As String
    Dim Thickness As Double
    Dim Color As Integer
```

Introduction

```
Dim Notes As String
Dim GUID As String
Dim SlabDepth As Double
Dim RibDepth As Double
Dim RibWidthTop As Double
Dim RibWidthBot As Double
Dim RibSpacing As Double
Dim ShearThickness As Double
Dim UnitWeight As Double

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new area property
ret = SapModel.PropArea.SetDeck("MyShellProp1A", eDeckType.Unfilled, eShellType.ShellThin,

'set new area property
ret = SapModel.PropArea.SetDeckUnfilled(Name:="MyShellProp1A", RibDepth:=1.1, RibWidthTop:

'get area property data
ret = SapModel.PropArea.GetDeck("MyShellProp1A", DeckType, ShellType, MatProp, Thickness,

'get area property data
ret = SapModel.PropArea.GetDeckUnfilled("MyShellProp1A", RibDepth, RibWidthTop, RibWidthBo

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

[cPropArea Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LST5D60D387_0") Method

Retrieves the modifier assignments for an area property.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetModifiers(
    string Name,
    ref double[] Value
)
```

```
Function GetModifiers (
    Name As String,
    ByRef Value As Double()
) As Integer
```

```
Dim instance As cPropArea
Dim Name As String
Dim Value As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetModifiers(Name,
    Value)
```

```
int GetModifiers(
    String^ Name,
    array<double>^% Value
)
```

```
abstract GetModifiers :
    Name : string *
    Value : float[] byref -> int
```

Parameters

Name

Type: [System.String](#)

The name of an existing area property.

Value

Type: [System.Double](#)

An array of 10 unitless modifiers.

Value	Modifier
Value(0)	Membrane f11 modifier

Value(1) Membrane f22 modifier
 Value(2) Membrane f12 modifier
 Value(3) Bending m11 modifier
 Value(4) Bending m22 modifier
 Value(5) Bending m12 modifier
 Value(6) Shear v13 modifier
 Value(7) Shear v23 modifier
 Value(8) Mass modifier
 Value(9) Weight modifier

Return Value

Type: $\hat{A}$ Int32

Returns zero if the modifier assignments are successfully retrieved; otherwise it returns a nonzero value.

Remarks

The default value for all modifiers is one.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim i As Integer
    Dim MyValue() As Double
    Dim Value() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'assign modifiers
    ReDim MyValue(9)
    For i = 0 To 9
        MyValue(i) = 1
    Next i
    MyValue(0) = 0.1
    ret = SapModel.PropArea.SetModifiers("SLAB1", MyValue)

    'get modifiers
    ret = SapModel.PropArea.GetModifiers("SLAB1", Value)
```

Introduction

```
'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPropArea Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LST50E855E Method

Retrieves the names of all defined area properties of the specified type.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNameList(
    ref int NumberNames,
    ref string[] MyName,
    int PropType = 0
)
```

```
Function GetNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String(),
    PropType As Integer = 0
) As Integer
```

```
Dim instance As cPropArea
Dim NumberNames As Integer
Dim MyName As String()
Dim PropType As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetNameList(NumberNames,
    MyName, PropType)
```

```
int GetNameList(
    int% NumberNames,
    array<String^>^% MyName,
    int PropType = 0
)
```

```
abstract GetNameList :
    NumberNames : int byref *
    MyName : string[] byref *
    ?PropType : int
(* Defaults:
    let _PropType = defaultArg PropType 0
*)
-> int
```

Parameters

NumberNames

Type: SystemInt32

The number of area property names retrieved by the program.

MyName

Type: SystemString

One-dimensional array of area property names. The MyName array is created as a dynamic, zero-based, array by the API user:

```
Dim MyName() as String
```

The array is dimensioned to (NumberNames - 1) inside the program, filled with the names, and returned to the API user.

PropType (Optional)

Type: SystemInt32

This optional value is 0, 1, 2 or 3, indicating the type of area properties included in the name list.

Value Type

0	All
1	Shell
2	Plane
3	Asolid

Return Value

Type: Int32

Returns zero if the names are successfully retrieved; otherwise it returns nonzero

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim i As Integer
    Dim NumberNames As Integer
    Dim MyName() As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'get area property names
```

Introduction

```
ret = SapModel.PropArea.GetNameList(NumberNames, MyName)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

📌 See Also

Reference

[cPropArea Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LSTACD90FBB_0") Method

Retrieves area property design parameters for a shell-type area section.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetShellDesign(
    string Name,
    ref string MatProp,
    ref int SteelLayoutOption,
    ref double DesignCoverTopDir1,
    ref double DesignCoverTopDir2,
    ref double DesignCoverBotDir1,
    ref double DesignCoverBotDir2
)
```

```
Function GetShellDesign (
    Name As String,
    ByRef MatProp As String,
    ByRef SteelLayoutOption As Integer,
    ByRef DesignCoverTopDir1 As Double,
    ByRef DesignCoverTopDir2 As Double,
    ByRef DesignCoverBotDir1 As Double,
    ByRef DesignCoverBotDir2 As Double
) As Integer
```

```
Dim instance As cPropArea
Dim Name As String
Dim MatProp As String
Dim SteelLayoutOption As Integer
Dim DesignCoverTopDir1 As Double
Dim DesignCoverTopDir2 As Double
Dim DesignCoverBotDir1 As Double
Dim DesignCoverBotDir2 As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetShellDesign(Name,
    MatProp, SteelLayoutOption, DesignCoverTopDir1,
    DesignCoverTopDir2, DesignCoverBotDir1,
    DesignCoverBotDir2)
```

```
int GetShellDesign(
    String^ Name,
    String^% MatProp,
    int% SteelLayoutOption,
    double% DesignCoverTopDir1,
    double% DesignCoverTopDir2,
```

cPropAreaspan id="LSTACD90FBB_0"AddLanguageSpecificTextSet("LSTACD90FBB_0?cpp=2781-.");Get

```
double% DesignCoverBotDir1,
double% DesignCoverBotDir2
)
```

```
abstract GetShellDesign :
    Name : string *
    MatProp : string byref *
    SteelLayoutOption : int byref *
    DesignCoverTopDir1 : float byref *
    DesignCoverTopDir2 : float byref *
    DesignCoverBotDir1 : float byref *
    DesignCoverBotDir2 : float byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing shell-type area property.

MatProp

Type:Â SystemString

The name of the material property for the area property.

SteelLayoutOption

Type:Â SystemInt32

This is 0, 1 or 2 indicating, the rebar layout option.

Value Layout

0 Default

1 One layer

2 Two layers

DesignCoverTopDir1

Type:Â SystemDouble

The cover to the centroid of the top reinforcing steel running in the local 1 axis direction of the area object. [L]

This item applies only when SteelLayoutOption = 1 or 2.

DesignCoverTopDir2

Type:Â SystemDouble

The cover to the centroid of the top reinforcing steel running in the local 2 axis direction of the area object. [L]

This item applies only when SteelLayoutOption = 1 or 2.

DesignCoverBotDir1

Type:Â SystemDouble

The cover to the centroid of the bottom reinforcing steel running in the local 1 axis direction of the area object. [L]

This item applies only when SteelLayoutOption = 2.

DesignCoverBotDir2

Type:Â SystemDouble

The cover to the centroid of the bottom reinforcing steel running in the local 2 axis direction of the area object. [L]

This item applies only when SteelLayoutOption = 2.

Return Value

Type: Int32

Returns zero if the parameters are successfully retrieved; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MatProp As String
    Dim SteelLayoutOption As integer
    Dim DesignCoverTopDir1 As Double
    Dim DesignCoverTopDir2 As Double
    Dim DesignCoverBotDir1 As Double
    Dim DesignCoverBotDir2 As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'get area property design parameters
    ret = SapModel.PropArea.GetShellDesign("SLAB1", MatProp, SteelLayoutOption, DesignCoverTopDir1, DesignCoverTopDir2, DesignCoverBotDir1, DesignCoverBotDir2)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cPropArea Interface

ETABSv1 Namespace

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Introduction

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CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LSTA371DD10_0") Method

Retrieves area property layer parameters for a shell-type area section.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetShellLayer(
    string Name,
    ref int NumberLayers,
    ref string[] LayerName,
    ref double[] Dist,
    ref double[] Thickness,
    ref string[] MatProp,
    ref bool[] Nonlinear,
    ref double[] MatAng,
    ref int[] NumIntegrationPts
)
```

```
Function GetShellLayer (
    Name As String,
    ByRef NumberLayers As Integer,
    ByRef LayerName As String(),
    ByRef Dist As Double(),
    ByRef Thickness As Double(),
    ByRef MatProp As String(),
    ByRef Nonlinear As Boolean(),
    ByRef MatAng As Double(),
    ByRef NumIntegrationPts As Integer()
) As Integer
```

```
Dim instance As cPropArea
Dim Name As String
Dim NumberLayers As Integer
Dim LayerName As String()
Dim Dist As Double()
Dim Thickness As Double()
Dim MatProp As String()
Dim Nonlinear As Boolean()
Dim MatAng As Double()
Dim NumIntegrationPts As Integer()
Dim returnValue As Integer

returnValue = instance.GetShellLayer(Name,
    NumberLayers, LayerName, Dist, Thickness,
    MatProp, Nonlinear, MatAng, NumIntegrationPts)
```

```
int GetShellLayer(
```

cPropAreaspan id="LSTA371DD10_0"AddLanguageSpecificTextSet("LSTA371DD10_0?cpp=::2785");GetS

```
String^ Name,
int% NumberLayers,
array<String^>^% LayerName,
array<double>^% Dist,
array<double>^% Thickness,
array<String^>^% MatProp,
array<bool>^% Nonlinear,
array<double>^% MatAng,
array<int>^% NumIntegrationPts
)
```

```
abstract GetShellLayer :
    Name : string *
    NumberLayers : int byref *
    LayerName : string[] byref *
    Dist : float[] byref *
    Thickness : float[] byref *
    MatProp : string[] byref *
    Nonlinear : bool[] byref *
    MatAng : float[] byref *
    NumIntegrationPts : int[] byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing shell-type area property that is specified to be a layered shell property.

NumberLayers

Type:Â SystemInt32

The number of layers in the area property.

LayerName

Type:Â SystemString

This is an array that includes the name of each layer.

Dist

Type:Â SystemDouble

This is an array that includes the distance from the area reference surface (area object joint location plus offsets) to the mid-height of the layer. [L]

Thickness

Type:Â SystemDouble

This is an array that includes the thickness of each layer. [L]

MatProp

Type:Â SystemString

This is an array that includes the name of the material property for the layer.

Nonlinear

Type:Â SystemBoolean

MatAng

Type:Â SystemDouble

This is an array that includes the material angle for the layer. [deg]

NumIntegrationPts

Type:Â SystemInt32

The number of integration points in the thickness direction for the layer. The locations are determined by the program using standard Gauss-quadrature rules.

Return Value

Type: Int32

Returns zero if the parameters are successfully retrieved, otherwise it returns a nonzero value.

The function returns an error if the specified area property is not a shell-type property specified to be a layered shell.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim MyNumberLayers As Integer
    Dim MyLayerName() As String
    Dim MyDist() As Double
    Dim MyThickness() As Double
    Dim MyNumIntegrationPts() As Integer
    Dim MyMatProp() As String
    Dim MyNonLinear() As Boolean
    Dim MyMatAng() As Double
    Dim NumberLayers As Integer
    Dim LayerName() As String
    Dim Dist() As Double
    Dim Thickness() As Double
    Dim MatProp() As String
    Dim NonLinear() As Boolean
    Dim MatAng() As Double
    Dim NumIntegrationPts() As Integer

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new area property
    ret = SapModel.PropArea.SetWall("A1", eWallPropType.Specified, eShellType.Layered, "", 0)

    'add A615Gr60 rebar material
    ret = SapModel.PropMaterial.AddMaterial(Name, eMatType.Rebar, "United States", "ASTM A615")

    'set area property layer parameters
    MyNumberLayers = 5
end Sub
```

Introduction

```
ReDim MyLayerName(MyNumberLayers - 1)
ReDim MyDist(MyNumberLayers - 1)
ReDim MyThickness(MyNumberLayers - 1)
ReDim MyNumIntegrationPts(MyNumberLayers - 1)
ReDim MyMatProp(MyNumberLayers - 1)
ReDim MyNonLinear(MyNumberLayers - 1)
ReDim MyMatAng(MyNumberLayers - 1)

MyLayerName(0) = "Concrete"
MyDist(0) = 0
MyThickness(0) = 16
MyNumIntegrationPts(0) = 2
MyMatProp(0) = "4000Psi"
MyMatAng(0) = 0

MyLayerName(1) = "Top Bar 1"
MyDist(1) = 6
MyThickness(1) = 0.03
MyNumIntegrationPts(1) = 1
MyMatProp(1) = Name
MyMatAng(1) = 0

MyLayerName(2) = "Top Bar 2"
MyDist(2) = 6
MyThickness(2) = 0.03
MyNumIntegrationPts(2) = 1
MyMatProp(2) = Name
MyMatAng(2) = 90

MyLayerName(3) = "Bot Bar 1"
MyDist(3) = -6
MyThickness(3) = 0.03
MyNumIntegrationPts(3) = 1
MyMatProp(3) = Name
MyMatAng(3) = 0

MyLayerName(4) = "Bot Bar 2"
MyDist(4) = -6
MyThickness(4) = 0.03
MyNumIntegrationPts(4) = 1
MyMatProp(4) = Name
MyMatAng(4) = 90

ret = SapModel.PropArea.SetShellLayer("A1", MyNumberLayers, MyLayerName, MyDist, MyThickness)

'get area property layer parameters
ret = SapModel.PropArea.GetShellLayer("A1", NumberLayers, LayerName, Dist, Thickness, MatProp)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

See Also

Reference

cPropArea Interface

ETABSv1 Namespace

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»

CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LSTFEFC4B4C_0") Method

Retrieves area property layer parameters for a shell-type area section.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetShellLayer_1(
    string Name,
    ref int NumberLayers,
    ref string[] LayerName,
    ref double[] Dist,
    ref double[] Thickness,
    ref int[] MyType,
    ref int[] NumIntegrationPts,
    ref string[] MatProp,
    ref double[] MatAng,
    ref int[] S11Type,
    ref int[] S22Type,
    ref int[] S12Type
)
```

```
Function GetShellLayer_1 (
    Name As String,
    ByRef NumberLayers As Integer,
    ByRef LayerName As String(),
    ByRef Dist As Double(),
    ByRef Thickness As Double(),
    ByRef MyType As Integer(),
    ByRef NumIntegrationPts As Integer(),
    ByRef MatProp As String(),
    ByRef MatAng As Double(),
    ByRef S11Type As Integer(),
    ByRef S22Type As Integer(),
    ByRef S12Type As Integer()
) As Integer
```

```
Dim instance As cPropArea
Dim Name As String
Dim NumberLayers As Integer
Dim LayerName As String()
Dim Dist As Double()
Dim Thickness As Double()
Dim MyType As Integer()
Dim NumIntegrationPts As Integer()
Dim MatProp As String()
Dim MatAng As Double()
Dim S11Type As Integer()
```

cPropAreaid="LSTFEFC4B4C_0"AddLanguageSpecificTextSet("LSTFEFC4B4C_0?cpp=2790.");GetS

```

Dim S22Type As Integer()
Dim S12Type As Integer()
Dim returnValue As Integer

returnValue = instance.GetShellLayer_1(Name,
NumberLayers, LayerName, Dist, Thickness,
MyType, NumIntegrationPts, MatProp,
MatAng, S11Type, S22Type, S12Type)

int GetShellLayer_1(
String^ Name,
int% NumberLayers,
array<String^>^% LayerName,
array<double>^% Dist,
array<double>^% Thickness,
array<int>^% MyType,
array<int>^% NumIntegrationPts,
array<String^>^% MatProp,
array<double>^% MatAng,
array<int>^% S11Type,
array<int>^% S22Type,
array<int>^% S12Type
)

abstract GetShellLayer_1 :
    Name : string *
    NumberLayers : int byref *
    LayerName : string[] byref *
    Dist : float[] byref *
    Thickness : float[] byref *
    MyType : int[] byref *
    NumIntegrationPts : int[] byref *
    MatProp : string[] byref *
    MatAng : float[] byref *
    S11Type : int[] byref *
    S22Type : int[] byref *
    S12Type : int[] byref -> int

```

Parameters

Name

Type:Â SystemString

The name of an existing shell-type area property that is specified to be a layered shell property.

NumberLayers

Type:Â SystemInt32

The number of layers in the area property.

LayerName

Type:Â SystemString

This is an array that includes the name of each layer.

Dist

Type:Â SystemDouble

This is an array that includes the distance from the area reference surface (area object joint location plus offsets) to the mid-height of the layer. [L]

Thickness

Type:Â SystemDouble

This is an array that includes the thickness of each layer. [L]

MyTypeType:Â SystemInt32

This is an array that includes 1, 2 or 3, indicating the layer type.

Value Type

1	Shell
2	Membrane
3	Plate

NumIntegrationPtsType:Â SystemInt32

The number of integration points in the thickness direction for the layer. The locations are determined by the program using standard Guass-quadrature rules.

MatPropType:Â SystemString

This is an array that includes the name of the material property for the layer.

MatAngType:Â SystemDouble

This is an array that includes the material angle for the layer. [deg]

S11TypeType:Â SystemInt32

This is an array that includes 0, 1 or 2, indicating the material component behavior.

Value S11Type

0	Inactive
1	Linear
2	Nonlinear

S22TypeType:Â SystemInt32

This is an array that includes 0, 1 or 2, indicating the material component behavior.

Value S22Type

0	Inactive
1	Linear
2	Nonlinear

S12TypeType:Â SystemInt32

This is an array that includes 0, 1 or 2, indicating the material component behavior.

Value S12Type

0	Inactive
1	Linear
2	Nonlinear

Return ValueType:Â Int32

Returns zero if the parameters are successfully retrieved, otherwise it returns a

nonzero value.

The function returns an error if the specified area property is not a shell-type property specified to be a layered shell.

Remarks

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim MyNumberLayers As Integer
    Dim MyLayerName() As String
    Dim MyDist() As Double
    Dim MyThickness() As Double
    Dim MyType() As Integer
    Dim MyNumIntegrationPts() As Integer
    Dim MyMatProp() As String
    Dim MyMatAng() As Double
    Dim MyS11Type() As Integer
    Dim MyS22Type() As Integer
    Dim MyS12Type() As Integer
    Dim NumberLayers As Integer
    Dim LayerName() As String
    Dim Dist() As Double
    Dim Thickness() As Double
    Dim SType() As Integer
    Dim MatProp() As String
    Dim MatAng() As Double
    Dim S11Type() As Integer
    Dim S22Type() As Integer
    Dim S12Type() As Integer
    Dim NumIntegrationPts() As Integer

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new area property
    ret = SapModel.PropArea.SetWall("A1", eWallPropType.Specified, eShellType.Layered, "", 0)

    'add A615Gr60 rebar material
    ret = SapModel.PropMaterial.AddMaterial(Name, eMatType.Rebar, "United States", "ASTM A615")

    'set area property layer parameters
```

Introduction

```
MyNumberLayers = 5
ReDim MyLayerName(MyNumberLayers - 1)
ReDim MyDist(MyNumberLayers - 1)
ReDim MyThickness(MyNumberLayers - 1)
ReDim MyType(MyNumberLayers - 1)
ReDim MyNumIntegrationPts(MyNumberLayers - 1)
ReDim MyMatProp(MyNumberLayers - 1)
ReDim MyMatAng(MyNumberLayers - 1)
ReDim MyS11Type(MyNumberLayers - 1)
ReDim MyS22Type(MyNumberLayers - 1)
ReDim MyS12Type(MyNumberLayers - 1)
```

```
MyLayerName(0) = "Concrete"
MyDist(0) = 0
MyThickness(0) = 16
MyType(0) = 1
MyNumIntegrationPts(0) = 2
MyMatProp(0) = "4000Psi"
MyMatAng(0) = 0
MyS11Type(0) = 1
MyS22Type(0) = 1
MyS12Type(0) = 1
```

```
MyLayerName(1) = "Top Bar 1"
MyDist(1) = 6
MyThickness(1) = 0.03
MyType(1) = 1
MyNumIntegrationPts(1) = 1
MyMatProp(1) = Name
MyMatAng(1) = 0
MyS11Type(1) = 1
MyS22Type(1) = 1
MyS12Type(1) = 1
```

```
MyLayerName(2) = "Top Bar 2"
MyDist(2) = 6
MyThickness(2) = 0.03
MyType(2) = 1
MyNumIntegrationPts(2) = 1
MyMatProp(2) = Name
MyMatAng(2) = 90
MyS11Type(2) = 1
MyS22Type(2) = 1
MyS12Type(2) = 1
```

```
MyLayerName(3) = "Bot Bar 1"
MyDist(3) = -6
MyThickness(3) = 0.03
MyType(3) = 1
MyNumIntegrationPts(3) = 1
MyMatProp(3) = Name
MyMatAng(3) = 0
MyS11Type(3) = 1
MyS22Type(3) = 1
MyS12Type(3) = 1
```

```
MyLayerName(4) = "Bot Bar 2"
MyDist(4) = -6
MyThickness(4) = 0.03
MyType(4) = 1
MyNumIntegrationPts(4) = 1
MyMatProp(4) = Name
```

Introduction

```
MyMatAng(4) = 90
MyS11Type(4) = 1
MyS22Type(4) = 1
MyS12Type(4) = 1

ret = SapModel.PropArea.SetShellLayer_1("A1", MyNumberLayers, MyLayerName, MyDist, MyThick

'get area property layer parameters
ret = SapModel.PropArea.GetShellLayer_1("A1", NumberLayers, LayerName, Dist, Thickness, St

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPropArea Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LSTFCB34432_0?cpp=::2796");GetShellLayer_2 Method

Retrieves area property layer parameters for a shell-type area section.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetShellLayer_2(
    string Name,
    ref int NumberLayers,
    ref string[] LayerName,
    ref double[] Dist,
    ref double[] Thickness,
    ref int[] MyType,
    ref int[] NumIntegrationPts,
    ref string[] MatProp,
    ref double[] MatAng,
    ref int[] MatBehavior,
    ref int[] S11Type,
    ref int[] S22Type,
    ref int[] S12Type
)
```

```
Function GetShellLayer_2 (
    Name As String,
    ByRef NumberLayers As Integer,
    ByRef LayerName As String(),
    ByRef Dist As Double(),
    ByRef Thickness As Double(),
    ByRef MyType As Integer(),
    ByRef NumIntegrationPts As Integer(),
    ByRef MatProp As String(),
    ByRef MatAng As Double(),
    ByRef MatBehavior As Integer(),
    ByRef S11Type As Integer(),
    ByRef S22Type As Integer(),
    ByRef S12Type As Integer()
) As Integer
```

```
Dim instance As cPropArea
Dim Name As String
Dim NumberLayers As Integer
Dim LayerName As String()
Dim Dist As Double()
Dim Thickness As Double()
Dim MyType As Integer()
Dim NumIntegrationPts As Integer()
Dim MatProp As String()
```

cPropAreaspan id="LSTFCB34432_0"AddLanguageSpecificTextSet("LSTFCB34432_0?cpp=::2796");GetShellLayer_2


```

Dim MatAng As Double()
Dim MatBehavior As Integer()
Dim S11Type As Integer()
Dim S22Type As Integer()
Dim S12Type As Integer()
Dim returnValue As Integer

returnValue = instance.GetShellLayer_2(Name,
NumberLayers, LayerName, Dist, Thickness,
MyType, NumIntegrationPts, MatProp,
MatAng, MatBehavior, S11Type, S22Type,
S12Type)

int GetShellLayer_2(
String^ Name,
int% NumberLayers,
array<String^>^% LayerName,
array<double>^% Dist,
array<double>^% Thickness,
array<int>^% MyType,
array<int>^% NumIntegrationPts,
array<String^>^% MatProp,
array<double>^% MatAng,
array<int>^% MatBehavior,
array<int>^% S11Type,
array<int>^% S22Type,
array<int>^% S12Type
)

abstract GetShellLayer_2 :
    Name : string *
    NumberLayers : int byref *
    LayerName : string[] byref *
    Dist : float[] byref *
    Thickness : float[] byref *
    MyType : int[] byref *
    NumIntegrationPts : int[] byref *
    MatProp : string[] byref *
    MatAng : float[] byref *
    MatBehavior : int[] byref *
    S11Type : int[] byref *
    S22Type : int[] byref *
    S12Type : int[] byref -> int

```

Parameters

Name

Type:Â [SystemString](#)

The name of an existing shell-type area property that is specified to be a layered shell property.

NumberLayers

Type:Â [SystemInt32](#)

The number of layers in the area property.

LayerName

Type:Â [SystemString](#)

This is an array that includes the name of each layer.

Dist

Type: SystemDouble

This is an array that includes the distance from the area reference surface (area object joint location plus offsets) to the mid-height of the layer. [L]

Thickness

Type: SystemDouble

This is an array that includes the thickness of each layer. [L]

MyType

Type: SystemInt32

This is an array that includes 1, 2 or 3, indicating the layer type.

Value Type

1	Shell
2	Membrane
3	Plate

NumIntegrationPts

Type: SystemInt32

The number of integration points in the thickness direction for the layer. The locations are determined by the program using standard Guass-quadrature rules.

MatProp

Type: SystemString

This is an array that includes the name of the material property for the layer.

MatAng

Type: SystemDouble

This is an array that includes the material angle for the layer. [deg]

MatBehavior

Type: SystemInt32

This is an array that includes 0 or 1 indicating the material behavior type of the layer.

Value Type

0	Directional
1	Coupled

S11Type

Type: SystemInt32

This is an array that includes 0, 1 or 2, indicating the material component behavior.

Value S11Type

0	Inactive
1	Linear
2	Nonlinear

S22Type

Type: SystemInt32

This is an array that includes 0, 1 or 2, indicating the material component behavior.

Value S22Type

0	Inactive
1	Linear
2	Nonlinear

S12TypeType: SystemInt32

This is an array that includes 0, 1 or 2, indicating the material component behavior.

Value S12Type

0	Inactive
1	Linear
2	Nonlinear

Return ValueType: Int32

Returns zero if the parameters are successfully retrieved, otherwise it returns a nonzero value.

The function returns an error if the specified area property is not a shell-type property specified to be a layered shell.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim MyNumberLayers As Integer
    Dim MyLayerName() As String
    Dim MyDist() As Double
    Dim MyThickness() As Double
    Dim MyType() As Integer
    Dim MyNumIntegrationPts() As Integer
    Dim MyMatProp() As String
    Dim MyMatAng() As Double
    Dim MyMatBehavior() As Integer
    Dim MyS11Type() As Integer
    Dim MyS22Type() As Integer
    Dim MyS12Type() As Integer
    Dim NumberLayers As Integer
    Dim LayerName() As String
    Dim Dist() As Double
    Dim Thickness() As Double
    Dim SType() As Integer
    Dim MatProp() As String
    Dim MatAng() As Double
    Dim MatBehavior() As Integer
    Dim S11Type() As Integer
    Dim S22Type() As Integer
    Dim S12Type() As Integer
    Dim NumIntegrationPts() As Integer

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
End Sub
```

Introduction

```
'start ETABS application
    ret = EtabsObject.ApplicationStart()

'create SapModel object
    SapModel = EtabsObject.SapModel

'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new area property
    ret = SapModel.PropArea.SetWall("A1", eWallPropType.Specified, eShellType.Layered, "", 0)

'add A615Gr60 rebar material
    ret = SapModel.PropMaterial.AddMaterial(Name, eMatType.Rebar, "United States", "ASTM A615"

'set area property layer parameters
    MyNumberLayers = 5
    ReDim MyLayerName(MyNumberLayers - 1)
    ReDim MyDist(MyNumberLayers - 1)
    ReDim MyThickness(MyNumberLayers - 1)
    ReDim MyType(MyNumberLayers - 1)
    ReDim MyNumIntegrationPts(MyNumberLayers - 1)
    ReDim MyMatProp(MyNumberLayers - 1)
    ReDim MyMatAng(MyNumberLayers - 1)
    ReDim MyMatBehavior(MyNumberLayers - 1)
    ReDim MyS11Type(MyNumberLayers - 1)
    ReDim MyS22Type(MyNumberLayers - 1)
    ReDim MyS12Type(MyNumberLayers - 1)

    MyLayerName(0) = "Concrete"
    MyDist(0) = 0
    MyThickness(0) = 16
    MyType(0) = 1
    MyNumIntegrationPts(0) = 2
    MyMatProp(0) = "4000Psi"
    MyMatAng(0) = 0
    MyMatBehavior(0) = 1
    MyS11Type(0) = 1
    MyS22Type(0) = 1
    MyS12Type(0) = 1

    MyLayerName(1) = "Top Bar 1"
    MyDist(1) = 6
    MyThickness(1) = 0.03
    MyType(1) = 1
    MyNumIntegrationPts(1) = 1
    MyMatProp(1) = Name
    MyMatAng(1) = 0
    MyMatBehavior(1) = 1
    MyS11Type(1) = 1
    MyS22Type(1) = 1
    MyS12Type(1) = 1

    MyLayerName(2) = "Top Bar 2"
    MyDist(2) = 6
    MyThickness(2) = 0.03
    MyType(2) = 1
    MyNumIntegrationPts(2) = 1
    MyMatProp(2) = Name
```

Introduction

```
MyMatAng(2) = 90
MyMatBehavior(2) = 1
MyS11Type(2) = 1
MyS22Type(2) = 1
MyS12Type(2) = 1

MyLayerName(3) = "Bot Bar 1"
MyDist(3) = -6
MyThickness(3) = 0.03
MyType(3) = 1
MyNumIntegrationPts(3) = 1
MyMatProp(3) = Name
MyMatAng(3) = 0
MyMatBehavior(3) = 1
MyS11Type(3) = 1
MyS22Type(3) = 1
MyS12Type(3) = 1

MyLayerName(4) = "Bot Bar 2"
MyDist(4) = -6
MyThickness(4) = 0.03
MyType(4) = 1
MyNumIntegrationPts(4) = 1
MyMatProp(4) = Name
MyMatAng(4) = 90
MyMatBehavior(4) = 1
MyS11Type(4) = 1
MyS22Type(4) = 1
MyS12Type(4) = 1

ret = SapModel.PropArea.SetShellLayer_2("A1", MyNumberLayers, MyLayerName, MyDist, MyThick

'get area property layer parameters
ret = SapModel.PropArea.GetShellLayer_2("A1", NumberLayers, LayerName, Dist, Thickness, St

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

See Also

Reference

cPropArea Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LSTA8B7E7F Method

Retrieves property data for a slab section.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetSlab(
    string Name,
    ref eSlabType SlabType,
    ref eShellType ShellType,
    ref string MatProp,
    ref double Thickness,
    ref int color,
    ref string notes,
    ref string GUID
)
```

```
Function GetSlab (
    Name As String,
    ByRef SlabType As eSlabType,
    ByRef ShellType As eShellType,
    ByRef MatProp As String,
    ByRef Thickness As Double,
    ByRef color As Integer,
    ByRef notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropArea
Dim Name As String
Dim SlabType As eSlabType
Dim ShellType As eShellType
Dim MatProp As String
Dim Thickness As Double
Dim color As Integer
Dim notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.GetSlab(Name, SlabType,
    ShellType, MatProp, Thickness, color,
    notes, GUID)
```

```
int GetSlab(
    String^ Name,
    eSlabType% SlabType,
    eShellType% ShellType,
```

cPropAreaid="LSTA8B7E7FB_0"AddLanguageSpecificTextSet("LSTA8B7E7FB_0?cpp=:2002");GetS

```
String^% MatProp,
double% Thickness,
int% color,
String^% notes,
String^% GUID
)
```

```
abstract GetSlab :
    Name : string *
    SlabType : eSlabType byref *
    ShellType : eShellType byref *
    MatProp : string byref *
    Thickness : float byref *
    color : int byref *
    notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing slab property.

SlabType

Type:Â ETABSv1eSlabType

This is one of the items in the eSlabType enumeration.

If this item is Ribbed, use the GetSlabRibbed(String, Double, Double, Double, Double, Double, Int32) function to get additional parameters.

If this item is Waffle, use the GetSlabWaffle(String, Double, Double, Double, Double, Double, Double) function to get additional parameters.

ShellType

Type:Â ETABSv1eShellType

This is one of the items in the eShellType enumeration.

MatProp

Type:Â SystemString

The name of the material property for the area property.

This item does not apply when ShellType is Layered.

Thickness

Type:Â SystemDouble

The membrane thickness. [L]

This item does not apply when ShellType is Layered.

color

Type:Â SystemInt32

The display color assigned to the property.

notes

Type:Â SystemString

The notes, if any, assigned to the property.

GUID

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the property.

Return ValueType: Int32

The function returns zero if the property data is successfully retrieved; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim SlabType As eSlabType
    Dim ShellType As eShellType
    Dim MatProp As String
    Dim Thickness As Double
    Dim Color As Integer
    Dim Notes As String
    Dim GUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new area property
    ret = SapModel.PropArea.SetSlab("MyShellProp1A", eSlabType.Drop, eShellType.ShellThin, "40")

    'get area property data
    ret = SapModel.PropArea.GetSlab("MyShellProp1A", SlabType, ShellType, MatProp, Thickness, Color, Notes, GUID)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cPropArea Interface

ETABSv1 Namespace

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»

CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LST7841CAF8_0") Method

Retrieves property data for a ribbed slab section.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetSlabRibbed(
    string Name,
    ref double OverallDepth,
    ref double SlabThickness,
    ref double StemWidthTop,
    ref double StemWidthBot,
    ref double RibSpacing,
    ref int RibsParallelTo
)
```

```
Function GetSlabRibbed (
    Name As String,
    ByRef OverallDepth As Double,
    ByRef SlabThickness As Double,
    ByRef StemWidthTop As Double,
    ByRef StemWidthBot As Double,
    ByRef RibSpacing As Double,
    ByRef RibsParallelTo As Integer
) As Integer
```

```
Dim instance As cPropArea
Dim Name As String
Dim OverallDepth As Double
Dim SlabThickness As Double
Dim StemWidthTop As Double
Dim StemWidthBot As Double
Dim RibSpacing As Double
Dim RibsParallelTo As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetSlabRibbed(Name,
    OverallDepth, SlabThickness, StemWidthTop,
    StemWidthBot, RibSpacing, RibsParallelTo)
```

```
int GetSlabRibbed(
    String^ Name,
    double% OverallDepth,
    double% SlabThickness,
    double% StemWidthTop,
    double% StemWidthBot,
    double% RibSpacing,
```

cPropAreaspan id="LST7841CAF8_0"AddLanguageSpecificTextSet("LST7841CAF8_0?cpp=::2006");GetSl

```

int% RibsParallelTo
)

abstract GetSlabRibbed :
    Name : string *
    OverallDepth : float byref *
    SlabThickness : float byref *
    StemWidthTop : float byref *
    StemWidthBot : float byref *
    RibSpacing : float byref *
    RibsParallelTo : int byref -> int

```

Parameters

Name

Type:Â [SystemString](#)

The name of an existing ribbed slab property.

OverallDepth

Type:Â [SystemDouble](#)

Overall Depth

SlabThickness

Type:Â [SystemDouble](#)

Slab Thickness

StemWidthTop

Type:Â [SystemDouble](#)

Stem Width at Top

StemWidthBot

Type:Â [SystemDouble](#)

Stem Width at Bottom

RibSpacing

Type:Â [SystemDouble](#)

Rib Spacing (Perpendicular to Rib Direction)

RibsParallelTo

Type:Â [SystemInt32](#)

This is the Local Axis that the Rib Direction is Parallel to

This value is 1 for the Local 1 Axis, and 2 for the Local 2 Axis.

Return Value

Type:Â [Int32](#)

The function returns zero if the property data is successfully retrieved; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

[Copy](#)

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim SlabType As eSlabType
    Dim ShellType As eShellType
    Dim MatProp As String

```

Introduction

```
Dim Thickness As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim OverallDepth As Double
Dim SlabThickness As Double
Dim StemWidthTop As Double
Dim StemWidthBot As Double
Dim RibSpacing As Double
Dim RibsParallelTo As Integer

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new area property
ret = SapModel.PropArea.SetSlab("MyShellProp1A", eSlabType.Ribbed, eShellType.ShellThin, '

'set new area property
ret = SapModel.PropArea.SetSlabRibbed(Name:="MyShellProp1A", OverallDepth:=11.1, SlabThick

'get area property data
ret = SapModel.PropArea.GetSlab("MyShellProp1A", SlabType, ShellType, MatProp, Thickness,

'get area property data
ret = SapModel.PropArea.GetSlabRibbed("MyShellProp1A", OverallDepth, SlabThickness, StemW

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPropArea Interface

ETABSv1 Namespace

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»

CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LSTCC511DE3_0") Method

Retrieves property data for a waffle slab section.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetSlabWaffle(
    string Name,
    ref double OverallDepth,
    ref double SlabThickness,
    ref double StemWidthTop,
    ref double StemWidthBot,
    ref double RibSpacingDir1,
    ref double RibSpacingDir2
)
```

```
Function GetSlabWaffle (
    Name As String,
    ByRef OverallDepth As Double,
    ByRef SlabThickness As Double,
    ByRef StemWidthTop As Double,
    ByRef StemWidthBot As Double,
    ByRef RibSpacingDir1 As Double,
    ByRef RibSpacingDir2 As Double
) As Integer
```

```
Dim instance As cPropArea
Dim Name As String
Dim OverallDepth As Double
Dim SlabThickness As Double
Dim StemWidthTop As Double
Dim StemWidthBot As Double
Dim RibSpacingDir1 As Double
Dim RibSpacingDir2 As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetSlabWaffle(Name,
    OverallDepth, SlabThickness, StemWidthTop,
    StemWidthBot, RibSpacingDir1, RibSpacingDir2)
```

```
int GetSlabWaffle(
    String^ Name,
    double% OverallDepth,
    double% SlabThickness,
    double% StemWidthTop,
    double% StemWidthBot,
    double% RibSpacingDir1,
```

cPropAreaspan id="LSTCC511DE3_0"AddLanguageSpecificTextSet("LSTCC511DE3_0?cpp=2009.");GetS

```
double% RibSpacingDir2
)
```

```
abstract GetSlabWaffle :
    Name : string *
    OverallDepth : float byref *
    SlabThickness : float byref *
    StemWidthTop : float byref *
    StemWidthBot : float byref *
    RibSpacingDir1 : float byref *
    RibSpacingDir2 : float byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing waffle slab property.

OverallDepth

Type:Â SystemDouble

Overall Depth

SlabThickness

Type:Â SystemDouble

Slab Thickness

StemWidthTop

Type:Â SystemDouble

Stem Width at Top

StemWidthBot

Type:Â SystemDouble

Stem Width at Bottom

RibSpacingDir1

Type:Â SystemDouble

Spacing of Ribs that are Parallel to Slab 1-Axis

RibSpacingDir2

Type:Â SystemDouble

Spacing of Ribs that are Parallel to Slab 2-Axis

Return Value

Type:Â Int32

The function returns zero if the property data is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim SlabType As eSlabType
    Dim ShellType As eShellType
    Dim MatProp As String
    Dim Thickness As Double
    Dim Color As Integer
```

Introduction

```
Dim Notes As String
Dim GUID As String
Dim OverallDepth As Double
Dim SlabThickness As Double
Dim StemWidthTop As Double
Dim StemWidthBot As Double
Dim RibSpacing1 As Double
Dim RibSpacing2 As Double

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new area property
ret = SapModel.PropArea.SetSlab("MyShellProp1A", eSlabType.Waffle, eShellType.ShellThin, '

'set new area property
ret = SapModel.PropArea.SetSlabWaffle(Name:="MyShellProp1A", OverallDepth:=11.1, SlabThick

'get area property data
ret = SapModel.PropArea.GetSlab("MyShellProp1A", SlabType, ShellType, MatProp, Thickness,

'get area property data
ret = SapModel.PropArea.GetSlabWaffle("MyShellProp1A", OverallDepth, SlabThickness, StemW

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

[cPropArea Interface](#)

[ETABSv1 Namespace](#)

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Namespace: ETABSv1

▲ Syntax

C#

VBC++F#

Copy

```
Function GetTypeOAPI (
  Name As String,
  ByRef PropType As Integer
) As Integer
```

```
returnValue = instance.GetType0API(Name,
PropType)
```

```
abstract GetTypeOAPI :
  Name : string *
  PropType : int byref -> int
```

Name

Type:Â SystemString

The name of an existing area property.

PropType

Type: SystemInt32

This is 1, 2 or 3, indicating the type of area property.

Value Type

1 Shell

- 2 Plane
- 3 Asolid

Return Value

Type: Int32

Returns zero if the type is successfully retrieved; otherwise it returns nonzero.

Remarks

Plane and Asolid area properties are not supported in ETABS.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim PropType As Integer

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'get area property type
    ret = SapModel.PropArea.GetTypeOAPI("SLAB1", PropType)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

cPropArea Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LSTA88ADD15_0?cpp=:2814.");GetWall Method

Retrieves property data for a wall section.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetWall(
    string Name,
    ref eWallPropType WallPropType,
    ref eShellType ShellType,
    ref string MatProp,
    ref double Thickness,
    ref int color,
    ref string notes,
    ref string GUID
)
```

```
Function GetWall (
    Name As String,
    ByRef WallPropType As eWallPropType,
    ByRef ShellType As eShellType,
    ByRef MatProp As String,
    ByRef Thickness As Double,
    ByRef color As Integer,
    ByRef notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropArea
Dim Name As String
Dim WallPropType As eWallPropType
Dim ShellType As eShellType
Dim MatProp As String
Dim Thickness As Double
Dim color As Integer
Dim notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.GetWall(Name, WallPropType,
    ShellType, MatProp, Thickness, color,
    notes, GUID)
```

```
int GetWall(
    String^ Name,
    eWallPropType% WallPropType,
    eShellType% ShellType,
```

cPropAreaspan id="LSTA88ADD15_0"AddLanguageSpecificTextSet("LSTA88ADD15_0?cpp=:2814.");GetWall

```
String^% MatProp,
double% Thickness,
int% color,
String^% notes,
String^% GUID
)
```

```
abstract GetWall :
    Name : string *
    WallPropType : eWallPropType byref *
    ShellType : eShellType byref *
    MatProp : string byref *
    Thickness : float byref *
    color : int byref *
    notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing wall property.

WallPropType

Type:Â ETABSv1eWallPropType

This is one of the items in the eWallPropType enumeration.

If this item is AutoSelectList, use the GetWallAutoSelectList(String,String,String) function to get additional parameters.

ShellType

Type:Â ETABSv1eShellType

This is one of the items in the eShellType enumeration.

MatProp

Type:Â SystemString

The name of the material property for the area property.

This item does not apply when ShellType is Layered.

Thickness

Type:Â SystemDouble

The membrane thickness. [L]

This item does not apply when ShellType is Layered.

color

Type:Â SystemInt32

The display color assigned to the property.

notes

Type:Â SystemString

The notes, if any, assigned to the property.

GUID

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the property.

Return Value

Type: Int32

The function returns zero if the property data is successfully retrieved; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim WallPropType As eWallPropType
    Dim ShellType As eShellType
    Dim MatProp As String
    Dim Thickness As Double
    Dim Color As Integer
    Dim Notes As String
    Dim GUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new area property
    ret = SapModel.PropArea.SetWall("MyShellProp1A", eWallPropType.Specified, eShellType.Shell

    'get area property data
    ret = SapModel.PropArea.GetWall("MyShellProp1A", WallPropType, ShellType, MatProp, Thickne

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cPropArea Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LSTBD1458B Method

Retrieves property data for an auto select list wall section.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetWallAutoSelectList(
    string Name,
    ref string[] AutoSelectList,
    ref string StartingProperty
)
```

```
Function GetWallAutoSelectList (
    Name As String,
    ByRef AutoSelectList As String(),
    ByRef StartingProperty As String
) As Integer
```

```
Dim instance As cPropArea
Dim Name As String
Dim AutoSelectList As String()
Dim StartingProperty As String
Dim returnValue As Integer
```

```
returnValue = instance.GetWallAutoSelectList(Name,
    AutoSelectList, StartingProperty)
```

```
int GetWallAutoSelectList(
    String^ Name,
    array<String^>^% AutoSelectList,
    String^% StartingProperty
)
```

```
abstract GetWallAutoSelectList :
    Name : string *
    AutoSelectList : string[] byref *
    StartingProperty : string byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing auto select list wall property.

AutoSelectList

Type: SystemString

This is an array of the names of the wall properties included in the auto select list.

StartingProperty

Type: SystemString

This is Median or the name of a wall property in the AutoSelectList array. It is the starting section for the auto select list.

Median indicates the Median Property by Thickness

Return Value

Type: Int32

The function returns zero if the property data is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim WallPropType As eWallPropType
    Dim ShellType As eShellType
    Dim MatProp As String
    Dim Thickness As Double
    Dim Color As Integer
    Dim Notes As String
    Dim GUID As String
    Dim AutoSelectList As String()
    Dim StartingProp As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new area property
    ret = SapModel.PropArea.SetWall("MyShellProp1A", eWallPropType.Specified, eShellType.Shell1)

    'set new area property
    ret = SapModel.PropArea.SetWall("MyShellProp1B", eWallPropType.Specified, eShellType.Shell1)

    'set new area property
    ret = SapModel.PropArea.SetWall("MyShellProp1C", eWallPropType.AutoSelectList, eShellType.Shell1)
```

Introduction

```
'set new area property
    ret = SapModel.PropArea.SetWall("MyShellProp1D", eWallPropType.AutoSelectList, eShellType.

'get new area property
    Dim tmpList As String()
    ReDim tmpList(2)
    tmpList(0) = "MyShellProp1A"
    tmpList(1) = "MyShellProp1B"
    tmpList(2) = "MyShellProp1C"
    ret = SapModel.PropArea.SetWallAutoSelectList("MyShellProp1D", tmpList, "MyShellProp1B")

'get area property data
    ret = SapModel.PropArea.GetWall("MyShellProp1D", WallPropType, ShellType, MatProp, Thickne

'get area property data
    ret = SapModel.PropArea.GetWallAutoSelectList("MyShellProp1D", AutoSelectList, StartingPro

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

🔗 See Also

Reference

[cPropArea Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LST3CB46314_0?cpp=::|282.1");SetDeck

Method

Initializes a deck property.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetDeck(
    string Name,
    eDeckType DeckType,
    eShellType ShellType,
    string MatProp,
    double Thickness,
    int color = -1,
    string notes = "",
    string GUID = ""
)
```

```
Function SetDeck (
    Name As String,
    DeckType As eDeckType,
    ShellType As eShellType,
    MatProp As String,
    Thickness As Double,
    Optional color As Integer = -1,
    Optional notes As String = "",
    Optional GUID As String = ""
) As Integer
```

```
Dim instance As cPropArea
Dim Name As String
Dim DeckType As eDeckType
Dim ShellType As eShellType
Dim MatProp As String
Dim Thickness As Double
Dim color As Integer
Dim notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.SetDeck(Name, DeckType,
    ShellType, MatProp, Thickness, color,
    notes, GUID)
```

```
int SetDeck(
    String^ Name,
    eDeckType DeckType,
    eShellType ShellType,
```

cPropAreaAddLanguageSpecificTextSet("LST3CB46314_0?cpp=::|282.1");SetDeck

```

String^ MatProp,
double Thickness,
int color = -1,
String^ notes = L"",
String^ GUID = L""
)

abstract SetDeck :
    Name : string *
    DeckType : eDeckType *
    ShellType : eShellType *
    MatProp : string *
    Thickness : float *
    ?color : int *
    ?notes : string *
    ?GUID : string
(* Defaults:
    let _color = defaultArg color -1
    let _notes = defaultArg notes ""
    let _GUID = defaultArg GUID ""
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of a deck property. If this is an existing property, that property is modified; otherwise, a new property is added.

DeckType

Type:Â ETABSv1eDeckType

This is one of the following items in the eDeckType enumeration.

If this item is Filled, use the SetDeckFilled(String, Double, Double, Double, Double, Double, Double, Double, Double, Double) function to get additional parameters.

If this item is Unfilled, use the SetDeckUnfilled(String, Double, Double, Double, Double, Double, Double) function to get additional parameters.

If this item is SolidSlab, use the SetDeckSolidSlab(String, Double, Double, Double, Double) function to get additional parameters.

ShellType

Type:Â ETABSv1eShellType

This is one of the items in the eShellType enumeration.

Please note that for deck properties, this is always Membrane

MatProp

Type:Â SystemString

The name of the material property for the area property.

This item does not apply when ShellType = 6.

Thickness

Type:Â SystemDouble

The membrane thickness. [L]

Introduction

This item does not apply when ShellType = 6.

color (Optional)

Type:Â SystemInt32

The display color assigned to the property.

notes (Optional)

Type:Â SystemString

The notes, if any, assigned to the property.

GUID (Optional)

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the property.

Return Value

Type:Â Int32

Returns zero if the property is successfully initialized; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim DeckType As eDeckType  
    Dim ShellType As eShellType  
    Dim MatProp As String  
    Dim Thickness As Double  
    Dim Color As Integer  
    Dim Notes As String  
    Dim GUID As String  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)  
  
    'set new area property  
    ret = SapModel.PropArea.SetDeck("MyShellProp1A", eDeckType.Filled, eShellType.ShellThin, '  
  
    'get area property data  
    ret = SapModel.PropArea.GetDeck("MyShellProp1A", DeckType, ShellType, MatProp, Thickness, '  
  
    'close ETABS  
    EtabsObject.ApplicationExit(False)  
  
    'clean up variables
```

Introduction

```
SapModel = Nothing  
EtabsObject = Nothing  
End Sub
```

▲ See Also

Reference

cPropArea Interface

ETABSv1 Namespace

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»

CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LST35129312_0?cpp=::;SetDeckFilled")

Method

Initializes a filled deck property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetDeckFilled(
    string Name,
    double SlabDepth,
    double RibDepth,
    double RibWidthTop,
    double RibWidthBot,
    double RibSpacing,
    double ShearThickness,
    double UnitWeight,
    double ShearStudDia,
    double ShearStudHt,
    double ShearStudFu
)
```

```
Function SetDeckFilled (
    Name As String,
    SlabDepth As Double,
    RibDepth As Double,
    RibWidthTop As Double,
    RibWidthBot As Double,
    RibSpacing As Double,
    ShearThickness As Double,
    UnitWeight As Double,
    ShearStudDia As Double,
    ShearStudHt As Double,
    ShearStudFu As Double
) As Integer
```

```
Dim instance As cPropArea
Dim Name As String
Dim SlabDepth As Double
Dim RibDepth As Double
Dim RibWidthTop As Double
Dim RibWidthBot As Double
Dim RibSpacing As Double
Dim ShearThickness As Double
Dim UnitWeight As Double
Dim ShearStudDia As Double
Dim ShearStudHt As Double
Dim ShearStudFu As Double
Dim returnValue As Integer
```

cPropAreaspan id="LST35129312_0"AddLanguageSpecificTextSet("LST35129312_0?cpp=::;SetDeckFilled")

```
returnValue = instance.SetDeckFilled(Name,
SlabDepth, RibDepth, RibWidthTop,
RibWidthBot, RibSpacing, ShearThickness,
UnitWeight, ShearStudDia, ShearStudHt,
ShearStudFu)
```

```
int SetDeckFilled(
String^ Name,
double SlabDepth,
double RibDepth,
double RibWidthTop,
double RibWidthBot,
double RibSpacing,
double ShearThickness,
double UnitWeight,
double ShearStudDia,
double ShearStudHt,
double ShearStudFu
)
```

```
abstract SetDeckFilled :
    Name : string *
    SlabDepth : float *
    RibDepth : float *
    RibWidthTop : float *
    RibWidthBot : float *
    RibSpacing : float *
    ShearThickness : float *
    UnitWeight : float *
    ShearStudDia : float *
    ShearStudHt : float *
    ShearStudFu : float -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing deck property created with the SetDeck(String, eDeckType, eShellType, String, Double, Int32, String, String) function.

SlabDepth

Type:Â SystemDouble

Slab Depth, tc

RibDepth

Type:Â SystemDouble

Rib Depth, hr

RibWidthTop

Type:Â SystemDouble

Rib Width Top, wrt

RibWidthBot

Type:Â SystemDouble

Rib Width Bottom, wrb

RibSpacing

Type:Â SystemDouble

Rib Spacing, sr

ShearThickness

Type: SystemDouble
 Deck Shear Thickness
 UnitWeight
 Type: SystemDouble
 Deck Unit Weight
 ShearStudDia
 Type: SystemDouble
 Shear Stud Diameter
 ShearStudHt
 Type: SystemDouble
 Shear Stud Height, hs
 ShearStudFu
 Type: SystemDouble
 Shear Stud Tensile Strength, Fu

Return Value

Type: Int32

Returns zero if the property is successfully initialized; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim DeckType As eDeckType
    Dim ShellType As eShellType
    Dim MatProp As String
    Dim Thickness As Double
    Dim Color As Integer
    Dim Notes As String
    Dim GUID As String
    Dim SlabDepth As Double
    Dim RibDepth As Double
    Dim RibWidthTop As Double
    Dim RibWidthBot As Double
    Dim RibSpacing As Double
    Dim DeckShearThickness As Double
    Dim DeckUnitWeight As Double
    Dim ShearStudDia As Double
    Dim ShearStudHt As Double
    Dim ShearStudFu As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
```

Return Value

Introduction

```
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new area property
ret = SapModel.PropArea.SetDeck("MyShellProp1A", eDeckType.Filled, eShellType.ShellThin, '

'set new area property
ret = SapModel.PropArea.SetDeckFilled(Name:="MyShellProp1A", SlabDepth:=15.6, RibDepth:=1.
    RibSpacing:=7, ShearThickness:=5, UnitWeight:=3.1, S

'get area property data
ret = SapModel.PropArea.GetDeck("MyShellProp1A", DeckType, ShellType, MatProp, Thickness,

'get area property data
ret = SapModel.PropArea.GetDeckFilled("MyShellProp1A", SlabDepth, RibDepth, RibWidthTop, F
    RibSpacing, DeckShearThickness, DeckUnitWeight, Shea

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

See Also

Reference

[cPropArea Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LSTCAD936C1_0?cpp=2029.");SetD

Method

Initializes a solid-slab deck property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetDeckSolidSlab(
    string Name,
    double SlabDepth,
    double ShearStudDia,
    double ShearStudHt,
    double ShearStudFu
)
```

```
Function SetDeckSolidSlab (
    Name As String,
    SlabDepth As Double,
    ShearStudDia As Double,
    ShearStudHt As Double,
    ShearStudFu As Double
) As Integer
```

```
Dim instance As cPropArea
Dim Name As String
Dim SlabDepth As Double
Dim ShearStudDia As Double
Dim ShearStudHt As Double
Dim ShearStudFu As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetDeckSolidSlab(Name,
    SlabDepth, ShearStudDia, ShearStudHt,
    ShearStudFu)
```

```
int SetDeckSolidSlab(
    String^ Name,
    double SlabDepth,
    double ShearStudDia,
    double ShearStudHt,
    double ShearStudFu
)
```

```
abstract SetDeckSolidSlab :
    Name : string *
    SlabDepth : float *
    ShearStudDia : float *
    ShearStudHt : float *
```

cPropAreaspan id="LSTCAD936C1_0"AddLanguageSpecificTextSet("LSTCAD936C1_0?cpp=2029.");SetD

ShearStudFu : float -> int

Parameters

Name

Type:Â SystemString

The name of an existing deck property created with the SetDeck(String, eDeckType, eShellType, String, Double, Int32, String, String) function.

SlabDepth

Type:Â SystemDouble

Slab Depth, tc

ShearStudDia

Type:Â SystemDouble

Shear Stud Diameter

ShearStudHt

Type:Â SystemDouble

Shear Stud Height, hs

ShearStudFu

Type:Â SystemDouble

Shear Stud Tensile Strength, Fu

Return Value

Type:Â Int32

Returns zero if the property is successfully initialized; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim DeckType As eDeckType
    Dim ShellType As eShellType
    Dim MatProp As String
    Dim Thickness As Double
    Dim Color As Integer
    Dim Notes As String
    Dim GUID As String
    Dim SlabDepth As Double
    Dim ShearStudDia As Double
    Dim ShearStudHt As Double
    Dim ShearStudFu As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel
```

Introduction

```
'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new area property
    ret = SapModel.PropArea.SetDeck("MyShellProp1A", eDeckType.SolidSlab, eShellType.ShellThi

'set new area property
    ret = SapModel.PropArea.SetDeckSolidSlab(Name:="MyShellProp1A", SlabDepth:=3.5, ShearStudD

'get area property data
    ret = SapModel.PropArea.GetDeck("MyShellProp1A", DeckType, ShellType, MatProp, Thickness,

'get area property data
    ret = SapModel.PropArea.GetDeckSolidSlab("MyShellProp1A", SlabDepth, ShearStudDia, ShearSt

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

🔗 See Also

Reference

[cPropArea Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LSTFD7E4414_0?cpp=::2032");SetDeckUnfilled Method

Initializes an unfilled deck property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetDeckUnfilled(
    string Name,
    double RibDepth,
    double RibWidthTop,
    double RibWidthBot,
    double RibSpacing,
    double ShearThickness,
    double UnitWeight
)
```

```
Function SetDeckUnfilled (
    Name As String,
    RibDepth As Double,
    RibWidthTop As Double,
    RibWidthBot As Double,
    RibSpacing As Double,
    ShearThickness As Double,
    UnitWeight As Double
) As Integer
```

```
Dim instance As cPropArea
Dim Name As String
Dim RibDepth As Double
Dim RibWidthTop As Double
Dim RibWidthBot As Double
Dim RibSpacing As Double
Dim ShearThickness As Double
Dim UnitWeight As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetDeckUnfilled(Name,
    RibDepth, RibWidthTop, RibWidthBot,
    RibSpacing, ShearThickness, UnitWeight)
```

```
int SetDeckUnfilled(
    String^ Name,
    double RibDepth,
    double RibWidthTop,
    double RibWidthBot,
    double RibSpacing,
    double ShearThickness,
```

cPropAreaspan id="LSTFD7E4414_0"AddLanguageSpecificTextSet("LSTFD7E4414_0?cpp=::2032");SetDeckUnfilled

```

double UnitWeight
)

abstract SetDeckUnfilled :
    Name : string *
    RibDepth : float *
    RibWidthTop : float *
    RibWidthBot : float *
    RibSpacing : float *
    ShearThickness : float *
    UnitWeight : float -> int

```

Parameters

Name

Type:Â SystemString

The name of an existing deck property created with the SetDeck(String, eDeckType, eShellType, String, Double, Int32, String, String) function.

RibDepth

Type:Â SystemDouble

Rib Depth, hr

RibWidthTop

Type:Â SystemDouble

Rib Width Top, wrt

RibWidthBot

Type:Â SystemDouble

Rib Width Bottom, wrb

RibSpacing

Type:Â SystemDouble

Rib Spacing, sr

ShearThickness

Type:Â SystemDouble

Deck Shear Thickness

UnitWeight

Type:Â SystemDouble

Deck Unit Weight

Return Value

Type:Â Int32

Returns zero if the property is successfully initialized; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim DeckType As eDeckType
    Dim ShellType As eShellType
    Dim MatProp As String
    Dim Thickness As Double

```

Introduction

```
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim SlabDepth As Double
Dim RibDepth As Double
Dim RibWidthTop As Double
Dim RibWidthBot As Double
Dim RibSpacing As Double
Dim ShearThickness As Double
Dim UnitWeight As Double

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new area property
ret = SapModel.PropArea.SetDeck("MyShellProp1A", eDeckType.Unfilled, eShellType.ShellThin,

'set new area property
ret = SapModel.PropArea.SetDeckUnfilled(Name:="MyShellProp1A", RibDepth:=1.1, RibWidthTop:

'get area property data
ret = SapModel.PropArea.GetDeck("MyShellProp1A", DeckType, ShellType, MatProp, Thickness,

'get area property data
ret = SapModel.PropArea.GetDeckUnfilled("MyShellProp1A", RibDepth, RibWidthTop, RibWidthBot

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

➤ See Also

Reference

cPropArea Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LST45589621_0") Method

Assigns property modifiers to an area property.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetModifiers(
    string Name,
    ref double[] Value
)
```

```
Function SetModifiers (
    Name As String,
    ByRef Value As Double()
) As Integer
```

```
Dim instance As cPropArea
Dim Name As String
Dim Value As Double()
Dim returnValue As Integer
```

```
returnValue = instance.SetModifiers(Name,
    Value)
```

```
int SetModifiers(
    String^ Name,
    array<double>^% Value
)
```

```
abstract SetModifiers :
    Name : string *
    Value : float[] byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing area property.

Value

Type: SystemDouble

An array of 10 unitless modifiers.

Value	Modifier
Value(0)	Membrane f11 modifier

Value(1) Membrane f22 modifier
 Value(2) Membrane f12 modifier
 Value(3) Bending m11 modifier
 Value(4) Bending m22 modifier
 Value(5) Bending m12 modifier
 Value(6) Shear v13 modifier
 Value(7) Shear v23 modifier
 Value(8) Mass modifier
 Value(9) Weight modifier

Return Value

Type: Int32

Returns zero if the modifiers are successfully assigned; otherwise it returns a nonzero value.

Remarks

The default value for all modifiers is one.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim i As Integer
    Dim MyValue() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'assign modifiers
    ReDim MyValue(9)
    For i = 0 To 9
        MyValue(i) = 1
    Next i
    MyValue(0) = 0.1
    ret = SapModel.PropArea.SetModifiers("SLAB1", MyValue)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
```


Introduction

```
SapModel = Nothing  
EtabsObject = Nothing  
End Sub
```

▲ See Also

Reference

cPropArea Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LSTEA7F57E8_0?cpp=::2808");SetShellDesign

Method

Assigns the design parameters for shell-type area properties.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetShellDesign(
    string Name,
    string MatProp,
    int SteelLayoutOption,
    double DesignCoverTopDir1,
    double DesignCoverTopDir2,
    double DesignCoverBotDir1,
    double DesignCoverBotDir2
)
```

```
Function SetShellDesign (
    Name As String,
    MatProp As String,
    SteelLayoutOption As Integer,
    DesignCoverTopDir1 As Double,
    DesignCoverTopDir2 As Double,
    DesignCoverBotDir1 As Double,
    DesignCoverBotDir2 As Double
) As Integer
```

```
Dim instance As cPropArea
Dim Name As String
Dim MatProp As String
Dim SteelLayoutOption As Integer
Dim DesignCoverTopDir1 As Double
Dim DesignCoverTopDir2 As Double
Dim DesignCoverBotDir1 As Double
Dim DesignCoverBotDir2 As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetShellDesign(Name,
    MatProp, SteelLayoutOption, DesignCoverTopDir1,
    DesignCoverTopDir2, DesignCoverBotDir1,
    DesignCoverBotDir2)
```

```
int SetShellDesign(
    String^ Name,
    String^ MatProp,
    int SteelLayoutOption,
    double DesignCoverTopDir1,
    double DesignCoverTopDir2,
```

cPropAreaspan id="LSTEA7F57E8_0"AddLanguageSpecificTextSet("LSTEA7F57E8_0?cpp=::2808");SetShellDesign

```

double DesignCoverBotDir1,
double DesignCoverBotDir2
)

abstract SetShellDesign :
    Name : string *
    MatProp : string *
    SteelLayoutOption : int *
    DesignCoverTopDir1 : float *
    DesignCoverTopDir2 : float *
    DesignCoverBotDir1 : float *
    DesignCoverBotDir2 : float -> int

```

Parameters

Name

Type:Â SystemString

The name of an existing shell-type area property.

MatProp

Type:Â SystemString

The name of the material property for the area property.

SteelLayoutOption

Type:Â SystemInt32

This is 0, 1 or 2 indicating, the rebar layout option.

Value Layout

0 Default

1 One layer

2 Two layers

DesignCoverTopDir1

Type:Â SystemDouble

The cover to the centroid of the top reinforcing steel running in the local 1 axis direction of the area object. [L]

This item applies only when SteelLayoutOption = 1 or 2.

DesignCoverTopDir2

Type:Â SystemDouble

The cover to the centroid of the top reinforcing steel running in the local 2 axis direction of the area object. [L]

This item applies only when SteelLayoutOption = 1 or 2.

DesignCoverBotDir1

Type:Â SystemDouble

The cover to the centroid of the bottom reinforcing steel running in the local 1 axis direction of the area object. [L]

This item applies only when SteelLayoutOption = 2.

DesignCoverBotDir2

Type:Â SystemDouble

The cover to the centroid of the bottom reinforcing steel running in the local 2 axis direction of the area object. [L]

This item applies only when SteelLayoutOption = 2.

Return Value

Type: Int32

Returns zero if the parameters are successfully assigned; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set area property design parameters
    ret = SapModel.PropArea.SetShellDesign("SLAB1", "A615Gr60", 2, 2, 3, 2.5, 3.5)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cPropArea Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LST54ADE729_0") Method

Assigns the layer parameters for shell-type area properties

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetShellLayer(
    string Name,
    int NumberLayers,
    ref string[] LayerName,
    ref double[] Dist,
    ref double[] Thickness,
    ref string[] MatProp,
    ref bool[] Nonlinear,
    ref double[] MatAng,
    ref int[] NumIntegrationPts
)
```

```
Function SetShellLayer (
    Name As String,
    NumberLayers As Integer,
    ByRef LayerName As String(),
    ByRef Dist As Double(),
    ByRef Thickness As Double(),
    ByRef MatProp As String(),
    ByRef Nonlinear As Boolean(),
    ByRef MatAng As Double(),
    ByRef NumIntegrationPts As Integer()
) As Integer
```

```
Dim instance As cPropArea
Dim Name As String
Dim NumberLayers As Integer
Dim LayerName As String()
Dim Dist As Double()
Dim Thickness As Double()
Dim MatProp As String()
Dim Nonlinear As Boolean()
Dim MatAng As Double()
Dim NumIntegrationPts As Integer()
Dim returnValue As Integer

returnValue = instance.SetShellLayer(Name,
    NumberLayers, LayerName, Dist, Thickness,
    MatProp, Nonlinear, MatAng, NumIntegrationPts)
```

```
int SetShellLayer(
```

cPropAreaid="LST54ADE729_0"AddLanguageSpecificTextSet("LST54ADE729_0?cpp=::2841");SetS

```
String^ Name,
int NumberLayers,
array<String^>^% LayerName,
array<double>^% Dist,
array<double>^% Thickness,
array<String^>^% MatProp,
array<bool>^% Nonlinear,
array<double>^% MatAng,
array<int>^% NumIntegrationPts
)
```

```
abstract SetShellLayer :
    Name : string *
    NumberLayers : int *
    LayerName : string[] byref *
    Dist : float[] byref *
    Thickness : float[] byref *
    MatProp : string[] byref *
    Nonlinear : bool[] byref *
    MatAng : float[] byref *
    NumIntegrationPts : int[] byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing shell-type area property that is specified to be a layered shell property.

NumberLayers

Type:Â SystemInt32

The number of layers in the area property.

LayerName

Type:Â SystemString

This is an array that includes the name of each layer.

Dist

Type:Â SystemDouble

This is an array that includes the distance from the area reference surface (area object joint location plus offsets) to the mid-height of the layer. [L]

Thickness

Type:Â SystemDouble

This is an array that includes the thickness of each layer. [L]

MatProp

Type:Â SystemString

This is an array that includes the name of the material property for the layer.

Nonlinear

Type:Â SystemBoolean

MatAng

Type:Â SystemDouble

This is an array that includes the material angle for the layer. [deg]

NumIntegrationPts

Type:Â SystemInt32

The number of integration points in the thickness direction for the layer. The locations are determined by the program using standard Gauss-quadrature rules.

Return Value

Type: Int32

Returns zero if the parameters are successfully assigned; otherwise it returns a nonzero value.

The function returns an error if the specified area property is not a shell-type property specified to be a layered shell.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim MyNumberLayers As Integer
    Dim MyLayerName() As String
    Dim MyDist() As Double
    Dim MyThickness() As Double
    Dim MyNumIntegrationPts() As Integer
    Dim MyMatProp() As String
    Dim MyNonlinear() As Boolean
    Dim MyMatAng() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new area property
    ret = SapModel.PropArea.SetWall("A1", eWallPropType.Specified, eShellType.Layered, "", 0)

    'add A615Gr60 rebar material
    ret = SapModel.PropMaterial.AddMaterial(Name, eMatType.Rebar, "United States", "ASTM A615")

    'set area property layer parameters
    MyNumberLayers = 5
    ReDim MyLayerName(MyNumberLayers - 1)
    ReDim MyDist(MyNumberLayers - 1)
    ReDim MyThickness(MyNumberLayers - 1)
    ReDim MyNumIntegrationPts(MyNumberLayers - 1)
    ReDim MyMatProp(MyNumberLayers - 1)
    ReDim MyNonLinear(MyNumberLayers - 1)
    ReDim MyMatAng(MyNumberLayers - 1)
```

Introduction

```
MyLayerName(0) = "Concrete"
MyDist(0) = 0
MyThickness(0) = 16
MyNumIntegrationPts(0) = 2
MyMatProp(0) = "4000Psi"
MyMatAng(0) = 0

MyLayerName(1) = "Top Bar 1"
MyDist(1) = 6
MyThickness(1) = 0.03
MyNumIntegrationPts(1) = 1
MyMatProp(1) = Name
MyMatAng(1) = 0

MyLayerName(2) = "Top Bar 2"
MyDist(2) = 6
MyThickness(2) = 0.03
MyNumIntegrationPts(2) = 1
MyMatProp(2) = Name
MyMatAng(2) = 90

MyLayerName(3) = "Bot Bar 1"
MyDist(3) = -6
MyThickness(3) = 0.03
MyNumIntegrationPts(3) = 1
MyMatProp(3) = Name
MyMatAng(3) = 0

MyLayerName(4) = "Bot Bar 2"
MyDist(4) = -6
MyThickness(4) = 0.03
MyNumIntegrationPts(4) = 1
MyMatProp(4) = Name
MyMatAng(4) = 90

ret = SapModel.PropArea.SetShellLayer("A1", MyNumberLayers, MyLayerName, MyDist, MyThickness)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

📌 See Also

Reference

cPropArea Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LST65FE8AC0") Method

Assigns the layer parameters for shell-type area properties

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetShellLayer_1(
    string Name,
    ref int NumberLayers,
    ref string[] LayerName,
    ref double[] Dist,
    ref double[] Thickness,
    ref int[] MyType,
    ref int[] NumIntegrationPts,
    ref string[] MatProp,
    ref double[] MatAng,
    ref int[] S11Type,
    ref int[] S22Type,
    ref int[] S12Type
)
```

```
Function SetShellLayer_1 (
    Name As String,
    ByRef NumberLayers As Integer,
    ByRef LayerName As String(),
    ByRef Dist As Double(),
    ByRef Thickness As Double(),
    ByRef MyType As Integer(),
    ByRef NumIntegrationPts As Integer(),
    ByRef MatProp As String(),
    ByRef MatAng As Double(),
    ByRef S11Type As Integer(),
    ByRef S22Type As Integer(),
    ByRef S12Type As Integer()
) As Integer
```

```
Dim instance As cPropArea
Dim Name As String
Dim NumberLayers As Integer
Dim LayerName As String()
Dim Dist As Double()
Dim Thickness As Double()
Dim MyType As Integer()
Dim NumIntegrationPts As Integer()
Dim MatProp As String()
Dim MatAng As Double()
Dim S11Type As Integer()
```

```

Dim S22Type As Integer()
Dim S12Type As Integer()
Dim returnValue As Integer

returnValue = instance.SetShellLayer_1(Name,
NumberLayers, LayerName, Dist, Thickness,
MyType, NumIntegrationPts, MatProp,
MatAng, S11Type, S22Type, S12Type)

int SetShellLayer_1(
String^ Name,
int% NumberLayers,
array<String^>^% LayerName,
array<double>^% Dist,
array<double>^% Thickness,
array<int>^% MyType,
array<int>^% NumIntegrationPts,
array<String^>^% MatProp,
array<double>^% MatAng,
array<int>^% S11Type,
array<int>^% S22Type,
array<int>^% S12Type
)

abstract SetShellLayer_1 :
    Name : string *
    NumberLayers : int byref *
    LayerName : string[] byref *
    Dist : float[] byref *
    Thickness : float[] byref *
    MyType : int[] byref *
    NumIntegrationPts : int[] byref *
    MatProp : string[] byref *
    MatAng : float[] byref *
    S11Type : int[] byref *
    S22Type : int[] byref *
    S12Type : int[] byref -> int

```

Parameters

Name

Type:Â SystemString

The name of an existing shell-type area property that is specified to be a layered shell property.

NumberLayers

Type:Â SystemInt32

The number of layers in the area property.

LayerName

Type:Â SystemString

This is an array that includes the name of each layer.

Dist

Type:Â SystemDouble

This is an array that includes the distance from the area reference surface (area object joint location plus offsets) to the mid-height of the layer. [L]

Thickness

Type:Â SystemDouble

This is an array that includes the thickness of each layer. [L]

MyTypeType:Â SystemInt32

This is an array that includes 1, 2 or 3, indicating the layer type.

Value Type

1	Shell
2	Membrane
3	Plate

NumIntegrationPtsType:Â SystemInt32

The number of integration points in the thickness direction for the layer. The locations are determined by the program using standard Guass-quadrature rules.

MatPropType:Â SystemString

This is an array that includes the name of the material property for the layer.

MatAngType:Â SystemDouble

This is an array that includes the material angle for the layer. [deg]

S11TypeType:Â SystemInt32

This is an array that includes 0, 1 or 2, indicating the material component behavior.

Value S11Type

0	Inactive
1	Linear
2	Nonlinear

S22TypeType:Â SystemInt32

This is an array that includes 0, 1 or 2, indicating the material component behavior.

Value S22Type

0	Inactive
1	Linear
2	Nonlinear

S12TypeType:Â SystemInt32

This is an array that includes 0, 1 or 2, indicating the material component behavior.

Value S12Type

0	Inactive
1	Linear
2	Nonlinear

Return ValueType:Â Int32

Returns zero if the parameters are successfully assigned; otherwise it returns a

nonzero value.

The function returns an error if the specified area property is not a shell-type property specified to be a layered shell.

Remarks

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim MyNumberLayers As Integer
    Dim MyLayerName() As String
    Dim MyDist() As Double
    Dim MyThickness() As Double
    Dim MyType() As Integer
    Dim MyNumIntegrationPts() As Integer
    Dim MyMatProp() As String
    Dim MyMatAng() As Double
    Dim MyS11Type() As Integer
    Dim MyS22Type() As Integer
    Dim MyS12Type() As Integer

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new area property
    ret = SapModel.PropArea.SetWall("A1", eWallPropType.Specified, eShellType.Layered, "", 0)

    'add A615Gr60 rebar material
    ret = SapModel.PropMaterial.AddMaterial(Name, eMatType.Rebar, "United States", "ASTM A615")

    'set area property layer parameters
    MyNumberLayers = 5
    ReDim MyLayerName(MyNumberLayers - 1)
    ReDim MyDist(MyNumberLayers - 1)
    ReDim MyThickness(MyNumberLayers - 1)
    ReDim MyType(MyNumberLayers - 1)
    ReDim MyNumIntegrationPts(MyNumberLayers - 1)
    ReDim MyMatProp(MyNumberLayers - 1)
    ReDim MyMatAng(MyNumberLayers - 1)
    ReDim MyS11Type(MyNumberLayers - 1)
    ReDim MyS22Type(MyNumberLayers - 1)
    ReDim MyS12Type(MyNumberLayers - 1)
```

Introduction

```
MyLayerName(0) = "Concrete"
MyDist(0) = 0
MyThickness(0) = 16
MyType(0) = 1
MyNumIntegrationPts(0) = 2
MyMatProp(0) = "4000Psi"
MyMatAng(0) = 0
MyS11Type(0) = 1
MyS22Type(0) = 1
MyS12Type(0) = 1

MyLayerName(1) = "Top Bar 1"
MyDist(1) = 6
MyThickness(1) = 0.03
MyType(1) = 1
MyNumIntegrationPts(1) = 1
MyMatProp(1) = Name
MyMatAng(1) = 0
MyS11Type(1) = 1
MyS22Type(1) = 1
MyS12Type(1) = 1

MyLayerName(2) = "Top Bar 2"
MyDist(2) = 6
MyThickness(2) = 0.03
MyType(2) = 1
MyNumIntegrationPts(2) = 1
MyMatProp(2) = Name
MyMatAng(2) = 90
MyS11Type(2) = 1
MyS22Type(2) = 1
MyS12Type(2) = 1

MyLayerName(3) = "Bot Bar 1"
MyDist(3) = -6
MyThickness(3) = 0.03
MyType(3) = 1
MyNumIntegrationPts(3) = 1
MyMatProp(3) = Name
MyMatAng(3) = 0
MyS11Type(3) = 1
MyS22Type(3) = 1
MyS12Type(3) = 1

MyLayerName(4) = "Bot Bar 2"
MyDist(4) = -6
MyThickness(4) = 0.03
MyType(4) = 1
MyNumIntegrationPts(4) = 1
MyMatProp(4) = Name
MyMatAng(4) = 90
MyS11Type(4) = 1
MyS22Type(4) = 1
MyS12Type(4) = 1

ret = SapModel.PropArea.SetShellLayer_1("A1", MyNumberLayers, MyLayerName, MyDist, MyThickn

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
```

Introduction

```
SapModel = Nothing  
EtabsObject = Nothing  
End Sub
```

▲ See Also

Reference

cPropArea Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LSTD9C00619_0?cpp=::1");SetSh

Method

Assigns the layer parameters for shell-type area properties

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetShellLayer_2(
    string Name,
    ref int NumberLayers,
    ref string[] LayerName,
    ref double[] Dist,
    ref double[] Thickness,
    ref int[] MyType,
    ref int[] NumIntegrationPts,
    ref string[] MatProp,
    ref double[] MatAng,
    ref int[] MatBehavior,
    ref int[] S11Type,
    ref int[] S22Type,
    ref int[] S12Type
)
```

```
Function SetShellLayer_2 (
    Name As String,
    ByRef NumberLayers As Integer,
    ByRef LayerName As String(),
    ByRef Dist As Double(),
    ByRef Thickness As Double(),
    ByRef MyType As Integer(),
    ByRef NumIntegrationPts As Integer(),
    ByRef MatProp As String(),
    ByRef MatAng As Double(),
    ByRef MatBehavior As Integer(),
    ByRef S11Type As Integer(),
    ByRef S22Type As Integer(),
    ByRef S12Type As Integer()
) As Integer
```

```
Dim instance As cPropArea
Dim Name As String
Dim NumberLayers As Integer
Dim LayerName As String()
Dim Dist As Double()
Dim Thickness As Double()
Dim MyType As Integer()
Dim NumIntegrationPts As Integer()
Dim MatProp As String()
```

cPropAreaspan id="LSTD9C00619_0"AddLanguageSpecificTextSet("LSTD9C00619_0?cpp=::1");SetSh

```

Dim MatAng As Double()
Dim MatBehavior As Integer()
Dim S11Type As Integer()
Dim S22Type As Integer()
Dim S12Type As Integer()
Dim returnValue As Integer

returnValue = instance.SetShellLayer_2(Name,
NumberLayers, LayerName, Dist, Thickness,
MyType, NumIntegrationPts, MatProp,
MatAng, MatBehavior, S11Type, S22Type,
S12Type)

int SetShellLayer_2(
String^ Name,
int% NumberLayers,
array<String^>^% LayerName,
array<double>^% Dist,
array<double>^% Thickness,
array<int>^% MyType,
array<int>^% NumIntegrationPts,
array<String^>^% MatProp,
array<double>^% MatAng,
array<int>^% MatBehavior,
array<int>^% S11Type,
array<int>^% S22Type,
array<int>^% S12Type
)

abstract SetShellLayer_2 :
    Name : string *
    NumberLayers : int byref *
    LayerName : string[] byref *
    Dist : float[] byref *
    Thickness : float[] byref *
    MyType : int[] byref *
    NumIntegrationPts : int[] byref *
    MatProp : string[] byref *
    MatAng : float[] byref *
    MatBehavior : int[] byref *
    S11Type : int[] byref *
    S22Type : int[] byref *
    S12Type : int[] byref -> int

```

Parameters

Name

Type:Â SystemString

The name of an existing shell-type area property that is specified to be a layered shell property.

NumberLayers

Type:Â SystemInt32

The number of layers in the area property.

LayerName

Type:Â SystemString

This is an array that includes the name of each layer.

Dist

Type: SystemDouble

This is an array that includes the distance from the area reference surface (area object joint location plus offsets) to the mid-height of the layer. [L]

Thickness

Type: SystemDouble

This is an array that includes the thickness of each layer. [L]

MyType

Type: SystemInt32

This is an array that includes 1, 2 or 3, indicating the layer type.

Value Type

1 Shell

2 Membrane

3 Plate

NumIntegrationPts

Type: SystemInt32

The number of integration points in the thickness direction for the layer. The locations are determined by the program using standard Guass-quadrature rules.

MatProp

Type: SystemString

This is an array that includes the name of the material property for the layer.

MatAng

Type: SystemDouble

This is an array that includes the material angle for the layer. [deg]

MatBehavior

Type: SystemInt32

This is an array that includes 0 or 1 indicating the material behavior type of the layer.

Value Type

0 Directional

1 Coupled

S11Type

Type: SystemInt32

This is an array that includes 0, 1 or 2, indicating the material component behavior.

Value S11Type

0 Inactive

1 Linear

2 Nonlinear

S22Type

Type: SystemInt32

This is an array that includes 0, 1 or 2, indicating the material component behavior.

Value S22Type

0 Inactive

1 Linear

2 Nonlinear

S12TypeType: SystemInt32

This is an array that includes 0, 1 or 2, indicating the material component behavior.

Value S12Type

0	Inactive
1	Linear
2	Nonlinear

Return ValueType: Int32

Returns zero if the parameters are successfully assigned; otherwise it returns a nonzero value.

The function returns an error if the specified area property is not a shell-type property specified to be a layered shell.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim MyNumberLayers As Integer
    Dim MyLayerName() As String
    Dim MyDist() As Double
    Dim MyThickness() As Double
    Dim MyType() As Integer
    Dim MyNumIntegrationPts() As Integer
    Dim MyMatProp() As String
    Dim MyMatAng() As Double
    Dim MyMatBehavior() As Integer
    Dim MyS11Type() As Integer
    Dim MyS22Type() As Integer
    Dim MyS12Type() As Integer

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)
```

Introduction

```
'set new area property
  ret = SapModel.PropArea.SetWall("A1", eWallPropType.Specified, eShellType.Layered, "", 0)

'add A615Gr60 rebar material
  ret = SapModel.PropMaterial.AddMaterial(Name, eMatType.Rebar, "United States", "ASTM A615"

'set area property layer parameters
  MyNumberLayers = 5
  ReDim MyLayerName(MyNumberLayers - 1)
  ReDim MyDist(MyNumberLayers - 1)
  ReDim MyThickness(MyNumberLayers - 1)
  ReDim MyType(MyNumberLayers - 1)
  ReDim MyNumIntegrationPts(MyNumberLayers - 1)
  ReDim MyMatProp(MyNumberLayers - 1)
  ReDim MyMatAng(MyNumberLayers - 1)
  ReDim MyMatBehavior(MyNumberLayers - 1)
  ReDim MyS11Type(MyNumberLayers - 1)
  ReDim MyS22Type(MyNumberLayers - 1)
  ReDim MyS12Type(MyNumberLayers - 1)

  MyLayerName(0) = "Concrete"
  MyDist(0) = 0
  MyThickness(0) = 16
  MyType(0) = 1
  MyNumIntegrationPts(0) = 2
  MyMatProp(0) = "4000Psi"
  MyMatAng(0) = 0
  MyMatBehavior(0) = 1
  MyS11Type(0) = 1
  MyS22Type(0) = 1
  MyS12Type(0) = 1

  MyLayerName(1) = "Top Bar 1"
  MyDist(1) = 6
  MyThickness(1) = 0.03
  MyType(1) = 1
  MyNumIntegrationPts(1) = 1
  MyMatProp(1) = Name
  MyMatAng(1) = 0
  MyMatBehavior(1) = 1
  MyS11Type(1) = 1
  MyS22Type(1) = 1
  MyS12Type(1) = 1

  MyLayerName(2) = "Top Bar 2"
  MyDist(2) = 6
  MyThickness(2) = 0.03
  MyType(2) = 1
  MyNumIntegrationPts(2) = 1
  MyMatProp(2) = Name
  MyMatAng(2) = 90
  MyMatBehavior(2) = 1
  MyS11Type(2) = 1
  MyS22Type(2) = 1
  MyS12Type(2) = 1

  MyLayerName(3) = "Bot Bar 1"
  MyDist(3) = -6
  MyThickness(3) = 0.03
  MyType(3) = 1
  MyNumIntegrationPts(3) = 1
  MyMatProp(3) = Name
```

Introduction

```
MyMatAng(3) = 0
MyMatBehavior(3) = 1
MyS11Type(3) = 1
MyS22Type(3) = 1
MyS12Type(3) = 1

MyLayerName(4) = "Bot Bar 2"
MyDist(4) = -6
MyThickness(4) = 0.03
MyType(4) = 1
MyNumIntegrationPts(4) = 1
MyMatProp(4) = Name
MyMatAng(4) = 90
MyMatBehavior(4) = 1
MyS11Type(4) = 1
MyS22Type(4) = 1
MyS12Type(4) = 1

ret = SapModel.PropArea.SetShellLayer_2("A1", MyNumberLayers, MyLayerName, MyDist, MyThick

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

See Also

Reference

cPropArea Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LST7371AC9F_0?cpp=::2057");SetSlabMethod

Initializes a slab property.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetSlab(
    string Name,
    eSlabType SlabType,
    eShellType ShellType,
    string MatProp,
    double Thickness,
    int color = -1,
    string notes = "",
    string GUID = ""
)
```

```
Function SetSlab (
    Name As String,
    SlabType As eSlabType,
    ShellType As eShellType,
    MatProp As String,
    Thickness As Double,
    Optional color As Integer = -1,
    Optional notes As String = "",
    Optional GUID As String = ""
) As Integer
```

```
Dim instance As cPropArea
Dim Name As String
Dim SlabType As eSlabType
Dim ShellType As eShellType
Dim MatProp As String
Dim Thickness As Double
Dim color As Integer
Dim notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.SetSlab(Name, SlabType,
    ShellType, MatProp, Thickness, color,
    notes, GUID)
```

```
int SetSlab(
    String^ Name,
    eSlabType SlabType,
    eShellType ShellType,
```

cPropAreaspan id="LST7371AC9F_0"AddLanguageSpecificTextSet("LST7371AC9F_0?cpp=::2057");SetSlab

```

String^ MatProp,
double Thickness,
int color = -1,
String^ notes = L"",
String^ GUID = L""
)

abstract SetSlab :
    Name : string *
    SlabType : eSlabType *
    ShellType : eShellType *
    MatProp : string *
    Thickness : float *
    ?color : int *
    ?notes : string *
    ?GUID : string
(* Defaults:
    let _color = defaultArg color -1
    let _notes = defaultArg notes ""
    let _GUID = defaultArg GUID ""
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of a slab property. If this is an existing property, that property is modified; otherwise, a new property is added.

SlabType

Type:Â ETABSv1eSlabType

This is one of the items in the eSlabType enumeration.

If this item is Ribbed, use the GetSlabRibbed(String, Double, Double, Double, Double, Double, Int32) function to get additional parameters.

If this item is Waffle, use the GetSlabWaffle(String, Double, Double, Double, Double, Double, Double) function to get additional parameters.

ShellType

Type:Â ETABSv1eShellType

This is one of the items in the eShellType enumeration.

MatProp

Type:Â SystemString

The name of the material property for the area property.

This item does not apply when ShellType is Layered.

Thickness

Type:Â SystemDouble

The membrane thickness. [L]

This item does not apply when ShellType is Layered.

color (Optional)

Type:Â SystemInt32

The display color assigned to the property.

notes (Optional)

Type:Â SystemString

The notes, if any, assigned to the property.

GUID (Optional)

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the property.

Return Value

Type:Â Int32

Returns zero if the property is successfully initialized; otherwise it returns a nonzero value.

▀ Remarks

▀ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim SlabType As eSlabType
    Dim ShellType As eShellType
    Dim MatProp As String
    Dim Thickness As Double
    Dim Color As Integer
    Dim Notes As String
    Dim GUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new area property
    ret = SapModel.PropArea.SetSlab("MyShellProp1A", eSlabType.Drop, eShellType.ShellThin, "40")

    'get area property data
    ret = SapModel.PropArea.GetSlab("MyShellProp1A", SlabType, ShellType, MatProp, Thickness, Color, Notes, GUID)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▀ See Also

Reference

cPropArea Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LST41A3D52E_0") Method

Initializes a ribbed slab section.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetSlabRibbed(
    string Name,
    double OverallDepth,
    double SlabThickness,
    double StemWidthTop,
    double StemWidthBot,
    double RibSpacing,
    int RibsParallelTo
)
```

```
Function SetSlabRibbed (
    Name As String,
    OverallDepth As Double,
    SlabThickness As Double,
    StemWidthTop As Double,
    StemWidthBot As Double,
    RibSpacing As Double,
    RibsParallelTo As Integer
) As Integer
```

```
Dim instance As cPropArea
Dim Name As String
Dim OverallDepth As Double
Dim SlabThickness As Double
Dim StemWidthTop As Double
Dim StemWidthBot As Double
Dim RibSpacing As Double
Dim RibsParallelTo As Integer
Dim returnValue As Integer
```

```
returnValue = instance.SetSlabRibbed(Name,
    OverallDepth, SlabThickness, StemWidthTop,
    StemWidthBot, RibSpacing, RibsParallelTo)
```

```
int SetSlabRibbed(
    String^ Name,
    double OverallDepth,
    double SlabThickness,
    double StemWidthTop,
    double StemWidthBot,
    double RibSpacing,
```

cPropAreaspan id="LST41A3D52E_0"AddLanguageSpecificTextSet("LST41A3D52E_0?cpp=::2861");SetSl

```

int RibsParallelTo
)

abstract SetSlabRibbed :
    Name : string *
    OverallDepth : float *
    SlabThickness : float *
    StemWidthTop : float *
    StemWidthBot : float *
    RibSpacing : float *
    RibsParallelTo : int -> int

```

Parameters

Name

Type:Â SystemString

The name of an existing ribbed slab property.

OverallDepth

Type:Â SystemDouble

Overall Depth

SlabThickness

Type:Â SystemDouble

Slab Thickness

StemWidthTop

Type:Â SystemDouble

Stem Width at Top

StemWidthBot

Type:Â SystemDouble

Stem Width at Bottom

RibSpacing

Type:Â SystemDouble

Rib Spacing (Perpendicular to Rib Direction)

RibsParallelTo

Type:Â SystemInt32

This is the Local Axis that the Rib Direction is Parallel to

This value is 1 for the Local 1 Axis, and 2 for the Local 2 Axis.

Return Value

Type:Â Int32

The function returns zero if the property data is successfully initialized; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim SlabType As eSlabType
    Dim ShellType As eShellType
    Dim MatProp As String

```

Introduction

```
Dim Thickness As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim OverallDepth As Double
Dim SlabThickness As Double
Dim StemWidthTop As Double
Dim StemWidthBot As Double
Dim RibSpacing As Double
Dim RibsParallelTo As Integer

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new area property
ret = SapModel.PropArea.SetSlab("MyShellProp1A", eSlabType.Ribbed, eShellType.ShellThin, '

'set new area property
ret = SapModel.PropArea.SetSlabRibbed(Name:="MyShellProp1A", OverallDepth:=11.1, SlabThick

'get area property data
ret = SapModel.PropArea.GetSlab("MyShellProp1A", SlabType, ShellType, MatProp, Thickness,

'get area property data
ret = SapModel.PropArea.GetSlabRibbed("MyShellProp1A", OverallDepth, SlabThickness, StemW

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPropArea Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LST942D09FB_0") Method

Initializes a waffle slab section.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetSlabWaffle(
    string Name,
    double OverallDepth,
    double SlabThickness,
    double StemWidthTop,
    double StemWidthBot,
    double RibSpacingDir1,
    double RibSpacingDir2
)
```

```
Function SetSlabWaffle (
    Name As String,
    OverallDepth As Double,
    SlabThickness As Double,
    StemWidthTop As Double,
    StemWidthBot As Double,
    RibSpacingDir1 As Double,
    RibSpacingDir2 As Double
) As Integer
```

```
Dim instance As cPropArea
Dim Name As String
Dim OverallDepth As Double
Dim SlabThickness As Double
Dim StemWidthTop As Double
Dim StemWidthBot As Double
Dim RibSpacingDir1 As Double
Dim RibSpacingDir2 As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetSlabWaffle(Name,
    OverallDepth, SlabThickness, StemWidthTop,
    StemWidthBot, RibSpacingDir1, RibSpacingDir2)
```

```
int SetSlabWaffle(
    String^ Name,
    double OverallDepth,
    double SlabThickness,
    double StemWidthTop,
    double StemWidthBot,
    double RibSpacingDir1,
```

cPropAreaspan id="LST942D09FB_0"AddLanguageSpecificTextSet("LST942D09FB_0?cpp=::2064");SetSlabWaffle

```
double RibSpacingDir2
)

abstract SetSlabWaffle :
    Name : string *
    OverallDepth : float *
    SlabThickness : float *
    StemWidthTop : float *
    StemWidthBot : float *
    RibSpacingDir1 : float *
    RibSpacingDir2 : float -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing waffle slab property.

OverallDepth

Type:Â SystemDouble

Overall Depth

SlabThickness

Type:Â SystemDouble

Slab Thickness

StemWidthTop

Type:Â SystemDouble

Stem Width at Top

StemWidthBot

Type:Â SystemDouble

Stem Width at Bottom

RibSpacingDir1

Type:Â SystemDouble

Spacing of Ribs that are Parallel to Slab 1-Axis

RibSpacingDir2

Type:Â SystemDouble

Spacing of Ribs that are Parallel to Slab 2-Axis

Return Value

Type:Â Int32

The function returns zero if the property data is successfully initialized; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim SlabType As eSlabType
    Dim ShellType As eShellType
    Dim MatProp As String
    Dim Thickness As Double
    Dim Color As Integer
```

Introduction

```
Dim Notes As String
Dim GUID As String
Dim OverallDepth As Double
Dim SlabThickness As Double
Dim StemWidthTop As Double
Dim StemWidthBot As Double
Dim RibSpacing1 As Double
Dim RibSpacing2 As Double

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new area property
ret = SapModel.PropArea.SetSlab("MyShellProp1A", eSlabType.Waffle, eShellType.ShellThin, '

'set new area property
ret = SapModel.PropArea.SetSlabWaffle(Name:="MyShellProp1A", OverallDepth:=11.1, SlabThick

'get area property data
ret = SapModel.PropArea.GetSlab("MyShellProp1A", SlabType, ShellType, MatProp, Thickness,

'get area property data
ret = SapModel.PropArea.GetSlabWaffle("MyShellProp1A", OverallDepth, SlabThickness, StemW

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

[cPropArea Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LSTD953E8C2_0?cpp=::2867");SetWall

Method

Initializes property data for a wall section.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetWall(
    string Name,
    eWallPropType WallPropType,
    eShellType ShellType,
    string MatProp,
    double Thickness,
    int color = -1,
    string notes = "",
    string GUID = ""
)
```

```
Function SetWall (
    Name As String,
    WallPropType As eWallPropType,
    ShellType As eShellType,
    MatProp As String,
    Thickness As Double,
    Optional color As Integer = -1,
    Optional notes As String = "",
    Optional GUID As String = ""
) As Integer
```

```
Dim instance As cPropArea
Dim Name As String
Dim WallPropType As eWallPropType
Dim ShellType As eShellType
Dim MatProp As String
Dim Thickness As Double
Dim color As Integer
Dim notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.SetWall(Name, WallPropType,
    ShellType, MatProp, Thickness, color,
    notes, GUID)
```

```
int SetWall(
    String^ Name,
    eWallPropType WallPropType,
    eShellType ShellType,
```

cPropAreaid="LSTD953E8C2_0"AddLanguageSpecificTextSet("LSTD953E8C2_0?cpp=::2867");SetWall

```

String^ MatProp,
double Thickness,
int color = -1,
String^ notes = L"",
String^ GUID = L""
)

abstract SetWall :
    Name : string *
    WallPropType : eWallPropType *
    ShellType : eShellType *
    MatProp : string *
    Thickness : float *
    ?color : int *
    ?notes : string *
    ?GUID : string
(* Defaults:
    let _color = defaultArg color -1
    let _notes = defaultArg notes ""
    let _GUID = defaultArg GUID ""
*)
-> int

```

Parameters

Name

Type:Â [SystemString](#)
The name of an existing wall property.

WallPropType

Type:Â [ETABSv1eWallPropType](#)
This is one of the items in the [eWallPropType](#) enumeration.

If this item is [AutoSelectList](#), use the [GetWallAutoSelectList\(String,String,String\)](#) function to get additional parameters.

ShellType

Type:Â [ETABSv1eShellType](#)
This is one of the items in the [eShellType](#) enumeration.

MatProp

Type:Â [SystemString](#)
The name of the material property for the area property.

This item does not apply when ShellType is [Layered](#).

Thickness

Type:Â [SystemDouble](#)
The membrane thickness. [L]

This item does not apply when ShellType is [Layered](#).

color (Optional)

Type:Â [SystemInt32](#)
The display color assigned to the property.

notes (Optional)

Type:Â [SystemString](#)
The notes, if any, assigned to the property.

GUID (Optional)

Type: SystemString

The GUID (global unique identifier), if any, assigned to the property.

Return Value

Type: Int32

The function returns zero if the property data is successfully initialized; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim WallPropType As eWallPropType
    Dim ShellType As eShellType
    Dim MatProp As String
    Dim Thickness As Double
    Dim Color As Integer
    Dim Notes As String
    Dim GUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new area property
    ret = SapModel.PropArea.SetWall("MyShellProp1A", eWallPropType.Specified, eShellType.ShellType)

    'get area property data
    ret = SapModel.PropArea.GetWall("MyShellProp1A", WallPropType, ShellType, MatProp, Thickness)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPropArea Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropAreaAddLanguageSpecificTextSet("LSTFEB33746_0?cpp=::|287.1");SetWallAutoSelectList

Method

Initializes property data for an auto select list wall section.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetWallAutoSelectList(
    string Name,
    string[] AutoSelectList,
    string StartingProperty = "Median"
)
```

```
Function SetWallAutoSelectList (
    Name As String,
    AutoSelectList As String(),
    StartingProperty As String = "Median"
) As Integer
```

```
Dim instance As cPropArea
Dim Name As String
Dim AutoSelectList As String()
Dim StartingProperty As String
Dim returnValue As Integer
```

```
returnValue = instance.SetWallAutoSelectList(Name,
    AutoSelectList, StartingProperty)
```

```
int SetWallAutoSelectList(
    String^ Name,
    array<String^>^ AutoSelectList,
    String^ StartingProperty = L"Median"
)
```

```
abstract SetWallAutoSelectList :
    Name : string *
    AutoSelectList : string[] *
    ?StartingProperty : string
(* Defaults:
    let _StartingProperty = defaultArg StartingProperty "Median"
*)
-> int
```

Parameters

Name

cPropAreaAddLanguageSpecificTextSet("LSTFEB33746_0?cpp=::|287.1");SetWallAutoSelectList

Type:Â SystemString

The name of an existing auto select list wall property.

AutoSelectList

Type:Â SystemString

This is an array of the names of the wall properties included in the auto select list.

StartingProperty (Optional)

Type:Â SystemString

This is Median or the name of a wall property in the AutoSelectList array. It is the starting section for the auto select list.

Median indicates the Median Property by Thickness

Return Value

Type:Â Int32

The function returns zero if the property data is successfully initialized; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim WallPropType As eWallPropType
    Dim ShellType As eShellType
    Dim MatProp As String
    Dim Thickness As Double
    Dim Color As Integer
    Dim Notes As String
    Dim GUID As String
    Dim AutoSelectList As String()
    Dim StartingProp As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new area property
    ret = SapModel.PropArea.SetWall("MyShellProp1A", eWallPropType.Specified, eShellType.Shell1)

    'set new area property
    ret = SapModel.PropArea.SetWall("MyShellProp1B", eWallPropType.Specified, eShellType.Shell1)
```

Introduction

```
'set new area property
    ret = SapModel.PropArea.SetWall("MyShellProp1C", eWallPropType.AutoSelectList, eShellType.

'set new area property
    ret = SapModel.PropArea.SetWall("MyShellProp1D", eWallPropType.AutoSelectList, eShellType.

'set new area property
    Dim tmpList As String()
    ReDim tmpList(2)
    tmpList(0) = "MyShellProp1A"
    tmpList(1) = "MyShellProp1B"
    tmpList(2) = "MyShellProp1C"
    ret = SapModel.PropArea.SetWallAutoSelectList("MyShellProp1D", tmpList, "MyShellProp1B")

'get area property data
    ret = SapModel.PropArea.GetWall("MyShellProp1D", WallPropType, ShellType, MatProp, Thickne

'get area property data
    ret = SapModel.PropArea.GetWallAutoSelectList("MyShellProp1D", AutoSelectList, StartingPro

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

🔗 See Also

Reference

[cPropArea Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropAreaSpring Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cPropAreaSpring
```

```
Public Interface cPropAreaSpring
```

```
Dim instance As cPropAreaSpring
```

```
public interface class cPropAreaSpring
```

```
type cPropAreaSpring = interface end
```

The cPropAreaSpring type exposes the following members.

▲ Methods

▲	Name	Description
◆	ChangeName	Changes the name of a defined area spring property
◆	Delete	Deletes the specified area spring property
◆	GetAreaSpringProp	Retrieves an existing named area spring property
◆	GetNameList	Retrieves the names of all defined area spring property
◆	SetAreaSpringProp	Creates a new named area spring property, or modifies an existing named area spring property

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropAreaSpring Methods

The [cPropAreaSpring](#) type exposes the following members.

Methods

	Name	Description
✚	ChangeName	Changes the name of a defined area spring property
✚	Delete	Deletes the specified area spring property
✚	GetAreaSpringProp	Retrieves an existing named area spring property
✚	GetNameList	Retrieves the names of all defined area spring property
✚	SetAreaSpringProp	Creates a new named area spring property, or modifies an existing named area spring property

[Top](#)

See Also

Reference

[cPropAreaSpring Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropAreaSpringAddLanguageSpecificTextSet("LST885D47BF_0") Method

Changes the name of a defined area spring property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int ChangeName(
    string Name,
    string NewName
)
```

```
Function ChangeName (
    Name As String,
    NewName As String
) As Integer
```

```
Dim instance As cPropAreaSpring
Dim Name As String
Dim NewName As String
Dim returnValue As Integer
```

```
returnValue = instance.ChangeName(Name,
    NewName)
```

```
int ChangeName(
    String^ Name,
    String^ NewName
)
```

```
abstract ChangeName :
    Name : string *
    NewName : string -> int
```

Parameters

Name

Type: SystemString

The existing name of a defined area spring property

NewName

Type: SystemString

The new name for the area spring property

Return Value

Type: Int32

Returns zero if the new name is successfully applied, otherwise it returns a nonzero value.

▀ **Remarks**

▀ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new spring property
    ret = SapModel.PropAreaSpring.SetAreaSpringProp("mySpringProp1", 0, 0, 100, 0)

    'change property name
    ret = SapModel.PropAreaSpring.ChangeName("mySpringProp1", "prop1")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▀ **See Also**

Reference

cPropAreaSpring Interface

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CSI API ETABS v1

cPropAreaSpringAddLanguageSpecificTextSet("LSTBC2F20AD_02878") Method

Deletes the specified area spring property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int Delete(
    string Name
)
```

```
Function Delete (
    Name As String
) As Integer
```

```
Dim instance As cPropAreaSpring
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.Delete(Name)
```

```
int Delete(
    String^ Name
)
```

```
abstract Delete :
    Name : string -> int
```

Parameters

Name

Type: SystemString

The name of an existing area spring property

Return Value

Type: Int32

Returns zero if the area spring property is successfully deleted, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
```

cPropAreaSpringid="LSTBC2F20AD_02878"AddLanguageSpecificTextSet("LSTBC2F20AD_02878::|nu=.

Introduction

```
Dim SapModel As cSapModel
Dim EtabsObject As cOAPI
Dim ret As Integer = -1

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'define new spring property
ret = SapModel.PropAreaSpring.SetAreaSpringProp("mySpringProp1", 0, 0, 100, 0)

'delete property
ret = SapModel.PropAreaSpring.Delete("mySpringProp1")

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

📌 See Also

Reference

[cPropAreaSpring Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropAreaSpringAddLanguageSpecificTextSet("LST99955FCF_0") Method

Retrieves an existing named area spring property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetAreaSpringProp(
    string Name,
    ref double U1,
    ref double U2,
    ref double U3,
    ref int NonlinearOption3,
    ref int SpringOption,
    ref string SoilProfile,
    ref double EndLengthRatio,
    ref double Period,
    ref int color,
    ref string notes,
    ref string iGUID
)
```

```
Function GetAreaSpringProp (
    Name As String,
    ByRef U1 As Double,
    ByRef U2 As Double,
    ByRef U3 As Double,
    ByRef NonlinearOption3 As Integer,
    ByRef SpringOption As Integer,
    ByRef SoilProfile As String,
    ByRef EndLengthRatio As Double,
    ByRef Period As Double,
    ByRef color As Integer,
    ByRef notes As String,
    ByRef iGUID As String
) As Integer
```

```
Dim instance As cPropAreaSpring
Dim Name As String
Dim U1 As Double
Dim U2 As Double
Dim U3 As Double
Dim NonlinearOption3 As Integer
Dim SpringOption As Integer
Dim SoilProfile As String
Dim EndLengthRatio As Double
Dim Period As Double
Dim color As Integer
```

cPropAreaSpringid="LST99955FCF_0"AddLanguageSpecificTextSet("LST99955FCF_0?2880:|nu=.")

Introduction

```
Dim notes As String
Dim iGUID As String
Dim returnValue As Integer

returnValue = instance.GetAreaSpringProp(Name,
U1, U2, U3, NonlinearOption3, SpringOption,
SoilProfile, EndLengthRatio, Period,
color, notes, iGUID)

int GetAreaSpringProp(
String^ Name,
double% U1,
double% U2,
double% U3,
int% NonlinearOption3,
int% SpringOption,
String^% SoilProfile,
double% EndLengthRatio,
double% Period,
int% color,
String^% notes,
String^% iGUID
)

abstract GetAreaSpringProp :
    Name : string *
    U1 : float byref *
    U2 : float byref *
    U3 : float byref *
    NonlinearOption3 : int byref *
    SpringOption : int byref *
    SoilProfile : string byref *
    EndLengthRatio : float byref *
    Period : float byref *
    color : int byref *
    notes : string byref *
    iGUID : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of the area spring property.

U1

Type:Â SystemDouble

The spring stiffness per unit area in the local 1 direction [F/L3]

U2

Type:Â SystemDouble

The spring stiffness per unit area in the local 2 direction [F/L3]

U3

Type:Â SystemDouble

The spring stiffness per unit area in the local 3 direction [F/L3]

NonlinearOption3

Type:Â SystemInt32

The nonlinear option for the local 3 direction, this is one of the following values:

- ◊ 0 = (None) Linear - Resists Tension and Compression
- ◊ 1 = Resists Compression only

◇ 2 = Resists Tension only

SpringOption

Type:Â SystemInt32

This argument is for a future release, however a placeholder of type Integer is required

SoilProfile

Type:Â SystemString

This argument is for a future release, however a placeholder of type String is required

EndLengthRatio

Type:Â SystemDouble

This argument is for a future release, however a placeholder of type Double is required

Period

Type:Â SystemDouble

This argument is for a future release, however a placeholder of type Double is required

color

Type:Â SystemInt32

The display color for the property specified as an integer

notes

Type:Â SystemString

The notes, if any, assigned to the property

iGUID

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the property

Return Value

Type:Â Int32

Returns zero if the property is successfully set, otherwise returns nonzero

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
```

Introduction

```
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'define new spring property
ret = SapModel.PropAreaSpring.SetAreaSpringProp("mySpringProp1", 0, 0, 100, 0)

'get property
Dim U1 As Double
Dim U2 As Double
Dim U3 As Double
Dim NonlinearOption2 As Integer
Dim SpringOption As Integer
Dim SoilProfile As String
Dim EndLengthRatio As Double
Dim Period As Double
Dim color As Integer
Dim notes As String
Dim iGuid As String
ret = SapModel.PropAreaSpring.GetAreaSpringProp("mySpringProp1", U1, U2, U3, NonlinearOption3, SpringOption, SoilProfile, EndLengthRatio, Period, color, notes, iGuid)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

[cPropAreaSpring Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropAreaSpringAddLanguageSpecificTextSet("LSTA5 Method

Retrieves the names of all defined area spring property

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetNameList(
    ref int NumberNames,
    ref string[] MyName
)
```

```
Function GetNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cPropAreaSpring
Dim NumberNames As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetNameList(NumberNames,
    MyName)
```

```
int GetNameList(
    int% NumberNames,
    array<String^>^% MyName
)
```

```
abstract GetNameList :
    NumberNames : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberNames

Type: SystemInt32

The number of area spring property defined in the program

MyName

Type: SystemString

This is a one-dimensional array of area spring property names. The MyName array is created as a dynamic, zero-based, array by the API user:

```
Dim MyName() as String
```


The array is dimensioned to (NumberNames + 1) inside the program, filled with the names, and returned to the API user.

Return Value

Type: Int32

Returns zero if the names are successfully retrieved, otherwise it returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new spring property
    ret = SapModel.PropAreaSpring.SetAreaSpringProp("mySpringProp1", 0, 0, 100, 0)

    'get name list of properties
    Dim NumberNames as Integer
    Dim MyName() as String
    ret = SapModel.PropAreaSpring.GetNameList(NumberNames, MyName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cPropAreaSpring Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropAreaSpringAddLanguageSpecificTextSet("LST49 Method

Creates a new named area spring property, or modifies an existing named area spring property

Namespace: Â ETABSV1

Assembly: Â ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetAreaSpringProp(
    string Name,
    double U1,
    double U2,
    double U3,
    int NonlinearOption3,
    int SpringOption = 1,
    string SoilProfile = "",
    double EndLengthRatio = 0,
    double Period = 0,
    int color = 0,
    string notes = "",
    string iGUID = ""
)
```

```
Function SetAreaSpringProp (
    Name As String,
    U1 As Double,
    U2 As Double,
    U3 As Double,
    NonlinearOption3 As Integer,
    SpringOption As Integer = 1,
    SoilProfile As String = "",
    EndLengthRatio As Double = 0,
    Period As Double = 0,
    Optional color As Integer = 0,
    Optional notes As String = "",
    Optional iGUID As String = ""
) As Integer
```

```
Dim instance As cPropAreaSpring
Dim Name As String
Dim U1 As Double
Dim U2 As Double
Dim U3 As Double
Dim NonlinearOption3 As Integer
Dim SpringOption As Integer
Dim SoilProfile As String
Dim EndLengthRatio As Double
Dim Period As Double
```

cPropAreaSpringspan id="LST49423610_0"AddLanguageSpecificTextSet("LST49423610_0?cpe=2887nu=.");;

Introduction

```
Dim color As Integer
Dim notes As String
Dim iGUID As String
Dim returnValue As Integer

returnValue = instance.SetAreaSpringProp(Name,
U1, U2, U3, NonlinearOption3, SpringOption,
SoilProfile, EndLengthRatio, Period,
color, notes, iGUID)

int SetAreaSpringProp(
String^ Name,
double U1,
double U2,
double U3,
int NonlinearOption3,
int SpringOption = 1,
String^ SoilProfile = L"",
double EndLengthRatio = 0,
double Period = 0,
int color = 0,
String^ notes = L"",
String^ iGUID = L""
)

abstract SetAreaSpringProp :
    Name : string *
    U1 : float *
    U2 : float *
    U3 : float *
    NonlinearOption3 : int *
    ?SpringOption : int *
    ?SoilProfile : string *
    ?EndLengthRatio : float *
    ?Period : float *
    ?color : int *
    ?notes : string *
    ?iGUID : string
(* Defaults:
    let _SpringOption = defaultArg SpringOption 1
    let _SoilProfile = defaultArg SoilProfile ""
    let _EndLengthRatio = defaultArg EndLengthRatio 0
    let _Period = defaultArg Period 0
    let _color = defaultArg color 0
    let _notes = defaultArg notes ""
    let _iGUID = defaultArg iGUID ""
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of the area spring property. If this is an existing property, that property is modified; otherwise, a new property is added.

U1

Type:Â SystemDouble

The spring stiffness per unit area in the local 1 direction [F/L3]

U2

Type:Â SystemDouble
 The spring stiffness per unit area in the local 2 direction [F/L3]

U3
 Type:Â SystemDouble
 The spring stiffness per unit area in the local 3 direction [F/L3]

NonlinearOption3
 Type:Â SystemInt32
 The nonlinear option for the local 3 direction, this is one of the following values:
 ◊ 0 = (None) Linear - Resists Tension and Compression
 ◊ 1 = Resists Compression only
 ◊ 2 = Resists Tension only

SpringOption (Optional)
 Type:Â SystemInt32
 This optional argument is for a future release

SoilProfile (Optional)
 Type:Â SystemString
 This optional argument is for a future release

EndLengthRatio (Optional)
 Type:Â SystemDouble
 This optional argument is for a future release

Period (Optional)
 Type:Â SystemDouble
 This optional argument is for a future release

color (Optional)
 Type:Â SystemInt32
 The display color for the property specified as an integer

notes (Optional)
 Type:Â SystemString
 The notes, if any, assigned to the property

iGUID (Optional)
 Type:Â SystemString
 The GUID (global unique identifier), if any, assigned to the property

Return Value

Type:Â Int32
 Returns zero if the property is successfully set, otherwise returns nonzero

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()
```

Introduction

```
'create SapModel object
  SapModel = EtabsObject.SapModel

'initialize model
  ret = SapModel.InitializeNewModel()

'create steel deck template model
  ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'define new spring property
  ret = SapModel.PropAreaSpring.SetAreaSpringProp("mySpringProp1", 0, 0, 100, 0)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

[cPropAreaSpring Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropFrame Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cPropFrame
```

```
Public Interface cPropFrame
```

```
Dim instance As cPropFrame
```

```
public interface class cPropFrame
```

```
type cPropFrame = interface end
```

The cPropFrame type exposes the following members.

▲ Properties

Â Name

Description

 [SDShape](#) Gives access to API calls related to Section Designer sections.

[Top](#)

▲ Methods

Â Name

 ChangeName	Changes the name of an existing frame section property.
 Count	Retrieves the total number of defined frame section properties
 Delete	Deletes a specified frame section property.
 GetAllFrameProperties	Retrieves select data for all frame properties in the model
 GetAllFrameProperties_2	Retrieves select data for all frame properties in the model, super
 GetAngle	Retrieves frame section property data for an angle-type frame s
 GetAutoSelectSteel	Retrieves frame section property data for a steel auto select list
 GetChannel	Retrieves frame section property data for an channel-type frame
 GetCircle	Retrieves frame section property data for a circular frame secti
 GetConcreteL	Retrieves frame section property data for a concrete L frame se
 GetConcreteTee	Retrieves frame section property data for a concrete tee frame s
 GetCoverPlatedI	retrieves frame section property data for a cover plated I-type f
 GetDbAngle	Retrieves frame section property data for a double angle-type fr
 GetDbChannel	Retrieves frame section property data for a double channel-type
 GetGeneral	Retrieves frame section property data for a general frame secti
 GetISection	Retrieves frame section property data for an I-type frame section

⇒ <u>GetMaterial</u>	Retrieves the material property of a defined frame section
⇒ <u>GetModifiers</u>	Retrieves the modifier assignments for a frame section property
⇒ <u>GetNameInPropFile</u>	Retrieves the names of the section property file from which an i
⇒ <u>GetNameList</u>	Retrieves the names of all defined frame section properties of th
⇒ <u>GetNonPrismatic</u>	Assigns data to a nonprismatic frame section property.
⇒ <u>GetPipe</u>	Retrieves frame section property data for a pipe-type frame sec
⇒ <u>GetPlate</u>	Retrieves frame section property data for a plate frame section.
⇒ <u>GetPropFileNameList</u>	Retrieves the names of all defined frame section properties of a
⇒ <u>GetRebarBeam</u>	Retrieves beam rebar data for frame sections.
⇒ <u>GetRebarColumn</u>	Retrieves column rebar data for frame sections.
⇒ <u>GetRebarColumn_1</u>	
⇒ <u>GetRectangle</u>	Retrieves frame section property data for a rectangular frame s
⇒ <u>GetRod</u>	Retrieves frame section property data for a rod frame section.
⇒ <u>GetSDSection</u>	Retrieves section property data for a section designer section
⇒ <u>GetSectProps</u>	
⇒ <u>GetSteelAngle</u>	Retrieves frame section property data for a steel angle frame se
⇒ <u>GetSteelTee</u>	Retrieves frame section property data for a steel tee frame sect
⇒ <u>GetTee</u>	Retrieves frame section property data for a tee-type frame secti
⇒ <u>GetTrapezoidal</u>	
⇒ <u>GetTube</u>	Retrieves frame section property data for a tube-type frame sec
⇒ <u>GetTypeOAPI</u>	Retrieves the property type for the specified frame section prop
⇒ <u>GetTypeRebar</u>	Retrieves the rebar design type for the specified frame section p
⇒ <u>ImportProp</u>	Imports a frame section property from a property file.
⇒ <u>SetAngle</u>	Initializes an angle-type frame section property. If this function
⇒ <u>SetAutoSelectSteel</u>	Assigns frame section properties to an auto select list. If this fun
⇒ <u>SetChannel</u>	Initializes an channel-type frame section property. If this functio
⇒ <u>SetCircle</u>	initializes a solid circular frame section property.If this function
⇒ <u>SetConcreteL</u>	Initializes a concrete L frame section property. If this function i
⇒ <u>SetConcreteTee</u>	Initializes a concrete tee frame section property. If this function
⇒ <u>SetCoverPlatedI</u>	Initializes a cover plated I-type frame section property. If this fu
⇒ <u>SetDblAngle</u>	Initializes a double angle-type frame section property. If this fun
⇒ <u>SetDblChannel</u>	Initializes a double channel-type frame section property. If this
⇒ <u>SetGeneral</u>	Initializes a general frame section property. If this function is ca
⇒ <u>SetISection</u>	Initializes an I-type frame section property. If this function is ca
⇒ <u>SetMaterial</u>	Sets the material property of a defined frame section
⇒ <u>SetModifiers</u>	Assigns property modifiers to a frame section property. The def
⇒ <u>SetNonPrismatic</u>	Assigns data to a nonprismatic frame section property.
⇒ <u>SetPipe</u>	Initializes a pipe-type frame section property. If this function is
⇒ <u>SetPlate</u>	Initializes a solid plate frame section property. If this function is
⇒ <u>SetRebarBeam</u>	Assigns beam rebar data to frame sections.
⇒ <u>SetRebarColumn</u>	Assigns column rebar data to frame sections.
⇒ <u>SetRectangle</u>	Initializes a solid rectangular frame section property. If this fun

Introduction

- ⇒ [SetRod](#) initializes a solid rod frame section property. If this function is c
- ⇒ [SetSDSection](#)
- ⇒ [SetSteelAngle](#) Initializes a steel angle frame section property. If this function i
- ⇒ [SetSteelTee](#) Initializes a steel tee frame section property. If this function is c
- ⇒ [SetTee](#) Initializes a tee-type frame section property. If this function is c
- ⇒ [SetTrapezoidal](#)
- ⇒ [SetTube](#) Initializes a tube-type frame section property. If this function is

[Top](#)

🔍 See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropFrame Properties

The [cPropFrame](#) type exposes the following members.

Properties

Â	Name	Description
	SDShape	Gives access to API calls related to Section Designer sections.

[Top](#)

See Also

Reference

[cPropFrame Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST655087") Property

Gives access to API calls related to Section Designer sections.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cPropFrameSDShape SDShape { get; }
```

```
ReadOnly Property SDShape As cPropFrameSDShape
Get
```

```
Dim instance As cPropFrame
Dim value As cPropFrameSDShape
```

```
value = instance.SDShape
```

```
property cPropFrameSDShape^ SDShape {
cPropFrameSDShape^ get ();
}
```

```
abstract SDShape : cPropFrameSDShape with get
```

Property Value

Type: [cPropFrameSDShape](#)

Return Value

Type: [cPropFrameSDShape](#)

▲ Remarks

▲ See Also

Reference

[cPropFrame Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropFrame Methods

The cPropFrame type exposes the following members.

Methods

Á	Name	
⇒	<u>ChangeName</u>	Changes the name of an existing frame section property.
⇒	<u>Count</u>	Retrieves the total number of defined frame section properties i
⇒	<u>Delete</u>	Deletes a specified frame section property.
⇒	<u>GetAllFrameProperties</u>	Retrieves select data for all frame properties in the model
⇒	<u>GetAllFrameProperties_2</u>	Retrieves select data for all frame properties in the model, supe
⇒	<u>GetAngle</u>	Retrieves frame section property data for an angle-type frame s
⇒	<u>GetAutoSelectSteel</u>	Retrieves frame section property data for a steel auto select list
⇒	<u>GetChannel</u>	Retrieves frame section property data for an channel-type frame
⇒	<u>GetCircle</u>	Retrieves frame section property data for a circular frame secti
⇒	<u>GetConcreteL</u>	Retrieves frame section property data for a concrete L frame se
⇒	<u>GetConcreteTee</u>	Retrieves frame section property data for a concrete tee frame :
⇒	<u>GetCoverPlatedI</u>	retrieves frame section property data for a cover plated I-type f
⇒	<u>GetDblAngle</u>	Retrieves frame section property data for a double angle-type fr
⇒	<u>GetDblChannel</u>	Retrieves frame section property data for a double channel-type
⇒	<u>GetGeneral</u>	Retrieves frame section property data for a general frame secti
⇒	<u>GetISection</u>	Retrieves frame section property data for an I-type frame section
⇒	<u>GetMaterial</u>	Retrieves the material property of a defined frame section
⇒	<u>GetModifiers</u>	Retrieves the modifier assignments for a frame section property
⇒	<u>GetNameInPropFile</u>	Retrieves the names of the section property file from which an i
⇒	<u>GetNameList</u>	Retrieves the names of all defined frame section properties of th
⇒	<u>GetNonPrismatic</u>	Assigns data to a nonprismatic frame section property.
⇒	<u>GetPipe</u>	Retrieves frame section property data for a pipe-type frame sec
⇒	<u>GetPlate</u>	Retrieves frame section property data for a plate frame section.
⇒	<u>GetPropFileNameList</u>	Retrieves the names of all defined frame section properties of a
⇒	<u>GetRebarBeam</u>	Retrieves beam rebar data for frame sections.
⇒	<u>GetRebarColumn</u>	Retrieves column rebar data for frame sections.
⇒	<u>GetRebarColumn_1</u>	
⇒	<u>GetRectangle</u>	Retrieves frame section property data for a rectangular frame s
⇒	<u>GetRod</u>	Retrieves frame section property data for a rod frame section.
⇒	<u>GetSDSection</u>	Retrieves section property data for a section designer section
⇒	<u>GetSectProps</u>	
⇒	<u>GetSteelAngle</u>	Retrieves frame section property data for a steel angle frame se
⇒	<u>GetSteelTee</u>	Retrieves frame section property data for a steel tee frame sect
⇒	<u>GetTee</u>	Retrieves frame section property data for a tee-type frame secti
⇒	<u>GetTrapezoidal</u>	

⇒ <u>GetTube</u>	Retrieves frame section property data for a tube-type frame section.
⇒ <u>GetTypeOAPI</u>	Retrieves the property type for the specified frame section property.
⇒ <u>GetTypeRebar</u>	Retrieves the rebar design type for the specified frame section property.
⇒ <u>ImportProp</u>	Imports a frame section property from a property file.
⇒ <u>SetAngle</u>	Initializes an angle-type frame section property. If this function is called with no arguments, it initializes a solid angle frame section property.
⇒ <u>SetAutoSelectSteel</u>	Assigns frame section properties to an auto select list. If this function is called with no arguments, it assigns all frame section properties to the list.
⇒ <u>SetChannel</u>	Initializes an channel-type frame section property. If this function is called with no arguments, it initializes a solid channel frame section property.
⇒ <u>SetCircle</u>	Initializes a solid circular frame section property. If this function is called with no arguments, it initializes a solid circular frame section property.
⇒ <u>SetConcreteL</u>	Initializes a concrete L frame section property. If this function is called with no arguments, it initializes a concrete L frame section property.
⇒ <u>SetConcreteTee</u>	Initializes a concrete tee frame section property. If this function is called with no arguments, it initializes a concrete tee frame section property.
⇒ <u>SetCoverPlatedI</u>	Initializes a cover plated I-type frame section property. If this function is called with no arguments, it initializes a cover plated I-type frame section property.
⇒ <u>SetDblAngle</u>	Initializes a double angle-type frame section property. If this function is called with no arguments, it initializes a double angle-type frame section property.
⇒ <u>SetDblChannel</u>	Initializes a double channel-type frame section property. If this function is called with no arguments, it initializes a double channel-type frame section property.
⇒ <u>SetGeneral</u>	Initializes a general frame section property. If this function is called with no arguments, it initializes a general frame section property.
⇒ <u>SetISection</u>	Initializes an I-type frame section property. If this function is called with no arguments, it initializes an I-type frame section property.
⇒ <u>SetMaterial</u>	Sets the material property of a defined frame section.
⇒ <u>SetModifiers</u>	Assigns property modifiers to a frame section property. The default modifiers are: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100.
⇒ <u>SetNonPrismatic</u>	Assigns data to a nonprismatic frame section property.
⇒ <u>SetPipe</u>	Initializes a pipe-type frame section property. If this function is called with no arguments, it initializes a solid pipe frame section property.
⇒ <u>SetPlate</u>	Initializes a solid plate frame section property. If this function is called with no arguments, it initializes a solid plate frame section property.
⇒ <u>SetRebarBeam</u>	Assigns beam rebar data to frame sections.
⇒ <u>SetRebarColumn</u>	Assigns column rebar data to frame sections.
⇒ <u>SetRectangle</u>	Initializes a solid rectangular frame section property. If this function is called with no arguments, it initializes a solid rectangular frame section property.
⇒ <u>SetRod</u>	Initializes a solid rod frame section property. If this function is called with no arguments, it initializes a solid rod frame section property.
⇒ <u>SetSDSection</u>	Initializes a solid section frame section property. If this function is called with no arguments, it initializes a solid section frame section property.
⇒ <u>SetSteelAngle</u>	Initializes a steel angle frame section property. If this function is called with no arguments, it initializes a steel angle frame section property.
⇒ <u>SetSteelTee</u>	Initializes a steel tee frame section property. If this function is called with no arguments, it initializes a steel tee frame section property.
⇒ <u>SetTee</u>	Initializes a tee-type frame section property. If this function is called with no arguments, it initializes a tee-type frame section property.
⇒ <u>SetTrapezoidal</u>	Initializes a trapezoidal frame section property. If this function is called with no arguments, it initializes a trapezoidal frame section property.
⇒ <u>SetTube</u>	Initializes a tube-type frame section property. If this function is called with no arguments, it initializes a tube-type frame section property.

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▲ See Also

Reference

[cPropFrame Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LSTB3D7E23D_0?cpp2898=.");Ch

Method

Changes the name of an existing frame section property.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int ChangeName(
    string Name,
    string NewName
)
```

```
Function ChangeName (
    Name As String,
    NewName As String
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim NewName As String
Dim returnValue As Integer
```

```
returnValue = instance.ChangeName(Name,
    NewName)
```

```
int ChangeName(
    String^ Name,
    String^ NewName
)
```

```
abstract ChangeName :
    Name : string *
    NewName : string -> int
```

Parameters

Name

Type: SystemString

The existing name of a defined frame section property.

NewName

Type: SystemString

The new name for the frame section property.

Return ValueType: Int32

Returns zero if the new name is successfully applied; otherwise it returns a nonzero value.

▀ **Remarks**

▀ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new frame section property
    ret = SapModel.PropFrame.SetAngle("ANGLE1", "A992Fy50", 6, 4, 0.5, 0.5)

    'change name of frame section property
    ret = SapModel.PropFrame.ChangeName("ANGLE1", "MySection")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▀ **See Also**

Reference

cPropFrame Interface

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LSTA30C5BC8_0?cpp2900=.")

Method

Retrieves the total number of defined frame section properties in the model.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int Count(
    eFramePropType PropType =
)
```

```
Function Count (
    ?PropType As eFramePropType =
) As Integer
```

```
Dim instance As cPropFrame
Dim PropType As eFramePropType
Dim returnValue As Integer
```

```
returnValue = instance.Count(PropType)
```

```
int Count(
    eFramePropType PropType =
)
```

```
abstract Count :
    ?PropType : eFramePropType
(* Defaults:
    let _PropType = defaultArg PropType
*)
-> int
```

Parameters

PropType (Optional)

Type: ETABSv1eFramePropType

This optional value is one of the items in the eFramePropType enumeration.

If no value is input for PropType, a count is returned for all frame section properties in the model regardless of type.

Return Value

Type: Int32

Returns the total number of defined frame section properties in the model. If desired,

cPropFramespan id="LSTA30C5BC8_0"AddLanguageSpecificTextSet("LSTA30C5BC8_0?cpp2900=.");Co

counts can be returned for all frame section properties of a specified type in the model.

Remarks

Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim Count As Integer  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)  
  
    'return number of defined frame section properties of all types  
    Count = SapModel.PropFrame.Count  
  
    'close ETABS  
    EtabsObject.ApplicationExit(False)  
  
    'clean up variables  
    SapModel = Nothing  
    EtabsObject = Nothing  
End Sub
```

See Also

Reference

cPropFrame Interface

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST2432E0D4_0") Method

Deletes a specified frame section property.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int Delete(
    string Name
)
```

```
Function Delete (
    Name As String
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.Delete(Name)
```

```
int Delete(
    String^ Name
)
```

```
abstract Delete :
    Name : string -> int
```

Parameters

Name

Type: SystemString

The name of an existing frame section property.

Return Value

Type: Int32

Returns zero if the frame section property is successfully deleted; otherwise it returns a nonzero value. It returns an error if the specified frame section property can not be deleted; for example, if it is being used by a frame object.

▲ Remarks

▲ Examples

VB

Copy

cPropFramespan id="LST2432E0D4_0"AddLanguageSpecificTextSet("LST2432E0D4_0?cpp=2902.");Delete

Introduction

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)  
  
    'set new frame section property  
    ret = SapModel.PropFrame.SetAngle("ANGLE1", "A992Fy50", 6, 4, 0.5, 0.5)  
  
    'delete frame property  
    ret = SapModel.PropFrame.Delete("ANGLE1")  
  
    'close ETABS  
    EtabsObject.ApplicationExit(False)  
  
    'clean up variables  
    SapModel = Nothing  
    EtabsObject = Nothing  
End Sub
```

🔗 See Also

Reference

[cPropFrame Interface](#)

[ETABSv1 Namespace](#)

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Namespace: ETABSv1

▲ Syntax

C#

VBC++

F#

Copy

```
Function GetAllFrameProperties (
ByRef NumberNames As Integer,
ByRef MyName As String(),
ByRef PropType As eFramePropType(),
ByRef t3 As Double(),
ByRef t2 As Double(),
ByRef tf As Double(),
ByRef tw As Double(),
ByRef t2b As Double(),
ByRef tfb As Double()
) As Integer
```

```
returnValue = instance.GetAllFrameProperties(NumberNames,
MyName, PropType, t3, t2, tf, tw, t2b,
tfb)
```

cPropFramespan id="LSTD478515D 0"AddLanguageSpecificTextSet("LSTD478515D 0?cpp=2904-");Get.

```

int% NumberNames,
array<String^>^% MyName,
array<eFramePropType>^% PropType,
array<double>^% t3,
array<double>^% t2,
array<double>^% tf,
array<double>^% tw,
array<double>^% t2b,
array<double>^% tfb
)

abstract GetAllFrameProperties :
    NumberNames : int byref *
    MyName : string[] byref *
    PropType : eFramePropType[] byref *
    t3 : float[] byref *
    t2 : float[] byref *
    tf : float[] byref *
    tw : float[] byref *
    t2b : float[] byref *
    tfb : float[] byref -> int

```

Parameters

NumberNames
Type:Â [SystemInt32](#)

MyName
Type:Â [SystemString](#)

PropType
Type:Â [ETABSv1eFramePropType](#)

t3
Type:Â [SystemDouble](#)

t2
Type:Â [SystemDouble](#)

tf
Type:Â [SystemDouble](#)

tw
Type:Â [SystemDouble](#)

t2b
Type:Â [SystemDouble](#)

tfb
Type:Â [SystemDouble](#)

Return Value

Type:Â [Int32](#)

▲ Remarks

▲ See Also

Reference

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GetAllFrameProperties(Int32,String,eFramePropType,Double,Double,Double,Double,Double,I

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

C#

VB

C++

F#

Copy

```
returnValue = instance.GetAllFrameProperties 2(NumberNames,
```

cPropFramespan id="LST2945378F 0"AddLanguageSpecificTextSet("LST2945378F 0?cpp=::2907");GetA

MyName, PropType, t3, t2, tf, tw, t2b,
tfb, Area)

```
int GetAllFrameProperties_2(
int% NumberNames,
array<String^>^% MyName,
array<eFramePropType>^% PropType,
array<double>^% t3,
array<double>^% t2,
array<double>^% tf,
array<double>^% tw,
array<double>^% t2b,
array<double>^% tfb,
array<double>^% Area
)
```

```
abstract GetAllFrameProperties_2 :
    NumberNames : int byref *
    MyName : string[] byref *
    PropType : eFramePropType[] byref *
    t3 : float[] byref *
    t2 : float[] byref *
    tf : float[] byref *
    tw : float[] byref *
    t2b : float[] byref *
    tfb : float[] byref *
    Area : float[] byref -> int
```

Parameters

NumberNames

Type:Â [SystemInt32](#)

MyName

Type:Â [SystemString](#)

PropType

Type:Â [ETABSv1eFramePropType](#)

t3

Type:Â [SystemDouble](#)

t2

Type:Â [SystemDouble](#)

tf

Type:Â [SystemDouble](#)

tw

Type:Â [SystemDouble](#)

t2b

Type:Â [SystemDouble](#)

tfb

Type:Â [SystemDouble](#)

Area

Type:Â [SystemDouble](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

cPropFrame Interface

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LSTFA31F02_0") Method

Retrieves frame section property data for an angle-type frame section.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetAngle(
    string Name,
    ref string FileName,
    ref string MatProp,
    ref double T3,
    ref double T2,
    ref double Tf,
    ref double Tw,
    ref int Color,
    ref string Notes,
    ref string GUID
)
```

```
Function GetAngle (
    Name As String,
    ByRef FileName As String,
    ByRef MatProp As String,
    ByRef T3 As Double,
    ByRef T2 As Double,
    ByRef Tf As Double,
    ByRef Tw As Double,
    ByRef Color As Integer,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim FileName As String
Dim MatProp As String
Dim T3 As Double
Dim T2 As Double
Dim Tf As Double
Dim Tw As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.GetAngle(Name,
    FileName, MatProp, T3, T2, Tf, Tw, Color,
```

cPropFrameid="LSTFA31F02_0"AddLanguageSpecificTextSet("LSTFA31F02_0?cpp=::|n2910;GetAng

Notes, GUID)

```
int GetAngle(
String^ Name,
String^% FileName,
String^% MatProp,
double% T3,
double% T2,
double% Tf,
double% Tw,
int% Color,
String^% Notes,
String^% GUID
)
```

```
abstract GetAngle :
    Name : string *
    FileName : string byref *
    MatProp : string byref *
    T3 : float byref *
    T2 : float byref *
    Tf : float byref *
    Tw : float byref *
    Color : int byref *
    Notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame section property.

FileName

Type:Â SystemString

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

T2

Type:Â SystemDouble

Tf

Type:Â SystemDouble

Tw

Type:Â SystemDouble

Color

Type:Â SystemInt32

Notes

Type:Â SystemString

GUID

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the section.

Return Value

Type: Int32

returns zero if the section property data is successfully retrieved; otherwise it returns a nonzero value.

▲ **Remarks**

This function is obsolete and has been superseded by GetSteelAngle(String, String, String, Double, Double, Double, Double, Double, Boolean, Boolean, Int32, String, String) and GetConcreteL(String, String, String, Double, Double, Double, Double, Double, Boolean, Boolean, Int32, String, String) as of version 16.0.0. This function is maintained for backwards compatibility.

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim FileName As String
    Dim MatProp As String
    Dim t3 As Double
    Dim t2 As Double
    Dim tf As Double
    Dim tw As Double
    Dim Color As Integer
    Dim Notes As String
    Dim GUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new frame section property
    ret = SapModel.PropFrame.SetAngle("ANGLE1", "A992Fy50", 6, 4, 0.5, 0.5)

    'get frame section property data
    ret = SapModel.PropFrame.GetAngle("ANGLE1", FileName, MatProp, t3, t2, tf, tw, Color, Notes)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST6F8C2137_0?cpp=2014.");GetAutoSelectSteel Method

Retrieves frame section property data for a steel auto select lists.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetAutoSelectSteel(
    string Name,
    ref int NumberItems,
    ref string[] SectName,
    ref string AutoStartSection,
    ref string Notes,
    ref string GUID
)
```

```
Function GetAutoSelectSteel (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef SectName As String(),
    ByRef AutoStartSection As String,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim NumberItems As Integer
Dim SectName As String()
Dim AutoStartSection As String
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.GetAutoSelectSteel(Name,
    NumberItems, SectName, AutoStartSection,
    Notes, GUID)
```

```
int GetAutoSelectSteel(
    String^ Name,
    int% NumberItems,
    array<String^>^% SectName,
    String^% AutoStartSection,
    String^% Notes,
    String^% GUID
)
```

abstract GetAutoSelectSteel :

cPropFrameid="LST6F8C2137_0"AddLanguageSpecificTextSet("LST6F8C2137_0?cpp=2014.");GetAutoSelectSteel

```

Name : string *
NumberItems : int byref *
SectName : string[] byref *
AutoStartSection : string byref *
Notes : string byref *
GUID : string byref -> int

```

Parameters

Name

Type:Â SystemString

The name of an existing auto select section listâ type frame section property.

NumberItems

Type:Â SystemInt32

The number of frame section properties included in the auto select list.

SectName

Type:Â SystemString

This is an array of the names of the frame section properties included in the auto select list.

AutoStartSection

Type:Â SystemString

This is either Median or the name of a frame section property in the SectName array. It is the starting section for the auto select list.

Notes

Type:Â SystemString

The notes, if any, assigned to the section.

GUID

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the section.

Return Value

Type:Â Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyName() As String
    Dim NumberItems As Integer
    Dim SectName() As String
    Dim AutoStartSection As String
    Dim Notes As String
    Dim GUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

```

Introduction

```
'create SapModel object
    SapModel = EtabsObject.SapModel

'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new frame section properties
    ret = SapModel.PropFrame.SetAngle("ANGLE1", "A992Fy50", 6, 4, 0.5, 0.5)
    ret = SapModel.PropFrame.SetAngle("ANGLE2", "A992Fy50", 5, 3, 0.4, 0.4)
    ret = SapModel.PropFrame.SetAngle("ANGLE3", "A992Fy50", 4, 2, 0.3, 0.3)

'set new auto select list frame section property
    ReDim MyName(2)
    MyName(0) = "ANGLE1"
    MyName(1) = "ANGLE2"
    MyName(2) = "ANGLE3"
    ret = SapModel.PropFrame.SetAutoSelectSteel("AUT01", 3, MyName, "ANGLE2")

'get auto select list data
    ret = SapModel.PropFrame.GetAutoSelectSteel("AUT01", NumberItems, SectName, AutoStartSection)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPropFrame Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST9FF3A02C_0?cpp=2017-")

Method

Retrieves frame section property data for an channel-type frame section.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetChannel(
    string Name,
    ref string FileName,
    ref string MatProp,
    ref double T3,
    ref double T2,
    ref double Tf,
    ref double Tw,
    ref int Color,
    ref string Notes,
    ref string GUID
)
```

```
Function GetChannel (
    Name As String,
    ByRef FileName As String,
    ByRef MatProp As String,
    ByRef T3 As Double,
    ByRef T2 As Double,
    ByRef Tf As Double,
    ByRef Tw As Double,
    ByRef Color As Integer,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim FileName As String
Dim MatProp As String
Dim T3 As Double
Dim T2 As Double
Dim Tf As Double
Dim Tw As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.GetChannel(Name,
    FileName, MatProp, T3, T2, Tf, Tw, Color,
```

cPropFramespan id="LST9FF3A02C_0"AddLanguageSpecificTextSet("LST9FF3A02C_0?cpp=2017-");Get

Notes, GUID)

```
int GetChannel(
String^ Name,
String^% FileName,
String^% MatProp,
double% T3,
double% T2,
double% Tf,
double% Tw,
int% Color,
String^% Notes,
String^% GUID
)
```

```
abstract GetChannel :
    Name : string *
    FileName : string byref *
    MatProp : string byref *
    T3 : float byref *
    T2 : float byref *
    Tf : float byref *
    Tw : float byref *
    Color : int byref *
    Notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame section property.

FileName

Type:Â SystemString

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

T2

Type:Â SystemDouble

Tf

Type:Â SystemDouble

Tw

Type:Â SystemDouble

Color

Type:Â SystemInt32

Notes

Type:Â SystemString

GUID

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the section.

Return Value

Type: $\hat{A}$ Int32

returns zero if the section property data is successfully retrieved; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim FileName As String
    Dim MatProp As String
    Dim t3 As Double
    Dim t2 As Double
    Dim tf As Double
    Dim tw As Double
    Dim Color As Integer
    Dim Notes As String
    Dim GUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new frame section property
    ret = SapModel.PropFrame.SetChannel("CHN1", "A992Fy50", 24, 6, 0.5, 0.3)

    'get frame section property data
    ret = SapModel.PropFrame.GetChannel("CHN1", FileName, MatProp, t3, t2, tf, tw, Color, Notes)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cPropFrame Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LSTB04A5B84_0?cpp=2021-")

Method

Retrieves frame section property data for a circular frame section.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetCircle(
    string Name,
    ref string FileName,
    ref string MatProp,
    ref double T3,
    ref int Color,
    ref string Notes,
    ref string GUID
)
```

```
Function GetCircle (
    Name As String,
    ByRef FileName As String,
    ByRef MatProp As String,
    ByRef T3 As Double,
    ByRef Color As Integer,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim FileName As String
Dim MatProp As String
Dim T3 As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.GetCircle(Name,
    FileName, MatProp, T3, Color, Notes,
    GUID)
```

```
int GetCircle(
    String^ Name,
    String^% FileName,
    String^% MatProp,
    double% T3,
    int% Color,
    String^% Notes,
```

cPropFramespan id="LSTB04A5B84_0"AddLanguageSpecificTextSet("LSTB04A5B84_0?cpp=2021-");Get

```
String^% GUID
)
```

```
abstract GetCircle :
    Name : string *
    FileName : string byref *
    MatProp : string byref *
    T3 : float byref *
    Color : int byref *
    Notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type:Â [SystemString](#)

The name of an existing circular frame section property.

FileName

Type:Â [SystemString](#)

If the section property was imported from a property file, this is the name of that file. If the section property was not imported, this item is blank.

MatProp

Type:Â [SystemString](#)

The name of the material property for the section.

T3

Type:Â [SystemDouble](#)

Color

Type:Â [SystemInt32](#)

Notes

Type:Â [SystemString](#)

GUID

Type:Â [SystemString](#)

The GUID (global unique identifier), if any, assigned to the section.

Return Value

Type:Â [Int32](#)

Returns zero if the section property data is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim FileName As String
    Dim MatProp As String
    Dim t3 As Double
    Dim Color As Integer
    Dim Notes As String
    Dim GUID As String
```

Introduction

```
'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
    ret = EtabsObject.ApplicationStart()

'create SapModel object
    SapModel = EtabsObject.SapModel

'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new frame section property
    ret = SapModel.PropFrame.SetCircle("C1", "4000Psi", 20)

'get frame section property data
    ret = SapModel.PropFrame.GetCircle("C1", FileName, MatProp, t3, Color, Notes, GUID)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

🔗 See Also

Reference

[cPropFrame Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST248801B2_0?cpp=:2024.") Method

Retrieves frame section property data for a concrete L frame section.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetConcreteL(
    string Name,
    ref string FileName,
    ref string MatProp,
    ref double T3,
    ref double T2,
    ref double Tf,
    ref double TwC,
    ref double TwT,
    ref bool MirrorAbout2,
    ref bool MirrorAbout3,
    ref int Color,
    ref string Notes,
    ref string GUID
)
```

```
Function GetConcreteL (
    Name As String,
    ByRef FileName As String,
    ByRef MatProp As String,
    ByRef T3 As Double,
    ByRef T2 As Double,
    ByRef Tf As Double,
    ByRef TwC As Double,
    ByRef TwT As Double,
    ByRef MirrorAbout2 As Boolean,
    ByRef MirrorAbout3 As Boolean,
    ByRef Color As Integer,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim FileName As String
Dim MatProp As String
Dim T3 As Double
Dim T2 As Double
Dim Tf As Double
Dim TwC As Double
Dim TwT As Double
```

cPropFrameid="LST248801B2_0"AddLanguageSpecificTextSet("LST248801B2_0?cpp=:2024.");GetC

Introduction

```
Dim MirrorAbout2 As Boolean
Dim MirrorAbout3 As Boolean
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer

returnValue = instance.GetConcreteL(Name,
FileName, MatProp, T3, T2, Tf, TwC,
TwT, MirrorAbout2, MirrorAbout3,
Color, Notes, GUID)

int GetConcreteL(
String^ Name,
String^% FileName,
String^% MatProp,
double% T3,
double% T2,
double% Tf,
double% TwC,
double% TwT,
bool% MirrorAbout2,
bool% MirrorAbout3,
int% Color,
String^% Notes,
String^% GUID
)

abstract GetConcreteL :
    Name : string *
    FileName : string byref *
    MatProp : string byref *
    T3 : float byref *
    T2 : float byref *
    Tf : float byref *
    TwC : float byref *
    TwT : float byref *
    MirrorAbout2 : bool byref *
    MirrorAbout3 : bool byref *
    Color : int byref *
    Notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame section property.

FileName

Type:Â SystemString

If the section property was imported from a property file, this is the name of that file. If the section property was not imported, this item is blank.

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

T2

Tf	Type:Â <u>SystemDouble</u>
TwC	Type:Â <u>SystemDouble</u> The vertical leg thickness at the corner. [L]
TwT	Type:Â <u>SystemDouble</u> The vertical leg thickness at the tip. [L]
MirrorAbout2	Type:Â <u>SystemBoolean</u> Indicates whether the section is mirrored about the local 2-axis.
MirrorAbout3	Type:Â <u>SystemBoolean</u> Indicates whether the section is mirrored about the local 3-axis.
Color	Type:Â <u>SystemInt32</u>
Notes	Type:Â <u>SystemString</u>
GUID	Type:Â <u>SystemString</u> The GUID (global unique identifier), if any, assigned to the section.

Return Value

Type:Â Int32
returns zero if the section property data is successfully retrieved; otherwise it returns a nonzero value.

▲ **Remarks**

This function supersedes GetAngle(String, String, String, Double, Double, Double, Double, Int32, String, String).

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim FileName As String
    Dim MatProp As String
    Dim t3 As Double
    Dim t2 As Double
    Dim tf As Double
    Dim twc As Double
    Dim twt As Double
    Dim mirror2 As Boolean
    Dim mirror3 As Boolean
    Dim Color As Integer
    Dim Notes As String
    Dim GUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

Return Value

Introduction

```
'start ETABS application
    ret = EtabsObject.ApplicationStart()

'create SapModel object
    SapModel = EtabsObject.SapModel

'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new frame section property
    ret = SapModel.PropFrame.SetConcreteL("ANGLE1", "4000Psi", 30, 36, 8, 7, 5, False, True)

'get frame section property data
    ret = SapModel.PropFrame.GetConcreteL("ANGLE1", FileName, MatProp, t3, t2, tf, twc, twt, m

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPropFrame Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST2B1CD3F6_0?cpp=2928=.");Ge

Method

Retrieves frame section property data for a concrete tee frame section.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetConcreteTee(
    string Name,
    ref string FileName,
    ref string MatProp,
    ref double T3,
    ref double T2,
    ref double Tf,
    ref double TwF,
    ref double TwT,
    ref bool MirrorAbout3,
    ref int Color,
    ref string Notes,
    ref string GUID
)
```

```
Function GetConcreteTee (
    Name As String,
    ByRef FileName As String,
    ByRef MatProp As String,
    ByRef T3 As Double,
    ByRef T2 As Double,
    ByRef Tf As Double,
    ByRef TwF As Double,
    ByRef TwT As Double,
    ByRef MirrorAbout3 As Boolean,
    ByRef Color As Integer,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim FileName As String
Dim MatProp As String
Dim T3 As Double
Dim T2 As Double
Dim Tf As Double
Dim TwF As Double
Dim TwT As Double
Dim MirrorAbout3 As Boolean
Dim Color As Integer
```

cPropFrameAddLanguageSpecificTextSet("LST2B1CD3F6_0?cpp=2928=.");Ge

Introduction

```
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer

returnValue = instance.GetConcreteTee(Name,
FileName, MatProp, T3, T2, Tf, TwF,
TwT, MirrorAbout3, Color, Notes, GUID)

int GetConcreteTee(
String^ Name,
String^% FileName,
String^% MatProp,
double% T3,
double% T2,
double% Tf,
double% TwF,
double% TwT,
bool% MirrorAbout3,
int% Color,
String^% Notes,
String^% GUID
)

abstract GetConcreteTee :
    Name : string *
    FileName : string byref *
    MatProp : string byref *
    T3 : float byref *
    T2 : float byref *
    Tf : float byref *
    TwF : float byref *
    TwT : float byref *
    MirrorAbout3 : bool byref *
    Color : int byref *
    Notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame section property.

FileName

Type:Â SystemString

If the section property was imported from a property file, this is the name of that file. If the section property was not imported, this item is blank.

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

T2

Type:Â SystemDouble

Tf

Type:Â SystemDouble

TwF

Type:Â SystemDouble
 The web thickness at the flange. [L]

TwT
 Type:Â SystemDouble
 The web thickness at the tip. [L]

MirrorAbout3
 Type:Â SystemBoolean
 Indicates whether the section is mirrored about the local 3-axis.

Color
 Type:Â SystemInt32

Notes
 Type:Â SystemString

GUID
 Type:Â SystemString
 The GUID (global unique identifier), if any, assigned to the section.

Return Value

Type:Â Int32
 Returns zero if the section property is successfully retrieved; otherwise it returns a nonzero value.

Remarks

This function supersedes GetTee(String, String, String, Double, Double, Double, Double, Int32, String, String).

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim FileName As String
    Dim MatProp As String
    Dim t3 As Double
    Dim t2 As Double
    Dim tf As Double
    Dim twf As Double
    Dim twt As Double
    Dim mirror3
    Dim Color As Integer
    Dim Notes As String
    Dim GUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()
```

Introduction

```
'create steel deck template model
  ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new frame section property
  ret = SapModel.PropFrame.SetConcreteTee("TEE1", "4000Psi", 30, 36, 8, 7, 5, False)

'get frame section property data
  ret = SapModel.PropFrame.GetConcreteTee("TEE1", FileName, MatProp, t3, t2, tf, twf, twt, m

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPropFrame Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LSTC5E24F52_0?cpp=2932-");Get

Method

retrieves frame section property data for a cover plated I-type frame section.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetCoverPlatedI(
    string Name,
    ref string SectName,
    ref double FyTopFlange,
    ref double FyWeb,
    ref double FyBotFlange,
    ref double Tc,
    ref double Bc,
    ref string MatPropTop,
    ref double Tcb,
    ref double Bcb,
    ref string MatPropBot,
    ref int Color,
    ref string Notes,
    ref string GUID
)
```

```
Function GetCoverPlatedI (
    Name As String,
    ByRef SectName As String,
    ByRef FyTopFlange As Double,
    ByRef FyWeb As Double,
    ByRef FyBotFlange As Double,
    ByRef Tc As Double,
    ByRef Bc As Double,
    ByRef MatPropTop As String,
    ByRef Tcb As Double,
    ByRef Bcb As Double,
    ByRef MatPropBot As String,
    ByRef Color As Integer,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim SectName As String
Dim FyTopFlange As Double
Dim FyWeb As Double
Dim FyBotFlange As Double
Dim Tc As Double
```

cPropFramespan id="LSTC5E24F52_0"AddLanguageSpecificTextSet("LSTC5E24F52_0?cpp=2932-");Get

Introduction

```
Dim Bc As Double
Dim MatPropTop As String
Dim Tcb As Double
Dim Bcb As Double
Dim MatPropBot As String
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer

returnValue = instance.GetCoverPlatedI(Name,
SectName, FyTopFlange, FyWeb, FyBotFlange,
Tc, Bc, MatPropTop, Tcb, Bcb, MatPropBot,
Color, Notes, GUID)
```

```
int GetCoverPlatedI(
String^ Name,
String^% SectName,
double% FyTopFlange,
double% FyWeb,
double% FyBotFlange,
double% Tc,
double% Bc,
String^% MatPropTop,
double% Tcb,
double% Bcb,
String^% MatPropBot,
int% Color,
String^% Notes,
String^% GUID
)
```

```
abstract GetCoverPlatedI :
    Name : string *
    SectName : string byref *
    FyTopFlange : float byref *
    FyWeb : float byref *
    FyBotFlange : float byref *
    Tc : float byref *
    Bc : float byref *
    MatPropTop : string byref *
    Tcb : float byref *
    Bcb : float byref *
    MatPropBot : string byref *
    Color : int byref *
    Notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame section property.

SectName

Type:Â SystemString

FyTopFlange

Type:Â SystemDouble

The yield strength of the top flange of the I-section. [F/L2]

Introduction

If this item is 0, the yield strength of the I-section specified by the SectName item is used.

FyWeb

Type:Â SystemDouble

The yield strength of the web of the I-section. [F/L2]

If this item is 0, the yield strength of the I-section specified by the SectName item is used.

FyBotFlange

Type:Â SystemDouble

The yield strength of the bottom flange of the I-section. [F/L2]

If this item is 0, the yield strength of the I-section specified by the SectName item is used.

Tc

Type:Â SystemDouble

Bc

Type:Â SystemDouble

MatPropTop

Type:Â SystemString

The name of the material property for the bottom cover plate.

This item applies only if both the tcb and the bcb items are greater than 0.

Tcb

Type:Â SystemDouble

Bcb

Type:Â SystemDouble

MatPropBot

Type:Â SystemString

The name of the material property for the bottom cover plate.

This item applies only if both the tcb and the bcb items are greater than 0.

Color

Type:Â SystemInt32

Notes

Type:Â SystemString

GUID

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the section.

Return Value

Type:Â Int32

Returns zero if the section property is successfully initialized; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel
```

Introduction

```
Dim EtabsObject As cOAPI
Dim ret As Integer = -1
Dim SectName As String
Dim FyTopFlange As Double
Dim FyWeb As Double
Dim FyBotFlange As Double
Dim tc As Double
Dim bc As Double
Dim MatPropTop As String
Dim tcb As Double
Dim bcb As Double
Dim MatPropBot As String
Dim Color As Integer
Dim Notes As String
Dim GUID As String

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new I-type frame section property
ret = SapModel.PropFrame.SetISection("ISEC", "A992Fy50", 24, 8, 0.5, 0.3, 8, 0.5)

'set new cover plated I-type frame section property
ret = SapModel.PropFrame.SetCoverPlatedI("CPI1", "ISEC", 0, 36, 0, 0.75, 14, "A992Fy50", 0)

'get frame section property data for cover plated I
ret = SapModel.PropFrame.GetCoverPlatedI("CPI1", SectName, FyTopFlange, FyWeb, FyBotFlange)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

🔗 See Also

Reference

[cPropFrame Interface](#)

[ETABSv1 Namespace](#)

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Introduction

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST9A6CAED6_0?cpp2917u=.");Get

Method

Retrieves frame section property data for a double angle-type frame section.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetDblAngle(
    string Name,
    ref string FileName,
    ref string MatProp,
    ref double T3,
    ref double T2,
    ref double Tf,
    ref double Tw,
    ref double Dis,
    ref int Color,
    ref string Notes,
    ref string GUID
)
```

```
Function GetDblAngle (
    Name As String,
    ByRef FileName As String,
    ByRef MatProp As String,
    ByRef T3 As Double,
    ByRef T2 As Double,
    ByRef Tf As Double,
    ByRef Tw As Double,
    ByRef Dis As Double,
    ByRef Color As Integer,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim FileName As String
Dim MatProp As String
Dim T3 As Double
Dim T2 As Double
Dim Tf As Double
Dim Tw As Double
Dim Dis As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

cPropFrameid="LST9A6CAED6_0"AddLanguageSpecificTextSet("LST9A6CAED6_0?cpp2917u=.");Get

```
returnValue = instance.GetDblAngle(Name,
FileName, MatProp, T3, T2, Tf, Tw, Dis,
Color, Notes, GUID)
```

```
int GetDblAngle(
String^ Name,
String^% FileName,
String^% MatProp,
double% T3,
double% T2,
double% Tf,
double% Tw,
double% Dis,
int% Color,
String^% Notes,
String^% GUID
)
```

```
abstract GetDblAngle :
    Name : string *
    FileName : string byref *
    MatProp : string byref *
    T3 : float byref *
    T2 : float byref *
    Tf : float byref *
    Tw : float byref *
    Dis : float byref *
    Color : int byref *
    Notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame section property.

FileName

Type:Â SystemString

If the section property was imported from a property file, this is the name of that file. If the section property was not imported, this item is blank.

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

T2

Type:Â SystemDouble

Tf

Type:Â SystemDouble

Tw

Type:Â SystemDouble

Dis

Type:Â SystemDouble

Color

Type:Â SystemInt32

Notes

Type:Â SystemString

GUID

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the section.

Return ValueType:Â Int32

returns zero if the section property data is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim FileName As String
    Dim MatProp As String
    Dim t3 As Double
    Dim t2 As Double
    Dim tf As Double
    Dim tw As Double
    Dim dis As Double
    Dim Color As Integer
    Dim Notes As String
    Dim GUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new frame section property
    ret = SapModel.PropFrame.SetDblAngle("DBANG1", "A992Fy50", 6, 9, 0.5, 0.5, 1)

    'get frame section property data
    ret = SapModel.PropFrame.GetDblAngle("DBANG1", FileName, MatProp, t3, t2, tf, tw, dis, Col

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
```

Return Value

End Sub

▲ See Also

Reference

cPropFrame Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST2AE866F3_0?cpp=2011-")

Method

Retrieves frame section property data for a double channel-type frame section.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetDblChannel(
    string Name,
    ref string FileName,
    ref string MatProp,
    ref double T3,
    ref double T2,
    ref double Tf,
    ref double Tw,
    ref double Dis,
    ref int Color,
    ref string Notes,
    ref string GUID
)
```

```
Function GetDblChannel (
    Name As String,
    ByRef FileName As String,
    ByRef MatProp As String,
    ByRef T3 As Double,
    ByRef T2 As Double,
    ByRef Tf As Double,
    ByRef Tw As Double,
    ByRef Dis As Double,
    ByRef Color As Integer,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim FileName As String
Dim MatProp As String
Dim T3 As Double
Dim T2 As Double
Dim Tf As Double
Dim Tw As Double
Dim Dis As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

cPropFrameid="LST2AE866F3_0"AddLanguageSpecificTextSet("LST2AE866F3_0?cpp=2011-");GetL

```
returnValue = instance.GetDblChannel(Name,
FileName, MatProp, T3, T2, Tf, Tw, Dis,
Color, Notes, GUID)
```

```
int GetDblChannel(
String^ Name,
String^% FileName,
String^% MatProp,
double% T3,
double% T2,
double% Tf,
double% Tw,
double% Dis,
int% Color,
String^% Notes,
String^% GUID
)
```

```
abstract GetDblChannel :
    Name : string *
    FileName : string byref *
    MatProp : string byref *
    T3 : float byref *
    T2 : float byref *
    Tf : float byref *
    Tw : float byref *
    Dis : float byref *
    Color : int byref *
    Notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame section property.

FileName

Type:Â SystemString

If the section property was imported from a property file, this is the name of that file. If the section property was not imported, this item is blank.

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

T2

Type:Â SystemDouble

Tf

Type:Â SystemDouble

Tw

Type:Â SystemDouble

Dis

Type:Â SystemDouble

Color

Type:Â SystemInt32

Notes

Type:Â SystemString

GUID

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the section.

Return ValueType:Â Int32

returns zero if the section property data is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim FileName As String
    Dim MatProp As String
    Dim t3 As Double
    Dim t2 As Double
    Dim tf As Double
    Dim tw As Double
    Dim dis As Double
    Dim Color As Integer
    Dim Notes As String
    Dim GUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new frame section property
    ret = SapModel.PropFrame.SetDbfChannel("DBCHN1", "A992Fy50", 12, 6.5, 0.5, 0.3, 0.5)

    'get frame section property data
    ret = SapModel.PropFrame.GetDbfChannel("DBCHN1", FileName, MatProp, t3, t2, tf, tw, dis, C

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
```

Return Value

End Sub

▲ See Also
Reference

cPropFrame Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LSTCC3D5420_0?cpp=2945=.");Ge

Method

Retrieves frame section property data for a general frame section

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetGeneral(
    string Name,
    ref string FileName,
    ref string MatProp,
    ref double T3,
    ref double T2,
    ref double Area,
    ref double As2,
    ref double As3,
    ref double Torsion,
    ref double I22,
    ref double I33,
    ref double S22,
    ref double S33,
    ref double Z22,
    ref double Z33,
    ref double R22,
    ref double R33,
    ref int Color,
    ref string Notes,
    ref string GUID
)
```

```
Function GetGeneral (
    Name As String,
    ByRef FileName As String,
    ByRef MatProp As String,
    ByRef T3 As Double,
    ByRef T2 As Double,
    ByRef Area As Double,
    ByRef As2 As Double,
    ByRef As3 As Double,
    ByRef Torsion As Double,
    ByRef I22 As Double,
    ByRef I33 As Double,
    ByRef S22 As Double,
    ByRef S33 As Double,
    ByRef Z22 As Double,
    ByRef Z33 As Double,
    ByRef R22 As Double,
    ByRef R33 As Double,
```

cPropFramespan id="LSTCC3D5420_0"AddLanguageSpecificTextSet("LSTCC3D5420_0?cpp=2945=.");Ge

Introduction

```
ByRef Color As Integer,  
ByRef Notes As String,  
ByRef GUID As String  
) As Integer
```

```
Dim instance As cPropFrame  
Dim Name As String  
Dim FileName As String  
Dim MatProp As String  
Dim T3 As Double  
Dim T2 As Double  
Dim Area As Double  
Dim As2 As Double  
Dim As3 As Double  
Dim Torsion As Double  
Dim I22 As Double  
Dim I33 As Double  
Dim S22 As Double  
Dim S33 As Double  
Dim Z22 As Double  
Dim Z33 As Double  
Dim R22 As Double  
Dim R33 As Double  
Dim Color As Integer  
Dim Notes As String  
Dim GUID As String  
Dim returnValue As Integer
```

```
returnValue = instance.GetGeneral(Name,  
FileName, MatProp, T3, T2, Area, As2,  
As3, Torsion, I22, I33, S22, S33, Z22,  
Z33, R22, R33, Color, Notes, GUID)
```

```
int GetGeneral(  
String^ Name,  
String^% FileName,  
String^% MatProp,  
double% T3,  
double% T2,  
double% Area,  
double% As2,  
double% As3,  
double% Torsion,  
double% I22,  
double% I33,  
double% S22,  
double% S33,  
double% Z22,  
double% Z33,  
double% R22,  
double% R33,  
int% Color,  
String^% Notes,  
String^% GUID  
)
```

```
abstract GetGeneral :  
    Name : string *  
    FileName : string byref *  
    MatProp : string byref *  
    T3 : float byref *  
    T2 : float byref *
```

```

Area : float byref *
As2 : float byref *
As3 : float byref *
Torsion : float byref *
I22 : float byref *
I33 : float byref *
S22 : float byref *
S33 : float byref *
Z22 : float byref *
Z33 : float byref *
R22 : float byref *
R33 : float byref *
Color : int byref *
Notes : string byref *
GUID : string byref -> int

```

Parameters

Name

Type:Â SystemString

The name of an existing frame section property.

FileName

Type:Â SystemString

If the section property was imported from a property file, this is the name of that file. If the section property was not imported, this item is blank.

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

T2

Type:Â SystemDouble

Area

Type:Â SystemDouble

The cross-sectional area. [L2]

As2

Type:Â SystemDouble

The shear area for forces in the section local 2-axis direction. [L2]

As3

Type:Â SystemDouble

The shear area for forces in the section local 3-axis direction. [L2]

Torsion

Type:Â SystemDouble

The torsional constant. [L4]

I22

Type:Â SystemDouble

The moment of inertia for bending about the local 2 axis. [L4]

I33

Type:Â SystemDouble

The moment of inertia for bending about the local 3 axis. [L4]

S22

Type:Â SystemDouble

The section modulus for bending about the local 2 axis. [L3]

S33	Type:Â <u>SystemDouble</u> The section modulus for bending about the local 3 axis. [L3]
Z22	Type:Â <u>SystemDouble</u> The plastic modulus for bending about the local 2 axis. [L3]
Z33	Type:Â <u>SystemDouble</u> The plastic modulus for bending about the local 3 axis. [L3]
R22	Type:Â <u>SystemDouble</u> The radius of gyration about the local 2 axis. [L]
R33	Type:Â <u>SystemDouble</u> The radius of gyration about the local 3 axis. [L]
Color	Type:Â <u>SystemInt32</u>
Notes	Type:Â <u>SystemString</u>
GUID	Type:Â <u>SystemString</u> The GUID (global unique identifier), if any, assigned to the section.

Return ValueType:Â Int32

Returns zero if the section property data is successfully retrieved; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim FileName As String
    Dim MatProp As String
    Dim t3 As Double
    Dim t2 As Double
    Dim Area As Double
    Dim as2 As Double
    Dim as3 As Double
    Dim Torsion As Double
    Dim I22 As Double
    Dim I33 As Double
    Dim S22 As Double
    Dim S33 As Double
    Dim Z22 As Double
    Dim Z33 As Double
    Dim R22 As Double
    Dim R33 As Double
    Dim Color As Integer
    Dim Notes As String
    Dim GUID As String
```


Introduction

```
'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
    ret = EtabsObject.ApplicationStart()

'create SapModel object
    SapModel = EtabsObject.SapModel

'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new frame section property
    ret = SapModel.PropFrame.SetGeneral("GEN1", "A992Fy50", 24, 14, 100, 80, 80, 4, 1000, 2400)

'get frame section property data
    ret = SapModel.PropFrame.GetGeneral("GEN1", FileName, MatProp, t3, t2, Area, As2, As3, Tor)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

[cPropFrame Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST450FA939_0") Method

Retrieves frame section property data for an I-type frame section.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetISection(
    string Name,
    ref string FileName,
    ref string MatProp,
    ref double T3,
    ref double T2,
    ref double Tf,
    ref double Tw,
    ref double T2b,
    ref double Tfb,
    ref int Color,
    ref string Notes,
    ref string GUID
)
```

```
Function GetISection (
    Name As String,
    ByRef FileName As String,
    ByRef MatProp As String,
    ByRef T3 As Double,
    ByRef T2 As Double,
    ByRef Tf As Double,
    ByRef Tw As Double,
    ByRef T2b As Double,
    ByRef Tfb As Double,
    ByRef Color As Integer,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim FileName As String
Dim MatProp As String
Dim T3 As Double
Dim T2 As Double
Dim Tf As Double
Dim Tw As Double
Dim T2b As Double
Dim Tfb As Double
Dim Color As Integer
```

cPropFrameid="LST450FA939_0"AddLanguageSpecificTextSet("LST450FA939_0?cpp=:2050.");GetI

Introduction

```
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer

returnValue = instance.GetISection(Name,
FileName, MatProp, T3, T2, Tf, Tw, T2b,
Tfb, Color, Notes, GUID)

int GetISection(
String^ Name,
String^% FileName,
String^% MatProp,
double% T3,
double% T2,
double% Tf,
double% Tw,
double% T2b,
double% Tfb,
int% Color,
String^% Notes,
String^% GUID
)

abstract GetISection :
    Name : string *
    FileName : string byref *
    MatProp : string byref *
    T3 : float byref *
    T2 : float byref *
    Tf : float byref *
    Tw : float byref *
    T2b : float byref *
    Tfb : float byref *
    Color : int byref *
    Notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing I-type frame section property.

FileName

Type:Â SystemString

If the section property was imported from a property file, this is the name of that file. If the section property was not imported, this item is blank.

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

T2

Type:Â SystemDouble

Tf

Type:Â SystemDouble

Tw

Type:Â SystemDouble

T2b

Type:Â SystemDouble

Tfb

Type:Â SystemDouble

Color

Type:Â SystemInt32

Notes

Type:Â SystemString

GUID

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the section.

Return ValueType:Â Int32

Returns zero if the section property data is successfully retrieved; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim FileName As String
    Dim MatProp As String
    Dim t3 As Double
    Dim t2 As Double
    Dim tf As Double
    Dim tw As Double
    Dim t2b As Double
    Dim tfb As Double
    Dim Color As Integer
    Dim Notes As String
    Dim GUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new frame section property
    ret = SapModel.PropFrame.SetISection("ISEC1", "A992Fy50", 24, 10, 0.5, 0.3, 14, 0.6)

    'get frame section property data
    ret = SapModel.PropFrame.GetISection("ISEC1", FileName, MatProp, t3, t2, tf, tw, t2b, tfb,
```

Return Value

```
'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPropFrame Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST33C5694E_0?cpp=2954.");GetMaterial

Retrieves the material property of a defined frame section

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetMaterial(
    string Name,
    ref string MatProp
)
```

```
Function GetMaterial (
    Name As String,
    ByRef MatProp As String
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim MatProp As String
Dim returnValue As Integer
```

```
returnValue = instance.GetMaterial(Name,
    MatProp)
```

```
int GetMaterial(
    String^ Name,
    String^% MatProp
)
```

```
abstract GetMaterial :
    Name : string *
    MatProp : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing frame section property.

MatProp

Type: [SystemString](#)

The name of the material property for the section.

Return Value

Type: Int32

Returns zero if the material property is successfully retrieved; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim FileName As String
    Dim MatProp As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new frame section property
    ret = SapModel.PropFrame.SetTee("TEE1", "A992Fy50", 12, 10, 0.6, 0.3)

    'get frame section material property data
    ret = SapModel.PropFrame.GetMaterial("TEE1", MatProp)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cPropFrame Interface

ETABSv1 Namespace

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i»j

CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LSTA5C13085_0")

Method

Retrieves the modifier assignments for a frame section property. The default value for all modifiers is one.

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetModifiers(
    string Name,
    ref double[] Value
)
```

```
Function GetModifiers (
    Name As String,
    ByRef Value As Double()
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim Value As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetModifiers(Name,
    Value)
```

```
int GetModifiers(
    String^ Name,
    array<double>^% Value
)
```

```
abstract GetModifiers :
    Name : string *
    Value : float[] byref -> int
```

Parameters

Name

Type: $\hat{A}$ SystemString

The name of an existing frame section property.

Value

Type: $\hat{A}$ SystemDouble

This is an array of eight unitless modifiers:

Value

Modifier

Value(0) Cross sectional area modifier

cPropFramespan id="LSTA5C13085_0"AddLanguageSpecificTextSet("LSTA5C13085_0?cpp=2956.");GetL

Introduction

Value(1) Shear area in local 2 direction modifier
Value(2) Shear area in local 3 direction modifier
Value(3) Torsional constant modifier
Value(4) Moment of inertia about local 2 axis modifier
Value(5) Moment of inertia about local 3 axis modifier
Value(6) Mass modifier
Value(7) Weight modifier

Return Value

Type: Int32

Returns zero if the modifier assignments are successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim i As Integer  
    Dim Value() As Double  
    Dim MyValue() As Double  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)  
  
    'set new frame section property  
    ret = SapModel.PropFrame.SetISection("ISEC1", "A992Fy50", 24, 10, 0.5, 0.3, 14, 0.6)  
  
    'assign modifiers  
    ReDim Value(7)  
    For i = 0 To 7  
        Value(i) = 1  
    Next i  
    Value(5) = 100  
    ret = SapModel.PropFrame.SetModifiers("ISEC1", Value)  
  
    'get modifiers  
    ret = SapModel.PropFrame.GetModifiers("ISEC1", MyValue)  
  
    'close ETABS  
    EtabsObject.ApplicationExit(False)
```

```
'clean up variables  
  SapModel = Nothing  
  EtabsObject = Nothing  
End Sub
```

📌 See Also

Reference

cPropFrame Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LSTFA77A12_0") Method

Retrieves the names of the section property file from which an imported frame section originated, and it also retrieves the section name used in the property file.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNameInPropFile(
    string Name,
    ref string NameInFile,
    ref string FileName,
    ref string MatProp,
    ref eFramePropType PropType
)
```

```
Function GetNameInPropFile (
    Name As String,
    ByRef NameInFile As String,
    ByRef FileName As String,
    ByRef MatProp As String,
    ByRef PropType As eFramePropType
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim NameInFile As String
Dim FileName As String
Dim MatProp As String
Dim PropType As eFramePropType
Dim returnValue As Integer
```

```
returnValue = instance.GetNameInPropFile(Name,
    NameInFile, FileName, MatProp, PropType)
```

```
int GetNameInPropFile(
    String^ Name,
    String^% NameInFile,
    String^% FileName,
    String^% MatProp,
    eFramePropType% PropType
)
```

```
abstract GetNameInPropFile :
    Name : string *
    NameInFile : string byref *
    FileName : string byref *
    MatProp : string byref *
```

cPropFrameAddLanguageSpecificTextSet("LSTFA77A12_0?cpp=::|n2959;GetNa

PropType : eFramePropType byref -> int

Parameters

Name

Type:Â SystemString

The name of an existing frame section property.

NameInFile

Type:Â SystemString

The name of the specified frame section property in the frame section property file.

FileName

Type:Â SystemString

The name of the frame section property file from which the specified frame section property was obtained.

MatProp

Type:Â SystemString

The name of the material property for the section.

PropType

Type:Â ETABSv1eFramePropType

This is one of the items in the eFramePropType enumeration.

Return Value

Type:Â Int32

Returns zero if the names are successfully retrieved; otherwise it returns nonzero.

▲Remarks

If the specified frame section property was not imported, blank strings are returned for NameInFile and FileName.

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NameInFile As String
    Dim FileName As String
    Dim MatProp As String
    Dim PropType As eFramePropType

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
```

Introduction

```
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'import new frame section property
ret = SapModel.PropFrame.ImportProp("MyFrame", "A992Fy50", "Sections8.pro", "W18X35")

'get frame property name in file
ret = SapModel.PropFrame.GetNameInPropFile("MyFrame", NameInFile, FileName, MatProp, PropT

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

🔗 See Also

Reference

[cPropFrame Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LSTD11EF676_0") Method

Retrieves the names of all defined frame section properties of the specified type.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNameList(
    ref int NumberNames,
    ref string[] MyName,
    eFramePropType PropType =
)
```

```
Function GetNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String(),
    Optional PropType As eFramePropType =
) As Integer
```

```
Dim instance As cPropFrame
Dim NumberNames As Integer
Dim MyName As String()
Dim PropType As eFramePropType
Dim returnValue As Integer
```

```
returnValue = instance.GetNameList(NumberNames,
MyName, PropType)
```

```
int GetNameList(
    int% NumberNames,
    array<String^>^% MyName,
    eFramePropType PropType =
)
```

```
abstract GetNameList :
    NumberNames : int byref *
    MyName : string[] byref *
    ?PropType : eFramePropType
(* Defaults:
    let _PropType = defaultArg PropType
*)
-> int
```

Parameters

NumberNames

Type: SystemInt32

The number of frame section property names retrieved by the program.

MyName

Type: SystemString

This is a one-dimensional array of frame section property names. The MyName array is created as a dynamic, zero-based, array by the API user:

```
Dim MyName() as String
```

The array is dimensioned to (NumberNames - 1) inside the program, filled with the names, and returned to the API user.

PropType (Optional)

Type: ETABSv1eFramePropType

This optional value is one of the items in the eFramePropType enumeration.

If no value is input for PropType, names are returned for all frame section properties in the model regardless of type.

Return Value

Type: Int32

Returns zero if the names are successfully retrieved; otherwise it returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberNames As Integer
    Dim MyName() As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'get frame property names
    ret = SapModel.PropFrame.GetNameList(NumberNames, MyName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
```

```
SapModel = Nothing  
EtabsObject = Nothing  
End Sub
```

▲ See Also

Reference

cPropFrame Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST25ADC0B8_0?cpp2965=.");Ge

Method

Assigns data to a nonprismatic frame section property.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetNonPrismatic(
    string Name,
    ref int NumberItems,
    ref string[] StartSec,
    ref string[] EndSec,
    ref double[] MyLength,
    ref int[] MyType,
    ref int[] EI33,
    ref int[] EI22,
    ref int Color,
    ref string Notes,
    ref string GUID
)
```

```
Function GetNonPrismatic (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef StartSec As String(),
    ByRef EndSec As String(),
    ByRef MyLength As Double(),
    ByRef MyType As Integer(),
    ByRef EI33 As Integer(),
    ByRef EI22 As Integer(),
    ByRef Color As Integer,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim NumberItems As Integer
Dim StartSec As String()
Dim EndSec As String()
Dim MyLength As Double()
Dim MyType As Integer()
Dim EI33 As Integer()
Dim EI22 As Integer()
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

cPropFrameAddLanguageSpecificTextSet("LST25ADC0B8_0?cpp2965=.");Ge

```
returnValue = instance.GetNonPrismatic(Name,
NumberItems, StartSec, EndSec, MyLength,
MyType, EI33, EI22, Color, Notes, GUID)
```

```
int GetNonPrismatic(
String^ Name,
int% NumberItems,
array<String^>^% StartSec,
array<String^>^% EndSec,
array<double>^% MyLength,
array<int>^% MyType,
array<int>^% EI33,
array<int>^% EI22,
int% Color,
String^% Notes,
String^% GUID
)
```

```
abstract GetNonPrismatic :
    Name : string *
    NumberItems : int byref *
    StartSec : string[] byref *
    EndSec : string[] byref *
    MyLength : float[] byref *
    MyType : int[] byref *
    EI33 : int[] byref *
    EI22 : int[] byref *
    Color : int byref *
    Notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type:Â [SystemString](#)

The name of an existing frame section property.

NumberItems

Type:Â [SystemInt32](#)

The number of segments assigned to the nonprismatic section.

StartSec

Type:Â [SystemString](#)

This is an array of the names of the frame section properties at the start of each segment.

EndSec

Type:Â [SystemString](#)

This is an array of the names of the frame section properties at the end of each segment.

MyLength

Type:Â [SystemDouble](#)

This is an array that includes the length of each segment. The length may be variable or absolute as indicated by the MyType item. [L] when length is absolute.

MyType

Type:Â [SystemInt32](#)

This is an array of either 1 or 2, indicating the length type for each segment:

	Value	MyType
	1	Variable (relative length)
	2	Absolute
EI33	Type:Â <u>SystemInt32</u> This is an array of 1, 2 or 3, indicating the variation type for EI33 in each segment:	
	Value	EI33
	1	Linear
	2	Parabolic
	2	Cubic
EI22	Type:Â <u>SystemInt32</u> This is an array of 1, 2 or 3, indicating the variation type for EI22 in each segment:	
	Value	EI22
	1	Linear
	2	Parabolic
	2	Cubic
Color	Type:Â <u>SystemInt32</u>	
Notes	Type:Â <u>SystemString</u>	
GUID	Type:Â <u>SystemString</u> The GUID (global unique identifier), if any, assigned to the section.	

Return ValueType:Â Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim StartSec() As String
    Dim EndSec() As String
    Dim Length() As Double
    Dim SegType() As Integer
    Dim EI33() As Integer
    Dim EI22() As Integer
    Dim NumberItems As Integer
    Dim MyStartSec() As String
    Dim MyEndSec() As String
    Dim MyLength() As Double
    Dim MySegType() As Integer
    Dim MyEI33() As Integer
    Dim MyEI22() As Integer
```

Introduction

```
Dim Color As Integer
Dim Notes As String
Dim GUID As String

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new I-type frame section property
ret = SapModel.PropFrame.SetISection("ISEC1", "A992Fy50", 24, 10, 0.5, 0.3, 14, 0.6)

'set new I-type frame section property
ret = SapModel.PropFrame.SetISection("ISEC2", "A992Fy50", 36, 15, 1, 0.5, 121, 0.9)

'set new nonprismatic frame section property
ReDim StartSec(2)
ReDim EndSec(2)
ReDim Length(2)
ReDim SegType(2)
ReDim EI33(2)
ReDim EI22(2)
StartSec(0) = "ISEC2"
EndSec(0) = "ISEC1"
Length(0) = 60
SegType(0) = 2
EI33(0)= 2
EI22(0)= 1

StartSec(1) = "ISEC1"
EndSec(1) = "ISEC1"
Length(1) = 1
SegType(1) = 1
EI33(1)= 2
EI22(1)= 1

StartSec(2) = "ISEC1"
EndSec(2) = "ISEC2"
Length(2) = 60
SegType(2) = 2
EI33(2)= 2
EI22(2)= 1

ret = SapModel.PropFrame.SetNonPrismatic("NP1", 3, StartSec, EndSec, Length, SegType, EI33)

'get nonprismatic section data
ret = SapModel.PropFrame.GetNonPrismatic("NP1", MyNumberItems, MyStartSec, MyEndSec, MyLength)

'close ETABS
EtabsObject.ApplicationExit(False)
```

```
'clean up variables  
  SapModel = Nothing  
  EtabsObject = Nothing  
End Sub
```

See Also

Reference

cPropFrame Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST59882A01_0?cpp=:2079.");GetPipe

Method

Retrieves frame section property data for a pipe-type frame section.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPipe(
    string Name,
    ref string FileName,
    ref string MatProp,
    ref double T3,
    ref double TW,
    ref int Color,
    ref string Notes,
    ref string GUID
)
```

```
Function GetPipe (
    Name As String,
    ByRef FileName As String,
    ByRef MatProp As String,
    ByRef T3 As Double,
    ByRef TW As Double,
    ByRef Color As Integer,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim FileName As String
Dim MatProp As String
Dim T3 As Double
Dim TW As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.GetPipe(Name, FileName,
    MatProp, T3, TW, Color, Notes, GUID)
```

```
int GetPipe(
    String^ Name,
    String^% FileName,
    String^% MatProp,
    double% T3,
```

cPropFrameid="LST59882A01_0"AddLanguageSpecificTextSet("LST59882A01_0?cpp=:2079.");GetP

```
double% TW,
int% Color,
String^% Notes,
String^% GUID
)

abstract GetPipe :
    Name : string *
    FileName : string byref *
    MatProp : string byref *
    T3 : float byref *
    TW : float byref *
    Color : int byref *
    Notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame section.

FileName

Type:Â SystemString

If the section property was imported from a property file, this is the name of that file. If the section property was not imported, this item is blank.

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

TW

Type:Â SystemDouble

The wall thickness. [L]

Color

Type:Â SystemInt32

Notes

Type:Â SystemString

GUID

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the section.

Return Value

Type:Â Int32

Returns zero if the section property data is successfully retrieved; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
```

Introduction

```
Dim FileName As String
Dim MatProp As String
Dim t3 As Double
Dim tw As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new frame section property
ret = SapModel.PropFrame.SetPipe("PIPE1", "A992Fy50", 8, 0.375)

'get frame section property data
ret = SapModel.PropFrame.GetPipe("PIPE1", FileName, MatProp, t3, tw, Color, Notes, GUID)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPropFrame Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST1F7EA024_0") Method

Retrieves frame section property data for a plate frame section.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPlate(
    string Name,
    ref string FileName,
    ref string MatProp,
    ref double T3,
    ref double T2,
    ref int Color,
    ref string Notes,
    ref string GUID
)
```

```
Function GetPlate (
    Name As String,
    ByRef FileName As String,
    ByRef MatProp As String,
    ByRef T3 As Double,
    ByRef T2 As Double,
    ByRef Color As Integer,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim FileName As String
Dim MatProp As String
Dim T3 As Double
Dim T2 As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.GetPlate(Name,
    FileName, MatProp, T3, T2, Color, Notes,
    GUID)
```

```
int GetPlate(
    String^ Name,
    String^% FileName,
    String^% MatProp,
```

cPropFramespan id="LST1F7EA024_0"AddLanguageSpecificTextSet("LST1F7EA024_0?cpp=2070.");GetL

```
double% T3,
double% T2,
int% Color,
String^% Notes,
String^% GUID
)
```

```
abstract GetPlate :
    Name : string *
    FileName : string byref *
    MatProp : string byref *
    T3 : float byref *
    T2 : float byref *
    Color : int byref *
    Notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type:Â [SystemString](#)

The name of an existing plate frame section property.

FileName

Type:Â [SystemString](#)

If the section property was imported from a property file, this is the name of that file. If the section property was not imported, this item is blank.

MatProp

Type:Â [SystemString](#)

The name of the material property for the section.

T3

Type:Â [SystemDouble](#)

T2

Type:Â [SystemDouble](#)

Color

Type:Â [SystemInt32](#)

Notes

Type:Â [SystemString](#)

GUID

Type:Â [SystemString](#)

The GUID (global unique identifier), if any, assigned to the section.

Return Value

Type:Â [Int32](#)

Returns zero if the section property is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
```

Introduction

```
Dim FileName As String
Dim MatProp As String
Dim t3 As Double
Dim t2 As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new frame section property
ret = SapModel.PropFrame.SetPlate("R1", "A992Fy50", 20, 12)

'get frame section property data
ret = SapModel.PropFrame.GetPlate("R1", FileName, MatProp, t3, t2, Color, Notes, GUID)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPropFrame Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LSTFAD1E58F_0?cpp=2976=.");Get

Method

Retrieves the names of all defined frame section properties of a specified type in a specified frame section property file.

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetPropFileNameList(
    string FileName,
    ref int NumberNames,
    ref string[] MyName,
    ref eFramePropType[] MyPropType,
    eFramePropType PropType =
)
```

```
Function GetPropFileNameList (
    FileName As String,
    ByRef NumberNames As Integer,
    ByRef MyName As String(),
    ByRef MyPropType As eFramePropType(),
    PropType As eFramePropType =
) As Integer
```

```
Dim instance As cPropFrame
Dim FileName As String
Dim NumberNames As Integer
Dim MyName As String()
Dim MyPropType As eFramePropType()
Dim PropType As eFramePropType
Dim returnValue As Integer
```

```
returnValue = instance.GetPropFileNameList(FileName,
    NumberNames, MyName, MyPropType,
    PropType)
```

```
int GetPropFileNameList(
    String^ FileName,
    int% NumberNames,
    array<String^>^% MyName,
    array<eFramePropType>^% MyPropType,
    eFramePropType PropType =
)
```

```
abstract GetPropFileNameList :
    FileName : string *
    NumberNames : int byref *
    MyName : string[] byref *
```

cPropFramespan id="LSTFAD1E58F_0"AddLanguageSpecificTextSet("LSTFAD1E58F_0?cpp=2976=.");Get

Introduction

```
MyPropType : eFramePropType[] byref *
?PropType : eFramePropType
(* Defaults:
    let _PropType = defaultArg PropType
*)
-> int
```

Parameters

FileName

Type:Â SystemString

The name of the frame section property file from which to get the name list.

In most cases, inputting only the name of the property file (e.g. Sections8.pro) is required, and the program will be able to find it. In some cases, inputting the full path to the property file may be necessary.

NumberNames

Type:Â SystemInt32

The number of frame section property names retrieved by the program.

MyName

Type:Â SystemString

This is an array that includes the property names obtained from the frame section property file.

MyPropType

Type:Â ETABSv1eFramePropType

This is an array that includes the property type for each property obtained from the frame section property file.

PropType (Optional)

Type:Â ETABSv1eFramePropType

This optional value is one of the items in the eFramePropType enumeration.

If no value is input for PropType, names are returned for all frame section properties in the specified file regardless of type.

Return Value

Type:Â Int32

Returns zero if the names are successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberNames As Integer
    Dim MyName() As String
    Dim MyPropType() As eFramePropType

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

Introduction

```
'start ETABS application
    ret = EtabsObject.ApplicationStart()

'create SapModel object
    SapModel = EtabsObject.SapModel

'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'get frame section property names
    ret = SapModel.PropFrame.GetPropFileNameList("Sections8.pro", NumberNames, MyName, MyPropT

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPropFrame Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST55DAC1F3_0?cpp=2979=");GetRebarBeam Method

Retrieves beam rebar data for frame sections.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetRebarBeam(
    string Name,
    ref string MatPropLong,
    ref string MatPropConfine,
    ref double CoverTop,
    ref double CoverBot,
    ref double TopLeftArea,
    ref double TopRightArea,
    ref double BotLeftArea,
    ref double BotRightArea
)
```

```
Function GetRebarBeam (
    Name As String,
    ByRef MatPropLong As String,
    ByRef MatPropConfine As String,
    ByRef CoverTop As Double,
    ByRef CoverBot As Double,
    ByRef TopLeftArea As Double,
    ByRef TopRightArea As Double,
    ByRef BotLeftArea As Double,
    ByRef BotRightArea As Double
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim MatPropLong As String
Dim MatPropConfine As String
Dim CoverTop As Double
Dim CoverBot As Double
Dim TopLeftArea As Double
Dim TopRightArea As Double
Dim BotLeftArea As Double
Dim BotRightArea As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetRebarBeam(Name,
    MatPropLong, MatPropConfine, CoverTop,
    CoverBot, TopLeftArea, TopRightArea,
    BotLeftArea, BotRightArea)
```

```
int GetRebarBeam(
String^ Name,
String^% MatPropLong,
String^% MatPropConfine,
double% CoverTop,
double% CoverBot,
double% TopLeftArea,
double% TopRightArea,
double% BotLeftArea,
double% BotRightArea
)
```

```
abstract GetRebarBeam :
    Name : string *
    MatPropLong : string byref *
    MatPropConfine : string byref *
    CoverTop : float byref *
    CoverBot : float byref *
    TopLeftArea : float byref *
    TopRightArea : float byref *
    BotLeftArea : float byref *
    BotRightArea : float byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame section property.

MatPropLong

Type:Â SystemString

The name of the rebar material property for the longitudinal rebar.

MatPropConfine

Type:Â SystemString

The name of the rebar material property for the confinement rebar.

CoverTop

Type:Â SystemDouble

The distance from the top of the beam to the centroid of the top longitudinal reinforcement. [L]

CoverBot

Type:Â SystemDouble

The distance from the bottom of the beam to the centroid of the bottom longitudinal reinforcement. [L]

TopLeftArea

Type:Â SystemDouble

The total area of longitudinal reinforcement at the top left end of the beam. [L2]

TopRightArea

Type:Â SystemDouble

The total area of longitudinal reinforcement at the top right end of the beam. [L2]

BotLeftArea

Type:Â SystemDouble

The total area of longitudinal reinforcement at the bottom left end of the beam. [L2]

BotRightArea

Type: SystemDouble

The total area of longitudinal reinforcement at the bottom right end of the beam. [L2]

Return Value

Type: Int32

Returns zero if the rebar data is successfully retrieved; otherwise it returns a nonzero value.

Remarks

This function applies only to the following section types. Calling this function for any other type of frame section property returns an error.

- I
- Angle
- Rectangular
- Circle

The material assigned to the specified frame section property must be concrete or this function returns an error.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim RebarName As String
    Dim MatPropLong As String
    Dim MatPropConfine As String
    Dim CoverTop As Double
    Dim CoverBot As Double
    Dim TopLeftArea As Double
    Dim TopRightArea As Double
    Dim BotLeftArea As Double
    Dim BotRightArea As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new frame section property
    ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)
```

Introduction

```
'add ASTM A706 rebar material
    ret = SapModel.PropMaterial.AddQuick(RebarName, eMatType.Rebar, , , , eMatTypeRebar.ASTM

'set beam rebar data
    ret = SapModel.PropFrame.SetRebarBeam("R1", RebarName, RebarName, 3.5, 3, 4.1, 4.2, 4.3, 4

'get beam rebar data
    ret = SapModel.PropFrame.GetRebarBeam("R1", MatPropLong, MatPropConfine, CoverTop, CoverBo

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPropFrame Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST73D2E74D_0?cpp=2980=.");Get Method

Retrieves column rebar data for frame sections.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetRebarColumn(
    string Name,
    ref string MatPropLong,
    ref string MatPropConfine,
    ref int Pattern,
    ref int ConfineType,
    ref double Cover,
    ref int NumberCBars,
    ref int NumberR3Bars,
    ref int NumberR2Bars,
    ref string RebarSize,
    ref string TieSize,
    ref double TieSpacingLongit,
    ref int Number2DirTieBars,
    ref int Number3DirTieBars,
    ref bool ToBeDesigned
)
```

```
Function GetRebarColumn (
    Name As String,
    ByRef MatPropLong As String,
    ByRef MatPropConfine As String,
    ByRef Pattern As Integer,
    ByRef ConfineType As Integer,
    ByRef Cover As Double,
    ByRef NumberCBars As Integer,
    ByRef NumberR3Bars As Integer,
    ByRef NumberR2Bars As Integer,
    ByRef RebarSize As String,
    ByRef TieSize As String,
    ByRef TieSpacingLongit As Double,
    ByRef Number2DirTieBars As Integer,
    ByRef Number3DirTieBars As Integer,
    ByRef ToBeDesigned As Boolean
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim MatPropLong As String
Dim MatPropConfine As String
Dim Pattern As Integer
```

cPropFramespan id="LST73D2E74D_0"AddLanguageSpecificTextSet("LST73D2E74D_0?cpp=2980=.");Get

```

Dim ConfineType As Integer
Dim Cover As Double
Dim NumberCBars As Integer
Dim NumberR3Bars As Integer
Dim NumberR2Bars As Integer
Dim RebarSize As String
Dim TieSize As String
Dim TieSpacingLongit As Double
Dim Number2DirTieBars As Integer
Dim Number3DirTieBars As Integer
Dim ToBeDesigned As Boolean
Dim returnValue As Integer

returnValue = instance.GetRebarColumn(Name,
MatPropLong, MatPropConfine, Pattern,
ConfineType, Cover, NumberCBars,
NumberR3Bars, NumberR2Bars, RebarSize,
TieSize, TieSpacingLongit, Number2DirTieBars,
Number3DirTieBars, ToBeDesigned)

int GetRebarColumn(
String^ Name,
String^% MatPropLong,
String^% MatPropConfine,
int% Pattern,
int% ConfineType,
double% Cover,
int% NumberCBars,
int% NumberR3Bars,
int% NumberR2Bars,
String^% RebarSize,
String^% TieSize,
double% TieSpacingLongit,
int% Number2DirTieBars,
int% Number3DirTieBars,
bool% ToBeDesigned
)

abstract GetRebarColumn :
    Name : string *
    MatPropLong : string byref *
    MatPropConfine : string byref *
    Pattern : int byref *
    ConfineType : int byref *
    Cover : float byref *
    NumberCBars : int byref *
    NumberR3Bars : int byref *
    NumberR2Bars : int byref *
    RebarSize : string byref *
    TieSize : string byref *
    TieSpacingLongit : float byref *
    Number2DirTieBars : int byref *
    Number3DirTieBars : int byref *
    ToBeDesigned : bool byref -> int

```

Parameters

Name

Type: SystemString

The name of an existing frame section property.

MatPropLongType: SystemString

The name of the rebar material property for the longitudinal rebar.

MatPropConfineType: SystemString

The name of the rebar material property for the confinement rebar.

PatternType: SystemInt32

This is either 1 or 2, indicating the rebar configuration.

Value Pattern

1 Rectangular

2 Circular

For circular frame section properties this item must be 2; otherwise an error is returned.

ConfineTypeType: SystemInt32

This is either 1 or 2, indicating the confinement bar type.

Value Type

1 Ties

2 Spiral

This item applies only when Pattern = 2. If Pattern = 1, the confinement bar type is assumed to be ties.

CoverType: SystemDouble

The clear cover for the confinement steel (ties). In the special case of circular reinforcement in a rectangular column, this is the minimum clear cover. [L]

NumberCBarsType: SystemInt32

This item applies to a circular rebar configuration, Pattern = 2. It is the total number of longitudinal reinforcing bars in the column.

NumberR3BarsType: SystemInt32

This item applies to a rectangular rebar configuration, Pattern = 1. It is the number of longitudinal bars (including the corner bar) on each face of the column that is parallel to the local 3-axis of the column.

NumberR2BarsType: SystemInt32

This item applies to a rectangular rebar configuration, Pattern = 1. It is the number of longitudinal bars (including the corner bar) on each face of the column that is parallel to the local 2-axis of the column.

RebarSizeType: SystemString

The rebar name for the longitudinal rebar in the column.

TieSizeType: SystemString

The rebar name for the confinement rebar in the column.

TieSpacingLongitType: SystemDouble

The longitudinal spacing of the confinement bars (ties). [L]

Number2DirTieBarsType: SystemInt32

This item applies to a rectangular reinforcing configuration, Pattern = 1. It is the number of confinement bars (tie legs) running in the local 2-axis direction of the column.

Number3DirTieBarsType: SystemInt32

This item applies to a rectangular reinforcing configuration, Pattern = 1. It is the number of confinement bars (tie legs) running in the local 3-axis direction of the column.

ToBeDesignedType: SystemBoolean

If this item is True, the column longitudinal rebar is to be designed; otherwise it is to be checked.

Return ValueType: Int32

Returns zero if the rebar data is successfully retrieved; otherwise it returns a nonzero value.

Remarks

This function applies only to the following section types. Calling this function for any other type of frame section property returns an error.

- Rectangular
- Circle

The material assigned to the specified frame section property must be concrete or this function returns an error.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim RebarName As String
    Dim MatPropLong As String
    Dim MatPropConfine As String
    Dim Pattern As Integer
    Dim ConfineType As Integer
    Dim Cover As Double
    Dim NumberCBars As Integer
    Dim NumberR3Bars As Integer
    Dim NumberR2Bars As Integer
    Dim RebarSize As String
    Dim TieSize As String
    Dim TieSpacingLongit As Double
    Dim Number2DirTieBars As Integer
    Dim Number3DirTieBars As Integer
    Dim ToBeDesigned As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
```

Introduction

```
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new frame section property
ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

'add ASTM A706 rebar material
ret = SapModel.PropMaterial.AddQuick(RebarName, eMatType.Rebar, , , , eMatTypeRebar.ASTM

'set column rebar data
ret = SapModel.PropFrame.SetRebarColumn("R1", RebarName, RebarName, 2, 2, 2, 10, 0, 0, "#1

'get column rebar data
ret = SapModel.PropFrame.GetRebarColumn("R1", MatPropLong, MatPropConfine, Pattern, Confir

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPropFrame Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST3F94DE Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetRebarColumn_1(
    string Name,
    ref string MatPropLong,
    ref string MatPropConfine,
    ref int Pattern,
    ref int ConfineType,
    ref double Cover,
    ref int NumberCBars,
    ref int NumberR3Bars,
    ref int NumberR2Bars,
    ref string RebarSize,
    ref string TieSize,
    ref double TieSpacingLongit,
    ref int Number2DirTieBars,
    ref int Number3DirTieBars,
    ref bool ToBeDesigned,
    ref string LongitCornerRebarSize,
    ref double LongitRebarArea,
    ref double LongitCornerRebarArea
)
```

```
Function GetRebarColumn_1 (
    Name As String,
    ByRef MatPropLong As String,
    ByRef MatPropConfine As String,
    ByRef Pattern As Integer,
    ByRef ConfineType As Integer,
    ByRef Cover As Double,
    ByRef NumberCBars As Integer,
    ByRef NumberR3Bars As Integer,
    ByRef NumberR2Bars As Integer,
    ByRef RebarSize As String,
    ByRef TieSize As String,
    ByRef TieSpacingLongit As Double,
    ByRef Number2DirTieBars As Integer,
    ByRef Number3DirTieBars As Integer,
    ByRef ToBeDesigned As Boolean,
    ByRef LongitCornerRebarSize As String,
    ByRef LongitRebarArea As Double,
    ByRef LongitCornerRebarArea As Double
) As Integer
```

Dim instance As cPropFrame

cPropFramespan id="LST3F94DB50_0"AddLanguageSpecificTextSet("LST3F94DB50_0?cpp=2988-.");Get

Introduction

```
Dim Name As String
Dim MatPropLong As String
Dim MatPropConfine As String
Dim Pattern As Integer
Dim ConfineType As Integer
Dim Cover As Double
Dim NumberCBars As Integer
Dim NumberR3Bars As Integer
Dim NumberR2Bars As Integer
Dim RebarSize As String
Dim TieSize As String
Dim TieSpacingLongit As Double
Dim Number2DirTieBars As Integer
Dim Number3DirTieBars As Integer
Dim ToBeDesigned As Boolean
Dim LongitCornerRebarSize As String
Dim LongitRebarArea As Double
Dim LongitCornerRebarArea As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetRebarColumn_1(Name,
MatPropLong, MatPropConfine, Pattern,
ConfineType, Cover, NumberCBars,
NumberR3Bars, NumberR2Bars, RebarSize,
TieSize, TieSpacingLongit, Number2DirTieBars,
Number3DirTieBars, ToBeDesigned,
LongitCornerRebarSize, LongitRebarArea,
LongitCornerRebarArea)
```

```
int GetRebarColumn_1(
String^ Name,
String^% MatPropLong,
String^% MatPropConfine,
int% Pattern,
int% ConfineType,
double% Cover,
int% NumberCBars,
int% NumberR3Bars,
int% NumberR2Bars,
String^% RebarSize,
String^% TieSize,
double% TieSpacingLongit,
int% Number2DirTieBars,
int% Number3DirTieBars,
bool% ToBeDesigned,
String^% LongitCornerRebarSize,
double% LongitRebarArea,
double% LongitCornerRebarArea
)
```

```
abstract GetRebarColumn_1 :
    Name : string *
    MatPropLong : string byref *
    MatPropConfine : string byref *
    Pattern : int byref *
    ConfineType : int byref *
    Cover : float byref *
    NumberCBars : int byref *
    NumberR3Bars : int byref *
    NumberR2Bars : int byref *
    RebarSize : string byref *
    TieSize : string byref *
```

```

TieSpacingLongit : float byref *
Number2DirTieBars : int byref *
Number3DirTieBars : int byref *
ToBeDesigned : bool byref *
LongitCornerRebarSize : string byref *
LongitRebarArea : float byref *
LongitCornerRebarArea : float byref -> int

```

Parameters

Name

Type: SystemString

MatPropLong

Type: SystemString

MatPropConfine

Type: SystemString

Pattern

Type: SystemInt32

ConfineType

Type: SystemInt32

Cover

Type: SystemDouble

NumberCBars

Type: SystemInt32

NumberR3Bars

Type: SystemInt32

NumberR2Bars

Type: SystemInt32

RebarSize

Type: SystemString

TieSize

Type: SystemString

TieSpacingLongit

Type: SystemDouble

Number2DirTieBars

Type: SystemInt32

Number3DirTieBars

Type: SystemInt32

ToBeDesigned

Type: SystemBoolean

LongitCornerRebarSize

Type: SystemString

LongitRebarArea

Type: SystemDouble

LongitCornerRebarArea

Type: SystemDouble

Return Value

Type: Int32

▲ See Also

Reference

cPropFrame Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LSTEF9C9D3E_0") Method

Retrieves frame section property data for a rectangular frame section.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetRectangle(
    string Name,
    ref string FileName,
    ref string MatProp,
    ref double T3,
    ref double T2,
    ref int Color,
    ref string Notes,
    ref string GUID
)
```

```
Function GetRectangle (
    Name As String,
    ByRef FileName As String,
    ByRef MatProp As String,
    ByRef T3 As Double,
    ByRef T2 As Double,
    ByRef Color As Integer,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim FileName As String
Dim MatProp As String
Dim T3 As Double
Dim T2 As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.GetRectangle(Name,
    FileName, MatProp, T3, T2, Color, Notes,
    GUID)
```

```
int GetRectangle(
    String^ Name,
    String^% FileName,
    String^% MatProp,
```

cPropFramespan id="LSTEF9C9D3E_0"AddLanguageSpecificTextSet("LSTEF9C9D3E_0?cpp2992u=.");Ge

```
double% T3,
double% T2,
int% Color,
String^% Notes,
String^% GUID
)

abstract GetRectangle :
    Name : string *
    FileName : string byref *
    MatProp : string byref *
    T3 : float byref *
    T2 : float byref *
    Color : int byref *
    Notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing or new frame section.

FileName

Type:Â SystemString

If the section property was imported from a property file, this is the name of that file. If the section property was not imported, this item is blank.

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

T2

Type:Â SystemDouble

Color

Type:Â SystemInt32

Notes

Type:Â SystemString

GUID

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the section.

Return Value

Type:Â Int32

Returns zero if the section property is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
```

Introduction

```
Dim FileName As String
Dim MatProp As String
Dim t3 As Double
Dim t2 As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new frame section property
ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

'get frame section property data
ret = SapModel.PropFrame.GetRectangle("R1", FileName, MatProp, t3, t2, Color, Notes, GUID)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPropFrame Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST40E89F23_0?cpp=:2005.");GetF

Method

Retrieves frame section property data for a rod frame section.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetRod(
    string Name,
    ref string FileName,
    ref string MatProp,
    ref double T3,
    ref int Color,
    ref string Notes,
    ref string GUID
)
```

```
Function GetRod (
    Name As String,
    ByRef FileName As String,
    ByRef MatProp As String,
    ByRef T3 As Double,
    ByRef Color As Integer,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim FileName As String
Dim MatProp As String
Dim T3 As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.GetRod(Name, FileName,
    MatProp, T3, Color, Notes, GUID)
```

```
int GetRod(
    String^ Name,
    String^% FileName,
    String^% MatProp,
    double% T3,
    int% Color,
    String^% Notes,
    String^% GUID
```

cPropFramespan id="LST40E89F23_0"AddLanguageSpecificTextSet("LST40E89F23_0?cpp=:2005.");GetF

)

```

abstract GetRod :
    Name : string *
    FileName : string byref *
    MatProp : string byref *
    T3 : float byref *
    Color : int byref *
    Notes : string byref *
    GUID : string byref -> int

```

Parameters

Name

Type:Â SystemString

The name of an existing rod frame section property.

FileName

Type:Â SystemString

If the section property was imported from a property file, this is the name of that file. If the section property was not imported, this item is blank.

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

Color

Type:Â SystemInt32

Notes

Type:Â SystemString

GUID

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the section.

Return Value

Type:Â Int32

Returns zero if the section property data is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim FileName As String
    Dim MatProp As String
    Dim t3 As Double
    Dim Color As Integer
    Dim Notes As String
    Dim GUID As String

```

'create ETABS object

Introduction

```
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new frame section property
ret = SapModel.PropFrame.SetRod("R1", "A992Fy50", 20)

'get frame section property data
ret = SapModel.PropFrame.GetRod("R1", FileName, MatProp, t3, Color, Notes, GUID)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPropFrame Interface](#)

[ETABSv1 Namespace](#)

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»

CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST8760023D_0") Method

Retrieves section property data for a section designer section

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetSDSection(
    string Name,
    ref string MatProp,
    ref int NumberItems,
    ref string[] ShapeName,
    ref int[] MyType,
    ref int DesignType,
    ref int Color,
    ref string Notes,
    ref string GUID
)
```

```
Function GetSDSection (
    Name As String,
    ByRef MatProp As String,
    ByRef NumberItems As Integer,
    ByRef ShapeName As String(),
    ByRef MyType As Integer(),
    ByRef DesignType As Integer,
    ByRef Color As Integer,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim MatProp As String
Dim NumberItems As Integer
Dim ShapeName As String()
Dim MyType As Integer()
Dim DesignType As Integer
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.GetSDSection(Name,
    MatProp, NumberItems, ShapeName,
    MyType, DesignType, Color, Notes,
    GUID)
```

cPropFrameid="LST8760023D_0"AddLanguageSpecificTextSet("LST8760023D_0?cpp=:2008.");GetS

```

int GetSDSection(
String^ Name,
String^% MatProp,
int% NumberItems,
array<String^>^% ShapeName,
array<int>^% MyType,
int% DesignType,
int% Color,
String^% Notes,
String^% GUID
)

abstract GetSDSection :
    Name : string *
    MatProp : string byref *
    NumberItems : int byref *
    ShapeName : string[] byref *
    MyType : int[] byref *
    DesignType : int byref *
    Color : int byref *
    Notes : string byref *
    GUID : string byref -> int

```

Parameters

Name

Type:Â SystemString

The name of an existing section designer property

MatProp

Type:Â SystemString

The name of the base material property for the section

NumberItems

Type:Â SystemInt32

The number of shapes defined in the section designer section

ShapeName

Type:Â SystemString

This is an array that includes the name of each shape in the section designer section

MyType

Type:Â SystemInt32

This is an array that includes the type of each shape in the section designer section

- 1 I-Section
- 2 Channel
- 3 Tee
- 4 Angle
- 5 Double Angle
- 6 Box
- 7 Pipe
- 8 Plate
- 61 Concrete Tee
- 62 Concrete L

- 63 Concrete Box
- 64 Concrete Pipe
- 65 Concrete Cross
- 101 Solid Rectangle
- 102 Solid Circle
- 103 Solid Segment
- 104 Solid Sector
- 201 Polygon
- 301 Reinforcing Single
- 302 Reinforcing Line
- 303 Reinforcing Rectangle
- 304 Reinforcing Circle
- 401 Reference Line
- 402 Reference Circle

DesignType

Type: SystemInt32

This is 0, 1, 2 or 3, indicating the design option for the section

- 0 No design
- 1 Design as general steel section
- 2 Design as a concrete column; check the reinforcing
- 3 Design as a concrete column; design the reinforcing

Color

Type: SystemInt32

The display color assigned to the section

Notes

Type: SystemString

The notes, if any, assigned to the section

GUID

Type: SystemString

The GUID (global unique identifier), if any, assigned to the section

Return Value

Type: Int32

Returns zero if the section property data is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ See Also

Reference

cPropFrame Interface

ETABSv1 Namespace

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Introduction

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST2BC179DE_0?cpp3002=.");Ge Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetSectProps(
    string Name,
    ref double Area,
    ref double As2,
    ref double As3,
    ref double Torsion,
    ref double I22,
    ref double I33,
    ref double S22,
    ref double S33,
    ref double Z22,
    ref double Z33,
    ref double R22,
    ref double R33
)
```

```
Function GetSectProps (
    Name As String,
    ByRef Area As Double,
    ByRef As2 As Double,
    ByRef As3 As Double,
    ByRef Torsion As Double,
    ByRef I22 As Double,
    ByRef I33 As Double,
    ByRef S22 As Double,
    ByRef S33 As Double,
    ByRef Z22 As Double,
    ByRef Z33 As Double,
    ByRef R22 As Double,
    ByRef R33 As Double
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim Area As Double
Dim As2 As Double
Dim As3 As Double
Dim Torsion As Double
Dim I22 As Double
Dim I33 As Double
Dim S22 As Double
Dim S33 As Double
Dim Z22 As Double
```

cPropFramespan id="LST2BC179DE_0"AddLanguageSpecificTextSet("LST2BC179DE_0?cpp3002=.");Ge

Introduction

```
Dim Z33 As Double
Dim R22 As Double
Dim R33 As Double
Dim returnValue As Integer

returnValue = instance.GetSectProps(Name,
Area, As2, As3, Torsion, I22, I33, S22,
S33, Z22, Z33, R22, R33)

int GetSectProps(
String^ Name,
double% Area,
double% As2,
double% As3,
double% Torsion,
double% I22,
double% I33,
double% S22,
double% S33,
double% Z22,
double% Z33,
double% R22,
double% R33
)

abstract GetSectProps :
    Name : string *
    Area : float byref *
    As2 : float byref *
    As3 : float byref *
    Torsion : float byref *
    I22 : float byref *
    I33 : float byref *
    S22 : float byref *
    S33 : float byref *
    Z22 : float byref *
    Z33 : float byref *
    R22 : float byref *
    R33 : float byref -> int
```

Parameters

Name	Type:Â <u>SystemString</u>
Area	Type:Â <u>SystemDouble</u>
As2	Type:Â <u>SystemDouble</u>
As3	Type:Â <u>SystemDouble</u>
Torsion	Type:Â <u>SystemDouble</u>
I22	Type:Â <u>SystemDouble</u>
I33	Type:Â <u>SystemDouble</u>
S22	Type:Â <u>SystemDouble</u>

S33 Type:Â SystemDouble
Z22 Type:Â SystemDouble
Z33 Type:Â SystemDouble
R22 Type:Â SystemDouble
R33 Type:Â SystemDouble

Return Value

Type:Â Int32

▲ See Also

Reference

cPropFrame Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST517414B_0?cpp=::|n3905GetSteMethod

Retrieves frame section property data for a steel angle frame section.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetSteelAngle(
    string Name,
    ref string FileName,
    ref string MatProp,
    ref double T3,
    ref double T2,
    ref double Tf,
    ref double Tw,
    ref double r,
    ref bool MirrorAbout2,
    ref bool MirrorAbout3,
    ref int Color,
    ref string Notes,
    ref string GUID
)
```

```
Function GetSteelAngle (
    Name As String,
    ByRef FileName As String,
    ByRef MatProp As String,
    ByRef T3 As Double,
    ByRef T2 As Double,
    ByRef Tf As Double,
    ByRef Tw As Double,
    ByRef r As Double,
    ByRef MirrorAbout2 As Boolean,
    ByRef MirrorAbout3 As Boolean,
    ByRef Color As Integer,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim FileName As String
Dim MatProp As String
Dim T3 As Double
Dim T2 As Double
Dim Tf As Double
Dim Tw As Double
Dim r As Double
```

cPropFrameid="LST517414B_0"AddLanguageSpecificTextSet("LST517414B_0?cpp=::|n3905GetSte

Introduction

```
Dim MirrorAbout2 As Boolean
Dim MirrorAbout3 As Boolean
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer

returnValue = instance.GetSteelAngle(Name,
FileName, MatProp, T3, T2, Tf, Tw, r,
MirrorAbout2, MirrorAbout3, Color,
Notes, GUID)

int GetSteelAngle(
String^ Name,
String^% FileName,
String^% MatProp,
double% T3,
double% T2,
double% Tf,
double% Tw,
double% r,
bool% MirrorAbout2,
bool% MirrorAbout3,
int% Color,
String^% Notes,
String^% GUID
)

abstract GetSteelAngle :
    Name : string *
    FileName : string byref *
    MatProp : string byref *
    T3 : float byref *
    T2 : float byref *
    Tf : float byref *
    Tw : float byref *
    r : float byref *
    MirrorAbout2 : bool byref *
    MirrorAbout3 : bool byref *
    Color : int byref *
    Notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame section property.

FileName

Type:Â SystemString

If the section property was imported from a property file, this is the name of that file. If the section property was not imported, this item is blank.

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

T2

Tf	Type:Â <u>SystemDouble</u>
Tw	Type:Â <u>SystemDouble</u>
r	Type:Â <u>SystemDouble</u> The fillet radius. [L]
MirrorAbout2	Type:Â <u>SystemBoolean</u> Indicates whether the section is mirrored about the local 2-axis.
MirrorAbout3	Type:Â <u>SystemBoolean</u> Indicates whether the section is mirrored about the local 3-axis.
Color	Type:Â <u>SystemInt32</u>
Notes	Type:Â <u>SystemString</u>
GUID	Type:Â <u>SystemString</u> The GUID (global unique identifier), if any, assigned to the section.

Return Value

Type:Â Int32

returns zero if the section property data is successfully retrieved; otherwise it returns a nonzero value.

Remarks

This function supersedes GetAngle(String, String, String, Double, Double, Double, Double, Int32, String, String).

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim FileName As String
    Dim MatProp As String
    Dim t3 As Double
    Dim t2 As Double
    Dim tf As Double
    Dim tw As Double
    Dim r As Double
    Dim mirror2 As Boolean
    Dim mirror3 As Boolean
    Dim Color As Integer
    Dim Notes As String
    Dim GUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

Introduction

```
'start ETABS application
    ret = EtabsObject.ApplicationStart()

'create SapModel object
    SapModel = EtabsObject.SapModel

'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new frame section property
    ret = SapModel.PropFrame.SetSteelAngle("ANGLE1", "A992Fy50", 6, 4, 0.5, 0.5, 0.5, False, F

'get frame section property data
    ret = SapModel.PropFrame.GetSteelAngle("ANGLE1", FileName, MatProp, t3, t2, tf, tw, r, mi

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

🔗 See Also

Reference

[cPropFrame Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LSTAAA83B95_0?cpp=3000=.");GetSteelTee Method

Retrieves frame section property data for a steel tee frame section.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetSteelTee(
    string Name,
    ref string FileName,
    ref string MatProp,
    ref double T3,
    ref double T2,
    ref double Tf,
    ref double Tw,
    ref double r,
    ref bool MirrorAbout3,
    ref int Color,
    ref string Notes,
    ref string GUID
)
```

```
Function GetSteelTee (
    Name As String,
    ByRef FileName As String,
    ByRef MatProp As String,
    ByRef T3 As Double,
    ByRef T2 As Double,
    ByRef Tf As Double,
    ByRef Tw As Double,
    ByRef r As Double,
    ByRef MirrorAbout3 As Boolean,
    ByRef Color As Integer,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim FileName As String
Dim MatProp As String
Dim T3 As Double
Dim T2 As Double
Dim Tf As Double
Dim Tw As Double
Dim r As Double
Dim MirrorAbout3 As Boolean
Dim Color As Integer
```

cPropFrameid="LSTAAA83B95_0"AddLanguageSpecificTextSet("LSTAAA83B95_0?cpp=3000=.");GetSteelTee

Introduction

```
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer

returnValue = instance.GetSteelTee(Name,
FileName, MatProp, T3, T2, Tf, Tw, r,
MirrorAbout3, Color, Notes, GUID)

int GetSteelTee(
String^ Name,
String^% FileName,
String^% MatProp,
double% T3,
double% T2,
double% Tf,
double% Tw,
double% r,
bool% MirrorAbout3,
int% Color,
String^% Notes,
String^% GUID
)

abstract GetSteelTee :
    Name : string *
    FileName : string byref *
    MatProp : string byref *
    T3 : float byref *
    T2 : float byref *
    Tf : float byref *
    Tw : float byref *
    r : float byref *
    MirrorAbout3 : bool byref *
    Color : int byref *
    Notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame section property.

FileName

Type:Â SystemString

If the section property was imported from a property file, this is the name of that file. If the section property was not imported, this item is blank.

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

T2

Type:Â SystemDouble

Tf

Type:Â SystemDouble

Tw

Type:Â SystemDouble

r

Type:Â SystemDouble

The fillet radius. [L]

MirrorAbout3

Type:Â SystemBoolean

Indicates whether the section is mirrored about the local 3-axis.

Color

Type:Â SystemInt32

Notes

Type:Â SystemString

GUID

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the section.

Return ValueType:Â Int32

Returns zero if the section property is successfully retrieved; otherwise it returns a nonzero value.

▲ **Remarks**This function supersedes GetTee(String, String, String, Double, Double, Double, Double, Int32, String, String).▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim FileName As String
    Dim MatProp As String
    Dim t3 As Double
    Dim t2 As Double
    Dim tf As Double
    Dim tw As Double
    Dim r As Double
    Dim mirror3
    Dim Color As Integer
    Dim Notes As String
    Dim GUID As String

    'create ETABS object
    Dim myHelper As cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)
```

Introduction

```
'set new frame section property
  ret = SapModel.PropFrame.SetSteelTee("TEE1", "A992Fy50", 12, 10, 0.6, 0.3, 0.3, False)

'get frame section property data
  ret = SapModel.PropFrame.GetSteelTee("TEE1", FileName, MatProp, t3, t2, tf, tw, r, mirror3

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPropFrame Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST9C653166_0?cpp=:3013.");GetT

Method

Retrieves frame section property data for a tee-type frame section.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetTee(
    string Name,
    ref string FileName,
    ref string MatProp,
    ref double T3,
    ref double T2,
    ref double Tf,
    ref double Tw,
    ref int Color,
    ref string Notes,
    ref string GUID
)
```

```
Function GetTee (
    Name As String,
    ByRef FileName As String,
    ByRef MatProp As String,
    ByRef T3 As Double,
    ByRef T2 As Double,
    ByRef Tf As Double,
    ByRef Tw As Double,
    ByRef Color As Integer,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim FileName As String
Dim MatProp As String
Dim T3 As Double
Dim T2 As Double
Dim Tf As Double
Dim Tw As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.GetTee(Name, FileName,
    MatProp, T3, T2, Tf, Tw, Color, Notes,
```

cPropFramespan id="LST9C653166_0"AddLanguageSpecificTextSet("LST9C653166_0?cpp=:3013.");GetT

GUID)

```
int GetTee(
String^ Name,
String^% FileName,
String^% MatProp,
double% T3,
double% T2,
double% Tf,
double% Tw,
int% Color,
String^% Notes,
String^% GUID
)
```

```
abstract GetTee :
    Name : string *
    FileName : string byref *
    MatProp : string byref *
    T3 : float byref *
    T2 : float byref *
    Tf : float byref *
    Tw : float byref *
    Color : int byref *
    Notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame section property.

FileName

Type:Â SystemString

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

T2

Type:Â SystemDouble

Tf

Type:Â SystemDouble

Tw

Type:Â SystemDouble

Color

Type:Â SystemInt32

Notes

Type:Â SystemString

GUID

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the section.

Return Value

Type: Int32

Returns zero if the section property is successfully retrieved; otherwise it returns a nonzero value.

Remarks

This function is obsolete and has been superseded by GetSteelTee(String, String, String, Double, Double, Double, Double, Double, Boolean, Int32, String, String) and GetConcreteTee(String, String, String, Double, Double, Double, Double, Double, Boolean, Int32, String, String) as of version 16.0.0. This function is maintained for backwards compatibility.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim FileName As String
    Dim MatProp As String
    Dim t3 As Double
    Dim t2 As Double
    Dim tf As Double
    Dim tw As Double
    Dim Color As Integer
    Dim Notes As String
    Dim GUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new frame section property
    ret = SapModel.PropFrame.SetTee("TEE1", "A992Fy50", 12, 10, 0.6, 0.3)

    'get frame section property data
    ret = SapModel.PropFrame.GetTee("TEE1", FileName, MatProp, t3, t2, tf, tw, Color, Notes, GUID)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

cPropFrame Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST69539D2B_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetTrapezoidal(
    string Name,
    ref string FileName,
    ref string MatProp,
    ref double T3,
    ref double T2,
    ref double T2b,
    ref int Color,
    ref string Notes,
    ref string GUID
)
```

```
Function GetTrapezoidal (
    Name As String,
    ByRef FileName As String,
    ByRef MatProp As String,
    ByRef T3 As Double,
    ByRef T2 As Double,
    ByRef T2b As Double,
    ByRef Color As Integer,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim FileName As String
Dim MatProp As String
Dim T3 As Double
Dim T2 As Double
Dim T2b As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.GetTrapezoidal(Name,
    FileName, MatProp, T3, T2, T2b, Color,
    Notes, GUID)
```

```
int GetTrapezoidal(
    String^ Name,
```

cPropFramespan id="LST69539D2B_0"AddLanguageSpecificTextSet("LST69539D2B_0?cpp=3017.");GetT

```

String^% FileName,
String^% MatProp,
double% T3,
double% T2,
double% T2b,
int% Color,
String^% Notes,
String^% GUID
)

abstract GetTrapezoidal :
    Name : string *
    FileName : string byref *
    MatProp : string byref *
    T3 : float byref *
    T2 : float byref *
    T2b : float byref *
    Color : int byref *
    Notes : string byref *
    GUID : string byref -> int

```

Parameters

Name
Type: [SystemString](#)

FileName
Type: [SystemString](#)

MatProp
Type: [SystemString](#)

T3
Type: [SystemDouble](#)

T2
Type: [SystemDouble](#)

T2b
Type: [SystemDouble](#)

Color
Type: [SystemInt32](#)

Notes
Type: [SystemString](#)

GUID
Type: [SystemString](#)

Return Value

Type: [Int32](#)

See Also

Reference

[cPropFrame Interface](#)

[ETABSv1 Namespace](#)

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Introduction

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST66EB16C3_0?cpp=3020-");Get

Method

Retrieves frame section property data for a tube-type frame section.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetTube(
    string Name,
    ref string FileName,
    ref string MatProp,
    ref double T3,
    ref double T2,
    ref double Tf,
    ref double Tw,
    ref int Color,
    ref string Notes,
    ref string GUID
)
```

```
Function GetTube (
    Name As String,
    ByRef FileName As String,
    ByRef MatProp As String,
    ByRef T3 As Double,
    ByRef T2 As Double,
    ByRef Tf As Double,
    ByRef Tw As Double,
    ByRef Color As Integer,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim FileName As String
Dim MatProp As String
Dim T3 As Double
Dim T2 As Double
Dim Tf As Double
Dim Tw As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.GetTube(Name, FileName,
    MatProp, T3, T2, Tf, Tw, Color, Notes,
```

cPropFramespan id="LST66EB16C3_0"AddLanguageSpecificTextSet("LST66EB16C3_0?cpp=3020-");Get

GUID)

```
int GetTube(
String^ Name,
String^% FileName,
String^% MatProp,
double% T3,
double% T2,
double% Tf,
double% Tw,
int% Color,
String^% Notes,
String^% GUID
)
```

```
abstract GetTube :
    Name : string *
    FileName : string byref *
    MatProp : string byref *
    T3 : float byref *
    T2 : float byref *
    Tf : float byref *
    Tw : float byref *
    Color : int byref *
    Notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame section property.

FileName

Type:Â SystemString

If the section property was imported from a property file, this is the name of that file. If the section property was not imported, this item is blank.

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

T2

Type:Â SystemDouble

Tf

Type:Â SystemDouble

Tw

Type:Â SystemDouble

Color

Type:Â SystemInt32

Notes

Type:Â SystemString

GUID

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the section.

Return Value

Type: Int32

Returns zero if the section property data is successfully retrieved; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new frame section property
    ret = SapModel.PropFrame.SetTube("TUBE1", "A992Fy50", 8, 6, 0.5, 0.5)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cPropFrame Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LSTD11E5A Method

Retrieves the property type for the specified frame section property.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetType0API(
    string Name,
    ref eFramePropType PropType
)
```

```
Function GetType0API (
    Name As String,
    ByRef PropType As eFramePropType
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim PropType As eFramePropType
Dim returnValue As Integer
```

```
returnValue = instance.GetType0API(Name,
    PropType)
```

```
int GetType0API(
    String^ Name,
    eFramePropType% PropType
)
```

```
abstract GetType0API :
    Name : string *
    PropType : eFramePropType byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing frame section property.

PropType

Type: ETABSv1eFramePropType

This is one of the items in the eFramePropType enumeration.

Return ValueType: Int32

Returns zero if the type is successfully retrieved; otherwise it returns nonzero.

▀ **Remarks**▀ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim PropType As eFramePropType

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'get frame property type
    ret = SapModel.PropFrame.GetTypeOAPI("FSEC1", PropType)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▀ **See Also****Reference**cPropFrame InterfaceETABSv1 Namespace

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST50AA3E9_0") Method

Retrieves the rebar design type for the specified frame section property

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetTypeRebar(
    string Name,
    ref int MyType
)
```

```
Function GetTypeRebar (
    Name As String,
    ByRef MyType As Integer
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim MyType As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetTypeRebar(Name,
    MyType)
```

```
int GetTypeRebar(
    String^ Name,
    int% MyType
)
```

```
abstract GetTypeRebar :
    Name : string *
    MyType : int byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing frame section property

MyType

Type: [SystemInt32](#)

This is 0, 1 or 2, indicating the rebar design type.

0 None

1 Column

2 Beam

Return Value

Type: Int32

Returns zero if the type is successfully retrieved; otherwise it returns nonzero

▀ **Remarks**

This function applies only to the following section property types. Calling this function for any other type of frame section property returns an error.

- Concrete Tee
- Concrete Angle
- Concrete Rectangle
- Concrete Circle

A nonzero rebar type is returned only if the frame section property has a concrete material.

▀ **See Also**

Reference

cPropFrame Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LSTD6581C63_0?cpp=3027-")

Method

Imports a frame section property from a property file.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int ImportProp(
    string Name,
    string MatProp,
    string FileName,
    string PropName,
    int Color = -1,
    string Notes = "",
    string GUID = ""
)
```

```
Function ImportProp (
    Name As String,
    MatProp As String,
    FileName As String,
    PropName As String,
    Optional As Integer = -1,
    Optional As String = "",
    Optional As String = ""
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim MatProp As String
Dim FileName As String
Dim PropName As String
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.ImportProp(Name,
    MatProp, FileName, PropName, Color,
    Notes, GUID)
```

```
int ImportProp(
    String^ Name,
    String^ MatProp,
    String^ FileName,
    String^ PropName,
    int Color = -1,
    String^ Notes = L"",
```

cPropFrameid="LSTD6581C63_0"AddLanguageSpecificTextSet("LSTD6581C63_0?cpp=3027-");Imp

```

String^ GUID = L""
)

abstract ImportProp :
    Name : string *
    MatProp : string *
    FileName : string *
    PropName : string *
    ?Color : int *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _Color = defaultArg Color -1
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of an existing or new frame section property. If this is an existing property, that property is modified; otherwise, a new property is added. This name does not need to be the same as the PropName item.

MatProp

Type:Â SystemString

The name of the material property for the section.

FileName

Type:Â SystemString

The name of the frame section property file from which to get the frame section property specified by the PropName item.

In most cases you can input just the name of the property file (e.g.

"AISC14.xml") and the program will be able to find it. In some cases you may have to input the full path to the property file.

PropName

Type:Â SystemString

The name of the frame section property, inside the property file specified by the FileName item, that is to be imported.

Color (Optional)

Type:Â SystemInt32

Notes (Optional)

Type:Â SystemString

GUID (Optional)

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the section. If this item is input as Default, the program assigns a GUID to the section.

Return Value

Type:Â Int32

Returns zero if the section property is successfully initialized; otherwise it returns a nonzero value.

Remarks

If the property file is not found, or the specified property name is not found in the property file, the section is set to be a general section with default properties.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'import new frame section property
    ret = SapModel.PropFrame.ImportProp("W18X35", "A992Fy50", "Sections8.pro", "W18X35")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

cPropFrame Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST8B4B1A3B_0?cpp=3060=.");Set

Method

Initializes an angle-type frame section property. If this function is called for an existing frame section property, all items for the section are reset to their default value.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetAngle(
    string Name,
    string MatProp,
    double T3,
    double T2,
    double Tf,
    double Tw,
    int Color = -1,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetAngle (
    Name As String,
    MatProp As String,
    T3 As Double,
    T2 As Double,
    Tf As Double,
    Tw As Double,
    Optional Color As Integer = -1,
    Optional Notes As String = "",
    Optional GUID As String = ""
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim MatProp As String
Dim T3 As Double
Dim T2 As Double
Dim Tf As Double
Dim Tw As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.SetAngle(Name,
    MatProp, T3, T2, Tf, Tw, Color, Notes,
    GUID)
```

cPropFramespan id="LST8B4B1A3B_0"AddLanguageSpecificTextSet("LST8B4B1A3B_0?cpp=3060=.");Set

```

int SetAngle(
String^ Name,
String^ MatProp,
double T3,
double T2,
double Tf,
double Tw,
int Color = -1,
String^ Notes = L"",
String^ GUID = L""
)

```

```

abstract SetAngle :
    Name : string *
    MatProp : string *
    T3 : float *
    T2 : float *
    Tf : float *
    Tw : float *
    ?Color : int *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _Color = defaultArg Color -1
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of an existing or new frame section property. If this is an existing property, that property is modified; otherwise, a new property is added.

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

T2

Type:Â SystemDouble

Tf

Type:Â SystemDouble

Tw

Type:Â SystemDouble

Color (Optional)

Type:Â SystemInt32

Notes (Optional)

Type:Â SystemString

GUID (Optional)

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the section. If this item is input as Default, then the program assigns a GUID to the section.

Return Value

Type: Int32

returns zero if the section property is successfully initialized; otherwise it returns a nonzero value.

▲ **Remarks**

This function is obsolete and has been superseded by SetSteelAngle(String, String, Double, Double, Double, Double, Double, Boolean, Boolean, Int32, String, String) and SetConcreteL(String, String, Double, Double, Double, Double, Double, Boolean, Boolean, Int32, String, String) as of version 16.0.0. This function is maintained for backwards compatibility.

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new frame section property
    ret = SapModel.PropFrame.SetAngle("ANGLE1", "A992Fy50", 6, 4, 0.5, 0.5)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cPropFrame Interface

ETABSv1 Namespace

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Introduction

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST2B03D61B_0?cpp=3034") Method

Assigns frame section properties to an auto select list. If this function is called for an existing frame section property, all items for the section are reset to their default value.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetAutoSelectSteel(
    string Name,
    int NumberItems,
    ref string[] SectName,
    string AutoStartSection = "Median",
    string Notes = "",
    string GUID = ""
)

Function SetAutoSelectSteel (
    Name As String,
    NumberItems As Integer,
    ByRef SectName As String(),
    AutoStartSection As String = "Median",
    Notes As String = "",
    GUID As String = ""
) As Integer

Dim instance As cPropFrame
Dim Name As String
Dim NumberItems As Integer
Dim SectName As String()
Dim AutoStartSection As String
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer

returnValue = instance.SetAutoSelectSteel(Name,
    NumberItems, SectName, AutoStartSection,
    Notes, GUID)

int SetAutoSelectSteel(
    String^ Name,
    int NumberItems,
    array<String^>^% SectName,
    String^ AutoStartSection = L"Median",
    String^ Notes = L"",
    String^ GUID = L""
)
```

cPropFrameAddLanguageSpecificTextSet("LST2B03D61B_0?cpp=3034");Set

```

abstract SetAutoSelectSteel :
    Name : string *
    NumberItems : int *
    SectName : string[] byref *
    ?AutoStartSection : string *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _AutoStartSection = defaultArg AutoStartSection "Median"
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int

```

Parameters

Name

Type:Â [SystemString](#)

The name of an existing or new frame section property. If this is an existing property, that property is modified; otherwise, a new property is added.

NumberItems

Type:Â [SystemInt32](#)

The number of frame section properties included in the auto select list.

SectName

Type:Â [SystemString](#)

This is an array of the names of the frame section properties included in the auto select list.

Auto select lists and nonprismatic (variable) sections are not allowed in this array.

AutoStartSection (Optional)

Type:Â [SystemString](#)

This is either Median or the name of a frame section property in the SectName array. It is the starting section for the auto select list.

Notes (Optional)

Type:Â [SystemString](#)

The notes, if any, assigned to the section.

GUID (Optional)

Type:Â [SystemString](#)

The GUID (global unique identifier), if any, assigned to the section. If this item is input as Default, the program assigns a GUID to the section.

Return Value

Type:Â [Int32](#)

Returns zero if the auto select list is successfully filled; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

[Copy](#)

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI

```

Introduction

```
Dim ret As Integer = -1
Dim MyName() As String

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new frame section properties
ret = SapModel.PropFrame.SetAngle("ANGLE1", "A992Fy50", 6, 4, 0.5, 0.5)
ret = SapModel.PropFrame.SetAngle("ANGLE2", "A992Fy50", 5, 3, 0.4, 0.4)
ret = SapModel.PropFrame.SetAngle("ANGLE3", "A992Fy50", 4, 2, 0.3, 0.3)

'set new auto select list frame section property
ReDim MyName(2)
MyName(0) = "ANGLE1"
MyName(1) = "ANGLE2"
MyName(2) = "ANGLE3"
ret = SapModel.PropFrame.SetAutoSelectSteel("AUT01", 3, MyName, "ANGLE2")

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

See Also

Reference

[cPropFrame Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST2BE68B71_0?cpp=3087-.");Set

Method

Initializes an channel-type frame section property. If this function is called for an existing frame section property, all items for the section are reset to their default value.

Namespace: Â ETABSv1

Assembly: Â ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetChannel(
    string Name,
    string MatProp,
    double T3,
    double T2,
    double Tf,
    double Tw,
    int Color = -1,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetChannel (
    Name As String,
    MatProp As String,
    T3 As Double,
    T2 As Double,
    Tf As Double,
    Tw As Double,
    Optional Color As Integer = -1,
    Optional Notes As String = "",
    Optional GUID As String = ""
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim MatProp As String
Dim T3 As Double
Dim T2 As Double
Dim Tf As Double
Dim Tw As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.SetChannel(Name,
    MatProp, T3, T2, Tf, Tw, Color, Notes,
    GUID)
```

cPropFramespan id="LST2BE68B71_0"AddLanguageSpecificTextSet("LST2BE68B71_0?cpp=3087-.");Set

```
int SetChannel(
String^ Name,
String^ MatProp,
double T3,
double T2,
double Tf,
double Tw,
int Color = -1,
String^ Notes = L"",
String^ GUID = L""
)
```

```
abstract SetChannel :
    Name : string *
    MatProp : string *
    T3 : float *
    T2 : float *
    Tf : float *
    Tw : float *
    ?Color : int *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _Color = defaultArg Color -1
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing or new frame section property. If this is an existing property, that property is modified; otherwise, a new property is added.

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

T2

Type:Â SystemDouble

Tf

Type:Â SystemDouble

Tw

Type:Â SystemDouble

Color (Optional)

Type:Â SystemInt32

Notes (Optional)

Type:Â SystemString

GUID (Optional)

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the section. If this item is input as Default, then the program assigns a GUID to the section.

Return Value

Type: Int32

returns zero if the section property is successfully initialized; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new frame section property
    ret = SapModel.PropFrame.SetChannel("CHN1", "A992Fy50", 24, 6, 0.5, 0.3)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cPropFrame Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LSTB7B01A Method

initializes a solid circular frame section property.If this function is called for an existing frame section property, all items for the section are reset to their default value.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetCircle(
    string Name,
    string MatProp,
    double T3,
    int Color = -1,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetCircle (
    Name As String,
    MatProp As String,
    T3 As Double,
    Optional As Integer = -1,
    Optional As String = "",
    Optional As String = ""
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim MatProp As String
Dim T3 As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.SetCircle(Name,
    MatProp, T3, Color, Notes, GUID)
```

```
int SetCircle(
    String^ Name,
    String^ MatProp,
    double T3,
    int Color = -1,
    String^ Notes = L"",
    String^ GUID = L""
)
```

cPropFrameAddLanguageSpecificTextSet("LSTB7B01AAA_0?cpp3040=.");Se

```

abstract SetCircle :
    Name : string *
    MatProp : string *
    T3 : float *
    ?Color : int *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _Color = defaultArg Color -1
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int

```

Parameters

Name

Type: SystemString

The name of an existing or new frame section property. If this is an existing property, that property is modified; otherwise, a new property is added.

MatProp

Type: SystemString

The name of the material property for the section.

T3

Type: SystemDouble

Color (Optional)

Type: SystemInt32

Notes (Optional)

Type: SystemString

GUID (Optional)

Type: SystemString

The GUID (global unique identifier), if any, assigned to the section. If this item is input as Default, the program assigns a GUID to the section.

Return Value

Type: Int32

Returns zero if the section property is successfully initialized; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object

```

Introduction

```
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new frame section property
ret = SapModel.PropFrame.SetCircle("C1", "4000Psi", 20)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPropFrame Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LSTE85ECEA2_0?cpp3048u=.");Set

Method

Initializes a concrete L frame section property. If this function is called for an existing frame section property, all items for the section are reset to their default value.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetConcreteL(
    string Name,
    string MatProp,
    double T3,
    double T2,
    double Tf,
    double TwC,
    double TwT,
    bool MirrorAbout2,
    bool MirrorAbout3,
    int Color = -1,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetConcreteL (
    Name As String,
    MatProp As String,
    T3 As Double,
    T2 As Double,
    Tf As Double,
    TwC As Double,
    TwT As Double,
    MirrorAbout2 As Boolean,
    MirrorAbout3 As Boolean,
    Optional Color As Integer = -1,
    Optional Notes As String = "",
    Optional GUID As String = ""
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim MatProp As String
Dim T3 As Double
Dim T2 As Double
Dim Tf As Double
Dim TwC As Double
Dim TwT As Double
Dim MirrorAbout2 As Boolean
Dim MirrorAbout3 As Boolean
```

cPropFramespan id="LSTE85ECEA2_0"AddLanguageSpecificTextSet("LSTE85ECEA2_0?cpp3048u=.");Set

Introduction

```
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer

returnValue = instance.SetConcreteL(Name,
MatProp, T3, T2, Tf, TwC, TwT, MirrorAbout2,
MirrorAbout3, Color, Notes, GUID)

int SetConcreteL(
String^ Name,
String^ MatProp,
double T3,
double T2,
double Tf,
double TwC,
double TwT,
bool MirrorAbout2,
bool MirrorAbout3,
int Color = -1,
String^ Notes = L"",
String^ GUID = L""
)

abstract SetConcreteL :
    Name : string *
    MatProp : string *
    T3 : float *
    T2 : float *
    Tf : float *
    TwC : float *
    TwT : float *
    MirrorAbout2 : bool *
    MirrorAbout3 : bool *
    ?Color : int *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _Color = defaultArg Color -1
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing or new frame section property. If this is an existing property, that property is modified; otherwise, a new property is added.

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

T2

Type:Â SystemDouble

Tf

Type:Â SystemDouble

TwC
Type:Â SystemDouble
The vertical leg thickness at the corner. [L]

TwT
Type:Â SystemDouble
The vertical leg thickness at the tip. [L]

MirrorAbout2
Type:Â SystemBoolean
Indicates whether the section is mirrored about the local 2-axis.

MirrorAbout3
Type:Â SystemBoolean
Indicates whether the section is mirrored about the local 3-axis.

Color (Optional)
Type:Â SystemInt32

Notes (Optional)
Type:Â SystemString

GUID (Optional)
Type:Â SystemString
The GUID (global unique identifier), if any, assigned to the section. If this item is input as Default, then the program assigns a GUID to the section.

Return Value

Type:Â Int32
returns zero if the section property is successfully initialized; otherwise it returns a nonzero value.

Remarks

This function supersedes SetAngle(String, String, Double, Double, Double, Double, Int32, String, String).

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new frame section property
```

Introduction

```
ret = SapModel.PropFrame.SetConcreteL("ANGLE1", "4000Psi", 30, 36, 8, 7, 5, False, True)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

📌 See Also

Reference

cPropFrame Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LSTEC9B8B78_0?cpp=3047=.");Set

Method

Initializes a concrete tee frame section property. If this function is called for an existing frame section property, all items for the section are reset to their default value.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetConcreteTee(
    string Name,
    string MatProp,
    double T3,
    double T2,
    double Tf,
    double TwF,
    double TwT,
    bool MirrorAbout3,
    int Color = -1,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetConcreteTee (
    Name As String,
    MatProp As String,
    T3 As Double,
    T2 As Double,
    Tf As Double,
    TwF As Double,
    TwT As Double,
    MirrorAbout3 As Boolean,
    Optional Color As Integer = -1,
    Optional Notes As String = "",
    Optional GUID As String = ""
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim MatProp As String
Dim T3 As Double
Dim T2 As Double
Dim Tf As Double
Dim TwF As Double
Dim TwT As Double
Dim MirrorAbout3 As Boolean
Dim Color As Integer
Dim Notes As String
```

cPropFramespan id="LSTEC9B8B78_0"AddLanguageSpecificTextSet("LSTEC9B8B78_0?cpp=3047=.");Set

Introduction

```
Dim GUID As String
Dim returnValue As Integer

returnValue = instance.SetConcreteTee(Name,
MatProp, T3, T2, Tf, TwF, TwT, MirrorAbout3,
Color, Notes, GUID)

int SetConcreteTee(
String^ Name,
String^ MatProp,
double T3,
double T2,
double Tf,
double TwF,
double TwT,
bool MirrorAbout3,
int Color = -1,
String^ Notes = L"",
String^ GUID = L""
)

abstract SetConcreteTee :
    Name : string *
    MatProp : string *
    T3 : float *
    T2 : float *
    Tf : float *
    TwF : float *
    TwT : float *
    MirrorAbout3 : bool *
    ?Color : int *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _Color = defaultArg Color -1
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing or new frame section property. If this is an existing property, that property is modified; otherwise, a new property is added.

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

T2

Type:Â SystemDouble

Tf

Type:Â SystemDouble

TwF

Type:Â SystemDouble

The web thickness at the flange. [L]
TwT
 Type:Â SystemDouble
 The web thickness at the tip. [L]
MirrorAbout3
 Type:Â SystemBoolean
 Indicates whether the section is mirrored about the local 3-axis.
Color (Optional)
 Type:Â SystemInt32
Notes (Optional)
 Type:Â SystemString
GUID (Optional)
 Type:Â SystemString
 The GUID (global unique identifier), if any, assigned to the section. If this item is input as Default, the program assigns a GUID to the section.

Return Value

Type:Â Int32
 Returns zero if the section property is successfully initialized; otherwise it returns a nonzero value.

Remarks

This function supersedes SetTee(String, String, Double, Double, Double, Double, Int32, String, String).

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new frame section property
    ret = SapModel.PropFrame.SetConcreteTee("TEE1", "4000Psi", 30, 36, 8, 7, 5, False)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
```

```
    EtabsObject = Nothing  
End Sub
```

▲ See Also

Reference

cPropFrame Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LSTC2179D76_0?cpp=3051-")

Method

Initializes a cover plated I-type frame section property. If this function is called for an existing frame section property, all items for the section are reset to their default value.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetCoverPlatedI(
    string Name,
    string SectName,
    double FyTopFlange,
    double FyWeb,
    double FyBotFlange,
    double Tc,
    double Bc,
    string MatPropTop,
    double Tcb,
    double Bcb,
    string MatPropBot,
    int Color = -1,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetCoverPlatedI (
    Name As String,
    SectName As String,
    FyTopFlange As Double,
    FyWeb As Double,
    FyBotFlange As Double,
    Tc As Double,
    Bc As Double,
    MatPropTop As String,
    Tcb As Double,
    Bcb As Double,
    MatPropBot As String,
    Color As Integer = -1,
    Notes As String = "",
    GUID As String = ""
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim SectName As String
Dim FyTopFlange As Double
Dim FyWeb As Double
```

cPropFramespan id="LSTC2179D76_0"AddLanguageSpecificTextSet("LSTC2179D76_0?cpp=3051-");Set

Introduction

```
Dim FyBotFlange As Double
Dim Tc As Double
Dim Bc As Double
Dim MatPropTop As String
Dim Tcb As Double
Dim Bcb As Double
Dim MatPropBot As String
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.SetCoverPlatedI(Name,
SectName, FyTopFlange, FyWeb, FyBotFlange,
Tc, Bc, MatPropTop, Tcb, Bcb, MatPropBot,
Color, Notes, GUID)
```

```
int SetCoverPlatedI(
String^ Name,
String^ SectName,
double FyTopFlange,
double FyWeb,
double FyBotFlange,
double Tc,
double Bc,
String^ MatPropTop,
double Tcb,
double Bcb,
String^ MatPropBot,
int Color = -1,
String^ Notes = L"",
String^ GUID = L""
)
```

```
abstract SetCoverPlatedI :
    Name : string *
    SectName : string *
    FyTopFlange : float *
    FyWeb : float *
    FyBotFlange : float *
    Tc : float *
    Bc : float *
    MatPropTop : string *
    Tcb : float *
    Bcb : float *
    MatPropBot : string *
    ?Color : int *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _Color = defaultArg Color -1
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int
```

Parameters

Name

Type:Â SystemString

Introduction

The name of an existing or new frame section property. If this is an existing property, that property is modified; otherwise, a new property is added.

SectName

Type:Â SystemString

FyTopFlange

Type:Â SystemDouble

The yield strength of the top flange of the I-section. [F/L2]

If this item is 0, the yield strength of the I-section specified by the SectName item is used.

FyWeb

Type:Â SystemDouble

The yield strength of the web of the I-section. [F/L2]

If this item is 0, the yield strength of the I-section specified by the SectName item is used.

FyBotFlange

Type:Â SystemDouble

The yield strength of the bottom flange of the I-section. [F/L2]

If this item is 0, the yield strength of the I-section specified by the SectName item is used.

Tc

Type:Â SystemDouble

Bc

Type:Â SystemDouble

MatPropTop

Type:Â SystemString

The name of the material property for the bottom cover plate.

This item applies only if both the tcb and the bcb items are greater than 0.

Tcb

Type:Â SystemDouble

Bcb

Type:Â SystemDouble

MatPropBot

Type:Â SystemString

The name of the material property for the bottom cover plate.

This item applies only if both the tcb and the bcb items are greater than 0.

Color (Optional)

Type:Â SystemInt32

Notes (Optional)

Type:Â SystemString

GUID (Optional)

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the section. If this item is input as Default, the program assigns a GUID to the section.

Return ValueType: Int32

Returns zero if the section property is successfully initialized; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new I-type frame section property
    ret = SapModel.PropFrame.SetISection("ISEC", "A992Fy50", 24, 8, 0.5, 0.3, 8, 0.5)

    'set new cover plated I-type frame section property
    ret = SapModel.PropFrame.SetCoverPlatedI("CPI1", "ISEC", 0, 36, 0, 0.75, 14, "A992Fy50", 0)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cPropFrame Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST4BDA4D3C_0?cpp3055u=.");Se

Method

Initializes a double angle-type frame section property. If this function is called for an existing frame section property, all items for the section are reset to their default value.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetDblAngle(
    string Name,
    string MatProp,
    double T3,
    double T2,
    double Tf,
    double Tw,
    double Dis,
    int Color = -1,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetDblAngle (
    Name As String,
    MatProp As String,
    T3 As Double,
    T2 As Double,
    Tf As Double,
    Tw As Double,
    Dis As Double,
    Optional As Integer = -1,
    Optional As String = "",
    Optional As String = ""
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim MatProp As String
Dim T3 As Double
Dim T2 As Double
Dim Tf As Double
Dim Tw As Double
Dim Dis As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

cPropFramespan id="LST4BDA4D3C_0"AddLanguageSpecificTextSet("LST4BDA4D3C_0?cpp3055u=.");Se

Introduction

```
returnValue = instance.SetDbAngle(Name,  
MatProp, T3, T2, Tf, Tw, Dis, Color,  
Notes, GUID)
```

```
int SetDbAngle(  
String^ Name,  
String^ MatProp,  
double T3,  
double T2,  
double Tf,  
double Tw,  
double Dis,  
int Color = -1,  
String^ Notes = L"",  
String^ GUID = L"  
)
```

```
abstract SetDbAngle :  
    Name : string *  
    MatProp : string *  
    T3 : float *  
    T2 : float *  
    Tf : float *  
    Tw : float *  
    Dis : float *  
    ?Color : int *  
    ?Notes : string *  
    ?GUID : string  
(* Defaults:  
    let _Color = defaultArg Color -1  
    let _Notes = defaultArg Notes ""  
    let _GUID = defaultArg GUID ""  
*)  
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing or new frame section property. If this is an existing property, that property is modified; otherwise, a new property is added.

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

T2

Type:Â SystemDouble

Tf

Type:Â SystemDouble

Tw

Type:Â SystemDouble

Dis

Type:Â SystemDouble

Color (Optional)

Type:Â SystemInt32

Notes (Optional)

Parameters

Type: SystemString
 GUID (Optional)

Type: SystemString

The GUID (global unique identifier), if any, assigned to the section. If this item is input as Default, the program assigns a GUID to the section.

Return Value

Type: Int32

Returns zero if the section property is successfully initialized; otherwise it returns a nonzero value.

▀ Remarks

▀ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new frame section property
    ret = SapModel.PropFrame.SetDbAngle("DBANG1", "A992Fy50", 6, 9, 0.5, 0.5, 1)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▀ See Also

Reference

cPropFrame Interface

ETABSv1 Namespace

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Introduction

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»

CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST1B5E8061_0?cpp=3059.")

Method

Initializes a double channel-type frame section property. If this function is called for an existing frame section property, all items for the section are reset to their default value.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetDblChannel(
    string Name,
    string MatProp,
    double T3,
    double T2,
    double Tf,
    double Tw,
    double Dis,
    int Color = -1,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetDblChannel (
    Name As String,
    MatProp As String,
    T3 As Double,
    T2 As Double,
    Tf As Double,
    Tw As Double,
    Dis As Double,
    Optional As Integer = -1,
    Optional As String = "",
    Optional As String = ""
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim MatProp As String
Dim T3 As Double
Dim T2 As Double
Dim Tf As Double
Dim Tw As Double
Dim Dis As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

cPropFramespan id="LST1B5E8061_0"AddLanguageSpecificTextSet("LST1B5E8061_0?cpp=3059.");SetD

Introduction

```
returnValue = instance.SetDblChannel(Name,  
MatProp, T3, T2, Tf, Tw, Dis, Color,  
Notes, GUID)
```

```
int SetDblChannel(  
String^ Name,  
String^ MatProp,  
double T3,  
double T2,  
double Tf,  
double Tw,  
double Dis,  
int Color = -1,  
String^ Notes = L"",  
String^ GUID = L"  
)
```

```
abstract SetDblChannel :  
    Name : string *  
    MatProp : string *  
    T3 : float *  
    T2 : float *  
    Tf : float *  
    Tw : float *  
    Dis : float *  
    ?Color : int *  
    ?Notes : string *  
    ?GUID : string  
(* Defaults:  
    let _Color = defaultArg Color -1  
    let _Notes = defaultArg Notes ""  
    let _GUID = defaultArg GUID ""  
*)  
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing or new frame section property. If this is an existing property, that property is modified; otherwise, a new property is added.

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

T2

Type:Â SystemDouble

Tf

Type:Â SystemDouble

Tw

Type:Â SystemDouble

Dis

Type:Â SystemDouble

Color (Optional)

Type:Â SystemInt32

Notes (Optional)

Parameters

Type: SystemString
 GUID (Optional)

Type: SystemString

The GUID (global unique identifier), if any, assigned to the section. If this item is input as Default, the program assigns a GUID to the section.

Return Value

Type: Int32

Returns zero if the section property is successfully initialized; otherwise it returns a nonzero value.

▀ Remarks

▀ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new frame section property
    ret = SapModel.PropFrame.SetDblChannel("DBCHN1", "A992Fy50", 12, 6.5, 0.5, 0.3, 0.5)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▀ See Also

Reference

cPropFrame Interface

ETABSv1 Namespace

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Introduction

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST9B7B0FD5_0?cpp=3060=.");Set

Method

Initializes a general frame section property. If this function is called for an existing frame section property, all items for the section are reset to their default value.

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetGeneral(
    string Name,
    string MatProp,
    double T3,
    double T2,
    double Area,
    double As2,
    double As3,
    double Torsion,
    double I22,
    double I33,
    double S22,
    double S33,
    double Z22,
    double Z33,
    double R22,
    double R33,
    int Color = -1,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetGeneral (
    Name As String,
    MatProp As String,
    T3 As Double,
    T2 As Double,
    Area As Double,
    As2 As Double,
    As3 As Double,
    Torsion As Double,
    I22 As Double,
    I33 As Double,
    S22 As Double,
    S33 As Double,
    Z22 As Double,
    Z33 As Double,
    R22 As Double,
    R33 As Double,
    Optional As Integer = -1,
```

cPropFramespan id="LST9B7B0FD5_0"AddLanguageSpecificTextSet("LST9B7B0FD5_0?cpp=3060=.");Set

Introduction

```
Notes As String = "",
GUID As String = ""
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim MatProp As String
Dim T3 As Double
Dim T2 As Double
Dim Area As Double
Dim As2 As Double
Dim As3 As Double
Dim Torsion As Double
Dim I22 As Double
Dim I33 As Double
Dim S22 As Double
Dim S33 As Double
Dim Z22 As Double
Dim Z33 As Double
Dim R22 As Double
Dim R33 As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.SetGeneral(Name,
MatProp, T3, T2, Area, As2, As3, Torsion,
I22, I33, S22, S33, Z22, Z33, R22, R33,
Color, Notes, GUID)
```

```
int SetGeneral(
String^ Name,
String^ MatProp,
double T3,
double T2,
double Area,
double As2,
double As3,
double Torsion,
double I22,
double I33,
double S22,
double S33,
double Z22,
double Z33,
double R22,
double R33,
int Color = -1,
String^ Notes = L"",
String^ GUID = L""
)
```

```
abstract SetGeneral :
    Name : string *
    MatProp : string *
    T3 : float *
    T2 : float *
    Area : float *
    As2 : float *
    As3 : float *
    Torsion : float *
```

cPropFramespan id="LST9B7B0FD5_0"AddLanguageSpecificTextSet("LST9B7B0FD5_0?cpp=3064=");Set

```

I22 : float *
I33 : float *
S22 : float *
S33 : float *
Z22 : float *
Z33 : float *
R22 : float *
R33 : float *
?Color : int *
?Notes : string *
?GUID : string
(* Defaults:
  let _Color = defaultArg Color -1
  let _Notes = defaultArg Notes ""
  let _GUID = defaultArg GUID ""
*)
-> int

```

Parameters

Name

Type: SystemString

The name of an existing or new frame section property. If this is an existing property, that property is modified; otherwise, a new property is added.

MatProp

Type: SystemString

The name of the material property for the section.

T3

Type: SystemDouble

T2

Type: SystemDouble

Area

Type: SystemDouble

The cross-sectional area. [L2]

As2

Type: SystemDouble

The shear area for forces in the section local 2-axis direction. [L2]

As3

Type: SystemDouble

The shear area for forces in the section local 3-axis direction. [L2]

Torsion

Type: SystemDouble

The torsional constant. [L4]

I22

Type: SystemDouble

The moment of inertia for bending about the local 2 axis. [L4]

I33

Type: SystemDouble

The moment of inertia for bending about the local 3 axis. [L4]

S22

Type: SystemDouble

The section modulus for bending about the local 2 axis. [L3]

S33

Type:Â SystemDouble
 The section modulus for bending about the local 3 axis. [L3]

Z22
 Type:Â SystemDouble
 The plastic modulus for bending about the local 2 axis. [L3]

Z33
 Type:Â SystemDouble
 The plastic modulus for bending about the local 3 axis. [L3]

R22
 Type:Â SystemDouble
 The radius of gyration about the local 2 axis. [L]

R33
 Type:Â SystemDouble
 The radius of gyration about the local 3 axis. [L]

Color (Optional)
 Type:Â SystemInt32

Notes (Optional)
 Type:Â SystemString

GUID (Optional)
 Type:Â SystemString
 The GUID (global unique identifier), if any, assigned to the section. If this item is input as Default, the program assigns a GUID to the section.

Return Value

Type:Â Int32
 Returns zero if the section property is successfully initialized; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new frame section property
    ret = SapModel.PropFrame.SetGeneral("GEN1", "A992Fy50", 24, 14, 100, 80, 80, 4, 1000, 2400)
```

```
'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPropFrame Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST27309C93_0") Method

Initializes an I-type frame section property. If this function is called for an existing frame section property, all items for the section are reset to their default value.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetISection(
    string Name,
    string MatProp,
    double T3,
    double T2,
    double Tf,
    double Tw,
    double T2b,
    double Tfb,
    int Color = -1,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetISection (
    Name As String,
    MatProp As String,
    T3 As Double,
    T2 As Double,
    Tf As Double,
    Tw As Double,
    T2b As Double,
    Tfb As Double,
    Optional Color As Integer = -1,
    Optional Notes As String = "",
    Optional GUID As String = ""
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim MatProp As String
Dim T3 As Double
Dim T2 As Double
Dim Tf As Double
Dim Tw As Double
Dim T2b As Double
Dim Tfb As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
```

cPropFramespan id="LST27309C93_0"AddLanguageSpecificTextSet("LST27309C93_0?cpp=:3068.");SetI


```

Dim returnValue As Integer

returnValue = instance.SetISection(Name,
MatProp, T3, T2, Tf, Tw, T2b, Tfb, Color,
Notes, GUID)

int SetISection(
String^ Name,
String^ MatProp,
double T3,
double T2,
double Tf,
double Tw,
double T2b,
double Tfb,
int Color = -1,
String^ Notes = L"",
String^ GUID = L""
)

abstract SetISection :
    Name : string *
    MatProp : string *
    T3 : float *
    T2 : float *
    Tf : float *
    Tw : float *
    T2b : float *
    Tfb : float *
    ?Color : int *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _Color = defaultArg Color -1
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of an existing or new frame section property. If this is an existing property, that property is modified; otherwise, a new property is added.

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

T2

Type:Â SystemDouble

Tf

Type:Â SystemDouble

Tw

Type:Â SystemDouble

T2b

Type:Â SystemDouble

Tfb

Type:Â SystemDouble

Color (Optional)

Type:Â SystemInt32

Notes (Optional)

Type:Â SystemString

GUID (Optional)

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the section. If this item is input as Default, the program assigns a GUID to the section.

Return Value

Type:Â Int32

Returns zero if the section property is successfully initialized; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new frame section property
    ret = SapModel.PropFrame.SetISection("ISEC1", "A992Fy50", 24, 10, 0.5, 0.3, 14, 0.6)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPropFrame Interface

ETABSv1 Namespace

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»

CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LSTB87C2D9D_0?cpp3072u=.");Se

Method

Sets the material property of a defined frame section

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetMaterial(
    string Name,
    string MatProp
)
```

```
Function SetMaterial (
    Name As String,
    MatProp As String
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim MatProp As String
Dim returnValue As Integer
```

```
returnValue = instance.SetMaterial(Name,
    MatProp)
```

```
int SetMaterial(
    String^ Name,
    String^ MatProp
)
```

```
abstract SetMaterial :
    Name : string *
    MatProp : string -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing frame section property.

MatProp

Type: [SystemString](#)

The name of the material property for the section.

Return Value

Type: Int32

Returns zero if the material property is successfully set; otherwise it returns a nonzero value.

▀ **Remarks**

▀ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim FileName As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new frame section property
    ret = SapModel.PropFrame.SetTee("TEE1", "A992Fy50", 12, 10, 0.6, 0.3)

    'set frame section material property data
    ret = SapModel.PropFrame.SetMaterial("TEE1", "A992Fy50")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▀ **See Also**

Reference

cPropFrame Interface

ETABSv1 Namespace

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i»j

CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LSTD67065E7_0?cpp=3074.");SetM

Method

Assigns property modifiers to a frame section property. The default value for all modifiers is one.

Namespace: Â ETABSv1

Assembly: Â ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetModifiers(
    string Name,
    ref double[] Value
)
```

```
Function SetModifiers (
    Name As String,
    ByRef Value As Double()
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim Value As Double()
Dim returnValue As Integer
```

```
returnValue = instance.SetModifiers(Name,
    Value)
```

```
int SetModifiers(
    String^ Name,
    array<double>^% Value
)
```

```
abstract SetModifiers :
    Name : string *
    Value : float[] byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame section property.

Value

Type:Â SystemDouble

This is an array of eight unitless modifiers:

Value

Modifier

Value(0) Cross sectional area modifier

cPropFramespan id="LSTD67065E7_0"AddLanguageSpecificTextSet("LSTD67065E7_0?cpp=3074.");SetM

Introduction

Value(1) Shear area in local 2 direction modifier
Value(2) Shear area in local 3 direction modifier
Value(3) Torsional constant modifier
Value(4) Moment of inertia about local 2 axis modifier
Value(5) Moment of inertia about local 3 axis modifier
Value(6) Mass modifier
Value(7) Weight modifier

Return Value

Type: Int32

Returns zero if the modifiers are successfully assigned; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim i As Integer  
    Dim Value() As Double  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)  
  
    'set new frame section property  
    ret = SapModel.PropFrame.SetISection("ISEC1", "A992Fy50", 24, 10, 0.5, 0.3, 14, 0.6)  
  
    'assign modifiers  
    ReDim Value(7)  
    For i = 0 To 7  
        Value(i) = 1  
    Next i  
    Value(5) = 100  
    ret = SapModel.PropFrame.SetModifiers("ISEC1", Value)  
  
    'close ETABS  
    EtabsObject.ApplicationExit(False)  
  
    'clean up variables  
    SapModel = Nothing  
    EtabsObject = Nothing
```

Parameters

End Sub

▲ See Also
Reference

cPropFrame Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LSTECB48E7D_0?cpp3077u=.");SetNonPrismatic

Method

Assigns data to a nonprismatic frame section property.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetNonPrismatic(
    string Name,
    int NumberItems,
    ref string[] StartSec,
    ref string[] EndSec,
    ref double[] MyLength,
    ref int[] MyType,
    ref int[] EI33,
    ref int[] EI22,
    int Color = -1,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetNonPrismatic (
    Name As String,
    NumberItems As Integer,
    ByRef StartSec As String(),
    ByRef EndSec As String(),
    ByRef MyLength As Double(),
    ByRef MyType As Integer(),
    ByRef EI33 As Integer(),
    ByRef EI22 As Integer(),
    Optional Color As Integer = -1,
    Optional Notes As String = "",
    Optional GUID As String = ""
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim NumberItems As Integer
Dim StartSec As String()
Dim EndSec As String()
Dim MyLength As Double()
Dim MyType As Integer()
Dim EI33 As Integer()
Dim EI22 As Integer()
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

cPropFramespan id="LSTECB48E7D_0"AddLanguageSpecificTextSet("LSTECB48E7D_0?cpp3077u=.");SetNonPrismatic

```
returnValue = instance.SetNonPrismatic(Name,
NumberItems, StartSec, EndSec, MyLength,
MyType, EI33, EI22, Color, Notes, GUID)
```

```
int SetNonPrismatic(
String^ Name,
int NumberItems,
array<String^>^% StartSec,
array<String^>^% EndSec,
array<double>^% MyLength,
array<int>^% MyType,
array<int>^% EI33,
array<int>^% EI22,
int Color = -1,
String^ Notes = L"",
String^ GUID = L""
)
```

```
abstract SetNonPrismatic :
    Name : string *
    NumberItems : int *
    StartSec : string[] byref *
    EndSec : string[] byref *
    MyLength : float[] byref *
    MyType : int[] byref *
    EI33 : int[] byref *
    EI22 : int[] byref *
    ?Color : int *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _Color = defaultArg Color -1
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing or new frame section property. If this is an existing property, that property is modified; otherwise, a new property is added.

NumberItems

Type: [SystemInt32](#)

The number of segments assigned to the nonprismatic section.

StartSec

Type: [SystemString](#)

This is an array of the names of the frame section properties at the start of each segment.

Auto select lists and nonprismatic sections are not allowed in this array.

EndSec

Type: [SystemString](#)

This is an array of the names of the frame section properties at the end of each segment.

Auto select lists and nonprismatic sections are not allowed in this array.

MyLength

Type:Â SystemDouble

This is an array that includes the length of each segment. The length may be variable or absolute as indicated by the MyType item. [L] when length is absolute.

MyType

Type:Â SystemInt32

This is an array of either 1 or 2, indicating the length type for each segment:

Value	MyType
1	Variable (relative length)
2	Absolute

EI33

Type:Â SystemInt32

This is an array of 1, 2 or 3, indicating the variation type for EI33 in each segment:

Value	EI33
1	Linear
2	Parabolic
2	Cubic

EI22

Type:Â SystemInt32

This is an array of 1, 2 or 3, indicating the variation type for EI22 in each segment:

Value	EI22
1	Linear
2	Parabolic
2	Cubic

Color (Optional)

Type:Â SystemInt32

Notes (Optional)

Type:Â SystemString

GUID (Optional)

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the section. If this item is input as Default, the program assigns a GUID to the section.

Return Value

Type:Â Int32

Returns zero if the data is successfully filled; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
```

Introduction

```
Dim StartSec() As String
Dim EndSec() As String
Dim Length() As Double
Dim SegType() As Integer
Dim EI33() As Integer
Dim EI22() As Integer

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new I-type frame section property
ret = SapModel.PropFrame.SetISection("ISEC1", "A992Fy50", 24, 10, 0.5, 0.3, 14, 0.6)

'set new I-type frame section property
ret = SapModel.PropFrame.SetISection("ISEC2", "A992Fy50", 36, 15, 1, 0.5, 121, 0.9)

'set new nonprismatic frame section property
ReDim StartSec(2)
ReDim EndSec(2)
ReDim Length(2)
ReDim SegType(2)
ReDim EI33(2)
ReDim EI22(2)
StartSec(0) = "ISEC2"
EndSec(0) = "ISEC1"
Length(0) = 60
SegType(0) = 2
EI33(0)= 2
EI22(0)= 1

StartSec(1) = "ISEC1"
EndSec(1) = "ISEC1"
Length(1) = 1
SegType(1) = 1
EI33(1)= 2
EI22(1)= 1

StartSec(2) = "ISEC1"
EndSec(2) = "ISEC2"
Length(2) = 60
SegType(2) = 2
EI33(2)= 2
EI22(2)= 1

ret = SapModel.PropFrame.SetNonPrismatic("NP1", 3, StartSec, EndSec, Length, SegType, EI33)

'close ETABS
EtabsObject.ApplicationExit(False)
```

```
'clean up variables  
  SapModel = Nothing  
  EtabsObject = Nothing  
End Sub
```

🔍 See Also

Reference

cPropFrame Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST1A057E

Method

Initializes a pipe-type frame section property. If this function is called for an existing frame section property, all items for the section are reset to their default value.

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetPipe(
    string Name,
    string MatProp,
    double T3,
    double TW,
    int Color = -1,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetPipe (
    Name As String,
    MatProp As String,
    T3 As Double,
    TW As Double,
    Optional As Integer = -1,
    Optional As String = "",
    Optional As String = ""
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim MatProp As String
Dim T3 As Double
Dim TW As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.SetPipe(Name, MatProp,
    T3, TW, Color, Notes, GUID)
```

```
int SetPipe(
    String^ Name,
    String^ MatProp,
    double T3,
    double TW,
    int Color = -1,
    String^ Notes = L"",
```

cPropFramespan id="LST1A057EF5_0"AddLanguageSpecificTextSet("LST1A057EF5_0?cpp=3082.");SetF

```

String^ GUID = L""
)

abstract SetPipe :
    Name : string *
    MatProp : string *
    T3 : float *
    TW : float *
    ?Color : int *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _Color = defaultArg Color -1
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of an existing or new frame section property. If this is an existing property, that property is modified; otherwise, a new property is added.

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

TW

Type:Â SystemDouble

The wall thickness. [L]

Color (Optional)

Type:Â SystemInt32

Notes (Optional)

Type:Â SystemString

GUID (Optional)

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the section. If this item is input as Default, the program assigns a GUID to the section.

Return Value

Type:Â Int32

Returns zero if the section property is successfully initialized; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

```

Introduction

```
'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
    ret = EtabsObject.ApplicationStart()

'create SapModel object
    SapModel = EtabsObject.SapModel

'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new frame section property
    ret = SapModel.PropFrame.SetPipe("PIPE1", "A992Fy50", 8, 0.375)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPropFrame Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST63093E6F_0?cpp=:3085.");SetP

Method

Initializes a solid plate frame section property. If this function is called for an existing frame section property, all items for the section are reset to their default value.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetPlate(
    string Name,
    string MatProp,
    double T3,
    double T2,
    int Color = -1,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetPlate (
    Name As String,
    MatProp As String,
    T3 As Double,
    T2 As Double,
    Optional As Integer = -1,
    Optional As String = "",
    Optional As String = ""
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim MatProp As String
Dim T3 As Double
Dim T2 As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.SetPlate(Name,
    MatProp, T3, T2, Color, Notes, GUID)
```

```
int SetPlate(
    String^ Name,
    String^ MatProp,
    double T3,
    double T2,
    int Color = -1,
    String^ Notes = L"",
```

cPropFramespan id="LST63093E6F_0"AddLanguageSpecificTextSet("LST63093E6F_0?cpp=:3085.");SetP

```

String^ GUID = L""
)

abstract SetPlate :
    Name : string *
    MatProp : string *
    T3 : float *
    T2 : float *
    ?Color : int *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _Color = defaultArg Color -1
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of an existing or new frame section property. If this is an existing property, that property is modified; otherwise, a new property is added.

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

T2

Type:Â SystemDouble

Color (Optional)

Type:Â SystemInt32

Notes (Optional)

Type:Â SystemString

GUID (Optional)

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the section. If this item is input as Default, the program assigns a GUID to the section.

Return Value

Type:Â Int32

Returns zero if the section property is successfully initialized; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

```

```

    'create ETABS object

```

Introduction

```
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new frame section property
ret = SapModel.PropFrame.SetPlate("R1", "A992Fy50", 20, 12)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

🔗 See Also

Reference

[cPropFrame Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LSTCCF527FD_0?cpp308u=.");Se

Method

Assigns beam rebar data to frame sections.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetRebarBeam(
    string Name,
    string MatPropLong,
    string MatPropConfine,
    double CoverTop,
    double CoverBot,
    double TopLeftArea,
    double TopRightArea,
    double BotLeftArea,
    double BotRightArea
)
```

```
Function SetRebarBeam (
    Name As String,
    MatPropLong As String,
    MatPropConfine As String,
    CoverTop As Double,
    CoverBot As Double,
    TopLeftArea As Double,
    TopRightArea As Double,
    BotLeftArea As Double,
    BotRightArea As Double
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim MatPropLong As String
Dim MatPropConfine As String
Dim CoverTop As Double
Dim CoverBot As Double
Dim TopLeftArea As Double
Dim TopRightArea As Double
Dim BotLeftArea As Double
Dim BotRightArea As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetRebarBeam(Name,
    MatPropLong, MatPropConfine, CoverTop,
    CoverBot, TopLeftArea, TopRightArea,
    BotLeftArea, BotRightArea)
```

```

int SetRebarBeam(
String^ Name,
String^ MatPropLong,
String^ MatPropConfine,
double CoverTop,
double CoverBot,
double TopLeftArea,
double TopRightArea,
double BotLeftArea,
double BotRightArea
)

```

```

abstract SetRebarBeam :
    Name : string *
    MatPropLong : string *
    MatPropConfine : string *
    CoverTop : float *
    CoverBot : float *
    TopLeftArea : float *
    TopRightArea : float *
    BotLeftArea : float *
    BotRightArea : float -> int

```

Parameters

Name

Type:Â SystemString

The name of an existing frame section property.

MatPropLong

Type:Â SystemString

The name of the rebar material property for the longitudinal rebar.

MatPropConfine

Type:Â SystemString

The name of the rebar material property for the confinement rebar.

CoverTop

Type:Â SystemDouble

The distance from the top of the beam to the centroid of the top longitudinal reinforcement. [L]

CoverBot

Type:Â SystemDouble

The distance from the bottom of the beam to the centroid of the bottom longitudinal reinforcement. [L]

TopLeftArea

Type:Â SystemDouble

The total area of longitudinal reinforcement at the top left end of the beam. [L2]

TopRightArea

Type:Â SystemDouble

The total area of longitudinal reinforcement at the top right end of the beam. [L2]

BotLeftArea

Type:Â SystemDouble

The total area of longitudinal reinforcement at the bottom left end of the beam. [L2]

BotRightArea

Type: SystemDouble

The total area of longitudinal reinforcement at the bottom right end of the beam. [L2]

Return Value

Type: Int32

Returns zero if the rebar data is successfully assigned; otherwise it returns a nonzero value.

Remarks

This function applies only to the following section types. Calling this function for any other type of frame section property returns an error.

- I
- Angle
- Rectangular
- Circle

The material assigned to the specified frame section property must be concrete or this function returns an error.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim RebarName As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new frame section property
    ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

    'add ASTM A706 rebar material
    ret = SapModel.PropMaterial.AddQuick(RebarName, eMatType.Rebar, , , , eMatTypeRebar.ASTM)

    'set beam rebar data
    ret = SapModel.PropFrame.SetRebarBeam("R1", RebarName, RebarName, 3.5, 3, 4.1, 4.2, 4.3, 4)

    'close ETABS
    EtabsObject.ApplicationExit(False)
```

```
'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

📌 See Also

Reference

cPropFrame Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST2F6507 Method

Assigns column rebar data to frame sections.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetRebarColumn(
    string Name,
    string MatPropLong,
    string MatPropConfine,
    int Pattern,
    int ConfineType,
    double Cover,
    int NumberCBars,
    int NumberR3Bars,
    int NumberR2Bars,
    string RebarSize,
    string TieSize,
    double TieSpacingLongit,
    int Number2DirTieBars,
    int Number3DirTieBars,
    bool ToBeDesigned
)
```

```
Function SetRebarColumn (
    Name As String,
    MatPropLong As String,
    MatPropConfine As String,
    Pattern As Integer,
    ConfineType As Integer,
    Cover As Double,
    NumberCBars As Integer,
    NumberR3Bars As Integer,
    NumberR2Bars As Integer,
    RebarSize As String,
    TieSize As String,
    TieSpacingLongit As Double,
    Number2DirTieBars As Integer,
    Number3DirTieBars As Integer,
    ToBeDesigned As Boolean
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim MatPropLong As String
Dim MatPropConfine As String
Dim Pattern As Integer
```

cPropFramespan id="LST2F650716_0"AddLanguageSpecificTextSet("LST2F650716_0?cpp=::3092");SetR

Introduction

```
Dim ConfineType As Integer
Dim Cover As Double
Dim NumberCBars As Integer
Dim NumberR3Bars As Integer
Dim NumberR2Bars As Integer
Dim RebarSize As String
Dim TieSize As String
Dim TieSpacingLongit As Double
Dim Number2DirTieBars As Integer
Dim Number3DirTieBars As Integer
Dim ToBeDesigned As Boolean
Dim returnValue As Integer

returnValue = instance.SetRebarColumn(Name,
MatPropLong, MatPropConfine, Pattern,
ConfineType, Cover, NumberCBars,
NumberR3Bars, NumberR2Bars, RebarSize,
TieSize, TieSpacingLongit, Number2DirTieBars,
Number3DirTieBars, ToBeDesigned)

int SetRebarColumn(
String^ Name,
String^ MatPropLong,
String^ MatPropConfine,
int Pattern,
int ConfineType,
double Cover,
int NumberCBars,
int NumberR3Bars,
int NumberR2Bars,
String^ RebarSize,
String^ TieSize,
double TieSpacingLongit,
int Number2DirTieBars,
int Number3DirTieBars,
bool ToBeDesigned
)

abstract SetRebarColumn :
    Name : string *
    MatPropLong : string *
    MatPropConfine : string *
    Pattern : int *
    ConfineType : int *
    Cover : float *
    NumberCBars : int *
    NumberR3Bars : int *
    NumberR2Bars : int *
    RebarSize : string *
    TieSize : string *
    TieSpacingLongit : float *
    Number2DirTieBars : int *
    Number3DirTieBars : int *
    ToBeDesigned : bool -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame section property.

MatPropLongType:Â SystemString

The name of the rebar material property for the longitudinal rebar.

MatPropConfineType:Â SystemString

The name of the rebar material property for the confinement rebar.

PatternType:Â SystemInt32

This is either 1 or 2, indicating the rebar configuration.

Value Pattern

1 Rectangular

2 Circular

For circular frame section properties this item must be 2; otherwise an error is returned.

ConfineTypeType:Â SystemInt32

This is either 1 or 2, indicating the confinement bar type.

Value Type

1 Ties

2 Spiral

This item applies only when Pattern = 2. If Pattern = 1, the confinement bar type is assumed to be ties.

CoverType:Â SystemDouble

The clear cover for the confinement steel (ties). In the special case of circular reinforcement in a rectangular column, this is the minimum clear cover. [L]

NumberCBarsType:Â SystemInt32

This item applies to a circular rebar configuration, Pattern = 2. It is the total number of longitudinal reinforcing bars in the column.

NumberR3BarsType:Â SystemInt32

This item applies to a rectangular rebar configuration, Pattern = 1. It is the number of longitudinal bars (including the corner bar) on each face of the column that is parallel to the local 3-axis of the column.

NumberR2BarsType:Â SystemInt32

This item applies to a rectangular rebar configuration, Pattern = 1. It is the number of longitudinal bars (including the corner bar) on each face of the column that is parallel to the local 2-axis of the column.

RebarSizeType:Â SystemString

The rebar name for the longitudinal rebar in the column.

TieSizeType:Â SystemString

The rebar name for the confinement rebar in the column.

TieSpacingLongitType:Â SystemDouble

The longitudinal spacing of the confinement bars (ties). [L]

Number2DirTieBarsType: SystemInt32

This item applies to a rectangular reinforcing configuration, Pattern = 1. It is the number of confinement bars (tie legs) running in the local 2-axis direction of the column.

Number3DirTieBarsType: SystemInt32

This item applies to a rectangular reinforcing configuration, Pattern = 1. It is the number of confinement bars (tie legs) running in the local 3-axis direction of the column.

ToBeDesignedType: SystemBoolean

If this item is True, the column longitudinal rebar is to be designed; otherwise it is to be checked.

Return ValueType: Int32

Returns zero if the rebar data is successfully assigned; otherwise it returns a nonzero value.

Remarks

This function applies only to the following section types. Calling this function for any other type of frame section property returns an error.

- Rectangular
- Circle

The material assigned to the specified frame section property must be concrete or this function returns an error.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim RebarName As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)
```

Introduction

```
'set new frame section property
    ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

'add ASTM A706 rebar material
    ret = SapModel.PropMaterial.AddQuick(RebarName, eMatType.Rebar, , , , eMatTypeRebar.ASTM

'set column rebar data
    ret = SapModel.PropFrame.SetRebarColumn("R1", RebarName, RebarName, 2, 2, 2, 10, 0, 0, "#1

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPropFrame Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST4209B9A5_0?cpp=3007.")

Method

Initializes a solid rectangular frame section property. If this function is called for an existing frame section property, all items for the section are reset to their default value.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetRectangle(
    string Name,
    string MatProp,
    double T3,
    double T2,
    int Color = -1,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetRectangle (
    Name As String,
    MatProp As String,
    T3 As Double,
    T2 As Double,
    Optional As Integer = -1,
    Optional Notes As String = "",
    Optional GUID As String = ""
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim MatProp As String
Dim T3 As Double
Dim T2 As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.SetRectangle(Name,
    MatProp, T3, T2, Color, Notes, GUID)
```

```
int SetRectangle(
    String^ Name,
    String^ MatProp,
    double T3,
    double T2,
    int Color = -1,
```

cPropFramespan id="LST4209B9A5_0"AddLanguageSpecificTextSet("LST4209B9A5_0?cpp=3007.");SetF

```

String^ Notes = L"",
String^ GUID = L""
)

abstract SetRectangle :
    Name : string *
    MatProp : string *
    T3 : float *
    T2 : float *
    ?Color : int *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _Color = defaultArg Color -1
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of an existing or new frame section property. If this is an existing property, that property is modified; otherwise, a new property is added.

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

T2

Type:Â SystemDouble

Color (Optional)

Type:Â SystemInt32

Notes (Optional)

Type:Â SystemString

GUID (Optional)

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the section. If this item is input as Default, the program assigns a GUID to the section.

Return Value

Type:Â Int32

Returns zero if the section property is successfully initialized; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

```

Introduction

```
'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
    ret = EtabsObject.ApplicationStart()

'create SapModel object
    SapModel = EtabsObject.SapModel

'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new frame section property
    ret = SapModel.PropFrame.SetRectangle("R1", "4000Psi", 20, 12)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPropFrame Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST9E56D40_0") Method

initializes a solid rod frame section property. If this function is called for an existing frame section property, all items for the section are reset to their default value.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetRod(
    string Name,
    string MatProp,
    double T3,
    int Color = -1,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetRod (
    Name As String,
    MatProp As String,
    T3 As Double,
    Optional Color As Integer = -1,
    Optional Notes As String = "",
    Optional GUID As String = ""
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim MatProp As String
Dim T3 As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.SetRod(Name, MatProp,
    T3, Color, Notes, GUID)
```

```
int SetRod(
    String^ Name,
    String^ MatProp,
    double T3,
    int Color = -1,
    String^ Notes = L"",
    String^ GUID = L""
)
```

abstract SetRod :

cPropFrameAddLanguageSpecificTextSet("LST9E56D40_0?cpp=::|n=100");SetRod


```

    Name : string *
    MatProp : string *
    T3 : float *
    ?Color : int *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _Color = defaultArg Color -1
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of an existing or new frame section property. If this is an existing property, that property is modified; otherwise, a new property is added.

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

Color (Optional)

Type:Â SystemInt32

Notes (Optional)

Type:Â SystemString

GUID (Optional)

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the section. If this item is input as Default, the program assigns a GUID to the section.

Return Value

Type:Â Int32

Returns zero if the section property is successfully initialized; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

```

Introduction

```
'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set new frame section property
    ret = SapModel.PropFrame.SetRod("R1", "A992Fy50", 20)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPropFrame Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST8E0DE823_0?cpp=3103-");SetMethod

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetSDSection(
    string Name,
    string MatProp,
    int DesignType = 0,
    int Color = -1,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetSDSection (
    Name As String,
    MatProp As String,
    DesignType As Integer = 0,
    Color As Integer = -1,
    Notes As String = "",
    GUID As String = ""
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim MatProp As String
Dim DesignType As Integer
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.SetSDSection(Name,
    MatProp, DesignType, Color, Notes,
    GUID)
```

```
int SetSDSection(
    String^ Name,
    String^ MatProp,
    int DesignType = 0,
    int Color = -1,
    String^ Notes = L"",
    String^ GUID = L""
)
```

```
abstract SetSDSection :
    Name : string *
```

cPropFrameid="LST8E0DE823_0"AddLanguageSpecificTextSet("LST8E0DE823_0?cpp=3103-");Set

```

    MatProp : string *
    ?DesignType : int *
    ?Color : int *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _DesignType = defaultArg DesignType 0
    let _Color = defaultArg Color -1
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int

```

Parameters

Name

Type: [SystemString](#)

MatProp

Type: [SystemString](#)

DesignType (Optional)

Type: [SystemInt32](#)

Color (Optional)

Type: [SystemInt32](#)

Notes (Optional)

Type: [SystemString](#)

GUID (Optional)

Type: [SystemString](#)

Return Value

Type: [Int32](#)

See Also

Reference

[cPropFrame Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LSTD81BF7

Method

Initializes a steel angle frame section property. If this function is called for an existing frame section property, all items for the section are reset to their default value.

Namespace: $\hat{A}$ ETABSv1

Assembly: $\hat{A}$ ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetSteelAngle(
    string Name,
    string MatProp,
    double T3,
    double T2,
    double Tf,
    double Tw,
    double r,
    bool MirrorAbout2,
    bool MirrorAbout3,
    int Color = -1,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetSteelAngle (
    Name As String,
    MatProp As String,
    T3 As Double,
    T2 As Double,
    Tf As Double,
    Tw As Double,
    r As Double,
    MirrorAbout2 As Boolean,
    MirrorAbout3 As Boolean,
    Optional As Integer = -1,
    Optional As String = "",
    Optional As String = ""
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim MatProp As String
Dim T3 As Double
Dim T2 As Double
Dim Tf As Double
Dim Tw As Double
Dim r As Double
Dim MirrorAbout2 As Boolean
Dim MirrorAbout3 As Boolean
```

cPropFramespan id="LSTD81BF7F7_0"AddLanguageSpecificTextSet("LSTD81BF7F7_0?cpp=3105.");Set

Introduction

```
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer

returnValue = instance.SetSteelAngle(Name,
MatProp, T3, T2, Tf, Tw, r, MirrorAbout2,
MirrorAbout3, Color, Notes, GUID)

int SetSteelAngle(
String^ Name,
String^ MatProp,
double T3,
double T2,
double Tf,
double Tw,
double r,
bool MirrorAbout2,
bool MirrorAbout3,
int Color = -1,
String^ Notes = L"",
String^ GUID = L""
)

abstract SetSteelAngle :
    Name : string *
    MatProp : string *
    T3 : float *
    T2 : float *
    Tf : float *
    Tw : float *
    r : float *
    MirrorAbout2 : bool *
    MirrorAbout3 : bool *
    ?Color : int *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _Color = defaultArg Color -1
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing or new frame section property. If this is an existing property, that property is modified; otherwise, a new property is added.

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

T2

Type:Â SystemDouble

Tf

Tw Type:Â SystemDouble

r Type:Â SystemDouble

Type:Â SystemDouble
The fillet radius. [L]

MirrorAbout2
Type:Â SystemBoolean
Indicates whether the section is mirrored about the local 2-axis.

MirrorAbout3
Type:Â SystemBoolean
Indicates whether the section is mirrored about the local 3-axis.

Color (Optional)
Type:Â SystemInt32

Notes (Optional)
Type:Â SystemString

GUID (Optional)
Type:Â SystemString
The GUID (global unique identifier), if any, assigned to the section. If this item is input as Default, then the program assigns a GUID to the section.

Return Value

Type:Â Int32
returns zero if the section property is successfully initialized; otherwise it returns a nonzero value.

Remarks

This function supersedes SetAngle(String, String, Double, Double, Double, Double, Int32, String, String).

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new frame section property
    ret = SapModel.PropFrame.SetSteelAngle("ANGLE1", "A992Fy50", 6, 4, 0.5, 0.5, 0.5, False, F
```

```
'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPropFrame Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST590A3216_0?cpp=:3109.");SetS

Method

Initializes a steel tee frame section property. If this function is called for an existing frame section property, all items for the section are reset to their default value.

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetSteelTee(
    string Name,
    string MatProp,
    double T3,
    double T2,
    double Tf,
    double Tw,
    double r,
    bool MirrorAbout3,
    int Color = -1,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetSteelTee (
    Name As String,
    MatProp As String,
    T3 As Double,
    T2 As Double,
    Tf As Double,
    Tw As Double,
    r As Double,
    MirrorAbout3 As Boolean,
    Optional Color As Integer = -1,
    Optional Notes As String = "",
    Optional GUID As String = ""
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim MatProp As String
Dim T3 As Double
Dim T2 As Double
Dim Tf As Double
Dim Tw As Double
Dim r As Double
Dim MirrorAbout3 As Boolean
Dim Color As Integer
Dim Notes As String
Dim GUID As String
```

cPropFramespan id="LST590A3216_0"AddLanguageSpecificTextSet("LST590A3216_0?cpp=:3109.");SetS

```

Dim returnValue As Integer

returnValue = instance.SetSteelTee(Name,
MatProp, T3, T2, Tf, Tw, r, MirrorAbout3,
Color, Notes, GUID)

int SetSteelTee(
String^ Name,
String^ MatProp,
double T3,
double T2,
double Tf,
double Tw,
double r,
bool MirrorAbout3,
int Color = -1,
String^ Notes = L"",
String^ GUID = L""
)

abstract SetSteelTee :
    Name : string *
    MatProp : string *
    T3 : float *
    T2 : float *
    Tf : float *
    Tw : float *
    r : float *
    MirrorAbout3 : bool *
    ?Color : int *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _Color = defaultArg Color -1
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of an existing or new frame section property. If this is an existing property, that property is modified; otherwise, a new property is added.

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

T2

Type:Â SystemDouble

Tf

Type:Â SystemDouble

Tw

Type:Â SystemDouble

r

Type:Â SystemDouble
 The fillet radius. [L]
 MirrorAbout3
 Type:Â SystemBoolean
 Indicates whether the section is mirrored about the local 3-axis.
 Color (Optional)
 Type:Â SystemInt32
 Notes (Optional)
 Type:Â SystemString
 GUID (Optional)
 Type:Â SystemString
 The GUID (global unique identifier), if any, assigned to the section. If this item is input as Default, the program assigns a GUID to the section.

Return Value

Type:Â Int32
 Returns zero if the section property is successfully initialized; otherwise it returns a nonzero value.

▲ Remarks

This function supersedes SetTee(String, String, Double, Double, Double, Double, Int32, String, String).

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new frame section property
    ret = SapModel.PropFrame.SetSteelTee("TEE1", "A992Fy50", 12, 10, 0.6, 0.3, 0.3, False)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPropFrame Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST4BE223A5_0?cpp=3113-");SetT

Method

Initializes a tee-type frame section property. If this function is called for an existing frame section property, all items for the section are reset to their default value.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetTee(
    string Name,
    string MatProp,
    double T3,
    double T2,
    double Tf,
    double Tw,
    int Color = -1,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetTee (
    Name As String,
    MatProp As String,
    T3 As Double,
    T2 As Double,
    Tf As Double,
    Tw As Double,
    Optional As Integer = -1,
    Optional As String = "",
    Optional As String = ""
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim MatProp As String
Dim T3 As Double
Dim T2 As Double
Dim Tf As Double
Dim Tw As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.SetTee(Name, MatProp,
    T3, T2, Tf, Tw, Color, Notes, GUID)
```

```
int SetTee(
```

cPropFramespan id="LST4BE223A5_0"AddLanguageSpecificTextSet("LST4BE223A5_0?cpp=3113-");SetT

```

String^ Name,
String^ MatProp,
double T3,
double T2,
double Tf,
double Tw,
int Color = -1,
String^ Notes = L"",
String^ GUID = L""
)

abstract SetTee :
    Name : string *
    MatProp : string *
    T3 : float *
    T2 : float *
    Tf : float *
    Tw : float *
    ?Color : int *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _Color = defaultArg Color -1
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of an existing or new frame section property. If this is an existing property, that property is modified; otherwise, a new property is added.

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

T2

Type:Â SystemDouble

Tf

Type:Â SystemDouble

Tw

Type:Â SystemDouble

Color (Optional)

Type:Â SystemInt32

Notes (Optional)

Type:Â SystemString

GUID (Optional)

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the section. If this item is input as Default, the program assigns a GUID to the section.

Return Value

Type: Int32

Returns zero if the section property is successfully initialized; otherwise it returns a nonzero value.

▲ **Remarks**

This function is obsolete and has been superseded by SetSteelTee(String, String, Double, Double, Double, Double, Double, Boolean, Int32, String, String) and SetConcreteTee(String, String, Double, Double, Double, Double, Double, Boolean, Int32, String, String) as of version 16.0.0. This function is maintained for backwards compatibility.

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new frame section property
    ret = SapModel.PropFrame.SetTee("TEE1", "A992Fy50", 12, 10, 0.6, 0.3)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cPropFrame Interface

ETABSv1 Namespace

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Introduction

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST122EE4E1_0?cpp=3117-");SetT

Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetTrapezoidal(
    string Name,
    string MatProp,
    double T3,
    double T2,
    double T2b,
    int Color = -1,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetTrapezoidal (
    Name As String,
    MatProp As String,
    T3 As Double,
    T2 As Double,
    T2b As Double,
    Optional As Integer = -1,
    Optional As String = "",
    Optional As String = ""
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim MatProp As String
Dim T3 As Double
Dim T2 As Double
Dim T2b As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.SetTrapezoidal(Name,
    MatProp, T3, T2, T2b, Color, Notes,
    GUID)
```

```
int SetTrapezoidal(
    String^ Name,
    String^ MatProp,
    double T3,
    double T2,
```

cPropFramespan id="LST122EE4E1_0"AddLanguageSpecificTextSet("LST122EE4E1_0?cpp=3117-");SetT

```

double T2b,
int Color = -1,
String^ Notes = L"",
String^ GUID = L""
)

abstract SetTrapezoidal :
    Name : string *
    MatProp : string *
    T3 : float *
    T2 : float *
    T2b : float *
    ?Color : int *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _Color = defaultArg Color -1
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int

```

Parameters

Name

Type: $\hat{A}$ [SystemString](#)

MatProp

Type: $\hat{A}$ [SystemString](#)

T3

Type: $\hat{A}$ [SystemDouble](#)

T2

Type: $\hat{A}$ [SystemDouble](#)

T2b

Type: $\hat{A}$ [SystemDouble](#)

Color (Optional)

Type: $\hat{A}$ [SystemInt32](#)

Notes (Optional)

Type: $\hat{A}$ [SystemString](#)

GUID (Optional)

Type: $\hat{A}$ [SystemString](#)

Return Value

Type: $\hat{A}$ [Int32](#)

▲ See Also

Reference

[cPropFrame Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropFrameAddLanguageSpecificTextSet("LST9817C21C_0?cpp=3120-");SetT

Method

Initializes a tube-type frame section property. If this function is called for an existing frame section property, all items for the section are reset to their default value.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetTube(
    string Name,
    string MatProp,
    double T3,
    double T2,
    double Tf,
    double Tw,
    int Color = -1,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetTube (
    Name As String,
    MatProp As String,
    T3 As Double,
    T2 As Double,
    Tf As Double,
    Tw As Double,
    Optional As Integer = -1,
    Optional As String = "",
    Optional As String = ""
) As Integer
```

```
Dim instance As cPropFrame
Dim Name As String
Dim MatProp As String
Dim T3 As Double
Dim T2 As Double
Dim Tf As Double
Dim Tw As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.SetTube(Name, MatProp,
    T3, T2, Tf, Tw, Color, Notes, GUID)
```

```
int SetTube(
```

cPropFramespan id="LST9817C21C_0"AddLanguageSpecificTextSet("LST9817C21C_0?cpp=3120-");SetT

```

String^ Name,
String^ MatProp,
double T3,
double T2,
double Tf,
double Tw,
int Color = -1,
String^ Notes = L"",
String^ GUID = L""
)

abstract SetTube :
    Name : string *
    MatProp : string *
    T3 : float *
    T2 : float *
    Tf : float *
    Tw : float *
    ?Color : int *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _Color = defaultArg Color -1
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of an existing or new frame section property. If this is an existing property, that property is modified; otherwise, a new property is added.

MatProp

Type:Â SystemString

The name of the material property for the section.

T3

Type:Â SystemDouble

T2

Type:Â SystemDouble

Tf

Type:Â SystemDouble

Tw

Type:Â SystemDouble

Color (Optional)

Type:Â SystemInt32

Notes (Optional)

Type:Â SystemString

GUID (Optional)

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the section. If this item is input as Default, the program assigns a GUID to the section.

Return Value

Type: Int32

Returns zero if the section property is successfully initialized; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim FileName As String
    Dim MatProp As String
    Dim t3 As Double
    Dim t2 As Double
    Dim tf As Double
    Dim tw As Double
    Dim Color As Integer
    Dim Notes As String
    Dim GUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set new frame section property
    ret = SapModel.PropFrame.SetTube("TUBE1", "A992Fy50", 8, 6, 0.5, 0.5)

    'get frame section property data
    ret = SapModel.PropFrame.GetTube("TUBE1", FileName, MatProp, t3, t2, tf, tw, Color, Notes)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

[cPropFrame Interface](#)

[ETABSv1 Namespace](#)

Introduction

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i>>i

CSI API ETABS v1

cPropFrameSDShape Interface

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
public interface cPropFrameSDShape
```

```
Public Interface cPropFrameSDShape
```

```
Dim instance As cPropFrameSDShape
```

```
public interface class cPropFrameSDShape
```

```
type cPropFrameSDShape = interface end
```

The cPropFrameSDShape type exposes the following members.

▲ Methods

$\hat{A}$	Name	Description
	<u>GetAngle</u>	Retrieves property data for an Angle shape in a section designer section
	<u>GetConcreteL</u>	Retrieves property data for a concrete L shape in a section designer section
	<u>GetConcreteTee</u>	Retrieves property data for a concrete tee shape in a section designer section
	<u>GetISection</u>	Retrieves property data for an I-section shape in a section designer section
	<u>GetReinfCircle</u>	Retrieves property data for a circular reinforcing shape in a section designer section
	<u>GetReinfCorner</u>	Retrieves corner point reinforcing data for solid rectangle, circle and polygon shapes in a section designer property
	<u>GetReinfEdge</u>	Retrieves edge reinforcing data for solid rectangle, circle, polygon, and rectangular reinforcing shapes in a section designer property
	<u>GetReinfLine</u>	Retrieves property data for a line reinforcing shape in a section designer section
	<u>GetReinfRectangular</u>	Retrieves property data for a rectangular reinforcing shape in a section designer section
	<u>GetReinfSingle</u>	Retrieves property data for a single bar reinforcing shape in a section designer section

- [GetSolidCircle](#) Retrieves property data for a solid circle shape in a section designer section
- [GetSolidRect](#) Retrieves property data for a solid rectangular shape in a section designer section
- [GetTee](#) Retrieves property data for a Tee shape in a section designer section

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropFrameSDShape Methods

The cPropFrameSDShape type exposes the following members.

Methods

Â	Name	Description
◆	<u>GetAngle</u>	Retrieves property data for an Angle shape in a section designer section
◆	<u>GetConcreteL</u>	Retrieves property data for a concrete L shape in a section designer section
◆	<u>GetConcreteTee</u>	Retrieves property data for a concrete tee shape in a section designer section
◆	<u>GetISection</u>	Retrieves property data for an I-section shape in a section designer section
◆	<u>GetReinfCircle</u>	Retrieves property data for a circular reinforcing shape in a section designer section
◆	<u>GetReinfCorner</u>	Retrieves corner point reinforcing data for solid rectangle, circle and polygon shapes in a section designer property
◆	<u>GetReinfEdge</u>	Retrieves edge reinforcing data for solid rectangle, circle, polygon, and rectangular reinforcing shapes in a section designer property
◆	<u>GetReinfLine</u>	Retrieves property data for a line reinforcing shape in a section designer section
◆	<u>GetReinfRectangular</u>	Retrieves property data for a rectangular reinforcing shape in a section designer section
◆	<u>GetReinfSingle</u>	Retrieves property data for a single bar reinforcing shape in a section designer section
◆	<u>GetSolidCircle</u>	Retrieves property data for a solid circle shape in a section designer section
◆	<u>GetSolidRect</u>	Retrieves property data for a solid rectangular shape in a section designer section
◆	<u>GetTee</u>	Retrieves property data for a Tee shape in a section designer section

Top

◆ See Also

Reference

cPropFrameSDShape Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameSDShapeAddLanguageSpecificTextSet("LSTC52780CB_0") Method

Retrieves property data for an Angle shape in a section designer section

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetAngle(
    string Name,
    string ShapeName,
    ref string MatProp,
    ref string PropName,
    ref int Color,
    ref double XCenter,
    ref double YCenter,
    ref double H,
    ref double Bf,
    ref double Tf,
    ref double Tw,
    ref double Rotation
)
```

```
Function GetAngle (
    Name As String,
    ShapeName As String,
    ByRef MatProp As String,
    ByRef PropName As String,
    ByRef Color As Integer,
    ByRef XCenter As Double,
    ByRef YCenter As Double,
    ByRef H As Double,
    ByRef Bf As Double,
    ByRef Tf As Double,
    ByRef Tw As Double,
    ByRef Rotation As Double
) As Integer
```

```
Dim instance As cPropFrameSDShape
Dim Name As String
Dim ShapeName As String
Dim MatProp As String
Dim PropName As String
Dim Color As Integer
Dim XCenter As Double
Dim YCenter As Double
Dim H As Double
Dim Bf As Double
Dim Tf As Double
```

cPropFrameSDShapesAddLanguageSpecificTextSet("LSTC52780CB_0")cpp=::|

Introduction

```
Dim Tw As Double
Dim Rotation As Double
Dim returnValue As Integer

returnValue = instance.GetAngle(Name,
ShapeName, MatProp, PropName, Color,
XCenter, YCenter, H, Bf, Tf, Tw, Rotation)

int GetAngle(
String^ Name,
String^ ShapeName,
String^% MatProp,
String^% PropName,
int% Color,
double% XCenter,
double% YCenter,
double% H,
double% Bf,
double% Tf,
double% Tw,
double% Rotation
)

abstract GetAngle :
    Name : string *
    ShapeName : string *
    MatProp : string byref *
    PropName : string byref *
    Color : int byref *
    XCenter : float byref *
    YCenter : float byref *
    H : float byref *
    Bf : float byref *
    Tf : float byref *
    Tw : float byref *
    Rotation : float byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame section property that is a section designer section

ShapeName

Type:Â SystemString

The name of an existing Angle shape in the specified frame section property

MatProp

Type:Â SystemString

The name of the material property for the shape

PropName

Type:Â SystemString

This is a blank string or the name of a defined Angle property that has been imported from a section property file. If it is the name of a defined Angle property, the section dimensions are taken from that property

Color

Type:Â SystemInt32

The fill color assigned to the shape

XCenter

Type: SystemDouble

The X-coordinate of the center of the shape in the section designer coordinate system. [L]

YCenter

Type: SystemDouble

The Y-coordinate of the center of the shape in the section designer coordinate system. [L]

H

Type: SystemDouble

The section depth. [L]

Bf

Type: SystemDouble

The flange width. [L]

Tf

Type: SystemDouble

The flange thickness. [L]

Tw

Type: SystemDouble

The web thickness. [L]

Rotation

Type: SystemDouble

The counter clockwise rotation of the shape from its default orientation. [deg]

Return Value

Type: Int32

Returns zero if the property data is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ See Also

Reference

cPropFrameSDShape Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameSDShapeAddLanguageSpecificTextSet("LSD00E4A8C_0") Method

Retrieves property data for a concrete L shape in a section designer section

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetConcreteL(
    string Name,
    string ShapeName,
    ref string MatProp,
    ref string PropName,
    ref int Color,
    ref double XCenter,
    ref double YCenter,
    ref double H,
    ref double Bf,
    ref double Tf,
    ref double Tw,
    ref double Rotation,
    ref bool MirrorAbout2,
    ref bool MirrorAbout3
)
```

```
Function GetConcreteL (
    Name As String,
    ShapeName As String,
    ByRef MatProp As String,
    ByRef PropName As String,
    ByRef Color As Integer,
    ByRef XCenter As Double,
    ByRef YCenter As Double,
    ByRef H As Double,
    ByRef Bf As Double,
    ByRef Tf As Double,
    ByRef Tw As Double,
    ByRef Rotation As Double,
    ByRef MirrorAbout2 As Boolean,
    ByRef MirrorAbout3 As Boolean
) As Integer
```

```
Dim instance As cPropFrameSDShape
Dim Name As String
Dim ShapeName As String
Dim MatProp As String
Dim PropName As String
Dim Color As Integer
Dim XCenter As Double
```

cPropFrameSDShapesid="LSTD00E4A8C_0"AddLanguageSpecificTextSet("LSTD00E4A8C_0")[?cpp=::1](#)

```

Dim YCenter As Double
Dim H As Double
Dim Bf As Double
Dim Tf As Double
Dim Tw As Double
Dim Rotation As Double
Dim MirrorAbout2 As Boolean
Dim MirrorAbout3 As Boolean
Dim returnValue As Integer

returnValue = instance.GetConcreteL(Name,
ShapeName, MatProp, PropName, Color,
XCenter, YCenter, H, Bf, Tf, Tw, Rotation,
MirrorAbout2, MirrorAbout3)

int GetConcreteL(
String^ Name,
String^ ShapeName,
String^% MatProp,
String^% PropName,
int% Color,
double% XCenter,
double% YCenter,
double% H,
double% Bf,
double% Tf,
double% Tw,
double% Rotation,
bool% MirrorAbout2,
bool% MirrorAbout3
)

abstract GetConcreteL :
    Name : string *
    ShapeName : string *
    MatProp : string byref *
    PropName : string byref *
    Color : int byref *
    XCenter : float byref *
    YCenter : float byref *
    H : float byref *
    Bf : float byref *
    Tf : float byref *
    Tw : float byref *
    Rotation : float byref *
    MirrorAbout2 : bool byref *
    MirrorAbout3 : bool byref -> int

```

Parameters

Name

Type:Â SystemString

The name of an existing frame section property that is a section designer section

ShapeName

Type:Â SystemString

The name of an existing concrete L shape in the specified frame section property

MatProp

	Type:Â <u>SystemString</u> The name of the material property for the shape
PropName	Type:Â <u>SystemString</u> This is a blank string or the name of a defined concrete L property that has been imported from a section property file. If it is the name of a defined concrete L property, the section dimensions are taken from that property
Color	Type:Â <u>SystemInt32</u> The fill color assigned to the shape
XCenter	Type:Â <u>SystemDouble</u> The X-coordinate of the center of the shape in the section designer coordinate system. [L]
YCenter	Type:Â <u>SystemDouble</u> The Y-coordinate of the center of the shape in the section designer coordinate system. [L]
H	Type:Â <u>SystemDouble</u> The section depth. [L]
Bf	Type:Â <u>SystemDouble</u> The flange width. [L]
Tf	Type:Â <u>SystemDouble</u> The flange thickness. [L]
Tw	Type:Â <u>SystemDouble</u> The web thickness. [L]
Rotation	Type:Â <u>SystemDouble</u> The counter clockwise rotation of the shape from its default orientation. [deg]
MirrorAbout2	Type:Â <u>SystemBoolean</u> Indicates whether the section is mirrored about the local 2-axis.
MirrorAbout3	Type:Â <u>SystemBoolean</u> Indicates whether the section is mirrored about the local 3-axis.

Return Value

Type:Â Int32
Returns zero if the property data is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ See Also

Reference

cPropFrameSDShape Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameSDShapeAddLanguageSpecificTextSet("LSTF8F82F9_03") Method

Retrieves property data for a concrete tee shape in a section designer section

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetConcreteTee(
    string Name,
    string ShapeName,
    ref string MatProp,
    ref string PropName,
    ref int Color,
    ref double XCenter,
    ref double YCenter,
    ref double H,
    ref double Bf,
    ref double Tf,
    ref double Tw,
    ref double Rotation,
    ref bool MirrorAbout3
)
```

```
Function GetConcreteTee (
    Name As String,
    ShapeName As String,
    ByRef MatProp As String,
    ByRef PropName As String,
    ByRef Color As Integer,
    ByRef XCenter As Double,
    ByRef YCenter As Double,
    ByRef H As Double,
    ByRef Bf As Double,
    ByRef Tf As Double,
    ByRef Tw As Double,
    ByRef Rotation As Double,
    ByRef MirrorAbout3 As Boolean
) As Integer
```

```
Dim instance As cPropFrameSDShape
Dim Name As String
Dim ShapeName As String
Dim MatProp As String
Dim PropName As String
Dim Color As Integer
Dim XCenter As Double
Dim YCenter As Double
Dim H As Double
```

Introduction

```
Dim Bf As Double
Dim Tf As Double
Dim Tw As Double
Dim Rotation As Double
Dim MirrorAbout3 As Boolean
Dim returnValue As Integer

returnValue = instance.GetConcreteTee(Name,
ShapeName, MatProp, PropName, Color,
XCenter, YCenter, H, Bf, Tf, Tw, Rotation,
MirrorAbout3)
```

```
int GetConcreteTee(
String^ Name,
String^ ShapeName,
String^% MatProp,
String^% PropName,
int% Color,
double% XCenter,
double% YCenter,
double% H,
double% Bf,
double% Tf,
double% Tw,
double% Rotation,
bool% MirrorAbout3
)
```

```
abstract GetConcreteTee :
    Name : string *
    ShapeName : string *
    MatProp : string byref *
    PropName : string byref *
    Color : int byref *
    XCenter : float byref *
    YCenter : float byref *
    H : float byref *
    Bf : float byref *
    Tf : float byref *
    Tw : float byref *
    Rotation : float byref *
    MirrorAbout3 : bool byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame section property that is a section designer section

ShapeName

Type:Â SystemString

The name of an existing concrete tee shape in the specified frame section property

MatProp

Type:Â SystemString

The name of the material property for the shape

PropName

Type: SystemString

This is a blank string or the name of a defined concrete tee property that has been imported from a section property file. If it is the name of a defined concrete tee property, the section dimensions are taken from that property.

Color

Type: SystemInt32

The fill color assigned to the shape

XCenter

Type: SystemDouble

The X-coordinate of the center of the shape in the section designer coordinate system. [L]

YCenter

Type: SystemDouble

The Y-coordinate of the center of the shape in the section designer coordinate system. [L]

H

Type: SystemDouble

The section depth. [L]

Bf

Type: SystemDouble

The section width. [L]

Tf

Type: SystemDouble

The flange thickness. [L]

Tw

Type: SystemDouble

The web thickness. [L]

Rotation

Type: SystemDouble

The counter clockwise rotation of the shape from its default orientation. [deg]

MirrorAbout3

Type: SystemBoolean

Indicates whether the section is mirrored about the local 3-axis.

Return Value

Type: Int32

Returns zero if the property data is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ See Also

Reference

cPropFrameSDShape Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameSDShapeAddLanguageSpecificTextSet("LST939DC4E6_0") Method

Retrieves property data for an I-section shape in a section designer section

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetISection(
    string Name,
    string ShapeName,
    ref string MatProp,
    ref string PropName,
    ref int Color,
    ref double XCenter,
    ref double YCenter,
    ref double H,
    ref double Bf,
    ref double Tf,
    ref double Tw,
    ref double Bfb,
    ref double Tfb,
    ref double Rotation
)
```

```
Function GetISection (
    Name As String,
    ShapeName As String,
    ByRef MatProp As String,
    ByRef PropName As String,
    ByRef Color As Integer,
    ByRef XCenter As Double,
    ByRef YCenter As Double,
    ByRef H As Double,
    ByRef Bf As Double,
    ByRef Tf As Double,
    ByRef Tw As Double,
    ByRef Bfb As Double,
    ByRef Tfb As Double,
    ByRef Rotation As Double
) As Integer
```

```
Dim instance As cPropFrameSDShape
Dim Name As String
Dim ShapeName As String
Dim MatProp As String
Dim PropName As String
Dim Color As Integer
Dim XCenter As Double
```

cPropFrameSDShapesid="LST939DC4E6_0"AddLanguageSpecificTextSet("LST939DC4E6_0")cpp=::|

Introduction

```
Dim YCenter As Double
Dim H As Double
Dim Bf As Double
Dim Tf As Double
Dim Tw As Double
Dim Bfb As Double
Dim Tfb As Double
Dim Rotation As Double
Dim returnValue As Integer

returnValue = instance.GetISection(Name,
ShapeName, MatProp, PropName, Color,
XCenter, YCenter, H, Bf, Tf, Tw, Bfb,
Tfb, Rotation)

int GetISection(
String^ Name,
String^ ShapeName,
String^% MatProp,
String^% PropName,
int% Color,
double% XCenter,
double% YCenter,
double% H,
double% Bf,
double% Tf,
double% Tw,
double% Bfb,
double% Tfb,
double% Rotation
)

abstract GetISection :
    Name : string *
    ShapeName : string *
    MatProp : string byref *
    PropName : string byref *
    Color : int byref *
    XCenter : float byref *
    YCenter : float byref *
    H : float byref *
    Bf : float byref *
    Tf : float byref *
    Tw : float byref *
    Bfb : float byref *
    Tfb : float byref *
    Rotation : float byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing frame section property that is a section designer section

ShapeName

Type: SystemString

The name of an existing I-section shape in the specified frame section property

MatProp

	Type:Â <u>SystemString</u> The name of the material property for the shape
PropName	Type:Â <u>SystemString</u> This is a blank string or the name of a defined I-section property that has been imported from a section property file. If it is the name of a defined I-section property, the section dimensions are taken from that property
Color	Type:Â <u>SystemInt32</u> The fill color assigned to the shape
XCenter	Type:Â <u>SystemDouble</u> The X-coordinate of the center of the shape in the section designer coordinate system. [L]
YCenter	Type:Â <u>SystemDouble</u> The Y-coordinate of the center of the shape in the section designer coordinate system. [L]
H	Type:Â <u>SystemDouble</u> The section depth. [L]
Bf	Type:Â <u>SystemDouble</u> The top flange width. [L]
Tf	Type:Â <u>SystemDouble</u> The top flange thickness. [L]
Tw	Type:Â <u>SystemDouble</u> The web thickness. [L]
Bfb	Type:Â <u>SystemDouble</u> The bottom flange width. [L]
Tfb	Type:Â <u>SystemDouble</u> The bottom flange thickness. [L]
Rotation	Type:Â <u>SystemDouble</u> The counter clockwise rotation of the shape from its default orientation. [deg]

Return Value

Type:Â Int32
Returns zero if the property data is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ See Also

Reference

cPropFrameSDShape Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameSDShapeAddLanguageSpecificTextSet("LST2873575F_0") Method

Retrieves property data for a circular reinforcing shape in a section designer section

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetReinfCircle(
    string Name,
    string ShapeName,
    ref double XCenter,
    ref double YCenter,
    ref double Diameter,
    ref int NumBars,
    ref double Rotation,
    ref string RebarSize,
    ref string MatRebar
)
```

```
Function GetReinfCircle (
    Name As String,
    ShapeName As String,
    ByRef XCenter As Double,
    ByRef YCenter As Double,
    ByRef Diameter As Double,
    ByRef NumBars As Integer,
    ByRef Rotation As Double,
    ByRef RebarSize As String,
    ByRef MatRebar As String
) As Integer
```

```
Dim instance As cPropFrameSDShape
Dim Name As String
Dim ShapeName As String
Dim XCenter As Double
Dim YCenter As Double
Dim Diameter As Double
Dim NumBars As Integer
Dim Rotation As Double
Dim RebarSize As String
Dim MatRebar As String
Dim returnValue As Integer
```

```
returnValue = instance.GetReinfCircle(Name,
    ShapeName, XCenter, YCenter, Diameter,
    NumBars, Rotation, RebarSize, MatRebar)
```

```
int GetReinfCircle(
```

```

String^ Name,
String^ ShapeName,
double% XCenter,
double% YCenter,
double% Diameter,
int% NumBars,
double% Rotation,
String^% RebarSize,
String^% MatRebar
)

abstract GetReinfCircle :
    Name : string *
    ShapeName : string *
    XCenter : float byref *
    YCenter : float byref *
    Diameter : float byref *
    NumBars : int byref *
    Rotation : float byref *
    RebarSize : string byref *
    MatRebar : string byref -> int

```

Parameters

Name

Type:Â SystemString

The name of an existing frame section property that is a section designer section

ShapeName

Type:Â SystemString

The name of a circular reinforcing shape in the section designer section

XCenter

Type:Â SystemDouble

The X-coordinate of the center of the shape in the section designer coordinate system. [L]

YCenter

Type:Â SystemDouble

The Y-coordinate of the center of the shape in the section designer coordinate system. [L]

Diameter

Type:Â SystemDouble

The diameter of the circular shape. [L]

NumBars

Type:Â SystemInt32

The number of equally spaced bars for the circular reinforcing

Rotation

Type:Â SystemDouble

The counter clockwise rotation of the shape from its default orientation. [deg]

RebarSize

Type:Â SystemString

The size of the reinforcing bar

MatRebar

Type:Â SystemString

The material property for the reinforcing steel

Return Value

Type: Int32

Returns zero if the property data is successfully retrieved; otherwise it returns a nonzero value

▀ Remarks

▀ See Also

Reference

cPropFrameSDShape Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameSDShapeAddLanguageSpecificTextSet("LST7A93A89_0") Method

Retrieves corner point reinforcing data for solid rectangle, circle and polygon shapes in a section designer property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetReinfCorner(
    string Name,
    string ShapeName,
    ref int NumberItems,
    ref int[] PointNum,
    ref string[] RebarSize
)
```

```
Function GetReinfCorner (
    Name As String,
    ShapeName As String,
    ByRef NumberItems As Integer,
    ByRef PointNum As Integer(),
    ByRef RebarSize As String()
) As Integer
```

```
Dim instance As cPropFrameSDShape
Dim Name As String
Dim ShapeName As String
Dim NumberItems As Integer
Dim PointNum As Integer()
Dim RebarSize As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetReinfCorner(Name,
    ShapeName, NumberItems, PointNum,
    RebarSize)
```

```
int GetReinfCorner(
    String^ Name,
    String^ ShapeName,
    int% NumberItems,
    array<int>^% PointNum,
    array<String>^% RebarSize
)
```

```
abstract GetReinfCorner :
    Name : string *
    ShapeName : string *
    NumberItems : int byref *
```

cPropFrameSDShapesid="LST7A93A89_0"AddLanguageSpecificTextSet("LST7A93A89_0")

```
PointNum : int[] byref *  
RebarSize : string[] byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing frame section property that is a section designer section

ShapeName

Type: SystemString

The name of an existing solid rectangle shape in the specified frame section property

NumberItems

Type: SystemInt32

The number of edges in the shape

PointNum

Type: SystemInt32

This is an array that includes the corner point number in the shape

RebarSize

Type: SystemString

This is an array that includes None or the name of a defined rebar, indicating the rebar assignment to the considered corner point

Return Value

Type: Int32

Returns zero if the property data is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ See Also

Reference

cPropFrameSDShape Interface

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CSI API ETABS v1

cPropFrameSDShapeAddLanguageSpecificTextSet("LST866BF791_0") Method

Retrieves edge reinforcing data for solid rectangle, circle, polygon, and rectangular reinforcing shapes in a section designer property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetReinfEdge(
    string Name,
    string ShapeName,
    ref int NumberItems,
    ref int[] EdgeNum,
    ref string[] RebarSize,
    ref double[] Spacing,
    ref double[] Cover
)
```

```
Function GetReinfEdge (
    Name As String,
    ShapeName As String,
    ByRef NumberItems As Integer,
    ByRef EdgeNum As Integer(),
    ByRef RebarSize As String(),
    ByRef Spacing As Double(),
    ByRef Cover As Double()
) As Integer
```

```
Dim instance As cPropFrameSDShape
Dim Name As String
Dim ShapeName As String
Dim NumberItems As Integer
Dim EdgeNum As Integer()
Dim RebarSize As String()
Dim Spacing As Double()
Dim Cover As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetReinfEdge(Name,
    ShapeName, NumberItems, EdgeNum,
    RebarSize, Spacing, Cover)
```

```
int GetReinfEdge(
    String^ Name,
    String^ ShapeName,
    int% NumberItems,
    array<int>^% EdgeNum,
    array<String>^% RebarSize,
```

cPropFrameSDShapesspan id="LST866BF791_0"AddLanguageSpecificTextSet("LST866BF7913_046pp=::|n

```
array<double>^% Spacing,
array<double>^% Cover
)
```

```
abstract GetReinfEdge :
    Name : string *
    ShapeName : string *
    NumberItems : int byref *
    EdgeNum : int[] byref *
    RebarSize : string[] byref *
    Spacing : float[] byref *
    Cover : float[] byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame section property that is a section designer section

ShapeName

Type:Â SystemString

The name of an existing solid rectangle shape in the specified frame section property

NumberItems

Type:Â SystemInt32

The number of edges in the shape

EdgeNum

Type:Â SystemInt32

This is an array that includes the edge number in the shape

RebarSize

Type:Â SystemString

This is an array that includes None or the name of a defined rebar, indicating the rebar assignment to the considered edge

Spacing

Type:Â SystemDouble

This is an array that includes the rebar maximum center-to-center along the considered edge. [L]

Cover

Type:Â SystemDouble

This is an array that includes the rebar clear cover along the considered edge. [L]

Return Value

Type:Â Int32

Returns zero if the property data is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ See Also

Reference

cPropFrameSDShape Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameSDShapeAddLanguageSpecificTextSet("LST73B270A9_0") Method

Retrieves property data for a line reinforcing shape in a section designer section

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetReinfLine(
    string Name,
    string ShapeName,
    ref double X1,
    ref double Y1,
    ref double X2,
    ref double Y2,
    ref double Spacing,
    ref string RebarSize,
    ref bool EndBars,
    ref string MatRebar
)
```

```
Function GetReinfLine (
    Name As String,
    ShapeName As String,
    ByRef X1 As Double,
    ByRef Y1 As Double,
    ByRef X2 As Double,
    ByRef Y2 As Double,
    ByRef Spacing As Double,
    ByRef RebarSize As String,
    ByRef EndBars As Boolean,
    ByRef MatRebar As String
) As Integer
```

```
Dim instance As cPropFrameSDShape
Dim Name As String
Dim ShapeName As String
Dim X1 As Double
Dim Y1 As Double
Dim X2 As Double
Dim Y2 As Double
Dim Spacing As Double
Dim RebarSize As String
Dim EndBars As Boolean
Dim MatRebar As String
Dim returnValue As Integer
```

```
returnValue = instance.GetReinfLine(Name,
    ShapeName, X1, Y1, X2, Y2, Spacing,
```

cPropFrameSDShapesspan id="LST73B270A9_0"AddLanguageSpecificTextSet("LST73B270A9_0")App=::n

RebarSize, EndBars, MatRebar)

```
int GetReinfLine(
String^ Name,
String^ ShapeName,
double% X1,
double% Y1,
double% X2,
double% Y2,
double% Spacing,
String^% RebarSize,
bool% EndBars,
String^% MatRebar
)
```

```
abstract GetReinfLine :
    Name : string *
    ShapeName : string *
    X1 : float byref *
    Y1 : float byref *
    X2 : float byref *
    Y2 : float byref *
    Spacing : float byref *
    RebarSize : string byref *
    EndBars : bool byref *
    MatRebar : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame section property that is a section designer section

ShapeName

Type:Â SystemString

The name of a line reinforcing shape in the section designer section

X1

Type:Â SystemDouble

The section designer X coordinate of the first drawn end point of the line reinforcing. [L]

Y1

Type:Â SystemDouble

The section designer Y coordinate of the first drawn end point of the line reinforcing. [L]

X2

Type:Â SystemDouble

The section designer X coordinate of the second drawn end point of the line reinforcing. [L]

Y2

Type:Â SystemDouble

The section designer Y coordinate of the second drawn end point of the line reinforcing. [L]

Spacing

Type:Â SystemDouble

The center-to-center spacing of the bars in the line pattern shape. [L]

RebarSize

Type: SystemString

The size of the reinforcing bars used in the line reinforcing shape

EndBars

Type: SystemBoolean

This item is True when there are bars at the end points of the line reinforcing

MatRebar

Type: SystemString

The material property for the reinforcing steel

Return Value

Type: Int32

Returns zero if the property data is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ See Also

Reference

cPropFrameSDShape Interface

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CSI API ETABS v1

cPropFrameSDShapeAddLanguageSpecificTextSet("LSTC4152F2D_0") Method

Retrieves property data for a rectangular reinforcing shape in a section designer section

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetReinfRectangular(
    string Name,
    string ShapeName,
    ref double XCenter,
    ref double YCenter,
    ref double H,
    ref double W,
    ref double Rotation,
    ref string MatRebar
)
```

```
Function GetReinfRectangular (
    Name As String,
    ShapeName As String,
    ByRef XCenter As Double,
    ByRef YCenter As Double,
    ByRef H As Double,
    ByRef W As Double,
    ByRef Rotation As Double,
    ByRef MatRebar As String
) As Integer
```

```
Dim instance As cPropFrameSDShape
Dim Name As String
Dim ShapeName As String
Dim XCenter As Double
Dim YCenter As Double
Dim H As Double
Dim W As Double
Dim Rotation As Double
Dim MatRebar As String
Dim returnValue As Integer
```

```
returnValue = instance.GetReinfRectangular(Name,
    ShapeName, XCenter, YCenter, H, W,
    Rotation, MatRebar)
```

```
int GetReinfRectangular(
    String^ Name,
    String^ ShapeName,
```

cPropFrameSDShapesid="LSTC4152F2D_0"AddLanguageSpecificTextSet("LSTC4152F2D_0")cpp=::r

```
double% XCenter,
double% YCenter,
double% H,
double% W,
double% Rotation,
String^% MatRebar
)
```

```
abstract GetReinfRectangular :
    Name : string *
    ShapeName : string *
    XCenter : float byref *
    YCenter : float byref *
    H : float byref *
    W : float byref *
    Rotation : float byref *
    MatRebar : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame section property that is a section designer section

ShapeName

Type:Â SystemString

The name of a rectangular reinforcing shape in the section designer section

XCenter

Type:Â SystemDouble

The X-coordinate of the center of the shape in the section designer coordinate system. [L]

YCenter

Type:Â SystemDouble

The Y-coordinate of the center of the shape in the section designer coordinate system. [L]

H

Type:Â SystemDouble

The section depth. [L]

W

Type:Â SystemDouble

The top flange width. [L]

Rotation

Type:Â SystemDouble

The counter clockwise rotation of the shape from its default orientation. [deg

MatRebar

Type:Â SystemString

The material property for the reinforcing steel

Return Value

Type:Â Int32

Returns zero if the property data is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ See Also

Reference

cPropFrameSDShape Interface

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CSI API ETABS v1

cPropFrameSDShapeAddLanguageSpecificTextSet("LST710BEFB5_0") Method

Retrieves property data for a single bar reinforcing shape in a section designer section

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetReinfSingle(
    string Name,
    string ShapeName,
    ref double XCenter,
    ref double YCenter,
    ref string RebarSize,
    ref string MatRebar
)
```

```
Function GetReinfSingle (
    Name As String,
    ShapeName As String,
    ByRef XCenter As Double,
    ByRef YCenter As Double,
    ByRef RebarSize As String,
    ByRef MatRebar As String
) As Integer
```

```
Dim instance As cPropFrameSDShape
Dim Name As String
Dim ShapeName As String
Dim XCenter As Double
Dim YCenter As Double
Dim RebarSize As String
Dim MatRebar As String
Dim returnValue As Integer
```

```
returnValue = instance.GetReinfSingle(Name,
    ShapeName, XCenter, YCenter, RebarSize,
    MatRebar)
```

```
int GetReinfSingle(
    String^ Name,
    String^ ShapeName,
    double% XCenter,
    double% YCenter,
    String^% RebarSize,
    String^% MatRebar
)
```

abstract GetReinfSingle :

cPropFrameSDShapesAddLanguageSpecificTextSet("LST710BEFB5_0")
 cpp=::r

```
Name : string *
ShapeName : string *
XCenter : float byref *
YCenter : float byref *
RebarSize : string byref *
MatRebar : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame section property that is a section designer section

ShapeName

Type:Â SystemString

The name of a single bar reinforcing shape in the section designer section

XCenter

Type:Â SystemDouble

The X-coordinate of the center of the shape in the section designer coordinate system. [L]

YCenter

Type:Â SystemDouble

The Y-coordinate of the center of the shape in the section designer coordinate system. [L]

RebarSize

Type:Â SystemString

The size of the reinforcing bar

MatRebar

Type:Â SystemString

The material property for the reinforcing steel

Return Value

Type:Â Int32

Returns zero if the property data is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ See Also

Reference

cPropFrameSDShape Interface

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CSI API ETABS v1

cPropFrameSDShapeAddLanguageSpecificTextSet("LST4032826E_0") Method

Retrieves property data for a solid circle shape in a section designer section

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetSolidCircle(
    string Name,
    string ShapeName,
    ref string MatProp,
    ref string SSOverwrite,
    ref int Color,
    ref double XCenter,
    ref double YCenter,
    ref double Diameter,
    ref bool Reinf,
    ref int NumberBars,
    ref double Rotation,
    ref double Cover,
    ref string RebarSize,
    ref string MatRebar
)
```

```
Function GetSolidCircle (
    Name As String,
    ShapeName As String,
    ByRef MatProp As String,
    ByRef SSOverwrite As String,
    ByRef Color As Integer,
    ByRef XCenter As Double,
    ByRef YCenter As Double,
    ByRef Diameter As Double,
    ByRef Reinf As Boolean,
    ByRef NumberBars As Integer,
    ByRef Rotation As Double,
    ByRef Cover As Double,
    ByRef RebarSize As String,
    ByRef MatRebar As String
) As Integer
```

```
Dim instance As cPropFrameSDShape
Dim Name As String
Dim ShapeName As String
Dim MatProp As String
Dim SSOverwrite As String
Dim Color As Integer
Dim XCenter As Double
```

```

Dim YCenter As Double
Dim Diameter As Double
Dim Reinf As Boolean
Dim NumberBars As Integer
Dim Rotation As Double
Dim Cover As Double
Dim RebarSize As String
Dim MatRebar As String
Dim returnValue As Integer

returnValue = instance.GetSolidCircle(Name,
ShapeName, MatProp, SSOverwrite,
Color, XCenter, YCenter, Diameter,
Reinf, NumberBars, Rotation, Cover,
RebarSize, MatRebar)

```

```

int GetSolidCircle(
String^ Name,
String^ ShapeName,
String^% MatProp,
String^% SSOverwrite,
int% Color,
double% XCenter,
double% YCenter,
double% Diameter,
bool% Reinf,
int% NumberBars,
double% Rotation,
double% Cover,
String^% RebarSize,
String^% MatRebar
)

```

```

abstract GetSolidCircle :
    Name : string *
    ShapeName : string *
    MatProp : string byref *
    SSOverwrite : string byref *
    Color : int byref *
    XCenter : float byref *
    YCenter : float byref *
    Diameter : float byref *
    Reinf : bool byref *
    NumberBars : int byref *
    Rotation : float byref *
    Cover : float byref *
    RebarSize : string byref *
    MatRebar : string byref -> int

```

Parameters

Name

Type:Â SystemString

The name of an existing frame section property that is a section designer section

ShapeName

Type:Â SystemString

The name of an existing solid circle shape in the specified frame section property

MatProp

Type:Â SystemString

The name of the material property for the shape

SSOverwrite

Type:Â SystemString

This is a blank string, Default, or the name of a defined stress-strain curve.

If this item is a blank string or Default, the shape stress-strain curve is based on the assigned material property.

Color

Type:Â SystemInt32

The fill color assigned to the shape

XCenter

Type:Â SystemDouble

The X-coordinate of the center of the shape in the section designer coordinate system. [L]

YCenter

Type:Â SystemDouble

The Y-coordinate of the center of the shape in the section designer coordinate system. [L]

Diameter

Type:Â SystemDouble

The diameter of the circle.[L]

Reinf

Type:Â SystemBoolean

This item is True when there is edge and corner reinforcing steel associated with the shape. The MatProp item must refer to a concrete material for this item to be True.

NumberBars

Type:Â SystemInt32

This item is visible only if the Reinf item is set to True. It is the number of equally spaced bars for the circular reinforcing.

Rotation

Type:Â SystemDouble

The counter clockwise rotation of the shape from its default orientation. [deg]

Cover

Type:Â SystemDouble

This item is visible only if the Reinf item is set to True. It is the clear cover for the specified rebar

RebarSize

Type:Â SystemString

This item is visible only if the Reinf item is set to True. It is the size of the reinforcing bar.

MatRebar

Type:Â SystemString

The material property for the edge and corner reinforcing steel associated with the shape. This item applies only when the MatProp item is a concrete material and the Reinf item is True.

Return Value

Type: Int32

Returns zero if the property data is successfully retrieved; otherwise it returns a nonzero value

▀ Remarks

▀ See Also

Reference

cPropFrameSDShape Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameSDShapeAddLanguageSpecificTextSet("LST7557A898_0") Method

Retrieves property data for a solid rectangular shape in a section designer section

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetSolidRect(
    string Name,
    string ShapeName,
    ref string MatProp,
    ref string SSOverwrite,
    ref int Color,
    ref double XCenter,
    ref double YCenter,
    ref double H,
    ref double W,
    ref double Rotation,
    ref bool Reinf,
    ref string MatRebar
)
```

```
Function GetSolidRect (
    Name As String,
    ShapeName As String,
    ByRef MatProp As String,
    ByRef SSOverwrite As String,
    ByRef Color As Integer,
    ByRef XCenter As Double,
    ByRef YCenter As Double,
    ByRef H As Double,
    ByRef W As Double,
    ByRef Rotation As Double,
    ByRef Reinf As Boolean,
    ByRef MatRebar As String
) As Integer
```

```
Dim instance As cPropFrameSDShape
Dim Name As String
Dim ShapeName As String
Dim MatProp As String
Dim SSOverwrite As String
Dim Color As Integer
Dim XCenter As Double
Dim YCenter As Double
Dim H As Double
Dim W As Double
Dim Rotation As Double
```

Introduction

```
Dim Reinf As Boolean
Dim MatRebar As String
Dim returnValue As Integer

returnValue = instance.GetSolidRect(Name,
ShapeName, MatProp, SSOverwrite,
Color, XCenter, YCenter, H, W, Rotation,
Reinf, MatRebar)

int GetSolidRect(
String^ Name,
String^ ShapeName,
String^% MatProp,
String^% SSOverwrite,
int% Color,
double% XCenter,
double% YCenter,
double% H,
double% W,
double% Rotation,
bool% Reinf,
String^% MatRebar
)

abstract GetSolidRect :
    Name : string *
    ShapeName : string *
    MatProp : string byref *
    SSOverwrite : string byref *
    Color : int byref *
    XCenter : float byref *
    YCenter : float byref *
    H : float byref *
    W : float byref *
    Rotation : float byref *
    Reinf : bool byref *
    MatRebar : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing frame section property that is a section designer section

ShapeName

Type:Â SystemString

The name of an existing solid rectangle shape in the specified frame section property

MatProp

Type:Â SystemString

The name of the material property for the shape

SSOverwrite

Type:Â SystemString

This is a blank string, Default, or the name of a defined stress-strain curve.

If this item is a blank string or Default, the shape stress-strain curve is based on the assigned material property.

Color

Type: SystemInt32

The fill color assigned to the shape

XCenter

Type: SystemDouble

The X-coordinate of the center of the shape in the section designer coordinate system. [L]

YCenter

Type: SystemDouble

The Y-coordinate of the center of the shape in the section designer coordinate system. [L]

H

Type: SystemDouble

The section depth. [L]

W

Type: SystemDouble

The section width. [L]

Rotation

Type: SystemDouble

The counter clockwise rotation of the shape from its default orientation. [deg]

Reinf

Type: SystemBoolean

This item is True when there is edge and corner reinforcing steel associated with the shape. The MatProp item must refer to a concrete material for this item to be True.

MatRebar

Type: SystemString

The material property for the edge and corner reinforcing steel associated with the shape. This item applies only when the MatProp item is a concrete material and the Reinf item is True.

Return Value

Type: Int32

Returns zero if the property data is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ See Also

Reference

cPropFrameSDShape Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropFrameSDShapeAddLanguageSpecificTextSet("LSTCD87C83D_0") Method

Retrieves property data for a Tee shape in a section designer section

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetTee(
    string Name,
    string ShapeName,
    ref string MatProp,
    ref string PropName,
    ref int Color,
    ref double XCenter,
    ref double YCenter,
    ref double H,
    ref double Bf,
    ref double Tf,
    ref double Tw,
    ref double Rotation
)
```

```
Function GetTee (
    Name As String,
    ShapeName As String,
    ByRef MatProp As String,
    ByRef PropName As String,
    ByRef Color As Integer,
    ByRef XCenter As Double,
    ByRef YCenter As Double,
    ByRef H As Double,
    ByRef Bf As Double,
    ByRef Tf As Double,
    ByRef Tw As Double,
    ByRef Rotation As Double
) As Integer
```

```
Dim instance As cPropFrameSDShape
Dim Name As String
Dim ShapeName As String
Dim MatProp As String
Dim PropName As String
Dim Color As Integer
Dim XCenter As Double
Dim YCenter As Double
Dim H As Double
Dim Bf As Double
Dim Tf As Double
```

cPropFrameSDShapeAddLanguageSpecificTextSet("LSTCD87C83D_0")[?cpp=:::3D60](#)

Introduction

```
Dim Tw As Double
Dim Rotation As Double
Dim returnValue As Integer

returnValue = instance.GetTee(Name, ShapeName,
MatProp, PropName, Color, XCenter,
YCenter, H, Bf, Tf, Tw, Rotation)

int GetTee(
String^ Name,
String^ ShapeName,
String^% MatProp,
String^% PropName,
int% Color,
double% XCenter,
double% YCenter,
double% H,
double% Bf,
double% Tf,
double% Tw,
double% Rotation
)

abstract GetTee :
    Name : string *
    ShapeName : string *
    MatProp : string byref *
    PropName : string byref *
    Color : int byref *
    XCenter : float byref *
    YCenter : float byref *
    H : float byref *
    Bf : float byref *
    Tf : float byref *
    Tw : float byref *
    Rotation : float byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing frame section property that is a section designer section

ShapeName

Type: SystemString

The name of an existing Tee shape in the specified frame section property

MatProp

Type: SystemString

The name of the material property for the shape

PropName

Type: SystemString

This is a blank string or the name of a defined Tee property that has been imported from a section property file. If it is the name of a defined Tee property, the section dimensions are taken from that property.

Color

Type: SystemInt32

The fill color assigned to the shape

XCenter

Type: SystemDouble

The X-coordinate of the center of the shape in the section designer coordinate system. [L]

YCenter

Type: SystemDouble

The Y-coordinate of the center of the shape in the section designer coordinate system. [L]

H

Type: SystemDouble

The section depth. [L]

Bf

Type: SystemDouble

The section width. [L]

Tf

Type: SystemDouble

The flange thickness. [L]

Tw

Type: SystemDouble

The web thickness. [L]

Rotation

Type: SystemDouble

The counter clockwise rotation of the shape from its default orientation. [deg]

Return Value

Type: Int32

Returns zero if the property data is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

▲ See Also

Reference

cPropFrameSDShape Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropLineSpring Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cPropLineSpring
```

```
Public Interface cPropLineSpring
```

```
Dim instance As cPropLineSpring
```

```
public interface class cPropLineSpring
```

```
type cPropLineSpring = interface end
```

The cPropLineSpring type exposes the following members.

▲ Methods

▲	Name	Description
◆	ChangeName	Changes the name of a defined line spring property
◆	Delete	Deletes the specified line spring property
◆	GetLineSpringProp	Retrieves an existing named line spring property
◆	GetNameList	Retrieves the names of all defined line spring property
◆	SetLineSpringProp	Creates a new named line spring property, or modifies an existing named line spring property

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropLineSpring Methods

The [cPropLineSpring](#) type exposes the following members.

Methods

	Name	Description
✚	ChangeName	Changes the name of a defined line spring property
✚	Delete	Deletes the specified line spring property
✚	GetLineSpringProp	Retrieves an existing named line spring property
✚	GetNameList	Retrieves the names of all defined line spring property
✚	SetLineSpringProp	Creates a new named line spring property, or modifies an existing named line spring property

[Top](#)

See Also

Reference

[cPropLineSpring Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropLineSpringAddLanguageSpecificTextSet("LST3D Method

Changes the name of a defined line spring property

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int ChangeName(
    string Name,
    string NewName
)
```

```
Function ChangeName (
    Name As String,
    NewName As String
) As Integer
```

```
Dim instance As cPropLineSpring
Dim Name As String
Dim NewName As String
Dim returnValue As Integer
```

```
returnValue = instance.ChangeName(Name,
    NewName)
```

```
int ChangeName(
    String^ Name,
    String^ NewName
)
```

```
abstract ChangeName :
    Name : string *
    NewName : string -> int
```

Parameters

Name

Type: SystemString

The existing name of a defined line spring property

NewName

Type: SystemString

The new name for the line spring property

Return Value

Type: Int32

Returns zero if the new name is successfully applied, otherwise it returns a nonzero value.

▀ **Remarks**

▀ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new spring property
    ret = SapModel.PropLineSpring.SetLineSpringProp("mySpringProp1", 0, 10, 20, 0, 0, 1)

    'change property name
    ret = SapModel.PropLineSpring.ChangeName("mySpringProp1", "prop1")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▀ **See Also**

Reference

cPropLineSpring Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropLineSpringAddLanguageSpecificTextSet("LSTF6 Method

Deletes the specified line spring property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int Delete(
    string Name
)
```

```
Function Delete (
    Name As String
) As Integer
```

```
Dim instance As cPropLineSpring
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.Delete(Name)
```

```
int Delete(
    String^ Name
)
```

```
abstract Delete :
    Name : string -> int
```

Parameters

Name

Type: SystemString

The name of an existing line spring property

Return Value

Type: Int32

Returns zero if the line spring property is successfully deleted, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
```

cPropLineSpringspan id="LSTF610305E_0"AddLanguageSpecificTextSet("LSTF610305E_0?c171|nu=.");

Introduction

```
Dim SapModel As cSapModel
Dim EtabsObject As cOAPI
Dim ret As Integer = -1

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'define new spring property
ret = SapModel.PropLineSpring.SetLineSpringProp("mySpringProp1", 0, 10, 20, 0, 0, 1)

'delete property
ret = SapModel.PropLineSpring.Delete("mySpringProp1")

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

🔗 See Also

Reference

[cPropLineSpring Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropLineSpringAddLanguageSpecificTextSet("LST19B2BB86_0?app=|nu=.") Method

Retrieves an existing named line spring property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetLineSpringProp(
    string Name,
    ref double U1,
    ref double U2,
    ref double U3,
    ref double R1,
    ref int NonlinearOption2,
    ref int NonlinearOption3,
    ref int color,
    ref string notes,
    ref string iGUID
)
```

```
Function GetLineSpringProp (
    Name As String,
    ByRef U1 As Double,
    ByRef U2 As Double,
    ByRef U3 As Double,
    ByRef R1 As Double,
    ByRef NonlinearOption2 As Integer,
    ByRef NonlinearOption3 As Integer,
    ByRef color As Integer,
    ByRef notes As String,
    ByRef iGUID As String
) As Integer
```

```
Dim instance As cPropLineSpring
Dim Name As String
Dim U1 As Double
Dim U2 As Double
Dim U3 As Double
Dim R1 As Double
Dim NonlinearOption2 As Integer
Dim NonlinearOption3 As Integer
Dim color As Integer
Dim notes As String
Dim iGUID As String
Dim returnValue As Integer
```

```
returnValue = instance.GetLineSpringProp(Name,
    U1, U2, U3, R1, NonlinearOption2, NonlinearOption3,
```

cPropLineSpringspan id="LST19B2BB86_0"AddLanguageSpecificTextSet("LST19B2BB86_0?app=|nu=.")

color, notes, iGUID)

```
int GetLineSpringProp(
String^ Name,
double% U1,
double% U2,
double% U3,
double% R1,
int% NonlinearOption2,
int% NonlinearOption3,
int% color,
String^% notes,
String^% iGUID
)
```

```
abstract GetLineSpringProp :
    Name : string *
    U1 : float byref *
    U2 : float byref *
    U3 : float byref *
    R1 : float byref *
    NonlinearOption2 : int byref *
    NonlinearOption3 : int byref *
    color : int byref *
    notes : string byref *
    iGUID : string byref -> int
```

Parameters

Name

Type: SystemString

The name of the line spring property.

U1

Type: SystemDouble

The spring stiffness per unit length in the local 1 direction [F/L2]

U2

Type: SystemDouble

The spring stiffness per unit length in the local 2 direction [F/L2]

U3

Type: SystemDouble

The spring stiffness per unit length in the local 3 direction [F/L2]

R1

Type: SystemDouble

The rotational spring stiffness about local 1 [F/rad]

NonlinearOption2

Type: SystemInt32

The nonlinear option for the local 2 direction, this is one of the following values:

◊ 0 = (None) Linear - Resists Tension and Compression

◊ 1 = Resists Compression only

◊ 2 = Resists Tension only

NonlinearOption3

Type: SystemInt32

The nonlinear option for the local 3 direction, this is one of the following values:

◊ 0 = (None) Linear - Resists Tension and Compression

◊ 1 = Resists Compression only

◇ 2 = Resists Tension only

color

Type:Â SystemInt32

The display color for the property specified as an integer

notes

Type:Â SystemString

The notes, if any, assigned to the property

iGUID

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the property

Return Value

Type:Â Int32

Returns zero if the property is successfully set, otherwise returns nonzero

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new spring property
    ret = SapModel.PropLineSpring.SetLineSpringProp("mySpringProp1", 0, 10, 20, 0, 0, 1)

    'get property
    Dim U1 As Double
    Dim U2 As Double
    Dim U3 As Double
    Dim R1 As Double
    Dim NonlinearOption2 As Integer
    Dim NonlinearOption3 As Integer
    Dim color As Integer
    Dim notes As String
    Dim iGuid As String

    ret = SapModel.PropLineSpring.GetLineSpringProp("mySpringProp1", U1, U2, U3, R1, NonlinearOpti

    'close ETABS
    EtabsObject.ApplicationExit(False)
```

Return Value

```
'clean up variables  
  SapModel = Nothing  
  EtabsObject = Nothing
```

End Sub

▀ See Also

Reference

cPropLineSpring Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropLineSpringAddLanguageSpecificTextSet("LST33 Method

Retrieves the names of all defined line spring property

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetNameList(
    ref int NumberNames,
    ref string[] MyName
)
```

```
Function GetNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cPropLineSpring
Dim NumberNames As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetNameList(NumberNames,
    MyName)
```

```
int GetNameList(
    int% NumberNames,
    array<String^>^% MyName
)
```

```
abstract GetNameList :
    NumberNames : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberNames

Type: System.Int32

The number of line spring property defined in the program

MyName

Type: System.String

This is a one-dimensional array of line spring property names. The MyName array is created as a dynamic, zero-based, array by the API user:

```
Dim MyName() as String
```

The array is dimensioned to (NumberNames + 1) inside the program, filled with the names, and returned to the API user.

Return Value

Type: Int32

Returns zero if the names are successfully retrieved, otherwise it returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new spring property
    ret = SapModel.PropLineSpring.SetLineSpringProp("mySpringProp1", 0, 10, 20, 0, 0, 1)

    'get name list of properties
    Dim NumberNames as Integer
    Dim MyName() as String
    ret = SapModel.PropLineSpring.GetNameList(NumberNames, MyName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cPropLineSpring Interface

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CSI API ETABS v1

cPropLineSpringAddLanguageSpecificTextSet("LSTD7 Method

Creates a new named line spring property, or modifies an existing named line spring property

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetLineSpringProp(
    string Name,
    double U1,
    double U2,
    double U3,
    double R1,
    int NonlinearOption2,
    int NonlinearOption3,
    int color = 0,
    string notes = "",
    string iGUID = ""
)
```

```
Function SetLineSpringProp (
    Name As String,
    U1 As Double,
    U2 As Double,
    U3 As Double,
    R1 As Double,
    NonlinearOption2 As Integer,
    NonlinearOption3 As Integer,
    Optional color As Integer = 0,
    Optional notes As String = "",
    Optional iGUID As String = ""
) As Integer
```

```
Dim instance As cPropLineSpring
Dim Name As String
Dim U1 As Double
Dim U2 As Double
Dim U3 As Double
Dim R1 As Double
Dim NonlinearOption2 As Integer
Dim NonlinearOption3 As Integer
Dim color As Integer
Dim notes As String
Dim iGUID As String
Dim returnValue As Integer
```

```
returnValue = instance.SetLineSpringProp(Name,
```

cPropLineSpringspan id="LSTD7259FA1_0"AddLanguageSpecificTextSet("LSTD7259FA1_0?app=|nu=.")

Introduction

U1, U2, U3, R1, NonlinearOption2, NonlinearOption3,
color, notes, iGUID)

```
int SetLineSpringProp(  
String^ Name,  
double U1,  
double U2,  
double U3,  
double R1,  
int NonlinearOption2,  
int NonlinearOption3,  
int color = 0,  
String^ notes = L"",  
String^ iGUID = L""  
)
```

```
abstract SetLineSpringProp :  
    Name : string *  
    U1 : float *  
    U2 : float *  
    U3 : float *  
    R1 : float *  
    NonlinearOption2 : int *  
    NonlinearOption3 : int *  
    ?color : int *  
    ?notes : string *  
    ?iGUID : string  
(* Defaults:  
    let _color = defaultArg color 0  
    let _notes = defaultArg notes ""  
    let _iGUID = defaultArg iGUID ""  
*)  
-> int
```

Parameters

Name

Type:Â SystemString

The name of the line spring property. If this is an existing property, that property is modified; otherwise, a new property is added.

U1

Type:Â SystemDouble

The spring stiffness per unit length in the local 1 direction [F/L2]

U2

Type:Â SystemDouble

The spring stiffness per unit length in the local 2 direction [F/L2]

U3

Type:Â SystemDouble

The spring stiffness per unit length in the local 3 direction [F/L2]

R1

Type:Â SystemDouble

The rotational spring stiffness about local 1 [F/rad]

NonlinearOption2

Type:Â SystemInt32

The nonlinear option for the local 2 direction, this is one of the following values:
◊ 0 = (None) Linear - Resists Tension and Compression

◇ 1 = Resists Compression only

◇ 2 = Resists Tension only

NonlinearOption3

Type:Â SystemInt32

The nonlinear option for the local 3 direction, this is one of the following values:

◇ 0 = (None) Linear - Resists Tension and Compression

◇ 1 = Resists Compression only

◇ 2 = Resists Tension only

color (Optional)

Type:Â SystemInt32

The display color for the property specified as an integer

notes (Optional)

Type:Â SystemString

The notes, if any, assigned to the property

iGUID (Optional)

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the property

Return Value

Type:Â Int32

Returns zero if the property is successfully set, otherwise returns nonzero

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new spring property
    ret = SapModel.PropLineSpring.SetLineSpringProp("mySpringProp1", 0, 10, 20, 0, 0, 1)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
end Sub
```

End Sub

▲ See Also

Reference

cPropLineSpring Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropLink Interface

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
public interface cPropLink
```

Public Interface cPropLink

```
Dim instance As cPropLink
```

```
public interface class cPropLink
```

```
type cPropLink = interface end
```

The cPropLink type exposes the following members.

▲ Methods

<u>Ä</u>	Name	Description
⇒	<u>ChangeName</u>	Changes the name of an existing link property.
⇒	<u>Count</u>	Returns the total number of defined link properties in the model. If desired, counts can be returned for all link properties of a specified type in the model.
⇒	<u>Delete</u>	Deletes a specified link property.
⇒	<u>GetAcceptanceCriteria</u>	Retrieves the acceptance criteria data for a link property
⇒	<u>GetDamper</u>	retrieves link property data for a damper-type link property.
⇒	<u>GetDamperBilinear</u>	
⇒	<u>GetDamperFrictionSpring</u>	
⇒	<u>GetFrictionIsolator</u>	Retrieves link property data for a friction isolator-type link property.
⇒	<u>GetGap</u>	Retrieves link property data for a gap-type link property.
⇒	<u>GetHook</u>	Retrieves link property data for a hook-type link property.
⇒	<u>GetLinear</u>	Retrieves link property data for a linear-type link property.
⇒	<u>GetMultiLinearElastic</u>	retrieves link property data for a multilinear elastic-type link property.

⇒ <u>GetMultiLinearPlastic</u>	retrieves link property data for a multilinear plastic-type link property.
⇒ <u>GetMultiLinearPoints</u>	Retrieves the force-deformation data for a specified degree of freedom in multilinear elastic and multilinear plastic link properties.
⇒ <u>GetNameList</u>	Retrieves the names of all defined link properties of the specified type.
⇒ <u>GetPDelta</u>	retrieves P-delta parameters for a link property.
⇒ <u>GetPlasticWen</u>	retrieves link property data for a plastic Wen-type link property.
⇒ <u>GetRubberIsolator</u>	retrieves link property data for a rubber isolator-type link property.
⇒ <u>GetSpringData</u>	retrieves length and area values for a link property that are used if the link property is specified in line and area spring assignments.
⇒ <u>GetTCFrictionIsolator</u>	Retrieves link property data for a T/C friction isolator-type link property.
⇒ <u>GetTypeOAPI</u>	Retrieves the property type for the specified link property.
⇒ <u>GetWeightAndMass</u>	Assigns weight and mass data to a link property.
⇒ <u>SetAcceptanceCriteria</u>	Assigns the acceptance criteria data for a link property
⇒ <u>SetDamper</u>	Initializes a damper-type link property. If this function is called for an existing link property, all items for the property are reset to their default values.
⇒ <u>SetDamperBilinear</u>	
⇒ <u>SetDamperFrictionSpring</u>	
⇒ <u>SetFrictionIsolator</u>	Initializes a friction isolator-type link property. If this function is called for an existing link property, all items for the property are reset to their default value.
⇒ <u>SetGap</u>	Initializes a gap-type link property. If this function is called for an existing link property, all items for the property are reset to their default value.
⇒ <u>SetHook</u>	Initializes a hook-type link property. If this function is called for an existing link property, all items for the property are reset to their default value.
⇒ <u>SetLinear</u>	Initializes a linear-type link property. If this function is called for an existing link property, all items for the property are reset to their default value.
⇒ <u>SetMultiLinearElastic</u>	Initializes a multilinear elastic-type link property. If this function is called for an existing link property, all items for the property are reset to their default values.
⇒ <u>SetMultiLinearPlastic</u>	Initializes a multilinear plastic-type link property. If this function is called for an existing link property, all items for the property are reset to their default values.
⇒ <u>SetMultiLinearPoints</u>	Sets the force-deformation data for a specified degree of freedom in multilinear elastic and multilinear plastic

link properties.

⇒ [SetPDelta](#)

Assigns P-delta parameters to a link property.

⇒ [SetPlasticWen](#)

Initializes a plastic Wen-type link property. If this function is called for an existing link property, all items for the property are reset to their default values.

⇒ [SetRubberIsolator](#)

Initializes a rubber isolator-type link property. If this function is called for an existing link property, all items for the property are reset to their default value.

⇒ [SetSpringData](#)

Assigns length and area values to a link property that are used if the link property is specified in line and area spring assignments.

⇒ [SetTCFrictionIsolator](#)

Initializes a T/C friction isolator-type link property. If this function is called for an existing link property, all items for the property are reset to their default value.

⇒ [SetWeightAndMass](#)

Retrieves weight and mass data for a link property.

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropLink Methods

The cPropLink type exposes the following members.

Methods

Á	Name	Description
⇒	<u>ChangeName</u>	Changes the name of an existing link property.
⇒	<u>Count</u>	Returns the total number of defined link properties in the model. If desired, counts can be returned for all link properties of a specified type in the model.
⇒	<u>Delete</u>	Deletes a specified link property.
⇒	<u>GetAcceptanceCriteria</u>	Retrieves the acceptance criteria data for a link property
⇒	<u>GetDamper</u>	retrieves link property data for a damper-type link property.
⇒	<u>GetDamperBilinear</u>	
⇒	<u>GetDamperFrictionSpring</u>	
⇒	<u>GetFrictionIsolator</u>	Retrieves link property data for a friction isolator-type link property.
⇒	<u>GetGap</u>	Retrieves link property data for a gap-type link property.
⇒	<u>GetHook</u>	Retrieves link property data for a hook-type link property.
⇒	<u>GetLinear</u>	Retrieves link property data for a linear-type link property.
⇒	<u>GetMultiLinearElastic</u>	retrieves link property data for a multilinear elastic-type link property.
⇒	<u>GetMultiLinearPlastic</u>	retrieves link property data for a multilinear plastic-type link property.
⇒	<u>GetMultiLinearPoints</u>	Retrieves the force-deformation data for a specified degree of freedom in multilinear elastic and multilinear plastic link properties.
⇒	<u>GetNameList</u>	Retrieves the names of all defined link properties of the specified type.
⇒	<u>GetPDelta</u>	retrieves P-delta parameters for a link property.
⇒	<u>GetPlasticWen</u>	retrieves link property data for a plastic Wen-type link property.
⇒	<u>GetRubberIsolator</u>	retrieves link property data for a rubber isolator-type link property.
⇒	<u>GetSpringData</u>	retrieves length and area values for a link property that are used if the link property is specified in line and area spring assignments.
⇒	<u>GetTCFrictionIsolator</u>	Retrieves link property data for a T/C friction isolator-type link property.

⇒ <u>GetTypeOAPI</u>	Retrieves the property type for the specified link property.
⇒ <u>GetWeightAndMass</u>	Assigns weight and mass data to a link property.
⇒ <u>SetAcceptanceCriteria</u>	Assigns the acceptance criteria data for a link property
⇒ <u>SetDamper</u>	Initializes a damper-type link property. If this function is called for an existing link property, all items for the property are reset to their default values.
⇒ <u>SetDamperBilinear</u>	
⇒ <u>SetDamperFrictionSpring</u>	
⇒ <u>SetFrictionIsolator</u>	Initializes a friction isolator-type link property. If this function is called for an existing link property, all items for the property are reset to their default value.
⇒ <u>SetGap</u>	Initializes a gap-type link property. If this function is called for an existing link property, all items for the property are reset to their default value.
⇒ <u>SetHook</u>	Initializes a hook-type link property. If this function is called for an existing link property, all items for the property are reset to their default value.
⇒ <u>SetLinear</u>	Initializes a linear-type link property. If this function is called for an existing link property, all items for the property are reset to their default value.
⇒ <u>SetMultiLinearElastic</u>	Initializes a multilinear elastic-type link property. If this function is called for an existing link property, all items for the property are reset to their default values.
⇒ <u>SetMultiLinearPlastic</u>	Initializes a multilinear plastic-type link property. If this function is called for an existing link property, all items for the property are reset to their default values.
⇒ <u>SetMultiLinearPoints</u>	Sets the force-deformation data for a specified degree of freedom in multilinear elastic and multilinear plastic link properties.
⇒ <u>SetPDelta</u>	Assigns P-delta parameters to a link property.
⇒ <u>SetPlasticWen</u>	Initializes a plastic Wen-type link property. If this function is called for an existing link property, all items for the property are reset to their default values.
⇒ <u>SetRubberIsolator</u>	Initializes a rubber isolator-type link property. If this function is called for an existing link property, all items for the property are reset to their default value.
⇒ <u>SetSpringData</u>	Assigns length and area values to a link property that are used if the link property is specified in line and area spring assignments.
⇒ <u>SetTCFrictionIsolator</u>	Initializes a T/C friction isolator-type link property. If this function is called for an existing link property, all items for the property are reset to their default value.
⇒ <u>SetWeightAndMass</u>	Retrieves weight and mass data for a link property.

[Top](#)

▲ See Also

Reference

cPropLink Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LSTE01B4387_0") Method

Changes the name of an existing link property.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int ChangeName(
    string Name,
    string NewName
)
```

```
Function ChangeName (
    Name As String,
    NewName As String
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim NewName As String
Dim returnValue As Integer
```

```
returnValue = instance.ChangeName(Name,
    NewName)
```

```
int ChangeName(
    String^ Name,
    String^ NewName
)
```

```
abstract ChangeName :
    Name : string *
    NewName : string -> int
```

Parameters

Name

Type: SystemString

The existing name of a defined link property.

NewName

Type: SystemString

The new name for the link property.

Return ValueType: $\hat{A}$ Int32

Returns zero if the new name is successfully applied; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim DOF() As Boolean
    Dim Fixed() As Boolean
    Dim Ke() As Double
    Dim Ce() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link property
    ReDim DOF(5)
    ReDim Fixed(5)
    ReDim Ke(5)
    ReDim Ce(5)
    DOF(0) = True
    Ke(0) = 12
    ret = SapModel.PropLink.SetLinear("L1", DOF, Fixed, Ke, Ce, 0, 0)

    'change name of link property
    ret = SapModel.PropLink.ChangeName("L1", "MyLink")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cPropLink Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LST6FE95469_0")

Method

Returns the total number of defined link properties in the model. If desired, counts can be returned for all link properties of a specified type in the model.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int Count(
    eLinkPropType PropType =
)
```

```
Function Count (
    Optional PropType As eLinkPropType =
) As Integer
```

```
Dim instance As cPropLink
Dim PropType As eLinkPropType
Dim returnValue As Integer
```

```
returnValue = instance.Count(PropType)
```

```
int Count(
    eLinkPropType PropType =
)
```

```
abstract Count :
    ?PropType : eLinkPropType
(* Defaults:
    let _PropType = defaultArg PropType
*)
-> int
```

Parameters

PropType (Optional)

Type: ETABSv1eLinkPropType

This optional value is one of the items in the eLinkPropType enumeration. If no value is input for PropType, a count is returned for all link properties in the model regardless of type.

Return Value

Type: Int32

Returns the total number of defined link properties in the model.

cPropLinkspan id="LST6FE95469_0"AddLanguageSpecificTextSet("LST6FE95469_0?cpp=::|n8493;Count I

Remarks

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim DOF() As Boolean
    Dim Fixed() As Boolean
    Dim Ke() As Double
    Dim Ce() As Double
    Dim Count As Integer

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link property
    ReDim DOF(5)
    ReDim Fixed(5)
    ReDim Ke(5)
    ReDim Ce(5)
    DOF(0) = True
    Ke(0) = 12
    ret = SapModel.PropLink.SetLinear("L1", DOF, Fixed, Ke, Ce, 0, 0)

    'return number of defined links of all types
    Count = SapModel.PropLink.Count

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

cPropLink Interface

ETABSv1 Namespace

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Introduction

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LST4C322B9_0") Delete Method

Deletes a specified link property.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int Delete(
    string Name
)
```

```
Function Delete (
    Name As String
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.Delete(Name)
```

```
int Delete(
    String^ Name
)
```

```
abstract Delete :
    Name : string -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing link property.

Return Value

Type: [Int32](#)

Returns zero if the link property is successfully deleted; otherwise it returns a nonzero value. It returns an error if the specified link property can not be deleted, for example, if it is being used by an existing link object.

▲ Remarks

▲ Examples

[VB](#)

[Copy](#)

Introduction

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim DOF() As Boolean  
    Dim Fixed() As Boolean  
    Dim Ke() As Double  
    Dim Ce() As Double  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)  
  
    'add link property  
    ReDim DOF(5)  
    ReDim Fixed(5)  
    ReDim Ke(5)  
    ReDim Ce(5)  
    DOF(0) = True  
    Ke(0) = 12  
    ret = SapModel.PropLink.SetLinear("L1", DOF, Fixed, Ke, Ce, 0, 0)  
  
    'delete link property  
    ret = SapModel.PropLink.Delete("L1")  
  
    'close ETABS  
    EtabsObject.ApplicationExit(False)  
  
    'clean up variables  
    SapModel = Nothing  
    EtabsObject = Nothing  
End Sub
```

🔗 See Also

Reference

[cPropLink Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LST62826EB0") Method

Retrieves the acceptance criteria data for a link property

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetAcceptanceCriteria(
    string Name,
    ref int AcceptanceType,
    ref bool Symmetric,
    ref bool[] Active,
    ref double[] IOPos,
    ref double[] LSPos,
    ref double[] CPPos,
    ref double[] IONeg,
    ref double[] LSNeg,
    ref double[] CPNeg
)
```

```
Function GetAcceptanceCriteria (
    Name As String,
    ByRef AcceptanceType As Integer,
    ByRef Symmetric As Boolean,
    ByRef Active As Boolean(),
    ByRef IOPos As Double(),
    ByRef LSPos As Double(),
    ByRef CPPos As Double(),
    ByRef IONeg As Double(),
    ByRef LSNeg As Double(),
    ByRef CPNeg As Double()
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim AcceptanceType As Integer
Dim Symmetric As Boolean
Dim Active As Boolean()
Dim IOPos As Double()
Dim LSPos As Double()
Dim CPPos As Double()
Dim IONeg As Double()
Dim LSNeg As Double()
Dim CPNeg As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetAcceptanceCriteria(Name,
    AcceptanceType, Symmetric, Active,
```

IOPos, LSPos, CPPos, IONeg, LSNeg, CPNeg)

```
int GetAcceptanceCriteria(
String^ Name,
int% AcceptanceType,
bool% Symmetric,
array<bool>^% Active,
array<double>^% IOPos,
array<double>^% LSPos,
array<double>^% CPPos,
array<double>^% IONeg,
array<double>^% LSNeg,
array<double>^% CPNeg
)
```

```
abstract GetAcceptanceCriteria :
    Name : string *
    AcceptanceType : int byref *
    Symmetric : bool byref *
    Active : bool[] byref *
    IOPos : float[] byref *
    LSPos : float[] byref *
    CPPos : float[] byref *
    IONeg : float[] byref *
    LSNeg : float[] byref *
    CPNeg : float[] byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing link property

AcceptanceType

Type: [SystemInt32](#)

This item is either 1 or 2. 1 means Force and 2 means Displacement.

Symmetric

Type: [SystemBoolean](#)

Indicates if the acceptance criteria is symmetric, that is, the negative values are the same as the positive values.

Active

Type: [SystemBoolean](#)

This is an array dimensioned to 5 indicating if the acceptance criteria is active for each. Items 0 thru 5 in the array refer to F1, F2, F3, M1, M2, M3 or U1, U2, U3, R1, R2, R3, respectively.

IOPos

Type: [SystemDouble](#)

This is an array dimensioned to 5 indicating the positive immediate occupancy acceptance criteria values for each DOF. Items 0 thru 5 in the array refer to F1, F2, F3, M1, M2, M3 or U1, U2, U3, R1, R2, R3, respectively.

LSPos

Type: [SystemDouble](#)

This is an array dimensioned to 5 indicating the positive life safety acceptance criteria values for each DOF. Items 0 thru 5 in the array refer to F1, F2, F3, M1, M2, M3 or U1, U2, U3, R1, R2, R3, respectively.

CPPosType: SystemDouble

This is an array dimensioned to 5 indicating the positive collapse prevention acceptance criteria values for each DOF. Items 0 thru 5 in the array refer to F1, F2, F3, M1, M2, M3 or U1, U2, U3, R1, R2, R3, respectively.

IONegType: SystemDouble

This is an array dimensioned to 5 indicating the negative immediate occupancy acceptance criteria values for each DOF. Items 0 thru 5 in the array refer to F1, F2, F3, M1, M2, M3 or U1, U2, U3, R1, R2, R3, respectively.

LSNegType: SystemDouble

This is an array dimensioned to 5 indicating the negative life safety acceptance criteria values for each DOF. Items 0 thru 5 in the array refer to F1, F2, F3, M1, M2, M3 or U1, U2, U3, R1, R2, R3, respectively.

CPNegType: SystemDouble

This is an array dimensioned to 5 indicating the negative collapse prevention acceptance criteria values for each DOF. Items 0 thru 5 in the array refer to F1, F2, F3, M1, M2, M3 or U1, U2, U3, R1, R2, R3, respectively.

Return ValueType: Int32

returns zero if the data is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyDOF() As Boolean
    Dim MyFixed() As Boolean
    Dim MyNonLinear() As Boolean
    Dim MyKe() As Double
    Dim MyCe() As Double
    Dim Active() As Boolean
    Dim IOPos() As Double
    Dim LSPos() As Double
    Dim CPPos() As Double
    Dim IONeg() As Double
    Dim LSNeg() As Double
    Dim CPNeg() As Double
    Dim AcceptanceType As Integer
    Dim Symmetric As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
```

Introduction

```
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'add a link property
ReDim MyDOF(5)
ReDim MyFixed(5)
ReDim MyNonLinear(5)
ReDim MyKe(5)
ReDim MyCe(5)

MyDOF(0) = True
MyKe(0) = 12
MyCe(0) = 0.01

MyDOF(1) = True
MyNonLinear(1) = True
MyKe(1) = 12
MyCe(1) = 0.01

MyDOF(2) = True
MyFixed(2) = True

ret = SapModel.PropLink.SetMultiLinearElastic("MLE1", MyDOF, MyFixed, MyNonLinear, MyKe, MyCe)

'set link property acceptance criteria
ReDim Active(5)
ReDim IOPos(5)
ReDim LSPos(5)
ReDim CPPos(5)
ReDim IONeg(5)
ReDim LSNeg(5)
ReDim CPNeg(5)
Active(0) = True
Active(1) = True
IOPos(0) = 100
LSPos(0) = 200
CPPos(0) = 300
IONeg(0) = 120
LSNeg(0) = 220
CPNeg(0) = 320
IOPos(1) = 150
LSPos(1) = 250
CPPos(1) = 350
IONeg(1) = 130
LSNeg(1) = 230
CPNeg(1) = 330
ret = SapModel.PropLink.SetAcceptanceCriteria("MLE1", 1, False, Active, IOPos, LSPos, CPPos, IONeg, LSNeg, CPNeg)

'get link property acceptance criteria
ReDim Active(5)
ReDim IOPos(5)
ReDim LSPos(5)
ReDim CPPos(5)
ReDim IONeg(5)
ReDim LSNeg(5)
ReDim CPNeg(5)
ret = SapModel.PropLink.GetAcceptanceCriteria("MLE1", AcceptanceType, Symmetric, Active, IOPos, LSPos, CPPos, IONeg, LSNeg, CPNeg)
```

Introduction

```
'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPropLink Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LST1EB4321D_0?cpp=:::3293");GetDa Method

retrieves link property data for a damper-type link property.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetDamper(
    string Name,
    ref bool[] DOF,
    ref bool[] Fixed,
    ref bool[] Nonlinear,
    ref double[] Ke,
    ref double[] Ce,
    ref double[] K,
    ref double[] C,
    ref double[] CExp,
    ref double DJ2,
    ref double DJ3,
    ref string Notes,
    ref string GUID
)
```

```
Function GetDamper (
    Name As String,
    ByRef DOF As Boolean(),
    ByRef Fixed As Boolean(),
    ByRef Nonlinear As Boolean(),
    ByRef Ke As Double(),
    ByRef Ce As Double(),
    ByRef K As Double(),
    ByRef C As Double(),
    ByRef CExp As Double(),
    ByRef DJ2 As Double,
    ByRef DJ3 As Double,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim DOF As Boolean()
Dim Fixed As Boolean()
Dim Nonlinear As Boolean()
Dim Ke As Double()
Dim Ce As Double()
Dim K As Double()
Dim C As Double()
```

cPropLinkspan id="LST1EB4321D_0"AddLanguageSpecificTextSet("LST1EB4321D_0?cpp=:::3293");GetDa

```

Dim CExp As Double()
Dim DJ2 As Double
Dim DJ3 As Double
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer

returnValue = instance.GetDamper(Name,
DOF, Fixed, Nonlinear, Ke, Ce, K, C,
CExp, DJ2, DJ3, Notes, GUID)

int GetDamper(
String^ Name,
array<bool>^% DOF,
array<bool>^% Fixed,
array<bool>^% Nonlinear,
array<double>^% Ke,
array<double>^% Ce,
array<double>^% K,
array<double>^% C,
array<double>^% CExp,
double% DJ2,
double% DJ3,
String^% Notes,
String^% GUID
)

abstract GetDamper :
    Name : string *
    DOF : bool[] byref *
    Fixed : bool[] byref *
    Nonlinear : bool[] byref *
    Ke : float[] byref *
    Ce : float[] byref *
    K : float[] byref *
    C : float[] byref *
    CExp : float[] byref *
    DJ2 : float byref *
    DJ3 : float byref *
    Notes : string byref *
    GUID : string byref -> int

```

Parameters

Name

Type: SystemString

The name of an existing damper-type link property.

DOF

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if properties exist for a specified degree of freedom.

Value DOF

DOF(0) U1

DOF(1) U2

DOF(2) U3

DOF(3) R1

DOF(4) R2

DOF(5) R3

Fixed

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if the specified degree of freedom is fixed (restrained).

Value	Fixity
--------------	---------------

Fixed(0) U1 fixity if DOF(0) = True

Fixed(1) U2 fixity if DOF(1) = True

Fixed(2) U3 fixity if DOF(2) = True

Fixed(3) R1 fixity if DOF(3) = True

Fixed(4) R2 fixity if DOF(4) = True

Fixed(5) R3 fixity if DOF(5) = True

The term Fixed(n) applies only when DOF(n) = True.

Nonlinear

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if nonlinear properties exist for a specified degree of freedom.

Value	Nonlinear
--------------	------------------

Nonlinear(0) U1 has nonlinear properties

Nonlinear(1) U2 has nonlinear properties

Nonlinear(2) U3 has nonlinear properties

Nonlinear(3) R1 has nonlinear properties

Nonlinear(4) R2 has nonlinear properties

Nonlinear(5) R3 has nonlinear properties

The term Nonlinear(n) applies only when DOF(n) = True and Fixed(n) = False.

Ke

Type: SystemDouble

This is an array of effective stiffness terms for the link property. The effective stiffness applies for linear analyses.

Value Stiffness

Ke(0) U1 [F/L]

Ke(1) U2 [F/L]

Ke(2) U3 [F/L]

Ke(3) R1 [FL]

Ke(4) R2 [FL]

Ke(5) R3 [FL]

The term ke(n) applies only when DOF(n) = True and Fixed(n) = False.

Ce

Type: SystemDouble

This is an array of effective damping terms for the link property. The effective stiffness applies for linear analyses.

Value Damping

Ce(0) U1 [F/L]

Ce(1) U2 [F/L]

Ce(2) U3 [F/L]

Ce(3) R1 [FL]

Ce(4) R2 [FL]

Ce(5) R3 [FL]

The term ce(n) applies only when DOF(n) = True and Fixed(n) = False.

K

Type: SystemDouble

This is an array of initial stiffness terms for the link property. The initial stiffness applies for nonlinear analyses.

Value Stiffness

K(0) U1 [F/L]

K(1) U2 [F/L]

K(2) U3 [F/L]

K(3) R1 [FL]

K(4) R2 [FL]

K(5) R3 [FL]

The term k(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

C

Type: SystemDouble

This is an array of nonlinear damping coefficient terms for the link property. The nonlinear damping coefficient applies for nonlinear analyses.

Value Damping

C(0) U1 [F/(L^{cexp})]

C(1) U2 [F/(L^{cexp})]

C(2) U3 [F/(L^{cexp})]

C(3) R1 [FL]

C(4) R2 [FL]

C(5) R3 [FL]

The term c(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

CExp

Type: SystemDouble

This is an array of the nonlinear damping exponent terms. The nonlinear damping exponent applies for nonlinear analyses. It is applied to the velocity across the damper in the equation of motion.

Value cexp

cexp(0) U1

cexp(1) U2

cexp(2) U3

cexp(3) R1

cexp(4) R2

cexp(5) R3

The term cexp(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

DJ2

Type:Â SystemDouble

The distance from the J-End of the link to the U2 shear spring. This item applies only when DOF(1) = True. [L]

DJ3

Type:Â SystemDouble

The distance from the J-End of the link to the U3 shear spring. This item applies only when DOF(2) = True. [L]

Notes

Type:Â SystemString

The notes, if any, assigned to the property.

GUID

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the property.

Return Value

Type:Â Int32

Returns zero if the property is successfully retrieved; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyDOF() As Boolean
    Dim MyFixed() As Boolean
    Dim MyNonLinear() As Boolean
    Dim MyKe() As Double
    Dim MyCe() As Double
    Dim MyK() As Double
    Dim MyC() As Double
    Dim MyCexp() As Double
    Dim DOF() As Boolean
    Dim Fixed() As Boolean
    Dim MyNonLinear() As Boolean
    Dim Ke() As Double
    Dim Ce() As Double
    Dim K() As Double
    Dim C() As Double
    Dim Cexp() As Double
    Dim dj2 As Double
    Dim dj3 As Double
    Dim Notes As String
    Dim GUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel
```

Introduction

```
'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'add link property
    ReDim MyDOF(5)
    ReDim MyFixed(5)
    ReDim MyNonLinear(5)
    ReDim MyKe(5)
    ReDim MyCe(5)
    ReDim MyK(5)
    ReDim MyC(5)
    ReDim MyCexp(5)

    MyDOF(0) = True
    MyKe(0) = 12
    MyCe(0) = 0.01

    MyDOF(1) = True
    MyNonLinear(1) = True
    MyKe(1) = 12
    MyCe(1) = 0.01
    MyK(1) = 20
    MyC(1)=0.08
    MyCexp(1) = 1.2

    MyDOF(2) = True
    MyFixed(2) = True

    ret = SapModel.PropLink.SetDamper("D1", MyDOF, MyFixed, MyNonLinear, MyKe, MyCe, MyK, MyC, MyCexp)

'get link property data
    ret = SapModel.PropLink.GetDamper("D1", DOF, Fixed, NonLinear, Ke, Ce, k, c, cexp, dj2, dj3)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

📌 See Also

Reference

[cPropLink Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LSTB82A5B20_0?cpp=::|3209);GetDa Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetDamperBilinear(
    string Name,
    ref bool[] dof,
    ref bool[] Fixed,
    ref bool[] Nonlinear,
    ref double[] ke,
    ref double[] ce,
    ref double[] k,
    ref double[] c,
    ref double[] CY,
    ref double[] ForceLimit,
    ref double dj2,
    ref double dj3,
    ref string notes,
    ref string GUID
)
```

```
Function GetDamperBilinear (
    Name As String,
    ByRef dof As Boolean(),
    ByRef Fixed As Boolean(),
    ByRef Nonlinear As Boolean(),
    ByRef ke As Double(),
    ByRef ce As Double(),
    ByRef k As Double(),
    ByRef c As Double(),
    ByRef CY As Double(),
    ByRef ForceLimit As Double(),
    ByRef dj2 As Double,
    ByRef dj3 As Double,
    ByRef notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim dof As Boolean()
Dim Fixed As Boolean()
Dim Nonlinear As Boolean()
Dim ke As Double()
Dim ce As Double()
Dim k As Double()
Dim c As Double()
```

cPropLinkspan id="LSTB82A5B20_0"AddLanguageSpecificTextSet("LSTB82A5B20_0?cpp=::|3209);GetDa

Introduction

```
Dim CY As Double()
Dim ForceLimit As Double()
Dim dj2 As Double
Dim dj3 As Double
Dim notes As String
Dim GUID As String
Dim returnValue As Integer

returnValue = instance.GetDamperBilinear(Name,
dof, Fixed, Nonlinear, ke, ce, k, c,
CY, ForceLimit, dj2, dj3, notes, GUID)
```

```
int GetDamperBilinear(
String^ Name,
array<bool>^% dof,
array<bool>^% Fixed,
array<bool>^% Nonlinear,
array<double>^% ke,
array<double>^% ce,
array<double>^% k,
array<double>^% c,
array<double>^% CY,
array<double>^% ForceLimit,
double% dj2,
double% dj3,
String^% notes,
String^% GUID
)
```

```
abstract GetDamperBilinear :
    Name : string *
    dof : bool[] byref *
    Fixed : bool[] byref *
    Nonlinear : bool[] byref *
    ke : float[] byref *
    ce : float[] byref *
    k : float[] byref *
    c : float[] byref *
    CY : float[] byref *
    ForceLimit : float[] byref *
    dj2 : float byref *
    dj3 : float byref *
    notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type:Â SystemString

dof

Type:Â SystemBoolean

Fixed

Type:Â SystemBoolean

Nonlinear

Type:Â SystemBoolean

ke

Type:Â SystemDouble

ce

k Type:Â [SystemDouble](#)
 c Type:Â [SystemDouble](#)
 CY Type:Â [SystemDouble](#)
 ForceLimit Type:Â [SystemDouble](#)
 dj2 Type:Â [SystemDouble](#)
 dj3 Type:Â [SystemDouble](#)
 notes Type:Â [SystemString](#)
 GUID Type:Â [SystemString](#)

Return Value

Type:Â [Int32](#)

🔍 See Also

Reference

[cPropLink Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LSTD1258EDF_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetDamperFrictionSpring(
    string Name,
    ref bool[] dof,
    ref bool[] Fixed,
    ref bool[] Nonlinear,
    ref double[] ke,
    ref double[] ce,
    ref double[] k,
    ref double[] K1,
    ref double[] K2,
    ref double[] u0,
    ref double[] us,
    ref int[] direction,
    ref double dj2,
    ref double dj3,
    ref string notes,
    ref string GUID
)
```

```
Function GetDamperFrictionSpring (
    Name As String,
    ByRef dof As Boolean(),
    ByRef Fixed As Boolean(),
    ByRef Nonlinear As Boolean(),
    ByRef ke As Double(),
    ByRef ce As Double(),
    ByRef k As Double(),
    ByRef K1 As Double(),
    ByRef K2 As Double(),
    ByRef u0 As Double(),
    ByRef us As Double(),
    ByRef direction As Integer(),
    ByRef dj2 As Double,
    ByRef dj3 As Double,
    ByRef notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim dof As Boolean()
Dim Fixed As Boolean()
Dim Nonlinear As Boolean()
```

cPropLinkspan id="LSTD1258EDF_0"AddLanguageSpecificTextSet("LSTD1258EDF_0?cpp=::3212");GetD


```

Dim ke As Double()
Dim ce As Double()
Dim k As Double()
Dim K1 As Double()
Dim K2 As Double()
Dim u0 As Double()
Dim us As Double()
Dim direction As Integer()
Dim dj2 As Double
Dim dj3 As Double
Dim notes As String
Dim GUID As String
Dim returnValue As Integer

returnValue = instance.GetDamperFrictionSpring(Name,
dof, Fixed, Nonlinear, ke, ce, k, K1,
K2, u0, us, direction, dj2, dj3, notes,
GUID)

```

```

int GetDamperFrictionSpring(
String^ Name,
array<bool>^% dof,
array<bool>^% Fixed,
array<bool>^% Nonlinear,
array<double>^% ke,
array<double>^% ce,
array<double>^% k,
array<double>^% K1,
array<double>^% K2,
array<double>^% u0,
array<double>^% us,
array<int>^% direction,
double% dj2,
double% dj3,
String^% notes,
String^% GUID
)

```

```

abstract GetDamperFrictionSpring :
    Name : string *
    dof : bool[] byref *
    Fixed : bool[] byref *
    Nonlinear : bool[] byref *
    ke : float[] byref *
    ce : float[] byref *
    k : float[] byref *
    K1 : float[] byref *
    K2 : float[] byref *
    u0 : float[] byref *
    us : float[] byref *
    direction : int[] byref *
    dj2 : float byref *
    dj3 : float byref *
    notes : string byref *
    GUID : string byref -> int

```

Parameters

Name

Type: SystemString

dof	Type:Â SystemBoolean
Fixed	Type:Â SystemBoolean
Nonlinear	Type:Â SystemBoolean
ke	Type:Â SystemDouble
ce	Type:Â SystemDouble
k	Type:Â SystemDouble
K1	Type:Â SystemDouble
K2	Type:Â SystemDouble
u0	Type:Â SystemDouble
us	Type:Â SystemDouble
direction	Type:Â SystemInt32
dj2	Type:Â SystemDouble
dj3	Type:Â SystemDouble
notes	Type:Â SystemString
GUID	Type:Â SystemString

Return ValueType:Â [Int32](#)

▲ See Also

Reference[cPropLink Interface](#)[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LST64EC8328") Method

Retrieves link property data for a friction isolator-type link property.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetFrictionIsolator(
    string Name,
    ref bool[] DOF,
    ref bool[] Fixed,
    ref bool[] Nonlinear,
    ref double[] Ke,
    ref double[] Ce,
    ref double[] K,
    ref double[] Slow,
    ref double[] Fast,
    ref double[] Rate,
    ref double[] Radius,
    ref double Damping,
    ref double DJ2,
    ref double DJ3,
    ref string Notes,
    ref string GUID
)
```

```
Function GetFrictionIsolator (
    Name As String,
    ByRef DOF As Boolean(),
    ByRef Fixed As Boolean(),
    ByRef Nonlinear As Boolean(),
    ByRef Ke As Double(),
    ByRef Ce As Double(),
    ByRef K As Double(),
    ByRef Slow As Double(),
    ByRef Fast As Double(),
    ByRef Rate As Double(),
    ByRef Radius As Double(),
    ByRef Damping As Double,
    ByRef DJ2 As Double,
    ByRef DJ3 As Double,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim DOF As Boolean()
```

cPropLinkspan id="LST64EC8328_0"AddLanguageSpecificTextSet("LST64EC8328_0?cpp=::|rs=15;GetFri

Introduction

```
Dim Fixed As Boolean()
Dim Nonlinear As Boolean()
Dim Ke As Double()
Dim Ce As Double()
Dim K As Double()
Dim Slow As Double()
Dim Fast As Double()
Dim Rate As Double()
Dim Radius As Double()
Dim Damping As Double
Dim DJ2 As Double
Dim DJ3 As Double
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.GetFrictionIsolator(Name,
DOF, Fixed, Nonlinear, Ke, Ce, K, Slow,
Fast, Rate, Radius, Damping, DJ2, DJ3,
Notes, GUID)
```

```
int GetFrictionIsolator(
String^ Name,
array<bool>^% DOF,
array<bool>^% Fixed,
array<bool>^% Nonlinear,
array<double>^% Ke,
array<double>^% Ce,
array<double>^% K,
array<double>^% Slow,
array<double>^% Fast,
array<double>^% Rate,
array<double>^% Radius,
double% Damping,
double% DJ2,
double% DJ3,
String^% Notes,
String^% GUID
)
```

```
abstract GetFrictionIsolator :
    Name : string *
    DOF : bool[] byref *
    Fixed : bool[] byref *
    Nonlinear : bool[] byref *
    Ke : float[] byref *
    Ce : float[] byref *
    K : float[] byref *
    Slow : float[] byref *
    Fast : float[] byref *
    Rate : float[] byref *
    Radius : float[] byref *
    Damping : float byref *
    DJ2 : float byref *
    DJ3 : float byref *
    Notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing friction isolator-type link property.

DOF

Type:Â SystemBoolean

This is a boolean array, dimensioned to 5, indicating if properties exist for a specified degree of freedom.

Value DOF

DOF(0) U1

DOF(1) U2

DOF(2) U3

DOF(3) R1

DOF(4) R2

DOF(5) R3

Fixed

Type:Â SystemBoolean

This is a boolean array, dimensioned to 5, indicating if the specified degree of freedom is fixed (restrained).

Value Fixity

Fixed(0) U1 fixity if DOF(0) = True

Fixed(1) U2 fixity if DOF(1) = True

Fixed(2) U3 fixity if DOF(2) = True

Fixed(3) R1 fixity if DOF(3) = True

Fixed(4) R2 fixity if DOF(4) = True

Fixed(5) R3 fixity if DOF(5) = True

Nonlinear

Type:Â SystemBoolean

This is a boolean array, dimensioned to 5, indicating if nonlinear properties exist for a specified degree of freedom.

Value Nonlinear

Nonlinear(0) U1 has nonlinear properties

Nonlinear(1) U2 has nonlinear properties

Nonlinear(2) U3 has nonlinear properties

Nonlinear(3) R1, not used

Nonlinear(4) R2, not used

Nonlinear(5) R3, not used

Note that this item is applicable only for degrees of freedom U1, U2 and U3. For those degrees of freedom, the term Nonlinear(n) applies only when DOF(n) = True and Fixed(n) = False.

Ke

Type:Â SystemDouble

This is an array of effective stiffness terms for the link property. The effective stiffness applies for linear analyses.

Value Stiffness

Ke(0) U1 [F/L]

Ke(1) U2 [F/L]

Ke(2) U3 [F/L]

Ke(3) R1 [FL]

Ke(4) R2 [FL]

Ke(5) R3 [FL]

The term ke(n) applies only when DOF(n) = True and Fixed(n) = False.

Ce

Type: SystemDouble

This is an array of effective damping terms for the link property. The effective stiffness applies for linear analyses.

Value Damping

Ce(0) U1 [F/L]

Ce(1) U2 [F/L]

Ce(2) U3 [F/L]

Ce(3) R1 [FL]

Ce(4) R2 [FL]

Ce(5) R3 [FL]

The term ce(n) applies only when DOF(n) = True and Fixed(n) = False.

K

Type: SystemDouble

This is an array of initial stiffness terms for the link property. The initial stiffness applies for nonlinear analyses.

Value Stiffness

K(0) U1 [F/L]

K(1) U2 [F/L]

K(2) U3 [F/L]

K(3) R1, not used

K(4) R2, not used

K(5) R3, not used

Note that this item is applicable only for degrees of freedom U1, U2 and U3. For those degrees of freedom, the term k(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Slow

Type: SystemDouble

This is an array of the friction coefficient at zero velocity terms for the link property. This coefficient applies for nonlinear analyses.

Value Slow

Slow(0) U1, not used

Slow(1) U2

Slow(2) U3

Slow(3) R1, not used

Slow(4) R2, not used

Slow(5) R3, not used

Note that this item is applicable only for degrees of freedom U2 and U3. For those degrees of freedom, the term Slow(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Fast

Type: SystemDouble

This is an array of the friction coefficient at fast velocity terms for the link property. This coefficient applies for nonlinear analyses.

Value	Fast
--------------	-------------

Fast(0) U1, not used

Fast(1) U2

Fast(2) U3

Fast(3) R1, not used

Fast(4) R2, not used

Fast(5) R3, not used

Note that this item is applicable only for degrees of freedom U2 and U3. For those degrees of freedom, the term Fast(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Rate

Type: SystemDouble

This is an array of the inverse of the characteristic sliding velocity terms for the link property. This item applies for nonlinear analyses.

Value	Rate
--------------	-------------

Rate(0) U1, not used

Rate(1) U2 [s/L]

Rate(2) U3 [s/L]

Rate(3) R1, not used

Rate(4) R2, not used

Rate(5) R3, not used

Note that this item is applicable only for degrees of freedom U2 and U3. For those degrees of freedom, the term Rate(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Radius

Type: SystemDouble

This is an array of the radius of the sliding contact surface terms for the link property. Inputting 0 means there is an infinite radius, that is, the slider is flat. This item applies for nonlinear analyses.

Value	Radius
--------------	---------------

Radius(0) U1, not used

Radius(1) U2 [L]

Radius(2) U3 [L]

Radius(3) R1, not used

Radius(4) R2, not used

Radius(5) R3, not used

Note that this item is applicable only for degrees of freedom U2 and U3. For those degrees of freedom, the term Radius(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

DampingType: SystemDouble

This is the nonlinear damping coefficient used for the axial translational degree of freedom, U1. This item applies for nonlinear analyses. [F/L]

DJ2Type: SystemDouble

The distance from the J-End of the link to the U2 shear spring. This item applies only when DOF(1) = True. [L]

DJ3Type: SystemDouble

The distance from the J-End of the link to the U3 shear spring. This item applies only when DOF(2) = True. [L]

NotesType: SystemString

The notes, if any, assigned to the property.

GUIDType: SystemString

The GUID (global unique identifier), if any, assigned to the property.

Return ValueType: Int32

Returns zero if the property is successfully retrieved; otherwise it returns a nonzero value.

▲ **Remarks**▲ **Examples****VB**Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyDOF() As Boolean
    Dim MyFixed() As Boolean
    Dim MyNonLinear() As Boolean
    Dim MyKe() As Double
    Dim MyCe() As Double
    Dim MyK() As Double
    Dim MySlow() As Double
    Dim MyFast() As Double
    Dim MyRate() As Double
    Dim MyRadius() As Double
    Dim DOF() As Boolean
    Dim Fixed() As Boolean
    Dim NonLinear() As Boolean
    Dim Ke() As Double
    Dim Ce() As Double
    Dim K() As Double
    Dim Slow() As Double
    Dim Fast() As Double
    Dim Rate() As Double
    Dim Radius() As Double
    Dim Damping As Double
    Dim dj2 As Double
    Dim dj3 As Double
    Dim Notes As String
```


Introduction

```
Dim GUID As String

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'add link property
ReDim MyDOF(5)
ReDim MyFixed(5)
ReDim MyNonLinear(5)
ReDim MyKe(5)
ReDim MyCe(5)
ReDim MyK(5)
ReDim MySlow(5)
ReDim MyFast(5)
ReDim MyRate(5)
ReDim MyRadius(5)

MyDOF(0) = True
MyKe(0) = 12
MyCe(0) = 0.01

MyDOF(1) = True
MyNonLinear(1) = True
MyKe(1) = 12
MyCe(1) = 0.01
MyK(1) = 20
MySlow(1)= 0.6
MyFast(1)= 0.5
MyRate(1)= 10
MyRadius(1)= 80

MyDOF(2) = True
MyNonLinear(2) = True
MyKe(2) = 14
MyCe(2) = 0.008
MyK(2) = 22
MySlow(2)= 0.66
MyFast(2)= 0.55
MyRate(2)= 12
MyRadius(2)= 75

MyDOF(3) = True
MyKe(3) = 15
MyCe(3) = 0

MyDOF(4) = True
MyFixed(4) = True

ret = SapModel.PropLink.SetFrictionIsolator("FI1", MyDOF, MyFixed, MyNonLinear, MyKe, MyCe
```

Introduction

```
'get link property data
    ret = SapModel.PropLink.GetFrictionIsolator("FI1", DOF, Fixed, NonLinear, Ke, Ce, k, Slow,
'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

➤ See Also

Reference

cPropLink Interface

ETABSv1 Namespace

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Namespace: ETABSv1

▲ Syntax

VBF#

```
int GetGap(
string Name,
ref bool[] DOF,
ref bool[] Fixed,
ref bool[] Nonlinear,
ref double[] Ke,
ref double[] Ce,
ref double[] K,
ref double[] Dis,
ref double DJ2,
ref double DJ3,
ref string Notes,
ref string GUID
)
```

```
Function GetGap (
Name As String,
ByRef DOF As Boolean(),
ByRef Fixed As Boolean(),
ByRef Nonlinear As Boolean(),
ByRef Ke As Double(),
ByRef Ce As Double(),
ByRef K As Double(),
ByRef Dis As Double(),
ByRef DJ2 As Double,
ByRef DJ3 As Double,
ByRef Notes As String,
ByRef GUID As String
) As Integer
```

cPropLinkspan id="LST735C165C_0"AddLanguageSpecificTextSet("LST735C165C_0?cpp=::|0223);GetGa

Introduction

```
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer

returnValue = instance.GetGap(Name, DOF,
Fixed, Nonlinear, Ke, Ce, K, Dis, DJ2,
DJ3, Notes, GUID)

int GetGap(
String^ Name,
array<bool>^% DOF,
array<bool>^% Fixed,
array<bool>^% Nonlinear,
array<double>^% Ke,
array<double>^% Ce,
array<double>^% K,
array<double>^% Dis,
double% DJ2,
double% DJ3,
String^% Notes,
String^% GUID
)

abstract GetGap :
    Name : string *
    DOF : bool[] byref *
    Fixed : bool[] byref *
    Nonlinear : bool[] byref *
    Ke : float[] byref *
    Ce : float[] byref *
    K : float[] byref *
    Dis : float[] byref *
    DJ2 : float byref *
    DJ3 : float byref *
    Notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing gap-type link property.

DOF

Type:Â SystemBoolean

This is a boolean array, dimensioned to 5, indicating if properties exist for a specified degree of freedom.

Value DOF

DOF(0) U1

DOF(1) U2

DOF(2) U3

DOF(3) R1

DOF(4) R2

DOF(5) R3

Fixed

Type:Â SystemBoolean

This is a boolean array, dimensioned to 5, indicating if the specified degree of freedom is fixed (restrained).

Value Fixity

Fixed(0) U1 fixity if DOF(0) = True
 Fixed(1) U2 fixity if DOF(1) = True
 Fixed(2) U3 fixity if DOF(2) = True
 Fixed(3) R1 fixity if DOF(3) = True
 Fixed(4) R2 fixity if DOF(4) = True
 Fixed(5) R3 fixity if DOF(5) = True

Nonlinear

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if nonlinear properties exist for a specified degree of freedom.

Value Nonlinear

Nonlinear(0) U1 has nonlinear properties
 Nonlinear(1) U2 has nonlinear properties
 Nonlinear(2) U3 has nonlinear properties
 Nonlinear(3) R1 has nonlinear properties
 Nonlinear(4) R2 has nonlinear properties
 Nonlinear(5) R3 has nonlinear properties

The term Nonlinear(n) applies only when DOF(n) = True and Fixed(n) = False.

Ke

Type: SystemDouble

This is an array of effective stiffness terms for the link property. The effective stiffness applies for linear analyses.

Value Stiffness

Ke(0) U1 [F/L]
 Ke(1) U2 [F/L]
 Ke(2) U3 [F/L]
 Ke(3) R1 [FL]
 Ke(4) R2 [FL]
 Ke(5) R3 [FL]

The term ke(n) applies only when DOF(n) = True and Fixed(n) = False.

Ce

Type: SystemDouble

This is an array of effective damping terms for the link property. The effective stiffness applies for linear analyses.

Value Damping

Ce(0) U1 [F/L]
 Ce(1) U2 [F/L]
 Ce(2) U3 [F/L]
 Ce(3) R1 [FL]
 Ce(4) R2 [FL]
 Ce(5) R3 [FL]

The term ce(n) applies only when DOF(n) = True and Fixed(n) = False.

K

Type: SystemDouble

This is an array of initial stiffness terms for the link property. The initial stiffness applies for nonlinear analyses.

Value Stiffness

K(0) U1 [F/L]

K(1) U2 [F/L]

K(2) U3 [F/L]

K(3) R1 [FL]

K(4) R2 [FL]

K(5) R3 [FL]

The term k(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Dis

Type: SystemDouble

This is an array of initial gap opening terms for the link property. The initial gap opening applies for nonlinear analyses.

Value Dis

dis(0) U1 [L]

dis(1) U2 [L]

dis(2) U3 [L]

dis(3) R1 [rad]

dis(4) R2 [rad]

dis(5) R3 [rad]

The term dis(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

DJ2

Type: SystemDouble

The distance from the J-End of the link to the U2 shear spring. This item applies only when DOF(1) = True. [L]

DJ3

Type: SystemDouble

The distance from the J-End of the link to the U3 shear spring. This item applies only when DOF(2) = True. [L]

Notes

Type: SystemString

The notes, if any, assigned to the property.

GUID

Type: SystemString

The GUID (global unique identifier), if any, assigned to the property. If this item is input as Default, the program assigns a GUID to the property.

Return ValueType: Int32

Returns zero if the property is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

Return Value

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyDOF() As Boolean
    Dim MyFixed() As Boolean
    Dim MyNonLinear() As Boolean
    Dim MyKe() As Double
    Dim MyCe() As Double
    Dim MyK() As Double
    Dim MyDis() As Double
    Dim DOF() As Boolean
    Dim Fixed() As Boolean
    Dim NonLinear() As Boolean
    Dim Ke() As Double
    Dim Ce() As Double
    Dim K() As Double
    Dim Dis() As Double
    Dim dj2 As Double
    Dim dj3 As Double
    Dim Notes As String
    Dim GUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link property
    ReDim MyDOF(5)
    ReDim MyFixed(5)
    ReDim MyNonLinear(5)
    ReDim MyKe(5)
    ReDim MyCe(5)
    ReDim MyK(5)
    ReDim MyDis(5)

    MyDOF(0) = True
    MyKe(0) = 12
    MyCe(0) = 0.01

    MyDOF(1) = True
    MyNonLinear(1) = True
    MyKe(1) = 12
    MyCe(1) = 0.01
    MyK(1) = 20
    MyDis(1) = 1.2
```

Introduction

```
MyDOF(2) = True
MyFixed(2) = True

ret = SapModel.PropLink.SetGap("G1", MyDOF, MyFixed, MyNonLinear, MyKe, MyCe, MyK, MyDis,
'get link property data
ret = SapModel.PropLink.GetGap("G1", DOF, Fixed, NonLinear, Ke, Ce, k, dis, dj2, dj3, Note
'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

See Also

Reference

cPropLink Interface

ETABSV1 Namespace

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LSTF29A6E3B_0?cpp=::|6229);GetHook

Method

Retrieves link property data for a hook-type link property.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetHook(
    string Name,
    ref bool[] DOF,
    ref bool[] Fixed,
    ref bool[] Nonlinear,
    ref double[] Ke,
    ref double[] Ce,
    ref double[] K,
    ref double[] Dis,
    ref double DJ2,
    ref double DJ3,
    ref string Notes,
    ref string GUID
)
```

```
Function GetHook (
    Name As String,
    ByRef DOF As Boolean(),
    ByRef Fixed As Boolean(),
    ByRef Nonlinear As Boolean(),
    ByRef Ke As Double(),
    ByRef Ce As Double(),
    ByRef K As Double(),
    ByRef Dis As Double(),
    ByRef DJ2 As Double,
    ByRef DJ3 As Double,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim DOF As Boolean()
Dim Fixed As Boolean()
Dim Nonlinear As Boolean()
Dim Ke As Double()
Dim Ce As Double()
Dim K As Double()
Dim Dis As Double()
Dim DJ2 As Double
Dim DJ3 As Double
```

cPropLinkspan id="LSTF29A6E3B_0"AddLanguageSpecificTextSet("LSTF29A6E3B_0?cpp=::|6229);GetHook

Introduction

```
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer

returnValue = instance.GetHook(Name, DOF,
Fixed, Nonlinear, Ke, Ce, K, Dis, DJ2,
DJ3, Notes, GUID)

int GetHook(
String^ Name,
array<bool>^% DOF,
array<bool>^% Fixed,
array<bool>^% Nonlinear,
array<double>^% Ke,
array<double>^% Ce,
array<double>^% K,
array<double>^% Dis,
double% DJ2,
double% DJ3,
String^% Notes,
String^% GUID
)

abstract GetHook :
    Name : string *
    DOF : bool[] byref *
    Fixed : bool[] byref *
    Nonlinear : bool[] byref *
    Ke : float[] byref *
    Ce : float[] byref *
    K : float[] byref *
    Dis : float[] byref *
    DJ2 : float byref *
    DJ3 : float byref *
    Notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing hook-type link property.

DOF

Type:Â SystemBoolean

This is a boolean array, dimensioned to 5, indicating if properties exist for a specified degree of freedom.

Value DOF

DOF(0) U1

DOF(1) U2

DOF(2) U3

DOF(3) R1

DOF(4) R2

DOF(5) R3

Fixed

Type:Â SystemBoolean

This is a boolean array, dimensioned to 5, indicating if the specified degree of freedom is fixed (restrained).

Value Fixity

Fixed(0) U1 fixity if DOF(0) = True

Fixed(1) U2 fixity if DOF(1) = True

Fixed(2) U3 fixity if DOF(2) = True

Fixed(3) R1 fixity if DOF(3) = True

Fixed(4) R2 fixity if DOF(4) = True

Fixed(5) R3 fixity if DOF(5) = True

Nonlinear

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if nonlinear properties exist for a specified degree of freedom.

Value Nonlinear

Nonlinear(0) U1 has nonlinear properties

Nonlinear(1) U2 has nonlinear properties

Nonlinear(2) U3 has nonlinear properties

Nonlinear(3) R1 has nonlinear properties

Nonlinear(4) R2 has nonlinear properties

Nonlinear(5) R3 has nonlinear properties

The term Nonlinear(n) applies only when DOF(n) = True and Fixed(n) = False.

Ke

Type: SystemDouble

This is an array of effective stiffness terms for the link property. The effective stiffness applies for linear analyses.

Value Stiffness

Ke(0) U1 [F/L]

Ke(1) U2 [F/L]

Ke(2) U3 [F/L]

Ke(3) R1 [FL]

Ke(4) R2 [FL]

Ke(5) R3 [FL]

The term ke(n) applies only when DOF(n) = True and Fixed(n) = False.

Ce

Type: SystemDouble

This is an array of effective damping terms for the link property. The effective stiffness applies for linear analyses.

Value Damping

Ce(0) U1 [F/L]

Ce(1) U2 [F/L]

Ce(2) U3 [F/L]

Ce(3) R1 [FL]

Ce(4) R2 [FL]

Ce(5) R3 [FL]

The term ce(n) applies only when DOF(n) = True and Fixed(n) = False.

K

Type: SystemDouble

This is an array of initial stiffness terms for the link property. The initial stiffness applies for nonlinear analyses.

Value Stiffness

K(0) U1 [F/L]

K(1) U2 [F/L]

K(2) U3 [F/L]

K(3) R1 [FL]

K(4) R2 [FL]

K(5) R3 [FL]

The term k(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Dis

Type: SystemDouble

This is an array of initial hook opening terms for the link property. The initial hook opening applies for nonlinear analyses.

Value Dis

dis(0) U1 [L]

dis(1) U2 [L]

dis(2) U3 [L]

dis(3) R1 [rad]

dis(4) R2 [rad]

dis(5) R3 [rad]

The term dis(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

DJ2

Type: SystemDouble

The distance from the J-End of the link to the U2 shear spring. This item applies only when DOF(1) = True. [L]

DJ3

Type: SystemDouble

The distance from the J-End of the link to the U3 shear spring. This item applies only when DOF(2) = True. [L]

Notes

Type: SystemString

The notes, if any, assigned to the property.

GUID

Type: SystemString

The GUID (global unique identifier), if any, assigned to the property. If this item is input as Default, the program assigns a GUID to the property.

Return ValueType: Int32

Returns zero if the property is successfully retrieved; otherwise it returns a nonzero value

▲ Remarks

Return Value

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyDOF() As Boolean
    Dim MyFixed() As Boolean
    Dim MyNonLinear() As Boolean
    Dim MyKe() As Double
    Dim MyCe() As Double
    Dim MyK() As Double
    Dim MyDis() As Double
    Dim DOF() As Boolean
    Dim Fixed() As Boolean
    Dim NonLinear() As Boolean
    Dim Ke() As Double
    Dim Ce() As Double
    Dim K() As Double
    Dim Dis() As Double
    Dim dj2 As Double
    Dim dj3 As Double
    Dim Notes As String
    Dim GUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link property
    ReDim MyDOF(5)
    ReDim MyFixed(5)
    ReDim MyNonLinear(5)
    ReDim MyKe(5)
    ReDim MyCe(5)
    ReDim MyK(5)
    ReDim MyDis(5)

    MyDOF(0) = True
    MyKe(0) = 12
    MyCe(0) = 0.01

    MyDOF(1) = True
    MyNonLinear(1) = True
    MyKe(1) = 12
    MyCe(1) = 0.01
    MyK(1) = 20
    MyDis(1) = 1.2
```

Introduction

```
MyDOF(2) = True
MyFixed(2) = True

ret = SapModel.PropLink.SetGap("G1", MyDOF, MyFixed, MyNonLinear, MyKe, MyCe, MyK, MyDis,
'get link property data
ret = SapModel.PropLink.GetGap("G1", DOF, Fixed, NonLinear, Ke, Ce, k, dis, dj2, dj3, Note
'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

See Also

Reference

cPropLink Interface

ETABSV1 Namespace

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LST286F22C6_0") Method

Retrieves link property data for a linear-type link property.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetLinear(
    string Name,
    ref bool[] DOF,
    ref bool[] Fixed,
    ref double[] Ke,
    ref double[] Ce,
    ref double DJ2,
    ref double DJ3,
    ref bool KeCoupled,
    ref bool CeCoupled,
    ref string Notes,
    ref string GUID
)
```

```
Function GetLinear (
    Name As String,
    ByRef DOF As Boolean(),
    ByRef Fixed As Boolean(),
    ByRef Ke As Double(),
    ByRef Ce As Double(),
    ByRef DJ2 As Double,
    ByRef DJ3 As Double,
    ByRef KeCoupled As Boolean,
    ByRef CeCoupled As Boolean,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim DOF As Boolean()
Dim Fixed As Boolean()
Dim Ke As Double()
Dim Ce As Double()
Dim DJ2 As Double
Dim DJ3 As Double
Dim KeCoupled As Boolean
Dim CeCoupled As Boolean
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

cPropLinkAddLanguageSpecificTextSet("LST286F22C6_0?cpp=::|n3235;GetLin

```
returnValue = instance.GetLinear(Name,
DOF, Fixed, Ke, Ce, DJ2, DJ3, KeCoupled,
CeCoupled, Notes, GUID)
```

```
int GetLinear(
String^ Name,
array<bool>^% DOF,
array<bool>^% Fixed,
array<double>^% Ke,
array<double>^% Ce,
double% DJ2,
double% DJ3,
bool% KeCoupled,
bool% CeCoupled,
String^% Notes,
String^% GUID
)
```

```
abstract GetLinear :
    Name : string *
    DOF : bool[] byref *
    Fixed : bool[] byref *
    Ke : float[] byref *
    Ce : float[] byref *
    DJ2 : float byref *
    DJ3 : float byref *
    KeCoupled : bool byref *
    CeCoupled : bool byref *
    Notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing linear-type link property.

DOF

Type:Â SystemBoolean

This is a boolean array, dimensioned to 5, indicating if properties exist for a specified degree of freedom.

Value DOF

DOF(0) U1

DOF(1) U2

DOF(2) U3

DOF(3) R1

DOF(4) R2

DOF(5) R3

Fixed

Type:Â SystemBoolean

This is a boolean array, dimensioned to 5, indicating if the specified degree of freedom is fixed (restrained).

Value

Fixity

Fixed(0) U1 fixity if DOF(0) = True

Fixed(1) U2 fixity if DOF(1) = True
 Fixed(2) U3 fixity if DOF(2) = True
 Fixed(3) R1 fixity if DOF(3) = True
 Fixed(4) R2 fixity if DOF(4) = True
 Fixed(5) R3 fixity if DOF(5) = True

Ke

Type: SystemDouble

This is an array of stiffness terms for the link property. There are 6 terms in the array if the stiffness is uncoupled and 21 if it is coupled. The KeCoupled item indicates if the stiffness is coupled.

If the stiffness is uncoupled:

Value Stiffness

Ke(0) U1 [F/L]
 Ke(1) U2 [F/L]
 Ke(2) U3 [F/L]
 Ke(3) R1 [FL]
 Ke(4) R2 [FL]
 Ke(5) R3 [FL]

If the stiffness is coupled:

Value Stiffness

Ke(0) U1U1 [F/L]
 Ke(1) U1U2 [F/L]
 Ke(2) U2U2 [F/L]
 Ke(3) U1U3 [F/L]
 Ke(4) U2U3 [F/L]
 Ke(5) U3U3 [F/L]
 Ke(6) U1R1 [F]
 Ke(7) U2R1 [F]
 Ke(8) U3R1 [F]
 Ke(9) R1R1 [FL]
 Ke(10) U1R2 [F]
 Ke(11) U2R2 [F]
 Ke(12) U3R2 [F]
 Ke(14) R2R2 [FL]
 Ke(15) U1R3 [F]
 Ke(16) U2R3 [F]
 Ke(17) U3R3 [F]
 Ke(18) R1R3 [FL]
 Ke(19) R2R3 [FL]
 Ke(20) R3R3 [FL]

Ce

Type: SystemDouble

This is an array of damping terms for the link property. There are 6 terms in the array if the damping is uncoupled and 21 if it is coupled. The CeCoupled item indicates if the damping is coupled.

If the damping is uncoupled:

Value Damping

Ce(0) U1 [F/L]

Ce(1) U2 [F/L]

Ce(2) U3 [F/L]

Ce(3) R1 [FL]

Ce(4) R2 [FL]

Ce(5) R3 [FL]

If the damping is coupled:

Value Damping

Ce(0) U1U1 [F/L]

Ce(1) U1U2 [F/L]

Ce(2) U2U2 [F/L]

Ce(3) U1U3 [F/L]

Ce(4) U2U3 [F/L]

Ce(5) U3U3 [F/L]

Ce(6) U1R1 [F]

Ce(7) U2R1 [F]

Ce(8) U3R1 [F]

Ce(9) R1R1 [FL]

Ce(10) U1R2 [F]

Ce(11) U2R2 [F]

Ce(12) U3R2 [F]

Ce(14) R2R2 [FL]

Ce(15) U1R3 [F]

Ce(16) U2R3 [F]

Ce(17) U3R3 [F]

Ce(18) R1R3 [FL]

Ce(19) R2R3 [FL]

Ce(20) R3R3 [FL]

DJ2

Type: SystemDouble

The distance from the J-End of the link to the U2 shear spring. This item applies only when DOF(1) = True. [L]

DJ3

Type: SystemDouble

The distance from the J-End of the link to the U3 shear spring. This item applies only when DOF(2) = True. [L]

KeCoupledType: SystemBoolean

This item is True if the link stiffness, Ke, is coupled. There are 21 terms in the Ke array if Ke is coupled; otherwise there are 6 terms

CeCoupledType: SystemBoolean

This item is True if the link damping, Ce, is coupled. There are 21 terms in the Ce array if Ce is coupled; otherwise there are 6 terms

NotesType: SystemString

The notes, if any, assigned to the property.

GUIDType: SystemString

The GUID (global unique identifier), if any, assigned to the property.

Return ValueType: Int32

Returns zero if the property data is successfully retrieved; otherwise it returns a nonzero value.

▲ **Remarks**▲ **Examples****VB**Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyDOF() As Boolean
    Dim MyFixed() As Boolean
    Dim MyKe() As Double
    Dim MyCe() As Double
    Dim DOF() As Boolean
    Dim Fixed() As Boolean
    Dim Ke() As Double
    Dim Ce() As Double
    Dim dj2 As Double
    Dim dj3 As Double
    Dim KeCoupled As Boolean
    Dim CeCoupled As Boolean
    Dim Notes As String
    Dim GUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
```

Introduction

```
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'add link property
ReDim MyDOF(5)
ReDim MyFixed(5)
ReDim MyKe(5)
ReDim MyCe(5)
MyDOF(0) = True
MyKe(0) = 12
ret = SapModel.PropLink.SetLinear("L1", MyDOF, MyFixed, MyKe, MyCe, 0, 0)

'get link property data
ret = SapModel.PropLink.GetLinear("L1", DOF, Fixed, Ke, Ce, dj2, dj3, KeCoupled, CeCoupled)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPropLink Interface

ETABSv1 Namespace

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»

CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LST499D38A7_0") Method

retrieves link property data for a multilinear elastic-type link property.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetMultiLinearElastic(
    string Name,
    ref bool[] D0F,
    ref bool[] Fixed,
    ref bool[] Nonlinear,
    ref double[] Ke,
    ref double[] Ce,
    ref double DJ2,
    ref double DJ3,
    ref string Notes,
    ref string GUID
)
```

```
Function GetMultiLinearElastic (
    Name As String,
    ByRef D0F As Boolean(),
    ByRef Fixed As Boolean(),
    ByRef Nonlinear As Boolean(),
    ByRef Ke As Double(),
    ByRef Ce As Double(),
    ByRef DJ2 As Double,
    ByRef DJ3 As Double,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim D0F As Boolean()
Dim Fixed As Boolean()
Dim Nonlinear As Boolean()
Dim Ke As Double()
Dim Ce As Double()
Dim DJ2 As Double
Dim DJ3 As Double
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.GetMultiLinearElastic(Name,
D0F, Fixed, Nonlinear, Ke, Ce, DJ2,
```

cPropLinkid="LST499D38A7_0"AddLanguageSpecificTextSet("LST499D38A7_0?cpp=::|rs241");GetMu

DJ3, Notes, GUID)

```
int GetMultiLinearElastic(
String^ Name,
array<bool>^% DOF,
array<bool>^% Fixed,
array<bool>^% Nonlinear,
array<double>^% Ke,
array<double>^% Ce,
double% DJ2,
double% DJ3,
String^% Notes,
String^% GUID
)
```

```
abstract GetMultiLinearElastic :
    Name : string *
    DOF : bool[] byref *
    Fixed : bool[] byref *
    Nonlinear : bool[] byref *
    Ke : float[] byref *
    Ce : float[] byref *
    DJ2 : float byref *
    DJ3 : float byref *
    Notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing damper-type link property.

DOF

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if properties exist for a specified degree of freedom.

Value DOF

DOF(0) U1

DOF(1) U2

DOF(2) U3

DOF(3) R1

DOF(4) R2

DOF(5) R3

Fixed

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if the specified degree of freedom is fixed (restrained).

Value Fixity

Fixed(0) U1 fixity if DOF(0) = True

Fixed(1) U2 fixity if DOF(1) = True

Fixed(2) U3 fixity if DOF(2) = True

Fixed(3) R1 fixity if DOF(3) = True

Fixed(4) R2 fixity if DOF(4) = True

Fixed(5) R3 fixity if DOF(5) = True

The term Fixed(n) applies only when DOF(n) = True.

Nonlinear

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if nonlinear properties exist for a specified degree of freedom.

Value

Nonlinear

Nonlinear(0) U1 has nonlinear properties

Nonlinear(1) U2 has nonlinear properties

Nonlinear(2) U3 has nonlinear properties

Nonlinear(3) R1 has nonlinear properties

Nonlinear(4) R2 has nonlinear properties

Nonlinear(5) R3 has nonlinear properties

The term Nonlinear(n) applies only when DOF(n) = True and Fixed(n) = False.

Ke

Type: SystemDouble

This is an array of effective stiffness terms for the link property. The effective stiffness applies for linear analyses.

Value Stiffness

Ke(0) U1 [F/L]

Ke(1) U2 [F/L]

Ke(2) U3 [F/L]

Ke(3) R1 [FL]

Ke(4) R2 [FL]

Ke(5) R3 [FL]

The term ke(n) applies only when DOF(n) = True and Fixed(n) = False.

Ce

Type: SystemDouble

This is an array of effective damping terms for the link property. The effective stiffness applies for linear analyses.

Value Damping

Ce(0) U1 [F/L]

Ce(1) U2 [F/L]

Ce(2) U3 [F/L]

Ce(3) R1 [FL]

Ce(4) R2 [FL]

Ce(5) R3 [FL]

The term ce(n) applies only when DOF(n) = True and Fixed(n) = False.

DJ2

Type: SystemDouble

The distance from the J-End of the link to the U2 shear spring. This item applies only when DOF(1) = True. [L]

DJ3

Type: SystemDouble

The distance from the J-End of the link to the U3 shear spring. This item applies

only when DOF(2) = True. [L]

Notes

Type: SystemString

The notes, if any, assigned to the property.

GUID

Type: SystemString

The GUID (global unique identifier), if any, assigned to the property.

Return Value

Type: Int32

Returns zero if the property is successfully retrieved; otherwise it returns a nonzero value.

Remarks

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyDOF() As Boolean
    Dim MyFixed() As Boolean
    Dim MyNonLinear() As Boolean
    Dim MyKe() As Double
    Dim MyCe() As Double
    Dim DOF() As Boolean
    Dim Fixed() As Boolean
    Dim MyNonLinear() As Boolean
    Dim Ke() As Double
    Dim Ce() As Double
    Dim dj2 As Double
    Dim dj3 As Double
    Dim Notes As String
    Dim GUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link property
    ReDim MyDOF(5)
    ReDim MyFixed(5)
    ReDim MyNonLinear(5)
    ReDim MyKe(5)
    ReDim MyCe(5)
```


Introduction

```
MyDOF(0) = True
MyKe(0) = 12
MyCe(0) = 0.01

MyDOF(1) = True
MyNonLinear(1) = True
MyKe(1) = 12
MyCe(1) = 0.01

MyDOF(2) = True
MyFixed(2) = True

ret = SapModel.PropLink.SetMultiLinearElastic("MLE1", MyDOF, MyFixed, MyNonLinear, MyKe, M

'get link property data
ret = SapModel.PropLink.GetMultiLinearElastic("MLE1", DOF, Fixed, NonLinear, Ke, Ce, dj2,

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

[cPropLink Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LST60845292_0") Method

retrieves link property data for a multilinear plastic-type link property.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetMultiLinearPlastic(
    string Name,
    ref bool[] D0F,
    ref bool[] Fixed,
    ref bool[] Nonlinear,
    ref double[] Ke,
    ref double[] Ce,
    ref double DJ2,
    ref double DJ3,
    ref string Notes,
    ref string GUID
)
```

```
Function GetMultiLinearPlastic (
    Name As String,
    ByRef D0F As Boolean(),
    ByRef Fixed As Boolean(),
    ByRef Nonlinear As Boolean(),
    ByRef Ke As Double(),
    ByRef Ce As Double(),
    ByRef DJ2 As Double,
    ByRef DJ3 As Double,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim D0F As Boolean()
Dim Fixed As Boolean()
Dim Nonlinear As Boolean()
Dim Ke As Double()
Dim Ce As Double()
Dim DJ2 As Double
Dim DJ3 As Double
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.GetMultiLinearPlastic(Name,
D0F, Fixed, Nonlinear, Ke, Ce, DJ2,
```

cPropLinkAddLanguageSpecificTextSet("LST60845292_0?cpp=::|nu3246GetMult

DJ3, Notes, GUID)

```
int GetMultiLinearPlastic(
String^ Name,
array<bool>^% DOF,
array<bool>^% Fixed,
array<bool>^% Nonlinear,
array<double>^% Ke,
array<double>^% Ce,
double% DJ2,
double% DJ3,
String^% Notes,
String^% GUID
)
```

```
abstract GetMultiLinearPlastic :
    Name : string *
    DOF : bool[] byref *
    Fixed : bool[] byref *
    Nonlinear : bool[] byref *
    Ke : float[] byref *
    Ce : float[] byref *
    DJ2 : float byref *
    DJ3 : float byref *
    Notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing damper-type link property.

DOF

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if properties exist for a specified degree of freedom.

Value DOF

DOF(0) U1

DOF(1) U2

DOF(2) U3

DOF(3) R1

DOF(4) R2

DOF(5) R3

Fixed

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if the specified degree of freedom is fixed (restrained).

Value Fixity

Fixed(0) U1 fixity if DOF(0) = True

Fixed(1) U2 fixity if DOF(1) = True

Fixed(2) U3 fixity if DOF(2) = True

Fixed(3) R1 fixity if DOF(3) = True

Fixed(4) R2 fixity if DOF(4) = True

Fixed(5) R3 fixity if DOF(5) = True

The term Fixed(n) applies only when DOF(n) = True.

Nonlinear

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if nonlinear properties exist for a specified degree of freedom.

Value

Nonlinear

Nonlinear(0) U1 has nonlinear properties

Nonlinear(1) U2 has nonlinear properties

Nonlinear(2) U3 has nonlinear properties

Nonlinear(3) R1 has nonlinear properties

Nonlinear(4) R2 has nonlinear properties

Nonlinear(5) R3 has nonlinear properties

The term Nonlinear(n) applies only when DOF(n) = True and Fixed(n) = False.

Ke

Type: SystemDouble

This is an array of effective stiffness terms for the link property. The effective stiffness applies for linear analyses.

Value Stiffness

Ke(0) U1 [F/L]

Ke(1) U2 [F/L]

Ke(2) U3 [F/L]

Ke(3) R1 [FL]

Ke(4) R2 [FL]

Ke(5) R3 [FL]

The term ke(n) applies only when DOF(n) = True and Fixed(n) = False.

Ce

Type: SystemDouble

This is an array of effective damping terms for the link property. The effective stiffness applies for linear analyses.

Value Damping

Ce(0) U1 [F/L]

Ce(1) U2 [F/L]

Ce(2) U3 [F/L]

Ce(3) R1 [FL]

Ce(4) R2 [FL]

Ce(5) R3 [FL]

The term ce(n) applies only when DOF(n) = True and Fixed(n) = False.

DJ2

Type: SystemDouble

The distance from the J-End of the link to the U2 shear spring. This item applies only when DOF(1) = True. [L]

DJ3

Type: SystemDouble

The distance from the J-End of the link to the U3 shear spring. This item applies

only when DOF(2) = True. [L]

Notes

Type: SystemString

The notes, if any, assigned to the property.

GUID

Type: SystemString

The GUID (global unique identifier), if any, assigned to the property.

Return Value

Type: Int32

Returns zero if the property is successfully retrieved; otherwise it returns a nonzero value.

Remarks

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyDOF() As Boolean
    Dim MyFixed() As Boolean
    Dim MyNonLinear() As Boolean
    Dim MyKe() As Double
    Dim MyCe() As Double
    Dim DOF() As Boolean
    Dim Fixed() As Boolean
    Dim MyNonLinear() As Boolean
    Dim Ke() As Double
    Dim Ce() As Double
    Dim dj2 As Double
    Dim dj3 As Double
    Dim Notes As String
    Dim GUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link property
    ReDim MyDOF(5)
    ReDim MyFixed(5)
    ReDim MyNonLinear(5)
    ReDim MyKe(5)
    ReDim MyCe(5)
```

Introduction

```
MyDOF(0) = True
MyKe(0) = 12
MyCe(0) = 0.01

MyDOF(1) = True
MyNonLinear(1) = True
MyKe(1) = 12
MyCe(1) = 0.01

MyDOF(2) = True
MyFixed(2) = True

ret = SapModel.PropLink.SetMultiLinearPlastic("MLP1", MyDOF, MyFixed, MyNonLinear, MyKe, M

'get link property data
ret = SapModel.PropLink.GetMultiLinearPlastic("MLP1", DOF, Fixed, NonLinear, Ke, Ce, dj2,

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

[cPropLink Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LSTDA08C876_0?cpp=::|8251");GetM Method

Retrieves the force-deformation data for a specified degree of freedom in multilinear elastic and multilinear plastic link properties.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetMultiLinearPoints(
    string Name,
    int DOF,
    ref int NumberPoints,
    ref double[] F,
    ref double[] D,
    ref int MyType,
    ref double A1,
    ref double A2,
    ref double B1,
    ref double B2,
    ref double Eta
)
```

```
Function GetMultiLinearPoints (
    Name As String,
    DOF As Integer,
    ByRef NumberPoints As Integer,
    ByRef F As Double(),
    ByRef D As Double(),
    ByRef MyType As Integer,
    ByRef A1 As Double,
    ByRef A2 As Double,
    ByRef B1 As Double,
    ByRef B2 As Double,
    ByRef Eta As Double
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim DOF As Integer
Dim NumberPoints As Integer
Dim F As Double()
Dim D As Double()
Dim MyType As Integer
Dim A1 As Double
Dim A2 As Double
Dim B1 As Double
Dim B2 As Double
Dim Eta As Double
```

cPropLinkspan id="LSTDA08C876_0"AddLanguageSpecificTextSet("LSTDA08C876_0?cpp=::|8251");GetM

```

Dim returnValue As Integer

returnValue = instance.GetMultiLinearPoints(Name,
DOF, NumberPoints, F, D, MyType, A1,
A2, B1, B2, Eta)

int GetMultiLinearPoints(
String^ Name,
int DOF,
int% NumberPoints,
array<double>^% F,
array<double>^% D,
int% MyType,
double% A1,
double% A2,
double% B1,
double% B2,
double% Eta
)

abstract GetMultiLinearPoints :
    Name : string *
    DOF : int *
    NumberPoints : int byref *
    F : float[] byref *
    D : float[] byref *
    MyType : int byref *
    A1 : float byref *
    A2 : float byref *
    B1 : float byref *
    B2 : float byref *
    Eta : float byref -> int

```

Parameters

Name

Type: SystemString

The name of an existing multilinear elastic or multilinear plastic link property.

DOF

Type: SystemInt32

This is 1, 2, 3, 4, 5 or 6, indicating the degree of freedom to which the multilinear points apply.

Value DOF

1	U1
2	U2
3	U3
4	R1
5	R2
6	R3

NumberPoints

Type: SystemInt32

The number of force-deformation points for the specified degree of freedom.

F

Type: SystemDouble

This is an array, dimensioned to NumberPoints - 1, that includes the force at

each point. When DOF is U1, U2 or U3, this is a force. When DOF is R1, R2 or R3, this is a moment. [F] if DOF is less than or equal to 3, and [FL] if DOF is greater than 3.

D

Type: SystemDouble

This is an array, dimensioned to NumberPoints - 1, that includes the displacement at each point. When DOF is U1, U2 or U3, this is a translation. When DOF is R1, R2 or R3, this is a rotation. [L] if DOF is less than or equal to 3, and [rad] if DOF is greater than 3.

MyType

Type: SystemInt32

This item applies only to multilinear plastic link properties. It is 1, 2 or 3, indicating the hysteresis type.

MyType Hysteresis Type

- | | |
|---|-----------|
| 1 | Kinematic |
| 2 | Takeda |
| 3 | Pivot |

A1

Type: SystemDouble

This item applies only to multilinear plastic link properties that have a pivot hysteresis type (MyType = 3). It is the Alpha1 hysteresis parameter.

A2

Type: SystemDouble

This item applies only to multilinear plastic link properties that have a pivot hysteresis type (MyType = 3). It is the Alpha2 hysteresis parameter.

B1

Type: SystemDouble

This item applies only to multilinear plastic link properties that have a pivot hysteresis type (MyType = 3). It is the Beta1 hysteresis parameter.

B2

Type: SystemDouble

This item applies only to multilinear plastic link properties that have a pivot hysteresis type (MyType = 3). It is the Beta2 hysteresis parameter.

Eta

Type: SystemDouble

This item applies only to multilinear plastic link properties that have a pivot hysteresis type (MyType = 3). It is the Eta hysteresis parameter.

Return Value

Type: Int32

Returns zero if the data is successfully assigned; otherwise it returns a nonzero value.

Remarks

To successfully retrieve this data from the indicated link property, the following conditions must be met:

1. The link property must be multilinear elastic or multilinear plastic.
2. The specified DOF must be active.
3. The specified DOF must not be fixed.
4. The specified DOF must be nonlinear.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyDOF() As Boolean
    Dim MyFixed() As Boolean
    Dim MyNonLinear() As Boolean
    Dim MyKe() As Double
    Dim MyCe() As Double
    Dim MyF() As Double
    Dim MyD() As Double
    Dim NumberPoints As Integer
    Dim F() As Double
    Dim D() As Double
    Dim MyType As Integer
    Dim a1 As Double
    Dim a2 As Double
    Dim b1 As Double
    Dim b2 As Double
    Dim eta As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link property
    ReDim MyDOF(5)
    ReDim MyFixed(5)
    ReDim MyNonLinear(5)
    ReDim MyKe(5)
    ReDim MyCe(5)

    MyDOF(0) = True
    MyKe(0) = 12
    MyCe(0) = 0.01

    MyDOF(1) = True
    MyNonLinear(1) = True
    MyKe(1) = 12
    MyCe(1) = 0.01

    MyDOF(2) = True
    MyFixed(2) = True

    ret = SapModel.PropLink.SetMultiLinearPlastic("MLP1", MyDOF, MyFixed, MyNonLinear, MyKe, MyCe)

    'set multilinear force-defomation data
```

Return Value

Introduction

```
ReDim MyF(4)
ReDim MyD(4)

MyF(0) = -12
MyF(1) = -10
MyF(2) = 0
MyF(3) = 8
MyF(4) = 9

MyD(0) = -8
MyD(1) = -0.6
MyD(2) = 0
MyD(3) = 0.2
MyD(4) = 6

ret = SapModel.PropLink.SetMultiLinearPoints("MLP1", 2, 5, MyF, MyD, 3, 9, 12, 0.75, 0.8,

'get multilinear force-defomation data
ret = SapModel.PropLink.GetMultiLinearPoints("MLP1", 2, NumberPoints, F, D, MyType, a1, a2

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

See Also

Reference

cPropLink Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LST86F940BC0") Method

Retrieves the names of all defined link properties of the specified type.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNameList(
    ref int NumberNames,
    ref string[] MyName,
    eLinkPropType PropType =
)
```

```
Function GetNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String(),
    Optional PropType As eLinkPropType =
) As Integer
```

```
Dim instance As cPropLink
Dim NumberNames As Integer
Dim MyName As String()
Dim PropType As eLinkPropType
Dim returnValue As Integer
```

```
returnValue = instance.GetNameList(NumberNames,
MyName, PropType)
```

```
int GetNameList(
    int% NumberNames,
    array<String^>^% MyName,
    eLinkPropType PropType =
)
```

```
abstract GetNameList :
    NumberNames : int byref *
    MyName : string[] byref *
    ?PropType : eLinkPropType
(* Defaults:
    let _PropType = defaultArg PropType
*)
-> int
```

Parameters

NumberNames

Type: SystemInt32

The number of link property names retrieved by the program.

MyName

Type: SystemString

This is a one-dimensional array of link property names. The MyName array is created as a dynamic, zero-based, array by the API user: Dim MyName() as String

The array is dimensioned to (NumberNames - 1) inside the program, filled with the names, and returned to the API user.

PropType (Optional)

Type: ETABSv1eLinkPropType

This optional value is one of the items in the eLinkPropType enumeration.

If no value is input for PropType, names are returned for all link properties in the model regardless of type.

Return Value

Type: Int32

Returns zero if the names are successfully retrieved; otherwise it returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim DOF() As Boolean
    Dim Fixed() As Boolean
    Dim Ke() As Double
    Dim Ce() As Double
    Dim NumberNames As Integer
    Dim MyName() As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link property
    ReDim DOF(5)
    ReDim Fixed(5)
    ReDim Ke(5)
    ReDim Ce(5)
```

Introduction

```
D0F(0) = True
Ke(0) = 12
ret = SapModel.PropLink.SetLinear("L1", D0F, Fixed, Ke, Ce, 0, 0)

'get link property names
ret = SapModel.PropLink.GetNameList(NumberNames, MyName)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

See Also

Reference

cPropLink Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LST62CA0E7C_0?cpp=::3259");GetP Method

retrieves P-delta parameters for a link property.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetPDelta(
    string Name,
    ref double[] Value
)
```

```
Function GetPDelta (
    Name As String,
    ByRef Value As Double()
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim Value As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetPDelta(Name,
    Value)
```

```
int GetPDelta(
    String^ Name,
    array<double>^% Value
)
```

```
abstract GetPDelta :
    Name : string *
    Value : float[] byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing link property.

Value

Type: SystemDouble

This is an array of P-delta parameters

Value

PDelta Parameter

Value(0) M2 P-delta to I-end of link as moment, M2I

Introduction

Value(1) M2 P-delta to J-end of link as moment, M2J

Value(2) M3 P-delta to I-end of link as moment, M3I

Value(3) M3 P-delta to J-end of link as moment, M3J

Return Value

Type: Int32

Returns zero if the values are successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim DOF() As Boolean  
    Dim Fixed() As Boolean  
    Dim Ke() As Double  
    Dim Ce() As Double  
    Dim MyValue() As Double  
    Dim Value() As Double  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)  
  
    'add link property  
    ReDim DOF(5)  
    ReDim Fixed(5)  
    ReDim Ke(5)  
    ReDim Ce(5)  
    DOF(0) = True  
    Ke(0) = 12  
    ret = SapModel.PropLink.SetLinear("L1", DOF, Fixed, Ke, Ce, 0, 0)  
  
    'set link property P-delta parameters  
    ReDim MyValue(3)  
    MyValue(0) = 0.6  
    MyValue(1) = 0.4  
    MyValue(2) = 0.3  
    MyValue(3) = 0.2  
    ret = SapModel.PropLink.SetPDelta("L1", MyValue)  
  
    'get link property P-delta parameters  
    ret = SapModel.PropLink.GetPDelta("L1", Value)
```



```
'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPropLink Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LST7F940A32_0?cpp=::|n8262;GetPla

Method

retrieves link property data for a plastic Wen-type link property.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetPlasticWen(
    string Name,
    ref bool[] DOF,
    ref bool[] Fixed,
    ref bool[] Nonlinear,
    ref double[] Ke,
    ref double[] Ce,
    ref double[] K,
    ref double[] Yield,
    ref double[] Ratio,
    ref double[] Exp,
    ref double DJ2,
    ref double DJ3,
    ref string Notes,
    ref string GUID
)
```

```
Function GetPlasticWen (
    Name As String,
    ByRef DOF As Boolean(),
    ByRef Fixed As Boolean(),
    ByRef Nonlinear As Boolean(),
    ByRef Ke As Double(),
    ByRef Ce As Double(),
    ByRef K As Double(),
    ByRef Yield As Double(),
    ByRef Ratio As Double(),
    ByRef Exp As Double(),
    ByRef DJ2 As Double,
    ByRef DJ3 As Double,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim DOF As Boolean()
Dim Fixed As Boolean()
Dim Nonlinear As Boolean()
Dim Ke As Double()
Dim Ce As Double()
```

cPropLinkspan id="LST7F940A32_0"AddLanguageSpecificTextSet("LST7F940A32_0?cpp=::|n8262;GetPla

Introduction

```
Dim K As Double()
Dim Yield As Double()
Dim Ratio As Double()
Dim Exp As Double()
Dim DJ2 As Double
Dim DJ3 As Double
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer

returnValue = instance.GetPlasticWen(Name,
DOF, Fixed, Nonlinear, Ke, Ce, K, Yield,
Ratio, Exp, DJ2, DJ3, Notes, GUID)
```

```
int GetPlasticWen(
String^ Name,
array<bool>^% DOF,
array<bool>^% Fixed,
array<bool>^% Nonlinear,
array<double>^% Ke,
array<double>^% Ce,
array<double>^% K,
array<double>^% Yield,
array<double>^% Ratio,
array<double>^% Exp,
double% DJ2,
double% DJ3,
String^% Notes,
String^% GUID
)
```

```
abstract GetPlasticWen :
    Name : string *
    DOF : bool[] byref *
    Fixed : bool[] byref *
    Nonlinear : bool[] byref *
    Ke : float[] byref *
    Ce : float[] byref *
    K : float[] byref *
    Yield : float[] byref *
    Ratio : float[] byref *
    Exp : float[] byref *
    DJ2 : float byref *
    DJ3 : float byref *
    Notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing plastic Wen-type link property.

DOF

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if properties exist for a specified degree of freedom.

Value DOF

DOF(0) U1

DOF(1) U2
 DOF(2) U3
 DOF(3) R1
 DOF(4) R2
 DOF(5) R3

Fixed

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if the specified degree of freedom is fixed (restrained).

Value Fixity

Fixed(0) U1 fixity if DOF(0) = True
 Fixed(1) U2 fixity if DOF(1) = True
 Fixed(2) U3 fixity if DOF(2) = True
 Fixed(3) R1 fixity if DOF(3) = True
 Fixed(4) R2 fixity if DOF(4) = True
 Fixed(5) R3 fixity if DOF(5) = True

Nonlinear

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if nonlinear properties exist for a specified degree of freedom.

Value Nonlinear

Nonlinear(0) U1 has nonlinear properties
 Nonlinear(1) U2 has nonlinear properties
 Nonlinear(2) U3 has nonlinear properties
 Nonlinear(3) R1 has nonlinear properties
 Nonlinear(4) R2 has nonlinear properties
 Nonlinear(5) R3 has nonlinear properties

The term Nonlinear(n) applies only when DOF(n) = True and Fixed(n) == False.

Ke

Type: SystemDouble

This is an array of effective stiffness terms for the link property. The effective stiffness applies for linear analyses.

Value Stiffness

Ke(0) U1 [F/L]
 Ke(1) U2 [F/L]
 Ke(2) U3 [F/L]
 Ke(3) R1 [FL]
 Ke(4) R2 [FL]
 Ke(5) R3 [FL]

The term ke(n) applies only when DOF(n) = True and Fixed(n) = False.

Ce

Type: SystemDouble

This is an array of effective damping terms for the link property. The effective stiffness applies for linear analyses.

Value Damping

Ce(0) U1 [F/L]

Ce(1) U2 [F/L]

Ce(2) U3 [F/L]

Ce(3) R1 [FL]

Ce(4) R2 [FL]

Ce(5) R3 [FL]

The term ce(n) applies only when DOF(n) = True and Fixed(n) = False.

K

Type: SystemDouble

This is an array of initial stiffness terms for the link property. The initial stiffness applies for nonlinear analyses.

Value Stiffness

K(0) U1 [F/L]

K(1) U2 [F/L]

K(2) U3 [F/L]

K(3) R1 [F]

K(4) R2 [F]

K(5) R3 [F]

The term k(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Yield

Type: SystemDouble

This is an array of yield force terms for the link property. The yield force applies for nonlinear analyses.

Value Yield

yield(0) U1 [F]

yield(1) U2 [F]

yield(2) U3 [F]

yield(3) R1 [F/L]

yield(4) R2 [F/L]

yield(5) R3 [F/L]

The term Yield(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Ratio

Type: SystemDouble

This is an array of post-yield stiffness ratio terms for the link property. The post-yield stiffness ratio applies for nonlinear analyses. It is the post-yield stiffness divided by the initial stiffness.

Value Ratio

Ratio(0) U1

Ratio(1) U2

Ratio(2) U3

Ratio(3) R1

Ratio(4) R2

Ratio(5) R3

The term Ratio(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Exp

Type:Â SystemDouble

This is an array of yield exponent terms for the link property. The yield exponent applies for nonlinear analyses. The yielding exponent that controls the sharpness of the transition from the initial stiffness to the yielded stiffness.

Value Exp

Exp(0) U1

Exp(1) U2

Exp(2) U3

Exp(3) R1

Exp(4) R2

Exp(5) R3

The term Exp(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

DJ2

Type:Â SystemDouble

The distance from the J-End of the link to the U2 shear spring. This item applies only when DOF(1) = True. [L]

DJ3

Type:Â SystemDouble

The distance from the J-End of the link to the U3 shear spring. This item applies only when DOF(2) = True. [L]

Notes

Type:Â SystemString

The notes, if any, assigned to the property.

GUID

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the property. If this item is input as Default, the program assigns a GUID to the property.

Return Value

Type:Â Int32

Returns zero if the property is successfully initialized; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim MyDOF() As Boolean  
    Dim MyFixed() As Boolean  
    Dim MyNonLinear() As Boolean  
    Dim MyKe() As Double  
    Dim MyCe() As Double  
    Dim MyK() As Double
```

Introduction

```
Dim MyYield() As Double
Dim MyRatio() As Double
Dim MyExp() As Double
Dim D0F() As Boolean
Dim Fixed() As Boolean
Dim NonLinear() as Boolean
Dim Ke() As Double
Dim Ce() As Double
Dim K() As Double
Dim Yield() As Double
Dim Ratio() As Double
Dim Exp() As Double
Dim dj2 As Double
Dim dj3 As Double
Dim Notes As String
Dim GUID As String

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'add link property
ReDim MyD0F(5)
ReDim MyFixed(5)
ReDim MyNonLinear(5)
ReDim MyKe(5)
ReDim MyCe(5)
ReDim MyK(5)
ReDim MyYield(5)
ReDim MyRatio(5)
ReDim MyExp(5)

MyD0F(0) = True
MyKe(0) = 12
MyCe(0) = 0.01

MyD0F(1) = True
MyNonLinear(1) = True
MyKe(1) = 12
MyCe(1) = 0.01
MyK(1) = 20
MyYield(1)= 50
MyRatio(1)= 0.1
MyExp(1)= 3

MyD0F(2) = True
MyFixed(2) = True

ret = SapModel.PropLink.SetPlasticWen("PW1", MyD0F, MyFixed, MyNonLinear, MyKe, MyCe, MyK,
```

Introduction

```
'get link property data
    ret = SapModel.PropLink.GetPlasticWen("PW1", D0F, Fixed, NonLinear, Ke, Ce, k, Yield, Rati

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

📌 See Also

Reference

cPropLink Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LSTB94D2099_0") Method

retrieves link property data for a rubber isolator-type link property.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetRubberIsolator(
    string Name,
    ref bool[] DOF,
    ref bool[] Fixed,
    ref bool[] Nonlinear,
    ref double[] Ke,
    ref double[] Ce,
    ref double[] K,
    ref double[] Yield,
    ref double[] Ratio,
    ref double DJ2,
    ref double DJ3,
    ref string Notes,
    ref string GUID
)
```

```
Function GetRubberIsolator (
    Name As String,
    ByRef DOF As Boolean(),
    ByRef Fixed As Boolean(),
    ByRef Nonlinear As Boolean(),
    ByRef Ke As Double(),
    ByRef Ce As Double(),
    ByRef K As Double(),
    ByRef Yield As Double(),
    ByRef Ratio As Double(),
    ByRef DJ2 As Double,
    ByRef DJ3 As Double,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim DOF As Boolean()
Dim Fixed As Boolean()
Dim Nonlinear As Boolean()
Dim Ke As Double()
Dim Ce As Double()
Dim K As Double()
Dim Yield As Double()
```

Introduction

```
Dim Ratio As Double()
Dim DJ2 As Double
Dim DJ3 As Double
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer

returnValue = instance.GetRubberIsolator(Name,
DOF, Fixed, Nonlinear, Ke, Ce, K, Yield,
Ratio, DJ2, DJ3, Notes, GUID)

int GetRubberIsolator(
String^ Name,
array<bool>^% DOF,
array<bool>^% Fixed,
array<bool>^% Nonlinear,
array<double>^% Ke,
array<double>^% Ce,
array<double>^% K,
array<double>^% Yield,
array<double>^% Ratio,
double% DJ2,
double% DJ3,
String^% Notes,
String^% GUID
)

abstract GetRubberIsolator :
    Name : string *
    DOF : bool[] byref *
    Fixed : bool[] byref *
    Nonlinear : bool[] byref *
    Ke : float[] byref *
    Ce : float[] byref *
    K : float[] byref *
    Yield : float[] byref *
    Ratio : float[] byref *
    DJ2 : float byref *
    DJ3 : float byref *
    Notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing rubber isolator-type link property.

DOF

Type:Â SystemBoolean

This is a boolean array, dimensioned to 5, indicating if properties exist for a specified degree of freedom.

Value DOF

DOF(0) U1

DOF(1) U2

DOF(2) U3

DOF(3) R1

DOF(4) R2

DOF(5) R3

FixedType: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if the specified degree of freedom is fixed (restrained).

Value	Fixity
--------------	---------------

Fixed(0) U1 fixity if DOF(0) = True

Fixed(1) U2 fixity if DOF(1) = True

Fixed(2) U3 fixity if DOF(2) = True

Fixed(3) R1 fixity if DOF(3) = True

Fixed(4) R2 fixity if DOF(4) = True

Fixed(5) R3 fixity if DOF(5) = True

NonlinearType: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if nonlinear properties exist for a specified degree of freedom.

Value	Nonlinear
--------------	------------------

Nonlinear(0) U1 has nonlinear properties

Nonlinear(1) U2 has nonlinear properties

Nonlinear(2) U3 has nonlinear properties

Nonlinear(3) R1, not used

Nonlinear(4) R2, not used

Nonlinear(5) R3, not used

Note that this item is applicable only for degrees of freedom U1, U2 and U3. For those degrees of freedom, the term Nonlinear(n) applies only when DOF(n) = True and Fixed(n) = False.

KeType: SystemDouble

This is an array of effective stiffness terms for the link property. The effective stiffness applies for linear analyses.

Value Stiffness

Ke(0) U1 [F/L]

Ke(1) U2 [F/L]

Ke(2) U3 [F/L]

Ke(3) R1 [FL]

Ke(4) R2 [FL]

Ke(5) R3 [FL]

The term ke(n) applies only when DOF(n) = True and Fixed(n) = False.

CeType: SystemDouble

This is an array of effective damping terms for the link property. The effective stiffness applies for linear analyses.

Value Damping

Ce(0) U1 [F/L]

Ce(1) U2 [F/L]

Ce(2) U3 [F/L]

Ce(3) R1 [FL]

Ce(4) R2 [FL]

Ce(5) R3 [FL]

The term ce(n) applies only when DOF(n) = True and Fixed(n) = False.

K

Type: SystemDouble

This is an array of initial stiffness terms for the link property. The initial stiffness applies for nonlinear analyses.

Value Stiffness

K(0) U1, not used

K(1) U2 [F/L]

K(2) U3 [F/L]

K(3) R1, not used

K(4) R2, not used

K(5) R3, not used

Note that this item is applicable only for degrees of freedom U2 and U3. For those degrees of freedom, the term k(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Yield

Type: SystemDouble

This is an array of yield force terms for the link property. The yield force applies for nonlinear analyses.

Value Yield

yield(0) U1, not used

yield(1) U2 [F]

yield(2) U3 [F]

yield(3) R1, not used

yield(4) R2, not used

yield(5) R3, not used

Note that this item is applicable only for degrees of freedom U2 and U3. For those degrees of freedom, the term Yield(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Ratio

Type: SystemDouble

This is an array of post-yield stiffness ratio terms for the link property. The post-yield stiffness ratio applies for nonlinear analyses. It is the post-yield stiffness divided by the initial stiffness.

Value Ratio

Ratio(0) U1, not used

Ratio(1) U2

Ratio(2) U3

Ratio(3) R1, not used

Ratio(4) R2, not used

Ratio(5) R3, not used

Introduction

Note that this item is applicable only for degrees of freedom U2 and U3. For those degrees of freedom, the term Ratio(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

DJ2

Type:Â SystemDouble

The distance from the J-End of the link to the U2 shear spring. This item applies only when DOF(1) = True. [L]

DJ3

Type:Â SystemDouble

The distance from the J-End of the link to the U3 shear spring. This item applies only when DOF(2) = True. [L]

Notes

Type:Â SystemString

The notes, if any, assigned to the property.

GUID

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the property.

Return Value

Type:Â Int32

Returns zero if the property is successfully retrieved; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim MyDOF() As Boolean  
    Dim MyFixed() As Boolean  
    Dim MyNonLinear() As Boolean  
    Dim MyKe() As Double  
    Dim MyCe() As Double  
    Dim MyK() As Double  
    Dim MyYield() As Double  
    Dim MyRatio() As Double  
    Dim DOF() As Boolean  
    Dim Fixed() As Boolean  
    Dim NonLinear() As Boolean  
    Dim Ke() As Double  
    Dim Ce() As Double  
    Dim K() As Double  
    Dim Yield() As Double  
    Dim Ratio() As Double  
    Dim dj2 As Double  
    Dim dj3 As Double  
    Dim Notes As String  
    Dim GUID As String  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

Introduction

```
'start ETABS application
    ret = EtabsObject.ApplicationStart()

'create SapModel object
    SapModel = EtabsObject.SapModel

'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'add link property
    ReDim MyDOF(5)
    ReDim MyFixed(5)
    ReDim MyNonLinear(5)
    ReDim MyKe(5)
    ReDim MyCe(5)
    ReDim MyK(5)
    ReDim MyYield(5)
    ReDim MyRatio(5)

    MyDOF(0) = True
    MyKe(0) = 12
    MyCe(0) = 0.01

    MyDOF(1) = True
    MyNonLinear(1) = True
    MyKe(1) = 12
    MyCe(1) = 0.01
    MyK(1) = 20
    MyYield(1)= 50
    MyRatio(1)= 0.1

    MyDOF(2) = True
    MyNonLinear(2) = True
    MyKe(2) = 15
    MyCe(2) = 0.008
    MyK(2) = 22
    MyYield(2)= 60
    MyRatio(2)= 0.15

    MyDOF(3) = True
    MyFixed(3) = True

    ret = SapModel.PropLink.SetRubberIsolator("RI1", MyDOF, MyFixed, MyNonLinear, MyKe, MyCe,

'get link property data
    ret = SapModel.PropLink.GetRubberIsolator("RI1", DOF, Fixed, NonLinear, Ke, Ce, k, Yield,

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPropLink Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LSTB434E2FF_0") Method

retrieves length and area values for a link property that are used if the link property is specified in line and area spring assignments.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetSpringData(
    string Name,
    ref double DefinedForThisLength,
    ref double DefinedForThisArea
)
```

```
Function GetSpringData (
    Name As String,
    ByRef DefinedForThisLength As Double,
    ByRef DefinedForThisArea As Double
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim DefinedForThisLength As Double
Dim DefinedForThisArea As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetSpringData(Name,
    DefinedForThisLength, DefinedForThisArea)
```

```
int GetSpringData(
    String^ Name,
    double% DefinedForThisLength,
    double% DefinedForThisArea
)
```

```
abstract GetSpringData :
    Name : string *
    DefinedForThisLength : float byref *
    DefinedForThisArea : float byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing link property.

DefinedForThisLength

cPropLinkspan id="LSTB434E2FF_0"AddLanguageSpecificTextSet("LSTB434E2FF_0?cpp=::|0276);GetSp

Type: SystemDouble

The link property is defined for this length in a line (frame) spring. [L]
DefinedForThisArea

Type: SystemDouble

The link property is defined for this area in an area spring. [L2]

Return Value

Type: Int32

Returns zero if the values are successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim DOF() As Boolean
    Dim Fixed() As Boolean
    Dim Ke() As Double
    Dim Ce() As Double
    Dim DefinedForThisLength As Double
    Dim DefinedForThisArea As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link property
    ReDim DOF(5)
    ReDim Fixed(5)
    ReDim Ke(5)
    ReDim Ce(5)
    DOF(0) = True
    Ke(0) = 12
    ret = SapModel.PropLink.SetLinear("L1", DOF, Fixed, Ke, Ce, 0, 0)

    'set link property spring data
    ret = SapModel.PropLink.SetSpringData("L1", 12, 2)

    'get link property spring data
    ret = SapModel.PropLink.GetSpringData("L1", DefinedForThisLength, DefinedForThisArea)

    'close ETABS
    EtabsObject.ApplicationExit(False)
```

```
'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

📌 See Also

Reference

cPropLink Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LSTC31C57C Method

Retrieves link property data for a T/C friction isolator-type link property.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetTCFrictionIsolator(
    string Name,
    ref bool[] DOF,
    ref bool[] Fixed,
    ref bool[] Nonlinear,
    ref double[] Ke,
    ref double[] Ce,
    ref double[] K,
    ref double[] Slow,
    ref double[] Fast,
    ref double[] Rate,
    ref double[] Radius,
    ref double[] SlowT,
    ref double[] FastT,
    ref double[] RateT,
    ref double Kt,
    ref double Dis,
    ref double Dist,
    ref double Damping,
    ref double DJ2,
    ref double DJ3,
    ref string Notes,
    ref string GUID
)
```

```
Function GetTCFrictionIsolator (
    Name As String,
    ByRef DOF As Boolean(),
    ByRef Fixed As Boolean(),
    ByRef Nonlinear As Boolean(),
    ByRef Ke As Double(),
    ByRef Ce As Double(),
    ByRef K As Double(),
    ByRef Slow As Double(),
    ByRef Fast As Double(),
    ByRef Rate As Double(),
    ByRef Radius As Double(),
    ByRef SlowT As Double(),
    ByRef FastT As Double(),
    ByRef RateT As Double(),
    ByRef Kt As Double,
```

Introduction

```
ByRef Dis As Double,  
ByRef Dist As Double,  
ByRef Damping As Double,  
ByRef DJ2 As Double,  
ByRef DJ3 As Double,  
ByRef Notes As String,  
ByRef GUID As String  
) As Integer
```

```
Dim instance As cPropLink  
Dim Name As String  
Dim DOF As Boolean()  
Dim Fixed As Boolean()  
Dim Nonlinear As Boolean()  
Dim Ke As Double()  
Dim Ce As Double()  
Dim K As Double()  
Dim Slow As Double()  
Dim Fast As Double()  
Dim Rate As Double()  
Dim Radius As Double()  
Dim SlowT As Double()  
Dim FastT As Double()  
Dim RateT As Double()  
Dim Kt As Double  
Dim Dis As Double  
Dim Dist As Double  
Dim Damping As Double  
Dim DJ2 As Double  
Dim DJ3 As Double  
Dim Notes As String  
Dim GUID As String  
Dim returnValue As Integer
```

```
returnValue = instance.GetTCFrictionIsolator(Name,  
DOF, Fixed, Nonlinear, Ke, Ce, K, Slow,  
Fast, Rate, Radius, SlowT, FastT, RateT,  
Kt, Dis, Dist, Damping, DJ2, DJ3, Notes,  
GUID)
```

```
int GetTCFrictionIsolator(  
String^ Name,  
array<bool>^% DOF,  
array<bool>^% Fixed,  
array<bool>^% Nonlinear,  
array<double>^% Ke,  
array<double>^% Ce,  
array<double>^% K,  
array<double>^% Slow,  
array<double>^% Fast,  
array<double>^% Rate,  
array<double>^% Radius,  
array<double>^% SlowT,  
array<double>^% FastT,  
array<double>^% RateT,  
double% Kt,  
double% Dis,  
double% Dist,  
double% Damping,  
double% DJ2,  
double% DJ3,  
String^% Notes,
```

cPropLinkspan id="LSTC31C57CF_0"AddLanguageSpecificTextSet("LSTC31C57CF_0?cpp=::3280");GetT

```
String^% GUID
)
```

```
abstract GetTCFrictionIsolator :
    Name : string *
    DOF : bool[] byref *
    Fixed : bool[] byref *
    Nonlinear : bool[] byref *
    Ke : float[] byref *
    Ce : float[] byref *
    K : float[] byref *
    Slow : float[] byref *
    Fast : float[] byref *
    Rate : float[] byref *
    Radius : float[] byref *
    SlowT : float[] byref *
    FastT : float[] byref *
    RateT : float[] byref *
    Kt : float byref *
    Dis : float byref *
    Dist : float byref *
    Damping : float byref *
    DJ2 : float byref *
    DJ3 : float byref *
    Notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing T/C friction isolator-type link property.

DOF

Type:Â SystemBoolean

This is a boolean array, dimensioned to 5, indicating if properties exist for a specified degree of freedom.

Value DOF

DOF(0) U1

DOF(1) U2

DOF(2) U3

DOF(3) R1

DOF(4) R2

DOF(5) R3

Fixed

Type:Â SystemBoolean

This is a boolean array, dimensioned to 5, indicating if the specified degree of freedom is fixed (restrained).

Value Fixity

Fixed(0) U1 fixity if DOF(0) = True

Fixed(1) U2 fixity if DOF(1) = True

Fixed(2) U3 fixity if DOF(2) = True

Fixed(3) R1 fixity if DOF(3) = True

Fixed(4) R2 fixity if DOF(4) = True

Fixed(5) R3 fixity if DOF(5) = True

Nonlinear

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if nonlinear properties exist for a specified degree of freedom.

Value

Nonlinear

Nonlinear(0) U1 has nonlinear properties

Nonlinear(1) U2 has nonlinear properties

Nonlinear(2) U3 has nonlinear properties

Nonlinear(3) R1, not used

Nonlinear(4) R2, not used

Nonlinear(5) R3, not used

Note that this item is applicable only for degrees of freedom U1, U2 and U3. For those degrees of freedom, the term Nonlinear(n) applies only when DOF(n) = True and Fixed(n) = False.

Ke

Type: SystemDouble

This is an array of effective stiffness terms for the link property. The effective stiffness applies for linear analyses.

Value Stiffness

Ke(0) U1 [F/L]

Ke(1) U2 [F/L]

Ke(2) U3 [F/L]

Ke(3) R1 [FL]

Ke(4) R2 [FL]

Ke(5) R3 [FL]

The term ke(n) applies only when DOF(n) = True and Fixed(n) = False.

Ce

Type: SystemDouble

This is an array of effective damping terms for the link property. The effective stiffness applies for linear analyses.

Value Damping

Ce(0) U1 [F/L]

Ce(1) U2 [F/L]

Ce(2) U3 [F/L]

Ce(3) R1 [FL]

Ce(4) R2 [FL]

Ce(5) R3 [FL]

The term ce(n) applies only when DOF(n) = True and Fixed(n) = False.

K

Type: SystemDouble

This is an array of initial stiffness terms for the link property. The initial stiffness applies for nonlinear analyses.

Value Stiffness

K(0) U1 [F/L]
 K(1) U2 [F/L]
 K(2) U3 [F/L]
 K(3) R1, not used
 K(4) R2, not used
 K(5) R3, not used

Note that this item is applicable only for degrees of freedom U1, U2 and U3. For those degrees of freedom, the term k(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Slow

Type: SystemDouble

This is an array of the friction coefficient at zero velocity terms when U1 is in compression for the link property. This coefficient applies for nonlinear analyses.

Value Slow

Slow(0) U1, not used
 Slow(1) U2
 Slow(2) U3
 Slow(3) R1, not used
 Slow(4) R2, not used
 Slow(5) R3, not used

Note that this item is applicable only for degrees of freedom U2 and U3. For those degrees of freedom, the term Slow(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Fast

Type: SystemDouble

This is an array of the friction coefficient at fast velocity terms when U1 is in compression for the link property. This coefficient applies for nonlinear analyses.

Value Fast

Fast(0) U1, not used
 Fast(1) U2
 Fast(2) U3
 Fast(3) R1, not used
 Fast(4) R2, not used
 Fast(5) R3, not used

Note that this item is applicable only for degrees of freedom U2 and U3. For those degrees of freedom, the term Fast(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Rate

Type: SystemDouble

This is an array of the inverse of the characteristic sliding velocity terms when U1 is in compression for the link property. This item applies for nonlinear analyses.

Value Rate

Rate(0) U1, not used

Rate(1) U2 [s/L]

Rate(2) U3 [s/L]

Rate(3) R1, not used

Rate(4) R2, not used

Rate(5) R3, not used

Note that this item is applicable only for degrees of freedom U2 and U3. For those degrees of freedom, the term Fast(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Radius

Type: SystemDouble

This is an array of the radius of the sliding contact surface terms for the link property. Inputting 0 means there is an infinite radius, that is, the slider is flat. This item applies for nonlinear analyses.

Value Radius

Radius(0) U1, not used

Radius(1) U2 [L]

Radius(2) U3 [L]

Radius(3) R1, not used

Radius(4) R2, not used

Radius(5) R3, not used

Note that this item is applicable only for degrees of freedom U2 and U3. For those degrees of freedom, the term Fast(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

SlowT

Type: SystemDouble

This is an array of the friction coefficient at zero velocity terms when U1 is in tension for the link property. This coefficient applies for nonlinear analyses.

Value Slow

SlowT(0) U1, not used

SlowT(1) U2

SlowT(2) U3

SlowT(3) R1, not used

SlowT(4) R2, not used

SlowT(5) R3, not used

Note that this item is applicable only for degrees of freedom U2 and U3. For those degrees of freedom, the term SlowT(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

FastT

Type: SystemDouble

This is an array of the friction coefficient at fast velocity terms when U1 is in tension for the link property. This coefficient applies for nonlinear analyses.

Value Fast

FastT(0) U1, not used

FastT(1) U2

FastT(2) U3

FastT(3) R1, not used

FastT(4) R2, not used

FastT(5) R3, not used

Note that this item is applicable only for degrees of freedom U2 and U3. For those degrees of freedom, the term FastT(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

RateT

Type: SystemDouble

This is an array of the inverse of the characteristic sliding velocity terms when U1 is in tension for the link property. This item applies for nonlinear analyses.

Value Rate

RateT(0) U1, not used

RateT(1) U2 [s/L]

RateT(2) U3 [s/L]

RateT(3) R1, not used

RateT(4) R2, not used

RateT(5) R3, not used

Note that this item is applicable only for degrees of freedom U2 and U3. For those degrees of freedom, the term RateT(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Kt

Type: SystemDouble

The axial translational tension stiffness for the U1 degree of freedom. This item applies for nonlinear analyses. [F/L]

Dis

Type: SystemDouble

The U1 degree of freedom gap opening for compression. This item applies for nonlinear analyses. [L]

Dist

Type: SystemDouble

The U1 degree of freedom gap opening for tension. This item applies for nonlinear analyses. [L]

Damping

Type: SystemDouble

This is the nonlinear damping coefficient used for the axial translational degree of freedom, U1. This item applies for nonlinear analyses. [F/L]

DJ2

Type: SystemDouble

The distance from the J-End of the link to the U2 shear spring. This item applies only when DOF(1) = True. [L]

DJ3

Type: SystemDouble

The distance from the J-End of the link to the U3 shear spring. This item applies only when DOF(2) = True. [L]

Notes

Type: SystemString

The notes, if any, assigned to the property.

GUID

Type: SystemString

The GUID (global unique identifier), if any, assigned to the property.

Return ValueType: Int32

Returns zero if the property is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyDOF() As Boolean
    Dim MyFixed() As Boolean
    Dim MyNonLinear() As Boolean
    Dim MyKe() As Double
    Dim MyCe() As Double
    Dim MyK() As Double
    Dim MySlow() As Double
    Dim MyFast() As Double
    Dim MyRate() As Double
    Dim MyRadius() As Double
    Dim MySlowT() As Double
    Dim MyFastT() As Double
    Dim MyRateT() As Double
    Dim DOF() As Boolean
    Dim Fixed() As Boolean
    Dim NonLinear() As Boolean
    Dim Ke() As Double
    Dim Ce() As Double
    Dim K() As Double
    Dim Slow() As Double
    Dim Fast() As Double
    Dim Rate() As Double
    Dim Radius() As Double
    Dim SlowT() As Double
    Dim FastT() As Double
    Dim RateT() As Double
    Dim kt As Double
    Dim dis As Double
    Dim dist As Double
    Dim Damping As Double
    Dim dj2 As Double
    Dim dj3 As Double
    Dim Notes As String
    Dim GUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()
```

Introduction

```
'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'add link property
    ReDim MyDOF(5)
    ReDim MyFixed(5)
    ReDim MyNonLinear(5)
    ReDim MyKe(5)
    ReDim MyCe(5)
    ReDim MyK(5)
    ReDim MySlow(5)
    ReDim MyFast(5)
    ReDim MyRate(5)
    ReDim MyRadius(5)
    ReDim MySlowT(5)
    ReDim MyFastT(5)
    ReDim MyRateT(5)

    MyDOF(0) = True
    MyNonLinear(0) = True
    MyKe(0) = 12
    MyCe(0) = 0.01
    MyK(0) = 1000

    MyDOF(1) = True
    MyNonLinear(1) = True
    MyKe(1) = 12
    MyCe(1) = 0.01
    MyK(1) = 20
    MySlow(1)= 0.6
    MyFast(1)= 0.5
    MyRate(1)= 10
    MyRadius(1)= 80
    MySlowT(1)= 0.61
    MyFastT(1)= 0.51
    MyRateT(1)= 10.1

    MyDOF(2) = True
    MyNonLinear(2) = True
    MyKe(2) = 14
    MyCe(2) = 0.008
    MyK(2) = 22
    MySlow(2)= 0.66
    MyFast(2)= 0.55
    MyRate(2)= 12
    MyRadius(2)= 75
    MySlowT(2)= 0.67
    MyFastT(2)= 0.56
    MyRateT(2)= 12.1

    MyDOF(3) = True
    MyKe(3) = 15
    MyCe(3) = 0

    MyDOF(4) = True
    MyFixed(4) = True

    ret = SapModel.PropLink.SetTCFrictionIsolator("TCFI1", MyDOF, MyFixed, MyNonLinear, MyKe,

'get link property data
    ret = SapModel.PropLink.GetTCFrictionIsolator("TCFI1", DOF, Fixed, NonLinear, Ke, Ce, k, S
```

```
'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPropLink Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LST6F93598D_0") Method

Retrieves the property type for the specified link property.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetType0API(
    string Name,
    ref eLinkPropType PropType
)
```

```
Function GetType0API (
    Name As String,
    ByRef PropType As eLinkPropType
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim PropType As eLinkPropType
Dim returnValue As Integer
```

```
returnValue = instance.GetType0API(Name,
    PropType)
```

```
int GetType0API(
    String^ Name,
    eLinkPropType% PropType
)
```

```
abstract GetType0API :
    Name : string *
    PropType : eLinkPropType byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing link property.

PropType

Type: ETABSv1eLinkPropType

This is one of the items in the eLinkPropType enumeration.

Return ValueType: $\hat{A}$ Int32

Returns zero if the type is successfully retrieved; otherwise it returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim DOF() As Boolean
    Dim Fixed() As Boolean
    Dim Ke() As Double
    Dim Ce() As Double
    Dim PropType As eLinkPropType

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link property
    ReDim DOF(5)
    ReDim Fixed(5)
    ReDim Ke(5)
    ReDim Ce(5)
    DOF(0) = True
    Ke(0) = 12
    ret = SapModel.PropLink.SetLinear("L1", DOF, Fixed, Ke, Ce, 0, 0)

    'get link property type
    ret = SapModel.PropLink.GetTypeOAPI("L1", PropType)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPropLink Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LSTB86F4B20_0") Method

Assigns weight and mass data to a link property.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetWeightAndMass(
    string Name,
    ref double W,
    ref double M,
    ref double R1,
    ref double R2,
    ref double R3
)
```

```
Function GetWeightAndMass (
    Name As String,
    ByRef W As Double,
    ByRef M As Double,
    ByRef R1 As Double,
    ByRef R2 As Double,
    ByRef R3 As Double
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim W As Double
Dim M As Double
Dim R1 As Double
Dim R2 As Double
Dim R3 As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetWeightAndMass(Name,
    W, M, R1, R2, R3)
```

```
int GetWeightAndMass(
    String^ Name,
    double% W,
    double% M,
    double% R1,
    double% R2,
    double% R3
)
```

```
abstract GetWeightAndMass :
    Name : string *
```

cPropLinkspan id="LSTB86F4B20_0"AddLanguageSpecificTextSet("LSTB86F4B20_0?cpp=::|rsid=3292;GetWe


```

W : float byref *
M : float byref *
R1 : float byref *
R2 : float byref *
R3 : float byref -> int

```

Parameters

Name

Type:Â SystemString
The name of an existing link property.

W

Type:Â SystemDouble
The weight of the link. [F]

M

Type:Â SystemDouble
The translational mass of the link. [M]

R1

Type:Â SystemDouble
The rotational inertia of the link about its local 1 axis. [ML2]

R2

Type:Â SystemDouble
The rotational inertia of the link about its local 2 axis. [ML2]

R3

Type:Â SystemDouble
The rotational inertia of the link about its local 3 axis. [ML2]

Return Value

Type:Â Int32

returns zero if the data is successfully assigned; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim DOF() As Boolean
    Dim Fixed() As Boolean
    Dim Ke() As Double
    Dim Ce() As Double
    Dim w As Double
    Dim m As Double
    Dim R1 As Double
    Dim R2 As Double
    Dim R3 As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

```

Introduction

```
'create SapModel object
    SapModel = EtabsObject.SapModel

'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'add link property
    ReDim DOF(5)
    ReDim Fixed(5)
    ReDim Ke(5)
    ReDim Ce(5)
    DOF(0) = True
    Ke(0) = 12
    ret = SapModel.PropLink.SetLinear("L1", DOF, Fixed, Ke, Ce, 0, 0)

'set link property weight and mass
    ret = SapModel.PropLink.SetWeightAndMass("L1", 10, 0.26, 0.0012, 0.0014, 0.0016)

'get link property weight and mass
    ret = SapModel.PropLink.GetWeightAndMass("L1", w, m, R1, R2, R3)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPropLink Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LST8D107537_0") Method

Assigns the acceptance criteria data for a link property

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetAcceptanceCriteria(
    string Name,
    int AcceptanceType,
    bool Symmetric,
    ref bool[] Active,
    ref double[] IOPos,
    ref double[] LSPos,
    ref double[] CPPos,
    ref double[] IONeg,
    ref double[] LSNeg,
    ref double[] CPNeg
)
```

```
Function SetAcceptanceCriteria (
    Name As String,
    AcceptanceType As Integer,
    Symmetric As Boolean,
    ByRef Active As Boolean(),
    ByRef IOPos As Double(),
    ByRef LSPos As Double(),
    ByRef CPPos As Double(),
    ByRef IONeg As Double(),
    ByRef LSNeg As Double(),
    ByRef CPNeg As Double()
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim AcceptanceType As Integer
Dim Symmetric As Boolean
Dim Active As Boolean()
Dim IOPos As Double()
Dim LSPos As Double()
Dim CPPos As Double()
Dim IONeg As Double()
Dim LSNeg As Double()
Dim CPNeg As Double()
Dim returnValue As Integer
```

```
returnValue = instance.SetAcceptanceCriteria(Name,
    AcceptanceType, Symmetric, Active,
```

IOPos, LSPos, CPPos, IONeg, LSNeg, CPNeg)

```
int SetAcceptanceCriteria(
String^ Name,
int AcceptanceType,
bool Symmetric,
array<bool>^% Active,
array<double>^% IOPos,
array<double>^% LSPos,
array<double>^% CPPos,
array<double>^% IONeg,
array<double>^% LSNeg,
array<double>^% CPNeg
)
```

```
abstract SetAcceptanceCriteria :
    Name : string *
    AcceptanceType : int *
    Symmetric : bool *
    Active : bool[] byref *
    IOPos : float[] byref *
    LSPos : float[] byref *
    CPPos : float[] byref *
    IONeg : float[] byref *
    LSNeg : float[] byref *
    CPNeg : float[] byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing link property

AcceptanceType

Type:Â SystemInt32

This item is either 1 or 2. 1 means Force and 2 means Displacement.

Symmetric

Type:Â SystemBoolean

Indicates if the acceptance criteria is symmetric, that is, the negative values are the same as the positive values.

Active

Type:Â SystemBoolean

This is an array dimensioned to 5 indicating if the acceptance criteria is active for each. Items 0 thru 5 in the array refer to F1, F2, F3, M1, M2, M3 or U1, U2, U3, R1, R2, R3, respectively.

IOPos

Type:Â SystemDouble

This is an array dimensioned to 5 indicating the positive immediate occupancy acceptance criteria values for each DOF. Items 0 thru 5 in the array refer to F1, F2, F3, M1, M2, M3 or U1, U2, U3, R1, R2, R3, respectively.

LSPos

Type:Â SystemDouble

This is an array dimensioned to 5 indicating the positive life safety acceptance criteria values for each DOF. Items 0 thru 5 in the array refer to F1, F2, F3, M1, M2, M3 or U1, U2, U3, R1, R2, R3, respectively.

CPPosType: SystemDouble

This is an array dimensioned to 5 indicating the positive collapse prevention acceptance criteria values for each DOF. Items 0 thru 5 in the array refer to F1, F2, F3, M1, M2, M3 or U1, U2, U3, R1, R2, R3, respectively.

IONegType: SystemDouble

This is an array dimensioned to 5 indicating the negative immediate occupancy acceptance criteria values for each DOF. Items 0 thru 5 in the array refer to F1, F2, F3, M1, M2, M3 or U1, U2, U3, R1, R2, R3, respectively.

LSNegType: SystemDouble

This is an array dimensioned to 5 indicating the negative life safety acceptance criteria values for each DOF. Items 0 thru 5 in the array refer to F1, F2, F3, M1, M2, M3 or U1, U2, U3, R1, R2, R3, respectively.

CPNegType: SystemDouble

This is an array dimensioned to 5 indicating the negative collapse prevention acceptance criteria values for each DOF. Items 0 thru 5 in the array refer to F1, F2, F3, M1, M2, M3 or U1, U2, U3, R1, R2, R3, respectively.

Return ValueType: Int32

returns zero if the data is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyDOF() As Boolean
    Dim MyFixed() As Boolean
    Dim MyNonLinear() As Boolean
    Dim MyKe() As Double
    Dim MyCe() As Double
    Dim Active() As Boolean
    Dim IOPos() As Double
    Dim LSPos() As Double
    Dim CPPos() As Double
    Dim IONeg() As Double
    Dim LSNeg() As Double
    Dim CPNeg() As Double
    Dim AcceptanceType As Integer
    Dim Symmetric As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
```

Introduction

```
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'add a link property
ReDim MyDOF(5)
ReDim MyFixed(5)
ReDim MyNonLinear(5)
ReDim MyKe(5)
ReDim MyCe(5)

MyDOF(0) = True
MyKe(0) = 12
MyCe(0) = 0.01

MyDOF(1) = True
MyNonLinear(1) = True
MyKe(1) = 12
MyCe(1) = 0.01

MyDOF(2) = True
MyFixed(2) = True

ret = SapModel.PropLink.SetMultiLinearElastic("MLE1", MyDOF, MyFixed, MyNonLinear, MyKe, M

'set link property acceptance criteria
ReDim Active(5)
ReDim IOPos(5)
ReDim LSPos(5)
ReDim CPPos(5)
ReDim IONeg(5)
ReDim LSNeg(5)
ReDim CPNeg(5)
Active(0) = True
Active(1) = True
IOPos(0) = 100
LSPos(0) = 200
CPPos(0) = 300
IONeg(0) = 120
LSNeg(0) = 220
CPNeg(0) = 320
IOPos(1) = 150
LSPos(1) = 250
CPPos(1) = 350
IONeg(1) = 130
LSNeg(1) = 230
CPNeg(1) = 330
ret = SapModel.PropLink.SetAcceptanceCriteria("MLE1", 1, False, Active, IOPos, LSPos, CPPo

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

See Also

Return Value

3298

Reference

cPropLink Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LST28E5CE9D_0?cpp=::300") Method

Initializes a damper-type link property. If this function is called for an existing link property, all items for the property are reset to their default values.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetDamper(
    string Name,
    ref bool[] DOF,
    ref bool[] Fixed,
    ref bool[] Nonlinear,
    ref double[] Ke,
    ref double[] Ce,
    ref double[] K,
    ref double[] C,
    ref double[] CExp,
    double DJ2,
    double DJ3,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetDamper (
    Name As String,
    ByRef DOF As Boolean(),
    ByRef Fixed As Boolean(),
    ByRef Nonlinear As Boolean(),
    ByRef Ke As Double(),
    ByRef Ce As Double(),
    ByRef K As Double(),
    ByRef C As Double(),
    ByRef CExp As Double(),
    DJ2 As Double,
    DJ3 As Double,
    Notes As String = "",
    GUID As String = ""
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim DOF As Boolean()
Dim Fixed As Boolean()
Dim Nonlinear As Boolean()
Dim Ke As Double()
Dim Ce As Double()
Dim K As Double()
```

cPropLinkspan id="LST28E5CE9D_0"AddLanguageSpecificTextSet("LST28E5CE9D_0?cpp=::300");SetD


```

Dim C As Double()
Dim CExp As Double()
Dim DJ2 As Double
Dim DJ3 As Double
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer

returnValue = instance.SetDamper(Name,
DOF, Fixed, Nonlinear, Ke, Ce, K, C,
CExp, DJ2, DJ3, Notes, GUID)

```

```

int SetDamper(
String^ Name,
array<bool>^% DOF,
array<bool>^% Fixed,
array<bool>^% Nonlinear,
array<double>^% Ke,
array<double>^% Ce,
array<double>^% K,
array<double>^% C,
array<double>^% CExp,
double DJ2,
double DJ3,
String^ Notes = L"",
String^ GUID = L""
)

```

```

abstract SetDamper :
    Name : string *
    DOF : bool[] byref *
    Fixed : bool[] byref *
    Nonlinear : bool[] byref *
    Ke : float[] byref *
    Ce : float[] byref *
    K : float[] byref *
    C : float[] byref *
    CExp : float[] byref *
    DJ2 : float *
    DJ3 : float *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int

```

Parameters

Name

Type: SystemString

The name of an existing or new link property. If this is an existing property, that property is modified; otherwise, a new property is added

DOF

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if properties exist for a specified degree of freedom.

Value DOF

DOF(0) U1
 DOF(1) U2
 DOF(2) U3
 DOF(3) R1
 DOF(4) R2
 DOF(5) R3

Fixed

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if the specified degree of freedom is fixed (restrained).

Value	Fixity
--------------	---------------

Fixed(0) U1	fixity if DOF(0) = True
-------------	-------------------------

Fixed(1) U2	fixity if DOF(1) = True
-------------	-------------------------

Fixed(2) U3	fixity if DOF(2) = True
-------------	-------------------------

Fixed(3) R1	fixity if DOF(3) = True
-------------	-------------------------

Fixed(4) R2	fixity if DOF(4) = True
-------------	-------------------------

Fixed(5) R3	fixity if DOF(5) = True
-------------	-------------------------

The term Fixed(n) applies only when DOF(n) = True.

Nonlinear

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if nonlinear properties exist for a specified degree of freedom.

Value	Nonlinear
--------------	------------------

Nonlinear(0) U1	has nonlinear properties
-----------------	--------------------------

Nonlinear(1) U2	has nonlinear properties
-----------------	--------------------------

Nonlinear(2) U3	has nonlinear properties
-----------------	--------------------------

Nonlinear(3) R1	has nonlinear properties
-----------------	--------------------------

Nonlinear(4) R2	has nonlinear properties
-----------------	--------------------------

Nonlinear(5) R3	has nonlinear properties
-----------------	--------------------------

The term Nonlinear(n) applies only when DOF(n) = True and Fixed(n) = False.

Ke

Type: SystemDouble

This is an array of effective stiffness terms for the link property. The effective stiffness applies for linear analyses.

Value Stiffness

Ke(0) U1	[F/L]
----------	-------

Ke(1) U2	[F/L]
----------	-------

Ke(2) U3	[F/L]
----------	-------

Ke(3) R1	[FL]
----------	------

Ke(4) R2	[FL]
----------	------

Ke(5) R3	[FL]
----------	------

The term ke(n) applies only when DOF(n) = True and Fixed(n) = False.

Ce

Type: SystemDouble

This is an array of effective damping terms for the link property. The effective

stiffness applies for linear analyses.

Value Damping

Ce(0) U1 [F/L]

Ce(1) U2 [F/L]

Ce(2) U3 [F/L]

Ce(3) R1 [FL]

Ce(4) R2 [FL]

Ce(5) R3 [FL]

The term ce(n) applies only when DOF(n) = True and Fixed(n) = False.

K

Type: SystemDouble

This is an array of initial stiffness terms for the link property. The initial stiffness applies for nonlinear analyses.

Value Stiffness

K(0) U1 [F/L]

K(1) U2 [F/L]

K(2) U3 [F/L]

K(3) R1 [FL]

K(4) R2 [FL]

K(5) R3 [FL]

The term k(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

C

Type: SystemDouble

This is an array of nonlinear damping coefficient terms for the link property. The nonlinear damping coefficient applies for nonlinear analyses.

Value Damping

C(0) U1 [F/(L^{cexp})]

C(1) U2 [F/(L^{cexp})]

C(2) U3 [F/(L^{cexp})]

C(3) R1 [FL]

C(4) R2 [FL]

C(5) R3 [FL]

The term c(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

CExp

Type: SystemDouble

This is an array of the nonlinear damping exponent terms. The nonlinear damping exponent applies for nonlinear analyses. It is applied to the velocity across the damper in the equation of motion.

Value CExp

cexp(0) U1

cexp(1) U2

cexp(2) U3

cexp(3) R1

cexp(4) R2

cexp(5) R3

The term cexp(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

DJ2

Type:Â SystemDouble

The distance from the J-End of the link to the U2 shear spring. This item applies only when DOF(1) = True. [L]

DJ3

Type:Â SystemDouble

The distance from the J-End of the link to the U3 shear spring. This item applies only when DOF(2) = True. [L]

Notes (Optional)

Type:Â SystemString

The notes, if any, assigned to the property.

GUID (Optional)

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the property. If this item is input as Default, the program assigns a GUID to the property.

Return Value

Type:Â Int32

Returns zero if the property is successfully initialized; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyDOF() As Boolean
    Dim MyFixed() As Boolean
    Dim MyNonLinear() As Boolean
    Dim MyKe() As Double
    Dim MyCe() As Double
    Dim MyK() As Double
    Dim MyC() As Double
    Dim MyCexp() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()
```

Introduction

```
'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'add link property
    ReDim MyDOF(5)
    ReDim MyFixed(5)
    ReDim MyNonLinear(5)
    ReDim MyKe(5)
    ReDim MyCe(5)
    ReDim MyK(5)
    ReDim MyC(5)
    ReDim MyCexp(5)

    MyDOF(0) = True
    MyKe(0) = 12
    MyCe(0) = 0.01

    MyDOF(1) = True
    MyNonLinear(1) = True
    MyKe(1) = 12
    MyCe(1) = 0.01
    MyK(1) = 20
    MyC(1)=0.08
    MyCexp(1) = 1.2

    MyDOF(2) = True
    MyFixed(2) = True

    ret = SapModel.PropLink.SetDamper("D1", MyDOF, MyFixed, MyNonLinear, MyKe, MyCe, MyK, MyC, MyCexp)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPropLink Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LST178F1924_0?cpp=::|num=306)SetDamperBilinear") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetDamperBilinear(
    string Name,
    ref bool[] dof,
    ref bool[] Fixed,
    ref bool[] Nonlinear,
    ref double[] ke,
    ref double[] ce,
    ref double[] k,
    ref double[] c,
    ref double[] CY,
    ref double[] ForceLimit,
    double dj2,
    double dj3,
    string notes = "",
    string GUID = ""
)
```

```
Function SetDamperBilinear (
    Name As String,
    ByRef dof As Boolean(),
    ByRef Fixed As Boolean(),
    ByRef Nonlinear As Boolean(),
    ByRef ke As Double(),
    ByRef ce As Double(),
    ByRef k As Double(),
    ByRef c As Double(),
    ByRef CY As Double(),
    ByRef ForceLimit As Double(),
    dj2 As Double,
    dj3 As Double,
    Optional notes As String = "",
    Optional GUID As String = ""
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim dof As Boolean()
Dim Fixed As Boolean()
Dim Nonlinear As Boolean()
Dim ke As Double()
Dim ce As Double()
Dim k As Double()
Dim c As Double()
```

cPropLinkspan id="LST178F1924_0"AddLanguageSpecificTextSet("LST178F1924_0?cpp=::|num=306)SetDamperBilinear")

```

Dim CY As Double()
Dim ForceLimit As Double()
Dim dj2 As Double
Dim dj3 As Double
Dim notes As String
Dim GUID As String
Dim returnValue As Integer

returnValue = instance.SetDamperBilinear(Name,
dof, Fixed, Nonlinear, ke, ce, k, c,
CY, ForceLimit, dj2, dj3, notes, GUID)

```

```

int SetDamperBilinear(
String^ Name,
array<bool>^% dof,
array<bool>^% Fixed,
array<bool>^% Nonlinear,
array<double>^% ke,
array<double>^% ce,
array<double>^% k,
array<double>^% c,
array<double>^% CY,
array<double>^% ForceLimit,
double dj2,
double dj3,
String^ notes = L"",
String^ GUID = L""
)

```

```

abstract SetDamperBilinear :
    Name : string *
    dof : bool[] byref *
    Fixed : bool[] byref *
    Nonlinear : bool[] byref *
    ke : float[] byref *
    ce : float[] byref *
    k : float[] byref *
    c : float[] byref *
    CY : float[] byref *
    ForceLimit : float[] byref *
    dj2 : float *
    dj3 : float *
    ?notes : string *
    ?GUID : string
(* Defaults:
    let _notes = defaultArg notes ""
    let _GUID = defaultArg GUID ""
*)
-> int

```

Parameters

Name

Type:Â SystemString

dof

Type:Â SystemBoolean

Fixed

Type:Â SystemBoolean

Nonlinear

Type:Â [SystemBoolean](#)
 ke
 Type:Â [SystemDouble](#)
 ce
 Type:Â [SystemDouble](#)
 k
 Type:Â [SystemDouble](#)
 c
 Type:Â [SystemDouble](#)
 CY
 Type:Â [SystemDouble](#)
 ForceLimit
 Type:Â [SystemDouble](#)
 dj2
 Type:Â [SystemDouble](#)
 dj3
 Type:Â [SystemDouble](#)
 notes (Optional)
 Type:Â [SystemString](#)
 GUID (Optional)
 Type:Â [SystemString](#)

Return Value

Type:Â [Int32](#)

▲ See Also

Reference

[cPropLink Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LST738B04F9_0?cpp=::|n8309;SetDamperFrictionSpring") Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetDamperFrictionSpring(
    string Name,
    ref bool[] dof,
    ref bool[] Fixed,
    ref bool[] Nonlinear,
    ref double[] ke,
    ref double[] ce,
    ref double[] k,
    ref double[] K1,
    ref double[] K2,
    ref double[] u0,
    ref double[] us,
    ref int[] direction,
    double dj2,
    double dj3,
    string notes = "",
    string GUID = ""
)
```

```
Function SetDamperFrictionSpring (
    Name As String,
    ByRef dof As Boolean(),
    ByRef Fixed As Boolean(),
    ByRef Nonlinear As Boolean(),
    ByRef ke As Double(),
    ByRef ce As Double(),
    ByRef k As Double(),
    ByRef K1 As Double(),
    ByRef K2 As Double(),
    ByRef u0 As Double(),
    ByRef us As Double(),
    ByRef direction As Integer(),
    dj2 As Double,
    dj3 As Double,
    Optional notes As String = "",
    Optional GUID As String = ""
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim dof As Boolean()
Dim Fixed As Boolean()
Dim Nonlinear As Boolean()
```

cPropLinkspan id="LST738B04F9_0"AddLanguageSpecificTextSet("LST738B04F9_0?cpp=::|n8309;SetDamperFrictionSpring")

Introduction

```
Dim ke As Double()
Dim ce As Double()
Dim k As Double()
Dim K1 As Double()
Dim K2 As Double()
Dim u0 As Double()
Dim us As Double()
Dim direction As Integer()
Dim dj2 As Double
Dim dj3 As Double
Dim notes As String
Dim GUID As String
Dim returnValue As Integer

returnValue = instance.SetDamperFrictionSpring(Name,
dof, Fixed, Nonlinear, ke, ce, k, K1,
K2, u0, us, direction, dj2, dj3, notes,
GUID)
```

```
int SetDamperFrictionSpring(
String^ Name,
array<bool>^% dof,
array<bool>^% Fixed,
array<bool>^% Nonlinear,
array<double>^% ke,
array<double>^% ce,
array<double>^% k,
array<double>^% K1,
array<double>^% K2,
array<double>^% u0,
array<double>^% us,
array<int>^% direction,
double dj2,
double dj3,
String^ notes = L"",
String^ GUID = L""
)
```

```
abstract SetDamperFrictionSpring :
    Name : string *
    dof : bool[] byref *
    Fixed : bool[] byref *
    Nonlinear : bool[] byref *
    ke : float[] byref *
    ce : float[] byref *
    k : float[] byref *
    K1 : float[] byref *
    K2 : float[] byref *
    u0 : float[] byref *
    us : float[] byref *
    direction : int[] byref *
    dj2 : float *
    dj3 : float *
    ?notes : string *
    ?GUID : string
(* Defaults:
    let _notes = defaultArg notes ""
    let _GUID = defaultArg GUID ""
*)
-> int
```

Parameters

Name

Type: SystemString

dof

Type: SystemBoolean

Fixed

Type: SystemBoolean

Nonlinear

Type: SystemBoolean

ke

Type: SystemDouble

ce

Type: SystemDouble

k

Type: SystemDouble

K1

Type: SystemDouble

K2

Type: SystemDouble

u0

Type: SystemDouble

us

Type: SystemDouble

direction

Type: SystemInt32

dj2

Type: SystemDouble

dj3

Type: SystemDouble

notes (Optional)

Type: SystemString

GUID (Optional)

Type: SystemString**Return Value**Type: Int32

▲ See Also

ReferencecPropLink InterfaceETABSv1 Namespace

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LSTF362BD1E_0?cpp=::|00#2");SetFrictionIsolator

Method

Initializes a friction isolator-type link property. If this function is called for an existing link property, all items for the property are reset to their default value.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetFrictionIsolator(
    string Name,
    ref bool[] DOF,
    ref bool[] Fixed,
    ref bool[] Nonlinear,
    ref double[] Ke,
    ref double[] Ce,
    ref double[] K,
    ref double[] Slow,
    ref double[] Fast,
    ref double[] Rate,
    ref double[] Radius,
    double Damping,
    double DJ2,
    double DJ3,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetFrictionIsolator (
    Name As String,
    ByRef DOF As Boolean(),
    ByRef Fixed As Boolean(),
    ByRef Nonlinear As Boolean(),
    ByRef Ke As Double(),
    ByRef Ce As Double(),
    ByRef K As Double(),
    ByRef Slow As Double(),
    ByRef Fast As Double(),
    ByRef Rate As Double(),
    ByRef Radius As Double(),
    Damping As Double,
    DJ2 As Double,
    DJ3 As Double,
    Notes As String = "",
    GUID As String = ""
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
```

cPropLinkspan id="LSTF362BD1E_0"AddLanguageSpecificTextSet("LSTF362BD1E_0?cpp=::|00#2");SetFrictionIsolator

Introduction

```
Dim DOF As Boolean()
Dim Fixed As Boolean()
Dim Nonlinear As Boolean()
Dim Ke As Double()
Dim Ce As Double()
Dim K As Double()
Dim Slow As Double()
Dim Fast As Double()
Dim Rate As Double()
Dim Radius As Double()
Dim Damping As Double
Dim DJ2 As Double
Dim DJ3 As Double
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.SetFrictionIsolator(Name,
DOF, Fixed, Nonlinear, Ke, Ce, K, Slow,
Fast, Rate, Radius, Damping, DJ2, DJ3,
Notes, GUID)
```

```
int SetFrictionIsolator(
String^ Name,
array<bool>^% DOF,
array<bool>^% Fixed,
array<bool>^% Nonlinear,
array<double>^% Ke,
array<double>^% Ce,
array<double>^% K,
array<double>^% Slow,
array<double>^% Fast,
array<double>^% Rate,
array<double>^% Radius,
double Damping,
double DJ2,
double DJ3,
String^ Notes = L"",
String^ GUID = L""
)
```

```
abstract SetFrictionIsolator :
    Name : string *
    DOF : bool[] byref *
    Fixed : bool[] byref *
    Nonlinear : bool[] byref *
    Ke : float[] byref *
    Ce : float[] byref *
    K : float[] byref *
    Slow : float[] byref *
    Fast : float[] byref *
    Rate : float[] byref *
    Radius : float[] byref *
    Damping : float *
    DJ2 : float *
    DJ3 : float *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
```

cPropLinkspan id="LSTF362BD1E_0"AddLanguageSpecificTextSet("LSTF362BD1E_0?cpp=::|0013");SetFr

-> int

Parameters

Name

Type:Â SystemString

The name of an existing or new link property. If this is an existing property, that property is modified; otherwise, a new property is added

DOF

Type:Â SystemBoolean

This is a boolean array, dimensioned to 5, indicating if properties exist for a specified degree of freedom.

Value DOF

DOF(0) U1

DOF(1) U2

DOF(2) U3

DOF(3) R1

DOF(4) R2

DOF(5) R3

Fixed

Type:Â SystemBoolean

This is a boolean array, dimensioned to 5, indicating if the specified degree of freedom is fixed (restrained).

Value Fixity

Fixed(0) U1 fixity if DOF(0) = True

Fixed(1) U2 fixity if DOF(1) = True

Fixed(2) U3 fixity if DOF(2) = True

Fixed(3) R1 fixity if DOF(3) = True

Fixed(4) R2 fixity if DOF(4) = True

Fixed(5) R3 fixity if DOF(5) = True

Nonlinear

Type:Â SystemBoolean

This is a boolean array, dimensioned to 5, indicating if nonlinear properties exist for a specified degree of freedom.

Value Nonlinear

Nonlinear(0) U1 has nonlinear properties

Nonlinear(1) U2 has nonlinear properties

Nonlinear(2) U3 has nonlinear properties

Nonlinear(3) R1, not used

Nonlinear(4) R2, not used

Nonlinear(5) R3, not used

Note that this item is applicable only for degrees of freedom U1, U2 and U3. For those degrees of freedom, the term Nonlinear(n) applies only when DOF(n) = True and Fixed(n) = False.

Ke

Type:Â SystemDouble

This is an array of effective stiffness terms for the link property. The effective stiffness applies for linear analyses.

Value Stiffness

Ke(0) U1 [F/L]

Ke(1) U2 [F/L]

Ke(2) U3 [F/L]

Ke(3) R1 [FL]

Ke(4) R2 [FL]

Ke(5) R3 [FL]

The term ke(n) applies only when DOF(n) = True and Fixed(n) = False.

Ce

Type: SystemDouble

This is an array of effective damping terms for the link property. The effective stiffness applies for linear analyses.

Value Damping

Ce(0) U1 [F/L]

Ce(1) U2 [F/L]

Ce(2) U3 [F/L]

Ce(3) R1 [FL]

Ce(4) R2 [FL]

Ce(5) R3 [FL]

The term ce(n) applies only when DOF(n) = True and Fixed(n) = False.

K

Type: SystemDouble

This is an array of initial stiffness terms for the link property. The initial stiffness applies for nonlinear analyses.

Value Stiffness

K(0) U1 [F/L]

K(1) U2 [F/L]

K(2) U3 [F/L]

K(3) R1, not used

K(4) R2, not used

K(5) R3, not used

Note that this item is applicable only for degrees of freedom U1, U2 and U3. For those degrees of freedom, the term k(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Slow

Type: SystemDouble

This is an array of the friction coefficient at zero velocity terms for the link property. This coefficient applies for nonlinear analyses.

Value Slow

Slow(0) U1, not used

Slow(1) U2

Slow(2) U3

Slow(3) R1, not used

Slow(4) R2, not used

Slow(5) R3, not used

Note that this item is applicable only for degrees of freedom U2 and U3. For those degrees of freedom, the term Slow(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Fast

Type: SystemDouble

This is an array of the friction coefficient at fast velocity terms for the link property. This coefficient applies for nonlinear analyses.

Value Fast

Fast(0) U1, not used

Fast(1) U2

Fast(2) U3

Fast(3) R1, not used

Fast(4) R2, not used

Fast(5) R3, not used

Note that this item is applicable only for degrees of freedom U2 and U3. For those degrees of freedom, the term Fast(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Rate

Type: SystemDouble

This is an array of the inverse of the characteristic sliding velocity terms for the link property. This item applies for nonlinear analyses.

Value Rate

Rate(0) U1, not used

Rate(1) U2 [s/L]

Rate(2) U3 [s/L]

Rate(3) R1, not used

Rate(4) R2, not used

Rate(5) R3, not used

Note that this item is applicable only for degrees of freedom U2 and U3. For those degrees of freedom, the term Rate(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Radius

Type: SystemDouble

This is an array of the radius of the sliding contact surface terms for the link property. Inputting 0 means there is an infinite radius, that is, the slider is flat. This item applies for nonlinear analyses.

Value Radius

Radius(0) U1, not used

Radius(1) U2 [L]

Radius(2) U3 [L]

Radius(3) R1, not used

Radius(4) R2, not used

Radius(5) R3, not used

Introduction

Note that this item is applicable only for degrees of freedom U2 and U3. For those degrees of freedom, the term Fast(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Damping

Type: SystemDouble

This is the nonlinear damping coefficient used for the axial translational degree of freedom, U1. This item applies for nonlinear analyses. [F/L]

DJ2

Type: SystemDouble

The distance from the J-End of the link to the U2 shear spring. This item applies only when DOF(1) = True. [L]

DJ3

Type: SystemDouble

The distance from the J-End of the link to the U3 shear spring. This item applies only when DOF(2) = True. [L]

Notes (Optional)

Type: SystemString

The notes, if any, assigned to the property.

GUID (Optional)

Type: SystemString

The GUID (global unique identifier), if any, assigned to the property. If this item is input as Default, the program assigns a GUID to the property.

Return Value

Type: Int32

Returns zero if the property is successfully initialized; otherwise it returns a nonzero value.

Remarks

Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim MyDOF() As Boolean  
    Dim MyFixed() As Boolean  
    Dim MyNonLinear() As Boolean  
    Dim MyKe() As Double  
    Dim MyCe() As Double  
    Dim MyK() As Double  
    Dim MySlow() As Double  
    Dim MyFast() As Double  
    Dim MyRate() As Double  
    Dim MyRadius() As Double  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object
```

Return Value

Introduction

```
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'add link property
ReDim MyDOF(5)
ReDim MyFixed(5)
ReDim MyNonLinear(5)
ReDim MyKe(5)
ReDim MyCe(5)
ReDim MyK(5)
ReDim MySlow(5)
ReDim MyFast(5)
ReDim MyRate(5)
ReDim MyRadius(5)

MyDOF(0) = True
MyKe(0) = 12
MyCe(0) = 0.01

MyDOF(1) = True
MyNonLinear(1) = True
MyKe(1) = 12
MyCe(1) = 0.01
MyK(1) = 20
MySlow(1)= 0.6
MyFast(1)= 0.5
MyRate(1)= 10
MyRadius(1)= 80

MyDOF(2) = True
MyNonLinear(2) = True
MyKe(2) = 14
MyCe(2) = 0.008
MyK(2) = 22
MySlow(2)= 0.66
MyFast(2)= 0.55
MyRate(2)= 12
MyRadius(2)= 75

MyDOF(3) = True
MyKe(3) = 15
MyCe(3) = 0

MyDOF(4) = True
MyFixed(4) = True

ret = SapModel.PropLink.SetFrictionIsolator("FI1", MyDOF, MyFixed, MyNonLinear, MyKe, MyCe)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

📌 See Also

Return Value

3318

Reference

cPropLink Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LSTD59D44B Method

Initializes a gap-type link property. If this function is called for an existing link property, all items for the property are reset to their default value.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetGap(
    string Name,
    ref bool[] DOF,
    ref bool[] Fixed,
    ref bool[] Nonlinear,
    ref double[] Ke,
    ref double[] Ce,
    ref double[] K,
    ref double[] Dis,
    double DJ2,
    double DJ3,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetGap (
    Name As String,
    ByRef DOF As Boolean(),
    ByRef Fixed As Boolean(),
    ByRef Nonlinear As Boolean(),
    ByRef Ke As Double(),
    ByRef Ce As Double(),
    ByRef K As Double(),
    ByRef Dis As Double(),
    DJ2 As Double,
    DJ3 As Double,
    Notes As String = "",
    GUID As String = ""
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim DOF As Boolean()
Dim Fixed As Boolean()
Dim Nonlinear As Boolean()
Dim Ke As Double()
Dim Ce As Double()
Dim K As Double()
Dim Dis As Double()
Dim DJ2 As Double
```

cPropLinkspan id="LSTD59D44BD_0"AddLanguageSpecificTextSet("LSTD59D44BD_0?cpp=::3329");SetG

```

Dim DJ3 As Double
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer

returnValue = instance.SetGap(Name, DOF,
Fixed, Nonlinear, Ke, Ce, K, Dis, DJ2,
DJ3, Notes, GUID)

int SetGap(
String^ Name,
array<bool>^% DOF,
array<bool>^% Fixed,
array<bool>^% Nonlinear,
array<double>^% Ke,
array<double>^% Ce,
array<double>^% K,
array<double>^% Dis,
double DJ2,
double DJ3,
String^ Notes = L"",
String^ GUID = L""
)

abstract SetGap :
    Name : string *
    DOF : bool[] byref *
    Fixed : bool[] byref *
    Nonlinear : bool[] byref *
    Ke : float[] byref *
    Ce : float[] byref *
    K : float[] byref *
    Dis : float[] byref *
    DJ2 : float *
    DJ3 : float *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int

```

Parameters

Name

Type: SystemString

The name of an existing or new link property. If this is an existing property, that property is modified; otherwise, a new property is added

DOF

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if properties exist for a specified degree of freedom.

Value DOF

DOF(0) U1

DOF(1) U2

DOF(2) U3

DOF(3) R1

DOF(4) R2

DOF(5) R3

FixedType: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if the specified degree of freedom is fixed (restrained).

Value	Fixity
--------------	---------------

Fixed(0) U1 fixity if DOF(0) = True

Fixed(1) U2 fixity if DOF(1) = True

Fixed(2) U3 fixity if DOF(2) = True

Fixed(3) R1 fixity if DOF(3) = True

Fixed(4) R2 fixity if DOF(4) = True

Fixed(5) R3 fixity if DOF(5) = True

NonlinearType: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if nonlinear properties exist for a specified degree of freedom.

Value	Nonlinear
--------------	------------------

Nonlinear(0) U1 has nonlinear properties

Nonlinear(1) U2 has nonlinear properties

Nonlinear(2) U3 has nonlinear properties

Nonlinear(3) R1 has nonlinear properties

Nonlinear(4) R2 has nonlinear properties

Nonlinear(5) R3 has nonlinear properties

The term Nonlinear(n) applies only when DOF(n) = True and Fixed(n) = False.

KeType: SystemDouble

This is an array of effective stiffness terms for the link property. The effective stiffness applies for linear analyses.

Value Stiffness

Ke(0) U1 [F/L]

Ke(1) U2 [F/L]

Ke(2) U3 [F/L]

Ke(3) R1 [FL]

Ke(4) R2 [FL]

Ke(5) R3 [FL]

The term ke(n) applies only when DOF(n) = True and Fixed(n) = False.

CeType: SystemDouble

This is an array of effective damping terms for the link property. The effective stiffness applies for linear analyses.

Value Damping

Ce(0) U1 [F/L]

Ce(1) U2 [F/L]

Ce(2) U3 [F/L]

Ce(3) R1 [FL]

Ce(4) R2 [FL]

Ce(5) R3 [FL]

The term ce(n) applies only when DOF(n) = True and Fixed(n) = False.

K

Type: SystemDouble

This is an array of initial stiffness terms for the link property. The initial stiffness applies for nonlinear analyses.

Value Stiffness

K(0) U1 [F/L]

K(1) U2 [F/L]

K(2) U3 [F/L]

K(3) R1 [FL]

K(4) R2 [FL]

K(5) R3 [FL]

The term k(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Dis

Type: SystemDouble

This is an array of initial gap opening terms for the link property. The initial gap opening applies for nonlinear analyses.

Value Dis

dis(0) U1 [L]

dis(1) U2 [L]

dis(2) U3 [L]

dis(3) R1 [rad]

dis(4) R2 [rad]

dis(5) R3 [rad]

The term dis(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

DJ2

Type: SystemDouble

The distance from the J-End of the link to the U2 shear spring. This item applies only when DOF(1) = True. [L]

DJ3

Type: SystemDouble

The distance from the J-End of the link to the U3 shear spring. This item applies only when DOF(2) = True. [L]

Notes (Optional)

Type: SystemString

The notes, if any, assigned to the property.

GUID (Optional)

Type: SystemString

The GUID (global unique identifier), if any, assigned to the property. If this item is input as Default, the program assigns a GUID to the property.

Return ValueType: Int32

Returns zero if the property is successfully initialized; otherwise it returns a nonzero value

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyDOF() As Boolean
    Dim MyFixed() As Boolean
    Dim MyNonLinear() As Boolean
    Dim MyKe() As Double
    Dim MyCe() As Double
    Dim MyK() As Double
    Dim MyDis() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link property
    ReDim MyDOF(5)
    ReDim MyFixed(5)
    ReDim MyNonLinear(5)
    ReDim MyKe(5)
    ReDim MyCe(5)
    ReDim MyK(5)
    ReDim MyDis(5)

    MyDOF(0) = True
    MyKe(0) = 12
    MyCe(0) = 0.01

    MyDOF(1) = True
    MyNonLinear(1) = True
    MyKe(1) = 12
    MyCe(1) = 0.01
    MyK(1) = 20
    MyDis(1) = 1.2

    MyDOF(2) = True
    MyFixed(2) = True
```


Introduction

```
ret = SapModel.PropLink.SetGap("G1", MyDOF, MyFixed, MyNonLinear, MyKe, MyCe, MyK, MyDis,  
'close ETABS  
  EtabsObject.ApplicationExit(False)  
  
'clean up variables  
  SapModel = Nothing  
  EtabsObject = Nothing  
End Sub
```

📌 See Also

Reference

[cPropLink Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LSTF734EA72_0") Method

Initializes a hook-type link property. If this function is called for an existing link property, all items for the property are reset to their default value.

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetHook(
    string Name,
    ref bool[] DOF,
    ref bool[] Fixed,
    ref bool[] Nonlinear,
    ref double[] Ke,
    ref double[] Ce,
    ref double[] K,
    ref double[] Dis,
    double DJ2,
    double DJ3,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetHook (
    Name As String,
    ByRef DOF As Boolean(),
    ByRef Fixed As Boolean(),
    ByRef Nonlinear As Boolean(),
    ByRef Ke As Double(),
    ByRef Ce As Double(),
    ByRef K As Double(),
    ByRef Dis As Double(),
    DJ2 As Double,
    DJ3 As Double,
    Notes As String = "",
    GUID As String = ""
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim DOF As Boolean()
Dim Fixed As Boolean()
Dim Nonlinear As Boolean()
Dim Ke As Double()
Dim Ce As Double()
Dim K As Double()
Dim Dis As Double()
Dim DJ2 As Double
```

cPropLinkspan id="LSTF734EA72_0"AddLanguageSpecificTextSet("LSTF734EA72_0?cpp=::|rob=25;SetHo

Introduction

```
Dim DJ3 As Double
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer

returnValue = instance.SetHook(Name, DOF,
Fixed, Nonlinear, Ke, Ce, K, Dis, DJ2,
DJ3, Notes, GUID)

int SetHook(
String^ Name,
array<bool>^% DOF,
array<bool>^% Fixed,
array<bool>^% Nonlinear,
array<double>^% Ke,
array<double>^% Ce,
array<double>^% K,
array<double>^% Dis,
double DJ2,
double DJ3,
String^ Notes = L"",
String^ GUID = L""
)

abstract SetHook :
    Name : string *
    DOF : bool[] byref *
    Fixed : bool[] byref *
    Nonlinear : bool[] byref *
    Ke : float[] byref *
    Ce : float[] byref *
    K : float[] byref *
    Dis : float[] byref *
    DJ2 : float *
    DJ3 : float *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing or new link property. If this is an existing property, that property is modified; otherwise, a new property is added

DOF

Type:Â SystemBoolean

This is a boolean array, dimensioned to 5, indicating if properties exist for a specified degree of freedom.

Value DOF

DOF(0) U1

DOF(1) U2

DOF(2) U3

DOF(3) R1

DOF(4) R2

DOF(5) R3

FixedType: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if the specified degree of freedom is fixed (restrained).

Value	Fixity
--------------	---------------

Fixed(0) U1 fixity if DOF(0) = True

Fixed(1) U2 fixity if DOF(1) = True

Fixed(2) U3 fixity if DOF(2) = True

Fixed(3) R1 fixity if DOF(3) = True

Fixed(4) R2 fixity if DOF(4) = True

Fixed(5) R3 fixity if DOF(5) = True

NonlinearType: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if nonlinear properties exist for a specified degree of freedom.

Value	Nonlinear
--------------	------------------

Nonlinear(0) U1 has nonlinear properties

Nonlinear(1) U2 has nonlinear properties

Nonlinear(2) U3 has nonlinear properties

Nonlinear(3) R1 has nonlinear properties

Nonlinear(4) R2 has nonlinear properties

Nonlinear(5) R3 has nonlinear properties

The term Nonlinear(n) applies only when DOF(n) = True and Fixed(n) = False.

KeType: SystemDouble

This is an array of effective stiffness terms for the link property. The effective stiffness applies for linear analyses.

Value Stiffness

Ke(0) U1 [F/L]

Ke(1) U2 [F/L]

Ke(2) U3 [F/L]

Ke(3) R1 [FL]

Ke(4) R2 [FL]

Ke(5) R3 [FL]

The term ke(n) applies only when DOF(n) = True and Fixed(n) = False.

CeType: SystemDouble

This is an array of effective damping terms for the link property. The effective stiffness applies for linear analyses.

Value Damping

Ce(0) U1 [F/L]

Ce(1) U2 [F/L]

Ce(2) U3 [F/L]

Ce(3) R1 [FL]

Ce(4) R2 [FL]

Ce(5) R3 [FL]

The term ce(n) applies only when DOF(n) = True and Fixed(n) = False.

K

Type: SystemDouble

This is an array of initial stiffness terms for the link property. The initial stiffness applies for nonlinear analyses.

Value Stiffness

K(0) U1 [F/L]

K(1) U2 [F/L]

K(2) U3 [F/L]

K(3) R1 [FL]

K(4) R2 [FL]

K(5) R3 [FL]

The term k(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Dis

Type: SystemDouble

This is an array of initial hook opening terms for the link property. The initial hook opening applies for nonlinear analyses.

Value Dis

dis(0) U1 [L]

dis(1) U2 [L]

dis(2) U3 [L]

dis(3) R1 [rad]

dis(4) R2 [rad]

dis(5) R3 [rad]

The term dis(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

DJ2

Type: SystemDouble

The distance from the J-End of the link to the U2 shear spring. This item applies only when DOF(1) = True. [L]

DJ3

Type: SystemDouble

The distance from the J-End of the link to the U3 shear spring. This item applies only when DOF(2) = True. [L]

Notes (Optional)

Type: SystemString

The notes, if any, assigned to the property.

GUID (Optional)

Type: SystemString

The GUID (global unique identifier), if any, assigned to the property. If this item is input as Default, the program assigns a GUID to the property.

Return ValueType: Int32

Returns zero if the property is successfully initialized; otherwise it returns a nonzero value

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyDOF() As Boolean
    Dim MyFixed() As Boolean
    Dim MyNonLinear() As Boolean
    Dim MyKe() As Double
    Dim MyCe() As Double
    Dim MyK() As Double
    Dim MyDis() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link property
    ReDim MyDOF(5)
    ReDim MyFixed(5)
    ReDim MyNonLinear(5)
    ReDim MyKe(5)
    ReDim MyCe(5)
    ReDim MyK(5)
    ReDim MyDis(5)

    MyDOF(0) = True
    MyKe(0) = 12
    MyCe(0) = 0.01

    MyDOF(1) = True
    MyNonLinear(1) = True
    MyKe(1) = 12
    MyCe(1) = 0.01
    MyK(1) = 20
    MyDis(1) = 1.2

    MyDOF(2) = True
    MyFixed(2) = True
```

Introduction

```
ret = SapModel.PropLink.SetGap("G1", MyDOF, MyFixed, MyNonLinear, MyKe, MyCe, MyK, MyDis,  
'close ETABS  
  EtabsObject.ApplicationExit(False)  
  
'clean up variables  
  SapModel = Nothing  
  EtabsObject = Nothing  
End Sub
```

📌 See Also

Reference

[cPropLink Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LSTCD825F9_0") Method

Initializes a linear-type link property. If this function is called for an existing link property, all items for the property are reset to their default value.

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetLinear(
    string Name,
    ref bool[] DOF,
    ref bool[] Fixed,
    ref double[] Ke,
    ref double[] Ce,
    double DJ2,
    double DJ3,
    bool KeCoupled = false,
    bool CeCoupled = false,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetLinear (
    Name As String,
    ByRef DOF As Boolean(),
    ByRef Fixed As Boolean(),
    ByRef Ke As Double(),
    ByRef Ce As Double(),
    DJ2 As Double,
    DJ3 As Double,
    KeCoupled As Boolean = false,
    CeCoupled As Boolean = false,
    Notes As String = "",
    GUID As String = ""
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim DOF As Boolean()
Dim Fixed As Boolean()
Dim Ke As Double()
Dim Ce As Double()
Dim DJ2 As Double
Dim DJ3 As Double
Dim KeCoupled As Boolean
Dim CeCoupled As Boolean
Dim Notes As String
Dim GUID As String
```

cPropLinkspan id="LSTCD825F9_0"AddLanguageSpecificTextSet("LSTCD825F9_0?cpp=::|nu338SetLinea


```
Dim returnValue As Integer
```

```
returnValue = instance.SetLinear(Name,  
DOF, Fixed, Ke, Ce, DJ2, DJ3, KeCoupled,  
CeCoupled, Notes, GUID)
```

```
int SetLinear(  
String^ Name,  
array<bool>^% DOF,  
array<bool>^% Fixed,  
array<double>^% Ke,  
array<double>^% Ce,  
double DJ2,  
double DJ3,  
bool KeCoupled = false,  
bool CeCoupled = false,  
String^ Notes = L"",  
String^ GUID = L""  
)
```

```
abstract SetLinear :  
    Name : string *  
    DOF : bool[] byref *  
    Fixed : bool[] byref *  
    Ke : float[] byref *  
    Ce : float[] byref *  
    DJ2 : float *  
    DJ3 : float *  
    ?KeCoupled : bool *  
    ?CeCoupled : bool *  
    ?Notes : string *  
    ?GUID : string  
(* Defaults:  
    let _KeCoupled = defaultArg KeCoupled false  
    let _CeCoupled = defaultArg CeCoupled false  
    let _Notes = defaultArg Notes ""  
    let _GUID = defaultArg GUID ""  
)  
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing or new link property. If this is an existing property, that property is modified; otherwise, a new property is added.

DOF

Type:Â SystemBoolean

This is a boolean array, dimensioned to 5, indicating if properties exist for a specified degree of freedom.

Value DOF

DOF(0) U1

DOF(1) U2

DOF(2) U3

DOF(3) R1

DOF(4) R2

DOF(5) R3

Fixed

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if the specified degree of freedom is fixed (restrained).

Value	Fixity
--------------	---------------

Fixed(0) U1 fixity if DOF(0) = True

Fixed(1) U2 fixity if DOF(1) = True

Fixed(2) U3 fixity if DOF(2) = True

Fixed(3) R1 fixity if DOF(3) = True

Fixed(4) R2 fixity if DOF(4) = True

Fixed(5) R3 fixity if DOF(5) = True

The term Fixed(n) applies only when DOF(n) = True

Ke

Type: SystemDouble

This is an array of stiffness terms for the link property. There are 6 terms in the array if the stiffness is uncoupled and 21 if it is coupled. The KeCoupled item indicates if the stiffness is coupled.

If the stiffness is uncoupled:

Value	Stiffness
--------------	------------------

Ke(0) U1 [F/L]

Ke(1) U2 [F/L]

Ke(2) U3 [F/L]

Ke(3) R1 [FL]

Ke(4) R2 [FL]

Ke(5) R3 [FL]

If the stiffness is coupled:

Value	Stiffness
--------------	------------------

Ke(0) U1U1 [F/L]

Ke(1) U1U2 [F/L]

Ke(2) U2U2 [F/L]

Ke(3) U1U3 [F/L]

Ke(4) U2U3 [F/L]

Ke(5) U3U3 [F/L]

Ke(6) U1R1 [F]

Ke(7) U2R1 [F]

Ke(8) U3R1 [F]

Ke(9) R1R1 [FL]

Ke(10) U1R2 [F]

Ke(11) U2R2 [F]

Ke(12) U3R2 [F]

Ke(14) R2R2 [FL]
 Ke(15) U1R3 [F]
 Ke(16) U2R3 [F]
 Ke(17) U3R3 [F]
 Ke(18) R1R3 [FL]
 Ke(19) R2R3 [FL]
 Ke(20) R3R3 [FL]

Ce

Type: SystemDouble

This is an array of damping terms for the link property. There are 6 terms in the array if the damping is uncoupled and 21 if it is coupled. The CeCoupled item indicates if the damping is coupled.

If the damping is uncoupled:

Value Damping

Ce(0) U1 [F/L]
 Ce(1) U2 [F/L]
 Ce(2) U3 [F/L]
 Ce(3) R1 [FL]
 Ce(4) R2 [FL]
 Ce(5) R3 [FL]

If the damping is coupled:

Value Damping

Ce(0) U1U1 [F/L]
 Ce(1) U1U2 [F/L]
 Ce(2) U2U2 [F/L]
 Ce(3) U1U3 [F/L]
 Ce(4) U2U3 [F/L]
 Ce(5) U3U3 [F/L]
 Ce(6) U1R1 [F]
 Ce(7) U2R1 [F]
 Ce(8) U3R1 [F]
 Ce(9) R1R1 [FL]
 Ce(10) U1R2 [F]
 Ce(11) U2R2 [F]
 Ce(12) U3R2 [F]
 Ce(14) R2R2 [FL]
 Ce(15) U1R3 [F]
 Ce(16) U2R3 [F]
 Ce(17) U3R3 [F]
 Ce(18) R1R3 [FL]

Ce(19) R2R3 [FL]

Ce(20) R3R3 [FL]

DJ2

Type:Â SystemDouble

The distance from the J-End of the link to the U2 shear spring. This item applies only when DOF(1) = True. [L]

DJ3

Type:Â SystemDouble

The distance from the J-End of the link to the U3 shear spring. This item applies only when DOF(2) = True. [L]

KeCoupled (Optional)

Type:Â SystemBoolean

This item is True if the link stiffness, Ke, is coupled. There are 21 terms in the Ke array if Ke is coupled; otherwise there are 6 terms

CeCoupled (Optional)

Type:Â SystemBoolean

This item is True if the link damping, Ce, is coupled. There are 21 terms in the Ce array if Ce is coupled; otherwise there are 6 terms

Notes (Optional)

Type:Â SystemString

The notes, if any, assigned to the property.

GUID (Optional)

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the property. If this item is input as Default, the program assigns a GUID to the property.

Return Value

Type:Â Int32

returns zero if the property is successfully initialized; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim DOF() As Boolean
    Dim Fixed() As Boolean
    Dim Ke() As Double
    Dim Ce() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel
```

Introduction

```
'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'add link property
    ReDim D0F(5)
    ReDim Fixed(5)
    ReDim Ke(5)
    ReDim Ce(5)
    D0F(0) = True
    Ke(0) = 12
    ret = SapModel.PropLink.SetLinear("L1", D0F, Fixed, Ke, Ce, 0, 0)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPropLink Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LST5AE59294_0") Method

Initializes a multilinear elastic-type link property. If this function is called for an existing link property, all items for the property are reset to their default values.

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetMultiLinearElastic(
    string Name,
    ref bool[] DOF,
    ref bool[] Fixed,
    ref bool[] Nonlinear,
    ref double[] Ke,
    ref double[] Ce,
    double DJ2,
    double DJ3,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetMultiLinearElastic (
    Name As String,
    ByRef DOF As Boolean(),
    ByRef Fixed As Boolean(),
    ByRef Nonlinear As Boolean(),
    ByRef Ke As Double(),
    ByRef Ce As Double(),
    DJ2 As Double,
    DJ3 As Double,
    Notes As String = "",
    GUID As String = ""
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim DOF As Boolean()
Dim Fixed As Boolean()
Dim Nonlinear As Boolean()
Dim Ke As Double()
Dim Ce As Double()
Dim DJ2 As Double
Dim DJ3 As Double
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.SetMultiLinearElastic(Name,
```

cPropLinkspan id="LST5AE59294_0"AddLanguageSpecificTextSet("LST5AE59294_0?cpp=::|n338;SetMul

DOF, Fixed, Nonlinear, Ke, Ce, DJ2,
DJ3, Notes, GUID)

```
int SetMultiLinearElastic(
String^ Name,
array<bool>^% DOF,
array<bool>^% Fixed,
array<bool>^% Nonlinear,
array<double>^% Ke,
array<double>^% Ce,
double DJ2,
double DJ3,
String^ Notes = L"",
String^ GUID = L""
)

abstract SetMultiLinearElastic :
    Name : string *
    DOF : bool[] byref *
    Fixed : bool[] byref *
    Nonlinear : bool[] byref *
    Ke : float[] byref *
    Ce : float[] byref *
    DJ2 : float *
    DJ3 : float *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing or new link property. If this is an existing property, that property is modified; otherwise, a new property is added

DOF

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if properties exist for a specified degree of freedom.

Value DOF

DOF(0) U1

DOF(1) U2

DOF(2) U3

DOF(3) R1

DOF(4) R2

DOF(5) R3

Fixed

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if the specified degree of freedom is fixed (restrained).

Value	Fixity
--------------	---------------

Fixed(0) U1 fixity if DOF(0) = True

Fixed(1) U2 fixity if DOF(1) = True

Fixed(2) U3 fixity if DOF(2) = True

Fixed(3) R1 fixity if DOF(3) = True

Fixed(4) R2 fixity if DOF(4) = True

Fixed(5) R3 fixity if DOF(5) = True

The term Fixed(n) applies only when DOF(n) = True.

NonlinearType: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if nonlinear properties exist for a specified degree of freedom.

Value	Nonlinear
--------------	------------------

Nonlinear(0) U1 has nonlinear properties

Nonlinear(1) U2 has nonlinear properties

Nonlinear(2) U3 has nonlinear properties

Nonlinear(3) R1 has nonlinear properties

Nonlinear(4) R2 has nonlinear properties

Nonlinear(5) R3 has nonlinear properties

The term Nonlinear(n) applies only when DOF(n) = True and Fixed(n) = False.

KeType: SystemDouble

This is an array of effective stiffness terms for the link property. The effective stiffness applies for linear analyses.

Value Stiffness

Ke(0) U1 [F/L]

Ke(1) U2 [F/L]

Ke(2) U3 [F/L]

Ke(3) R1 [FL]

Ke(4) R2 [FL]

Ke(5) R3 [FL]

The term ke(n) applies only when DOF(n) = True and Fixed(n) = False.

CeType: SystemDouble

This is an array of effective damping terms for the link property. The effective stiffness applies for linear analyses.

Value Damping

Ce(0) U1 [F/L]

Ce(1) U2 [F/L]

Ce(2) U3 [F/L]

Ce(3) R1 [FL]

Ce(4) R2 [FL]

Ce(5) R3 [FL]

The term ce(n) applies only when DOF(n) = True and Fixed(n) = False.

DJ2

Type: SystemDouble

The distance from the J-End of the link to the U2 shear spring. This item applies only when DOF(1) = True. [L]

DJ3

Type: SystemDouble

The distance from the J-End of the link to the U3 shear spring. This item applies only when DOF(2) = True. [L]

Notes (Optional)

Type: SystemString

The notes, if any, assigned to the property.

GUID (Optional)

Type: SystemString

The GUID (global unique identifier), if any, assigned to the property. If this item is input as Default, the program assigns a GUID to the property.

Return Value

Type: Int32

Returns zero if the property is successfully initialized; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyDOF() As Boolean
    Dim MyFixed() As Boolean
    Dim MyNonLinear() As Boolean
    Dim MyKe() As Double
    Dim MyCe() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link property
    ReDim MyDOF(5)
    ReDim MyFixed(5)
    ReDim MyNonLinear(5)
    ReDim MyKe(5)
    ReDim MyCe(5)
```

Introduction

```
MyDOF(0) = True
MyKe(0) = 12
MyCe(0) = 0.01

MyDOF(1) = True
MyNonLinear(1) = True
MyKe(1) = 12
MyCe(1) = 0.01

MyDOF(2) = True
MyFixed(2) = True

ret = SapModel.PropLink.SetMultiLinearElastic("MLE1", MyDOF, MyFixed, MyNonLinear, MyKe, M

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPropLink Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LST362B53BF_0") Method

Initializes a multilinear plastic-type link property. If this function is called for an existing link property, all items for the property are reset to their default values.

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetMultiLinearPlastic(
    string Name,
    ref bool[] DOF,
    ref bool[] Fixed,
    ref bool[] Nonlinear,
    ref double[] Ke,
    ref double[] Ce,
    double DJ2,
    double DJ3,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetMultiLinearPlastic (
    Name As String,
    ByRef DOF As Boolean(),
    ByRef Fixed As Boolean(),
    ByRef Nonlinear As Boolean(),
    ByRef Ke As Double(),
    ByRef Ce As Double(),
    DJ2 As Double,
    DJ3 As Double,
    Notes As String = "",
    GUID As String = ""
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim DOF As Boolean()
Dim Fixed As Boolean()
Dim Nonlinear As Boolean()
Dim Ke As Double()
Dim Ce As Double()
Dim DJ2 As Double
Dim DJ3 As Double
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.SetMultiLinearPlastic(Name,
```

cPropLinkspan id="LST362B53BF_0"AddLanguageSpecificTextSet("LST362B53BF_0?cpp=::|rob43;SetMu

DOF, Fixed, Nonlinear, Ke, Ce, DJ2,
DJ3, Notes, GUID)

```
int SetMultiLinearPlastic(
String^ Name,
array<bool>^% DOF,
array<bool>^% Fixed,
array<bool>^% Nonlinear,
array<double>^% Ke,
array<double>^% Ce,
double DJ2,
double DJ3,
String^ Notes = L"",
String^ GUID = L""
)

abstract SetMultiLinearPlastic :
    Name : string *
    DOF : bool[] byref *
    Fixed : bool[] byref *
    Nonlinear : bool[] byref *
    Ke : float[] byref *
    Ce : float[] byref *
    DJ2 : float *
    DJ3 : float *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing or new link property. If this is an existing property, that property is modified; otherwise, a new property is added

DOF

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if properties exist for a specified degree of freedom.

Value DOF

DOF(0) U1

DOF(1) U2

DOF(2) U3

DOF(3) R1

DOF(4) R2

DOF(5) R3

Fixed

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if the specified degree of freedom is fixed (restrained).

Value	Fixity
--------------	---------------

Fixed(0) U1 fixity if DOF(0) = True

Fixed(1) U2 fixity if DOF(1) = True

Fixed(2) U3 fixity if DOF(2) = True

Fixed(3) R1 fixity if DOF(3) = True

Fixed(4) R2 fixity if DOF(4) = True

Fixed(5) R3 fixity if DOF(5) = True

The term Fixed(n) applies only when DOF(n) = True.

NonlinearType: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if nonlinear properties exist for a specified degree of freedom.

Value	Nonlinear
--------------	------------------

Nonlinear(0) U1 has nonlinear properties

Nonlinear(1) U2 has nonlinear properties

Nonlinear(2) U3 has nonlinear properties

Nonlinear(3) R1 has nonlinear properties

Nonlinear(4) R2 has nonlinear properties

Nonlinear(5) R3 has nonlinear properties

The term Nonlinear(n) applies only when DOF(n) = True and Fixed(n) = False.

KeType: SystemDouble

This is an array of effective stiffness terms for the link property. The effective stiffness applies for linear analyses.

Value Stiffness

Ke(0) U1 [F/L]

Ke(1) U2 [F/L]

Ke(2) U3 [F/L]

Ke(3) R1 [FL]

Ke(4) R2 [FL]

Ke(5) R3 [FL]

The term ke(n) applies only when DOF(n) = True and Fixed(n) = False.

CeType: SystemDouble

This is an array of effective damping terms for the link property. The effective stiffness applies for linear analyses.

Value Damping

Ce(0) U1 [F/L]

Ce(1) U2 [F/L]

Ce(2) U3 [F/L]

Ce(3) R1 [FL]

Ce(4) R2 [FL]

Ce(5) R3 [FL]

The term ce(n) applies only when DOF(n) = True and Fixed(n) = False.

DJ2

Type:Â SystemDouble

The distance from the J-End of the link to the U2 shear spring. This item applies only when DOF(1) = True. [L]

DJ3

Type:Â SystemDouble

The distance from the J-End of the link to the U3 shear spring. This item applies only when DOF(2) = True. [L]

Notes (Optional)

Type:Â SystemString

The notes, if any, assigned to the property.

GUID (Optional)

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the property. If this item is input as Default, the program assigns a GUID to the property.

Return Value

Type:Â Int32

Returns zero if the property is successfully initialized; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyDOF() As Boolean
    Dim MyFixed() As Boolean
    Dim MyNonLinear() As Boolean
    Dim MyKe() As Double
    Dim MyCe() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link property
    ReDim MyDOF(5)
    ReDim MyFixed(5)
    ReDim MyNonLinear(5)
    ReDim MyKe(5)
    ReDim MyCe(5)
```

Introduction

```
MyDOF(0) = True
MyKe(0) = 12
MyCe(0) = 0.01

MyDOF(1) = True
MyNonLinear(1) = True
MyKe(1) = 12
MyCe(1) = 0.01

MyDOF(2) = True
MyFixed(2) = True

ret = SapModel.PropLink.SetMultiLinearPlastic("MLP1", MyDOF, MyFixed, MyNonLinear, MyKe, M

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPropLink Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LST945C8452_0?cpp=::|n8348;SetMul

Method

Sets the force-deformation data for a specified degree of freedom in multilinear elastic and multilinear plastic link properties.

Namespace: Â ETABSV1

Assembly: Â ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetMultiLinearPoints(
    string Name,
    int DOF,
    int NumberPoints,
    ref double[] F,
    ref double[] D,
    int MyType = 1,
    double A1 = 0,
    double A2 = 0,
    double B1 = 0,
    double B2 = 0,
    double Eta = 0
)
```

```
Function SetMultiLinearPoints (
    Name As String,
    DOF As Integer,
    NumberPoints As Integer,
    ByRef F As Double(),
    ByRef D As Double(),
    MyType As Integer = 1,
    A1 As Double = 0,
    A2 As Double = 0,
    B1 As Double = 0,
    B2 As Double = 0,
    Eta As Double = 0
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim DOF As Integer
Dim NumberPoints As Integer
Dim F As Double()
Dim D As Double()
Dim MyType As Integer
Dim A1 As Double
Dim A2 As Double
Dim B1 As Double
Dim B2 As Double
Dim Eta As Double
```

cPropLinkspan id="LST945C8452_0"AddLanguageSpecificTextSet("LST945C8452_0?cpp=::|n8348;SetMul


```

Dim returnValue As Integer

returnValue = instance.SetMultiLinearPoints(Name,
DOF, NumberPoints, F, D, MyType, A1,
A2, B1, B2, Eta)

int SetMultiLinearPoints(
String^ Name,
int DOF,
int NumberPoints,
array<double>^% F,
array<double>^% D,
int MyType = 1,
double A1 = 0,
double A2 = 0,
double B1 = 0,
double B2 = 0,
double Eta = 0
)

abstract SetMultiLinearPoints :
    Name : string *
    DOF : int *
    NumberPoints : int *
    F : float[] byref *
    D : float[] byref *
    ?MyType : int *
    ?A1 : float *
    ?A2 : float *
    ?B1 : float *
    ?B2 : float *
    ?Eta : float
(* Defaults:
    let _MyType = defaultArg MyType 1
    let _A1 = defaultArg A1 0
    let _A2 = defaultArg A2 0
    let _B1 = defaultArg B1 0
    let _B2 = defaultArg B2 0
    let _Eta = defaultArg Eta 0
*)
-> int

```

Parameters

Name

Type: SystemString

The name of an existing multilinear elastic or multilinear plastic link property.

DOF

Type: SystemInt32

This is 1, 2, 3, 4, 5 or 6, indicating the degree of freedom to which the multilinear points apply.

Value DOF

1	U1
2	U2
3	U3
4	R1
5	R2

6 R3

NumberPointsType: SystemInt32

The number of force-deformation points for the specified degree of freedom.

FType: SystemDouble

This is an array, dimensioned to NumberPoints - 1, that includes the force at each point. When DOF is U1, U2 or U3, this is a force. When DOF is R1, R2 or R3, this is a moment. [F] if DOF is less than or equal to 3, and [FL] if DOF is greater than 3.

DType: SystemDouble

This is an array, dimensioned to NumberPoints - 1, that includes the displacement at each point. When DOF is U1, U2 or U3, this is a translation. When DOF is R1, R2 or R3, this is a rotation. [L] if DOF is less than or equal to 3, and [rad] if DOF is greater than 3.

MyType (Optional)Type: SystemInt32

This item applies only to multilinear plastic link properties. It is 1, 2 or 3, indicating the hysteresis type.

MyType Hysteresis Type

- 1 Kinematic
- 2 Takeda
- 3 Pivot

A1 (Optional)Type: SystemDouble

This item applies only to multilinear plastic link properties that have a pivot hysteresis type (MyType = 3). It is the Alpha1 hysteresis parameter.

A2 (Optional)Type: SystemDouble

This item applies only to multilinear plastic link properties that have a pivot hysteresis type (MyType = 3). It is the Alpha2 hysteresis parameter.

B1 (Optional)Type: SystemDouble

This item applies only to multilinear plastic link properties that have a pivot hysteresis type (MyType = 3). It is the Beta1 hysteresis parameter.

B2 (Optional)Type: SystemDouble

This item applies only to multilinear plastic link properties that have a pivot hysteresis type (MyType = 3). It is the Beta2 hysteresis parameter.

Eta (Optional)Type: SystemDouble

This item applies only to multilinear plastic link properties that have a pivot hysteresis type (MyType = 3). It is the Eta hysteresis parameter.

Return ValueType: Int32

Returns zero if the data is successfully assigned; otherwise it returns a nonzero value.

Remarks

Return Value

3350

To successfully apply this data to the indicated link property, the following conditions must be met:

1. The link property must be multilinear elastic or multilinear plastic.
2. The specified DOF must be active.
3. The specified DOF must not be fixed.
4. The specified DOF must be nonlinear.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyDOF() As Boolean
    Dim MyFixed() As Boolean
    Dim MyNonLinear() As Boolean
    Dim MyKe() As Double
    Dim MyCe() As Double
    Dim MyF() As Double
    Dim MyD() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link property
    ReDim MyDOF(5)
    ReDim MyFixed(5)
    ReDim MyNonLinear(5)
    ReDim MyKe(5)
    ReDim MyCe(5)

    MyDOF(0) = True
    MyKe(0) = 12
    MyCe(0) = 0.01

    MyDOF(1) = True
    MyNonLinear(1) = True
    MyKe(1) = 12
    MyCe(1) = 0.01

    MyDOF(2) = True
    MyFixed(2) = True

    ret = SapModel.PropLink.SetMultiLinearPlastic("MLP1", MyDOF, MyFixed, MyNonLinear, MyKe, MyCe)
```

Introduction

```
'set multilinear force-defomation data
  ReDim MyF(4)
  ReDim MyD(4)

  MyF(0) = -12
  MyF(1) = -10
  MyF(2) = 0
  MyF(3) = 8
  MyF(4) = 9

  MyD(0) = -8
  MyD(1) = -0.6
  MyD(2) = 0
  MyD(3) = 0.2
  MyD(4) = 6

  ret = SapModel.PropLink.SetMultiLinearPoints("MLP1", 2, 5, MyF, MyD, 3, 9, 12, 0.75, 0.8,

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

[cPropLink Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LST9F8101300") Method

Assigns P-delta parameters to a link property.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetPDelta(
    string Name,
    ref double[] Value
)
```

```
Function SetPDelta (
    Name As String,
    ByRef Value As Double()
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim Value As Double()
Dim returnValue As Integer
```

```
returnValue = instance.SetPDelta(Name,
    Value)
```

```
int SetPDelta(
    String^ Name,
    array<double>^% Value
)
```

```
abstract SetPDelta :
    Name : string *
    Value : float[] byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing link property.

Value

Type: SystemDouble

This is an array of P-delta parameters

Value

PDelta Parameter

Value(0) M2 P-delta to I-end of link as moment, M2I

Value(1) M2 P-delta to J-end of link as moment, M2J

Value(2) M3 P-delta to I-end of link as moment, M3I

Value(3) M3 P-delta to J-end of link as moment, M3J

Return Value

Type: Int32

Returns zero if the values are successfully assigned; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim DOF() As Boolean
    Dim Fixed() As Boolean
    Dim Ke() As Double
    Dim Ce() As Double
    Dim MyValue() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link property
    ReDim DOF(5)
    ReDim Fixed(5)
    ReDim Ke(5)
    ReDim Ce(5)
    DOF(0) = True
    Ke(0) = 12
    ret = SapModel.PropLink.SetLinear("L1", DOF, Fixed, Ke, Ce, 0, 0)

    'set link property P-delta parameters
    ReDim MyValue(3)
    MyValue(0) = 0.6
    MyValue(1) = 0.4
    MyValue(2) = 0.3
    MyValue(3) = 0.2
    ret = SapModel.PropLink.SetPDelta("L1", MyValue)

    'close ETABS
    EtabsObject.ApplicationExit(False)
```

```
'clean up variables  
  SapModel = Nothing  
  EtabsObject = Nothing  
End Sub
```

See Also

Reference

cPropLink Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LSTCE28CF05_0") Method

Initializes a plastic Wen-type link property. If this function is called for an existing link property, all items for the property are reset to their default values.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetPlasticWen(
    string Name,
    ref bool[] DOF,
    ref bool[] Fixed,
    ref bool[] Nonlinear,
    ref double[] Ke,
    ref double[] Ce,
    ref double[] K,
    ref double[] Yield,
    ref double[] Ratio,
    ref double[] Exp,
    double DJ2,
    double DJ3,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetPlasticWen (
    Name As String,
    ByRef DOF As Boolean(),
    ByRef Fixed As Boolean(),
    ByRef Nonlinear As Boolean(),
    ByRef Ke As Double(),
    ByRef Ce As Double(),
    ByRef K As Double(),
    ByRef Yield As Double(),
    ByRef Ratio As Double(),
    ByRef Exp As Double(),
    DJ2 As Double,
    DJ3 As Double,
    Notes As String = "",
    GUID As String = ""
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim DOF As Boolean()
Dim Fixed As Boolean()
Dim Nonlinear As Boolean()
Dim Ke As Double()
```

cPropLinkspan id="LSTCE28CF05_0"AddLanguageSpecificTextSet("LSTCE28CF05_0?cpp=::0056");SetPl


```

Dim Ce As Double()
Dim K As Double()
Dim Yield As Double()
Dim Ratio As Double()
Dim Exp As Double()
Dim DJ2 As Double
Dim DJ3 As Double
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer

returnValue = instance.SetPlasticWen(Name,
DOF, Fixed, Nonlinear, Ke, Ce, K, Yield,
Ratio, Exp, DJ2, DJ3, Notes, GUID)

```

```

int SetPlasticWen(
String^ Name,
array<bool>^% DOF,
array<bool>^% Fixed,
array<bool>^% Nonlinear,
array<double>^% Ke,
array<double>^% Ce,
array<double>^% K,
array<double>^% Yield,
array<double>^% Ratio,
array<double>^% Exp,
double DJ2,
double DJ3,
String^ Notes = L"",
String^ GUID = L""
)

```

```

abstract SetPlasticWen :
    Name : string *
    DOF : bool[] byref *
    Fixed : bool[] byref *
    Nonlinear : bool[] byref *
    Ke : float[] byref *
    Ce : float[] byref *
    K : float[] byref *
    Yield : float[] byref *
    Ratio : float[] byref *
    Exp : float[] byref *
    DJ2 : float *
    DJ3 : float *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int

```

Parameters

Name

Type: SystemString

The name of an existing or new link property. If this is an existing property, that property is modified; otherwise, a new property is added.

DOF

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if properties exist for a specified degree of freedom.

Value DOF

DOF(0) U1

DOF(1) U2

DOF(2) U3

DOF(3) R1

DOF(4) R2

DOF(5) R3

Fixed

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if the specified degree of freedom is fixed (restrained).

Value Fixity

Fixed(0) U1 fixity if DOF(0) = True

Fixed(1) U2 fixity if DOF(1) = True

Fixed(2) U3 fixity if DOF(2) = True

Fixed(3) R1 fixity if DOF(3) = True

Fixed(4) R2 fixity if DOF(4) = True

Fixed(5) R3 fixity if DOF(5) = True

Nonlinear

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if nonlinear properties exist for a specified degree of freedom.

Value Nonlinear

Nonlinear(0) U1 has nonlinear properties

Nonlinear(1) U2 has nonlinear properties

Nonlinear(2) U3 has nonlinear properties

Nonlinear(3) R1 has nonlinear properties

Nonlinear(4) R2 has nonlinear properties

Nonlinear(5) R3 has nonlinear properties

The term Nonlinear(n) applies only when DOF(n) = True and Fixed(n) = False.

Ke

Type: SystemDouble

This is an array of effective stiffness terms for the link property. The effective stiffness applies for linear analyses.

Value Stiffness

Ke(0) U1 [F/L]

Ke(1) U2 [F/L]

Ke(2) U3 [F/L]

Ke(3) R1 [FL]

Ke(4) R2 [FL]

Ke(5) R3 [FL]

The term ke(n) applies only when DOF(n) = True and Fixed(n) = False.

Ce

Type: SystemDouble

This is an array of effective damping terms for the link property. The effective stiffness applies for linear analyses.

Value Damping

Ce(0) U1 [F/L]

Ce(1) U2 [F/L]

Ce(2) U3 [F/L]

Ce(3) R1 [FL]

Ce(4) R2 [FL]

Ce(5) R3 [FL]

The term ce(n) applies only when DOF(n) = True and Fixed(n) = False.

K

Type: SystemDouble

This is an array of initial stiffness terms for the link property. The initial stiffness applies for nonlinear analyses.

Value Stiffness

K(0) U1 [F/L]

K(1) U2 [F/L]

K(2) U3 [F/L]

K(3) R1 [F]

K(4) R2 [F]

K(5) R3 [F]

The term k(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Yield

Type: SystemDouble

This is an array of yield force terms for the link property. The yield force applies for nonlinear analyses.

Value Yield

yield(0) U1 [F]

yield(1) U2 [F]

yield(2) U3 [F]

yield(3) R1 [F/L]

yield(4) R2 [F/L]

yield(5) R3 [F/L]

The term Yield(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Ratio

Type: SystemDouble

This is an array of post-yield stiffness ratio terms for the link property. The post-yield stiffness ratio applies for nonlinear analyses. It is the post-yield stiffness divided by the initial stiffness.

Value Ratio

Ratio(0) U1

Ratio(1) U2

Ratio(2) U3

Ratio(3) R1

Ratio(4) R2

Ratio(5) R3

The term Ratio(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Exp

Type: SystemDouble

This is an array of yield exponent terms for the link property. The yield exponent applies for nonlinear analyses. The yielding exponent that controls the sharpness of the transition from the initial stiffness to the yielded stiffness.

Value Exp

Exp(0) U1

Exp(1) U2

Exp(2) U3

Exp(3) R1

Exp(4) R2

Exp(5) R3

The term Exp(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

DJ2

Type: SystemDouble

The distance from the J-End of the link to the U2 shear spring. This item applies only when DOF(1) = True. [L]

DJ3

Type: SystemDouble

The distance from the J-End of the link to the U3 shear spring. This item applies only when DOF(2) = True. [L]

Notes (Optional)

Type: SystemString

The notes, if any, assigned to the property.

GUID (Optional)

Type: SystemString

The GUID (global unique identifier), if any, assigned to the property. If this item is input as Default, the program assigns a GUID to the property.

Return Value

Type: Int32

Returns zero if the property is successfully initialized; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1
```

Introduction

```
Dim MyDOF() As Boolean
Dim MyFixed() As Boolean
Dim MyNonLinear() as Boolean
Dim MyKe() As Double
Dim MyCe() As Double
Dim MyK() As Double
Dim MyYield() As Double
Dim MyRatio() As Double
Dim MyExp() As Double

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'add link property
ReDim MyDOF(5)
ReDim MyFixed(5)
ReDim MyNonLinear(5)
ReDim MyKe(5)
ReDim MyCe(5)
ReDim MyK(5)
ReDim MyYield(5)
ReDim MyRatio(5)
ReDim MyExp(5)

MyDOF(0) = True
MyKe(0) = 12
MyCe(0) = 0.01

MyDOF(1) = True
MyNonLinear(1) = True
MyKe(1) = 12
MyCe(1) = 0.01
MyK(1) = 20
MyYield(1)= 50
MyRatio(1)= 0.1
MyExp(1)= 3

MyDOF(2) = True
MyFixed(2) = True

ret = SapModel.PropLink.SetPlasticWen("PW1", MyDOF, MyFixed, MyNonLinear, MyKe, MyCe, MyK, MyYield, MyRatio, MyExp)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPropLink Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LST1F9BA53_0") Method

Initializes a rubber isolator-type link property. If this function is called for an existing link property, all items for the property are reset to their default value.

Namespace: $\hat{A}$ ETABSv1

Assembly: $\hat{A}$ ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetRubberIsolator(
    string Name,
    ref bool[] DOF,
    ref bool[] Fixed,
    ref bool[] Nonlinear,
    ref double[] Ke,
    ref double[] Ce,
    ref double[] K,
    ref double[] Yield,
    ref double[] Ratio,
    double DJ2,
    double DJ3,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetRubberIsolator (
    Name As String,
    ByRef DOF As Boolean(),
    ByRef Fixed As Boolean(),
    ByRef Nonlinear As Boolean(),
    ByRef Ke As Double(),
    ByRef Ce As Double(),
    ByRef K As Double(),
    ByRef Yield As Double(),
    ByRef Ratio As Double(),
    DJ2 As Double,
    DJ3 As Double,
    Notes As String = "",
    GUID As String = ""
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim DOF As Boolean()
Dim Fixed As Boolean()
Dim Nonlinear As Boolean()
Dim Ke As Double()
Dim Ce As Double()
Dim K As Double()
```

cPropLinkspan id="LST1F9BA53_0"AddLanguageSpecificTextSet("LST1F9BA53_0?cpp=::|nu=3363SetRubber

Introduction

```
Dim Yield As Double()
Dim Ratio As Double()
Dim DJ2 As Double
Dim DJ3 As Double
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer

returnValue = instance.SetRubberIsolator(Name,
DOF, Fixed, Nonlinear, Ke, Ce, K, Yield,
Ratio, DJ2, DJ3, Notes, GUID)

int SetRubberIsolator(
String^ Name,
array<bool>^% DOF,
array<bool>^% Fixed,
array<bool>^% Nonlinear,
array<double>^% Ke,
array<double>^% Ce,
array<double>^% K,
array<double>^% Yield,
array<double>^% Ratio,
double DJ2,
double DJ3,
String^ Notes = L"",
String^ GUID = L""
)

abstract SetRubberIsolator :
    Name : string *
    DOF : bool[] byref *
    Fixed : bool[] byref *
    Nonlinear : bool[] byref *
    Ke : float[] byref *
    Ce : float[] byref *
    K : float[] byref *
    Yield : float[] byref *
    Ratio : float[] byref *
    DJ2 : float *
    DJ3 : float *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing or new link property. If this is an existing property, that property is modified; otherwise, a new property is added.

DOF

Type:Â SystemBoolean

This is a boolean array, dimensioned to 5, indicating if properties exist for a specified degree of freedom.

Value DOF

DOF(0) U1
 DOF(1) U2
 DOF(2) U3
 DOF(3) R1
 DOF(4) R2
 DOF(5) R3

Fixed

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if the specified degree of freedom is fixed (restrained).

Value	Fixity
--------------	---------------

Fixed(0) U1	fixity if DOF(0) = True
Fixed(1) U2	fixity if DOF(1) = True
Fixed(2) U3	fixity if DOF(2) = True
Fixed(3) R1	fixity if DOF(3) = True
Fixed(4) R2	fixity if DOF(4) = True
Fixed(5) R3	fixity if DOF(5) = True

Nonlinear

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if nonlinear properties exist for a specified degree of freedom.

Value	Nonlinear
--------------	------------------

Nonlinear(0) U1	has nonlinear properties
Nonlinear(1) U2	has nonlinear properties
Nonlinear(2) U3	has nonlinear properties
Nonlinear(3) R1	not used
Nonlinear(4) R2	not used
Nonlinear(5) R3	not used

Note that this item is applicable only for degrees of freedom U1, U2 and U3. For those degrees of freedom, the term Nonlinear(n) applies only when DOF(n) = True and Fixed(n) = False.

Ke

Type: SystemDouble

This is an array of effective stiffness terms for the link property. The effective stiffness applies for linear analyses.

Value Stiffness

Ke(0) U1	[F/L]
Ke(1) U2	[F/L]
Ke(2) U3	[F/L]
Ke(3) R1	[FL]
Ke(4) R2	[FL]
Ke(5) R3	[FL]

The term ke(n) applies only when DOF(n) = True and Fixed(n) = False.

Ce

Type: SystemDouble

This is an array of effective damping terms for the link property. The effective stiffness applies for linear analyses.

Value Damping

Ce(0) U1 [F/L]

Ce(1) U2 [F/L]

Ce(2) U3 [F/L]

Ce(3) R1 [FL]

Ce(4) R2 [FL]

Ce(5) R3 [FL]

The term ce(n) applies only when DOF(n) = True and Fixed(n) = False.

K

Type: SystemDouble

This is an array of initial stiffness terms for the link property. The initial stiffness applies for nonlinear analyses.

Value Stiffness

K(0) U1, not used

K(1) U2 [F/L]

K(2) U3 [F/L]

K(3) R1, not used

K(4) R2, not used

K(5) R3, not used

Note that this item is applicable only for degrees of freedom U2 and U3. For those degrees of freedom, the term k(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Yield

Type: SystemDouble

This is an array of yield force terms for the link property. The yield force applies for nonlinear analyses.

Value Yield

yield(0) U1, not used

yield(1) U2 [F]

yield(2) U3 [F]

yield(3) R1, not used

yield(4) R2, not used

yield(5) R3, not used

Note that this item is applicable only for degrees of freedom U2 and U3. For those degrees of freedom, the term Yield(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Ratio

Type: SystemDouble

This is an array of post-yield stiffness ratio terms for the link property. The post-yield stiffness ratio applies for nonlinear analyses. It is the post-yield stiffness divided by the initial stiffness.

Value Ratio

Ratio(0) U1, not used

Ratio(1) U2

Ratio(2) U3

Ratio(3) R1, not used

Ratio(4) R2, not used

Ratio(5) R3, not used

Note that this item is applicable only for degrees of freedom U2 and U3. For those degrees of freedom, the term Ratio(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

DJ2

Type:Â SystemDouble

The distance from the J-End of the link to the U2 shear spring. This item applies only when DOF(1) = True. [L]

DJ3

Type:Â SystemDouble

The distance from the J-End of the link to the U3 shear spring. This item applies only when DOF(2) = True. [L]

Notes (Optional)

Type:Â SystemString

The notes, if any, assigned to the property.

GUID (Optional)

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the property. If this item is input as Default, the program assigns a GUID to the property.

Return Value

Type:Â Int32

Returns zero if the property is successfully initialized; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyDOF() As Boolean
    Dim MyFixed() As Boolean
    Dim MyNonLinear() As Boolean
    Dim MyKe() As Double
    Dim MyCe() As Double
    Dim MyK() As Double
    Dim MyYield() As Double
    Dim MyRatio() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
```

Introduction

```
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'add link property
ReDim MyDOF(5)
ReDim MyFixed(5)
ReDim MyNonLinear(5)
ReDim MyKe(5)
ReDim MyCe(5)
ReDim MyK(5)
ReDim MyYield(5)
ReDim MyRatio(5)

MyDOF(0) = True
MyKe(0) = 12
MyCe(0) = 0.01

MyDOF(1) = True
MyNonLinear(1) = True
MyKe(1) = 12
MyCe(1) = 0.01
MyK(1) = 20
MyYield(1)= 50
MyRatio(1)= 0.1

MyDOF(2) = True
MyNonLinear(2) = True
MyKe(2) = 15
MyCe(2) = 0.008
MyK(2) = 22
MyYield(2)= 60
MyRatio(2)= 0.15

MyDOF(3) = True
MyFixed(3) = True

ret = SapModel.PropLink.SetRubberIsolator("RI1", MyDOF, MyFixed, MyNonLinear, MyKe, MyCe,

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

See Also

Reference

cPropLink Interface

ETABSv1 Namespace

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Introduction

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LSTBA583B1") Method

Assigns length and area values to a link property that are used if the link property is specified in line and area spring assignments.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetSpringData(
    string Name,
    double DefinedForThisLength,
    double DefinedForThisArea
)
```

```
Function SetSpringData (
    Name As String,
    DefinedForThisLength As Double,
    DefinedForThisArea As Double
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim DefinedForThisLength As Double
Dim DefinedForThisArea As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetSpringData(Name,
    DefinedForThisLength, DefinedForThisArea)
```

```
int SetSpringData(
    String^ Name,
    double DefinedForThisLength,
    double DefinedForThisArea
)
```

```
abstract SetSpringData :
    Name : string *
    DefinedForThisLength : float *
    DefinedForThisArea : float -> int
```

Parameters

Name

Type: SystemString

The name of an existing link property.

DefinedForThisLength

cPropLinkspan id="LSTBA583B1_0"AddLanguageSpecificTextSet("LSTBA583B1_0?cpp=::|nu=3376SetSpring

Type: SystemDouble

The link property is defined for this length in a line (frame) spring. [L]
DefinedForThisArea

Type: SystemDouble

The link property is defined for this area in an area spring. [L2]

Return Value

Type: Int32

Returns zero if the values are successfully assigned; otherwise it returns a nonzero value.

▀ Remarks

▀ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim DOF() As Boolean
    Dim Fixed() As Boolean
    Dim Ke() As Double
    Dim Ce() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link property
    ReDim DOF(5)
    ReDim Fixed(5)
    ReDim Ke(5)
    ReDim Ce(5)
    DOF(0) = True
    Ke(0) = 12
    ret = SapModel.PropLink.SetLinear("L1", DOF, Fixed, Ke, Ce, 0, 0)

    'set link property spring data
    ret = SapModel.PropLink.SetSpringData("L1", 12, 2)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPropLink Interface

ETABSv1 Namespace

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»

CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LST75D5A857_0")

Method

Initializes a T/C friction isolator-type link property. If this function is called for an existing link property, all items for the property are reset to their default value.

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetTCFrictionIsolator(
    string Name,
    ref bool[] DOF,
    ref bool[] Fixed,
    ref bool[] Nonlinear,
    ref double[] Ke,
    ref double[] Ce,
    ref double[] K,
    ref double[] Slow,
    ref double[] Fast,
    ref double[] Rate,
    ref double[] Radius,
    ref double[] SlowT,
    ref double[] FastT,
    ref double[] RateT,
    double Kt,
    double Dis,
    double Dist,
    double Damping,
    double DJ2,
    double DJ3,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetTCFrictionIsolator (
    Name As String,
    ByRef DOF As Boolean(),
    ByRef Fixed As Boolean(),
    ByRef Nonlinear As Boolean(),
    ByRef Ke As Double(),
    ByRef Ce As Double(),
    ByRef K As Double(),
    ByRef Slow As Double(),
    ByRef Fast As Double(),
    ByRef Rate As Double(),
    ByRef Radius As Double(),
    ByRef SlowT As Double(),
    ByRef FastT As Double(),
    ByRef RateT As Double(),
```

Introduction

```
Kt As Double,  
Dis As Double,  
Dist As Double,  
Damping As Double,  
DJ2 As Double,  
DJ3 As Double,  
    Notes As String = "",  
    GUID As String = ""  
) As Integer
```

```
Dim instance As cPropLink  
Dim Name As String  
Dim DOF As Boolean()  
Dim Fixed As Boolean()  
Dim Nonlinear As Boolean()  
Dim Ke As Double()  
Dim Ce As Double()  
Dim K As Double()  
Dim Slow As Double()  
Dim Fast As Double()  
Dim Rate As Double()  
Dim Radius As Double()  
Dim SlowT As Double()  
Dim FastT As Double()  
Dim RateT As Double()  
Dim Kt As Double  
Dim Dis As Double  
Dim Dist As Double  
Dim Damping As Double  
Dim DJ2 As Double  
Dim DJ3 As Double  
Dim Notes As String  
Dim GUID As String  
Dim returnValue As Integer
```

```
returnValue = instance.SetTCFrictionIsolator(Name,  
DOF, Fixed, Nonlinear, Ke, Ce, K, Slow,  
Fast, Rate, Radius, SlowT, FastT, RateT,  
Kt, Dis, Dist, Damping, DJ2, DJ3, Notes,  
GUID)
```

```
int SetTCFrictionIsolator(  
String^ Name,  
array<bool>^% DOF,  
array<bool>^% Fixed,  
array<bool>^% Nonlinear,  
array<double>^% Ke,  
array<double>^% Ce,  
array<double>^% K,  
array<double>^% Slow,  
array<double>^% Fast,  
array<double>^% Rate,  
array<double>^% Radius,  
array<double>^% SlowT,  
array<double>^% FastT,  
array<double>^% RateT,  
double Kt,  
double Dis,  
double Dist,  
double Damping,  
double DJ2,  
double DJ3,
```

cPropLinkAddLanguageSpecificTextSet("LST75D5A857_0?cpp=::|robust");SetTC

```

String^ Notes = L"",
String^ GUID = L""
)

abstract SetTCFrictionIsolator :
    Name : string *
    DOF : bool[] byref *
    Fixed : bool[] byref *
    Nonlinear : bool[] byref *
    Ke : float[] byref *
    Ce : float[] byref *
    K : float[] byref *
    Slow : float[] byref *
    Fast : float[] byref *
    Rate : float[] byref *
    Radius : float[] byref *
    SlowT : float[] byref *
    FastT : float[] byref *
    RateT : float[] byref *
    Kt : float *
    Dis : float *
    Dist : float *
    Damping : float *
    DJ2 : float *
    DJ3 : float *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of an existing or new link property. If this is an existing property, that property is modified; otherwise, a new property is added.

DOF

Type:Â SystemBoolean

This is a boolean array, dimensioned to 5, indicating if properties exist for a specified degree of freedom.

Value DOF

DOF(0) U1

DOF(1) U2

DOF(2) U3

DOF(3) R1

DOF(4) R2

DOF(5) R3

Fixed

Type:Â SystemBoolean

This is a boolean array, dimensioned to 5, indicating if the specified degree of freedom is fixed (restrained).

Value

Fixity

Fixed(0) U1 fixity if DOF(0) = True
 Fixed(1) U2 fixity if DOF(1) = True
 Fixed(2) U3 fixity if DOF(2) = True
 Fixed(3) R1 fixity if DOF(3) = True
 Fixed(4) R2 fixity if DOF(4) = True
 Fixed(5) R3 fixity if DOF(5) = True

Nonlinear

Type: SystemBoolean

This is a boolean array, dimensioned to 5, indicating if nonlinear properties exist for a specified degree of freedom.

Value	Nonlinear
--------------	------------------

Nonlinear(0) U1 has nonlinear properties
 Nonlinear(1) U2 has nonlinear properties
 Nonlinear(2) U3 has nonlinear properties
 Nonlinear(3) R1, not used
 Nonlinear(4) R2, not used
 Nonlinear(5) R3, not used

Note that this item is applicable only for degrees of freedom U1, U2 and U3. For those degrees of freedom, the term Nonlinear(n) applies only when DOF(n) = True and Fixed(n) = False.

Ke

Type: SystemDouble

This is an array of effective stiffness terms for the link property. The effective stiffness applies for linear analyses.

Value Stiffness

Ke(0) U1 [F/L]
 Ke(1) U2 [F/L]
 Ke(2) U3 [F/L]
 Ke(3) R1 [FL]
 Ke(4) R2 [FL]
 Ke(5) R3 [FL]

The term ke(n) applies only when DOF(n) = True and Fixed(n) = False.

Ce

Type: SystemDouble

This is an array of effective damping terms for the link property. The effective stiffness applies for linear analyses.

Value Damping

Ce(0) U1 [F/L]
 Ce(1) U2 [F/L]
 Ce(2) U3 [F/L]
 Ce(3) R1 [FL]
 Ce(4) R2 [FL]
 Ce(5) R3 [FL]

The term ce(n) applies only when DOF(n) = True and Fixed(n) = False.

K

Type: SystemDouble

This is an array of initial stiffness terms for the link property. The initial stiffness applies for nonlinear analyses.

Value Stiffness

K(0) U1 [F/L]

K(1) U2 [F/L]

K(2) U3 [F/L]

K(3) R1, not used

K(4) R2, not used

K(5) R3, not used

Note that this item is applicable only for degrees of freedom U1, U2 and U3. For those degrees of freedom, the term k(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Slow

Type: SystemDouble

This is an array of the friction coefficient at zero velocity terms when U1 is in compression for the link property. This coefficient applies for nonlinear analyses.

Value Slow

Slow(0) U1, not used

Slow(1) U2

Slow(2) U3

Slow(3) R1, not used

Slow(4) R2, not used

Slow(5) R3, not used

Note that this item is applicable only for degrees of freedom U2 and U3. For those degrees of freedom, the term Slow(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Fast

Type: SystemDouble

This is an array of the friction coefficient at fast velocity terms when U1 is in compression for the link property. This coefficient applies for nonlinear analyses.

Value Fast

Fast(0) U1, not used

Fast(1) U2

Fast(2) U3

Fast(3) R1, not used

Fast(4) R2, not used

Fast(5) R3, not used

Note that this item is applicable only for degrees of freedom U2 and U3. For those degrees of freedom, the term Fast(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Rate

Type: SystemDouble

This is an array of the inverse of the characteristic sliding velocity terms when

U1 is in compression for the link property. This item applies for nonlinear analyses.

Value Rate

Rate(0) U1, not used

Rate(1) U2 [s/L]

Rate(2) U3 [s/L]

Rate(3) R1, not used

Rate(4) R2, not used

Rate(5) R3, not used

Note that this item is applicable only for degrees of freedom U2 and U3. For those degrees of freedom, the term Fast(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Radius

Type: SystemDouble

This is an array of the radius of the sliding contact surface terms for the link property. Inputting 0 means there is an infinite radius, that is, the slider is flat. This item applies for nonlinear analyses.

Value Radius

Radius(0) U1, not used

Radius(1) U2 [L]

Radius(2) U3 [L]

Radius(3) R1, not used

Radius(4) R2, not used

Radius(5) R3, not used

Note that this item is applicable only for degrees of freedom U2 and U3. For those degrees of freedom, the term Fast(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

SlowT

Type: SystemDouble

This is an array of the friction coefficient at zero velocity terms when U1 is in tension for the link property. This coefficient applies for nonlinear analyses.

Value Slow

SlowT(0) U1, not used

SlowT(1) U2

SlowT(2) U3

SlowT(3) R1, not used

SlowT(4) R2, not used

SlowT(5) R3, not used

Note that this item is applicable only for degrees of freedom U2 and U3. For those degrees of freedom, the term SlowT(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

FastT

Type: SystemDouble

This is an array of the friction coefficient at fast velocity terms when U1 is in tension for the link property. This coefficient applies for nonlinear analyses.

Value Fast

FastT(0) U1, not used

FastT(1) U2

FastT(2) U3

FastT(3) R1, not used

FastT(4) R2, not used

FastT(5) R3, not used

Note that this item is applicable only for degrees of freedom U2 and U3. For those degrees of freedom, the term FastT(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

RateT

Type: SystemDouble

This is an array of the inverse of the characteristic sliding velocity terms when U1 is in tension for the link property. This item applies for nonlinear analyses.

Value Rate

RateT(0) U1, not used

RateT(1) U2 [s/L]

RateT(2) U3 [s/L]

RateT(3) R1, not used

RateT(4) R2, not used

RateT(5) R3, not used

Note that this item is applicable only for degrees of freedom U2 and U3. For those degrees of freedom, the term RateT(n) applies only when DOF(n) = True, Fixed(n) = False, and Nonlinear(n) = True.

Kt

Type: SystemDouble

The axial translational tension stiffness for the U1 degree of freedom. This item applies for nonlinear analyses. [F/L]

Dis

Type: SystemDouble

The U1 degree of freedom gap opening for compression. This item applies for nonlinear analyses. [L]

Dist

Type: SystemDouble

The U1 degree of freedom gap opening for tension. This item applies for nonlinear analyses. [L]

Damping

Type: SystemDouble

This is the nonlinear damping coefficient used for the axial translational degree of freedom, U1. This item applies for nonlinear analyses. [F/L]

DJ2

Type: SystemDouble

The distance from the J-End of the link to the U2 shear spring. This item applies only when DOF(1) = True. [L]

DJ3

Type: SystemDouble

The distance from the J-End of the link to the U3 shear spring. This item applies only when DOF(2) = True. [L]

Notes (Optional)

Type:Â SystemString

The notes, if any, assigned to the property.

GUID (Optional)

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the property. If this item is input as Default, the program assigns a GUID to the property.

Return Value

Type:Â Int32

Returns zero if the property is successfully initialized; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyDOF() As Boolean
    Dim MyFixed() As Boolean
    Dim MyNonLinear() As Boolean
    Dim MyKe() As Double
    Dim MyCe() As Double
    Dim MyK() As Double
    Dim MySlow() As Double
    Dim MyFast() As Double
    Dim MyRate() As Double
    Dim MyRadius() As Double
    Dim MySlowT() As Double
    Dim MyFastT() As Double
    Dim MyRateT() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add link property
    ReDim MyDOF(5)
    ReDim MyFixed(5)
    ReDim MyNonLinear(5)
    ReDim MyKe(5)
    ReDim MyCe(5)
    ReDim MyK(5)
    ReDim MySlow(5)
    ReDim MyFast(5)
```


Introduction

```
ReDim MyRate(5)
ReDim MyRadius(5)
ReDim MySlowT(5)
ReDim MyFastT(5)
ReDim MyRateT(5)

MyDOF(0) = True
MyNonLinear(0) = True
MyKe(0) = 12
MyCe(0) = 0.01
MyK(0) = 1000

MyDOF(1) = True
MyNonLinear(1) = True
MyKe(1) = 12
MyCe(1) = 0.01
MyK(1) = 20
MySlow(1)= 0.6
MyFast(1)= 0.5
MyRate(1)= 10
MyRadius(1)= 80
MySlowT(1)= 0.61
MyFastT(1)= 0.51
MyRateT(1)= 10.1

MyDOF(2) = True
MyNonLinear(2) = True
MyKe(2) = 14
MyCe(2) = 0.008
MyK(2) = 22
MySlow(2)= 0.66
MyFast(2)= 0.55
MyRate(2)= 12
MyRadius(2)= 75
MySlowT(2)= 0.67
MyFastT(2)= 0.56
MyRateT(2)= 12.1

MyDOF(3) = True
MyKe(3) = 15
MyCe(3) = 0

MyDOF(4) = True
MyFixed(4) = True

ret = SapModel.PropLink.SetTCFrictionIsolator("TCFI1", MyDOF, MyFixed, MyNonLinear, MyKe,
'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

🔗 See Also

Reference

[cPropLink Interface](#)
[ETABSv1 Namespace](#)

Introduction

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CSI API ETABS v1

cPropLinkAddLanguageSpecificTextSet("LST3AA3A17B_0?cpp=::|8083");SetW

Method

Retrieves weight and mass data for a link property.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetWeightAndMass(
    string Name,
    double W,
    double M,
    double R1,
    double R2,
    double R3
)
```

```
Function SetWeightAndMass (
    Name As String,
    W As Double,
    M As Double,
    R1 As Double,
    R2 As Double,
    R3 As Double
) As Integer
```

```
Dim instance As cPropLink
Dim Name As String
Dim W As Double
Dim M As Double
Dim R1 As Double
Dim R2 As Double
Dim R3 As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetWeightAndMass(Name,
    W, M, R1, R2, R3)
```

```
int SetWeightAndMass(
    String^ Name,
    double W,
    double M,
    double R1,
    double R2,
    double R3
)
```

```
abstract SetWeightAndMass :
    Name : string *
```

cPropLinkspan id="LST3AA3A17B_0"AddLanguageSpecificTextSet("LST3AA3A17B_0?cpp=::|8083");SetW

```

W : float *
M : float *
R1 : float *
R2 : float *
R3 : float -> int

```

Parameters

Name

Type:Â SystemString
The name of an existing link property.

W

Type:Â SystemDouble
The weight of the link. [F]

M

Type:Â SystemDouble
The translational mass of the link. [M]

R1

Type:Â SystemDouble
The rotational inertia of the link about its local 1 axis. [ML2]

R2

Type:Â SystemDouble
The rotational inertia of the link about its local 2 axis. [ML2]

R3

Type:Â SystemDouble
The rotational inertia of the link about its local 3 axis. [ML2]

Return Value

Type:Â Int32

returns zero if the data is successfully retrieved; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim DOF() As Boolean
    Dim Fixed() As Boolean
    Dim Ke() As Double
    Dim Ce() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

```

Introduction

```
'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'add link property
    ReDim DOF(5)
    ReDim Fixed(5)
    ReDim Ke(5)
    ReDim Ce(5)
    DOF(0) = True
    Ke(0) = 12
    ret = SapModel.PropLink.SetLinear("L1", DOF, Fixed, Ke, Ce, 0, 0)

'set link property weight and mass
    ret = SapModel.PropLink.SetWeightAndMass("L1", 10, 0.26, 0.0012, 0.0014, 0.0016)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPropLink Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropMaterial Interface

Namespace: $\hat{A}$ ETABSv1

Assembly: $\hat{A}$ ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
public interface cPropMaterial
```

```
Public Interface cPropMaterial
```

```
Dim instance As cPropMaterial
```

```
public interface class cPropMaterial
```

```
type cPropMaterial = interface end
```

The cPropMaterial type exposes the following members.

▲ Properties

$\hat{A}$ Name	Description
----------------	-------------

 <u>TimeDep</u>	API calls for time dependent material properties.
--	---

Top

▲ Methods

$\hat{A}$ Name	Description
----------------	-------------

 <u>AddMaterial</u>	Adds a new material property to the model.
 <u>ChangeName</u>	Changes the name of an existing material property.
 <u>Count</u>	Gets the total number of defined material properties in the model. If desired, counts can be returned for all material properties of a specified type in the model.
 <u>Delete</u>	Deletes a specified material property.
 <u>GetDamping</u>	Retrieves the additional material damping data for the material.
 <u>GetMassSource</u>	DEPRECATED Retrieves the mass source for the model.
 <u>GetMassSource_1</u>	Retrieves the mass source for the model.
 <u>GetMaterial</u>	Retrieves some basic material property data.
 <u>GetMPAnisotropic</u>	Retrieves the mechanical properties for a material with an anisotropic directional symmetry type.
 <u>GetMPIsotropic</u>	Retrieves the mechanical properties for a material with an isotropic directional symmetry type.
 <u>GetMPOrthotropic</u>	Retrieves the mechanical properties for a material with an orthotropic directional symmetry type.

Introduction

⇒ <u>GetMPUniaxial</u>	Retrieves the mechanical properties for a material with an uniaxial directional symmetry type.
⇒ <u>GetNameList</u>	Retrieves the names of all defined material properties of the specified type.
⇒ <u>GetOConcrete</u>	(NEWER FUNCTION AVAILABLE) Retrieves the other material property data for concrete materials.
⇒ <u>GetOConcrete_1</u>	Retrieves the other material property data for concrete materials.
⇒ <u>GetONoDesign</u>	Retrieves the other material property data for no design type materials.
⇒ <u>GetORebar</u>	(NEWER FUNCTION AVAILABLE) Retrieves the other material property data for rebar materials.
⇒ <u>GetORebar_1</u>	Retrieves the other material property data for rebar materials.
⇒ <u>GetOSteel</u>	(NEWER FUNCTION AVAILABLE) Retrieves the other material property data for steel materials.
⇒ <u>GetOSteel_1</u>	Retrieves the other material property data for steel materials.
⇒ <u>GetOTendon</u>	(NEWER FUNCTION AVAILABLE) Retrieves the other material property data for tendon materials.
⇒ <u>GetOTendon_1</u>	Retrieves the other material property data for tendon materials.
⇒ <u>GetSSCurve</u>	Retrieves the material stress-strain curve.
⇒ <u>GetTemp</u>	Retrieves the temperatures at which properties are specified for a material.
⇒ <u>GetTypeOAPI</u>	
⇒ <u>GetWeightAndMass</u>	Retrieves the weight per unit volume and mass per unit volume of the material.
⇒ <u>SetDamping</u>	Sets the additional material damping data for the material.
⇒ <u>SetMassSource</u>	DEPRECATED Sets the mass source for the model.
⇒ <u>SetMassSource_1</u>	Sets the mass source for the model.
⇒ <u>SetMaterial</u>	DEPRECATED. Initializes a material property. If this function is called for an existing material property, all items for the material are reset to their default value.
⇒ <u>SetMPAnisotropic</u>	Sets the material directional symmetry type to anisotropic, and assigns the anisotropic mechanical properties.
⇒ <u>SetMPIsotropic</u>	Sets the material directional symmetry type to isotropic, and assigns the isotropic mechanical properties.
⇒ <u>SetMPOrthotropic</u>	Sets the material directional symmetry type to orthotropic, and assigns the orthotropic mechanical properties.
⇒ <u>SetMPUniaxial</u>	Sets the material directional symmetry type to uniaxial, and assigns the uniaxial mechanical properties.
⇒ <u>SetOConcrete</u>	(NEWER FUNCTION AVAILABLE) Sets the other material property data for concrete materials.
⇒ <u>SetOConcrete_1</u>	Sets the other material property data for concrete materials.
⇒ <u>SetONoDesign</u>	

Introduction

Sets the other material property data for no design type materials.

- ⇒ [SetORebar](#) (NEWER FUNCTION AVAILABLE) Sets the other material property data for rebar materials.
- ⇒ [SetORebar_1](#) Sets the other material property data for rebar materials.
- ⇒ [SetOSteel](#) (NEWER FUNCTION AVAILABLE) Sets the other material property data for steel materials.
- ⇒ [SetOSteel_1](#) Sets the other material property data for steel materials.
- ⇒ [SetOTendon](#) (NEWER FUNCTION AVAILABLE) Sets the other material property data for tendon materials.
- ⇒ [SetOTendon_1](#) Sets the other material property data for tendon materials.
- ⇒ [SetSSCurve](#) Sets the material stress-strain curve for materials that are specified to have user-defined stress-strain curves.
- ⇒ [SetTemp](#) assigns the temperatures at which properties are specified for a material. This data is required only for materials whose properties are temperature dependent.
- ⇒ [SetWeightAndMass](#) Assigns weight per unit volume or mass per unit volume to a material property.

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropMaterial Properties

The cPropMaterial type exposes the following members.

Properties

Name	Description
------	-------------

 <u>TimeDep</u>	API calls for time dependent material properties.
--	---

Top

See Also

Reference

cPropMaterial Interface

ETABSV1 Namespace

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LSTEBF75596_0") Property

API calls for time dependent material properties.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
cPropMaterialTD TimeDep { get; }
```

```
ReadOnly Property TimeDep As cPropMaterialTD
Get
```

```
Dim instance As cPropMaterial
Dim value As cPropMaterialTD
```

```
value = instance.TimeDep
```

```
property cPropMaterialTD^ TimeDep {
cPropMaterialTD^ get ();
}
```

```
abstract TimeDep : cPropMaterialTD with get
```

Property Value

Type: cPropMaterialTD

■ See Also

Reference

cPropMaterial Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropMaterial Methods

The cPropMaterial type exposes the following members.

Methods

Â	Name	Description
⇒	<u>AddMaterial</u>	Adds a new material property to the model.
⇒	<u>ChangeName</u>	Changes the name of an existing material property.
⇒	<u>Count</u>	Gets the total number of defined material properties in the model. If desired, counts can be returned for all material properties of a specified type in the model.
⇒	<u>Delete</u>	Deletes a specified material property.
⇒	<u>GetDamping</u>	Retrieves the additional material damping data for the material.
⇒	<u>GetMassSource</u>	DEPRECATED Retrieves the mass source for the model.
⇒	<u>GetMassSource_1</u>	Retrieves the mass source for the model.
⇒	<u>GetMaterial</u>	Retrieves some basic material property data.
⇒	<u>GetMPAnisotropic</u>	Retrieves the mechanical properties for a material with an anisotropic directional symmetry type.
⇒	<u>GetMPIsotropic</u>	Retrieves the mechanical properties for a material with an isotropic directional symmetry type.
⇒	<u>GetMPOrthotropic</u>	Retrieves the mechanical properties for a material with an orthotropic directional symmetry type.
⇒	<u>GetMPUniaxial</u>	Retrieves the mechanical properties for a material with an uniaxial directional symmetry type.
⇒	<u>GetNameList</u>	Retrieves the names of all defined material properties of the specified type.
⇒	<u>GetOConcrete</u>	(NEWER FUNCTION AVAILABLE) Retrieves the other material property data for concrete materials.
⇒	<u>GetOConcrete_1</u>	Retrieves the other material property data for concrete materials.
⇒	<u>GetONoDesign</u>	Retrieves the other material property data for no design type materials.
⇒	<u>GetORebar</u>	(NEWER FUNCTION AVAILABLE) Retrieves the other material property data for rebar materials.
⇒	<u>GetORebar_1</u>	Retrieves the other material property data for rebar materials.
⇒	<u>GetOSteel</u>	(NEWER FUNCTION AVAILABLE) Retrieves the other material property data for steel materials.
⇒	<u>GetOSteel_1</u>	Retrieves the other material property data for steel materials.
⇒	<u>GetOTendon</u>	(NEWER FUNCTION AVAILABLE) Retrieves the other material property data for tendon materials.
⇒	<u>GetOTendon_1</u>	Retrieves the other material property data for tendon materials.

⇒ <u>GetSSCurve</u>	Retrieves the material stress-strain curve.
⇒ <u>GetTemp</u>	Retrieves the temperatures at which properties are specified for a material.
⇒ <u>GetTypeOAPI</u>	
⇒ <u>GetWeightAndMass</u>	Retrieves the weight per unit volume and mass per unit volume of the material.
⇒ <u>SetDamping</u>	Sets the additional material damping data for the material.
⇒ <u>SetMassSource</u>	DEPRECATED Sets the mass source for the model.
⇒ <u>SetMassSource_1</u>	Sets the mass source for the model.
⇒ <u>SetMaterial</u>	DEPRECATED. Initializes a material property. If this function is called for an existing material property, all items for the material are reset to their default value.
⇒ <u>SetMPAnisotropic</u>	Sets the material directional symmetry type to anisotropic, and assigns the anisotropic mechanical properties.
⇒ <u>SetMPIsotropic</u>	Sets the material directional symmetry type to isotropic, and assigns the isotropic mechanical properties.
⇒ <u>SetMPOrthotropic</u>	Sets the material directional symmetry type to orthotropic, and assigns the orthotropic mechanical properties.
⇒ <u>SetMPUniaxial</u>	Sets the material directional symmetry type to uniaxial, and assigns the uniaxial mechanical properties.
⇒ <u>SetOConcrete</u>	(NEWER FUNCTION AVAILABLE) Sets the other material property data for concrete materials.
⇒ <u>SetOConcrete_1</u>	Sets the other material property data for concrete materials.
⇒ <u>SetONoDesign</u>	Sets the other material property data for no design type materials.
⇒ <u>SetORebar</u>	(NEWER FUNCTION AVAILABLE) Sets the other material property data for rebar materials.
⇒ <u>SetORebar_1</u>	Sets the other material property data for rebar materials.
⇒ <u>SetOSteel</u>	(NEWER FUNCTION AVAILABLE) Sets the other material property data for steel materials.
⇒ <u>SetOSteel_1</u>	Sets the other material property data for steel materials.
⇒ <u>SetOTendon</u>	(NEWER FUNCTION AVAILABLE) Sets the other material property data for tendon materials.
⇒ <u>SetOTendon_1</u>	Sets the other material property data for tendon materials.
⇒ <u>SetSSCurve</u>	Sets the material stress-strain curve for materials that are specified to have user-defined stress-strain curves.
⇒ <u>SetTemp</u>	assigns the temperatures at which properties are specified for a material. This data is required only for materials whose properties are temperature dependent.
⇒ <u>SetWeightAndMass</u>	Assigns weight per unit volume or mass per unit volume to a material property.

Top

▲ See Also

Reference

cPropMaterial Interface

ETABSv1 Namespace

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Namespace: ETABSV1

▲ Syntax

C#

VBC++

F#

Copy

```
Function AddMaterial (
ByRef Name As String,
MatType As eMatType,
Region As String,
Standard As String,
Grade As String,
Optional Name As String = ""
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim MatType As eMatType
Dim Region As String
Dim Standard As String
Dim Grade As String
Dim UserName As String
Dim returnValue As Integer
```

```
returnValue = instance.AddMaterial(Name,
MatType, Region, Standard, Grade,
UserName)
```

```
int AddMaterial(
String^% Name,
eMatType MatType,
String^ Region,
String^ Standard,
String^ Grade,
String^ UserName = L""
)
```

cPropMaterialsAddLanguageSpecificTextSet("LST7113B64A_0?cpp=3394=.");Ad

```

    Name : string byref *
    MatType : eMatType *
    Region : string *
    Standard : string *
    Grade : string *
    ?UserName : string
(* Defaults:
    let _UserName = defaultArg UserName ""
*)
-> int

```

Parameters

Name

Type:Â SystemString

This item is returned by the program. It is the name that the program ultimately assigns for the material property. If no UserName is specified, the program assigns a default name to the material property. If a UserName is specified and that name is not used for another material property, the UserName is assigned to the material property.

MatType

Type:Â ETABSV1eMatType

This is one of the items in the eMatType enumeration.

Region

Type:Â SystemString

The region name of the material property that is user-predefined in the file "CSiMaterialLibrary*.xml" located in subfolder "Property Libraries" under the program installation.

Standard

Type:Â SystemString

The Standard name of the material property with the specified MatType within the specified region.

Grade

Type:Â SystemString

UserName (Optional)

Type:Â SystemString

Return Value

Type:Â Int32

Returns zero if the material property is successfully added, otherwise it returns a nonzero value.

Remarks

This function adds a new material property to the model based on the Code-specified and other pre-defined material properties defined in the installed file "CSiMaterialLibrary*.xml" located in subfolder "Property Libraries" under the program installation folder.

Examples

VB

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI

```

Introduction

```
Dim ret As Integer = -1

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'add ASTM A706 rebar material property in United states Region
ret = SapModel.PropMaterial.AddMaterial(Name, eMatType.Rebar, "United States", "ASTM A706")

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPropMaterial Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LST977B32A8_0?cpp3397=.");Ch

Method

Changes the name of an existing material property.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int ChangeName(
    string Name,
    string NewName
)
```

```
Function ChangeName (
    Name As String,
    NewName As String
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim NewName As String
Dim returnValue As Integer
```

```
returnValue = instance.ChangeName(Name,
    NewName)
```

```
int ChangeName(
    String^ Name,
    String^ NewName
)
```

```
abstract ChangeName :
    Name : string *
    NewName : string -> int
```

Parameters

Name

Type: SystemString

The existing name of a defined material property.

NewName

Type: SystemString

The new name for the material property.

Return ValueType: Int32

Returns zero if the new name is successfully applied; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'change name of material property
    ret = SapModel.PropMaterial.ChangeName("4000Psi", "MatConc")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cPropMaterial Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LST479E5573_0?cpp=3300=.");Count

Method

Gets the total number of defined material properties in the model. If desired, counts can be returned for all material properties of a specified type in the model.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int Count(
    eMatType MatType =
)
```

```
Function Count (
    Optional MatType As eMatType =
) As Integer
```

```
Dim instance As cPropMaterial
Dim MatType As eMatType
Dim returnValue As Integer
```

```
returnValue = instance.Count(MatType)
```

```
int Count(
    eMatType MatType =
)
```

```
abstract Count :
    ?MatType : eMatType
(* Defaults:
    let _MatType = defaultArg MatType
*)
-> int
```

Parameters

MatType (Optional)

Type: ETABSv1eMatType

This optional value is one of the items in the eMatType enumeration.

Return Value

Type: Int32

Returns the total number of defined material properties in the model. If desired, counts can be returned for all material properties of a specified type in the model.

Remarks

cPropMaterialsid="LST479E5573_0"AddLanguageSpecificTextSet("LST479E5573_0?cpp=3300=.");Count

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Count as Integer

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'return number of defined material properties of all types
    Count = SapModel.PropMaterial.Count

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

[cPropMaterial Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LST62E55B45_0?cpp3401u=.");Delete Method

Deletes a specified material property.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int Delete(
    string Name
)
```

```
Function Delete (
    Name As String
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.Delete(Name)
```

```
int Delete(
    String^ Name
)
```

```
abstract Delete :
    Name : string -> int
```

Parameters

Name

Type: SystemString

The name of an existing material property.

Return Value

Type: Int32

returns zero if the material property is successfully deleted; otherwise it returns a nonzero value. It returns an error if the specified material property can not be deleted, for example, if it is being used in a section property.

▲ Remarks

▲ Examples

VB

Copy

cPropMaterialsspan id="LST62E55B45_0"AddLanguageSpecificTextSet("LST62E55B45_0?cpp3401u=.");Delete

Introduction

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'add ASTM A706 rebar material  
    ret = SapModel.PropMaterial.AddQuick(Name, eMatType.Rebar, , , , eMatTypeRebar.ASTM_A706)  
  
    'delete material  
    ret = SapModel.PropMaterial.Delete(Name)  
  
    'close ETABS  
    EtabsObject.ApplicationExit(False)  
  
    'clean up variables  
    SapModel = Nothing  
    EtabsObject = Nothing  
End Sub
```

See Also

Reference

[cPropMaterial Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LST180D5C27_0?cpp3400=");GetDamping

Method

Retrieves the additional material damping data for the material.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetDamping(
    string Name,
    ref double ModalRatio,
    ref double ViscousMassCoeff,
    ref double ViscousStiffCoeff,
    ref double HystereticMassCoeff,
    ref double HystereticStiffCoeff,
    double Temp = 0
)
```

```
Function GetDamping (
    Name As String,
    ByRef ModalRatio As Double,
    ByRef ViscousMassCoeff As Double,
    ByRef ViscousStiffCoeff As Double,
    ByRef HystereticMassCoeff As Double,
    ByRef HystereticStiffCoeff As Double,
    Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim ModalRatio As Double
Dim ViscousMassCoeff As Double
Dim ViscousStiffCoeff As Double
Dim HystereticMassCoeff As Double
Dim HystereticStiffCoeff As Double
Dim Temp As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetDamping(Name,
    ModalRatio, ViscousMassCoeff, ViscousStiffCoeff,
    HystereticMassCoeff, HystereticStiffCoeff,
    Temp)
```

```
int GetDamping(
    String^ Name,
    double% ModalRatio,
    double% ViscousMassCoeff,
    double% ViscousStiffCoeff,
    double% HystereticMassCoeff,
```

cPropMaterialspan id="LST180D5C27_0"AddLanguageSpecificTextSet("LST180D5C27_0?cpp3400=");GetDamping

Introduction

```
double% HystereticStiffCoeff,  
double Temp = 0  
)  
  
abstract GetDamping :  
    Name : string *  
    ModalRatio : float byref *  
    ViscousMassCoeff : float byref *  
    ViscousStiffCoeff : float byref *  
    HystereticMassCoeff : float byref *  
    HystereticStiffCoeff : float byref *  
    ?Temp : float  
(* Defaults:  
    let _Temp = defaultArg Temp 0  
*)  
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing material property.

ModalRatio

Type:Â SystemDouble

The modal damping ratio.

ViscousMassCoeff

Type:Â SystemDouble

The mass coefficient for viscous proportional damping.

ViscousStiffCoeff

Type:Â SystemDouble

The stiffness coefficient for viscous proportional damping.

HystereticMassCoeff

Type:Â SystemDouble

The mass coefficient for hysteretic proportional damping.

HystereticStiffCoeff

Type:Â SystemDouble

The stiffness coefficient for hysteretic proportional damping.

Temp (Optional)

Type:Â SystemDouble

The temperature at which the specified data is to be retrieved. The temperature must have been previously defined for the material

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature

Return Value

Type:Â Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim ModalRatio As Double
    Dim ViscousMassCoeff As Double
    Dim ViscousStiffCoeff As Double
    Dim HystereticMassCoeff As Double
    Dim HystereticStiffCoeff As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'assign material damping data
    ret = SapModel.PropMaterial.SetDamping("4000Psi", 0.04, 0, 0, 0, 0)

    'get material damping data
    ret = SapModel.PropMaterial.GetDamping("4000Psi", ModalRatio, ViscousMassCoeff, ViscousStiffCoeff)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

ReferencecPropMaterial InterfaceETABSv1 Namespace

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»

CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LSTA86B Method

DEPRECATED Retrieves the mass source for the model.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetMassSource(
    ref int MyOption,
    ref int NumberLoads,
    ref string[] LoadPat,
    ref double[] SF
)
```

```
Function GetMassSource (
    ByRef MyOption As Integer,
    ByRef NumberLoads As Integer,
    ByRef LoadPat As String(),
    ByRef SF As Double()
) As Integer
```

```
Dim instance As cPropMaterial
Dim MyOption As Integer
Dim NumberLoads As Integer
Dim LoadPat As String()
Dim SF As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetMassSource(MyOption,
    NumberLoads, LoadPat, SF)
```

```
int GetMassSource(
    int% MyOption,
    int% NumberLoads,
    array<String^>^% LoadPat,
    array<double>^% SF
)
```

```
abstract GetMassSource :
    MyOption : int byref *
    NumberLoads : int byref *
    LoadPat : string[] byref *
    SF : float[] byref -> int
```

Parameters**MyOption**Type: SystemInt32

This is 1, 2 or 3, indicating the mass source option.

Value**Option**

- | | |
|---|--|
| 1 | From element self mass and additional masses |
| 2 | From loads |
| 3 | From element self mass and additional masses and loads |

NumberLoadsType: SystemInt32

The number of load patterns from which mass is obtained. This item applies only when MyOption is 2 or 3.

LoadPatType: SystemString

This is an array of the names of the load patterns from which mass is obtained. This item applies only when MyOption is 2 or 3.

SFType: SystemDouble

This is an array of load patterns multipliers used to calculate the mass. This item applies only when MyOption is 2 or 3

Return ValueType: Int32

Returns zero if the mass source is successfully retrieved; otherwise it returns a nonzero value.

▲ **Remarks**This function is DEPRECATED. Please use GetMassSource_1(Boolean, Boolean, Boolean, Int32, String, Double)▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyLoadPat() As String
    Dim MySF() As Double
    Dim MyOption As Integer
    Dim NumberLoads As Integer
    Dim LoadPat() As String
    Dim sf() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel
```

Introduction

```
'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set mass source
    ReDim MyLoadPat(1)
    ReDim MySF(1)
    MyLoadPat(0) = "DEAD"
    MyLoadPat(1) = "LIVE"
    MySF(0) = 1
    MySF(1) = 0.2
    ret = SapModel.PropMaterial.SetMassSource(3, 2, MyLoadPat, MySF)

'get mass source
    ret = SapModel.PropMaterial.GetMassSource(MyOption, NumberLoads, LoadPat, sf)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

[cPropMaterial Interface](#)

[ETABSv1 Namespace](#)

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»

CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LSTCCAC Method

Retrieves the mass source for the model.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetMassSource_1(
    ref bool IncludeElements,
    ref bool IncludeAddedMass,
    ref bool IncludeLoads,
    ref int NumberLoads,
    ref string[] LoadPat,
    ref double[] sf
)
```

```
Function GetMassSource_1 (
    ByRef IncludeElements As Boolean,
    ByRef IncludeAddedMass As Boolean,
    ByRef IncludeLoads As Boolean,
    ByRef NumberLoads As Integer,
    ByRef LoadPat As String(),
    ByRef sf As Double()
) As Integer
```

```
Dim instance As cPropMaterial
Dim IncludeElements As Boolean
Dim IncludeAddedMass As Boolean
Dim IncludeLoads As Boolean
Dim NumberLoads As Integer
Dim LoadPat As String()
Dim sf As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetMassSource_1(IncludeElements,
    IncludeAddedMass, IncludeLoads,
    NumberLoads, LoadPat, sf)
```

```
int GetMassSource_1(
    bool% IncludeElements,
    bool% IncludeAddedMass,
    bool% IncludeLoads,
    int% NumberLoads,
    array<String^>^% LoadPat,
    array<double>^% sf
)
```

```
abstract GetMassSource_1 :
```

cPropMaterialAddLanguageSpecificTextSet("LSTCCACA565_0?cp409nu=".");

```

IncludeElements : bool byref *
IncludeAddedMass : bool byref *
IncludeLoads : bool byref *
NumberLoads : int byref *
LoadPat : string[] byref *
sf : float[] byref -> int

```

Parameters

IncludeElements

Type:Â SystemBoolean
 Include element self mass

IncludeAddedMass

Type:Â SystemBoolean
 Include additional masses

IncludeLoads

Type:Â SystemBoolean
 Include specified loads

NumberLoads

Type:Â SystemInt32
 The number of load patterns from which mass is obtained. This item applies only when IncludeLoads is True.

LoadPat

Type:Â SystemString
 This is an array of the names of the load patterns from which mass is obtained. This item applies only when IncludeLoads is True.

sf

Type:Â SystemDouble
 This is an array of load patterns multipliers used to calculate the mass. This item applies only when IncludeLoads is True.

Return Value

Type:Â Int32

Returns zero if the mass source is successfully retrieved; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyLoadPat() As String
    Dim MySF() As Double
    Dim MyOption As Integer
    Dim NumberLoads As Integer
    Dim LoadPat() As String
    Dim sf() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

```

Introduction

```
'start ETABS application
    ret = EtabsObject.ApplicationStart()

'create SapModel object
    SapModel = EtabsObject.SapModel

'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set mass source
    ReDim MyLoadPat(1)
    ReDim MySF(1)
    MyLoadPat(0) = "DEAD"
    MyLoadPat(1) = "LIVE"
    MySF(0) = 1
    MySF(1) = 0.2
    ret = SapModel.PropMaterial.SetMassSource(3, 2, MyLoadPat, MySF)

'get mass source
    ret = SapModel.PropMaterial.GetMassSource(MyOption, NumberLoads, LoadPat, sf)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPropMaterial Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LSTD35A Method

Retrieves some basic material property data.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetMaterial(
    string Name,
    ref eMatType MatType,
    ref int Color,
    ref string Notes,
    ref string GUID
)
```

```
Function GetMaterial (
    Name As String,
    ByRef MatType As eMatType,
    ByRef Color As Integer,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim MatType As eMatType
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.GetMaterial(Name,
    MatType, Color, Notes, GUID)
```

```
int GetMaterial(
    String^ Name,
    eMatType% MatType,
    int% Color,
    String^% Notes,
    String^% GUID
)
```

```
abstract GetMaterial :
    Name : string *
    MatType : eMatType byref *
    Color : int byref *
    Notes : string byref *
    GUID : string byref -> int
```

cPropMaterialspan id="LSTD35A4C12_0"AddLanguageSpecificTextSet("LSTD35A4C12_0?cpp3412u=.");G

Parameters

Name

Type:Â SystemString

The name of an existing material property.

MatType

Type:Â ETABSv1eMatTypeThis is one of the items in the eMatType enumeration.

Color

Type:Â SystemInt32

Notes

Type:Â SystemString

GUID

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the material.

Return ValueType:Â Int32

Returns zero if the material is successfully retrieved; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MatType As eMatType
    Dim Color As Integer
    Dim Notes As String
    Dim GUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'initialize new material property
    ret = SapModel.PropMaterial.SetMaterial("Steel", eMatType.Steel, -1, "API example test", '

    'get basic material property data
    ret = SapModel.PropMaterial.GetMaterial("Steel", MatType, Color, Notes, GUID)

    'close ETABS
```

Introduction

```
EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

cPropMaterial Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LST699F1505_0?cpp=3415.")

Method

Retrieves the mechanical properties for a material with an anisotropic directional symmetry type.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetMPAnisotropic(
    string Name,
    ref double[] E,
    ref double[] U,
    ref double[] A,
    ref double[] G,
    double Temp = 0
)
```

```
Function GetMPAnisotropic (
    Name As String,
    ByRef E As Double(),
    ByRef U As Double(),
    ByRef A As Double(),
    ByRef G As Double(),
    Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim E As Double()
Dim U As Double()
Dim A As Double()
Dim G As Double()
Dim Temp As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetMPAnisotropic(Name,
    E, U, A, G, Temp)
```

```
int GetMPAnisotropic(
    String^ Name,
    array<double>^% E,
    array<double>^% U,
    array<double>^% A,
    array<double>^% G,
    double Temp = 0
)
```

abstract GetMPAnisotropic :

cPropMaterialsid="LST699F1505_0"AddLanguageSpecificTextSet("LST699F1505_0?cpp=3415.");Get

```

    Name : string *
    E : float[] byref *
    U : float[] byref *
    A : float[] byref *
    G : float[] byref *
    ?Temp : float
(* Defaults:
    let _Temp = defaultArg Temp 0
*)
-> int

```

Parameters

Name

Type: SystemString

The name of an existing material property.

E

Type: SystemDouble

This is an array that includes the modulus of elasticity.

Value Term

e(0) E1 [F/L2]

e(1) E2 [F/L2]

e(2) E3 [F/L2]

U

Type: SystemDouble

This is an array that includes poisson's ratio.

Value Term

u(0) U12

u(1) U13

u(2) U23

u(3) U14

u(4) U24

u(5) U34

u(6) U15

u(7) U25

u(8) U35

u(9) U45

u(10) U16

u(11) U26

u(12) U36

u(13) U46

u(14) U56

A

Type: SystemDouble

This is an array that includes the thermal coefficient.

Value Term

a(0) A1 [1/T]

a(1) A2 [1/T]

a(2) A3 [1/T]
 a(3) A12 [1/T]
 a(4) A13 [1/T]
 a(5) A23 [1/T]

G

Type: SystemDouble

This is an array that includes the shear modulus.

Value Term

g(0) G12 [F/L2]
 g(1) G13 [F/L2]
 g(2) G23 [F/L2]

Temp (Optional)

Type: SystemDouble

The temperature at which the specified data is to be retrieved. The temperature must have been previously defined for the material.

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return Value

Type: Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value. The function returns an error if the symmetry type of the specified material is not anisotropic.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyE() As Double
    Dim MyU() As Double
    Dim MyA() As Double
    Dim MyG() As Double
    Dim e() As Double
    Dim u() As Double
    Dim a() As Double
    Dim g() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel
```

Introduction

```
'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'initialize new material property
    ret = SapModel.PropMaterial.SetMaterial("Steel", eMatType.Steel)

'assign anisotropic mechanical properties
    ReDim MyE(2)
    ReDim MyU(14)
    ReDim MyA(6)
    ReDim MyG(2)
    MyE(0)=30000
    MyE(1)=10000
    MyE(2)=2000
    MyU(0)=0.2
    MyU(1)=0.05
    MyU(2)=0.1
    MyU(3)=0
    MyU(4)=0
    MyU(5)=0
    MyU(6)=0
    MyU(7)=0
    MyU(8)=0.01
    MyU(9)=0
    MyU(10)=0
    MyU(11)=0
    MyU(12)=0
    MyU(13)=0
    MyU(14)=0
    MyA(0)=6.5E-6
    MyA(1)=6.5E-6
    MyA(2)=6.5E-6
    MyA(3)=6.5E-6
    MyA(4)=6.5E-6
    MyA(5)=6.5E-6
    MyG(0)=1500
    MyG(1)=2500
    MyG(2)=8700
    ret = SapModel.PropMaterial.SetMPAnisotropic("Steel", MyE, MyU, MyA, MyG)

'get anisotropic mechanical properties
    ret = SapModel.PropMaterial.GetMPAnisotropic("Steel", e, u, a, g)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPropMaterial Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LST20773246_0") Method

Retrieves the mechanical properties for a material with an isotropic directional symmetry type.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetMPIsotropic(
    string Name,
    ref double E,
    ref double U,
    ref double A,
    ref double G,
    double Temp = 0
)
```

```
Function GetMPIsotropic (
    Name As String,
    ByRef E As Double,
    ByRef U As Double,
    ByRef A As Double,
    ByRef G As Double,
    Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim E As Double
Dim U As Double
Dim A As Double
Dim G As Double
Dim Temp As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetMPIsotropic(Name,
    E, U, A, G, Temp)
```

```
int GetMPIsotropic(
    String^ Name,
    double% E,
    double% U,
    double% A,
    double% G,
    double% Temp = 0
)
```

abstract GetMPIsotropic :

cPropMaterialspan id="LST20773246_0"AddLanguageSpecificTextSet("LST20773246_0?cpp=3420-");Get


```

    Name : string *
    E : float byref *
    U : float byref *
    A : float byref *
    G : float byref *
    ?Temp : float
(* Defaults:
    let _Temp = defaultArg Temp 0
*)
-> int

```

Parameters

Name

Type:Â SystemString
The name of an existing material property.

E

Type:Â SystemDouble
The modulus of elasticity. [F/L2]

U

Type:Â SystemDouble
Poissonâs ratio.

A

Type:Â SystemDouble
The thermal coefficient. [1/T]

G

Type:Â SystemDouble
The shear modulus. For isotropic materials this value is program calculated from the modulus of elasticity and poissonâs ratio. [F/L2]

Temp (Optional)

Type:Â SystemDouble
the temperature at which the specified data is to be retrieved. The temperature must have been defined previously for the material.

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return Value

Type:Â Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value. The function returns an error if the symmetry type of the specified material is not isotropic.

▲ Remarks

▲ Examples

VB

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim e As Double
    Dim u As Double

```

Introduction

```
Dim a As Double
Dim g As Double

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'initialize new material property
ret = SapModel.PropMaterial.SetMaterial("Steel", eMatType.Steel)

'assign isotropic mechanical properties
ret = SapModel.PropMaterial.SetMPIsotropic("Steel", 29500, 0.25, 6E-06)

'get isotropic mechanical properties
ret = SapModel.PropMaterial.GetMPIsotropic("Steel", e, u, a, g)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

🔗 See Also

Reference

[cPropMaterial Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LST4EC6DEEA_0?cprop=.") Method

Retrieves the mechanical properties for a material with an orthotropic directional symmetry type.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetMP0rthotropic(
    string Name,
    ref double[] E,
    ref double[] U,
    ref double[] A,
    ref double[] G,
    double Temp = 0
)
```

```
Function GetMP0rthotropic (
    Name As String,
    ByRef E As Double(),
    ByRef U As Double(),
    ByRef A As Double(),
    ByRef G As Double(),
    Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim E As Double()
Dim U As Double()
Dim A As Double()
Dim G As Double()
Dim Temp As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetMP0rthotropic(Name,
    E, U, A, G, Temp)
```

```
int GetMP0rthotropic(
    String^ Name,
    array<double>^% E,
    array<double>^% U,
    array<double>^% A,
    array<double>^% G,
    double Temp = 0
)
```

```
abstract GetMP0rthotropic :
```

```
cPropMaterialspan id="LST4EC6DEEA_0"AddLanguageSpecificTextSet("LST4EC6DEEA_0?cprop=.")>
```

```

    Name : string *
    E : float[] byref *
    U : float[] byref *
    A : float[] byref *
    G : float[] byref *
    ?Temp : float
(* Defaults:
    let _Temp = defaultArg Temp 0
*)
-> int

```

Parameters

Name

Type: SystemString

The name of an existing material property.

E

Type: SystemDouble

This is an array that includes the modulus of elasticity.

Value Term

e(0) E1 [F/L2]

e(1) E2 [F/L2]

e(2) E3 [F/L2]

U

Type: SystemDouble

This is an array that includes poisson's ratio.

Value Term

u(0) U12

u(1) U13

u(2) U23

A

Type: SystemDouble

This is an array that includes the thermal coefficient.

Value Term

a(0) A1 [1/T]

a(1) A2 [1/T]

a(2) A3 [1/T]

G

Type: SystemDouble

This is an array that includes the shear modulus.

Value Term

g(0) G12 [F/L2]

g(1) G13 [F/L2]

g(2) G23 [F/L2]

Temp (Optional)

Type: SystemDouble

The temperature at which the specified data is to be retrieved. The temperature must have been previously defined for the material.

Introduction

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return Value

Type: $\hat{A}$ Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value. The function returns an error if the symmetry type of the specified material is not orthotropic.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim MyE() As Double  
    Dim MyU() As Double  
    Dim MyA() As Double  
    Dim MyG() As Double  
    Dim e() As Double  
    Dim u() As Double  
    Dim a() As Double  
    Dim g() As Double  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)  
  
    'initialize new material property  
    ret = SapModel.PropMaterial.SetMaterial("Steel", eMatType.Steel)  
  
    'assign anisotropic mechanical properties  
    ReDim MyE(2)  
    ReDim MyU(2)  
    ReDim MyA(2)  
    ReDim MyG(2)  
    MyE(0)=30000  
    MyE(1)=10000  
    MyE(2)=2000  
    MyU(0)=0.2  
    MyU(1)=0.05  
    MyU(2)=0.1  
    MyA(0)=6.5E-6  
    MyA(1)=6.5E-6
```

Introduction

```
MyA(2)=6.5E-6
MyG(0)=1500
MyG(1)=2500
MyG(2)=8700
ret = SapModel.PropMaterial.SetMPOrthotropic("Steel", MyE, MyU, MyA, MyG)

'get anisotropic mechanical properties
ret = SapModel.PropMaterial.GetMPOrthotropic("Steel", e, u, a, g)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPropMaterial Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LST95DD0319_0?cpp3427u=.");Ge

Method

Retrieves the mechanical properties for a material with an uniaxial directional symmetry type.

Namespace: Â ETABSv1

Assembly: Â ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

⚡ Syntax

C#

VB

C++

F#

Copy

```
int GetMPUniaxial(
    string Name,
    ref double E,
    ref double A,
    double Temp = 0
)
```

```
Function GetMPUniaxial (
    Name As String,
    ByRef E As Double,
    ByRef A As Double,
    Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim E As Double
Dim A As Double
Dim Temp As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetMPUniaxial(Name,
    E, A, Temp)
```

```
int GetMPUniaxial(
    String^ Name,
    double% E,
    double% A,
    double Temp = 0
)
```

```
abstract GetMPUniaxial :
    Name : string *
    E : float byref *
    A : float byref *
    ?Temp : float
(* Defaults:
    let _Temp = defaultArg Temp 0
*)
-> int
```

cPropMaterialsspan id="LST95DD0319_0"AddLanguageSpecificTextSet("LST95DD0319_0?cpp3427u=.");Ge

Parameters

Name

Type: SystemString

The name of an existing material property.

E

Type: SystemDouble

The modulus of elasticity. [F/L2]

A

Type: SystemDouble

The thermal coefficient. [1/T]

Temp (Optional)

Type: SystemDouble

the temperature at which the specified data is to be retrieved. The temperature must have been defined previously for the material.

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return ValueType: Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value. The function returns an error if the symmetry type of the specified material is not isotropic.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim e As Double
    Dim a As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'initialize new material property
    ret = SapModel.PropMaterial.SetMaterial("Steel", eMatType.Steel)
```


Introduction

```
'assign isotropic mechanical properties
    ret = SapModel.PropMaterial.SetMPUniaxial("Steel", 29500, 6E-06)

'get isotropic mechanical properties
    ret = SapModel.PropMaterial.GetMPUniaxial("Steel", e, a)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPropMaterial Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LSTB0368426_0") Method

Retrieves the names of all defined material properties of the specified type.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNameList(
    ref int NumberNames,
    ref string[] MyName,
    eMatType MatType =
)
```

```
Function GetNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String(),
    Optional MatType As eMatType =
) As Integer
```

```
Dim instance As cPropMaterial
Dim NumberNames As Integer
Dim MyName As String()
Dim MatType As eMatType
Dim returnValue As Integer
```

```
returnValue = instance.GetNameList(NumberNames,
MyName, MatType)
```

```
int GetNameList(
    int% NumberNames,
    array<String^>^% MyName,
    eMatType MatType =
)
```

```
abstract GetNameList :
    NumberNames : int byref *
    MyName : string[] byref *
    ?MatType : eMatType
(* Defaults:
    let _MatType = defaultArg MatType
*)
-> int
```

Parameters

NumberNames

Type: SystemInt32

The number of material property names retrieved by the program.

MyName

Type: SystemString

One-dimensional array of material property names. The MyName array is created as a dynamic, zero-based, array by the API user:

```
Dim MyName() as String
```

The array is dimensioned to (NumberNames - 1) inside the program, filled with the names, and returned to the API user.

MatType (Optional)

Type: ETABSv1eMatType

This optional value is one of the items in the eMatType enumeration.

If no value is input for MatType, names are returned for all material properties in the model regardless of type.

Return Value

Type: Int32

Returns zero if the names are successfully retrieved; otherwise it returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberNames As Integer
    Dim MyName() As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'get material property names
    ret = SapModel.PropMaterial.GetNameList(NumberNames, MyName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
```

Introduction

```
SapModel = Nothing  
EtabsObject = Nothing  
End Sub
```

▲ See Also

Reference

cPropMaterial Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LST8F917615_0?cpp=3400.") Method

(NEWER FUNCTION AVAILABLE) Retrieves the other material property data for concrete materials.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int Get0Concrete(
    string Name,
    ref double Fc,
    ref bool IsLightweight,
    ref double FcsFactor,
    ref int SStype,
    ref int SSHysType,
    ref double StrainAtFc,
    ref double StrainUltimate,
    ref double FrictionAngle,
    ref double DilatationalAngle,
    double Temp = 0
)
```

```
Function Get0Concrete (
    Name As String,
    ByRef Fc As Double,
    ByRef IsLightweight As Boolean,
    ByRef FcsFactor As Double,
    ByRef SStype As Integer,
    ByRef SSHysType As Integer,
    ByRef StrainAtFc As Double,
    ByRef StrainUltimate As Double,
    ByRef FrictionAngle As Double,
    ByRef DilatationalAngle As Double,
    Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim Fc As Double
Dim IsLightweight As Boolean
Dim FcsFactor As Double
Dim SStype As Integer
Dim SSHysType As Integer
Dim StrainAtFc As Double
Dim StrainUltimate As Double
Dim FrictionAngle As Double
Dim DilatationalAngle As Double
Dim Temp As Double
```

cPropMaterialsid="LST8F917615_0"AddLanguageSpecificTextSet("LST8F917615_0?cpp=3400.");Get

```
Dim returnValue As Integer
```

```
returnValue = instance.Get0Concrete(Name,  
Fc, IsLightweight, FcsFactor, SSType,  
SSHysType, StrainAtFc, StrainUltimate,  
FrictionAngle, DilatationalAngle,  
Temp)
```

```
int Get0Concrete(  
String^ Name,  
double% Fc,  
bool% IsLightweight,  
double% FcsFactor,  
int% SSType,  
int% SSHysType,  
double% StrainAtFc,  
double% StrainUltimate,  
double% FrictionAngle,  
double% DilatationalAngle,  
double Temp = 0  
)
```

```
abstract Get0Concrete :  
    Name : string *  
    Fc : float byref *  
    IsLightweight : bool byref *  
    FcsFactor : float byref *  
    SSType : int byref *  
    SSHysType : int byref *  
    StrainAtFc : float byref *  
    StrainUltimate : float byref *  
    FrictionAngle : float byref *  
    DilatationalAngle : float byref *  
    ?Temp : float  
(* Defaults:  
    let _Temp = defaultArg Temp 0  
*)  
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing concrete material property.

Fc

Type:Â SystemDouble

IsLightweight

Type:Â SystemBoolean

If this item is True, the concrete is assumed to be lightweight concrete.

FcsFactor

Type:Â SystemDouble

SSType

Type:Â SystemInt32

This is 0, 1 or 2, indicating the stress-strain hysteresis type.

Value **SSType**

0 User defined

1 Parametric - Simple

2 Parametric - Mander

SSHysType

Type:Â SystemInt32

This is 0 through 7, indicating the stress-strain curve type.

Value SSHysType

- 0 Elastic
- 1 Kinematic
- 2 Takeda
- 3 Pivot
- 4 Concrete
- 5 BRB Hardening
- 6 Degrading
- 7 Isotropic

StrainAtFc

Type:Â SystemDouble

StrainUltimate

Type:Â SystemDouble

This item applies only to parametric stress-strain curves. It is the ultimate unconfined strain capacity. This item must be larger than the StrainAtfc item

FrictionAngle

Type:Â SystemDouble

The Drucker-Prager friction angle. This item must be smaller or equal to 0 and less than 90. [deg]

DilatationalAngle

Type:Â SystemDouble

The Drucker-Prager dilatational angle. This item must be smaller or equal to 0 and less than 90. [deg]

Temp (Optional)

Type:Â SystemDouble

The temperature at which the specified data is to be retrieved. The temperature must have been defined previously for the material.

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return ValueType:Â Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim fc As Double
    Dim IsLightweight As Boolean
```

Introduction

```
Dim fcsfactor As Double
Dim SStype As Integer
Dim SSHystype As Integer
Dim StrainAtfc As Double
Dim StrainUltimate As Double
Dim FrictionAngle As Double
Dim DilatationalAngle As Double

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'initialize new material property
ret = SapModel.PropMaterial.SetMaterial("Concrete", eMatType.Concrete)

'assign other properties
ret = SapModel.PropMaterial.Set0Concrete("Concrete", 5, False, 0, 1, 2, 0.0022, 0.0052)

'get other properties
ret = SapModel.PropMaterial.Get0Concrete("Concrete", fc, IsLightweight, fcsfactor, SStype,

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

[cPropMaterial Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LST2DF3D814_0?cpp3437u=.");G

Method

Retrieves the other material property data for concrete materials.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int Get0Concrete_1(
    string Name,
    ref double Fc,
    ref bool IsLightweight,
    ref double FcsFactor,
    ref int SStype,
    ref int SSHysType,
    ref double StrainAtFc,
    ref double StrainUltimate,
    ref double FinalSlope,
    ref double FrictionAngle,
    ref double DilatationalAngle,
    double Temp = 0
)
```

```
Function Get0Concrete_1 (
    Name As String,
    ByRef Fc As Double,
    ByRef IsLightweight As Boolean,
    ByRef FcsFactor As Double,
    ByRef SStype As Integer,
    ByRef SSHysType As Integer,
    ByRef StrainAtFc As Double,
    ByRef StrainUltimate As Double,
    ByRef FinalSlope As Double,
    ByRef FrictionAngle As Double,
    ByRef DilatationalAngle As Double,
    Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim Fc As Double
Dim IsLightweight As Boolean
Dim FcsFactor As Double
Dim SStype As Integer
Dim SSHysType As Integer
Dim StrainAtFc As Double
Dim StrainUltimate As Double
Dim FinalSlope As Double
Dim FrictionAngle As Double
```

cPropMaterialsid="LST2DF3D814_0"AddLanguageSpecificTextSet("LST2DF3D814_0?cpp3437u=.");G

Introduction

```
Dim DilatationalAngle As Double
Dim Temp As Double
Dim returnValue As Integer

returnValue = instance.Get0Concrete_1(Name,
Fc, IsLightweight, FcsFactor, SSType,
SSHysType, StrainAtFc, StrainUltimate,
FinalSlope, FrictionAngle, DilatationalAngle,
Temp)

int Get0Concrete_1(
String^ Name,
double% Fc,
bool% IsLightweight,
double% FcsFactor,
int% SSType,
int% SSHysType,
double% StrainAtFc,
double% StrainUltimate,
double% FinalSlope,
double% FrictionAngle,
double% DilatationalAngle,
double Temp = 0
)

abstract Get0Concrete_1 :
    Name : string *
    Fc : float byref *
    IsLightweight : bool byref *
    FcsFactor : float byref *
    SSType : int byref *
    SSHysType : int byref *
    StrainAtFc : float byref *
    StrainUltimate : float byref *
    FinalSlope : float byref *
    FrictionAngle : float byref *
    DilatationalAngle : float byref *
    ?Temp : float
(* Defaults:
    let _Temp = defaultArg Temp 0
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing concrete material property.

Fc

Type:Â SystemDouble

IsLightweight

Type:Â SystemBoolean

If this item is True, the concrete is assumed to be lightweight concrete.

FcsFactor

Type:Â SystemDouble

SSType

Type:Â SystemInt32

This is 0, 1 or 2, indicating the stress-strain curve type.

Value	SSType
0	User defined
1	Parametric - Simple
2	Parametric - Mander

SSHysTypeType:Â SystemInt32

This is 0 through 7, indicating the stress-strain curve type.

Value	SSHysType
--------------	------------------

0	Elastic
1	Kinematic
2	Takeda
3	Pivot
4	Concrete
5	BRB Hardening
6	Degrading
7	Isotropic

StrainAtFcType:Â SystemDouble**StrainUltimate**Type:Â SystemDouble

This item applies only to parametric stress-strain curves. It is the ultimate unconfined strain capacity. This item must be larger than the StrainAtfc item

FinalSlopeType:Â SystemDouble

This item applies only to parametric stress-strain curves. It is a multiplier on the material modulus of elasticity, E. This value multiplied times E gives the final slope on the compression side of the curve.

FrictionAngleType:Â SystemDouble

The Drucker-Prager friction angle. This item must be smaller or equal to 0 and less than 90. [deg]

DilatationalAngleType:Â SystemDouble

The Drucker-Prager dilatational angle. This item must be smaller or equal to 0 and less than 90. [deg]

Temp (Optional)Type:Â SystemDouble

The temperature at which the specified data is to be retrieved. The temperature must have been defined previously for the material.

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return ValueType:Â Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value.

Remarks

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim fc As Double
    Dim IsLightweight As Boolean
    Dim fcsfactor As Double
    Dim SSType As Integer
    Dim SSHysType As Integer
    Dim StrainAtfc As Double
    Dim StrainUltimate As Double
    Dim FinalSlope As Double
    Dim FrictionAngle As Double
    Dim DilatationalAngle As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'initialize new material property
    ret = SapModel.PropMaterial.SetMaterial("Concrete", eMatType.Concrete)

    'assign other properties
    ret = SapModel.PropMaterial.Set0Concrete_1("Concrete", 5, False, 0, 1, 2, 0.0022, 0.0052,

    'get other properties
    ret = SapModel.PropMaterial.Get0Concrete_1("Concrete", fc, IsLightweight, fcsfactor, SSType, SSHysType, StrainAtfc, StrainUltimate, FinalSlope, FrictionAngle, DilatationalAngle)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

cPropMaterial Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LSTF463FAC4_0") Method

Retrieves the other material property data for no design type materials.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int Get0NoDesign(
    string Name,
    ref double FrictionAngle,
    ref double DilatationalAngle,
    double Temp = 0
)
```

```
Function Get0NoDesign (
    Name As String,
    ByRef FrictionAngle As Double,
    ByRef DilatationalAngle As Double,
    Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim FrictionAngle As Double
Dim DilatationalAngle As Double
Dim Temp As Double
Dim returnValue As Integer
```

```
returnValue = instance.Get0NoDesign(Name,
    FrictionAngle, DilatationalAngle,
    Temp)
```

```
int Get0NoDesign(
    String^ Name,
    double% FrictionAngle,
    double% DilatationalAngle,
    double Temp = 0
)
```

```
abstract Get0NoDesign :
    Name : string *
    FrictionAngle : float byref *
    DilatationalAngle : float byref *
    ?Temp : float
(* Defaults:
    let _Temp = defaultArg Temp 0
*)
-> int
```

cPropMaterialsid="LSTF463FAC4_0"AddLanguageSpecificTextSet("LSTF463FAC4_0?cpp3442u=.");G

Parameters

Name

Type:Â SystemString

The name of an existing concrete material property.

FrictionAngle

Type:Â SystemDouble

The Drucker-Prager friction angle. This item must be smaller or equal to 0 and less than 90. [deg]

DilatationalAngle

Type:Â SystemDouble

The Drucker-Prager dilatational angle. This item must be smaller or equal to 0 and less than 90. [deg]

Temp (Optional)

Type:Â SystemDouble

The temperature at which the specified data is to be retrieved. The temperature must have been defined previously for the material.

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return ValueType:Â Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim FrictionAngle As Double
    Dim DilatationalAngle As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'initialize new material property
    ret = SapModel.PropMaterial.SetMaterial("NoDesign", eMatType.NoDesign)
```

Introduction

```
'assign other properties
    ret = SapModel.PropMaterial.Set0NoDesign("NoDesign", 10, 15)

'get other properties
    ret = SapModel.PropMaterial.Get0NoDesign("NoDesign", FrictionAngle, DilatationalAngle)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPropMaterial Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LSTB8D6 Method

(NEWER FUNCTION AVAILABLE) Retrieves the other material property data for rebar materials.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int Get0Rebar(
    string Name,
    ref double Fy,
    ref double Fu,
    ref double EFy,
    ref double EFu,
    ref int SStype,
    ref int SSHysType,
    ref double StrainAtHardening,
    ref double StrainUltimate,
    ref bool UseCaltransSSDefaults,
    double Temp = 0
)
```

```
Function Get0Rebar (
    Name As String,
    ByRef Fy As Double,
    ByRef Fu As Double,
    ByRef EFy As Double,
    ByRef EFu As Double,
    ByRef SStype As Integer,
    ByRef SSHysType As Integer,
    ByRef StrainAtHardening As Double,
    ByRef StrainUltimate As Double,
    ByRef UseCaltransSSDefaults As Boolean,
    Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim Fy As Double
Dim Fu As Double
Dim EFy As Double
Dim EFu As Double
Dim SStype As Integer
Dim SSHysType As Integer
Dim StrainAtHardening As Double
Dim StrainUltimate As Double
Dim UseCaltransSSDefaults As Boolean
Dim Temp As Double
```

cPropMaterialsspan id="LSTB8D61DF1_0"AddLanguageSpecificTextSet("LSTB8D61DF1_0?cpp3445Fu=.");G

```

Dim returnValue As Integer

returnValue = instance.Get0Rebar(Name,
    Fy, Fu, EFy, EFu, SStype, SSHysType,
    StrainAtHardening, StrainUltimate,
    UseCaltransSSDefaults, Temp)

int Get0Rebar(
    String^ Name,
    double% Fy,
    double% Fu,
    double% EFy,
    double% EFu,
    int% SStype,
    int% SSHysType,
    double% StrainAtHardening,
    double% StrainUltimate,
    bool% UseCaltransSSDefaults,
    double Temp = 0
)

abstract Get0Rebar :
    Name : string *
    Fy : float byref *
    Fu : float byref *
    EFy : float byref *
    EFu : float byref *
    SStype : int byref *
    SSHysType : int byref *
    StrainAtHardening : float byref *
    StrainUltimate : float byref *
    UseCaltransSSDefaults : bool byref *
    ?Temp : float
(* Defaults:
    let _Temp = defaultArg Temp 0
*)
-> int

```

Parameters

Name

Type:Â [SystemString](#)

The name of an existing rebar material property.

Fy

Type:Â [SystemDouble](#)

Fu

Type:Â [SystemDouble](#)

The minimum tensile stress. [F/L2]

EFy

Type:Â [SystemDouble](#)

EFu

Type:Â [SystemDouble](#)

SStype

Type:Â [SystemInt32](#)

This is 0, 1 or 2, indicating the stress-strain curve type.

Value **SStype**

0 User defined

- 1 Parametric - Simple
- 2 Parametric - Park

SSHysTypeType:Â SystemInt32

This is 0 through 7, indicating the stress-strain curve type.

Value SSHysType

- 0 Elastic
- 1 Kinematic
- 2 Takeda
- 3 Pivot
- 4 Concrete
- 5 BRB Hardening
- 6 Degradation
- 7 Isotropic

StrainAtHardeningType:Â SystemDouble

This item applies only when parametric stress-strain curves are used and when UseCaltransSSDefaults is False. It is the strain at the onset of strain hardening.

StrainUltimateType:Â SystemDouble

This item applies only when parametric stress-strain curves are used and when UseCaltransSSDefaults is False. It is the ultimate strain capacity. This item must be larger than the StrainAtHardening item.

UseCaltransSSDefaultsType:Â SystemBoolean

If this item is True, the program uses Caltrans default controlling strain values, which are bar size dependent.

Temp (Optional)Type:Â SystemDouble

The temperature at which the specified data is to be retrieved. The temperature must have been defined previously for the material.

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return ValueType:Â Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value.

▲ **Remarks**▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Fy As Double
    Dim Fu As Double
```

Introduction

```
Dim eFy As Double
Dim eFu As Double
Dim SSType As Integer
Dim SSHysType As Integer
Dim StrainAtHardening As Double
Dim StrainUltimate As Double
Dim UseCaltransSSDefaults As Boolean

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'initialize new material property
ret = SapModel.PropMaterial.SetMaterial("Rebar", eMatType.Rebar)

'assign other properties
ret = SapModel.PropMaterial.SetORebar("Rebar", 62, 93, 70, 102, 2, 2, 0.02, 0.1, False)

'get other properties
ret = SapModel.PropMaterial.GetORebar("Rebar", Fy, Fu, eFy, eFu, SSType, SSHysType, StrainAtHardening, StrainUltimate)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

[cPropMaterial Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LST19B2A Method

Retrieves the other material property data for rebar materials.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int Get0Rebar_1(
    string Name,
    ref double Fy,
    ref double Fu,
    ref double EFy,
    ref double EFu,
    ref int SStype,
    ref int SSHysType,
    ref double StrainAtHardening,
    ref double StrainUltimate,
    ref double FinalSlope,
    ref bool UseCaltransSSDefaults,
    double Temp = 0
)
```

```
Function Get0Rebar_1 (
    Name As String,
    ByRef Fy As Double,
    ByRef Fu As Double,
    ByRef EFy As Double,
    ByRef EFu As Double,
    ByRef SStype As Integer,
    ByRef SSHysType As Integer,
    ByRef StrainAtHardening As Double,
    ByRef StrainUltimate As Double,
    ByRef FinalSlope As Double,
    ByRef UseCaltransSSDefaults As Boolean,
    Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim Fy As Double
Dim Fu As Double
Dim EFy As Double
Dim EFu As Double
Dim SStype As Integer
Dim SSHysType As Integer
Dim StrainAtHardening As Double
Dim StrainUltimate As Double
Dim FinalSlope As Double
```

cPropMaterialspan id="LST19B2ADDA_0"AddLanguageSpecificTextSet("LST19B2ADDA_0?cp49nu=.");C

Introduction

```
Dim UseCaltransSSDefaults As Boolean
Dim Temp As Double
Dim returnValue As Integer

returnValue = instance.Get0Rebar_1(Name,
Fy, Fu, EFy, EFu, SStype, SSHysType,
StrainAtHardening, StrainUltimate,
FinalSlope, UseCaltransSSDefaults,
Temp)

int Get0Rebar_1(
String^ Name,
double% Fy,
double% Fu,
double% EFy,
double% EFu,
int% SStype,
int% SSHysType,
double% StrainAtHardening,
double% StrainUltimate,
double% FinalSlope,
bool% UseCaltransSSDefaults,
double Temp = 0
)

abstract Get0Rebar_1 :
    Name : string *
    Fy : float byref *
    Fu : float byref *
    EFy : float byref *
    EFu : float byref *
    SStype : int byref *
    SSHysType : int byref *
    StrainAtHardening : float byref *
    StrainUltimate : float byref *
    FinalSlope : float byref *
    UseCaltransSSDefaults : bool byref *
    ?Temp : float
(* Defaults:
    let _Temp = defaultArg Temp 0
*)
-> int
```

Parameters

Name	Type:Â <u>SystemString</u> The name of an existing rebar material property.
Fy	Type:Â <u>SystemDouble</u>
Fu	Type:Â <u>SystemDouble</u> The minimum tensile stress. [F/L2]
EFy	Type:Â <u>SystemDouble</u>
EFu	Type:Â <u>SystemDouble</u>
SStype	

Type: SystemInt32

This is 0, 1 or 2, indicating the stress-strain curve type.

Value SSType

- 0 User defined
- 1 Parametric - Simple
- 2 Parametric - Park

SSHysType

Type: SystemInt32

This is 0 through 7, indicating the stress-strain curve type.

Value SSHysType

- 0 Elastic
- 1 Kinematic
- 2 Takeda
- 3 Pivot
- 4 Concrete
- 5 BRB Hardening
- 6 Degrading
- 7 Isotropic

StrainAtHardening

Type: SystemDouble

This item applies only when parametric stress-strain curves are used and when UseCaltransSSDefaults is False. It is the strain at the onset of strain hardening.

StrainUltimate

Type: SystemDouble

This item applies only when parametric stress-strain curves are used and when UseCaltransSSDefaults is False. It is the ultimate strain capacity. This item must be larger than the StrainAtHardening item.

FinalSlope

Type: SystemDouble

This item applies only to parametric stress-strain curves. It is a multiplier on the material modulus of elasticity, E. This value multiplied times E gives the final slope of the curve.

UseCaltransSSDefaults

Type: SystemBoolean

If this item is True, the program uses Caltrans default controlling strain values, which are bar size dependent.

Temp (Optional)

Type: SystemDouble

The temperature at which the specified data is to be retrieved. The temperature must have been defined previously for the material.

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return ValueType: $\hat{A}$ Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Fy As Double
    Dim Fu As Double
    Dim eFy As Double
    Dim eFu As Double
    Dim SStype As Integer
    Dim SSHysType As Integer
    Dim StrainAtHardening As Double
    Dim StrainUltimate As Double
    Dim FinalSlope As Double
    Dim UseCaltransSSDefaults As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'initialize new material property
    ret = SapModel.PropMaterial.SetMaterial("Rebar", eMatType.Rebar)

    'assign other properties
    ret = SapModel.PropMaterial.Set0Rebar_1("Rebar", 62, 93, 70, 102, 2, 2, 0.02, 0.1, -0.1, F

    'get other properties
    ret = SapModel.PropMaterial.Get0Rebar_1("Rebar", Fy, Fu, eFy, eFu, SStype, SSHysType, Stra

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPropMaterial Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LST37C7F09A_0?cpp3454u=.");Get0Steel

Method

(NEWER FUNCTION AVAILABLE) Retrieves the other material property data for steel materials.

Namespace: `ETABSv1`

Assembly: `ETABSv1 (in ETABSv1.dll)` Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int Get0Steel(
    string Name,
    ref double Fy,
    ref double Fu,
    ref double EFy,
    ref double EFu,
    ref int SStype,
    ref int SSHysType,
    ref double StrainAtHardening,
    ref double StrainAtMaxStress,
    ref double StrainAtRupture,
    double Temp = 0
)
```

```
Function Get0Steel (
    Name As String,
    ByRef Fy As Double,
    ByRef Fu As Double,
    ByRef EFy As Double,
    ByRef EFu As Double,
    ByRef SStype As Integer,
    ByRef SSHysType As Integer,
    ByRef StrainAtHardening As Double,
    ByRef StrainAtMaxStress As Double,
    ByRef StrainAtRupture As Double,
    Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim Fy As Double
Dim Fu As Double
Dim EFy As Double
Dim EFu As Double
Dim SStype As Integer
Dim SSHysType As Integer
Dim StrainAtHardening As Double
Dim StrainAtMaxStress As Double
Dim StrainAtRupture As Double
Dim Temp As Double
```

cPropMaterialsAddLanguageSpecificTextSet("LST37C7F09A_0?cpp3454u=.");Get0Steel

```
Dim returnValue As Integer
```

```
returnValue = instance.Get0Steel(Name,  
Fu, EFy, EFu, SStype, SSHysType,  
StrainAtHardening, StrainAtMaxStress,  
StrainAtRupture, Temp)
```

```
int Get0Steel(  
String^ Name,  
double% Fy,  
double% Fu,  
double% EFy,  
double% EFu,  
int% SStype,  
int% SSHysType,  
double% StrainAtHardening,  
double% StrainAtMaxStress,  
double% StrainAtRupture,  
double Temp = 0  
)
```

```
abstract Get0Steel :  
    Name : string *  
    Fy : float byref *  
    Fu : float byref *  
    EFy : float byref *  
    EFu : float byref *  
    SStype : int byref *  
    SSHysType : int byref *  
    StrainAtHardening : float byref *  
    StrainAtMaxStress : float byref *  
    StrainAtRupture : float byref *  
    ?Temp : float  
(* Defaults:  
    let _Temp = defaultArg Temp 0  
)  
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing steel material property.

Fy

Type:Â SystemDouble

Fu

Type:Â SystemDouble

The minimum tensile stress. [F/L2]

EFy

Type:Â SystemDouble

EFu

Type:Â SystemDouble

SStype

Type:Â SystemInt32

This is 0 or 1, indicating the stress-strain curve type.

Value **SStype**

0 User defined

1 Parametric - Simple

SSHysType
 Type:Â SystemInt32
 This is 0 through 7, indicating the stress-strain curve type.
Value SSHysType

0	Elastic
1	Kinematic
2	Takeda
3	Pivot
4	Concrete
5	BRB Hardening
6	Degrading
7	Isotropic

StrainAtHardening
 Type:Â SystemDouble
 This item applies only when parametric stress-strain curves are used. It is the strain at the onset of strain hardening.

StrainAtMaxStress
 Type:Â SystemDouble
 This item applies only when parametric stress-strain curves are used. It is the strain at the maximum stress. This item must be larger than the StrainAtHardening item.

StrainAtRupture
 Type:Â SystemDouble
 This item applies only when parametric stress-strain curves are used. It is the strain at rupture. This item must be larger than the StrainAtMaxStress item.

Temp (Optional)
 Type:Â SystemDouble
 The temperature at which the specified data is to be retrieved. The temperature must have been defined previously for the material.

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return Value

Type:Â Int32
 Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Fy As Double
    Dim Fu As Double
    Dim eFy As Double
```

Introduction

```
Dim eFu As Double
Dim SStype As Integer
Dim SSHysType As Integer
Dim StrainAtHardening As Double
Dim StrainAtMaxStress As Double
Dim StrainAtRupture As Double

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'initialize new material property
ret = SapModel.PropMaterial.SetMaterial("Steel", eMatType.Steel)

'assign other properties
ret = SapModel.PropMaterial.Set0Steel_1("Steel", 55, 68, 60, 70, 1, 2, 0.02, 0.1, 0.2)

'get other properties
ret = SapModel.PropMaterial.Get0Steel_1("Steel", Fy, Fu, eFy, eFu, SStype, SSHysType, StrainAtHardening, StrainAtMaxStress, StrainAtRupture)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

See Also

Reference

[cPropMaterial Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LSTBDD52C06_0?cpu=.") Method

Retrieves the other material property data for steel materials.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int Get0Steel_1(
    string Name,
    ref double Fy,
    ref double Fu,
    ref double EFy,
    ref double EFu,
    ref int SStype,
    ref int SSHysType,
    ref double StrainAtHardening,
    ref double StrainAtMaxStress,
    ref double StrainAtRupture,
    ref double FinalSlope,
    double Temp = 0
)
```

```
Function Get0Steel_1 (
    Name As String,
    ByRef Fy As Double,
    ByRef Fu As Double,
    ByRef EFy As Double,
    ByRef EFu As Double,
    ByRef SStype As Integer,
    ByRef SSHysType As Integer,
    ByRef StrainAtHardening As Double,
    ByRef StrainAtMaxStress As Double,
    ByRef StrainAtRupture As Double,
    ByRef FinalSlope As Double,
    Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim Fy As Double
Dim Fu As Double
Dim EFy As Double
Dim EFu As Double
Dim SStype As Integer
Dim SSHysType As Integer
Dim StrainAtHardening As Double
Dim StrainAtMaxStress As Double
Dim StrainAtRupture As Double
```

cPropMaterialsspan id="LSTBDD52C06_0"AddLanguageSpecificTextSet("LSTBDD52C06_0?cpu=.");G

Introduction

```
Dim FinalSlope As Double
Dim Temp As Double
Dim returnValue As Integer
```

```
returnValue = instance.Get0Steel_1(Name,
Fy, Fu, EFy, EFu, SStype, SSHysType,
StrainAtHardening, StrainAtMaxStress,
StrainAtRupture, FinalSlope, Temp)
```

```
int Get0Steel_1(
String^ Name,
double% Fy,
double% Fu,
double% EFy,
double% EFu,
int% SStype,
int% SSHysType,
double% StrainAtHardening,
double% StrainAtMaxStress,
double% StrainAtRupture,
double% FinalSlope,
double Temp = 0
)
```

```
abstract Get0Steel_1 :
    Name : string *
    Fy : float byref *
    Fu : float byref *
    EFy : float byref *
    EFu : float byref *
    SStype : int byref *
    SSHysType : int byref *
    StrainAtHardening : float byref *
    StrainAtMaxStress : float byref *
    StrainAtRupture : float byref *
    FinalSlope : float byref *
    ?Temp : float
(* Defaults:
    let _Temp = defaultArg Temp 0
*)
-> int
```

Parameters

Name	Type:Â <u>SystemString</u> The name of an existing steel material property.
Fy	Type:Â <u>SystemDouble</u>
Fu	Type:Â <u>SystemDouble</u> The minimum tensile stress. [F/L2]
EFy	Type:Â <u>SystemDouble</u>
EFu	Type:Â <u>SystemDouble</u>
SStype	

Type: SystemInt32

This is 0 or 1, indicating the stress-strain curve type.

Value SStype

- 0 User defined
- 1 Parametric - Simple

SSHysType

Type: SystemInt32

This is 0 through 7, indicating the stress-strain curve type.

Value SSHysType

- 0 Elastic
- 1 Kinematic
- 2 Takeda
- 3 Pivot
- 4 Concrete
- 5 BRB Hardening
- 6 Degradation
- 7 Isotropic

StrainAtHardening

Type: SystemDouble

This item applies only when parametric stress-strain curves are used. It is the strain at the onset of strain hardening.

StrainAtMaxStress

Type: SystemDouble

This item applies only when parametric stress-strain curves are used. It is the strain at the maximum stress. This item must be larger than the StrainAtHardening item.

StrainAtRupture

Type: SystemDouble

This item applies only when parametric stress-strain curves are used. It is the strain at rupture. This item must be larger than the StrainAtMaxStress item.

FinalSlope

Type: SystemDouble

This item applies only to parametric stress-strain curves. It is a multiplier on the material modulus of elasticity, E. This value multiplied times E gives the final slope of the curve.

Temp (Optional)

Type: SystemDouble

The temperature at which the specified data is to be retrieved. The temperature must have been defined previously for the material.

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return Value

Type: Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value.

Remarks

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Fy As Double
    Dim Fu As Double
    Dim eFy As Double
    Dim eFu As Double
    Dim SStype As Integer
    Dim SSHysType As Integer
    Dim StrainAtHardening As Double
    Dim StrainAtMaxStress As Double
    Dim StrainAtRupture As Double
    Dim FinalSlope As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'initialize new material property
    ret = SapModel.PropMaterial.SetMaterial("Steel", eMatType.Steel)

    'assign other properties
    ret = SapModel.PropMaterial.Set0Steel_1("Steel", 55, 68, 60, 70, 1, 2, 0.02, 0.1, 0.2, -0.01)

    'get other properties
    ret = SapModel.PropMaterial.Get0Steel_1("Steel", Fy, Fu, eFy, eFu, SStype, SSHysType, StrainAtHardening, StrainAtMaxStress, StrainAtRupture, FinalSlope)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

cPropMaterial Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LST2E0B519_0?cpp=::3463");GetO

Method

(NEWER FUNCTION AVAILABLE) Retrieves the other material property data for tendon materials.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int Get0Tendon(
    string Name,
    ref double Fy,
    ref double Fu,
    ref int SStype,
    ref int SSHysType,
    double Temp = 0
)
```

```
Function Get0Tendon (
    Name As String,
    ByRef Fy As Double,
    ByRef Fu As Double,
    ByRef SStype As Integer,
    ByRef SSHysType As Integer,
    Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim Fy As Double
Dim Fu As Double
Dim SStype As Integer
Dim SSHysType As Integer
Dim Temp As Double
Dim returnValue As Integer
```

```
returnValue = instance.Get0Tendon(Name,
    Fy, Fu, SStype, SSHysType, Temp)
```

```
int Get0Tendon(
    String^ Name,
    double% Fy,
    double% Fu,
    int% SStype,
    int% SSHysType,
    double Temp = 0
)
```

abstract Get0Tendon :

cPropMaterialsid="LST2E0B519_0"AddLanguageSpecificTextSet("LST2E0B519_0?cpp=::3463");GetO

```

    Name : string *
    Fy : float byref *
    Fu : float byref *
    SStype : int byref *
    SSHysType : int byref *
    ?Temp : float
(* Defaults:
    let _Temp = defaultArg Temp 0
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of an existing tendon material property.

Fy

Type:Â SystemDouble

Fu

Type:Â SystemDouble

The minimum tensile stress. [F/L2]

SStype

Type:Â SystemInt32

This is 0, 1 or 2, indicating the stress-strain curve type.

Value	SStype
0	User defined
1	Parametric - 250 ksi strand
1	Parametric - 270 ksi strand

SSHysType

Type:Â SystemInt32

This is 0 through 7, indicating the stress-strain curve type.

Value	SSHysType
0	Elastic
1	Kinematic
2	Takeda
3	Pivot
4	Concrete
5	BRB Hardening
6	Degrading
7	Isotropic

Temp (Optional)

Type:Â SystemDouble

The temperature at which the specified data to be retrieved. The temperature must have been defined previously for the material.

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return Value

Type: Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value. The function returns an error if the specified material is not tendon.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Fy As Double
    Dim Fu As Double
    Dim SStype As Integer
    Dim SSHysType As Integer

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'initialize new material property
    ret = SapModel.PropMaterial.SetMaterial("Tendon", eMatType.Tendon)

    'assign other properties
    ret = SapModel.PropMaterial.SetOTendon("Tendon", 230, 255, 1, 1)

    'get other properties
    ret = SapModel.PropMaterial.GetOTendon("Tendon", Fy, Fu, SStype, SSHysType)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cPropMaterial Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LSTD5D0 Method

Retrieves the other material property data for tendon materials.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int Get0Tendon_1(
    string Name,
    ref double Fy,
    ref double Fu,
    ref int SStype,
    ref int SSHysType,
    ref double FinalSlope,
    double Temp = 0
)
```

```
Function Get0Tendon_1 (
    Name As String,
    ByRef Fy As Double,
    ByRef Fu As Double,
    ByRef SStype As Integer,
    ByRef SSHysType As Integer,
    ByRef FinalSlope As Double,
    Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim Fy As Double
Dim Fu As Double
Dim SStype As Integer
Dim SSHysType As Integer
Dim FinalSlope As Double
Dim Temp As Double
Dim returnValue As Integer
```

```
returnValue = instance.Get0Tendon_1(Name,
    Fy, Fu, SStype, SSHysType, FinalSlope,
    Temp)
```

```
int Get0Tendon_1(
    String^ Name,
    double% Fy,
    double% Fu,
    int% SStype,
    int% SSHysType,
    double% FinalSlope,
```

cPropMaterialsAddLanguageSpecificTextSet("LSTD5D0172F_0?cpp3467u=.");G

```

double Temp = 0
)

abstract GetOTendon_1 :
    Name : string *
    Fy : float byref *
    Fu : float byref *
    SSType : int byref *
    SSHysType : int byref *
    FinalSlope : float byref *
    ?Temp : float
(* Defaults:
    let _Temp = defaultArg Temp 0
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of an existing tendon material property.

Fy

Type:Â SystemDouble

Fu

Type:Â SystemDouble

The minimum tensile stress. [F/L2]

SSType

Type:Â SystemInt32

This is 0, 1 or 2, indicating the stress-strain curve type.

Value	SSType
0	User defined
1	Parametric - 250 ksi strand
1	Parametric - 270 ksi strand

SSHysType

Type:Â SystemInt32

This is 0 through 7, indicating the stress-strain curve type.

Value	SSHysType
0	Elastic
1	Kinematic
2	Takeda
3	Pivot
4	Concrete
5	BRB Hardening
6	Degrading
7	Isotropic

FinalSlope

Type:Â SystemDouble

This item applies only to parametric stress-strain curves. It is a multiplier on the material modulus of elasticity, E. This value multiplied times E gives the final slope of the curve.

Temp (Optional)Type: SystemDouble

The temperature at which the specified data to be retrieved. The temperature must have been defined previously for the material.

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return ValueType: Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value. The function returns an error if the specified material is not tendon.

▲ **Remarks**▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Fy As Double
    Dim Fu As Double
    Dim SStype As Integer
    Dim SSHysType As Integer
    Dim FinalSlope As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'initialize new material property
    ret = SapModel.PropMaterial.SetMaterial("Tendon", eMatType.Tendon)

    'assign other properties
    ret = SapModel.PropMaterial.SetOTendon_1("Tendon", 230, 255, 1, 1, -0.1)

    'get other properties
    ret = SapModel.PropMaterial.GetOTendon_1("Tendon", Fy, Fu, SStype, SSHysType, FinalSlope)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
```

```
    EtabsObject = Nothing  
End Sub
```

▲ See Also

Reference

cPropMaterial Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LSTFAA7 Method

Retrieves the material stress-strain curve.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetSSCurve(
    string Name,
    ref int NumberPoints,
    ref int[] PointID,
    ref double[] Strain,
    ref double[] Stress,
    string SectName = "",
    double RebarArea = 0,
    double Temp = 0
)
```

```
Function GetSSCurve (
    Name As String,
    ByRef NumberPoints As Integer,
    ByRef PointID As Integer(),
    ByRef Strain As Double(),
    ByRef Stress As Double(),
    SectName As String = "",
    RebarArea As Double = 0,
    Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim NumberPoints As Integer
Dim PointID As Integer()
Dim Strain As Double()
Dim Stress As Double()
Dim SectName As String
Dim RebarArea As Double
Dim Temp As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetSSCurve(Name,
    NumberPoints, PointID, Strain, Stress,
    SectName, RebarArea, Temp)
```

```
int GetSSCurve(
    String^ Name,
    int% NumberPoints,
    array<int>^% PointID,
```

cPropMaterialsid="LSTFAA7CA0A_0"AddLanguageSpecificTextSet("LSTFAA7CA0A_0?cp471nu=.");C

```

array<double>^% Strain,
array<double>^% Stress,
String^ SectName = L"",
double RebarArea = 0,
double Temp = 0
)

abstract GetSSCurve :
    Name : string *
    NumberPoints : int byref *
    PointID : int[] byref *
    Strain : float[] byref *
    Stress : float[] byref *
    ?SectName : string *
    ?RebarArea : float *
    ?Temp : float
(* Defaults:
    let _SectName = defaultArg SectName ""
    let _RebarArea = defaultArg RebarArea 0
    let _Temp = defaultArg Temp 0
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of an existing material property.

NumberPoints

Type:Â SystemInt32

The number of points in the stress-strain curve.

PointID

Type:Â SystemInt32

This is one of the following integers which sets the point ID. The point ID controls the color that will be displayed for hinges in a deformed shape plot.

Value Point ID

-5	-E
-4	-D
-3	-C
-2	-B
-1	-A
0	None
1	A
2	B
3	C
4	D
5	E

Strain

Type:Â SystemDouble

This is an array that includes the strain at each point on the stress strain curve.

Stress

Type: SystemDouble

This is an array that includes the stress at each point on the stress strain curve.
[F/L2]

SectName (Optional)

Type: SystemString

This item applies only if the specified material is concrete with a Mander concrete type.

This is the frame section property for which the Mander stress-strain curve is retrieved.

The section must be round or rectangular

RebarArea (Optional)

Type: SystemDouble

This is the area of the rebar for which the stress-strain curve is retrieved.

This item applies only if the specified material is rebar, which does not have a user-defined stress-strain curve and is specified to use Caltrans default controlling strain values, which are bar size dependent.

Temp (Optional)

Type: SystemDouble

The temperature at which the specified data is to be retrieved. The temperature must have been defined previously for the material.

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return Value

Type: Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberPoints As Integer
    Dim PointID() As Integer
    Dim Strain() As Double
    Dim Stress() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel
```

Introduction

```
'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'get stress strain curve
    ret = SapModel.PropMaterial.GetSSCurve("A992Fy50", NumberPoints, PointID, Strain, Stress)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPropMaterial Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LST1DF01788_0?cpp3475=.");Ge

Method

Retrieves the temperatures at which properties are specified for a material.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetTemp(
    string Name,
    ref int NumberItems,
    ref double[] Temp
)
```

```
Function GetTemp (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef Temp As Double()
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim NumberItems As Integer
Dim Temp As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetTemp(Name, NumberItems,
    Temp)
```

```
int GetTemp(
    String^ Name,
    int% NumberItems,
    array<double>^% Temp
)
```

```
abstract GetTemp :
    Name : string *
    NumberItems : int byref *
    Temp : float[] byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of a material property.

NumberItems

Type:Â SystemInt32

The number of different temperatures at which properties are specified for the material.

Temp

Type:Â SystemDouble

This is an array that includes the different temperatures at which properties are specified for the material.

Return Value

Type:Â Int32

Returns zero if the temperatures are successfully retrieved; otherwise it returns a nonzero value

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberItems As Integer
    Dim MyTemp() As Double
    Dim Temp() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'initialize new material property
    ret = SapModel.PropMaterial.SetMaterial("Steel", eMatType.Steel)

    'specify temps at which properties will be provided
    ReDim MyTemp(2)
    MyTemp(0) = 0
    MyTemp(1) = 50
    MyTemp(2) = 100
    ret = SapModel.PropMaterial.SetTemp("Steel", 3, MyTemp)

    'get temps at which properties are provided
    ret = SapModel.PropMaterial.GetTemp("Steel", NumberItems, Temp)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
```


Introduction

```
SapModel = Nothing  
EtabsObject = Nothing  
End Sub
```

▲ See Also

Reference

cPropMaterial Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LST744F99CC_0?cpp3476u=.");G Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetType0API(
    string Name,
    ref eMatType MatType,
    ref int SymType
)
```

```
Function GetType0API (
    Name As String,
    ByRef MatType As eMatType,
    ByRef SymType As Integer
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim MatType As eMatType
Dim SymType As Integer
Dim returnValue As Integer
```

```
returnValue = instance.GetType0API(Name,
    MatType, SymType)
```

```
int GetType0API(
    String^ Name,
    eMatType% MatType,
    int% SymType
)
```

```
abstract GetType0API :
    Name : string *
    MatType : eMatType byref *
    SymType : int byref -> int
```

Parameters

Name

Type: SystemString

MatType

Type: ETABSv1eMatType

SymType

Type: SystemInt32

cPropMaterialsspan id="LST744F99CC_0"AddLanguageSpecificTextSet("LST744F99CC_0?cpp3476u=.");G

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cPropMaterial Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LST74489A94_0?cpp=3460=.");GetMaterialMethod

Retrieves the weight per unit volume and mass per unit volume of the material.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetWeightAndMass(
    string Name,
    ref double W,
    ref double M,
    double Temp = 0
)
```

```
Function GetWeightAndMass (
    Name As String,
    ByRef W As Double,
    ByRef M As Double,
    Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim W As Double
Dim M As Double
Dim Temp As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetWeightAndMass(Name,
W, M, Temp)
```

```
int GetWeightAndMass(
    String^ Name,
    double% W,
    double% M,
    double Temp = 0
)
```

```
abstract GetWeightAndMass :
    Name : string *
    W : float byref *
    M : float byref *
    ?Temp : float
(* Defaults:
    let _Temp = defaultArg Temp 0
*)
-> int
```

cPropMaterialsid="LST74489A94_0"AddLanguageSpecificTextSet("LST74489A94_0?cpp=3460=.");GetMaterial

Parameters

Name

Type:Â SystemString

The name of an existing material property.

W

Type:Â SystemDouble

The weight per unit volume for the material. [F/L3]

M

Type:Â SystemDouble

The mass per unit volume for the material. [M/L3]

Temp (Optional)

Type:Â SystemDouble

the temperature at which the specified data is to be retrieved.

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return ValueType:Â Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim w As Double
    Dim m As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'initialize new material property
    ret = SapModel.PropMaterial.SetMaterial("Steel", eMatType.Steel)

    'assign material property weight per unit volume
    ret = SapModel.PropMaterial.SetWeightAndMass("Steel", 1, 0.00029)
```

Introduction

```
'get material weight and mass per unit volume
  ret = SapModel.PropMaterial.GetWeightAndMass("Steel", w, m)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

cPropMaterial Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LSTC5043B69_0?cpp3488u=.");Se

Method

Sets the additional material damping data for the material.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetDamping(
    string Name,
    double ModalRatio,
    double ViscousMassCoeff,
    double ViscousStiffCoeff,
    double HystereticMassCoeff,
    double HystereticStiffCoeff,
    double Temp = 0
)
```

```
Function SetDamping (
    Name As String,
    ModalRatio As Double,
    ViscousMassCoeff As Double,
    ViscousStiffCoeff As Double,
    HystereticMassCoeff As Double,
    HystereticStiffCoeff As Double,
    Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim ModalRatio As Double
Dim ViscousMassCoeff As Double
Dim ViscousStiffCoeff As Double
Dim HystereticMassCoeff As Double
Dim HystereticStiffCoeff As Double
Dim Temp As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetDamping(Name,
    ModalRatio, ViscousMassCoeff, ViscousStiffCoeff,
    HystereticMassCoeff, HystereticStiffCoeff,
    Temp)
```

```
int SetDamping(
    String^ Name,
    double ModalRatio,
    double ViscousMassCoeff,
    double ViscousStiffCoeff,
    double HystereticMassCoeff,
```

cPropMaterialsspan id="LSTC5043B69_0"AddLanguageSpecificTextSet("LSTC5043B69_0?cpp3488u=.");Se

```

double HystereticStiffCoeff,
double Temp = 0
)

abstract SetDamping :
    Name : string *
    ModalRatio : float *
    ViscousMassCoeff : float *
    ViscousStiffCoeff : float *
    HystereticMassCoeff : float *
    HystereticStiffCoeff : float *
    ?Temp : float
(* Defaults:
    let _Temp = defaultArg Temp 0
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of an existing material property.

ModalRatio

Type:Â SystemDouble

The modal damping ratio.

ViscousMassCoeff

Type:Â SystemDouble

The mass coefficient for viscous proportional damping.

ViscousStiffCoeff

Type:Â SystemDouble

The stiffness coefficient for viscous proportional damping.

HystereticMassCoeff

Type:Â SystemDouble

The mass coefficient for hysteretic proportional damping.

HystereticStiffCoeff

Type:Â SystemDouble

The stiffness coefficient for hysteretic proportional damping.

Temp (Optional)

Type:Â SystemDouble

The temperature at which the specified data applies. The temperature must have been previously defined for the material.

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return Value

Type:Â Int32

Returns zero if the data is successfully assigned; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'assign material damping data
    ret = SapModel.PropMaterial.SetDamping("4000Psi", 0.04, 0, 0, 0, 0)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

ReferencecPropMaterial InterfaceETABSv1 Namespace

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»

CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LST672212B7_0?cpp=3486=.");SetMethod

DEPRECATED Sets the mass source for the model.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetMassSource(
    int MyOption,
    int NumberLoads,
    ref string[] LoadPat,
    ref double[] SF
)
```

```
Function SetMassSource (
    MyOption As Integer,
    NumberLoads As Integer,
    ByRef LoadPat As String(),
    ByRef SF As Double()
) As Integer
```

```
Dim instance As cPropMaterial
Dim MyOption As Integer
Dim NumberLoads As Integer
Dim LoadPat As String()
Dim SF As Double()
Dim returnValue As Integer
```

```
returnValue = instance.SetMassSource(MyOption,
    NumberLoads, LoadPat, SF)
```

```
int SetMassSource(
    int MyOption,
    int NumberLoads,
    array<String^>^% LoadPat,
    array<double>^% SF
)
```

```
abstract SetMassSource :
    MyOption : int *
    NumberLoads : int *
    LoadPat : string[] byref *
    SF : float[] byref -> int
```

Parameters**MyOption**Type: SystemInt32

This is 1, 2 or 3, indicating the mass source option.

Value**Option**

- | | |
|---|--|
| 1 | From element self mass and additional masses |
| 2 | From loads |
| 3 | From element self mass and additional masses and loads |

NumberLoadsType: SystemInt32

The number of load patterns from which mass is obtained. This item applies only when MyOption is 2 or 3.

LoadPatType: SystemString

This is an array of the names of the load patterns from which mass is obtained. This item applies only when MyOption is 2 or 3.

SFType: SystemDouble

This is an array of load patterns multipliers used to calculate the mass. This item applies only when MyOption is 2 or 3

Return ValueType: Int32

Returns zero if the mass source is successfully set; otherwise it returns a nonzero value.

▲ **Remarks**▲ **Examples****VB**Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyLoadPat() As String
    Dim MySF() As Double
    Dim MyOption As Integer
    Dim NumberLoads As Integer
    Dim LoadPat() As String
    Dim sf() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()
```

Introduction

```
'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set mass source
    ReDim MyLoadPat(1)
    ReDim MySF(1)
    MyLoadPat(0) = "DEAD"
    MyLoadPat(1) = "LIVE"
    MySF(0) = 1
    MySF(1) = 0.2
    ret = SapModel.PropMaterial.SetMassSource(3, 2, MyLoadPat, MySF)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

cPropMaterial Interface

ETABSv1 Namespace

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»

CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LST57AF93F1_0?cpp3480=.");Set

Method

Sets the mass source for the model.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int SetMassSource_1(
    ref bool IncludeElements,
    ref bool IncludeAddedMass,
    ref bool IncludeLoads,
    int NumberLoads,
    ref string[] LoadPat,
    ref double[] sf
)
```

```
Function SetMassSource_1 (
    ByRef IncludeElements As Boolean,
    ByRef IncludeAddedMass As Boolean,
    ByRef IncludeLoads As Boolean,
    NumberLoads As Integer,
    ByRef LoadPat As String(),
    ByRef sf As Double()
) As Integer
```

```
Dim instance As cPropMaterial
Dim IncludeElements As Boolean
Dim IncludeAddedMass As Boolean
Dim IncludeLoads As Boolean
Dim NumberLoads As Integer
Dim LoadPat As String()
Dim sf As Double()
Dim returnValue As Integer
```

```
returnValue = instance.SetMassSource_1(IncludeElements,
    IncludeAddedMass, IncludeLoads,
    NumberLoads, LoadPat, sf)
```

```
int SetMassSource_1(
    bool% IncludeElements,
    bool% IncludeAddedMass,
    bool% IncludeLoads,
    int NumberLoads,
    array<String^>^% LoadPat,
    array<double>^% sf
)
```

abstract SetMassSource_1 :

cPropMaterialAddLanguageSpecificTextSet("LST57AF93F1_0?cpp3480=.");Set

```

IncludeElements : bool byref *
IncludeAddedMass : bool byref *
IncludeLoads : bool byref *
NumberLoads : int *
LoadPat : string[] byref *
sf : float[] byref -> int

```

Parameters

IncludeElements

Type:Â SystemBoolean
Include element self mass

IncludeAddedMass

Type:Â SystemBoolean
Include additional masses

IncludeLoads

Type:Â SystemBoolean
Include specified loads

NumberLoads

Type:Â SystemInt32
The number of load patterns from which mass is obtained. This item applies only when IncludeLoads is True.

LoadPat

Type:Â SystemString
This is an array of the names of the load patterns from which mass is obtained. This item applies only when IncludeLoads is True.

sf

Type:Â SystemDouble
This is an array of load patterns multipliers used to calculate the mass. This item applies only when IncludeLoads is True.

Return Value

Type:Â Int32

Returns zero if the mass source is successfully set; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyLoadPat() As String
    Dim MySF() As Double
    Dim MyOption As Integer
    Dim NumberLoads As Integer
    Dim LoadPat() As String
    Dim sf() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

```

Introduction

```
'start ETABS application
    ret = EtabsObject.ApplicationStart()

'create SapModel object
    SapModel = EtabsObject.SapModel

'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'set mass source
    ReDim MyLoadPat(1)
    ReDim MySF(1)
    MyLoadPat(0) = "DEAD"
    MyLoadPat(1) = "LIVE"
    MySF(0) = 1
    MySF(1) = 0.2
    ret = SapModel.PropMaterial.SetMassSource(3, 2, MyLoadPat, MySF)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPropMaterial Interface](#)

[ETABSv1 Namespace](#)

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»

CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LSTA5E5EC2D_0?cpnu=")

Method

DEPRECATED. Initializes a material property. If this function is called for an existing material property, all items for the material are reset to their default value.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetMaterial(
    string Name,
    eMatType MatType,
    int Color = -1,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetMaterial (
    Name As String,
    MatType As eMatType,
    Optional As Integer = -1,
    Optional As String = "",
    Optional As String = ""
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim MatType As eMatType
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.SetMaterial(Name,
    MatType, Color, Notes, GUID)
```

```
int SetMaterial(
    String^ Name,
    eMatType MatType,
    int Color = -1,
    String^ Notes = L"",
    String^ GUID = L""
)
```

```
abstract SetMaterial :
    Name : string *
    MatType : eMatType *
    ?Color : int *
    ?Notes : string *
```

cPropMaterialspan id="LSTA5E5EC2D_0"AddLanguageSpecificTextSet("LSTA5E5EC2D_0?cpnu=");S


```

        ?GUID : string
(* Defaults:
    let _Color = defaultArg Color -1
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of an existing or new material property. If this is an existing property, that property is modified; otherwise, a new property is added.

MatType

Type:Â ETABSV1eMatType

This is one of the items in the eMatType enumeration.

Color (Optional)

Type:Â SystemInt32

Notes (Optional)

Type:Â SystemString

GUID (Optional)

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the material. If this item is input as Default, the program assigns a GUID to the material.

Return Value

Type:Â Int32

Returns zero if the material is successfully initialized; otherwise it returns a nonzero value.

Remarks

This function is DEPRECATED. Please use AddMaterial(String, eMatType, String, String, String, String) instead.

Examples

VB

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model

```

Introduction

```
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'initialize new material property
ret = SapModel.PropMaterial.SetMaterial("Steel", eMatType.Steel)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPropMaterial Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LST5662E Method

Sets the material directional symmetry type to anisotropic, and assigns the anisotropic mechanical properties.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetMPAnisotropic(
    string Name,
    ref double[] E,
    ref double[] U,
    ref double[] A,
    ref double[] G,
    double Temp = 0
)
```

```
Function SetMPAnisotropic (
    Name As String,
    ByRef E As Double(),
    ByRef U As Double(),
    ByRef A As Double(),
    ByRef G As Double(),
    Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim E As Double()
Dim U As Double()
Dim A As Double()
Dim G As Double()
Dim Temp As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetMPAnisotropic(Name,
    E, U, A, G, Temp)
```

```
int SetMPAnisotropic(
    String^ Name,
    array<double>^% E,
    array<double>^% U,
    array<double>^% A,
    array<double>^% G,
    double Temp = 0
)
```

```
abstract SetMPAnisotropic :
```

cPropMaterialspan id="LST5662BE34_0"AddLanguageSpecificTextSet("LST5662BE34_0?cpp3495=.");Se

```

    Name : string *
    E : float[] byref *
    U : float[] byref *
    A : float[] byref *
    G : float[] byref *
    ?Temp : float
(* Defaults:
    let _Temp = defaultArg Temp 0
*)
-> int

```

Parameters

Name

Type: SystemString

The name of an existing material property.

E

Type: SystemDouble

This is an array that includes the modulus of elasticity.

Value Term

e(0) E1 [F/L2]

e(1) E2 [F/L2]

e(2) E3 [F/L2]

U

Type: SystemDouble

This is an array that includes poisson's ratio.

Value Term

u(0) U12

u(1) U13

u(2) U23

u(3) U14

u(4) U24

u(5) U34

u(6) U15

u(7) U25

u(8) U35

u(9) U45

u(10) U16

u(11) U26

u(12) U36

u(13) U46

u(14) U56

A

Type: SystemDouble

This is an array that includes the thermal coefficient.

Value Term

a(0) A1 [1/T]

a(1) A2 [1/T]

a(2) A3 [1/T]
 a(3) A12 [1/T]
 a(4) A13 [1/T]
 a(5) A23 [1/T]

G

Type: SystemDouble

This is an array that includes the shear modulus.

Value Term

g(0) G12 [F/L2]
 g(1) G13 [F/L2]
 g(2) G23 [F/L2]

Temp (Optional)

Type: SystemDouble

The temperature at which the specified data applies. The temperature must have been defined previously for the material.

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return Value

Type: Int32

Returns zero if the data is successfully assigned; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyE() As Double
    Dim MyU() As Double
    Dim MyA() As Double
    Dim MyG() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'initialize new material property
```

Introduction

```
ret = SapModel.PropMaterial.SetMaterial("Steel", eMatType.Steel)

'assign anisotropic mechanical properties
ReDim MyE(2)
ReDim MyU(14)
ReDim MyA(6)
ReDim MyG(2)
MyE(0)=30000
MyE(1)=10000
MyE(2)=2000
MyU(0)=0.2
MyU(1)=0.05
MyU(2)=0.1
MyU(3)=0
MyU(4)=0
MyU(5)=0
MyU(6)=0
MyU(7)=0
MyU(8)=0.01
MyU(9)=0
MyU(10)=0
MyU(11)=0
MyU(12)=0
MyU(13)=0
MyU(14)=0
MyA(0)=6.5E-6
MyA(1)=6.5E-6
MyA(2)=6.5E-6
MyA(3)=6.5E-6
MyA(4)=6.5E-6
MyA(5)=6.5E-6
MyG(0)=1500
MyG(1)=2500
MyG(2)=8700
ret = SapModel.PropMaterial.SetMPAnisotropic("Steel", MyE, MyU, MyA, MyG)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

See Also

Reference

[cPropMaterial Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LST188A5F3E_0") Method

Sets the material directional symmetry type to isotropic, and assigns the isotropic mechanical properties.

Namespace: `ETABSv1`

Assembly: `ETABSv1` (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetMPIsotropic(
    string Name,
    double E,
    double U,
    double A,
    double Temp = 0
)
```

```
Function SetMPIsotropic (
    Name As String,
    E As Double,
    U As Double,
    A As Double,
    Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim E As Double
Dim U As Double
Dim A As Double
Dim Temp As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetMPIsotropic(Name,
    E, U, A, Temp)
```

```
int SetMPIsotropic(
    String^ Name,
    double E,
    double U,
    double A,
    double Temp = 0
)
```

```
abstract SetMPIsotropic :
    Name : string *
    E : float *
    U : float *
    A : float *
```

cPropMaterialspan id="LST188A5F3E_0"AddLanguageSpecificTextSet("LST188A5F3E_0?cpp3499u=.");Se

```

        ?Temp : float
(* Defaults:
    let _Temp = defaultArg Temp 0
*)
-> int

```

Parameters

Name

Type:Â SystemString
The name of an existing material property.

E

Type:Â SystemDouble
The modulus of elasticity. [F/L2]

U

Type:Â SystemDouble
Poissonâs ratio.

A

Type:Â SystemDouble
The thermal coefficient. [1/T]

Temp (Optional)

Type:Â SystemDouble
The temperature at which the specified data applies. The temperature must have been defined previously for the material.

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return Value

Type:Â Int32

Returns zero if the data is successfully assigned; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

```


Introduction

```
'create steel deck template model
  ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'initialize new material property
  ret = SapModel.PropMaterial.SetMaterial("Steel", eMatType.Steel)

'assign isotropic mechanical properties
  ret = SapModel.PropMaterial.SetMPIsotropic("Steel", 29500, 0.25, 6E-06)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPropMaterial Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LST6CEB210_0?cpp=:3502.");SetMaterial

Method

Sets the material directional symmetry type to orthotropic, and assigns the orthotropic mechanical properties.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetMP0rthotropic(
    string Name,
    ref double[] E,
    ref double[] U,
    ref double[] A,
    ref double[] G,
    double Temp = 0
)
```

```
Function SetMP0rthotropic (
    Name As String,
    ByRef E As Double(),
    ByRef U As Double(),
    ByRef A As Double(),
    ByRef G As Double(),
    Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim E As Double()
Dim U As Double()
Dim A As Double()
Dim G As Double()
Dim Temp As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetMP0rthotropic(Name,
    E, U, A, G, Temp)
```

```
int SetMP0rthotropic(
    String^ Name,
    array<double>^% E,
    array<double>^% U,
    array<double>^% A,
    array<double>^% G,
    double Temp = 0
)
```

abstract SetMP0rthotropic :

cPropMaterialsid="LST6CEB210_0"AddLanguageSpecificTextSet("LST6CEB210_0?cpp=:3502.");SetMaterial

```

    Name : string *
    E : float[] byref *
    U : float[] byref *
    A : float[] byref *
    G : float[] byref *
    ?Temp : float
(* Defaults:
    let _Temp = defaultArg Temp 0
*)
-> int

```

Parameters

Name

Type: SystemString

The name of an existing material property.

E

Type: SystemDouble

This is an array that includes the modulus of elasticity.

Value Term

e(0) E1 [F/L2]

e(1) E2 [F/L2]

e(2) E3 [F/L2]

U

Type: SystemDouble

This is an array that includes poisson's ratio.

Value Term

u(0) U12

u(1) U13

u(2) U23

A

Type: SystemDouble

This is an array that includes the thermal coefficient.

Value Term

a(0) A1 [1/T]

a(1) A2 [1/T]

a(2) A3 [1/T]

G

Type: SystemDouble

This is an array that includes the shear modulus.

Value Term

g(0) G12 [F/L2]

g(1) G13 [F/L2]

g(2) G23 [F/L2]

Temp (Optional)

Type: SystemDouble

The temperature at which the specified data applies. The temperature must have been defined previously for the material.

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return Value

Type: $\hat{A}$ Int32

Returns zero if the data is successfully assigned; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MyE() As Double
    Dim MyU() As Double
    Dim MyA() As Double
    Dim MyG() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'initialize new material property
    ret = SapModel.PropMaterial.SetMaterial("Steel", eMatType.Steel)

    'assign anisotropic mechanical properties
    ReDim MyE(2)
    ReDim MyU(2)
    ReDim MyA(2)
    ReDim MyG(2)
    MyE(0)=30000
    MyE(1)=10000
    MyE(2)=2000
    MyU(0)=0.2
    MyU(1)=0.05
    MyU(2)=0.1
    MyA(0)=6.5E-6
    MyA(1)=6.5E-6
    MyA(2)=6.5E-6
    MyG(0)=1500
    MyG(1)=2500
    MyG(2)=8700
    ret = SapModel.PropMaterial.SetMPOrthotropic("Steel", MyE, MyU, MyA, MyG)

    'close ETABS
```

Return Value

Introduction

```
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

cPropMaterial Interface

ETABSv1 Namespace

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»

CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LST7E11A511_0?cpp3506=.");SetMPUniaxial

Method

Sets the material directional symmetry type to uniaxial, and assigns the uniaxial mechanical properties.

Namespace: `ETABSv1`

Assembly: `ETABSv1 (in ETABSv1.dll)` Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetMPUniaxial(
    string Name,
    double E,
    double A,
    double Temp = 0
)
```

```
Function SetMPUniaxial (
    Name As String,
    E As Double,
    A As Double,
    Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim E As Double
Dim A As Double
Dim Temp As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetMPUniaxial(Name,
    E, A, Temp)
```

```
int SetMPUniaxial(
    String^ Name,
    double E,
    double A,
    double Temp = 0
)
```

```
abstract SetMPUniaxial :
    Name : string *
    E : float *
    A : float *
    ?Temp : float
(* Defaults:
    let _Temp = defaultArg Temp 0
*)
-> int
```

cPropMaterialspan id="LST7E11A511_0"AddLanguageSpecificTextSet("LST7E11A511_0?cpp3506=.");SetMPUniaxial

Parameters

Name

Type:Â SystemString

The name of an existing material property.

E

Type:Â SystemDouble

The modulus of elasticity. [F/L2]

A

Type:Â SystemDouble

The thermal coefficient. [1/T]

Temp (Optional)

Type:Â SystemDouble

The temperature at which the specified data applies. The temperature must have been defined previously for the material.

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return ValueType:Â Int32

Returns zero if the data is successfully assigned; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'initialize new material property
    ret = SapModel.PropMaterial.SetMaterial("Steel", eMatType.Steel)

    'assign isotropic mechanical properties
    ret = SapModel.PropMaterial.SetMPUniaxial("Steel", 29500, 6E-06)

    'close ETABS
```

Introduction

```
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

cPropMaterial Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LST9148D40F_0?cpp3500=.");Set

Method

(NEWER FUNCTION AVAILABLE) Sets the other material property data for concrete materials.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int Set0Concrete(
    string Name,
    double Fc,
    bool IsLightweight,
    double FcsFactor,
    int SStype,
    int SSHysType,
    double StrainAtFc,
    double StrainUltimate,
    double FrictionAngle = 0,
    double DilatationalAngle = 0,
    double Temp = 0
)
```

```
Function Set0Concrete (
    Name As String,
    Fc As Double,
    IsLightweight As Boolean,
    FcsFactor As Double,
    SStype As Integer,
    SSHysType As Integer,
    StrainAtFc As Double,
    StrainUltimate As Double,
    FrictionAngle As Double = 0,
    DilatationalAngle As Double = 0,
    Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim Fc As Double
Dim IsLightweight As Boolean
Dim FcsFactor As Double
Dim SStype As Integer
Dim SSHysType As Integer
Dim StrainAtFc As Double
Dim StrainUltimate As Double
Dim FrictionAngle As Double
Dim DilatationalAngle As Double
Dim Temp As Double
```

cPropMaterialsid="LST9148D40F_0"AddLanguageSpecificTextSet("LST9148D40F_0?cpp3500=.");Set

Introduction

Dim returnValue As Integer

```
returnValue = instance.Set0Concrete(Name,  
Fc, IsLightweight, FcsFactor, SSType,  
SSHysType, StrainAtFc, StrainUltimate,  
FrictionAngle, DilatationalAngle,  
Temp)
```

```
int Set0Concrete(  
String^ Name,  
double Fc,  
bool IsLightweight,  
double FcsFactor,  
int SSType,  
int SSHysType,  
double StrainAtFc,  
double StrainUltimate,  
double FrictionAngle = 0,  
double DilatationalAngle = 0,  
double Temp = 0  
)
```

```
abstract Set0Concrete :  
    Name : string *  
    Fc : float *  
    IsLightweight : bool *  
    FcsFactor : float *  
    SSType : int *  
    SSHysType : int *  
    StrainAtFc : float *  
    StrainUltimate : float *  
    ?FrictionAngle : float *  
    ?DilatationalAngle : float *  
    ?Temp : float  
(* Defaults:  
    let _FrictionAngle = defaultArg FrictionAngle 0  
    let _DilatationalAngle = defaultArg DilatationalAngle 0  
    let _Temp = defaultArg Temp 0  
)  
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing concrete material property.

Fc

Type:Â SystemDouble

IsLightweight

Type:Â SystemBoolean

If this item is True, the concrete is assumed to be lightweight concrete.

FcsFactor

Type:Â SystemDouble

SSType

Type:Â SystemInt32

This is 0, 1 or 2, indicating the stress-strain curve type.

Value **SSType**

- 0 User defined
- 1 Parametric - Simple
- 2 Parametric - Mander

SSHysTypeType: SystemInt32

This is 0 through 7, indicating the stress-strain curve type.

Value SSHysType

- 0 Elastic
- 1 Kinematic
- 2 Takeda
- 3 Pivot
- 4 Concrete
- 5 BRB Hardening
- 6 Degradation
- 7 Isotropic

StrainAtFcType: SystemDouble**StrainUltimate**Type: SystemDouble

This item applies only to parametric stress-strain curves. It is the ultimate unconfined strain capacity. This item must be larger than the StrainAtfc item

FrictionAngle (Optional)Type: SystemDouble

The Drucker-Prager friction angle. This item must be smaller or equal to 0 and less than 90. [deg]

DilatationalAngle (Optional)Type: SystemDouble

The Drucker-Prager dilatational angle. This item must be smaller or equal to 0 and less than 90. [deg]

Temp (Optional)Type: SystemDouble

The temperature at which the specified data applies. The temperature must have been defined previously for the material.

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return ValueType: Int32

Returns zero if the data is successfully assigned; otherwise it returns a nonzero value.

▲ **Remarks**▲ **Examples**

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI
```

Introduction

```
Dim ret As Integer = -1

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'initialize new material property
ret = SapModel.PropMaterial.SetMaterial("Concrete", eMatType.Concrete)

'assign other properties
ret = SapModel.PropMaterial.Set0Concrete("Concrete", 5, False, 0, 1, 2, 0.0022, 0.0052)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

See Also

Reference

cPropMaterial Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LSTFA6A Method

Sets the other material property data for concrete materials.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int Set0Concrete_1(
    string Name,
    double Fc,
    bool IsLightweight,
    double FcsFactor,
    int SSType,
    int SSHysType,
    double StrainAtFc,
    double StrainUltimate,
    double FinalSlope,
    double FrictionAngle = 0,
    double DilatationalAngle = 0,
    double Temp = 0
)
```

```
Function Set0Concrete_1 (
    Name As String,
    Fc As Double,
    IsLightweight As Boolean,
    FcsFactor As Double,
    SSType As Integer,
    SSHysType As Integer,
    StrainAtFc As Double,
    StrainUltimate As Double,
    FinalSlope As Double,
    FrictionAngle As Double = 0,
    DilatationalAngle As Double = 0,
    Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim Fc As Double
Dim IsLightweight As Boolean
Dim FcsFactor As Double
Dim SSType As Integer
Dim SSHysType As Integer
Dim StrainAtFc As Double
Dim StrainUltimate As Double
Dim FinalSlope As Double
Dim FrictionAngle As Double
```

cPropMaterialspan id="LSTFA6AB5C5_0"AddLanguageSpecificTextSet("LSTFA6AB5C5_0?cp513u=");S

Introduction

```
Dim DilatationalAngle As Double
Dim Temp As Double
Dim returnValue As Integer

returnValue = instance.Set0Concrete_1(Name,
Fc, IsLightweight, FcsFactor, SSType,
SSHysType, StrainAtFc, StrainUltimate,
FinalSlope, FrictionAngle, DilatationalAngle,
Temp)

int Set0Concrete_1(
String^ Name,
double Fc,
bool IsLightweight,
double FcsFactor,
int SSType,
int SSHysType,
double StrainAtFc,
double StrainUltimate,
double FinalSlope,
double FrictionAngle = 0,
double DilatationalAngle = 0,
double Temp = 0
)

abstract Set0Concrete_1 :
    Name : string *
    Fc : float *
    IsLightweight : bool *
    FcsFactor : float *
    SSType : int *
    SSHysType : int *
    StrainAtFc : float *
    StrainUltimate : float *
    FinalSlope : float *
    ?FrictionAngle : float *
    ?DilatationalAngle : float *
    ?Temp : float
(* Defaults:
    let _FrictionAngle = defaultArg FrictionAngle 0
    let _DilatationalAngle = defaultArg DilatationalAngle 0
    let _Temp = defaultArg Temp 0
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing concrete material property.

Fc

Type:Â SystemDouble

IsLightweight

Type:Â SystemBoolean

If this item is True, the concrete is assumed to be lightweight concrete.

FcsFactor

Type:Â SystemDouble

SSType

Type: SystemInt32

This is 0, 1 or 2, indicating the stress-strain curve type.

Value SSType

- 0 User defined
- 1 Parametric - Simple
- 2 Parametric - Mander

SSHysType

Type: SystemInt32

This is 0 through 7, indicating the stress-strain curve type.

Value SSHysType

- 0 Elastic
- 1 Kinematic
- 2 Takeda
- 3 Pivot
- 4 Concrete
- 5 BRB Hardening
- 6 Degradation
- 7 Isotropic

StrainAtFc

Type: SystemDouble

StrainUltimate

Type: SystemDouble

This item applies only to parametric stress-strain curves. It is the ultimate unconfined strain capacity. This item must be larger than the StrainAtfc item

FinalSlope

Type: SystemDouble

This item applies only to parametric stress-strain curves. It is a multiplier on the material modulus of elasticity, E. This value multiplied times E gives the final slope on the compression side of the curve.

FrictionAngle (Optional)

Type: SystemDouble

The Drucker-Prager friction angle. This item must be smaller or equal to 0 and less than 90. [deg]

DilatationalAngle (Optional)

Type: SystemDouble

The Drucker-Prager dilatational angle. This item must be smaller or equal to 0 and less than 90. [deg]

Temp (Optional)

Type: SystemDouble

The temperature at which the specified data applies. The temperature must have been defined previously for the material.

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return Value

Type: Int32

Returns zero if the data is successfully assigned; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'initialize new material property
    ret = SapModel.PropMaterial.SetMaterial("Concrete", eMatType.Concrete)

    'assign other properties
    ret = SapModel.PropMaterial.Set0Concrete_1("Concrete", 5, False, 0, 1, 2, 0.0022, 0.0052,

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cPropMaterial Interface

ETABSv1 Namespace

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i>¿

CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LSTCFBEF908_0?cp3517u=.");S

Method

Sets the other material property data for no design type materials.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int Set0NoDesign(
    string Name,
    double FrictionAngle = 0,
    double DilatationalAngle = 0,
    double Temp = 0
)
```

```
Function Set0NoDesign (
    Name As String,
    Optional FrictionAngle As Double = 0,
    Optional DilatationalAngle As Double = 0,
    Optional Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim FrictionAngle As Double
Dim DilatationalAngle As Double
Dim Temp As Double
Dim returnValue As Integer
```

```
returnValue = instance.Set0NoDesign(Name,
    FrictionAngle, DilatationalAngle,
    Temp)
```

```
int Set0NoDesign(
    String^ Name,
    double FrictionAngle = 0,
    double DilatationalAngle = 0,
    double Temp = 0
)
```

```
abstract Set0NoDesign :
    Name : string *
    ?FrictionAngle : float *
    ?DilatationalAngle : float *
    ?Temp : float
(* Defaults:
    let _FrictionAngle = defaultArg FrictionAngle 0
    let _DilatationalAngle = defaultArg DilatationalAngle 0
    let _Temp = defaultArg Temp 0
```

cPropMaterialsid="LSTCFBEF908_0"AddLanguageSpecificTextSet("LSTCFBEF908_0?cp3517u=.");S

*)
-> int

Parameters

Name

Type:Â SystemString

The name of an existing concrete material property.

FrictionAngle (Optional)

Type:Â SystemDouble

The Drucker-Prager friction angle. This item must be smaller or equal to 0 and less than 90. [deg]

DilatationalAngle (Optional)

Type:Â SystemDouble

The Drucker-Prager dilatational angle. This item must be smaller or equal to 0 and less than 90. [deg]

Temp (Optional)

Type:Â SystemDouble

The temperature at which the specified data applies. The temperature must have been defined previously for the material.

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return Value

Type:Â Int32

Returns zero if the data is successfully assigned; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'initialize new material property
```

Introduction

```
ret = SapModel.PropMaterial.SetMaterial("NoDesign", eMatType.NoDesign)

'assign other properties
ret = SapModel.PropMaterial.Set0NoDesign("NoDesign", 10, 15)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPropMaterial Interface

ETABSv1 Namespace

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Namespace: ETABSV1

▲ Syntax

C#

VB

C++

F#

Copy

```
Function SetORebar (
Name As String,
Fy As Double,
Fu As Double,
EFy As Double,
EFu As Double,
SSType As Integer,
SSHysType As Integer,
StrainAtHardening As Double,
StrainUltimate As Double,
UseCaltransSSDefaults As Boolean,
OpenAS Double = 0
) As Integer
```

cPropMaterials;S

```

Dim returnValue As Integer

returnValue = instance.Set0Rebar(Name,
    Fy, Fu, EFy, EFu, SStype, SSHysType,
    StrainAtHardening, StrainUltimate,
    UseCaltransSSDefaults, Temp)

int Set0Rebar(
    String^ Name,
    double Fy,
    double Fu,
    double EFy,
    double EFu,
    int SStype,
    int SSHysType,
    double StrainAtHardening,
    double StrainUltimate,
    bool UseCaltransSSDefaults,
    double Temp = 0
)

abstract Set0Rebar :
    Name : string *
    Fy : float *
    Fu : float *
    EFy : float *
    EFu : float *
    SStype : int *
    SSHysType : int *
    StrainAtHardening : float *
    StrainUltimate : float *
    UseCaltransSSDefaults : bool *
    ?Temp : float
(* Defaults:
    let _Temp = defaultArg Temp 0
*)
-> int

```

Parameters

Name

Type: SystemString

The name of an existing rebar material property.

Fy

Type: SystemDouble

Fu

Type: SystemDouble

The minimum tensile stress. [F/L2]

EFy

Type: SystemDouble

EFu

Type: SystemDouble

SStype

Type: SystemInt32

This is 0, 1 or 2, indicating the stress-strain curve type.

Value **SStype**

0 User defined

- 1 Parametric - Simple
- 2 Parametric - Park

SSHysType

Type: SystemInt32

This is 0 through 7, indicating the stress-strain curve type.

Value SSHysType

- 0 Elastic
- 1 Kinematic
- 2 Takeda
- 3 Pivot
- 4 Concrete
- 5 BRB Hardening
- 6 Degrading
- 7 Isotropic

StrainAtHardening

Type: SystemDouble

This item applies only when parametric stress-strain curves are used and when UseCaltransSSDefaults is False. It is the strain at the onset of strain hardening.

StrainUltimate

Type: SystemDouble

This item applies only when parametric stress-strain curves are used and when UseCaltransSSDefaults is False. It is the ultimate strain capacity. This item must be larger than the StrainAtHardening item.

UseCaltransSSDefaults

Type: SystemBoolean

If this item is True, the program uses Caltrans default controlling strain values, which are bar size dependent.

Temp (Optional)

Type: SystemDouble

The temperature at which the specified data applies. The temperature must have been defined previously for the material.

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return Value

Type: Int32

Returns zero if the data is successfully assigned; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
```

Introduction

```
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'initialize new material property
ret = SapModel.PropMaterial.SetMaterial("Rebar", eMatType.Rebar)

'assign other properties
ret = SapModel.PropMaterial.Set0Rebar("Rebar", 62, 93, 70, 102, 2, 2, 0.02, 0.1, False)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPropMaterial Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LST9218D0BD_0") Method

Sets the other material property data for rebar materials.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int Set0Rebar_1(
    string Name,
    double Fy,
    double Fu,
    double EFy,
    double EFu,
    int SStype,
    int SSHysType,
    double StrainAtHardening,
    double StrainUltimate,
    double FinalSlope,
    bool UseCaltransSSDefaults,
    double Temp = 0
)
```

```
Function Set0Rebar_1 (
    Name As String,
    Fy As Double,
    Fu As Double,
    EFy As Double,
    EFu As Double,
    SStype As Integer,
    SSHysType As Integer,
    StrainAtHardening As Double,
    StrainUltimate As Double,
    FinalSlope As Double,
    UseCaltransSSDefaults As Boolean,
    Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim Fy As Double
Dim Fu As Double
Dim EFy As Double
Dim EFu As Double
Dim SStype As Integer
Dim SSHysType As Integer
Dim StrainAtHardening As Double
Dim StrainUltimate As Double
Dim FinalSlope As Double
```

cPropMaterialspan id="LST9218D0BD_0"AddLanguageSpecificTextSet("LST9218D0BD_0?cpp3524u=.");S

Introduction

```
Dim UseCaltransSSDefaults As Boolean
Dim Temp As Double
Dim returnValue As Integer

returnValue = instance.Set0Rebar_1(Name,
Fy, Fu, EFy, EFu, SStype, SSHysType,
StrainAtHardening, StrainUltimate,
FinalSlope, UseCaltransSSDefaults,
Temp)

int Set0Rebar_1(
String^ Name,
double Fy,
double Fu,
double EFy,
double EFu,
int SStype,
int SSHysType,
double StrainAtHardening,
double StrainUltimate,
double FinalSlope,
bool UseCaltransSSDefaults,
double Temp = 0
)

abstract Set0Rebar_1 :
    Name : string *
    Fy : float *
    Fu : float *
    EFy : float *
    EFu : float *
    SStype : int *
    SSHysType : int *
    StrainAtHardening : float *
    StrainUltimate : float *
    FinalSlope : float *
    UseCaltransSSDefaults : bool *
    ?Temp : float
(* Defaults:
    let _Temp = defaultArg Temp 0
*)
-> int
```

Parameters

Name	Type:Â <u>SystemString</u> The name of an existing rebar material property.
Fy	Type:Â <u>SystemDouble</u>
Fu	Type:Â <u>SystemDouble</u> The minimum tensile stress. [F/L2]
EFy	Type:Â <u>SystemDouble</u>
EFu	Type:Â <u>SystemDouble</u>
SStype	

Type: SystemInt32

This is 0, 1 or 2, indicating the stress-strain curve type.

Value SSType

- 0 User defined
- 1 Parametric - Simple
- 2 Parametric - Park

SSHysType

Type: SystemInt32

This is 0 through 7, indicating the stress-strain curve type.

Value SSHysType

- 0 Elastic
- 1 Kinematic
- 2 Takeda
- 3 Pivot
- 4 Concrete
- 5 BRB Hardening
- 6 Degrading
- 7 Isotropic

StrainAtHardening

Type: SystemDouble

This item applies only when parametric stress-strain curves are used and when UseCaltransSSDefaults is False. It is the strain at the onset of strain hardening.

StrainUltimate

Type: SystemDouble

This item applies only when parametric stress-strain curves are used and when UseCaltransSSDefaults is False. It is the ultimate strain capacity. This item must be larger than the StrainAtHardening item.

FinalSlope

Type: SystemDouble

This item applies only to parametric stress-strain curves. It is a multiplier on the material modulus of elasticity, E. This value multiplied times E gives the final slope of the curve.

UseCaltransSSDefaults

Type: SystemBoolean

If this item is True, the program uses Caltrans default controlling strain values, which are bar size dependent.

Temp (Optional)

Type: SystemDouble

The temperature at which the specified data applies. The temperature must have been defined previously for the material.

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return Value

Type: Int32

Returns zero if the data is successfully assigned; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'initialize new material property
    ret = SapModel.PropMaterial.SetMaterial("Rebar", eMatType.Rebar)

    'assign other properties
    ret = SapModel.PropMaterial.SetORebar_1("Rebar", 62, 93, 70, 102, 2, 2, 0.02, 0.1, -0.1, F

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cPropMaterial Interface

ETABSv1 Namespace

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»

CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LST582D7044_0?cpp=3528=.");Set

Method

(NEWER FUNCTION AVAILABLE) Sets the other material property data for steel materials.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int Set0Steel(
    string Name,
    double Fy,
    double Fu,
    double EFy,
    double EFu,
    int SStype,
    int SSHysType,
    double StrainAtHardening,
    double StrainAtMaxStress,
    double StrainAtRupture,
    double Temp = 0
)
```

```
Function Set0Steel (
    Name As String,
    Fy As Double,
    Fu As Double,
    EFy As Double,
    EFu As Double,
    SStype As Integer,
    SSHysType As Integer,
    StrainAtHardening As Double,
    StrainAtMaxStress As Double,
    StrainAtRupture As Double,
    Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim Fy As Double
Dim Fu As Double
Dim EFy As Double
Dim EFu As Double
Dim SStype As Integer
Dim SSHysType As Integer
Dim StrainAtHardening As Double
Dim StrainAtMaxStress As Double
Dim StrainAtRupture As Double
Dim Temp As Double
```

cPropMaterialsid="LST582D7044_0"AddLanguageSpecificTextSet("LST582D7044_0?cpp=3528=.");Set

```

Dim returnValue As Integer

returnValue = instance.Set0Steel(Name,
    Fy, Fu, EFy, EFu, SStype, SSHysType,
    StrainAtHardening, StrainAtMaxStress,
    StrainAtRupture, Temp)

int Set0Steel(
    String^ Name,
    double Fy,
    double Fu,
    double EFy,
    double EFu,
    int SStype,
    int SSHysType,
    double StrainAtHardening,
    double StrainAtMaxStress,
    double StrainAtRupture,
    double Temp = 0
)

abstract Set0Steel :
    Name : string *
    Fy : float *
    Fu : float *
    EFy : float *
    EFu : float *
    SStype : int *
    SSHysType : int *
    StrainAtHardening : float *
    StrainAtMaxStress : float *
    StrainAtRupture : float *
    ?Temp : float
(* Defaults:
    let _Temp = defaultArg Temp 0
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of an existing steel material property.

Fy

Type:Â SystemDouble

Fu

Type:Â SystemDouble

The minimum tensile stress. [F/L2]

EFy

Type:Â SystemDouble

EFu

Type:Â SystemDouble

SStype

Type:Â SystemInt32

This is 0 or 1, indicating the stress-strain curve type.

Value **SStype**

0 User defined

1 Parametric - Simple

SSHysType

Type:Â SystemInt32

This is 0 through 7, indicating the stress-strain curve type.

Value SSHysType

- 0 Elastic
- 1 Kinematic
- 2 Takeda
- 3 Pivot
- 4 Concrete
- 5 BRB Hardening
- 6 Degrading
- 7 Isotropic

StrainAtHardening

Type:Â SystemDouble

This item applies only when parametric stress-strain curves are used. It is the strain at the onset of strain hardening.

StrainAtMaxStress

Type:Â SystemDouble

This item applies only when parametric stress-strain curves are used. It is the strain at maximum stress. This item must be larger than the StrainAtHardening item.

StrainAtRupture

Type:Â SystemDouble

This item applies only when parametric stress-strain curves are used. It is the strain at rupture. This item must be larger than the StrainAtMaxStress item.

Temp (Optional)

Type:Â SystemDouble

The temperature at which the specified data applies. The temperature must have been defined previously for the material.

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return ValueType:Â Int32

Returns zero if the data is successfully assigned; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
```

Introduction

```
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'initialize new material property
ret = SapModel.PropMaterial.SetMaterial("Steel", eMatType.Steel)

'assign other properties
ret = SapModel.PropMaterial.Set0Steel("Steel", 55, 68, 60, 70, 1, 2, 0.02, 0.1, 0.2)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPropMaterial Interface](#)

[ETABSv1 Namespace](#)

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Namespace: ETABSv1

▲ Syntax

C#

VBC++F#

Copy

```
Function Set0Steel_1 (
  Name As String,
  Fy As Double,
  Fu As Double,
  EFy As Double,
  EFu As Double,
  SStype As Integer,
  SSHysType As Integer,
  StrainAtHardening As Double,
  StrainAtMaxStress As Double,
  StrainAtRupture As Double,
  FinalSlope As Double,
  OpennAs Double = 0
) As Integer
```

cPropMaterialsAddLanguageSpecificTextSet("LSTBAF03588_0?cpp3582u=");Se

Introduction

```
Dim FinalSlope As Double
Dim Temp As Double
Dim returnValue As Integer

returnValue = instance.Set0Steel_1(Name,
Fy, Fu, EFy, EFu, SStype, SSHysType,
StrainAtHardening, StrainAtMaxStress,
StrainAtRupture, FinalSlope, Temp)

int Set0Steel_1(
String^ Name,
double Fy,
double Fu,
double EFy,
double EFu,
int SStype,
int SSHysType,
double StrainAtHardening,
double StrainAtMaxStress,
double StrainAtRupture,
double FinalSlope,
double Temp = 0
)

abstract Set0Steel_1 :
    Name : string *
    Fy : float *
    Fu : float *
    EFy : float *
    EFu : float *
    SStype : int *
    SSHysType : int *
    StrainAtHardening : float *
    StrainAtMaxStress : float *
    StrainAtRupture : float *
    FinalSlope : float *
    ?Temp : float
(* Defaults:
    let _Temp = defaultArg Temp 0
*)
-> int
```

Parameters

Name	Type:Â <u>SystemString</u> The name of an existing steel material property.
Fy	Type:Â <u>SystemDouble</u>
Fu	Type:Â <u>SystemDouble</u> The minimum tensile stress. [F/L2]
EFy	Type:Â <u>SystemDouble</u>
EFu	Type:Â <u>SystemDouble</u>
SStype	

Type: SystemInt32

This is 0 or 1, indicating the stress-strain curve type.

Value SSType

- 0 User defined
- 1 Parametric - Simple

SSHysType

Type: SystemInt32

This is 0 through 7, indicating the stress-strain curve type.

Value SSHysType

- 0 Elastic
- 1 Kinematic
- 2 Takeda
- 3 Pivot
- 4 Concrete
- 5 BRB Hardening
- 6 Degradation
- 7 Isotropic

StrainAtHardening

Type: SystemDouble

This item applies only when parametric stress-strain curves are used. It is the strain at the onset of strain hardening.

StrainAtMaxStress

Type: SystemDouble

This item applies only when parametric stress-strain curves are used. It is the strain at maximum stress. This item must be larger than the StrainAtHardening item.

StrainAtRupture

Type: SystemDouble

This item applies only when parametric stress-strain curves are used. It is the strain at rupture. This item must be larger than the StrainAtMaxStress item.

FinalSlope

Type: SystemDouble

This item applies only to parametric stress-strain curves. It is a multiplier on the material modulus of elasticity, E. This value multiplied times E gives the final slope of the curve.

Temp (Optional)

Type: SystemDouble

The temperature at which the specified data applies. The temperature must have been defined previously for the material.

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return Value

Type: Int32

Returns zero if the data is successfully assigned; otherwise it returns a nonzero value.

Remarks

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'initialize new material property
    ret = SapModel.PropMaterial.SetMaterial("Steel", eMatType.Steel)

    'assign other properties
    ret = SapModel.PropMaterial.Set0Steel_1("Steel", 55, 68, 60, 70, 1, 2, 0.02, 0.1, 0.2, -0.01)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

cPropMaterial Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LSTDCD33F6_0") Method

(NEWER FUNCTION AVAILABLE) Sets the other material property data for tendon materials.

Namespace: $\hat{A}$ ETABSV1

Assembly: $\hat{A}$ ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int Set0Tendon(
    string Name,
    double Fy,
    double Fu,
    int SStype,
    int SSHysType,
    double Temp = 0
)
```

```
Function Set0Tendon (
    Name As String,
    Fy As Double,
    Fu As Double,
    SStype As Integer,
    SSHysType As Integer,
    Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim Fy As Double
Dim Fu As Double
Dim SStype As Integer
Dim SSHysType As Integer
Dim Temp As Double
Dim returnValue As Integer
```

```
returnValue = instance.Set0Tendon(Name,
    Fy, Fu, SStype, SSHysType, Temp)
```

```
int Set0Tendon(
    String^ Name,
    double Fy,
    double Fu,
    int SStype,
    int SSHysType,
    double Temp = 0
)
```

abstract Set0Tendon :

cPropMaterialsspan id="LSTDCD33F6_0"AddLanguageSpecificTextSet("LSTDCD33F6_0?cpp=3586-");Set0Tendon("LSTDCD33F6_0", 0, 0, 0, 0, 0);

```

    Name : string *
    Fy : float *
    Fu : float *
    SStype : int *
    SSHysType : int *
    ?Temp : float
(* Defaults:
    let _Temp = defaultArg Temp 0
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of an existing tendon material property.

Fy

Type:Â SystemDouble

Fu

Type:Â SystemDouble

The minimum tensile stress. [F/L2]

SStype

Type:Â SystemInt32

This is 0, 1 or 2, indicating the stress-strain curve type.

Value	SStype
0	User defined
1	Parametric - 250 ksi strand
1	Parametric - 270 ksi strand

SSHysType

Type:Â SystemInt32

This is 0 through 7, indicating the stress-strain curve type.

Value	SSHysType
0	Elastic
1	Kinematic
2	Takeda
3	Pivot
4	Concrete
5	BRB Hardening
6	Degrading
7	Isotropic

Temp (Optional)

Type:Â SystemDouble

The temperature at which the specified data applies. The temperature must have been defined previously for the material.

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return Value

Type: Int32

Returns zero if the data is successfully assigned; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'initialize new material property
    ret = SapModel.PropMaterial.SetMaterial("Tendon", eMatType.Tendon)

    'assign other properties
    ret = SapModel.PropMaterial.SetOTendon("Tendon", 230, 255, 1, 1)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cPropMaterial Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LSTEF67C5E_0") Method

Sets the other material property data for tendon materials.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int Set0Tendon_1(
    string Name,
    double Fy,
    double Fu,
    int SStype,
    int SSHysType,
    double FinalSlope,
    double Temp = 0
)
```

```
Function Set0Tendon_1 (
    Name As String,
    Fy As Double,
    Fu As Double,
    SStype As Integer,
    SSHysType As Integer,
    FinalSlope As Double,
    Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim Fy As Double
Dim Fu As Double
Dim SStype As Integer
Dim SSHysType As Integer
Dim FinalSlope As Double
Dim Temp As Double
Dim returnValue As Integer
```

```
returnValue = instance.Set0Tendon_1(Name,
    Fy, Fu, SStype, SSHysType, FinalSlope,
    Temp)
```

```
int Set0Tendon_1(
    String^ Name,
    double Fy,
    double Fu,
    int SStype,
    int SSHysType,
    double FinalSlope,
```

cPropMaterialspan id="LSTEF67C5E_0"AddLanguageSpecificTextSet("LSTEF67C5E_0?cp539u=");S

```

double Temp = 0
)

abstract Set0Tendon_1 :
  Name : string *
  Fy : float *
  Fu : float *
  SStype : int *
  SSHysType : int *
  FinalSlope : float *
  ?Temp : float
(* Defaults:
    let _Temp = defaultArg Temp 0
*)
-> int

```

Parameters

Name

Type: SystemString

The name of an existing tendon material property.

Fy

Type: SystemDouble

Fu

Type: SystemDouble

The minimum tensile stress. [F/L2]

SStype

Type: SystemInt32

This is 0, 1 or 2, indicating the stress-strain curve type.

Value	SStype
0	User defined
1	Parametric - 250 ksi strand
1	Parametric - 270 ksi strand

SSHysType

Type: SystemInt32

This is 0 through 7, indicating the stress-strain curve type.

Value	SSHysType
0	Elastic
1	Kinematic
2	Takeda
3	Pivot
4	Concrete
5	BRB Hardening
6	Degrading
7	Isotropic

FinalSlope

Type: SystemDouble

This item applies only to parametric stress-strain curves. It is a multiplier on the material modulus of elasticity, E. This value multiplied times E gives the final slope of the curve.

Temp (Optional)Type: SystemDouble

The temperature at which the specified data applies. The temperature must have been defined previously for the material.

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return ValueType: Int32

Returns zero if the data is successfully assigned; otherwise it returns a nonzero value.

▲ **Remarks**▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'initialize new material property
    ret = SapModel.PropMaterial.SetMaterial("Tendon", eMatType.Tendon)

    'assign other properties
    ret = SapModel.PropMaterial.SetOTendon_1("Tendon", 230, 255, 1, 1, -0.1)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cPropMaterial Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LST8E4EAB9D_0?cpu=.") Method

Sets the material stress-strain curve for materials that are specified to have user-defined stress-strain curves.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetSSCurve(
    string Name,
    int NumberPoints,
    ref int[] PointID,
    ref double[] Strain,
    ref double[] Stress,
    double Temp = 0
)
```

```
Function SetSSCurve (
    Name As String,
    NumberPoints As Integer,
    ByRef PointID As Integer(),
    ByRef Strain As Double(),
    ByRef Stress As Double(),
    Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim NumberPoints As Integer
Dim PointID As Integer()
Dim Strain As Double()
Dim Stress As Double()
Dim Temp As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetSSCurve(Name,
    NumberPoints, PointID, Strain, Stress,
    Temp)
```

```
int SetSSCurve(
    String^ Name,
    int NumberPoints,
    array<int>^% PointID,
    array<double>^% Strain,
    array<double>^% Stress,
    double Temp = 0
)
```

```

abstract SetSSCurve :
    Name : string *
    NumberPoints : int *
    PointID : int[] byref *
    Strain : float[] byref *
    Stress : float[] byref *
    ?Temp : float
(* Defaults:
    let _Temp = defaultArg Temp 0
*)
-> int

```

Parameters

Name

Type: SystemString

The name of an existing material property.

NumberPoints

Type: SystemInt32

The number of points in the stress-strain curve. This item must be at least 3.

PointID

Type: SystemInt32

This is one of the following integers which sets the point ID. The point ID controls the color that will be displayed for hinges in a deformed shape plot.

Value Point ID

-5	-E
-4	-D
-3	-C
-2	-B
-1	-A
0	None
1	A
2	B
3	C
4	D
5	E

The point IDs must be input in numerically increasing order, except that 0 (None) values are allowed anywhere. No duplicate values are allowed except for 0 (None).

Strain

Type: SystemDouble

This is an array that includes the strain at each point on the stress strain curve. The strains must increase monotonically.

Stress

Type: SystemDouble

This is an array that includes the stress at each point on the stress strain curve. [F/L2]

Points that have a negative strain must have a zero or negative stress. Similarly, points that have a positive strain must have a zero or positive stress.

There must be one point defined that has zero strain and zero stress.

Temp (Optional)

Type: SystemDouble

The temperature at which the specified data applies. The temperature must have been defined previously for the material.

This item applies only if the specified material has properties that are temperature dependent. That is, it applies only if properties are specified for the material at more than one temperature.

Return Value

Type: Int32

Returns zero if the data is successfully assigned; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim PointID() As Integer
    Dim Strain() As Double
    Dim Stress() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'initialize new material property
    ret = SapModel.PropMaterial.SetMaterial("Steel", eMatType.Steel)

    'assign other properties
    ret = SapModel.PropMaterial.Set0Steel("Steel", 55, 68, 60, 70, 0, 1, 0, 0, 0)

    'assign user SS curve
    ReDim PointID(4)
    ReDim Strain(4)
    ReDim Stress(4)
    Strain(0) = -0.003: Stress(0) = -50: PointID(0) = -3
    Strain(1) = -0.001: Stress(1) = -25: PointID(1) = 0
    Strain(2) = 0: Stress(2) = -0: PointID(2) = 1
    Strain(3) = 0.003: Stress(3) = 40: PointID(3) = 0
    Strain(4) = 0.008: Stress(4) = 80: PointID(4) = 5
    ret = SapModel.PropMaterial.SetSSCurve("Steel", 5, PointID, Strain, Stress)
```

```
'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPropMaterial Interface

ETABSv1 Namespace

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»

CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LST3625F9C9_0?cpp3547=.");SetTemp

Method

assigns the temperatures at which properties are specified for a material. This data is required only for materials whose properties are temperature dependent.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetTemp(
    string Name,
    int NumberItems,
    ref double[] Temp
)
```

```
Function SetTemp (
    Name As String,
    NumberItems As Integer,
    ByRef Temp As Double()
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim NumberItems As Integer
Dim Temp As Double()
Dim returnValue As Integer
```

```
returnValue = instance.SetTemp(Name, NumberItems,
    Temp)
```

```
int SetTemp(
    String^ Name,
    int NumberItems,
    array<double>^% Temp
)
```

```
abstract SetTemp :
    Name : string *
    NumberItems : int *
    Temp : float[] byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing material property.

NumberItems

cPropMaterialspan id="LST3625F9C9_0"AddLanguageSpecificTextSet("LST3625F9C9_0?cpp3547=.");SetTemp

Type:Â SystemInt32

The number of different temperatures at which properties are specified for the material.

Temp

Type:Â SystemDouble

This is an array that includes the different temperatures at which properties are specified for the material.

Return Value

Type:Â Int32

Returns zero if the temperatures are successfully set; otherwise it returns a nonzero value

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'initialize new material property
    ret = SapModel.PropMaterial.SetMaterial("Steel", eMatType.Steel)

    'specify temps at which properties will be provided
    ReDim MyTemp(2)
    MyTemp(0) = 0
    MyTemp(1) = 50
    MyTemp(2) = 100
    ret = SapModel.PropMaterial.SetTemp("Steel", 3, MyTemp)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPropMaterial Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropMaterialAddLanguageSpecificTextSet("LST36C6A362_0?cpp3550u=.");Set

Method

Assigns weight per unit volume or mass per unit volume to a material property.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetWeightAndMass(
    string Name,
    int MyOption,
    double Value,
    double Temp = 0
)
```

```
Function SetWeightAndMass (
    Name As String,
    MyOption As Integer,
    Value As Double,
    Temp As Double = 0
) As Integer
```

```
Dim instance As cPropMaterial
Dim Name As String
Dim MyOption As Integer
Dim Value As Double
Dim Temp As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetWeightAndMass(Name,
    MyOption, Value, Temp)
```

```
int SetWeightAndMass(
    String^ Name,
    int MyOption,
    double Value,
    double Temp = 0
)
```

```
abstract SetWeightAndMass :
    Name : string *
    MyOption : int *
    Value : float *
    ?Temp : float
(* Defaults:
    let _Temp = defaultArg Temp 0
*)
-> int
```

cPropMaterialspan id="LST36C6A362_0"AddLanguageSpecificTextSet("LST36C6A362_0?cpp3550u=.");Set

Parameters

Name

Type:Â SystemString

The name of an existing material property.

MyOption

Type:Â SystemInt32

This is either 1 or 2, indicating what is specified by the Value item.

Value**Option**

1 Weight per unit volume is specified

2 Mass per unit volume is specified

Value

Type:Â SystemDouble

This is either the weight per unit volume or the mass per unit volume, depending on the value of the MyOption item. [F/L3] for MyOption = 1 (weight), and [M/L3] for MyOption = 2 (mass)

Temp (Optional)

Type:Â SystemDouble**Return Value**Type:Â Int32

Returns zero if the data is successfully assigned; otherwise it returns a nonzero value.

▲Remarks

▲Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'initialize new material property
    ret = SapModel.PropMaterial.SetMaterial("Steel", eMatType.Steel)

    'assign material property weight per unit volume
    ret = SapModel.PropMaterial.SetWeightAndMass("Steel", 1, 0.00029)

    'close ETABS
```

Introduction

```
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

cPropMaterial Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropPointSpring Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cPropPointSpring
```

```
Public Interface cPropPointSpring
```

```
Dim instance As cPropPointSpring
```

```
public interface class cPropPointSpring
```

```
type cPropPointSpring = interface end
```

The cPropPointSpring type exposes the following members.

▲ Methods

▲	Name	Description
⇒	ChangeName	Changes the name of a defined point spring property
⇒	Delete	Deletes the specified point spring property
⇒	GetLinks	Retrieves the single joint links for a named point spring property
⇒	GetNameList	Retrieves the names of all defined point spring property
⇒	GetPointSpringProp	Retrieves an existing named point spring property
⇒	SetLinks	Sets single joint links for a named point spring property
⇒	SetPointSpringProp	Creates a new named point spring property, or modifies an existing named point spring property

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropPointSpring Methods

The [cPropPointSpring](#) type exposes the following members.

Methods

⌚	Name	Description
⇒	ChangeName	Changes the name of a defined point spring property
⇒	Delete	Deletes the specified point spring property
⇒	GetLinks	Retrieves the single joint links for a named point spring property
⇒	GetNameList	Retrieves the names of all defined point spring property
⇒	GetPointSpringProp	Retrieves an existing named point spring property
⇒	SetLinks	Sets single joint links for a named point spring property
⇒	SetPointSpringProp	Creates a new named point spring property, or modifies an existing named point spring property

[Top](#)

See Also

Reference

[cPropPointSpring Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cPropPointSpringAddLanguageSpecificTextSet("LST5D274749_0") Method

Changes the name of a defined point spring property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int ChangeName(
    string Name,
    string NewName
)
```

```
Function ChangeName (
    Name As String,
    NewName As String
) As Integer
```

```
Dim instance As cPropPointSpring
Dim Name As String
Dim NewName As String
Dim returnValue As Integer
```

```
returnValue = instance.ChangeName(Name,
    NewName)
```

```
int ChangeName(
    String^ Name,
    String^ NewName
)
```

```
abstract ChangeName :
    Name : string *
    NewName : string -> int
```

Parameters

Name

Type: SystemString

The existing name of a defined point spring property

NewName

Type: SystemString

The new name for the point spring property

Return Value

Type: Int32

Returns zero if the new name is successfully applied, otherwise it returns a nonzero value.

▀ Remarks

▀ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new spring property
    Dim k() as Double
    ReDim k(5)
    k(2) = 1
    ret = SapModel.PropPointSpring.SetPointSpringProp("mySpringProp1", 1, k)

    'change property name
    ret = SapModel.PropPointSpring.ChangeName("mySpringProp1", "prop1")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▀ See Also

Reference

cPropPointSpring Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropPointSpringAddLanguageSpecificTextSet("LST6297AEB2_0") Method

Deletes the specified point spring property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int Delete(
    string Name
)
```

```
Function Delete (
    Name As String
) As Integer
```

```
Dim instance As cPropPointSpring
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.Delete(Name)
```

```
int Delete(
    String^ Name
)
```

```
abstract Delete :
    Name : string -> int
```

Parameters

Name

Type: SystemString

The name of an existing point spring property

Return Value

Type: Int32

Returns zero if the point spring property is successfully deleted, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
```

cPropPointSpringspan id="LST6297AEB2_0"AddLanguageSpecificTextSet("LST6297AEB2_0?3558::|nu=".

Introduction

```
Dim SapModel As cSapModel
Dim EtabsObject As cOAPI
Dim ret As Integer = -1

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'define new spring property
Dim k() as Double
ReDim k(5)
k(2) = 1
ret = SapModel.PropPointSpring.SetPointSpringProp("mySpringProp1", 1, k)

'delete property
ret = SapModel.PropPointSpring.Delete("mySpringProp1")

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

[cPropPointSpring Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cPropPointSpringAddLanguageSpecificTextSet("LST317A7AA4_0?3560::|nu="

Method

Retrieves the single joint links for a named point spring property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetLinks(
    string Name,
    ref int NumberLinks,
    ref string[] LinkNames,
    ref int[] LinkAxialDirs,
    ref double[] LinkAngles
)
```

```
Function GetLinks (
    Name As String,
    ByRef NumberLinks As Integer,
    ByRef LinkNames As String(),
    ByRef LinkAxialDirs As Integer(),
    ByRef LinkAngles As Double()
) As Integer
```

```
Dim instance As cPropPointSpring
Dim Name As String
Dim NumberLinks As Integer
Dim LinkNames As String()
Dim LinkAxialDirs As Integer()
Dim LinkAngles As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetLinks(Name,
    NumberLinks, LinkNames, LinkAxialDirs,
    LinkAngles)
```

```
int GetLinks(
    String^ Name,
    int% NumberLinks,
    array<String^>^% LinkNames,
    array<int>^% LinkAxialDirs,
    array<double>^% LinkAngles
)
```

```
abstract GetLinks :
    Name : string *
    NumberLinks : int byref *
    LinkNames : string[] byref *
    LinkAxialDirs : int[] byref *
```

cPropPointSpringspan id="LST317A7AA4_0"AddLanguageSpecificTextSet("LST317A7AA4_0?3560::|nu="

LinkAngles : float[] byref -> int

Parameters

Name

Type:Â SystemString

The name of an existing point spring property. This spring property must be of type option 1

NumberLinks

Type:Â SystemInt32

The number of links

LinkNames

Type:Â SystemString

This array contains the names of existing link properties

LinkAxialDirs

Type:Â SystemInt32

This array contains integers representing the axial direction of their respective link properties and can take one of six values:

◇ -1 : -X

◇ -2 : -Y

◇ -3 : -Z

◇ 1 : +X

◇ 2 : +Y

◇ 3 : +Z

LinkAngles

Type:Â SystemDouble

This array contains the axis 2 angles of their respective link properties [deg]

Return Value

Type:Â Int32

Returns zero if the links are successfully retrieved, otherwise returns nonzero

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
```

```
'create ETABS object
```

```
    Dim myHelper as cHelper = New Helper
```

```
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

```
'start ETABS application
```

```
    ret = EtabsObject.ApplicationStart()
```

```
'create SapModel object
```

```
    SapModel = EtabsObject.SapModel
```

```
'initialize model
```

```
    ret = SapModel.InitializeNewModel()
```

Introduction

```
'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'define new link property
    Dim DOF() As Boolean
    Dim Fixed() As Boolean
    Dim Ke() As Double
    Dim Ce() As Double
    ReDim DOF(5)
    ReDim Fixed(5)
    ReDim Ke(5)
    ReDim Ce(5)
    DOF(0) = True
    Ke(0) = 12
    ret = SapModel.PropLink.SetLinear("myLink1", DOF, Fixed, Ke, Ce, 0, 0)

'define new spring property
    Dim k() as Double
    ReDim k(5)
    k(2) = 1
    ret = SapModel.PropPointSpring.SetPointSpringProp("myPtSpringProp", 1, k)

'set links in spring property
    Dim NumberLinks As Integer
    Dim LinkNames() As String
    Dim LinkAxialDirs() As Integer
    Dim LinkAngles() As Double

    NumberLinks = 1
    ReDim LinkNames(NumberLinks - 1)
    ReDim LinkAxialDirs(NumberLinks - 1)
    ReDim LinkAngles(NumberLinks - 1)
    LinkNames(0) = "myLink1"
    LinkAxialDirs(0) = -2
    LinkAngles(0) = 45

    ret = SapModel.PropPointSpring.SetLinks("myPtSpringProp", NumberLinks, LinkNames, LinkAxialDirs, LinkAngles)

'get links
    Dim outNumberLinks As Integer
    Dim outLinkNames() As String
    Dim outLinkAxialDirs() As Integer
    Dim outLinkAngles() As Double

    ret = SapModel.PropPointSpring.GetLinks("myPtSpringProp", outNumberLinks, outLinkNames, outLinkAxialDirs, outLinkAngles)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

🔗 See Also

Reference

[cPropPointSpring Interface](#)
[ETABSv1 Namespace](#)

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Namespace: ETABSv1

▲ Syntax

VB

F#

Copy

```
Function GetNameList (
ByRef NumberNames As Integer,
ByRef MyName As String()
) As Integer
```

```
int GetNameList(
int% NumberNames,
array<String^>^% MyName
)
```

Parameters

Type: SystemInt32

MyName

Type: SystemString

```
Dim MyName() as String
```


Introduction

The array is dimensioned to (NumberNames + 1) inside the program, filled with the names, and returned to the API user.

Return Value

Type: Int32

Returns zero if the names are successfully retrieved, otherwise it returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)  
  
    'define new spring property  
    Dim k() as Double  
    ReDim k(5)  
    k(2) = 1  
    ret = SapModel.PropPointSpring.SetPointSpringProp("mySpringProp1", 1, k)  
  
    'get name list of properties  
    Dim NumberNames as Integer  
    Dim MyName() as String  
    ret = SapModel.PropPointSpring.GetNameList(NumberNames, MyName)  
  
    'close ETABS  
    EtabsObject.ApplicationExit(False)  
  
    'clean up variables  
    SapModel = Nothing  
    EtabsObject = Nothing  
  
End Sub
```

▲ See Also

Reference

cPropPointSpring Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropPointSpringAddLanguageSpecificTextSet("LSTB Method

Retrieves an existing named point spring property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetPointSpringProp(
    string Name,
    ref int SpringOption,
    ref double[] k,
    ref string CSys,
    ref string SoilProfile,
    ref string Footing,
    ref double Period,
    ref int color,
    ref string notes,
    ref string iGUID
)
```

```
Function GetPointSpringProp (
    Name As String,
    ByRef SpringOption As Integer,
    ByRef k As Double(),
    ByRef CSys As String,
    ByRef SoilProfile As String,
    ByRef Footing As String,
    ByRef Period As Double,
    ByRef color As Integer,
    ByRef notes As String,
    ByRef iGUID As String
) As Integer
```

```
Dim instance As cPropPointSpring
Dim Name As String
Dim SpringOption As Integer
Dim k As Double()
Dim CSys As String
Dim SoilProfile As String
Dim Footing As String
Dim Period As Double
Dim color As Integer
Dim notes As String
Dim iGUID As String
Dim returnValue As Integer
```

```
returnValue = instance.GetPointSpringProp(Name,
    SpringOption, k, CSys, SoilProfile,
```

cPropPointSpringspan id="LSTB46CEFDE_0"AddLanguageSpecificTextSet("LSTB46CEFDE_0567=::|nu=

Footing, Period, color, notes, iGUID)

```
int GetPointSpringProp(
String^ Name,
int% SpringOption,
array<double>^% k,
String^% CSys,
String^% SoilProfile,
String^% Footing,
double% Period,
int% color,
String^% notes,
String^% iGUID
)
```

```
abstract GetPointSpringProp :
    Name : string *
    SpringOption : int byref *
    k : float[] byref *
    CSys : string byref *
    SoilProfile : string byref *
    Footing : string byref *
    Period : float byref *
    color : int byref *
    notes : string byref *
    iGUID : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of the point spring property

SpringOption

Type:Â SystemInt32

The spring stiffness option, either 1 or 2:

1. User Specified / Link Properties
2. Based on Soil Profile and Footing Dimensions

k

Type:Â SystemDouble

This is an array of six spring stiffness values:

- ◊ Value(0) = U1 [F/L]
- ◊ Value(1) = U2 [F/L]
- ◊ Value(2) = U3 [F/L]
- ◊ Value(3) = R1 [FL/rad]
- ◊ Value(4) = R2 [FL/rad]
- ◊ Value(5) = R3 [FL/rad]

This argument only applies when SpringOption = 1

CSys

Type:Â SystemString

The name of the coordinate system in which the property is defined.

This argument only applies when SpringOption = 1

SoilProfile

Type:Â SystemString

The name of an existing Soil Profile

This argument only applies when SpringOption = 2

Footring

Type:Â SystemString

The name of an existing Isolated Column Footring

This argument only applies when SpringOption = 2

Period

Type:Â SystemDouble

The first-mode time period [sec]

This argument only applies when SpringOption = 2

color

Type:Â SystemInt32

The display color for the property specified as an integer

notes

Type:Â SystemString

The notes, if any, assigned to the property

iGUID

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the property

Return Value

Type:Â Int32

Returns zero if the property is retrieved successfully, otherwise returns nonzero

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new spring property
    Dim k() as Double
    ReDim k(5)
    k(2) = 1
    ret = SapModel.PropPointSpring.SetPointSpringProp("mySpringProp1", 1, k)
```

Introduction

```
'get spring property
  Dim SpringOption As Integer
  Dim k1() As Double
  Dim CSys As String
  Dim SoilProfile As String
  Dim Footing As String
  Dim Period As Double
  Dim color As Integer
  Dim notes As String
  Dim iGuid As String
  ret = SapModel.PropPointSpring.GetPointSpringProp("mySpringProp1", SpringOption, k1, CSys, SoilProfile, Footing, Period, color, notes, iGuid)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

🔍 See Also

Reference

[cPropPointSpring Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cPropPointSpringAddLanguageSpecificTextSet("LSTE Method

Sets single joint links for a named point spring property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetLinks(
    string Name,
    int NumberLinks,
    ref string[] LinkNames,
    ref int[] LinkAxialDirs,
    ref double[] LinkAngles
)
```

```
Function SetLinks (
    Name As String,
    NumberLinks As Integer,
    ByRef LinkNames As String(),
    ByRef LinkAxialDirs As Integer(),
    ByRef LinkAngles As Double()
) As Integer
```

```
Dim instance As cPropPointSpring
Dim Name As String
Dim NumberLinks As Integer
Dim LinkNames As String()
Dim LinkAxialDirs As Integer()
Dim LinkAngles As Double()
Dim returnValue As Integer
```

```
returnValue = instance.SetLinks(Name,
    NumberLinks, LinkNames, LinkAxialDirs,
    LinkAngles)
```

```
int SetLinks(
    String^ Name,
    int NumberLinks,
    array<String^>^% LinkNames,
    array<int>^% LinkAxialDirs,
    array<double>^% LinkAngles
)
```

```
abstract SetLinks :
    Name : string *
    NumberLinks : int *
    LinkNames : string[] byref *
    LinkAxialDirs : int[] byref *
```

LinkAngles : float[] byref -> int

Parameters

Name

Type:Â SystemString

The name of an existing point spring property. This spring property must be of type option 1

NumberLinks

Type:Â SystemInt32

The number of links

LinkNames

Type:Â SystemString

This array contains the names of existing link properties

LinkAxialDirs

Type:Â SystemInt32

This array contains integers representing the axial direction of their respective link properties and can take one of six values:

◇ -1 : -X

◇ -2 : -Y

◇ -3 : -Z

◇ 1 : +X

◇ 2 : +Y

◇ 3 : +Z

LinkAngles

Type:Â SystemDouble

This array contains the axis 2 angles of their respective link properties [deg]

Return Value

Type:Â Int32

Returns zero if the links are successfully set, otherwise returns nonzero

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()
```


Introduction

```
'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'define new link property
Dim DOF() As Boolean
Dim Fixed() As Boolean
Dim Ke() As Double
Dim Ce() As Double
ReDim DOF(5)
ReDim Fixed(5)
ReDim Ke(5)
ReDim Ce(5)
DOF(0) = True
Ke(0) = 12
ret = SapModel.PropLink.SetLinear("myLink1", DOF, Fixed, Ke, Ce, 0, 0)

'define new spring property
Dim k() as Double
ReDim k(5)
k(2) = 1
ret = SapModel.PropPointSpring.SetPointSpringProp("myPtSpringProp", 1, k)

'set links in spring property
Dim NumberLinks As Integer
Dim LinkNames() As String
Dim LinkAxialDirs() As Integer
Dim LinkAngles() As Double

NumberLinks = 1
ReDim LinkNames(NumberLinks - 1)
ReDim LinkAxialDirs(NumberLinks - 1)
ReDim LinkAngles(NumberLinks - 1)
LinkNames(0) = "myLink1"
LinkAxialDirs(0) = -2
LinkAngles(0) = 45

ret = SapModel.PropPointSpring.SetLinks("myPtSpringProp", NumberLinks, LinkNames, LinkAxialDirs)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

cPropPointSpring Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropPointSpringAddLanguageSpecificTextSet("LST4

Method

Creates a new named point spring property, or modifies an existing named point spring property

Namespace: Â ETABSV1

Assembly: Â ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetPointSpringProp(
    string Name,
    int SpringOption,
    ref double[] k,
    string CSys = "",
    string SoilProfile = "",
    string Footing = "",
    double Period = 0,
    int color = 0,
    string notes = "",
    string iGUID = ""
)
```

```
Function SetPointSpringProp (
    Name As String,
    SpringOption As Integer,
    ByRef k As Double(),
    CSys As String = "",
    SoilProfile As String = "",
    Footing As String = "",
    Period As Double = 0,
    color As Integer = 0,
    notes As String = "",
    iGUID As String = ""
) As Integer
```

```
Dim instance As cPropPointSpring
Dim Name As String
Dim SpringOption As Integer
Dim k As Double()
Dim CSys As String
Dim SoilProfile As String
Dim Footing As String
Dim Period As Double
Dim color As Integer
Dim notes As String
Dim iGUID As String
Dim returnValue As Integer
```

```
returnValue = instance.SetPointSpringProp(Name,
```

cPropPointSpringspan id="LST4D95EDF0_0"AddLanguageSpecificTextSet("LST4D95EDF0_03574=::|nu=.

```
SpringOption, k, CSys, SoilProfile,
Footing, Period, color, notes, iGUID)
```

```
int SetPointSpringProp(
String^ Name,
int SpringOption,
array<double>^% k,
String^ CSys = L"",
String^ SoilProfile = L"",
String^ Footing = L"",
double Period = 0,
int color = 0,
String^ notes = L"",
String^ iGUID = L""
)
```

```
abstract SetPointSpringProp :
    Name : string *
    SpringOption : int *
    k : float[] byref *
    ?CSys : string *
    ?SoilProfile : string *
    ?Footing : string *
    ?Period : float *
    ?color : int *
    ?notes : string *
    ?iGUID : string
(* Defaults:
    let _CSys = defaultArg CSys ""
    let _SoilProfile = defaultArg SoilProfile ""
    let _Footing = defaultArg Footing ""
    let _Period = defaultArg Period 0
    let _color = defaultArg color 0
    let _notes = defaultArg notes ""
    let _iGUID = defaultArg iGUID ""
*)
-> int
```

Parameters

Name

Type: [SystemString](#)

The name of the point spring property. If this is an existing property, that property is modified; otherwise, a new property is added.

SpringOption

Type: [SystemInt32](#)

The spring stiffness option, either 1 or 2:

1. User Specified / Link Properties
2. Based on Soil Profile and Footing Dimensions

k

Type: [SystemDouble](#)

This is an array of six spring stiffness values:

- ◊ Value(0) = U1 [F/L]
- ◊ Value(1) = U2 [F/L]
- ◊ Value(2) = U3 [F/L]
- ◊ Value(3) = R1 [FL/rad]
- ◊ Value(4) = R2 [FL/rad]

◇ Value(5) = R3 [FL/rad]

This argument only applies when SpringOption = 1

CSys (Optional)

Type:Â SystemString

The name of the coordinate system in which the property is defined. If this item not defined or left blank, the Global coordinate system is assumed.

This argument only applies when SpringOption = 1

SoilProfile (Optional)

Type:Â SystemString

The name of an existing Soil Profile

This argument only applies when SpringOption = 2

Footing (Optional)

Type:Â SystemString

The name of an existing Isolated Column Footing

This argument only applies when SpringOption = 2

Period (Optional)

Type:Â SystemDouble

The first-mode time period [sec]

This argument only applies when SpringOption = 2

color (Optional)

Type:Â SystemInt32

The display color for the property specified as an integer

notes (Optional)

Type:Â SystemString

The notes, if any, assigned to the property

iGUID (Optional)

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the property

Return Value

Type:Â Int32

Returns zero if the property is successfully set, otherwise returns nonzero

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1
```

'create ETABS object

```
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
```

'start ETABS application

```
    ret = EtabsObject.ApplicationStart()
```

Introduction

```
'create SapModel object
  SapModel = EtabsObject.SapModel

'initialize model
  ret = SapModel.InitializeNewModel()

'create steel deck template model
  ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'define new spring property
  Dim k() as Double
  ReDim k(5)
  k(2) = 1
  ret = SapModel.PropPointSpring.SetPointSpringProp("mySpringProp1", 1, k)

'close ETABS
  EtabsObject.ApplicationExit(False)

'clean up variables
  SapModel = Nothing
  EtabsObject = Nothing

End Sub
```

▲ See Also

Reference

[cPropPointSpring Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropRebar Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cPropRebar
```

```
Public Interface cPropRebar
```

```
Dim instance As cPropRebar
```

```
public interface class cPropRebar
```

```
type cPropRebar = interface end
```

The cPropRebar type exposes the following members.

▲ Methods

▲	Name	Description
🔗	GetNameList	Retrieves the names of all defined rebar properties
🔗	GetNameListWithData	Retrieves the names and data of all defined rebar properties
🔗	GetRebarProps	Retrieves dimension data for a rebar property
🔗	GetRebarPropsWithGUID	Retrieves dimension data and GUID for a rebar property

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropRebar Methods

The [cPropRebar](#) type exposes the following members.

Methods

	Name	Description
➤	GetNameList	Retrieves the names of all defined rebar properties
➤	GetNameListWithData	Retrieves the names and data of all defined rebar properties
➤	GetRebarProps	Retrieves dimension data for a rebar property
➤	GetRebarPropsWithGUID	Retrieves dimension data and GUID for a rebar property

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➤ See Also

Reference

[cPropRebar Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cPropRebarAddLanguageSpecificTextSet("LST4762D4 Method

Retrieves the names of all defined rebar properties

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetNameList(
    ref int NumberNames,
    ref string[] MyName
)
```

```
Function GetNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cPropRebar
Dim NumberNames As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetNameList(NumberNames,
    MyName)
```

```
int GetNameList(
    int% NumberNames,
    array<String^>^% MyName
)
```

```
abstract GetNameList :
    NumberNames : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberNames

Type: System.Int32

The number of rebar properties

MyName

Type: System.String

This is a one-dimensional array of rebar property names. The MyName array is created as a dynamic, zero-based, array by the API user:

```
Dim MyName() as String
```

cPropRebarspan id="LST4762D4D3_0"AddLanguageSpecificTextSet("LST4762D4D3_0?cpp=3580.");Getf

Introduction

The array is dimensioned to (NumberNames + 1) inside the ETABS program, filled with the names, and returned to the API user.

Return Value

Type: Int32

Returns zero if the data is successfully retrieved, otherwise it returns nonzero

▲ Remarks

▲ See Also

Reference

cPropRebar Interface

ETABSV1 Namespace

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CSI API ETABS v1

cPropRebarAddLanguageSpecificTextSet("LST7CBF9789_0?cpp=3582.");GetTextSet("LST7CBF9789_0?cpp=3582.");

Method

Retrieves the names and data of all defined rebar properties

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNameListWithData(
    ref int NumberNames,
    ref string[] MyName,
    ref double[] Areas,
    ref double[] Diameters,
    ref string[] MyGUID
)
```

```
Function GetNameListWithData (
    ByRef NumberNames As Integer,
    ByRef MyName As String(),
    ByRef Areas As Double(),
    ByRef Diameters As Double(),
    ByRef MyGUID As String()
) As Integer
```

```
Dim instance As cPropRebar
Dim NumberNames As Integer
Dim MyName As String()
Dim Areas As Double()
Dim Diameters As Double()
Dim MyGUID As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetNameListWithData(NumberNames,
    MyName, Areas, Diameters, MyGUID)
```

```
int GetNameListWithData(
    int% NumberNames,
    array<String^>^% MyName,
    array<double>^% Areas,
    array<double>^% Diameters,
    array<String^>^% MyGUID
)
```

```
abstract GetNameListWithData :
    NumberNames : int byref *
    MyName : string[] byref *
    Areas : float[] byref *
    Diameters : float[] byref *
    MyGUID : string[] byref -> int
```

cPropRebarspan id="LST7CBF9789_0"AddLanguageSpecificTextSet("LST7CBF9789_0?cpp=3582.");GetTextSet("LST7CBF9789_0?cpp=3582.");

Parameters

NumberNames

Type: SystemInt32

The number of rebar properties

MyName

Type: SystemString

This is a one-dimensional array of rebar property names. The MyName array is created as a dynamic, zero-based, array by the API user:

```
Dim MyName() as String
```

The array is dimensioned to (NumberNames + 1) inside the ETABS program, filled with the names, and returned to the API user.

Areas

Type: SystemDouble

Diameters

Type: SystemDouble

MyGUID

Type: SystemString

This is a one-dimensional array of rebar property GUIDs.

Return Value

Type: Int32

Returns zero if the data is successfully retrieved, otherwise it returns nonzero

▲ Remarks

▲ See Also

Reference

cPropRebar Interface

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CSI API ETABS v1

cPropRebarAddLanguageSpecificTextSet("LSTB2EED583_0?cpp=3584=.");GetRebarProps Method

Retrieves dimension data for a rebar property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetRebarProps(
    string Name,
    ref double Area,
    ref double Diameter
)
```

```
Function GetRebarProps (
    Name As String,
    ByRef Area As Double,
    ByRef Diameter As Double
) As Integer
```

```
Dim instance As cPropRebar
Dim Name As String
Dim Area As Double
Dim Diameter As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetRebarProps(Name,
    Area, Diameter)
```

```
int GetRebarProps(
    String^ Name,
    double% Area,
    double% Diameter
)
```

```
abstract GetRebarProps :
    Name : string *
    Area : float byref *
    Diameter : float byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing rebar property

Area

Type: [SystemDouble](#)

Diameter

Type: SystemDouble

Return Value

Type: Int32

Returns zero if the data is successfully retrieved, otherwise it returns nonzero

▀ Remarks

▀ See Also

Reference

cPropRebar Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropRebarAddLanguageSpecificTextSet("LST958CC7E7_0?cpp=3586-");GetRebarPropsWithGUID Method

Retrieves dimension data and GUID for a rebar property

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetRebarPropsWithGUID(
    string Name,
    ref double Area,
    ref double Diameter,
    ref string MyGUID
)
```

```
Function GetRebarPropsWithGUID (
    Name As String,
    ByRef Area As Double,
    ByRef Diameter As Double,
    ByRef MyGUID As String
) As Integer
```

```
Dim instance As cPropRebar
Dim Name As String
Dim Area As Double
Dim Diameter As Double
Dim MyGUID As String
Dim returnValue As Integer
```

```
returnValue = instance.GetRebarPropsWithGUID(Name,
    Area, Diameter, MyGUID)
```

```
int GetRebarPropsWithGUID(
    String^ Name,
    double% Area,
    double% Diameter,
    String^% MyGUID
)
```

```
abstract GetRebarPropsWithGUID :
    Name : string *
    Area : float byref *
    Diameter : float byref *
    MyGUID : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing rebar property

Area

Type:Â SystemDouble

Diameter

Type:Â SystemDouble

MyGUID

Type:Â SystemString

Return Value

Type:Â Int32

Returns zero if the data is successfully retrieved, otherwise it returns nonzero

▲ Remarks

▲ See Also

Reference

cPropRebar Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropTendon Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cPropTendon
```

```
Public Interface cPropTendon
```

```
Dim instance As cPropTendon
```

```
public interface class cPropTendon
```

```
type cPropTendon = interface end
```

The cPropTendon type exposes the following members.

▲ Methods

▲	Name	Description
◆	ChangeName	Changes the name of an existing tendon property.
◆	Count	Retrieves the total number of defined tendon properties in the model.
◆	Delete	Deletes a specified tendon property.
◆	GetNameList	Retrieves the names of all defined tendon properties in the model.
◆	GetProp	Retrieves tendon property definition data.
◆	SetProp	Defines a tendon property

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▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropTendon Methods

The [cPropTendon](#) type exposes the following members.

Methods

⌚	Name	Description
🔗	ChangeName	Changes the name of an existing tendon property.
🔗	Count	Retrieves the total number of defined tendon properties in the model.
🔗	Delete	Deletes a specified tendon property.
🔗	GetNameList	Retrieves the names of all defined tendon properties in the model.
🔗	GetProp	Retrieves tendon property definition data.
🔗	SetProp	Defines a tendon property

[Top](#)

⚡ See Also

Reference

[cPropTendon Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cPropTendonAddLanguageSpecificTextSet("LST27762C55_0") Method

Changes the name of an existing tendon property.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int ChangeName(
    string Name,
    string NewName
)
```

```
Function ChangeName (
    Name As String,
    NewName As String
) As Integer
```

```
Dim instance As cPropTendon
Dim Name As String
Dim NewName As String
Dim returnValue As Integer
```

```
returnValue = instance.ChangeName(Name,
    NewName)
```

```
int ChangeName(
    String^ Name,
    String^ NewName
)
```

```
abstract ChangeName :
    Name : string *
    NewName : string -> int
```

Parameters

Name

Type: SystemString

The existing name of a defined tendon property.

NewName

Type: SystemString

The new name for the tendon property.

Return Value

Type: Int32

Returns zero if the new name is successfully applied; otherwise it returns a nonzero value.

▀ Remarks

▀ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add tendon material
    ret = SapModel.PropMaterial.AddQuick(Name, eMatType.Tendon, , , , , eMatTypeTendon.ASTM_)

    'set new tendon property
    ret = SapModel.PropTendon.SetProp("T1", Name, 1, 2.25)

    'change name of tendon property
    ret = SapModel.PropTendon.ChangeName("T1", "MyTendon")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▀ See Also

Reference

cPropTendon Interface

ETABSv1 Namespace

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Introduction

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CSI API ETABS v1

cPropTendonAddLanguageSpecificTextSet("LST42378532_0") Method

Retrieves the total number of defined tendon properties in the model.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int Count()
```

Function Count As Integer

```
Dim instance As cPropTendon
```

```
Dim returnValue As Integer
```

```
returnValue = instance.Count()
```

```
int Count()
```

```
abstract Count : unit -> int
```

Return Value

Type: Int32

Returns the total number of defined tendon properties in the model.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim Name As String  
    Dim Count As Integer  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model
```

cPropTendonspan id="LST42378532_0"AddLanguageSpecificTextSet("LST42378532_0?cpp=3593.");Cou

Introduction

```
ret = SapModel.InitializeNewModel()  
  
'create steel deck template model  
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)  
  
'add tendon material  
ret = SapModel.PropMaterial.AddQuick(Name, eMatType.Tendon, , , , , eMatTypeTendon.ASTM_  
  
'set new tendon property  
ret = SapModel.PropTendon.SetProp("T1", Name, 1, 2.25)  
  
'return number of defined tendon properties  
Count = SapModel.PropTendon.Count  
  
'close ETABS  
EtabsObject.ApplicationExit(False)  
  
'clean up variables  
SapModel = Nothing  
EtabsObject = Nothing  
End Sub
```

▲ See Also

Reference

cPropTendon Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropTendonAddLanguageSpecificTextSet("LSTA520F9AF_0?cpp3595") Method

Deletes a specified tendon property.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int Delete(
    string Name
)
```

```
Function Delete (
    Name As String
) As Integer
```

```
Dim instance As cPropTendon
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.Delete(Name)
```

```
int Delete(
    String^ Name
)
```

```
abstract Delete :
    Name : string -> int
```

Parameters

Name

Type: SystemString

The name of an existing tendon property.

Return Value

Type: Int32

Returns zero if the property is successfully deleted; otherwise it returns a nonzero value. It returns an error if the specified property can not be deleted, for example, if it is assigned to an existing object.

▲ Remarks

▲ Examples

VB

Copy

cPropTendonspan id="LSTA520F9AF_0"AddLanguageSpecificTextSet("LSTA520F9AF_0?cpp3595");De

Introduction

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim Name As String  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)  
  
    'add tendon material  
    ret = SapModel.PropMaterial.AddQuick(Name, eMatType.Tendon, , , , , eMatTypeTendon.ASTM_  
  
    'set new tendon properties  
    ret = SapModel.PropTendon.SetProp("T1", Name, 1, 2.25)  
    ret = SapModel.PropTendon.SetProp("T2", Name, 1, 3.25)  
  
    'delete tendon property  
    ret = SapModel.PropTendon.Delete("T1")  
  
    'close ETABS  
    EtabsObject.ApplicationExit(False)  
  
    'clean up variables  
    SapModel = Nothing  
    EtabsObject = Nothing  
End Sub
```

🔗 See Also

Reference

[cPropTendon Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cPropTendonAddLanguageSpecificTextSet("LST1B4BB78F_0?cpp3597u=.");Get

Method

Retrieves the names of all defined tendon properties in the model.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNameList(
    ref int NumberNames,
    ref string[] MyName
)
```

```
Function GetNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cPropTendon
Dim NumberNames As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetNameList(NumberNames,
    MyName)
```

```
int GetNameList(
    int% NumberNames,
    array<String^>^% MyName
)
```

```
abstract GetNameList :
    NumberNames : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberNames

Type: [System.Int32](#)

The number of tendon property names retrieved by the program.

MyName

Type: [System.String](#)

This is a one-dimensional array of tendon property names. The MyName array is created as a dynamic, zero-based, array by the API user: Dim MyName() as String

The array is dimensioned to (NumberNames - 1) inside the program, filled with the names, and returned to the API user.

Return Value

Type: Int32

Returns zero if the names are successfully retrieved; otherwise it returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim NumberNames As Integer
    Dim MyName() As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add tendon material
    ret = SapModel.PropMaterial.AddQuick(Name, eMatType.Tendon, , , , , eMatTypeTendon.ASTM_

    'set new tendon property
    ret = SapModel.PropTendon.SetProp("T1", Name, 1, 2.25)

    'get tendon property names
    ret = SapModel.PropTendon.GetNameList(NumberNames, MyName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cPropTendon Interface

ETABSv1 Namespace

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CSI API ETABS v1

cPropTendonAddLanguageSpecificTextSet("LST7D6B9386_0?cpp=.") Method

Retrieves tendon property definition data.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetProp(
    string Name,
    ref string MatProp,
    ref int ModelingOption,
    ref double Area,
    ref int Color,
    ref string Notes,
    ref string GUID
)
```

```
Function GetProp (
    Name As String,
    ByRef MatProp As String,
    ByRef ModelingOption As Integer,
    ByRef Area As Double,
    ByRef Color As Integer,
    ByRef Notes As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cPropTendon
Dim Name As String
Dim MatProp As String
Dim ModelingOption As Integer
Dim Area As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.GetProp(Name, MatProp,
    ModelingOption, Area, Color, Notes,
    GUID)
```

```
int GetProp(
    String^ Name,
    String^% MatProp,
    int% ModelingOption,
    double% Area,
    int% Color,
    String^% Notes,
```

cPropTendonid="LST7D6B9386_0"AddLanguageSpecificTextSet("LST7D6B9386_0?cpp=.");Ge

```
String^% GUID
)
```

```
abstract GetProp :
    Name : string *
    MatProp : string byref *
    ModelingOption : int byref *
    Area : float byref *
    Color : int byref *
    Notes : string byref *
    GUID : string byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing tendon property.

MatProp

Type:Â SystemString

The name of the material property assigned to the tendon property.

ModelingOption

Type:Â SystemInt32

This is either 1 or 2, indicating the tendon modeling option.

Value	Modeling Option
1	Model tendon as loads
2	Model tendon as elements

Area

Type:Â SystemDouble

The cross-sectional area of the tendon. [L2]

Color

Type:Â SystemInt32

Notes

Type:Â SystemString

GUID

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the property.

Return Value

Type:Â Int32

Returns zero if the property is successfully retrieved; otherwise it returns a nonzero value.

Remarks

Modeling tendons as elements (ModelingOption = 2) is not supported in ETABS.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim MatProp As String
```

Introduction

```
Dim ModelingOption As Integer
Dim Area As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'add tendon material
ret = SapModel.PropMaterial.AddQuick(Name, eMatType.Tendon, , , , , eMatTypeTendon.ASTM_)

'set new tendon property
ret = SapModel.PropTendon.SetProp("T1", Name, 1, 2.25)

'get tendon property data
ret = SapModel.PropTendon.GetProp("T1", MatProp, ModelingOption, Area, Color, Notes, GUID)

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cPropTendon Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cPropTendonAddLanguageSpecificTextSet("LST69569 Method

Defines a tendon property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetProp(
    string Name,
    string MatProp,
    int ModelingOption,
    double Area,
    int Color = -1,
    string Notes = "",
    string GUID = ""
)
```

```
Function SetProp (
    Name As String,
    MatProp As String,
    ModelingOption As Integer,
    Area As Double,
    Optional As Integer = -1,
    Optional As String = "",
    Optional As String = ""
) As Integer
```

```
Dim instance As cPropTendon
Dim Name As String
Dim MatProp As String
Dim ModelingOption As Integer
Dim Area As Double
Dim Color As Integer
Dim Notes As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.SetProp(Name, MatProp,
    ModelingOption, Area, Color, Notes,
    GUID)
```

```
int SetProp(
    String^ Name,
    String^ MatProp,
    int ModelingOption,
    double Area,
    int Color = -1,
    String^ Notes = L"",
```

cPropTendonspan id="LST6956910D_0"AddLanguageSpecificTextSet("LST6956910D_0?cpp=3606.");Set

```
String^ GUID = L""
)

abstract SetProp :
    Name : string *
    MatProp : string *
    ModelingOption : int *
    Area : float *
    ?Color : int *
    ?Notes : string *
    ?GUID : string
(* Defaults:
    let _Color = defaultArg Color -1
    let _Notes = defaultArg Notes ""
    let _GUID = defaultArg GUID ""
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing or new tendon property. If this is an existing property, that property is modified; otherwise, a new property is added.

MatProp

Type:Â SystemString

The name of the material property assigned to the tendon property.

ModelingOption

Type:Â SystemInt32

This is either 1 or 2, indicating the tendon modeling option.

Value Modeling Option

1 Model tendon as loads

2 Model tendon as elements

Area

Type:Â SystemDouble

The cross-sectional area of the tendon. [L2]

Color (Optional)

Type:Â SystemInt32

Notes (Optional)

Type:Â SystemString

GUID (Optional)

Type:Â SystemString

The GUID (global unique identifier), if any, assigned to the property. If this item is input as Default, the program assigns a GUID to the property

Return Value

Type:Â Int32

Returns zero if the property is successfully defined; otherwise it returns a nonzero value.

Remarks

Modeling tendons as elements (ModelingOption = 2) is not supported in ETABS.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'add tendon material
    ret = SapModel.PropMaterial.AddQuick(Name, eMatType.Tendon, , , , , eMatTypeTendon.ASTM_

    'set new tendon property
    ret = SapModel.PropTendon.SetProp("T1", Name, 1, 2.25)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

[cPropTendon Interface](#)

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cSapModel Interface

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
public interface cSapModel
```

```
Public Interface cSapModel
```

```
Dim instance As cSapModel
```

```
public interface class cSapModel
```

```
type cSapModel = interface end
```

The cSapModel type exposes the following members.

Properties

<u>ETABSv1</u>	Name	Description
	<u>Analyze</u>	
	<u>AreaElm</u>	
	<u>AreaObj</u>	
	<u>ConstraintDef</u>	
	<u>DatabaseTables</u>	
	<u>DesignCompositeBeam</u>	
	<u>DesignConcrete</u>	
	<u>DesignConcreteSlab</u>	
	<u>DesignResults</u>	
	<u>DesignShearWall</u>	
	<u>DesignSteel</u>	
	<u>Detailing</u>	
	<u>Diaphragm</u>	
	<u>EditArea</u>	
	<u>EditFrame</u>	
	<u>EditGeneral</u>	
	<u>EditPoint</u>	
	<u>File</u>	
	<u>FrameObj</u>	

	<u>Func</u>
	<u>GDispl</u>
	<u>GridSys</u>
	<u>GroupDef</u>
	<u>LineElm</u>
	<u>LinkElm</u>
	<u>LinkObj</u>
	<u>LoadCases</u>
	<u>LoadPatterns</u>
	<u>Options</u>
	<u>PatternDef</u>
	<u>PierLabel</u>
	<u>PointElm</u>
	<u>PointObj</u>
	<u>PropArea</u>
	<u>PropAreaSpring</u>
	<u>PropFrame</u>
	<u>PropLineSpring</u>
	<u>PropLink</u>
	<u>PropMaterial</u>
	<u>PropPointSpring</u>
	<u>PropRebar</u>
	<u>PropTendon</u>
	<u>RespCombo</u>
	<u>Results</u>
	<u>SelectObj</u>
	<u>SpandrelLabel</u>
	<u>Story</u>
	<u>TendonObj</u>
	<u>Tower</u>
	<u>View</u>

[Top](#)

📁 Methods

⌂

Name

Description

💡

[GetDatabaseUnits](#)

Returns a value from the [eUnits](#) enumeration indicating the database units for the model. All data is internally stored in the model in these units and converted to the present units as needed.

💡

[GetDatabaseUnits_2](#)

Retrieves the database units for the model. All data is internally stored in the model in these units and converted to the present units as needed.

💡

[GetMergeTol](#)

💡

[GetModelFilename](#)

Introduction

Returns a string that represents the filename of the current model, with or without the full path.

⇒ [GetModelFilepath](#)

Returns the filepath of the current model

⇒ [GetModelIsLocked](#)

⇒ [GetPresentCoordSystem](#)

⇒ [GetPresentUnits](#)

Returns a value from the [eUnits](#) enumeration indicating the units presently specified for the model.

⇒ [GetPresentUnits_2](#)

Retrieves the units presently specified for the model

⇒ [GetProgramInfo](#)

Retrieve information about the program

⇒ [GetProjectInfo](#)

Retrieves the project information data.

⇒ [GetVersion](#)

Returns the program version.

⇒ [InitializeNewModel](#)

This function clears the previous model and initializes the program for a new model. If it is later needed, you should save your previous model prior to calling this function.

⇒ [SetMergeTol](#)

⇒ [SetModelIsLocked](#)

Locks or unlocks the model.

⇒ [SetPresentUnits](#)

Sets the display (present) units.

⇒ [SetPresentUnits_2](#)

Specifies the units for the model

⇒ [SetProjectInfo](#)

Sets the data for an item in the project information.

⇒ [TreeIsUpdateSuspended](#)

⇒ [TreeResumeUpdateData](#)

⇒ [TreeSuspendUpdateData](#)

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▀ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModel Properties

The cSapModel type exposes the following members.

Properties

⌕	Name	Description
	<u>Analyze</u>	
	<u>AreaElm</u>	
	<u>AreaObj</u>	
	<u>ConstraintDef</u>	
	<u>DatabaseTables</u>	
	<u>DesignCompositeBeam</u>	
	<u>DesignConcrete</u>	
	<u>DesignConcreteSlab</u>	
	<u>DesignResults</u>	
	<u>DesignShearWall</u>	
	<u>DesignSteel</u>	
	<u>Detailing</u>	
	<u>Diaphragm</u>	
	<u>EditArea</u>	
	<u>EditFrame</u>	
	<u>EditGeneral</u>	
	<u>EditPoint</u>	
	<u>File</u>	
	<u>FrameObj</u>	
	<u>Func</u>	
	<u>GDispl</u>	
	<u>GridSys</u>	
	<u>GroupDef</u>	
	<u>LineElm</u>	
	<u>LinkElm</u>	
	<u>LinkObj</u>	
	<u>LoadCases</u>	
	<u>LoadPatterns</u>	
	<u>Options</u>	
	<u>PatternDef</u>	
	<u>PierLabel</u>	
	<u>PointElm</u>	
	<u>PointObj</u>	
	<u>PropArea</u>	
	<u>PropAreaSpring</u>	

-  [PropFrame](#)
-  [PropLineSpring](#)
-  [PropLink](#)
-  [PropMaterial](#)
-  [PropPointSpring](#)
-  [PropRebar](#)
-  [PropTendon](#)
-  [RespCombo](#)
-  [Results](#)
-  [SelectObj](#)
-  [SpandrelLabel](#)
-  [Story](#)
-  [TendonObj](#)
-  [Tower](#)
-  [View](#)

Gives access to API calls related to analysis results.

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▀ See Also

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[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LSTA4E9E015_0?cpp=::3614.");Analy

Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cAnalyze Analyze { get; }
```

```
ReadOnly Property Analyze As cAnalyze
Get
```

```
Dim instance As cSapModel
Dim value As cAnalyze
```

```
value = instance.Analyze
```

```
property cAnalyze^ Analyze {
cAnalyze^ get ();
}
```

```
abstract Analyze : cAnalyze with get
```

Property Value

Type: [cAnalyze](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST6B6EC045_0") Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cAreaElm AreaElm { get; }
```

```
ReadOnly Property AreaElm As cAreaElm
Get
```

```
Dim instance As cSapModel
Dim value As cAreaElm
```

```
value = instance.AreaElm
```

```
property cAreaElm^ AreaElm {
cAreaElm^ get ();
}
```

```
abstract AreaElm : cAreaElm with get
```

Property Value

Type: [cAreaElm](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LSTC8C54BA0_0?cpp=3p13-");AreaObj Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cAreaObj AreaObj { get; }
```

```
ReadOnly Property AreaObj As cAreaObj
Get
```

```
Dim instance As cSapModel
Dim value As cAreaObj
```

```
value = instance.AreaObj
```

```
property cAreaObj^ AreaObj {
cAreaObj^ get ();
}
```

```
abstract AreaObj : cAreaObj with get
```

Property Value

Type: [cAreaObj](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST8A8B2CAF_0?cpp=3p14.");Con Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cConstraint ConstraintDef { get; }
```

```
ReadOnly Property ConstraintDef As cConstraint  
Get
```

```
Dim instance As cSapModel  
Dim value As cConstraint
```

```
value = instance.ConstraintDef
```

```
property cConstraint^ ConstraintDef {  
    cConstraint^ get ();  
}
```

```
abstract ConstraintDef : cConstraint with get
```

Property Value

Type: [cConstraint](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST548F4320_0?cpp=::|DatabaseTables") Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cDatabaseTables DatabaseTables { get; }
```

```
ReadOnly Property DatabaseTables As cDatabaseTables  
Get
```

```
Dim instance As cSapModel  
Dim value As cDatabaseTables
```

```
value = instance.DatabaseTables
```

```
property cDatabaseTables^ DatabaseTables {  
    cDatabaseTables^ get ();  
}
```

```
abstract DatabaseTables : cDatabaseTables with get
```

Property Value

Type: [cDatabaseTables](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST96CF27F0") Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cDesignCompositeBeam DesignCompositeBeam { get; }
```

```
ReadOnly Property DesignCompositeBeam As cDesignCompositeBeam
Get
```

```
Dim instance As cSapModel
Dim value As cDesignCompositeBeam
```

```
value = instance.DesignCompositeBeam
```

```
property cDesignCompositeBeam^ DesignCompositeBeam {
cDesignCompositeBeam^ get ();
}
```

```
abstract DesignCompositeBeam : cDesignCompositeBeam with get
```

Property Value

Type: [cDesignCompositeBeam](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST34DEF035_0?cpp=::3617.");DesignConcrete Property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
cDesignConcrete DesignConcrete { get; }
```

```
ReadOnly Property DesignConcrete As cDesignConcrete
Get
```

```
Dim instance As cSapModel
Dim value As cDesignConcrete
```

```
value = instance.DesignConcrete
```

```
property cDesignConcrete^ DesignConcrete {
cDesignConcrete^ get ();
}
```

```
abstract DesignConcrete : cDesignConcrete with get
```

Property Value

Type: cDesignConcrete

▲ See Also

Reference

cSapModel Interface

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST33BA2650_0?cpp=::18");DesignConcreteSlab Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cDesignConcreteSlab DesignConcreteSlab { get; }
```

```
ReadOnly Property DesignConcreteSlab As cDesignConcreteSlab
Get
```

```
Dim instance As cSapModel
Dim value As cDesignConcreteSlab
```

```
value = instance.DesignConcreteSlab
```

```
property cDesignConcreteSlab^ DesignConcreteSlab {
cDesignConcreteSlab^ get ();
}
```

```
abstract DesignConcreteSlab : cDesignConcreteSlab with get
```

Property Value

Type: [cDesignConcreteSlab](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LSTC381F7E2_0?cpp=::3619");DesignResults Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cDesignResults DesignResults { get; }
```

```
ReadOnly Property DesignResults As cDesignResults
Get
```

```
Dim instance As cSapModel
Dim value As cDesignResults
```

```
value = instance.DesignResults
```

```
property cDesignResults^ DesignResults {
cDesignResults^ get ();
}
```

```
abstract DesignResults : cDesignResults with get
```

Property Value

Type: [cDesignResults](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LSTB1FBEC Property

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
cDesignShearWall DesignShearWall { get; }
```

```
ReadOnly Property DesignShearWall As cDesignShearWall  
Get
```

```
Dim instance As cSapModel  
Dim value As cDesignShearWall
```

```
value = instance.DesignShearWall
```

```
property cDesignShearWall^ DesignShearWall {  
    cDesignShearWall^ get ();  
}
```

```
abstract DesignShearWall : cDesignShearWall with get
```

Property Value

Type: cDesignShearWall

▲ See Also

Reference

cSapModel Interface

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST2B08CA0B_0") Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cDesignSteel DesignSteel { get; }
```

```
ReadOnly Property DesignSteel As cDesignSteel
Get
```

```
Dim instance As cSapModel
Dim value As cDesignSteel
```

```
value = instance.DesignSteel
```

```
property cDesignSteel^ DesignSteel {
cDesignSteel^ get ();
}
```

```
abstract DesignSteel : cDesignSteel with get
```

Property Value

Type: [cDesignSteel](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST507C6C32_0?cpp=::3622");Detailing Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cDetailing Detailing { get; }
```

```
ReadOnly Property Detailing As cDetailing
Get
```

```
Dim instance As cSapModel
Dim value As cDetailing
```

```
value = instance.Detailing
```

```
property cDetailing^ Detailing {
cDetailing^ get ();
}
```

```
abstract Detailing : cDetailing with get
```

Property Value

Type: [cDetailing](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LSTAADCDO Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cDiaphragm Diaphragm { get; }
```

```
ReadOnly Property Diaphragm As cDiaphragm  
Get
```

```
Dim instance As cSapModel  
Dim value As cDiaphragm
```

```
value = instance.Diaphragm
```

```
property cDiaphragm^ Diaphragm {  
    cDiaphragm^ get ();  
}
```

```
abstract Diaphragm : cDiaphragm with get
```

Property Value

Type: [cDiaphragm](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LSTB16DF9C9_0") Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cEditArea EditArea { get; }
```

```
ReadOnly Property EditArea As cEditArea
Get
```

```
Dim instance As cSapModel
Dim value As cEditArea
```

```
value = instance.EditArea
```

```
property cEditArea^ EditArea {
cEditArea^ get ();
}
```

```
abstract EditArea : cEditArea with get
```

Property Value

Type: [cEditArea](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST8FC6E8D Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cEditFrame EditFrame { get; }
```

```
ReadOnly Property EditFrame As cEditFrame
Get
```

```
Dim instance As cSapModel
Dim value As cEditFrame
```

```
value = instance.EditFrame
```

```
property cEditFrame^ EditFrame {
cEditFrame^ get ();
}
```

```
abstract EditFrame : cEditFrame with get
```

Property Value

Type: [cEditFrame](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LSTE6126E_0") Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cEditGeneral EditGeneral { get; }
```

```
ReadOnly Property EditGeneral As cEditGeneral
Get
```

```
Dim instance As cSapModel
Dim value As cEditGeneral
```

```
value = instance.EditGeneral
```

```
property cEditGeneral^ EditGeneral {
cEditGeneral^ get ();
}
```

```
abstract EditGeneral : cEditGeneral with get
```

Property Value

Type: [cEditGeneral](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST468C90C6_0?cpp=::3627.");EditP Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cEditPoint EditPoint { get; }
```

```
ReadOnly Property EditPoint As cEditPoint
Get
```

```
Dim instance As cSapModel
Dim value As cEditPoint
```

```
value = instance.EditPoint
```

```
property cEditPoint^ EditPoint {
cEditPoint^ get ();
}
```

```
abstract EditPoint : cEditPoint with get
```

Property Value

Type: [cEditPoint](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LSTD60D8C Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cFile File { get; }
```

```
ReadOnly Property File As cFile
Get
```

```
Dim instance As cSapModel
Dim value As cFile
```

```
value = instance.File
```

```
property cFile^ File {
cFile^ get ();
}
```

```
abstract File : cFile with get
```

Property Value

Type: [cFile](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST6B421029_0?cpp=::|6629);FrameObj Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cFrameObj FrameObj { get; }
```

```
ReadOnly Property FrameObj As cFrameObj  
Get
```

```
Dim instance As cSapModel  
Dim value As cFrameObj
```

```
value = instance.FrameObj
```

```
property cFrameObj^ FrameObj {  
    cFrameObj^ get ();  
}
```

```
abstract FrameObj : cFrameObj with get
```

Property Value

Type: [cFrameObj](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST320A761D_0?cpp=:::3630");Func Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cFunction Func { get; }
```

```
ReadOnly Property Func As cFunction
Get
```

```
Dim instance As cSapModel
Dim value As cFunction
```

```
value = instance.Func
```

```
property cFunction^ Func {
cFunction^ get ();
}
```

```
abstract Func : cFunction with get
```

Property Value

Type: [cFunction](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LSTAE9D8E37_0") Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cGenDispl GDispl { get; }
```

```
ReadOnly Property GDispl As cGenDispl  
Get
```

```
Dim instance As cSapModel  
Dim value As cGenDispl
```

```
value = instance.GDispl
```

```
property cGenDispl^ GDispl {  
    cGenDispl^ get ();  
}
```

```
abstract GDispl : cGenDispl with get
```

Property Value

Type: [cGenDispl](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST6EE1FE6B_0") Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cGridSys GridSys { get; }
```

```
ReadOnly Property GridSys As cGridSys
Get
```

```
Dim instance As cSapModel
Dim value As cGridSys
```

```
value = instance.GridSys
```

```
property cGridSys^ GridSys {
cGridSys^ get ();
}
```

```
abstract GridSys : cGridSys with get
```

Property Value

Type: [cGridSys](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST7299405A_0?cpp=:::3633");GroupProperty

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cGroup GroupDef { get; }
```

```
ReadOnly Property GroupDef As cGroup
Get
```

```
Dim instance As cSapModel
Dim value As cGroup
```

```
value = instance.GroupDef
```

```
property cGroup^ GroupDef {
cGroup^ get ();
}
```

```
abstract GroupDef : cGroup with get
```

Property Value

Type: [cGroup](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST5F55EEE3_0") Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cLineElm LineElm { get; }
```

```
ReadOnly Property LineElm As cLineElm  
Get
```

```
Dim instance As cSapModel  
Dim value As cLineElm
```

```
value = instance.LineElm
```

```
property cLineElm^ LineElm {  
    cLineElm^ get ();  
}
```

```
abstract LineElm : cLineElm with get
```

Property Value

Type: [cLineElm](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST8F37EEE3_0?cpp=:3605.");LinkE Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cLinkElm LinkElm { get; }
```

```
ReadOnly Property LinkElm As cLinkElm
Get
```

```
Dim instance As cSapModel
Dim value As cLinkElm
```

```
value = instance.LinkElm
```

```
property cLinkElm^ LinkElm {
cLinkElm^ get ();
}
```

```
abstract LinkElm : cLinkElm with get
```

Property Value

Type: [cLinkElm](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST1CADBEAC_0?cpp3636u=.");LinkObj Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cLinkObj LinkObj { get; }
```

```
ReadOnly Property LinkObj As cLinkObj
Get
```

```
Dim instance As cSapModel
Dim value As cLinkObj
```

```
value = instance.LinkObj
```

```
property cLinkObj^ LinkObj {
cLinkObj^ get ();
}
```

```
abstract LinkObj : cLinkObj with get
```

Property Value

Type: [cLinkObj](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST3BEB8D Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cLoadCases LoadCases { get; }
```

```
ReadOnly Property LoadCases As cLoadCases
Get
```

```
Dim instance As cSapModel
Dim value As cLoadCases
```

```
value = instance.LoadCases
```

```
property cLoadCases^ LoadCases {
cLoadCases^ get ();
}
```

```
abstract LoadCases : cLoadCases with get
```

Property Value

Type: [cLoadCases](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST4059333F_0?cpp=::|LoadP Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cLoadPatterns LoadPatterns { get; }
```

```
ReadOnly Property LoadPatterns As cLoadPatterns  
Get
```

```
Dim instance As cSapModel  
Dim value As cLoadPatterns
```

```
value = instance.LoadPatterns
```

```
property cLoadPatterns^ LoadPatterns {  
    cLoadPatterns^ get ();  
}
```

```
abstract LoadPatterns : cLoadPatterns with get
```

Property Value

Type: [cLoadPatterns](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST6D84272E_0?cpp=:::3639");Options

Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cOptions Options { get; }
```

```
ReadOnly Property Options As cOptions
Get
```

```
Dim instance As cSapModel
Dim value As cOptions
```

```
value = instance.Options
```

```
property cOptions^ Options {
cOptions^ get ();
}
```

```
abstract Options : cOptions with get
```

Property Value

Type: [cOptions](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST56404673_0") Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cPattern PatternDef { get; }
```

```
ReadOnly Property PatternDef As cPattern
Get
```

```
Dim instance As cSapModel
Dim value As cPattern
```

```
value = instance.PatternDef
```

```
property cPattern^ PatternDef {
cPattern^ get ();
}
```

```
abstract PatternDef : cPattern with get
```

Property Value

Type: [cPattern](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST9E51A41A_0?cpp=::3641.");PierL Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cPierLabel PierLabel { get; }
```

```
ReadOnly Property PierLabel As cPierLabel  
Get
```

```
Dim instance As cSapModel  
Dim value As cPierLabel
```

```
value = instance.PierLabel
```

```
property cPierLabel^ PierLabel {  
    cPierLabel^ get ();  
}
```

```
abstract PierLabel : cPierLabel with get
```

Property Value

Type: [cPierLabel](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST32770FD Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cPointElm PointElm { get; }
```

```
ReadOnly Property PointElm As cPointElm
Get
```

```
Dim instance As cSapModel
Dim value As cPointElm
```

```
value = instance.PointElm
```

```
property cPointElm^ PointElm {
cPointElm^ get ();
}
```

```
abstract PointElm : cPointElm with get
```

Property Value

Type: [cPointElm](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST342926A6_0?cpp=:::3643");PointC Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cPointObj PointObj { get; }
```

```
ReadOnly Property PointObj As cPointObj  
Get
```

```
Dim instance As cSapModel  
Dim value As cPointObj
```

```
value = instance.PointObj
```

```
property cPointObj^ PointObj {  
    cPointObj^ get ();  
}
```

```
abstract PointObj : cPointObj with get
```

Property Value

Type: [cPointObj](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST61AD3DF8_0?cpp=3644.");Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cPropArea PropArea { get; }
```

```
ReadOnly Property PropArea As cPropArea
Get
```

```
Dim instance As cSapModel
Dim value As cPropArea
```

```
value = instance.PropArea
```

```
property cPropArea^ PropArea {
cPropArea^ get ();
}
```

```
abstract PropArea : cPropArea with get
```

Property Value

Type: [cPropArea](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST83D55242_0?cpp=::|8645");PropA Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cPropAreaSpring PropAreaSpring { get; }
```

```
ReadOnly Property PropAreaSpring As cPropAreaSpring
Get
```

```
Dim instance As cSapModel
Dim value As cPropAreaSpring
```

```
value = instance.PropAreaSpring
```

```
property cPropAreaSpring^ PropAreaSpring {
cPropAreaSpring^ get ();
}
```

```
abstract PropAreaSpring : cPropAreaSpring with get
```

Property Value

Type: [cPropAreaSpring](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST8A62231B_0?cpp=::3646");Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cPropFrame PropFrame { get; }
```

```
ReadOnly Property PropFrame As cPropFrame
Get
```

```
Dim instance As cSapModel
Dim value As cPropFrame
```

```
value = instance.PropFrame
```

```
property cPropFrame^ PropFrame {
cPropFrame^ get ();
}
```

```
abstract PropFrame : cPropFrame with get
```

Property Value

Type: [cPropFrame](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST31BCF721_0?cpp=::3647.");Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cPropLineSpring PropLineSpring { get; }
```

```
ReadOnly Property PropLineSpring As cPropLineSpring
Get
```

```
Dim instance As cSapModel
Dim value As cPropLineSpring
```

```
value = instance.PropLineSpring
```

```
property cPropLineSpring^ PropLineSpring {
cPropLineSpring^ get ();
}
```

```
abstract PropLineSpring : cPropLineSpring with get
```

Property Value

Type: [cPropLineSpring](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST8FF86E30_0?cpp=::3648");PropL Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cPropLink PropLink { get; }
```

```
ReadOnly Property PropLink As cPropLink  
Get
```

```
Dim instance As cSapModel  
Dim value As cPropLink
```

```
value = instance.PropLink
```

```
property cPropLink^ PropLink {  
    cPropLink^ get ();  
}
```

```
abstract PropLink : cPropLink with get
```

Property Value

Type: [cPropLink](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST16B8E378_0?cpp=::3649");Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cPropMaterial PropMaterial { get; }
```

```
ReadOnly Property PropMaterial As cPropMaterial
Get
```

```
Dim instance As cSapModel
Dim value As cPropMaterial
```

```
value = instance.PropMaterial
```

```
property cPropMaterial^ PropMaterial {
cPropMaterial^ get ();
}
```

```
abstract PropMaterial : cPropMaterial with get
```

Property Value

Type: [cPropMaterial](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST88261ACB_0?cpp=:3659.");Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cPropPointSpring PropPointSpring { get; }
```

```
ReadOnly Property PropPointSpring As cPropPointSpring
Get
```

```
Dim instance As cSapModel
Dim value As cPropPointSpring
```

```
value = instance.PropPointSpring
```

```
property cPropPointSpring^ PropPointSpring {
cPropPointSpring^ get ();
}
```

```
abstract PropPointSpring : cPropPointSpring with get
```

Property Value

Type: [cPropPointSpring](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST1677AC1F_0?cpp=::3651.");Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cPropRebar PropRebar { get; }
```

```
ReadOnly Property PropRebar As cPropRebar
Get
```

```
Dim instance As cSapModel
Dim value As cPropRebar
```

```
value = instance.PropRebar
```

```
property cPropRebar^ PropRebar {
cPropRebar^ get ();
}
```

```
abstract PropRebar : cPropRebar with get
```

Property Value

Type: [cPropRebar](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST6FAC2388_0?cpp=::3652");Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cPropTendon PropTendon { get; }
```

```
ReadOnly Property PropTendon As cPropTendon
Get
```

```
Dim instance As cSapModel
Dim value As cPropTendon
```

```
value = instance.PropTendon
```

```
property cPropTendon^ PropTendon {
cPropTendon^ get ();
}
```

```
abstract PropTendon : cPropTendon with get
```

Property Value

Type: [cPropTendon](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LSTA05A1371_0?cpp=::3653");Resp

Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cCombo RespCombo { get; }
```

```
ReadOnly Property RespCombo As cCombo
Get
```

```
Dim instance As cSapModel
Dim value As cCombo
```

```
value = instance.RespCombo
```

```
property cCombo^ RespCombo {
cCombo^ get ();
}
```

```
abstract RespCombo : cCombo with get
```

Property Value

Type: [cCombo](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LSTC6FB2B Property

Gives access to API calls related to analysis results.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cAnalysisResults Results { get; }
```

```
ReadOnly Property Results As cAnalysisResults
Get
```

```
Dim instance As cSapModel
Dim value As cAnalysisResults
```

```
value = instance.Results
```

```
property cAnalysisResults^ Results {
cAnalysisResults^ get ();
}
```

```
abstract Results : cAnalysisResults with get
```

Property Value

Type: [cAnalysisResults](#)

Return Value

Type: [cAnalysisResults](#)

▲ Remarks

The analysis results are returned in a collection of one-dimensional result arrays. Each result array is created as a dynamic array by the API user. As an example, in VB .NET a dynamic string array is defined by:

VB

[Copy](#)

```
Dim MyArray() as String
```

The array is dimensioned to (NumberResults + 1) inside the program, filled with the result values, and returned to the API user.

The arrays are zero-based. Thus the first result is at array index 0, and the last result is at array index (NumberResults- 1). For example, the StepType() array is filled as:

cSapModelspan id="LSTC6FB2BE3_0"AddLanguageSpecificTextSet("LSTC6FB2BE3_0?cpp=354.");Res

VB

Copy

StepType(0) = Step type for first result returned

VB

Copy

StepType(1) = Step type for second result

VB

Copy

StepType(NumberResults- 1) = Step type for last result

Immediately before requesting results data, it is a good idea to clear the SelectedForOutput flag for all load cases and response combinations and then to set the flag True for those cases and combos for which output is to be generated. This avoids confusion as to which cases and combos are currently selected for output.

▲ See Also

Reference

cSapModel Interface

ETABSv1 Namespace

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST87C934F4_0?cpp=::3656");SelectProperty

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cSelect SelectObj { get; }
```

```
ReadOnly Property SelectObj As cSelect
Get
```

```
Dim instance As cSapModel
Dim value As cSelect
```

```
value = instance.SelectObj
```

```
property cSelect^ SelectObj {
cSelect^ get ();
}
```

```
abstract SelectObj : cSelect with get
```

Property Value

Type: [cSelect](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LSTBAA05274_0") Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cSpandrelLabel SpandrelLabel { get; }
```

```
ReadOnly Property SpandrelLabel As cSpandrelLabel
Get
```

```
Dim instance As cSapModel
Dim value As cSpandrelLabel
```

```
value = instance.SpandrelLabel
```

```
property cSpandrelLabel^ SpandrelLabel {
cSpandrelLabel^ get ();
}
```

```
abstract SpandrelLabel : cSpandrelLabel with get
```

Property Value

Type: [cSpandrelLabel](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LSTEED7A284_0") Story Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cStory Story { get; }
```

```
ReadOnly Property Story As cStory
Get
```

```
Dim instance As cSapModel
Dim value As cStory
```

```
value = instance.Story
```

```
property cStory^ Story {
cStory^ get ();
}
```

```
abstract Story : cStory with get
```

Property Value

Type: [cStory](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LSTB61830C9_0?cpp=::p659");TendonObj Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cTendonObj TendonObj { get; }
```

```
ReadOnly Property TendonObj As cTendonObj
Get
```

```
Dim instance As cSapModel
Dim value As cTendonObj
```

```
value = instance.TendonObj
```

```
property cTendonObj^ TendonObj {
cTendonObj^ get ();
}
```

```
abstract TendonObj : cTendonObj with get
```

Property Value

Type: [cTendonObj](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LSTAE4947C8_0?cpp=:3669.");Tower

Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cTower Tower { get; }
```

```
ReadOnly Property Tower As cTower
Get
```

```
Dim instance As cSapModel
Dim value As cTower
```

```
value = instance.Tower
```

```
property cTower^ Tower {
cTower^ get ();
}
```

```
abstract Tower : cTower with get
```

Property Value

Type: [cTower](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST2FF5C9A_0?cpp=::|n365");View Property

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
cView View { get; }
```

```
ReadOnly Property View As cView
Get
```

```
Dim instance As cSapModel
Dim value As cView
```

```
value = instance.View
```

```
property cView^ View {
cView^ get ();
}
```

```
abstract View : cView with get
```

Property Value

Type: [cView](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModel Methods

The cSapModel type exposes the following members.

Methods

Á	Name	Description
⇒	<u>GetDatabaseUnits</u>	Returns a value from the <u>eUnits</u> enumeration indicating the database units for the model. All data is internally stored in the model in these units and converted to the present units as needed.
⇒	<u>GetDatabaseUnits_2</u>	Retrieves the database units for the model. All data is internally stored in the model in these units and converted to the present units as needed.
⇒	<u>GetMergeTol</u>	
⇒	<u>GetModelFilename</u>	Returns a string that represents the filename of the current model, with or without the full path.
⇒	<u>GetModelFilepath</u>	Returns the filepath of the current model
⇒	<u>GetModelIsLocked</u>	
⇒	<u>GetPresentCoordSystem</u>	
⇒	<u>GetPresentUnits</u>	Returns a value from the <u>eUnits</u> enumeration indicating the units presently specified for the model.
⇒	<u>GetPresentUnits_2</u>	Retrieves the units presently specified for the model
⇒	<u>GetProgramInfo</u>	Retrieve information about the program
⇒	<u>GetProjectInfo</u>	Retrieves the project information data.
⇒	<u>GetVersion</u>	Returns the program version.
⇒	<u>InitializeNewModel</u>	This function clears the previous model and initializes the program for a new model. If it is later needed, you should save your previous model prior to calling this function.
⇒	<u>SetMergeTol</u>	
⇒	<u>SetModelIsLocked</u>	Locks or unlocks the model.
⇒	<u>SetPresentUnits</u>	Sets the display (present) units.
⇒	<u>SetPresentUnits_2</u>	Specifies the units for the model
⇒	<u>SetProjectInfo</u>	Sets the data for an item in the project information.
⇒	<u>TreeIsUpdateSuspended</u>	
⇒	<u>TreeResumeUpdateData</u>	
⇒	<u>TreeSuspendUpdateData</u>	

Top

See Also

Reference

cSapModel Interface

ETABSV1 Namespace

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST2F5F3A17_0?cpp=::3664");GetDatabaseUnits() Method

Returns a value from the [eUnits](#) enumeration indicating the database units for the model. All data is internally stored in the model in these units and converted to the present units as needed.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
eUnits GetDatabaseUnits()
```

```
Function GetDatabaseUnits As eUnits
```

```
Dim instance As cSapModel
```

```
Dim returnValue As eUnits
```

```
returnValue = instance.GetDatabaseUnits()
```

```
eUnits GetDatabaseUnits()
```

```
abstract GetDatabaseUnits : unit -> eUnits
```

Return Value

Type: [eUnits](#)

Returns a value from the [eUnits](#) enumeration if successful, otherwise returns zero.

Remarks

Examples

[VB](#)

[Copy](#)

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
```

cSapModelspan id="LST2F5F3A17_0"AddLanguageSpecificTextSet("LST2F5F3A17_0?cpp=::3664");GetDatabaseUnits()

Introduction

```
ret = SapModel.InitializeNewModel()  
  
'create steel deck template model  
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)  
  
'get database units  
Dim dbUnits as eUnits  
dbUnits = SapModel.GetDatabaseUnits()  
  
'run model (this will create the analysis model)  
ret = SapModel.Analyze.RunAnalysis  
  
'close ETABS  
EtabsObject.ApplicationExit(False)  
  
'clean up variables  
SapModel = Nothing  
EtabsObject = Nothing  
End Sub
```

📌 See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LSTFD197EB8_0")

Method

Retrieves the database units for the model. All data is internally stored in the model in these units and converted to the present units as needed.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetDatabaseUnits_2(
    ref eForce forceUnits,
    ref eLength lengthUnits,
    ref eTemperature temperatureUnits
)
```

```
Function GetDatabaseUnits_2 (
    ByRef forceUnits As eForce,
    ByRef lengthUnits As eLength,
    ByRef temperatureUnits As eTemperature
) As Integer
```

```
Dim instance As cSapModel
Dim forceUnits As eForce
Dim lengthUnits As eLength
Dim temperatureUnits As eTemperature
Dim returnValue As Integer
```

```
returnValue = instance.GetDatabaseUnits_2(forceUnits,
    lengthUnits, temperatureUnits)
```

```
int GetDatabaseUnits_2(
    eForce% forceUnits,
    eLength% lengthUnits,
    eTemperature% temperatureUnits
)
```

```
abstract GetDatabaseUnits_2 :
    forceUnits : eForce byref *
    lengthUnits : eLength byref *
    temperatureUnits : eTemperature byref -> int
```

Parameters

forceUnits

Type: [ETABSv1eForce](#)

A value from the [eForce](#) enumeration

lengthUnits

cSapModelspan id="LSTFD197EB8_0"AddLanguageSpecificTextSet("LSTFD197EB8_0?cpp=:3666.");GetD

Type: ETABSv1eLength

A value from the eLength enumeration

temperatureUnits

Type: ETABSv1eTemperature

A value from the eTemperature enumeration

Return Value

Type: Int32

Returns zero if the units are successfully retrieved, otherwise returns nonzero

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'get database units
    Dim forceUnits As eForce
    Dim lengthUnits As eLength
    Dim temperatureUnits As eTemperature
    ret = SapModel.GetDatabaseUnits_2(forceUnits, lengthUnits, temperatureUnits)

    'run model (this will create the analysis model)
    ret = SapModel.Analyze.RunAnalysis

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cSapModel Interface

ETABSv1 Namespace

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LSTF809D2C Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetMergeTol(
    ref double MergeTol
)
```

```
Function GetMergeTol (
    ByRef MergeTol As Double
) As Integer
```

```
Dim instance As cSapModel
Dim MergeTol As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetMergeTol(MergeTol)
```

```
int GetMergeTol(
    double% MergeTol
)
```

```
abstract GetMergeTol :
    MergeTol : float byref -> int
```

Parameters

MergeTol
Type: [System.Double](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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Introduction

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST1C30AEFA_0?cpp=3p71=");GetModelFilename

Method

Returns a string that represents the filename of the current model, with or without the full path.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
string GetModelFilename(
    bool IncludePath = true
)
```

```
Function GetModelFilename (
    Optional IncludePath As Boolean = true
) As String
```

```
Dim instance As cSapModel
Dim IncludePath As Boolean
Dim returnValue As String
```

```
returnValue = instance.GetModelFilename(IncludePath)
```

```
String^ GetModelFilename(
    bool IncludePath = true
)
```

```
abstract GetModelFilename :
    ?IncludePath : bool
(* Defaults:
    let _IncludePath = defaultArg IncludePath true
*)
-> string
```

Parameters

IncludePath (Optional)

Type: System.Boolean

A boolean (True or False) value. When this item is True, the returned filename includes the full path where the file is located.

Return Value

Type: String

Returns a string that represents the filename of the current model, with or without the full path.

cSapModelspan id="LST1C30AEFA_0"AddLanguageSpecificTextSet("LST1C30AEFA_0?cpp=3p71=");GetModelFilename

Remarks

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'save the model
    ret = SapModel.File.Save("C:\CSI_API_temp\API_1-001.edb")

    'display the filename of the model
    MsgBox(SapModel.GetModelFilename())

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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Namespace: ETABSv1

▲ Syntax

C#

VBC++

F#

Copy

```
Dim instance As cSapModel
```

```
returnValue = instance.GetModelFilepath()
```

```
String^ GetModelFilepath()
```

```
abstract GetModelFilepath : unit -> string
```

Type: String

Remarks

▲ See Also

cSapModel Interface

ETABSv1 Namespace

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST49D606B0_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

bool GetModelIsLocked()

Function GetModelIsLocked As Boolean

Dim instance As cSapModel

Dim returnValue As Boolean

returnValue = instance.GetModelIsLocked()

bool GetModelIsLocked()

abstract GetModelIsLocked : unit -> bool

Return Value

Type: [Boolean](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LSTA42ADD Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

string GetPresentCoordSystem()

Function GetPresentCoordSystem As String

Dim instance As cSapModel

Dim returnValue As String

returnValue = instance.GetPresentCoordSystem()

String^ GetPresentCoordSystem()

abstract GetPresentCoordSystem : unit -> string

Return Value

Type: [String](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LSTD9DFA6A0")

Method

Returns a value from the [eUnits](#) enumeration indicating the units presently specified for the model.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

eUnits GetPresentUnits()

Function GetPresentUnits As eUnits

Dim instance As cSapModel

Dim returnValue As eUnits

returnValue = instance.GetPresentUnits()

eUnits GetPresentUnits()

abstract GetPresentUnits : unit -> eUnits

Return Value

Type: [eUnits](#)

Returns a value from the [eUnits](#) enumeration if successful, otherwise returns zero.

Remarks

Examples

[VB](#)

[Copy](#)

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()
```

cSapModelspan id="LSTD9DFA6A_0"AddLanguageSpecificTextSet("LSTD9DFA6A_0?cpp=::|3676);GetPr

Introduction

```
'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'get present units
    Dim presentUnits as eUnits
    presentUnits = SapModel.GetPresentUnits()

'run model (this will create the analysis model)
    ret = SapModel.Analyze.RunAnalysis

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LSTC41F0893_0")

Method

Retrieves the units presently specified for the model

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetPresentUnits_2(
    ref eForce forceUnits,
    ref eLength lengthUnits,
    ref eTemperature temperatureUnits
)
```

```
Function GetPresentUnits_2 (
    ByRef forceUnits As eForce,
    ByRef lengthUnits As eLength,
    ByRef temperatureUnits As eTemperature
) As Integer
```

```
Dim instance As cSapModel
Dim forceUnits As eForce
Dim lengthUnits As eLength
Dim temperatureUnits As eTemperature
Dim returnValue As Integer
```

```
returnValue = instance.GetPresentUnits_2(forceUnits,
    lengthUnits, temperatureUnits)
```

```
int GetPresentUnits_2(
    eForce% forceUnits,
    eLength% lengthUnits,
    eTemperature% temperatureUnits
)
```

```
abstract GetPresentUnits_2 :
    forceUnits : eForce byref *
    lengthUnits : eLength byref *
    temperatureUnits : eTemperature byref -> int
```

Parameters

forceUnits

Type: [ETABSv1eForce](#)

A value from the [eForce](#) enumeration

lengthUnits

Type: ETABSv1eLength

A value from the eLength enumeration

temperatureUnits

Type: ETABSv1eTemperature

A value from the eTemperature enumeration

Return Value

Type: Int32

Returns zero if the units are successfully retrieved, otherwise returns nonzero

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'get present units
    Dim forceUnits As eForce
    Dim lengthUnits As eLength
    Dim temperatureUnits As eTemperature
    ret = SapModel.GetPresentUnits_2(forceUnits, lengthUnits, temperatureUnits)

    'run model (this will create the analysis model)
    ret = SapModel.Analyze.RunAnalysis

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cSapModel Interface

ETABSv1 Namespace

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LSTD49F9566_0?cpp=::1");GetProgramInfo

Method

Retrieve information about the program

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetProgramInfo(
    ref string ProgramName,
    ref string ProgramVersion,
    ref string ProgramLevel
)
```

```
Function GetProgramInfo (
    ByRef ProgramName As String,
    ByRef ProgramVersion As String,
    ByRef ProgramLevel As String
) As Integer
```

```
Dim instance As cSapModel
Dim ProgramName As String
Dim ProgramVersion As String
Dim ProgramLevel As String
Dim returnValue As Integer
```

```
returnValue = instance.GetProgramInfo(ProgramName,
    ProgramVersion, ProgramLevel)
```

```
int GetProgramInfo(
    String^% ProgramName,
    String^% ProgramVersion,
    String^% ProgramLevel
)
```

```
abstract GetProgramInfo :
    ProgramName : string byref *
    ProgramVersion : string byref *
    ProgramLevel : string byref -> int
```

Parameters

ProgramName

Type: [SystemString](#)

The program name

ProgramVersion

Type:Â SystemString
The program version
ProgramLevel
Type:Â SystemString
The program level

Return Value

Type:Â Int32
Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value.
▲ See Also

Reference

cSapModel Interface

ETABSv1 Namespace

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST29D46237_0?cpp=::|8083");GetProjectInfo

Method

Retrieves the project information data.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetProjectInfo(
    ref int NumberItems,
    ref string[] Item,
    ref string[] Data
)
```

```
Function GetProjectInfo (
    ByRef NumberItems As Integer,
    ByRef Item As String(),
    ByRef Data As String()
) As Integer
```

```
Dim instance As cSapModel
Dim NumberItems As Integer
Dim Item As String()
Dim Data As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetProjectInfo(NumberItems,
    Item, Data)
```

```
int GetProjectInfo(
    int% NumberItems,
    array<String^>^% Item,
    array<String^>^% Data
)
```

```
abstract GetProjectInfo :
    NumberItems : int byref *
    Item : string[] byref *
    Data : string[] byref -> int
```

Parameters

NumberItems

Type: [System.Int32](#)

The number of project info items returned.

Item

Type: SystemString

This is an array that includes the name of the project information item.

Data

Type: SystemString

Return Value

Type: Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

▲ See Also

Reference

cSapModel Interface

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST148C3CA Method

Returns the program version.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetVersion(
    ref string Version,
    ref double MyVersionNumber
)
```

```
Function GetVersion (
    ByRef Version As String,
    ByRef MyVersionNumber As Double
) As Integer
```

```
Dim instance As cSapModel
Dim Version As String
Dim MyVersionNumber As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetVersion(Version,
MyVersionNumber)
```

```
int GetVersion(
    String^% Version,
    double% MyVersionNumber
)
```

```
abstract GetVersion :
    Version : string byref *
    MyVersionNumber : float byref -> int
```

Parameters

Version

Type: SystemString

MyVersionNumber

Type: SystemDouble

The program version number that is used internally by the program and not displayed to the user.

Return Value

Type: Int32

Returns zero if the information is successfully retrieved; otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Version As String
    Dim MyVersionNumber As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'get program version
    ret = SapModel.GetVersion(Version, MyVersionNumber)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cSapModel Interface

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LSTB8665F52_0")

Method

This function clears the previous model and initializes the program for a new model. If it is later needed, you should save your previous model prior to calling this function.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int InitializeNewModel(
    eUnits Units = eUnits.kip_in_F
)
```

```
Function InitializeNewModel (
    Optional As eUnits = eUnits.kip_in_F
) As Integer
```

```
Dim instance As cSapModel
Dim Units As eUnits
Dim returnValue As Integer
```

```
returnValue = instance.InitializeNewModel(Units)
```

```
int InitializeNewModel(
    eUnits Units = eUnits::kip_in_F
)
```

```
abstract InitializeNewModel :
    ?Units : eUnits
(* Defaults:
    let _Units = defaultArg Units eUnits.kip_in_F
*)
-> int
```

Parameters

Units (Optional)

Type: ETABSv1eUnits

This is the display (present) units for the new model. Possible items in the eUnits enumeration.

All numerical items are shown in the GUI in display units. All API calls use display units.

All data stored internally in the model is in database units. If the length unit in the display units is inch or feet, the database units will be set to lb_in_F.

cSapModelspan id="LSTB8665F52_0"AddLanguageSpecificTextSet("LSTB8665F52_0?cpp=::|0007");Initiali

Otherwise, the database units are set to N_mm_C.

Return Value

Type: Int32

The function returns zero if a new model is successfully initialized, otherwise it returns a nonzero value.

Remarks

After calling the InitializeNewModel function, it is not necessary to also call the ApplicationStart function because the functionality of the ApplicationStart function is included in the InitializeNewModel function.

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'run model (this will create the analysis model)
    ret = SapModel.Analyze.RunAnalysis

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

cSapModel Interface

ETABSv1 Namespace

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST9495AFC7_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetMergeTol(
double MergeTol
)
```

```
Function SetMergeTol (
MergeTol As Double
) As Integer
```

```
Dim instance As cSapModel
Dim MergeTol As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetMergeTol(MergeTol)
```

```
int SetMergeTol(
double MergeTol
)
```

```
abstract SetMergeTol :
    MergeTol : float -> int
```

Parameters

MergeTol
Type: [System.Double](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LSTF76E04B6_0?cpp=::3692");SetM

Method

Locks or unlocks the model.

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int SetModelIsLocked(
    bool Lockit
)
```

```
Function SetModelIsLocked (
    Lockit As Boolean
) As Integer
```

```
Dim instance As cSapModel
Dim Lockit As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetModelIsLocked(Lockit)
```

```
int SetModelIsLocked(
    bool Lockit
)
```

```
abstract SetModelIsLocked :
    Lockit : bool -> int
```

Parameters

Lockit

Type: SystemBoolean

The item is True if the model is to be locked and False if it is to be unlocked.

Return Value

Type: Int32

Returns zero if the locked status of the model is successfully set. Otherwise it returns a nonzero value.

▲ Remarks

With some exceptions, definitions and assignments can not be changed in a model while the model is locked. If an attempt is made to change a definition or assignment while the model is locked and that change is not allowed in a locked model, an error will be returned.

cSapModelspan id="LSTF76E04B6_0"AddLanguageSpecificTextSet("LSTF76E04B6_0?cpp=::3692");SetM

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'Lock model
    ret = SapModel.SetModelIsLocked(True)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LSTB3AE38D_0?cpp=::|n3694;SetPre

Method

Sets the display (present) units.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetPresentUnits(
    eUnits Units
)
```

```
Function SetPresentUnits (
    Units As eUnits
) As Integer
```

```
Dim instance As cSapModel
Dim Units As eUnits
Dim returnValue As Integer
```

```
returnValue = instance.SetPresentUnits(Units)
```

```
int SetPresentUnits(
    eUnits Units
)
```

```
abstract SetPresentUnits :
    Units : eUnits -> int
```

Parameters

Units

Type: [ETABSv1eUnits](#)

One of the items in the [eUnits](#) enumeration

Return Value

Type: [Int32](#)

Returns zero if the units are successfully set and nonzero if they are not set.

▲ Remarks

▲ Examples

VB

[Copy](#)

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As c0API
```

cSapModelspan id="LSTB3AE38D_0"AddLanguageSpecificTextSet("LSTB3AE38D_0?cpp=::|n3694;SetPre

Introduction

```
Dim ret As Integer = -1

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'set present units to KN-m
ret = SapModel.SetPresentUnits(KN_m_C)

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'get present units
Dim presentUnits as eUnits
presentUnits = SapModel.GetPresentUnits()

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST6A74209_0?cpp=::|nu3696SetPres Method

Specifies the units for the model

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetPresentUnits_2(
    eForce forceUnits,
    eLength lengthUnits,
    eTemperature temperatureUnits
)
```

```
Function SetPresentUnits_2 (
    forceUnits As eForce,
    lengthUnits As eLength,
    temperatureUnits As eTemperature
) As Integer
```

```
Dim instance As cSapModel
Dim forceUnits As eForce
Dim lengthUnits As eLength
Dim temperatureUnits As eTemperature
Dim returnValue As Integer
```

```
returnValue = instance.SetPresentUnits_2(forceUnits,
    lengthUnits, temperatureUnits)
```

```
int SetPresentUnits_2(
    eForce forceUnits,
    eLength lengthUnits,
    eTemperature temperatureUnits
)
```

```
abstract SetPresentUnits_2 :
    forceUnits : eForce *
    lengthUnits : eLength *
    temperatureUnits : eTemperature -> int
```

Parameters

forceUnits

Type: [ETABSv1eForce](#)

A value from the [eForce](#) enumeration

lengthUnits

Type: ETABSv1eLength

A value from the eLength enumeration

temperatureUnits

Type: ETABSv1eTemperature

A value from the eTemperature enumeration

Return Value

Type: Int32

Returns zero if the present units are successfully set, otherwise returns nonzero

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'set present units
    ret = SapModel.SetPresentUnits_2(eForce.kN, eLength.m, eTemperature.C)

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'get present units
    Dim forceUnits As eForce
    Dim lengthUnits As eLength
    Dim temperatureUnits As eTemperature
    ret = SapModel.GetPresentUnits_2(forceUnits, lengthUnits, temperatureUnits)

    'run model (this will create the analysis model)
    ret = SapModel.Analyze.RunAnalysis

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cSapModel Interface

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST4C893D5D_0")

Method

Sets the data for an item in the project information.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetProjectInfo(
    string Item,
    string Data
)
```

```
Function SetProjectInfo (
    Item As String,
    Data As String
) As Integer
```

```
Dim instance As cSapModel
Dim Item As String
Dim Data As String
Dim returnValue As Integer
```

```
returnValue = instance.SetProjectInfo(Item,
    Data)
```

```
int SetProjectInfo(
    String^ Item,
    String^ Data
)
```

```
abstract SetProjectInfo :
    Item : string *
    Data : string -> int
```

Parameters

Item

Type: SystemString

The name of the project information item to be set.

Data

Type: SystemString

Return ValueType: Int32

Returns zero if the data is successfully set; otherwise it returns a nonzero value.

▲ **Remarks**▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set project information data
    ret = SapModel.SetProjectInfo("Company Name", "Computers and Structures, Inc.")
    ret = SapModel.SetProjectInfo("Project Name", "API Testing")
    ret = SapModel.SetProjectInfo("My Item", "My Data")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also****Reference**cSapModel InterfaceETABSV1 Namespace

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LSTE658AA83_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int TreeIsUpdateSuspended(
    ref bool IsSuspended
)
```

```
Function TreeIsUpdateSuspended (
    ByRef IsSuspended As Boolean
) As Integer
```

```
Dim instance As cSapModel
Dim IsSuspended As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.TreeIsUpdateSuspended(IsSuspended)
```

```
int TreeIsUpdateSuspended(
    bool% IsSuspended
)
```

```
abstract TreeIsUpdateSuspended :
    IsSuspended : bool byref -> int
```

Parameters

IsSuspended
Type: [System.Boolean](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LST9395E97F_0?cpp=::|8703");TreeF Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

int TreeResumeUpdateData()

Function TreeResumeUpdateData As Integer

Dim instance As cSapModel

Dim returnValue As Integer

returnValue = instance.TreeResumeUpdateData()

int TreeResumeUpdateData()

abstract TreeResumeUpdateData : unit -> int

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSapModelAddLanguageSpecificTextSet("LSTE6FCB898_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int TreeSuspendUpdateData(
    bool updateAtResume
)
```

```
Function TreeSuspendUpdateData (
    updateAtResume As Boolean
) As Integer
```

```
Dim instance As cSapModel
Dim updateAtResume As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.TreeSuspendUpdateData(updateAtResume)
```

```
int TreeSuspendUpdateData(
    bool updateAtResume
)
```

```
abstract TreeSuspendUpdateData :
    updateAtResume : bool -> int
```

Parameters

updateAtResume
Type: [System.Boolean](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cSapModel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSelect Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cSelect
```

```
Public Interface cSelect
```

```
Dim instance As cSelect
```

```
public interface class cSelect
```

```
type cSelect = interface end
```

The cSelect type exposes the following members.

▲ Methods

▲	Name	Description
◆	All	Selects or deselects all objects in the model.
◆	ClearSelection	Deselects all objects in the model.
◆	GetSelected	Retrieves a list of selected objects
◆	Group	Selects or deselects all objects in the specified group.
◆	InvertSelection	Deselects all selected objects and selects all unselected objects; that is, it inverts the selection
◆	PreviousSelection	Restores the previous selection

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSelect Methods

The cSelect type exposes the following members.

Methods

⌕	Name	Description
⌕	<u>All</u>	Selects or deselects all objects in the model.
⌕	<u>ClearSelection</u>	Deselects all objects in the model.
⌕	<u>GetSelected</u>	Retrieves a list of selected objects
⌕	<u>Group</u>	Selects or deselects all objects in the specified group.
⌕	<u>InvertSelection</u>	Deselects all selected objects and selects all unselected objects; that is, it inverts the selection
⌕	<u>PreviousSelection</u>	Restores the previous selection

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⌕ See Also

Reference

cSelect Interface

ETABSv1 Namespace

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CSI API ETABS v1

cSelectAddLanguageSpecificTextSet("LSTC26FCA50_0") Method

Selects or deselects all objects in the model.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int All(
    bool Deselect = false
)
```

```
Function All (
    Optional Deselect As Boolean = false
) As Integer
```

```
Dim instance As cSelect
Dim Deselect As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.All(Deselect)
```

```
int All(
    bool Deselect = false
)
```

```
abstract All :
    ?Deselect : bool
(* Defaults:
    let _Deselect = defaultArg Deselect false
*)
-> int
```

Parameters

Deselect (Optional)

Type: [System.Boolean](#)

The item is False if objects are to be selected and True if they are to be deselected.

Return Value

Type: [Int32](#)

Returns zero if the selection is successfully completed, otherwise it returns nonzero.

▲ Remarks

▲ Examples

cSelectAddLanguageSpecificTextSet("LSTC26FCA50_0?cpp=::|nu3708")All Meth

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'select all
    ret = SapModel.SelectObj.All()

    'deselect all
    ret = SapModel.SelectObj.All(True)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

ReferencecSelect InterfaceETABSv1 Namespace

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CSI API ETABS v1

cSelectAddLanguageSpecificTextSet("LSTBB59FA20_0")

Method

Deselects all objects in the model.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int ClearSelection()
```

```
Function ClearSelection As Integer
```

```
Dim instance As cSelect
```

```
Dim returnValue As Integer
```

```
returnValue = instance.ClearSelection()
```

```
int ClearSelection()
```

```
abstract ClearSelection : unit -> int
```

Return Value

Type: Int32

Returns zero if the selection status is successfully set, otherwise it returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()
```

Introduction

```
'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

'select all
    ret = SapModel.SelectObj.All

'clear selection
    ret = SapModel.SelectObj.ClearSelection()

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

🔍 See Also

Reference

cSelect Interface

ETABSv1 Namespace

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CSI API ETABS v1

cSelectAddLanguageSpecificTextSet("LST586E065E_0 Method

Retrieves a list of selected objects

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetSelected(
    ref int NumberItems,
    ref int[] ObjectType,
    ref string[] ObjectName
)
```

```
Function GetSelected (
    ByRef NumberItems As Integer,
    ByRef ObjectType As Integer(),
    ByRef ObjectName As String()
) As Integer
```

```
Dim instance As cSelect
Dim NumberItems As Integer
Dim ObjectType As Integer()
Dim ObjectName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetSelected(NumberItems,
    ObjectType, ObjectName)
```

```
int GetSelected(
    int% NumberItems,
    array<int>^% ObjectType,
    array<String>^% ObjectName
)
```

```
abstract GetSelected :
    NumberItems : int byref *
    ObjectType : int[] byref *
    ObjectName : string[] byref -> int
```

Parameters

NumberItems

Type: System.Int32

The number of selected objects

ObjectType

Type: SystemInt32

This is an array that includes the object type of each selected object.

1. Point object
2. Frame object
3. Cable object
4. Tendon object
5. Area object
6. Solid object
7. Link object

ObjectName

Type: SystemString

This is an array that includes the name of each selected object

Return Value

Type: Int32

Returns zero if the selection list is successfully retrieved, otherwise it returns a nonzero value

▲ Remarks

▲ See Also

Reference

cSelect Interface

ETABSv1 Namespace

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CSI API ETABS v1

cSelectAddLanguageSpecificTextSet("LST7509C626_0 Method

Selects or deselects all objects in the specified group.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int Group(
    string Name,
    bool Deselect = false
)
```

```
Function Group (
    Name As String,
    Deselect As Boolean = false
) As Integer
```

```
Dim instance As cSelect
Dim Name As String
Dim Deselect As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.Group(Name, Deselect)
```

```
int Group(
    String^ Name,
    bool Deselect = false
)
```

```
abstract Group :
    Name : string *
    ?Deselect : bool
(* Defaults:
    let _Deselect = defaultArg Deselect false
*)
-> int
```

Parameters

Name

Type: SystemString

The name of an existing group.

Deselect (Optional)

Type: SystemBoolean

Return Value

Type: Int32

Returns zero if the selection is successfully completed, otherwise it returns nonzero.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'select group
    ret = SapModel.SelectObj.Group("ALL")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also**

Reference

cSelect Interface

ETABSv1 Namespace

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CSI API ETABS v1

cSelectAddLanguageSpecificTextSet("LSTE1F6273C_0") Method

Deselects all selected objects and selects all unselected objects; that is, it inverts the selection

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int InvertSelection()
```

```
Function InvertSelection As Integer
```

```
Dim instance As cSelect
```

```
Dim returnValue As Integer
```

```
returnValue = instance.InvertSelection()
```

```
int InvertSelection()
```

```
abstract InvertSelection : unit -> int
```

Return Value

Type: [Int32](#)

Returns zero if the selection is successfully inverted, otherwise it returns nonzero

▲ Remarks

▲ See Also

Reference

[cSelect Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSelectAddLanguageSpecificTextSet("LSTBD87E41_0? Method

Restores the previous selection

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▀ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int PreviousSelection()
```

```
Function PreviousSelection As Integer
```

```
Dim instance As cSelect
```

```
Dim returnValue As Integer
```

```
returnValue = instance.PreviousSelection()
```

```
int PreviousSelection()
```

```
abstract PreviousSelection : unit -> int
```

Return Value

Type: [Int32](#)

Returns zero if the selection is successfully restored, otherwise it returns nonzero

▀ Remarks

▀ See Also

Reference

[cSelect Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSpandrelLabel Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cSpandrelLabel
```

```
Public Interface cSpandrelLabel
```

```
Dim instance As cSpandrelLabel
```

```
public interface class cSpandrelLabel
```

```
type cSpandrelLabel = interface end
```

The cSpandrelLabel type exposes the following members.

▲ Methods

▲	Name	Description
◆	ChangeName	Changes the name of a defined Spandrel Label
◆	Delete	Deletes the specified Spandrel Label
◆	GetNameList	Retrieves the names of all defined Spandrel Labels
◆	GetSectionProperties	Retrieves the section properties for a specified spandrel
◆	GetSpandrel	Retrieves the specified Spandrel Label
◆	SetSpandrel	Adds a new Spandrel Label, or modifies an existing Spandrel Label

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▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSpandrelLabel Methods

The [cSpandrelLabel](#) type exposes the following members.

Methods

⌕	Name	Description
⌕	ChangeName	Changes the name of a defined Spandrel Label
⌕	Delete	Deletes the specified Spandrel Label
⌕	GetNameList	Retrieves the names of all defined Spandrel Labels
⌕	GetSectionProperties	Retrieves the section properties for a specified spandrel
⌕	GetSpandrel	Retrieves the specified Spandrel Label
⌕	SetSpandrel	Adds a new Spandrel Label, or modifies an existing Spandrel Label

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⌕ See Also

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CSI API ETABS v1

cSpandrelLabelAddLanguageSpecificTextSet("LST11B Method

Changes the name of a defined Spandrel Label

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int ChangeName(
    string Name,
    string NewName
)
```

```
Function ChangeName (
    Name As String,
    NewName As String
) As Integer
```

```
Dim instance As cSpandrelLabel
Dim Name As String
Dim NewName As String
Dim returnValue As Integer
```

```
returnValue = instance.ChangeName(Name,
    NewName)
```

```
int ChangeName(
    String^ Name,
    String^ NewName
)
```

```
abstract ChangeName :
    Name : string *
    NewName : string -> int
```

Parameters

Name

Type: SystemString

The existing name of a defined Spandrel Label

NewName

Type: SystemString

The new name for the Spandrel Label

Return Value

Type: Int32

Returns zero if the new name is successfully applied, otherwise it returns a nonzero value.

▀ **Remarks**

▀ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new Spandrel label
    ret = SapModel.SpandrelLabel.SetSpandrel("MySpandrel", False)

    'change Spandrel label
    ret = SapModel.SpandrelLabel.ChangeName("MySpandrel", "Spandrel1A")

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▀ **See Also**

Reference

cSpandrelLabel Interface

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CSI API ETABS v1

cSpandrelLabelAddLanguageSpecificTextSet("LSTE4E Method

Deletes the specified Spandrel Label

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int Delete(
    string Name
)
```

```
Function Delete (
    Name As String
) As Integer
```

```
Dim instance As cSpandrelLabel
Dim Name As String
Dim returnValue As Integer
```

```
returnValue = instance.Delete(Name)
```

```
int Delete(
    String^ Name
)
```

```
abstract Delete :
    Name : string -> int
```

Parameters

Name

Type: SystemString

The name of an existing Spandrel Label

Return Value

Type: Int32

Returns zero if the Spandrel Label is successfully deleted, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
```

cSpandrelLabelspan id="LSTE4E46838_0"AddLanguageSpecificTextSet("LSTE4E46838_0?cp=22nu=");D

Introduction

```
Dim SapModel As cSapModel
Dim EtabsObject As cOAPI
Dim ret As Integer = -1

'create ETABS object
Dim myHelper as cHelper = New Helper
EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

'start ETABS application
ret = EtabsObject.ApplicationStart()

'create SapModel object
SapModel = EtabsObject.SapModel

'initialize model
ret = SapModel.InitializeNewModel()

'create steel deck template model
ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'define new Spandrel label
ret = SapModel.SpandrelLabel.SetSpandrel("MySpandrel", False)

'change Spandrel name
ret = SapModel.SpandrelLabel.Delete("MySpandrel")

'close ETABS
EtabsObject.ApplicationExit(False)

'clean up variables
SapModel = Nothing
EtabsObject = Nothing

End Sub
```

📌 See Also

Reference

[cSpandrelLabel Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cSpandrelLabelAddLanguageSpecificTextSet("LSTE1D Method

Retrieves the names of all defined Spandrel Labels

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetNameList(
    ref int NumberNames,
    ref string[] MyName,
    ref bool[] IsMultiStory
)
```

```
Function GetNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String(),
    ByRef IsMultiStory As Boolean()
) As Integer
```

```
Dim instance As cSpandrelLabel
Dim NumberNames As Integer
Dim MyName As String()
Dim IsMultiStory As Boolean()
Dim returnValue As Integer
```

```
returnValue = instance.GetNameList(NumberNames,
    MyName, IsMultiStory)
```

```
int GetNameList(
    int% NumberNames,
    array<String^>^% MyName,
    array<bool>^% IsMultiStory
)
```

```
abstract GetNameList :
    NumberNames : int byref *
    MyName : string[] byref *
    IsMultiStory : bool[] byref -> int
```

Parameters

NumberNames

Type: SystemInt32

The number of Spandrel Labels defined in the program

MyName

Type: SystemString

This is a one-dimensional array of coordinate system names. The MyName array is created as a dynamic, zero-based, array by the API user:

```
Dim MyName() as String
```

The array is dimensioned to (NumberNames + 1) inside the ETABS program, filled with the names, and returned to the API user.

IsMultiStory

Type: SystemBoolean

This values of this array are True or False, indicating whether the associated Spandrel Label spans multiple story levels

Return Value

Type: Int32

Returns zero if the names are successfully retrieved, otherwise it returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberNames As Integer
    Dim MyName As String()
    Dim IsMultiStory As Boolean()

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new Spandrel label
    ret = SapModel.SpandrelLabel.SetSpandrel("MySpandrel", False)

    'get names
    ret = SapModel.SpandrelLabel.GetNameList(NumberNames, MyName)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
```

End Sub

▲ See Also

Reference

cSpandrelLabel Interface

ETABSv1 Namespace

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CSI API ETABS v1

cSpandrelLabelAddLanguageSpecificTextSet("LST3F3 Method

Retrieves the section properties for a specified spandrel

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetSectionProperties(
    string Name,
    ref int NumberStories,
    ref string[] StoryName,
    ref int[] NumAreaObj,
    ref int[] NumLineObj,
    ref double[] Length,
    ref double[] DepthLeft,
    ref double[] ThickLeft,
    ref double[] DepthRight,
    ref double[] ThickRight,
    ref string[] MatProp,
    ref double[] CGLeftX,
    ref double[] CGLeftY,
    ref double[] CGLeftZ,
    ref double[] CGRightX,
    ref double[] CGRightY,
    ref double[] CGRightZ
)
```

```
Function GetSectionProperties (
    Name As String,
    ByRef NumberStories As Integer,
    ByRef StoryName As String(),
    ByRef NumAreaObj As Integer(),
    ByRef NumLineObj As Integer(),
    ByRef Length As Double(),
    ByRef DepthLeft As Double(),
    ByRef ThickLeft As Double(),
    ByRef DepthRight As Double(),
    ByRef ThickRight As Double(),
    ByRef MatProp As String(),
    ByRef CGLeftX As Double(),
    ByRef CGLeftY As Double(),
    ByRef CGLeftZ As Double(),
    ByRef CGRightX As Double(),
    ByRef CGRightY As Double(),
    ByRef CGRightZ As Double()
) As Integer
```

```
Dim instance As cSpandrelLabel
```

```
cSpandrelLabelspan id="LST3F3A9E8B_0"AddLanguageSpecificTextSet("LST3F3A9E8B_0?c0727|nu=.");
```

Introduction

```
Dim Name As String
Dim NumberStories As Integer
Dim StoryName As String()
Dim NumAreaObj As Integer()
Dim NumLineObj As Integer()
Dim Length As Double()
Dim DepthLeft As Double()
Dim ThickLeft As Double()
Dim DepthRight As Double()
Dim ThickRight As Double()
Dim MatProp As String()
Dim CGLeftX As Double()
Dim CGLeftY As Double()
Dim CGLeftZ As Double()
Dim CGRightX As Double()
Dim CGRightY As Double()
Dim CGRightZ As Double()
Dim returnValue As Integer

returnValue = instance.GetSectionProperties(Name,
NumberStories, StoryName, NumAreaObj,
NumLineObj, Length, DepthLeft, ThickLeft,
DepthRight, ThickRight, MatProp,
CGLeftX, CGLeftY, CGLeftZ, CGRightX,
CGRightY, CGRightZ)
```

```
int GetSectionProperties(
String^ Name,
int% NumberStories,
array<String^>^% StoryName,
array<int>^% NumAreaObj,
array<int>^% NumLineObj,
array<double>^% Length,
array<double>^% DepthLeft,
array<double>^% ThickLeft,
array<double>^% DepthRight,
array<double>^% ThickRight,
array<String^>^% MatProp,
array<double>^% CGLeftX,
array<double>^% CGLeftY,
array<double>^% CGLeftZ,
array<double>^% CGRightX,
array<double>^% CGRightY,
array<double>^% CGRightZ
)
```

```
abstract GetSectionProperties :
    Name : string *
    NumberStories : int byref *
    StoryName : string[] byref *
    NumAreaObj : int[] byref *
    NumLineObj : int[] byref *
    Length : float[] byref *
    DepthLeft : float[] byref *
    ThickLeft : float[] byref *
    DepthRight : float[] byref *
    ThickRight : float[] byref *
    MatProp : string[] byref *
    CGLeftX : float[] byref *
    CGLeftY : float[] byref *
    CGLeftZ : float[] byref *
    CGRightX : float[] byref *
```

```
CGRightY : float[] byref *
CGRightZ : float[] byref -> int
```

Parameters

Name

Type:Â SystemString

The name of an existing spandrel

NumberStories

Type:Â SystemInt32

The number of stories that the spandrel exists on and for which section properties are provided

StoryName

Type:Â SystemString

This is an array that includes the story names at which the spandrel exists

NumAreaObj

Type:Â SystemInt32

This is an array that includes the number of area objects in the spandrel at each story

NumLineObj

Type:Â SystemInt32

This is an array that includes the number of line objects in the spandrel at each story

Length

Type:Â SystemDouble

This is an array that includes the spandrel length

DepthLeft

Type:Â SystemDouble

This is an array that includes the spandrel depth at the left end

ThickLeft

Type:Â SystemDouble

This is an array that includes the spandrel thickness at the left end

DepthRight

Type:Â SystemDouble

This is an array that includes the spandrel depth at the right end

ThickRight

Type:Â SystemDouble

This is an array that includes the spandrel thickness at the right end

MatProp

Type:Â SystemString

This is an array that includes the pier material property

CGLeftX

Type:Â SystemDouble

This is an array that includes the x-coordinate of the center of gravity at the left end of the spandrel

CGLeftY

Type:Â SystemDouble

This is an array that includes the y-coordinate of the center of gravity at the left end of the spandrel

CGLeftZ

Type: SystemDouble

This is an array that includes the z-coordinate of the center of gravity at the left end of the spandrel

CGRightX

Type: SystemDouble

This is an array that includes the x-coordinate of the center of gravity at the right end of the spandrel

CGRightY

Type: SystemDouble

This is an array that includes the y-coordinate of the center of gravity at the right end of the spandrel

CGRightZ

Type: SystemDouble

This is an array that includes the z-coordinate of the center of gravity at the right end of the spandrel

Return Value

Type: Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value

■ See Also

Reference

cSpandrelLabel Interface

ETABSv1 Namespace

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CSI API ETABS v1

cSpandrelLabelAddLanguageSpecificTextSet("LST5B1 Method

Retrieves the specified Spandrel Label

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetSpandrel(
    string Name,
    ref bool IsMultiStory
)
```

```
Function GetSpandrel (
    Name As String,
    ByRef IsMultiStory As Boolean
) As Integer
```

```
Dim instance As cSpandrelLabel
Dim Name As String
Dim IsMultiStory As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetSpandrel(Name,
    IsMultiStory)
```

```
int GetSpandrel(
    String^ Name,
    bool% IsMultiStory
)
```

```
abstract GetSpandrel :
    Name : string *
    IsMultiStory : bool byref -> int
```

Parameters

Name

Type: SystemString

The name of an existing Spandrel Label

IsMultiStory

Type: SystemBoolean

Whether the Spandrel Label spans multiple story levels

Return ValueType: Int32

Returns zero if the Spandrel Label exists, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim IsMultiStory As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new Spandrel label
    ret = SapModel.SpandrelLabel.SetSpandrel("MySpandrel", False)

    'check Spandrel label
    ret = SapModel.SpandrelLabel.GetSpandrel("MySpandrel", IsMultiStory)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ See Also

ReferencecSpandrelLabel InterfaceETABSv1 Namespace

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CSI API ETABS v1

cSpandrelLabelAddLanguageSpecificTextSet("LST85D Method

Adds a new Spandrel Label, or modifies an existing Spandrel Label

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetSpandrel(
    string Name,
    bool IsMultiStory
)
```

```
Function SetSpandrel (
    Name As String,
    IsMultiStory As Boolean
) As Integer
```

```
Dim instance As cSpandrelLabel
Dim Name As String
Dim IsMultiStory As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetSpandrel(Name,
    IsMultiStory)
```

```
int SetSpandrel(
    String^ Name,
    bool IsMultiStory
)
```

```
abstract SetSpandrel :
    Name : string *
    IsMultiStory : bool -> int
```

Parameters

Name

Type: SystemString

The name of a Spandrel Label. If this is the name of an existing spandrel label, that spandrel label is modified, otherwise a new spandrel label is added.

IsMultiStory

Type: SystemBoolean

Whether the Spandrel Label spans multiple story levels

Return Value

Type: Int32

Returns zero if the Spandrel Label is successfully added or modified, otherwise it returns a nonzero value.

▲ **Remarks**

▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberNames As Integer
    Dim MyName As String()

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'define new Spandrel label
    ret = SapModel.SpandrelLabel.SetSpandrel("MySpandrel", False)

    'modify Spandrel label
    ret = SapModel.SpandrelLabel.SetSpandrel("MySpandrel", True)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing

End Sub
```

▲ **See Also**

Reference

[cSpandrelLabel Interface](#)

[ETABSv1 Namespace](#)

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Introduction

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CSI API ETABS v1

cStory Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cStory
```

Public Interface cStory

```
Dim instance As cStory
```

```
public interface class cStory
```

```
type cStory = interface end
```

The cStory type exposes the following members.

Methods

A

Name

-  [GetElevation](#) Retrieves the elevation of a defined story
-  [GetGUID](#) Retrieves the GUID of a defined story
-  [GetHeight](#) Retrieves the height of a defined story
-  [GetMasterStory](#) Retrieves whether a defined story is a master story
-  [GetNameList](#) Retrieves the names of all defined stories
-  [GetSimilarTo](#) Retrieves whether a story is a master story, and if not, which master story it is similar to
-  [GetSplice](#) Retrieves the story splice height, if applicable
This function is DEPRECATED. Please refer to [GetStories_2\(Double, Int32\)](#)
-  [GetStories](#)
Retrieves the story information for the current tower
Retrieves the story information for the current tower.
-  [GetStories_2](#)
This function supersedes GetStories
-  [SetElevation](#) Sets the elevation of a defined story
-  [SetGUID](#) Sets the GUID of a defined story
-  [SetHeight](#) Sets the height of a defined story
-  [SetMasterStory](#) Sets whether a defined story is a master story
-  [SetSimilarTo](#) Sets the master story that a defined story should be similar to
-  [SetSplice](#) Sets the splice height of a defined story
-  [SetStories](#) This function is DEPRECATED. Please refer to [SetStories_2\(Double, Int32\)](#)

Introduction

Sets the stories for the current tower

Sets the stories for the current tower.

SetStories_2

This function supersedes SetStories. Note: this function can only be used

[Top](#)

 [See Also](#)

Reference

ETABSv1 Namespace

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CSI API ETABS v1

cStory Methods

The cStory type exposes the following members.

Methods

Â Name

- ⇒ GetElevation Retrieves the elevation of a defined story
- ⇒ GetGUID Retrieves the GUID of a defined story
- ⇒ GetHeight Retrieves the height of a defined story
- ⇒ GetMasterStory Retrieves whether a defined story is a master story
- ⇒ GetNameList Retrieves the names of all defined stories
- ⇒ GetSimilarTo Retrieves whether a story is a master story, and if not, which master stor
- ⇒ GetSplice Retrieves the story splice height, if applicable
This function is DEPRECATED. Please refer to GetStories_2(Double, Int3
- ⇒ GetStories
Retrieves the story information for the current tower
Retrieves the story information for the current tower.
- ⇒ GetStories_2
This function supersedes GetStories
- ⇒ SetElevation Sets the elevation of a defined story
- ⇒ SetGUID Sets the GUID of a defined story
- ⇒ SetHeight Sets the height of a defined story
- ⇒ SetMasterStory Sets whether a defined story is a master story
- ⇒ SetSimilarTo Sets the master story that a defined story should be similar to
- ⇒ SetSplice Sets the splice height of a defined story
This function is DEPRECATED. Please refer to SetStories_2(Double, Int3
- ⇒ SetStories
Sets the stories for the current tower
Sets the stories for the current tower.
- ⇒ SetStories_2
This function supersedes SetStories. Note: this function can only be used

Top

See Also

Reference

cStory Interface

ETABSv1 Namespace

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CSI API ETABS v1

cStoryAddLanguageSpecificTextSet("LSTC8216A13_0?Method

Retrieves the elevation of a defined story

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetElevation(
    string Name,
    ref double Elevation
)
```

```
Function GetElevation (
    Name As String,
    ByRef Elevation As Double
) As Integer
```

```
Dim instance As cStory
Dim Name As String
Dim Elevation As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetElevation(Name,
    Elevation)
```

```
int GetElevation(
    String^ Name,
    double% Elevation
)
```

```
abstract GetElevation :
    Name : string *
    Elevation : float byref -> int
```

Parameters

Name

Type: SystemString

The name of a defined story

Elevation

Type: SystemDouble

The elevation of the story

Return ValueType: Int32

Returns zero if the value is successfully retrieved, otherwise returns nonzero.

▲ **Remarks**▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Elevation As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'get story elevation
    ret = SapModel.Story.GetElevation("Story2", Elevation)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also****Reference**cStory InterfaceETABSv1 Namespace

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CSI API ETABS v1

cStoryAddLanguageSpecificTextSet("LSTF2339D18_0?Method

Retrieves the GUID of a defined story

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetGUID(
    string Name,
    ref string GUID
)
```

```
Function GetGUID (
    Name As String,
    ByRef GUID As String
) As Integer
```

```
Dim instance As cStory
Dim Name As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.GetGUID(Name, GUID)
```

```
int GetGUID(
    String^ Name,
    String^% GUID
)
```

```
abstract GetGUID :
    Name : string *
    GUID : string byref -> int
```

Parameters

Name

Type: SystemString

The name of a defined story

GUID

Type: SystemString

The GUID of the story

Return ValueType: Int32

Returns zero if the value is successfully retrieved, otherwise returns nonzero.

▲ **Remarks**▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim iGUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'get story GUID
    ret = SapModel.Story.GetGUID("Story2", iGUID)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also****Reference**cStory InterfaceETABSv1 Namespace

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CSI API ETABS v1

cStoryAddLanguageSpecificTextSet("LST139C5DC_0?Method

Retrieves the height of a defined story

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetHeight(
    string Name,
    ref double Height
)
```

```
Function GetHeight (
    Name As String,
    ByRef Height As Double
) As Integer
```

```
Dim instance As cStory
Dim Name As String
Dim Height As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetHeight(Name,
    Height)
```

```
int GetHeight(
    String^ Name,
    double% Height
)
```

```
abstract GetHeight :
    Name : string *
    Height : float byref -> int
```

Parameters

Name

Type: SystemString

The name of a defined story

Height

Type: SystemDouble

The height of the story

Return ValueType: Int32

Returns zero if the value is successfully retrieved, otherwise returns nonzero.

▀ Remarks

▀ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Height As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'get story height
    ret = SapModel.Story.GetHeight("Story2", Height)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▀ See Also

ReferencecStory InterfaceETABSv1 Namespace

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CSI API ETABS v1

cStoryAddLanguageSpecificTextSet("LSTCD37E3FF_0 Method

Retrieves whether a defined story is a master story

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetMasterStory(
    string Name,
    ref bool IsMasterStory
)
```

```
Function GetMasterStory (
    Name As String,
    ByRef IsMasterStory As Boolean
) As Integer
```

```
Dim instance As cStory
Dim Name As String
Dim IsMasterStory As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetMasterStory(Name,
    IsMasterStory)
```

```
int GetMasterStory(
    String^ Name,
    bool% IsMasterStory
)
```

```
abstract GetMasterStory :
    Name : string *
    IsMasterStory : bool byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of a defined story

IsMasterStory

Type: [SystemBoolean](#)

True if the story is a master story, False otherwise

Return ValueType: Int32

Returns zero if the value is successfully retrieved, otherwise returns nonzero.

▀ **Remarks**▀ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MasterStory As Boolean

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'get story master story
    ret = SapModel.Story.GetMasterStory("Story4", MasterStory)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▀ **See Also****Reference**cStory InterfaceETABSv1 Namespace

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Namespace: ETABSv1

▲ Syntax

C#

VBC++

F#

Copy

```
Function GetNameList (
ByRef NumberNames As Integer,
ByRef MyName As String()
) As Integer
```

```
returnValue = instance.GetNameList(NumberNames,
MyName)
```

```
int GetNameList(
int% NumberNames,
array<String^>% MyName
)
```

Parameters

Type: SystemInt32

MyName

Type: SystemString

```
Dim MyName() as String
```

Introduction

The array is dimensioned to (NumberNames + 1) inside the program, filled with the names, and returned to the API user.

Return Value

Type: Int32

Returns zero if the names are successfully retrieved, otherwise returns nonzero.

Remarks

Examples

VB

Copy

```
Public Sub Example()  
    Dim SapModel As cSapModel  
    Dim EtabsObject As cOAPI  
    Dim ret As Integer = -1  
    Dim NumberNames As Integer  
    Dim MyName() As String  
  
    'create ETABS object  
    Dim myHelper as cHelper = New Helper  
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")  
  
    'start ETABS application  
    ret = EtabsObject.ApplicationStart()  
  
    'create SapModel object  
    SapModel = EtabsObject.SapModel  
  
    'initialize model  
    ret = SapModel.InitializeNewModel()  
  
    'create steel deck template model  
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)  
  
    'get story names  
    ret = SapModel.Story.GetNameList(NumberNames, MyName)  
  
    'close ETABS  
    EtabsObject.ApplicationExit(False)  
  
    'clean up variables  
    SapModel = Nothing  
    EtabsObject = Nothing  
End Sub
```

See Also

Reference

cStory Interface

ETABSv1 Namespace

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CSI API ETABS v1

cStoryAddLanguageSpecificTextSet("LST734F977E_0?Method

Retrieves whether a story is a master story, and if not, which master story it is similar to

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetSimilarTo(
    string Name,
    ref bool IsMasterStory,
    ref string SimilarToStory
)
```

```
Function GetSimilarTo (
    Name As String,
    ByRef IsMasterStory As Boolean,
    ByRef SimilarToStory As String
) As Integer
```

```
Dim instance As cStory
Dim Name As String
Dim IsMasterStory As Boolean
Dim SimilarToStory As String
Dim returnValue As Integer
```

```
returnValue = instance.GetSimilarTo(Name,
    IsMasterStory, SimilarToStory)
```

```
int GetSimilarTo(
    String^ Name,
    bool% IsMasterStory,
    String^% SimilarToStory
)
```

```
abstract GetSimilarTo :
    Name : string *
    IsMasterStory : bool byref *
    SimilarToStory : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of a defined story

IsMasterStory

cStoryid="LST734F977E_0"AddLanguageSpecificTextSet("LST734F977E_0?cpp=::|nu=.3740tSimilar

Type: SystemBoolean

True if the story is a master story, False otherwise

SimilarToStory

Type: SystemString

If IsMasterStory is False then this is the master story that the requested story is similar to.

Return Value

Type: Int32

Returns zero if the value is successfully retrieved, otherwise returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim MasterStory As Boolean
    Dim SimilarToStory As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'get story similar data
    ret = SapModel.Story.GetSimilarTo("Story2", MasterStory, SimilarToStory)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cStory Interface

ETABSv1 Namespace

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CSI API ETABS v1

cStoryAddLanguageSpecificTextSet("LST80A07BCF_0 Method

Retrieves the story splice height, if applicable

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetSplice(
    string Name,
    ref bool SpliceAbove,
    ref double SpliceHeight
)
```

```
Function GetSplice (
    Name As String,
    ByRef SpliceAbove As Boolean,
    ByRef SpliceHeight As Double
) As Integer
```

```
Dim instance As cStory
Dim Name As String
Dim SpliceAbove As Boolean
Dim SpliceHeight As Double
Dim returnValue As Integer
```

```
returnValue = instance.GetSplice(Name,
    SpliceAbove, SpliceHeight)
```

```
int GetSplice(
    String^ Name,
    bool% SpliceAbove,
    double% SpliceHeight
)
```

```
abstract GetSplice :
    Name : string *
    SpliceAbove : bool byref *
    SpliceHeight : float byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of a defined story

SpliceAbove

Type: SystemBoolean

This is True if the story has a splice height, and False otherwise

SpliceHeight

Type: SystemDouble

The story splice height

Return Value

Type: Int32

Returns zero if the value is successfully retrieved, otherwise returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim SpliceAbove As Boolean
    Dim SpliceHeight As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'get story splice information
    ret = SapModel.Story.GetSplice("Story2", SpliceAbove, SpliceHeight)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cStory Interface

ETABSv1 Namespace

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Introduction

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»

CSI API ETABS v1

cStoryAddLanguageSpecificTextSet("LST8137A449_0?Method

This function is DEPRECATED. Please refer to [GetStories_2\(Double, Int32, String, Double, Double, Boolean, String, Boolean, Double, Int32\)](#)

Retrieves the story information for the current tower

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetStories(
    ref int NumberStories,
    ref string[] StoryNames,
    ref double[] StoryElevations,
    ref double[] StoryHeights,
    ref bool[] IsMasterStory,
    ref string[] SimilarToStory,
    ref bool[] SpliceAbove,
    ref double[] SpliceHeight
)
```

```
Function GetStories (
    ByRef NumberStories As Integer,
    ByRef StoryNames As String(),
    ByRef StoryElevations As Double(),
    ByRef StoryHeights As Double(),
    ByRef IsMasterStory As Boolean(),
    ByRef SimilarToStory As String(),
    ByRef SpliceAbove As Boolean(),
    ByRef SpliceHeight As Double()
) As Integer
```

```
Dim instance As cStory
Dim NumberStories As Integer
Dim StoryNames As String()
Dim StoryElevations As Double()
Dim StoryHeights As Double()
Dim IsMasterStory As Boolean()
Dim SimilarToStory As String()
Dim SpliceAbove As Boolean()
Dim SpliceHeight As Double()
Dim returnValue As Integer
```

```
returnValue = instance.GetStories(NumberStories,
    StoryNames, StoryElevations, StoryHeights,
    IsMasterStory, SimilarToStory, SpliceAbove,
    SpliceHeight)
```

```

int GetStories(
int% NumberStories,
array<String^>% StoryNames,
array<double>% StoryElevations,
array<double>% StoryHeights,
array<bool>% IsMasterStory,
array<String^>% SimilarToStory,
array<bool>% SpliceAbove,
array<double>% SpliceHeight
)

abstract GetStories :
    NumberStories : int byref *
    StoryNames : string[] byref *
    StoryElevations : float[] byref *
    StoryHeights : float[] byref *
    IsMasterStory : bool[] byref *
    SimilarToStory : string[] byref *
    SpliceAbove : bool[] byref *
    SpliceHeight : float[] byref -> int

```

Parameters

NumberStories

Type:Â SystemInt32

The number of stories. All output arrays have length (NumberStories + 1), because they include values for "Base" , which is not a story.

StoryNames

Type:Â SystemString

The names of the stories. This array will include "Base" .

StoryElevations

Type:Â SystemDouble

The story elevations. The first value is the "Base" elevation.

StoryHeights

Type:Â SystemDouble

The story heights. The first value is the "Base" height.

IsMasterStory

Type:Â SystemBoolean

Whether the story is a master story

SimilarToStory

Type:Â SystemString

If the story is not a master story, which master story the story is similar to

SpliceAbove

Type:Â SystemBoolean

This is True if the story has a splice height, and False otherwise

SpliceHeight

Type:Â SystemDouble

The story splice height

Return Value

Type:Â Int32

Returns zero if the story data is successfully retrieved, otherwise returns nonzero.

▲ Remarks

To change the current tower, use [SetActiveTower\(String\)](#)

Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberStories As Integer
    Dim StoryNames() As String
    Dim StoryHeights() As Double
    Dim StoryElevations() As Double
    Dim IsMasterStory() As Boolean
    Dim SimilarToStory() As String
    Dim SpliceAbove() As Boolean
    Dim SpliceHeight() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'get stories
    ret = SapModel.Story.GetStories(NumberStories, StoryNames, StoryHeights, StoryElevations, _
                                    IsMasterStory, SimilarToStory, SpliceAbove, SpliceHeight)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

See Also

Reference

[cStory Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cStoryAddLanguageSpecificTextSet("LSTFE7C54C0_0 Method

Retrieves the story information for the current tower.

This function supersedes GetStories

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetStories_2(
    ref double BaseElevation,
    ref int NumberStories,
    ref string[] StoryNames,
    ref double[] StoryElevations,
    ref double[] StoryHeights,
    ref bool[] IsMasterStory,
    ref string[] SimilarToStory,
    ref bool[] SpliceAbove,
    ref double[] SpliceHeight,
    ref int[] color
)
```

```
Function GetStories_2 (
    ByRef BaseElevation As Double,
    ByRef NumberStories As Integer,
    ByRef StoryNames As String(),
    ByRef StoryElevations As Double(),
    ByRef StoryHeights As Double(),
    ByRef IsMasterStory As Boolean(),
    ByRef SimilarToStory As String(),
    ByRef SpliceAbove As Boolean(),
    ByRef SpliceHeight As Double(),
    ByRef color As Integer()
) As Integer
```

```
Dim instance As cStory
Dim BaseElevation As Double
Dim NumberStories As Integer
Dim StoryNames As String()
Dim StoryElevations As Double()
Dim StoryHeights As Double()
Dim IsMasterStory As Boolean()
Dim SimilarToStory As String()
Dim SpliceAbove As Boolean()
Dim SpliceHeight As Double()
Dim color As Integer()
Dim returnValue As Integer
```

Introduction

```
returnValue = instance.GetStories_2(BaseElevation,  
NumberStories, StoryNames, StoryElevations,  
StoryHeights, IsMasterStory, SimilarToStory,  
SpliceAbove, SpliceHeight, color)
```

```
int GetStories_2(  
double% BaseElevation,  
int% NumberStories,  
array<String^>% StoryNames,  
array<double>% StoryElevations,  
array<double>% StoryHeights,  
array<bool>% IsMasterStory,  
array<String^>% SimilarToStory,  
array<bool>% SpliceAbove,  
array<double>% SpliceHeight,  
array<int>% color  
)
```

```
abstract GetStories_2 :  
    BaseElevation : float byref *  
    NumberStories : int byref *  
    StoryNames : string[] byref *  
    StoryElevations : float[] byref *  
    StoryHeights : float[] byref *  
    IsMasterStory : bool[] byref *  
    SimilarToStory : string[] byref *  
    SpliceAbove : bool[] byref *  
    SpliceHeight : float[] byref *  
    color : int[] byref -> int
```

Parameters

BaseElevation

Type:Â SystemDouble

The elevation of the base [L]

NumberStories

Type:Â SystemInt32

The number of stories

StoryNames

Type:Â SystemString

The names of the stories

StoryElevations

Type:Â SystemDouble

The story elevations [L]

StoryHeights

Type:Â SystemDouble

The story heights [L]

IsMasterStory

Type:Â SystemBoolean

Whether the story is a master story

SimilarToStory

Type:Â SystemString

If the story is not a master story, which master story the story is similar to

SpliceAbove

Type:Â SystemBoolean

This is True if the story has a splice height, and False otherwise

SpliceHeight

Type: SystemDouble

The story splice height [L]

color

Type: SystemInt32

The display color for the story specified as an Integer

Return Value

Type: Int32

Returns zero if the story data is successfully retrieved, otherwise returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim BaseElevation As Double
    Dim NumberStories As Integer
    Dim StoryNames() As String
    Dim StoryElevations() As Double
    Dim StoryHeights() As Double
    Dim IsMasterStory() As Boolean
    Dim SimilarToStory() As String
    Dim SpliceAbove() As Boolean
    Dim SpliceHeight() As Double
    Dim color As Integer()

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'get stories
    ret = SapModel.Story.GetStories_2(BaseElevation, NumberStories, StoryNames, StoryElevation,
    StoryHeights, IsMasterStory, SimilarToStory, SpliceAbove)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cStory Interface

ETABSv1 Namespace

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CSI API ETABS v1

cStoryAddLanguageSpecificTextSet("LST8CB7C44E_0 Method

Sets the elevation of a defined story

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetElevation(
    string Name,
    double Elevation
)
```

```
Function SetElevation (
    Name As String,
    Elevation As Double
) As Integer
```

```
Dim instance As cStory
Dim Name As String
Dim Elevation As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetElevation(Name,
    Elevation)
```

```
int SetElevation(
    String^ Name,
    double Elevation
)
```

```
abstract SetElevation :
    Name : string *
    Elevation : float -> int
```

Parameters

Name

Type: [SystemString](#)

The name of a defined story

Elevation

Type: [SystemDouble](#)

The elevation of the story

Return ValueType: Int32

Returns zero if the value is successfully set, otherwise returns nonzero.

▲ **Remarks**▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set story elevation
    ret = SapModel.Story.SetElevation("Story2", 25)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also****Reference**cStory InterfaceETABSv1 Namespace

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»

CSI API ETABS v1

cStoryAddLanguageSpecificTextSet("LST6B17491B_0?GUID=00000000-0000-0000-0000-00000000")

Method

Sets the GUID of a defined story

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetGUID(
    string Name,
    string GUID = ""
)
```

```
Function SetGUID (
    Name As String,
    ?GUID As String = ""
) As Integer
```

```
Dim instance As cStory
Dim Name As String
Dim GUID As String
Dim returnValue As Integer
```

```
returnValue = instance.SetGUID(Name, GUID)
```

```
int SetGUID(
    String^ Name,
    String^ GUID = L""
)
```

```
abstract SetGUID :
    Name : string *
    ?GUID : string
(* Defaults:
    let _GUID = defaultArg GUID ""
*)
-> int
```

Parameters

Name

Type: [SystemString](#)

The name of a defined story

GUID (Optional)

Type: [SystemString](#)

The GUID of the story

Return ValueType: Int32

Returns zero if the value is successfully set, otherwise returns nonzero.

▲ **Remarks**▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim iGUID As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set story GUID
    Dim g As String = Guid.NewGuid.ToString
    ret = SapModel.Story.SetGUID("Story2", g)

    'get story GUID
    ret = SapModel.Story.GetGUID("Story2", iGUID)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also****Reference**cStory InterfaceETABSv1 Namespace

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»

CSI API ETABS v1

cStoryAddLanguageSpecificTextSet("LSTA8BAD203_0 Method

Sets the height of a defined story

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetHeight(
    string Name,
    double Height
)
```

```
Function SetHeight (
    Name As String,
    Height As Double
) As Integer
```

```
Dim instance As cStory
Dim Name As String
Dim Height As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetHeight(Name,
    Height)
```

```
int SetHeight(
    String^ Name,
    double Height
)
```

```
abstract SetHeight :
    Name : string *
    Height : float -> int
```

Parameters

Name

Type: SystemString

The name of a defined story

Height

Type: SystemDouble

The height of the story

Return ValueType: Int32

Returns zero if the value is successfully set, otherwise returns nonzero.

▲ **Remarks**▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set story height
    ret = SapModel.Story.SetHeight("Story3", 100)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also****Reference**cStory InterfaceETABSV1 Namespace

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»

CSI API ETABS v1

cStoryAddLanguageSpecificTextSet("LST532227AD_0?Method

Sets whether a defined story is a master story

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetMasterStory(
    string Name,
    bool IsMasterStory
)
```

```
Function SetMasterStory (
    Name As String,
    IsMasterStory As Boolean
) As Integer
```

```
Dim instance As cStory
Dim Name As String
Dim IsMasterStory As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.SetMasterStory(Name,
    IsMasterStory)
```

```
int SetMasterStory(
    String^ Name,
    bool IsMasterStory
)
```

```
abstract SetMasterStory :
    Name : string *
    IsMasterStory : bool -> int
```

Parameters

Name

Type: [SystemString](#)

The name of a defined story

IsMasterStory

Type: [SystemBoolean](#)

True if the story is a master story, False otherwise

Return ValueType: Int32

Returns zero if the value is successfully set, otherwise returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim IsMasterStory As Boolean
    Dim SimilarToStory As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'unset story as master story
    ret = SapModel.Story.SetMasterStory("Story4", False)

    'check if story is a master story
    ret = SapModel.Story.GetMasterStory("Story4", IsMasterStory)

    'check other stories no longer using Story 4 as master
    ret = SapModel.Story.GetSimilarTo("Story2", IsMasterStory, SimilarToStory)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

ReferencecStory InterfaceETABSv1 Namespace

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Introduction

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CSI API ETABS v1

cStoryAddLanguageSpecificTextSet("LST7063729E_0?Method

Sets the master story that a defined story should be similar to

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetSimilarTo(
    string Name,
    string SimilarToStory
)
```

```
Function SetSimilarTo (
    Name As String,
    SimilarToStory As String
) As Integer
```

```
Dim instance As cStory
Dim Name As String
Dim SimilarToStory As String
Dim returnValue As Integer
```

```
returnValue = instance.SetSimilarTo(Name,
    SimilarToStory)
```

```
int SetSimilarTo(
    String^ Name,
    String^ SimilarToStory
)
```

```
abstract SetSimilarTo :
    Name : string *
    SimilarToStory : string -> int
```

Parameters

Name

Type: [SystemString](#)

The name of a defined story which is not a master story

SimilarToStory

Type: [SystemString](#)

The name of a defined master story that the requested story should be similar to

Return ValueType: Int32

Returns zero if the value is successfully set, otherwise returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim IsMasterStory As Boolean
    Dim SimilarToStory As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set story as master story
    ret = SapModel.Story.SetMasterStory("Story2", True)

    'set story similar to
    ret = SapModel.Story.SetSimilarTo("Story3", "Story2")

    'check
    ret = SapModel.Story.GetSimilarTo("Story3", IsMasterStory, SimilarToStory)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

ReferencecStory InterfaceETABSv1 Namespace

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Introduction

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CSI API ETABS v1

cStoryAddLanguageSpecificTextSet("LST7C2FEEB0_0 Method

Sets the splice height of a defined story

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetSplice(
    string Name,
    bool SpliceAbove,
    double SpliceHeight
)
```

```
Function SetSplice (
    Name As String,
    SpliceAbove As Boolean,
    SpliceHeight As Double
) As Integer
```

```
Dim instance As cStory
Dim Name As String
Dim SpliceAbove As Boolean
Dim SpliceHeight As Double
Dim returnValue As Integer
```

```
returnValue = instance.SetSplice(Name,
    SpliceAbove, SpliceHeight)
```

```
int SetSplice(
    String^ Name,
    bool SpliceAbove,
    double SpliceHeight
)
```

```
abstract SetSplice :
    Name : string *
    SpliceAbove : bool *
    SpliceHeight : float -> int
```

Parameters

Name

Type: SystemString

The name of a defined story

SpliceAbove

Type: SystemBoolean

This is True if the story has a splice height, and False otherwise

SpliceHeight

Type: SystemDouble

The story splice height

Return Value

Type: Int32

Returns zero if the value is successfully set, otherwise returns nonzero.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim IsMasterStory As Boolean
    Dim SimilarToStory As String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set story splice information
    ret = SapModel.Story.SetSplice("Story2", True, 2)

    'get story splice information
    ret = SapModel.Story.GetSplice("Story2", SpliceAbove, SpliceHeight)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cStory Interface

ETABSv1 Namespace

Introduction

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CSI API ETABS v1

cStoryAddLanguageSpecificTextSet("LST7DD6422E_00") Method

This function is DEPRECATED. Please refer to [SetStories_2\(Double, Int32, String, Double, Boolean, String, Boolean, Double, Int32\)](#)

Sets the stories for the current tower

Namespace: Â [ETABSV1](#)

Assembly: Â ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetStories(
    string[] StoryNames,
    double[] StoryElevations,
    double[] StoryHeights,
    bool[] IsMasterStory,
    string[] SimilarToStory,
    bool[] SpliceAbove,
    double[] SpliceHeight
)
```

```
Function SetStories (
    StoryNames As String(),
    StoryElevations As Double(),
    StoryHeights As Double(),
    IsMasterStory As Boolean(),
    SimilarToStory As String(),
    SpliceAbove As Boolean(),
    SpliceHeight As Double()
) As Integer
```

```
Dim instance As cStory
Dim StoryNames As String()
Dim StoryElevations As Double()
Dim StoryHeights As Double()
Dim IsMasterStory As Boolean()
Dim SimilarToStory As String()
Dim SpliceAbove As Boolean()
Dim SpliceHeight As Double()
Dim returnValue As Integer
```

```
returnValue = instance.SetStories(StoryNames,
    StoryElevations, StoryHeights, IsMasterStory,
    SimilarToStory, SpliceAbove, SpliceHeight)
```

```
int SetStories(
    array<String>^ StoryNames,
    array<double>^ StoryElevations,
    array<double>^ StoryHeights,
```

cStoryspan id="LST7DD6422E_0"AddLanguageSpecificTextSet("LST7DD6422E_0?cpp=::|nu=377;SetStorie

```

array<bool>^ IsMasterStory,
array<String>^ SimilarToStory,
array<bool>^ SpliceAbove,
array<double>^ SpliceHeight
)

abstract SetStories :
    StoryNames : string[] *
    StoryElevations : float[] *
    StoryHeights : float[] *
    IsMasterStory : bool[] *
    SimilarToStory : string[] *
    SpliceAbove : bool[] *
    SpliceHeight : float[] -> int

```

Parameters

StoryNames

Type:Â SystemString

The names of the stories

StoryElevations

Type:Â SystemDouble

The story elevations. This array has length (Number of stories + 1). The first value in the array is the "Base" elevation, commonly 0.

StoryHeights

Type:Â SystemDouble

The story heights. Although this argument is not optional, an empty array may be used, and the heights will be inferred from the StoryElevations array. This array has length (Number of stories), with no value for "Base"

IsMasterStory

Type:Â SystemBoolean

Whether the story is a master story. This array has length (Number of stories), with no value for "Base"

SimilarToStory

Type:Â SystemString

If the story is not a master story, which master story the story is similar to. This array has length (Number of stories), with no value for "Base"

SpliceAbove

Type:Â SystemBoolean

This is True if the story has a splice height, and False otherwise. This array has length (Number of stories), with no value for "Base"

SpliceHeight

Type:Â SystemDouble

The story splice heights, if applicable. This array has length (Number of stories), with no value for "Base"

Return Value

Type:Â Int32

Returns zero if the story data is successfully set, otherwise returns nonzero.

▲ Remarks

To change the current tower, use SetActiveTower(String)

▲ Examples

VB

Copy

```

Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim NumberStories As Integer
    Dim StoryNames() As String
    Dim StoryHeights() As Double
    Dim StoryElevations() As Double
    Dim IsMasterStory() As Boolean
    Dim SimilarToStory() As String
    Dim SpliceAbove() As Boolean
    Dim SpliceHeight() As Double

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'set stories
    Dim inStoryNames As String() = {"MyStory1", "MyStory2", "MyStory3", "MyStory4", "MyStory5"}
    Dim inStoryElevations As Double() = {0, 5, 15, 23, 23.5, 40, 50}
    Dim inStoryHeights As Double() = {2, 5, 7, 33, 1, 9}
    Dim inIsMasterStory As Boolean() = {False, False, False, False, False, True}
    Dim inSimilarToStory As String() = {"None", "", "MyStory6", "MyStory6", "MyStory6", "MyStory6"}
    Dim inSpliceAbove As Boolean() = {False, True, False, True, False, True}
    Dim inSpliceHeight As Double() = {0, 0, 2, 2, 0, 1}

    ret = SapModel.Story.SetStories(inStoryNames, inStoryElevations, inStoryHeights, _
                                   inIsMasterStory, inSimilarToStory, inSpliceAbove, inSpliceHeight)

    'get stories
    ret = SapModel.Story.GetStories(NumberStories, StoryNames, StoryHeights, StoryElevations, _
                                   IsMasterStory, SimilarToStory, SpliceAbove, SpliceHeight)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub

```

 See Also

Reference

cStory Interface

ETABSv1 Namespace

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CSI API ETABS v1

cStoryAddLanguageSpecificTextSet("LST23D9E85D_0") Method

Sets the stories for the current tower.

This function supersedes SetStories. Note: this function can only be used when no objects exist in the model.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetStories_2(
double BaseElevation,
int NumberStories,
ref string[] StoryNames,
ref double[] StoryHeights,
ref bool[] IsMasterStory,
ref string[] SimilarToStory,
ref bool[] SpliceAbove,
ref double[] SpliceHeight,
ref int[] color
)
```

```
Function SetStories_2 (
BaseElevation As Double,
NumberStories As Integer,
ByRef StoryNames As String(),
ByRef StoryHeights As Double(),
ByRef IsMasterStory As Boolean(),
ByRef SimilarToStory As String(),
ByRef SpliceAbove As Boolean(),
ByRef SpliceHeight As Double(),
ByRef color As Integer()
) As Integer
```

```
Dim instance As cStory
Dim BaseElevation As Double
Dim NumberStories As Integer
Dim StoryNames As String()
Dim StoryHeights As Double()
Dim IsMasterStory As Boolean()
Dim SimilarToStory As String()
Dim SpliceAbove As Boolean()
Dim SpliceHeight As Double()
Dim color As Integer()
Dim returnValue As Integer
```

```
returnValue = instance.SetStories_2(BaseElevation,
NumberStories, StoryNames, StoryHeights,
```

cStoryspan id="LST23D9E85D_0"AddLanguageSpecificTextSet("LST23D9E85D_0?cpp=::|nu=378SetStorie

Introduction

IsMasterStory, SimilarToStory, SpliceAbove,
SpliceHeight, color)

```
int SetStories_2(  
double BaseElevation,  
int NumberStories,  
array<String^>^% StoryNames,  
array<double>^% StoryHeights,  
array<bool>^% IsMasterStory,  
array<String^>^% SimilarToStory,  
array<bool>^% SpliceAbove,  
array<double>^% SpliceHeight,  
array<int>^% color  
)
```

```
abstract SetStories_2 :  
    BaseElevation : float *  
    NumberStories : int *  
    StoryNames : string[] byref *  
    StoryHeights : float[] byref *  
    IsMasterStory : bool[] byref *  
    SimilarToStory : string[] byref *  
    SpliceAbove : bool[] byref *  
    SpliceHeight : float[] byref *  
    color : int[] byref -> int
```

Parameters

BaseElevation

Type:Â SystemDouble

The elevation of the base [L]

NumberStories

Type:Â SystemInt32

The number of stories

StoryNames

Type:Â SystemString

The names of the stories

StoryHeights

Type:Â SystemDouble

The story heights [L]

IsMasterStory

Type:Â SystemBoolean

Whether the story is a master story

SimilarToStory

Type:Â SystemString

If the story is not a master story, which master story the story is similar to

SpliceAbove

Type:Â SystemBoolean

This is True if the story has a splice height, and False otherwise

SpliceHeight

Type:Â SystemDouble

The story splice height [L]

color

Type:Â SystemInt32

The display color for the story specified as an Integer

Return ValueType: Int32

Returns zero if the story data is successfully set, otherwise returns nonzero.

▲ **Remarks**▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewGridOnly(4, 12, 12, 4, 4, 24, 24)
    ret = SapModel.SetPresentUnits(ETABSv1.eUnits.kip_ft_F)

    'set stories
    Dim inStoryNames As String() = {"MyStory1", "MyStory2", "MyStory3", "MyStory4", "MyStory5"}
    Dim inStoryHeights As Double() = {10, 10, 15, 10, 10, 20}
    Dim inIsMasterStory As Boolean() = {False, False, True, False, False, True}
    Dim inSimilarToStory As String() = {"MyStory3", "MyStory3", "None", "MyStory6", "", "None"}
    Dim inSpliceAbove As Boolean() = {False, True, False, True, False, False}
    Dim inSpliceHeight As Double() = {0, 1, 0, 2, 0, 0}
    Dim inColor as Integer() = {65535, 0, 255, 16711935, 16711680, 0}

    ret = SapModel.Story.SetStories_2(2.0, 6, inStoryNames, inStoryHeights, inIsMasterStory,
                                     inSimilarToStory, inSpliceAbove, inSpliceHeight, inColor)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ **See Also****Reference**cStory InterfaceETABSv1 Namespace

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CSI API ETABS v1

cTendonObj Interface

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cTendonObj
```

```
Public Interface cTendonObj
```

```
Dim instance As cTendonObj
```

```
public interface class cTendonObj
```

```
type cTendonObj = interface end
```

The cTendonObj type exposes the following members.

▲ Methods

A	Name	Description
✚	ChangeName	
✚	Count	
✚	GetDatumOffset	Retrieves the datum offset - intended for internal use
✚	GetDrawingPoint	Retrieves drawing point data - intended for internal use
✚	GetGroupAssign	Retrieves the groups to which a tendon object is assigned.
✚	GetLoadForceStress_1	Retrieves tendon load data
✚	GetLossesDetailed	Retrieves tendon loss data - detailed
✚	GetLossesFixed	Retrieves tendon loss data - fixed
✚	GetLossesPercent	Retrieves tendon loss data - percentage
✚	GetNameList	
✚	GetNameListOnStory	
✚	GetNumberStrands	Retrieves the number of strands
✚	GetProperty	
✚	GetSelected	
✚	GetTendonGeometry	
✚	SetGroupAssign	Adds/removes tendon objects to/from a specified group.
✚	SetSelected	

[Top](#)

▲ See Also

Reference

ETABSv1 Namespace

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CSI API ETABS v1

cTendonObj Methods

The cTendonObj type exposes the following members.

Methods

Â	Name	Description
⇒	<u>ChangeName</u>	
⇒	<u>Count</u>	
⇒	<u>GetDatumOffset</u>	Retrieves the datum offset - intended for internal use
⇒	<u>GetDrawingPoint</u>	Retrieves drawing point data - intended for internal use
⇒	<u>GetGroupAssign</u>	Retrieves the groups to which a tendon object is assigned.
⇒	<u>GetLoadForceStress_1</u>	Retrieves tendon load data
⇒	<u>GetLossesDetailed</u>	Retrieves tendon loss data - detailed
⇒	<u>GetLossesFixed</u>	Retrieves tendon loss data - fixed
⇒	<u>GetLossesPercent</u>	Retrieves tendon loss data - percentage
⇒	<u>GetNameList</u>	
⇒	<u>GetNameListOnStory</u>	
⇒	<u>GetNumberStrands</u>	Retrieves the number of strands
⇒	<u>GetProperty</u>	
⇒	<u>GetSelected</u>	
⇒	<u>GetTendonGeometry</u>	
⇒	<u>SetGroupAssign</u>	Adds/removes tendon objects to/from a specified group.
⇒	<u>SetSelected</u>	

Top

See Also

Reference

cTendonObj Interface

ETABSV1 Namespace

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CSI API ETABS v1

cTendonObjAddLanguageSpecificTextSet("LST7AF57A41_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int ChangeName(
    string Name,
    string NewName
)
```

```
Function ChangeName (
    Name As String,
    NewName As String
) As Integer
```

```
Dim instance As cTendonObj
Dim Name As String
Dim NewName As String
Dim returnValue As Integer
```

```
returnValue = instance.ChangeName(Name,
    NewName)
```

```
int ChangeName(
    String^ Name,
    String^ NewName
)
```

```
abstract ChangeName :
    Name : string *
    NewName : string -> int
```

Parameters

Name

Type: [SystemString](#)

NewName

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cTendonObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cTendonObjAddLanguageSpecificTextSet("LST5A43E3C4_0?cpp=3790-");Cou Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

int Count()

Function Count As Integer

Dim instance As cTendonObj

Dim returnValue As Integer

returnValue = instance.Count()

int Count()

abstract Count : unit -> int

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cTendonObj Interface](#)

[ETABSv1 Namespace](#)

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Namespace: ETABSv1

▲ Syntax

C#

VBC++

F#

Copy

```
Function GetDatumOffset (
Name As String,
ByRef NumberItems As Integer,
ByRef TendonName As String(),
ByRef DatumOffset As Double(),
Optional eItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cTendonObj
Dim Name As String
Dim NumberItems As Integer
Dim TendonName As String()
Dim DatumOffset As Double()
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.GetDatumOffset(Name,
NumberItems, TendonName, DatumOffset,
ItemType)
```

```
int GetDatumOffset(  
String^ Name,  
int% NumberItems,  
array<String^>% TendonName,  
array<double>% DatumOffset,  
eItemType ItemType = eItemType::Objects  
)
```

cTendonObjspan id="LST23791B1_0"AddLanguageSpecificTextSet("LST23791B1_0?cpp=::|nu3791tGetData

```

        ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of an existing tendon object or group, depending on the value of the ItemType item.

NumberItems

Type:Â SystemInt32

The total number of items retrieved for the specified tendon objects, all the retrieved data arrays are dimensioned to this length

TendonName

Type:Â SystemString

The name of the tendon object associated with each value.

DatumOffset

Type:Â SystemDouble

Global Z dist from model datum to tendon layout datum, positive in positive global Z direction [L]

ItemType (Optional)

Type:Â ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, data is retrieved for the tendon object specified by the Name item.

If this item is Group, data is retrieved for all tendon objects in the group specified by the Name item.

If this item is SelectedObjects, data is retrieved for all selected tendon objects, and the Name item is ignored.

Return Value

Type:Â Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value.

▲ Remarks

This function is intended for internal use

▲ See Also

Reference

cTendonObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cTendonObjAddLanguageSpecificTextSet("LSTA608DA Method

Retrieves drawing point data - intended for internal use

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetDrawingPoint(
    string Name,
    ref int NumberItems,
    ref string[] TendonName,
    ref string[] DrawingPointID,
    ref double[] gx,
    ref double[] gy,
    ref double[] gz,
    eItemType ItemType = eItemType.Objects
)
```

```
Function GetDrawingPoint (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef TendonName As String(),
    ByRef DrawingPointID As String(),
    ByRef gx As Double(),
    ByRef gy As Double(),
    ByRef gz As Double(),
    Optional ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cTendonObj
Dim Name As String
Dim NumberItems As Integer
Dim TendonName As String()
Dim DrawingPointID As String()
Dim gx As Double()
Dim gy As Double()
Dim gz As Double()
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.GetDrawingPoint(Name,
    NumberItems, TendonName, DrawingPointID,
    gx, gy, gz, ItemType)
```

```
int GetDrawingPoint(
    String^ Name,
    int% NumberItems,
    array<String^>% TendonName,
```

cTendonObjspan id="LSTA608DAB9_0"AddLanguageSpecificTextSet("LSTA608DAB9_0?cpp=3704=");Get

```

array<String>^% DrawingPointID,
array<double>^% gx,
array<double>^% gy,
array<double>^% gz,
eItemType ItemType = eItemType::Objects
)

abstract GetDrawingPoint :
    Name : string *
    NumberItems : int byref *
    TendonName : string[] byref *
    DrawingPointID : string[] byref *
    gx : float[] byref *
    gy : float[] byref *
    gz : float[] byref *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int

```

Parameters

Name

Type: SystemString

The name of an existing tendon object or group, depending on the value of the ItemType item.

NumberItems

Type: SystemInt32

The total number of items retrieved for the specified tendon objects, all the retrieved data arrays are dimensioned to this length

TendonName

Type: SystemString

The name of the tendon object associated with each value.

DrawingPointID

Type: SystemString

The drawing point ID associated with each value

gx

Type: SystemDouble

Global x value [L]

gy

Type: SystemDouble

Global y value [L]

gz

Type: SystemDouble

Global z value [L]

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, data is retrieved for the tendon object specified by the Name item.

Introduction

If this item is Group, data is retrieved for all tendon objects in the group specified by the Name item.

If this item is SelectedObjects, data is retrieved for all selected tendon objects, and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value.

Remarks

Drawing point data may include points from the meshed model not available through GetTendonGeometry(String, Int32, Double, Double, Double, String)

See Also

Reference

cTendonObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cTendonObjAddLanguageSpecificTextSet("LSTCD63ABBC_0?cpp=977u=.");G

Method

Retrieves the groups to which a tendon object is assigned.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetGroupAssign(
    string Name,
    ref int NumberGroups,
    ref string[] Groups
)
```

```
Function GetGroupAssign (
    Name As String,
    ByRef NumberGroups As Integer,
    ByRef Groups As String()
) As Integer
```

```
Dim instance As cTendonObj
Dim Name As String
Dim NumberGroups As Integer
Dim Groups As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetGroupAssign(Name,
    NumberGroups, Groups)
```

```
int GetGroupAssign(
    String^ Name,
    int% NumberGroups,
    array<String^>^% Groups
)
```

```
abstract GetGroupAssign :
    Name : string *
    NumberGroups : int byref *
    Groups : string[] byref -> int
```

Parameters

Name

Type: [SystemString](#)

The name of an existing tendon object.

NumberGroups

Type:Â SystemInt32

The number of group names retrieved.

Groups

Type:Â SystemString

This is an array of the names of the groups to which the tendon object is assigned.

Return Value

Type:Â Int32

Returns zero if the group assignments are successfully retrieved, otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim NumberGroups As Integer
    Dim Groups() as String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

    'add tendon object manually with name "TendonObject1"
    Name = "TendonObject1"

    'define new groups
    ret = SapModel.GroupDef.SetGroup("Group1")
    ret = SapModel.GroupDef.SetGroup("Group2")

    'add tendon object to groups
    ret = SapModel.TendonObj.SetGroupAssign(Name, "Group1")
    ret = SapModel.TendonObj.SetGroupAssign(Name, "Group2")

    'get tendon object groups
    ret = SapModel.TendonObj.GetGroupAssign(Name, NumberGroups, Groups)

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
```

```
    EtabsObject = Nothing  
End Sub
```

▲ See Also

Reference

cTendonObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cTendonObjAddLanguageSpecificTextSet("LSTF60C30E8_0") Method

Retrieves tendon load data

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetLoadForceStress_1(
    string Name,
    ref int NumberItems,
    ref string[] TendonName,
    ref string[] LoadPatFinal,
    ref string[] LoadPatTransfer,
    ref int[] JackFrom,
    ref int[] LoadType,
    ref double[] LoadValue,
    ref int[] LossSpecification,
    eItemType ItemType = eItemType.Objects
)
```

```
Function GetLoadForceStress_1 (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef TendonName As String(),
    ByRef LoadPatFinal As String(),
    ByRef LoadPatTransfer As String(),
    ByRef JackFrom As Integer(),
    ByRef LoadType As Integer(),
    ByRef LoadValue As Double(),
    ByRef LossSpecification As Integer(),
    ByRef ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cTendonObj
Dim Name As String
Dim NumberItems As Integer
Dim TendonName As String()
Dim LoadPatFinal As String()
Dim LoadPatTransfer As String()
Dim JackFrom As Integer()
Dim LoadType As Integer()
Dim LoadValue As Double()
Dim LossSpecification As Integer()
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.GetLoadForceStress_1(Name,
    NumberItems, TendonName, LoadPatFinal,
```

cTendonObjAddLanguageSpecificTextSet("LSTF60C30E8_0?cpp=3B00-");GetL

Introduction

LoadPatTransfer, JackFrom, LoadType,
LoadValue, LossSpecification, ItemType)

```
int GetLoadForceStress_1(  
String^ Name,  
int% NumberItems,  
array<String^>^% TendonName,  
array<String^>^% LoadPatFinal,  
array<String^>^% LoadPatTransfer,  
array<int>^% JackFrom,  
array<int>^% LoadType,  
array<double>^% LoadValue,  
array<int>^% LossSpecification,  
eItemType ItemType = eItemType::Objects  
)
```

```
abstract GetLoadForceStress_1 :  
    Name : string *  
    NumberItems : int byref *  
    TendonName : string[] byref *  
    LoadPatFinal : string[] byref *  
    LoadPatTransfer : string[] byref *  
    JackFrom : int[] byref *  
    LoadType : int[] byref *  
    LoadValue : float[] byref *  
    LossSpecification : int[] byref *  
    ?ItemType : eItemType  
(* Defaults:  
    let _ItemType = defaultArg ItemType eItemType.Objects  
*)  
-> int
```

Parameters

Name

Type:Â [SystemString](#)

The name of an existing tendon object or group, depending on the value of the ItemType item.

NumberItems

Type:Â [SystemInt32](#)

The total number of items retrieved for the specified tendon objects

TendonName

Type:Â [SystemString](#)

This is an array that includes the name of the tendon object associated with each value.

LoadPatFinal

Type:Â [SystemString](#)

This is an array that includes the final load pattern name

LoadPatTransfer

Type:Â [SystemString](#)

This is an array that includes the transfer load pattern name

JackFrom

Type:Â [SystemInt32](#)

The jacking location, an item from the following list

1. Start Of Tendon
2. End Of Tendon

3. Both Ends

LoadType

Type: $\hat{A}$ SystemInt32

This is an array that includes either 0 or 1, indicating the type of load

◊ 0 = Force

◊ 1 = Stress

LoadValue

Type: $\hat{A}$ SystemDouble

This is an array that includes the load value. [F] when LoadType is 0, and [F/L2] when LoadType is 1

LossSpecification

Type: $\hat{A}$ SystemInt32

The tendon loss type, an item from the following list

1. Percent

2. Fixed

3. Detailed

Once the loss type is known, the functions GetLossesDetailed(String, Int32, String, Double, Double, Double, Double, Double, Double, eItemType) , GetLossesFixed(String, Int32, String, Double, Double, eItemType), and GetLossesPercent(String, Int32, String, Double, Double, eItemType) can be called to retrieve additional data

ItemType (Optional)

Type: $\hat{A}$ ETABSv1eItemTypeThis is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, data is retrieved for the tendon object specified by the Name item.

If this item is Group, data is retrieved for all tendon objects in the group specified by the Name item.

If this item is SelectedObjects, data is retrieved for all selected tendon objects, and the Name item is ignored.

Return ValueType: $\hat{A}$ Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value.

▲ See Also

ReferencecTendonObj InterfaceETABSv1 Namespace

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CSI API ETABS v1

cTendonObjAddLanguageSpecificTextSet("LST13F79B1D_0?cpp=3803-");GetL

Method

Retrieves tendon loss data - detailed

Namespace: ETABSV1**Assembly:** ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#VBC++F#Copy

```
int GetLossesDetailed(
    string Name,
    ref int NumberItems,
    ref string[] TendonName,
    ref double[] CurvatureCoeff,
    ref double[] WobbleCoeff,
    ref double[] LossAnchorage,
    ref double[] LossShortening,
    ref double[] LossCreep,
    ref double[] LossShrinkage,
    ref double[] LossSteelRelax,
    eItemType ItemType = eItemType.Objects
)
```

```
Function GetLossesDetailed (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef TendonName As String(),
    ByRef CurvatureCoeff As Double(),
    ByRef WobbleCoeff As Double(),
    ByRef LossAnchorage As Double(),
    ByRef LossShortening As Double(),
    ByRef LossCreep As Double(),
    ByRef LossShrinkage As Double(),
    ByRef LossSteelRelax As Double(),
    ByRef ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cTendonObj
Dim Name As String
Dim NumberItems As Integer
Dim TendonName As String()
Dim CurvatureCoeff As Double()
Dim WobbleCoeff As Double()
Dim LossAnchorage As Double()
Dim LossShortening As Double()
Dim LossCreep As Double()
Dim LossShrinkage As Double()
Dim LossSteelRelax As Double()
Dim ItemType As eItemType
Dim returnValue As Integer
```

cTendonObjspan id="LST13F79B1D_0"AddLanguageSpecificTextSet("LST13F79B1D_0?cpp=3803-");GetL

```
returnValue = instance.GetLossesDetailed(Name,
NumberItems, TendonName, CurvatureCoeff,
WobbleCoeff, LossAnchorage, LossShortening,
LossCreep, LossShrinkage, LossSteelRelax,
ItemType)
```

```
int GetLossesDetailed(
String^ Name,
int% NumberItems,
array<String^>^% TendonName,
array<double>^% CurvatureCoeff,
array<double>^% WobbleCoeff,
array<double>^% LossAnchorage,
array<double>^% LossShortening,
array<double>^% LossCreep,
array<double>^% LossShrinkage,
array<double>^% LossSteelRelax,
eItemType ItemType = eItemType::Objects
)
```

```
abstract GetLossesDetailed :
    Name : string *
    NumberItems : int byref *
    TendonName : string[] byref *
    CurvatureCoeff : float[] byref *
    WobbleCoeff : float[] byref *
    LossAnchorage : float[] byref *
    LossShortening : float[] byref *
    LossCreep : float[] byref *
    LossShrinkage : float[] byref *
    LossSteelRelax : float[] byref *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type:Â [SystemString](#)

The name of an existing tendon object or group, depending on the value of the ItemType item.

NumberItems

Type:Â [SystemInt32](#)

The total number of items retrieved for the specified tendon objects

TendonName

Type:Â [SystemString](#)

This is an array that includes the name of the tendon object associated with each value.

CurvatureCoeff

Type:Â [SystemDouble](#)

This is an array that includes the curvature coefficient

WobbleCoeff

Type:Â [SystemDouble](#)

This is an array that includes the wobble coefficient [1/L]

LossAnchorage

Type: SystemDouble

This is an array that includes the anchorage set slip [L]

LossShortening

Type: SystemDouble

This is an array that includes the elastic shortening stress [F/L2]

LossCreep

Type: SystemDouble

This is an array that includes the creep stress [F/L2]

LossShrinkage

Type: SystemDouble

This is an array that includes the shrinkage stress [F/L2]

LossSteelRelax

Type: SystemDouble

This is an array that includes the steel relaxation stress [F/L2]

ItemType (Optional)

Type: ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, data is retrieved for the tendon object specified by the Name item.

If this item is Group, data is retrieved for all tendon objects in the group specified by the Name item.

If this item is SelectedObjects, data is retrieved for all selected tendon objects, and the Name item is ignored.

Return Value

Type: Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value.

▲ See Also

Reference

cTendonObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cTendonObjAddLanguageSpecificTextSet("LSTF6C80904_0?cpp=:3806.");GetL

Method

Retrieves tendon loss data - fixed

Namespace: ETABSV1

Assembly: ETABSV1 (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetLossesFixed(
    string Name,
    ref int NumberItems,
    ref string[] TendonName,
    ref double[] StressingFixed,
    ref double[] LongTermFixed,
    eItemType ItemType = eItemType.Objects
)
```

```
Function GetLossesFixed (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef TendonName As String(),
    ByRef StressingFixed As Double(),
    ByRef LongTermFixed As Double(),
    ByRef ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cTendonObj
Dim Name As String
Dim NumberItems As Integer
Dim TendonName As String()
Dim StressingFixed As Double()
Dim LongTermFixed As Double()
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.GetLossesFixed(Name,
    NumberItems, TendonName, StressingFixed,
    LongTermFixed, ItemType)
```

```
int GetLossesFixed(
    String^ Name,
    int% NumberItems,
    array<String^>^% TendonName,
    array<double>^% StressingFixed,
    array<double>^% LongTermFixed,
    eItemType ItemType = eItemType::Objects
)
```

abstract GetLossesFixed :

cTendonObjAddLanguageSpecificTextSet("LSTF6C80904_0?cpp=:3806.");GetL

Introduction

```
Name : string *
NumberItems : int byref *
TendonName : string[] byref *
StressingFixed : float[] byref *
LongTermFixed : float[] byref *
?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing tendon object or group, depending on the value of the ItemType item.

NumberItems

Type:Â SystemInt32

The total number of items retrieved for the specified tendon objects

TendonName

Type:Â SystemString

This is an array that includes the name of the tendon object associated with each value.

StressingFixed

Type:Â SystemDouble

This is an array that includes the stressing losses [F/L2]

LongTermFixed

Type:Â SystemDouble

This is an array that includes the long term losses [F/L2]

ItemType (Optional)

Type:Â ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, data is retrieved for the tendon object specified by the Name item.

If this item is Group, data is retrieved for all tendon objects in the group specified by the Name item.

If this item is SelectedObjects, data is retrieved for all selected tendon objects, and the Name item is ignored.

Return Value

Type:Â Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value.

▲ See Also

Reference

cTendonObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cTendonObjAddLanguageSpecificTextSet("LST5B621E05_0?cpp=:3809.");GetL

Method

Retrieves tendon loss data - percentage

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

■ Syntax

C#

VB

C++

F#

Copy

```
int GetLossesPercent(
    string Name,
    ref int NumberItems,
    ref string[] TendonName,
    ref double[] StressingPercent,
    ref double[] LongTermPercent,
    eItemType ItemType = eItemType.Objects
)
```

```
Function GetLossesPercent (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef TendonName As String(),
    ByRef StressingPercent As Double(),
    ByRef LongTermPercent As Double(),
    ByRef ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cTendonObj
Dim Name As String
Dim NumberItems As Integer
Dim TendonName As String()
Dim StressingPercent As Double()
Dim LongTermPercent As Double()
Dim ItemType As eItemType
Dim returnValue As Integer

returnValue = instance.GetLossesPercent(Name,
    NumberItems, TendonName, StressingPercent,
    LongTermPercent, ItemType)
```

```
int GetLossesPercent(
    String^ Name,
    int% NumberItems,
    array<String^>^% TendonName,
    array<double>^% StressingPercent,
    array<double>^% LongTermPercent,
    eItemType ItemType = eItemType::Objects
)
```

abstract GetLossesPercent :

cTendonObjAddLanguageSpecificTextSet("LST5B621E05_0?cpp=:3809.");GetL

Introduction

```
Name : string *
NumberItems : int byref *
TendonName : string[] byref *
StressingPercent : float[] byref *
LongTermPercent : float[] byref *
?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type:Â SystemString

The name of an existing tendon object or group, depending on the value of the ItemType item.

NumberItems

Type:Â SystemInt32

The total number of items retrieved for the specified tendon objects

TendonName

Type:Â SystemString

This is an array that includes the name of the tendon object associated with each value.

StressingPercent

Type:Â SystemDouble

This is an array that includes the stressing losses

LongTermPercent

Type:Â SystemDouble

This is an array that includes the long term losses

ItemType (Optional)

Type:Â ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◇ Object = 0

◇ Group = 1

◇ SelectedObjects = 2

If this item is Objects, data is retrieved for the tendon object specified by the Name item.

If this item is Group, data is retrieved for all tendon objects in the group specified by the Name item.

If this item is SelectedObjects, data is retrieved for all selected tendon objects, and the Name item is ignored.

Return Value

Type:Â Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value.

▲ See Also

Reference

cTendonObj Interface

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CSI API ETABS v1

cTendonObjAddLanguageSpecificTextSet("LSTB014DA Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int GetNameList(
    ref int NumberNames,
    ref string[] MyName
)
```

```
Function GetNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cTendonObj
Dim NumberNames As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetNameList(NumberNames,
    MyName)
```

```
int GetNameList(
    int% NumberNames,
    array<String^>^% MyName
)
```

```
abstract GetNameList :
    NumberNames : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberNames

Type: SystemInt32

MyName

Type: SystemString

Return Value

Type: Int32

▲ See Also

Reference

cTendonObj Interface

ETABSv1 Namespace

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Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

C#

C++

F#

Copy

```
abstract GetNameListOnStory :
  StoryName : string *
  NumberNames : int byref *
  MyName : string[] byref -> int
```

Type: SystemString

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cTendonObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cTendonObjAddLanguageSpecificTextSet("LSTC5D7A42E_0?cpp=3816")

Method

Retrieves the number of strands

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int GetNumberStrands(
    string Name,
    ref int NumberItems,
    ref string[] TendonName,
    ref int[] NumberStrands,
    eItemType ItemType = eItemType.Objects
)
```

```
Function GetNumberStrands (
    Name As String,
    ByRef NumberItems As Integer,
    ByRef TendonName As String(),
    ByRef NumberStrands As Integer(),
    Optional ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cTendonObj
Dim Name As String
Dim NumberItems As Integer
Dim TendonName As String()
Dim NumberStrands As Integer()
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.GetNumberStrands(Name,
    NumberItems, TendonName, NumberStrands,
    ItemType)
```

```
int GetNumberStrands(
    String^ Name,
    int% NumberItems,
    array<String^>^% TendonName,
    array<int>^% NumberStrands,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract GetNumberStrands :
    Name : string *
    NumberItems : int byref *
    TendonName : string[] byref *
    NumberStrands : int[] byref *
```

cTendonObjspan id="LSTC5D7A42E_0"AddLanguageSpecificTextSet("LSTC5D7A42E_0?cpp=3816");Ge

```

        ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int

```

Parameters

Name

Type:Â SystemString

The name of an existing tendon object or group, depending on the value of the ItemType item.

NumberItems

Type:Â SystemInt32

The total number of items retrieved for the specified tendon objects

TendonName

Type:Â SystemString

This is an array that includes the name of the tendon object associated with each value.

NumberStrands

Type:Â SystemInt32

This is an array that includes the number of strands

ItemType (Optional)

Type:Â ETABSv1eItemType

This is one of the following items in the eItemType enumeration:

◊ Object = 0

◊ Group = 1

◊ SelectedObjects = 2

If this item is Objects, data is retrieved for the tendon object specified by the Name item.

If this item is Group, data is retrieved for all tendon objects in the group specified by the Name item.

If this item is SelectedObjects, data is retrieved for all selected tendon objects, and the Name item is ignored.

Return Value

Type:Â Int32

Returns zero if the data is successfully retrieved; otherwise it returns a nonzero value.

▲ See Also

Reference

cTendonObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cTendonObjAddLanguageSpecificTextSet("LST4C157D Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetProperty(
    string Name,
    ref string PropName
)
```

```
Function GetProperty (
    Name As String,
    ByRef PropName As String
) As Integer
```

```
Dim instance As cTendonObj
Dim Name As String
Dim PropName As String
Dim returnValue As Integer
```

```
returnValue = instance.GetProperty(Name,
    PropName)
```

```
int GetProperty(
    String^ Name,
    String^% PropName
)
```

```
abstract GetProperty :
    Name : string *
    PropName : string byref -> int
```

Parameters

Name

Type: [SystemString](#)

PropName

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cTendonObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cTendonObjAddLanguageSpecificTextSet("LSTC0B6A Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetSelected(
    string Name,
    ref bool Selected
)
```

```
Function GetSelected (
    Name As String,
    ByRef Selected As Boolean
) As Integer
```

```
Dim instance As cTendonObj
Dim Name As String
Dim Selected As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.GetSelected(Name,
    Selected)
```

```
int GetSelected(
    String^ Name,
    bool% Selected
)
```

```
abstract GetSelected :
    Name : string *
    Selected : bool byref -> int
```

Parameters

Name

Type: [SystemString](#)

Selected

Type: [SystemBoolean](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cTendonObj Interface

ETABSv1 Namespace

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CSI API ETABS v1

cTendonObjAddLanguageSpecificTextSet("LST9D45546E_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetTendonGeometry(
    string Name,
    ref int NumberPoints,
    ref double[] X,
    ref double[] Y,
    ref double[] Z,
    string CSys = "Global"
)
```

```
Function GetTendonGeometry (
    Name As String,
    ByRef NumberPoints As Integer,
    ByRef X As Double(),
    ByRef Y As Double(),
    ByRef Z As Double(),
    CSys As String = "Global"
) As Integer
```

```
Dim instance As cTendonObj
Dim Name As String
Dim NumberPoints As Integer
Dim X As Double()
Dim Y As Double()
Dim Z As Double()
Dim CSys As String
Dim returnValue As Integer
```

```
returnValue = instance.GetTendonGeometry(Name,
    NumberPoints, X, Y, Z, CSys)
```

```
int GetTendonGeometry(
    String^ Name,
    int% NumberPoints,
    array<double>^% X,
    array<double>^% Y,
    array<double>^% Z,
    String^ CSys = L"Global"
)
```

```
abstract GetTendonGeometry :
    Name : string *
    NumberPoints : int byref *
```

cTendonObjspan id="LST9D45546E_0"AddLanguageSpecificTextSet("LST9D45546E_0?cpp=3822.");GetT

```

    X : float[] byref *
    Y : float[] byref *
    Z : float[] byref *
    ?CSys : string
(* Defaults:
    let _CSys = defaultArg CSys "Global"
*)
-> int

```

Parameters

Name

Type: [SystemString](#)

NumberPoints

Type: [SystemInt32](#)

X

Type: [SystemDouble](#)

Y

Type: [SystemDouble](#)

Z

Type: [SystemDouble](#)

CSys (Optional)

Type: [SystemString](#)

Return Value

Type: [Int32](#)

See Also

Reference

[cTendonObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cTendonObjAddLanguageSpecificTextSet("LSTFC82FA4A_0?cpp=38204=");SetMethod

Adds/removes tendon objects to/from a specified group.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int SetGroupAssign(
    string Name,
    string GroupName,
    bool Remove = false,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetGroupAssign (
    Name As String,
    GroupName As String,
    Remove As Boolean = false,
    ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cTendonObj
Dim Name As String
Dim GroupName As String
Dim Remove As Boolean
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetGroupAssign(Name,
    GroupName, Remove, ItemType)
```

```
int SetGroupAssign(
    String^ Name,
    String^ GroupName,
    bool Remove = false,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetGroupAssign :
    Name : string *
    GroupName : string *
    ?Remove : bool *
    ?ItemType : eItemType
(* Defaults:
    let _Remove = defaultArg Remove false
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

cTendonObjid="LSTFC82FA4A_0"AddLanguageSpecificTextSet("LSTFC82FA4A_0?cpp=38204=");Set

Parameters**Name**

Type:Â SystemString

The name of an existing tendon object or group, depending on the value of the ItemType item.

GroupName

Type:Â SystemString

The name of an existing group to which the assignment is made.

Remove (Optional)

Type:Â SystemBoolean

If this item is False, the specified tendon objects are added to the group specified by the GroupName item. If it is True, the tendon objects are removed from the group.

ItemType (Optional)

Type:Â ETABSv1eItemType

This is one of the items in the eItemType enumeration.

If this item is Objects, the tendon object specified by the Name item is added to/removed from the group specified by the GroupName item.

If this item is Group, the tendon objects in the group specified by the Name item are added to/removed from the group specified by the GroupName item.

If this item is SelectedObjects, all selected tendon objects are added to/removed from the group specified by the GroupName item, and the Name item is ignored.

Return Value

Type:Â Int32

Returns zero if the group assignment is successful; otherwise it returns a nonzero value.

▲ **Remarks**▲ **Examples**

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1
    Dim Name As String
    Dim NumberGroups As Integer
    Dim Groups() as String

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel
```

Introduction

```
'initialize model
    ret = SapModel.InitializeNewModel()

'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4, 12, 12, 4, 4, 24, 24)

'add tendon object manually with name "TendonObject1"
    Name = "TendonObject1"

'define new groups
    ret = SapModel.GroupDef.SetGroup("Group1")
    ret = SapModel.GroupDef.SetGroup("Group2")

'add tendon object to groups
    ret = SapModel.TendonObj.SetGroupAssign(Name, "Group1")
    ret = SapModel.TendonObj.SetGroupAssign(Name, "Group2")

'get tendon object groups
    ret = SapModel.TendonObj.GetGroupAssign(Name, NumberGroups, Groups)

'close ETABS
    EtabsObject.ApplicationExit(False)

'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

[cTendonObj Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cTendonObjAddLanguageSpecificTextSet("LSTFE5E29FB_0?cpp=3827=.");SetMethod

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetSelected(
    string Name,
    bool Selected,
    eItemType ItemType = eItemType.Objects
)
```

```
Function SetSelected (
    Name As String,
    Selected As Boolean,
    ?ItemType As eItemType = eItemType.Objects
) As Integer
```

```
Dim instance As cTendonObj
Dim Name As String
Dim Selected As Boolean
Dim ItemType As eItemType
Dim returnValue As Integer
```

```
returnValue = instance.SetSelected(Name,
    Selected, ItemType)
```

```
int SetSelected(
    String^ Name,
    bool Selected,
    eItemType ItemType = eItemType::Objects
)
```

```
abstract SetSelected :
    Name : string *
    Selected : bool *
    ?ItemType : eItemType
(* Defaults:
    let _ItemType = defaultArg ItemType eItemType.Objects
*)
-> int
```

Parameters

Name

Type: [SystemString](#)

Selected

cTendonObjAddLanguageSpecificTextSet("LSTFE5E29FB_0?cpp=3827=.");Set

Type: SystemBoolean
ItemType (Optional)
Type: ETABSv1eItemType

Return Value

Type: Int32

See Also

Reference

cTendonObj Interface

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CSI API ETABS v1

cTower Interface

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cTower
```

```
Public Interface cTower
```

```
Dim instance As cTower
```

```
public interface class cTower
```

```
type cTower = interface end
```

The cTower type exposes the following members.

▲ Methods

▲	Name	Description
⇒	AddCopyOfTower	
⇒	AddNewTower	
⇒	AllowMultipleTowers	
⇒	DeleteTower	
⇒	GetActiveTower	
⇒	GetNameList	
⇒	RenameTower	
⇒	SetActiveTower	

[Top](#)

▲ See Also

Reference

[ETABSV1 Namespace](#)

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CSI API ETABS v1

cTower Methods

The cTower type exposes the following members.

Methods

	Name	Description
⇒	<u>AddCopyOfTower</u>	
⇒	<u>AddNewTower</u>	
⇒	<u>AllowMultipleTowers</u>	
⇒	<u>DeleteTower</u>	
⇒	<u>GetActiveTower</u>	
⇒	<u>GetNameList</u>	
⇒	<u>RenameTower</u>	
⇒	<u>SetActiveTower</u>	

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See Also

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cTower Interface

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CSI API ETABS v1

cTowerAddLanguageSpecificTextSet("LST9B2D92F5_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int AddCopyOfTower(
    string TowerName,
    string NewTowerName
)
```

```
Function AddCopyOfTower (
    TowerName As String,
    NewTowerName As String
) As Integer
```

```
Dim instance As cTower
Dim TowerName As String
Dim NewTowerName As String
Dim returnValue As Integer
```

```
returnValue = instance.AddCopyOfTower(TowerName,
    NewTowerName)
```

```
int AddCopyOfTower(
    String^ TowerName,
    String^ NewTowerName
)
```

```
abstract AddCopyOfTower :
    TowerName : string *
    NewTowerName : string -> int
```

Parameters

TowerName

Type: [SystemString](#)

NewTowerName

Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cTower Interface

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CSI API ETABS v1

cTowerAddLanguageSpecificTextSet("LST6E281390_0 Method

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
int AddNewTower(
    string TowerName,
    int NumberStories,
    double TypicalStoryHeight,
    double BotttomStoryHeight
)
```

```
Function AddNewTower (
    TowerName As String,
    NumberStories As Integer,
    TypicalStoryHeight As Double,
    BotttomStoryHeight As Double
) As Integer
```

```
Dim instance As cTower
Dim TowerName As String
Dim NumberStories As Integer
Dim TypicalStoryHeight As Double
Dim BotttomStoryHeight As Double
Dim returnValue As Integer
```

```
returnValue = instance.AddNewTower(TowerName,
    NumberStories, TypicalStoryHeight,
    BotttomStoryHeight)
```

```
int AddNewTower(
    String^ TowerName,
    int NumberStories,
    double TypicalStoryHeight,
    double BotttomStoryHeight
)
```

```
abstract AddNewTower :
    TowerName : string *
    NumberStories : int *
    TypicalStoryHeight : float *
    BotttomStoryHeight : float -> int
```

Parameters

TowerName

Type: SystemString

NumberStories

Type: SystemInt32

TypicalStoryHeight

Type: SystemDouble

BottomStoryHeight

Type: SystemDouble

Return Value

Type: Int32

▲ See Also

Reference

cTower Interface

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CSI API ETABS v1

cTowerAddLanguageSpecificTextSet("LST2CA3AB0A_0?cpp=::|nub35AllowM

Method

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int AllowMultipleTowers(
    bool AllowMultTowers,
    string RetainedTower = "",
    bool Combine = true
)
```

```
Function AllowMultipleTowers (
    AllowMultTowers As Boolean,
    RetainedTower As String = "",
    Combine As Boolean = true
) As Integer
```

```
Dim instance As cTower
Dim AllowMultTowers As Boolean
Dim RetainedTower As String
Dim Combine As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.AllowMultipleTowers(AllowMultTowers,
    RetainedTower, Combine)
```

```
int AllowMultipleTowers(
    bool AllowMultTowers,
    String^ RetainedTower = L"",
    bool Combine = true
)
```

```
abstract AllowMultipleTowers :
    AllowMultTowers : bool *
    ?RetainedTower : string *
    ?Combine : bool
(* Defaults:
    let _RetainedTower = defaultArg RetainedTower ""
    let _Combine = defaultArg Combine true
*)
-> int
```

Parameters

AllowMultTowers

Type: [System.Boolean](#)

cTowersAddLanguageSpecificTextSet("LST2CA3AB0A_0?cpp=::|nub35AllowM

RetainedTower (Optional)

Type: SystemString

Combine (Optional)

Type: SystemBoolean

Return Value

Type: Int32

▲ See Also

Reference

cTower Interface

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CSI API ETABS v1

cTowerAddLanguageSpecificTextSet("LST6FF7338F_0 Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int DeleteTower(
    string TowerName,
    bool Associate,
    string AssocWithTower = ""
)
```

```
Function DeleteTower (
    TowerName As String,
    Associate As Boolean,
    AssocWithTower As String = ""
) As Integer
```

```
Dim instance As cTower
Dim TowerName As String
Dim Associate As Boolean
Dim AssocWithTower As String
Dim returnValue As Integer
```

```
returnValue = instance.DeleteTower(TowerName,
    Associate, AssocWithTower)
```

```
int DeleteTower(
    String^ TowerName,
    bool Associate,
    String^ AssocWithTower = L""
)
```

```
abstract DeleteTower :
    TowerName : string *
    Associate : bool *
    ?AssocWithTower : string
(* Defaults:
    let _AssocWithTower = defaultArg AssocWithTower ""
*)
-> int
```

Parameters

TowerName

Type: [SystemString](#)

Associate

cTowerspan id="LST6FF7338F_0"AddLanguageSpecificTextSet("LST6FF7338F_0?cpp=::|nu=38,DeleteTov

Type: SystemBoolean
AssocWithTower (Optional)
Type: SystemString

Return Value

Type: Int32

■ See Also

Reference

cTower Interface

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CSI API ETABS v1

cTowerAddLanguageSpecificTextSet("LST19861515_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetActiveTower(
    ref string TowerName
)
```

```
Function GetActiveTower (
    ByRef TowerName As String
) As Integer
```

```
Dim instance As cTower
Dim TowerName As String
Dim returnValue As Integer
```

```
returnValue = instance.GetActiveTower(TowerName)
```

```
int GetActiveTower(
    String^% TowerName
)
```

```
abstract GetActiveTower :
    TowerName : string byref -> int
```

Parameters

TowerName
Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cTower Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cTowerAddLanguageSpecificTextSet("LST5F2D88EB_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int GetNameList(
    ref int NumberNames,
    ref string[] MyName
)
```

```
Function GetNameList (
    ByRef NumberNames As Integer,
    ByRef MyName As String()
) As Integer
```

```
Dim instance As cTower
Dim NumberNames As Integer
Dim MyName As String()
Dim returnValue As Integer
```

```
returnValue = instance.GetNameList(NumberNames,
    MyName)
```

```
int GetNameList(
    int% NumberNames,
    array<String^>^% MyName
)
```

```
abstract GetNameList :
    NumberNames : int byref *
    MyName : string[] byref -> int
```

Parameters

NumberNames

Type: [System.Int32](#)

MyName

Type: [System.String](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cTower Interface

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CSI API ETABS v1

cTowerAddLanguageSpecificTextSet("LSTCDCE358B_0") Method

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int RenameTower(
    string TowerName,
    string NewTowerName
)
```

```
Function RenameTower (
    TowerName As String,
    NewTowerName As String
) As Integer
```

```
Dim instance As cTower
Dim TowerName As String
Dim NewTowerName As String
Dim returnValue As Integer
```

```
returnValue = instance.RenameTower(TowerName,
    NewTowerName)
```

```
int RenameTower(
    String^ TowerName,
    String^ NewTowerName
)
```

```
abstract RenameTower :
    TowerName : string *
    NewTowerName : string -> int
```

Parameters

TowerName
Type: [SystemString](#)

NewTowerName
Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

cTower Interface

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CSI API ETABS v1

cTowerAddLanguageSpecificTextSet("LSTB5CAE001_0") Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int SetActiveTower(
    string TowerName
)
```

```
Function SetActiveTower (
    TowerName As String
) As Integer
```

```
Dim instance As cTower
Dim TowerName As String
Dim returnValue As Integer
```

```
returnValue = instance.SetActiveTower(TowerName)
```

```
int SetActiveTower(
    String^ TowerName
)
```

```
abstract SetActiveTower :
    TowerName : string -> int
```

Parameters

TowerName
Type: [SystemString](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cTower Interface](#)

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CSI API ETABS v1

cView Interface

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public interface cView
```

```
Public Interface cView
```

```
Dim instance As cView
```

```
public interface class cView
```

```
type cView = interface end
```

The cView type exposes the following members.

▲ Methods

▲	Name	Description
🔗	RefreshView	Refreshes the view for the specified window(s).
🔗	RefreshWindow	

[Top](#)

▲ See Also

Reference

[ETABSv1 Namespace](#)

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CSI API ETABS v1

cView Methods

The cView type exposes the following members.

Methods

⌂	Name	Description
🔍	<u>RefreshView</u>	Refreshes the view for the specified window(s).
🔍	<u>RefreshWindow</u>	

Top

🔍 See Also

Reference

cView Interface

ETABSv1 Namespace

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CSI API ETABS v1

cViewAddLanguageSpecificTextSet("LST5BDB3BD_0? Method

Refreshes the view for the specified window(s).

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
int RefreshView(
    int Window = 0,
    bool Zoom = true
)
```

```
Function RefreshView (
    Window As Integer = 0,
    Zoom As Boolean = true
) As Integer
```

```
Dim instance As cView
Dim Window As Integer
Dim Zoom As Boolean
Dim returnValue As Integer
```

```
returnValue = instance.RefreshView(Window,
Zoom)
```

```
int RefreshView(
    int Window = 0,
    bool Zoom = true
)
```

```
abstract RefreshView :
    ?Window : int *
    ?Zoom : bool
(* Defaults:
    let _Window = defaultArg Window 0
    let _Zoom = defaultArg Zoom true
*)
-> int
```

Parameters

Window (Optional)

Type: SystemInt32

This is 0 meaning all windows or an existing window number. It indicates the window(s) to have its view refreshed.

Zoom (Optional)

cViewspan id="LST5BDB3BD_0"AddLanguageSpecificTextSet("LST5BDB3BD_0?cpp=::|nu=."3849RefreshView

Type: System.Boolean

If this item is True, the window zoom is maintained when the view is refreshed.
If it is False, the zoom returns to a default zoom.

Return Value

Type: Int32

Returns zero if the window views are successfully refreshed; otherwise it returns a nonzero value.

▲ Remarks

▲ Examples

VB

Copy

```
Public Sub Example()
    Dim SapModel As cSapModel
    Dim EtabsObject As cOAPI
    Dim ret As Integer = -1

    'create ETABS object
    Dim myHelper as cHelper = New Helper
    EtabsObject = myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")

    'start ETABS application
    ret = EtabsObject.ApplicationStart()

    'create SapModel object
    SapModel = EtabsObject.SapModel

    'initialize model
    ret = SapModel.InitializeNewModel()

    'create steel deck template model
    ret = SapModel.File.NewSteelDeck(4,12,12,4,4,24,24)

    'refresh view
    ret = SapModel.View.RefreshView()

    'close ETABS
    EtabsObject.ApplicationExit(False)

    'clean up variables
    SapModel = Nothing
    EtabsObject = Nothing
End Sub
```

▲ See Also

Reference

cView Interface

ETABSv1 Namespace

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CSI API ETABS v1

cViewAddLanguageSpecificTextSet("LST569F6A72_0? Method

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
int RefreshWindow(
    int Window = 0
)
```

```
Function RefreshWindow (
    Optional Window As Integer = 0
) As Integer
```

```
Dim instance As cView
Dim Window As Integer
Dim returnValue As Integer
```

```
returnValue = instance.RefreshWindow(Window)
```

```
int RefreshWindow(
    int Window = 0
)
```

```
abstract RefreshWindow :
    ?Window : int
(* Defaults:
    let _Window = defaultArg Window 0
*)
-> int
```

Parameters

Window (Optional)
Type: [SystemInt32](#)

Return Value

Type: [Int32](#)

▲ See Also

Reference

[cView Interface](#)

[ETABSv1 Namespace](#)

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CSI API ETABS v1

e2DFrameType Enumeration

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum e2DFrameType
```

```
Public Enumeration e2DFrameType
```

```
Dim instance As e2DFrameType
```

```
public enum class e2DFrameType
```

```
type e2DFrameType
```

▲ Members

Member name	Value	Description
PortalFrame	0	
ConcentricBraced	1	
EccentricBraced	2	

▲ See Also

Reference

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CSI API ETABS v1

e3DFrameType Enumeration

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum e3DFrameType
```

```
Public Enumeration e3DFrameType
```

```
Dim instance As e3DFrameType
```

```
public enum class e3DFrameType
```

```
type e3DFrameType
```

▲ Members

Â Member name Value Description

OpenFrame	0
PerimeterFrame	1
BeamSlab	2
FlatPlate	3

▲ See Also

Reference

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CSI API ETABS v1

eAreaDesignOrientation Enumeration

The possible design orientations for a shell.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eAreaDesignOrientation
```

```
Public Enumeration eAreaDesignOrientation
```

```
Dim instance As eAreaDesignOrientation
```

```
public enum class eAreaDesignOrientation
```

```
type eAreaDesignOrientation
```

Members

	Member name	Value	Description
	Wall	1	
	Floor	2	
	Ramp_DO_NOT_USE	3	
	Null	4	
	Other	5	

Remarks

See Also

Reference

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CSI API ETABS v1

eCNameType Enumeration

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eCNameType
```

```
Public Enumeration eCNameType
```

```
Dim instance As eCNameType
```

```
public enum class eCNameType
```

```
type eCNameType
```

▲ Members

Â Member name Value Description

LoadCase	0
----------	---

LoadCombo	1
-----------	---

▲ See Also

Reference

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CSI API ETABS v1

eConstraintAxis Enumeration

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eConstraintAxis
```

```
Public Enumeration eConstraintAxis
```

```
Dim instance As eConstraintAxis
```

```
public enum class eConstraintAxis
```

```
type eConstraintAxis
```

▲ Members

Â Member name Value Description

X	1	
Y	2	
Z	3	
AutoAxis	4	

▲ See Also

Reference

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CSI API ETABS v1

eConstraintType Enumeration

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eConstraintType
```

Public Enumeration eConstraintType

```
Dim instance As eConstraintType
```

```
public enum class eConstraintType
```

```
type eConstraintType
```

▲ Members

Â Member name Value Description

Body	1
Diaphragm	2
Plate	3
Rod	4
Beam	5
Equal	6
Local	7
Weld	8
Line	13

▲ See Also

Reference

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CSI API ETABS v1

eDeckType Enumeration

The possible deck types.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eDeckType
```

```
Public Enumeration eDeckType
```

```
Dim instance As eDeckType
```

```
public enum class eDeckType
```

```
type eDeckType
```

▲ Members

▲ Member name Value Description

Filled	1
Unfilled	2
SolidSlab	3

▲ Remarks

▲ See Also

Reference

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CSI API ETABS v1

eDesignActionType Enumeration

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eDesignActionType
```

Public Enumeration eDesignActionType

```
Dim instance As eDesignActionType
```

```
public enum class eDesignActionType
```

```
type eDesignActionType
```

▲ Members

Member name	Value	Description
NonComposite	1	
ShortTermComposite	2	
LongTermComposite	3	
Staged	4	
Other	5	

▲ See Also

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CSI API ETABS v1

eDiaphragmOption Enumeration

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eDiaphragmOption
```

```
Public Enumeration eDiaphragmOption
```

```
Dim instance As eDiaphragmOption
```

```
public enum class eDiaphragmOption
```

```
type eDiaphragmOption
```

▲ Members

Member name	Value	Description
Disconnect	1	
FromShellObject	2	
DefinedDiaphragm	3	

▲ See Also

Reference

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CSI API ETABS v1

eForce Enumeration

The force units that can be specified for the model.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
public enum eForce
```

```
Public Enumeration eForce
```

```
Dim instance As eForce
```

```
public enum class eForce
```

```
type eForce
```

Members

Member name Value Description

NotApplicable	0	
lb	1	
kip	2	
N	3	
kN	4	
kgf	5	
tonf	6	

Remarks

See Also

Reference

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CSI API ETABS v1

eFrameDesignOrientation Enumeration

The possible design orientations for a frame.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eFrameDesignOrientation
```

```
Public Enumeration eFrameDesignOrientation
```

```
Dim instance As eFrameDesignOrientation
```

```
public enum class eFrameDesignOrientation
```

```
type eFrameDesignOrientation
```

▲ Members

Â Member name Value Description

Column	1	
Beam	2	
Brace	3	
Null	4	
Other	5	

▲ Remarks

▲ See Also

Reference

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CSI API ETABS v1

eFramePropType Enumeration

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eFramePropType
```

```
Public Enumeration eFramePropType
```

```
Dim instance As eFramePropType
```

```
public enum class eFramePropType
```

```
type eFramePropType
```

Members

	Member name	Value	Description
I		1	
Channel		2	
T		3	
Angle		4	
DblAngle		5	
Box		6	
Pipe		7	
Rectangular		8	
Circle		9	
General		10	
DbChannel		11	
Auto		12	
SD		13	
Variable		14	
Joist		15	
Bridge		16	
Cold_C		17	
Cold_2C		18	
Cold_Z		19	
Cold_L		20	
Cold_2L		21	

Cold_Hat	22
BuiltupICoverplate	23
PCCGirderI	24
PCCGirderU	25
BuiltupIHybrid	26
BuiltupUHybrid	27
Concrete_L	28
FilledTube	29
FilledPipe	30
EncasedRectangle	31
EncasedCircle	32
BucklingRestrainedBrace	33
CoreBrace_BRB	34
ConcreteTee	35
ConcreteBox	36
ConcretePipe	37
ConcreteCross	38
SteelPlate	39
SteelRod	40
PCCGirderSuperT	41
Cold_Box	42
Cold_I	43
Cold_Pipe	44
Cold_T	45
Trapezoidal	46

▲ See Also

Reference

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CSI API ETABS v1

eHingeLocationType Enumeration

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eHingeLocationType
```

```
Public Enumeration eHingeLocationType
```

```
Dim instance As eHingeLocationType
```

```
public enum class eHingeLocationType
```

```
type eHingeLocationType
```

▲ Members

Â Member name Value Description

RelativeDistance 1

OffsetFromIEnd 2

OffsetFromJEnd 3

▲ See Also

Reference

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CSI API ETABS v1

eltemType Enumeration

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eItemType
```

```
Public Enumeration eItemType
```

```
Dim instance As eItemType
```

```
public enum class eItemType
```

```
type eItemType
```

▲ Members

Â Member name Value Description

Objects	0
Group	1
SelectedObjects	2

▲ See Also

Reference

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CSI API ETABS v1

eltemTypeElm Enumeration

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eItemTypeElm
```

```
Public Enumeration eItemTypeElm
```

```
Dim instance As eItemTypeElm
```

```
public enum class eItemTypeElm
```

```
type eItemTypeElm
```

▲ Members

Â Member name Value Description

ObjectElm	0
Element	1
GroupElm	2
SelectionElm	3

▲ See Also

Reference

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CSI API ETABS v1

eLength Enumeration

The length units that can be specified for the model.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
public enum eLength
```

```
Public Enumeration eLength
```

```
Dim instance As eLength
```

```
public enum class eLength
```

```
type eLength
```

▲ Members

Â Member name Value Description

NotApplicable	0	
inch	1	
ft	2	
micron	3	
mm	4	
cm	5	
m	6	

▲ Remarks

▲ See Also

Reference

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CSI API ETABS v1

eLinkPropType Enumeration

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eLinkPropType
```

Public Enumeration eLinkPropType

Dim instance As eLinkPropType

```
public enum class eLinkPropType
```

```
type eLinkPropType
```

▲ Members

Member name	Value	Description
Linear	1	
Damper	2	
Gap	3	
Hook	4	
PlasticWen	5	
Isolator1	6	
Isolator2	7	
MultilinearElastic	8	
MultilinearPlastic	9	
Isolator3	10	

▲ See Also

Reference

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CSI API ETABS v1

eLoadCaseType Enumeration

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eLoadCaseType
```

```
Public Enumeration eLoadCaseType
```

```
Dim instance As eLoadCaseType
```

```
public enum class eLoadCaseType
```

```
type eLoadCaseType
```

▲ Members

Member name	Value	Description
LinearStatic	1	
NonlinearStatic	2	
Modal	3	
ResponseSpectrum	4	
LinearHistory	5	
NonlinearHistory	6	
LinearDynamic	7	
NonlinearDynamic	8	
MovingLoad	9	
Buckling	10	
SteadyState	11	
PowerSpectralDensity	12	
LinearStaticMultiStep	13	
HyperStatic	14	

▲ See Also

Reference

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CSI API ETABS v1

eLoadPatternType Enumeration

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

C#

VB

C++

F#

Copy

```
public enum eLoadPatternType
```

```
Public Enumeration eLoadPatternType
```

```
Dim instance As eLoadPatternType
```

```
public enum class eLoadPatternType
```

```
type eLoadPatternType
```

Members

Member name	Value	Description
Dead	1	
SuperDead	2	
Live	3	
ReduceLive	4	
Quake	5	
Wind	6	
Snow	7	
Other	8	
Move	9	Not valid for ETABS and SAFE
Temperature	10	
Rooflive	11	
Notional	12	
PatternLive	13	
Wave	14	Not valid for ETABS and SAFE
Braking	15	Not valid for ETABS and SAFE
Centrifugal	16	Not valid for ETABS and SAFE
Friction	17	Not valid for ETABS and SAFE
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VesselCollision	25	Not valid for ETABS and SAFE
TemperatureGradient	26	Not valid for ETABS and SAFE
Settlement	27	Not valid for ETABS and SAFE
Shrinkage	28	Not valid for ETABS and SAFE
Creep	29	Not valid for ETABS and SAFE
WaterloadPressure	30	Not valid for ETABS and SAFE
LiveLoadSurcharge	31	Not valid for ETABS and SAFE
LockedInForces	32	Not valid for ETABS and SAFE
PedestrianLL	33	Not valid for ETABS and SAFE
Prestress	34	
Hyperstatic	35	Not valid for ETABS and SAFE
Bouyancy	36	Not valid for ETABS and SAFE
StreamFlow	37	Not valid for ETABS and SAFE
Impact	38	Not valid for ETABS and SAFE
Construction	39	
DeadWearing	40	Not valid for ETABS and SAFE
DeadWater	41	Not valid for ETABS and SAFE
DeadManufacture	42	Not valid for ETABS and SAFE
EarthHydrostatic	43	Not valid for ETABS and SAFE
PassiveEarthPressure	44	Not valid for ETABS and SAFE
ActiveEarthPressure	45	Not valid for ETABS and SAFE
PedestrianLLReduced	46	Not valid for ETABS and SAFE
SnowHighAltitude	47	Not valid for ETABS and SAFE
EuroLm1Char	48	Not valid for ETABS and SAFE
EuroLm1Freq	49	Not valid for ETABS and SAFE
EuroLm2	50	Not valid for ETABS and SAFE
EuroLm3	51	Not valid for ETABS and SAFE
EuroLm4	52	Not valid for ETABS and SAFE
SeaState	53	Not valid for ETABS and SAFE
Permit	54	Not valid for ETABS and SAFE
MoveFatigue	55	Not valid for ETABS and SAFE
MoveFatiguePermit	56	Not valid for ETABS and SAFE
MoveDeflection	57	Not valid for ETABS and SAFE
MoveTrain	58	Not valid for ETABS and SAFE
PrestressTransfer	59	
PatternAuto	60	
QuakeDrift	61	
QuakeVerticalOnly	62	Not valid for ETABS and SAFE

▲ See Also

Reference

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CSI API ETABS v1

eMatCoupledType Enumeration

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eMatCoupledType
```

```
Public Enumeration eMatCoupledType
```

```
Dim instance As eMatCoupledType
```

```
public enum class eMatCoupledType
```

```
type eMatCoupledType
```

▲ Members

Member name	Value	Description
None	1	
VonMisesPlasticity	2	
ModifiedDarwinPecknoldConcrete	3	

▲ See Also

Reference

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CSI API ETABS v1

eMatType Enumeration

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eMatType
```

```
Public Enumeration eMatType
```

```
Dim instance As eMatType
```

```
public enum class eMatType
```

```
type eMatType
```

▲ Members

Â Member name Value Description

Steel	1
Concrete	2
NoDesign	3
Aluminum	4
ColdFormed	5
Rebar	6
Tendon	7
Masonry	8

▲ See Also

Reference

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CSI API ETABS v1

eMatTypeAluminum Enumeration

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eMatTypeAluminum
```

```
Public Enumeration eMatTypeAluminum
```

```
Dim instance As eMatTypeAluminum
```

```
public enum class eMatTypeAluminum
```

```
type eMatTypeAluminum
```

▲ Members

Member name	Value	Description
SubType_6061_T6	1	
SubType_6063_T6	2	
SubType_5052_H34	3	

▲ See Also

Reference

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CSI API ETABS v1

eMatTypeColdFormed Enumeration

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eMatTypeColdFormed
```

```
Public Enumeration eMatTypeColdFormed
```

```
Dim instance As eMatTypeColdFormed
```

```
public enum class eMatTypeColdFormed
```

```
type eMatTypeColdFormed
```

▲ Members

Member name	Value	Description
ASTM_A653SQGr33	1	
ASTM_A653SQGr50	2	

▲ See Also

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CSI API ETABS v1

eMatTypeConcrete Enumeration

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eMatTypeConcrete
```

```
Public Enumeration eMatTypeConcrete
```

```
Dim instance As eMatTypeConcrete
```

```
public enum class eMatTypeConcrete
```

```
type eMatTypeConcrete
```

▲ Members

▲	Member name	Value	Description
	FC3000_NormalWeight	1	
	FC4000_NormalWeight	2	
	FC5000_NormalWeight	3	
	FC6000_NormalWeight	4	
	FC3000_LightWeight	5	
	FC4000_LightWeight	6	
	FC5000_LightWeight	7	
	FC6000_LightWeight	8	
	Chinese_C20_NormalWeight	9	
	Chinese_C30_NormalWeight	10	
	Chinese_C40_NormalWeight	11	
	Indian_M15_NormalWeight	12	
	Indian_M20_NormalWeight	13	
	Indian_M25_NormalWeight	14	
	Indian_M30_NormalWeight	15	
	Indian_M35_NormalWeight	16	
	Indian_M40_NormalWeight	17	
	Indian_M45_NormalWeight	18	
	Indian_M50_NormalWeight	19	
	Indian_M55_NormalWeight	20	
	Indian_M60_NormalWeight	21	

EN_C12_NormalWeight	22
EN_C16_NormalWeight	23
EN_C20_NormalWeight	24
EN_C25_NormalWeight	25
EN_C30_NormalWeight	26
EN_C35_NormalWeight	27
EN_C40_NormalWeight	28
EN_C45_NormalWeight	29
EN_C50_NormalWeight	30
EN_C55_NormalWeight	31
EN_C60_NormalWeight	32
EN_C70_NormalWeight	33
EN_C80_NormalWeight	34
EN_C90_NormalWeight	35

▲ See Also

Reference

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CSI API ETABS v1

eMatTypeRebar Enumeration

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eMatTypeRebar
```

```
Public Enumeration eMatTypeRebar
```

```
Dim instance As eMatTypeRebar
```

```
public enum class eMatTypeRebar
```

```
type eMatTypeRebar
```

▲ Members

Â Member name Value Description

ASTM_A615Gr40	1	
ASTM_A615Gr60	2	
ASTM_A615Gr75	3	
ASTM_A706	4	
Chinese_HP235	5	
Chinese_HRB335	6	
Chinese_HRB400	7	
Indian_Mild250	8	
Indian_HYSD415	9	
Indian_HYSD500	10	
Indian_HYSD550	11	

▲ See Also

Reference

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CSI API ETABS v1

eMatTypeSteel Enumeration

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eMatTypeSteel
```

```
Public Enumeration eMatTypeSteel
```

```
Dim instance As eMatTypeSteel
```

```
public enum class eMatTypeSteel
```

```
type eMatTypeSteel
```

▲ Members

Member name	Value	Description
ASTM_A36	1	
ASTM_A53GrB	2	
ASTM_A500GrB_Fy42	3	
ASTM_A500GrB_Fy46	4	
ASTM_A572Gr50	5	
ASTM_A913Gr50	6	
ASTM_A992_Fy50	7	
Chinese_Q235	8	
Chinese_Q345	9	
Indian_Fe250	10	
Indian_Fe345	11	
EN100252_S235	12	
EN100252_S275	13	
EN100252_S355	14	
EN100252_S450	15	
Chinese_Q355	16	

▲ See Also

Reference

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CSI API ETABS v1

eMatTypeTendon Enumeration

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eMatTypeTendon
```

```
Public Enumeration eMatTypeTendon
```

```
Dim instance As eMatTypeTendon
```

```
public enum class eMatTypeTendon
```

```
type eMatTypeTendon
```

▲ Members

Member name	Value	Description
ASTM_A416Gr250	1	
ASTM_A416Gr270	2	

▲ See Also

Reference

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CSI API ETABS v1

eNamedSetType Enumeration

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eNamedSetType
```

```
Public Enumeration eNamedSetType
```

```
Dim instance As eNamedSetType
```

```
public enum class eNamedSetType
```

```
type eNamedSetType
```

▲ Members

▲	Member name	Value	Description
	All	0	
	UpdateBridgeObject	1	
	RunAnalysis	2	
	RunBridgeDesignSuperstructure	3	
	RunBridgeDesignSubstructure	4	
	RunBridgeDesignSeismic	5	
	RunBridgeRatingSuperstructure	6	
	RunMemberRating	7	
	JointTHResponseSpectra	8	
	NamedDisplay	9	
	PlotFunctionTraces	10	
	PushoverCurve	11	
	VirtualWork	12	
	TableSet	13	
	TableGroupSuperset	14	
	BridgeSeismicReport	15	
	BridgeSuperstructureResponse	16	
	BridgeCalculationReport	17	

▲ See Also

Reference

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CSI API ETABS v1

eObjType Enumeration

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eObjType
```

```
Public Enumeration eObjType
```

```
Dim instance As eObjType
```

```
public enum class eObjType
```

```
type eObjType
```

▲ Members

Â Member name Value Description

Point	1
Frame	2
Area	3
Solid	6

▲ See Also

Reference

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CSI API ETABS v1

eReturnCode Enumeration

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eReturnCode
```

```
Public Enumeration eReturnCode
```

```
Dim instance As eReturnCode
```

```
public enum class eReturnCode
```

```
type eReturnCode
```

▲ Members

Member name	Value	Description
NotApplicable	-100	
NotImplemented	-99	
NoError	0	
UnspecifiedError	1	
Deprecated	-98	
TableIsObsolete	-97	
TableDoesNotExist	-96	

▲ See Also

Reference

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CSI API ETABS v1

eShellType Enumeration

The possible shell types.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eShellType
```

```
Public Enumeration eShellType
```

```
Dim instance As eShellType
```

```
public enum class eShellType
```

```
type eShellType
```

▲ Members

Member name	Value	Description
ShellThin	1	
ShellThick	2	
Membrane	3	
PlateThin_DO_NOT_USE	4	
PlateThick_DO_NOT_USE	5	
Layered	6	

▲ Remarks

▲ See Also

Reference

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CSI API ETABS v1

eSlabType Enumeration

The possible slab types.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eSlabType
```

```
Public Enumeration eSlabType
```

```
Dim instance As eSlabType
```

```
public enum class eSlabType
```

```
type eSlabType
```

▲ Members

▲	Member name	Value	Description
	Slab	0	
	Drop	1	
	Stiff_DO_NOT_USE	2	
	Ribbed	3	
	Waffle	4	
	Mat	5	
	Footing	6	

▲ Remarks

▲ See Also

Reference

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CSI API ETABS v1

eSuperObjectClass Enumeration

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eSuperObjectClass
```

```
Public Enumeration eSuperObjectClass
```

```
Dim instance As eSuperObjectClass
```

```
public enum class eSuperObjectClass
```

```
type eSuperObjectClass
```

▲ Members

Member name	Value	Description
None	0	
SuperObject	1	
Foundation	2	
BridgeFoundation	3	

▲ See Also

Reference

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CSI API ETABS v1

eTemperature Enumeration

The temperature units that can be specified for the model.

Namespace: ETABSv1

Assembly: ETABSv1 (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

C#

VB

C++

F#

Copy

```
public enum eTemperature
```

```
Public Enumeration eTemperature
```

```
Dim instance As eTemperature
```

```
public enum class eTemperature
```

```
type eTemperature
```

▲ Members

Â Member name Value Description

NotApplicable	0
F	1
C	2

▲ Remarks

▲ See Also

Reference

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CSI API ETABS v1

eTemplateType Enumeration

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eTemplateType
```

```
Public Enumeration eTemplateType
```

```
Dim instance As eTemplateType
```

```
public enum class eTemplateType
```

```
type eTemplateType
```

▲ Members

Member name	Value	Description
Grid	0	
Clear	1	
Beam	2	
SlopedTruss	3	
VerticalTruss	4	
SpaceTruss	5	
PortalFrame	6	
BracedFrame	7	
EccentricFrame	8	
PerimeterFrame	9	
SpaceFrame	10	
Bridge	11	
Barrel	12	
Cylinder	13	
Dome	14	
ShearWall	15	
Floor	16	
Advanced	17	
UndergroundConcrete	18	
Truss2D	19	
Truss3D	20	

Frame2D	21
Frame3D	22
BridgeWizard	23
PipesAndPlates	24
Shells	25
SolidModels	26
StorageStructures	27
Staircases	28
CableBridges	29

📌 See Also

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CSI API ETABS v1

eUnits Enumeration

The units that can be specified for the model.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eUnits
```

```
Public Enumeration eUnits
```

```
Dim instance As eUnits
```

```
public enum class eUnits
```

```
type eUnits
```

Members

Member name Value Description

lb_in_F	1	
lb_ft_F	2	
kip_in_F	3	
kip_ft_F	4	
kN_mm_C	5	
kN_m_C	6	
kgf_mm_C	7	
kgf_m_C	8	
N_mm_C	9	
N_m_C	10	
Ton_mm_C	11	
Ton_m_C	12	
kN_cm_C	13	
kgf_cm_C	14	
N_cm_C	15	
Ton_cm_C	16	

Remarks

See Also

Reference

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CSI API ETABS v1

eWallPierRebarLayerType Enumeration

The pier rebar types.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eWallPierRebarLayerType
```

```
Public Enumeration eWallPierRebarLayerType
```

```
Dim instance As eWallPierRebarLayerType
```

```
public enum class eWallPierRebarLayerType
```

```
type eWallPierRebarLayerType
```

Members

	Member name	Value	Description
	Vertical_Distributed_MiddleZone_Eachface	1	
	Horizontal_Distributed_MiddleZone_Eachface	2	
	Vertical_Distributed_EndZoneI_Total	3	
	Vertical_Distributed_EndZoneJ_Total	4	
	Confinement_EndZoneI	5	
	Confinement_EndZoneJ	6	
	Diagonal_Each	7	

Remarks

See Also

Reference

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CSI API ETABS v1

eWallPropType Enumeration

The wall property type.

Namespace: [ETABSV1](#)

Assembly: [ETABSV1](#) (in ETABSV1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eWallPropType
```

```
Public Enumeration eWallPropType
```

```
Dim instance As eWallPropType
```

```
public enum class eWallPropType
```

```
type eWallPropType
```

▲ Members

Â Member name Value Description

Specified	1
-----------	---

AutoSelectList	2
----------------	---

▲ Remarks

▲ See Also

Reference

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CSI API ETABS v1

eWallSpandrelRebarLayerType Enumeration

The pier rebar types.

Namespace: [ETABSv1](#)

Assembly: [ETABSv1](#) (in ETABSv1.dll) Version: 1.0.0.0 (1.24.0.0)

▲ Syntax

[C#](#)

[VB](#)

[C++](#)

[F#](#)

[Copy](#)

```
public enum eWallSpandrelRebarLayerType
```

```
Public Enumeration eWallSpandrelRebarLayerType
```

```
Dim instance As eWallSpandrelRebarLayerType
```

```
public enum class eWallSpandrelRebarLayerType
```

```
type eWallSpandrelRebarLayerType
```

▲ Members

Member name	Value	Description
Horizontal_Top_Total	1	
Horizontal_Bottom_Total	2	
Horizontal_Distributed_Eachface	3	
Vertical_Ties_Distributed	4	
Diagonal_Each	5	

▲ Remarks

▲ See Also

Reference

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CSI API ETABS v1

Breaking changes in v19.0.0, v19.0.1, and v19.0.2

The following inadvertent changes broke compatibility with earlier API versions:

- In ETABS v19.0.0 (ETABSV1.dll version 1.10 and CSiAPIv1.dll version 1.10), eMatTypeSteel_Chinese_Q345 enumeration got renamed to eMatTypeSteel_Chinese_Q355.
 - ◆ Affected programs:
 - ◊ ETABS v19.0.0, v19.0.1, v19.0.2
 - ◆ Affected API clients:
 - ◊ Compiled COM clients (e.g. VB6, Delphi) failed to start.
 - ◊ Interpreted COM clients (e.g. VBA) failed to compile/run if the affected enumeration was used.
 - ◆ Fix:
 - ◊ eMatTypeSteel_Chinese_Q345 enumeration was reinstated
- ETABS versions 19.0.0, 19.0.1, and 19.0.2 and corresponding ETABSV1.dll versions 1.10, 1.11, and 1.12 and CSiAPIv1.dll versions between 1.9 and 1.13 should not to be used for developing plug-ins and/or API scripts to ensure full compatibility with past and future API versions.

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